



Managing wine quality

Volume 2: Oenology and wine quality

Edited by Andrew G. Reynolds

Managing wine quality

Related titles:

Managing wine quality Volume 1: Viticulture and wine quality
(ISBN 978-1-84569-484-5)

Many aspects of both grape production and winemaking influence wine flavour, aroma, colour and mouthfeel. Factors affecting wine sensory attributes have not always been properly understood, limiting the ability of winemakers to manage these variables. In recent years, though, advances in research have provided new understanding of the scientific basis of quality variation in wine, promoting developments in viticulture and oenology. With chapters by leading experts, *Managing wine quality Volume 1: Viticulture and wine quality* reviews major wine quality attributes, such as colour and mouthfeel, the measurement of grape and wine properties and the effects of viticultural practices on the final product. The focus is on recent developments, advanced methods and likely future technologies.

Winemaking problems solved
(ISBN 978-1-84569-475-3)

Arranged in practical question-and-answer format, *Winemaking problems solved* provides brief, quickly accessible solutions to issues of frequent concern to winemaking professionals. The contributions, which are written by leading experts from industry and academia, span major aspects of the winemaking process from grape handling and fermentation processes to filtration, bottling and winery sanitation.

Brewing: Science and practice
(ISBN 978-1-85573-490-6)

Brewing: Science and practice updates and revises the previous work of this distinguished team of authors, producing the standard work in its field. The book covers all stages of brewing from raw materials, including the chemistry of hops and the biology of yeasts, through individual processes – such as mashing and wort separation – to packaging, storage and distribution. Key quality issues such as flavour and the chemical and physical properties of finished beers are discussed.

Details of these books and a complete list of Woodhead titles can be obtained by:

- visiting our web site at www.woodheadpublishing.com
- contacting Customer Services (e-mail: sales@woodheadpublishing.com; fax: +44 (0) 1223 893694; tel.: +44 (0) 1223 891358 ext. 130; address: Woodhead Publishing Limited, Abington Hall, Granta Park, Great Abington, Cambridge CB21 6AH, UK)

Woodhead Publishing Series in Food Science, Technology and Nutrition:
Number 192

Managing wine quality

Volume 2: Oenology and wine quality

Edited by
Andrew G. Reynolds



Oxford Cambridge Philadelphia New Delhi

© Woodhead Publishing Limited, 2010

Published by Woodhead Publishing Limited, Abington Hall, Granta Park,
Great Abington, Cambridge CB21 6AH, UK
www.woodheadpublishing.com

Woodhead Publishing, 525 South 4th Street #241, Philadelphia, PA 19147, USA

Woodhead Publishing India Private Limited, G-2, Vardaan House, 7/28 Ansari Road,
Daryaganj, New Delhi – 110002, India
www.woodheadpublishingindia.com

First published 2010, Woodhead Publishing Limited

© Woodhead Publishing Limited, 2010

The authors have asserted their moral rights.

This book contains information obtained from authentic and highly regarded sources. Reprinted material is quoted with permission, and sources are indicated. Reasonable efforts have been made to publish reliable data and information, but the authors and the publisher cannot assume responsibility for the validity of all materials. Neither the authors nor the publisher, nor anyone else associated with this publication, shall be liable for any loss, damage or liability directly or indirectly caused or alleged to be caused by this book.

Neither this book nor any part may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, microfilming and recording, or by any information storage or retrieval system, without permission in writing from Woodhead Publishing Limited.

The consent of Woodhead Publishing Limited does not extend to copying for general distribution, for promotion, for creating new works, or for resale. Specific permission must be obtained in writing from Woodhead Publishing Limited for such copying.

Trademark notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation, without intent to infringe.

British Library Cataloguing in Publication Data

A catalogue record for this book is available from the British Library.

ISBN 978-1-84569-798-3 (print)

ISBN 978-1-84569-998-7 (online)

ISSN 2042-8049 Woodhead Publishing Series in Food Science, Technology and Nutrition (print)

ISSN 2042-8057 Woodhead Publishing Series in Food Science, Technology and Nutrition (online)

The publisher's policy is to use permanent paper from mills that operate a sustainable forestry policy, and which has been manufactured from pulp which is processed using acid-free and elemental chlorine-free practices. Furthermore, the publisher ensures that the text paper and cover board used have met acceptable environmental accreditation standards.

Typeset by Ann Buchan (Typesetters), Middlesex, UK

Printed by TJI Digital, Padstow, Cornwall, UK

Contents

Contributor contact details xiii

Woodhead Publishing Series in Food Science, Technology and Nutrition . . xvii

Preface xxv

Part I Winemaking technologies and wine quality

1 Yeast fermentation management for improved wine quality 3

G. Specht, Lallemant, USA

1.1	Introduction	3
1.2	Yeast and fermentation management and wine quality	4
1.3	Yeast rehydration and handling	5
1.4	Yeast inoculation	8
1.5	Yeast inoculation rate	8
1.6	Yeast inoculation timing	9
1.7	Sequential yeast inoculation strategies	10
1.8	Yeast storage	10
1.9	Nutrient strategies	11
1.10	Difficult fermentation conditions	19
1.11	Sulphur compounds and their management	21
1.12	Preventing stuck and sluggish fermentations	23
1.13	Restarting stuck and sluggish fermentations	27
1.14	Conclusion	29
1.15	Acknowledgements	29
1.16	References	29

2	Metabolic engineering of wine yeast and advances in yeast selection methods for improved wine quality	34
	<i>B. Divol and F. F. Bauer, Stellenbosch University, South Africa</i>	
2.1	Introduction	34
2.2	Improving wine yeasts: current targets	37
2.3	A systems biology approach to wine yeast studies	41
2.4	Biotechnology, systems biology and the generation of new yeast strains	44
2.5	Molecular biology and systems biology in the identification of wine yeasts	47
2.6	Future trends	51
2.7	References	52
3	Effects of malolactic fermentation on wine quality	60
	<i>A. Lonvaud-Funel, University Victor Segalen Bordeaux II, France</i>	
3.1	Introduction	60
3.2	Spontaneous growth of lactic acid bacteria in wine	61
3.3	Variations in the diversity of lactic acid bacteria species during winemaking	71
3.4	Lactic acid bacteria and improving wine quality	74
3.5	Lactic acid bacteria and wine spoilage: undesirable lactic acid bacteria strains	79
3.6	Controlling malolactic fermentation by malolactic starters	85
3.7	Conclusions and future trends	88
3.8	References	89
4	Enzymes and wine quality	93
	<i>R.-M. Canal-Llaubères, Novozymes, France</i>	
4.1	Introduction	93
4.2	Definitions and production methods	94
4.3	Regulatory aspects	98
4.4	Enzyme applications in winemaking	99
4.5	Advances in enzyme discovery	104
4.6	Enzyme use in pre-fermentation stages	109
4.7	Enzyme use in post-fermentation stages	118
4.8	Monitoring enzyme performance	124
4.9	Future trends	126
4.10	Conclusions	127
4.11	Sources of further information	128
4.12	Acknowledgements	128
4.13	References	128

5	Membrane and other techniques for the management of wine composition	133
	<i>D. Wollan, Memstar Pty Ltd, Australia</i>	
5.1	Introduction	133
5.2	Some caveats	134
5.3	Some perspective – convention and intervention	134
5.4	Next-generation tools – phase change techniques	135
5.5	Membrane separation techniques	139
5.6	Membrane separation treatment and recombination	145
5.7	Volatile acidity removal	146
5.8	The problem of excess alcohol	148
5.9	Taint removal	154
5.10	Ultrafiltration	157
5.11	Electrodialysis	158
5.12	References	162
6	Ageing on lees (<i>sur lies</i>) and the use of speciality inactive yeasts during wine fermentation	164
	<i>C. Charpentier, Université de Bourgogne, France</i>	
6.1	Introduction	164
6.2	Definition and composition of lees	164
6.3	Yeast autolysis	166
6.4	Ageing of white wines on lees	170
6.5	Ageing of red wines on lees	172
6.6	Ageing of sparkling wines	176
6.7	Removal of undesirable compounds from wine	178
6.8	Yeast specialities mimicking lees	180
6.9	Conclusions	182
6.10	References	182
7	New directions in stabilization, clarification and fining of white wines	188
	<i>R. Marchal, University of Reims, France; and E. J. Waters, The Australian Wine Research Institute, Australia</i>	
7.1	Introduction	188
7.2	White wines, proteins and haze	189
7.3	The origin of wine proteins	190
7.4	Characterization of wine proteins	191
7.5	Protein levels in white wines	193
7.6	Protein haze formation in wine	197
7.7	Bentonite fining	198
7.8	Use of gelatine in white wine fining	206
7.9	Wine fining with plant proteins	207
7.10	Must clarification using the flotation technique	209

7.11	Other fining agents	214
7.12	Equipment for the addition of fining agents to wine	216
7.13	Wine fining: general conclusion and practical recommendations	216
7.14	Acknowledgements	217
7.15	References	217
8	Micro-oxygenation, oak alternatives and added tannins and wine quality	226
	<i>W. J. du Toit, Stellenbosch University, South Africa</i>	
8.1	Introduction	226
8.2	Basic oxidation reactions and substrates of oxidation in wine	227
8.3	Basic phenolic reactions in red wine involving oxygen	228
8.4	When does oxygen come into contact with wine?	229
8.5	Micro-oxygenation	230
8.6	Recent micro-oxygenation research at the Department of Viticulture and Oenology, Stellenbosch University	235
8.7	A few recommendations when using micro-oxygenation	243
8.8	Alternative oak treatments	246
8.9	Exogenous tannins in winemaking	249
8.10	Future trends	251
8.11	References	251
9	Alternatives to cork in wine bottle closures	255
	<i>J. Goode, www.wineanorak.com, UK</i>	
9.1	Introduction	255
9.2	The key property of closures: oxygen transmission	256
9.3	The various closure types	259
9.4	Conclusions and future trends	269
9.5	References	269
10	Current issues in organic winemaking: consumer expectations, producer attitudes and oenological innovation	271
	<i>D. Rauhut, Forschungsanstalt Geisenheim (Geisenheim Research Center (GRC)), Germany; and C. Micheloni, AIAB – Italian Association for Organic Agriculture, Italy</i>	
10.1	Introduction	271
10.2	Organic wine: a synthesis attempt	276
10.3	Harmonisation process	286
10.4	Future trends	287
10.5	Acknowledgements	288
10.6	References	288

Part II Managing sensory quality

11 Yeast selection for wine flavour modulation	293
<i>P. Marullo, SARCO Laffort Inc., France; and D. Dubourdieu, INRA-Université Bordeaux, France</i>	
11.1 Introduction	293
11.2 Key issues in efficient wine yeast selection	294
11.3 Selection of natural yeast isolates: methods and limits	295
11.4 Metabolic engineering	296
11.5 Conventional genetic strategies	298
11.6 Mixed cultures as an alternative strategy	303
11.7 Yeast by-products affecting wine aromas: glycerol	304
11.8 Yeast by-products affecting wine aromas: acetic acid	307
11.9 Yeast by-products affecting wine aromas: hydrogen sulphide	309
11.10 Yeast by-products affecting wine aromas: higher alcohols	313
11.11 Yeast by-products affecting wine aromas: esters	318
11.12 Varietal aromas resulting from grape precursor biotransformation	323
11.13 Conclusions and future trends	333
11.14 References	334
12 <i>Brettanomyces/Dekkera</i> off-flavours and other wine faults associated with microbial spoilage	346
<i>L. Conterno, Fondazione E. Mach – IASMA Research and Innovation Centre, Italy; and T. Henick-Kling, Washington State University, USA</i>	
12.1 Introduction	346
12.2 <i>Brettanomyces/Dekkera</i> off-flavours and their related metabolism	349
12.3 <i>Brettanomyces/Dekkera</i> taxonomy and phylogenetic relationships with other wine yeasts	352
12.4 <i>Brettanomyces/Dekkera</i> physiology	355
12.5 Other defects associated with the presence of <i>Brettanomyces/Dekkera</i>	358
12.6 Other faults associated with microbial spoilage	361
12.7 Detection and methods to prevent and defeat microbial spoilage faults	367
12.8 Conclusions	372
12.9 References	372
13 Reducing cork taint in wine	388
<i>R. Jung and V. Schaefer, Forschungsanstalt Geisenheim (Geisenheim Research Center (GRC)), Germany</i>	
13.1 Introduction: cork taint	388
13.2 Compounds causing musty–mouldy off-flavours	389

13.3	Quality management and control methods for wine corks: introduction	394
13.4	Test procedures to evaluate the quality of cork stoppers	396
13.5	Standard test procedures	397
13.6	Additional test procedures	401
13.7	Handling and processing of corks and bottles during bottling and storage	402
13.8	Prevention of musty–mouldy off-flavours in the cellar environment	405
13.9	Methods to reduce musty off-flavours in contaminated wines . .	410
13.10	References	414
14	Ladybug (Coccinellidae) taint in wine	418
	<i>A. Botezatu and G. Pickering, Brock University, Canada</i>	
14.1	Introduction	418
14.2	Quality implications	419
14.3	Causal compound(s)	421
14.4	Threshold and tolerances	424
14.5	Other Coccinellidae species	424
14.6	Post-harvest prevention and remediation	425
14.7	Conclusions and future trends	428
14.8	Abbreviations	428
14.9	References	429
15	Understanding and controlling non-enzymatic wine oxidation	432
	<i>P. A. Kilmartin, The University of Auckland, New Zealand</i>	
15.1	Introduction	432
15.2	Oxygen in wine	433
15.3	Polyphenol oxidation	434
15.4	Oxidation of aroma compounds	438
15.5	Measures of wine oxidation status	440
15.6	White wine oxidation	442
15.7	Red wine oxidation	444
15.8	Influence of wine antioxidants	446
15.9	Conclusion	450
15.10	References	451
16	Ageing and flavour deterioration in wine	459
	<i>A. W. Linsenmeier, D. Rauhut and W. R. Sponholz, Forschungs- anstalt Geisenheim (Geisenheim Research Center (GRC)), Germany</i>	
16.1	Introduction: ageing	459
16.2	Sensory changes during storage/ageing	460
16.3	Aromatic compounds related to flavour deterioration	461

16.4	Chemical reactions of ageing	469
16.5	Factors influencing the ageing process and future trends in research	473
16.6	Untypical ageing (UTA) off-flavour	476
16.7	References	486
17	Biogenic amines and the winemaking process	494
	<i>M. V. Moreno-Arribas, Institute of Industrial Fermentations (CSIC), Spain; and A. Y. Smit and M. du Toit, Stellenbosch University, South Africa</i>	
17.1	Introduction	494
17.2	Incidence of biogenic amines in wines and health effects	495
17.3	Formation of biogenic amines during the winemaking process .	502
17.4	Methods of detection and quantification of biogenic amines in wines	511
17.5	Methods and tools to prevent the presence of biogenic amines in wines	513
17.6	Future trends	514
17.7	References	515
18	Managing the quality of icewines	523
	<i>A. J. Bowen, Brock University, Canada</i>	
18.1	Introduction	523
18.2	Definitions of icewine	524
18.3	Viticulture	526
18.4	Harvest considerations	531
18.5	Oenology	534
18.6	Chemical analysis of icewines	540
18.7	Sensory properties of icewine	545
18.8	Authentication	548
18.9	Future trends	549
18.10	References	550
19	Managing the quality of sparkling wines	553
	<i>S. Buxaderas and E. López-Tamames, University of Barcelona, Spain</i>	
19.1	Types of sparkling wines: definitions and characteristics	553
19.2	Description of the organoleptic characteristics of sparkling wines	554
19.3	Factors affecting sensory quality	570
19.4	Quality control	578
19.5	Conclusions	580
19.6	References	582

20 Extraction technologies and wine quality	589
<i>A. Razungles, Montpellier SupAgro, France</i>	
20.1 Introduction	589
20.2 Extraction factors	592
20.3 Techniques applied to white wine vinification	596
20.4 Techniques applied to rosé wine vinification	598
20.5 Techniques applied to traditional red wine vinification	599
20.6 Very hot, short maceration applied to red wine vinification	609
20.7 Vinification of red wines by carbonic maceration	617
20.8 Traditional vinification of red wines with whole berries	620
20.9 Draining and pressing	621
20.10 Conclusion	625
20.11 References	625
<i>Index</i>	<i>631</i>

Contributor contact details

(* = main contact)

Editor

Dr Andrew G. Reynolds
500 Glenridge Avenue
Brock University
St Catharines
Ontario
LS2 3A1
Canada

Email: areynolds@brocku.ca

Chapter 1

G. Specht
Lallemand
P.O. Box 5512
Petaluma
California 94955
USA

Email: gspecht@lallemand.com

Chapter 2

B. Divol* and F. F. Bauer
Institute for Wine Biotechnology
Stellenbosch University
Private Bag X1

Matieland 7602
South Africa

Email: divol@sun.ac.za

Chapter 3

A. Lonvaud-Funel,
UMR Oenologie- ISVV
University Victor Segalen
Bordeaux 2.
CS 50008.
33882 Villenave D'Ornon Cedex
France

Email: aline.lonvaud@u-bordeaux2.fr

Chapter 4

Dr R.-M. Canal-Llaubères
Global Application Manager Wine
Novozymes France S.A. (Bordeaux
Cedex)
23, Parvis des Chartrons
33074 Bordeaux Cedex
France

Email: rcl@novozymes.com

Chapter 5

D. Wollan
Memstar Pty Ltd
29 Dalgety Street
Oakleigh
Victoria 3166
Australia
Email: dwollan@memstar.com.au

Chapter 6

C. Charpentier
UMR Santé de la Vigne et Qualité du
Vin
University of Burgundy
Campus universitaire
BP 27877
21078 Dijon Cedex
France
Email: claudine.charpentier@u-bourgogne.fr

Chapter 7

R. Marchal*
Laboratory of Enology and Applied
Chemistry
University of Reims
BP 1039
51687 Reims Cedex 02
France
Email: richard.marchal@univ-reims.fr

E. J. Waters
The Australian Wine Research
Institute
P.O. Box 197
Glen Osmond
South Australia 5064
Australia

Chapter 8

W. J. du Toit
Department of Viticulture and
Oenology
Stellenbosch University
Private Bag X1
Matieland 7602
South Africa
Email: wdutoit@sun.ac.za

Chapter 9

Jamie Goode
6 The Green
High Street
Feltham
TW13 4AF
UK
Email: jamie@wineanorak.com

Chapter 10

D. Rauhut*
Forschungsanstalt Geisenheim
(Geisenheim Research Center
(GRC))
Section of Microbiology and Bio-
chemistry
Von-Lade Straße 1
D-65366 Geisenheim
Germany
Email: Doris.Rauhut@fa-gm.de

C. Micheloni
AIAB – Italian Association for
Organic Agriculture
via Piave 14
I-00187 Rome
Italy
Email: c.micheloni@aiab.it

Chapter 11

P. Marullo*

LAFFORT

BP 40

F-33 072 Bordeaux

France

Email: philippe.marullo@
u-bordeaux2.fr

D. Dubourdieu

Institut des Sciences de la Vigne et du
Vin

UMR Oenologie; INRA-Université
Bordeaux 2,

210 Chemin de Leysotte CS 50008

F-33 882 Villenave d'Ornon

France

Chapter 12

L. Conterno*

Fondazione E. Mach

IASMA Research and Innovation
Centre

Food Quality and Nutrition

Via E. Mach 1

38010 San Michele a/Adige (TN)

Italy

Email: lorenza.conterno@iasma.it

T. Henick-Kling

Viticulture and Enology Program

Washington State University

2710 University Drive

Richland

WA 99354

USA

Email: thk@wsu.edu

Chapter 13

R. Jung and V. Schaefer*

Forschungsanstalt Geisenheim

(Geisenheim Research Center
(GRC))

Section of Oenology and Wine
Technology

Blaubachstraße 19

D-65366 Geisenheim

Germany

Email: V.Schaefer@fa-gm.de;
r.jung@fa-gm.de

Chapter 14

A. Botezatu and G. Pickering*

Brock University

500 Glenridge Avenue

St Catharines

Ontario

LS2 3A1

Canada

Email: ab08dj@brocku.ca;
gpickering@brocku.ca

Chapter 15

P. A. Kilmartin

Wine Science Programme

The University of Auckland

Private Bag 92019

Auckland

New Zealand

Email: p.kilmartin@auckland.ac.nz

Chapter 16

A. W. Linsenmeier*
Section of Soil Science and Plant
Nutrition
Forschungsanstalt Geisenheim
(Geisenheim Research Center
(GRC))
Von-Lade Straße 1
D-65366 Geisenheim
Germany

Email: linsenmeier@fa-gm.de

D. Rauhut and W. R. Sponholz
Section of Microbiology and Bio-
chemistry
Forschungsanstalt Geisenheim
Von-Lade Straße 1
D-65366 Geisenheim
Germany

Chapter 17

M. V. Moreno-Arribas*
Institute of Industrial Fermentations
(CSIC)
Juan de la Cierva 3
28006 Madrid
Spain

Email: mvmoreno@ifi.csic.es

A. Y. Smit and M. du Toit
Institute for Wine Biotechnology
Stellenbosch University
Private Bag X1
Matieland 7602
South Africa

Email: aysmit@sun.ac.za;
mdt@sun.ac.za

Chapter 18

A. J. Bowen
Cool Climate Oenology and Viticul-
ture Institute
Brock University
500 Glenridge Avenue
St Catharines
Ontario
LS2 3A1
Canada

Email: amyjbowen@hotmail.com

Chapter 19

S. Buxaderas* and E. López-
Tamames
Department of Nutrition and Food
Science
University of Barcelona
Faculty of Pharmacy
Av. Joan XXIII, s/n
08028 Barcelona
Spain

Email: susanabuxaderas@ub.edu

Chapter 20

A. Razungles
Montpellier SupAgro
Institut des Hautes Etudes de la Vigne
et du Vin
UMR Sciences pour l'Oenologie
2 Place Pierre Viala
34060 Montpellier Cédex 1
France

Email: razungle@supagro.inra.fr

Woodhead Publishing Series in Food Science, Technology and Nutrition

- 1 **Chilled foods: a comprehensive guide** *Edited by C. Dennis and M. Stringer*
- 2 **Yoghurt: science and technology** *A. Y. Tamime and R. K. Robinson*
- 3 **Food processing technology: principles and practice** *P. J. Fellows*
- 4 **Bender's dictionary of nutrition and food technology Sixth edition** *D. A. Bender*
- 5 **Determination of veterinary residues in food** *Edited by N. T. Crosby*
- 6 **Food contaminants: sources and surveillance** *Edited by C. Creaser and R. Purchase*
- 7 **Nitrates and nitrites in food and water** *Edited by M. J. Hill*
- 8 **Pesticide chemistry and bioscience: the food-environment challenge** *Edited by G. T. Brooks and T. Roberts*
- 9 **Pesticides: developments, impacts and controls** *Edited by G. A. Best and A. D. Ruthven*
- 10 **Dietary fibre: chemical and biological aspects** *Edited by D. A. T. Southgate, K. W. Waldron, I. T. Johnson and G. R. Fenwick*
- 11 **Vitamins and minerals in health and nutrition** *M. Tolonen*
- 12 **Technology of biscuits, crackers and cookies Second edition** *D. Manley*
- 13 **Instrumentation and sensors for the food industry** *Edited by E. Kress-Rogers*
- 14 **Food and cancer prevention: chemical and biological aspects** *Edited by K. W. Waldron, I. T. Johnson and G. R. Fenwick*
- 15 **Food colloids: proteins, lipids and polysaccharides** *Edited by E. Dickinson and B. Bergenstahl*
- 16 **Food emulsions and foams** *Edited by E. Dickinson*
- 17 **Maillard reactions in chemistry, food and health** *Edited by T. P. Labuza, V. Monnier, J. Baynes and J. O'Brien*
- 18 **The Maillard reaction in foods and medicine** *Edited by J. O'Brien, H. E. Nursten, M. J. Crabbe and J. M. Ames*
- 19 **Encapsulation and controlled release** *Edited by D. R. Karsa and R. A. Stephenson*
- 20 **Flavours and fragrances** *Edited by A. D. Swift*
- 21 **Feta and related cheeses** *Edited by A. Y. Tamime and R. K. Robinson*
- 22 **Biochemistry of milk products** *Edited by A. T. Andrews and J. R. Varley*
- 23 **Physical properties of foods and food processing systems** *M. J. Lewis*

- 24 **Food irradiation: a reference guide** *V. M. Wilkinson and G. Gould*
- 25 **Kent's technology of cereals: an introduction for students of food science and agriculture** Fourth edition *N. L. Kent and A. D. Evers*
- 26 **Biosensors for food analysis** Edited by *A. O. Scott*
- 27 **Separation processes in the food and biotechnology industries: principles and applications** Edited by *A.S. Grandison and M. J. Lewis*
- 28 **Handbook of indices of food quality and authenticity** *R.S. Singhal, P. K. Kulkarni and D. V. Rege*
- 29 **Principles and practices for the safe processing of foods** *D. A. Shapton and N. F. Shapton*
- 30 **Biscuit, cookie and cracker manufacturing manuals Volume 1: ingredients** *D. Manley*
- 31 **Biscuit, cookie and cracker manufacturing manuals Volume 2: biscuit doughs** *D. Manley*
- 32 **Biscuit, cookie and cracker manufacturing manuals Volume 3: biscuit dough piece forming** *D. Manley*
- 33 **Biscuit, cookie and cracker manufacturing manuals Volume 4: baking and cooling of biscuits** *D. Manley*
- 34 **Biscuit, cookie and cracker manufacturing manuals Volume 5: secondary processing in biscuit manufacturing** *D. Manley*
- 35 **Biscuit, cookie and cracker manufacturing manuals Volume 6: biscuit packaging and storage** *D. Manley*
- 36 **Practical dehydration** Second edition *M. Greensmith*
- 37 **Lawrie's meat science** Sixth edition *R. A. Lawrie*
- 38 **Yoghurt: science and technology** Second edition *A. Y Tamime and R. K. Robinson*
- 39 **New ingredients in food processing: biochemistry and agriculture** *G. Linden and D. Lorient*
- 40 **Benders' dictionary of nutrition and food technology** Seventh edition *D A Bender and A. E. Bender*
- 41 **Technology of biscuits, crackers and cookies** Third edition *D. Manley*
- 42 **Food processing technology: principles and practice** Second edition *P. J. Fellows*
- 43 **Managing frozen foods** Edited by *C. J. Kennedy*
- 44 **Handbook of hydrocolloids** Edited by *G. O. Phillips and P. A. Williams*
- 45 **Food labelling** Edited by *J. R. Blanchfield*
- 46 **Cereal biotechnology** Edited by *P. C. Morris and J. H. Bryce*
- 47 **Food intolerance and the food industry** Edited by *T. Dean*
- 48 **The stability and shelf-life of food** Edited by *D. Kilcast and P. Subramaniam*
- 49 **Functional foods: concept to product** Edited by *G. R. Gibson and C. M. Williams*
- 50 **Chilled foods: a comprehensive guide** Second edition Edited by *M. Stringer and C. Dennis*
- 51 **HACCP in the meat industry** Edited by *M. Brown*
- 52 **Biscuit, cracker and cookie recipes for the food industry** *D. Manley*
- 53 **Cereals processing technology** Edited by *G. Owens*
- 54 **Baking problems solved** *S. P. Cauvain and L. S. Young*
- 55 **Thermal technologies in food processing** Edited by *P. Richardson*
- 56 **Frying: improving quality** Edited by *J. B. Rossell*
- 57 **Food chemical safety Volume 1: contaminants** Edited by *D. Watson*
- 58 **Making the most of HACCP: learning from others' experience** Edited by *T. Mayes and S. Mortimore*
- 59 **Food process modelling** Edited by *L. M. M. Tijskens, M. L. A. T. M. Hertog and B. M. Nicolai*
- 60 **EU food law: a practical guide** Edited by *K. Goodburn*
- 61 **Extrusion cooking: technologies and applications** Edited by *R. Guy*

- 62 **Auditing in the food industry: from safety and quality to environmental and other audits** Edited by M. Dillon and C. Griffith
- 63 **Handbook of herbs and spices Volume 1** Edited by K. V. Peter
- 64 **Food product development: maximising success** M. Earle, R. Earle and A. Anderson
- 65 **Instrumentation and sensors for the food industry Second edition** Edited by E. Kress-Rogers and C. J. B. Brimelow
- 66 **Food chemical safety Volume 2: additives** Edited by D. Watson
- 67 **Fruit and vegetable biotechnology** Edited by V. Valpuesta
- 68 **Foodborne pathogens: hazards, risk analysis and control** Edited by C. de W. Blackburn and P. J. McClure
- 69 **Meat refrigeration** S. J. James and C. James
- 70 **Lockhart and Wiseman's crop husbandry Eighth edition** H. J. S. Finch, A. M. Samuel and G. P. F. Lane
- 71 **Safety and quality issues in fish processing** Edited by H. A. Bremner
- 72 **Minimal processing technologies in the food industries** Edited by T. Ohlsson and N. Bengtsson
- 73 **Fruit and vegetable processing: improving quality** Edited by W. Jongen
- 74 **The nutrition handbook for food processors** Edited by C. J. K. Henry and C. Chapman
- 75 **Colour in food: improving quality** Edited by D MacDougall
- 76 **Meat processing: improving quality** Edited by J. P. Kerry, J. F. Kerry and D. A. Ledward
- 77 **Microbiological risk assessment in food processing** Edited by M. Brown and M. Stringer
- 78 **Performance functional foods** Edited by D. Watson
- 79 **Functional dairy products Volume 1** Edited by T. Mattila-Sandholm and M. Saarela
- 80 **Taints and off-flavours in foods** Edited by B. Baigrie
- 81 **Yeasts in food** Edited by T. Boekhout and V. Robert
- 82 **Phytochemical functional foods** Edited by I. T. Johnson and G. Williamson
- 83 **Novel food packaging techniques** Edited by R. Ahvenainen
- 84 **Detecting pathogens in food** Edited by T. A. McMeekin
- 85 **Natural antimicrobials for the minimal processing of foods** Edited by S. Roller
- 86 **Texture in food Volume 1: semi-solid foods** Edited by B. M. McKenna
- 87 **Dairy processing: improving quality** Edited by G Smit
- 88 **Hygiene in food processing: principles and practice** Edited by H. L. M. Lelieveld, M. A. Mostert, B. White and J. Holah
- 89 **Rapid and on-line instrumentation for food quality assurance** Edited by I. Tothill
- 90 **Sausage manufacture: principles and practice** E. Essien
- 91 **Environmentally-friendly food processing** Edited by B. Mattsson and U. Sonesson
- 92 **Bread making: improving quality** Edited by S. P. Cauvain
- 93 **Food preservation techniques** Edited by P. Zeuthen and L. Bøgh-Sørensen
- 94 **Food authenticity and traceability** Edited by M. Lees
- 95 **Analytical methods for food additives** R. Wood, L. Foster, A. Damant and P. Key
- 96 **Handbook of herbs and spices Volume 2** Edited by K. V. Peter
- 97 **Texture in food Volume 2: solid foods** Edited by D. Kilcast
- 98 **Proteins in food processing** Edited by R. Yada
- 99 **Detecting foreign bodies in food** Edited by M. Edwards
- 100 **Understanding and measuring the shelf-life of food** Edited by R. Steele
- 101 **Poultry meat processing and quality** Edited by G. Mead
- 102 **Functional foods, ageing and degenerative disease** Edited by C. Remacle and B. Reusens

- 103 **Mycotoxins in food: detection and control** Edited by N. Magan and M. Olsen
- 104 **Improving the thermal processing of foods** Edited by P. Richardson
- 105 **Pesticide, veterinary and other residues in food** Edited by D. Watson
- 106 **Starch in food: structure, functions and applications** Edited by A-C Eliasson
- 107 **Functional foods, cardiovascular disease and diabetes** Edited by A. Arnoldi
- 108 **Brewing: science and practice** D. E. Briggs, P. A. Brookes, R. Stevens and C. A. Boulton
- 109 **Using cereal science and technology for the benefit of consumers: proceedings of the 12th International ICC Cereal and Bread Congress, 24-26th May, 2004, Harrogate, UK** Edited by S. P. Cauvain, L. S. Young and S. Salmon
- 110 **Improving the safety of fresh meat** Edited by J. Sofos
- 111 **Understanding pathogen behaviour in food: virulence, stress response and resistance** Edited by M. Griffiths
- 112 **The microwave processing of foods** Edited by H. Schubert and M. Regier
- 113 **Food safety control in the poultry industry** Edited by G. Mead
- 114 **Improving the safety of fresh fruit and vegetables** Edited by W. Jongen
- 115 **Food, diet and obesity** Edited by D. Mela
- 116 **Handbook of hygiene control in the food industry** Edited by H. L. M. Lelieveld, M. A. Mostert and J. Holah
- 117 **Detecting allergens in food** Edited by S. Koppelman and S. Hefle
- 118 **Improving the fat content of foods** Edited by C. Williams and J. Buttriss
- 119 **Improving traceability in food processing and distribution** Edited by I. Smith and A. Furness
- 120 **Flavour in food** Edited by A. Voilley and P. Etievant
- 121 **The Chorleywood bread process** S. P. Cauvain and L. S. Young
- 122 **Food spoilage microorganisms** Edited by C. de W. Blackburn
- 123 **Emerging foodborne pathogens** Edited by Y. Motarjemi and M. Adams
- 124 **Benders' dictionary of nutrition and food technology Eighth edition** D. A. Bender
- 125 **Optimising sweet taste in foods** Edited by W. J. Spillane
- 126 **Brewing: new technologies** Edited by C. Bamforth
- 127 **Handbook of herbs and spices Volume 3** Edited by K. V. Peter
- 128 **Lawrie's meat science Seventh edition** R. A. Lawrie in collaboration with D. A. Ledward
- 129 **Modifying lipids for use in food** Edited by F. Gunstone
- 130 **Meat products handbook: practical science and technology** G. Feiner
- 131 **Food consumption and disease risk: consumer-pathogen interactions** Edited by M. Potter
- 132 **Acrylamide and other hazardous compounds in heat-treated foods** Edited by K. Skog and J. Alexander
- 133 **Managing allergens in food** Edited by C. Mills, H. Wichers and K. Hoffman-Sommergruber
- 134 **Microbiological analysis of red meat, poultry and eggs** Edited by G. Mead
- 135 **Maximising the value of marine by-products** Edited by F. Shahidi
- 136 **Chemical migration and food contact materials** Edited by K. Barnes, R. Sinclair and D. Watson
- 137 **Understanding consumers of food products** Edited by L. Frewer and H. van Trijp
- 138 **Reducing salt in foods: practical strategies** Edited by D. Kilcast and F. Angus
- 139 **Modelling microorganisms in food** Edited by S. Brul, S. Van Gerwen and M. Zwietering
- 140 **Tamime and Robinson's Yoghurt: science and technology Third edition** A. Y. Tamime and R. K. Robinson
- 141 **Handbook of waste management and co-product recovery in food processing: Volume 1** Edited by K. W. Waldron

- 142 **Improving the flavour of cheese** *Edited by B. Weimer*
- 143 **Novel food ingredients for weight control** *Edited by C. J. K. Henry*
- 144 **Consumer-led food product development** *Edited by H. MacFie*
- 145 **Functional dairy products Volume 2** *Edited by M. Saarela*
- 146 **Modifying flavour in food** *Edited by A. J. Taylor and J. Hort*
- 147 **Cheese problems solved** *Edited by P. L. H. McSweeney*
- 148 **Handbook of organic food safety and quality** *Edited by J. Cooper, C. Leifert and U. Niggli*
- 149 **Understanding and controlling the microstructure of complex foods** *Edited by D. J. McClements*
- 150 **Novel enzyme technology for food applications** *Edited by R. Rastall*
- 151 **Food preservation by pulsed electric fields: from research to application** *Edited by H. L. M. Lelieveld and S. W. H. de Haan*
- 152 **Technology of functional cereal products** *Edited by B. R. Hamaker*
- 153 **Case studies in food product development** *Edited by M. Earle and R. Earle*
- 154 **Delivery and controlled release of bioactives in foods and nutraceuticals** *Edited by N. Garti*
- 155 **Fruit and vegetable flavour: recent advances and future prospects** *Edited by B. Brückner and S. G. Wyllie*
- 156 **Food fortification and supplementation: technological, safety and regulatory aspects** *Edited by P. Berry Ottaway*
- 157 **Improving the health-promoting properties of fruit and vegetable products** *Edited by F. A. Tomás-Barberán and M. I. Gil*
- 158 **Improving seafood products for the consumer** *Edited by T. Børresen*
- 159 **In-pack processed foods: improving quality** *Edited by P. Richardson*
- 160 **Handbook of water and energy management in food processing** *Edited by J. Klemeš, R. Smith and J-K Kim*
- 161 **Environmentally compatible food packaging** *Edited by E. Chiellini*
- 162 **Improving farmed fish quality and safety** *Edited by Ø. Lie*
- 163 **Carbohydrate-active enzymes** *Edited by K-H Park*
- 164 **Chilled foods: a comprehensive guide Third edition** *Edited by M. Brown*
- 165 **Food for the ageing population** *Edited by M. M. Raats, C. P. G. M. de Groot and W. A Van Staveren*
- 166 **Improving the sensory and nutritional quality of fresh meat** *Edited by J. P. Kerry and D. A. Ledward*
- 167 **Shellfish safety and quality** *Edited by S. E. Shumway and G. E. Rodrick*
- 168 **Functional and speciality beverage technology** *Edited by P. Paquin*
- 169 **Functional foods: principles and technology** *M. Guo*
- 170 **Endocrine-disrupting chemicals in food** *Edited by I. Shaw*
- 171 **Meals in science and practice: interdisciplinary research and business applications** *Edited by H. L. Meiselman*
- 172 **Food constituents and oral health: current status and future prospects** *Edited by M. Wilson*
- 173 **Handbook of hydrocolloids Second edition** *Edited by G. O. Phillips and P. A. Williams*
- 174 **Food processing technology: principles and practice Third edition** *P. J. Fellows*
- 175 **Science and technology of enrobed and filled chocolate, confectionery and bakery products** *Edited by G. Talbot*
- 176 **Foodborne pathogens: hazards, risk analysis and control Second edition** *Edited by C. de W. Blackburn and P. J. McClure*
- 177 **Designing functional foods: measuring and controlling food structure breakdown and absorption** *Edited by D. J. McClements and E. A. Decker*
- 178 **New technologies in aquaculture: improving production efficiency, quality and environmental management** *Edited by G. Burnell and G. Allan*

- 179 **More baking problems solved** *S. P. Cauvain and L. S. Young*
- 180 **Soft drink and fruit juice problems solved** *P. Ashurst and R. Hargitt*
- 181 **Biofilms in the food and beverage industries** *Edited by P. M. Fratamico, B. A. Annous and N. W. Gunther*
- 182 **Dairy-derived ingredients: food and nutraceutical uses** *Edited by M. Corredig*
- 183 **Handbook of waste management and co-product recovery in food processing Volume 2** *Edited by K. W. Waldron*
- 184 **Innovations in food labelling** *Edited by J. Albert*
- 185 **Delivering performance in food supply chains** *Edited by C. Mena and G. Stevens*
- 186 **Chemical deterioration and physical instability of food and beverages** *Edited by L. H. Skibsted, J. Risbo and M. L. Andersen*
- 187 **Managing wine quality Volume 1: viticulture and wine quality** *Edited by A.G. Reynolds*
- 188 **Improving the safety and quality of milk Volume 1: milk production and processing** *Edited by M. Griffiths*
- 189 **Improving the safety and quality of milk Volume 2: improving quality in milk products** *Edited by M. Griffiths*
- 190 **Cereal grains: assessing and managing quality** *Edited by C. Wrigley and I. Batey*
- 191 **Sensory analysis for food and beverage quality control: a practical guide** *Edited by D. Kilcast*
- 192 **Managing wine quality Volume 2: oenology and wine quality** *Edited by A. G. Reynolds*
- 193 **Winemaking problems solved** *Edited by C. E. Butzke*
- 194 **Environmental assessment and management in the food industry** *Edited by U. Sonesson, J. Berlin and F. Ziegler*
- 195 **Consumer-driven innovation in food and personal care products** *Edited by S.R. Jaeger and H. MacFie*
- 196 **Tracing pathogens in the food chain** *Edited by S. Brul, P.M. Fratamico and T.A. McMeekin*
- 197 **Case studies in novel food processing technologies: innovations in processing, packaging, and predictive modelling** *Edited by C. J. Doona, K Kustin and F. E. Feeherry*
- 198 **Freeze-drying of pharmaceutical and food products** *T-C Hua, B-L Liu and H Zhang*
- 199 **Oxidation in foods and beverages and antioxidant applications: Volume 1 Understanding mechanisms of oxidation and antioxidant activity** *Edited by E. A. Decker, R. J. Elias and D. J. McClements*
- 200 **Oxidation in foods and beverages and antioxidant applications: Volume 2 Management in different industry sectors** *Edited by E. A. Decker, R. J. Elias and D. J. McClements*
- 201 **Protective cultures, antimicrobial metabolites and bacteriophages for food and beverage biopreservation** *Edited by C. Lacroix*
- 202 **Separation, extraction and concentration processes in the food, beverage and nutraceutical industries** *Edited by S. S. H. Rizvi*
- 203 **Determining mycotoxins and mycotoxigenic fungi in food and feed** *Edited by S. De Saeger*
- 204 **Developing children's food products** *Edited by D. Kilcast and F. Angus*
- 205 **Functional foods: concept to profit Second edition** *Edited by M. Saarela*
- 206 **Postharvest biology and technology of tropical and subtropical fruits Volume 1** *Edited by E. M. Yahia*
- 207 **Postharvest biology and technology of tropical and subtropical fruits Volume 2** *Edited by E. M. Yahia*

- 208 **Postharvest biology and technology of tropical and subtropical fruits Volume 3**
Edited by E. M. Yahia
- 209 **Postharvest biology and technology of tropical and subtropical fruits Volume 4**
Edited by E. M. Yahia
- 210 **Food and beverage stability and shelf-life** *Edited by D. Kilcast and P. Subramaniam*
- 211 **Processed Meats: improving safety, nutrition and quality** *Edited by J. P. Kerry and J. F. Kerry*
- 212 **Food chain integrity: a holistic approach to food traceability, authenticity, safety and bioterrorism prevention** *Edited by J. Hoorfar, K. Jordan, F. Butler and R. Prugger*
- 213 **Improving the safety and quality of eggs and egg products Volume 1** *Edited by Y. Nys, M. Bain and F. Van Immerseel*
- 214 **Improving the safety and quality of eggs and egg products Volume 2** *Edited by Y. Nys, M. Bain and F. Van Immerseel*
- 215 **Feed and fodder contamination: effects on livestock and food safety** *Edited by J. Fink-Gremmels*
- 216 **Hygiene in the design, construction and renovation of food processing factories** *Edited by H. L. M. Lelieveld and J. Holah*
- 217 **Technology of biscuits, crackers and cookies Fourth edition** *Edited by D. Manley*
- 218 **Nanotechnology in the food, beverage and nutraceutical industries** *Edited by Q. Huang*
- 219 **Rice quality** *K. R. Bhattacharya*
- 220 **Meat, poultry and seafood packaging** *Edited by J. P. Kerry*
- 221 **Reducing saturated fats in foods** *Edited by G. Talbot*
- 222 **Handbook of food proteins** *Edited by G. O. Phillips and P. A. Williams*
- 223 **Lifetime nutritional influences on cognition, behaviour and psychiatric illness**
Edited by D. Benton

Preface

Our current knowledge of the chemistry, microbiology and technology of winemaking is astounding. Volume 2 of *Managing wine quality* provides 20 chapters on the science of winemaking from leading experts around the world. We anticipate that this book, and its companion volume dealing with viticultural technologies, will provide a valuable resource for students, scholars and members of the wine industry. It is our opinion that these two volumes could form the basis for a senior level undergraduate or graduate level course in wine science. Volume 2 is divided into two sections: Part I, Winemaking technologies and wine quality; and Part II, Managing sensory quality. Part I contains microbiological topics such as yeast and fermentation management (Specht), metabolic engineering of wine yeasts (Divol and Bauer), and malolactic fermentation (Lonvaud-Funel). General wine technology subjects range from the use of oenological enzymes (Canal-Llaubères), membrane separation and recombination (Wollan), ageing on lees (Charpentier), stabilisation and clarification (Marchal and Waters), and the use of micro-oxygenation and oak alternatives (du Toit). Specialty topics such as alternative wine closures (Goode) and ‘low input’ (e.g. organic) approaches to winemaking (Rauhut and Micheloni) are included. Part II of this volume, dealing with managing wine sensory quality, includes some exciting new information on the impact of various oenological practices on aroma and flavour. Some examples are yeast selection for wine flavour modulation (Marullo and Dubourdieu), *Brettanomyces/Dekkera* off-flavours (Conterno and Henick-Kling), reduction of cork taints (Jung and Schaefer), ladybug taint (Botezatu and Pickering), non-enzymatic wine oxidation (Kilmartin), atypical ageing (Linsenmeier, Rauhut and Sponholz), and the avoidance of biogenic amine production during winemaking (Moreno-Arribas, Smit and du Toit) are also included. The reader will additionally find two chapters on specialty wine products – icewines (Bowen) and sparkling wines (Buxaderas and López-Tamames). The final chapter discusses the impact of specific technologies on wine aroma and flavour such as extraction processes (Razungles). Volume 2 and its companion volume represent the state of the art on winemaking science

and technology from the vineyard to the glass. I hope that the reader finds *Managing wine quality* both edifying and enjoyable, and that it will be considered a valuable resource for years to come.

Andrew G. Reynolds
Editor

Part I

Winemaking technologies and wine quality

1

Yeast fermentation management for improved wine quality

G. Specht, Lallemant, USA

Abstract: The yeast and alcohol fermentation management practices reviewed in this chapter are applicable for most alcohol fermentations that winemakers will encounter. The way the yeast is prepared at the yeast production facility and managed during rehydration and throughout the early phases of alcohol fermentation by the winemaker will have a substantial influence on the yeast's ability to conduct a problem-free fermentation and the resulting wine quality. In spite of improved winemaking technology, there is still much to learn about properly managing yeasts and conducting healthy stress-free fermentations as sluggish and stuck fermentations are still a universal problem.

Key words: yeast, stuck fermentations, yeast nutrition, alcohol fermentation, yeast inoculation.

1.1 Introduction

Kilgore Trout once wrote a short story which was a dialogue between two pieces of yeast. They were discussing the possible purposes of life as they ate sugar and suffocated in their own excrement. Because of their limited intelligence, they never came close to guessing that they were making Champagne.

(Kurt Vonnegut, *Breakfast of Champions*)

Alcoholic fermentation is the process by which a microorganism (yeast) converts sugar into alcohol and carbon dioxide gas. Most winemakers can appreciate that

yeast is not just some unnecessary ingredient but that it is a living organism whose role is so critical that without yeast there is no wine. These diverse single-celled organisms can be divided into 60 broad genera such as *Saccharomyces* and over 700 more tightly defined groups called species such as *cerevisiae*. Meaningful selection of yeast was begun about a century and a half ago by Louis Pasteur who demonstrated the role of live yeast in the fermentation process.

The first *Saccharomyces* yeasts were selected for winemaking roughly 80 years ago; the main criteria for selection were their ability to perform reliably and complete the fermentation. It was through the persistence of the wine institutes that the starter culture technique was refined and liquid yeast starter cultures were accepted by winemakers. In the mid-1960s, active dried oenological yeast starter cultures were developed (Degré, 1993), and these have gained mainstream acceptance for use by winemakers in the majority of wine made today. Since the 1980s, universities and oenological research institutes have based their selection strategy of naturally-occurring wine yeast on their research focus and available resources. Some notable examples of these yeast selection strategies would include the competitive factor (Barre and Vézinhét, 1984), ecological origin (Delteil and Lozano, 1995), varietal volatile compounds (Tominaga *et al.*, 1998), breeding techniques and genetic engineering of the yeast, as covered in Chapter 2.

1.2 Yeast and fermentation management and wine quality

Among the selected *Saccharomyces* available as active dried yeast, there are substantial differences in their kinetics and aptitude to achieve a complete fermentation. These differences come from the yeast's nitrogen and oxygen needs, some of which have been characterized (Sablayrolles *et al.*, 2000), and their ability to tolerate juice conditions. Another important influence on the yeast fermentation kinetics is the individual yeast's ability to compete against the microbial soup of indigenous yeast present in the juice or must. Under winery conditions, competition between yeast for the uptake of nutrients and the influence of the competitive factor is a practical key point in the choice of yeast. The method of yeast biomass propagation prior to dehydration has also been demonstrated in winery scale fermentations to have a direct impact on the performance of the yeast during fermentation and on the resulting wine quality (Bohlscheid *et al.*, 2007b). More details on this will be shown in a later section of this chapter.

Oenological yeasts offer winemakers an additional tool for differentiating their wines (Eglinton *et al.*, 2003). Determining which yeast to use, and using them in a controlled manner, can help to achieve the subtle aroma and flavour diversity that many winemakers seek (Henschke, 1997b; Heard, 1999; Lambrechts and Pretorius, 2000; Howell *et al.*, 2005).

The secondary metabolism of the specific yeast produces compounds that have great importance in the wine analytical and sensory profiles. There are substantial differences among strains of *Saccharomyces* in terms of their production of esters, sulphur compounds, varietal compounds such as β -damascenone, and

polysaccharides, as well as the impact of their mannoproteins on the volatility of certain compounds and on the stability of pigments and polyphenols (Delteil and Jarry, 1992).

With such large possible analytical differences among yeasts, it is logical that correspondingly large sensory differences have been demonstrated as early as the mid-1980s. Sensory differences due to the yeasts may come from their interaction with grape flavour precursors or be due to the yeast metabolites (Pretorius *et al.*, 2008). Yeast mannoproteins released in the fermentation also have a sensory impact on the aromatic expression of varietal volatile compound (Wolz, 2005). In multivariate sensory studies with principal component analysis (PCA) of sensory analysis results, different yeast strains in red wines gave sensory differences with magnitudes as great as those obtained from grapes at different stages of ripeness, or varying maceration lengths; techniques generally recognized as having a big impact on the wine style (Delteil, 2001). These sensory differences may become even more apparent with ageing (Dumont *et al.*, 1994).

It is critical for the winemaker to consider using suitably selected yeast that can grow and express its metabolic activity under the given juice or must conditions. For example, a yeast showing good tolerances to low temperatures and highly clarified juices for making a lower alcohol aromatic white wine would probably not ferment very well under higher temperature and higher alcohol potential conditions in a concentrated red must and vice versa.

The yeast's impact on wine quality can also be associated with its production of off-flavours such as volatile acidity, acetaldehyde, ethyl acetate and negative sulphur compounds. The prevention of off-flavour formation will be addressed in later sections of this chapter that will cover proper yeast handling, feeding and other good fermentation practices that contribute to yeast health and vitality.

1.3 Yeast rehydration and handling

Proper rehydration is perhaps the most critical phase in using active dried yeast. When selected yeasts are produced in an active dried form, the goal is to get a very high and viable cell population prior to a series of water removal steps until 5–8% moisture remains in the yeast powder. This low moisture level is necessary to ensure a good shelf life in order to conserve the yeast's potential activity for more than 36 months. These drying stages remove not only extracellular water, but also most of the water within the cell and bound to the cell's organelles, causing the yeast cells to shrink and desiccate. With extremely low water activity, there is almost no metabolic activity. To be functional again, the dried yeast cells must reabsorb all of their water. When the dried yeast comes into contact with water (or any other liquid), the cells literally act like dried sponges and suck up the needed water in seconds.

In winemaking, the grape juice is a very hostile medium for *Saccharomyces*: high osmotic pressure, low pH and often high SO₂. Most selected yeasts resist these conditions when their membrane and their intracellular components are in good

physiological conditions. Therefore, before inoculating a grape juice, it is absolutely necessary to bring back vital water at the right temperature to the active dried yeast cells in order for their membrane and intracellular components to reorganize properly. Not only will yeast cells not disperse very well if not properly rehydrated, they can lose a large amount of cellular components, reducing the efficiency of oxygen and nutrient transfer to the cells (Henick-Kling, 1988). This impedes yeast growth and activity leading to sluggish and stuck fermentations. The use of clean potable water at the right temperature for the right length of time would appear to be the most simple and effective yeast rehydration protocol.

Proper yeast rehydration gets them off to a good start and helps to ensure that the yeast will stay healthy throughout the fermentation. Degré (1993) proposed a general recommendation procedure for active dried yeast rehydration which is as follows:

- sprinkle 500 g of dry yeast into 5 L of warm water (35–40 °C);
- stir the suspension after 5 minutes to re-suspend the yeast cells;
- leave the suspended yeast cells no more than 30 minutes;
- add yeast to 20–25 hL must to be fermented, which would correspond to a yeast inoculation level of dosage 25–30 g/hL *ca.* $2\text{--}5 \times 10^6$ cfu/mL.

Recent studies have investigated the effects of rehydration protocols on the recovery of fermentative activity of different yeasts. Some differences were observed among yeasts using different rehydration protocols; however, these

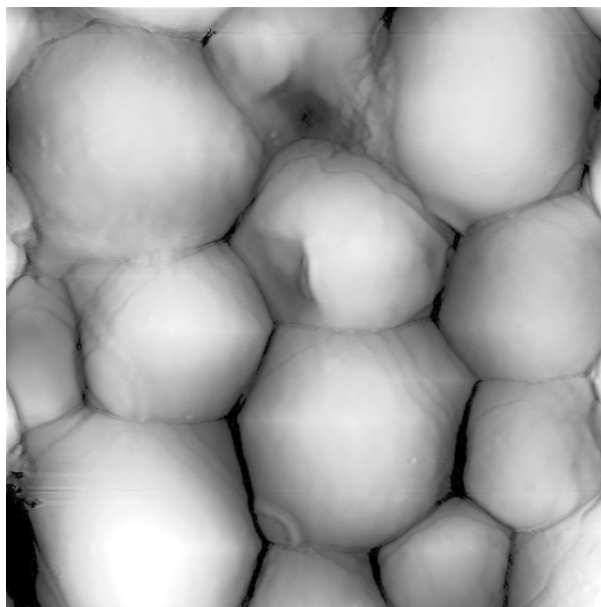


Fig. 1.1 Atomic force microscopy photograph of rehydrated active dried yeast after 20 minutes; taken at The University of Abertay Dundee, Scotland.

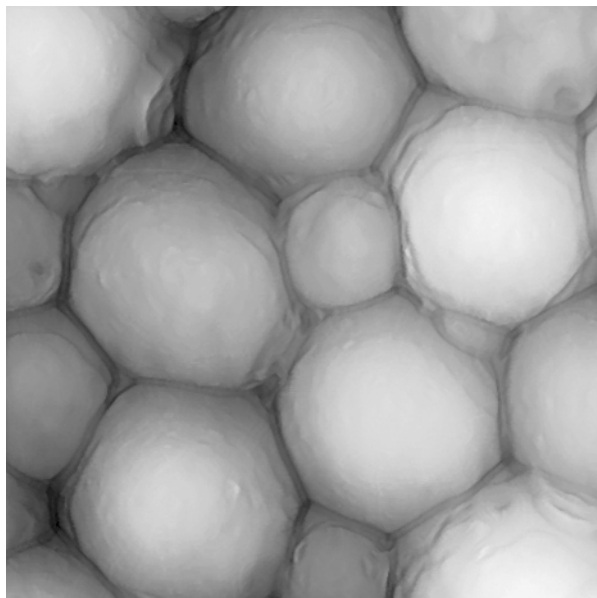


Fig. 1.2 Atomic force microscopy photograph of rehydrated active dried yeast after 20 minutes in the presence of supplemented micronutrients and sterols; taken at The University of Abertay Dundee, Scotland.

differences were not large enough, suggesting that the best advice is to follow the yeast producer's rehydration instructions (Sablayrolles *et al.*, 2006). Other recent studies have shown the influence on yeast viability and vitality of rehydrating active dried yeast in the presence of micronutrient and or sterol and unsaturated fatty acid-enriched inactivated yeast suspension during rehydration (Soubeyrand *et al.*, 2005). Yeast rehydration photographs taken at The University of Abertay Dundee using atomic force microscopy show rehydrated active dried yeasts after 20 minutes (Fig. 1.1) and under the same rehydration conditions with supplemented micronutrients and sterols (Fig. 1.2). Higher maximum yeast cell density and shorter overall fermentation lengths were observed when using these types of rehydration nutrients, especially under high sugar concentrations such as in ice wine musts (Kontkanen *et al.*, 2004).

Using grape juice or must in the initial yeast rehydration media may lead to the yeasts being directly exposed to residual agrochemicals, sulphur dioxide or other inhibitors. The addition of grape juice or must was found, however, to be beneficial if the total rehydration length exceeded 30 minutes (Radler *et al.*, 1985). In practice, grape juice or must should be added 15 to 30 minutes after the initial yeast rehydration in water. This will also help to adapt the rehydrated yeast suspension to the cooler juice or must temperature and avoid cold shocking the yeast starter culture.

1.4 Yeast inoculation

Once you have properly prepared the yeast starter culture, there are a few common-sense guidelines to keep in mind in order to help the yeast you want to dominate the fermentation and lower the possibility of problem fermentations. When inoculating the fermentor, there is a high probability that the inoculated *S. cerevisiae* will dominate the fermentation (Schütz and Gafner, 1993). However, inoculating with specific yeast will not guarantee the dominance of that yeast or their contribution to the fermentation. Significant factors that affect this outcome will be how the fermentation is managed and the population of the indigenous yeasts already acclimated to that juice or must.

Winery fermentation trials were made in many regions in the late 1980s and early 1990s using a selective plating technique in order to determine the influence of winemaking practices on the inoculated yeast implantation success. The results (Delteil and Aizac, 1998) showed that in the early part of the harvest the percentage of successful inoculated yeast implantation declined over the duration of the harvest. This is most likely the result of the indigenous flora becoming more established over the course of the harvest as cellar cleaning practices become less effective. Longer juice settling times also resulted in lower yeast implantation. The practice of juice centrifugation resulted in high implantation levels which supports the impact of the acclimated indigenous yeasts on the ability of the inoculated yeast to implant (Querol *et al.*, 1992).

1.5 Yeast inoculation rate

The juice or must sugar content and the hygienic conditions of the must are fundamental when choosing the inoculation rate; if indigenous yeasts or bacteria are present in high numbers, the more precarious the hygienic conditions and, therefore, a relatively large inoculation rate of the oenological yeast should be used. For potential alcohol wines around 14%, the number of healthy yeast cells should be around 5×10^6 cells/mL which is generally obtained by the addition of 25 g/hL of active dried yeast. The suggested rate of yeast inoculation is based on the fact that a proper initial cell density is required for the onset and completion of fermentation. Before the onset of fermentation, during the lag phase, it is ideal to keep this lag phase as short as possible to avoid potential microbial spoilage and production of undesirable compounds. This situation is directly correlated to the initial yeast cell density or inoculation rate. Increasing the inoculum density will result in a decreased lag time, up to certain point where the effect is not significant due to the crowding effect. During wine fermentation, the inoculated yeast cell population will usually undergo roughly five cell divisions or a doubling of the yeast population with every cell division until the juice is colonized. One of the most important factors in obtaining a steady and complete fermentation is the presence of an adequate cell population when the yeast has finished growing, which is normally when 30–50% of the sugars remain to be fermented. In order to

Table 1.1 Yeast inoculation rate comparison trial made at INRA Research Institute Pech Rouge, France

	10 g/hL	25 g/hL
Lag phase (hours)	50	32
Fermentation length (hours)	778	505
Residual sugar (g/L)	1.6	0.8
Volatile acidity (g/L H ₂ SO ₄)	0.54	0.46

complete fermentation at that stage where problems usually start occurring, approximately $120\text{--}150 \times 10^6$ cells/mL is needed. This population is more easily achieved when the yeast cell inoculum is approximately 5×10^6 cells/mL, which is obtained when the inoculation rate is 25 g/hL (Monk, 1997). It is important to note that some yeasts will require higher cell density to complete fermentation because their metabolism is different compared to others.

Table 1.1 shows the importance of an inoculum of 25 g/hL versus 10 g/hL. A yeast inoculation level trial done by Lallemant in collaboration with the INRA Pech Rouge, clearly showed the benefit of inoculating a Chardonnay must with 25 g/hL since it resulted in a substantial reduction of lag phase time, fermentation length, residual sugars and a reduction of volatile acidity.

1.6 Yeast inoculation timing

It is essential to inoculate the must as soon as possible to help ensure the dominance of the oenological yeast. In large fermentors that take a long time to fill, it is necessary to inoculate the yeast starter to the bottom of the fermentor (or as soon as possible during filling) to allow them to go through their lag phase and compete right away with the indigenous microbial population. Also, it is important to try to inoculate the must when the temperature is $>12^\circ\text{C}$, since lower temperatures may be stressful for most yeast and result in sluggish or stuck fermentations.

There are exceptions when winemakers may want to inoculate at lower temperatures. Cold soaking, also known as pre-fermentative cold maceration, populations of apiculated yeasts such as *Kloeckera apiculata* or *Hanseniaspora uvarum* can develop in substantial amounts and produce high amounts of ethyl acetate, which has a very pungent ascendent aroma. At low concentrations, ethyl acetate can emphasize dryness and burning sensations in the mouth during tasting. When the grapes are immediately inoculated with yeast at about 5×10^6 live cells/mL, the yeast will begin its lag phase and develop slowly. Even at cold soak temperatures of around 10°C , the selected *Saccharomyces* population will begin to use the nutrients in the must, making them unavailable to the indigenous apiculate yeast population. Because the juice is very cold for the recently-rehydrated *Saccharomyces*, it is recommended to rehydrate the yeast with inactivated yeast-based micronutrients, sterols and unsaturated fatty acids in order to help it support the low temperature and limit lag phase duration. In addition, the

yeast to be inoculated should be adapted to the cold must temperature by combining the rehydration suspension with the colder must. This will help the yeast adjust to the colder must temperature and avoid cold shocking the yeast. The adaptation step may need repeating in very-low-temperature musts.

1.7 Sequential yeast inoculation strategies

There are two scenarios where sequential yeast inoculation strategies might be considered by winemakers. The first situation would be addressing potential problems with fermentations at very high initial soluble solids concentrations [28 °Brix or density (specific gravity) of 1.110] for which the goal is to finish the alcoholic fermentation with very low residual sugar. This method is already in use by winemakers dealing with extremely mature grapes. The *Saccharomyces* yeast is inoculated in two stages: the first at the beginning of fermentation and the second once the density has reached a certain point but before symptoms of a stuck or sluggish fermentation are apparent. At first, 25 g/hL of the yeast is rehydrated in water supplemented with high yeast sterol-based inactivated yeast to help protect the yeast from osmotic shock, and then this protected yeast is inoculated. At a density of around 1.020 (~5 °Brix), another 25 g/hL is inoculated (also rehydrated in the same manner).

This sequential inoculation strategy is also well adapted for yeasts with particular qualities that winemakers would like to take advantage of but cannot due to the sensitivity of that yeast to very high potential alcohol situations. The proper way to carry out this method is to first inoculate the must with the type of this yeast in question and then during the alcoholic fermentation, sequentially inoculate with alcohol-tolerant yeast.

A second considered sequential inoculation strategy is the use of non-*Saccharomyces* yeast followed by a *Saccharomyces*. More knowledge is available on the non-*Saccharomyces* impact on the sensory profiles of the wines (Ciani and Ferraro, 1998). Some of these yeasts like *Pichia fermentans*, *Candida stellata* or *Torulaspora delbrueckii* have been studied for their interesting organoleptic contributions (Moreno *et al.*, 1991; Ferraro *et al.*, 2000; Clemente-Jimenez *et al.*, 2005). Although some of these yeasts will contribute aromatic compounds of interest in some styles of wines, most of them are not very reliable alcoholic fermentors. A recent project by Ortiz-Julien *et al.* (2005) evaluating sequential inoculation of first a non-*Saccharomyces* followed by a good fermenting *Saccharomyces* during a late stage of alcoholic fermentation have shown promise for some styles of wine; however, further investigation is needed.

1.8 Yeast storage

In the dry state, active dried yeasts contain <8% moisture, which does not allow very substantial metabolic activity, but it is just enough moisture to preserve the

viability for several years depending on the yeast and how it was produced. When preserved in their original vacuum-sealed packaging, active dried yeast may lose up to 10% of their viability per year when stored at an average ambient temperature of 20 °C. From a microbiological point of view, the loss of 10% of the yeasts cells would have a slight effect on the inoculum and could easily be compensated for by increasing the inoculation rate by 10%. For this reason, if they are still in their original packaging they can be used up to 2 to 4 years after their date of production without much loss in activity. Nevertheless, specific and simple methods exist to verify yeast activity, and these can be obtained upon request from the active dried yeast producer.

1.9 Nutrient strategies

In order to conduct a complete alcoholic fermentation with reliable kinetics, it is necessary for the grape must to have adequate nutrition in a tolerable environment for the development of healthy yeast. This will ensure an optimum production of yeast biomass and consequently reduces the risk of sluggish or stuck fermentations. Compared to 20 years ago, we now have a better understanding of the complex and often frustrating interactions in grape must and their influence on yeast growth and survival during alcoholic fermentation. However, it is still difficult to predict when there is a nutrient deficiency in a must so many winemakers now routinely add supplements using the initial must's yeast assimilable nitrogen (YAN) concentration, as a guideline. In addition to the YAN concentration the alcohol potential in a must should also be a guideline when making nutrient supplement decisions.

1.9.1 Nitrogen

Nitrogen is an important factor in oenology, playing a vital role in the kinetics of alcoholic fermentation. Yeast assimilable nitrogen, made up of ammonia nitrogen and α -amino nitrogen (free amino acids minus the proline that is not assimilated by the yeast), is the nutrient with the greatest effect on the rate and outcome of alcoholic fermentation (Bezenger and Navarro, 1987). During fermentation, the yeasts transport ammonium (NH_4^+) and amino acids from the must into the yeast cells, where they are kept for later use. These sources of nitrogen are used in the synthesis of the yeast membrane during cell multiplication and for production of the enzymes necessary for glycolysis, i.e., for the conversion of glucose and fructose to ethanol. In addition, NH_4^+ and amino acids are used to biosynthesize permease enzymes situated in the yeast membranes. These permeases are responsible for the transport of such components as other amino acids and sugars into the cell.

A YAN deficiency in the must limits the growth of yeast and the fermentation rate (Bely, 1990). During alcoholic fermentation, sugars are consumed during the stationary phase when the nitrogen gradually becomes less available. Since YAN is an essential nutrient involved in the transport of sugars into the cell via protein

synthesis, this partially explains why both the yeast metabolism and the fermentation activity slow down (Salmon, 1996). Thus, the lower the concentration of YAN in a must, the greater the risk of sluggish fermentation (Kunkee, 1991). Most musts contain between 80 and 400 mg/L of YAN; the threshold for YAN deficiency is about 150 mg/L for a must with an initial sugar concentration of about 200 g/L (Henschke and Jiranek, 1993). A nitrogen deficiency in the must is also responsible for stopping the protein synthesis in the yeast. The consequence of this inactivation will be a substantial decrease in the sugar transport activity, thereby increasing the risk of a stuck fermentation (Basturia and Lagunas, 1986).

The nitrogen supply, either from the grape or from the addition of nitrogen supplements, can influence yeast growth, fermentation time and the sensory properties of a wine. Different yeasts under similar conditions have different growth rates, nutrient requirements and abilities to produce and excrete esters and fusel alcohols. The interaction between yeast and grape must composition can influence wine quality and style (Pretorius *et al.*, 2008). When nitrogen is limiting in fermentation, the yeast – which needs more to continue building protein for sugar transport – will degrade its intracellular pool of amino acids and assimilate the nitrogen part. So, when sulphur-containing amino acids are degraded, the result is an accumulation of the SH fraction that will be released by the yeast as H_2S (Jiranek *et al.*, 1995; Erasmus *et al.*, 2003). Hydrogen sulphide, when present in concentrations $>10 \mu\text{g/L}$, adversely affects the sensory quality of wine. When YAN is coupled with an imbalance of pantothenic acid and/or biotin, it can also result in an increase in H_2S production (Wang *et al.*, 2003; Bohlscheid *et al.*, 2007a). When H_2S is produced in the early stage of fermentation, it can react with other compounds to produce sulphides, disulphides, thiols and mercaptans. Those sulphur compounds can develop during ageing or after bottling and also negatively affect wine quality (Rauhut *et al.*, 1993). Occurrence of H_2S can also be caused by improper application of agrochemicals containing elemental sulphur.

1.9.2 Organic nitrogen

The amino acids contained in grapes are predominantly proline, arginine and glutamine. Vineyard practices and climatic conditions influence the concentration of amino acids.

The amino acids and peptides coming from inactivated yeasts are the principal source of supplemented organic nitrogen. The importance of organic nitrogen from yeasts is well known as a highly efficient nutrient source for wine yeasts, especially when compared to inorganic nitrogen from diammonium phosphate (DAP) (Fig. 1.3). In addition, organic nitrogen helps avoid over-production of undesirable sulphur compounds, compared with use of DAP as a nitrogen source. Inactivated yeast-based supplements containing a high level of organic nitrogen can also help winemakers achieve steady fermentations while limiting temperature peaks.

The inactivated yeast-based supplements not only provide organic nitrogen, but they are also composed of yeast cell walls that are beneficial to the yeast during

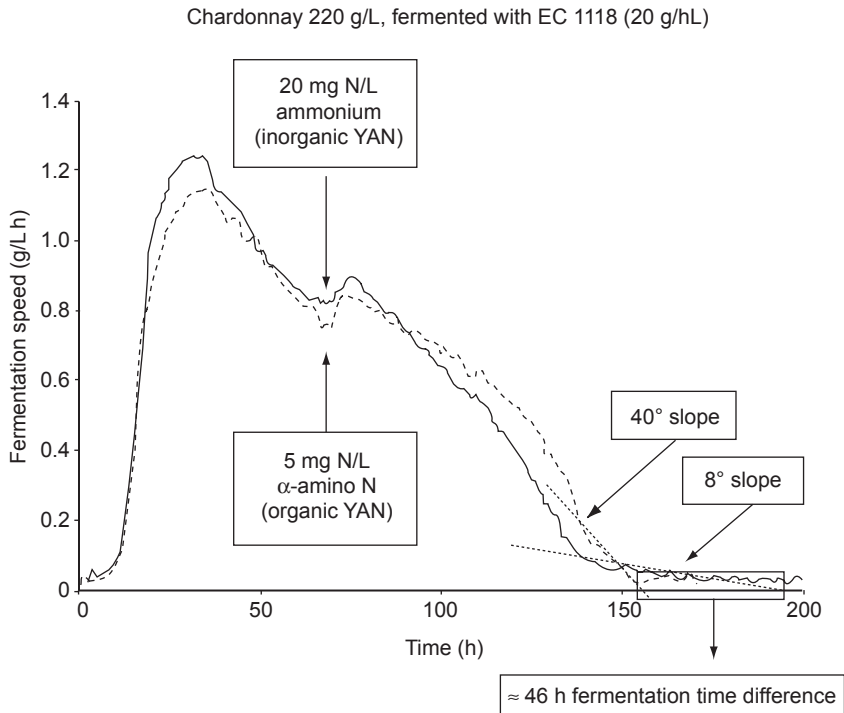


Fig. 1.3 Comparison of the effects of organic versus inorganic nitrogen on fermentation kinetics measured by CO₂ release at the INRA Research Institute Montpellier, France in 2002.

fermentation. Their cell wall fraction also supplies sterols and lipids which are survival factors helping to maintain the membrane fluidity and avoiding alcohol toxicity. Additionally, cell walls can be useful, as they can be rich in polysaccharides having a large surface area with sorption capacity to remove residual agrochemicals and other potential inhibitory substances resulting from yeast metabolism, such as octanoic (caprylic) and decanoic (lauric) fatty acids (Lafon-Lafourcade *et al.*, 1984). These compounds can adhere to the yeast cell wall and inhibit yeast activity in the last phase of the fermentation process, because they alter the permeability of the cell. Finally, yeast cell walls can act as elements of support, helping to keep the yeast in suspension during fermentation and avoiding their settling on the bottom of the fermentor.

1.9.3 Inorganic nitrogen

Inorganic nitrogen in the form of ammonia salts such as DAP or ammonia sulphate is used in winemaking to supplement the nitrogen availability, especially in musts high in potential alcohol and low in YAN. However, inorganic nitrogen supplemented on its own will not obtain the best fermentation results, and over-treatment can lead to the development of harsh characters in wine. Under nitrogen-limiting

conditions, the goal is to balance the level of inorganic nitrogen with organic nitrogen from α -amino acids to achieve optimal nitrogen concentrations. The result of a good balance of these products will be fermentation security and the best expression of fermentation and varietal characters.

1.9.4 How much nitrogen is needed?

The minimal quantity of YAN needed for yeasts is reported to be 150–200 mg/L, depending upon the must soluble solids concentration. The higher the initial soluble solids concentration, the more nitrogen will be required by the yeast. The amount of YAN needed will also vary according to the yeast used as they have different relative nitrogen demands (Sablayrolles *et al.*, 2000). The bioavailability of key vitamins that allow an optimal use of nitrogen also has an influence on the amount of YAN (Wang *et al.*, 2003).

1.9.5 When should you add nitrogen?

There are two optimal times to consider adding YAN to the fermenting must (Sablayrolles *et al.*, 1996). The first time for YAN addition is during the beginning of yeast cell growth as soon as the fermentation is active, to provide YAN for yeast growth when it is deficient in the must. If YAN concentrations are adequate, however, this first addition is unnecessary. A second optimum time for a YAN addition is at the end of yeast cell growth, which helps to maintain the yeast's vitality to ensure a satisfactory fermentation finish. This corresponds to the consumption of about one third of the sugar, which is at the end of the growth phase entering into the yeast's stationary phase or at maximum cell yeast density. Table 1.2 summarizes a good yeast inoculation and nutrient strategy based on must YAN and initial Brix level to help avoid fermentation problems.

1.9.6 Oxygen

During the early stages of alcoholic fermentation, *Saccharomyces* is able to use oxygen for the synthesis of sterols. These sterols help in keeping the yeast membrane fluid and in resisting osmotic shock and ethanol toxicity. The practical consequences are minimal acetic acid production during yeast growth and a healthier yeast population resulting in a more reliable fermentation finish. Even though there is an important variation among the different oenological yeasts in their response to oxygen addition, yeasts scavenge oxygen very quickly. There are no risks of juice oxidation when aerating or adding oxygen at the recommended times and quantities, even in very fragile, readily-oxidized juice from varieties such as Sauvignon blanc or Viognier.

There are two optimal times to aerate or add oxygen (4–6 mg/L) to the fermenting must (Sablayrolles *et al.*, 1996). First, during the beginning of yeast cell growth as soon as the fermentation is active to help the yeast resist the osmotic shock and limit acetic acid production. In practice, this corresponds to the consumption of

Table 1.2 Yeast inoculation and nutrient strategy summary under different must conditions

Must Brix	Yeast inoculation rate	Yeast stimulant or protector during yeast rehydration	Nutrient strategy depending on must YAN content during fermentation
<25 °Brix	25 g/hL	30 g/hL stimulant	YAN >200 mg/L: no addition
		30 g/hL stimulant	YAN 100–200 mg/L: add complex nutrient at inoculation and again at one-third sugar depletion
		30 g/hL stimulant	YAN <100 mg/L: add complex nutrient + 15 g/hL DAP at inoculation and again at one-third sugar depletion
25–32 °Brix	25–50 g/hL. The higher the sugar content, the higher the inoculation rate	30–60 g/hL protector	YAN >200 mg/L: add complex nutrient at one-third sugar depletion
			YAN 100–200 mg/L: add complex nutrient + 15 g/hL DAP at inoculation and again at one-third sugar depletion
			YAN <100 mg/L: add complex nutrient + 25g/hL DAP at inoculation and again at one-third sugar depletion
>32 °Brix	50 g/hL	60 g/hL protector	Regardless of must YAN content, add complex nutrient at inoculation and again at one-third sugar depletion

15–30 g/L sugar. It is recommended to aerate or add oxygen a second time at the end of yeast cell growth to ensure a satisfactory fermentation finish.

Use of a special device such as a macro-oxygenator to bring tiny enough bubbles of oxygen to the yeast at the right time and the right dosage is important. Pumping over can bring in some oxygen, but the juice is not homogeneous in the tank, which means that not all of the yeast population will have equal access to the oxygen. An additional recommended method of providing sterols to the yeast is during the yeast rehydration step by adding inactivated yeast-based supplements that are rich in sterols and polyunsaturated fatty acids to help protect the yeast from the initial osmotic shock. Finally, at one third of fermentation the addition of inactivated yeast-based complex nutrients provides additional sterols along with a balanced source of nitrogen. This helps the yeast to resist alcohol toxicity at the end of fermentation especially in high alcohol potential juices or musts.

In practice, there is a substantial lowering of the potential for sulphur off-flavours formation when oxygen addition and these inactivated yeast-based

protectant and nutrient supplements are used. This is due to maintaining a better yeast membrane physiology to help keep the yeast from becoming stressed.

1.9.7 Minerals

Insufficient attention is given to the bioavailability of metal ions in the physiology and fermentation performance of yeast in winemaking. On a dry weight basis, yeast contains an average of 2.6% phosphate (PO_4) as phosphorus pentoxide, 2.5% potassium (K^+) as potassium oxide, 0.4% magnesium (Mg^+) as magnesium oxide, 0.5% sulphur and trace amounts of calcium (Ca^{2+}), chlorine, copper, iron, zinc and manganese. The relative proportion of these elements in yeast indicates their importance to yeast metabolism.

Yeasts are unable to grow unless provided with a source of PO_4^- . This anion is involved in energy conservation and transfer, and is part of many organic compounds in the yeast cell. Fortunately, PO_4^- naturally present in must is usually adequate for yeast growth; however, limited growth stimulation can occur by increasing the PO_4^- concentration.

Potassium is considered a survival factor also required for yeast growth and fermentation, although it is not incorporated in the structural or organic components of the cell. Grapes must also contain adequate amounts of K^+ for the operation of transport systems and active yeast growth, although an imbalance between K^+ and H^+ concentration on the yeast cell viability could be critical especially $\text{pH} < 3.3$ (Kudo *et al.*, 1998).

Magnesium, which is the mineral with the highest intercellular concentration in the yeast cell, is essential for the growth and metabolism of the yeast (Walker, 1996) as well as for helping the yeast with ethanol tolerance (Dombeck and

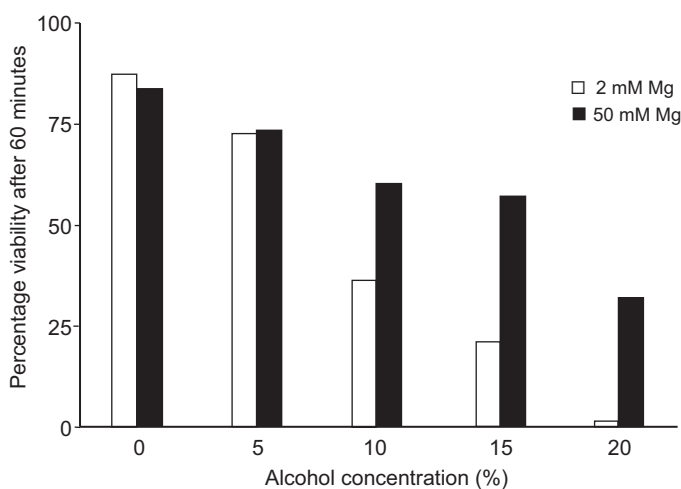


Fig. 1.4 Viability of *Saccharomyces cerevisiae* after 60 minutes exposure to increasing alcohol concentrations at low (2 mM) and high (50 mM) magnesium concentrations. Adapted from Birch and Walker (2000).

Ingram, 1986) as shown in Fig. 1.4. The bioavailability to the yeast is typically not high in these and other metal ions such as zinc, which is critical for many enzymes such as alcohol dehydrogenase (ADH) and has a direct influence on selected yeast physiology and fermentation performance, because they are often bound to must components such as proteins and polyphenols (De Nicola *et al.*, 2009). In addition, the concentration of these minerals is lower in extremely mature grapes and in those compromised by fungus or rot, which consume large quantities of minerals. The elemental concentration of the soil, irrigation, yield and canopy management also plays a role in mineral availability. One can increase the concentration of metal ions by making additions to the rehydration water and must, but their bioavailability and quantity must be balanced, because an overdose of Ca^{2+} , for example, can result in the inhibition of the effects of Mg^{+} due to uptake antagonism (Walker *et al.*, 2003).

1.9.8 Vitamins

Vitamins are organic compounds essential for the optimum growth of yeast cells and the capacity to survive in stressful conditions. A vitamin deficiency can induce sudden changes in the fermentation kinetics. The majority of vitamins act as enzymatic cofactors, intervene in energy transfers or support the integrity of the membrane.

Biotin, for example, favours the production of esters and allows for a better cellular viability for the fermentation. Nitrogen and biotin interactions were shown to influence fermentation length and volatile production by *Saccharomyces* depending on yeast (Bohlscheid *et al.*, 2007a). Another key vitamin pantothenic acid (vitamin B_5), which is involved in the metabolism of lipids, has a positive effect on the sensory qualities of the wine, diminishing the risk of H_2S production (Edwards and Bohlscheid, 2007) and volatile acidity. A survey of these vitamin concentrations in grapes indicated that many samples were deficient in biotin while pantothenic acid was normally found in adequate amounts (Edwards *et al.*, 2008). Because it is not easy to evaluate whether a must is deficient, it would be prudent to routinely supplement with a balanced vitamin supplement.

Also the subject of numerous studies, thiamine is another vitamin that can be limiting, and thiamine (vitamin B_1) deficiency can cause stuck fermentations. This temporary deficiency is often linked to certain pre-fermentation treatments and poorly controlled conditions (problems with SO_2 and temperature). Thiamine has the ability to combine with SO_2 , rendering it unavailable to the yeast. Indigenous flora such as *Kloeckera* will consume thiamine in just a few hours, which is fast enough to deprive the selected yeast of this key vitamin (Bataillon *et al.*, 1996).

With the potential for interference from must components or indigenous organisms, the optimal strategy for introducing micronutrients and vitamins, and to have them used by the targeted yeast, is by directly making them available to the yeast in the rehydration water. Development and assessment of the effectiveness of yeast stimulants, protectors and complex nutrients on yeast growth and fermentation performance is typically performed at laboratory scale using rigorously controlled small-scale fermentors like the ones shown in Fig. 1.5 (Sablayrolles *et al.*, 1987). These results can be validated in scaled-up trial fermentors under controlled conditions (Fig. 1.6).



Fig. 1.5 Photograph of a laboratory-scale fermentor as described by Sablayrolles *et al.* (1987).



Fig. 1.6 Photograph of a pilot-scale fermentor at INRA Research Institute Pech Rouge, France.

1.10 Difficult fermentation conditions

1.10.1 Difficult fermentation conditions – high potential alcohol

The *S. cerevisiae* strains used in winemaking are selected for particular characteristics such as rapid and regular fermentation rates, tolerance to ethanol, low H_2S production and specific sensory contributions. The use of oenological yeasts is an important tool in the proper management of fermentation, as well as in preparation for malolactic fermentation. However, even with the use of active dry yeasts, problems can occur during fermentation. For instance, sluggish or stuck alcoholic fermentations are still observed, in part due to longer ‘hang-times’ of grapes prior to harvest (delayed harvest). While producing beneficial flavours, this viticulture practice increases soluble solids concentrations in musts and results in higher potential alcohol levels, often close to inhibitory amounts. The kinetics of sugar utilization by *S. cerevisiae* during fermentation is largely driven by sugar transport, and glucose is typically consumed at a faster rate than fructose. In sluggish fermentations, the maximal rate of fermentation is reduced after most of the glucose is consumed, and fermentation can become stuck with a substantial concentration of fructose remaining. As higher concentrations of alcohol become toxic to the yeast cell, the use of fructose is compromised, resulting in stuck fermentations. Other problems resulting from high alcohol concentrations include the production of H_2S and other reduced sulphur aromas as well as excessive volatile acidity.

Aside from the genetic differences in yeast’s alcohol tolerance, the way the yeast is prepared at the yeast production facility and then managed during rehydration and through the yeast’s growth phase by the winemaker will have a substantial influence on the yeast’s ability to tolerate higher potential alcohols. A new method for preparing active dried yeast to better handle the trend of higher potential alcohols coming from higher maturity fruit was evaluated over several recent vintages. Both laboratory and field trials have demonstrated that under variable must nutrient conditions, active dried yeasts prepared using a different process had resulted in faster fermentations coupled with reductions in H_2S and volatile acidity production.

Yeast prepared using the different method completed the fermentation of a 2004 Syrah with adequate nitrogen (300 mg/L) more than 4 days earlier than the same yeast prepared by the traditional drying method (Bohlscheid *et al.*, 2007b). The effect was also seen in Syrah musts that had lower nitrogen concentrations (80 mg/L) in 2005 and 2006 where fermentations in both cases finished earlier. These observations support the contention that the same yeast prepared using different processing methods has an impact on how the yeast will ultimately perform under difficult must composition conditions.

The importance of aerobic survival factors on maintaining the yeast membrane fluidity and their resulting tolerance to high alcohol, good yeast nutrition and other good fermentation practice considerations when dealing with high potential alcohol musts are all keys to fermenting difficult musts. In high potential alcohol reds some additional good fermentation practices include:

- aeration or oxygen additions when the cap forms (usually when 15 g/L sugar is fermented) and again at one-third sugar depletion;
- temperature management during yeast rehydration, the initial phase of the fermentation and at the peak of fermentation;
- regular movement of the yeasts back up into suspension during their death phase towards the end of fermentation.

In higher potential alcohol white musts some additional good fermentation practices include:

- initial juice turbidity level optimum between 80 and 150 nephelometric turbidity units (NTU);
- aeration or oxygen addition as soon as the fermentation is active (usually when 15 g/L sugar is fermented) and again at one-third sugar depletion;
- temperature management during yeast rehydration, yeast inoculation at the end of the fermentation;
- regular movement of the yeasts during their death phase towards the end of fermentation.

1.10.2 Difficult fermentation conditions – under- or over-clarified white juice at low temperatures

Saccharomyces reacts to the presence of solids in the juice by a higher production of sulphur compounds such as 3-(methylthio)-propanol-1, also known as methionol. The higher the juice turbidity (expressed in NTU), the higher the production of this sulphur compound (Lavigne *et al.*, 1993). These sulphur compounds are of great importance in the aromatic profile as they have low sensory thresholds (methionol: 2 mg/L with cooked potato aromas). It is therefore recommended to clarify the juice below 150–200 NTU to help prevent their formation by the yeast and to allow the fruit character to develop.

Conversely, juice clarification, by eliminating grape solids, removes materials that may be a source of grape sterols and long-chain fatty acids that are of nutritional value for the yeast. A technique has been developed to lower turbidity to a balanced level leaving desirable grape solids from the pectic flakes formed during pectinase enzyme action during juice clarification. When this technique cannot be applied in the winery, it is recommended to supply yeast sterols to the yeast through the addition of inactivated yeast enriched in sterols during yeast rehydration.

When juice turbidity is <200 NTU and the juice is >23 °Brix with low organic nitrogen, it is also recommended to supplement the yeast with organic nitrogen, sterols and polyunsaturated fatty acids through the addition of a complex yeast nutrient at the beginning of fermentation. Otherwise, the yeast will produce high amounts of sulphur compounds due to the low availability of organic nitrogen (Delteil, 1998).

1.10.3 Difficult fermentation conditions – tall and narrow tanks

During fermentation, *Saccharomyces* produces CO₂. This CO₂ in high concentration produces bubbles that rise to the surface helping to keep some of the yeast cells in suspension. However, some cells tend to settle out, resulting in different yeast populations at different levels in the fermentor and, as soon as 1 mm of yeast cells have accumulated on the bottom of the tank, the cells underneath the pile have no access to sugar and die from starvation (Block and Vlassides, 2000). The yeast cells that are still in suspension have to work harder in order to assimilate more sugar per yeast cell until the fermentation is finished. The stressed and dying yeast cells produce very reductive conditions that favour the production of sulphur compounds.

As the fermentation slows down, more yeast cells will tend to settle on the bottom of the fermentor, leaving even fewer yeast cells in contact with sugar, amplifying the problem. This issue is exacerbated in tall and narrow fermentors when more yeast settles out than in fermentors with normal dimensions, and they are exposed to higher liquid column pressure, stressing the yeast even further. Under cooler fermentation conditions the CO₂ tends to stay in solution and it can build to toxic levels for the yeast causing stuck fermentations.

In practice, it is recommended to re-suspend the yeast frequently by different means: stirring by hand in barrel, deep punching down in open tanks, *delestage*, propeller agitator in bigger tanks, nitrogen or air injection, etc. Under these difficult fermentation conditions, the yeast will benefit by daily mixing of the settled yeast as soon as the fermentation is active. In practice, regular agitation gives more regular and more complete fermentation and lower negative sulphur compounds.

1.10.4 Difficult fermentation conditions – high sugar

Hostile yeast environments are also found in extremely high sugar musts intended for icewine or sweet wines (*vins liquoreux*). The hyper osmotic stress placed on the yeast can reduce yeast biomass, sugar consumption and ultimately generate a stuck or sluggish fermentation as well as an increased production of glycerol and acetic acid (see Chapter 18). When the soluble solids concentration is >25 °Brix, a higher yeast inoculation level than normal is required along with more emphasis placed on the initial handling and acclimation. A study of inoculation protocols for icewine fermentations was made to determine the best approach for preparing the yeast for these extreme winemaking conditions. A higher than normal yeast inoculation level of 50 g/hL, along with the use of a yeast rehydration stimulant in the rehydration water and a yeast conditioning step, reduced the fermentation length and the rate of volatile acidity production as a function of sugar consumed, and maintained higher yeast cell viability during the fermentation (Kontkanen *et al.*, 2004).

1.11 Sulphur compounds and their management

Sulphur compounds are aromas that are perceived through ortho-olfaction (smelling) or retro-olfaction (aromas perceived in the mouth). The long list of descriptors

includes burnt match, rubber, burnt rubber, garlic, onion, rotten egg, etc. They may also have unusual aromas like rancid fatty acids or meaty characters. Some of the chemical compounds implicated in sulphur odours include H_2S , ethanethiol, methanethiol, dimethyl-sulphide or dimethyl-disulphide. All of these compounds are produced at variable concentrations in wine by *Saccharomyces*. Since sulphur compounds are so prevalent, the risk of sulphur off-aromas ruining a wine is always present. At certain concentrations, some of the listed compounds combine with other sulphur compounds to contribute fruity aromas. Additionally, chemistry and sensory are not always the same because some compounds without the sulphur atom give the perception of 'sulphur odours' even though they usually have a different production pathway and may not be prevented the same way as sulphur compounds. These compounds include some fatty acids, 3-methyl-indole (also known as skatole) and 2-aminoacetophenone, which is a compound implicated in atypical ageing (Rapp *et al.*, 1993).

Sulphur compounds also have an impact on the mouthfeel as the percentage of compounds that stays in the wine has an aggressive impact on the mouth mucosa, just as they can be aggressive while smelling them. When you successfully treat a sulphur odour problem the mouthfeel is typically less aggressive than before.

The yeasts and their nutrition and fermentation temperature management have a direct impact on the occurrence of sulphur compound production during and after fermentation (Rauhut *et al.*, 1999). During the first third of fermentation, which is during yeast population growth phase, undesirable sulphur compounds can be dealt with by adding complex yeast nutrients. Organic nitrogen will rebalance the nitrogen-sulphur metabolism, helping to avoid further sulphur compound production. Polyunsaturated fatty acids and sterols from the complex yeast nutrient will improve the cell membrane physiology and diminish the yeast stress and the resulting sulphur compound production. Specific inactivated yeast cell structures from complex nutrients may also have a sorption capacity for some sulphur compounds, changing their volatility and diminishing their sensory impact. The addition of oxygen at the same time (4–6 mg/L in white juice; 6–10 mg/L in red must) will improve the effect due to yeast sterol synthesis and better physiology of the membrane as well as by acting directly on some sulphur compounds, stripping or oxidizing them to less odour-active forms. It is recommended to avoid the addition of inorganic nitrogen from ammonia salts such as DAP or di-ammonium sulphate, especially when organic nitrogen is limited. Otherwise the yeast will produce more hydrogen sulphide (H_2S) as soon as ammonium is absorbed.

After the second half of fermentation, if sulphur compounds are detected, it is recommended to add specific inactivated yeasts due to their sorption capacity 'sponge effect' by the yeast cell wall structure on sulphur compounds. The actively fermenting yeast will also take advantage of and react positively to the new source of fatty acids and sterols. At this time of fermentation, in white and rosé juices, as long as there is more than 30 g/L of sugar, a short 6–12 hour and small 0.5–2 mg/L/month addition of oxygen may help. As soon as alcoholic fermentation is finished, it is recommended to rack the wine off the lees trying to eliminate a part of the inactive yeast that had adsorbed sulphur compounds during the active

fermentation. Some of those cells agglomerate and sediment at the bottom of the tank within 12–24 hours.

1.12 Preventing stuck and sluggish fermentations

Stuck or sluggish fermentations have plagued many if not all winemakers at some point in their career. Sluggish and stuck alcoholic fermentations, long described in the literature, have not yet been relegated to history. If fermentation slows down or stops, some change in fermentation conditions has clearly curtailed the yeast's cellular activity. The causes of stuck and sluggish fermentations are complex and usually inter-related (Henschke, 1997a; Bisson and Butzke, 2000). The culprits include deficiencies (in nitrogen, oxygen, vitamins or minerals), and other inhibiting effects on the growth of yeasts (Bezenger and Navarro, 1987; Bely, 1990) or on their capacity to convert sugar to alcohol (Basturia and Lagunas, 1986), due to medium-chain fatty acids, ethanol, temperature, CO₂, by-products and co-metabolites such as acetic acid (Edwards *et al.*, 1996). Modern means now exist to help avoid such problems during fermentation.

1.12.1 Nutrition

Nutrient imbalances can be corrected through the good management of yeast nutrition. Nutrient complexes, combined with an oxygen addition at the optimal timing of additions, permit the controlled development of the yeast population that encourages first survival and fermentation metabolisms (Sablayrolles and Barre, 1986; Alexandre *et al.*, 1994; Feuillat, 1995; Sablayrolles *et al.*, 1996; Blateyron *et al.*, 1998). The principal nutrient deficiency impeding fermentation is lack of usable nitrogen. During growth, yeast cells must consume nitrogen to reproduce nuclear protein and cell protein (Basturia and Lagunas, 1986). They also need nitrogen to produce enzymes. Various sources of nitrogen are available in the must, but not all are accessible to the growing yeast cells. While the importance of nitrogen is known, it is not always given adequate attention by winemakers. Many winemakers fail to add a sufficient amount of usable nitrogen; indeed, nitrogen is often added only when problems arise. By that time, it is usually too late.

1.12.2 Aerobic survival factors

Lipids are an essential component of the yeast cell membrane. They are necessary for the budding and growth of yeast cells in the early stage of fermentation and to protect the yeast cell from alcohol toxicity in later stages (Kirsop, 1982; Larue and Lafon-Lafourcade, 1989). Once the yeast enters the anaerobic growth phase, each budding cycle depletes the amount of lipids by half. If insufficient lipids are available, the cell's semi-permeable membrane does not function properly. It cannot maintain the osmotic balance of nutrients (sugar) inside the yeast cell and by-products (alcohol) outside, so the growth cycle is restricted. Adding oxygen at

the beginning and at the end of the yeast exponential growth phase during fermentation helps the yeast produce its own lipids (Sablayrolles *et al.*, 1996). Another effective way to prevent the depletion of lipids is to add them during the yeast rehydration through the use of a sterol and polyunsaturated fatty acid enriched inactivated yeast. Stuck fermentations resulting from difficult fermentation conditions may be avoided through the utilization of inactivated yeast-based stimulant and protectant during the yeast rehydration phase. This not only helps the yeast avoid osmotic shock when added to high-sugar musts, but also helps to avoid alcohol toxicity during the later phase of fermentation (Soubeyrand *et al.*, 2005; Pillet *et al.*, 2006).

The cell walls of enriched inactivated yeast contain not only lipids but substantial amounts of polysaccharides, including chitin. Chitin increases the surface area in the must, which helps keep the yeast cells from settling to the bottom of a fermentor, where they may become weak and stressed. Improving the yeast's ability to stay in suspension is especially important if highly clarified juices or musts from concentrates are being fermented or after bentonite settling or heavily ameliorated or very cool fermentations.

1.12.3 Temperature management

In general, temperature management is very important throughout fermentation by avoiding sudden temperature shifts as well as temperature extremes beyond the tolerance of the yeast. Temperature also plays an important role during yeast multiplication early in the fermentation process. When yeast multiply rapidly, they produce ethanol faster than they can excrete it. As a result, alcohol accumulates within the cell and negatively alters the physiological state of the cellular membrane. This early excess of intracellular alcohol will limit the cell's resistance to alcohol toxicity toward the end of fermentation, contributing to stuck or sluggish fermentation. From a yeast's viewpoint, and depending on the initial sugar concentration of the must, the optimal fermentation temperature just under the cap of a red must begins under 25 °C and takes more than 3 days to increase to a maximum of 28 °C for a 22 °Brix juice or must. The more yeast is stressed early in fermentation, the lower the peak fermentation temperature it can tolerate. Different thresholds of temperature resistance have also been found among different yeast strains.

1.12.4 Oenological yeast selection

To avoid stuck and sluggish fermentations, several criteria should be taken into account when selecting yeast for their reliability to complete a given fermentation:

Lag phase and multiplication

Yeast's ability to compete with wild microflora and dominate fermentation depends on its lag phase, which is the initial phase where the yeast builds up its cell food reserves after rehydration and adding it to the fermentor before the cells start

to divide. This usually takes about 6–12 hours on average in a typical wine fermentation. The lag phase should be short since the longer the lag phase, the longer the contaminating organisms have an opportunity to flourish and cause potential fermentation problems as well as off-flavours.

Different yeasts demonstrate different fermentation kinetics. The speed of yeast multiplication during their exponential growth phase also depends on the fermentation conditions. Ideally the exponential growth phase should not occur so quickly that the alcohol accumulates in the yeast cell, causing problems later on; however, it should not be too slow either since the yeast might not dominate the fermentation. Roughly every 4–8 hours is a good average time period for every doubling of the selected yeast population. Usually about one-third of the sugars are used up by the end of the yeast's exponential growth phase.

Suitability for the limiting conditions in the must

Yeasts have widely varying capacities to withstand difficult fermentation conditions, particularly under high alcohol (14%) concentration, low pH and extreme temperatures (Pretorius, 2007). Fermentation performances in must with nitrogen and oxygen deficiencies also vary with different yeast strains.

Competitive factor

The K2 competitive-factor protein is most commonly found among the competitive-positive yeast that produces the competitive-factor protein (Barre, 1992). This K2 competitive-factor protein is also the only one showing any substantial activity at the pH conditions found in wines. Oenological yeasts fall into one of three categories when it comes to the influence of the competitive-factor proteins:

- 'competitive-positive', which produce this protein and may dominate competitive-sensitive yeasts;
- 'competitive-sensitive', which are dominated by competitive-positive yeasts and do not produce competitive-factor proteins;
- 'competitive-neutral', which are not affected by and do not produce the proteins.

The use of yeasts with resistance to the K2 protein is therefore strongly recommended, especially when it can be assumed that the must contains a large K2 yeast population.

1.12.5 Yeast rehydration and handling

Due to the lack of perceived importance, proper yeast rehydration and handling procedures are often overlooked. As mentioned previously, when using active dried yeast, proper rehydration is critical in order to avoid problem fermentations. Yeast cells will not disperse in must effectively if they are not properly rehydrated, and they can also lose a large amount of cytoplasm, reducing the efficiency of oxygen and nutrient transfer to the cells (Van Steveninck and Ledebor, 1974). This impedes growth and activity, leading to sluggish or stuck fermentations. Proper

rehydration can ensure healthy yeast cells and good fermentation characteristics. It is best to rehydrate in water rather than in must, even though must contains sugars that may help shorten the lag phase, because must may also contain SO₂ or residual fungicides that could be lethal during the rehydration stage. Once rehydrated, the yeast cells can resist SO₂ and low levels of agrochemicals, but not during liquid uptake. It is essential to avoid cold-shocking the yeast by adjusting the yeast suspension temperature. This is done by slowly (more than 5 minutes) mixing an equal amount of juice to be fermented with the rehydrated yeast suspension. The temperature difference between the rehydration medium and the must to be inoculated should never be >10 °C.

Finally, it is essential to inoculate with a high enough initial yeast population (5×10^6 cfu/mL for most wine fermentation, depending on the yeast and soluble solids concentration of the must). This is especially important because of the trend to harvest grapes at higher maturities. This helps overcome an over-dilution of the initial yeast cell inoculation, making it possible to reach the maximum yeast cell population necessary for a healthy fermentation.

1.12.6 Agrochemical and natural inhibitors

Agrochemical residues have frequently been implicated as a cause of fermentation problems. Residues can either act directly as fungicides do or indirectly as when the yeast has to detoxify the fermentation medium before it can start fermenting. In years of late-season drought, high concentrations of residues may remain on the grapes, resulting in higher rates of stuck and sluggish fermentations. Natural inhibitors, such as medium-chain fatty acids, contribute to stuck and sluggish fermentations by modifying the yeast sugar transport capacity and interfering with the membrane sugar transport proteins (Fugelsang *et al.*, 1992). The production of these saturated fatty acids by yeast is favoured by stressful fermentation conditions, which can arise with low juice turbidity, very high initial sugars, the condition of the yeast or extreme fermentation temperatures during the later phase of fermentation. The proactive use of inactivated yeast with very high sorption capacity for saturated short- and medium-chain fatty acids and/or residual agrochemicals is a good preventative approach to avoiding stuck and sluggish fermentations when these stressful conditions or improper use of agrochemicals are suspected (Lafon-Lafourcade *et al.*, 1984).

High concentrations of acetic acid produced by contaminating organisms or even some selected yeasts, especially under a deficiency of aerobic factors, have also been found to cause stuck fermentations (Smith *et al.*, 1995; Edwards *et al.*, 1996). In higher pH musts where contaminating organisms are an issue, winemakers may lower the pH through tartaric acid adjustments and add higher levels of SO₂ or lysozyme to help avoid the growth of potential spoilage *lactobacilli* that are known to produce acetic acid. When acetic acid levels become too high, some countries allow the use of acetic acid removal technology that will lower its concentration to levels that will allow the fermentation to proceed. Blending is another option for lowering the concentrations of acetic acid; however, caution is

advised with this approach as the high acetic acid-stuck wine should be filtered and stabilized prior to blending. This will help avoid any possibility of the spoilage organisms causing the high acetic acid problem to infect the wine that is used for blending.

Despite these vital precautions, stuck fermentations can still occur, although at lower frequency. Other factors can intervene, such as early contamination from undesirable microorganisms, over-clarification of must or the under-estimation of the alcohol potential.

1.13 Restarting stuck and sluggish fermentations

The best way to deal with stuck and sluggish fermentations is through prevention but, no matter what precautions are taken, problematic fermentations may occur. Whatever method is adopted for resolving stuck fermentations, it is necessary to react quickly and begin by finding out the basic numbers in order to adjust these variables if necessary – especially the pH, alcohol, residual sugars, volatile acidity, total and free SO_2 – of the stuck wine. It also helps to have a good microscope or access to a competent microbiology lab in order to see the microbial soup. That will help to determine the best approach for restarting the fermentation. The key is to find a restart approach that is effective yet relatively simple. In some cases, simply bringing the yeast culture back up into suspension, slowly warming the tank and/or re-inoculating with a fresh yeast culture prepared as if it were the initial inoculation may work to jump-start the sluggish fermentation. In most cases though, once a stuck fermentation occurs where the sugar has not dropped in several days, a more intensive intervention may be necessary.

Many different protocols for restarting sluggish or stuck alcoholic fermentations have been proposed and most of them are based on three basic steps. First, take the necessary precautions to avoid any growth of spoilage bacteria by adding SO_2 or lysozyme, which is a protein prepared from egg whites that will help to control lactic acid bacteria at higher wine pH. Many stuck fermentations are difficult to trace back to their root causes. Stuck fermentations caused by an obvious microbial contamination and/or high levels of inhibitors will require more emphasis on their removal during the stuck wine preparation. Yeast cell wall-based preparations with high sorption capacity can also be mixed into the wine and allowed to settle for 1 or 2 days and then racked off or filtered to help remove potential inhibitory substances in the stuck wine.

While the wine is settling or being filtered, the preparation and activation of the rescue yeast can begin which is the second step in a stuck fermentation restart protocol. It is recommended to select a rescue yeast that is alcohol tolerant, a vigorous fermentor and has a good capacity to uptake fructose. Under oenological conditions, the main sugars fermentable by *S. cerevisiae* are glucose and fructose. Both of these hexoses are generally present in musts in equivalent quantities, but the proportions may vary in some musts. *S. cerevisiae* prefers to consume glucose, which explains why, when fermentations become stuck, the remaining sugar is

mainly fructose. According to Gäfner and Schütz (1996), it is possible to predict stuck fermentation when the glucose/fructose ratio (GFR) is under 0.1. It is now established that genetic variations exist in the capacity of yeast to consume fructose (Luyten *et al.*, 2005; Guillaume *et al.*, 2007). A method developed to measure the 'fructophilic index', to help determine the ability of particular yeast to consume fructose, was used to evaluate the fermentation performance of selected yeasts under oenological conditions, paying attention to their capacity to consume fructose (Ortiz-Julien *et al.*, 2008). The yeasts evaluated differed in their capacity to consume fructose, even under different nitrogen, temperature or GFR levels which is an indicator of performance in potentially problematic must, where the GFR is lower and/or the must conditions are difficult.

The third step is to calculate the amount of yeast required for the total volume of stuck wine at 50 g/hL. Use twice this amount, 100 g/hL, if good temperature control in the cellar is lacking. Once the selected rescue yeast is properly rehydrated following the recommendations mentioned earlier in this chapter, activate the prepared rescue yeast with nutrients and sugar. The nutrient content of the stuck fermentation will most likely be quite low and unable to support adequate yeast growth. In addition, the rescue yeast culture will require adaptation to the alcohol content of the wine. This can be done by preparing an initial starter mixture at 25–30 °C and then adding the rescue yeast suspension. The starter mixture is made up with 2.5% of the stuck wine, an equal volume of water, 50 g/hL of a complex yeast nutrient to the wine and water mix. Adjust the sugar level of this mixture to 5 °Brix with juice, concentrate or sugar.

Once the rescue yeast is added to this initial starter culture mixture, the third step of starting the fermentation and adding the stuck wine in small increments can be monitored by checking the sugar level of the starter culture. When the sugar level has dropped by half (approximately 2.5 °Brix), begin to add the stuck wine to the starter and maintain between 20 and 25 °C. A very critical point is not to allow the sugar to be completely depleted. Some protocols suggest adding yeast cell walls at 20 g/hL to each batch prior to adding to the starter for providing nucleation sites as well as sorption of residual yeast inhibitors. The correct time to add a new batch of stuck wine is when the sugar from the previous addition has decreased by half. Only in the last batch of added stuck wine should the sugar be allowed to completely deplete.

1.13.1 Restarting using encapsulated yeast

An alternative method for restarting stuck fermentations has been used successfully for many years in secondary in-bottle fermentation. The use of encapsulated yeast technology reduces wine handling and helps to keep potential spoilage at a minimum. The technique of encapsulating yeast cells in alginate gel, a natural polysaccharide extracted from seaweed, allows the substrate and metabolites to move throughout the gel without releasing the *Saccharomyces* yeast cells into must. The encapsulated beads are made of two different layers: the inner layer containing the rescue yeast cells and the outer layer made of sterile alginate. The

rescue yeasts are acclimatized to a harsh wine environment during the encapsulation process. This treatment allows relatively easy direct inoculation into a stuck or sluggish fermentation. The best results for using encapsulated yeast technology in restarting stuck fermentations are when the instructions for use are strictly followed and when they are used to restart stuck wines that do not have very harsh limiting conditions such as high acetic acid or alcohol.

1.14 Conclusion

The yeast and fermentation management practices reviewed in this chapter should be applicable for most fermentation that winemakers will encounter. In spite of improved winemaking technology, there is still much to learn about properly managing yeasts and conducting healthy stress-free fermentations as sluggish and stuck fermentations are still a universal problem. The yeast used to ferment grape juice and the way that fermentation is managed is potentially a major factor which can influence wine quality. The beneficial contribution of yeast and how they are managed is becoming more significant, especially when viticultural practices have optimized grape quality and fermentation conditions allow the winemaker intervention and control. The future of yeasts is still untapped, and many yeast products selected from nature and developed have yet to be exploited.

1.15 Acknowledgements

Technical information was contributed to this text by: Clayton Cone, alcoholic fermentation and facility hygiene consultant; Dominique Delteil, private consultant and former scientific director at the Institut Cooperatif du Vin; Dr Anne Ortiz-Julien, Lallemend Oenology R&D director; Dr Sibylle Krieger-Weber, Lallemend Oenology MLF R&D director and Dr Graeme Walker, The University of Abertay Dundee.

1.16 References

- Alexandre H, Long TM, Feuillat M and Charpentier C (1994), Contribution à l'étude des bourbes: influence sur la fermentiscibilité des moûts, *Rev. Fr. Oenol.*, **146**, 11–19.
- Barre P (1992), Essai de mise en évidence de groupe d'activité killer parmi 55 souches de *Saccharomyces* isolées des vins, *Sci. Aliments*, **2**, 297–312.
- Barre P and Vézinhet F (1984), Evaluation towards fermentation with pure culture yeasts in winemaking, *Microbiol. Sci.*, **1**, 159–163.
- Basturia A and Lagunas B (1986), Catabolic inactivation of the glucose transport system in *Saccharomyces cerevisiae*, *J. Gen. Microbiol.*, **132**, 379–385.
- Bataillon M, Rico A, Sablayrolles JM, Salmon JM and Barre P (1996), Early thiamine assimilation by yeasts under enological conditions: impact on alcoholic fermentation kinetics, *J. Ferment. Bioeng.*, **82**, 145–150.
- Bely M (1990), Description of alcoholic fermentation kinetics: its variability and significance, *Am. J. Enol. Vitic.*, **41**, 319–324.

- Bezenger M and Navarro JM (1987), Grape juice fermentation: effects of initial nitrogen concentration on culture parameters, *Sci. Aliments*, **7**, 41–60.
- Birch RM and Walker G (2000), Influence of magnesium ions on heat shock and ethanol stress responses of *Saccharomyces cerevisiae*, *Enzym. Microb. Technol.*, **26**, 678–687.
- Bisson L and Butzke C (2000), Diagnosis and rectification of stuck and sluggish fermentations, *Am. J. Enol. Vitic.*, **51**, 168–177.
- Blateyron L, Aguera E, Dubois C, Gerland C and Sablayrolles JM (1998), Control of oxygen additions during alcoholic fermentations, *Vitic. Enol. Sci.*, **53**, 131–135.
- Block D, and Vlassides S (2000), Evaluation of cell concentration profiles and mixing in unagitated wine fermentors, *Am. J. Enol. Vitic.*, **51**, 73–80.
- Bohlscheid J, Fellman J, Wang X, Ansen D and Edwards C (2007a), The influence of nitrogen and biotin interactions on the performance of *Saccharomyces* in alcoholic fermentations, *J. Appl. Microbiol.*, **102**, 390–400.
- Bohlscheid J, Specht G, Ortiz-Julien A, Maloney J, Bertheau B, Ross C and Edwards C (2007b), Application of a new yeast preparation for problem grape musts, *J. Wine Res.*, **18**, 173–185.
- Ciani M and Ferraro L (1998), Combined use of immobilized *Candida stellata* cells and *Saccharomyces cerevisiae* to improve the quality of wines, *J. Appl. Microbiol.*, **85**, 247–254.
- Clemente-Jimenez JM, Mingorance-Cazorla L, Martínez-Rodríguez S, Las Heras-Vázquez F and Rodríguez-Vico F (2005), Influence of sequential yeast mixtures on wine fermentation, *Int. J. Food Microbiol.*, **98**, 301–308.
- Degré R (1993), Selection and commercial cultivation of wine yeast and bacteria, in Fleet G (ed.), *Wine Microbiology and Biotechnology*, Harwood Academic, Chur, Switzerland, 421–446.
- Delteil D (1998), Présentation d'une technique de débourage des jus blancs et rosés méditerranéens, *Rev. Fr. Oenol.*, **173**, 34–36.
- Delteil D (2001), Aspects pratiques du levurage en conditions méditerranéennes, technique d'inoculation et rapport entre population sélectionnée et population indigène, *Rev. Fr. Oenol.*, **189**, 14–19.
- Delteil D and Aizac T (1998), Comparison of yeast inoculation techniques by use of a 'marked' yeast strain, *Aust. NZ Wine Ind. J.*, **3**(3), 53–56.
- Delteil D and Jarry JM (1992), Characteristic effects of two commercial yeast strains on Chardonnay wine volatiles and polysaccharides composition, *Aust. NZ Wine Ind. J.*, **7**(1), 29–33.
- Delteil D and Lozano L (1995), Les critères de choix d'une souche de levure sélectionnée pour la vinification en blanc, *Rev. Fr. Oenol.*, **153**, 79–82.
- De Nicola R, Hall N, Bollag T, Thermogiannis G and Walker G (2009), Zinc accumulation and utilisation by wine yeasts, *Int. J. Wine Res.*, **1**, 1–10.
- Dombeck K and Ingram L (1986), Magnesium limitation and its role in apparent toxicity of ethanol in yeast fermentation, *Appl. Environ. Microbiol.*, **52**, 975–981.
- Dumont A (1994), *Sensory and chemical evaluation of Riesling, Chardonnay and Pinot Noir fermented by different strains of Saccharomyces cerevisiae*. Thesis. Oregon State University, Corvallis, OR.
- Edwards C and Bohlscheid J (2007), Impact of pantothenic acid addition on H₂S production by *Saccharomyces* under fermentative conditions, *Enzym. Microb. Technol.*, **41**, 1–4.
- Edwards C, Huang Y, Peterson J and Haag K (1996), Relationship between sluggish fermentations and the antagonism of yeast by lactic acid bacteria, *Am. J. Enol. Vitic.*, **47**, 1–10.
- Edwards C, Hagen K and Keller M (2008), Survey of biotin, pantothenic acid, and assimilable nitrogen in winegrapes from the Pacific Northwest, *Am. J. Enol. Vitic.*, **59**, 432–436.
- Eglinton J, Henschke P, Hoj P and Pretorius S (2003), Winemaking properties and potential of *Saccharomyces bayanus* wine yeast – harnessing the untapped potential of yeast biodiversity, *Aust. NZ Wine Ind. J.*, **18**(6), 16–19.

- Erasmus D, Van der Merwe G and Van Vuuren H (2003), Genome-wide expression analyses : metabolic adaptation of *Saccharomyces cerevisiae* to high sugar stress, *FEMS Yeast Res.*, **3**, 375–399.
- Ferraro L, Faticenti F and Ciani M (2000), Pilot scale vinification process using immobilized *Candida stellata* cells and *Saccharomyces cerevisiae*, *Process Biochem.*, **35**, 1125–1129.
- Fugelsang K, Osborn M and Muller C (1992), *Brettanomyces* and *Dekkera*: their implications in winemaking, in Gump B and Pruett D (eds), *Analysis, Characterization and Technological Advances in Wine and Beer Production*, American Chemical Society Symposium Series, Washington, DC, 110–129.
- Feuillat M (1995), D  bourbage des mo  ts blancs, Trouver le bon   quilibre, *Viti.*, **195**, 49–50.
- Gafner J and Sch  tz M (1996), Impact of glucose-fructose-ratio on stuck fermentations: practical experience to restart stuck fermentations, *Vitic. Enol. Sci.*, **51**, 214–218.
- Guillaume C, Delobel P, Sablayrolles JM and Blondin B (2007), Molecular basis of fructose utilization by the wine yeast *Saccharomyces cerevisiae*: a mutated HXT3 allele enhances fructose fermentation, *Appl. Environ. Microbiol.*, **73**, 2432–2439.
- Heard G (1999), Novel yeasts in winemaking – looking to the future, *Food Aust.*, **51**, 347–352.
- Henick-Kling T (1988), Yeast and bacteria control in winemaking, in Linskens HF and Jackson JF (eds), *Modern Methods of Plant Analysis. New Series, Volume 6 Wine Analysis*, Springer-Verlag, Berlin Heidelberg, 276–315.
- Henschke P and Jiranek V (1993), Yeasts – metabolism of nitrogen compounds, in: Fleet GH (ed.), *Wine Microbiology and Biotechnology*, Harwood Academic Publishers, Chur, Switzerland, 77–164.
- Henschke P (1997a), Stuck fermentation; causes, prevention and cure, in Allen M, Leske P and Baldwin G (eds), *Advances in Juice Clarification and Yeast Inoculation*, Australian Society of Viticulture and Oenology, Adelaide, SA, 30–38, 41.
- Henschke P (1997b), Wine yeast, in Zimmermann K and Entian D (eds), *Yeast Sugar Metabolism: Biochemistry, Genetics, Biotechnology, and Applications*, Technomic, Lancaster, PA, 527–560.
- Howell K, Swiegers J, Eglinton J, Bellon J, Henschke P, Fleet G, H  j P, Pretorius I and de Barros Lopes M (2005), Microbiological tuning of wine composition, in Blair R, Williams P and Pretorius I (eds), *Proceedings of the Twelfth Australian Wine Industry Technical Conference*, Australian Wine Industry Technical Conference Inc., Adelaide, SA, 141–144.
- Jiranek V, Langridge P and Henschke P (1995), Regulation of hydrogen sulfide liberation in wine-producing *Saccharomyces cerevisiae* strains by assimilable nitrogen, *Appl. Environ. Microbiol.*, **61**, 461–467.
- Kirsop B (1982), Developments in beer fermentation, *Topics Enzyme Ferment. Biotechnol.*, **6**, 79–105.
- Kontkanen D, Inglis D, Pickering G and Reynolds A (2004), Effect of yeast inoculation rate, acclimatization, and nutrient addition on icewine fermentation, *Am. J. Enol. Vitic.*, **55**, 363–370.
- Kudo M, Vagnoli P and Bisson L (1998), Imbalance of pH and potassium concentration as a cause of stuck fermentations, *Am. J. Enol. Vitic.*, **49**, 295–301.
- Kunkee R (1991), Relationship between nitrogen content of must and sluggish fermentation, in Rantz, J. (ed.), *Proceedings of the International Symposium on Nitrogen in Grapes and Wines*, American Society of Enology and Viticulture, Davis, CA, 148–155.
- Lafon-Lafourcade S, Geneix C and Rib  reau-Gayon P (1984), Inhibition of alcoholic fermentation of grape must by fatty acids produced by yeasts and their elimination by yeast ghosts, *Appl. Environ. Microbiol.*, **47**, 1246–1249.
- Lambrechts M and Pretorius I (2000), Yeast and its importance to wine aroma – a review, *SA J. Enol. Vitic.*, **21**, 97–129.

- Larue F and Lafon-Lafourcade S (1989), Survival factors in wine fermentation, in Van Uden N (ed.), *Alcohol Tolerance in Yeasts and Bacteria*, CRC Press, Boca Raton, FL, 193–215.
- Lavigne V, Chatonnet P and Dubourdieu D (1993), Les défauts olfactifs de réduction d'origine fermentaire. Connaissance aromatique des cépages et qualité des vins, *Rev. Fr. Oenol.*, **149**, 288–296.
- Luyten K, Perez M, Michel R, Riou C and Blondin B (2005), Analysis of *Saccharomyces cerevisiae* hexose carrier expression during wine fermentation: both low- and high-affinity Hxt transporters are expressed, *FEMS Yeast Res.*, **5**, 351–361.
- Monk P (1997), Optimum usage of active dry wine yeast, in Allen M, Leske P and Baldwin G (eds), *Advances in Juice Clarification and Yeast Inoculation*, Australian Society of Viticulture and Oenology, Adelaide, SA, 22–23.
- Moreno J, Millán C, Ortega J and Medina M (1991), Analytical differentiation of wine fermentations using pure and mixed yeast cultures, *J. Ind. Microbiol.*, **7**, 181–190.
- Ortiz-Julien A, Languet P, Aguera E, Samson A and Salmon JM (2005), Valorisation aromatique des moûts par l'utilisation séquentielle de levure d'espèces non-Saccharomyces et Saccharomyces, *Rev. F. Oenol.*, **117**, 31–33.
- Ortiz-Julien A, Dumont A, Raynal C, Pillet O and Raginel F (2008), La capacité de consommation du fructose par les levures œnologiques, *Rev. F. Oenol.*, **129**, 15–18.
- Pillet O, Ortiz-Julien A and Salmon JM (2006), Réduire les risques d'arrêt de fermentation alcoolique : le rôle protecteur des stérols pendant la rehydratation phase des levures sèches actives, *Rev. F. Oenol.*, **210**, 25–28.
- Pretorius I (2007), Not all wine yeast are equal, *Microbiology Australia*, **28**, 55–58.
- Pretorius I, Henschke P and Bartowsky E (2008), Yeast and bacterial modulation of wine aroma and flavour, *Aus. J. Grape Wine Res.*, **11**, 139–173.
- Querol A, Barrio E, Huerta T and Ramón D (1992), Molecular monitoring of wine fermentations conducted by active dry yeast, *Appl. Environ. Microbiol.*, **58**, 2948–2953.
- Radler F, Dietrich K and Schöning I (1985), Mikrobiologische prüfung von trockenhefepräparaten für die weinbereitung, *Dtsch. Lebensmitt. Rundsch.*, **3**, 73–77.
- Rapp A, Versini G and Ullemeyer H (1993), 2-Aminoacetophenon: verursachende komponente der 'Untypischen Alterungsnote' (Naphtalinton, Hybridton) bei Wein, *Vitis*, **32**, 61–62.
- Rauhut D, Kurbel H and Dittrich H (1993), Sulfur compounds and their influence on wine quality, *Die Wein-Wiss.*, **48**, 214–218.
- Rauhut D, Kurbel H, Ellwanger S, Löhnertz O and Großmann M (1999), Influence of yeast strain, assimilable nitrogen, fermentation temperature and sulfur residues on the occurrence of volatile sulfur compounds during and after fermentation, *Proceedings, Oenologie 99, 6e Symposium International d'Oenologie*, Bordeaux/France, 10–12 June, 305–308.
- Sablayrolles JM and Barre P (1986), Evaluation des besoins en oxygène de fermentations alcooliques en conditions œnologiques stimulées, *Sci. Aliments*, **6**, 373–383.
- Sablayrolles JM, Barre P and Grenier P (1987), Design of a laboratory automatic system for studying alcoholic fermentations in anisothermal enological conditions, *Biotechnol. Tech.*, **1**, 181–184.
- Sablayrolles JM, Dubois C, Manginot C, Roustan JL and Barre P (1996), Effectiveness of combined ammoniacal nitrogen and oxygen additions for completion of sluggish and stuck fermentations, *J. Ferment. Bioeng.*, **82**, 377–381.
- Sablayrolles, J.M., Julien, A., Roustan, J.L and Dulau, L. (2000). Comparison of nitrogen and oxygen demands of enological yeasts: technological consequences, *Am. J. Enol. Vitic.*, **51**, 200–222.
- Sablayrolles JM, Soubeyrand V and Julien A (2006), Rehydration protocols for active dry wine yeasts and the search for early indicators of yeast activity, *Am. J. Enol. Vitic.*, **57**, 474–480.
- Salmon JM (1996), Sluggish and stuck fermentations: some actual trends on their physiological basis, *Vitic. Enol. Sci.*, **51**, 137–140.

- Schütz M and Gafner J (1993), Analyses of yeast diversity during spontaneous and induced alcoholic fermentations, *J. Appl. Bacteriol.*, **75**, 551–558.
- Soubeyrand V, Salmon J, Luparia V, Williams P, Doco T, Vernhet H and Julien A (2005), Formation of micella containing solubilized sterols during rehydration of active dry yeasts improves their fermenting capacity, *J. Agric. Food Chem.*, **53**, 8025–8032.
- Smith C, Rasmussen J, Schultz E, Snyder R and Jones R (1995), Acetic acid as a causative agent in producing stuck fermentations, *Am. J. Enol. Vitic.*, **46**, 278–280.
- Tominaga T, Murat M and Dubourdiou D (1998), Development of a method for analyzing the volatile thiols involved in the characteristic aroma of wines made from *Vitis vinifera* L. cv. Sauvignon blanc, *J. Agric. Food Chem.*, **46**, 1044–1048.
- Van Stevenink J and Ledeboer JM (1974), Phase transitions in the yeast cell membrane, The influence of temperature on the reconstitution of active dry yeast, *Biochim. Biophys. Acta.*, **352**, 64–70.
- Walker G and Maynard AI (1996), Magnesium-limited growth of *Saccharomyces cerevisiae*, *Enz. Microb. Technol.*, **18**, 455–459.
- Walker G, Birch R and Ciani M (2003), Magnesium, calcium and fermentative metabolism in wine yeasts, *J. Wine Res.*, **14**, 3–15.
- Wang X, Bohlscheid J and Edwards C (2003), Fermentative activity and production of volatile compounds by *Saccharomyces* grown in synthetic grape juice media deficient in assimilable nitrogen and/or pantothenic acid, *J. Appl. Microbiol.*, **94**, 1–11.
- Wolz S (2005), Extraction of mannoproteins and polysaccharides and their effect on aroma. Improvement of mouthfeel, *Dtsch. Weinmagazin*, **22**, 21–25.

Metabolic engineering of wine yeast and advances in yeast selection methods for improved wine quality

B. Divol and F. F. Bauer, Stellenbosch University, South Africa

Abstract: Most alcoholic fermentations in winemaking are nowadays performed by commercially produced yeast strains that are inoculated into the fermentation vessel. Although many yeast strains are available on the global market, the generation of new strains that display different and/or new traits continues to attract significant industrial interest. Many traditional breeding and selection, as well as genetic engineering, approaches have been used to achieve specific targets. These approaches can now be complemented with powerful systems biology tools which provide new insights and opportunities for the generation of new yeast strains while holistically improving our knowledge of wine microbial ecology.

Key words: wine yeasts, biotechnology, systems biology, yeast improvement, yeast identification.

2.1 Introduction

Since the initial findings of Pasteur (Pasteur, 1866), it has been well established that grape must serves as a natural habitat for a rich microflora. Pasteur, indeed, had already identified yeasts and bacteria and realized that some of these microorganisms were responsible for conducting alcoholic fermentation. During alcoholic fermentation, the diverse microflora found in the grape must is usually out-competed by yeasts belonging to the genus *Saccharomyces*. Species belonging to this genus are generally more tolerant to high sugar concentrations and resist

Table 2.1 Examples of commercial strains selected for specific traits with regard to fermentation efficiency and contribution to the organoleptic profile of wine

Selected trait	Commercial strain	Providing company
<i>Fermentation efficiency</i>		
Quick onset of fermentation	522 Davis	Laffort Oenologie (Bordeaux, France)
	EC1118®, lalvin T73®	Lallemmand (Blagnac, France)
	Levuline CER	OenoFrance (Reims, France)
Sparkling wines (either for first or second fermentation)	Fermivin PDM	DSM (Heanor, UK)
	PM	Laffort Oenologie
	Lalvin C®	Lallemmand
	Levuline AGGLO,	OenoFrance
	Levuline CHP	
'Difficult fermentation', restarting stuck fermentations	Fermichamp	DSM
	BO213	Laffort Oenologie
	Lalvin EC1118®, Uvaferm 43™	Lallemmand
	Levuline FB	OenoFrance
Fermentation in low pH, high total SO ₂ , low temperature	Lalvin DV10®	Lallemmand
Fermentation in cold temperature	WE372	Anchor Yeast (Eppindust, South Africa)
Killer phenotype	F5	Laffort Oenologie
	Levuline Killer	OenoFrance
<i>Contribution to organoleptic profile</i>		
Enhancing cultivar typicity	'Collection cépage' range	DSM
	VL1, VL2, VL3C	Laffort Oenologie
	'Vitilevure' range	Martin Vialatte (Reims, France)
	Enoferm BDX®, BM4X4®,	Lallemmand
	Lalvin BM45®,	
	Lalvin CY3079®,	
	'Lalvin ICV®' range	
	Levuline Ribera,	OenoFrance
	Levuline Lumaú	
Improving secondary aroma	Lalvin Rhône 4600®,	Lallemmand
	Lalvin Rhône 2323®,	
	Lalvin Rhône 2056®,	
	Lalvin R2®,	
	Lalvin C®, Lalvin 71B®	
	Levuline Arpège, Levuline	OenoFrance
	Nov'Eline, Levuline Primeur	
	RB4	Laffort Oenologie
<i>Intraspecific hybrids</i>		
Enhancing the organoleptic potential of white/red wines	VIN 13/NT202	Anchor Yeast
Fermentations at low temperature, low nitrogen content and high potential alcohol	Cross Evolution	Lallemmand

cont.

Table 2.1 *cont.*

Selected trait	Commercial strain	Providing company
<i>Interspecific hybrids</i>		
Quick onset of fermentation, recommended for low temperatures	W27, W46 (both <i>S. cerevisiae</i> × <i>S. kudriavzevii</i>)	Lallemand
Enhancing the organoleptic potential of white wines	Exotics SPH (<i>S. cerevisiae</i> × <i>S. paradoxus</i>)	Anchor Yeast

increasing concentrations in ethanol more efficiently than most other organisms. Several species have been identified as being able to complete alcoholic fermentation including *S. cerevisiae*, *S. bayanus* and *S. paradoxus*.

Today, most wines are no longer fermented by the natural microflora. Alcoholic fermentation is rather conducted by selected commercial yeast strains that are inoculated into the must directly after pressing. These strains have been selected because they possess specific traits that can benefit wine production and/or product quality (Table 2.1). In particular, such strains generally provide for a better control of the fermentation process. Nevertheless, and although there are several hundred commercial wine yeast strains available on the global market, most, if not all characteristics of wine yeast strains can still be improved, and wine yeast producers continually search for strains that would help to better address specific production problems or quality parameters.

Traditionally, the selection of commercial wine yeast strains focused on generating strains that displayed satisfying fermentation profiles. Since then, many other aspects of wine yeast strains with an impact on wine processing and on wine quality have become the target of strain improvement programs. Many aspects of wine processing can be improved through yeast strains with enhanced or new abilities, such as the secretion of specific hydrolytic enzymes. Moreover, yeast strains impact significantly on the quality of the final product through traits such as the formation of secondary aroma compounds (e.g. higher alcohols and esters), the hydrolysis of grape components such as berries cell walls (e.g. through the secretion of pectinolytic enzymes) and the subsequent liberation of aroma precursors revealing the variety flavours. More recently, yeast selection has also focused on many additional traits that are related to new consumer demands with regard to ecologically sustainable production practices and the health-related properties of wine.

Initially, all commercial strains were strains directly isolated from natural or industrial habitats such as grape surface, must, wine, winery walls and equipment. For several decades, such direct isolation remained the preferred approach for finding new strains of commercial interest. However, in the 1990s, more focused strain development based on traditional biotechnological methods such as breeding and mutagenesis gained some popularity. Such methods allow for the improvement of existing strains in a random, non-directed way, and require significant efforts in terms of selection strategies. These methods remain popular to date, and many of the

new yeast strains released on the market in previous years have been the results of such projects. These methods have also been complemented with approaches that are based on 'directed' or 'adaptive' evolution that lead to the targeted improvement of specific traits that can be the subject of evolutionary selection pressure. Finally, the emergence of molecular biology and biotechnology has recently opened new horizons for the creation of new strains based on specific and targeted improvement of existing strains. Whereas the traditional methods, including directed evolution, do not require any *a priori* knowledge about the specific molecular features of a specific trait or phenotype (structural genes, proteins and their mode of action as well as regulation), molecular biology approaches are by definition knowledge-based. The improvement is targeted because the specific gene or function to be improved or newly added is well understood and therefore has become amenable to such an approach. As a matter of principle, such an approach should also allow for specifically targeted improvement, without changing other traits of the strain to be improved. This is different from any of the traditional approaches, where modifications cannot be targeted towards single traits. However, although significant scientific progress has been made in this field, the very limited acceptance of this technology in a conservative industry has thus far limited its impact. Indeed, only two GM wine yeast strains have to date received FDA (Food and Drug Administration) approval for the North American market.

An additional development over the past decade has been a significant regain of interest in natural non-*Saccharomyces* yeasts found in grape juice. Some of these strains have been shown to positively contribute to the organoleptic properties of wine, or to be efficient fermentation aids in sub-optimal conditions. In order to enrich wine complexity, the inoculation of such species into grape juice has been studied and successfully implemented in commercial wineries. As most of these strains are not able to complete alcoholic fermentation on their own, co-inoculation with *S. cerevisiae* is frequently necessary. Various species have been selected and reviewed recently (Fleet, 2008) to serve several purposes. Such species generally can provide certain traits that are absent in *S. cerevisiae* leading, for example, to the release of new flavour compounds, the formation of increased concentrations of glycerol or lactic acid, the production of less volatile acidity and the consumption of malic acid. Similar aims can of course also be achieved through recourse to biotechnology and genetic engineering by introducing these characters into *S. cerevisiae*.

This chapter summarizes the targets for improving wine yeasts, describes the newly available tools for gaining knowledge on *S. cerevisiae* and provides examples of outputs that can be generated from the traditional and cutting-edge microbiological and molecular techniques.

2.2 Improving wine yeasts: current targets

Ecological studies suggest that the natural ecological niche of *S. cerevisiae* strains is likely to be habitats associated with high sugar environments where rapid fermentation and rapidly rising ethanol levels can provide this yeast with a

significant selective advantage over other species. As a consequence, it is this yeast species that dominates most industrial processes that are linked to the fermentation of simple hexoses such as winemaking and brewing.

From a process perspective, the essential aim of winemaking is that alcoholic fermentation needs to be completed (the exception being wines intended to be naturally sweet). In wine, completion refers to achieving a very low level of residual sugar (the definition of a dry table wine requires less than 5 g of sugar/L) keeping in mind that the initial sugar level in an average grape must ranges from 180 g/L to 250 g/L. However, grape must is a highly variable substrate not only with regard to sugar content, and frequently does not provide for optimal growth conditions. Factors such as excessively high concentration of sugars, purposefully maintained cold temperature (to maintain fruity aroma), presence of inhibitory compounds such as botryticine, the lack of assimilable nitrogen, oxygen or of vitamins, can result in alcoholic fermentation becoming sluggish or even stuck, leading to the formation of off-flavours, the development of spoilage microorganisms and other serious problems. When the first yeast strains were commercialized as dry active yeast for direct addition in grape juice, the main objective of these strains was therefore to provide winemakers with a means to reliably complete alcoholic fermentations. Over the years, numerous strains have been isolated from indigenous fermentation for their particular oenological properties. While fast completion of alcoholic fermentation remained an essential criterion for selection, new criteria arose over time such as completion of fermentation in 'difficult conditions', the ability to out-compete indigenous strains, the tolerance to various stresses and formation of multiple aroma compounds thereby improving the overall quality of wine.

If all the above criteria related to fermentation performance are met, other important considerations relate to the various aroma compounds that contribute to the overall quality of wine and that are synthesized *de novo* by the yeast or are being released from precursors in the must through the action of various yeast enzymes. Wine yeast strains indeed differ significantly in their ability to produce such flavour and aroma-active compounds or enzymes, and specific yeast impart highly specific organoleptic characteristics to the wine.

Many of these specific traits were improved through strain selection and improvement programs based on hybridization techniques or random mutagenesis. Such methods have been recently reviewed (Cebollero and Gonzalez, 2004; Giudici *et al.*, 2005; Pretorius and Høj, 2005; Schuller and Casal, 2005; Donalies *et al.*, 2008). However, the potential of these techniques is limited to improving the natural genetic potential available within the species *S. cerevisiae*. With the emergence of molecular biology and the possibility of introducing new traits in yeast via inter- and intraspecific breeding or genetic engineering, yeasts have been regarded as the potential vehicle to extend the properties of strains to factors such as the reduction of processing problems, the biocontrol of spoilage microorganisms, the production of enhanced or new compounds responsible for increasing the organoleptic profile or the global wholesomeness of wine.

The other targets for improving wine yeast strains are numerous and have been

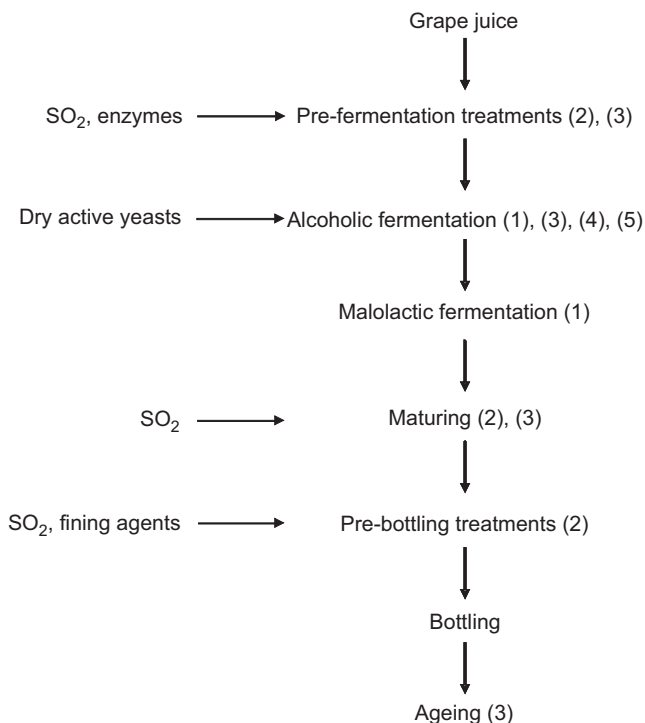


Fig. 2.1 Levels of action for potentially improved yeasts during winemaking: (1) improved fermentation performance; (2) enhanced processing efficiency; (3) contribution to biocontrol of spoilage microorganisms; (4) enhanced production of aroma compounds; (5) contribution to wine wholesomeness.

extensively reviewed (Ramon, 1997; Rainieri and Pretorius, 2000; Dequin, 2001; Pretorius and Bauer, 2002; Bisson, 2004; Giudici *et al.*, 2005; Pretorius and Høj, 2005; Schuller and Casal, 2005; Cebollero *et al.*, 2007). Besides improving fermentation performance and enhancing wine flavours and aroma as described above, they comprise several other broad target areas, including improving wine processing efficiency, producing healthier wines and improving microbial biocontrol. Depending on the target selected for yeast improvement, various levels of winemaking may be influenced, not only alcoholic fermentation (Fig. 2.1).

More specifically, the main focuses for improving fermentation performance are generally stress tolerance and especially ethanol tolerance, sugar consumption – especially of fructose (Berthels *et al.*, 2008; Santos *et al.*, 2008) – nitrogen assimilation (Salmon and Barre, 1998), resistance to antimicrobial compounds such as killer toxins and sulphur dioxide, and foam formation.

The key factors for improving wine processing efficiency are protein and polysaccharide clarification (Fernandez-Gonzalez *et al.*, 2005; Louw *et al.*, 2006; van Rensburg *et al.*, 2007), cell sedimentation and flocculation as well as cell flotation and flor formation.

Microbial biocontrol can be achieved by enhancing the production of sulphur dioxide and the production of killer toxins directed towards other strains of *S. cerevisiae* and, using genetic engineering, towards other species via the introduction of genes coding for antimicrobial enzymes or peptides such as lysozyme (Ibrahim *et al.*, 2001), glucanases, chitinases (Carstens *et al.*, 2003), bacteriocins (Schoeman *et al.*, 1999; van Reenen *et al.*, 2003) and zymocins (Yap *et al.*, 2000).

The production of healthier wine via the compounds produced by *S. cerevisiae* during alcoholic fermentation can be achieved by the production of resveratrol, a compound responsible for the so-called 'French paradox'. This compound could either be directly anabolized by the yeast cell via the introduction of the stilbene pathway (Becker *et al.*, 2003) or be released from the grape compounds by the production of β -glucosidases by *S. cerevisiae*. Reduced formation of ethyl carbamate, a potentially carcinogenic compound, can be achieved by manipulating the amino acid metabolism leading to urea formation (Coulon *et al.*, 2006). Biogenic amines are produced by lactic acid bacteria in wine. Their formation could be avoided if the bacterial strains responsible for their production were absent. This could be achieved by the production of bacteriocins by *S. cerevisiae*. Finally, there is a growing consumer demand for reduced-ethanol wines. Manipulating the metabolic pathways leading to the formation of ethanol (Remize *et al.*, 1999; De Barros Lopes *et al.*, 2000; Remize *et al.*, 2000; Henricsson *et al.*, 2005) or introducing an enzyme diverting the glucose flux from ethanol production (Malherbe *et al.*, 2003) might be the means to achieve this goal.

Enhancing wine flavours and aroma is one of the main goals for improving wine yeast. The current targets are the release of more grape terpenoids from their sugar moiety via the production and secretion of β -glucosidases (Arévalo Villena *et al.*, 2005; van Rensburg *et al.*, 2005) and the *de novo* production of terpenoids by *S. cerevisiae* (Herrero *et al.*, 2008), the increased production of desirable volatile esters and higher alcohols (Lilly *et al.*, 2006a,b), the increased production of glycerol (Remize and Dequin, 1998; Eglinton *et al.*, 2002), the diminution of wine acidity via the conversion of malic acid to lactic acid (Volschenk *et al.*, 1997; Husnik *et al.*, 2006) or to ethanol (Volschenk *et al.*, 1997), the optimization of phenolics metabolism, the reduced production of sulphite and sulphide (Romano *et al.*, 1985) and the release of volatile thiols (Howell *et al.*, 2004).

The total number of genetically modified industrial yeasts ready for application was reviewed some years ago (Akada, 2002; Schuller and Casal, 2005). Since then, the list has grown significantly. However, consumers are still very reluctant to accept the idea of GM food. Furthermore, winemaking is still considered as a traditional art, even if this conception can now be considered somewhat outdated. Nevertheless, the idea of 'transgenic wine' made by GM yeasts is not ready to be accepted but, as stated by Ceborello *et al.* (2007), 'not yet, but an evolution is to be expected during the next few years, especially in New World countries'. For proof, two GM wine strains have been granted the 'Generally Regarded As Safe' status in North America: ML01, a strain able to perform malolactic fermentation during alcoholic fermentation and 522^{EC}, a strain producing lower amounts of urea and consequently lower amounts of ethyl carbamate.

2.3 A systems biology approach to wine yeast studies

2.3.1 Systems biology: a revolutionary approach to understanding yeast

Over the past century, *S. cerevisiae* has been extensively studied. Its importance in the fermented food industry, such as the baking, brewing and wine industries, and its relative simplicity as a eukaryotic organism has made this yeast one of the central model systems for biological studies. The *S. cerevisiae* genome was the first eukaryotic genome to be sequenced (Goffeau *et al.*, 1996). Genetic and biochemical studies have allowed the unravelling of the functioning of individual parts of the cell, including for instance an in-depth understanding of glycolysis (the conversion of sugar to ethanol), the requirement for nitrogen and vitamins, the mechanisms involved in stress resistance and the formation of aroma compounds during alcoholic fermentation. This gene-by-gene or protein-by-protein approach has been very productive in the past and is still extensively used in laboratories worldwide. Nevertheless, if these mechanisms are today relatively well known, we do not fully understand how the different parameters integrate with each other. Attempts to model the overall fermentation system (Sheridan *et al.*, 2005) are limited to understanding the inputs to and outputs from the cell, while our knowledge of the integrated biological networks that constitute a living cell remains limited.

Recent developments in the biological sciences allow complementing the traditional biological sciences by more holistic or systems biology approaches that investigate biological processes on the whole cell level. These holistic approaches are referred to as 'omics', grouping individual fields such as genomics, transcriptomics, proteomics, metabolomics, fluxomics and interactomics (Fig. 2.2). These new approaches aim at analyzing and monitoring entire groups of macromolecules such as DNA, RNA, proteins and metabolites on a whole cell, helped by the emergence of new data analysis tools recently developed in the statistical sciences (e.g. multivariate data analysis) and in bioinformatics (Palsson, 2002; Lavine and Workman, 2006).

Genomics aims at understanding gene expression and interaction in the genome. Unravelling the response and adaptation of the cell to environmental stimuli at the genetic level is the main goal of this technology. It will allow comprehending the differences between strains and the metabolic pathways associated with stress response and environmental changes. Whole-genome studies of deletion mutants cultivated in the presence of various chemicals and in various environmental conditions can, for instance, be conducted in order to identify the phenotype associated with specific genes of unknown functions (Hillenmeyer *et al.*, 2008).

Transcriptomics aims at unravelling the effect of the global physiological and metabolic machinery on transcription by identifying simultaneously expressed genes. This can contribute to the characterization of the role of various genes. The major tools used for this type of study are DNA microarrays (Schena *et al.*, 1995) and derivatives (see Section 2.4.1).

Proteomics aims at identifying and characterizing complete sets of proteins as well as protein–protein interactions. This type of approach is usually combined with complete transcription data in order to establish an exhaustive cellular map.

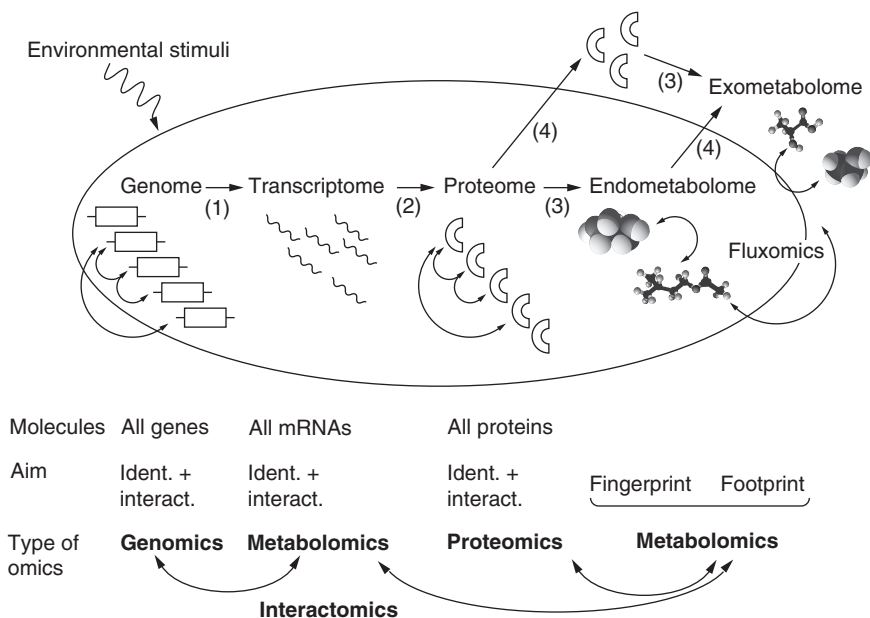


Fig. 2.2 The different ‘-omics’ involved in systems biology: (1) transcription; (2) translation; (3) biochemical reactions; (4) secretion. Ident. = identification (of function or new molecule); interact. = interactions with each other.

Various methods are available to conduct proteomic studies, such as two-dimensional gel electrophoresis (O’Farrell, 1997). A large number of proteins are detectable via this technique, but it suffers a poor reproducibility between gels. Moreover, all proteins cannot be retained on the gels and only proteins that are already identified can be isolated. High-throughput gas chromatography (GC) coupled with mass spectrometry (MS) has thus been set up for better accuracy of identification. In parallel, two-dimensional nano-liquid chromatography (LC)–mass spectroscopy is now used in order to separate peptides obtained from proteins digests. Powerful systems such as iTRAQ allow comparing two sets of proteins with such systems. Networks of protein–protein interactions can potentially also be identified (Han *et al.*, 2004).

Metabolomics aims at exhaustively identifying the entire set of metabolites present under various life conditions. Metabolites can be first separated and then identified by GC or LC and MS, respectively. In wine chemistry, tremendous progress has been made in the years since 2000 and laboratories are now able to detect and quantify a large number of metabolites present in the fermenting yeast and in wine. Complex phenotypes can be characterized through metabolomic studies, and complex molecular responses to environmental stimuli can therefore be unravelled and new metabolic pathways discovered (Villas-Boas *et al.*, 2007). These studies allow footprint or fingerprint processes occurring in response to developmental, genetic or environmental stimuli to be identified. The term foot-

print refers to the exometabolome (i.e. all the metabolites secreted outside the cell) whereas the term fingerprint refers to the endometabolome (i.e. all the metabolites within the cell). Metabolomics studies are challenged by extremely large data sets that are generated, aligned and compared. In wine, another challenge is created by the fact that the exometabolome is an amalgam of the metabolites originating the grapes, the yeasts and the bacteria (Borneman *et al.*, 2007).

Fluxomics aims at integrating the metabolites levels and their fluxes. This will provide a model of the cellular reaction networks. Flux can be measured by methods based on nuclear magnetic resonance (NMR) spectroscopy and mass spectrometry. In yeast, fluxomics studies can be conducted to understand phenotypes changes and carbon flux following any perturbation of the system (Nielsen, 2003).

The ultimate level of systems biology has been coined in the word interactomics. This aims to compare and integrate all the previously described levels. This is essential for gaining the final understanding of the whole system. Comparisons of yeast transcriptomes and proteomes cultivated under various conditions have been conducted (Kolkman *et al.*, 2006). In wine yeasts, a successful comparison between transcriptome and exometabolome has very recently enabled predicting the impact of changes in expression level of individual genes involved in the production of aroma compounds (Rossouw *et al.*, 2008).

In brief, systems biology aims at deciphering the complex network of molecular interactions in continuous evolution in response to environmental changes. Applied to wine yeast, the techniques associated with the different levels of systems biology contribute to our global understanding of the molecular adaptation of yeast throughout the course of alcoholic fermentation, the interactions between the grape juice/wine and the cells and between the living microorganisms present in the medium. Ultimately, the impact of the yeast on the final wine composition could be modelled.

2.3.2 Application of systems biology techniques in wine yeast studies

Oenology and the yeast sciences have benefited from the fact that *S. cerevisiae* is the favourite model system for biological sciences in general. Therefore, wine molecular biologists have immediately foreseen the potential of such new technologies (Borneman *et al.*, 2007; Pizarro *et al.*, 2007; Moore *et al.*, 2008) that are now used in oenology laboratories worldwide. Most of the oenologically relevant traits of wine yeast strains are of a complex nature, with many genes, proteins and metabolites interacting to create a final physiological output. To improve such systems in an efficient and targeted manner, an in-depth understanding of these networks is essential. The complexity of these networks, however, cannot be understood through the study of individual components. Many, if not all, of the properties of such networks indeed emerge from the system itself, and the impact of perturbations to such networks cannot be predicted based on the understanding of individual or modified components alone. Systems biology approaches, therefore, provide opportunities to understand and

model the entire system, and should therefore lead to new biotechnological approaches that will result in adjustments to the behaviour of the system as a whole and not just that of individual components.

For wine yeast biotechnologists, understanding the molecular changes occurring in *S. cerevisiae* in a dynamically changing environment (i.e. grape juice under alcoholic fermentation) is therefore currently the major challenge. Individual studies have already focused on the global cell responses to various stresses on the transcriptomics level, the global behaviour throughout alcoholic fermentation (Rossignol *et al.*, 2003; Varela *et al.*, 2005), as well as under high sugar stress (Erasmus *et al.*, 2003), high ethanol stress (Alexandre *et al.*, 2001), nitrogen-limitation stress (Mendes-Ferreira *et al.*, 2007) and finally under the stress of rehydration and inoculation (Rossignol *et al.*, 2006). These approaches have also been applied to genetically-engineered *S. cerevisiae* strains, such as the so-called malolactic yeast (Husnik *et al.*, 2006). These transcriptomic data have also been linked with proteome (Zuzuarregui *et al.*, 2006; Salvadó *et al.*, 2008) and metabolome data (Beltran *et al.*, 2006; Rossouw, *et al.*, 2008). While these studies are still preliminary, the data already indicate that predictive models of complex systems can be generated.

Other studies of the exometabolome provided a footprint of individual strains that were compared with each other and with blended yeast cultures (Howell *et al.*, 2006). Visible and near-infrared spectroscopies were also recently used in order to establish the fingerprint of various species. The latter authors predicted that this metabolomic study will allow identifying the function of genes producing unknown phenotypes (Cozzolino *et al.*, 2006).

2.4 Biotechnology, systems biology and the generation of new yeast strains

2.4.1 The breeding and other ‘traditional’ approaches

Breeding has long been regarded as a way to shuffle the genetic potential of two strains, while creating new strains combining the positive characteristics of both parents, after a long genotypic and/or phenotypic screening.

Since the mid-1980s, intraspecific breeding has been used to generate improved wine yeasts (see Table 2.1 for examples of commercialized intraspecific hybrids). For instance, a strain selected for its high flocculation phenotype, as well as its inability to form hydrogen sulphide together with a high fermentation rate and high ethanol production was generated (Romano *et al.*, 1985). More sophisticated breeding approaches are now used to create improved strains with a strong oenological potential. For example, successive hybridization and segregation steps can lead to the targeted isolation of clones that combine a large number of oenological traits derived from the two parents (Marullo *et al.*, 2006).

Interspecific breeding has also been implemented by wine microbiologists via mating, since it was demonstrated that natural hybrids within the genus *Saccharomyces* *sensu stricto* exists and that ‘directed’ mating was possible (Masneuf *et al.*,

1998; De Barros Lopes *et al.*, 2002; González *et al.*, 2006, 2007; Gangl *et al.*, 2009). Hybrids between *S. cerevisiae*, *S. bayanus* and *S. kudriavzevii* have been isolated by the latter authors (see strains W27 and W46 in Table 2.1). *S. cerevisiae* and *S. bayanus* have been bred in order to create hybrids with improved fermentation efficiency at low temperature (Sato *et al.*, 2002). Hybrids between *S. cerevisiae* and *S. paradoxus* have also been recently generated (Mocke, 2005) (see strain Exotics SPH in Table 2.1).

Random mutagenesis might also lead to the creation of strains potentially interesting from an oenological point of view. It is often performed by UV radiation or by contact with a chemical agent such as ethyl-methane sulphonate. A few wine strains have been generated through random mutagenesis. The new strains exhibited improved autolytic behaviour after second fermentation in sparkling winemaking (González *et al.*, 2003) or better nitrogen assimilation and fermentation performance (Salmon and Barre, 1998).

The use of DNA microarrays for the transcriptomics studies has now been tremendously diversified. DNA arrays are used to locate transcription factor binding sites (ChIP-chip), to identify protein–RNA interactions (RIP-chip) and to detect DNA duplicates, deletions and nucleotide polymorphism via comparative genome hybridization (CGH), as reviewed recently (Borneman *et al.*, 2007). This latter aspect is now used for the selection of wine yeast via intraspecific DNA polymorphism in given loci. These quantitative trait loci (QTL) can then be mapped and their phenotypic effects studied and exploited for selection purposes. The method has recently been applied to identify genetic markers of multigenic traits of oenological interest (Marullo *et al.*, 2007a). Coupled with DNA microarrays, the technique allows the generation of wine yeast strains through directed breeding. In a specific case, the generated strains exhibited reduced hydrogen sulphide and off-flavours production and improved fermentation kinetics (Marullo *et al.*, 2007b).

2.4.2 Potential of genetic engineering

The main hurdles in the creation of improved new yeast through these ‘traditional’ techniques come from the fact that the improvement is limited to the genetic potential of the yeasts and/or the fact that wine strains are diploid, aneuploid or even polyploid (Bakalinsky and Snow, 1990), but never haploid; therefore, the selection of a character carried by recessive alleles is difficult. Some of those limitations can be overcome by using genetic engineering. This technique is based on the introduction of specific DNA fragments such as genes encoding a desirable function or promoters allowing the up- or down-regulation of specific genes. The DNA fragment that is introduced can originate various sources: strains of the same species, strains of other more or less closely-related species and even organisms far from the recipient in the evolutionary tree. Genetic engineering is the only solution when the introduced trait is totally absent from *S. cerevisiae* and other closely-related species. The resource base is therefore nearly endless and the potential of genetic engineering surpasses the potential of traditional techniques. It is generally

assumed that the introduction of a small piece of DNA does not change the strain significantly beyond the targeted modification.

In order to control the success of an integrative transformation, a selectable marker is usually introduced alongside the gene of interest. In laboratory yeasts, deletion of certain amino acid synthesis-encoding genes allows selecting positive transformants through the recovery of the ability to grow without the specific amino acid added in the medium. Wine yeasts are usually prototrophic and thus the auxotrophic markers used for transformation of laboratory strains cannot be used for selection. To overcome this, a gene conferring resistance to a particular chemical compound is usually transformed alongside the gene of interest. Resistance to cyclohexamide has been used in transforming wine yeast (Pérez-González *et al.*, 1993). A more popular selectable marker has been the resistance to the antibiotic G418 (Jimenez and Davies, 1980; Wach *et al.*, 1994). Selectable markers other than antibiotic resistance have also been identified. These include resistance to sulphometuron (Petering *et al.*, 1991), blasticidin (Fukuda *et al.*, 1998) hygromycin B, nourseothricin and bialaphos (Goldstein and McCusker, 1999) and *p*-fluorophenylalanine (PFP) resistance (Cebollero and Gonzalez, 2004).

Nevertheless, the introduction of a gene conferring resistance for an antibiotic is only acceptable for research purposes, while the recombinant strain carrying this insertion cannot be used for industrial purposes. A system to loop out this undesirable gene has therefore been implemented (Güldener *et al.*, 1996).

Other types of cloning strategies have also been investigated. The construction of auxotrophic wine yeast strains through interruption of both (all) copies of the *URA3* gene has been suggested (Puig *et al.*, 1998). After transformation, selection can be done using *URA3* as would be done for laboratory strains. However, this would have to be carried out for every strain to be transformed and will be very time-consuming. Self-cloning has been recognized as a means to bypass this problem. In this case, the DNA introduced originates solely from yeast. This technique has especially been used for changing defective alleles into dominant or functional alleles, thereby conferring resistance to various chemical compounds. The dominant allele can also be directly introduced if the strain to be transformed does not possess this gene at all. These genes are then used as selectable markers. An example is the use of the K1 killer toxin gene as a dominant selectable marker (Boone *et al.*, 1990). Transcription of this gene produces the K1 killer toxin and also conveys resistance to the toxin to the yeast. In this case, the yeast K1 killer toxin gene is integrated into the genome of a wine yeast along with the gene of interest. Transformants are then selected on resistance to the killer toxin. This has additional benefits: firstly, it improves the yeast's anticontaminant properties against K1 killer toxin sensitive species, and secondly, it kills cells that lose the K1 killer toxin gene allowing, in theory, only the transformant to survive during the fermentation. In wine yeasts, genes such as *SSU1* and *FZF1* for enhanced sulphite resistance (Park *et al.*, 1999), *SFA1* for improved resistance to formaldehyde (Schmidt *et al.*, 1997), *CUP1* conferring resistance to copper (Hottiger *et al.*, 1995), etc. can also be used as selectable markers, as recently reviewed (Bisson, 2004).

One of the main fears of anti-GMO activists is the dissemination of these transgenic organisms in nature and their crossing with other related organisms, thereby creating an environmental threat. In the case of wine yeasts, this has been recently taken into great consideration and sterile wine yeasts have been created by deleting the *IME1* gene (Ramirez and Ambrona, 2008). The new strains are unable to form spores and therefore to survive after the end of alcoholic fermentation or to mate with wild yeasts.

2.4.3 Metabolic engineering

Considering the relatively recent development in systems biology and its exploitation by the wine molecular biologists, the future of improving wine yeast probably lies in metabolic engineering. This new technological ensemble is defined as 'the directed improvement of the cellular properties achieved from the interplay of theoretical analysis, replying on biochemical information and the application of genetic engineering' (Stephanopoulos *et al.*, 1998). A framework has been recently defined for integrating all the components of metabolic engineering (i.e. genomic information, bioinformatics tools and molecular biology) for improving wine yeasts (Pizarro *et al.*, 2007). Within this framework, the needs for new yeast strains are determined by the consumers' demand for new types and styles of wine and the producers' demand for improved process. The gene(s) subject to modification is (are) identified with the aid of bioinformatics tools. Biotechnology, directed evolution and/or mutagenesis are the key techniques for engineering the expected modification(s). The use of systems biology techniques, including the global analysis of the transcriptome, proteome and fluxome, is finally used to provide an overview of the effect of the modification within the entire cell system. This approach will allow a better and wider understanding of the potential side-effects of a specific genetic modification. Once validated, the new improved yeast can be used by the wine industry for creating new styles and better quality wines. A systems biology approach will probably facilitate the acceptance of GM yeasts by the public, as the global effect of genetic engineering will be revealed.

2.5 Molecular biology and systems biology in the identification of wine yeasts

Before the emergence of molecular biology, techniques to identify microorganisms in a complex microflora were based on their isolation and the determination of their phenotypic characteristics. However, these techniques only provide an incomplete representation of the real microflora (Brock, 1987). Firstly, the capacity of microorganisms to form colonies on solid medium can vary according to their physiological state. The identification is thus confined to microorganisms that were able to form colonies at a given time. It has indeed been recently shown that yeasts, as bacteria, can also enter a viable but non-culturable state (Del Mar Lleó *et al.*, 2000; Millet and Lonvaud-Funel, 2000; Mills, 2002; Blevé *et al.*, 2003;

Guillou *et al.*, 2003; Divol and Lonvaud-Funel, 2005). Secondly, it is also now well known that metabolic properties of individuals can also evolve through the loss of a metabolic function, the gaining of resistance, etc. The identification of microorganisms based on their ability to degrade various substrates or to resist to a few chemical compounds is thus biased. The use of molecular biology is now universally recognized as the most reliable means for yeast identification and, since the first phylogeny studies based on DNA (Zuckerlandl and Pauling, 1965), numerous techniques have been implemented, based on hybridization, karyotyping and polymerase chain reaction (PCR).

2.5.1 Hybridization and karyotyping

DNA–DNA hybridization is mostly used to identify wine lactic acid bacteria (Lonvaud-Funel *et al.*, 1991). Karyotyping is a powerful discriminatory technique based on pulsed field gel electrophoresis (PFGE) to discriminate between strains of *S. cerevisiae* (Frezier and Dubourdieu, 1992; Querol *et al.*, 1992b). Unlike baker's strains, it seems that wine strains have a relatively stable karyotype (Codón *et al.*, 1997), making this technique reliable in the study of wine yeast ecology. Nevertheless, recombination events between Ty transposons may occur, thereby creating a certain instability (Kidwell and Lisch, 2000). This method has recently been coupled with restriction enzyme analysis (REA) preceding the PFGE to discriminate between other yeast species such as *B. bruxellensis* in wine (Miot-Sertier and Lonvaud-Funel, 2007). Moreover, the recent development in systems biology techniques allowed to implement a new type of karyotyping through DNA microarray (Zuzuarregui *et al.*, 2006) and microarray karyotyping (Dunn *et al.*, 2005), which identify shared traits and unique signature between strains of *S. cerevisiae*.

Beyond hybridization and karyotyping, several PCR-based techniques have been developed in order to discriminate between yeast species and strains.

2.5.2 Polymerase chain reaction (PCR)-based interspecies discrimination

Species-specific PCRs have been set up in order to detect pre-selected yeast species in a given medium (Pearson and McKee, 1992). Nested-PCR has also been implemented in order to detect very low yeast populations such as those of *B. bruxellensis* during wine ageing (Ibeas *et al.*, 1996). More recently, real-time PCR has been used to couple the identification of a given yeast species and the direct quantification of its population, thereby avoiding plating and uncertainty linked to the potential viable but non-culturable (VBNC) state (Phister and Mills, 2003; Delaherche *et al.*, 2004; Rawsthorne and Phister, 2006). Universal primers allow amplifying a whole range of species (Casey and Dobson, 2004), but the use of real-time PCR for detecting many species at a time is still under investigation and only a few studies are available in the literature for the time being.

Nevertheless, this technique is not adapted to describing the full ecology of yeast in a complex living system. Other techniques have therefore been imple-

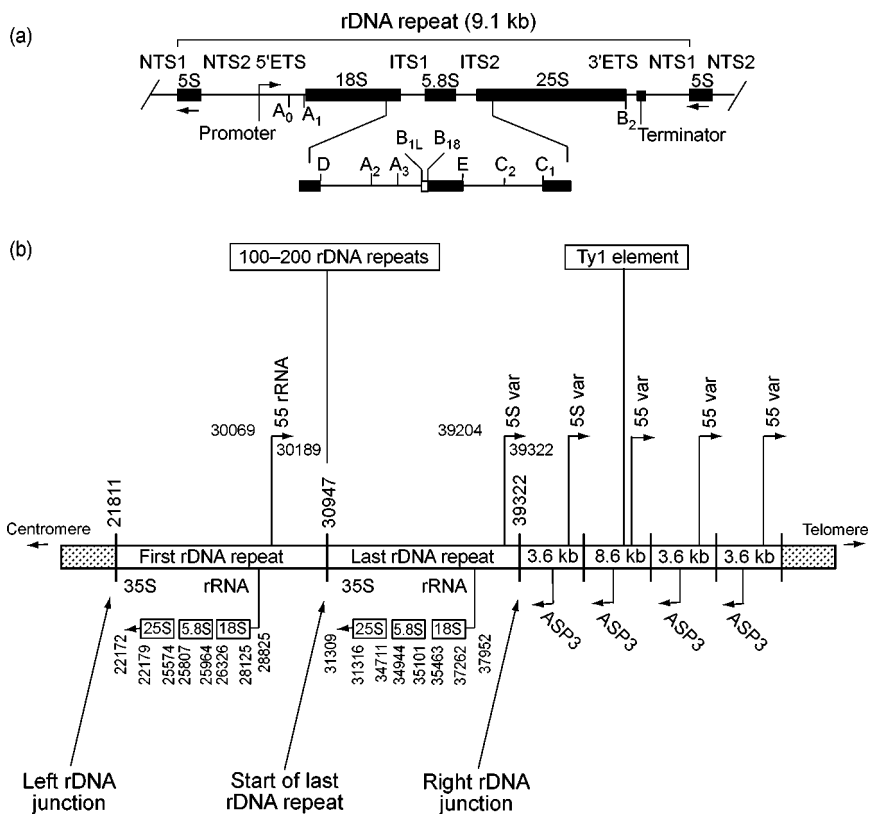


Fig. 2.3 RDN1 locus in *Saccharomyces cerevisiae*. (a) Pre-ribosomal RNA (before splicing) including one 35S rRNA, two non-transcribed spacers (NTS) regions interrupted by one 5S rRNA. The 35S rRNA comprises one 18S, one 5.8S and one 25S, separated by two internal transcribed spacers (ITS) regions. The 35S is flanked on each side by one external transcribed spacer (ETS) region (Venema and Tollervey, 1995). (b) Global representation of the RDN1 locus, located on the right strand of chromosome XII. This whole unit is repeated twice (Johnston *et al.*, 1997).

mented. They are mostly based on the amplification of various regions in the rDNA locus. This region of the genome is constituted by a highly repeated succession of hypervariable regions interrupted by highly conserved regions (Fig. 2.3).

PCR-restriction fragment length polymorphism (RFLP) is based on the amplification of the ITS1-5.8S rRNA-ITS2 region. The size of the amplicons obtained after PCR differs greatly between yeast species. Moreover, the discrimination is refined by a restriction analysis using different enzymes. The results are compared to those of databases (Esteve-Zarzoso *et al.*, 1999). This technique has been successfully applied to wine microbial ecology studies (Guillamón *et al.*, 1998; Granchi *et al.*, 1999; Pramateftaki *et al.*, 2000). It has even been implemented to differentiate between species within a mixture of species in wine (Espinosa, *et al.*, 2002; Divol and Lonvaud-Funel, 2005), but this technique is not highly efficient

for this particular purpose. PCR–RFLP is one of the simplest ways to identify colonies, but not yeast within mixed populations.

One of the most commonly applied techniques to serve this latter purpose is the PCR–denaturing gradient gel electrophoresis (DGGE). The principle of this technique lies behind the amplification of a hypervariable DNA region (most commonly in the rDNA locus) and its following separation on a denaturing gel; the migration length is dependent on the sequence, not on the length of the amplicon. This method allows separating fragments of identical size with varying sequences. By PCR focusing on the DNA extracted from a mixture of yeast species, the electrophoresis allows separation of the different species-specific bands. Denaturing can be achieved by a temperature (thermal gradient gel electrophoresis – TGGE) or a chemical gradient (DGGE). The application of TGGE is limited as it seems that DGGE is better suited for this type of study. TGGE was nonetheless used to discriminate wine yeast species (Fernández-González *et al.*, 2001). DGGE has been extensively applied to discriminate between wine yeast species (Cocolin, *et al.*, 2000, 2001, 2002; Mills, 2002; Divol and Lonvaud-Funel, 2005; Renouf *et al.*, 2006). The use of reverse-transcription-PCR–DGGE was recommended by Del Mar Lleó *et al.* (2000) in order to avoid the amplification of dead yeast and was applied in wine ecology (Divol, 2004; Hierro *et al.*, 2006).

2.5.3 Intraspecies discrimination

One of the main techniques used to discriminate between wine yeast strains is a PCR based on amplifying the region comprised between two δ elements (LTR regions of the Ty2 transposons in *S. cerevisiae*). It was first implemented (Ness *et al.*, 1993; Masneuf and Dubourdieu, 1994) and later improved (Legras and Karst, 2003). This is the most commonly used technique because of its simplicity and its robustness.

The PCR–RFLP technique has been applied to discriminate between *S. cerevisiae* strains on mitochondrial DNA (Querol *et al.*, 1992a). However, extracting mDNA is technically delicate and time-consuming.

A very powerful intraspecific discrimination tool has been implemented very recently. This technique is based on the amplification of simple sequence repeat (also known as microsatellites) loci dispersed throughout the genome. The length polymorphism of these repeats allows discriminating *S. cerevisiae* strains (Gallego *et al.*, 1998; Gonzalez-Techera *et al.*, 2001; Perez *et al.*, 2001; Legras *et al.*, 2005). This technique has even been recently made more discriminatory by coupling the amplifications of various microsatellites at the same time through multiplex PCR (Vaudano and Garcia-Moruno, 2008). It has thus become one of the most popular techniques for intraspecies discrimination.

The main drawback to the techniques previously described is that they allow discrimination of *S. cerevisiae* strain only. The presence of microsatellites and transposons highly repeated throughout the entire genome is indeed characteristic of *S. cerevisiae*. Other techniques can nevertheless be used for intraspecific differentiation.

Random amplified polymorphic DNA (RAPD) uses the existence of short repeated sequences dispersed throughout the genome. Only one primer is used and the elongation size is random. This technique has been used to discriminate wine strains regardless of the species (Quesada and Cenis, 1995).

Amplified fragment length polymorphism (AFLP) is based on the PCR amplification of DNA fragments after restriction digest. It has also been applied to differentiate wine strains regardless of the species (De Barros Lopes *et al.*, 1999; Flores Berrios *et al.*, 2005; Curtin *et al.*, 2007).

Very recently, multilocus sequence typing (MLST) has been proposed for discriminating between strains of the genus *Candida* (restricted to species of clinical interest). The highly resolutive method, first proposed for bacteria, has been applied to yeasts of clinical interest, including various species of *Candida* (Bougnoux *et al.*, 2002; Jacobsen *et al.*, 2007, 2008; Lin *et al.*, 2007). This technique is based on the analysis of nucleotide polymorphisms of the sequences of 450–500-bp of internal fragments of housekeeping genes. This highly discriminatory technique should be tried in wine strains, although ploidy might render the amplifications poorly repeatable and the MLST analysis difficult.

2.6 Future trends

As highlighted above, the implementation of cutting-edge technologies defines the future of wine yeast research. Although ‘traditional’ techniques such as isolation, breeding, gene-by-gene or protein-by-protein approaches to molecular engineering have been very insightful in providing a better knowledge of wine strains with respect to their physiology and cellular metabolisms and fruitful in providing new strains to the wine industry, high throughput technologies now provide the means to better pre-define and understand the impact of specific biotechnological approaches. Therefore, the immediate future should be directed towards better characterizing wine yeast strains as such, in comparison to domesticated laboratory strains. Many studies are nowadays based on the genome sequence of the strain S288C available online (*Saccharomyces* genome database: www.yeastgenome.org), but wine molecular biologists know well that numerous differences are observed between industrial and laboratory strains, beyond the difference in ploidy. Beyond the genomic studies performed on wine strains by comparing their genome to that of S288C or between each other (Hauser *et al.*, 2001; Carreto *et al.*, 2008), the genome of the wine strain AWRI1631 has been sequenced (Borneman *et al.*, 2008). This has allowed quantifying and locating numerous single nucleotide polymorphisms as well as confirming the Ty elements arrangements particular to wine strains as previously noticed (Rachidi *et al.*, 1999). Several DNA regions were also identified as specific to AWRI1631. The genes of these specific regions seem to encode for proteins specific to industrial strains. This will lead to the use of metabolic engineering, a technology combining traditional microbiology, molecular biology, genetic engineering, chemical sciences and cutting-edge analytical tools in

order to improve wine strains while fully understanding the consequences of the changes introduced.

Even beyond the use of modern emerging technologies such as systems biology, the future probably lies in a system such as the multidisciplinary integrated Wine Science Research Niche Area, as proposed by the Department of Viticulture and Oenology and the Institute for Wine Biotechnology at Stellenbosch University, South Africa (Moore *et al.*, 2008). These academic institutions suggest to fully integrate modern and traditional techniques into a holistic scheme: biology, chemistry, viticulture, oenology interconnected via (chemo)metric and metabolomic techniques. This ambitious approach aims at providing tools to the wine industry for increasing its competitiveness. Biotechnology can indeed be the solution for achieving the following final goals: (i) environmentally friendly production practices in the vineyard, (ii) reduced production costs (yeasts with improved fermentation performance and resistance to nutrient stress), (iii) production of high quality wines (better colour, flavour and aroma), (iv) providing increased health benefits. The risk associated with the release of GM organisms should also be assessed (Moore *et al.*, 2008).

2.7 References

- Akada R (2002), Genetically modified industrial yeast ready for application, *J Biosci Bioeng*, **94**, 536–544.
- Alexandre H, Ansaney-Galeote V, Dequin S and Blondin B (2001), Global gene expression during short-term ethanol stress in *Saccharomyces cerevisiae*, *FEBS Lett*, **498**, 98–103.
- Arévalo Villena M, Úbeda Iranza J F, Cordero Otero R R and Briones Pérez A I (2005), Optimization of a rapid method for studying the cellular location of β -glucosidase activity in wine yeasts, *J Appl Microbiol*, **99**, 558–564.
- Bakalinsky A and Snow R (1990), The chromosomal constitution of wine strains of *Saccharomyces cerevisiae*, *Yeast*, **6**, 367–382.
- Becker J v W, Armstrong G O, van der Merwe G K, Lambrechts M G, Vivier M A and Pretorius I S (2003), Metabolic engineering of *Saccharomyces cerevisiae* for the synthesis of the wine-related antioxidant resveratrol, *FEMS Yeast Res*, **4**, 79–85.
- Beltran G, Novo M, Leberre V, Sokol S, Labourdette D, Guillamón J M, Mas A, Francois J and Rozes N (2006), Integration of transcriptomic and metabolic analyses for understanding the global responses of low-temperature winemaking fermentations, *FEMS Yeast Res*, **6**, 1167–1183.
- Berthels N J, Cordero Otero R R, Bauer F F, Pretorius I S and Thevelein J M (2008), Correlation between glucose/fructose discrepancy and hexokinase kinetic properties in different *Saccharomyces cerevisiae* wine yeast strains, *Appl Microbiol Biotechnol*, **77**, 1083–1091.
- Bisson L F (2004), The biotechnology of wine yeast, *Food Biotechnol*, **18**, 63–96.
- Bleve G, Rizzotti L, Dellaglio F and Torriani S (2003), Development of reverse-transcription (RT)-PCR and real-time RT-PCR assays for rapid detection and quantification of viable yeasts and molds contaminating yogurts and pasteurized food products, *Appl Environ Microbiol*, **69**, 4116–4122.
- Boone C, Sdicu A M, Wagner J, Degré R, Sanchez C and Bussey H (1990), Integration of the yeast K1 killer toxin gene into the genome of marked wine yeasts and its effect on vinification, *Am J Enol Vitic*, **41**, 37–42.
- Borneman A R, Chambers P J and Pretorius I S (2007), Yeast systems biology: modelling the winemaker's art, *Trends Biotechnol*, **25**, 349–355.

- Borneman A R, Forgan A H, Pretorius I S and Chambers P J (2008), Comparative genome analysis of a *Saccharomyces cerevisiae* wine strain, *FEMS Yeast Res*, **8**, 1185–1195.
- Bougnoux M-E, Morand S and d'Enfert C (2002), Usefulness of multilocus sequence typing for characterization of clinical isolates of *Candida albicans*, *J Clin Microbiol*, **40**, 1290–1297.
- Brock T D (1987), The study of microorganisms in situ: progress and problems, *Symp Soc Gen Microbiol*, **41**, 1–17.
- Carreto L, Eiriz M F, Gomes A C, Pereira P M, Schuller D and Santos M A (2008), Comparative genomics of wild type yeast strains unveils important genome diversity, *BMC Genomics*, **9**, 524.
- Carstens M, Vivier M A, van Rensburg P and Pretorius I S (2003), Overexpression, secretion and antifungal activity of the *Saccharomyces cerevisiae* chitinase, *Ann Microbiol*, **50**, 15–28.
- Casey G D and Dobson A D (2004), Potential of using real-time PCR-based detection of spoilage yeast in fruit juice – a preliminary study, *Int J Food Microbiol*, **91**, 327–335.
- Cebollero E and Gonzalez R (2004), Comparison of two alternative dominant selectable markers for wine yeast transformation, *Appl Environ Microbiol*, **70**, 7018–7023.
- Cebollero E, Gonzalez-Ramos D, Tabera L and Gonzalez R (2007), Transgenic wine yeast technology comes of age: is it time for transgenic wine?, *Biotechnol Lett*, **29**, 191–200.
- Cocolin L, Bisson L F and Mills D A (2000), Direct profiling of the yeast dynamics in wine fermentations, *FEMS Microbiol Lett*, **189**, 81–87.
- Cocolin L, Heisey A and Mills D A (2001), Direct identification of the indigenous yeasts in commercial wine fermentations, *Am J Enol Vitic*, **52**, 49–53.
- Cocolin L, Manzano M, Rebecca S and Comi G (2002), Monitoring of yeast population changes during a continuous wine fermentation by molecular methods, *Am J Enol Vitic*, **53**, 24–27.
- Codón A C, Benitez T and Korhola M (1997), Chromosomal reorganization during meiosis of *Saccharomyces cerevisiae* baker's yeasts, *Curr Genet*, **32**, 247–259.
- Coulon J, Husnik J I, Inglis D L, van der Merwe G K, Lonvaud A, Erasmus D J and van Vuuren H J (2006), Metabolic engineering of *Saccharomyces cerevisiae* to minimize the production of ethyl carbamate in wine, *Am J Enol Vitic*, **57**, 113–124.
- Cozzolino D, Flood L, Bellon J, Gishen M and De Barros Lopes M (2006), Combining near infrared spectroscopy and multivariate analysis as a tool to differentiate different strains of *Saccharomyces cerevisiae*: a metabolomic study, *Yeast*, **23**, 1089–1096.
- Curtin C D, Bellon J R, Henschke P A, Godden P W and de Barros Lopes M (2007), Genetic diversity of *Dekkera bruxellensis* yeasts isolated from Australian wineries, *FEMS Yeast Res*, **7**, 471–481.
- De Barros Lopes M, Rainieri S, Henschke P A and Langridge P (1999), AFLP fingerprinting for analysis of yeast genetic variation, *Int J Syst Bacteriol*, **49**, 915–924.
- De Barros Lopes M, Rehman A-U, Gockowiak H, Heinrich A, Langridge P and Henschke P (2000), Fermentation properties of a wine yeast over-expressing the *Saccharomyces cerevisiae* glycerol 3-phosphate dehydrogenase gene (*GPD2*), *Aust J Grape Wine Res*, **6**, 208–215.
- De Barros Lopes M, Bellon J R, Shirley N J and Ganter P F (2002), Evidence for multiple interspecific hybridization in *Saccharomyces sensu stricto* species, *FEMS Yeast Res*, **1**, 323–331.
- Del Mar Lleó, Pierobon S, Tafi M C, Signoretto C and Canepari P (2000), mRNA detection by reverse-transcription-PCR for monitoring viability over time in an *Enterococcus faecalis* viable but nonculturable population maintained in a laboratory microcosm, *Appl Environ Microbiol*, **66**, 4564–4567.
- Delaherche A, Claisse O and Lonvaud-Funel A (2004), Detection and quantification of *Brettanomyces bruxellensis* and 'ropy' *Pediococcus damnosus* strains in wine by real-time polymerase chain reaction, *J Appl Microbiol*, **97**, 110–115.

- Dequin S (2001), The potential of genetic engineering for improving brewing, winemaking and baking yeasts, *Appl Microbiol Biotechnol*, **56**, 577–588.
- Divol B (2004) *Botrytis-affected wines microbiology during maturation. Genetic characterisation of Saccharomyces cerevisiae strains responsible for refermentations*, PhD thesis, Institut National Polytechnique de Toulouse, France.
- Divol B and Lonvaud-Funel A (2005), Evidence for viable but nonculturable yeasts in Botrytized wines, *J Appl Microbiol*, **99**, 85–93.
- Donalies U E, Nguyen H T, Stahl U and Nevoigt E (2008), Improvement of *Saccharomyces* yeast strains used in brewing, wine-making and baking, *Adv Biochem Eng Biotechnol*, **111**, 67–98.
- Dunn B, Levine R P and Sherlock G (2005), Microarray karyotyping of commercial wine yeast strains reveals shared, as well as unique, genomic signatures, *BMC Genomics*, **6**, 53.
- Eglinton J M, Heinrich A J, Pollnitz A P, Langridge P, Henschke P A and de Barros Lopes M (2002), Decreasing acetic acid accumulation by a glycerol overproducing strain of *Saccharomyces cerevisiae* by deleting the *ALD6* aldehyde dehydrogenase gene, *Yeast*, **19**, 295–301.
- Erasmus D J, van der Merwe G K and van Vuuren H J J (2003), Genome-wide expression analyses: metabolic adaptation of *Saccharomyces cerevisiae* to high sugar stress, *FEMS Yeast Res*, **3**, 375–399.
- Espinosa J C, Fernández-González M, Ubeda J and Briones A (2002), Identification of wine yeasts by PCR–RFLP, *Food Technol Biotechnol*, **40**, 157–160.
- Esteve-Zarzoso B, Belloch C, Uruburu F and Querol A (1999), Identification of yeasts by RFLP analysis of the 5.8S rRNA gene and the two ribosomal internal transcribed spacers, *Int J Syst Bacteriol*, **49**, 329–337.
- Fernández-González M, Espinosa J C, Úbeda J F and Briones A I (2001), Yeasts present during wine fermentation: comparative analysis of conventional plating and PCR–TTGE, *Syst Appl Microbiol*, **24**, 634–638.
- Fernández-González M, Ubeda J F, Cordero Otero R R, Thanvanthri Gururajan V and Briones A L (2005), Engineering of an oenological *Saccharomyces cerevisiae* strain with pectinolytic activity and its effect on wine, *Int J Food Microbiol*, **102**, 173–183.
- Fleet G H (2008), Wine yeasts for the future, *FEMS Yeast Res*, **8**, 979–995.
- Flores Berrios E P, Alba González J F, Arrizon Gaviño J P, Romano P, Capece A and Gschaedler Mathis A (2005), The uses of AFLP for detecting DNA polymorphism, genotype identification and genetic diversity between yeasts isolated from Mexican agave-distilled beverages and from grape musts, *Lett Appl Microbiol*, **41**, 147–152.
- Frezier V and Dubourdieu D (1992), Ecology of yeast strain *Saccharomyces cerevisiae* during spontaneous fermentation in a Bordeaux winery, *Am J Enol Vitic*, **43**, 375–380.
- Fukuda K, Yamamoto N, Kiyokawa Y, Yanagiuchi T, Wakai Y, Kitamoto Y, Inoue Y and Kimura A (1998), Balance of activities for alcohol acetyltransferase and esterase in *Saccharomyces cerevisiae* is important for production of isoamyl acetate, *Appl Environ Microbiol*, **64**, 4076–4078.
- Gallego F J, Perez M A, Martinez I and Hidalgo P (1998), Microsatellites obtained from database sequences are useful to characterize *Saccharomyces cerevisiae* strains, *Am J Enol Vitic*, **49**, 350–351.
- Gangl H, Batusic M, Tscheik G, Tiefenbrunner W, Hack C and Lopandic K (200p), Exceptional fermentation characteristics of natural hybrids from *Saccharomyces cerevisiae* and *S. kudriavzevii*, *N Biotechnol*, **25**, 244–251.
- Giudici P, Solieri L, Pulvirenti A M and Cassanelli S (2005), Strategies and perspectives for genetic improvement of wine yeasts, *Appl Microbiol Biotechnol*, **66**, 622–628.
- Goffeau A, Barrell B G, Bussey H, Davis R W, Dujon B, Feldmann H, Galibert F, Hoheisel J D, *et al.* (1996), Life with 6000 genes, *Science*, **274**, 546–567.
- Goldstein A L and McCusker J H (1999), Three new dominant drug resistance cassettes for gene disruption in *Saccharomyces cerevisiae*, *Yeast*, **15**, 1541–1553.

- Gonzalez-Techera A, Jubany S, Carrau F M and Caggero C (2001), Differentiation of industrial wine yeast strains using microsatellite markers, *Lett Appl Microbiol*, **33**, 71–75.
- Gonzalez R, Martinez-Rodriguez A and Carrascosa A (2003), Yeast autolysis mutants potentially useful for sparkling wine production, *Int J Food Microbiol*, **84**, 21–26.
- González S S, Barrio E, Gafner J and Querol A (2006), Natural hybrids from *Saccharomyces cerevisiae*, *Saccharomyces bayanus* and *Saccharomyces kudriavzevii* in wine fermentations, *FEMS Yeast Res*, **6**, 1221–1234.
- González S S, Gallo L, Climent M A, Barrio E and Querol A (2007), Enological characterization of natural hybrids from *Saccharomyces cerevisiae* and *S. kudriavzevii*, *Int J Food Microbiol*, **116**, 11–18.
- Granchi L, Bosco M, Messina A and Vincenzini M (1999), Rapid detection and quantification of yeast species during spontaneous wine fermentation by PCR–RFLP analysis of the rDNA ITS region, *J Appl Microbiol*, **89**, 949–956.
- Guillamón J M, Sabaté J, Barrio E, Cano J and Querol A (1998), Rapid identification of wine yeast species based on RFLP analysis of the ribosomal ITS regions, *Arch Microbiol*, **169**, 387–392.
- Guillou S, Besnard V, El Murr N and Federighi M (2003), Viability of *Saccharomyces cerevisiae* cells exposed to low-amperage electrolysis as assessed by staining procedure and ATP content, *Int J Food Microbiol*, **88**, 85–89.
- Guldener U, Heck S, Fiedler T, Beinhauer J and Hegemann J H (1996), A new efficient gene disruption cassette for repeated use in budding yeast, *Nucleic Acids Res*, **24**, 2519–2524.
- Han J, Bertin N, Hao T, Goldberg D S, Berriz G F, Zhang L V, Dupuy D, Walhout A J M, Cusick M E, Roth F P and Vidal M (2004), Evidence for dynamically organized modularity in the yeast protein-protein interaction network, *Nature*, **430**, 88–93.
- Hauser N C, Fellenberg K, Gil R, Bastuck S, Hoheisel J D and Pérez-Ortín J E (2001), Whole genome analysis of a wine yeast strain, *Comp Funct Genomics*, **2**, 69–79.
- Henricsson C, de Jesus Ferreira M C, Hedfalk K, Elbing K, Larsson C, Bill R M, Norbeck J, Hohmann S and Gustafsson L (2005), Engineering of a novel *Saccharomyces cerevisiae* wine strain with a respiratory phenotype at high external glucose concentrations, *Appl Environ Microbiol*, **71**, 6185–6182.
- Herrero O, Ramon D and Orejas M (2008), Engineering the *Saccharomyces cerevisiae* isoprenoid pathway for de novo production of aromatic monoterpenes in wine, *Metab Eng*, **10**, 78–86.
- Hierro N, Esteve-Zarzoso B, González A, Mas A and Guillamón J M (2006), Real-time quantitative PCR (QPCR) and reverse transcription-QPCR for detection and enumeration of total yeasts in wine, *Appl Environ Microbiol*, **72**, 7148–7155.
- Hillenmeyer M E, Fung E, Wildenhain J, Pierce S E, Hoon S, Lee W, Proctor M, St-Onge R P, Tyers M, Koller D, Altman R D, Davis R W, Nislow C and Giaever G (2008), The chemical genomic portrait of yeast: uncovering a phenotype for all genes, *Science*, **320**, 362–365.
- Hottiger T, Kuhla J, Fuerst P, Spielmann A, Garn M, Haemmerli S and Heim J (1995), 2- μ m vectors containing the *Saccharomyces cerevisiae* metallothionein gene as a selectable marker: excellent stability in complex media and high-level expression of a recombinant protein from a *CUP1*-promoter controlled expression cassette in cis, *Yeast*, **11**, 1–4.
- Howell K S, Swiegers J H, Elsey G M, Siebert T E, Bartowsky E, J., Fleet G H, Pretorius I S and De Barros Lopes M A (2004), Variation in 4-mercapto-4-methyl-pentan-2-one release by *Saccharomyces cerevisiae* commercial wine strains under different fermentation conditions, *FEMS Microbiol Lett*, **240**, 125–129.
- Howell K S, Cozzolino D, Bartowsky E J, Fleet G H and Henschke P A (2006), Metabolic profiling as a tool for revealing *Saccharomyces* interactions during wine fermentation, *FEMS Yeast Res*, **6**, 91–101.
- Husnik J I, Volschenk H, Bauer J, Colavizza D, Luo Z and van Vuuren H J J (2006), Metabolic engineering of malolactic wine yeast, *Metab Eng*, **8**, 315–323.

- Ibeas J I, Lozano I, Perdigonés F and Jiménez J (1996), Detection of *Dekkera-Brettanomyces* strains in sherry by a nested PCR method, *Appl Environ Microbiol*, **62**, 998–1003.
- Ibrahim H R, Matsuzaki T and Aoki T (2001), Genetic evidence that antibacterial activity of lysozyme is independent of its catalytic function, *FEMS Lett*, **506**, 27–32.
- Jacobsen M D, Gow N A, Maiden M C, Shaw D J and Odds F C (2007), Strain typing and determination of population of *Candida krusei* by multilocus sequence typing, *J Clin Microbiol*, **45**, 317–323.
- Jacobsen M D, Bougnoux M-E, d'Enfert C and Odds F C (2008), Multilocus sequence typing of *Candida albicans* isolates from animals, *Res Microbiol*, **159**, 436–440.
- Jiménez A and Davies J (1980), Expression of a transposable antibiotic resistance element in *Saccharomyces*, *Nature*, **287**, 869–871.
- Johnston M *et al.* (1997), The nucleotide sequence of *Saccharomyces cerevisiae* chromosome XII, *Nature*, **387** (suppl.), 87–90.
- Kidwell M G and Lisch D R (2000), Transposable elements and host genome evolution, *Trends Ecol Evol*, **15**, 95–99.
- Kolkman A, Daran-Lapujade P, A. F, Oltshoorn M M A, Pronk J T, Slijper M and Heck A J R (2006), Proteome analysis of yeast response to various nutrient limitations, *Mol Syst Biol*, **2**, 2006.0026.
- Lavine B and Workman J (2006), Chemometrics, *Anal Chem*, **78**, 4137–4145.
- Legras J-L and Karst F (2003), Optimization of interdelta analysis for *Saccharomyces cerevisiae* strain characterization, *FEMS Microbiol Lett*, **221**, 249–255.
- Legras J-L, Ruh O, Merdinoglu D and Karst F (2005), Selection of hypervariable microsatellite loci for the characterization of *Saccharomyces cerevisiae* strains, *Int J Food Microbiol*, **102**, 73–83.
- Lilly M, Bauer F F, Styger G, Lambrechts M G, Cozzolino D and Pretorius I S (2006a), The effect of increased yeast alcohol dehydrogenase and esterase activity on the flavour profiles of wine and distillates, *Yeast*, **23**, 641–659.
- Lilly M, Bauer F F, Styger G, Lambrechts M G and Pretorius I S (2006b), The effect of increased branched-chain amino acid transaminase activity in yeast on the production of higher alcohols and on the flavour of wine and distillates, *FEMS Yeast Res*, **6**, 726–743.
- Lin C Y, Chen Y C, Lo H J, Chen K W and Li S Y (2007), Assessment of *Candida glabrata* strain relatedness by pulsed field gel electrophoresis and multilocus sequence typing, *J Clin Microbiol*, **45**, 2452–2459.
- Lonvaud-Funel A, Joyeux A and Ledoux O (1991), Speciation of lactic acid bacteria in fermenting grape must and wine by colony hybridization with non-isotopic DNA probes, *J Appl Bacteriol*, **71**, 501–508.
- Louw C T, Pretorius I S and van Rensburg P (2006), The effect of polysaccharide-degrading wine yeast transformants on the efficiency of wine processing and wine flavour, *J Biotechnol*, **125**, 447–461.
- Malherbe D F, Du Toit M, Cordero Otero R R, van Rensburg P and Pretorius I S (2003), Expression of the *Aspergillus niger* glucose oxidase gene in *Saccharomyces cerevisiae* and its potential applications in wine production, *Appl Microbiol Biotechnol*, **61**, 502–551.
- Marullo P, Bely M, Masneuf-Pomarede I, Pons M, Aigle M and Dubourdieu D (2006), Breeding strategies for combining fermentative qualities and reducing off-flavor production in a wine yeast model, *FEMS Yeast Res*, **6**, 268–279.
- Marullo P, Aigle M, Bely M, Masneuf-Pomarede I, Durrrens P and Dubourdieu D (2007a), Single QTL mapping and nucleotide-level resolution of a physiologic trait in wine *Saccharomyces cerevisiae* strains, *FEMS Yeast Res*, **7**, 941–952.
- Marullo P, Yvert G, Bely M, Aigle M and Dubourdieu D (2007b), Efficient use of DNA molecular marker to construct industrial yeast strains, *FEMS Yeast Res*, **7**, 1295–1306.
- Masneuf I and Dubourdieu D (1994), Comparaison de deux techniques d'identification des souches de levures de vinification basées sur le polymorphisme de l'ADN génomique: réaction de polymérisation en chaîne (PCR) et analyse des caryotypes (électrophorèse en champ pulsé), *J Int Sci Vigne Vin*, **28**, 153–160.

- Masneuf I, Hansen J, Groth C, Piskur J and Dubourdieu D (1998), New hybrids between *Saccharomyces sensu stricto* yeast species found among wine and cider production strains, *Appl Environ Microbiol*, **64**, 3887–3892.
- Mendes-Ferreira A, Del Olmo M, Garcia-Martinez J, Jimenez-Marti E, Mendes-Faia A, Perez-Ortin J E and Leao C (2007), Transcriptional response of *Saccharomyces cerevisiae* to different nitrogen concentrations during alcoholic fermentation, *Appl Environ Microbiol*, **73**, 3049–3060.
- Millet V and Lonvaud-Funel A (2000), The viable but non-culturable state of wine micro-organisms during storage, *Lett Appl Microbiol*, **30**, 136–141.
- Mills D A (2002), Yeast diversity and persistence in *Botrytis*-affected wine fermentations, *Appl Environ Microbiol*, **68**, 4884–4893.
- Miot-Sertier C and Lonvaud-Funel A (2007), Development of a molecular method for the typing of *Brettanomyces bruxellensis* (*Dekkera bruxellensis*) at the strain level, *J Appl Microbiol*, **102**, 555–562.
- Moore J P, Divol B, Young P R, Nieuwoudt H H, Ramburan V, du Toit M, Bauer F F and Vivier M A (2008), Wine biotechnology in South Africa: towards a systems biology approach in wine science, *Biotechnol J*, **3**, 1–13.
- Ness F, Lavalley F, Dubourdieu D, Aigle M and Dulau L (1993), Identification of yeast strains using the polymerase chain reaction, *J Sci Food Agric*, **62**, 89–94.
- Nielsen J (2003), It's all about metabolic fluxes, *J Bacteriol*, **185**, 7031–7035.
- O'Farrell P H (1997), High resolution two-dimensional electrophoresis of proteins, *J Biol Chem*, **250**, 4007–4021.
- Palsson B O (2002), *In silico* biology through 'omics', *Nat Biotechnol*, **20**, 649–650.
- Park H, Lopez N I and Bakalinsky A (1999), Use of sulfite resistance in *Saccharomyces cerevisiae* as a dominant selectable marker, *Curr Genet*, **36**, 2801–2906.
- Pasteur L (1866), *Etudes sur le vin*, Paris, Imprimerie impériale.
- Pearson B M and McKee R A (1992), Rapid identification of *Saccharomyces cerevisiae*, *Zygosaccharomyces bailii* and *Zygosaccharomyces rouxii*, *Int J Food Microbiol*, **16**, 63–67.
- Pérez-González J A, González R, Querol A, Sendra J and Ramón D (1993), Construction of a recombinant wine yeast strain expressing β -(1,4)-endoglucanase and its use in microvinification processes, *Appl Environ Microbiol*, **59**, 2801–2806.
- Perez M A, Gallego F J, Martinez I and Hidalgo P (2001), Detection, distribution and selection of microsatellites (SSRs) in the genome of the yeast *Saccharomyces cerevisiae* as molecular markers, *Lett Appl Microbiol*, **33**, 461–466.
- Petering J E, Henschke P A and Langridge P (1991), The *Escherichia coli* β -glucuronidase gene as a marker for *Saccharomyces* yeast strain identification, *Am J Enol Vitic*, **42**, 6–12.
- Phister T G and Mills D A (2003), Real-time PCR assay for detection and enumeration of *Dekkera bruxellensis* in wine, *Appl Environ Microbiol*, **69**, 7430–7434.
- Pizarro F, Vargas F A and Agosin E (2007), A systems biology perspective of wine fermentations, *Yeast*, **24**, 977–991.
- Pramateftaki P V, Lanaridis P and Typas M A (2000), Molecular identification of wine yeasts at species or strain level: a case study with strains from two vine-growing areas of Greece, *J Appl Microbiol*, **89**, 236–248.
- Pretorius I S and Bauer F F (2002), Meeting the consumer challenge through genetically modified customized wine-yeast strains, *Trends Biotechnol*, **20**, 426–432.
- Pretorius I S and Høj P B (2005), Grape and wine biotechnology: challenges, opportunities and potential benefits, *Aust J Grape Wine Res*, **11**, 83–108.
- Puig S, Ramon D and Perez-Ortin J E (1998), Optimized method to obtain stable food-safe recombinant wine yeast strains, *J Agric Food Chem*, **46**, 1689–1693.
- Querol A, Barrio E, Huerta T and Ramón D (1992a), Molecular monitoring of wine fermentations conducted by active dry yeast strains, *Appl Environ Microbiol*, **58**, 2948–2953.

- Querol A, Barrio E and Ramón D (1992b), A comparative study of different methods of yeast strain characterization, *Syst Appl Microbiol*, **15**, 439–446.
- Quesada M P and Cenis J L (1995), Use of random amplified polymorphic DNA (RAPD–PCR) in the characterization of wine yeasts, *Am J Enol Vitic*, **46**, 204–208.
- Rachidi N, Barre P and Blondin B (1999), Multiple Ty-mediated chromosomal translocations lead to karyotype changes in a wine strain of *Saccharomyces cerevisiae*, *Mol Gen Genet*, **261**, 841–850.
- Rainieri S and Pretorius I S (2000), Selection and improvement of wine yeasts, *Ann Microbiol*, **50**, 15–32.
- Ramirez M and Ambrona J (2008), Construction of sterile *ime1Δ*-transgenic *Saccharomyces cerevisiae* wine yeasts unable to disseminate in nature, *Appl Environ Microbiol*, **74**, 2129–2134.
- Ramon D (1997), Trends in wine microbiology, *Microbiologia*, **13**, 405–411.
- Rawsthorne H and Phister T G (2006), A real-time PCR assay for the enumeration and detection of *Zygosaccharomyces bailii* from wine and fruit juices, *Int J Food Microbiol*, **112**, 1–7.
- Remize F and Dequin S (1998), Engineering of glycerol metabolism in wine yeast, in Osseweijer P and van Dijken J P (eds), *Yeast as a Cell Factory*, Vlaardingen, the Netherlands, EEC, 30–31.
- Remize F, Roustan J-L, Sablayrolles J, Barre P and Dequin S (1999), Glycerol overproduction by engineered *Saccharomyces cerevisiae* wine yeast strains leads to substantial changes in by-product formation and to a stimulation of fermentation rate in stationary phase, *Appl Environ Microbiol*, **65**, 143–149.
- Remize F, Sablayrolles J M and Dequin S (2000), Re-assessment of the influence of yeast strain and environmental factors on glycerol production in wine, *J Appl Microbiol*, **88**, 371–378.
- Renouf V, Falcou M, Miot-Sertier C, Perello M C, de Revel G and Lonvaud-Funel A (2006), Interactions between *Brettanomyces bruxellensis* and other yeast species during the initial stages of winemaking, *J Appl Microbiol*, **100**, 1208–1219.
- Romano P, Soli M G, Suzzi G, Grazia L and Zambonelli C (1985), Improvement of a wine *Saccharomyces cerevisiae* strain by a breeding program, *Appl Environ Microbiol*, **57**, 830–835.
- Rossignol T, Dulau L, Julien A and Blondin B (2003), Genome-wide monitoring of wine yeast gene expression during alcoholic fermentation, *Yeast*, **20**, 1369–1385.
- Rossignol T, Postaire P, Storaú J and Blondin B (2006), Analysis of the genomic response of a wine yeast to rehydration and inoculation, *Appl Microbiol Biotechnol*, **71**, 699–712.
- Rossouw D, Næs T and Bauer F F (2008), Linking gene regulation and the exo-metabolome: a comparative transcriptomics approach to identify genes that impact on the production of volatile aroma compounds in yeast, *BMC Genomics*, **9**, 530–548.
- Salmon J M and Barre P (1998), Improvement of nitrogen assimilation and fermentation kinetics under enological conditions by derepression of alternative nitrogen-assimilatory pathways in an industrial *Saccharomyces cerevisiae* strain, *Appl Environ Microbiol*, **64**, 3831–3837.
- Salvadó Z, Chiva R, Rodríguez-Vargas S, Rándež-Gil F, Mas A and Guillamón J M (2008), Proteomic evolution of a wine yeast during the first hours of fermentation, *FEMS Yeast Res*, **8**, 1137–1146.
- Santos J, Sousa M J, Cardoso H, Inácio J, Silva S, Spencer-Martins I and Leão C (2008), Ethanol tolerance of sugar transport, and the rectification of stuck wine fermentations, *Microbiology*, **154**, 422–430.
- Sato M, Kishimoto M, Watari J and Takashio M (2002), Breeding of brewer's yeast by hybridization between a top-fermenting yeast *Saccharomyces cerevisiae* and a cryophilic yeast *Saccharomyces bayanus*, *J Biosci Bioeng*, **93**, 509–511.
- Schena M, Shalon D, Davis R W and Brown P O (1995), Quantitative monitoring of gene expression patterns with a complementary DNA microarray, *Science*, **270**, 467–656.

- Schmidt M, Coemer A, Greg M and Brendel A (1997), Genetic engineering of baker's and wine yeasts using formaldehyde hyperresistance-mediating plasmids, *Brz J Med Biol Res*, **30**, 1407–1414.
- Schoeman H, Vivier M A, Du Toit M, Dicks L M T and Pretorius I S (1999), The development of bactericidal yeast strains expressing the *Pediococcus acidilactici* pediocin gene (*pedA*) in *Saccharomyces cerevisiae*, *Yeast*, **15**, 647–656.
- Schuller D and Casal M (2005), The use of genetically modified *Saccharomyces cerevisiae* strains in the wine industry, *Appl Microbiol Biotechnol*, **68**, 292–304.
- Sheridan C M, Bauer F F, Burton S and Lorenzen L (2005), A critical process analysis of wine production to improve cost, quality and environmental performance, *Water Sci Technol*, **51**, 39–46.
- Stephanopoulos G, Aristidou A and Nielsen J (1998), *Metabolic Engineering: Principles and Methodologies*, New York, Academic Press.
- van Reenen C A, Chikindas M L, van Zyl W H and Dicks L M T (2003), Characterization and heterologous expression of a class IIa bacteriocin, plantaricin 423 from *Lactobacillus plantarum* 423, in *Saccharomyces cerevisiae*, *Int J Food Microbiol*, **81**, 29–40.
- van Rensburg P, Stidwell T, Lambrechts M G, Cordero Otero R R and Pretorius I S (2005), Development and assessment of a recombinant *Saccharomyces cerevisiae* wine yeast producing two aroma enhancing β -glucosidases encoded by the *Saccharomycopsis fibuligera* *BGL1* and *BGL2* genes, *Ann Microbiol*, **55**, 33–42.
- van Rensburg P, Strauss M L A, Lambrechts M G, Cordero Otero R R and Pretorius I S (2007), The heterologous expression of polysaccharide-encoding genes with oenological relevance in *Saccharomyces cerevisiae*, *J Appl Microbiol*, **103**, 2228–2257.
- Varela C, Cardenas J, Melo F and Agosin E (2005), Quantitative analysis of wine yeast gene expression profiles under winemaking conditions, *Yeast*, **22**, 369–383.
- Vaudano E and Garcia-Moruno E (2008), Discrimination of *Saccharomyces cerevisiae* wine strains using microsatellite multiplex PCR and band pattern analysis, *Food Microbiol*, **25**, 56–64.
- Venema J and Tollervey D (1995), Processing of pre-ribosomal RNA in *Saccharomyces cerevisiae*, *Yeast*, **11**, 1629–1650.
- Villas-Boas S G, Roessner U, Hansen M A E, Smedsgaard J and Nielsen J (2007), Yeast metabolomics: the discovery of new metabolic pathways in *Saccharomyces cerevisiae*, in *Metabolome Analysis: an Introduction*, Hoboken, NJ, Wiley, 189–202.
- Volschenk H, Viljoen M, Grobler J, Petzold B, Bauer F, Subden R E, Young R A, Lonvaud A, Denayrolles M and van Vuuren H J (1997), Engineering pathways for malate degradation in *Saccharomyces cerevisiae*, *Nat Biotechnol*, **15**, 253–257.
- Wach A, Brachat A, Pöhlmann R and Philippsen P (1994), New heterologous modules for classical or PCR-based gene disruptions in *Saccharomyces cerevisiae*, *Yeast*, **10**, 1793–1808.
- Yap N A, de Barros Lopes M, Langridge P and Henschke P A (2000), The incidence of killer activity of non-*Saccharomyces* yeasts towards indigenous yeast species of grape must: potential application in wine fermentation, *J Appl Bacteriol*, **89**, 381–389.
- Zuckerkindl E and Pauling L (1965), Molecules as documents of evolutionary history, *J Theor Biol*, **8**, 357–366.
- Zuzuarregui A, Monteoliva L, Gil C and del Olmo M (2006), Transcriptomic and proteomic approach for understanding the molecular basis of adaptation of *Saccharomyces cerevisiae* to wine fermentation, *Appl Environ Microbiol*, **72**, 836–847.

3

Effects of malolactic fermentation on wine quality

A. Lonvaud-Funel, University Victor Segalen Bordeaux II, France

Abstract: Malolactic fermentation is an essential step in winemaking, resulting from the spontaneous growth of lactic acid bacteria (LAB) in the microbial ecosystem of grapes and wine. The first part of this chapter describes how LAB multiply and survive during winemaking, identifying the main factors in their development. It goes on to describe the metabolic pathways in operation during, and possibly after, fermentation, highlighting the beneficial and detrimental transformations that affect wine quality.

Key words: lactic acid bacteria, growth, sensory quality, spoilage.

3.1 Introduction

Malolactic fermentation, the second intense period in winemaking, takes place after alcoholic fermentation. The term ‘malolactic fermentation’ has been used since the early 20th century, when wine producers and the first oenologists discovered this secondary process following alcoholic fermentation, eventually recognizing that it was caused by lactic acid bacteria (LAB) activity. In fact, from a biochemical standpoint, it is not really a fermentation process but, like alcoholic fermentation, it does involve gas emissions, so winemakers called it ‘malolactic fermentation’.

During malolactic fermentation, the biomass of LAB that develops in wine, after yeast has declined and slowly disappeared, decarboxylates malic acid to form lactic acid. However, many other biochemical reactions occur at the same time. Many other grape must and wine compounds are involved as promoters of

bacterial growth or survival. Consequently, many other molecules besides lactic acid accumulate in the wine. Some enhance quality, while others may be detrimental to sensory qualities or undesirable for health. However, most of these 'secondary' reactions are still unknown. They concern very small amounts of substrates and products that will not be identified until the chemical changes before and after malolactic fermentation have been elucidated. Micro-, pico-, or even nanomolar concentrations of these compounds have an impact on quality.

LAB are part of the natural microflora that contaminates grapes and wine. Strains are selected by numerous factors involving interactions between microorganisms and physicochemical conditions. Winemakers need to promote the development of bacterial biomass to ensure complete malolactic fermentation, while keeping it under control. Spontaneous selection, ideally resulting in the predominance of *Oenococcus oeni*, is not failsafe and cannot reliably prevent spoilage, especially after malolactic fermentation.

3.2 Spontaneous growth of lactic acid bacteria in wine

3.2.1 Interactions between yeast and lactic acid bacteria: inhibition and activation of lactic acid bacteria by yeast

The extremely complex microbial system on the surface of the berries is dramatically simplified as soon as grapes are crushed, due to the change in the environment. Obviously, most of the microorganisms cannot adapt to the new conditions, initially dominated by high sugar concentration and acid pH. In spite of the elimination of dozens of species, the grape must is still contaminated by a very diverse microflora, including the yeast and bacteria involved in winemaking. The system then evolves as a result of interactions between yeast, bacteria, and the environment (Fig. 3.1).

Yeasts, the microorganisms best adapted to the acidity, sugar, and nutrient concentrations in grape must, multiply rapidly and predominate during the initial fermentation stages. Several genera are present, but *Saccharomyces cerevisiae* soon outnumber the others, becoming the main species responsible for the first phase in alcoholic fermentation and the key competitors of LAB.

The total initial population of LAB in must usually varies from 10^2 CFU/mL to 10^4 CFU/mL. This contamination mainly depends on the harvest conditions, including the weather during the last days of ripening. Due to their greater sensitivity to acid pH, bacteria are less adapted to growth in grape must than yeast. The bacteria multiply very briefly, for 1 or 2 days, until their growth is inhibited by the rapidly expanding yeast population and the addition of sulphur dioxide. This suits the winemaker's objectives by preventing the premature intervention of LAB. Indeed, the key concern is that the yeast population should multiply very actively, unhindered by interactions with bacteria, and metabolize all the fermentable sugars (mainly glucose and fructose). Many mechanisms are involved, but only a few have been fully characterized. Bacteria are inhibited not only by unsuccessful

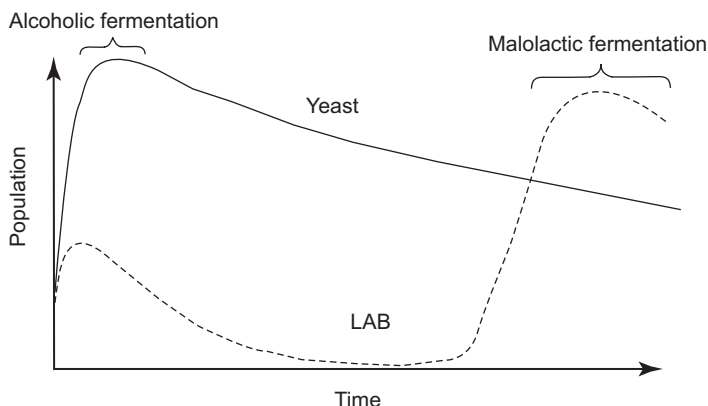


Fig. 3.1 General evolution of yeast and lactic acid bacteria populations during winemaking.

competition with yeast for nutrients, but also, more importantly, by the toxic compounds released (Ribéreau-Gayon *et al.*, 2006).

Indeed, at the beginning of alcoholic fermentation, the rapid consumption of the medium by yeast results in severe deficiencies in amino acids essential for bacteria. For example, arginine, one of the most important amino acids in the must, metabolized by many bacteria, may be entirely eliminated. Other growth factors may also be totally consumed. The decreasing sugar concentration has no effect, as bacteria only need small amounts of hexoses and may also ferment pentoses. In addition, the most significant obstacle to bacterial survival and growth is the increasing concentration of ethanol and other by-products of the yeast metabolism. Some, like fatty acids are toxic to bacteria (Section 3.2.3). Consequently, during alcoholic fermentation, bacteria are inhibited by both nutrient deficiencies and the increasing toxicity of the environment.

Once all the sugars have been fermented, yeast growth and survival deteriorates, with slower metabolic activity and a gradual decrease in total population. At that stage, the medium contains only a few hundred mg/L unfermented sugar, glucose, fructose, and pentoses. The alcohol concentration is usually around 12–14% (v/v), depending on the initial sugar concentration. The composition of the medium in terms of nutrients, vitamins, amino acids, and growth factors has also undergone a profound change. The balance between the nutritional conditions and toxicity of the medium finally encourages the growth of the LAB strains that survived alcoholic fermentation. Indeed, yeast autolysis releases their intracellular contents, providing nutrients to stimulate bacterial growth. Indeed, most nutrient broths for laboratory cultures are based on yeast extract. The yeast decline phase is also accelerated by the presence of bacteria, assumed to increase the autolysis rate via enzyme activities that break down the yeast cell walls, in a self-perpetuating cycle.

To summarize, interactions between *S. cerevisiae* and LAB play a key role in the evolution of the system: bacteria are initially inhibited, then stimulated by

yeast. Under normal, favourable conditions, they ensure the spontaneous succession of alcoholic and malolactic fermentation. Interactions between yeast and bacteria are surprisingly complex. Not only are the mechanisms not fully identified, but also they differ according to the strain. Some yeast strains are known to have a more positive effect on malolactic fermentation, probably due to the nature of the by-products secreted, but also to the composition of macromolecules released from yeast cell walls.

At the end of alcoholic fermentation, the microbial population consists of small quantities of *S. cerevisiae*, non-*Saccharomyces* yeasts, and LAB. The bacterial population varies from 10 CFU/mL to 10⁴ CFU/mL, depending on the environmental conditions. It remains at this concentration for a variable time period then increases rapidly to exceed the 10⁶ CFU/mL required for active malolactic fermentation.

3.2.2 The main nutrients for lactic acid bacteria in wine: sugars, amino acids, and organic acids

A huge diversity of molecules, e.g. sugars, amino acids, vitamins, and minerals, are necessary for LAB, like any viable cell, to multiply. These needs vary from one species and strain to another. Most of the energy necessary for the cell to function and produce a variety of essential macromolecules, such as nucleic acids, peptidoglycan, lipids, structural, and enzyme proteins, as well as membrane transporters of all kinds, is supplied by fermentation.

Lactic acid bacteria ferment sugars via either homofermentative or heterofermentative pathways. The energy available varies, depending on the type of sugars, hexoses or pentoses, as well as interactions with other pathways. The ability to ferment residual sugars varies according to the species and strain. In wine, the dominant bacteria are heterofermentative, both lactobacilli, such as *Lactobacillus hilgardii* and *L. brevis*, and cocci such as *O. oeni*. *O. oeni* is usually dominant, but lactobacilli may become the majority strain after malolactic fermentation (Ribéreau-Gayon *et al.*, 2006).

The main products of glucose heterofermentation are not only lactic acid, but also ethanol, carbon dioxide, and acetic acid. Fructose products are the same, but this hexose may also be reduced to mannitol. Even if part of the fructose is not fermented, this reaction has a positive effect on the bacteria, by re-oxidizing coenzymes, thus improving the balance between oxidized and reduced coenzymes required for their general metabolism. Consequently acetyl-phosphate is not reduced to acetaldehyde and ethanol, but produces acetic acid, while the phosphate group is used for ATP synthesis (Fig. 3.2).

Heterofermentative bacteria grow better if a mixture of glucose and fructose is available, as is the case in wine. If alcoholic fermentation is properly completed, the quantities of glucose and fructose left unfermented may vary, but usually remain below 2g/L. Even if there were no other sugars present in wine, these quantities would be sufficient to develop the biomass required to degrade malic acid.

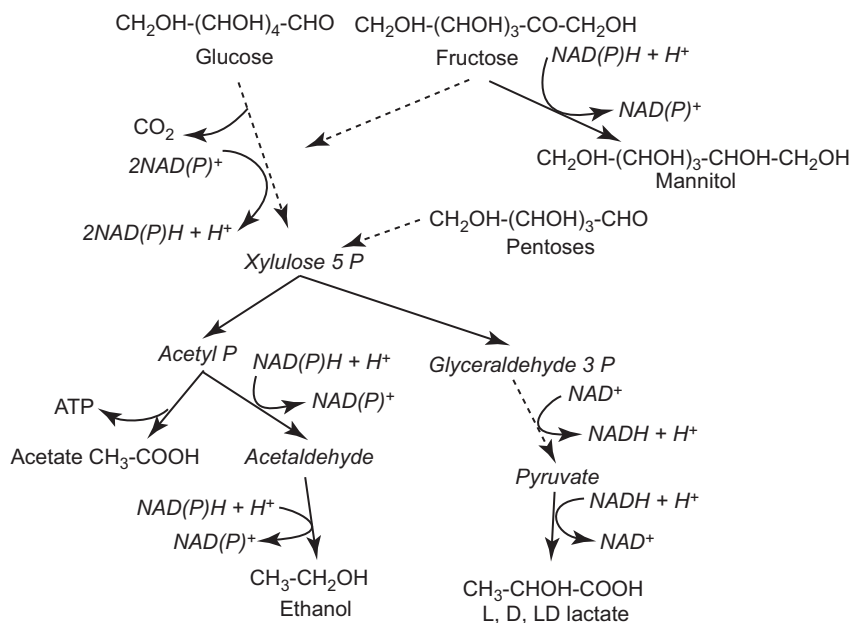


Fig. 3.2 Schematic pathway of heterofermentation and mannitol production.

The other free sugars used by LAB in wine are pentoses, arabinose and xylose, present in grape must but not fermented by yeast. Concentrations of several hundred mg/L provide a significant source of energy. Arabinose is generally the most abundant. These pentoses are fermented by heterofermentative bacteria via the same pathway as hexoses, joining the pentose phosphate pathway at ribulose-5-phosphate. After the breakdown of the xylulose-5-phosphate, the energy comes from the phosphorylation level of acetyl-P and the recovery of ATP (Fig. 3.2). However, all species and strains are not able to metabolize arabinose and xylose. Most lactobacilli in wine are heterofermentative, like *L. hilgardii* and *L. brevis*, or facultatively heterofermentative, like *L. plantarum*. Most strains of these species are capable of fermenting pentoses. As for *O. oeni*, even if their enzyme equipment for the pentose phosphate pathway functions, as shown by the fermentation of glucose and fructose, some strains do not ferment arabinose and/or xylose. The difference is probably due to the lack of a transport system in some strains. The ability to ferment pentoses represents an advantage in terms of energy supply. Overall, hexoses and pentoses in wine after fermentation supply adequate sugar to support a wide variety of lactic bacteria. In addition LAB are capable of metabolizing the osidic moiety of glycosylated molecules, such as wine aroma and colour compounds. This additional sugar source is probably more advantageous during storage after winemaking, when bacteria have to find unusual energy sources to survive.

Energy is also indirectly provided by the metabolism of organic acids, such as malic and citric acid, degraded during malolactic fermentation. Their contribution

is significantly lower than that obtained by fermentation of sugars, but they offer an appreciable energy source for lactic bacteria in wine, which develop mainly after alcoholic fermentation has consumed other nutrients. ATP is synthesized by the decarboxylation of malic acid (Section 3.4.1) at membrane level via a substrate/product and proton exchange, involving electrical charge transport on both sides of the membrane (Poolman *et al.*, 1991). In the more complex degradation of citric acid (Section 3.4.2), ATP is generated from pyruvate. A very simple experiment demonstrates the impact of citric acid on these bacteria: adding citric acid to a sample where malic and citric acid had been totally degraded by malolactic fermentation, results in growth of the bacteria, whereas they barely survive or even die in the control.

The nitrogen supply is another decisive factor for growth in any living cell. In winemaking, LAB are completely dependent on the amino acids available in wine, as, unlike yeast, they are unable to use ammonium nitrogen for amino acid synthesis. All amino acids are clearly needed to build proteins and some are also assimilated to provide energy. The routes by which they are synthesized from organic molecules vary in complexity and do not even exist in some bacteria. Consequently, amino acids that cannot be derived from organic acids and/or other amino acids must be absorbed from the environment. The requirement for these essential amino acids, used directly by the cells, varies from one species and strain to another, e.g. lactobacilli need less than cocci. Glutamic acid, valine, leucine, and isoleucine are essential for most species of *Lactobacillus*, with some strain-specific requirements. Cocci, including *O. oeni*, require a much wider range of essential acids, including methionine, phenylalanine, serine, arginine, tyrosine, and tryptophan. In addition, even if they are not indispensable, they generally need growth activators.

Besides their use in protein synthesis, amino acids may also maybe metabolized to release ATP. This is the case for arginine, often the most abundant amino acid in must and wine. Heterofermentative lactobacilli are well known to use the arginine deiminase pathway, but its use by *O. oeni* was not envisaged. However, it is now recognized that some strains of this species degrade arginine via the same pathway. The ADI pathway comprises three reactions, catalyzed by arginine deiminase (ADI; EC 3.5.3.6), ornithine transcarbamylase (OTC; EC 2.1.3.3), and carbamate kinase (CK; EC 2.7.2.2) (Fig. 3.3). The genes encoding these three proteins and arginine/ornithine exchangers (ArcD1 and ArcD2) have been sequenced for *O. oeni* and a regulator has been identified. The clearly-characterized *arc* operon of *O. oeni* comprises six genes: *arcR*, *A*, *B*, *C*, *D1*, *D2*. The *arc* locus also includes another gene, *argS2*, coding for a putative arginyl-tRNA synthase. Finally, the entire ADI locus (10542 bp) of *O. oeni* is thought to comprise seven genes. *ArgS2* is involved in regulation, funnelling the available arginine towards either degradation as an energy source or protein synthesis. Interestingly, the *arc* locus is either totally present or absent, according to the strain, and is related to the ability to assimilate arginine (Fig. 3.4) (Nehme *et al.*, 2006). Consequently, since ATP is released via this pathway, *O. oeni* strains that carry the locus probably have an additional advantage, especially in wine, where few energy sources are available (Tonon and Lonvaud-Funel, 2000).

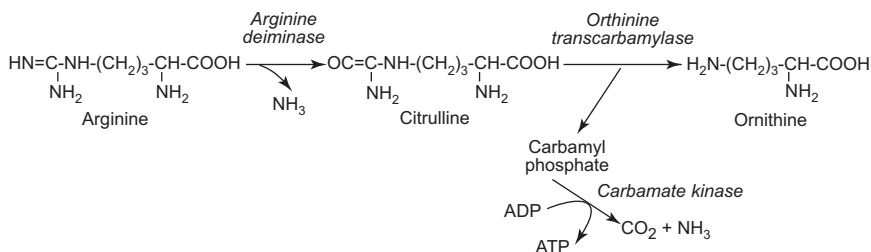


Fig. 3.3 The arginine deiminase (ADI) pathway.

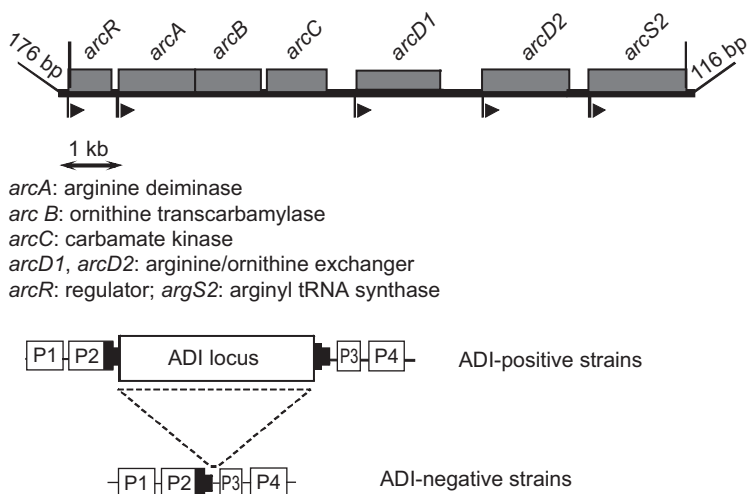


Fig. 3.4 The seven genes that comprise the arginine deiminase locus ‘ADI locus’ (upper part of the figure). The ADI locus is inserted in ADI-positive strains in the chromosome within a region that has been found in ADI-negative strains (between P2 and P3) (lower part of the figure).

Lactic acid bacteria in wine also have a variable capacity to decarboxylate some amino acids to form biogenic amines. This subject is discussed below, as the possible health effect of amines represent a real concern for winemakers (Section 3.5.2). However, it should also be noted that these reactions help the bacteria to grow, or even survive.

As a general rule, there is no real nitrogen deficiency in wine, after fermentation. The free amino acid composition is varied and seems sufficient to cover needs. In addition, bacterial protease activity hydrolyzes the abundant peptides and proteins in the medium. Nitrogen deficiency only occurs during the intense period of alcoholic fermentation, when, for example, the arginine concentration may virtually drop to zero after a few days of fermentation when the yeast biomass is growing very rapidly. The temporary deprivation of specific amino acids is probably a factor in the spontaneous elimination of certain bacterial strains present

in the must. However, these deficiencies are only temporary, thanks to synthesis, secretion, and yeast autolysis, enhancing the nitrogen concentration of the medium and both the variety and concentrations of amino acids. The nitrogen concentration of the medium prior to malolactic fermentation is adequate to support the expected increase in bacterial biomass, especially when the wine remains in contact with yeast lees that release their contents following autolysis. Substantial quantities of free amino acids, as well as peptides and proteins, are still available when malolactic fermentation has been completed (Remize *et al.*, 2006).

The purine and pyrimidine bases are not only compounds of nucleic acids, but also cofactors essential to cells. They are present in the must and remain in the wine after fermentation. They are also released by yeast during fermentation, as well as in the lees. Vitamins, generally involved in metabolism as cofactors, are naturally essential to bacteria. Thiamine, riboflavin, pantothenic acid, nicotinamide, nicotinic acid, and folic acid are the most necessary and concentrations in wine after malolactic fermentation still exceed requirements. These molecules are present in grape must and, although they are used by yeast, they are returned, at even higher concentrations in some cases. Finally, monovalent and divalent cations (K^+ , Na^+ , Mn^{2+} , and Mg^{2+}) are also necessary for the proper growth of bacteria. None of the few studies on this subject has shown any restriction on growth due to deficiencies of these elements.

Basically, all the evidence indicates that wine is easily capable of providing the large, complex nutrition requirements of LAB identified in laboratory experiments. Variations in nutrient availability depend not only on the initial composition of the grape must, but also on the changes caused by yeast, in terms of residual sugar and nitrogen-based substances. Moreover, different yeast strains, as well as, probably, the duration of autolysis and contact with the lees, are decisive factors.

3.2.3 The main factors that affect lactic acid bacteria growth in wine: pH, ethanol, temperature, sulphur dioxide, and other toxic compounds

Lactic acid bacteria growth should be as fast as possible, starting immediately after alcoholic fermentation. Indeed, the wine is still unstable from a microbial standpoint as *Brettanomyces bruxellensis* spoilage yeast, for example, selected together with the bacteria during alcoholic fermentation, may grow even faster than bacteria. The latency phase between alcoholic and malolactic fermentation usually lasts just a few days or weeks, but may be as long as several months in some cases. This latency phase usually ends with a sudden spurt of bacterial growth. Within a few days, populations that were initially at very low concentrations, about $10-10^2$ CFU/mL, may reach and exceed the 10^6 CFU/mL threshold that triggers malolactic fermentation, leading to significant degradation of L-malic acid. The length of the lag phase depends on many physical and chemical factors, including pH, alcohol concentration, temperature, and sulphur dioxide concentration. These crucial parameters determine the recommendations for malolactic fermentation control. To summarize, the higher the alcohol concentration, the lower the pH and temperature, the longer the latency phase, especially if free sulphur dioxide is also present.

pH

pH is probably the factor with the greatest influence on the development and fate of LAB in wine. Depending on several factors, such as the grape variety, climate, and ripeness, the minimum pH of wine before malolactic fermentation ranges from 2.9 to 3.6. Bacteria grow better at higher pH, usually resulting in a shorter time lag before malolactic fermentation. For example, all other conditions being similar, the biomass required to degrade malic acid is reached sooner at pH 3.4 than at pH 3.2. pH also has an impact on strain selection. Bacteria have to resist acidity and maintain a suitable intracellular pH to maintain the activity of all the systems required for viability. The membrane is the barrier that ensures this homeostasis. The sophisticated systems that prevent proton accumulation inside the cell comprise different membrane proteins according to the species, and even strain, of bacteria. Among the diverse mixture of strains in the grape must, those strains best adapted to the pH are selected. Obviously, at more acidic pH, fewer strains remain active. In general, *O. oeni* is best-suited to malolactic fermentation in acidic wines, but substantial populations of *Lactobacilli* and *Pediococci* remain. Although these species are not very active during malolactic fermentation, they survive well in wine, even after sulphiting, known to be less effective at high pH, and may thus impact the composition of the finished wine.

Sulphur dioxide

Its toxicity for microorganisms makes sulphur dioxide the most useful tool for microbial stabilization of wine. Lactic acid bacteria are more sensitive than yeast, especially to free sulphur dioxide, and its efficacy is strictly correlated with pH. Although grapes are generally sulphited on harvesting, free sulphur dioxide concentrations drop to around zero during fermentation, due to its aptitude to combine with many oxo compounds (ketonic compounds), including the acetaldehyde produced by yeast. However, some yeast strains also produce sulphur dioxide, which accumulates in the free form and may delay the beginning of malolactic fermentation. Afterwards, when malolactic fermentation is completed, sulphiting is essential, since no more biological activity is needed. The quantity of sulphur dioxide added at that time must not only take into account the fact that it combines with compounds in wine but also that free SO₂ is less effective at low pH. The lower the pH, the lower the dose of free SO₂ required to eliminate LAB. The concentration, expressed in molecular SO₂, should be around 0.4 mg/L.

Ethanol

Ethanol is toxic to all microorganisms, particularly LAB. It operates mainly on the cell membrane level, by modifying its fluidity and the environment of the proteins involved in both intracellular homeostasis and transporters. Excessive concentrations damage the structures built between lipids and proteins, eliminating the barrier between the inside of the bacterium and the wine. Membrane biological activities are crucial for controlling exchanges between outside and inside. Major damage reduces or even suppresses membrane functions and may result in cell death. However, the defence mechanisms of LAB enable them to withstand up to

12% ethanol. While ethanol concentration above 7–8% may have an inhibiting effect, depending on other environmental constraints, at lower concentrations, up to about 5%, it acts as an activator. It is assumed that its effect on fluidity facilitates the transport of certain substrates. Growth becomes more difficult at concentrations above 10%, but is still possible, even at 14–15%, and severely inhibited above that concentration. Tolerance varies among strains, with cocci, especially *O. oeni*, more resistant to high alcohol concentrations. In contrast, some lactobacilli, such as *L. plantarum*, are more easily eliminated by ethanol. The situation is further complicated by the fact that some strains of *O. oeni* are more sensitive than others. Lactic bacterial spoilage even occurs in fortified wines with ethanol concentrations exceeding 18% and heterofermentative *Lactobacilli* (*L. fructivorans* and *L. hilgardii*), as well as *L. plantarum* and *O. oeni*, have been isolated from spoiled wines. Actually, like many phenotypic traits, tolerance to ethanol depends on the strain and its genetic heritage. For several years, research has focused on identifying the mechanisms involved in ethanol resistance and its genetic determinants.

However, these three primary factors, pH, ethanol, and SO₂, interact and a favourable value of one factor may attenuate sensitivity to another (Ribéreau-Gayon *et al.*, 2006). For example, a higher pH may partially offset a high ethanol concentration. Alcohol concentration and pH depend on the raw material: very ripe grapes produce more alcohol, but less acidity. However, there are limits, and malolactic fermentation in red wine is difficult at ethanol concentrations above 15%, even if the pH is as high as 3.8–3.9. Acidity is a more significant variable than alcohol, as low acidity always entails a risk of microbiological deterioration and spoilage, mainly due to bacteria. During fermentation, low acidity hinders interactions between yeast and LAB, promoting early bacterial development and exacerbating competition with the yeast, potentially leading to a slow or stuck fermentation. This may be followed by lactic spoilage, characterized by volatile acidity resulting from the heterofermentation of sugars. During ageing, it decreases the effectiveness of SO₂ and facilitates bacterial survival.

Other compounds

Besides ethanol, the only molecules derived from the yeast metabolism currently known to have a toxic effect on LAB are medium-chain fatty acids and their esters, as well as sulphur dioxide, in addition to the amounts already present in wine following sulphiting of the grape must. Fatty acids are mainly synthesized by yeast for their membrane phospholipids, but some are secreted into the medium, at concentrations depending not only on the strain but also on the growing conditions. Consequently the toxicity of the environment is mainly due to yeast activity. In addition, due to specific metabolic features, the type of yeast strains dominant during alcoholic fermentation may impact the ‘malolactic fermentability’ of the resulting wine (Table 3.1).

C₁₀- and C₁₂-chain fatty acids, as well as the C₁₈ palmitic and palmitoleic acids, deteriorate membrane functions, thus inhibiting bacterial growth, malolactic fermentation, and other enzyme activities (Lonvaud-Funel *et al.*, 1988). Studies have

Table 3.1 Duration of the malolactic fermentation in eight wines resulting from the alcoholic fermentation of the same grape must by eight different yeast strains and inoculated with four different malolactic starters. Dodecanoic acid (C_{12}), the most toxic fatty acid, and SO_2 were determined after the alcoholic fermentation

Yeast	Average duration (days)	C_{12} (mg/L)	Free SO_2 (mg/L)
A	12	0.3	8
B	9.25	0.4	10
C	6.25	0.2	3
D	9.25	0.3	21
E	8.5	0.3	35
F	21	0.45	51
G	10.25	0.3	16
H	7	0.4	2

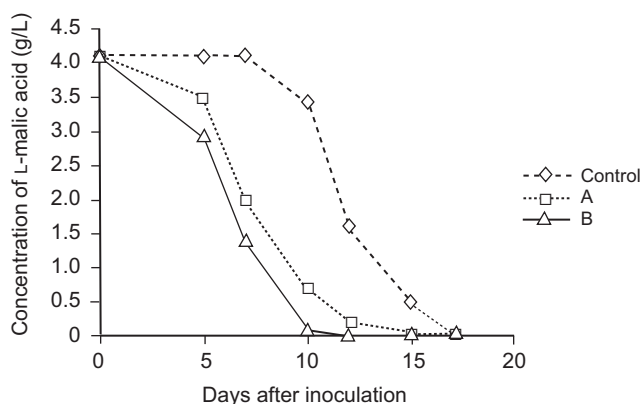


Fig. 3.5 Evolution of L-malic acid in a red wine. Wines A and B were, respectively, treated with 0.1 and 0.5 g/L of 'yeast cell walls'. Wines A, B and the control were inoculated at the same time with the same starter.

demonstrated not only individual but also synergistic effects. They affect interactions between yeast and LAB, and these phenomena are exacerbated by the presence of ethanol. However, like ethanol, the accumulation of fatty acids in wine is unavoidable. Knowledge of these inhibitors was used to develop a detoxifying pre-treatment with 'yeast hulls'. The treatment product, consisting of yeast cell wall, washed to eliminate lipids, fixes the fatty acids in the wine. This reduces toxicity, facilitating the growth and survival of bacteria and significantly accelerating malolactic fermentation (Fig. 3.5).

Temperature

Temperature is the easiest parameter to control during winemaking, thanks to temperature-controlled vats. The optimum temperature for growing LAB in the laboratory is 27–30 °C. In wine, however, it is around 20–22 °C, due to the very

different chemical composition of the medium. The difference is probably mainly due to effect of ethanol on plasma membrane as, like temperature, it affects membrane fluidity and activity. However, it is not uncommon for malolactic fermentation to continue at temperatures even lower than 18–19 °C, once it has already started, as the necessary biomass was already well established when conditions were more favourable. The degradation of malic acid simply continues at a slower rate. In any case, it is recommended to keep wine at a temperature of around 20 °C until the bacterial population has increased sufficiently for malic acid degradation to be effective.

To summarize, if temperature is properly controlled, the alcohol/pH balance determines the latency period between the two fermentations and the time required to complete malolactic fermentation.

3.3 Variations in the diversity of lactic acid bacteria species during winemaking

3.3.1 Lactic acid bacteria species in grape must and wine

A dozen species of LAB were identified by traditional methods, involving isolating bacteria cultured on nutrient agar medium, and determining their phenotypes by referring to the classification in Bergey's Manual. Today, molecular methods offer faster, more reliable identification. Earlier DNA/DNA hybridization methods have now been replaced by molecular tool sets for identifying these bacteria. Some require the isolation of bacteria on nutrient agar medium, while others, such as polymerase chain reaction–denaturing gradient gel electrophoresis (PCR–DGGE), use DNA directly extracted from grape must or wine and provide more comprehensive identification. The latest advances in identifying LAB species in grapes and wine target the *rpoB* gene that encodes the β subunit of RNA polymerase (Claisse *et al.*, 2007). These molecular methods have identified a larger number of species, particularly *Lactobacilli*.

Table 3.2 lists the lactic bacteria identified to date in grapes and wine, highlighting the great diversity of species in the lactobacillus genus. Most species are strictly heterofermentative. The only two species classified as facultatively

Table 3.2 List of the main lactic acid bacteria species in grapes and wine

Genera	Species
<i>Lactobacillus</i>	Optional heterofermentative: <i>L. plantarum</i> , <i>L. casei</i> Strict heterofermentative: <i>L. hilgardii</i> , <i>L. brevis</i> , <i>L. fructivorans</i> , <i>L. diolivorans</i> , <i>L. buchneri</i> , <i>L. fermentum</i> , <i>L. jensenii</i> , <i>L. kunkei</i> , <i>L. nagelii</i> , <i>L. vermiforme</i>
<i>Pediococcus</i>	<i>P. parvulus</i> , <i>P. damnose</i> , <i>P. pentosacens</i>
<i>Leuconostoc</i>	<i>L. mesenteroides</i>
<i>Oenococcus</i>	<i>O. oeni</i>

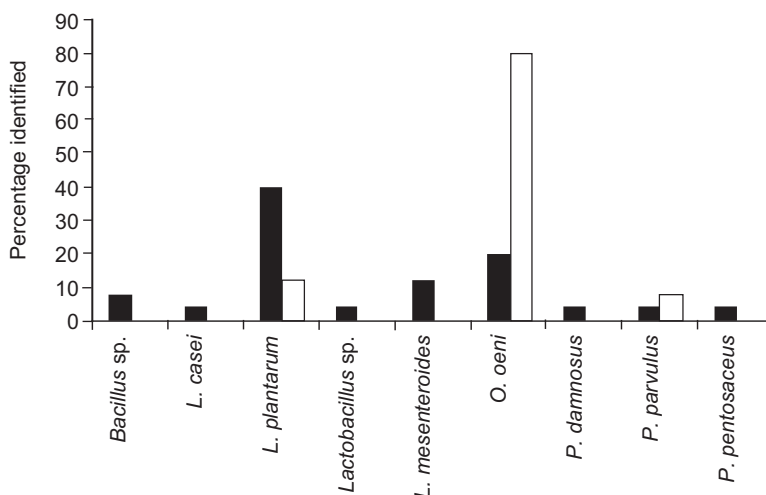


Fig. 3.6 Percentage of different lactic acid bacteria species identified at the beginning of alcoholic fermentation (filled columns) and at the end of alcoholic fermentation (open columns).

heterofermentative are homofermentative in relation to glucose as they only produce lactic acid under optimum conditions, although their heterofermentative metabolic pathway also enables them to ferment pentoses. Several species of homofermentative *Pediococcus* have been identified. Only two heterofermentative cocci species have been identified to date: *Leuconostoc mesenteroides* and *Oenococcus oeni*, the latter being considerably more prevalent.

3.3.2 Selection of *Oenococcus oeni* during alcoholic fermentation

Under normal winemaking conditions, the LAB already present in grape must are inhibited by antagonisms with yeast during alcoholic fermentation. In must, the initial population of 10^2 – 10^4 UFC/mL may include as many as eight or even nine species. The variety and concentration depend on climate, ripeness, and, possibly, other factors. During alcoholic fermentation, the total population grows during the first 2 or 3 days, not usually exceeding 10^5 UFC/mL, then declines. Under these conditions, by the end of alcoholic fermentation, the bacterial population is usually under 10^3 UFC/mL. Competition with yeast not only makes it difficult for the bacteria to multiply but also induces a significant selection among the species, in favour of *O. oeni* (Fig. 3.6). This species is the best adapted to the harsh conditions in wine and largely dominates the microbial system during malolactic fermentation. However, other *Lactobacillus* sp. or *Pediococcus* sp. may also survive, especially at higher pH, and it is not unusual to isolate them from wine during barrel ageing or in bottle.

3.3.3 Intra-specific diversity of *O. oeni* and variations during winemaking

While it is important to identify the LAB species, it has gradually become accepted that identifying specific strains is much more useful for winemakers. This is well known in the case of *S. cerevisiae* yeast, where specific strains are selected for use in alcoholic fermentation. Before this could be applied to bacteria, the first problem to solve was how to distinguish among different strains. There is obviously no interest in studying the properties of a strain if it cannot be reliably distinguished from other strains.

Thanks to molecular methods, it is now possible to type *O. oeni* strains by profile. Two methods are used. One, using PCR or random amplification of polymorphic DNA (RAPD), was described by (Zapparoli *et al.*, 2000) and also used by (Bartowsky *et al.*, 2003). This method is fast, but the profiles are sometimes difficult to distinguish from each other and reproducibility is poor. The other, much more reliable, robust method, is based on the hydrolysis of total DNA. The few cutting site enzymes are chosen to generate a suitable number of DNA fragments, which are then separated for analysis. However, these fragments are long and migration in a pulsed electric field is required to separate them. This process, commonly known as PFGE (pulse field gel electrophoresis), produces a different profile for each strain. In the rare cases where one enzyme produces the same pattern for different strains, they are distinguished by using different enzyme.

RAPD or PFGE typing of a collection of *O. oeni* collected during winemaking showed that diversity decreased from the beginning to the end of the process. Figure 3.7 shows the results of RAPD identification performed on isolates; the profiles give an approximation of diversity and show the dominant profiles. The upper part of the figure features different profiles, showing that several strains were present in similar proportions. The two profiles shown in the lower part indicated that only two strains dominated the system. Thus, both species diversity and strain diversity within the *O. oeni* species decrease during fermentation. The strains responsible for malolactic fermentation in wine are already present in the must and survive in an environment that becomes increasingly inhospitable as fermentation progresses.

Subtractive genome hybridization between two strains of opposite oenological potential showed many different DNA fragments in the two genomes. These fragments were used to screen a collection of high-performance and banal strains. A previous genomic comparison between two strains with opposite oenological potential revealed that 16 genetic regions were systematically found in the strains that performed best (Delaherche *et al.*, 2006). These regions were used as markers to characterize the *O. oeni* strains isolated from three wines, A, B, and C, during the winemaking process. Some strains had relatively few markers while others had all of them. In addition, the average number of markers per strain increased as winemaking progressed, reaching a maximum during malolactic fermentation (Table 3.3). Finally, the strains responsible for malolactic fermentation had the most markers. In wines A and C, these strains (A-I, A-II, A-III, C-I and C-II) had all 16 markers. On the grape surface and in the must, these strains were diluted among others that had fewer markers, but they became dominant by the end of

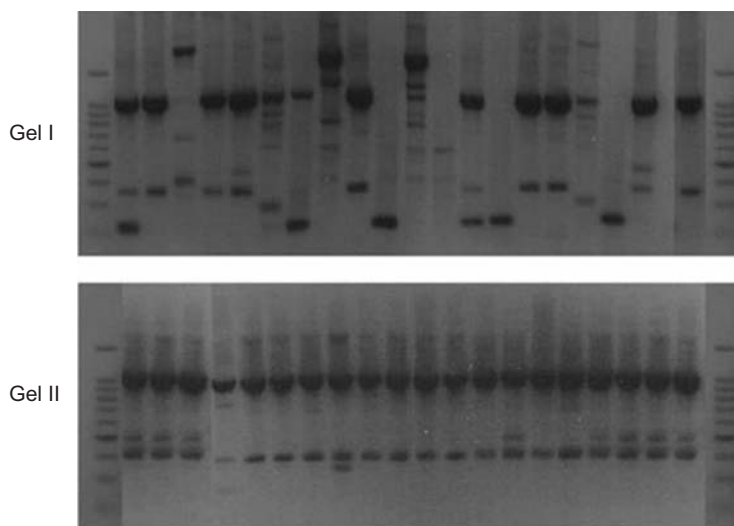


Fig. 3.7 RAPD profiles of *O. oeni* isolated at the beginning (Gel I) and at the end of the alcoholic fermentation (Gel II).

Table 3.3 Evolution of the number of *O.oeni* strains isolated from the grape to the wine during malolactic fermentation (typed by RAPD) and number of genetic marker regions in the strains

	Number of RAPD profiles			Number of markers ¹		
	Plot A	Plot B	Plot C	Plot A	Plot B	Plot C
Grape	12	7	8	10	9	10
Must	10	4	7	13	13	11
Mid AF	4	4	6	15	13	11
End AF	4	3	4	15	13	13
Mid MLF	3 (A-I, A-II, A-III)	3	2 (C-I, C-II)	16	14	16

¹ The three strains isolated in wine A (A-I, A-II and A-III) and the two strains isolated in wine C (C-I and C-II) had all the 16 genetic markers.

RAPD, random amplified polymorphic DNA. AF, alcoholic fermentation. MLF, malolactic fermentation.

alcoholic fermentation and were the only ones present during malolactic fermentation (Renouf *et al.*, 2008).

3.4 Lactic acid bacteria and improving wine quality

After alcoholic fermentation, the L-malic acid concentration in the wine is lower than in grapes, although malic enzyme activity in the most effective yeast strains degrades at most 25–30% of the initial total. Malolactic fermentation by bacteria consists of the decarboxylation of malic acid to L-lactic acid, catalyzed by the

malolactic enzyme (Ribéreau-Gayon *et al.*, 2006). Total acidity decreases, as the L-malic acid molecule is stoichiometrically replaced by L-lactic acid. The apparent drop in acidity is not only due to loss of the carboxylic function, but also reflects the fact that the strong green taste of malic acid is replaced by the smoother taste of lactic acid. The most obvious effect of malolactic fermentation is, therefore, biological deacidification. At the same time, pH increases. This transformation dominates all the other metabolisms. Increasingly, in regions where overripe grapes produce wines with low acidity, deacidification is not required. The initial malic acid concentration of these wines is so low that malolactic fermentation barely reduces their acidity. It is still important, however, as LAB degrade many other substrates in wine to produce new compounds. While the nature of these molecules is relatively little-known, sensory analysis detected unambiguous changes, showing that the same wine evolves differently depending on the conditions of malolactic fermentation. The ability to degrade or synthesize new molecules with sensory impact is also strain-dependent. However, some varietal aromas revealed by yeast during alcoholic fermentation disappear or change after malolactic fermentation. Some red and white wines are characterized by grape aromas, which are attenuated by malolactic fermentation. This explains why malolactic fermentation has a positive impact on most red and white wines where the fruitiness of the grape variety is not essential. The more complex flavours and mouthfeel of these wines are enhanced by malolactic fermentation. This is often followed by a long period of barrel-ageing, and these wines only reach full maturity after several years' bottle-ageing.

3.4.1 Deacidification by malic acid degradation: the malolactic reaction, the malolactic enzyme

Depending on the grape variety, region of production, or the ripeness of the grapes, the malolactic reaction transforms from 1.5 to over 8 g/L L-malic acid. The reaction is catalyzed by the malolactic enzyme, a protein that has been extracted from several species of LAB (Lonvaud-Funel, 1995). This enzyme is distinct from the malic enzyme, known in many other organisms and microorganisms, as it leads directly to L-lactate (Fig. 3.8). Although NAD⁺ is an essential cofactor, its role is not really understood. According to the reaction, it is not reduced, as is the case with the malic enzyme in the oxidative decarboxylation of L-malate to pyruvate. Although these two proteins have different functions, it is remarkable to note that the sequence of amino acids derived from their nucleic sequence coding is very similar (Groissillier and Lonvaud-Funel, 1999). So far, the difference in activity has not been explained comparing the protein sequences.

The malolactic activity of bacteria in wine depends not only on the concentration and specific activity of the protein but also on the integrity of the cells. Indeed, the reaction takes place inside the cell in an optimum environment, protected from the numerous inhibitors in wine. Any damage to the membrane reduces the malolactic activity of the entire cell. The key factor is that the optimum pH for enzyme activity is 5.7, whereas bacteria evolve in wine at a much lower pH.

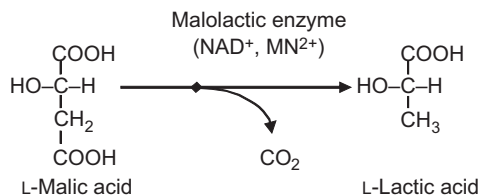


Fig. 3.8 The malolactic reaction.

Clearly, it is essential to preserve the integrity of the cell membrane activities that control internal pH, as well as the protein involved transporting the substrate and the efflux of the product. Undamaged cells also provide the necessary cofactors (Mn^{2+} , NAD^+) as well as protection from the inhibitory effects of many compounds in the medium (e.g. carboxylic acids and phenolic compounds). For these reasons, no enzyme preparation directly usable in wine, nor any enzyme reactor for wine treatment has ever been effective.

In practice, malolactic fermentation requires a large biomass of viable bacteria, in excess of the threshold value of 10^6 CFU/mL. This concentration is usually reached spontaneously after the growth phase, but is also the objective for malolactic starters, added to replace defective indigenous microflora.

3.4.2 Citric acid degradation: metabolic pathway

Unlike the malic acid concentration, citric acid concentrations remain relatively stable, at about 300 mg/L, in must and wine. Nevertheless, the degradation of this acid by LAB is important, as even low concentrations of the end products, acetic acid and diacetyl, impact the aromatic quality of wine (Bartowsky and Henschke, 2004). The first reaction of the metabolic pathway produces acetic acid and oxaloacetic acid by cleavage of citric acid, catalyzed by citrate lyase (Fig. 3.9). Oxaloacetic acid is then decarboxylated to pyruvate, which is further transformed in several ways. Some molecules form fatty acids and others become acetoinic compounds (diacetyl, acetoin, and butanediol), while lactic acid may only be formed if reduced coenzyme is available. The degradation of citric acid by LAB automatically leads to an increase in volatile acidity in wine, as an average of 1.2 molecules of acetic acid are produced from each molecule of citric acid. However, due to the small quantities concerned, this phenomenon is not detrimental to wine quality.

The pyruvate that is not directed towards lipids or reduced to lactate becomes involved in a series of reactions that produce diacetyl, acetoin, and butanediol. Diacetyl, the most highly-oxidized of these three molecules, is the most fragrant, giving wine a buttery aroma at concentrations above 5–8 mg/L. This effect is more or less acceptable and appreciated in different wines, but it generally enhances their sensory complexity.

O. oeni, the dominant species in malolactic fermentation, always degrades citric acid, producing variable amounts of diacetyl from the same initial concentration of

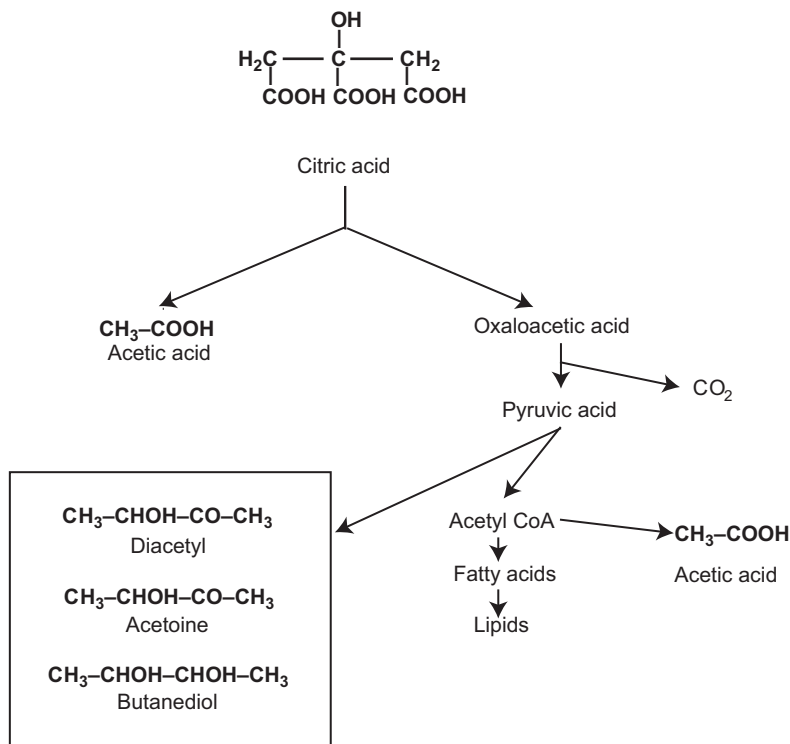


Fig. 3.9 Schematic representation of the citric acid degradation pathway by lactic acid bacteria.

citric. Diacetyl production increases when the need for pyruvate is low, especially when bacterial multiplication is slow, due to limitation by adverse environmental conditions. Thus, for a given strain of *O. oeni*, more diacetyl accumulates in wine at lower pH and temperatures. Theoretically, all other conditions being equal, wines where malolactic fermentation is slower have relatively high diacetyl concentrations. In fact, however, during the degradation of citric acid, the diacetyl concentration reaches a maximum, then drops as it is reduced to form much less fragrant acetoin. Moreover, this transformation is promoted by prolonged contact with bacterial biomass or yeast lees. Consequently, winemakers may, to some extent, maintain diacetyl concentrations to enhance aroma by eliminating the microorganisms earlier in the process or, in contrast, attenuate its impact by maintaining the wine on the lees (Nielsen and Richelieu, 1999).

In addition, diacetyl is a highly reactive molecule, due to its two ketonic functional groups (ketones). It reacts chemically with some amino acids and produces complex molecules with sulphur amino acids in wine, especially cysteine, resulting in a variety of floral, fruity, and toasty aromas that contribute to the final bouquet of wine (Pripis-Nicolau *et al.*, 2000).

3.4.3 Methionine metabolism: metabolic pathway and sulphur compounds

Methionine, a sulphur-containing amino acid, may be transformed into sulphur compounds that generally impact sensory quality. In laboratory cultures, bacteria isolated from wine degraded methionine, producing mainly methanethiol, dimethyl disulfide, 3-(methylsulfanyl)propan-1-ol, and 3-(methylsulfanyl)propionic acid. All strains of lactobacilli, especially *O. oeni*, form significant quantities of methanethiol and dimethyl disulfide. These low boiling point, volatile sulphur compounds are highly odoriferous and unpleasant-smelling. *O. oeni* also accumulates large amounts of 3-(methylsulfanyl)propan-1-ol and 3-(methylsulfanyl)propionic acid in wine. The former is the most significant volatile sulphur compound in wines, both in terms of the quantities present and its contribution to aromatic complexity. Its perception threshold in synthetic solution is 1200 µg/L. Only 3-(methylsulfanyl)propionic acid is systematically formed in significant amounts during malolactic fermentation in red wine, with a potential organoleptic impact when final concentrations exceed the perception threshold. In wine, the relevant aroma descriptor is red-berry fruit (Pripis-Nicolau *et al.*, 2004).

The first phase of this metabolic pathway in *O. oeni* is catalyzed by an aminotransferase similar to the aromatic amino acids and aspartate aminotransferase described in other LAB. This is the key reaction in *O. oeni*'s methionine metabolism, producing an intermediate compound, 2-oxo-4-(methylthio)butyric acid, which is reduced to methional, another important intermediate. Finally, methional is either reduced to 3-(methylsulfanyl)propan-1-ol or oxidized to 3-(methylsulfanyl)propionic acid. These rapid transformations are only detectable during the mid-exponential growth phase of *O. oeni*. Furthermore, 2-oxo-4-(methylthio)butyric acid probably forms 3-(methylsulfanyl)propionic acid via an oxidative decarboxylation pathway (Fig. 3.10).

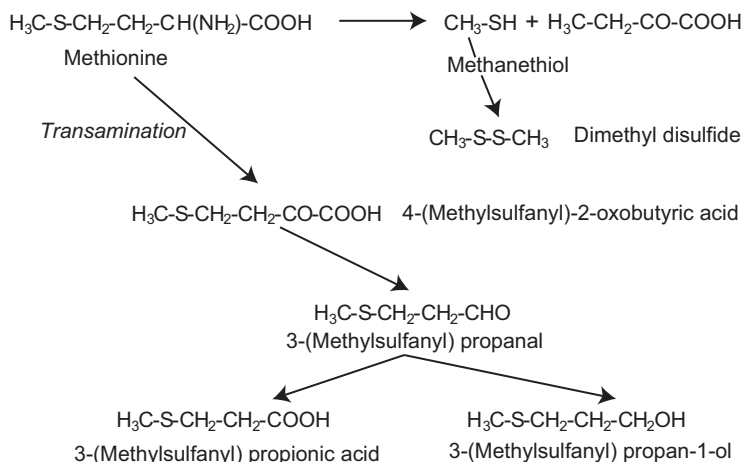


Fig. 3.10 The main steps of methionine conversion into odorous molecules by lactic acid bacteria.

3.4.4 Osidase activities and aroma compounds

Some transformations involving glycosidases, acting on various glycosidated derivatives in grape must and wine, have been demonstrated in LAB (Grimaldi *et al.*, 2005). These diverse reactions have attracted considerable interest, as they may clarify the role of bacteria in the sensory changes observed during malolactic fermentation. Hydrolysis of these molecules releases free aglycones, which contribute to colour or aroma. The sugar part of the molecule is metabolized by the fermentative pathway. Lactic acid bacteria probably benefit from this additional source of energy in the nutrient-depleted medium after alcoholic fermentation (Vivas *et al.*, 1997).

Grape mono- and diglucosides are precursors of non-volatile aromatic compounds, such as norisoprenoids, as well as highly fragrant terpenes. Aroma compounds, such as linalol, farnesol, and β -damascenone, are released during malolactic fermentation, with varying efficiency, depending on the bacteria strain and composition of the substrate. The release of variety-specific volatile compounds has been demonstrated for Tannat, Chardonnay, and Muscat. Ugliano *et al.* (2003) compared the impact of several malolactic starters on different Muscat aroma substrates. The grape variety aromas released by bacteria vary widely, depending on the strains and terpene substrates involved. The *O. oeni* strain studied by Indecco *et al.* (2004) exhibited glucosidase activity, but had low L-rhamnopyranosidase and L-arabinofuranosidase. It released the disaccharide bound by glycosidic linkages to the aglycone characteristic of Chardonnay. Finally, in their study on Tannat, Boido *et al.* (2002) suggested that the aromatic aglycones released were further metabolized, unless they were trapped by macromolecules, such as bacterial exopolysaccharides. Other glycosylated compounds that offer possible substrates for bacteria are anthocyanins, the red grape pigments also found in wine. These consist of monoglucosides and acylated monoglucosides of various anthocyanidins, mainly malvidin.

3.5 Lactic acid bacteria and wine spoilage: undesirable lactic acid bacteria strains

3.5.1 Increased volatile acidity due to the fermentation of sugars: 'lactic spoilage'

Heterofermentative LAB, including *O. oeni*, ferment glucose and fructose via the pentose phosphate pathway, to produce mainly D-lactic acid, acetic acid, ethanol, and CO₂. Fructose is also reduced to mannitol. The best energy substrates for *O. oeni* and other LAB are sugars – glucose, fructose, and pentoses – not fermented by yeast during alcoholic fermentation. As concentrations are normally below 1 g/L, if alcoholic fermentation was correctly completed, any increase in volatile acidity due to heterofermentative LAB is limited. The situation is totally different if the wine still contains several grams per litre of glucose and fructose. Under those conditions, even *O. oeni* may become a spoilage agent, as the hetero-

fermentative pathway produces significant amounts of acetic acid, resulting in 'lactic spoilage', one of the most challenging problems in winemaking.

This phenomenon occurs when the antagonism during alcoholic fermentation is inefficient and relatively high concentrations of bacteria, around 10^4 – 10^5 CFU/mL, remain in the medium. High pH, above 3.5–3.6, is the main factor responsible for this situation. If alcoholic fermentation slows down for any reason, yeast activity declines and the LAB benefit from the decrease in competition. The growing population of *O. oeni* at the end of alcoholic fermentation exacerbates this decline by direct hydrolysis activities on yeast cell walls, slowing alcoholic fermentation even more. Concentrations of unfermented sugars may be as high as several grams per litre, even exceeding 10 g/L in some cases.

The most efficient way to reduce the bacterial population during alcoholic fermentation is to sulphite the grapes as soon as they are harvested or crushed. Sulphur dioxide is more toxic to bacteria than yeast. At the end of alcoholic fermentation, free sulphur dioxide concentrations are very low or non-existent, as it combines with acetaldehyde produced by yeast and other compounds in wine. The bound form is much less toxic to bacteria, so it does not prevent the subsequent growth necessary for malolactic fermentation (Ribéreau-Gayon *et al.*, 2006).

3.5.2 Production of biogenic amines: metabolic pathways, enzymatic and genetic aspects

Wine may contain variable concentrations of several aliphatic, aromatic, and heterocyclic compounds, such as ethylamine, isoamylamine, diaminobutane (putrescine), diaminopentane (cadaverine), tyramine, phenylethylamine, and histamine. Some of these compounds are already present in grapes (isoamylamine, methylamine, and putrescine), while others are produced during or after malolactic fermentation (Lonvaud-Funel, 2001). In this case, they result from the decarboxylation of the corresponding amino acid (Fig. 3.11). Putrescine may also be produced by deamination of agmatine, produced by the decarboxylation of arginine.

The first study concerning histamine in winemaking produced conflicting results. The most common assumption was that *Pediococcus* spp. were responsible for this phenomenon, as they had not been properly controlled during winemaking. However, as analyses revealed that many wines contained biogenic amines, more detailed studies were carried out in the early 1990s. The first study examined the bacterial biomass harvested from a wine containing high concentrations of biogenic amines. It consisted mainly of *O. oeni* strains. They were inoculated into amine-free wines where they produce the amines. Further analyses revealed that more amines were produced in wines with high pH and those maintained on yeast lees (Lonvaud-Funel and Joyeux, 1994).

One *O. oeni* strain was isolated. Its histidine decarboxylase (HDC) activity was studied and the enzyme was purified to homogeneity and fully characterized (Coton *et al.*, 1998b). However, not all *O. oeni* produce histamine. The gene encoding *hdcA* decarboxylase was sequenced, followed by the entire operon.

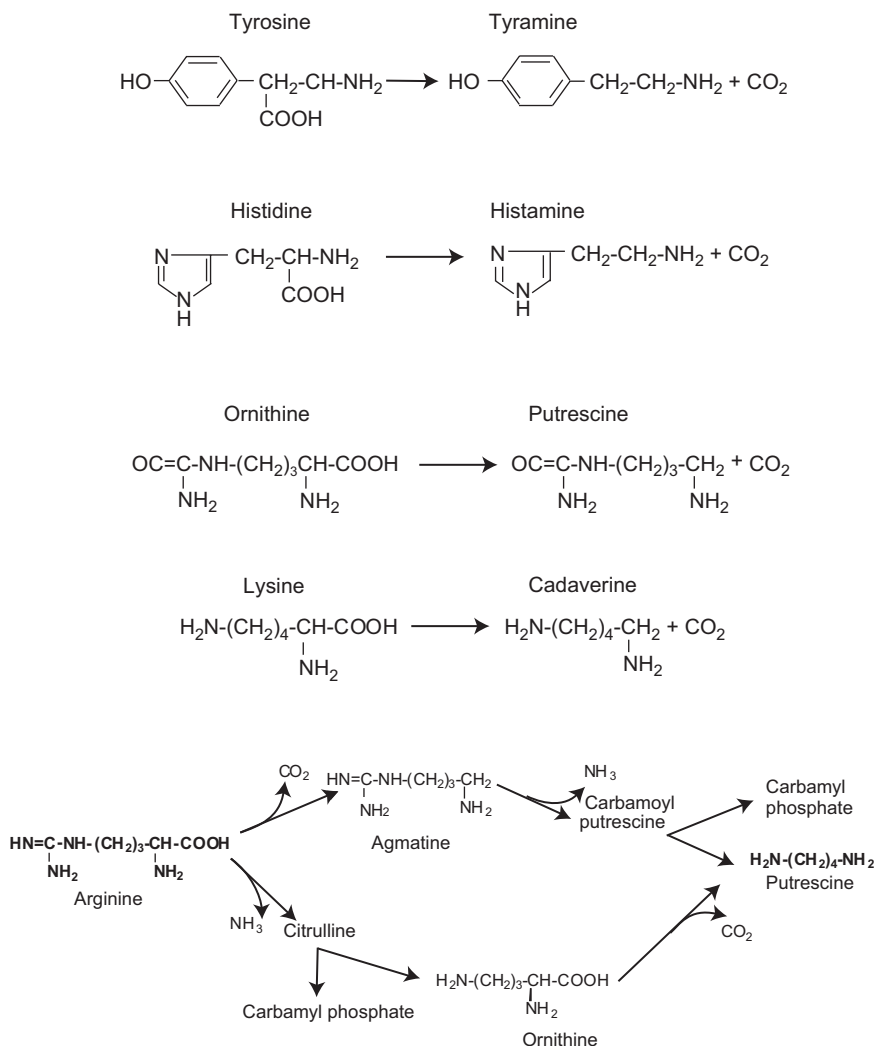


Fig. 3.11 Biogenic amine production from amino acids by lactic acid bacteria.

Tyramine production was investigated in the same way, including analysis of tyrosine decarboxylase (TDC), initially purified from a *L. brevis* isolated from wine. The *tdc* gene sequence and its entire genetic environment were identified using the purified enzyme sequence in a reverse genomic approach (Lucas *et al.*, 2003). Both HDC and TDC activities require a full sequence of four genes: one for decarboxylase, one for an unknown protein function, one for the corresponding aminocyl tRNA gene, and one for an antiport for exchanging the amino acid with the biogenic amine (Fig. 3.12). Putrescine is produced by the decarboxylation of ornithine, or the deamination of agmatine, produced by the decarboxylation of

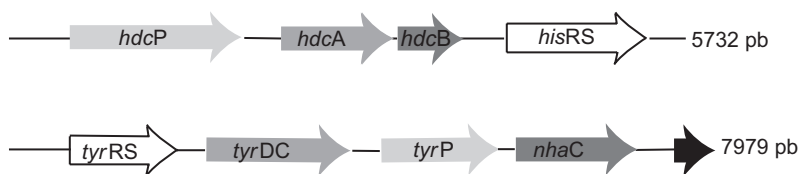


Fig. 3.12 Schematic representation of the gene clusters coding for the histidine and tyramine decarboxylation pathways. *hdcA* and *tyrDC*: histidine and tyramine decarboxylases. *hdcP* and *tyrP*: histidine/histamine and tyrosine/tyramine antiporters. *hisRS* and *tyrRS*: histidyl and tyrosyl tRNA synthases. *hdcB*: unknown. *nhaC*: Na⁺ exchanger.

arginine. The ornithine decarboxylase gene of one *O. oeni* strain is described as part of an entire cluster, also comprising a gene coding for a protein similar to a putrescine/ornithine antiporter (Marcobal *et al.*, 2004, 2006a). Interestingly the agmatine deiminase is a continuation of the *tdc* operon in the genome of an *L. brevis* strain (Lucas *et al.*, 2007). The coding cluster of genes suggests that the successive reactions are very similar to the arginine deiminase pathway. It comprises six putative genes coding for a putrescine transcarbamylase, agmatine/putrescine exchanger, agmatine deiminase, carbamate kinase, and a transcription regulator. This deiminase pathway, like the arginine deiminase pathway, generates metabolic energy by hydrolyzing carbamyl phosphate and produces ammonia, which is involved in pH homeostasis.

The most interesting finding concerning the genetics of the biogenic amine-producing strains is that they carry either the entire region necessary for the phenotype or none of it (Lucas *et al.*, 2005). Most publications suggest that this variability is due to horizontal gene transfers. Some clusters are stabilized on the chromosome and others on plasmids. It may depend on the strain and its history in the wine environment, as all these pathways are beneficial for their survival and growth in wine via energy conservation mechanisms in the electrogenic transport of substrate/product and its decarboxylation, and energy production via phosphorylation, in the case of agmatine deiminase (Lucas *et al.*, 2007). In fact, only some strains of LAB in any species may decarboxylate amino acids, so, for example, it is not true to say that *Pediococcus* spp., for example, are responsible for biogenic amines. However, although biogenic amine-producing strains of several species are very common in wine (Lucas *et al.*, 2008), all the affected wines do not contain biogenic amines and concentrations are not directly proportional to population. Many environmental factors are assumed to play a crucial role in determining the final amine concentrations (Marcobal *et al.*, 2006b).

3.5.3 Glycerol degradation to acrolein and ‘bitterness’

Besides ethanol, glycerol, an important factor in wine quality, is the most plentiful compound in wine produced by yeast during alcoholic fermentation. Wines contain an average of 8–9 g/L glycerol and some sweet wines have even higher values. Some LAB degrade glycerol via the glycerol dehydratase enzyme (Claisse

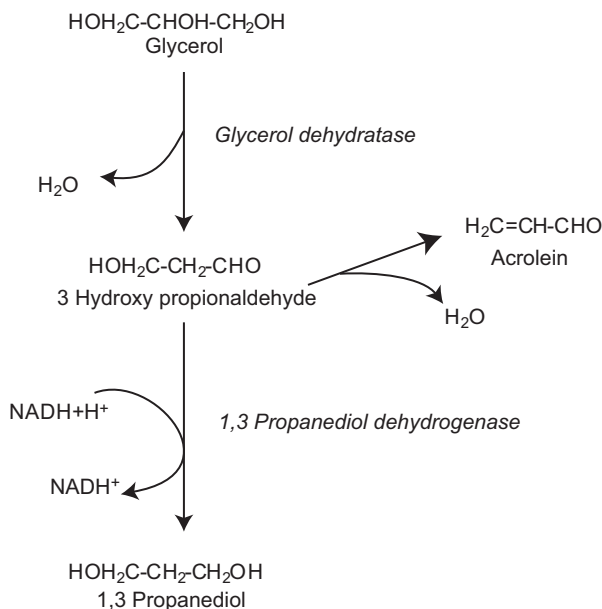


Fig. 3.13 Schematic representation of the production of acrolein from glycerol by lactic acid bacteria.

and Lonvaud-Funel, 2000). This is the first step in the reactions leading to 1,3-propanediol, via an intermediate product, 3-hydroxypropionaldehyde (3-HPA), which is easily dehydrated to form acrolein (Fig. 3.13). This pathway is not present in all LAB strains and has only been identified to date in several *Lactobacillus* species, such as *L. collinoides*, isolated from cider, as well as *L. hilgardii* and *L. diolivorans*, isolated from wine (Garai-Ibabe *et al.*, 2008). It is undesirable due to the subsequent production of acrolein, an irritant, carcinogenic compound, which is also detrimental to sensory quality, as acrolein/polyphenol complexes have a strong bitter taste.

The glycerol dehydratase pathway comprises glycerol dehydratase and 1,3 propanediol dehydrogenase (Sauvageot *et al.*, 2000). Thirteen genes encode the full pathway, coding for two glycerol dehydratase sub-units, propanediol dehydrogenase, and several other proteins whose functions are still unknown (Claisse, unpublished results). The few strains carrying this activity are often present in wines distilled to make brandy. In this case, acrolein produced during heating is very detrimental to the quality. The frequency of these strains is low. However, high-pH wines are more susceptible to their development and to bitterness.

3.5.4 Glucan synthesis and ‘ropy’ wines

Ropiness was probably one of the first types of spoilage described in wine and is still common today. Ropy wines have abnormally high viscosity and an oily or

slimy appearance. It occurs when LAB synthesize an exopolysaccharide, glucan, consisting of sophorose units ($\beta 1 \rightarrow 3$ glucose, $\beta 1 \rightarrow 2$ glucose) (Llauberes *et al.*, 1990). Even small quantities of glucan, as low as ≈ 100 mg/L, greatly increase viscosity by interacting with the wine matrix, and only low concentrations of residual sugar are required. This spoilage is generally detected in bottle, long after bottling, but may also occur in vats or barrels resulting in an unacceptable mouthfeel. The high viscosity may be corrected simply by handling the wine. In practice, if the ropy wine is in a vat, it is simply necessary to pump it into another vat. However, this problem often develops in bottle. In that case, the wine is drained into a vat and bottled again after suitable stabilization treatment, i.e. sterile filtration or heat treatment, in addition to proper sulphiting.

Pediococcus damnosus or *P. parvulus* have been identified as the main bacterial strains responsible for this type of spoilage. However, many wines contain high concentrations of *Pediococci* and are not spoiled, as only certain strains produce the glucan. They differ from non-ropy strains by the presence of an additional 5.5 kb plasmid. Cultivating ropy strains by successive transfers in a growth medium without ethanol resulted in the loss of this plasmid and the phenotype (Lonvaud-Funel *et al.*, 1993). Sequencing the plasmid revealed three putative genes encoding a replicating protein, a mobilization protein, and a glucosyl-transferase (GTF). The latter is a single transmembrane protein involved in polymerizing glucosyl units from UDP-glucose. The *gtf* gene (formerly known as '*dps*') was also found in an *O. oeni* strain isolated from wine, but it was integrated in the chromosome (Walling *et al.*, 2005; Dols-Lafargue *et al.*, 2008). Sequencing the region revealed that this gene possibly formed part of a genomic island. The expression of *gtf* was demonstrated in all the strains where it was detected by PCR. This *O. oeni* strain accumulates a β glucan in the medium along with several other heteropolysaccharides. Initial studies concerning *P. damnosus* revealed that the 'ropy' strains were generally better adapted to wine conditions than 'non-ropy' variants. One hypothesis was that the bacteria were protected by a capsule, especially as it has now been proved that the presence of *gtf* enhances the strains' tolerance to acidic pH and, thus, their capacity to survive in red and white wine. Nevertheless, at least for *O. oeni*, the presence of *gtf* does not, apparently, systematically induce ropiness. Limited availability of the UDP-glucose precursor may restrict polysaccharide formation.

3.5.5 New tools for detecting undesirable strains

It is always useful to detect and/or count spoilage strains in wineries where wines have already been spoiled. Indeed, specific spoilage strains are probably well-adapted to their ecological niche, so spoilage is frequently a recurrent problem. For example, wines from some wineries often contain biogenic amines, while others never do. The same is true of ropiness. Undesirable strains find more propitious conditions in certain wineries, so it is absolutely necessary to analyze the microflora as early as possible in the winemaking process.

Thanks to all the research carried out at the genetic level, which has investigated

wine spoilage caused by LAB, new effective, reliable tools are now available. The first molecular tool, designed to identify ropy strains, was based on knowledge of the plasmid. Initial efforts consisted of isolating a DNA probe, a plasmid fragment, only capable of detecting specific ropy *P. damnosus* strains by DNA/DNA hybridization (Lonvaud-Funel *et al.*, 1993). This probe may be used for colony-hybridization and provides a reliable, selective count of ropy strains among the bacterial microflora grown on culture plates. However, nowadays, PCR is the most useful early detection system (Gindreau *et al.*, 2001) and is even capable of counting the undesirable strains involved in the different spoilage described above. Primers, based on sequences of the key genes, encoding for essential activities, have been designed both for standard and quantitative PCR.

It is now easy to detect bacteria of bitterness, using the PCR test based on the nucleotide sequence of the gene encoding a subunit of glycerol dehydratase (Claisse and Lonvaud-Funel, 2001). Similarly, PCR tools have been designed to detect and quantify biogenic amine-producing strains in wine. In addition to histidine decarboxylase (Le Jeune *et al.*, 1995; Coton *et al.*, 1998a), other suitable probes have been deduced from diverse decarboxylase and agmatine deiminase gene sequences (Coton and Coton, 2005; Marcobal *et al.*, 2005; Torriani *et al.*, 2008). Quantitative PCR is now available for monitoring bacterial populations during winemaking and in finished wine (Nannelli *et al.*, 2008). As for ropiness, primers are chosen in the gene that encodes glycosyltransferase. Within approximately 8 hours, PCR is capable of detecting a population of ropy strains in wine at a concentration of 10^2 CFU/mL (Delaherche *et al.*, 2004).

3.6 Controlling malolactic fermentation by malolactic starters

3.6.1 Reasons for using malolactic starters

When alcoholic fermentation is completed, the LAB population in wine consists mainly of *O. oeni*, at concentrations of 10^3 – 10^4 CFU/mL. This bacterial population is smaller in acidic wines or those with a high alcohol concentration: only a few CFU/mL. In contrast, at high pH, the bacteria multiply during the last few days of alcoholic fermentation or in the post-fermentation period. In this case, malolactic fermentation starts quickly.

Generally, a few days or weeks elapse between the end of alcoholic fermentation and the beginning of malolactic fermentation, triggered when the bacteria count reaches 10^6 CFU/mL (Ribéreau-Gayon *et al.*, 2006). Meanwhile, the wine must be kept at temperatures in the vicinity of 20 °C. If the process does not start spontaneously, adding a malolactic starter is recommended, in order to avoid leaving the wine unprotected from the development of other, undesirable microorganisms. Indeed, even if *O. oeni* fail to multiply, other bacteria or yeasts may develop, or at least prepare to do so, including certain *Lactobacilli*, *Pediococci*, and *Brettanomyces bruxellensis* yeasts, with potentially devastating effects.

Furthermore, depending on cellar temperature, heating the vats for prolonged periods may be prohibitively expensive. Malolactic starter may be added to offset a deficiency in the indigenous microflora. The bacteria are added at high concentrations, above the threshold of 10^6 CFU/mL, eliminating the need for a multiplication phase (Lafon-Lafourcade *et al.*, 1983). In some wineries, the presence of biogenic amine-producing bacteria, wine composition, and winemaking conditions also justify the use of starters. This practice does not eliminate unwanted bacteria but prevents their proliferation.

3.6.2 Definition of malolactic starters

The degradation of malic acid is the most visible, desirable sign of lactic acid bacterial activity in wine. It is catalyzed by the malolactic enzyme inside the bacterium, which decarboxylates malic acid into L-lactic acid. To degrade all the malic acid requires a certain quantity of enzyme, contained in the bacterial biomass. The required quantity of bacteria is produced by spontaneous growth or added in the form of a starter. Using a malolactic starter amounts to adding malolactic enzyme to the wine. This enzyme only functions at precisely-controlled pH and requires cofactors: these conditions exist within living malolactic bacterial cells, but not in the wine (Lonvaud-Funel, 1995). Malolactic starters consist of concentrated LAB preparations of *O. oeni* species, which must remain viable to maintain their malolactic activity, even if they are not really able to multiply. Starters are added to the wine all at once and must find the resources to survive until the malic acid is totally degraded. They need energy and must synthesize proteins and lipids to help them resist the toxic environment in wine, notably by modifying their membrane. The malolactic reaction provides energy, as does the degradation of sugars, such as pentose, as well as citric acid, amino acids, and other molecules. However, if the cells are no longer able to withstand the toxicity of the environment, due to lack of energy, they die and malolactic activity stops. These factors make it difficult to control malolactic fermentation with starters and challenging to select the most suitable strains (Coucheney *et al.*, 2005). It is much more complex than selecting and using active dry yeast to control fermentation.

3.6.3 Isolation and selection of malolactic starters

Malolactic starters currently consist of pure cultures of *O. oeni*, with the exception of Viniflora LP, an *L. plantarum* preparation launched in the early 1980s specifically for partial deacidification of wine. Some starters mixed several *O. oeni* strains, but this did not enhance their efficiency, as the various strains compete rather than being activated in turn, so that one strain usually eliminates the others.

The first step is to isolate the bacteria during malolactic fermentation, generally from wines with a high ethanol concentration or low pH, on the principle that they must be high performers to thrive under these difficult conditions. The only way to isolate a bacterial strain is by cultivating wine samples on nutrient agar plates to obtain colonies for selection.

Strains are purified by repeated subcultures on nutrient agar plates. As many other bacterial species present in wine form colonies, isolated clones are identified to ensure that subcultures only contain *O. oeni*. Then the long process of selection begins. Strains are evaluated for their winemaking capacity, i.e. primarily their ability to tolerate acidic pH and high ethanol concentrations, as well as to degrade malic acid. Indeed, the main challenge for malolactic starters is to survive and grow once they have been added to the wine. Selection trials may be carried out in model wine solutions, or even real wines with an appropriate composition, focusing on pH and ethanol, as well as, occasionally, SO_2 . However, it is by no means certain that any strains are really more resistant to sulphites than others, and selecting on this criterion would be questionable, for two reasons. Firstly, when starters are added, there is usually no free SO_2 left in the medium. Secondly, if an SO_2 -resistant strain was added, stabilization problems should occur later.

For the moment, selection does not consider criteria other than survival and adaptation to wine. Although years of practice have shown that the sensory quality of wine is impacted by the strains responsible for malolactic fermentation, there is no serious basis or easy method for involving sensory criteria in the selection. Much more research is still needed on this point. The most complex issue to resolve is that, from a sensory standpoint, the same strain does not give the same result in different wines.

It may be advantageous to take into account other criteria, previously considered secondary. For example, following recent studies, care is now taken to ensure that candidate starter strains do not produce biogenic amines (histamine, tyramine, putrescine, or cadaverine).

Once starter strains have been isolated, purified, and selected, production involves the following stages: multiplication, biomass harvesting, freeze-drying or freezing, and packaging for shipment. Multiplication is achieved by a series of increasingly large-scale cultures. The composition of the medium and growing conditions are designed to enhance the performance of the biomass and minimize preparation costs. The cells are separated and concentrated by centrifugation, then prepared for freeze-drying or freezing. Most preparations are freeze-dried, as this is the most effective way to preserve viability for several months and also facilitates handling the product. The bacteria are packaged in bags containing the quantities required to inoculate wine in volumes ranging from a barrel up to 250 hL. Some cultures are frozen.

3.6.4 Using malolactic starters

Malolactic starters are usually added to wine after alcoholic fermentation, when the indigenous bacterial population is too low and does not grow, due to inhibiting conditions, such as low pH or high ethanol concentration. The massive addition of bacteria at a concentration above 10^6 UFC/mL replaces the indigenous population, reducing the time-lag between the two fermentations. There are two types of malolactic starters. The first starters, launched in the early 1980s and still, in some cases, available today, had to be 'reactivated'. The reactivation step is needed as

the bacteria have lost their original capacity to adapt to the wine environment during the isolation and preparation (multiplication, freeze-drying, etc.) phases. Consequently, lyophilized bacteria lose viability very quickly when they are put into wine. Within 4 or 5 hours, the initial population of 10^7 UFC/mL may have dropped as low as 10^5 UFC/mL, depending on the composition of the wine, and is no longer capable of degrading the malic acid. Reactivation helps the bacterial cells to maintain the membrane functions that are damaged by wine, thus enhancing their survival rate. Industrial preparations are reactivated in a medium consisting of dilute grape juice or wine, supplemented with yeast extract, and adjusted to a pH of around 4–4.5. The reactivation kit is distributed with the bacteria. The mixture is incubated for 24 or 48 hours at 20 to 25 °C, then added to wine at a minimum concentration of 5×10^6 CFU/mL, so the population in the reactivated starters must exceed 10^9 UFC/mL. During reactivation, at such high concentrations the biomass does not multiply, but recovers a physiological state that enables the bacteria to tolerate the toxicity of wine.

The first ready-to-use malolactic starter was released in the early 1990s. The preparation is formulated to ensure good survival in wine: the bacteria are able to form functional membranes, thus ensuring sustainability in spite of the stress (Nielsen *et al.*, 1996). Reactivated cultures may also be used when the conditions in wine are particularly stressful and survival may be enhanced by prior treatment with ‘yeast hulls’. This material (thoroughly cleaned yeast cell walls) fixes bacterial inhibitors, including toxic fatty acids released by yeast into the wine. The yeast hulls preparation is added to the wine about 2 days before the starter, and racking prior to inoculation with bacteria is usually beneficial.

Today, winemakers increasingly add starters during the early stages in fermentation, on the assumption that the freeze-dried LAB gradually re-adapt to conditions in the fermenting must and are ready to multiply more rapidly once the yeast decline. Many trials have reported the effectiveness of this practice, but it requires total, constant control of fermentation, to ensure that the bacteria do not ferment sugars left unfermented by yeasts, otherwise they produce acetic acid. Furthermore, it is vital to ensure that competition with bacteria does not reduce yeast activity, which is crucial to avoid a slowdown at the end of fermentation. Yeasts, however, may eliminate most of the malolactic starter. In fact, the balance between the two populations inoculated early in winemaking depends largely on the composition of the ecosystem, including pH, and trends are difficult to predict.

3.7 Conclusions and future trends

Malolactic fermentation was identified and recognized as an integral part of the winemaking process in the 1930s. It took many years before winemakers were finally convinced that it was beneficial for most wines. The isolation and identification of the bacteria involved were reported in numerous publications from various teams around the world in the 1950s. The physiological data accumulated by wine microbiologists provided winemakers with the information required to

control bacterial growth in wine and eliminate them after winemaking. Detailed data were obtained on the substrates metabolized by bacteria and the by-products formed. A considerable body of research on the malolactic reaction led to the purification of the malolactic enzyme in the early 1970s. However, although the rules for successful malolactic fermentation had been established, a number of issues had not been resolved. The first tests involving inoculating wines with laboratory-cultured bacterial biomass in the early 1970s encountered tremendous difficulties and ended in failure. Nevertheless, the first malolactic starters used after reactivation were marketed, followed by the first starter for direct inoculation in 1993. At the same, an increasing number of teams have worked on this subject in industrial and university research centres. Work in genomics and proteomics is now beginning to explain and interpret the difficulty of adapting *O. oeni* bacteria. Knowledge of the complete genomes and specificities of this species suggests that further progress is required, not only in understanding the mechanisms involved but also in improving the strains selected for industrial starters. Many developments are expected from research into metabolic changes, in *O. oeni*, as well as other bacteria species. Changes in wine flavour and aroma during malolactic fermentation that have been clearly detected by sensory analysis must be identified at molecular level. This is a prerequisite for later study of the metabolic pathways at the most fundamental level of genes and their regulators. Research should also consider LAB species other than *O. oeni*.

3.8 References

- Bartowsky E, Costello P and Henscke P (2003), Differentiation of Australian wine isolates of *Oenococcus oeni* using random amplified polymorphic DNA(RAPD). *Aust J Grape Wine Res*, **9**, 122–126.
- Bartowsky E J and Henschke P A (2004), The ‘buttery’ attribute of wine – diacetyl – desirability, spoilage and beyond. *Int J Food Microbiol*, **96**, 235–52.
- Boido E, Lloret A, Medina K, Carrau F and Dellacase E (2002), Effect of b glycosidase activity of *Oenococcus oeni* on the glycosylated flavour precursors of Tannat wine during malolactic fermentation. *J Agric Food Chem*, **50**, 2344–2349.
- Claissse O and Lonvaud-Funel A (2000), Assimilation of glycerol by a strain of *Lactobacillus collinoides* isolated from cider. *Food Microbiol*, **17**, 513–519.
- Claissse O and Lonvaud-Funel A (2001), Primers and a specific DNA probe for detecting lactic acid bacteria producing 3-hydroxypropionaldehyde from glycerol in spoiled ciders. *J Food Prot*, **64**, 833–837.
- Claissse O, Renouf V and Lonvaud-Funel A (2007), Differentiation of wine lactic acid bacteria species based on RFLP analysis of a partial sequence of rpoB gene. *J Microbiol Methods*, **69**, 387–390.
- Coton E and Coton M (2005), Multiplex PCR for colony direct detection of Gram-positive histamine- and tyramine-producing bacteria. *J Microbiol Methods*, **63**, 296–304.
- Coton E, Rolan G, Bertrand A and Lonvaud-Funel A (1998a), Histamine-producing lactic acid bacteria in wines. Early detection, frequency and distribution. *Am J Enol Vitic*, **49**, 199–204.
- Coton E, Rollan G C and Lonvaud-Funel A (1998b), Histidine decarboxylase of *Leuconostoc oenos* 9204: purification, kinetic properties, cloning and nucleotide sequence of the hdc gene. *J Appl Microbiol*, **84**, 143–51.

- Coucheney F, Desroche N, Bou M, Tourdot-Marechal R, Dulau L and Guzzo J (2005), A new approach for selection of *Oenococcus oeni* strains in order to produce malolactic starters. *Int J Food Microbiol*, **105**, 463–70.
- Delaherche A, Claisse O and Lonvaud-Funel A (2004), Detection and quantification of *Brettanomyces bruxellensis* and 'ropy' *Pediococcus damnosus* strains in wine by real-time polymerase chain reaction. *J Appl Microbiol*, **97**, 910–915.
- Delaherche A, Bon E, Dupe A, Lucas M, Arveiler B, De Daruvar A and Lonvaud-Funel A (2006), Intraspecific diversity of *Oenococcus oeni* strains determined by sequence analysis of target genes. *Appl Microbiol Biotechnol*, **73**, 394–403.
- Dols-Lafargue M, Lee H Y, Le Marrec C, Heyraud A, Chambat G and Lonvaud-Funel A. (2008), Characterization of gtf, a glucosyltransferase gene in the genomes of *Pediococcus parvulus* and *Oenococcus oeni*, two bacterial species commonly found in wine. *Appl Environ Microbiol*, **74**, 4079–4090.
- Garai-Ibabe G, Ibarburu I, Berregi I, Claisse O, Lonvaud-Funel A, Irastorza A and Duenas M T (2008), Glycerol metabolism and bitterness producing lactic acid bacteria in cidermaking. *Int J Food Microbiol*, **121**, 253–261.
- Gindreau E, Walling E and Lonvaud-Funel A (2001), Direct polymerase chain reaction detection of ropy *Pediococcus damnosus* strains in wine. *J Appl Microbiol*, **90**, 535–542.
- Grimaldi A, Bartowsky E and Jiranek V (2005), A survey of glycosidase activities of commercial wine strains of *Oenococcus oeni*. *Int J Food Microbiol*, **105**, 233–244.
- Groissillier A and Lonvaud-Funel A (1999), Comparison of partial malolactic enzyme gene sequences for phylogenetic analysis of some lactic acid bacteria species and relationships with the malic enzyme. *Int J Syst Bacteriol*, **49**(4), 1417–1428.
- Indecco N, Bartowsky E, Kassab S, Lante A, Spettoli P and Henscke P (2004), Release of glycosidically bound compounds of Chardonnay by *Oenococcus oeni* during malolactic fermentation. *Food Microbiol*, **21**, 257–265.
- Lafon-Lafourcade S, Lonvaud-Funel A and Carre E (1983), Lactic acid bacteria of wines: stimulation of growth and malolactic fermentation. *Antonie Van Leeuwenhoek*, **49**, 349–352.
- Le Jeune C, Lonvaud-Funel A, Ten Brink B, Hofstra H and Van Der Vossen J (1995), Development of a detection system for histidine decarboxylating lactic acid bacteria based on histidine decarboxylase activity test, DNA probes and PCR. *J Appl Bacteriol*, **78**, 316–326.
- Llauberes R M, Richard B, Lonvaud A, Dubourdieu D and Fournet B (1990), Structure of an exocellular beta-D-glucan from *Pediococcus sp.*, a wine lactic bacteria. *Carbohydr Res*, **203**, 103–107.
- Lonvaud-Funel A (1995), Microbiology of malolactic fermentation. Molecular aspects. . *FEMS Microbiol Lett*, **125**, 37–44.
- Lonvaud-Funel A (2001), Biogenic amines in wines: role of lactic acid bacteria. *FEMS Microbiol Lett*, **199**, 9–13.
- Lonvaud-Funel A and Joyeux A (1994), Histamine production by wine lactic acid bacteria: isolation of a histamine-producing strain of *Leuconostoc oenos*. *J Appl Bacteriol*, **77**, 401–407.
- Lonvaud-Funel A, Joyeux A and Desens C (1988), The inhibition of malolactic fermentation of wines by products of yeast metabolism. *J Sci Food Agric*, **44**, 183–191.
- Lonvaud-Funel A, Guilloux Y and Joyeux A (1993), Isolation of a DNA probe for identification of glucan producing *Pediococcus damnosus* in wines. *J Appl Bacteriol*, **74**, 41–47.
- Lucas P, Landete J, Coton M, Coton E and Lonvaud-Funel A (2003), The tyrosine decarboxylase operon of *Lactobacillus brevis* IOEB 9809: characterization and conservation in tyramine-producing bacteria. *FEMS Microbiol Lett*, **229**, 65–71.
- Lucas P M, Wolken W A, Claisse O, Lolkema J S and Lonvaud-Funel A (2005), Histamine-producing pathway encoded on an unstable plasmid in *Lactobacillus hilgardii* 0006. *Appl Environ Microbiol*, **71**, 1417–1424.

- Lucas P M, Blancato V S, Claisse O, Magni C, Lolkema J S and Lonvaud-Funel A (2007), Agmatine deiminase pathway genes in *Lactobacillus brevis* are linked to the tyrosine decarboxylation operon in a putative acid resistance locus. *Microbiology*, **153**, 2221–2230.
- Lucas P, Claisse O and Lonvaud-Funel A (2008), High frequency of histamine-producing bacteria in enological environment and instability of the phenotype. *Appl Environ Microbiol*, **74**, 811–817.
- Marcobal A, De Las Rivas B, Moreno-Arribas M V and Munoz R (2004), Identification of the ornithine decarboxylase gene in the putrescine-producer *Oenococcus oeni* BIFI-83. *FEMS Microbiol Lett*, **239**, 213–220.
- Marcobal A, De Las Rivas B, Moreno-Arribas M V and Munoz R (2005), Multiplex PCR method for the simultaneous detection of histamine-, tyramine-, and putrescine-producing lactic acid bacteria in foods. *J Food Prot*, **68**, 874–878.
- Marcobal A, De Las Rivas B., Moreno-Arribas M V and Munoz R (2006a), Evidence for horizontal gene transfer as origin of putrescine production in *Oenococcus oeni* RM83. *Appl Environ Microbiol*, **72**, 7954–7958.
- Marcobal A, Martin-Alvarez P J, Polo M C, Munoz R and Moreno-Arribas M V (2006b), Formation of biogenic amines throughout the industrial manufacture of red wine. *J Food Prot*, **69**, 397–404.
- Nannelli F, Claisse O, Gindreau E, Lonvaud-Funel A and Lucas P M (2008), Determination of lactic acid bacteria producing biogenic amines in wine by quantitative PCR methods. *Lett Appl Microbiol*, **47**, 594–599.
- Nehme B, Ganga M A and Lonvaud-Funel A (2006), The arginine deiminase locus of *Oenococcus oeni* includes a putative arginyl-tRNA synthetase ArgS2 at its 3'-end. *Appl Microbiol Biotechnol*, **70**, 590–597.
- Nielsen J C and Richelieu M (1999), Control of flavor development in wine during and after malolactic fermentation by *Oenococcus oeni*. *Appl Environ Microbiol*, **65**, 740–745.
- Nielsen J C, Pahl C and Lonvaud-Funel A (1996), Malolactic fermentation in wine by direct inoculation with freeze-dried *Leuconostoc oenos* cultures. *Am J Enol Vitic*, **47**, 42–48.
- Poolman B, Molenaar D, Smid E J, Ubbink T, Abee T, Renault P P and Konings W N (1991), Malolactic fermentation: electrogenic malate uptake and malate/lactate antiport generate metabolic energy. *J Bacteriol*, **173**, 6030–6037.
- Pripis-Nicolau L, De Revel G, Bertrand A and Maujean A (2000), Formation of flavor components by the reaction of amino acid and carbonyl compounds in mild conditions. *J Agric Food Chem*, **48**, 3761–3766.
- Pripis-Nicolau L, De Revel G, Bertrand A and Lonvaud-Funel A (2004), Methionine catabolism and production of volatile sulphur compounds by *Oenococcus oeni*. *J Appl Microbiol*, **96**, 1176–1184.
- Remize F, Gaudin A, Kong Y, Guzzo J, Alexandre H, Krieger S and Guilloux-Benatier M (2006), *Oenococcus oeni* preference for peptides: qualitative and quantitative analysis of nitrogen assimilation. *Arch Microbiol*, **185**, 459–469.
- Renouf V, Delaherche A, Claisse O and Lonvaud-Funel A (2008), Correlation between indigenous *Oenococcus oeni* strain resistance and the presence of genetic markers. *J Ind Microbiol Biotechnol*, **35**, 27–33.
- Ribéreau-Gayon P, Dubourdieu D, Donèche B and Lonvaud A (2006a), *Handbook of Enology. Vol 1. The microbiology of wine and vinifications*, Wiley, Chichester, UK.
- Sauvageot N, Gouffi K, Laplace J M and Auffray Y (2000), Glycerol metabolism in *Lactobacillus collinoides*: production of 3-hydroxypropionaldehyde, a precursor of acrolein. *Int J Food Microbiol*, **55**, 167–170.
- Tonon T and Lonvaud-Funel A (2000), Metabolism of arginine and its positive effect on growth and revival of *Oenococcus oeni*. *J Appl Microbiol*, **89**, 526–531.
- Torriani S, Gatto V, Sembeni S, Tofalo R, Suzzi G, Belletti N, Gardini F and Bover-Cid S (2008), Rapid detection and quantification of tyrosine decarboxylase gene (*tdc*) and its expression in gram-positive bacteria associated with fermented foods using PCR-based methods. *J Food Prot*, **71**, 93–101.

- Ugliano M, Genovese A and Moio L (2003), Hydrolysis of wine aroma precursors during malolactic fermentation with four commercial starter cultures of *Oenococcus oeni*. *J Agric Food Chem*, **51**, 5073–5078.
- Vivas N, Lonvaud-Funel A and Glories Y (1997), Effect of phenolic acids and anthocyanins on growth, viability and malolactic activity of a lactic acid bacterium. *Food Microbiol*, **14**, 291–300.
- Walling E, Gindreau E and Lonvaud-Funel A (2005), A putative glucan synthase gene dps detected in exopolysaccharide-producing *Pediococcus damnosus* and *Oenococcus oeni* strains isolated from wine and cider. *Int J Food Microbiol*, **98**, 53–62.
- Zapparoli G, Reguant C, Bordons A, Tarraini S and Dellaglio F (2000), Genomic DNA fingerprinting of *Oenococcus oeni* strains by pulse field gel electrophoresis and randomly amplified polymorphic DNA-PCR. *Curr Microbiol*, **40**, 351–355.

4

Enzymes and wine quality

R.-M. Canal-Llaubères, Novozymes, France

Abstract: Nowadays, biotechnology is gaining ground to make or modify products or processes for specific uses in many aspects of food production. In winemaking, enzymes are recognized as important processing tools. Derived from nature, enzymes of fungal origin are keys to many production steps: maceration–extraction, clarification, maturation and filtration. This chapter underlines the benefits of enzymes with respect to wine quality and also takes into account the economic aspects of the use of enzymes.

Key words: pectinases, β -glucanases, glycosidases, cinnamoyl esterases, *Aspergillus niger*, *Trichoderma harzianum*, maceration, extraction, clarification, maturation, ageing on lees, aroma liberation, filtration, pectin test.

4.1 Introduction

Since the 1970s, the use of enzymes has drastically increased in many fields of application, in particular the recent search for a new ‘carburant’ based on biotechnology for the production of bio-ethanol. The success of enzymes can be explained by their efficiency, their selectivity, their convenience in use and their limited carbon footprint. Enzymes help to streamline production processes, protect product quality and reduce production costs, thereby reducing the contribution to climate change.

In winemaking, the use of enzymes is today a common practice in all wine producing countries; however, research is still important to fully understand the role of these biocatalysts in the various processing steps. Moreover, their use is not always systematic or well mastered. Enzymes like yeast and bacteria have an important role to play in the search for wine quality, and innovation should focus on the new challenges the wine industry is facing.

This chapter, after some review of definitions, production methods, mode of use and regulatory aspects, will focus on the application of enzymes in the major wine processing steps – namely extraction, clarification, maturation and filtration – and will emphasize their benefits.

4.2 Definitions and production methods

4.2.1 Definitions

Enzymes are biochemicals. Many of the transformation processes used in various industries have inherent drawbacks from an economical and environmental point of view. In a number of cases, some of these drawbacks can be solved by using enzymes. Enzyme reactions may often be carried out under mild conditions; they are highly specific and involve high reaction rates.

Enzymes are biologically efficient catalysts in the form of proteins that catalyze biochemical reactions in the cells of living organisms. As such, they have evolved – along with the cells – under the same conditions that satisfy the metabolic requirements of an extensive range of cell types. In general, these metabolic requirements can be defined as:

- chemical reaction under mild conditions;
- highly specific action;
- very high reaction rates.

Industrial enzymes originate from biological systems. They contribute to sustainable development through being isolated from microorganisms which are fermented using primarily renewable resources. Among the six defined enzyme classes – oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases – hydrolases are by far the most widely used class of enzymes in industry; there is no use of ligases at present. Hydrolases catalyze hydrolysis, the cleavage of substrates by water. The reactions include the cleavage of peptide bonds in proteins, glycosidic bonds in carbohydrates and ester bonds in lipids. In general, larger molecules are broken down to smaller fragments. Table 4.1 shows a selection of enzymes currently used in industrial processes, listed according to class.

Historical background

Enzymes have always been used either as vegetal products rich in enzymes (papain or bromelain) or in the form of microorganisms in various applications (brewing, bread making, and alcohol production). The history of modern enzymology started in 1874 when the Danish chemist Christian Hansen produced the first sample of rennet from calves' stomachs. It was almost one century later that enzymes were introduced to oenology. The first enzyme ever used in winemaking was Ultrazym® 100 G, a pectinase developed by Ciba-Geigy to clarify grape must; today it is the property of Novozymes. At the end of the 1970s, enzymes were mainly used to improve pressing and clarification in white winemaking. During the 1980s,

Table 4.1 A selection of enzymes used in industrial processes

Enzyme class	Industrial enzymes	Applications
1. Oxidoreductases	Laccases	Chlorine-free denim bleaching process
2. Transferases	Fructosyltransferases	Production of functional sweeteners
3. Hydrolases	Amylases	Larger bread volume
	Lipases	Dough strengthening in bread making
	Pectinases	Extraction and clarification in juice production and winemaking
4. Lyases	Alpha-acetolactate decarboxylase	Beer maturation time reduction
5. Isomerases	Glucose isomerases	Sweetness increase of sugar syrup

Source: Enzymes at work, Novozymes A/S.

enzymes began to be applied to improve polyphenol extraction, filtration and aroma liberation. Pectinases purified from negative side activities were introduced in the 1990s thanks to research work conducted on esterases (see Section 4.5). New applications have included the use of urease to reduce the amount of ethyl carbamate, and Lysozyme to better control wine microbial stability. A review on enzymes in winemaking (Canal-Llaubères, 1992a) reports the significance of enzymes produced by the wine microorganisms as well as the use of exogenous enzymes for juice extraction, clarification and flavour development.

Although enzymes are present in grapes and microorganisms, industrial enzymes are often needed in order to help winemaking processes, to reduce time and to make better use of the valuable grape components for release into wine such as colour, aroma and structure. They also help to generate profit (yield increase, quality improvement) by streamlining the production processes. Currently, the use of enzymes in winemaking has become common, but their use is still not systematic. Application conditions need to be better understood, in particular the dose, the timing of addition and how to apply the enzymes.

4.2.2 Production methods

Although enzymes are probably the most complex chemicals commercially available in large quantities, the methods used in production of enzymes are in principle very simple: a microorganism is cultivated in a suitable medium from which the enzyme is subsequently recovered. Production, isolation and economics of extracellular enzymes were described by Aunstrup (1979). Industrial enzymes are produced using fermentation processes which can be either submerged in closed vessels or solid state fermentation on trays. Most of the enzymes are of extracellular origin – secreted from the cell. Cultivation of microorganisms is a simple way of obtaining enzymes in large quantities. In Asia, this method has been used since prehistoric times in the preparation of a variety of fermented foods and beverages. The main enzymes were amylolytic and proteolytic and the organisms used were predominantly of the genera *Aspergillus* or *Mucor*. Strain toxicology testing is part

of the development program, and standard quality insurance is based on microbe counts and heavy metals control to secure safety of food-grade enzymes (based on GRAS – generally recognized as safe – microorganisms).

Submerged liquid fermentation

Selected microorganisms (bacteria or fungi) are grown carefully in closed vessels containing a rich broth of nutrients – the fermentation medium – in aerobic and aseptic conditions. As the microorganisms break down the nutrients, they produce the desired enzymes. Most often, enzymes are secreted into the fermentation medium (extracellular enzymes). Thanks to the development of large-scale fermentation technologies, production of microbial enzymes accounts today for a significant proportion of the biotechnology industry's total output. Fermentation takes place in large vessels called fermentors, with volumes of up to 1000 m³. The fermentation media comprises nutrients based on renewable raw materials like corn starch, sugars, and soy grits. Various inorganic salts might also be added depending on the microorganisms being grown. Both fed-batch and continuous fermentation processes are common. In the fed-batch process, sterilized nutrients are added to the fermentor during the growth of the biomass. In the continuous process, sterilized liquid nutrients are fed into the vessel at the same flow rate as the fermentation broth leaving the system thereby achieving a steady-state production. Operational parameters such as temperature, pH, oxygen consumption and carbon dioxide formation are usually measured and carefully controlled to optimize the fermentation process (Fig. 4.1).

Solid state fermentation

The microorganism is grown on a moist substrate like wheat bran or agro-industrial by-products (sugar beet for example). The cultivation can be performed either on trays or in rotating drums tumbling the substrate to ensure aeration. The advantages of this system are the high aeration rate obtainable together with the low water concentration present. These advantages, and perhaps the fact that the mycelium is allowed to grow exposed to air, result in the formation of a broader variety of enzymes than would be possible in submerged cultures. Preferred enzymes produced in semi-solid fermentation are amylases, proteases and pectinases, for example.

Recovery

The purpose of the recovery process is to prepare a finished product of satisfactory purity and stability in as few steps as possible and with a minimum loss of enzyme activity. The first step in harvesting enzymes from the fermentation medium is to remove insoluble products, primarily microbial cells. This is done by centrifugation or microfiltration steps. Extracellular enzymes remain in the fermentation broth after the biomass has been removed. The biomass can be recycled as a fertilizer on local farms, as is done at all Novozyme's major production sites. Before use, the microorganisms are inactivated and the biomass is stabilized during storage.

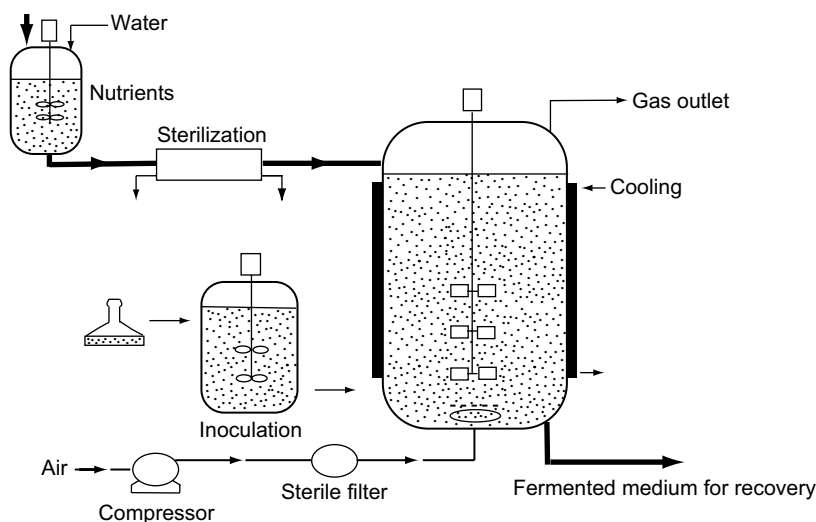


Fig. 4.1 The fermentation process for industrial enzyme production. Source: Enzymes at work, Novozymes A/S, www.novozymes.com.

The enzymes in the remaining broth are then concentrated by evaporation, membrane filtration or crystallization, depending on their intended application. If pure enzyme preparations are required, for example, for research purposes, they are usually isolated by gel or ion-exchange chromatography.

Certain applications need solid enzyme products so the crude enzyme is processed into granulates for convenient dust-free use (baking granulates to be mixed with flour). Liquid formulations are also available because they are easier to handle and dose along with other liquid ingredients. Glucose isomerases used in the starch industry to convert glucose into fructose are immobilized typically on the surface of particles of an inert carrier material held in reactor columns or towers. This is done to prolong enzyme working life; such immobilized enzymes may go on working for over a year.

Well-known enzyme producers' brand names for winemaking are: Ultrazym, Vinozym, VinoCrush, VinoClear, Novoclar, Novarom, VinoTaste, Vinoflow, Glucanex, Rapidase, Rohavin. In addition, a large number of formulators are promoting the use of enzymes in winemaking under numerous trademarks, including: Lafase, Lallzym, Endozym, Enovin, Dépectil, KZym, Panzym, Trenolin, and others.

Enzyme composition

Enzymes for winemaking are available in solid or liquid forms. The solid formulations are available as powder or micro-granulates. The shelf-life of a solid enzyme is longer than for a liquid formulation, respectively 3–4 years versus 1–2 years. They do not need cool storage like the liquids, which need to be stored between 0 and 10 °C. The carrier used for the enzyme formulation is often

maltodextrin. The liquid formulations are stabilized either with KCl, glycerol or NaCl and might contain preservatives such as potassium sorbate or sodium benzoate. Powdered enzymes should be manipulated with caution due to potential risk of allergy.

4.3 Regulatory aspects

Enzymes in winemaking are considered as processing aids and, as such, are no longer active in the finished wines due to either inhibition by the wine components (tannins for example) or removal during the process (bentonite fining for example). However, some enzymes might still have some activity in wine, such as the glycosidases which are enhanced by ethanol.

The use of enzymes is regulated in the various countries, based on lists like those in Europe, the USA or Australia. Enzyme preparations used in winemaking comply with the specifications from JECFA (Joint FAO/WHO Expert Committee on Food Additives) and from FCC (Food Chemical Codex) for food-grade enzymes.

4.3.1 European regulation

The European regulation has been updated through council regulation (EC) 479/2008 from April 29, 2008 on the common organization of the market in wine, amending regulation (EC) N° 1493/1999, and now refers to International Organisation of Vine and Wines (OIV) validated oenological treatments (L 148/2 EN Official Journal of the European Union 6.6.2008). Regulation 479/2008 has been amended by regulation 491/2009 from May 2009 introducing some changes to winemaking practices, by regulation 606/2009 from July 2009 related to oenological practices and applicable restrictions, and by regulation 607/2009 related to designations and labelling of wines. In Europe, the use of enzymes is restricted to pectinases from *Aspergillus niger*, β -glucanases from *Trichoderma harzianum*, urease from *Lactobacillus fermentum* and Lysozyme obtained from white hen eggs. The possibility to add new enzymes to the list is now greater than before 2008, because if the enzyme is covered by an OIV resolution it is already considered as evaluated both in terms of safety and efficacy. However the addition into the law must still go through the Common Market Organization approval procedure which lasts around 9 months.

4.3.2 Other regulations

Registrations are today covered by the presence of the enzyme on a list in the USA, Australia, New Zealand and Europe. It seems, in general, that the other wine-producing countries outside Europe are following the OIV recommendations. In the USA, the list is larger than that of the OIV, since the TTB (Alcohol and Tobacco Tax and Trade Bureau) has listed more enzymes for use in wine production, for

example amylases, catalase, glucose oxidase and protease. In Australia and New Zealand, there is a general list for food use. There is no specific wine law restricting any enzymes, but wineries observe the OIV oenological practices.

4.3.3 The association of enzyme producers: AMFEP

Major enzyme manufacturers and associated members are part of the AMFEP (the Association of Manufacturers of Fermentation Enzyme Products), a non-profit European industry association created in 1977. Its members produce and commercialize enzymes for food, feed and technical applications (for example, detergents, textiles, etc.). An updated list of commercial enzymes is available on their website and gives an indication of the enzyme family, producing host organisms and donor, I.U.B. number (a nomenclature given by the International Union of Biochemistry for recommended name and number of each enzyme) and application sector. More information can be found at www.amfep.org. The association is in constant dialogue with other international organizations to keep up with changing rules to provide safety and transparency. AMFEP fruitfully co-operates with European institutions and partner associations in the EU, Japan and the USA.

4.4 Enzyme applications in winemaking

4.4.1 Mode and conditions of use

Enzymes are used in many production steps in winemaking: on grapes, in the must and in the wine after alcoholic fermentation (Canal-Llaubères, 2000). Major applications are: extraction, clarification, maturation, either for aroma liberation or ageing on lees, and filtration.

Solid enzymes should be dissolved (1 part to 10) prior to use in water, must or wine. Liquid enzymes should also be diluted prior to addition. They can be incorporated in various ways:

- by spraying the grapes at the reception quay or into the press;
- by using a dosing pump at the start of the process;
- by adding the enzyme at the bottom of the clarification tank before filling it.

During maceration, enzymes applied to grapes will be well-mixed after a pumping over or a press turn. It is important that enzymes are added as soon as possible in the process to maximize efficiency. The use of SO₂ up to 500 mg/L will not affect the enzyme activity. Bentonite should be added only after enzymatic action to avoid enzyme inhibition by adsorption.

4.4.2 Dose recommendations

It is advisable to apply the right dose in order to obtain the best result from enzyme use. Under-dosing will require longer action time and benefits will be not achieved.

Table 4.2 OIV resolutions relevant to enzyme use in the wine industry

Resolution	Application	Scope
Oeno 11 2004	Use of enzymes for the clarification of musts	Facilitate the clarification of musts
Oeno 12 2004	Use of enzymes for the clarification of wines	Facilitate the clarification of wines
Oeno 13 2004	Use of enzymes for the maceration of grapes	(a) Facilitate operations for obtaining musts such as draining and pressing (b) Facilitate racking operations (c) Facilitate the extraction of colouring matter and polyphenols (d) Facilitate the extraction of aromas and aromatic precursors from grape berry skin
Oeno 14 2004	Use of enzymes for improving the filterability of musts	Improve the filterability of musts by specific hydrolysis of colloids
Oeno 15 2004	Use of enzymes for improving filterability of wines	Improve the filterability of wines by specific hydrolysis of colloids
Oeno 16 2004	Use of enzymes for the release of flavouring substances in must	Enhance the aromatic potential of grapes from glycosylated precursor substances
Oeno 17 2004	Use of enzymes for the release of flavouring substances in wine	Enhance the aromatic potential in wine, from glycosylated precursors from grapes
Oeno 18 2004	Use of enzymes for improving the solubilization of yeast compounds during winemaking on lees	(a) Facilitate the release of yeast-soluble constituents into wine (b) Improve the colloidal stability of wines

Source: www.oiv.org.

The dose is linked to: the strength of the enzyme preparation, must pH, temperature and contact time. For a given time, the enzyme dose should be increased if the pH and/or the temperature are low. Doses should be higher when applying enzymes on red grapes due to partial inhibition by the polyphenols. Pectinases are active from 5–60 °C. Below 5 °C, the action is very slow and at ≥ 60 °C the enzyme (which is an active protein) is inactivated by the temperature. The dose should be increased for pectin-rich varieties or for small berries with thick skins. Average doses are 3–5 g/100 kg in extraction and 1–3 g/hL in clarification. There is no technical risk in over-dosing the enzymes.

4.4.3 List of enzyme applications in wine

Enzyme applications in winemaking were listed in 2004 by OIV (oen 11–18, 2004). The documents can be consulted on-line at www.oiv.int. The major applications are summarized in Table 4.2. Lysozyme was previously added into the oenological codex (oen 15 2001) and urease monograph can be found in oen 5 2005.

4.4.4 Analytical methods

Analytical methods for measuring activities have been developed and are available on the OIV website under resolutions oenology 2007 (oen 06 2007 on cinnamoyl esterase, oen 05 2007 on glycosidases, oen 08 2007 on Lysozyme) and 2008 (oen 8, 9 and 10 2008 for cellulases, pectin esterase and polygalacturonases, respectively). New methods are under development for measuring endo-polygalacturonase and β -glucanase activities.

In an original study (Guérin *et al.*, 2009), 41 commercial enzyme preparations for winemaking were analyzed for their main enzymatic activities: pectinases, hemicellulases, cellulases and glycosidases. Major differences were noted between the enzymes labelled for aroma release and the enzyme used for colour extraction, clarification, sedimentation and pressing. Methods developed in this study have been referenced by OIV for measuring enzymatic activities.

4.4.5 Enzyme activities

Pectinases

This term covers at least six different enzymes which have a role in the hydrolysis of the pectin molecule (Fig. 4.2). Commercial products contain a mixture of these enzymes needed in the application. The enzymes are produced by cultivation of *Aspergillus* sp. strains in submerged or solid-state cultures. The major enzymes are polygalacturonases (exo, EC 3.2.1.67 and endo, EC 3.2.1.15), pectinesterase (EC 3.1.1.11), pectinlyases (EC 4.2.2.10), arabinanases, rhamnogalacturonases and galactanases. Many enzymes are present in the enzyme preparation as isoenzymes working under different pH, temperature optima and degree of esterification of the

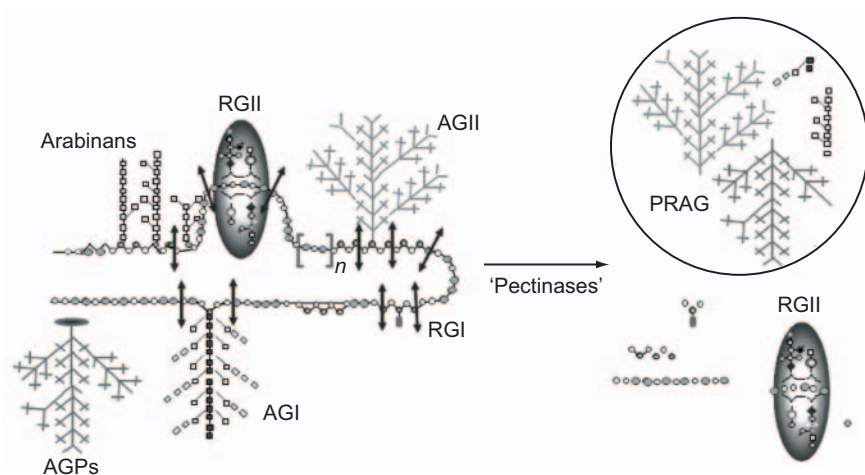


Fig. 4.2 The pectin substrate and the action of pectinases. AG, arabino-galactan; AGP, arabino-galactan protein; RG, rhamno-galacturonan; PRAG, polysaccharides rich in arabinose and galactose (Doco *et al.*, 1995).

pectin. For example, five polygalacturonases, two pectin lyases and two pectin esterases have been found in pectinases produced by *A. niger*. This explains why all pectinases do not give the same results in a given application. Flanzy (1998) summarized the enzyme activities that can be present in industrial pectinase preparations.

Hemicellulases

Hemicelluloses are made up of four major polysaccharides: arabinans, galactans, xylans and xyloglucans. Various enzymes (arabinanases, galactanases and xylanases) are involved in the degradation of these substrates (Brillouet and Hoebler, 1985; Brillouet *et al.*, 1991). One might consider arabinan and galactan as part of the pectin molecule because they constitute the side chains of this big polysaccharide molecule. Their oenological roles have not been clearly established.

Cellulases

These enzymes are involved in the depolymerization of the native cellulose. As for pectinases, various enzyme activities are needed to hydrolyze the cellulose. Endoglucanases (EC 3.2.1.4) seem to act first. Other enzymes, such as exo-cellobiohydrolases (EC 3.2.1.91), exo-glucohydrolases (EC 3.2.1.74) and β -glucosidases (EC 3.2.1.21) are also implicated in this degradation. As for hemicellulases, their oenological role has not been fully demonstrated.

Glycosidases

Many studies have focused on β -glucosidases because of their wide occurrence in plants, fungi and yeasts (Leclerc *et al.*, 1987; Laffort *et al.*, 1989). Grossmann *et al.* (1987) studied the β -glucosidase activity of *Hansenula* species found in must and Dubourdieu *et al.* (1988) studied it in *Saccharomyces cerevisiae*. Glycosidases are involved in the release of free aroma compounds which are linked to sugars in the grape must or wine, such as, for example, the glycosidic precursors of monoterpenes and norisoprenoids (see section on aroma liberation, p.119). β -glucosidase activity can be measured on p-nitrophenol-glucoside (Arnaldos *et al.*, 1999). A quicker modified assay has been proposed by Tate and Reynolds (2006).

Glucanases

Commercial β -glucanase preparations are used to hydrolyze barley gums in the brewing industry. They are produced by *Bacillus amyloliquefaciens*. Fungal β -glucanases are also made by fungi of the *Aspergillus* group. They are often formed as a side activity in pectinase preparation. The structure of the plant glucan differs from that of the fungal glucan: the former being composed of β 1,3–1,4 structures which are hydrolyzed by the β 1,4 glucanase and the latter – the fungal glucan – presenting a β 1,3 backbone being hydrolyzed by the β 1,3 glucanase. In winemaking, specific glucanases have been developed to hydrolyze *Botrytis* and yeast glucans in order to improve clarification and filterability of wines (Villettaz *et al.*, 1984; Dubourdieu *et al.*, 1985). The only specific industrial enzyme preparation available in winemaking applications derived from a selected strain of

Trichoderma harzianum. Developed in the 1980s, β -glucanases have been authorized for use in wine since 1995 in Europe and only recently in the USA. The best known enzyme product is Glucanex®.

Proteases

Proteases hydrolyze the peptide linkages between the amino acid units of proteins. Very little information is available on the efficiency of exogenous proteases in wine application. One important aspect is their potential role in protein haze reduction, but the effect on wine stabilization has not been established and other approaches have been taken by the industry (e.g. colloidal protection; Waters *et al.*, 1994, Moine-Ledoux and Dubourdieu, 1996). The search for fungal enzymes that could degrade wine proteins has so far remained unsuccessful (Modra and Williams, 1988). However, these activities may play a role in the release of peptides during the maturation phase (Humbert-Goffard, 2003). Takahashi *et al.* (1991) reported the presence of acid proteases in *A. niger*, and this activity might also be present as side activity in pectinase preparations. More research is needed to establish the possible role of proteases in winemaking applications.

Urease

The presence of ethyl carbamate in wine has been regulated since 1985. At that time, the Canadian government fixed its tolerance level in alcoholic beverages at 30 $\mu\text{g/L}$. According to Bertrand and Ingargiola (1993), 1% of white wines and 3% of red wines were above this limit in a survey analyzing 1800 wines from 16 different producing regions. A urease (EC 3.5.1.5) was developed and tested to hydrolyze urea (Fujinawa *et al.*, 1990). The recommended dose is 25 mg/L with a contact time of four weeks. By reducing the level of urea in wine containing $> 2 \text{ mg/L}$, it is possible to limit the formation of ethyl carbamate. Another solution is to monitor the level of nitrogen used in the vineyard in order to control the level of urea in wine.

Lysozyme

Lysozyme is a food preservative. In wine, microbial stability is usually obtained by addition of SO_2 . A large number of studies (e.g. Gerbaux *et al.*, 1999) have shown that it is possible to reach microbial stability by the addition of Lysozyme in wine (Gerbaux *et al.*, 1999). Therefore the level of SO_2 required to protect wine against microbial development can be reduced (Gerbaux *et al.*, 1997). Lysozyme, which is extracted from hen egg white, is used in the pharmaceutical and food industry. It degrades the cell walls of gram-positive bacteria such as lactic bacteria, *Oenococcus*, *Pediococcus* and *Lactobacillus*. It is used to limit malolactic bacterial growth during wine fermentation. It can greatly reduce the need for SO_2 , which poses a health hazard to individuals allergic to sulphites.

In contrast to SO_2 , the efficiency of Lysozyme increases when wine pH rises. The application rate varies between 250 and 500 mg/L. Treatment with Lysozyme applies in the following cases:

- to block malolactic fermentation (MLF) in white wines;
- to stabilize red wines after MLF;
- to protect wines from stuck fermentation.

Its use is authorized in Europe, in the USA and many other countries and described in the OIV oenological codex.

4.5 Advances in enzyme discovery

As most of the enzymes used in winemaking are not derived from genetically modified microorganisms, they contain many enzymatic activities besides the main enzymatic activities in each single preparation. This is the case regardless of the microorganism that is used for production of pectinases, glucanases or any other enzymes.

4.5.1 Enzyme purification

Commercial enzymes are typically crude fungal preparations. Even though they go through several filtration steps to remove small molecules such as salts from the fermentation broth, they contain various activities besides the main enzyme activity. For example, fine purification as ammonium sulphate precipitation is reserved for very pure enzymes for pharmaceutical use. It is known through literature that pectinases might contain side activities of β -glycosidases, β -galactosidases, esterases and cellulases, among others, and major activities of polygalacturonases, pectin lyases, pectin esterases, rhamnogalacturonases and arabinogalactanases (Burkhardt, 1976; Somers *et al.*, 1987; Wightman and Wrolstad, 1996). Nowadays, high-quality purified pectinase preparations are available as commercial enzymes for use in winemaking thanks to specific purification steps during production.

The presence of cinnamoyl esterase activity in pectinases produced by *Aspergillus* sp. can be detrimental to wine quality. This activity, defined as 'depsidase' by Burkhardt (1976), has been considered as responsible for the loss of freshness and fruitiness of white German wines. Studies conducted by Barbe (1995) and Chatonnet *et al.* (1992) have elucidated the role of cinnamoyl esterase in the formation of volatile phenols by *S. cerevisiae* in white wines (Fig. 4.3).

This activity is responsible for the hydrolysis of the hydroxycinnamic acids present as esters of tartaric acids in the pulp of the grape berry. The released acid phenols are then decarboxylated by the yeast to form the corresponding volatile phenols (vinyl-4-phenol, vinyl-4-guaiacol).

After characterization of the cinnamoyl esterase activity (Barbe and Dubourdieu, 1998), an acceptable enzyme level in pectinases was defined to avoid the production of volatile phenols above the aroma threshold (770 $\mu\text{g/L}$) when using a 'phenolic off flavour' ('POF') + yeast strain like EG8 C, thereby protecting the fruitiness of the wine (Fig. 4.4). Therefore, since 1993, this activity has been

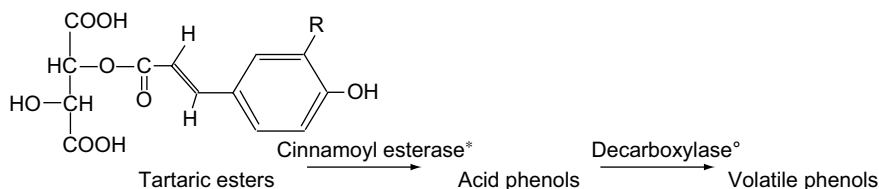


Fig. 4.3 The biochemical pathway leading to the formation of vinyl-4-phenol. *Originates from *Aspergillus* sp. pectinases or *Botrytis cinerea*. °Originates from *Saccharomyces cerevisiae* (POF+ strains) or *Brettanomyces*.

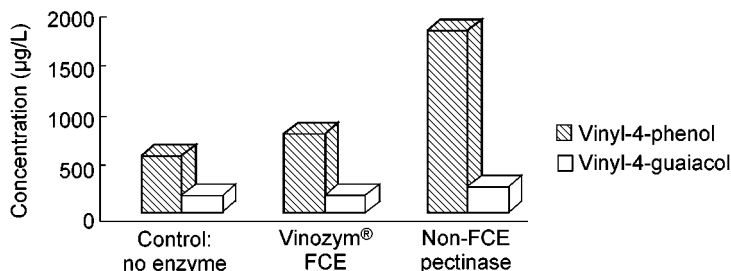


Fig. 4.4 Volatile phenols (vinyl-4-phenol and vinyl-4-guaiacol) concentration in Sauvignon blanc wines fermented with the yeast EG8C from grapes macerated without enzyme (control), with a FCE pectinase (Vinozym® FCE) and with a non-FCE pectinase (Chatonnet *et al.*, 1992).

removed from Novozymes' 'FCE' (almost 'free cinnamoyl esterase') enzymes, and now pectinases with a low level of cinnamoyl esterase activity are available to winemakers. Phenolic off-flavour (POF) is the name given to the gene coding the synthesis of cinnamate decarboxylase. Most of the yeasts produce this enzyme activity, responsible for the formation of volatile phenols.

The use of purified pectinase limits the concentration of volatile phenols to below aroma threshold (770 µg/L). Wines produced with non-purified pectinase are frequently rejected in tastings. Similar results have been reported on Semillon, Muscat and Riesling (Barbe, 1995).

Later, esterase activity was also found in pectinase preparations used in the making of red wine (Wightman *et al.*, 1997). The authors reported that, due to the presence of this activity, caffeic acid was released by the cleavage of tartaric acid from caffeoyl tartrate. These activities occurred in both Cabernet Sauvignon and Pinot noir. Gerbaux *et al.* (2002) demonstrated that cinnamoyl esterase side activity in pectinase preparations was responsible for the formation of an increased concentration of ethyl phenols (responsible for animal, leather, ink and horse stable off-odours) by *Brettanomyces* in Pinot noir wines and therefore recommended the use of purified enzymes in the production of red wine.

Recent studies have demonstrated that the cinnamoyl esterase present in pectinases used in the production of red wine is responsible for the formation of new anthocyanin-derived pigments (Ducasse, 2009). Acid phenols decarboxylated by

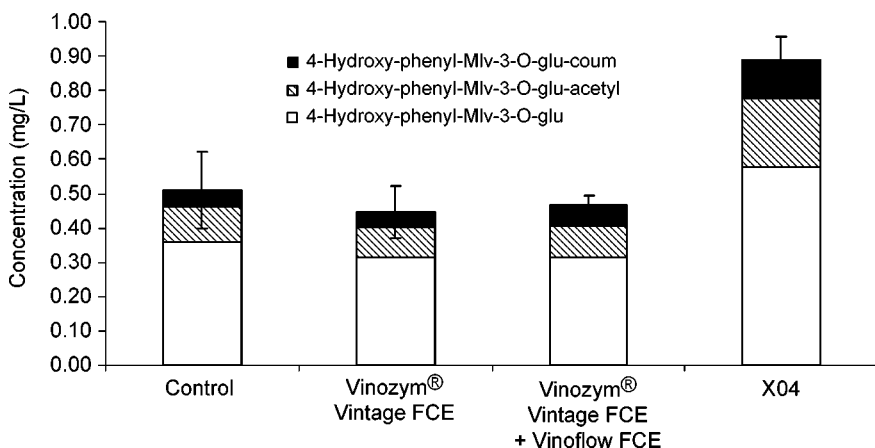


Fig. 4.5 Anthocyanin–vinylphenol adducts in response to three enzyme treatments. Merlot, 2004 harvest. X04, non-FCE pectinase 2004 (Ducasse, 2009) (Mlv, malvidine).

the yeast in vinylphenols react with anthocyanins. The anthocyanin–vinylphenol adducts were measured in 2004 Merlot wines treated with FCE and non-FCE pectinases. In the presence of cinnamoyl esterase activity, the concentration of these adducts increased substantially (Fig. 4.5).

The use of non-FCE pectinases increases the concentration of the anthocyanin–vinylphenol adducts by almost a factor of 2. These orange coloured compounds are present in low concentrations in red wines (1–5 mg/L), compared to anthocyanins (200–800 mg/L) and tannins (1000–5000 mg/L). It is doubtful that they have an influence on wine colour. The formation of these compounds reveals that cinnamate decarboxylases in yeasts are also active in red wines, releasing vinyl-4-phenol. These compounds can also be of pure chemical origin and could become an ageing indicator in wines (Schwarz *et al.*, 2003; Oliveira *et al.*, 2007).

Anthocyanin-derived pigments can be new markers for detecting the presence of cinnamoyl esterase activity in pectinases used on red grapes.

4.5.2 Wine polysaccharide composition

The wine macromolecules are mainly composed of polyphenols and polysaccharides often associated with proteins. Both families of compounds are important to wine because, on the one hand, the pigments and tannins play a significant role in colour and mouthfeel and, on the other hand, polysaccharides involved in complexes modify the colloidal stability of wines (Riou *et al.*, 2002) and the perception of tannin astringency (Vidal *et al.*, 2003).

The polysaccharidic fraction includes polysaccharides rich in arabinose and galactose (PRAG), such as type II arabinogalactan-proteins (Brillouet *et al.*, 1990), arabinans and rhamnogalacturonans (RGI and RGII) coming from the pecto-cellulosic cell walls of grape berries (Villataz *et al.*, 1981; Doco and

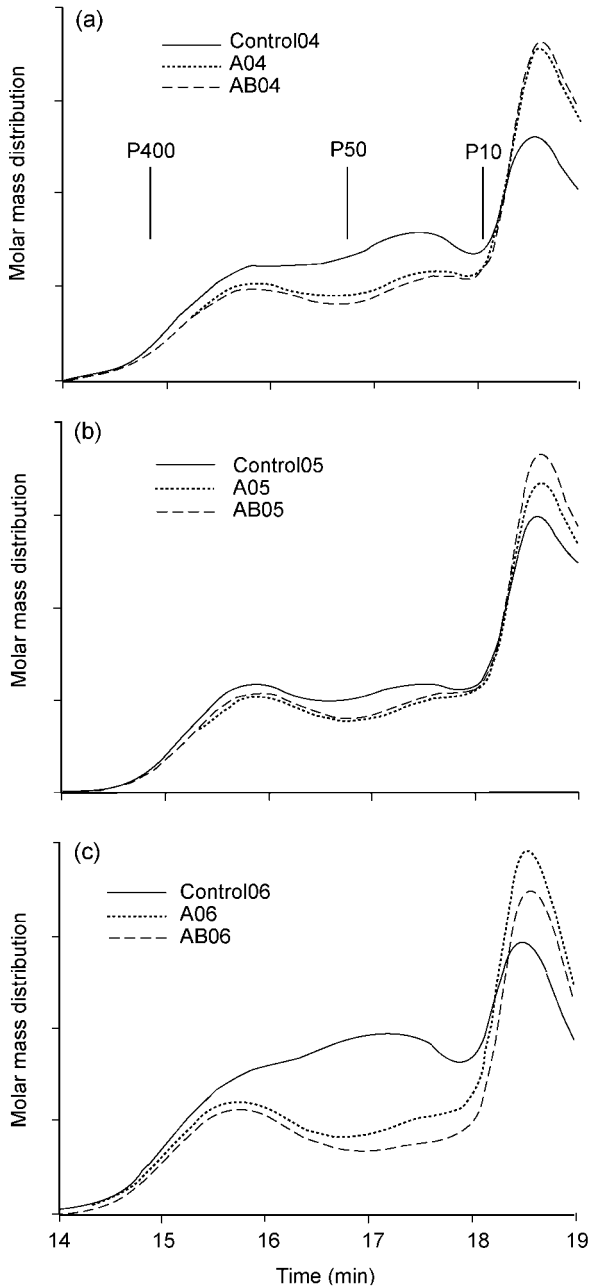


Fig. 4.6 HPLC polysaccharide profiles in molar mass distribution of Merlot wines from 2004 (a), 2005 (b) and 2006 (c) vintages macerated with (A, Vinozym® Vintage FCE; AB, Vinozym® Vintage FCE + Vinoflow FCE) and without (Control) pectinases. The molar masses are deduced from calibration with pullulans (P400, 380 kDa; P50, 50 kDa; P10, 10 kDa) (Ducasse, 2009).

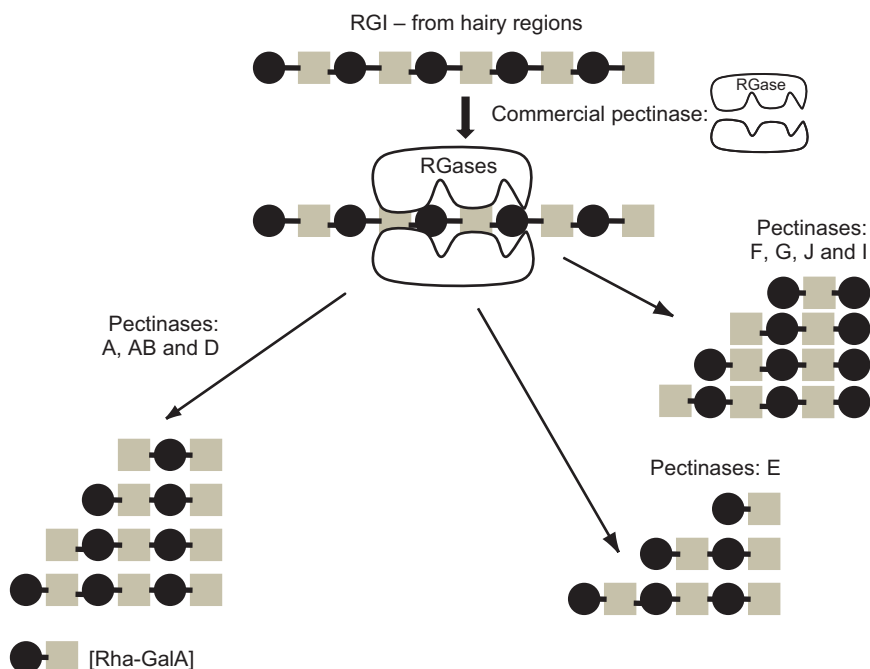


Fig. 4.7 Degradation patterns of rhamnogalacturonan (RG) I from pectin's hairy regions by rhamnogalacturonases (RGase) present in commercial pectinases (A, B, D, E, F, G, J and I). Liberation of pectic oligosaccharides composed of rhamnose and galacturonic acid units (Ducasse, 2009).

Brillouet, 1993; Pellerin *et al.*, 1995), oligosaccharides (Ducasse *et al.*, 2009) and mannoproteins (MP) released by yeasts during fermentation (Llaubères *et al.*, 1987), and during maturation of wine on lees (Doco *et al.*, 2003; Feuillat, 2003).

Ducasse (2009) has shown that pectinases used in red grape extraction modify the molecular weight distribution of polysaccharides in wine as previously described in the literature (Ayestarán *et al.*, 2004).

The profiles of molar mass distribution analyzed by liquid high-performance exclusion chromatography on a Shodex column are presented in Fig. 4.6. Calibration was done with pullulans, polysaccharide polymers consisting of maltotriose (3 glucose) units. The first peak eluting between 14 and 16.5 minutes contained mainly MP; the second peak eluting between 16.5 and 18 minutes contained mainly PRAGs and the third peak between 18 and 19 minutes was mostly composed of RGII.

Pectinases modify the polysaccharide profiles by reducing the molecular weight of grape polysaccharides (PRAGs and RGII) and by inducing the loss of the terminal arabinose residues of the PRAGs (Doco *et al.*, 2007). Ducasse *et al.* (2010) demonstrated that the concentration of PRAGs was reduced from 150 to 100 mg/L in 2004 and from 100 to 20 mg/L in 2006 (no major reduction in 2005). The concentration of RGII was increased from 100 to 170 mg/L for the three

vintages. The effect of pectinases on PRAG reduction (25–75%) varied according to vintages. This polysaccharide could be considered as a marker of grape maturity. Further observed, the degradation pattern of the pectin also included the release of oligosaccharides of lower molecular weight. The use of various pectinases led to the release of different oligomers (Fig. 4.7). These oligomers are released by rhamnogalacturonases present in the pectinase preparations. Oligosaccharide concentrations varied from 250 mg/L in Merlot wines to 330 mg/L in Carignan studied by Ducasse *et al.* (2010).

Further studies are needed to better understand the role of the oligosaccharide fragments as done previously for the PRAG or the RGII fractions.

4.6 Enzyme use in pre-fermentation stages

Enzymes are used in extraction in white, rosé and red wine processing as well as in must clarification. The use of pectinases on grapes allows reducing the mash viscosity and favours the release the grape must/wine at low pressure. Pectinases hydrolyze primarily the pecto-cellulosic cell wall of the pulp cells. As a result, the juice yield of fractions <1 bar is increased mostly as free-run must/wine, resulting in a greater volume of higher quality juice. Juice yield increase varies between 4–6% in reds and 6–11% in whites, depending on grape variety, maturity and extraction processes. Thanks to enzyme use on grapes, all processing steps are facilitated. Enzymes in solution (dissolution of enzyme in solid form is necessary to achieve good results: it is done by mixing one part of enzyme to 10 parts of either must or water) are applied directly to the grapes at the crusher-destemmer for white grapes or in the maceration tanks for red grapes.

4.6.1 Extraction in whites or rosés

The effect of pectinases used at the crusher-destemmer is quick and easy to monitor, as free-run and low-pressure juice fractions (<1 bar) are readily released from the press. Pilot-scale trials using a Speidel bladder press show that free-run juice volume is increased up to 50%, juice fraction at 0.6 bar by 22% and total juice yield by 10% (Canal-Llaubères, 2004) (Fig. 4.8).

Similar results were obtained in industrial direct pressing trials on Grenache grapes. In the experiment, pectinases of similar concentration were used at 3 g/100 kg (Vinozym® Process) and 1 g/100 kg (pectinase 1) of grapes at the crusher. Higher juice volumes were obtained on combined free-run and dynamic draining (0.1 bar) and juice fraction at 0.8 bar when applying the pectinase at 3 g/100 kg grapes. The total juice yield was increased by 11.9% (Table 4.3). In these conditions, the winery needed only 116 kg of grapes (instead of 133 kg) to produce 100 litres of wine. Furthermore, the juice resulting from Vinozym® Process processed grapes was quicker and easier to clarify. Volume of sediment was reduced compared to pectinase 1, generating an extra gain of 3% (Fig. 4.9).

The most common equipment includes pneumatic presses and draining tanks.

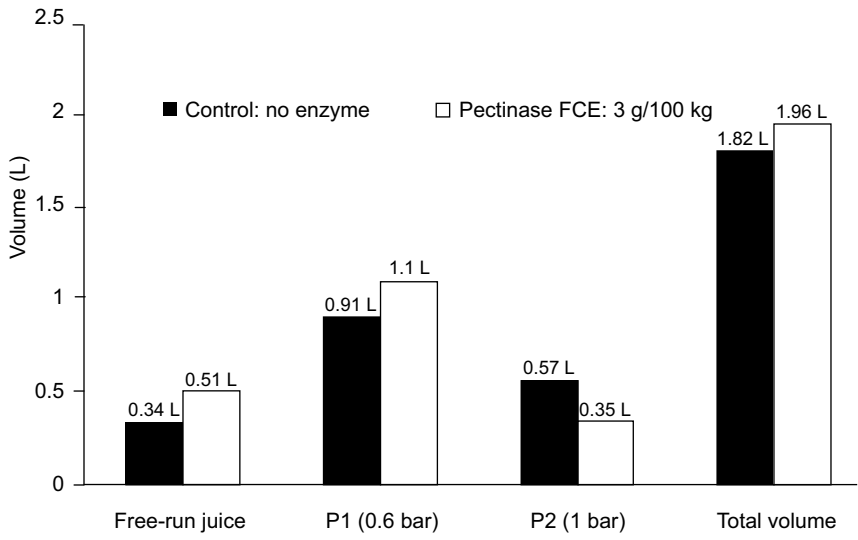


Fig. 4.8 Pressing trials on Sauvignon grapes, volumes of juice fractions without and with enzyme treatment (pectinase FCE at 3 g/100 kg). Contact time with enzyme: 30 minutes.

There is a trend towards using decanter technology as well. If the correct enzyme dose is applied, the total juice yield (free-run juice and pressure fractions) is increased by up to 6–11 % depending on the grape variety, maturity and equipment. The pressing time can be reduced by 30%. This rule also applies to the maceration time.

In addition, downstream processing is further improved. On average, enzymatic treatment resulted in 30% more clarified juice against a non-enzyme-treated lot after 24 hours clarification in trials conducted on Semillon (Provisor, 2008, personal communication).

Table 4.3 Comparison of two pectinases in terms of volume of juice fractions from 40 tons of Grenache grapes enzyme treated at the crusher (pH 3.3, 15 °C) on Bucher XPert 100–320 Inertis (Coopérative La Valdèze, 2009)

Enzymes	Volume (hL)		Yield					
	Free-run	Dynamic draining ^a	free-run	P1 (0.8 bar) volume (hL)	Yield P1	P2 (2.0 bar) volume (hL)	Yield P2	Total yield
Vinozym®								
Process (3 g/100 kg)	99	140	60.4%	71	17.9%	34	8.6%	86.9%
Pectinase 1 (1 g/100 kg)	102	115	50.2%	67	15.5%	40	9.3%	75.0%

^aGentle free-run extraction during 24 min at 0–0.12 bar.
P1 and P2, first and second pressing, respectively.

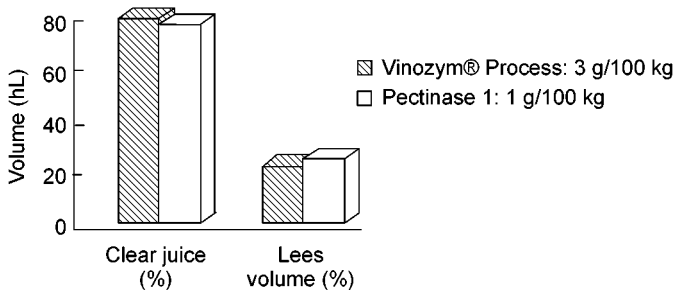


Fig. 4.9 Clarification of Grenache must in response to two pectinases used in extraction. Percentage of clear juice and sediment volume after settling 16 hours at 15 °C (direct pressing, 40 tons, pH 3.3, pressing cycle 3 h 30 min. Cooperative La Valdèze, 2009). The initial turbidity was: Vinoxym® Process, 175 NTU; pectinase 1, 433 NTU.

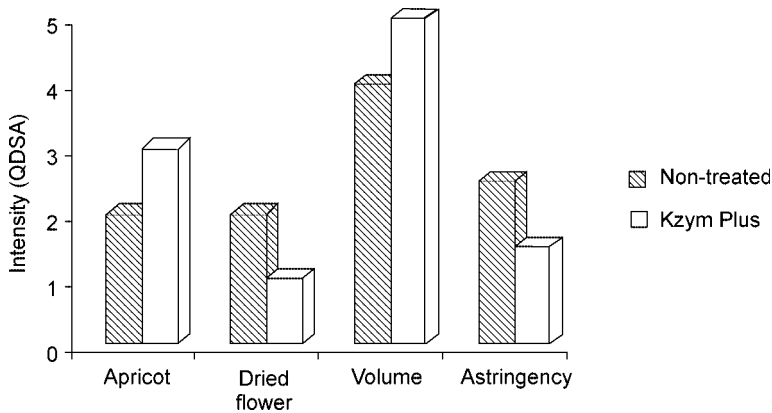


Fig. 4.10 Sensory attributes (apricot, dried flowers, volume, astringency) in response to enzyme treatment (Kzym Plus) on Muscat (1992). Pectinase was used during grapes maceration (QDSA: quantified descriptive sensory analysis) (Delteil, personal communication, 1992).

Even though the relationships between the enzymatic profile and the technology effect were difficult to demonstrate, the effect of pectinases on grape juice yield as well as on the clarification speed in studies on rosé was remarkable (Cayla *et al.*, 2009). Among eight commercial pectinases tested on extraction (addition of the enzymes to the grape), the mean increase of free-run juice was 17% compared to no enzyme use. The same is noted for the total juice yield which increased by 10%. These yield increases have a direct economic impact.

For white or rosé winemaking, pectinases can be used for two options. The first one is qualitative: the enzyme helps to achieve shorter maceration times and softer pressing, releasing higher quality juice. Oxidation is limited and free-run juice quality is better. Aroma extraction can be optimized as well; pectinases influence the wine sensory profile (Fig. 4.10; Delteil, personal communication, 1992). The second is quantitative: pectinases applied to grapes allow an increase in the

performance of the equipment (shorter time, more efficient pressing). In both cases, the use of enzymes on grapes will help the clarification and the settling of the must.

4.6.2 Clarification in whites, rosés and reds

Clarification of grape musts prior to fermentation or of young wines right after pressing, is an important step in winemaking. A good settling in white and rosé processing reduces the contact with herbaceous lees, thereby limiting the risk of formation of C-6 compounds responsible for vegetal aromas. In reds, it helps to finish alcoholic fermentation or to start malolactic fermentation with clean wines.

Enzyme preparations for clarification contain mainly pectinase activities as previously defined. The hydrolysis of the pectin allows reducing the must viscosity and eliminating the protective colloidal effects of macromolecules.

The clarification process consists of three steps: the first step is a depectinization whereby enzymes help to speed up the process; the second step is an agglomeration of the insoluble compounds; and the third step is the sedimentation of these compounds. As in the extraction process, it is advisable to use enzyme preparations with low level of cinnamoyl esterase activity to limit the risk of formation of volatile phenols.

Depectinization can easily be monitored with the alcohol test (also called pectin test), a powerful tool to monitor if pectin has been hydrolyzed (see Section 4.8). Only when pectin has been broken into smaller fragments can agglomeration start and sedimentation follow. The performance of a given pectinase can be controlled using an Imhoff cone to follow the clarification of must in laboratory experiments (see Section 4.8).

Operating mode: pectinases are added at the bottom of the clarification tank right after the press or with a dosing pump. The dose varies between 1 and 3 g/hL. Depectinization occurs very rapidly within 1 to 3 hours if the right enzyme dose is applied; it can be monitored with the pectin test. When the pectin test is negative, depectinization is accomplished. Depending on grape variety, maturity level and industrial extraction conditions, depectinization may last up to 8 hours. Non-enzyme-treated musts could require more than 24 hours before depectinization starts. Clarification can be achieved by static settling, centrifugation or flotation. With the last two procedures, the performance can be increased if the pectin test is negative. In order to increase the filtration of the lees, an extra enzyme addition can be made. If pectinases have been used on grapes prior to pressing, it is advisable to add enzymes only in the last press fractions (>1 bar).

In a trial on Semillon, enzymatic clarification resulted in clarified must yields from 64% to 85% after 24 hours contact time in 200 L tanks. Therefore the enzymes tested gave settling rates which reflected enzyme activity. Enzyme dose varied from 0.8–2 g/hL (in tested enzyme order: 0.8; 1; 2; 1.1; 1.1). Under these conditions, NS35032 was the best performing enzyme in relation to clarification speed, which occurred in 7 to 8 hours after enzyme addition. All other enzymes started must clarification within 14 hours after addition. First in yield and sediment compaction was NS35040 (Fig. 4.11). It is remarkable that the control juice with

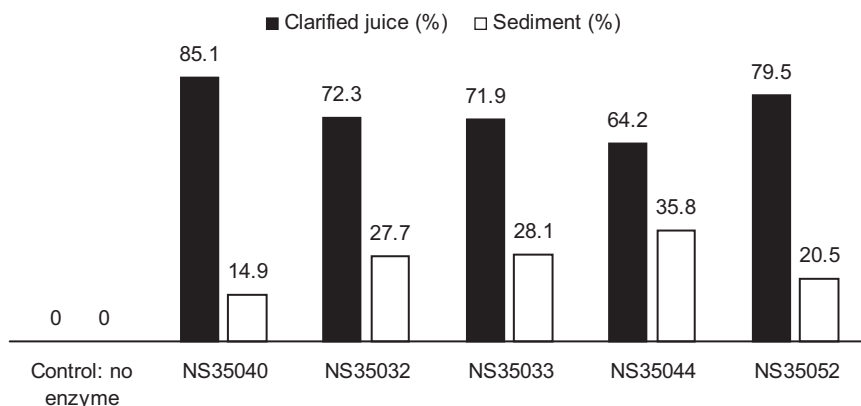


Fig. 4.11 Percentages of clarified juice and sediment volume on Semillon must non-enzyme-treated and enzyme-treated with various pectinases in pilot 200 hL tanks. Conditions: juice pH 3.2, 13 °C, initial turbidity > 1000 NTU, 24 hours contact time (Provisor, 2008).

no enzyme treatment had not started the clarification. The use of pectinases is clearly an advantage both in terms of speed and juice quality.

4.6.3 Extraction in red winemaking

The use of pectinases in red winemaking is becoming a common practice, even though some contradictory results have been reported on colour extraction. However more and more research studies are confirming the potential of enzymes in red winemaking. The grape skin cell walls are a limiting barrier to the release of polyphenols into the must during fermentation (Amrani Joutei and Glories, 1995). Among all the factors and techniques influencing extraction, the use of maceration enzymes helps in extracting and stabilizing the colour (Parley, 1997; Bautista-Ortin *et al.*, 2005). Pectinases are able to degrade the polysaccharides of the cell walls, releasing tannins and anthocyanins, but they also help release polysaccharides of lower molecular weight. Their effect can be visualized using electron microscopy. In the study conducted by Ducruet (2000), microperforations in the cell walls were only visible in enzyme-treated grapes (Vinozym®) (Fig. 4.12).

Recent studies have shown the positive role played by pectinases in the polyphenol and polysaccharide composition of Merlot wines in three vintages (Ducasse *et al.*, 2009). Wines were analyzed after 20 months. Enzyme-treated wines contained more RGII (60% increase in three vintages) and had lower amounts of PRAGs with lower molecular weight (Fig. 4.13).

An increase in colour intensity of derived pigments resistant to sulphite bleaching and in condensed tannins was observed by Ducasse *et al.* (2009). Figure 4.14 shows the effect during maceration of the release of pigments (measured as colour intensity – CI) and phenolics (measured as total polyphenol index – TPI – or OD280 nm). Grapes were maintained in cold soak for 3 days prior to first pumping over for cap management. The use of pectinases favoured a quicker and

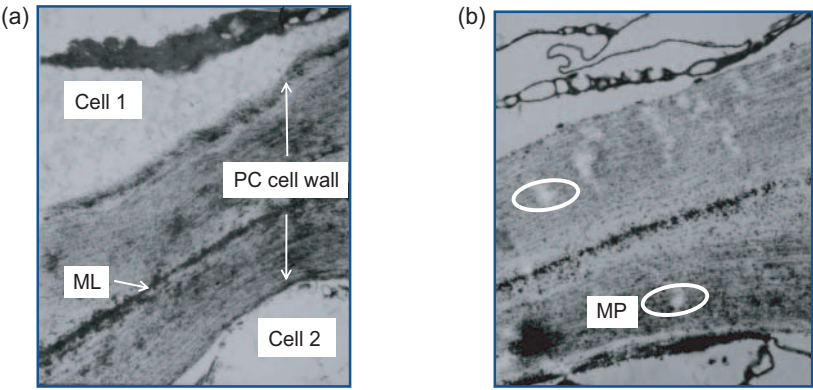


Fig. 4.12 Effect of pectinase hydrolysis (Vinozym®) on the pecto-cellulosic cell wall of Merlot grape berry showing selective extraction of intracellular compounds: (a) control, non-enzyme-treated; (b) enzyme (Vinozym®)-treated. Electron microscopy (Ducruet, 2000). ML, middle lamella; MP, micro-perforation; PC, pecto-cellulosic cell wall. White ellipses show microperforation.

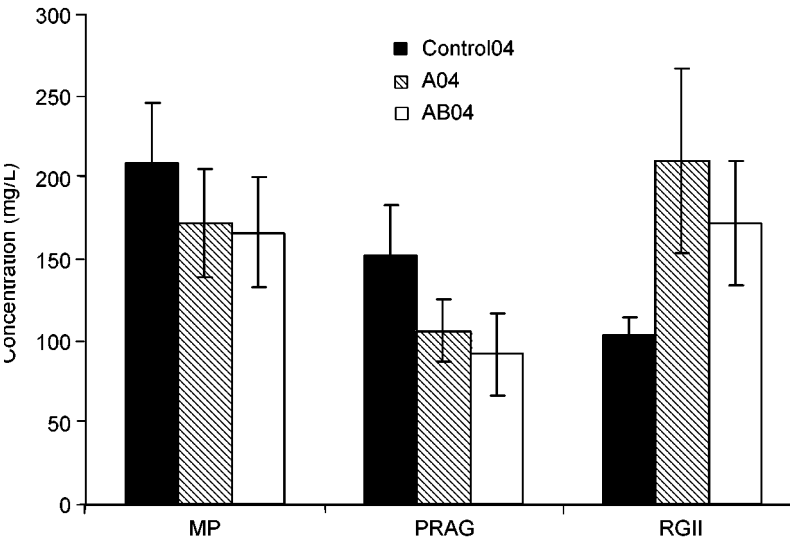


Fig. 4.13 Concentration (mg/L) of 2004 Merlot wines in mannoproteins (MPs), polysaccharides rich in arabinose and galactose (PRAGs) and rhamnogalacturon II (RGII) in absence and presence of pectinase treatment for extraction (control, no enzyme; A, Vinozym® Vintage FCE; AB, Vinozym® Vintage FCE + Vinoflow FCE) (Ducasse, 2009).

higher release of both colour and phenolics compared to no enzyme addition. After 20 months of ageing, the wines showed an increase in colour, tannins and other phenolic analytes (Table 4.4).

Variations in extraction are modulated by the grape maturity (vintage effect) and by the process. The colour is not related exclusively to anthocyanin concentration but depends on the nature and proportion of genuine anthocyanins and

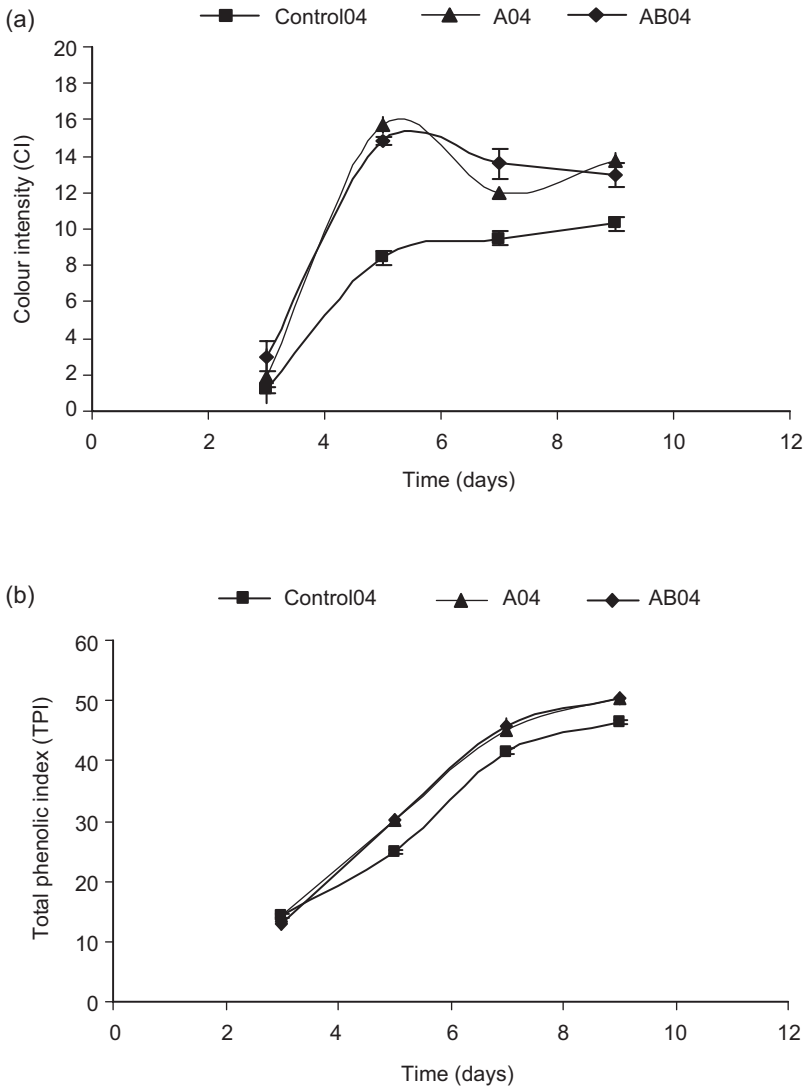


Fig. 4.14 Colour (CI) (a) and phenolics (TPI) (b) extraction kinetics during maceration of 2004 Merlot wines (Ducasse, 2009). Control, no enzyme; A, Vinoxym® Vintage FCE; AB, Vinoxym® Vintage FCE + Vinoflow FCE.

derivatives and on the copigmentation phenomenon. Positive effect on colour gain with exogenous pectinases has also been underlined by Main and Morris (2007) in newly bottled wines and wines stored for 11 or 20 months. Pectinases promoted more polymeric pigment formation and more ionized anthocyanin than the control wine. Tannins extracted by the enzyme treatment come from the grape skin as they are mainly present in the form of epi-3-O-gallate (%gal) and epigallocatechin (%egc) units. These data confirm the results of Ducruet (2000) who demonstrated

Table 4.4 Colour and tannin composition of Merlot wines in response to two pectinase enzyme treatments, 2004. Means are based on three measurements per treatment

Treatment	CI _{corr}	TPI	PR _{SO₂}	DP _m	%gal	%egc
Control	10.07	36.9	2.31	4.65	2.38	17.9
Vinozym® Vintage FCE	10.98	43.8	2.58	5.17	3.1	18.89
Vinozym® Vintage FCE + Vinoflow FCE	11.4	40.3	2.7	5.12	3.33	18.78

CI_{corr}, colour intensity corrected; TPI, total phenolic index; PR_{SO₂}, pigments resistant to bisulphite decoloration; DP_m, mean degree of polymerization; %gal, percentage epi-3-O-gallate units; %egc, percentage epigallocatechin units.

Source: Ducasse, 2009.

that pectinases do not extract grape seed tannins. Compounds extracted are slightly larger molecules as shown by the mean degree of polymerization. Changes in polysaccharide composition, induced by enzyme treatment, allow rapid clarification of young wine and improve filterability. Finally, these changes impact the colloidal stability and the mouthfeel properties of phenolic compounds in wine as shown earlier in model solutions (Vidal *et al.*, 2003). Other benefits include increased wine yield, quicker young wine clarification and higher filterability.

In a red wine extraction experiment conducted on Australian Merlot and Cabernet Sauvignon, the use of extractive enzymes during wine fermentation on short maceration process (rotary fermentors) aided in the extraction of colour, phenolics and tannins. In addition, wine yield increased and turbidity declined quicker in the treated Merlot. On average, the free-run fraction increased by 3.5% and the average overall yield increased by 5.5%. For Cabernet Sauvignon, the results showed less variation in yields compared to the Merlot trials. The highest overall yield was obtained with a pectinase which showed a 3% higher yield than the control with no enzyme addition. The quantity of marc produced with enzyme-treatment was on average 2% less in Merlot and 1% less in Cabernet Sauvignon compared to control treatments. Finally, a winemaker's assessment showed sensory differences amongst the treatments. Differences were found in terms of tannin levels, type of tannins, mouthfeel and aroma intensities. In Merlot, some pectinase-treated wines had more developed flavours, dark fruit characters and more tannin extraction; in Cabernet Sauvignon, some enzyme-treated wines were more intense, more fruit-driven, with velvety tannins (Giesbertz and Davis, personal communication, 2008).

4.6.4 Thermovinification

Nowadays, the use of heat treatment of grapes (thermo treatment, flash release) is becoming more widespread in large production operations in order to answer the market needs for fruity soft red wines. The technique has also been developed to gain time and tank capacity and has been the solution to inhibit laccases in *Botrytis* infected grapes. However, the use of heat is responsible for the production of red musts with very high turbidity, above 2000 NTU (nephelometric turbidity unit)

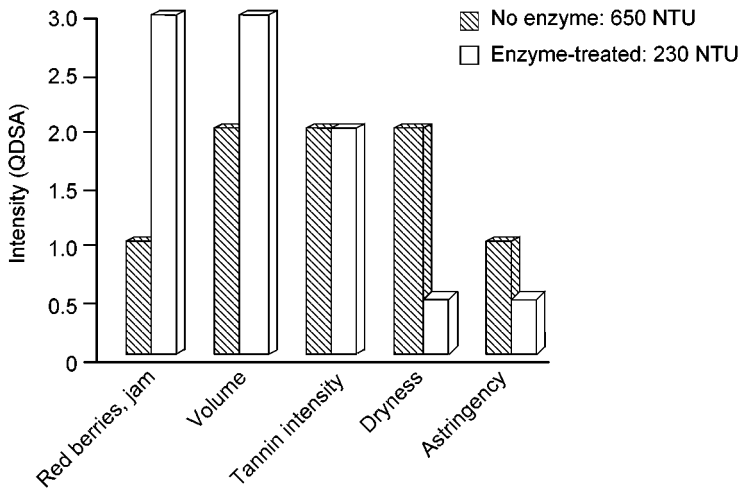


Fig. 4.15 Sensory attributes (red berries, jam, volume, tannin intensity, dryness, astringency) in response to enzyme treatment (Flashzym at the reception bin) prior to thermo-treatment of Carignan grapes (Aliot *et al.*, 2008).

using a calibrated nephelometer. Static decantation of these red juices heavily charged with vegetal solids prior to fermentation is difficult to achieve. In order to prevent sensory defects (sulphur, sauerkraut and garlic aromas) and to protect the aroma, pre-fermentation clarification techniques like flotation or rotary vacuum filtration are being used. As with white winemaking, the control of the turbidity of the red musts is very important to produce fruity red wines.

The use of pectinases is therefore mandatory to help depectinize the thermo-treated red musts with a positive impact on wine sensory attributes. Trials conducted by Institut Coopératif du Vin (ICV) demonstrated that enzyme-treated Carignan wines had a much lower turbidity (almost three times lower than control wine) and were fruitier, rounder with less dryness and astringency (Fig. 4.15) (Aliot *et al.*, 2008), applying the method developed by Delteil (2000).

With the use of thermal treatment, enzymes have a clear benefit on clarification and, whatever the techniques applied for juice separation (flotation, rotary vacuum filter), enzymes are key in the process to reduce turbidity and achieve an efficient clarification. In a series of trials in large-scale conditions (Languedoc's cooperatives), the use of pectinases allowed a reduction in the turbidity by 50–80% compared to a control. The use of rotary vacuum filter without enzyme is not advisable, as turbidity cannot be lowered below 500 NTU and some enzyme-treated juices remained with relatively high turbidity, indicating that this technique needs to be carefully monitored to be efficient (Table 4.5). Based on substantial practical experience, ICV advises monitoring turbidity around 100–150 NTU for crisp (acetate of isoamyle) wine production and around 250–350 NTU for fruitier wines. The enzyme performance varies according to the dose, contact time and temperature. Temperature is the limiting factor and enzymes are inhibited by high

Table 4.5 Effect of pectinase treatment on the clarification of Carignan thermo-treated must. Juice batches were monitored after rotary vacuum filter separation over two production days (one with enzyme-treated grapes with Flashzym at 15 g/T, one with no enzyme). Flash release, enzyme addition after cooling down

Carignan juice batch no.	Turbidity (NTU)	
	Enzyme-treated, day 1	No enzyme, day 2
1	169	520
2	143	592
3	107	659
4	104	990
5	121	1000
6	458	1000
7	459	1000
8	479	1000
9	610	1000

NTU, nephelometric turbidity unit.

Source: Aliot *et al.*, 2008.

temperatures above 60 °C. Therefore, it is recommended to add the enzyme when the temperature has been lowered. Enzymes can be added at grape reception or after heat treatment when the mash has been partially cooled down (Table 4.5).

From an oenological point of view, by modifying the concentration in tannins and polysaccharides, enzymes play a role in the colloidal stability of the wine. Thanks to tannin enrichment during extraction, pectinases can favour reactions between anthocyanins and tannins and the formation of derived pigments. Studies in model solutions have clearly demonstrated colour intensification by copigmentation with tannins (Ducasse, 2009).

4.7 Enzyme use in post-fermentation stages

4.7.1 Maturation

During the post-fermentation stage, wines will undergo changes in their colloidal structure to reach stability, limpidity and sensory development prior to bottling. Wines can benefit from ageing on their fermentation lees or from the release of grape aroma precursors.

Ageing on lees

The benefits of ageing on lees have been demonstrated in a large number of publications. One of the major mechanisms involved during this process is the yeast autolysis liberating intracellular substances such as amino acids (Feuillat and Charpentier, 1982), peptides (Bahayan and Bezrukov, 1985), nucleotides (Leroy *et al.*, 1990) and polysaccharides (Llaubères *et al.*, 1987) thanks to endogenous enzymes. The positive effects observed in white winemaking have motivated winemakers to apply this technique also in red winemaking. Enzymes containing

both pectinases and glucanases (β -1,3-1,6 glucanases) were developed in the mid-1990s to speed up the autolysis process, thereby improving clarification and filterability (Llaubères, 1988; Villettaz, 1990), enabling the release of savoury compounds that improve mouthfeel. The presence of peptides of low molecular weight (0.5–3 kDa) was demonstrated by Humbert-Goffard (2003). In white wines, their concentration is above the perception threshold, which was established at 16 mg/L leucine equivalents. Acid proteases present in the enzyme preparations are responsible for the release of these compounds.

During winemaking, enzyme preparations are added to the wine immediately after alcoholic fermentation either on fine or gross lees. Maturation can be conducted in barrel or in stainless steel vats. Besides the release of peptides, these enzymes show a clarification effect and an improvement in filterability (see Section 4.7.2 on filtration).

Aroma liberation

Enzyme preparations containing glycosidases (β -glucosidases, α -arabinosidases, α -rhamnosidases, β -apiosidases) were introduced to the market at the end of the 1980s. These enzymes are used to enhance wine aroma thanks to the liberation into the wine of free aroma compounds such as terpenes (Cordonnier and Bayonove, 1974; Williams *et al.*, 1981; Grossmann and Rapp, 1988; Gunata *et al.*, 1990a, b; Canal-Llaubères, 1994; Delcroix *et al.*, 1994; Francis *et al.*, 1996). Glycosidases also exist in grapes and yeasts, but their performance under winemaking conditions is not always adequate (Aryan *et al.*, 1987). Wine from grape varieties rich in glycosylated aroma precursors (non-odoriferous) such as Muscat, Gewürztraminer, Riesling or even Sauvignon blanc can be lifted to become fruitier or more floral. Aroma compounds are present in the form of mono or disaccharide glycosides (Williams *et al.*, 1982). They are located in the grape skin and pulp. Glycosidases are obtained from *Aspergillus* sp. strains under specific fermentation conditions. The aroma molecule is linked to glucose, and β -glucosidase activity is needed to free up the aroma. The hydrolysis of the aroma precursors is sequential and various glycosidases are involved in the release of the free aroma compounds (Günata *et al.*, 1988, 1997). Trials conducted with an experimental β -glucosidase from *Aspergillus* sp. have demonstrated that positive results can be obtained under winemaking conditions (Canal-Llaubères, 1989).

The mechanism of action involves enzymatic liberation of free, aromatic compounds, e.g. linalool, citronellol, nerol, geraniol, phenylethanol, etc., from their glycosidic precursors (Fig. 4.16). In grapes, the glycosidic aroma precursors are present in much higher concentrations than the free aroma compounds (Cordonnier and Bayonove, 1974). Therefore, there is an interest from a winemaking point of view in hydrolyzing the precursors to free up aroma. Even though they are liberated during maturation by acid hydrolysis, the process is very slow. Exogenous glycosidases allow a quick release of the aroma compounds. Contact time with the enzyme preparation may vary from some weeks to one month at a dose of 5–10 g/hL. The enzyme preparation is added into the wine immediately after the alcoholic fermentation on the freshly racked wine. Wine should preferably be free

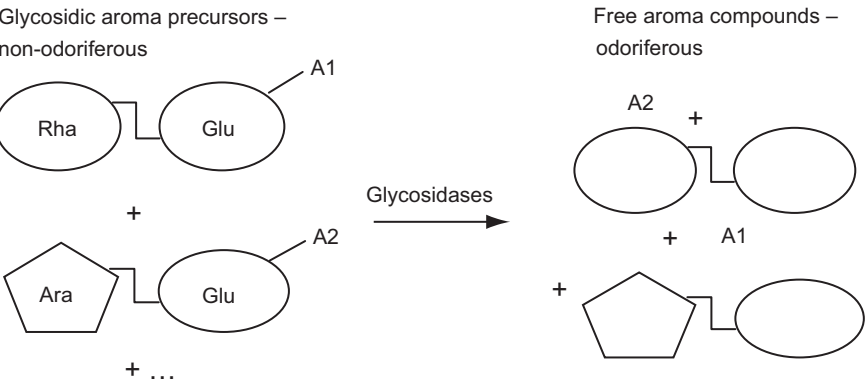


Fig. 4.16 The aroma liberation pathway catalyzed by glycosidases.

of bentonite to avoid enzyme inhibition. As glycosidases are inhibited by glucose, it is advisable to increase the enzyme dose up to 10 g/hL for treating sweet dessert wines from aromatic varieties. A side benefit is earlier post-fermentation clarification which helps protect the freshness and the cleanliness of the aroma of young wines. The SO₂ level should be monitored once a week during the treatment period because it has a tendency to combine. If desired, enzyme activity can be stopped with 20 g/hL bentonite.

Thanks to the treatment with an aroma liberating enzyme, the glycosylated aromas are freed, resulting in a double concentration of terpenes in the wine. In the example shown in Fig. 4.17, two of them, linalool and geraniol, were above the aroma threshold and contributed positively to increased floral character.

As for pectinases, it is possible to run laboratory tests in bottles to judge the

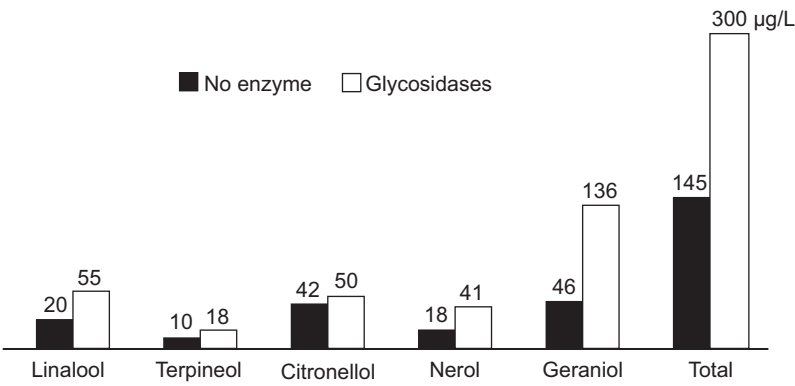


Fig. 4.17 Aroma enhancement in Gewürztraminer in response to glycosidase enzyme treatment. Monoterpene concentration given in µg/L. Details of treatment: pH 3.32; enzyme, Novarom @ 5 g/100 kg added at the end of alcoholic fermentation and at first racking.

effect before treating large volumes. For this, wines can be treated with glycosidases over time and aroma development followed by tasting.

The enzyme activity can be stopped by a bentonite fining. A low dosage, around 20 g/hL, is enough to stop enzyme activity. During enzyme treatment, the level of free SO₂ needs to be monitored as it has a tendency to combine, thereby limiting the protection against oxidation. According to the wine style, the winemaker can either treat a full lot of wine or prepare an aromatic reserve for blending purposes. The combined use of pectinases in extraction (release of aroma precursors) and β -glycosidases during maturation (release of free volatile aroma) can help to lift the overall wine aroma intensity.

4.7.2 Filtration

This step is one of the most important steps in winemaking, but it is certainly one of the most difficult to master due to the complexity of the colloidal matrix. Settling and decantation over time during the maturation phase help clarification by removal of remaining solids and microorganisms (yeast, bacteria), but not enough to reach the necessary clarity. The presence of protective colloids is responsible for difficulties in clarification; their presence limits sedimentation and contributes to clogging of filters. The nature of the particles contributing to turbidity influences both the quality of the clarification and the filtration flux. In young wines, macromolecules of various origins, in particular polysaccharides from grapes (Villettaz *et al.*, 1983; Belleville *et al.*, 1993; Doco and Brillouet, 1993; Doco *et al.*, 1995, 1999; Pellerin *et al.*, 1995) and microorganisms (Dubourdieu and Ribéreau-Gayon, 1981; Villettaz and Amado, 1981; Llaubères *et al.*, 1987, 1990; Agudelo *et al.*, 1994; Waters *et al.*, 1994; Doco *et al.*, 1996) are present and have an impact on filterability. Therefore, enzyme blends have been developed to improve filterability; they contain pectinases and β -glucanases (Villettaz *et al.*, 1984; Dubourdieu *et al.*, 1985). These enzyme activities help reduce the molecular weight of the grape pectins and yeast glucans present in wines and therefore greatly improve wine filterability (Humbert-Goffard *et al.*, 2003).

A demonstration of the role of the pectinase/ β -glucanase blend was established by Humbert-Goffard (2003) as well as the separate roles of pure β -1-3 glucanase and β -1-6 glucanase (Fig. 4.18). Similar results were obtained on a 2003 Merlot wine, but filterability was improved to a lesser extent than for white wines. As for whites, the pectinase/glucanase blend performs best (Fig. 4.19). These same authors also demonstrated the positive effect of these enzymes applied to white and red wine with lees (> 2000 NTU), on both clarification and sedimentation. In large-scale application trials, the use of pectinase/glucanase blend helped to improve the filtration flux rate, reducing the filtration time and consumption of Kieselguhr. Treatment with Vinoflow on Gamay wine allowed savings of 3 hours 30 min and 65 kg of filtration earth to filter 1140 hL of wine compared with the same wine with no enzyme treatment (Table 4.6).

As it is almost impossible to determine the filterability of gross wines, Romat and Reynou (2007) worked on the development of a new filterability index. This

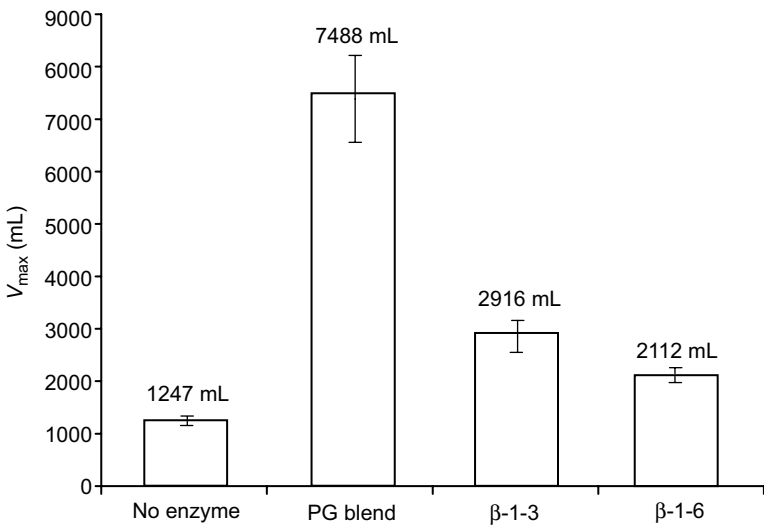


Fig. 4.18 Filterability of white Sauvignon wine in response to enzyme treatments. PG, pectinase/glucanase blend (Vinoflow); β -1-3 glucanase pure; β -1-6 glucanase pure (Humbert-Goffard, 2003).

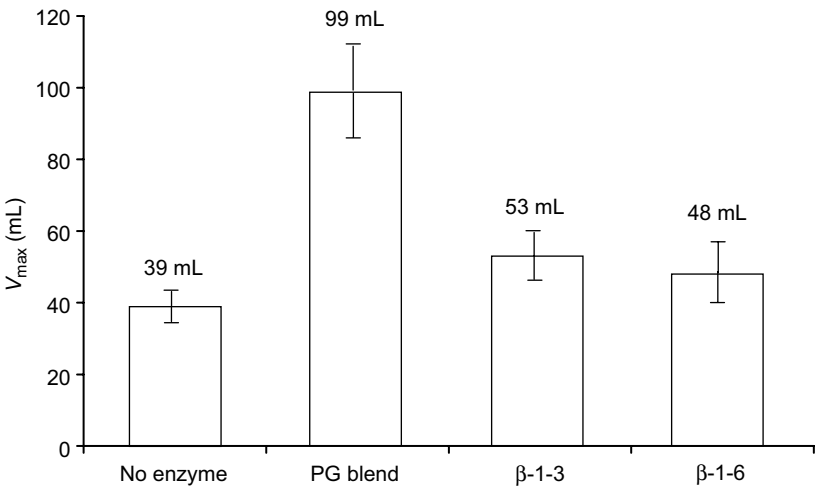


Fig. 4.19 Filterability of red Merlot wine 2003 in response to enzyme treatments. PG, pectinase/glucanase blend (Vinoflow); β -1-3 glucanase pure; β -1-6 glucanase pure (Humbert-Goffard, 2003).

Table 4.6 Comparison of Kieselguhr earth filtration cycles of a 2003 Gamay wine (pH 3.8, TPI – total phenolics measured by optical density at 280 nm) with and without pectinase/ β -glucanase enzyme blend treatment (Vinoflow G: 5 g/hL, 1 week contact time, 10 °C)

Cycle no.	Parameter	Control: no enzyme	Vinoflow G	Gain
Cycle 1	Volume filtered (hL)	360	410	50 hL
	Preparation (min)	25	25	
	Filtration (min)	225	245	
	Cleaning (min)	30	30	
	Earth flow (g/hL)	180	127	13 kg
Cycle 2	Volume filtered (hL)	250	290	40 hL
	Preparation (min)	30	25	
	Filtration (min)	195	180	
	Cleaning (min)	30	30	
	Earth flow (g/hL)	152	107	7 kg
Cycle 3	Volume filtered (hL)	340	440	100 hL
	Preparation (min)	30	20	
	Filtration (min)	245	255	
	Cleaning (min)	30	30	
	Earth flow (g/hL)	165	114	6 kg
Cycle 4	Volume filtered (hL)	200	0	
	Preparation (min)	30	0	
	Filtration (min)	155	0	
	Cleaning (min)	30	0	
	Earth flow (g/hL)	195	0	39 kg
Total	Volume filtered (hL)	1150	1140	
	Filtration time (min)	1055	840	215 min
	Earth consumption (kg)	198	133	65 kg

Source: Humbert-Goffard, personal communication.

test is similar to the clogging index measurement and uses membranes of various porosities (5, 1.2 and 0.65 μ); it is based on the progressive pore clogging law). This law is characterized by the equation $t/V: kt + 1/q_0$ where t is the filtration time, V the volume filtered at time t and q_0 the initial flux. Its representation is a straight line. At constant pressure, the volume filtered over time can be plotted as $t/V = f(t)$. The slope, k , depends on the clogging aptitude of the filtered wine. During filtration, the wine settles and the particles reduce the pore size. The lower the k value, the better the filterability.

Trials were conducted on many red wines to demonstrate the gain in filterability after enzyme treatment applying this law (Canal-Llaubères *et al.*, 2007; Romat, 2007). As shown by many studies, there is no correlation between turbidity and filterability. Wines with similar turbidity can have different filterability indices. In all the tests, the enzyme-treated wines had an increased filterability with lower k values (Table 4.7) (Kanev, 2006).

Thanks to trials conducted on a large number of wines, Romat and Reynou

Table 4.7 Changes in turbidity and filterability index in five red table wines in response to enzyme treatment (PG blend: pectinase/ β -glucanase; Vinoflow)

Trial wines	Turbidity (NTU)		Filterability index ($k \times 10^{-3}$) ^a	
	Control	Enzyme: PG blend	Control	Enzyme: PG blend
Red table wine	22	10	11	3
Bordeaux red	8	2.9	7.7	4.4
Bordeaux Médoc, tank maturation	7.9	7.6	3.5	2.6
Bordeaux Médoc, barrel maturation	8.6	8.7	8.3	6.6
Beaujolais	24	23	25	17

NTU, nephelometric turbidity units.

^a Filterability index, i.e. the slope of the filtration curve (progressive pure clogging law).

Source: Kanev, 2006.

(2007) defined a tool called CFLA (criteria for filtration Lamothe–Abiet) and a table which can predict the filtration of gross wines based on wine turbidity (NTU <3, 3 <NTU <15, 15 <NTU <50) and CFLA values (CFLA <10, 10 <CFLA <50, 50 <CFLA <200). Therefore, advice can be given for the selection of adequate filtration media in order to prepare the wine in the best conditions for final filtration. In order to evaluate gross wines filterability, it is recommended to measure both turbidity and CFLA.

4.8 Monitoring enzyme performance

Enzyme performance can be monitored thanks to simple tools which can be applied easily in wineries. The easiest is the pectin test or alcohol test. Other tests use turbidity measurement or filterability measurement. A simple clarification test can be conducted in the laboratory prior to harvest to determine the enzyme dose as it greatly depends on grape variety and maturity.

4.8.1 The pectin test

The pectin test or alcohol test is a simple tool to monitor enzyme performance. The presence of pectin is visualized by adding acidified alcohol (ethanol + 1% pure HCl) into a grape juice or a wine sample.

Preparing the test solution

Place 250 mL of alcohol (industrial ethanol) in the flask, add 2.5 mL of pure hydrochloric acid at 37%. Mix gently. The solution is ready and stable. This volume of test solution allows running 25 tests.

Running the test

The must or the wine samples should be taken from the tank. The sample should stand for 5 minutes prior to the test (sedimentation of particles).

In a test tube:

- add 5 mL of must or of wine;
- then add 10 mL of acidified alcohol (with 1% HCl);
- mix gently and let stand before reading.

Wait 5 minutes after mixing to read the results for must samples and 10 minutes for wine samples. The pectin precipitates in the presence of acidified alcohol.

Reading the scale

The presence of pectin is visualized by the presence of flakes (Fig. 4.20). When pectin is completely hydrolyzed, the liquid is clear (0). Slight presence is characterized by a very light haze (1). Medium to high presence shows flakes (2) which are immediately visible after alcohol addition (3). The presence of pectin indicates that pectinases need to be added in order to hydrolyze pectin to allow smooth processing. Only pectinases can help remove pectin.

4.8.2 Clarification test

Lab-scale trials can be performed using either a 1 L glass cylinder or a 1 L Imhoff cone. The settling cone is used to separate sediment from water. Graduated from 0 to 1000 mL, it has a leak-proof bottom closure for measuring and removing the settleable matter. The clarification test is performed in a thermostatted bath to match wine industry conditions, often at 10 °C. It allows determination of the enzyme dose, which varies according to the maturity level and the grape variety, or comparison of enzyme performance between products (Fig. 4.21).

Besides these laboratory trials, pilot- or large-scale trials can be conducted to demonstrate the efficiency of the enzyme treatment and evaluate the benefits.

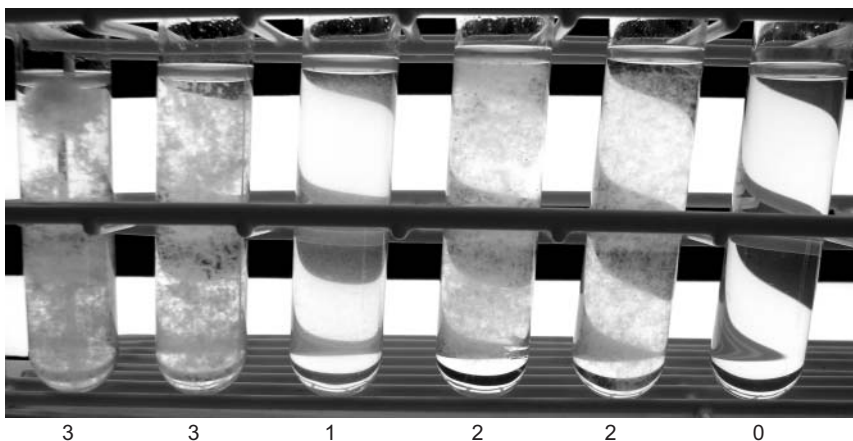


Fig. 4.20 The pectin test with acidified alcohol gives an indication of the presence of pectin: 3 corresponds to heavy flakes, 2 to fluffy flakes, 1 to haziness and 0 to clear juice.

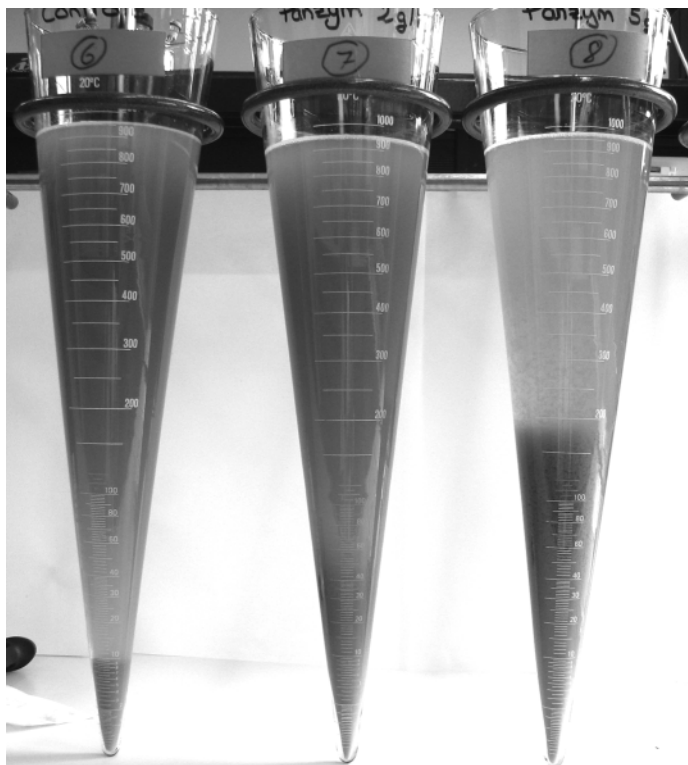


Fig. 4.21 Clarification test using 1 L Imhoff cones in a thermostatted bath (10 °C). From left to right: control with no enzyme addition – turbid juice; juice with 1 g/hL pectinase – start of depectinization; juice with 2 g/hL pectinase – start of clarification and settling.

4.8.3 Other tests

Enzyme performance can be monitored by measuring the turbidity, which is generally expressed in NTU. Major wineries are equipped to measure this variable, which can be used to illustrate enzyme performance and monitor the equipment (flotation, filtration).

Finally, by measuring the effect of an enzyme in simple ways (grape tonnage, juice or wine volume, time), wineries can easily calculate their return on investment. The increase in yield is enough to get a positive contribution whatever the wine sale prices. Other benefits can be evaluated through observations and sensory evaluation.

4.9 Future trends

New and better understanding of the mechanisms of enzymes for improvement of wine quality continues to emerge and help winemakers to reach various wine

style objectives. The wine matrix is complex and therefore a better understanding of the interactions between its major components is key to a mastery of wine stability.

Among the hydrolases, proteases should be further investigated as these enzymes might be involved in wine clarification and stabilization processes. A new Lysozyme from fungal origin could be available for winemakers to minimize the risk of allergy which could be inherent to enzyme extracted from hen egg white. Finally, the use of enzymes to reduce the alcohol level in wine could be revisited, not to produce low-alcohol products as previously tested (Villettaz, 1986) but to reduce by 1 or 2% the alcohol level of high-alcohol wines which is becoming a challenge for the industry. A new attempt has recently been initiated using a glucose oxidase (Biyela *et al.*, 2009). Further research is needed to check the stability and ageing potential of enzyme-treated wines. The impact of the enzyme on wine overall quality should also be evaluated.

4.10 Conclusions

Enzymes are recognized as useful processing aids in major winemaking application segments. Among the preparations offered by the enzyme producers, pectinases are by far the most widely used. Their use at an early stage in extraction on grapes and later on in clarification helps secure final wine quality. They are available as purified formulations with low cinnamoyl esterase activity. Enzymes help in extracting the potential originally present in the grapes. Since 2000, research and application studies have allowed better understanding and documentation of their benefits in terms of results and conditions of use. Wineries are less reluctant to use enzymes in red winemaking; however, there is still a great potential for generalizing their usage. At present, large producing operations need enzymes to run their processes in a more efficient way in order to avoid bottlenecks. Time is often an issue and enzymes, acting as catalysts, are the key to smooth running. Major benefits include yield increases at lower pressures; softer, higher and quicker extraction of colour, aroma and tannins; faster and more efficient clarification; better stabilization; positive contribution to wine sensory properties. The investment in enzymes always turns into positive earnings.

In association with pectinases, glucanases are mandatory in filtration application. Some polysaccharides like glucan from yeast or *Botrytis* can only be removed by enzymatic treatment. Thanks to their hydrolysis, other wine oenological treatments like fining can be more efficient. Enzymes ensure improved wine filterability and prevent losses in volume and quality.

Finally, enzymes should be one tool among others in the search for more natural wines and sustainable production. Innovation is needed in order to develop new biotechnology solutions for the wine industry.

4.11 Sources of further information

- Macrowine, Montpellier, France, 4–6 June 2008, Paris, France: INRA.
- The Third International Symposium on Pectins and Pectinases, Wageningen, the Netherlands, 21–23 April, 2008, Wageningen, the Netherlands: Wageningen Academic Publishers
- Organisation Internationale de la Vigne et du Vin (OIV): www.oiv.int.
- Association of Formulators and Manufacturers of Enzyme Products (AMFEP): www.amfep.org.
- Institut Cooperatif du Vin (ICV): www.icv.fr
- EC Regulation No. 479/2008 on the common organisation of the market in wine, *Official Journal of the European Union*, **L148**, 6 June 2008, 1–61: <http://faolex.fao.org/docs/pdf/eur80018.pdf>.
- References for comparison of worldwide regulations are part of a study conducted by Méhu (2001).

4.12 Acknowledgements

We would like to thank all the research students who have devoted their PhD work to a better understanding of enzyme mode of action and performance under winemaking conditions: Christine Barbe, Julien Ducruet, Anne Humbert-Goffard, Cécile Thibon, Marie-Agnès Ducasse, the project trainees and the students in oenology working in our Wine Experimental Cellar. Our recognition goes as well to the university professors having led these researches: Denis Dubourdieu, Yves Glories, Takatoshi-Tominaga, Martine Mietton-Peuchot at the University of Bordeaux and Véronique Cheynier, Thierry Doco and Hélène Fulcrand at INRA (National Institute of Agronomic Research), University of Montpellier. Thanks also to our partners Laffort and Lamothe-Abiet.

Finally, financial assistance from ARNT (Association Nationale de la Recherche et de la Technologie) and Conseil Régional d'Aquitaine are gratefully acknowledged.

4.13 References

- Agudelo J, Suarez JA, Olano A, Callejo MJ and Colomo M, (1994), Study of the glucan fraction in rosy wines. *Z. Lebensm. Unters. Forsch.*, **199**, 26–28.
- Aliot R, Blateyron L and Olivier B (2008), Thermovinification et turbidity: quel impact sur le fruité? *Rev. Œnol. Bourgogne*, **129**, 38–40.
- Amrani Joutei K and Glories Y (1995), Tannins and anthocyanins of grape berries: localization and extraction techniques. *Rev. Fr. Œnol.*, **153**, 28–31.
- Arnaldos T, Serrano M, Calderon A and Munoz R (1999), A rapid and continuous spectrophotometric method to measure β -glucosidase activity based on p-nitrophenol β -O-D-glucopyranoside hydrolysis. *Phytochem. Anal.*, **10**, 170–174.
- Aryan AP, Wilson B, Strauss CR and Williams PJ (1987), The properties of glycosidases of *Vitis vinifera* and a comparison of their β -glucosidase activity with that of exogenous

- enzymes. An assessment of possible applications in enology. *Am. J. Enol. Vitic.*, **38**, 182–188.
- Aunstrup K (1979), Production, isolation and economics of extracellular enzymes. *Appl. Biochem. Bioeng.*, **2**, 27–69.
- Ayestaran B, Guadalupe Z and Leon D (2004), Quantification of major grape polysaccharides (Tempranillo v.) released by maceration enzymes during the fermentation process. *Anal. Chim. Acta*, **513**, 29–39.
- Babayan TL and Bezrukov MG (1985), Autolysis in yeasts. *Acta Biotechnol.*, **2**, 129–136.
- Barbe C (1995), *Recherches sur les activités estérases contaminants des préparations pectolytiques. Applications technologiques*. Thèse de doctorat de l'université de Bordeaux 2, n° 348.
- Barbe C and Dubourdieu D (1998), Characterization and purification of a cinnamate esterase from *Aspergillus niger* industrial pectinase preparation. *J. Sci. Food Agric.*, **78**, 471–478.
- Bautista-Ortín AB, Martínez-Cutillas A, Ros-García JM, López-Roca JM and Gómez-Plaza E (2005), *Int. J. Food Sci. Technol.*, **40**, 867–878.
- Belleville M, Williams P and Brillouet J-M, (1993), A linear arabinan from red wine. *Phytochem.*, **33**, 227–229.
- Bertrand A and Ingargiola M-C (1993), Etude de l'efficacité de l'uréase pour l'élimination de l'urée, principal précurseur du carbamate d'éthyle au cours du vieillissement. *Bull. de l'OIV*.
- Biyela BNE, du Toit WJ, Malherbe DF and van Rensburg P (2009), The production of reduced-alcohol wines using Gluzyme Mono 10.000 BG treated grape juice. *S. Afr. J. Enol. Vitic.*, **30**(2), 32–39.
- Brillouet JM and Hoebler C (1985), Les hémicellulases. In Mouranche A and Costes C (eds), *Hydrolases et dépolymérase – enzymes d'intérêt industriel*. Gauthier Villars, Paris, France, 165–197.
- Brillouet J-M, Bosso C and Moutounet M (1990), Isolation, purification and characterization of an arabinogalactan from red wine. *Am. J. Enol. Vitic.*, **41**, 29–36.
- Brillouet JM, Williams P and Moutounet M (1991), Purification and some properties of a novel endo- β -galactanase from *Aspergillus niger*. *Agric. Biol. Chem.*, **55**, 1565–1571.
- Burkhardt VR, (1976), Depsidspaltende Nebenwirkung von Enzympräparaten für die Be- und Verarbeitung von Lebensmitteln, beobachtet bei der Aufbereitung von Obst- und Traubenmaischen. *Deutsch. Lebens.-Rundsch*, **72**(12), 417–420.
- Canal-Llaubères R-M (1989), Les enzymes industrielles dans la biotechnologie du vin. *Rev. Enolog.*, **53**, 17–22.
- Canal-Llaubères R-M (1992a), Enzymes in winemaking. In Fleet GH (ed.), *Wine Microbiology and Biotechnology*, Harwood Academic Publishers, Chur, Switzerland, 477–506.
- Canal-Llaubères R-M (1992b), New developments in industrial enzymes for wine treatment (aroma liberation). *Proceedings of the 3rd International Symposium Innovations in Wine Technology*, Stuttgart, Germany, 25–27 May, 147–157.
- Canal-Llaubères R.-M. (1994), Enhancing the aroma of wines. *Aust. Grapegrow. Winemak.*, **368**, 49–51.
- Canal-Llaubères R.-M. (2000), Les enzymes et leur application en oenologie. In *Produits de traitement et auxiliaires d'élaboration des moûts et des vins*, Editions Féret, Bordeaux, 43–60.
- Canal-Llaubères R-M (2004). Pre-fermentation stages in white wine making application of pectinases during pressing. Poster presentation, *7th International Symposium Innovations in Wine Technology*, Stuttgart, Germany, 10–11 May.
- Canal-Llaubères R-M, Kanev T, Romat H, Poupot C and Mietton-Peuchot M (2007), A new filterability index adapted to gross wines prior to filtration. Poster presentation, *8th International Symposium Innovations in Wine Technology*, Stuttgart, Germany, 22–26 April.
- Cayla L, Cottureau P, Masson G and Guérin L (2009), Les enzymes en oenologie. Intérêt dans les opérations préfermentaires sur vin rosé. *Rev. Fr. Oenol.*, **239**, 2–9.

- Chatonnet P, Barbe C, Canal-Llaubères R., Dubourdieu D and Boidron J-N (1992), Incidence de certaines préparations pectolytiques sur la teneur en phénols volatils des vins blancs. *J. Int. Sci. Vigne Vin*, **26**(4), 253–269.
- Cordonnier R and Bayonove C (1974), Mise en évidence dans la baie de raisin, variété muscat d'Alexandrie, de monoterpènes liés révélables par une ou plusieurs enzymes du fruit. *C.R. Acad. Sci. Ser D*, **278**, 3387–3390.
- Delcroix A, Günata Z, Sapis J, Salmon J and Bayonove C (1994), Glycosidase activities of three enological yeast strains during winemaking: effect on the terpenol content of muscat wine. *Am. J. Enol. Vitic.*, **45**, 291–296.
- Delteil D (2000), Evaluation sensorielle du profil gustatif des vins. *Rev. Oenol.*, **94**, 21–23.
- Doco T and Brillouet J-M (1993) Isolation and characterization of a rhamnogalacturonan II from red wine. *Carbohydr. Res.*, **243**, 333–343.
- Doco T, Lecas M, Pellerin P, Brillouet J-M and Moutounet M (1995), Les polysaccharides pectiques de la pulpe et de la pellicule de raisin. Quel devenir pendant la phase pré-fermentaire? *Rev. Fr. Oenol.*, **153**, 16–23.
- Doco T, Brillouet J-M and Moutounet M (1996), Evolution of grape (Carignan noir cv.) and yeast polysaccharides during fermentation and post-maceration. *Am. J. Enol. Vitic.*, **7**, 108–110.
- Doco T, Vuchot P and Cheynier V (2003), Structural modification of arabinogalactan-proteins during aging of red wines on lees. *Am. J. Enol. Vitic.*, **54**, 150–157.
- Doco T, Williams P and Cheynier V (2007), Effet de flash release and pectinolytic enzyme treatments on wine polysaccharide composition. *J. Agric. Food Chem.*, **55**, 6643–6649.
- Dubourdieu D and Ribéreau-Gayon P (1981), Structure of the exocellular beta-D-glucan from *Botrytis cinerea*. *Carbohydr. Res.*, **93**, 294–299.
- Dubourdieu D, Desplanques C, Villettaz J-C and Ribéreau-Gayon P (1985), Investigations of an industrial β -D-glucanase from *Trichoderma harzianum*. *Carbohydr. Res.*, **144**, 277–287.
- Dubourdieu D, Darriet P, Ollivier C, Boidron J-N and Ribéreau-Gayon P (1988), Rôle de la levure *Saccharomyces cerevisiae* dans l'hydrolyse enzymatique des hétérosides terpéniques du jus de raisin. *C. R. Acad. Sci. Paris*, **306**, 489–493.
- Ducasse M-A (2009), *Impact des enzymes de macération sur la composition en polysaccharides et en polyphénols des vins rouges. Etude de l'évolution de ces composés en solution modèle vin*. Thèse de l'université de Montpellier 2.
- Ducasse M-A, Canal-Llaubères R-M, de Lumley M, Williams P, Souquet J-M, Fulcrand H, Doco T and Cheynier V (2009), Effect of macerating enzyme treatment on polyphenol and polysaccharide composition of red wines. *Food Chemistry*, **118**, 369–376.
- Ducasse MA, Williams P, Meudec E, Cheynier V and Doco T (2010), Isolation of Carignan and Merlot red wine oligosaccharides and their characterization by ESI-MS. *Carbohydr. Polym.*, **79**, 747–754.
- Ducruet J (2000), *Etude des protéines du vin rouge. Mécanismes d'action d'enzymes de macération sur le raisin rouge*. Thèse de l'université de Bordeaux 2, n°742.
- Feuillat M and Charpentier C (1982), Autolysis of yeasts in Champagne. *Am. J. Enol. Vitic.*, **33**, 6–13.
- Flanzy M (1998), *Oenologie: fondements scientifiques et technologiques*. Tec & Doc, Lavoisier, Paris, France
- Francis IL, Tate ME and Williams (1996), The effect of hydrolysis conditions on the aroma released from Semillon grape glycosides. *Aust. J. Grape Wine Res.*, **2**, 70–76.
- Fujinawa S, Burns G and De La Teja P (1990), Application of acid urease to reduction of urea in commercial wines. *Am. J. Enol. Vitic.*, **41**, 350–354.
- Gerbaux V, Villa A, Monamy C and Bertand A (1997), Use of lysozyme to inhibit malolactic fermentation and to stabilize wine after malolactic fermentation. *Am. J. Enol. Vitic.*, **48**, 49–54.
- Gerbaux V, Meistermann E, Cottureau P, Barrière C, Cuinier C, Berger J-L and Villa A (1999), Utilisation de la lysozyme en œnologie. *Bull. OIV*, **72**, 348–373.

- Gerbaux V, Vincent B and Bertrand A (2002), Influence of maceration temperature and enzymes on the content of volatile phenols in Pinot noir wines. *Am. J. Enol. Vitic.*, **53**, 131–137.
- Grossmann M and Rapp A (1988), Steigerung des Sortentypischen Weinbuketts nach Enzymbehandlung. *Deutsch. Lebens.-Rundsch.*, **84**, 35–37.
- Grossman M, Rapp A and Rieth W (1987), Enzymatische Freisetzung gebundener Aromastoffe in Wein. *Deutsch. Lebens.-Rundsch.*, **83**, 7–12.
- Guérin L, Sutter D, Demois A, Chereau M and Trandafir G (2009), Determination of activity profiles of main commercial enzyme preparations used in winemaking. *Am. J. Enol. Vitic.*, **60**(3), 322–331.
- Günata Z, Bitteur S, Brillouet J-M, Bayonove C and Cordonnier R (1988), Sequential enzymatic hydrolysis of potentially aromatic glycosides from grapes. *Carbohydr. Res.*, **184**, 139–149.
- Günata Z, Bayonove C, Tapiero C and Cordonnier R (1990a), Hydrolysis of grape monoterpenyl β -D-glucosides by various β -glucosidases. *J. Agric. Food Chem.*, **38**, 1232–1236.
- Günata Z, Bayonove CL, Cordonnier RE, Arnaud A and Galzy P (1990b), Hydrolysis of grape monoterpenyl glycosides by *Candida molischiana* and *Candida wickerhamii* beta glucosidases. *J. Sci. Food Agric.*, **50**, 499–506.
- Günata Z, Dugelay I, Vallier J and Bayonove C (1997), Multiple forms of glycosidases in an enzyme preparation from *Aspergillus niger*: partial characterization of a β -apiosidase. *Enzyme Microb Technol.*, **21**, 39–44.
- Humbert-Goffard A (2003), *Recherches sur les phénomènes enzymatiques intervenant lors de l'élevage des vins*. Thèse de doctorat de l'université de Bordeaux 2, n° 1076.
- Humbert-Goffard A, Basque E, Vatin L and Canal-Llaubères R-M (2003), Rôle des préparations enzymatiques à base de β -glucanases sur la mise au propre et la filtration des vins. *Rev. Fr. Œnol.*, **201**, 28–31.
- Kanev T (2006), *Mise au point d'un nouvel indice de filtrabilité pour évaluer l'efficacité d'enzymes oenologiques à base de pectinases et de β -glucanases*. Rapport de stage pour le Master Professionnel Environnement Viti-Vinicole, faculté d'œnologie de Bordeaux, laboratoire Génie des Procédés.
- Laffort J-F, Romat H and Darriet P (1989), Les levures et l'expression aromatique des vins blancs. *Rev. Œnolog.*, **53**, 9–12.
- Leclerc M, Arnaud A, Ratomahenina R and Galzy P, (1987), Yeast β -glucosidases. *Biotechnol. Genet. Eng. Rev.*, **5**, 269–295.
- Llaubères R-M (1988), *Les polysaccharides sécrétés dans les vins par Saccharomyces cerevisiae et Pediococcus sp.* Thèse de doctorat de l'université de Bordeaux 2.
- Llaubères R-M, Dubourdieu D and Villettaz J-C (1987), Exocellular polysaccharides from *Saccharomyces* in wine. *J. Sci. Food Agric.*, **41**, 277–286.
- Llaubères R-M, Richard B, Lonvaud A, Dubourdieu D and Fournet B (1990), Structure of an exocellular beta-D-glucan from *Pediococcus* sp., a wine lactic bacteria. *Carbohydr. Res.*, **203**, 103–107.
- Main GL and Morris JR (2007), Effect of macerating enzymes and postfermentation grape-seed tannin on the color of Cynthiana wines. *Am. J. Enol. Vitic.*, **58**(3), 365–372.
- Modra EJ and Williams PJ (1988), Are proteases active in wines and juices? *Aust. Grapegrow. Winemak.*, **292**, 42–46.
- Moine-Ledoux V and Dubourdieu D (1996), Mise au point d'une préparation industrielle extraite de la levure, améliorant la stabilité protéique et inhibant les précipitations tartriques des vins. In Lonvaud-Funel A (ed). Œnologie 95. Tec & Doc, Lavoisier, Paris, France
- Oliviera J, de Freitas V, Silva AMS and Mateus N (2007), Reaction between hydroxycinnamic acids and anthocyanin-pyruvic acid adducts yielding new portisins. *J. Agric. Food Chem.*, **55**, 6349–6356.
- Parley A, (1997), *The effect of pre-fermentation enzyme maceration on extraction and colour stability in Pinot noir wine*. Thesis, Lincoln University.

- Riou V, Vernhet A, Doco T and Moutounet M (2002), Aggregation of grape seed tannins in model wine. Effect of wine polysaccharides. *Food Hydrocolloids*, **16**, 17–23.
- Romat H (2007), Coefficient de colmatage, une nouvelle approche de la filtration des vins. *Rev. Œnolog. Fr.*, **123**, 31–33.
- Romat H and Reynou G (2007), Proposition de critères de filtration en application du coefficient de colmatage. *Rev. Œnolog. Fr.*, **124**, 36–38.
- Schwarz M, Wabnitz TC and Wintherhalter P (2003), Pathway leading to the formation of anthocyanin-vinylphenol adducts and related pigments in red wines. *J. Agric. Food Chem.*, **51**, 3682–3687.
- Somers TC, Verette E and Pocock KF (1987), Hydroxycinnamates esters of *V. vinifera*: changes during white vinification and effects of exogenous enzyme hydrolysis. *J. Sci. Food Agric.*, **40**, 67–78.
- Takahashi K, Inoue H, Sakai K, Kohama T, Kitahara S, Takishima K, Tanji M, Athauda SB and Akanuma H (1991), The primary structure of *Aspergillus niger* acid proteinase. *A. J. Biol. Chem.*, **266**, 19480–19483.
- Tate D and Reynolds AG (2006), Validation of a rapid method for measuring β -glycosidase activity in fermenting muscat grape musts. *Am. J. Enol. Vitic.*, **57**(1), 60–68.
- Vidal S, Francis L, Guyot S, Marnet N, Kwiatkowski M, Gawel R, Cheynier V and Waters EJ (2003), The mouth-feel properties of grape and apple proanthocyanidins in a wine-like medium. *J. Sci. Food Agric.*, **83**, 564–573.
- Vidal S, Francis L, Williams P, Kwiatkowski M, Gawel R, Cheynier V and Waters E (2004), The mouth-feel properties of polysaccharides and anthocyanins in a wine like medium. *Food Chem.*, **85**(4), 519–525.
- Villettaz J-C (1986), A new method for the production of low alcohol wines and better balanced wines. In *Proceedings of 6th Australian Wine Industry Technical Conference*, Australian Wine Research Institutem Adelaide, SA, 125–128.
- Villettaz J-C (1990), Les colloïdes colmatants et la filtration des vins. *Rev. Fr. Oenol.*, **122**, 59–63.
- Villettaz J-C and Amado R (1981), Untersuchung der löslichen neutral polysaccharidfraktion von Traubensaft und Wein. *Lebensm. Wiss. Und Technol.*, **14**, 176–181.
- Villettaz J-C, Amado R and Neukom H (1981), Structural investigations of an arabinan from grape juice. *Carbohydr. Polym.*, **1**, 101–105.
- Villettaz J-C, Amado R and Neukom H (1982), Investigations of colloid substances in must and wines. In Baltes W, Czedik-Eysenberg P B and Pfannhauser W (eds), *Recent Developments in Food Analysis*, Verlag Chemie, Berlin, Germany, 476–481.
- Villettaz J-C, Steiner D and Trogus H (1984), The use of glucanase as an enzyme in wine clarification and filtration. *Am. J. Enol. Vitic.*, **35**, 253–256.
- Waters EJ, Pellerin P and Brillouet JM (1994), A *Saccharomyces* mannoprotein that protects wine from protein haze. *Carbohydr. Polym.*, **23**, 185–191.
- Williams P, Strauss C and Wilson B (1981), Classification of the monoterpenoid composition of muscat grapes. *Am. J. Enol. Vitic.*, **32**, 230.
- Williams P, Strauss C, Wilson B and Massy-Westropp R (1982), Novel monoterpene disaccharide glycosides of *Vitis vinifera* grapes and wines. *Phytochemistry*, **21**, 1013–1020.
- Wightman JD and Wrolstad RE (1996), Beta-glucosidase activity in juice processing enzymes based on anthocyanin analysis. *J. Food Sci.*, **61**, 544–552.
- Wightman JD, Price ST, Watson BT and Wrolstad RE (1997), Some effects of processing enzymes on anthocyanins and phenolics in Pinot noir and Cabernet Sauvignon wines. *Am. J. Enol. Vitic.*, **48**, 39–48.

5

Membrane and other techniques for the management of wine composition

D. Wollan, Memstar Pty Ltd, Australia

Abstract: Wine composition can now be successfully adjusted by a range of membrane separation and phase change techniques. This chapter explains the application of such processes as reverse osmosis, nanofiltration, ultrafiltration, distillation, freeze concentration, evaporative perstraction, adsorption and electro dialysis to the management of wine composition. Flavour concentration, alcohol adjustment, volatile acidity reduction, taint removal and tartrate stabilisation are typical treatments using this range of new winemaking technologies.

Keywords: wine membrane separation technology, reverse osmosis, alcohol adjustment, tartrate stabilisation, electro dialysis.

5.1 Introduction

Winemakers have been managing wine composition since the beginning of their art. This has involved utilisation of whatever technology – microbiological, chemical or mechanical – has been available at the time. This chapter will deal with a number of physico-chemical technologies which have an increasing role in modern winemaking. To be clear, these are defined as those methods which utilise or manipulate some physical chemical, particularly colligative, property of the grape must or wine (or, by exception, those methods that do not include viticultural techniques, the addition of selected microorganisms or permitted chemical adjuncts).

Such techniques are characterised by the application of energy – mechanical, electrical or thermal – to the winemaking process. They include:

- separation;
- clarification;
- phase change;
- heating and cooling.

More specifically, this chapter will briefly consider such ‘high-tech’ methods as low temperature/vacuum distillation and freeze concentration. However, the emphasis will be on the application to modern winemaking of such membrane- and resin-based procedures as:

- reverse osmosis;
- nanofiltration;
- ultrafiltration;
- evaporative perstraction;
- adsorption; and
- electrodialysis.

In particular, there will be close attention to the ability of these techniques to separate or fractionate wines with a view to removing certain undesirable components and, in so doing, adjusting wine composition.

5.2 Some caveats

Much of this technology is relatively new, with practical application in the winemaking process generally spanning less than 20 years. In many jurisdictions, local laws and regulations have not kept pace with these technological developments and they are not permitted winemaking practices.

In addition, a number of these processes are proprietary. The intellectual property embodied in some of these is closely guarded and, in some cases, subject to patents – granted or otherwise. For these reasons, at the time of writing, the full details of some of these technologies are not available and so their descriptions here are subject to some speculation by the author.

Finally, a declaration: the author, through his company, has been an active participant in these developments for some years and has a commercial interest in a number of these technologies.

5.3 Some perspective – convention and intervention

These technologies may sound exotic and manipulative, but some perspective is in order: winemaking has always been an act of human intervention to moderate and control the natural spoilage mechanisms that would otherwise make the product of fermentation undrinkable. Historically, the winemaker’s aim has been to produce

a satisfying and safe beverage characterised by its alcohol, acidity, tannin, flavour and colour. Left to 'nature' there would be no wine, only vinegar.

Winemakers are all familiar and generally comfortable with manipulating wine composition with such 'low-tech' non-additive options as:

- selection of harvest time;
- temperature of processing and fermentation;
- running off juice (saignée);
- press extraction rates;
- oak barrel fermentation and maturation;
- mechanical clarification of juice and wine;
- cold stabilisation;
- filtration.

These conventional techniques have little to offer with regard to more difficult compositional problems such as:

- lack of flavour, colour, tannin concentration;
- immature flavours and tannins;
- insufficient or excessive acidity;
- heat (protein) instability;
- high or low alcohol;
- excessive volatile acidity;
- extraneous taints, e.g. smoke and *Brettanomyces*.

For these issues, a new generation of tools is required.

5.4 Next-generation tools – phase change techniques

The first of these groups of next-generation tools can broadly be classified as 'phase change' techniques. These rely on the differing responses of grape, juice and wine components to heating and cooling. In particular, separation and fractionation of these components may be achieved by phase changes such as freezing or evaporation. Some 'natural' examples of these phenomena are the concentration effects of berry freezing as in ice wine; and dehydration as in Botrytis and Amarone wine styles. The technical responses to these phenomena are the techniques of freeze concentration, evaporation and distillation.

5.4.1 Freeze concentration

Over the years, freeze concentration has been used by winemakers to remove water from grape juice and, less extensively, from wine. At its crudest, a block of ice is allowed to form on an in-tank heat exchanger; the juice or wine is removed; the ice thaws and the water is drained away. This is then repeated as necessary to achieve the desired concentration.

More sophisticated, automatic systems such as from GEA Messo PT utilise a

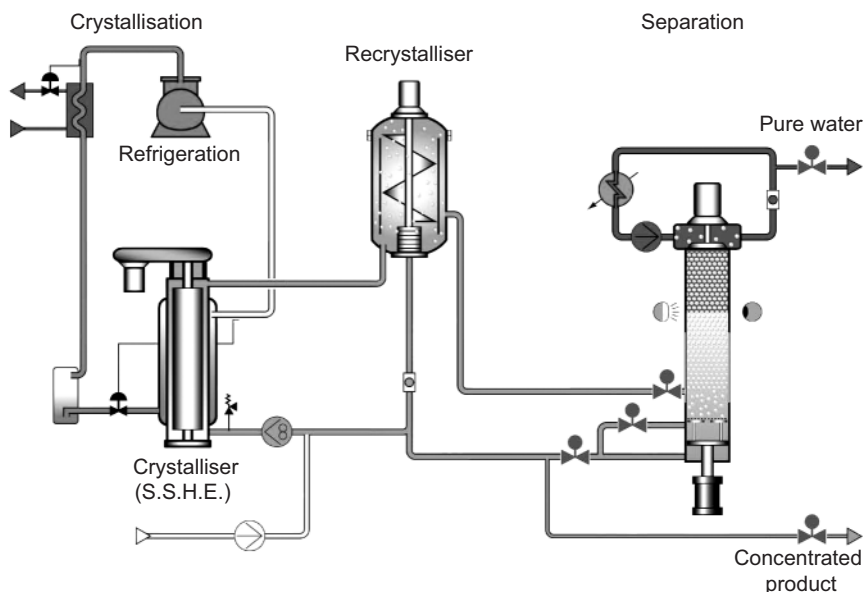


Fig. 5.1 Freeze concentration of fruit juice using the GEA crystallisation process (GEA Messo PT, 2009).

continuous feed of product through a crystalliser (scraped surface heat exchange) to a recrystalliser (Fig. 5.1). Small ice crystal nuclei form in the product which is recirculated through the heat exchange and recrystalliser where these ice crystals form and grow. When the crystals achieve a certain size, they transfer to a wash column where they are rinsed to remove concentrate and then thawed and removed as water. Concentrated product is removed through a filter in the separation stage (GEA Messo PT, 2009). This technique is reportedly successful in its stated purpose, but it is energy demanding.

5.4.2 Evaporation

Commercial evaporation systems have been used extensively for many years for fruit juice concentration. In the grape and wine industry, grape juice has been concentrated by such systems to remove water and make a high sugar (60–70 °Brix) material that is added back to wine for the purpose of sweetening the final product. This has not generally been considered a quality product because the process, unless carried out at lower temperature such as under vacuum, has tended to lead to some thermal degradation. Moreover, when conducted under vacuum, there was often a loss of volatile flavour components. The latest, and most successful, implementation of the principle is the Spinning Cone Column, which is described below.

5.4.3 Spinning Cone distillation

The Spinning Cone distillation system was developed by Australian company

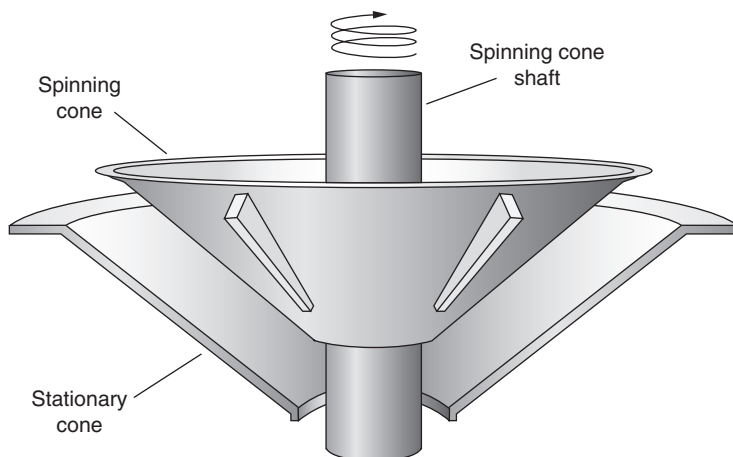


Fig. 5.2 Spinning Cone Column – consists of a rotating shaft with alternating spinning and stationary cones (FT Technologies, 2009b).

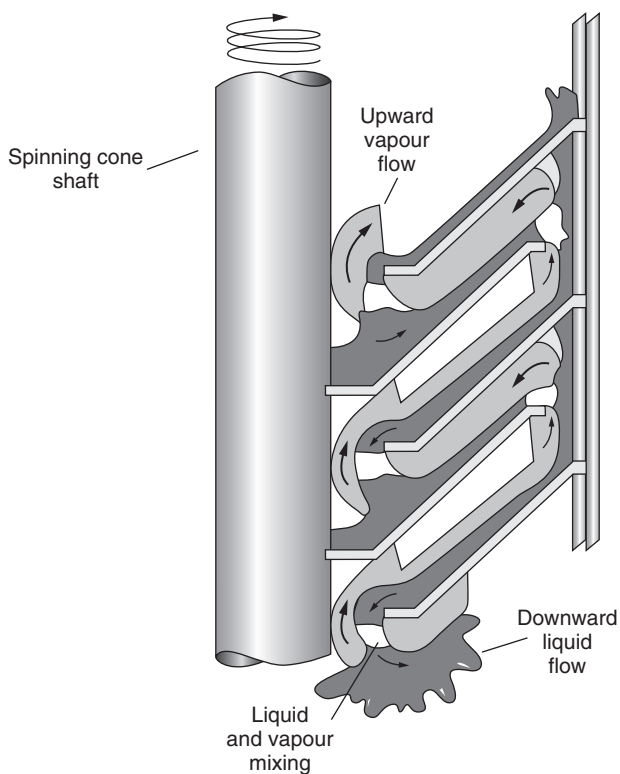


Fig. 5.3 Spinning Cone Column – a thin film of product descends the column of cones. Volatile components are stripped from this by an ascending stream of vapour (gas or steam) in counter-flow (FT Technologies, 2009b).



Fig. 5.4 Spinning Cone Column – Model SCC10,000 (FT Technologies, 2009c).

Flavourtech and is well established around the world for a range of food and beverage applications (FT Technologies, 2009a,b,c). Essentially, the process is based on a column with a stack of alternating static and spinning cones. Product enters at the top and a thin film forms on the cones as it descends. In counter-flow with this is a stripping stream of vapour (gas or steam) which ascends the column under vacuum. The more volatile components such as flavours and alcohol evaporate preferentially into this ascending stream and are thus removed. They are then condensed and collected at the top of the column for removal or secondary separation and recombination (Figs 5.2–5.4).

In the wine industry, the Spinning Cone Column has established a number of uses:

- *Concentration of grape juice* – it is considered to yield a superior product to the evaporator systems mentioned above – temperatures are lower and desirable volatiles are preserved.
- *Alcohol reduction* – this is a significant application for the technology and is discussed later.
- *Removal of SO_2 from grape juice* – where grape juice has been processed with high levels of SO_2 as a preservative, the Spinning Cone Column has been an

effective technique for removing this so that the juice can be consumed or fermented to wine.

- *Flavour recovery from marc and lees* – at the end of the winemaking process the waste products – marc and lees – still contain some desirable flavours. These may be extracted then concentrated with this technique.

5.5 Membrane separation techniques

Another group may be categorised as membrane separation techniques. With these, mechanical, thermal and/or electrical energy are applied to the material being treated. The aim is to create a pressure, concentration or electrical potential difference across a membrane and so provide a driving force for various wine components to pass through that membrane. These processes have in common the application of cross-flow filtration principles and rely on the properties of distinct membrane types to pass various wine components differentially, and so achieve separations at the molecular level. They include:

- reverse osmosis;
- nanofiltration;
- ultrafiltration;
- evaporative perstraction;
- electrodialysis.

These techniques have already found growing application in winemaking for:

- flavour concentration;
- alcohol adjustment;
- volatile acidity removal;
- taint treatment;
- tannin fractionation;
- tartrate stabilisation.

5.5.1 Some basic membrane terminology (as related to wine treatment)

- *Cross-flow*. In cross-flow filtration techniques, the flow of wine is parallel to the surface of the membrane, as opposed to the more familiar type of ‘dead-end’ filtration where the wine flow is perpendicular to the surface of the membrane. The cross-flow configuration greatly reduces the risk of membrane fouling but, for this to be effective, the ratio of cross-flow feed to the flow of permeate must be adequate. (As a rough guide, this is generally at least 4:1 for most of the applications discussed here.)
- *Osmotic pressure*. This is defined as the ‘excess pressure required to maintain osmotic equilibrium between a solution and the pure solvent separated by a membrane permeable only to the solvent’ (International Union of Pure and Applied Chemistry, 1996).

- *Permeate*. This is the portion of the wine feed that permeates the membrane. In effect, it is the 'filtrate'.
- *Retentate*. This is the portion of the wine feed that does not permeate the membrane. As such, it is the 'concentrate' form of the wine.
- *Membrane surface area*. This is the effective surface area of membrane that is exposed to pressurised wine feed and able to pass permeate.
- *Molecular weight cut-off*. In reverse osmosis, nanofiltration and ultrafiltration applications where separation occurs at the molecular level, a membrane is characterised by its molecular weight cut-off. This is generally accepted to be the molecular weight of a spherical, non-charged solute species at which more than 90% of that solute is retained by the membrane.
- *Flux*. Flux represents the volume of permeate flowing through a unit of membrane surface area during a given time.
- *Permeate flow*. Permeate flow = flux $\times$ membrane surface area. It is the effective measure of productivity of a given system.
- *Trans-membrane pressure*. This is the difference in pressure between the feed or retentate side of the membrane and the permeate side. It is the driving force for liquid to permeate the membrane. Generally, the higher the trans-membrane pressure, the greater is the flux.
- *Passage*. A measure of the extent to which a particular molecular or ionic species permeates the membrane and so passes into the permeate. It is the inverse of rejection ($= 1 - 1/\text{rejection}$) or the extent to which that species is rejected by the membrane. Passage is expressed in terms of the ratio of the concentration of the component (say 'A') in the permeate compared to its concentration in the feed:

$$\text{Passage 'A'} = [A_{\text{permeate}}]/[A_{\text{feed}}]$$

5.5.2 Reverse osmosis

Reverse osmosis: liquid-phase pressure-driven separation process in which applied transmembrane pressure causes selective movement of solvent against its osmotic pressure difference.

(International Union of Pure and Applied Chemistry, 1996)

Reverse osmosis (RO) has its origins and found its greatest industrial application in the purification of water, both wastewater treatment and desalination. It relies on the ability of certain semi-permeable polymeric membranes to allow the passage of solvents while retaining solutes. In situations where a high concentration solution is separated from a lower strength solution by such a membrane, an osmotic potential is generated such that solvent flows through the membrane from the lower concentration to higher concentration sides. This flow continues until such time as the solution concentrations on both sides of the membrane are the same or a pressure equal to or greater than the osmotic pressure difference of the two solutions is applied to the high-concentration side. Where the applied pressure is

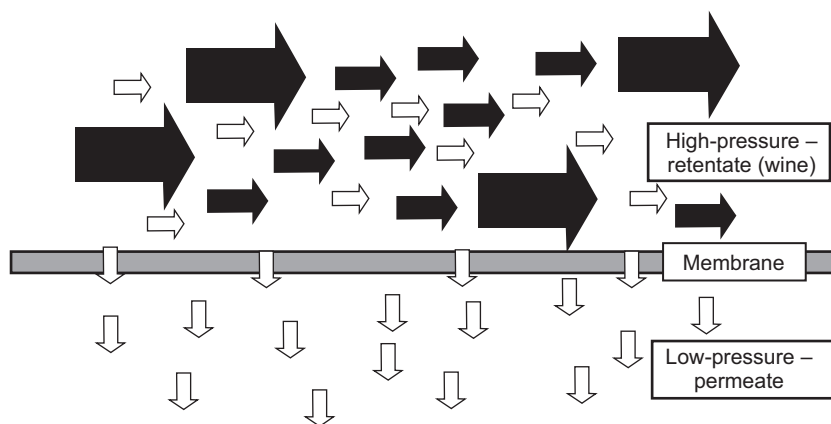


Fig. 5.5 Reverse osmosis – wine flows at high pressure past the ‘retentate’ side of the semi-permeable membrane where its vinous character is retained and becomes concentrated. Permeate, which has the appearance of water and contains little if any of the colour or flavour of the wine, consists of the lower molecular weight components which pass through the membrane.

greater than the osmotic pressure difference, solute flows in the reverse direction, hence ‘reverse osmosis’.

The widespread use of this process in water treatment, as well as food, dairy, pharmaceutical and general industrial applications, allowed the commercial development of a range of different membrane materials, formats and sizes. It was not long before some of these found their first application in the winemaking process.

In the case of wine treatment by reverse osmosis, the wine remains substantially on the high-pressure or ‘retentate’ side of the membrane and its vinous character is retained in the unfiltered portion which becomes concentrated. Some of the lower molecular weight fraction passes slowly through the filter membrane to the low-pressure, ‘permeate’ side. This permeate has the appearance of water and contains little if any of the colour or flavour of wine (Fig. 5.5)

A review of various wine components and their molecular weights shows that this ability to fractionate wine according to molecular weights in the range of 50–300 is a very powerful tool. For significant passage, typical RO membranes have a nominal molecular weight (MW) cut-off (MWCO) of 100 Daltons (Table 5.1). They have the ability to allow the passage of only those components to be removed: water (MW = 18) is the smallest molecule in wine and passes readily. Other constituents with MW <100, such as carbon dioxide (MW = 44), acetaldehyde (MW = 44), ethanol (MW = 46), acetic acid (MW = 60) and ethyl acetate (MW = 88), are passed into the permeate but at lower concentrations than are present in

Table 5.1 Some wine components and their molecular weights – separation by RO membrane with molecular weight cut off ~100 Daltons

Wine component	Molecular weight	
Water	18	
Carbon dioxide	44	
Acetaldehyde	44	
Ethanol	46	
Acetic acid	60	
Ethyl acetate	88	
Lactic acid	90	↑Components pass through RO membrane into permeate (but at differential rates in reverse order of molecular weight)
Malic acid	134	↓Components rejected by RO membrane – remain substantially in concentrate
Tartaric acid	150	
Volatile phenols	120–150	
Glucose/fructose	180	
Flavonoids	>300	

the wine. Of the other components, only minor quantities of lactic acid (MW = 90) pass into the permeate. There is virtually no passage of tartaric acid (MW = 150) or malic acid (MW = 134).

It should also be noted that in aqueous solutions such as wine, ions exist in a hydrated form. These ion–water complexes are generally too large to pass readily through normal RO membranes. Therefore, the extent to which a compound exists in an undissociated (non-ionised) form will affect its passage through any particular membrane type.

Almost all other constituents including sugars, esters, anthocyanins and other phenolics do not pass the membrane to any significant extent and are retained in the wine.

5.5.3 Nanofiltration

Nanofiltration: pressure-driven membrane-based separation process in which particles and dissolved molecules smaller than about 2 nm are rejected.

(International Union of Pure and Applied Chemistry, 1996)

Closely related to reverse osmosis is the process known as nanofiltration (NF). Practically, the demarcation between nanofiltration and RO is vague and related to molecular size. Arguably, the type of nanofiltration of interest in the wine industry is really a particular type of ‘loose’ RO where the MWCO of the membranes is between about 150 and 300 Daltons. There is also some selectivity due to electrical charge of the species. Notably, monovalent ions such as sodium, potassium, acetate and lactate are passed while divalent ions such as calcium and tartrate are not. This last feature is probably the more significant distinction.

Table 5.2 Separation by NF membranes with molecular weight cut-off ~150–200 Daltons

Wine component	Molecular weight (Da)	
Water	18	
Carbon dioxide	44	
Acetaldehyde	44	
Ethanol	46	
Acetic acid	60	
Ethyl acetate	88	
Lactic acid	90	
Malic acid	134	
Tartaric acid	150	
Volatile phenols	120–150	
Glucose/fructose	180	↑Components pass through NF membrane – into permeate (but at differential rates in reverse order of molecular weight)
Flavonoids	>300	↓Components rejected by NF membrane – remain substantially in concentrate

Practically, ‘looser’ NF membranes are used in preference to ‘tighter’ RO membranes when higher permeate flux rates and/or greater passage of higher molecular weight components are important. In all such choices, there is inevitably some compromise between the passage into the permeate (and potential loss) of desirable wine components as well as undesirable.

Table 5.2 shows the typical characteristics of looser NF-type membranes and the increased passage into the permeate of wine components with molecular weights in the range 150–200 Daltons. For some applications, this can be beneficial; in others this can lead to the greater loss of or damage to desirable wine components.

5.5.4 Concentration by reverse osmosis

This first winemaking application of RO was concentration. Grape must has been concentrated by RO since the 1980s. Clarified grape juice is circulated at high pressure through a membrane system which has been selected to retain most grape constituents but allow the passage of water. Figure 5.6 illustrates a typical grape juice or wine concentration configuration: Product is circulated by pump through a RO membrane. System pressure is controlled at between 25 and 80 bar (400–1200 psi). A stream of permeate containing predominantly water and alcohol is generated and removed to waste or some other use.

Juice concentration by this means is a valuable approach where grapes have become dilute just prior to harvest as a result of rain or over-irrigation. By removing water, other juice components including sugar, acids and flavours are concentrated. It is preferable to the other option of chaptalisation where only the

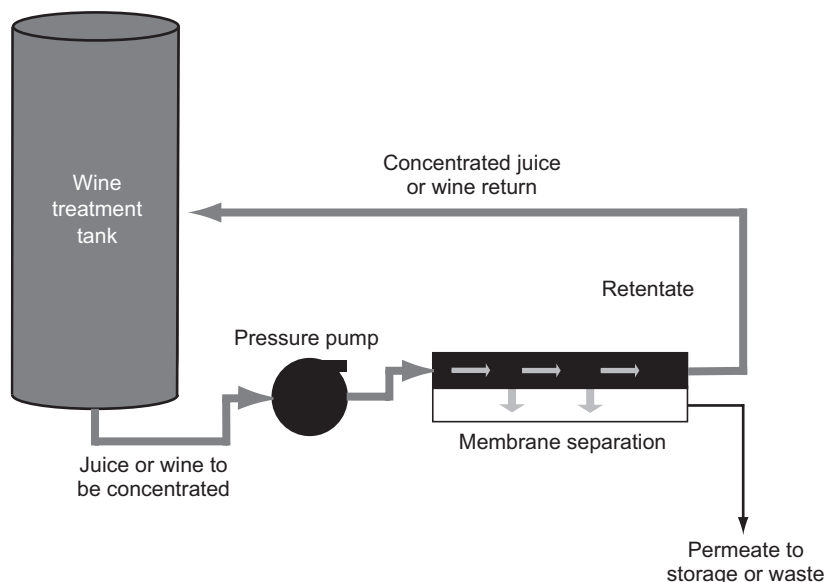


Fig. 5.6 Grape juice or wine concentration – the product being treated is circulated through a membrane and a permeate stream is separated and removed. The remaining wine or juice becomes concentrated in its higher molecular weight components which do not permeate the membrane.

sugar, and hence alcohol level, is adjusted. The technique is less successful with underripe fruit where green flavours, high tannin and high acidity are also concentrated.

In most of the warmer world grapegrowing areas, juice concentration is also problematic in that sugar levels are increased beyond already excessive levels. Concentration of the fermented wine is usually more appropriate. In this case, not just water but some alcohol is also removed, thus not adding to the general problem of excess alcohol.

As with all of these techniques, the choice of membranes, operating conditions and extent of processing can have a major effect on the final wine composition. To understand better the effects of typical concentration with different membrane types, Table 5.3 illustrates the key analytical variables, titratable acidity and alcohol concentration, for control wine, permeate and concentrate. As might be expected, the looser (NF-type) membranes allowed the passage into the permeate of more acid and more alcohol than the tighter (RO-type) membranes. Consequently the concentrate from tighter membranes was relatively enriched in acid and alcohol compared to the loose membrane concentrate, and both concentrates were higher in alcohol and acidity than the control wine. In effect, the major constituent that was preferentially removed from the concentrated wine was water.

Table 5.3 Comparison of the wine concentration effects of two different membrane types (trials performed by the author). In these trials, a volume of wine was concentrated as shown in Fig. 5.6 such that 8% of the volume was separated as permeate. That permeate, and the remaining concentrate (comprising 92% of the original volume), were analysed for two key parameters – titratable acidity and alcohol. This was repeated for NF- and RO-type membranes and the results illustrate the different permeabilities of these with regards to wine acids and alcohol

	Titratable acidity (g/l as tartaric acid)	Alcohol (% v/v)
Control wine	5.9	14.10%
8% Concentration with ‘loose’, NF-type membranes		
Permeate	2.4	13.35%
Concentrate	6.2	14.15%
8% Concentration with ‘tight’, RO-type membranes		
Permeate	0.9	8.00%
Concentrate	6.3	14.70%

5.6 Membrane separation treatment and recombination

The membrane application discussed earlier involves the permanent separation of the juice or wine into concentrate and permeate streams, the idea being to enrich the concentrate stream in one or more desirable components (while hopefully not concentrating less desirable components). It is important to note that, by themselves, RO and NF cannot preferentially remove anything from grape juice or wine except water (Table 5.3). A more elaborate strategy is required if other components are to be selectively removed. This is possible using the concept of membrane separation, treatment and recombination.

According to this general process, wine is passed through a membrane system to create concentrate (retentate) and permeate streams. The respective compositions of these streams will depend on the specific characteristics of the membrane, the initial composition of the wine being treated and processing conditions such as pressure and temperature (Table 5.3).

The permeate is then treated by some secondary, permeate treatment process to remove a particular, undesirable constituent and the treated permeate is recombined with the concentrate. (Fig. 5.7) The aim of this two-step procedure is to retain in the concentrate stream those desirable wine components (typically wine acids, flavours, colour, tannins) that could potentially be damaged or removed in the second, permeate treatment step. The more selective is the first membrane separation step, the less selective need be the second, treatment step – and vice versa.

The result is a wine with substantially the same volume and constituents except that a specific component has been reduced, enhancing the perception of desirable flavours. In general, these membrane applications (RO and NF) seek to ‘surgically’ remove a particular low-molecular-weight constituent of the wine with minimal changes in the rest of the wine’s composition. In the arrangement

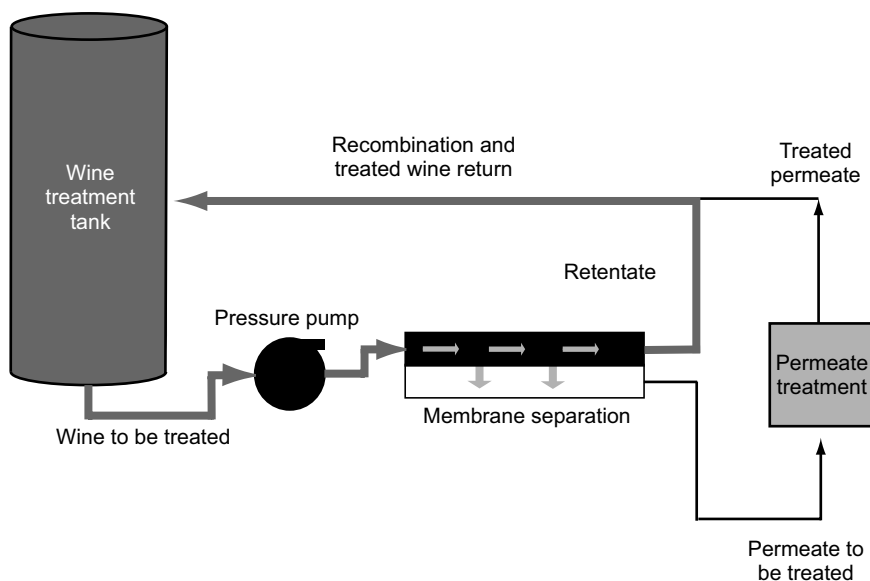


Fig. 5.7 Membrane separation, treatment and recombination – selective removal of undesirable wine components is possible with a two-stage process involving the separation of a permeate stream which is treated and then the stream, minus the offending component, is recombined with the wine concentrate.

illustrated, the treated permeate and concentrate are blended in line and this recombined wine stream is returned to the original treatment vessel (Fig. 5.7). The process then continues for as long as necessary to reduce the concentration of the undesirable constituents in the treatment vessel to the required level. Another, quite valid approach is to transfer the treated wine to a separate vessel.

5.7 Volatile acidity removal

The original RO membrane separation, treatment and recombination strategy was invented and patented by Clark Smith of Californian company, Vinovation, Inc. (Smith, 1996) The first, widespread use of this technology was to remove excess volatile acidity, typically acetic acid and ethyl acetate (Fig. 5.8).

In Smith's process, wine is circulated at high pressure through a RO membrane separation system to create a stream of permeate which contains the acetic acid and ethyl acetate to be removed. It is important to note that for this application, the selection of membrane type can have a significant effect on system performance and the final composition of the treated wine. Ideally, a membrane with a MWCO of about 100 is preferred (Table 5.1). This molecular size separation allows significant passage into the permeate of acetic acid (MW = 60) and ethyl acetate (MW = 88), while other important wine acids such as lactic, malic and tartaric which are larger are substantially rejected by the membrane and so remain in the

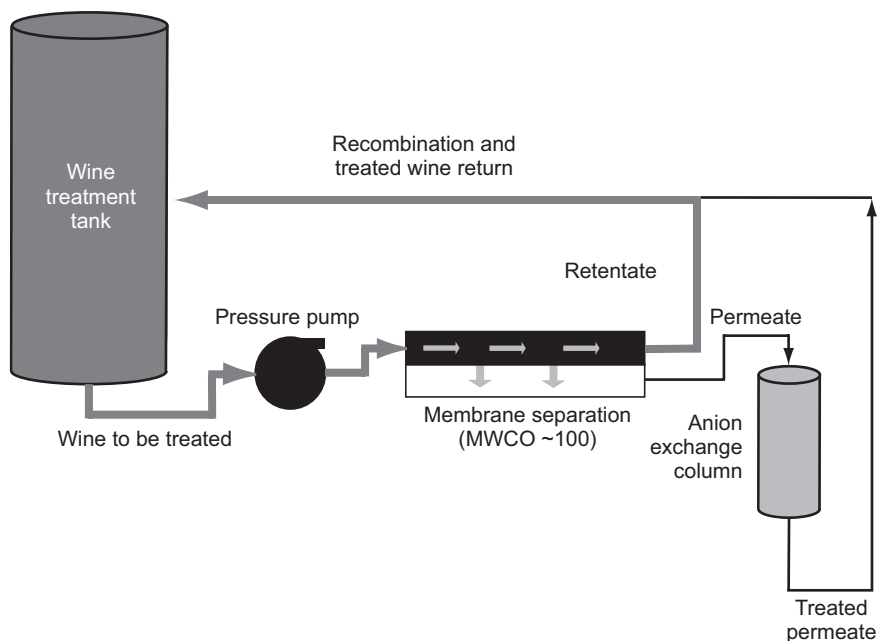


Fig. 5.8 Volatile acidity removal – permeate is separated by a RO stage then passed through an anion exchange column where the offending acetate is removed.

concentrate with most of the other desirable wine constituents such as flavours, tannins and colour.

This membrane separation of wine organic acids is further enhanced by the weak acidity of acetic acid compared to the other acids. Table 5.4 lists some common wine acids and their relative acidity as measured by pKa. (The lower the pKa value, the stronger is the acidity.) This means that at typical wine pH of 3.5, compared to acetic acid, much more of the stronger acids such as tartaric acid will be in the dissociated form and so much less likely to pass into the permeate.

Table 5.4 pKa values for some common wine acids. pKa is the measure of the pH at which the concentration of an acid in its undissociated or molecular form is equal to the concentration of the acid in its dissociated or ionised form. Where the acid is divalent, the numerals 1 and 2 refer to the first and second dissociation states. The lower the pKa, the 'stronger' is the acid

Acid	pKa
Tartaric	(1) = 3.01, (2) = 4.05
Malic	(1) = 3.46, (2) = 5.05
Lactic	3.81
Succinic	(1) = 4.18, (2) = 5.23
Acetic	4.73

Source: Ribéreau-Gayon *et al.*, 2006.

If a NF membrane were used in this application, more of the wine acids other than acetic would pass into the permeate and so would potentially be removed in the second step. This is partly because the typical NF membrane with a nominal MWCO of 150–300 would allow the larger molecular forms of the acids to permeate the membrane, and partly because it has a selective ability to pass monovalent compared to multivalent ions.

This permeate then passes into a charged anion exchange column where the acetic acid (as acetate) is adsorbed on the selected resin prior to the permeate recombining with the concentrate and returning to the rest of the wine. The high pH conditions in the column promote the hydrolysis of ethyl acetate to ethanol and acetate, this acetate also being adsorbed on the resin as above. As the process progresses, the anion exchange resin eventually becomes saturated; the pH reduces and the acetate in the permeate is no longer adsorbed. At that time the process is interrupted; the column is emptied of permeate; the resin is regenerated with a concentrated caustic solution then rinsed and refilled with permeate to continue processing. The process continues for as long as necessary to remove the required amount of volatile acidity.

5.8 The problem of excess alcohol

It is arguable that one of the major defects in many modern wines is excessive ethanol. For many winemakers, high levels of alcohol are deemed to be inevitably associated with fully ripe, higher quality fruit. This is a consequence of a purposeful striving for more mature fruit flavours and tannins and softer acidity rather than a desire for higher alcohol *per se*. Improved viticultural methods and a generally warming climate have aided this to the extent that this pursuit of greater ripeness by winemakers in many parts of the world has resulted in wines with excessive alcohol. Besides increasing the intoxicating and otherwise damaging health effects of the wine, this manifests itself in a reduced perception of wine aroma as well as an unpleasant hotness or bitterness on the palate.

A measure of the extent of this problem shows it is growing. Wine samples analysed by the Australian Wine Research Institute (AWRI) over the 20 years from 1984 to 2004 have shown a steady increase in alcohol level over this period (Godden and Gishen, 2005). For example, the mean for all red wine samples (total sample size, $N = 21\,752$) analysed in 2004 was 13.9% compared with 12.3% in 1984. For white wines the trend was similar but less extreme: 1984 mean was 12.2%, peaking at 13.2% in 2000 ($N = 15\,584$). These elevated alcohol levels can have other damaging effects on wine quality such as prolonged or arrested primary and secondary fermentations, leading to higher levels of residual sugar, with consequent microbiological spoilage, loss of SO_2 and oxidation.

5.8.1 Traditional options for lower wine alcohol

The simplest method for reducing wine alcohol is to pick the grapes earlier. This

has obvious limitations in that flavour and tannin development are retarded. Even if a winemaker chooses this path, this is becoming more difficult because the evident warming climate trend means that the window of opportunity for harvesting grapes is narrowing: grapes that are underripe then pass through optimum ripeness to being overripe within a shorter period than they can be harvested, processed and fermented. A method for removing some of this alcohol would allow winemakers to pick their grapes at optimum ripeness from the point of view of flavour maturity without suffering the negative effects of excessive alcohol.

One approach is to add water to the grape must or wine. While this dilution has been practised for centuries, it diminishes wine quality by reducing the overall concentration of the wine. It is also illegal in many jurisdictions.

5.8.2 Distillation-based techniques

The traditional means of extracting alcohol from wine is by distillation. Typically this involves boiling the wine so that the alcohol is vaporised and collected at the top of a distillation column. The dealcoholised portion of the wine is collected as 'bottoms'. As a means of concentrating alcohol to make a spirit or brandy, this technique is well known and accepted. It is not successful as a means of producing dealcoholised wine because the heat in the distillation process damages many wine components and the more volatile wine flavours are stripped from the wine and pass up the column where they are concentrated and removed in the distillate.

To some extent, the damage to heat fragile components can be mitigated by conducting the distillation at lower temperature and pressure (vacuum distillation). However, the more volatile wine flavours are still stripped from the wine and pass with the distillate so the problem remains to separate these from the alcoholic distillate and return them (hopefully undamaged) to the wine.

5.8.3 Spinning Cone Column

This low-temperature distillation technique is the fundamental approach of the Spinning Cone Column as described previously. In the case of alcohol adjustment, a portion of the wine is processed through the column twice. On the first pass at lower temperature (~25 °C), delicate flavours are removed and stored. On the second pass at higher temperature (~40–50 °C), alcohol is removed and the flavours and dealcoholised wine are then recombined with the bulk of the wine.

This system is complex, capital-intensive and relatively immobile. There is also some possibility of flavour loss or damage, and the alcohol is removed at relatively low strength (approximately 55–60% v/v) so overall volume loss from the wine is significant. On the other hand, it is capable of reducing the alcohol in wine to a very low level in just two passes so it is relatively quick and efficient.

5.8.4 Membrane separation and distillation of the permeate

Another technique was invented and commercialised by Clark Smith and is

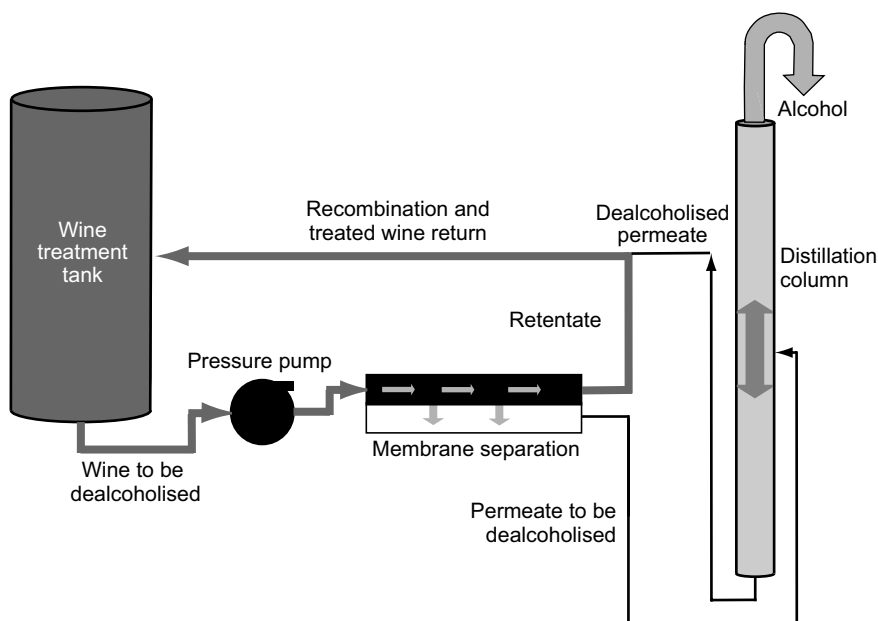


Fig. 5.9 Membrane separation and permeate distillation – in the second stage of this process, a stream of alcoholic permeate is dealcoholised in the distillation column and is then recombined with the wine concentrate. A stream of high-strength alcohol is removed as a by-product.

described by the Smith patent (Smith, 1996). It employs the strategy of membrane separation, treatment and recombination described previously. In this case, the permeate treatment step is a distillation. The reason this is not damaging to the labile wine components is that these remain very substantially in the retentate or concentrate and so are not exposed to heat or any stripping process.

Wine is processed through a RO system to generate a permeate stream which is then distilled in a reasonably conventional distillation column, comprising stripping and rectification stages. Typically, heat is applied to the still by a closed steam coil or other means so that no extraneous water is added to the product. The distillate, which consists very substantially of high-strength alcohol, is removed and sold. The residual material, being dealcoholised permeate, is returned to the wine, thus reducing its alcohol content (Fig. 5.9). This is an effective process for reducing wine alcohol with minimal if any damage to the wine. However, it is relatively immobile and costly in energy terms as well as infrastructure costs.

It is important to note that the use of conventional distillation as described presumes that the permeate being distilled has been generated by relatively 'tight' RO, and not NF, membranes. While these latter membranes are more productive because of their higher flux and higher alcohol passage, the higher MWCO of these means more heat-sensitive material (e.g. residual sugars, see Tables 5.1 and 5.2) will pass into the permeate and potentially be damaged at normal distillation

temperatures. This can be mitigated by using a vacuum distillation process, but this is more complex and costly.

The choice of strategies becomes a trade-off between a costly but highly selective RO membrane process coupled with conventional distillation and a more productive but less selective NF membrane process coupled with a more costly vacuum distillation step. If practised appropriately, both approaches can give quite satisfactory results.

5.8.5 'Diafiltration'

Another option practised in jurisdictions where this is allowed is to combine membrane separation and water addition. This is similar to the concept of diafiltration where a membrane-permeable component (in this case, alcohol) is 'washed' from a solution. This can take a couple of basic forms: a certain amount of alcoholic permeate is removed by RO or NF (as in Fig. 5.6) and it is replaced with the same amount of 'make-up' water. Alternatively that water can be added to the wine to attain a desired alcohol level after which the diluted mixture is concentrated to approximately the original volume by membrane concentration as described previously.

This make-up water could be from grape or non-grape sources according to the local regulations and preferably should be RO-treated to remove extraneous minerals and contaminants. It should also be noted that if NF is used for this, ultimate wine quality is compromised because there is arguably too great a loss of desirable wine components in the permeate being removed. In any case, in most wine producing countries this practice of adding water to wine is illegal or of dubious regulatory status.

5.8.6 Evaporative perstraction

Perstraction: separation process in which membrane permeation and extraction phenomena occur by contacting the downstream with an extracting solvent.

(International Union of Pure and Applied Chemistry, 1996)

An approach described by Michaels *et al.* (1998) uses the process of evaporative perstraction to remove alcohol from alcoholic beverages (Fig. 5.10). This technique involves passing a stream of wine through a membrane contactor. Separated from the wine stream by a hydrophobic membrane, a counter-flow of water is passed through the same contactor and alcohol passes through the membrane from the beverage to the water. This process is based on the principle that ethanol, as a volatile wine component, has a significant vapour pressure. This leads to its evaporation from the liquid phase into the gas phase in the porous matrix of the hydrophobic membrane. By virtue of the concentration difference of ethanol across the membrane, there is a tendency for the ethanol vapour to diffuse across the membrane pores and subsequently condense and dissolve into the water on the other side of the membrane.

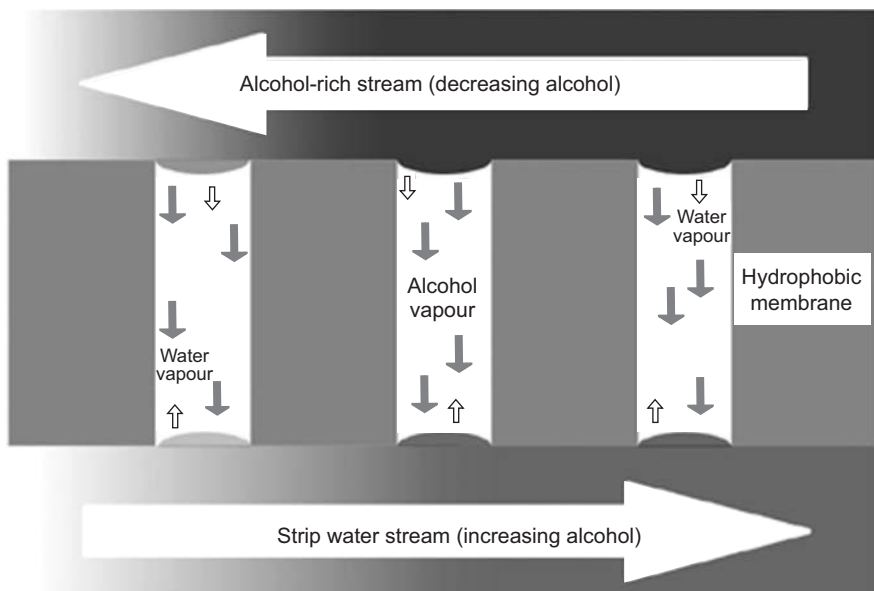


Fig. 5.10 Evaporative perstraction – an alcohol-rich stream (wine or permeate) is separated from a counter-flow of strip water by a hydrophobic membrane. Driven by the higher alcohol concentration, there is a passage of alcohol vapour into the strip water flow.

In practice, this process leads to high levels of extraction of other desirable volatile components from the wine, such as flavours, esters and SO_2 . One approach suggested by the developers of this technique was to dose the strip solution with these compounds so that no concentration gradient for the compound exists. This is complex and expensive and renders the by-product less useful. Other efforts to limit the extraction of desirable volatiles by recycling some of the strip stream reduce the efficiency of the process. Efficiency is also compromised by the presence of relatively large concentrations of CO_2 and other gases in wine. These cannot easily be removed directly from the wine without also removing desirable volatiles.

Another characteristic of this particular process is the undesirable tendency for water vapour to pass across the membrane from the strip water to the wine. This occurs because the vapour pressure of water above the strip water surface is slightly higher than the water vapour pressure on the wine side of the membrane as a result of the higher osmotic potential of the wine. The inventors of this technique were able to measure this phenomenon and suggested that the flow of water from the strip into the wine could be controlled by adding sodium chloride to the strip, and so manipulating the relative osmotic potentials of the wine and strip (Michaels, 1993).

As with the previously described dosing of other wine components to control unwanted trans-membrane migration, this limits the practical application of direct treatment of wine using evaporative perstraction.

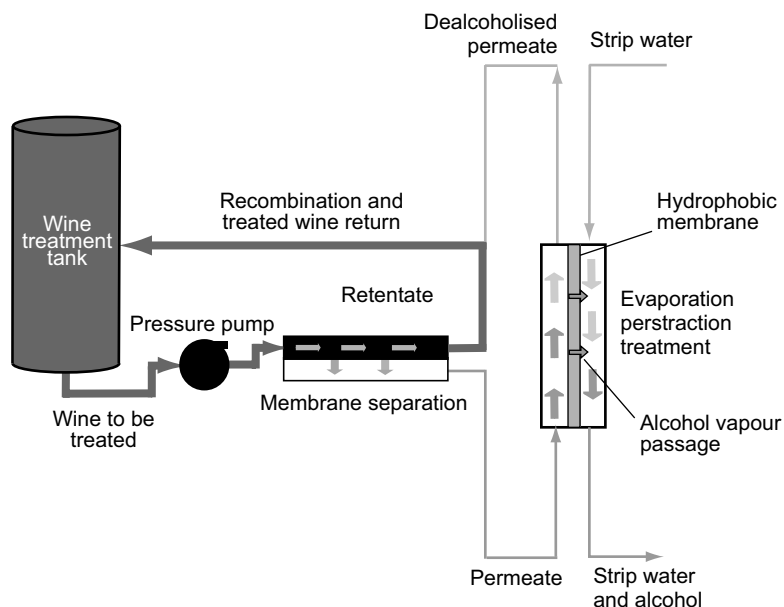


Fig. 5.11 Alcohol reduction by membrane separation and evaporative perstraction. Water is used to strip alcohol from the permeate stream of RO processed wine. This permeate is then recombined with the wine from which it was extracted, thus lowering the alcohol of the blend.



Fig. 5.12 Memstar alcohol adjustment system – this is an example of a large commercial installation that reduces wine alcohol by membrane separation and evaporative perstraction.

5.8.7 Membrane separation and evaporative perstraction (Memstar process)

In recent years, an invention by the author (Wollan, 2005) has applied the process of evaporative perstraction to the permeate stream of RO- or NF-processed wine (Smith, 2007). As with previously described membrane separation, treatment and recombination techniques, RO or NF separation of wine into concentrate and permeate streams is followed by a secondary treatment of the permeate to remove alcohol selectively (Fig. 5.11).

By treating membrane permeate and not wine, the extraction of volatiles is substantially reduced because of their limited passage from the wine into the permeate stream. As well as this, the permeate has a much lower osmotic potential than the original wine, so minimising the water vapour pressure difference between alcoholic permeate on one side of the membrane and strip water on the other. The result is that any tendency for water to move from the strip into the permeate (and hence the wine) is greatly reduced or eliminated.

This process is controlled and optimised by the selection of appropriate membranes and operating parameters such as temperature, pressure and flow rate to maximise the passage of ethanol while limiting the passage of other compounds. This technology is now well established and has been adopted commercially in many countries around the world. The equipment is mobile and efficient and allows winemakers very fine control of alcohol levels without the complications and expense of more elaborate distillation procedures (Fig. 5.12).

5.9 Taint removal

5.9.1 Smoke taint

In early 2003, large areas of south east Australia experienced massive bush- or wildfires. Intense smoke from these fires spread widely and lingered for many weeks. In many of the adjacent viticultural areas, grapes on the vine were heavily exposed to this smoke and it was suspected that they could be tainted. This was a problem of considerable economic consequence to affected vineyards and wineries and a solution needed to be found. A major crisis project was established by the industry and the AWRI to identify, quantify and propose remedies for this taint (AWRI, 2003). While smoke consists of numerous components, the research determined that the indicator compounds, guaiacol (2-methoxyphenol – Fig. 5.13) and 4-methylguaiacol (2-methoxy-4 methylphenol – Fig. 5.14) were strongly correlated with, but not necessarily solely responsible for, the perception of smoke taint in grapes and subsequently in the resultant wine. Given this, removing them without also removing more desirable wine components was going to be difficult, if not impossible, with conventional wine treatments. At about this time, the author speculated that some of the compounds responsible for taint could be selectively removed using the membrane separation treatment and recombination strategy described generically previously.

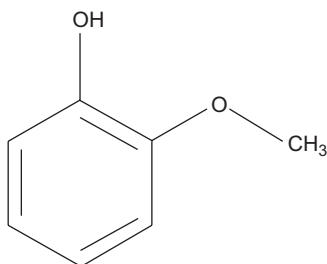


Fig. 5.13 Guaiacol (2-methoxyphenol).

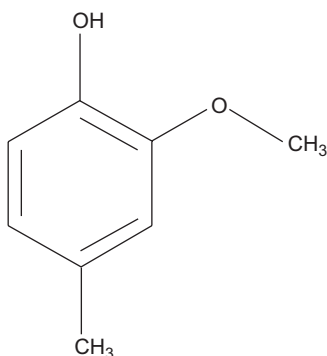


Fig. 5.14 4-Methylguaiacol (2-methoxy-4 methylphenol).

For this to be successful, it would be necessary to generate a permeate stream relatively rich in the compounds to be removed while most desirable wine components remained in the retentate. With molecular weights of 124.1 and 138.2, respectively, it was unlikely that guaiacol and 4-methylguaiacol would pass the standard RO membrane (MWCO ~100) in significant concentrations to be subsequently removed.

A looser NF-type membrane (MWCO ~150–200) was selected for this application. In addition, a means of removing the taint from the permeate had to be found. It was hypothesised that secondary treatment of the permeate with an adsorptive medium such as activated carbon could remove the offensive smoke taint compounds. Early trials were set up and both organoleptic and analytical data demonstrated the success of the concept. It was rapidly adapted for commercial application and an apparatus was set up as shown (Fig. 5.15). This was readily accepted as a service by winemakers, and in the following year (2004) millions of litres of tainted wine were successfully recovered in Australia and in the Okanagan Valley of British Columbia, Canada.

In the years since then, fires in or near viticultural areas have become tragically common occurrences and smoke taint has been experienced in wines from other regions from Australia, Greece, South Africa and California. This has been accompanied by extensive research into the problem, especially in Australia

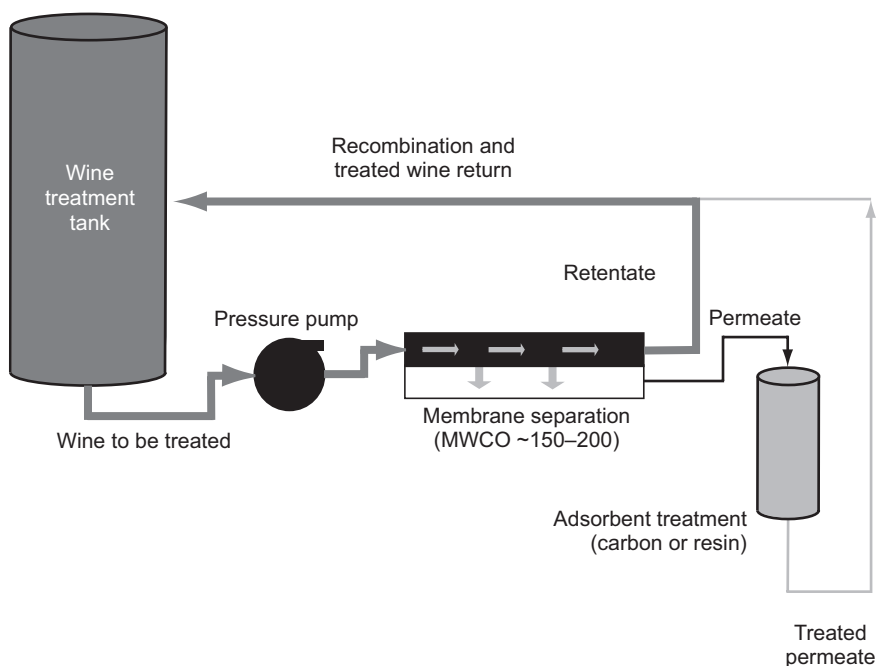


Fig. 5.15 Taint removal by membrane separation (NF) and adsorption – permeate containing taints such as the volatile phenols passes through the selected membranes and is then treated in a column containing an adsorbent medium such as activated carbon or selective resins to remove the taint prior to recombination.

(Kennison, 2009). As well as this industry-wide, institutional research, further work was done by Memstar to refine and improve the process, including changing the preferred, secondary adsorbent medium from activated carbon to a proprietary polymeric resin material with greater selectivity for the offending compounds. Trials of this improved material resulted in demonstrable and lasting taint reduction, as determined organoleptically and analytically (Wilkinson, 2009).

The process is now available commercially in Australia and other countries. Other suppliers have also adapted the technique and offered their own approach. In one case, the wine appears to be treated by direct contact with a proprietary adsorbent material (VA Filtration, 2009). For this strategy to be effective, the adsorbent material would need to be particularly selective for the taint compounds as there would be no preliminary membrane separation step. At the time of writing, no independent reports were available to the author to verify these claims.

5.9.2 *Brettanomyces* taint

It has been known for some years that both the absolute and relative concentrations of 4-ethylphenol (4-EP) (Fig. 5.16, MW = 122.2) and 4-ethylguaiacol (4-EG) (Fig. 5.17, MW = 152.2) are important indicators of the presence of *Brettanomyces* taint in wine. At the time that the smoke taint removal process was being developed

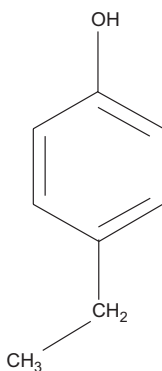


Fig. 5.16 4-Ethylphenol.

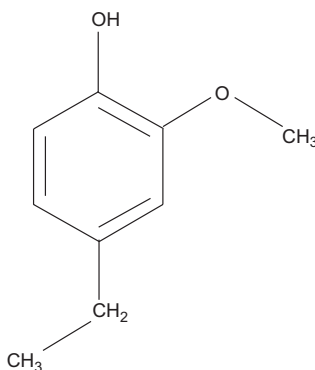


Fig. 5.17 4-Ethylguaiacol (4-ethyl-2-methoxyphenol).

and trialled, it seemed reasonable to expect that, if it could effectively reduce the concentration of guaiacol and 4-methylguaiacol, it should be capable of doing the same with 4-EP and 4-EG which are chemically very similar. The process as described previously was adapted in some minor ways and applied to wines that had distinct *Brettanomyces* taint characteristics, both analytically and organoleptically (Fig. 5.15). Early trials were successful. For example, in one trial, the 4-EP in one treated wine was reduced from 722 $\mu\text{g/L}$ to 353 $\mu\text{g/L}$ (Wollan, 2004). Taint removal by membrane technology has now been extensively tested in practice and is widely accepted by winemakers.

5.10 Ultrafiltration

Ultrafiltration: pressure-driven membrane-based separation process in which particles and dissolved macromolecules smaller than 0.1 μm and larger than about 2 nm are rejected.

(International Union of Pure and Applied Chemistry, 1996)

Ultrafiltration is another form of cross-flow filtration that has at various stages shown potential practical application in the wine industry. According to the IUPAC definition above, the molecular separation range proposed translates into a MWCO somewhere about 20 000 to 200 000 Daltons. It thus bridges the gap between nanofiltration and the microfiltration with which most winemakers are familiar. This places its separation capabilities in the realm of colloids and macromolecules – the area of proteins and the larger polymeric phenolics.

The technique undoubtedly suffered because of this. In the 1980s, it was investigated and suggested as an alternative to bentonite in wine processing – a means of removing heat-unstable proteins. While it may have shown some success in this regard, winemakers who tried this discovered that it removed other valuable components as well, with reports of diminished fruit flavour intensity (Hsu *et al.*, 1987).

Less well known is the technique's utility in fractionating tannins which allows their selective removal or concentration. This was considered in a patent covering the removal of harsh tannins from red wines (Shrikhande, 1989). While still relatively unappreciated, this technology appears to have achieved some successful winemaking application in the areas of: removal of browning and pink hues from sparkling wine pressings; the softening of harsh red wines, and the production of concentrated tannin extracts. (Tangent Membrane Systems, 2009)

5.11 Electrodialysis

Electrodialysis: membrane-based separation process in which ions are driven through an ion-selective membrane under the influence of an electric field.

(International Union of Pure and Applied Chemistry, 1996)

Electrodialysis (ED) is a powerful technique for selectively removing undesirable ionic components from wine. In particular, it has found its greatest winemaking application in the removal of excess potassium and bitartrate ions that are responsible for cold (tartrate) instability. The process was developed by French research organisation INRA and was commercialised in collaboration with Eurodia Industrie SA (Eurodia Industrie SA, 2009).

Traditionally, cold stabilisation has been achieved by chilling the wine so that, in effect, it becomes supersaturated with regards to the unstable species – potassium bitartrate. Crystallisation is encouraged and the wine is held cold until no further crystallisation takes place. The potassium bitartrate is then filtered out and the wine warmed to ambient temperature for bottling. The process is protracted, unreliable and very energy demanding. By comparison, cold stabilisation by ED is rapid, less wasteful and more energy efficient, using about 80% less power. The process takes place at ambient temperature and requires no further filtration to remove the potassium bitartrate crystals. It is generally regarded as having neutral or better quality impact on wines and minimal losses. Besides this, it is able to

reduce calcium levels which can lead to a very troublesome form of instability and which are not readily treated by conventional cold stabilisation.

While ED has many benefits, its wide commercial acceptance has been slow, possibly as a result of winemakers' lack of familiarity with the technology and possibly because of the relatively large capital cost of a typical installation. As they become more familiar with the process and the imperative to reduce power usage becomes stronger, ED will no doubt become an important winemaking tool.

5.11.1 The principle of electrodialysis

The principle of the ED process involves the use of special ion-selective membrane sheets. These are arranged in a stack comprising cells of alternating cation-selective (CSM) and anion-selective (ASM) membranes. These membranes are separated by mesh spacers which create narrow channels through which liquid can flow.

This stack of membranes is clamped between two large electrodes – an anode and a cathode. In the configuration for wine tartrate stabilisation, wine to be treated flows in an alternating set of channels with a conductant or 'brine' comprising a dilute acid solution of controlled pH and conductivity. This brine is circulated through the adjacent set of channels.

A controlled voltage is applied to the electrodes and an electric field is

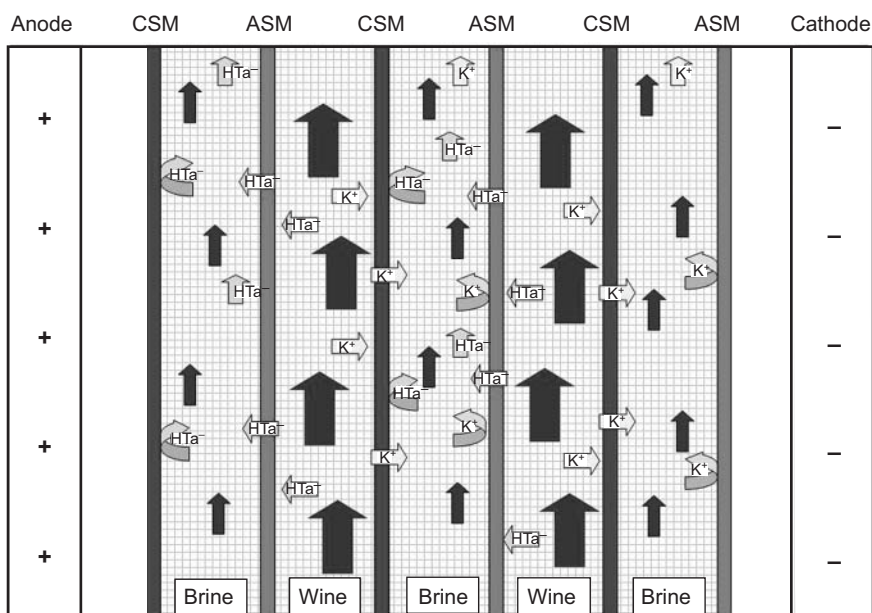


Fig. 5.18 Electrodialysis – wine to be treated passes through a stack of alternating ASM and CSM clamped between an anode and a cathode. On application of an electric field, anions and cations are selectively removed from the wine into a brine waste stream.

generated. Under the influence of this electric field, positively charged cations in the wine such as K^+ , Na^+ and Ca^{2+} are attracted to the negatively charged cathode. Conversely, negatively charged anions such as bitartrate (HTa^-) and chloride are attracted to the anode (While all the major wine cations – sodium, potassium and calcium – are drawn to the cathode, and all of the wine anions – tartrate, bitartrate, malate, acetate, chloride, sulphate – are attracted to the anode, only the potassium and bitartrate species are shown in the diagram.) (Fig. 5.18).

The CSM are formulated to allow the passage of positively charged cations which are driven towards the cathode by the applied electric field. Similarly, negatively charged anions pass through the ASM as they are attracted to the anode. Having said this, the various cations and anions permeate the membranes to different extents.

After the negatively charged anions pass through the ASM and into the brine channel, they continue to migrate towards the anion. However, they cannot pass through the CSM and remain in the brine channel. Similarly, the cations cannot pass through the ASM and also remain in the brine channel. The K^+ cations and HTa^- anions which pass from the wine channel into the brine channel are removed in the brine which leaves the system as a waste stream. The effective result of this process is a reduction in the concentration of K^+ and HTa^- in the wine, demonstrated by a reduction in its conductivity and an improvement in its tartrate stability.

5.11.2 Process description

A typical ED plant schematic is shown in Fig. 5.19 and described more fully below. Prior to treatment commencing, a sample of the wine to be stabilised is tested using a proprietary procedure to determine the degree of tartaric instability. This 4-hour test predicts the extent of deionisation required to achieve 'stability'. The result is expressed in terms of % conductivity reduction from the original untreated wine to the treated.

When the degree of treatment has been determined, wine to be treated is pumped or siphoned from its vessel in the cellar to the ED plant. Here it fills a small untreated wine buffer tank from which it is then pumped at a controlled rate through the membrane stack(s) where it is deionised as described previously. The treated wine then returns to a buffer tank via a conductivity meter.

The control system includes a DC power supply which applies to the stack a voltage which produces a flow of ions or current in the stack. This is set to achieve the required rate of deionisation as measured by the conductivity at the meter. Generally, the system achieves this in one pass of the wine through the stack. However, there are occasions when the voltage to generate the required current is higher than the system can safely deliver. Examples of this are low wine temperature, extremely unstable wine or fouled membranes. In these instances, if the conductivity as measured at the meter is still not low enough to meet the set point, the pump on the treated wine buffer tank does not activate and eventually the level in that tank rises so that partially treated wine flows over the weir separating it from the untreated wine buffer tank. It then passes through the stack and is treated again.

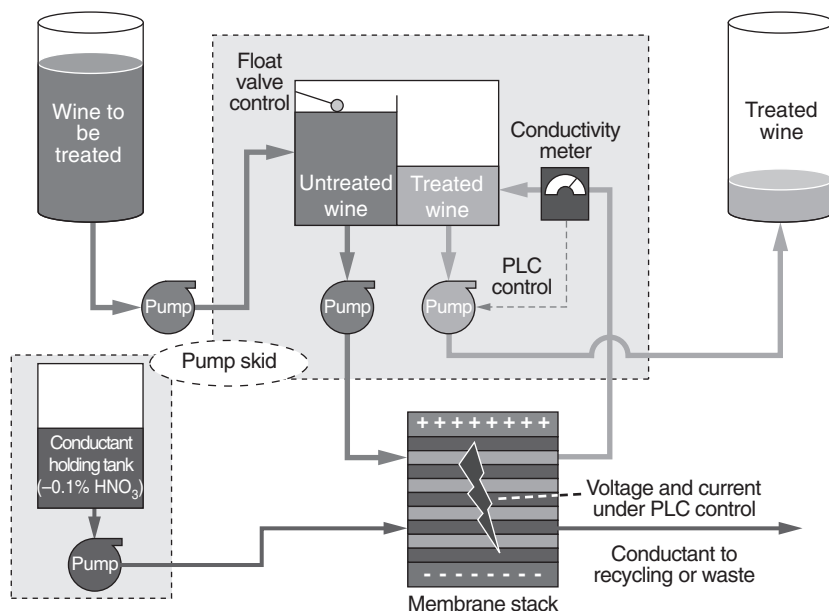


Fig. 5.19 Electrodesis – the typical plant configuration allows the controlled removal from wine of the excess ions responsible for cold instability.

Only when the measured conductivity is below the set point does the treated wine pump activate and transfer the wine to the receiving vessel in the cellar.

While wine flows through one set of channels in the stack(s), a dilute acidic conductant flows through the alternating set of channels. This watery stream receives the ions transferred from the wine through the membranes and becomes progressively more concentrated in these ions (predominantly potassium and bitartrate). Eventually, this flow of waste passes to storage or may be further treated by an optional RO stage to recover some of the water and produce a more concentrated effluent stream from which the tartrate may be recovered.

5.11.3 Salt reduction by electrodesis

Interestingly, ED technology has proven uniquely valuable in reducing excessive ‘saltiness’ in wine. High levels of sodium and chloride are an unfortunate feature of occasional wines produced in Australia and possibly other countries, particularly from vineyards with saline soils, poor quality irrigation water and in prolonged drought seasons.

Commercial-scale trials conducted by the author showed the results of different degrees of deionisation by ED on a wine (2008 South Australian Grenache Shiraz blend) known to have an unacceptably high chloride concentration (Table 5.5). The analyses show the changes in key parameters: conductivity, concentration of major anions, major cations, pH and titratable acidity. For example, sample 4

Table 5.5 Effect of different degrees of ED treatment on a wine with elevated chloride. A wine (2008 South Australian Grenache Shiraz blend) known to be high in chloride was treated using the ED process for deionisation. The analysis for the untreated, control wine is shown in column 1. During processing, the rate of deionisation treatment, expressed as the change (Δ) in conductivity, was varied. At each of these different rates, a sample was taken and analysed. The results are shown in columns 2 to 5.

	Sample				
	1	2	3	4	5
Treatment	Control				
(Δ conductivity)	0%	12%	21%	36%	43%
Conductivity ($\mu\text{S}/\text{cm}$)	3150	2770	2490	2020	1790
Cl as NaCl (mg/L)	1300	1200	910	530	420
Na (mg/L)	151	148	139	118	108
K (mg/L)	1186	1092	998	767	674
Ca (mg/L)	23	18	14	9	5
Cu (mg/L)	0.48	0.29	0.38	0.23	0.33
Fe (mg/L)	1.4	1.1	1.0	0.8	0.7
SO ₄ (mg/L)	600	600	630	520	550
PO ₄ (mg/L)	900	920	890	810	760
pH	3.49	3.48	3.48	3.44	3.40
Titrateable acidity (g/L as tartaric)	6.5	6.5	6.4	6.3	6.2

Source: Wollan, unpublished.

relates to a 36% deionisation as measured by the reduction in conductivity compared to the control (sample 1). Sample 4 shows a 59% reduction in chloride, 22% in sodium, 35% in potassium and 61% in calcium.

5.11.4 Acidity adjustment with bipolar membrane electrodialysis

More recently, the basic ED concept has been adapted with some modifications of the membrane configuration to offer a novel alternative to other methods of pH and acidity correction. This is known as bipolar membrane ED and it is in the process of commercialisation (Lutin and Bar, 2007).

5.12 References

- AWRI (2003), *49th Annual Report. Australian Wine Research Institute*. Australian Wine Research Institute, Adelaide, SA.
- Eurodia Industrie SA (2009), *Eurodia*, available at: <http://www.eurodia.com/html/ele/html> (accessed March 2010).
- FT Technologies (2009a), *How does the SCC work?*, available at: http://www.ft-tech.net/Technologies/Spinning_Cone_Column/how-does-scc-work.html (accessed March 2010).
- FT Technologies (2009b), *SCC 10,000 for coffee extract processing*, available at: http://www.ft-tech.net/Technologies/Spinning_Cone_Column/SCC%2010-000%20Coffee.html (accessed March 2010).
- FT Technologies (2009c), *Spinning Cone Column*, available at: http://www.ft-tech.net/Technologies/Spinning_Cone_Column/spinning-cone-column.html (accessed March 2010).

- 2010).
- GEA Messo PT (2009), *Freeze concentration of citrus and other fruit juices*, available at: [http://www.gea-crystallization.com/geacrystal/cmsresources.nsf/filenames/TSF02.052009H.citrus.pdf/\\$file/TSF02.052009H.citrus.pdf](http://www.gea-crystallization.com/geacrystal/cmsresources.nsf/filenames/TSF02.052009H.citrus.pdf/$file/TSF02.052009H.citrus.pdf) (accessed March 2010).
- Godden P and Gishen M (2005), AWRI Paper – trends in the composition of Australian wine (A Donaldson, ed.). *The Australian and New Zealand Wine Industry Journal*, **20**(5), 21–46.
- Hsu J, Heatherbell D, Flores J and Watson BT (1987), Heat-unstable proteins in grape juice and wine. II. Characterization and removal by ultrafiltration. *American Journal of Enology and Viticulture*, **38** (1), 17–22.
- International Union of Pure and Applied Chemistry (1996), Terminology for membranes and membrane processes (IUPAC Recommendations 1996), *Pure and Applied Chemistry*, **68**(7), 1479–1489.
- Kennison K (2009), *Bushfire Generated Smoke Taint in Grapes and Wine*, Final report to the GWRDC for project number RD05/02-3, Grape and Wine Research and Development Corporation, Adelaide, SA.
- Lutin F and Bar D (2007), Keep it natural ! Adjusting the pH of food products without chemical additives thanks to Bipolar Membrane Electrodialysis, North American Membrane Society, 15th May, Orlando, FL, available at: <http://www.ameridia.com/html/FoodpHAdjust.pdf> (accessed March 2010).
- Michaels AS (1993), *Membrane Extraction Process*. Patent No. WO/1993/022036. WIPO.
- Michaels AS, Canning RP and Hogan P (1998), *Methods for dealcoholization employing perstration (sic)* Patent No. 5817359. United States of America.
- Ribéreau-Gayon P, Glories Y, Maujean A and Dubourdieu D (2006), *Handbook of Enology* (2nd edn), Vol. 2, The Chemistry of Wine Stabilization and Treatments. Chichester, UK, Wiley.
- Shrikhande AJ (1989), *Ultrafiltration of red wines*. Patent No. 4834998. United States of America.
- Smith CR (1996), *Apparatus and method for removing compounds from a solution*. Patent No. 5480665. United States of America.
- Smith M (2007), Memstar responds to rising alcohol levels. *The Australian and New Zealand Wine Industry Journal*, **3**, 105–106.
- Tangent Membrane Systems (2009), *Ultrafiltration*, available at: <http://www.tangentmembranes.com/uftech.html> (accessed March 2010).
- VA Filtration (2009), *Smoke Taint Reduction*, available at: http://www.vafiltration.com/Smoke_taint_1.htm (accessed March 2010).
- Wilkinson KL (2009), *Amelioration of Smoke Derived Taint in Wine by Membrane Filtration and Solid Phase Adsorption*, Final report to the GWRDC for project number R006/04-2, Grape and Wine Research and Development Corporation, Adelaide, SA, available at: http://www.gwrdc.com.au/downloads/ResearchTopics/RD_06-04-2_Final_Report.pdf (accessed March 2010).
- Wollan D (2005), *Alcohol reduction in beverages*. Patent No. 2005252264. Australia.
- Wollan D (2004), Physico-chemical tuning of wine composition. In Blair R, Williams P and Pretorius S (eds), *Proceedings of the Twelfth Australian Wine Industry Technical Conference*, Australian Wine Industry Technical Conference Inc., Adelaide, SA, 135–140.

6

Ageing on lees (*sur lies*) and the use of speciality inactive yeasts during wine fermentation

C. Charpentier, Université de Bourgogne, France

Abstract: The chapter begins by describing the lees and autolysis phenomena: mechanism and released products. It then reviews the influence of ageing '*sur lies*' on the organoleptic and hygienic properties and the stability of white, red and sparkling wines. The chapter includes a description of different yeast preparations mimicking the lees and their use in winemaking.

Key words: ageing, lees, autolysis, wine quality and stability, yeast derivatives.

6.1 Introduction

Ageing of wines on lees is a traditional oenological practice used during the manufacture of wines. After alcoholic fermentation, the wines are placed in barrels or in stainless steel tanks with their yeasts for several months. Lees can be periodically stirred (*bâtonnage*). In Burgundy '*grands crus*' white wines are traditionally aged on lees and, in recent years, this technique has increased in popularity for the wines of other regions and for the ageing of red wines. The primary reasons for ageing on lees are usually based on stylistic goals: to enhance the structure and mouthfeel, give an extra body and increase the aromatic complexity.

6.2 Definition and composition of lees

The definition of wine lees given by EEC Regulation N° 337/79 states that

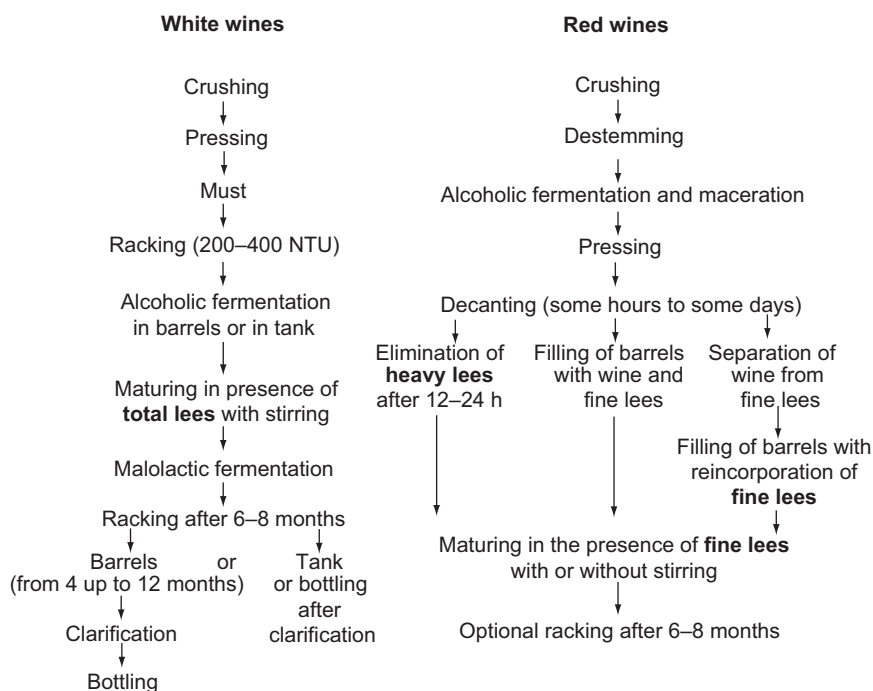
Table 6.1 Influence of decanting duration on the composition of lees

Decanting duration (hours)	Nitrogen (mg/g dry weight)	Polysaccharides (glucose mg/g dry weight)
0	13.35	569.72
24	14.8	570.98
48	16.01	597.95
72	26.02	635.64

Source: Charpentier and Feuillat, 2008.

wine lees is the residue that forms at the bottom of recipients containing wine, after fermentation during storage or after authorized treatments, as well as the residue obtained following the filtration or centrifugation of this product.

Wine lees consist mainly of yeast cells produced during alcoholic fermentation mixed with tartaric crystals, bacteria and some plant cell debris. The composition of lees is variable; it depends on the settling duration (Table 6.1). Heavy lees are the particles that settle within 24 hours when they come from wine without pectin. Their size is from 100 μm to 2 mm. Light lees are particles that remain in suspension 24 hours after the wine has been moved (draining, racking, stirring,

**Fig. 6.1** Scheme of white and red winemaking technology: intervention of lees.

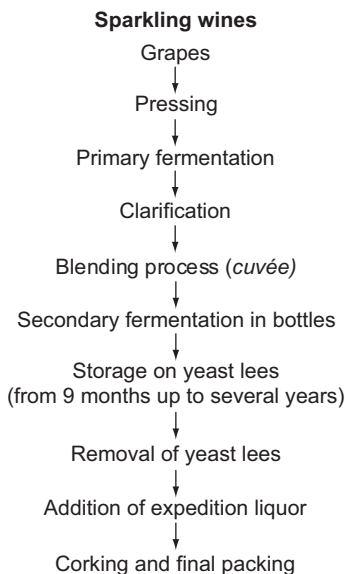


Fig. 6.2 Scheme of sparkling winemaking technology: intervention of lees.

pumping, etc.). The size of light lees is from 1 μm to <50 μm . Adding of 5% of healthy fresh lees, not diluted, containing yeasts from the recent fermentation of dry wine is authorised by EEC regulation (N° 822/87).

A wine is called 'on lees' when there is no clarification after alcoholic fermentation. Traditionally, only selected white wines, mainly from Burgundy, were left in contact with lees, as well as sparkling wines produced by the traditional method. Nowadays, ageing on lees is more widespread in all viticulture areas, in particular for red wines for periods ranging from 2 to 14 months (Trioné and Martinez, 2001).

The schemes of standard white, red and sparkling winemaking technology with intervention of lees are presented in Figs 6.1 and 6.2.

During contact of lees with wine, yeast autolysis occurs and results in the release of many substances that exert a major impact on the stability and on the sensory characteristics of wines. From an organoleptic point of view, several studies indicate that ageing wines on lees led to well-balanced and smooth wines and, more generally, to the modification of the aromatic intensity of wines (Feuillat and Charpentier, 1982; Lubbers *et al.*, 1994a). Moreover, it has been speculated that the presence of lees may confer, to the wine, a high cellaring capacity (Feuillat, 1987a).

6.3 Yeast autolysis

During wine ageing on lees, the yeast lees themselves deteriorate following cell death; this biological degradation is called autolysis. Autolysis represents an

enzymatic self-degradation of cellular constituents that takes place at the end of the stationary phase of growth and is usually associated with cell death (Babayan and Berzakov, 1985).

6.3.1 Structural and ultrastructural changes in the yeast cell

The modification in the organisation and the structure of yeast during storage in wine was studied by Avakyants (1982) and Martinez-Rodriguez *et al.* (2001). The following changes were noted: (i) decrease in cell size; (ii) progressive alteration of cytoplasm integrity and cellular organelles; (iii) progressive separation of the plasma membrane from the cell wall; and (iv) thickening of the cell wall without its disruption. The external surface of the cell became wrinkled. These changes were associated with an increase of the content of cell wall polysaccharides and of the ratio of cell wall mannose:glucose (Charpentier *et al.*, 1986).

6.3.2 Autolysis mechanisms

In winemaking, autolysis occurs during storage of wines on lees with or without periodic stirring (*bâtonnage*). However, wines impose particular conditions on the autolytic process, namely low pH (3.0–3.5), and low temperatures (often less than 15 °C). These conditions are very different from the optimal conditions for autolysis (pH 5.0, temperature 45 °C), but the decrease of the autolytic process is compensated by the length of ageing time of wine on lees: several months or years.

Studies on yeast autolysis under acid conditions can be considered under two categories: (i) proteolysis and (ii) degradation of the cell wall.

Proteolysis

The yeast cell contains a wide array of hydrolytic enzymes and, of the enzymes involved in autolysis, the proteases have been the most studied. Using specific proteases inhibitors, Lurton *et al.* (1989) showed that at wine pH, protease A was the principal enzyme involved in proteolysis during autolysis, releasing 85% of total nitrogen. This enzyme is an endopeptidase that produces high proportion of peptides (Alexandre *et al.*, 2001).

From a biochemical perspective, autolysis consists of five stages.

1. Disorganisation of the membranous system of the cell (e.g. lysosomes, cytoplasmic membrane, other organelles) permitting the liberation of hydrolytic enzymes in the cytoplasm.
2. Inactivation of specific inhibitors of these enzymes by the action of proteases or, alternatively, the proteolytic activation of these enzymes.
3. Enzymatic degradation of intracellular macromolecules, leading to an accumulation of the products of degradation in the space restricted by the cell wall.
4. An increase in the porosity of the cell wall, allowing release of the autolytic products into the extracellular environment.
5. Further autolytic degradation of substances in the extracellular environment by yeast enzymes.

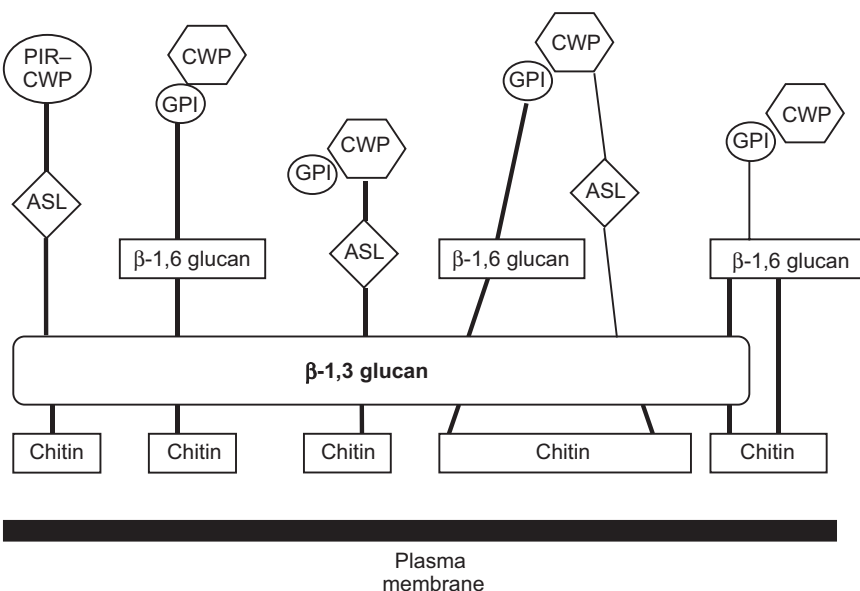


Fig. 6.3 Schematic representation of the different covalently linked proteins in the yeast cell wall.

Degradation of the cell wall

In *Saccharomyces cerevisiae*, the cell wall makes up 15–30% of the dry weight of the cell. The cell wall of *S. cerevisiae* is mainly composed of glucans (with β -1,2 and β -1,6 linkages), chitin and proteins (often highly O- and N-mannosylated) interconnected by covalent and non-covalent bonds (De Groot *et al.*, 2005). β -1,3 glucan, the major component of the cell wall, forms an elastic microfibrillar polysaccharide skeleton surrounding the cell to which other wall constituents are covalently anchored. Chitin is often cross-linked to the β -1,3 glucan microfibrillar backbone on its inner side and, to a lesser extent to short side-chains of β -1,6 glucan. Cell wall proteins (CWPs) (Fig. 6.3) can be:

- (a) Loosely associated either non-covalently or through disulphide bridges with other cell wall components.
- (b) Covalently linked to β -1,3 glucan:
 - directly via an alkali-sensitive linkage (ASL) such as PIR-CWPs (CWPs with internal repeats);
 - indirectly by a β -1,6 glucan moiety through a phosphodiester bridge in their glycosyl phosphatidylinositol remnants (GPI)-CWPs.
- (c) Covalently anchored to chitin by a β -1,6 glucan moiety via a GPI remnant.

Both exo- β -1,3 glucanase and endo β -1,3 glucanase activities have been detected in periplasmic and wall fractions of *S. cerevisiae* (Hien and Fleet 1983; Fleet, 1991). No mannanase activity has been detected in extracts of *S. cerevisiae*. This is consistent with findings that mannan rather than mannose is released as

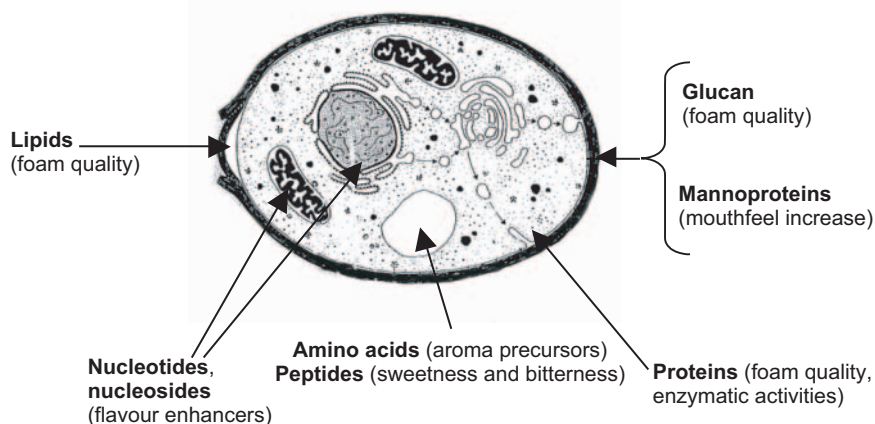


Fig. 6.4 Origin of the different compounds released by yeasts during autolysis and their potential impact on stability and organoleptic properties of wine.

major soluble product during the autolysis of cell walls of *S. cerevisiae*. Charpentier and Freyssinet (1989) and Freyssinet *et al.* (1989) showed that cell wall degradation could be summarised as follows. First, glucans are hydrolysed by glucanases, thus releasing mannoproteins entrapped or covalently linked to the glucans. Second, the glucans are released due to either residual activities of cell wall glucanases or solubilised glucanases in the medium. Finally, the protein fraction of the mannoproteins is degraded by proteases released by yeasts.

The release of polysaccharides varies according to yeast strain (Escot *et al.*, 2001; Caridi, 2007), physiological state of yeasts (Guilloux-Benatier and Chassagne, 2003), temperature (Babayan and Berzuka, 1985) and duration of storage on lees (Charpentier *et al.*, 1986).

Recently, autophagy was shown to play a possible role in release of yeast compounds in the sparkling wines during second fermentation (Cebollero and Gonzalez, 2006). Autophagy is a ubiquitous process in eukaryotic cells that involves bulk degradation of the cytoplasm and organelles in the vacuole or lysosome. In *S. cerevisiae*, autophagy is triggered by starvation conditions and is considered an adaptive response that allows survival of the cell due to recycling of the building blocks resulting from its own digestion.

6.3.3 Compounds released during autolysis

The result of autolysis (Fig. 6.4) is the release of proteins (Feuillat *et al.*, 1989; Martinez-Rodriguez and Polo, 2000), mannoproteins (Charpentier and Freyssinet, 1989; Charpentier, 2000; Fornairon-Bonnefond *et al.*, 2002), peptides (Moreno-Arribas *et al.*, 1996; Carnevillier *et al.*, 2000; Desportes *et al.*, 2001), nucleic acids (Leroy *et al.*, 1990), amino acids (Feuillat and Charpentier, 1982), nucleotides and nucleosides (Courtis *et al.*, 1998; Zhao and Fleet, 2003, 2005; Charpentier *et al.*,

2005), free fatty acids (Piton *et al.*, 1988; Troton *et al.*, 1989; Gallart *et al.*, 2002) and volatile compounds (Chung, 1986).

6.4 Ageing of white wines on lees

White wines ageing on lees is a traditional winemaking technique used in many countries. Typical wines aged on lees are Muscadet wines from the Loire Valley in France, wines from the Golfe du Lion region, wines from the Pays d'Oc and the famous wines from Burgundy, as well as wines from Italy, California, Australia and South Africa. In Burgundy, the white wines undergo alcoholic fermentation and are aged in 228 L oak barrels in the presence of their lees with periodic stirring. They are kept on lees until malolactic fermentation.

White wines can also be aged on lees in vats, but problems related to reduction can occur. The reducing power of lees is not counterbalanced by oxygen diffusion through the recipient and there is formation of reductive aromas. To prevent reduction, the winemakers take wines off the lees and put them in vats where they are stirred for 1 to 2 months and added back to the wine.

Wines aged on lees are generally described as having increased aromatic complexity resulting from release of nitrogenous compounds and extra body, due to polysaccharides from the yeast cell wall.

6.4.1 Nitrogenous substances and polysaccharides

Ferrari and Feuillat (1988) studied the evolution of nitrogen and amino acids in white wine aged in barrels in the presence of lees with or without periodical stirring (bâtonnage). The wine left on lees had a higher concentration of total nitrogen and amino acids, suggesting autolytic release from yeasts. This release was higher in the 'bâtonnés' wines after 6 months of ageing. For 'bâtonnés' wines, the increase was 60% and for 'non-bâtonnés' wines only 20%. Enrichment with polysaccharides (mannoproteins of yeast cell walls) was observed in the 'bâtonnés wines' during the first months of ageing but, after 3 months, there was a decrease in the polysaccharides and high molecular weight fractions disappeared (Feuillat *et al.*, 1989). This hydrolysis of polysaccharides was due to a β -(1,3) glucanase activity released by yeasts in the wine. Some degradation of proteins by acid protease A liberated during yeast autolysis was also evident.

6.4.2 Volatile compounds

Lees may either release to the wines or fix numerous volatile compounds which give rise to complex balances of aromas that impact their quality and contribute to the wine maturation. Lubbers *et al.* (1994b) noticed sensory impact of lees on the persistence of aromatic sensations, but contact with lees could also diminish the concentration of volatile compounds and their volatility. Light lees produce esters that improve the quality of the wines but also higher alcohols and volatile fatty acids detrimental to wine quality (Feuillat *et al.*, 1989).

Chung (1986) studied the release of volatile compounds by autolysis of yeasts in a model wine (ethanol 12% v/v, pH 3) at 13 °C over 4 months. Esters were the major family of volatile compounds released during autolysis both qualitatively and quantitatively. Ethyl esters (with fruity odours) increased all along the autolysis. The increase of long-chain fatty esters (C_{14} , C_{16} , and C_{18}) was faster than that of short-chain fatty esters.

The second family of volatile compounds studied was alcohols: isoamyl alcohol and phenyl-2-ethanol (rose odour) increased during the first 3 months. Terpene alcohols, geraniol, linalool, citronellol, α -terpineol and farnesol, were identified. They all increased during ageing except linalool, which decreased after 3 months. Six aldehydes were also identified, the most abundant being 3-methylbutanal (40% of the total aldehydes). Most of the aldehydes were present at concentrations close to or superior to their perception threshold levels. Aldehydes are generally described as having a grassy odour that negatively affects wine aroma. Most of aldehydes disappeared during ageing, and only benzaldehyde (bitter almond odour) increased. Eight lactones (peach and coconut odours) were identified; α -decalactone was the most abundant. Vitispirane (camphor or eucalyptus odour) was also produced. Sulphur compound production was observed, and six compounds were identified that increased during ageing.

Bautista *et al.* (2007) elaborated white wine from different Galician varieties and studied the evolution of aromatic composition of these wines aged on lees or not on lees. Generally they observed a significant increase of all volatile compounds (monoterpenes, ethyl esters, acetates, higher alcohols, fatty acids) after 2 months of contact with lees and also after 7 months. The sensory analysis demonstrated distinctive flavour profiles for the different treatments. The 7-month lees contact showed a loss of fruity and flora aromas typical of the grapes, and an increase in sulphurous and disagreeable aromas.

6.4.3 Interactions of lees with wine aroma compounds

Many factors have influence on wine aroma, such as environmental and management practices, the grape varieties employed, the winemaking techniques (including lees contact), the yeasts, etc. The contact of wine with the lees confers to the wine modifications in physico-chemical composition and variations in the volatile and organoleptic profile. Cell wall mannoproteins produced by yeasts are the second abundant family of polysaccharides in wine. It is known that aroma compounds are able to interact with polysaccharides and proteins, and the strength of interactions was found to depend on the nature of aroma compounds and on the nature of macromolecules.

The capacity of yeast cell walls to bind wine aroma compounds was investigated by Lubbers *et al.* (1994a,b) in a model wine. Yeast cell walls led to a decrease of volatility of all the aroma compounds. However, the level of binding increased with the hydrophobic nature of the aroma.

Chatonnet *et al.* (1992) found that wine ageing on lees diminished the impact of the wood taste and aroma, preventing the wood taste from having an excessive

predominance over the rest of the wine aromas. Jimenez Moreno and Ancín Azpilicueta (2007) studied the binding of wine volatile compounds from oak wood by the wine lees during simulation of wine ageing. The addition of lees produced a decrease of the oak volatile compounds. The compounds from wood which had the highest affinity for the lees were eugenol, 4-propylguaicol, 4-methylguaicol, furfural and hydroxymethylfurfural. Oak lactones and vanillin were also bound to the lees, although to a lesser degree. The lees did not bind guaicol and γ -nonalactone.

6.4.4 Protein and tartrate stabilisation

The stability of proteins in white wines systematically improves when they are aged on lees in barrels. Wine proteins, which cause hazes in wine, have been identified as the pathogenesis-related thaumatin-like proteins and chitinases, present in the pulp and skin of grapes. The haze-protective factors released by yeasts have been found to be mannoproteins (Ledoux *et al.*, 1992; Waters *et al.*, 1994). The mechanism of haze reduction by mannoproteins is attributed to lowering the size of the haze particles below 5 μm , invisible to naked eye. Ageing on lees also improves tartrate stability (Lubbers *et al.*, 1993; Moine-Ledoux *et al.*, 1997).

6.4.5 Biogenic amines and lees

Biogenic amines cause undesirable physiological effects for consumers (see Chapter 17). The role of lees in the presence of amines in wines is crucial, not only in their role of releasing amino acids, which can thereafter be decarboxylated, but also as a 'microorganism reservoir' (Perez-Seradilla and Luque de Castro, 2008).

Gonzales-Marco and Ancín-Azpículata (2006) studied the evolution of amines in Chardonnay wines, and Martin-Alvarez *et al.* (2006) studied their evolution in red wines; both authors concluded that biogenic amine concentration is higher in wines aged on lees than in those aged without lees. These results are in contrast to those of Marques *et al.* (2008), who showed that in red wines, 2 months after malolactic fermentation, the concentrations of biogenic amines: histamine, isoamylamine, phenylethylamine and putrescine were identical in wines aged with and without lees; only tyramine and cadaverine concentrations were higher for wines stored on lees. Six months after malolactic fermentation, it was possible to observe a slight increase of tyramine and a decrease of cadaverine in wines stored on lees. Therefore, more research in this field is needed to understand the influence of ageing on lees.

6.5 Ageing of red wines on lees

6.5.1 Effects on phenols and anthocyanins

Ageing red wines on lees allows a better wine structure and density – and therefore sensorial characteristics – to be obtained, a consequence of the release of

polysaccharides from the wall of the dead yeast cells. It also allows the monomeric anthocyanin content to be stabilised, favouring the stability of the wine's colour. The ageing of the wine in oak barrels in the presence of lees was a habitual procedure in many viticultural regions years ago. Red wines were usually introduced to the oak barrels immediately after the end of alcoholic fermentation to follow the malolactic fermentation in those containers. Oak ageing is typically carried out today in the presence of lees until the first racking (Rodriguez *et al.*, 2005).

Polyphenols play a major role in the quality of red wines. They contribute to the wine's organoleptic properties such as its colour, bitterness and astringency. This is the reason why the adsorption of anthocyanins by wine lees (Morata *et al.*, 2003, 2005; Caridi, 2007) has been a matter of research, since anthocyanins decrease in wine after ageing with lees.

Rodriguez *et al.* (2005) reported that the presence of the lees during oak ageing of red wine produces wines with substantial differences in colour, proanthocyanidin concentration and astringency. The presence of lees seemed to exert a significant role on wine colour; thus, colour intensity, saturation C^* and hue H^* were lower and lightness L^* higher in wine that aged in contact with lees. These results stated that the presence of lees would produce wines with somewhat less intense colour and saturation but slightly less evolved towards yellowish nuances.

The presence of lees seemed also to decrease the proanthocyanidin concentrations, their mDP (mean degree of polymerisation) and the astringency of the wines, and increase the mouthfeel sensation. This fact may be explained by reaction of proanthocyanidins with compounds released by yeast autolysis such as proteins and mannoproteins

Various studies have tried to determine the precise nature of the potential interactions between yeast lees and polyphenols. Vasserot *et al.* (1997) showed that adsorption of anthocyanins on lees involved the establishment of weak and reversible interactions and appeared to have at least two distinct phases, with the second phase a slow and incomplete uptake of anthocyanins which may correspond to a diffusion-limited phenomenon. Mazauric and Salmon (2005) studied the chemical composition of polyphenolic compounds remaining in solution or adsorbed on lees after various contact times during the simulation of wine ageing. The analysis of remnant polyphenols in the wine indicated that polyphenol adsorption on lees follows bi-phasic kinetics as shown by Vasserot *et al.* (1997). An initial and rapid fixation was followed by a slow, constant and saturating fixation that reached its maximum after about 1 week. Only very few monomeric compounds remained adsorbed on yeast lees, and no preferential adsorption of low or high polymeric size tannins occurred. The remnant condensed tannins in the wine contained fewer epigallocatechin units than the initial tannins, indicating that polar condensed tannins were preferentially adsorbed on yeast lees. From a sensory point of view, it was previously shown that the trihydroxylation of the B-ring led to a decrease of coarseness of wine. Therefore, it could be expected an increase of coarseness of wines aged on lees. Conversely, the efficiency of anthocyanin adsorption on yeast lees was unrelated to its polarity.

Moreno-Arribas *et al.* (2008) studied the changes that anthocyanins undergo during malolactic fermentation and ageing of wine on lees during 14 months in oak barrels. All the different anthocyanin-3-glucosides identified in the wines were influenced by both the time of ageing and the technological procedure investigated. Their concentrations progressively decreased during the first months of ageing; later, an important increase was observed at 14 months. Ageing on lees seemed to influence the quantity of anthocyanin-3-glucosides. Delphinidin-3-glucoside, petunidin-3-glucoside, peonidin-3-glucoside and malvidin-3-glucoside were more abundant in wines aged in presence of lees than in wines submitted to racking, filtration, clarification and cold stabilisation.

Hernandez *et al.* (2006) reported a comparative study of non-anthocyanin phenol compounds conducted on 47 red wines resulting from different technological conditions: malolactic fermentation in barrel or in tank; with or without wine clarification; ageing with or without lees; and stirring or no stirring of lees. The results indicated that after 6 months of ageing in barrels, the treatment used to make the wines did not influence the concentration of any of the phenolic compounds except for myricetin: its concentration was lower in wines aged without lees. There was also an increase of the hydroxycinnamic acids *trans*-caffeic and *trans*-p-coumaric acids. This increase during the ageing of wines on lees could be due to the enzymatic activities of yeast and lactic bacteria from the lees. Later, a progressive decrease was observed for all compounds, with lower values at 12 months and especially after 14 months of ageing in barrels. The comparison of the treatments with respect to composition of lower molecular weight phenolic compounds showed that the wines aged on lees had the same characteristics independent of the *bâtonnage* treatment. The non-anthocyanic compounds did not differ between treatments, since wines with the same phenolic characteristics were obtained after 14 months.

Sartini *et al.* (2007) compared the influence of separate or combined treatments (chips, chips and micro-oxygenation, chips and lees and micro-oxygenation) on the phenolic compounds of a red Sangiovese wine. The phenolic composition of wine aged with lees (jointly with chips and micro-oxygenation treatments) was slightly affected, but its colour and its red polymeric pigments were higher than those of wines submitted to the other treatments, confirming that yeast lees stabilise colour.

6.5.2 Interactions between mannoproteins and tannins

Wine balance can be described as an equilibrium between sweet and acid + phenolic compounds. Lees contact has a major impact on structural balance. Polysaccharides freed by lees have the ability to bind with phenolic compounds, thereby decreasing the perception of both the tannins and, therefore, acidity. Feuillat *et al.* (2001) reported that mannoproteins content of red wines were increased by ageing on lees with stirring. The same study showed an increase of combined anthocyanins by 20% (higher polyvinylpyrrolidone – PVVP – index) and of the amount of colloidal tannins (lower gelatine index and higher

ethanol index). The colour stability was better and astringency reduced. Addition of mannoproteins released by yeast autolysis (Escot *et al.*, 2001) gave the same results.

Several studies have been carried out to determine the interactions between yeast mannoproteins and tannins. The influence of mannoproteins, which differed by their origin or their mode of preparation, on tannin aggregation was tested by Charpentier *et al.* (2004). Among the mannoproteins tested, the most efficient was released by *S. cerevisiae* strain BM45 during alcoholic fermentation, and its mannose/glucose ratio was close to one.

Riou *et al.* (2002) studied the ability of wine polysaccharides to interfere with tannin aggregation. They demonstrated that it is mainly mannoproteins that interfere with tannin particle growth. With these protective polysaccharides, initial aggregation was not inhibited, but the mean apparent particle size remained constant. The mechanisms implicated in this 'protective effect' are still unclear. Two hypotheses can be put forward: (i) molecular association in solution between polysaccharides and polyphenols competing with tannin aggregation; (ii) polysaccharide adsorption on particles formed by tannins, preventing particle growth. More recently, results obtained by Poncet-Legrand *et al.* (2007) confirmed that the mechanism involved is probably steric stabilisation: medium and low molecular weight polymers are more efficient than high molecular weight polymers, and all polysaccharides tested prevented polyphenol precipitation when ionic strength was increased. That allowed dismissing an electrostatic stabilisation mechanism. This coating of tannins by mannoproteins was first proposed by Saucier *et al.* (1995).

6.5.3 Redox phenomena associated with the presence of lees

Yeast lees during wine ageing have a reductive effect by consuming dissolved oxygen and reducing the amount which will remain in the wine. The ability of lees to consume oxygen is not uniquely due to residual viable cells. From a technological point of view, the oxidation potential of red wines aged on lees was lower than that of a control wine, and the components released into the wine during autolysis slowed considerably the oxidation of several purified phenolic compounds (Vivas *et al.*, 2000). The low redox potential may be due to the oxygen consumption by yeast lees. Fornairon *et al.* (1999) showed that yeast lees remaining at the end of alcoholic fermentation exhibited specific oxygen utilisation rates ranging from $1\text{--}4\ \mu\text{mol O}_2\ \text{h}^{-1} \times 10^{-10}$ cells from the second month to the third year of wine ageing. During simulation of wine ageing, it was shown that some membrane lipids of yeast lees, in contact with dissolved oxygen, even at very low concentration, undergo a mild oxidation. These oxidation reactions lead to the production of reactive oxygen species (ROS) and to the production of unknown end-products (Salmon *et al.*, 2000). To find a relationship between oxygen consumption by yeast lees and the final wine composition after ageing, Fornairon-Bonnefond and Salmon (2003) followed different markers of yeast autolysis (proteins, nitrogen and polysaccharides); no differences were found. These same authors showed that

the oxygen consumption rates and the total capacity of oxygen consumption by the corresponding yeast lees clearly decreased when oxygen was added during fermentation. According to Salmon (2006), yeast lees have much higher potential oxygen consumption rates and affinities than wine polyphenols and compete with them for oxygen.

6.6 Ageing of sparkling wines

The 'Méthode Traditionnelle' used to produce Champagne wines involves several distinct steps: a first alcoholic fermentation of musts issued from different grape varieties such as Pinot noir, Pinot Meunier, Chardonnay, blending of the several different still wines (*assemblage*) then a second fermentation in bottle (*prise de mousse*). The second fermentation lasts approximately 6 weeks; then the yeasts die and settle on the inside of the bottle. The bottles are stored for a further period of time, sometimes up to 3 years or even longer. During this storage on lees, a 'toasty flavour', very much sought after, comes from breakdown of the dead yeast cells. Apart from these aroma and flavour changes, the wine can take on a creamy texture, giving a very pleasing sensation of the palate. The longer the sparkling wine ages, the richer this flavour becomes, and the more a sparkling wine is valued. The best and most expensive champagnes are aged for 5 or more years. In Australia, sparkling red wines are produced by a process similar to the 'méthode traditionnelle', but the wines are deep red in colour and possess different characteristics to other sparkling wines. Traditionally these wines have been called 'Sparkling Burgundy'. The varieties of grapes used to make this style of wine include Shiraz, Pinot noir, Cabernet Sauvignon, Merlot and Durif. Most commercial styles remain on yeast lees for 1–3 years. Yeast autolysis in sparkling wines only starts 2–4 months after completion of second fermentation (Feuillat and Charpentier, 1982; Connew, 1998; Alexandre and Guilloux-Benatier, 2006).

6.6.1 Aroma

The characteristic aroma of cava is acquired mainly during the ageing process when yeast autolysis and physico-chemical changes occur. Isoamyl and hexyl acetates, ethyl decanoate, 2-phenylethyl acetate and vitispiranes are markers of young sparkling wines (<9 months) (Riu-Aumatel *et al.*, 2006). There may be two explanations for this formation and decrease of esters. Yeasts contain enzymes simultaneously involved in the formation and degradation of esters. Alternatively, some esters, especially long-chain esters, could be adsorbed on the cell walls (Pozo-Bayon *et al.*, 2003). Vitispiranes and 1,2-dihydro-1,1,6-trimethylnaphthalene (TDN) concentrations were able to discriminate aged cavas (>20 months), while diethyl succinate could be an age marker throughout the ageing period (Francioli *et al.*, 2003; Riu-Aumatel *et al.*, 2006; Bosch-Fusté, 2007). Vitispiranes could originate from the hydrolysis of glycosidically-bound precursors, which are carotenoid-derived megastigmane compounds. TDN may be a direct degradation

product of carotene, although two possible glycosylated precursors have been reported in grape juice (Silva Ferreira and Guedes de Pinho, 2004).

Leroy *et al.* (1990) showed a loss of nucleic acids in yeast cells, and an increase in the concentration of nucleic acid materials in sparkling wine, which occurred mainly in the first 10–15 months of ageing of Champagne wines on lees. Thus, at the end of 15 months of storage, the nucleic acids in the yeast cells decreased by 16% of the original value. This slow release or nucleic acid degradation is due to the unfavourable conditions for the autolysis process of champagne making. Courtis *et al.* (1998) studied the total nucleic acid concentration of a sparkling wine matured on lees for a period of 5 months at 21 °C. A rapid release of nucleic acids was observed with the total nucleic acid concentration of the wine rising from 130 µg/mL to 340 µg/mL within 80 days. After this period, there was relatively little release of nucleic acids into the wine (a further 4% in 50 days). However, the formation of flavour nucleotides during this process was not studied.

Nucleotides, particularly 5'-nucleotides such as 5'-GMP (guanosine 5'-monophosphate) and 5'-IMP (inosine 5'-monophosphate), can be important flavouring agents. These compounds have little or no flavour or aroma themselves but can enhance the flavour and mouthfeel of other compounds. Ribonucleotides were only detected in yeast autolysate in a study on Champagnes (Zhao and Fleet, 2003, 2005). The maximum concentration of total nucleotides was very low, with a maximum of about 3 mg/L in wine aged on yeasts for 9 years. In young wines, the most important nucleotide was 5'-UMP but, after 2 years of ageing, its concentration decreased and the concentration of 5'-GMP slowly increased (Charpentier *et al.*, 2005). The threshold values of the most representative nucleotides in Champagne wines were higher than the concentrations found in the same wines. However, it is known that there is synergism between the different nucleotides and also in the presence of glutamic acid.

6.6.2 Foaming properties

Foam appearance is an organoleptic property of utmost importance for sparkling wine, since the first contact between wine and wine taster is visual. A few seconds after the sparkling wine has been poured, the abundant foam disappears. Then, a collar of fine bubbles of equal size is expected from the effervescence originating from the nucleation sites located on impurities that are stuck on the glass wall. This ring of bubbles is the result of a balance between formation by nucleation, growth, surface arrival, migration to the glass periphery on the one hand, and disappearance by reabsorption, bursting and coalescence on the other. Sparkling wines are not pure liquids. In addition to water, they contain alcohol, dissolved carbon dioxide molecules and many other organic compounds. The molecules of these organic compounds are both water-soluble and water-insoluble parts. Such substances are known as surfactants. Surfactants are attracted to the surface of bubbles and position their water-insoluble parts inside the gas-filled bubble so that they do not touch the liquid surrounding the bubble. Adsorbed surfactants stiffen the bubbles by forming a sort of shield – a rigid wall on their surface – and prevent the

bursting of bubbles, thus improving their stability. These surfactants are macromolecules such as proteins and glycoproteins. Two standard parameters are used to evaluate foam capacity: the maximum height reached by the foam after carbon dioxide injection through a glass frit (HM), which could represent the foamability, and the foam stability height (HS) during carbon dioxide injection, which could represent the wine's ability to produce stable foam or persistence of foam collar.

Maujean *et al.* (1990) observed a reduction in the values of foam height during the ageing of wine in the bottle with yeasts but an increase of foam stability. The same observation was made by Puyeo *et al.* (1995) and Andres-Lacueva *et al.* (1997). For Spanish sparkling wines, 18 months of ageing seemed to be the best for foamability and stability time, apparently due to the release of compounds such as proteins and polysaccharides by yeast autolysis.

The foaming properties of sparkling wines are related to the composition of base wine (Brissonnet and Maujean, 1991, 1993; Andres-Lacueva *et al.*, 1997) but also to the compounds released by yeast autolysis. Pueyo *et al.* (1995) determined the chemical composition and foaming characteristics of base wines and cavas, and applied multivariate analysis techniques to find the relationship between the two sets of data. Foam height was positively related to the total content of palmitic acid. Foam stability was positively related to the concentration of proteins, γ -butyrolactone and xylose in polysaccharides, and negatively related to the total concentration of SO_2 .

Moreno-Arribas *et al.* (2000) evaluated the possible relationship between the composition of wines with respect to those compounds that are the most modified during ageing with yeasts. It was observed that the height reached by the foam showed a positive correlation with most of the free amino acids, polysaccharides and protein nitrogen. Foam stability was highly correlated with total nitrogen, neutral polysaccharides and glutamic acid. No relationship was found between foam characteristics and concentration of wine peptides.

Other compounds released by yeast during bottle ageing are lipids (Piton *et al.*, 1988; Troton *et al.*, 1989; Puyeo *et al.*, 2000). Gallart *et al.* (2002) studied the influence of fatty acids (free and bound as ethyl esters) on wine foaming. The free fatty acids C_8 , C_{10} , and C_{12} were negatively correlated with foamability (HM) whereas the ethyl esters of hexanoic, octanoic and decanoic acids were positively related to HM. The value of HM was directly proportional to the ratio of esterified to unesterified fatty acids. No influence was observed on the stability time of the foam.

6.7 Removal of undesirable compounds from wine

Lees and dry yeasts are capable of adsorbing many compounds. Mannoproteins, located in the outermost layer of the yeast cell wall, are responsible for this property. In the genus *Saccharomyces*, the glycan parts of mannoproteins contain essentially neutral oligosaccharides composed of mannose and N-acetylglucosamine but also acidic oligosaccharides with mannosylphosphate. In the pH range

of wine, mannoproteins carry negative charges and may establish electrostatic and ionic interactions with the other compounds of the wine.

6.7.1 Volatile phenols

Chassagne *et al.* (2005) studied adsorption of 4-ethylguaiacol and 4-ethylphenol by yeast lees in a model wine and a Pinot noir wine. They showed that yeast lees were effective in removal of these volatile phenols. However, the effect of yeast lees on volatile phenols sorption in the model wines was highly influenced by yeast autolysis level and physico-chemical variables such as ethanol concentration, temperature, pH and other wine compounds adsorbed by lees that would increase the number of adsorption sites.

6.7.2 Ochratoxin A

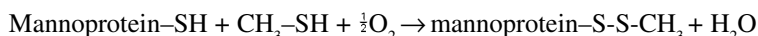
Ochratoxin A (OTA) (see Chapter 14 in Volume 1 of *Managing wine quality*) is a carcinogenic mycotoxin produced by several species of fungi belonging to the *Aspergillus* and the *Penicillium* genera. Since 2006, a new EEC regulation has imposed 0.2 µg/L as the upper limit for ochratoxin concentration in wine (CR 1881/2006). The recent investigations are therefore concentrated on the study of decontamination of OTA in musts and wines. Garcia Moruno *et al.* (2005) obtained very good results in OTA reduction by treatment of red wine with residual yeasts from the alcoholic fermentation. Consistently, reduction of OTA was obtained after only 90 minutes of lees contact, with a very small quantity (only 20 g/L). After 7 days of lees contact, the OTA reduction was higher than 70% with the white lees and ≈ 50% with the red lees. The differences found in the treatments with white and red lees were attributed to competition between phenolic compounds and OTA for the same binding sites on the cell surface.

More recently, Cecchini *et al.* (2006) showed that the percentage of OTA found in extracts of lees by absolute methanol was higher for lees of red wine (between 22.4% and 30.3%) than for lees of white wine (between 17.2% and 21.5%). They explained this by the fact, that besides an adsorption phenomenon between OTA and yeast cells, an independent interaction between OTA and phenolic compounds, or between these and other metabolic compounds produced by yeast, was possible. Caridi *et al.* (2006) confirmed the capability of lees to remove OTA from wines and considered yeast cell walls act as a sponge.

6.7.3 Thiols

Liberation of sulphur products occurs during the traditional on-lees maturation of wines, but lees are also able to remove some of the sulphur products found in wines. Lavigne and Dubourdieu (1996) reported that yeast lees had the capability to adsorb certain volatile thiols. Yeast cell walls prepared by mechanical disruption of the whole cells in the presence of a reducing agent were also able to adsorb the same thiols. On the other hand, after hydrolysis by a β-glucanase preparation able

to remove mannoproteins, cell walls lost their adsorption capacity. These results indicated that thiol consumption by yeast lees is mediated by the establishment of disulphide bridges between thiols and the cysteinyl residues of yeast cell wall mannoproteins according to the following mechanism:



Vasserot *et al.* (2003) did not agree with such a mechanism, and they showed the consumption of thiols did not require molecular oxygen and was inhibited by ethylene diamine tetraacetic acid (EDTA). They suggested that metallic cations in both phenomena were involved.

6.7.4 Phenolic compounds and browning pigments

In order to replace charcoal as an agent for reducing colour in coloured musts and wines, Vasserot *et al.* (1997) proposed using yeast lees. In spite of a greater affinity for anthocyanins, lees were less efficient than charcoal towards anthocyanin adsorption. They were sensitive to ethanol concentration, SO_2 and pH, which all affected adsorption capacity, and to anthocyanin polarity in such way that anthocyanins were far more adsorbed because they were more polar.

Colour darkening during storage of white wines results essentially from chemical reaction involving flavanol-3-derivatives. Bonilla *et al.* (2000) have proposed the use of yeasts in wine fining treatments due to the selectivity of their cell walls to brown polymers. In white wines of the Sherry type, yeasts delayed browning and decreased the concentration of brown pigments (Lopez-Toledano *et al.*, 2006).

6.8 Yeast specialities mimicking lees

In the past few years, given the fact that ageing on lees has been shown to have a favourable effect on stability and quality of wines but is slow and uneconomic, the use of yeast derivatives in winemaking has greatly increased. In the food industry, yeast extracts are widely used as flavouring agents to simulate meat-like, broth-like or cheese-like flavours and to aromatise snacks and soups. Such preparations cannot be used for winemaking, and special technologies have been developed to produce odourless and flavourless derivatives. They are obtained from oenological yeasts (*S. cerevisiae*) by autolytic, plasmolytic or hydrolytic processes, and then concentrated or dried to give the commercial formulates. These products are sold under different names (e.g. yeast cell walls, ageing agent, flavour enhancer, tartrate stabiliser). Their composition is not always well-known and, sometimes, they are a mixture of yeast products and enzymes (Pozo-Bayon *et al.*, 2009). Nevertheless, they can be classified into five groups:

- (a) inactive yeasts obtained by thermal inactivation;
- (b) yeast autolysates obtained by incubation of yeasts in conditions allowing autolysis by yeasts enzymes or in presence of exogenous enzymes – they contain the intracellular contents and the cell walls;

- (c) yeast extracts derived from intracellular contents only;
- (d) yeast ghosts or yeast hulls derived from the cell walls;
- (e) mannoproteins extracted from yeast cell walls and more or less purified.

Ammonium salts, thiamine, microcrystalline cellulose, glucanases can be added in these yeast preparations.

6.8.1 Inactive yeasts

Inactivated yeasts are used to serve as yeast nutrients, to react with tannins and reduce their astringency, to enhance the mouthfeel of wines, stabilise colour of red wine and prevent premature oxidation in white wines. Soubeyrand *et al.* (2005) found that addition of inactive yeasts to rehydration medium of LSA (*levures sèches actives*) was very efficient for the completion of fermentations. Inactive yeasts provided sterols by formation of micelles by membrane phospholipids and specific cell wall polysaccharides and sterols.

6.8.2 Yeast extracts

Yeast extracts or yeast autolysates used in the food industry give off-flavours when added to wine. For this reason, Feuillat (1986) developed a specific procedure for preparing autolysate of wine yeasts for addition to wine. Wines treated with this autolysate contained more nitrogen, proteins and amino acids, and polysaccharides. The maturation of sparkling wines was accelerated and they displayed better aromatic and effervescent quality. Feuillat (1987b) and Comuzzo *et al.* (2006) studied the volatile compounds of three commercial yeast extracts and autolysates. More than 160 volatile compounds were detected. Yeast derivatives added to wine modified its aroma composition, either affecting the volatility of wine aroma compounds or by releasing exogenous volatiles. The lowest additions (200 mg/L) generally determined the presence of flowery and fruity notes that were not detected in the odour profile of the control wine; these aromas were connected with esters (ethyl octanoate, 2-phenylethyl acetate, isoamyl octanoate, ethyl myristate) and alcohols and terpenes (linalool, 1-octanol, 1-decanol, β -citronellol). By increasing the dosage, yeast-like, cheese-like and unpleasant notes appeared.

Industrial products obtained by thermal lysis of yeasts cell walls were added to red wines with the aim to stabilise phenolic fraction (Comuzzo *et al.*, 2005). The effects on wine colour and astringency were strongly influenced by the characteristics of the wine and the dosage. They were limited to short times of bottle storage, and a loss in the positive effect on colour intensity was observed.

6.8.3 Yeast mannoproteins

Yeast mannoproteins extracted by yeast heating or by enzymatic activity, such as β -1,3 glucanase and β -1,6 glucanase and by ultrafiltration have been the objective of many studies (Feuillat *et al.*, 1985; Charpentier, 2000; Caridi, 2006). The use of

mannoproteins as oenological adjuvants was approved by the Organisation Internationale de la Vigne et du Vin in 2001 (resolution OENO 4/2001) but, under EU law, their use is currently permitted only for experiment trials (EU Regulation 1622/2000, art. 41). The following functions of mannoproteins have been described: activators of lactic bacteria growth (Guilloux-Benatier *et al.*, 1991); aroma enhancers (Lubbers *et al.*, 1994b; Chalier *et al.*, 2007); inhibitors of tartrate salt crystallisation (Lubbers *et al.*, 1993; Gerbaud *et al.*, 1996; Moine-Ledoux and Dubourdieu, 1999); preventors of protein haze (Waters *et al.*, 1994). Moine-Ledoux and Dubourdieu (1999) showed that an N-glycosylated 31.8 kDa that corresponds to a partial invertase fragment of *S. cerevisiae* improved protein stability of wines.

6.8.4 Yeast cell walls

Yeast cell walls were used for the first time by Lafon-Lafourcade (1984) to remove medium-chain fatty acid inhibitors of alcoholic and malolactic fermentations. Now, they are sold by different companies to remove cork and mouldy taints, to inactivate *Brettanomyces* and to eliminate volatile phenols. Some preparations contain a specific peptidic fraction released by autolysis (Moine *et al.*, 2007), that increases the ‘*sucrosité*’ of the wines.

6.9 Conclusions

The importance and benefits of lees for the quality and the stability of wines are well-known by winemakers. In Burgundy they say ‘lees are for wine like a mother for a child’. This dictum was verified by numerous scientific studies. Lees stirring schedules and total time spent by wine on lees must be adapted to the wine type and style. This ageing is long and needs a lot of controls. Some risks are associated with ageing: apparition of sulphur odours, development and survival of *Brettanomyces* (Guilloux-Benatier *et al.*, 2001). Until now, various ways of accelerating yeast autolysis effects have been examined. The use of yeast derivatives has been widely extended within the oenological industry, but there is a lack of scientific information about these products and on their effects on wine quality. Another way to shorten ageing could be use of yeast mutants obtained by UV mutagenesis (Nunez *et al.*, 2005) carrying out accelerated autolysis. Preliminary results have been obtained for production of sparkling wines but need to be confirmed.

6.10 References

- Alexandre H and Guilloux-Benatier M (2006), Yeast autolysis in sparkling wines. *Aust. J. Grape Wine Res.*, **12**, 119–127.
- Alexandre H, Heintz D, Chassagne D, Guilloux-Benatier M, Charpentier C and Feuillat M (2001), Protease A activity and nitrogen fractions released during alcoholic fermentation and autolysis in enological conditions. *J. Ind. Microbiol. Biotechnol.*, **26**, 255–240.

- Andres-Lacueva C, Lamuela-Raventos RM, Buxaderas S and del Carmen de la Torre-Boronat M (1997), Influence of variety and aging on foaming properties of Cava (sparkling wine). *J. Agric. Food Chem.*, **45**, 2520–2525.
- Avakyan SP (1982), Etudes au microscope électronique de l'autolyse des levures de vin, *Vinodelie i vinogradstvo*, **2**, 55–59.
- Babayan TL and Berzukov MG (1985), Autolysis in yeast. *Acta Biotechnologia*, **2**, 129–136.
- Bautista R, Fernandez E and Falque E (2007), Effect of the contact with fermentation-lees or commercial-lees on the volatile composition of white wines. *Eur. Food Res. Technol.*, **224**, 405–413.
- Bonilla F, Mayen M, Merida J and Medina M (2000), Yeast used as fining treatment to correct browning in white wines. *J. Agric. Food Chem.*, **49**, 1928–1933.
- Bosch-Fusté J, Riu-Aumatell M, Guadayaol J.M, Caixach J, Lopez-Tamames E and Buxaderas S (2007), Volatile profiles of sparkling wines obtained by three extraction methods and gas chromatography-mass spectrometry (CG-MS) analysis. *Food Chem.*, **105**, 428–435.
- Brissonnet F and Maujean A (1991), Identification of some foam-active compounds in Champagne base wine. *Am. J. Enol. Vitic.*, **42**, 97–102.
- Brissonnet F and Maujean A (1993), Characterization of foaming proteins in a Champagne base wine. *Am. J. Enol. Vitic.*, **44**, 297–301.
- Caridi A (2006), Enological functions of parietal yeast mannoproteins. *Antonie Van Leeuwenhoek*, **89**, 417–422.
- Caridi A (2007), New perspectives in safety and quality enhancement of wine through selection of yeast based on the parietal adsorption activity. *Int. J. Food Microbiol.*, **120**, 167–172.
- Caridi A, Galvano F, Tafuri A and Ritieni A (2006b), Ochratoxin A removal during winemaking. *Enzyme Microb. Technol.*, **40**(1), 122–126.
- Carneville V, Charpentier C and Feuillat M (2000), Production de peptides par *Saccharomyces cerevisiae* au cours de la fermentation et de l'autolyse sur moût synthétique. In Lonvaud-Funel A (ed.) *Oenologie 99, 6ème symposium d'oenologie*, Lavoisier Tec & Doc, Paris, France, 287–289.
- Cebollero E and Gonzalez R (2006), Induction of autophagy by second-fermentation yeasts during elaboration of sparkling wines. *Appl. Environ. Microbiol.*, **72**(6), 4121–4127.
- Cecchini F, Morassut M, Garcia Moruno E and di Stefano R (2006), Influence of yeast strain on ochratoxin A content during fermentation of white and red must. *Food Microbiol.*, **23**, 411–417.
- Chalier P, Angot B, Delteil D, Doco T and Gunata Z (2007), Interactions between aroma compounds and whole mannoprotein isolated from *Saccharomyces cerevisiae* strains. *Food Chem.*, **100**, 22–30.
- Charpentier C (2000), Yeast autolysis and yeast macromolecules? Their contribution to wine flavor and stability. *Am. J. Enol. Vitic.*, **51**, 271–277.
- Charpentier C and Freyssinet M (1989), The mechanism of yeast autolysis in wine. *Yeast*, **5**, S181–S186.
- Charpentier C and Feuillat M (2008), Elevage des vins rouges sur lies. Incidence d'une β -glucanase sur la composition en polysaccharides et leurs interactions avec les polyphénols. *Rev. Œnologues*, **129**, 31–35.
- Charpentier C, Nguyen Van Long T, Bonaly R and Feuillat M (1986), Alteration of cell wall structure in *Saccharomyces cerevisiae* and *Saccharomyces bayanus* during autolysis. *Appl. Microbiol. Biotechnol.*, **24**, 405–413.
- Charpentier C, Escot S, Gonzalez E and Feuillat M (2004), The influence of yeast glycosylated proteins on tannins aggregation in model solution. *J. Int. Sci. Vigne Vin*, **38**, 209–218.
- Charpentier C, Aussenac J, Charpentier M, Prome JC, Duteurtre B and Feuillat M (2005), Release of nucleotides and nucleosides during yeast autolysis: kinetics and potential impact on flavor. *J. Agric Food Chem*, **53**(8), 3000–3007.
- Chatonnet P, Dubourdieu D and Boidron JN (1992), Incidence de conditions et d'élevage des

- vins blancs secs en barrique sur leur composition en substances cédées par le bois de chêne. *Sci. Aliments*, **12**, 665–685.
- Chassagne D, Guilloux-Benatier M, Alexandre H and Voilley A (2005), Sorption of wine volatile phenols by yeast lees. *Food Chem.*, **91**(1), 39–44.
- Chung SH (1986), *Contribution à l'étude de la formation des composés volatils au cours de l'autolyse des levures de vinification*. Doctorat de 3ème cycle, University of Burgundy, Dijon, France.
- Comuzzo P, Tat L, Battistutta F and Tasso A (2005), Effet d'un lysat industriel de levure sur l'évolution des vins rouges en bouteille. *J. Int. Sc. Vigne Vin*, **39**(2), 83–90.
- Comuzzo P, Tat L, Tonizzo A and Battistutta F (2006), Yeast derivatives (extracts and autolysates) in winemaking: release of volatile compounds and effects on wine aroma volatility. *Food Chem.*, **99**(2), 217–230.
- Connew SJ (1998), Yeast autolysis: a review of current research. *Wine Industry J.*, **13**, 161–164.
- Courtis K, Todd B and Zhao J (1998), The potential role of nucleotides in wine flavour. *Aust. Grapegrow. Winemak.*, **409**, 51–53.
- De Groot PWJ, Ram AF and Klis FM (2005), Features and functions of covalently linked proteins in fungal cell walls. *Fungal Genet. Biol.*, **42**, 657–675.
- Desportes C, Charpentier M, Duteurtre B, Maujean A and Duchiron F (2001), Isolation, identification and organoleptic characterization of low molecular-weight peptides from wine. *Am. J. Enol. Vitic.*, **52**, 376–380.
- Escot S, Feuillat M, Dulau L and Charpentier C (2001), Release of polysaccharides by yeast and the influence of polysaccharides on colour stability and wine astringency. *Aust. J. Grape Wine Res.*, **7**, 153–159.
- Ferrari G and Feuillat M (1988), L'élevage sur lies des vins blancs de Bourgogne I. Etude des composés azotés, des acides gras et analyse sensorielle des vins. *Vitis* **27**, 183–197.
- Feuillat M (1986), *Autolysats de levures à usage œnologique et leur procédé de fabrication*. Patent ANVAR n° 86-16318. University of Burgundy, Dijon, France.
- Feuillat M (1987a), L'élevage des vins blancs de Bourgogne. Le bois et la qualité des vins et eaux-de-vie (special issue). *Conn. Vigne Vin*, 123–142.
- Feuillat M (1987b) Préparation d'autolysats de levures et addition dans les vins effervescents élaborés selon la méthode champenoise. *Rev. Fr. Oenol.*, **109**, 17–27.
- Feuillat M and Charpentier C (1982), Autolysis of yeasts in Champagne. *Am. J. Enol. Vitic.*, **38**, 6–13.
- Feuillat M, Charpentier C and Nguyen Van Long T (1985), Les mannoprotéines de levures: un adjuvant œnologique possible. *Bull. O.I.V.*, **71**, 945–967.
- Feuillat M, Freyssinet M and Charpentier C (1989), L'élevage sur lies des vins blancs de Bourgogne II Evolution des macromolécules: polysaccharides et protéines. *Vitis*, **28**, 161–176.
- Feuillat M, Escot S and Charpentier C (2001), Elevage des vins rouges sur lies fines. Intérêt des interactions entre polysaccharides de levures et polyphénols. *Rev. Oenolog.*, **98**, 17–18.
- Fleet GH (1991), Cell walls. In Rose AH and Harrison JS (eds), *The yeasts. Yeast organelles*, Academic Press, London, UK.
- Fornairon-Bonnefond C and Salmon JM (2003), Impact of oxygen consumption by yeast lees on the autolysis phenomenon during simulation of wine aging on lees. *J. Agric. Food Chem.*, **51**, 2584–2590.
- Fornairon C, Mazauric JP, Salmon JP and Moutounet M (1999), Observations on the oxygen consumption during maturation of wines on lees. *J. Int. Sci. Vigne Vin*, **33**, 79–86.
- Fornairon-Bonnefond C, Camarasa C, Moutounet M and Salmon JM (2002), New trends on yeast autolysis and wine ageing on lees: a bibliographic review. *J. Int. Sci. Vigne Vin*, **36**, 49–69.
- Francioli S, Torrens G, Riu-Amatel M, Lopez-Tamames E and Buxaderas S (2003), Volatile compounds by SPME-GC as age markers of sparkling wines. *Am. J. Enol. Vitic.*, **54**(3), 158–162.

- Freyssinet M, Feuillat M and Charpentier C (1989), Rôle de la paroi cellulaire dans l'autolyse des levures. Application oenologiques. In Ribéreau-Gayon P and Lonvaud-Funel A (eds), *Actualités oenologiques 89, Compte-rendus 4ème symposium International d'oenologie*, Bordeaux, Dunod, Paris, France, 160–168.
- Gallart M, Lopez-Tamanez E, Suberbiola G and Buxaderas S (2002), Influence of fatty acids on wine foaming. *J. Agric. Food Chem.*, **50**, 7042–7045.
- Garcia Moruno E, Sanlorenzo C, Boccaccino B and Di Stefano R (2005), Treatment with yeast to reduce the concentration of ochratoxin A in red wine. *Am. J. Enol. Vitic.*, **56**, 73–76.
- Gerbaud V, Gabas N, Laguerie C, Blouin J, Vidal S, Moutounet M and Pellerin P (1996), Effect of wine polysaccharides on the nucleation of potassium hydrogen tartrate in model solutions. *Trans. I. Chem.*, **74**, 782–790.
- Gonzalez-Marco A and Ancín Azpilicueta C (2006), Influence of lees contact on evolution of amines in Chardonnay wine. *J. Food Sci.*, **71**, 544–548.
- Guilloux-Benatier M and Chassagne D (2003), Comparison of components released by fermented and active dry yeasts after aging on lees in a model wine. *J. Agric. Food Chem.*, **51**, 746–751.
- Guilloux-Benatier M and Feuillat M (1991), Utilisation d'adjuvants d'origine levurienne pour améliorer l'ensemencement des vins en bactéries sélectionnées. *Rev. Fr. Oenol.*, **132**, 51–55.
- Guilloux-Benatier M, Chassagne D, Charpentier C and Feuillat M (2001), Influence of yeast autolysis after alcoholic fermentation on the development of *Brettanomyces/Dekkera* in wine. *J. Int. Sc. Vigne Vin*, **35**, 157–164.
- Hernandez T, Estrella I, Carlavilla D, Martin-Alvarez PJ and Moreno-Arribas MV (2006), Phenolic compounds in red wine subjected to industrial malolactic fermentation and ageing on lees. *Anal. Chim. Acta*, **563**(1–2), 116–125.
- Hien HGH and Fleet GH (1983), Separation and characterization of six (1,3beta-glucanase) from *Saccharomyces cerevisiae*. *J. Bacteriol.*, **156**, 1204–1213.
- Jimenez Moreno N and Ancín Azpilicueta C (2007), Binding of oak volatile compounds by wine lees during simulation of wine ageing. *LWT – Food Sci. Technol.*, **40**(4), 619–624.
- Lafon-Lafourcade S, Geneix C and Ribéreau-Gayon P (1984), Ways of initiating action of yeast ghosts during winemaking. *Conn. Vigne Vin*, **18**, 111–125.
- Lavigne V and Dubourdieu D (1996), Demonstration and interprétation of the yeast lees ability to absorb certain volatile thiols present in wine. *J. Inter. Sci. Vigne Vin*, **30**, 2860–2864.
- Ledoux V, Dulau L and Dubourdieu D (1992), Improvement of protein stability during storage of wine on its lees. *J. Int. Sci. Vigne Vin*, **26**, 239–251.
- Leroy MJ, Charpentier M, Duteurtre B, Feuillat M and Charpentier C (1990), Yeast autolysis in Champagne aging. *Am. J. Enol. Vitic.*, **41**, 21–28.
- Lopez-Toledano A, Mayen M, Merida J and Medina M (2006), Yeast used to delay browning in white wines. *Food Chem.*, **97**, 489–504.
- Lubbers S, Léger P, Feuillat M and Charpentier C (1993), Effet colloïdes protecteurs d'extraits de parois de levures sur la stabilité tartrique d'un vin modèle. *J. Int. Sci. Vigne Vin*, **27**, 13–22.
- Lubbers S, Charpentier C, Feuillat M and Voilley A (1994a), Influence of the yeast cell walls on the behaviour of aroma compounds in model wine. *Am. J. Enol. Vitic.*, **45**, 29–33.
- Lubbers S, Voilley A, Feuillat M and Charpentier C (1994b), Influence of mannoproteins from yeast on the aroma intensity of a model wine. *Lebensm. Wiss. Technol.*, **27**, 108–114.
- Lurton L, Seguin YP and Feuillat M (1989), Etude de la protéolyse au cours de l'autolyse de levures en milieu acide. *Sci. Aliments*, **9**, 111–124.
- Marques AP, Leitao M and SanRomao MV (2008), Biogenic amines in wines: influence of oenological factors. *Food Chem.*, **107**, 853–860.
- Martin-Alvarez PJ, Marcobal A, Polo C and Moreno-Arribas M-A (2006), Influence of technological practices on biogenic amine contents in red wines. *Eur. Food Res. Technol.*, **222**, 420–424.

- Martinez-Rodriguez AJ and Polo MC (2000), Characterization of the nitrogen compounds released during yeast autolysis in a model wine system. *J. Agric. Food Chem.*, **48**(4), 1081–1085.
- Martinez-Rodriguez AJ, Polo MC and Carrascosa AV (2001), Structural and ultrastructural changes in yeast cells during autolysis in a model wine system and in sparkling wines. *Int. J. Food Microbiol.*, **71**, 45–51.
- Maujean A., Poinseau P., Dantan H., Brissonnet F and Cossiez E. (1990), Etude de la tenue et de la qualité de mousse des vins effervescents II Mise au point d'une technique de mesure de la moussabilité, de la tenue, et de la stabilité de la mousse des vins effervescents. *Bull. Off. Int. Vigne Vins*, **63**, 405–427.
- Mazauric J.P and Salmon JM (2005), Interactions between yeast lees and wine polyphenols during simulation of wine aging: I. Analysis of remnant polyphenolic compounds in the resulting wine. *J. Agric. Food Chem.*, **53**, 5647–5653.
- Moine-Ledoux V and Dubourdieu D (1999), An invertase fragment responsible for improving the protein stability of dry white wines. *J. Sci. Food Agric.*, **79**, 537–543.
- Moine-Ledoux V, Perrin A, Paladu I and Dubourdieu D (1997), Premiers résultats de stabilisation tartrique des vins par addition de mannoprotéines purifiées (mannostab) *J. Int. Sci. Vigne Vin*, **31**, 163–169.
- Moine V, Marullo P and Arfeuille C (2007), Identification moléculaire d'une fraction peptidique sapide libérée au cours de l'élevage sur lies responsable d'une perception de sucrosité. *Oeno 2007*, 25–27 June, Bordeaux, France.
- Morata A, Gomez-Cordoves MC, Suberviola JBB, Colomo B and Suarez JA (2003), Adsorption of anthocyanins by yeast cell wall during the fermentation of red wines. *J. Agric. Food Chem.*, **51**(14), 4084–4088.
- Morata A, Gomez-Cordoves MC, Colomo B and Suarez JA (2005), Cell wall anthocyanin adsorption by different *Saccharomyces* strains during the fermentation of *Vitis vinifera* L. cv *Graciano* grapes. *Eur. Food Res. Technol.*, **220**(3–4), 341–346.
- Moreno-Arribas V, Pueyo E and Polo MC (1996), Peptides in musts and wines. Changes during the manufacture of Cava (sparkling wines). *J. Agric. Food Chem.*, **44**(12), 3783–3788.
- Moreno-Arribas MV, Pueyo E, Nieto FJ, Martin-Alvarez PJ and Polo MC (2000), Influence of the polysaccharides and nitrogen compounds on foaming properties of sparkling wines. *Food Chem.*, **70**, 309–317.
- Moreno-Arribas MV, Gomez-Cordoves C and Martin-Alvarez PJ (2008), Evolution of red wine anthocyanins during malolactic fermentation, postfermentative treatments and aging with lees. *Food Chem.*, **109**(1), 149–158.
- Nunez YP, Carrascosa AV, Gonzalez R, Polo MC and Martinez-Rodriguez AJ (2005), Effect of accelerated autolysis of yeast on the composition and foaming properties of sparkling wines elaborated by a champenoise method. *J. Agric. Food Chem.*, **53**(18), 7232–7237.
- Perez-Seradilla JA and Luque de Castro (2008), Role of lees in wine production: a review. *Food Chem.*, **111**, 447–456.
- Piton F, Charpentier M and Troton D (1988), Cell wall and lipid changes in *Saccharomyces cerevisiae* during aging of Champagne wine. *Am. J. Enol. Vitic.*, **39**(3), 221–226.
- Poncet-Legrand C, Doco T, Williams P and AV (2007), Inhibition of grape seed tannins aggregation by wine mannoproteins: effect of polysaccharide molecular weight. *Am. J. Enol. Vitic.*, **58**(1), 87–91.
- Pozo-Bayon MA, Pueyo E, Martin-Alvarez PJ, Martinez-Rodriguez AJ and Polo MC (2003), Influence of yeast strain, bentonite action, and aging time on volatile compounds of sparkling wines. *Am. J. Enol. Vitic.*, **54**(4), 273–278.
- Pozo-Bayon MA, Andujar-Ortiz A and Moreno-Arribas VM (2009), Scientific evidences beyond the application of inactive dry yeast preparations in winemaking. *Food Res. Int.*, **42**, 754–761.
- Pueyo E, Martinez-Alvarez PJ and Polo MC (1995), Relationship between foam characteristics and chemical composition in wines and Cava (sparkling wines). *Am. J. Enol. Vitic.*, **46**, 518–524.

- Pueyo E, Martinez-Rodriguez A, Polo MC, Santa-Maria G and Bartolome B (2000), Release of lipids during yeast autolysis in a model wine system. *J. Agric. Food Chem.*, **48**(1), 116–122.
- Riou V, Vernhet A, Doco T and M M (2002), Aggregation of grape seed tannins in model wine-effect of wine polysaccharides. *Food Hydrocolloids*, **16**, 17–23.
- Riu-Aumatell M, Bosch-Fuste J, Lopez-Tamames E and Buxaderas S (2006), Development of volatile compounds of cava (Spanish sparkling wine) during long ageing time in contact with lees. *Food Chem.*, **95**(2), 237–242.
- Rodriguez M, Lezaun J, Canals R, Llaudy MC, Canals JM and Zamora F (2005), Influence of the presence of the lees during oak ageing on colour and phenolic compounds composition of red wine. *Food Sci. Technol. Int.*, **11**(4), 289–295.
- Salmon J-M (2006), Interactions between yeast, oxygen and polyphenols during alcoholic fermentations: Practical implications. *LWT – Food Sci. Technol.*, **39**(9), 959–965.
- Salmon J-M, Fornairon-Bonnefond C, Mazauric J-P and Moutounet M (2000), Oxygen consumption by wine lees: impact on lees integrity during wine ageing. *Food Chem.*, **71**(4), 519–528.
- Sartini E, Arfelli G, Fabiani A and Piva A (2007), Influence of chips, lees and micro-oxygenation during aging on the phenolic composition of a red Sangiovese wine. *Food Chem.*, **104**(4), 1599–1604.
- Saucier C, Roux E and Glories Y (1995), Stabilité colloïdale des polymères catéchiques. Influence des polysaccharides. In Lonvaud-Funel A (ed.), *Oenologi e1995. Vème symposium International d'Oenologie*, Lavoisier Tec and Doc, Paris, France, 395–400.
- Saucier C, Glories Y and Roux D (2000), Interactions tanins-colloïdes: nouvelles avancées concernant la notion de 'bons' et de mauvais tanins. *Rev. Œnologues*, **94**, 9–10.
- Silva Ferreira AC and Guedes de Pinho (2004), Norisoprenoids profile during port wine ageing-influence of some technological parameters. *Anal. Chim. Acta*, **513**, 169–176.
- Soubeyrand V, Luparia V, Williams P, Doco T, Ortis-Julien A and Salmon JM (2005), Formation of micella containing solubilized sterols during rehydration of active dry yeasts improve their fermenting capacity. *J. Agric. Food Chem.*, **53**, 8025–8032.
- Trioné D and Martinez A (2001), Elevage sur lies des vins rouges *Rev. Œnologues*, **101**, 19–21.
- Troton D, Charpentier M, Robillard B, Calvayrac R and Duteurtre B (1989), Evolution of the lipid contents of Champagne wines during the second fermentation of *Saccharomyces cerevisiae*. *Am. J. Enol. Vitic.*, **40**, 175–182.
- Vasserot Y, Caillet S and Maujean A (1997), Study of anthocyanin adsorption by yeast lees. Effect of some physicochemical parameters. *Am. J. Enol. Vitic.*, **48**(4), 433–437.
- Vasserot Y, Steinmetz V and Jeandet P (2003), Study of thiol consumption by yeast lees. *Antonie van Leeuwenhoek*, **83**, 201–207.
- Vivas N and Saint Crieg de Gaulejac N (2000), L'enjeu œnologique de l'élevage sur lies des vins rouges. II. propriétés et modes de valorisation. Connaissances actuelles & Avenir de l'élevage en barriques *J. Sci. Tech. Tonnellerie*, N° spécial (43–46),
- Waters EJ, Pellerin P and Brillouet JM (1994), A *Saccharomyces* mannoprotein that protects wine from protein haze. *Carbohydr. Polym.*, **23**, 185–191.
- Zhao J and Fleet GH (2003), Degradation of DNA during the autolysis of *Sacharomyces cerevisiae*. *J. Ind. Microbiol. Biotechnol.*, **30**, 175–182.
- Zhao J and Fleet GH (2005), Degradation of RNA during the autolysis of *Saccharomyces cerevisiae* produces predominantly ribonucleotides. *J. Ind. Microbiol. Biotechnol.*, **9**, 415–423.

New directions in stabilization, clarification and fining of white wines

R. Marchal, University of Reims, France; and E. J. Waters, The Australian Wine Research Institute, Australia

Abstract: Both organic and inorganic fining agents (bentonite) are commonly used to clarify and stabilize white wines, thus avoiding the appearance of haze in the bottle. Fining involves the formation of a floccular precipitate in wine which will absorb or entrain the natural haze-forming constituents and colloidal particles while settling. The fining process must also be reasonably rapid, and the loss of saleable product in the sediment or lees should be minimal. Finally, the fining process should not have any undesirable effects, like removal of desired flavour or addition of undesired flavour components. It is thus our intention here to review the chemistry of fining and present some results of recent investigations.

Key words: white wine, protein, haze, bentonite, finings, lees, colloidal stability.

7.1 Introduction

After completion of the alcoholic fermentation in tanks or barrels, a white wine is a colloidal solution as well as a colloidal suspension. The density of the particles, which is close to that of the wine, electric repulsion forces and diffusion phenomena lead to very slow settling. Indeed, waiting for wines to settle on their own is not generally a commercially viable option and centrifugation and filtration also does not guarantee protection against colloidal haze.

For these reasons, both organic and inorganic fining agents are commonly used to clarify and stabilize white wines, thus avoiding the appearance of haze in the bottle. The winemaker must produce clear wines, and this lack of turbidity must be

stable over time. Fining involves the formation of a floccular precipitate in wine which will absorb or entrain the natural haze-forming constituents and/or colloidal particles while settling. After a settling period, the supernatant can be withdrawn and receive a polishing filtration prior to bottling and sale.

For the production of white sparkling wine with the 'traditional method', juice and must particles are eliminated using a flotation technique, but the objective is the same. The fining process must also be reasonably rapid, and the loss of saleable product in the sediment or lees should be minimal. Finally, the clarified beverage should remain clear and neither the clarification nor the fining process should have any undesirable effects, like removal of desired flavour or addition of undesired flavour components.

As described above, fining is generally used to clarify a wine. One particular aspect of fining that differs from this is bentonite fining to remove proteins. The proteins that are removed by absorption onto bentonite are soluble in wine; not present as insoluble particles. In this case, the fining is undertaken to prevent particle formation in the future, not to remove particles that have already formed.

At this time, the chemistry of the fining process is poorly understood, and hence it is not surprising that this process is often considered more like an art than a science. For this reason, the fining process can be inefficient. It is thus our intention here to review the chemistry of fining and present some results of recent investigations.

Fining is the addition to wine of a reactive or an adsorptive substance (in some cases, two and even three fining agents are used simultaneously) to remove (or reduce the concentration of) one or more undesirable constituents. Fining agents are primarily added for the purposes of achieving clarity and stability, but also for modifying colour and flavour of juices and wines. Fining agents are grouped according to their general nature:

- (a) earths: montmorillonite, bentonite, kaolinite (commonly named kaolin);
- (b) animal proteins: gelatine (pork, cow, fish), isinglass (fish), caseins;
- (c) plant proteins: wheat gluten, soya, lupine, (garden) pea;
- (d) wood charcoal (carbons);
- (e) synthetic polymers: polyvinylpyrrolidone (PVPP);
- (f) silicon dioxide (kieselsohl).

The most commonly used fining agents in the wine industry are 'bentonites' (mainly containing montmorillonite) and proteins that associate with tannins. Gelatine has been used in oenology since the 19th century but, due to the 'mad cow' crisis in 1999, plant proteins are now more and more used on an industrial scale. For this reason, this chapter will focus on bentonite and plant proteins, with respect to their roles in fining to achieve clarification. Because bentonite also has an important and specific role in preventing protein haze in wines, this instability and its prevention will also be described.

7.2 White wines, proteins and haze

Proteins have the ability to aggregate together in finished white wines and form

unattractive hazes and sediments. Published scientific studies of these proteins and of protein haze in wine began in earnest in the late 1950s and early 1960s with work by J. Koch in Geisenheim (Koch and Sajak, 1959), H.W Berg at the University of California, Davis (Berg and Akiyoshi, 1961; Bayly and Berg, 1967; Moretti and Berg, 1965) and B. Rankine at the Australian Wine Research Institute in Adelaide (Rankine, 1963).

Endogenous wine proteins, of course, also have other roles in wine and have been examined for their impact on the aroma and taste of wines (Peng *et al.*, 1997; Jones *et al.*, 2008), on sparkling wine foam characteristics (Senec *et al.*, 1999; Girbau-Sola *et al.*, 2002; Marchal *et al.*, 2006), on lacquer-like bottle deposit in red wines (Waters *et al.*, 1994; Peng *et al.*, 1996a,b) and as allergens in rare cases of grape and wine allergy (Pastorello *et al.*, 2003; Schad *et al.*, 2005; Sbornik *et al.*, 2007; Vassilopoulou *et al.*, 2007); these aspects are presented in other chapters.

In this chapter, the first part will focus on the role of endogenous wine proteins in white wine haze formation. The oenological practices used to avoid its formation in bottle are presented in the second part.

7.3 The origin of wine proteins

This question has occupied researchers since the 1950s. Bayly and Berg (1967) fermented a model juice solution and noted that protein levels contributed by the yeast were very low and probably negligible in most cases. Twenty years later, Hsu and Heatherbell, (1987a) concluded the same using polyacrylamide gel electrophoresis. More recently, Ferreira *et al.* (2000) and Dambrouck *et al.* (2003) used modern immunological techniques to confirm that wine proteins originate predominantly from the grape. A contrasting possibility was raised by Yokotsuka *et al.* (1991), who analyzed the protein profile of *Vitis vinifera* cv. Koshu grapes as well as the resulting wine made from the same grapes. In conflict with other authors, they found eight wine protein fractions not present in the juice and suggested they had come from yeast. Kwon (2004) demonstrated that there are multiple biological sources of wine proteins using nano-high-performance liquid chromatography/tandem mass spectrometry, although the relative levels of proteins from microbiological sources were not established by this methodology.

Waters and colleagues (Waters *et al.*, 1991, 1992) described two major wine protein fractions in *V. vinifera* cv Muscat Gordo Blanc wine. By sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE), these proteins had molecular masses of 24 and 32 kDa. By analyzing the amino acid sequence of the proteins, Waters *et al.* (1996) showed that the 24 kDa and 32 kDa proteins shared homology with thaumatin and chitinases, respectively, and were highly similar to other plant pathogenesis-related (PR) proteins. Tattersall *et al.* (1997) further characterized the 24 kDa protein, *V. vinifera* thaumatin-like protein 1 (VvTL1). This protein is highly expressed in conjunction with the onset of sugar accumulation and softening in the grape berry. It has also been observed that there is

relatively high expression after *véraison* of chitinase encoding genes in *V. vinifera* cv Shiraz grape berries (Robinson *et al.*, 1997).

7.4 Characterization of wine proteins

7.4.1 Molecular weight

In the 1960s, electrophoresis enabled separation of different sized wine proteins, with researchers at UC Davis, (Moretti and Berg, 1965; Bayly and Berg, 1967) finding four different protein bands, varying in concentration within and among wines from different cultivars of *V. vinifera*. These researchers were the first to raise the possibility that certain protein fractions, rather than total protein, might be responsible for protein instability. Somers and Ziemelis (1973) used gel filtration chromatography to separate wine proteins from other constituents and, by using the exclusion limit of their gel, concluded that the wine protein size was between 10 and 50 kDa. Hsu and Heatherbell (1987a) discovered many different protein fractions with a range of 11.2–65 kDa. In a subsequent study (Hsu and Heatherbell, 1987b), it was suggested that low molecular weight proteins (20–30 kDa) were important to heat instability of wines, rather than those with high molecular weights. This was later confirmed (Waters *et al.*, 1992). In an electrospray mass spectrometry study of the proteins in the juice of 19 cultivars of *Vitis vinifera* (Hayasaka *et al.*, 2001), a range of masses (13–33 kDa) was observed. The proteins were identified as mainly thaumatin-like proteins (range of 21 239–21 272 Da) and chitinases (range of 25 330–25 631 Da). Small variations in the masses of these protein due to small variation in their polypeptide sequence form the basis of a robust method of varietal identification based on mass spectrometry (Hayasaka *et al.*, 2001).

7.4.2 Isoelectric point of wine proteins

At a certain pH, the positive and negative charges of a protein are equal. This pH value is known as the isoelectric point, or isoionic point, of the protein. The lower the difference between a model solution pH and the isoelectric point of a protein, the lower the net charge on that protein fraction and the lower the solubility of that fraction. If the juice or wine pH values are quite different from that of the protein isoelectric point, then the protein charge is high and the higher will be the ability of that protein to electrostatically bind to fining agents. Therefore, the isoelectric properties of proteins influence not only their natural solubility in wine but also their affinity to be removed with various agents. The relationships between wine pH and isoelectric points can be illustrated in the following example. We consider a base wine with pH = 3.2. Then the protein fractions having a pI >3.2 will be all positively charged and those with a pI <3.2 will be negatively charged. Those protein fractions with isoelectric points closest to pH 3.2 have a limited charge.

Protein clouding in white wines seems to be a greater problem when the wine pH is close to the isoelectric point of the various protein fractions. This is due to the

fact that bentonite will remove, preferentially, the most positively charged proteins. The electrostatic charge of various protein fractions explains the observable phenomena of not being able to stabilize certain wines with the use of bentonite alone, or only with excessive amounts that can strip the wine character. But the pI of proteins only partially explains wine haze formation. It is also important to note that other factors, as yet not clearly identified, can intervene.

Research into the isoelectric point of wine proteins has often been concurrent with studies of wine protein size. Proteins with low pI were found to be significant contributors to total wine protein (Moretti and Berg, 1965) and to wine haze (Bayly and Berg, 1967). Hsu and Heatherbell, (1987a) confirmed this observation and suggested that the majority of wine proteins had a pI of 4.1–5.8, whilst Lee (1986) suggested the major protein fractions of wine had a pI of 4.8–5.7. Dawes *et al.* (1994) fractionated wine proteins on the basis of their pI and found that the five different fractions all produced haze after heat treatment. Haze particle formation was found to differ between the fractions, however, leading to a statement that other wine components, such as phenolic compounds, need to be considered to fully understand protein haze.

7.4.3 Glycosylation

A better comprehension of the wine behaviour (haze formation, but also foam properties and tartaric stability) depends on knowledge of the biochemical characteristics of protein. As a result, studying wine glycoproteins is very interesting. The presence of arabinogalactan protein (AGP) in must (Saulnier and Brillouet, 1989) and of yeast mannoproteins in wine (Waters *et al.*, 1994; Dupin *et al.*, 2000) has been clearly demonstrated for a long time. In both cases, these studies concern real glycoproteins, named proteoglycans by convention. The protein moiety represents less than 10% of the molecule, and the molecular weights are, respectively, in these examples 165 000 and 420 000. With regard to protein of less than 70 000, protein-polysaccharides complexes have also been suggested for a long time (Hsu and Heatherbell, 1987a; Paetzold *et al.*, 1990; Yokotstuka *et al.*, 1991; Waters *et al.*, 1993). However, for all of these proteins having molecular weights of less than 70 000 Da, the covalent link between the glycan and the polypeptidic moiety has never been clearly evidenced.

The existence of real glycoproteins in a Chardonnay wine was demonstrated using enzymatic deglycosylation (Marchal *et al.*, 1996). Seven proteins were isolated in a first step by concanavalin A affinity chromatography. The proteins of 24/25, 30 and 60/64 kDa (grape invertase) were then purified by preparative isoelectric focusing (pH gradient 2.5–5) and by preparative SDS–PAGE. The 30 kDa protein presents a low hydrophobicity (780 cal/amino acid residue), a homogeneous molecular weight, and an isoelectric point close to 2.5. It also has the characteristic of being retained by *Lens culinaris* agglutinin (LCA). Proteins of 24/25 and 60/64 kDa present heterogeneous MW, a pI close to 3.9 and a hydrophobicity 30% superior to that of the 30 kDa molecule. Moreover, these two proteins were not retained by LCA. The three analyzed proteins are not susceptible to *O*-

glycosidase activities. In return, the 24/25 kDa protein undergoes a 3100 Da variation after treatment with the peptide-N-glycanase F: it is then a true *N*-glycosyl protein. The comparison of the must and the corresponding wine proteic fractions isolated by concanavalin A shows that the two heterogeneous MW molecules (24/25 and 60/64 kDa), corresponding to a microheterogeneity of the polyaminoacid chain glycosylation, originate from the grape berry (Dambrouck *et al.*, 2003). In addition, they suffer no modification during the alcoholic fermentation.

7.4.4 Stability

The unusual aspect of wine protein instability is that the proteins responsible for protein haze in the long term are, paradoxically, very stable themselves in the short term and survive the vinification process. It is, therefore, not surprising that wine proteins are resistant to low pH and enzymatic or non-enzymatic proteolysis (Waters *et al.*, 1992). The mechanism for the resistance has not yet been fully elucidated, but resistance to low pH and proteolysis are characteristics of plant pathogenesis-related proteins in general. Experimental evidence suggests that it is also characteristic of the grape pathogenesis-related proteins involved, rather than phenolic association or glycosylation (Waters *et al.*, 1995). Limited proteolytic processing of the wine proteins can, however, occur during white table wine vinification (Waters *et al.*, 1998) and during the Champagne winemaking process (Manteau *et al.*, 2003). These data indicate there could be scope to exploit this susceptibility to proteolysis under certain conditions and thus prevent wine protein haze.

7.5 Protein levels in white wines

7.5.1 The problem of wine protein quantification

Protein levels in white wine have been reported by several authors and have been shown to differ by variety. Lee (1986) reported a range of protein concentration from 18–81 mg/L in 14 wines from different Australian regions and made from different varieties. Some of these wines appeared to have been fined with bentonite prior to analysis. Pocock *et al.* (1998) reported concentrations in unfined Australian wines up to several hundred mg/L. Hsu and Heatherbell (1987b) found a range of 19–44 mg/L in four different unfined white wines from Oregon, while a very large variation (20–260 mg/L) was noted by Bayly and Berg (1967).

The Bradford method is the most commonly used method in oenology for direct protein quantification because of its rapidity and its reproducibility. However, interferences falsify the estimation of protein in white base wines from Pinot noir and Chardonnay grapes. Ultrafiltration with a 10 kDa molecular weight cut-off (MWCO) membrane showed that non-protein compounds largely react with the Coomassie Brilliant Blue (CBB) and can produce an interference equal to 30% to

90% of the initial value (Marchal *et al.*, 1997). Ethanol and exogenous and endogenous phenolic compounds are principally responsible for these interferences. They act indirectly by favouring the neutral form of the CBB dye. However, interferences from ethanol and phenolics are not additive. Furthermore, the interaction is negative (Marchal *et al.*, 1997). Both bentonite and vegetable charcoal treatments partially eliminate proteins as well as phenolics. The phenolics/proteins ratios (in equivalents of bovine serum albumin – BSA – estimated by the direct Bradford method) increases on increasing bentonite fining and decreases on charcoal treatment. For these reasons, the deproteinization effects of oenological treatments are very difficult to estimate by direct measurement using the Bradford method.

A modified Bradford method (Marchal *et al.*, 1997) can be used as follows to correct for interferences: (i) estimation of wine reactivity with CBB (this is the direct Bradford method) = Wine Direct Estimation, noted as W_{DE} ; (ii) ultrafiltration of the wine with a 3 or 10 kDa MWCO membrane, and estimation of the ultrafiltrate reactivity with the CBB = Ultrafiltrate Direct Estimation, noted as UF_{DE} ; (iii) determination of the corrected Wine Protein Content, $W_{PC} = W_{DE} - UF_{DE}$. Nevertheless, white wines contain many glycoproteins, sometimes highly glycosylated (more than 90% for some yeast mannoproteins), and their reaction with the dye is likely compromised. This means that estimation of protein content (absolute concentration) in wine will remain a problem. Comparison between wines to estimate protein content differences (based on relative value), for example, is therefore preferred. Other assays of wine protein concentration such as the method described by Vincenzi and colleagues (2005) also appear to be suitable.

7.5.2 Protein concentration and berry development

In grapevines, the synthesis of the PR proteins is regulated in a developmental and tissue-specific manner. In *Vitis vinifera* cv Muscat Gordo Blanco berries, both the expression of the VvTL1 gene and the levels of the corresponding major thaumatin-like protein increased dramatically after the onset of berry softening (*véraison*) and continued throughout berry ripening (Tattersall *et al.*, 1997). Similar developmental patterns were also found in the expression of genes encoding chitinases, some identical to those involved in wine protein haze (Derckel *et al.*, 1996, 1998; Robinson *et al.*, 1997). An immunological study of *V. labruscana* cv Concord also showed that thaumatin-like proteins and chitinases accumulate during berry ripening (Salzman *et al.*, 1998).

In all cultivars of *V. vinifera* studied so far, thaumatin-like proteins and chitinases are the major soluble components of grapes (Peng *et al.*, 1997; Tattersall *et al.*, 1997; Pocock *et al.*, 1998, 2000). The predominance of these PR proteins was clear at all stages of berry development following *véraison* (Pocock *et al.*, 2000). Importantly, as the concentration of these proteins in the berries continually increases during ripening, it can be presumed that the haze-forming potential increases as ripening proceeds (Murphey *et al.*, 1989; Tattersall *et al.*, 1997; Pocock *et al.*, 2000).

7.5.3 Effect of grape fungal infection

While PR protein synthesis in healthy grapes appears to be triggered by *véraison*, this does not mean that the classical PR protein inducers – stress, wounding and pathogenic attack – cannot further modulate the levels of PR proteins in grapes. Grape PR proteins exhibit antifungal activity *in vitro* against common fungal pathogens of grapevines, including *Uncinula necator*, *Botrytis cinerea*, *Phomopsis viticola*, *Elsinoe ampelina* and *Trichoderma harzianum* (Giannakis *et al.*, 1998; Salzman *et al.*, 1998; Tattersall *et al.*, 2001; Jayasankar *et al.*, 2003; Monteiro *et al.*, 2003a). It would be tempting to speculate that the antifungal activity observed *in vitro* reflects the main function of the PR proteins *in vivo*, and that their expression in fruit after *véraison* represents a pre-emptive defense mechanism for fruit. Jayasankar *et al.* (2003) added further credence to this hypothesis by demonstrating that grapevines regenerated after *in vitro* selection with *E. ampelina* culture filtrates had greater disease resistance and high constitutive expression of PR proteins, including VvTL1. Studies in which the synthesis of PR proteins is modified by gene technology would allow us to investigate this hypothesis further.

Increased expression of some PR genes and enhanced concentrations of some PR proteins have been observed in leaves and berries from grapevines infected with pathogens (Renault *et al.*, 1996; Derckel *et al.*, 1998; Jacobs *et al.*, 1999; Bézier *et al.*, 2002; Robert *et al.*, 2002). In glasshouse experiments, Monteiro *et al.* (2003b) observed increased levels of thaumatin-like proteins in berries infected with *U. necator* compared to uninfected berries. Jacobs *et al.* (1999) demonstrated that chitinase and beta-1,3-glucanase activity increased in grape berries and leaves in response to powdery mildew infection, and that expression of genes, VvChi3, VvGlub and VvTL2, coding for PR proteins, was also strongly induced. Of these three putative gene products, only VvTL2 has been detected as a soluble protein in grape juices and wines (Waters *et al.*, 1996). Girbau *et al.* (2004) demonstrated that powdery mildew infection of *V. vinifera* cv. Chardonnay grape bunches resulted in increased levels of a grape minor thaumatin-like protein, VvTL2, in wine. At high levels of infection (> 30% of berries in a bunch infected), this had a significant impact on the level of haziness in the wine following a heat test.

Contrary to expectations that fungal diseases would lead to elevated levels of PR proteins in berries, Marchal *et al.* (1998) observed that juice from berries infected by *B. cinerea* showed reduced protein levels, and suggested that proteolytic enzymes from *B. cinerea* were responsible for this. Secretion of proteases by *B. cinerea* has been observed in culture media and on fruits such as apple (Zalewska-Sobczak *et al.*, 1981) and tomato (Brown and Adikaram, 1983). Girbau *et al.* (2004) also examined the impact of infection of grapes with *B. cinerea* in the vineyard and showed that infection resulted in marked decreases in the levels of PR proteins in the berries. Similar although less dramatic trends of reductions in protein levels were seen in laboratory experiments in which otherwise healthy berries were inoculated with *B. cinerea* (Girbau *et al.*, 2004). Protein levels were also reduced in juice when *B. cinerea* was grown in this medium (Girbau *et al.*, 2004).

The reduction in protein levels in the juice from *Botrytis*-infected grapes could,

as suggested by Marchal *et al.* (1998, 2006), be due to proteolytic degradation of grape PR proteins by enzymes of *B. cinerea*, although there are other plausible explanations for these observations. Other studies demonstrated changes in the electrophoretic characteristics of wine proteins originating from botrytized grapes (Cilindre *et al.*, 2007, 2008), although the cause of such alterations was not explained. In a recent study (Marchal *et al.*, 2006), total proteins released by *B. cinerea* in a synthetic liquid medium showed proteolytic activity against must proteins in a model wine. These results demonstrated that *B. cinerea* proteases were able to degrade must proteins and were, above all, active at must and wine pH and in the presence of ethanol and SO₂ (Marchal *et al.*, 2006). Such enzymes have the potential to replace bentonite fining for protein stabilization in oenology, a goal of much research effort worldwide.

7.5.4 Role of water stress

The effect of water stress on the expression of PR proteins in grapes has been examined by analyzing the PR protein content of *V. vinifera* cv Shiraz berries from a replicated irrigation trial (Pocock *et al.*, 2000). The lack of irrigation gave clear physiological signs of vine water stress but did not lead to elevated levels of PR proteins in the berries. At a fixed amount of protein per berry, however, the protein concentration in the juice from water stressed berries was higher than that from irrigated berries because berries from irrigated vines were larger and thus berry solutes were less concentrated. This effect of water stress on berry size is a general phenomenon (Smart and Coombe, 1983) and it is likely that anecdotal reports that haze problems are greater in drought years are due to changes in berry sizes in these years rather than a direct physiological response of the berries to water stress in the form of enhanced PR protein production. Studies of gene expression under carefully controlled experimental conditions involving water stress would add further support to this hypothesis.

7.5.5 Mechanical harvesting

Paetzold *et al.* (1990) observed that mechanically harvested grapes produced must with increased protein content compared to that of hand harvested fruit pressed as whole bunches. They suggested that the lack of stems during crushing led to lower polyphenolic content in the must from mechanically harvested fruit compared to that from hand harvested fruit and therefore more protein was lost in complexes with phenolics from must from hand harvested fruit. Dubourdieu and Canal-Llaubères (1989) showed that wine made after maceration of destalked grapes for 18 hours contained more protein than wine made after immediate pressing of whole bunches. Pocock and colleagues (Pocock and Waters, 1998; Pocock *et al.*, 1998) examined the impact of mechanical harvesting alone, not destemming, on the PR proteins in grapes and wine. Mechanical harvesting, with its associated prolonged transport of the fruit, resulted in higher PR protein levels in the juice and wine. Indeed mechanical harvesting of white grapes and subsequent transport was

found to double the amount of bentonite required for prevention of protein haze when compared to fruit harvested by hand and transported from the same vineyard before destemming and crushing (Pocock and Waters, 1998). This does not appear to be a result of increased protein synthesis in response to wounding as comparisons among hand harvested berries, mechanically harvested intact berries and the predominant form of mechanically harvested fruit – a mixture of broken fruit and juice – indicated little if any protein was produced as a result of stress caused by mechanical harvesting. Increases in protein content of juice from mechanically harvested fruit thus appear to be due to extraction or release of protein from or associated with pulp and skins rather than a physiological wounding response by the berry.

7.6 Protein haze formation in wine

The mechanism of protein haze formation in wines is not fully understood. Slow denaturation of wine proteins is thought to lead to protein aggregation, flocculation into a hazy suspension and, finally, formation of visual precipitates. The importance of non-proteinaceous factors, such as proanthocyanidins (Koch and Sajak, 1959; Yokotsuka *et al.*, 1991; Waters *et al.*, 1995a), in white wine protein haze formation has been suspected for some time. Other factors such as polysaccharides, alcohol levels and pH have also been implicated (Siebert *et al.*, 1996a; Mesquita *et al.*, 2001). It has been observed that grape protein added to model wine does not precipitate or produce haze when heated, whereas visually obvious hazes occur when the same protein is added to a commercial wine (Pocock *et al.*, 2006).

The current theory of the mechanism of haze formation formulated by Pocock *et al.* (2006) is as follows. Individual grape PR proteins probably exist as separate globular entities freely soluble in wine, tightly coiled and containing between six and eight disulphide bridges. The first step in the process of haze formation is to uncoil these proteins, or to denature them. This is accelerated by heating. The second step involves aggregation of the denatured proteins as haze particles.

The size and amount of protein haze formed in a wine is strongly influenced by other wine components. Pocock *et al.* (2006) has demonstrated that one wine component, the sulphate anion, previously referred to as factor X, is essential for haze formation. If the sulphate anion is not present, heating does not result in a visible haze.

Sulphate is one of the Hofmeister series of anions, a ranking of the ability of various ions to precipitate proteins (Kunz *et al.*, 2004). In simple terms, precipitation of proteins by kosmotropic anions occurs due to ‘salting out’ – a competition between the anion and the protein for water of solvation resulting in a loss of water from the protein surface. This process is classically applied in ammonium sulphate precipitation as the first step in many protein purification schemes, although the levels employed are several fold higher than those in wine. In the particular case of white wine, this loss of water of solvation, even by a relatively low amount of sulphate anion, by a protein in a solution containing a variety of cations and other

anions and between 9% and 13% (v/v) ethanol, may be enough to affect the tertiary structure and/or aggregation of proteins.

Whilst sulphate appears to be fundamental to haze formation, other wine components such as phenolic compounds remain as candidate haze modulators. One possibility is that white wine phenolic compounds affect the particle size of denatured aggregated proteins, possibly through crosslinking. Several researchers (Oh *et al.*, 1980; Siebert *et al.*, 1996b) have suggested a hydrophobic mechanism for the interaction between phenolic compounds and proteins, in which the protein has a fixed number of phenolic binding sites. More of these sites are exposed when the protein is denatured.

Protein haze in white wine thus differs in several aspects from protein haze in beer. It is well established that beer protein haze is due to interactions between proteins, derived from the barley storage protein hordein and rich in proline, and hop polyphenolic compounds (Bamforth, 1999; Siebert, 1999; Siebert and Lynn, 2003; Miedl *et al.*, 2005). White wine proteins are not derived from storage proteins of grape seed nor are they as rich in proline as hordein. In addition, wine protein haze formation cannot be eliminated by removing polyphenolic compounds by polyvinylpolypyrrolidone (PVPP) (Pocock *et al.*, 2006) while in beer this has been applied as a commercial strategy (Madigan *et al.*, 2000; Leiper *et al.*, 2005).

7.7 Bentonite fining

The major winemaking process to affect the levels of proteins in wine is bentonite fining since the works of Saywell (1934) in the USA and Ribéreau-Gayon and Peynaud (1935) in Europe. Authorization for the use of bentonite followed the contribution of Milisavljevic (1963). Today, bentonite is used almost universally throughout the wine industry for the prevention of wine protein haze through removal of proteins before bottling (Blade and Boulton, 1988; Ferreira *et al.*, 2002; Waters *et al.*, 2005).

7.7.1 What is a bentonite?

Bentonite is a volcanic material which was deposited millions of years ago in broad layers, which weathered and changed from a fragile glassy state into a mineral one. This mineral is classified as a montmorillonite, which refers to the small French town (Montmorillon) where it was first discovered. Bentonites of various types exist in different geographical locations throughout the world, are mined from different depths and are obtained in different levels of purity, particle size, adsorption capacity and swelling ability (Marchal *et al.*, 1995). For these reasons, the type and source of bentonite used by winemakers greatly affect protein removal and oenological efficiency. In the USA, bentonite is principally mined in Fort Benton (Wyoming) – hence the term ‘Wyoming clay’ or ‘Benton-ite’. Bentonite is a complex hydrated aluminium silicate with exchangeable cationic components:

(Al, Fe, Mg) $\text{Si}_4\text{O}_{10}(\text{OH})_2(\text{Na}, \text{Ca})$. The most commonly used bentonite form in oenology is the sodium bentonite. Sodium bentonites have enhanced protein binding capabilities over calcium bentonites (Marchal *et al.*, 1995). This is generally the result of variations in the clay cation exchange capacity (CEC).

Bentonite exists as small plates which, when hydrated, separate to form a colloidal suspension with enormous surface area (300–900 m²/g) (Marchal *et al.*, 1995). Its subsequent activity in solution is like that of a multi-plated negatively charged structure able to exchange its cations with positively charged components of the juice or the wine (not only proteins). Bentonite absorption of uncharged molecules also occurs if they are polar. Additionally, platelet edges are positively charged and some limited binding of negatively charged proteins may occur there.

There can be slight differences in bentonite from one shipment to another (this makes it imperative that the same lot of bentonite should be employed for both laboratory trials and cellar activity). Calcium bentonite platelets tend to clump together, thus reducing the exposed surface area and therefore, protein binding (Marchal *et al.*, 1995). Calcium bentonite produces more compact lees than sodium bentonite. Because of its compact lees, calcium bentonite is generally preferred to sodium bentonite as a riddling agent in the ‘*méthode traditionnelle*’, the so-called ‘*méthode champenoise*’.

7.7.2 Methods for bentonite preparation

The method of preparation significantly affects the ability of bentonite to remove wine or juice proteins. Bentonite is made up of small platelets that are separated by a layer of water molecules. During hydration, the charged platelets are thought to repel each other and ‘pop apart’. As this occurs, swelling begins. Water molecules partially neutralize the exposed surfaces holding them apart, thus exposing the large reactive surfaces. In most wineries, bentonites are hydrated by addition of water. This allows the platelets to disperse and to form a gel. Most winemakers prepare bentonite by simply adding it slowly to water and letting the slurry stand for a day or two prior to use. When properly dispersed, bentonite sets up a network commonly known as the ‘house of cards’. This network encases droplets of water, which protect the bentonite from coalescing or flocculating with itself. In order for bentonite to be effective in binding with proteins, bentonite platelets must be separated into a homogeneous suspension. The bentonite/water ratio in the slurry is usually 5–10% w/v. The total quantity of water should not exceed 1% of the wine volume treated. Because bentonite’s protein binding activity is due to its exposed surface area, slurries for laboratory trials must be prepared exactly in the same way as suspensions used for cellar fining.

Sometimes, bentonite is directly resuspended in wine just before being introduced in the tank, without time for swelling (Marchal *et al.*, 2002c) (see the section on white wine fining and volume of lees). This way of using bentonite has some economic benefits over the use of swollen clay in water (Marchal *et al.*, 2002d).

7.7.3 Protein adsorption by bentonite particles

A number of studies have indicated that different protein fractions require distinct bentonite concentrations for protein removal and consequent heat stabilization (Duncan, 1992; Ferreira *et al.*, 2002). Bentonite fining has been shown to remove higher pI (5.8–8.0) and intermediate molecular weight (MW; 32–45 kDa) proteins first (Hsu and Heatherbell, 1987b). However, these represent only a small portion of the soluble proteins. Proteins with a MW of 60–65 kDa, and with wide pI range (4.1–8.0) were highly resistant to removal by bentonite fining (i.e. required significant bentonite addition) and typically remained in protein-stabilized wine (Hsu and Heatherbell, 1987b). Hsu and Heatherbell (1987b) concluded that it is necessary to remove lower pI (4.1–5.8), lower MW (12.6 kDa and 20–30 kDa) proteins, which represent a major component of proteins present, to protein stabilize wines. In contrast to these findings, a study by Dawes *et al.* (1994) found that there was no bentonite selectivity based on isoelectric point, and that bentonite fining resulted in the removal of all the different protein fractions. Further, the amount of protein depletion (across all protein fractions) observed in this study corresponded linearly with the level of bentonite addition (percentage reduction in protein concentration per g/L of bentonite added ranged from 70%–89%).

The same conclusion was made by Moine-Ledoux and Dubourdieu (1999), with the exception of the protein fraction released by lees during ageing with stirring. In this study, bentonite particles are able to remove all of the grape protein fractions present in a white wine but the amount eliminated was different from one fraction to another. These results clearly show that the protein MW and pI only partially explain its behaviour in wine and during the fining process.

These different conclusions in the published literature might be attributed in part to the different methods used to fractionate proteins and assess their levels. The mechanisms by which polypeptides and proteins adsorb on clay minerals have been studied using homopolyamino-acids (polylysine and polyglutamic acid) and a synthetic montmorillonite (Gougeon *et al.*, 2002, 2003). The use of complementary approaches (^{13}C NMR, specific surface area by BET measurements, X-ray diffraction (XRD), adsorption/desorption isotherms) has led to a detailed description of the interactions between ‘positively’ charged proteins and ‘negatively’ charged plate surfaces in a synthetic wine. ^{13}C NMR spectra showed that these polypeptides, which exhibit mixture of helical and random coil conformations in the bulk, tend to unfold and adopt a more extended random coil structure upon adsorption on phyllosilicates. Values of the basal spacing measured by XRD on dehydrated samples clearly indicated that, in the presence of adsorbed polypeptides, the silicate layers do not collapse. Incorporation of polypeptide fragments within the interlayer spaces was also revealed by the decrease in the specific surface areas. All together, these results show that polypeptides are adsorbed at the periphery of the montmorillonite particles through specific interactions between the protein side chains and the silicate sheets, whereas the polypeptide backbones do not enter the interlayer spaces (Gougeon *et al.*, 2002, 2003).

The adsorption of the standard protein, BSA by bentonite in model wine solutions was studied by Blade and Boulton (1988). Adsorption was shown to be

independent of temperature, but varied slightly with protein content, pH and ethanol content. In another study (Achaerandio *et al.*, 2001), bentonite adsorption was evaluated with three proteins (BSA, ovalbumin, lysozyme) in a model wine solution. The effect of ethanol content and protein molecular weight on the adsorption capacity of bentonite was also studied. Adsorption capacity tended to increase with increasing ethanol concentrations with regard to adsorption of BSA and lysozyme; however, no change was observed for ovalbumin. Blade and Boulton (1988) showed that maximal absorption was reached rapidly, and was complete within 30 seconds of the addition. This is consistent with an earlier study (Lee, 1986), in which Gewürztraminer wine fined with bentonite was rendered stable 1 minute after bentonite addition and with later studies of in-line dosing described below (Muhlack *et al.*, 2006; Nordestgaard *et al.*, 2006). In comparison, bentonite fining in a winery setting typically takes one to two weeks, depending on the tank size and rate of bentonite addition used.

7.7.4 Predictive assays for heat stability

The amount of bentonite required to prevent protein haze formation differs from wine to wine and is usually determined by a laboratory fining trial and assessed by a heat stability test. The heat stability test most commonly employed is to observe whether a wine sample produces excessive (e.g. >2 NTU) turbidity following heating at 80 °C for 6 hours. Pocock and Waters (2006) have shown that this test does not always accurately predict haze formation during long-term storage trials of white wines. Furthermore, Pocock and Waters (2006) demonstrated that most of the current predictive assays used by winemakers could be biased towards over-fining and it is also likely that technical errors performing both the heating and the haze measurement steps add to their inaccuracy. This could lead to fining with more bentonite than is needed under most conditions encountered commercially by bottled wines during transport and storage. Clearly, further trials with more wines and with storage under a wider range of conditions would be necessary to confirm these results.

Therefore, an alternative test method, such as heating at 80 °C for 2 hours, an 'alkali'-modified Coomassie-dye assay (e.g. Boyes *et al.*, 1997), or the more recently available reagent-based test kits such as Proteotest or Prostab, could present an opportunity to decrease bentonite dose and reduce volume of wine occluded in bentonite lees if they could be shown to predict haze formation in wine more accurately. If these alternative test methods could replace the current heat test method of 80 °C for 6 hours, this would also reduce the amount of time and/or effort presently required by a winery's laboratory to conduct heat stability testing.

7.7.5 The sensory impact of bentonite fining

One of the perceived disadvantages of bentonite fining is that it is thought to reduce flavour and aroma of wines. This perception is not supported by strong evidence in the normal operational range of bentonite additions by winemakers. Sensory

evaluation of wines treated with bentonite showed no significant differences between the control and the fined samples (Leske *et al.*, 1995). Similarly, using difference testing, Pocock *et al.* (2003) reported that bentonite fining of a Chardonnay and Semillon wine had no effect on wine aroma and palate. This contrasts with previous findings (Miller *et al.*, 1985) that demonstrated reduced concentration of aroma compounds after bentonite addition to juice, must or wine. More recently, Pollnitz *et al.* (2003) elegantly confirmed that aroma compounds can be absorbed by bentonite, as did Cabaroglu *et al.* (2003), although the later study found no sensory effect of bentonite fining of *V. vinifera* cv Muscat Ottonel or Gewurztraminer wine. Rankine (1989) stated that bentonite fining results in the loss of aroma and flavour and Martínez-Rodríguez and Polo (2003) extended this conclusion to sparkling wines when bentonite is added to the tirage solution. This has led to the widespread conclusion, throughout both literature and industry, that bentonite fining at typical addition rates has a detrimental effect on wine aroma and flavour, despite the fact that the conclusions of Miller *et al.* (1985) were influenced by many factors and 'should not be considered general' (Miller *et al.*, 1985). Similarly, Leske *et al.* (1995), concluded that 'lack of significant differences observed in this trial suggests that the negative effect of fining may not be evident to a consumer in a commercial situation' and 'the magnitude of any such [negative sensory] effect will presumably depend on the individual wine, the bentonite chosen, and the rate of addition chosen'.

Bentonite treatment may also have detrimental effects on foam characteristics of sparkling wine. Then, care must be taken in consideration when a sparkling base wine is fined with bentonite in order to preserve the essential quality that is a visual aspect of the foam ring in the glass. Excessive use of bentonite for the fining of sparkling wine *cuvées* can produce a finished product that has a large bubble size and a poor bubble stability as a result of a reduction in protein content (Marchal *et al.*, 1999). In some wines, free tartaric acid can be complexed with proteins and polyphenolics, inhibiting potassium bitartrate crystal formation. Removal of a portion of these complexing compounds with bentonite can enhance potassium bitartrate stability. Additionally, bentonite fining of wines during cold stabilization allows potassium bitartrate crystals to compact the bentonite lees.

However, because the use of bentonite is thought to alter the organoleptic characteristics of wines (Miller *et al.*, 1985; Main and Morris, 1994; Lubbers *et al.*, 1996), other authors have searched for alternative treatments for the use of bentonites, particularly mannoproteins (Ledoux *et al.*, 1992; Lubbers *et al.*, 1993; Waters *et al.*, 1996; Moine-Ledoux and Dubourdieu 1999, 2007; Dupin *et al.*, 2000; Ferreira *et al.*, 2004).

7.7.6 White wine bentonite fining and volume of lees

The commonly expressed problem with sodium bentonite is indeed excessive lees production. Between 5 and 10% of the wine may be occluded in the lees (Tattersall *et al.*, 2001; Marchal *et al.*, 2002c). During recovery by rotary drum vacuum (RDV) filtration, this occluded wine may be downgraded in quality (Waters *et al.*,

2005). Sodium (Na) bentonites swell more than calcium (Ca) bentonites, and thus produce a greater volume of lees for an equivalent dose, but are more efficient at adsorbing proteins and, hence, require a lower dose than Ca bentonites to achieve heat stability (Zoecklein, 1988; Marchal *et al.*, 1995).

However, there are few quantitative data that show what these relationships actually are. Analysis of a study of bentonites from more than a decade ago by AWRI (Morant, 1990) suggests that Ca bentonites, whilst requiring (on average) a dose three times higher than Na bentonites to achieve heat stability, will produce one quarter of the lees. Furthermore, Na/Ca bentonites required a dose 50% greater than Na bentonites but generated one half the volume of lees. The data presented indicate that using Na/Ca or Ca bentonites could significantly reduce values losses arising from bentonite fining. For example, estimates from the above data suggest that a Na bentonite dose of 0.4 g/L produces 6% v/v lees whereas a Ca bentonite, by reducing lees volume, could regain up to 5% of the original volume of the wine that would ordinarily be occluded in Na bentonite lees.

However, if this is the case, why are Na/Ca or Ca bentonites not more widely used? There are a number of possible reasons. First, higher doses of Ca bentonite require greater volume of slurry or more concentrated slurry. Calculations assuming a 2% constraint on water addition to the wine suggest that for a Na bentonite dose of 1 g/L, 5% w/v slurry would be required. At the higher doses of Na/Ca and Ca bentonite necessary to achieve the same treatment effect, the slurry concentrations are 7.5 and 15% w/v, respectively. It may not be possible or practical to achieve these higher slurry concentrations and thus Na bentonites are favoured. However, Ca bentonites swell less and higher slurry concentrations may be possible when compared with Na bentonites. Investigations of the rheology of bentonite slurries would provide valuable data to answer this question. Second, Ca bentonites release Ca during bentonite fining and may contribute to tartrate instability. Third, Ca bentonites may contribute to heavy metal pick up (Boulton *et al.*, 1996) and, thus, their use may be undesirable at elevated bentonite dose rates. However, this was not observed by Morant (1990) when she measured trace metal concentrations for arsenic and copper in wine following fining. In fact, no difference in heavy metal content was observed between the Na, Na/Ca and Ca bentonites.

Centrifugation (followed by filtration) is a method employed to minimize the problem of wine loss after bentonite fining. In wineries, bentonite often stays in contact with wine for a prolonged period of time (depending on the capacity/height of the tank). Nevertheless, wine proteins react with bentonite within the first minute of contact (Blade and Boulton, 1988). Therefore, the winemaker does not necessarily need to let wine settle following bentonite addition but may remove bentonite and proteins 'in line' with the proper centrifugation/filtration equipment.

An additional method for avoiding excessive lees formation in wine is to 'hydrate' the bentonite in the wine to be fined rather than in water (Marchal *et al.*, 2002d). Although this can significantly reduce the bentonite's binding ability (~50%), it often produces about one half of the normal lees volume. Bentonite may

also be counterfined with kieselsool (aqueous silicon dioxide) to aid in lees compaction.

Gelatine is a positively charged protein which will bind with negatively charged species such as tannins, kieselsools and bentonite. Gelatine can be used to help bentonite flocculation and possibly aid in lees compaction.

Another method used to solve the problems of excessive lees (and flavour stripping) caused by fining wine with bentonite is to ferment in contact with bentonite. Fermentation in the presence of bentonite is an old practice used in Europe for protein stabilization. Such a practice avoids or minimizes the need for subsequent bentonite addition into wine. Fermentation in contact with bentonite has several advantages: (i) only juice components are adsorbed onto bentonite and not the fermentation by-products or barrel-ageing constituents; (ii) fermentation lees have a lower monetary value than do finished wine lees. Thus, protein stabilization or partial stabilization during fermentation may be an important economic consideration.

Nevertheless, addition of bentonite at the beginning of the alcoholic fermentation may deplete the assimilable nitrogen content of the grape juice due to nitrogen compounds binding/adsorption. This may result in fermentation sticking and/or hydrogen sulphide production. Addition of an exogenous source of nitrogen eliminates these potential problems.

7.7.7 Bentonite regeneration after wine fining

Bentonite regeneration refers to the desorption of adsorbed wine protein from the bentonite surface, and would permit bentonite to be reused. However, a commercial process for bentonite regeneration does not currently exist, and thus bentonite is only used once before being discarded. An early study (Armstrong and Chesters, 1964) investigated the effect of pH on the desorption of pepsin from bentonite. In this study, 62.8% of the pepsin (pI 2.8), which had been adsorbed onto the bentonite at pH 3.0, was desorbed by raising the pH to 5.2 using sodium hydroxide. In a more recent study (Churchman, 1999), the batch treatment of bentonite–protein complexes with a range of bases at a variety of different concentrations, durations and agitation methods was examined. The greatest degree of desorption was achieved with sodium carbonate at a low solid:solution ratio and with agitation. Similar results were obtained with sodium hydroxide at pH 12 and 13 (protein released per gram bentonite = 52 and 65 mg/g, respectively). However, from the study it was concluded that the use of alkaline pH and salts was ineffective for substantial removal of protein from bentonites. Instead, the study suggested protein degradation followed by a treatment to displace the products of protein breakdown from the bentonite was required. For example, a batchwise treatment that employed hydrogen peroxide, sodium carbonate and photo-oxidation with UV radiation was found to be effective in removing residual protein from bentonite.

As discussed by Waters and colleagues (Waters *et al.*, 2005), in both these studies, protein was desorbed from bentonite by increasing the pH, but only in a

batch system. As adsorption and desorption are equilibrium processes, it is hardly surprising that residual protein was retained on the bentonite surface. In contrast, in a continuous flow system, fresh reactant could be continually fed to the adsorption system, while ion exchange products could be continually transported away from the system, driving the exchange process to completion. Continuous flow contactors, such as continuous stirred tanks, packed bed columns and fluidized beds, are commonplace in many chemical industries, as is continuous catalyst regeneration (Fogler, 1992). Therefore, it was postulated that effective regeneration of bentonite might be achieved by treatment with a base such as sodium hydroxide in a simple continuous flow system (Waters *et al.*, 2005). However, the difficulty in regenerating bentonite by salts and alkalis could also suggest that protein adsorption on bentonite may involve other mechanisms than solely cation exchange; thus complete regeneration of bentonite by such a process may not be feasible.

7.7.8 Improving bentonite efficiency by in-line dosing

Gravity settling of the bentonite normally requires several days to 2 weeks, according to the capacity and the geometry of the tank. Protein adsorption during bentonite fining is rapid; the treatment effect is completed within several minutes (Blade and Boulton, 1988; Muhlack *et al.*, 2006). Hence, the long processing time required for bentonite fining is constrained by gravity settling of the lees and to maximize wine recovery during racking. Consequently, there is interest in developing more efficient ways of using bentonite.

The in-line dosing paradigm exploits the observation that protein adsorption by bentonite occurs rapidly. Furthermore, if combined with centrifugation, the bentonite and wine can be separated and the bentonite lees are simultaneously compacted to reduce value losses. In-line dosing methods for bentonite fining are already used by some wineries and anecdotal reports suggest that commercial applications are growing.

Results obtained from pilot-scale studies of in-line dosing by Colby and colleagues (Muhlack *et al.*, 2006; Nordestgaard *et al.*, 2006) confirmed that a contact time of only a few minutes was sufficient for adsorption of the haze-forming proteins by bentonite. There was no detectable sensory impact on the wine of in-line dosing compared to batch addition.

These studies also observed a carryover of bentonite from the centrifuge during in-line dosing of bentonite. This bentonite carryover could be reduced by decreasing the operating flow rate to the centrifuge. Furthermore, Na–Ca bentonite was separated more easily by centrifugation than Na bentonites, and thus produced less carryover at equivalent flow-rates.

This carryover problem, if centrifugation is used, is a disadvantage of the in-line dosing method because it contributes to downstream wine losses from tanks containing settled lees and/or constrains the throughput achievable. Some potential solutions that could be used to overcome this problem include: a more efficient centrifuge; multiple clarification steps; an alternative and more efficient clarification

method; an alternative bentonite that is more easily clarified; and ‘counter-fining’ with another fining agent to improve bentonite separation.

7.8 Use of gelatine in white wine fining

The primary reaction occurring with gelatine is a complex formation between polyphenols in the wine and the protein of gelatine to give the desired floccular precipitate. The second reaction, less well understood, but equally important, is the complex formation between the natural wine proteins and the added protein, i.e. gelatine.

7.8.1 The nature of oenological gelatines

Gelatine is prepared from derived sources of collagen. Gelatine is a protein, that is, a polymer of amino acids joined together by peptide bonds. Hence, proteins can be depicted as long molecules with many different side chains, which accounts for their varying properties. Proline, a very important amino acid of gelatine, imparts a twist to the chain and affects the shape of the protein molecule and its rigidity. The protein chain is amphoteric and can carry either a positive or a negative charge depending on the pH of the medium. In wine and beverages at a pH of 2.9–3.6, most of the amino groups are positively charged and most of the acidic groups are uncharged. Gelatine then behaves as a cation. It attracts and forms polar associates with anions in solution. In addition, proteins form associations due to hydrogen bonding using the negatively charged oxygen and nitrogen atoms in the molecules.

As described above, the isoelectric point (pI) of a protein is the pH at which the protein will not migrate in an electric field. At that pH, the molecule carries an equal number of positive and negative charges: the molecule is isoionic, in the absence of added ions other than hydrogen and hydroxyl ions in solution. Gelatine is rather unique in that it can have an isoelectric point anywhere between pH 9 and pH 5, depending upon the source and method of production. Gelatines used in oenology are usually derived from acid pre-treated pigskin (Type A) and have isoelectric points between 6 and 9. Gelatines with a high gel strength (Bloom strength) have the higher pI, and gelatines with a low Bloom strength have a pI closer to 6. Some oenological gelatines are derived from limed hide or limed ossein (Type B gelatines), and all of them have a pI close to 5. The significance of pI is, of course, that the higher the pI, the greater the cationic charge on the molecule at the wine pH. In other words, at pH 2.9–3.6, all gelatines would be positively charged, but the charge density would be far higher for high pI gelatines.

7.8.2 Polyphenol–protein reaction

Both tannins and anthocyanins in wines are molecules containing benzene rings with adjacent hydroxyl groups which are proposed as a major source of hydrogen bonds at the basis of complex formation between gelatine and tannins or

anthocyanins in wines. Gelatine is held to be particularly suited to hydrogen bonding because one third of the amino acids are glycine, where $R = H$, and hence steric hindrance to hydrogen bonding would be far less than with proteins containing less glycine. However, the tannin/gelatine complex is also very pH-dependent and disappears at approximately pH 8, which would be due to both molecules becoming negatively charged and hence mutually repulsive. Hence, the role of polar bonding between molecules of dissimilar charges must not be overlooked.

7.8.3 Gelatine fining

In Europe, most of the available gelatines are Type A pigskin gelatine (ossein gelatine, having a higher viscosity, is mainly employed in film forming applications). There is a wealth of European data which show that low Bloom strength gelatine (low molecular weight – LMW) is optimum for fining (Calderon *et al.*, 1968). This ‘dictate of the art’ is probably due to the facility of use of LMW gelatines when compared with high molecular weight (HMW) gelatines that need to be heated before incorporation in the wine. High Bloom strength (HMW) gelatine used in fining has to be prepared as a dilute solution of gelatine; one must remember that, on cooling, the solution will gel. Hence it is of prime importance to ensure that when a warm gelatine solution is added, it is added at a point of very intense agitation such that the small amount of gelatine is intimately mixed into a large bulk of beverages before any gelling can occur. In fact, there is really no detectable difference in performance between the use of HMW and LMW gelatines (Marchal *et al.*, 1993). This applied to all parameters tested, i.e. sediment volume, clarity of supernatant and protein stability. Hence, the gelatine to use is often (essentially!) determined by economics (the lowest Bloom strength gelatines command the lowest price) and facility of use.

- *Remark 1.* Gelatine solution gel is an excellent nutrient for most forms of microbiological life. Hence, a number of manufacturers produce a highly concentrated solution of non-gelling hydrolyzed gelatine preserved with SO_2 .
- *Remark 2.* Gelatine is primarily used to soften red wines, but it is also used to reduce the phenol level and brown colour in white juices before fermentation. The potential for overfining with gelatine is great. Kieselsol, a negatively charged silica compound, is recommended for white wines. Kieselsol helps to moderate the effect of gelatine on wine flavour; it reduces the amount of gelatine needed and the volume of lees produced. Kieselsol or tannins must be added to most white wines following gelatine fining or the gelatine will remain suspended in solution.

7.9 Wine fining with plant proteins

The influence of treatments with tannins complexed with gelatine on wine composition and sensory perception has largely been reported in scientific studies. These

have often focused on the factors contributing to protein–polyphenolic interactions, which are responsible for the expected flocculation and clarification. However, in 1999, the incidence of the bovine spongiform encephalopathy ('mad cow disease') led to grave concerns about the use of proteins derived from animal sources in the food supply, and winemakers are encouraged to discontinue use of gelatines (especially bovine gelatine) and animal proteins in general. In Europe, the concern of transmitting this disease to humans led to a ban on the use of bovine plasma and blood cells, commonly but incorrectly called blood albumin and glue (Regulation CE 2087/97. Council, 20 October 1997). Some winemakers are also hesitant to use caseins because of their bovine origin. The problem is the same for isinglass because of its animal origin too. Given the above, it was important to develop treatments that could replace gelatine and bentonite finings. Since 1999, many investigations have been carried out with wheat prolamins, commonly called gluten, as white musts and wines clarifying agents. Different experimental procedures were established to compare gluten efficaciousness with usual fining agents (Lefebvre *et al.*, 2000; Marchal *et al.*, 2002b,c).

7.9.1 Comparison of wheat gluten and other fining agents for white wine

The results presented here concern a Chardonnay base wine from the Champagne wine-growing area (Marchal *et al.*, 2002c). Preliminary laboratory tests showed that glutens (used alone) were not able to clarify white wines. This is certainly related to the low dispersion of the viscoelastic particles, leading to a small contact surface. Also, glutens composed of prolamins with a low molecular weight cannot lead to the formation of flocculates with wine particles. For all of the glutens tested, an addition of 5 g/hL led to an increase in the wine turbidity. Yet, the clarifying effect from higher additions of these proteins (10 g/hL) increased proportionally to the gluten quantity added to the wine, even if differences between glutens were observed. These variations can be explained by the different treatments applied to gluten, and to the protein composition which depends on the wheat species. As for classical fining treatments (bentonite, gelatine, casein, etc.), the oenology literature does not provide precise information (molecular mechanisms) able to explain the differences observed.

After fining, turbidities obtained for wines treated with glutens (and gluten + tannins) were between the minimal and maximal values obtained with the different associations of tannins–gelatines or bentonites. However, the best results were obtained with 1 g/hL fish glues (particularly when in combination with tannins).

7.9.2 Combination of gluten with other fining agents for Chardonnay wine clarification

Clarifications obtained with the combination of gluten and tannins were always better than with gluten used alone: there is a synergy between hydrolyzed gluten proteins and tannins (Marchal *et al.*, 2002c). However, when the dose of gluten increased (40 g/hL), the synergy between wheat proteins and tannins is less marked.

The combination of gluten–casein generates negative interactions. When compared to the untreated wine, gluten alone or casein alone, both at 10 g/hL, led respectively to a decrease of 24 and 44 NTU. However, the combination of both in the same conditions made the turbidity fall by only 41 NTU. Therefore, the combination of the two is not advisable, nor is the combination caseins + bentonite if we exclusively consider the decrease in turbidity. To sum up, these results show that gluten is an efficient fining agent for clarifying white wines after alcoholic fermentation. In laboratory-scale studies, differences between the efficiencies at 28 and 48 hours were quite small or even non-existent for numerous treatments. This makes it possible to accomplish small-scale fining tests with short reaction times. The influence of the fining type on lees volume has also to be taken into account because it is related to the loss of wine. All the glutens generate volumes of lees that are similar to the values observed for casein, tannin–gelatine and fish gelatine. However, these volumes are smaller than the ones obtained with bentonite–casein or with bentonites. Globally, glutens generate volumes of lees comparable to those obtained with animal proteins used as fining agents.

7.10 Must clarification using the flotation technique

Some sparkling wines such as ‘Clairette de Die’ (France), ‘Asti Spumante’ (Italy) and many others throughout the world (Spain, Portugal, Greece, Brazil for example) are sparkling wines with approximately 6–8% alcohol and 35–50 g/L residual sugars. Pectolytic enzymes are often added to the Muscat grapes in the crusher to increase the extraction of aroma compounds. The consequence of this treatment is that the must presents a high turbidity (1000–1500 NTU) before lees settling. During alcoholic fermentation, the must is bottled when the density is around 1040–1050 g/L; the fermentation continues in the bottle until the pressure reaches generally 6 bars, then stops spontaneously and the sediment is removed after 6 to 8 months bottle-ageing. Subsequent riddling and disgorging of the wine are all the more easy when the turbidity of the must is low. Centrifugation after pressing or before bottling gives poor clarification and filtration is difficult to achieve and expensive. For these reasons, organic and mineral fining agents are commonly used to clarify white musts using the flotation technique (Sahraoui, 1991; Ferrarini *et al.*, 1992; Davin and Sahraoui, 1993; Déchaudat, 1995; Marchal *et al.*, 2003).

7.10.1 Principle of flotation

Flotation is a solid–liquid separation process used when the density, ρ , of the particles is lower than that of the liquid containing them. In a must, most particles have a density close to that of the must. This density can be artificially reduced by using gas bubbles that are retained by and ‘stick’ to the particles (Fig. 7.1). When air bubbles are directly injected into the must the process is called assisted flotation (Fig. 7.1a). When air bubbles are injected into a must treated with a fining agent,

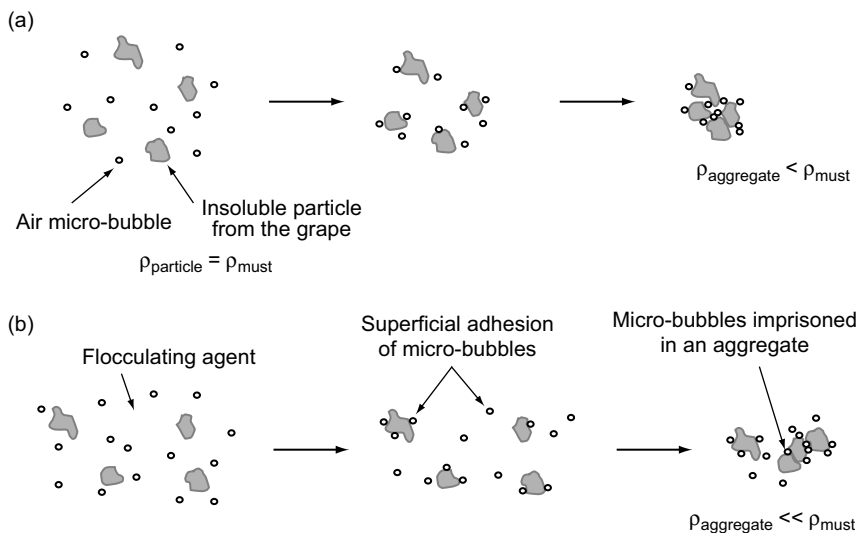


Fig. 7.1 Principle of (a) assisted flotation and (b) induced flotation (ρ , density).

the association between air bubbles and must flocculates give insoluble complexes with even lower density compared with that of the liquid. Then, these complexes easily reach the surface of the tank and the process is called induced flotation (Fig. 7.1b). When the diameter of air bubbles is between 40–70 μm , the process is called dissolved-air flotation. The speed at which the gas-particle complexes rise through the liquid to reach the liquid/air surface largely depends on the diameter of microbubbles, the mass of particles, the densities of the liquid, that of the gas and that of the particles. The temperature and the hydrophobicity of the particles are also important parameters for the must clarification using the flotation technique.

Density of must particles depends on the flocculation conditions following pressing. At first, mini-flocculates are formed. We observe after this an aggregation of these flocculates, leading to larger flocculates separable from the liquid. This second step is made easier when the must is treated with pectolytic enzymes, in the crusher or just after pressing. The vintage, the variety, the maturity of grape berry also influence flocculation. In oenology, it is possible to use flocculating agents to bind particles. These agents are mineral adjuvants (bentonite, silica gel) or natural polymers extracted from animal substances (pork, bovine and fish gelatines). Proteins isolated from plants (wheat gluten, lupine, pea) are suitable alternatives.

7.10.2 Flotation trials at a laboratory scale

A 'miniflotator' (1 to 2 litre flotation tank) can be used for laboratory experiments (Fig. 7.2) (Marchal *et al.*, 2003). The miniflotator containing the treated must is pressurized (6 bars) using the industrial pressurized air system. Using pressure, the

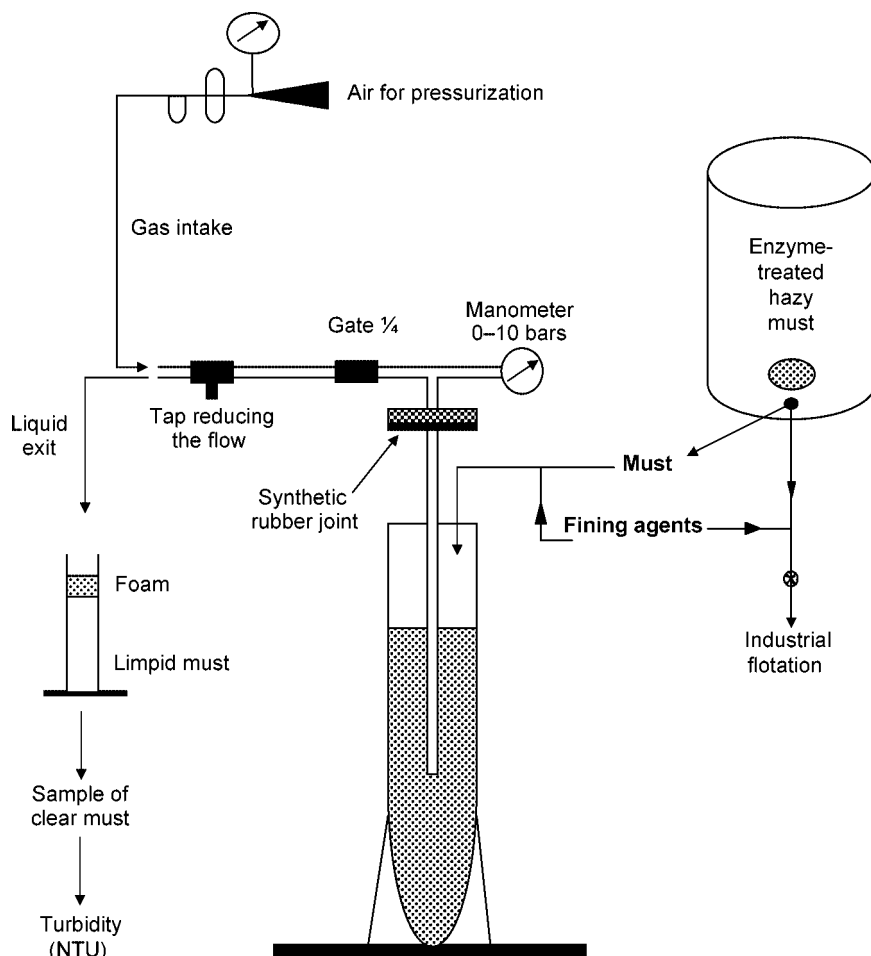


Fig. 7.2 Schematic of a laboratory flotation system.

must is poured into a graduated cylinder using a stainless pipe. After 3 minutes, the turbidity was measured. During the depressurization of the must (previously at 6 bars), microbubbles are formed because the liquid becomes supersaturated with air. This assisted flotation gave a turbidity generally equal to 30% of that of the non-treated must (without any fining agent), but the turbidity still remained around 300 NTU, too high a value for this wine process. When a mineral flocculating agent such as bentonite or silica gel is added, the turbidity of the must after flotation is nearly 80% of that of the control must (Marchal *et al.*, 2003). If we consider exclusively turbidities, these results show that mineral fining agents, when used alone, have poor efficiencies on grape juice. Bentonite, instead of its bad clarifying action for the flotation technique, is used to reduce the risk of protein haze that is often observed for Muscat wines if must proteins are not removed. In this case, it

is essential for the winemaker to consider two oenological parameters: the decrease in juice turbidity after fining, but also (if not essentially) the risk of protein haze.

In contrast, wheat gluten allows good clarifications (reduction to about 90% compared to the turbidity of the control must). When gluten is combined with bentonite, the turbidity decreases to 30–40%, compared to that of the must clarified with flotation without fining. The clarification is more marked if silica gel is combined with gluten and bentonite. On the basis of these results, one can consider that the use of silica gel significantly improved the efficiency of the flotation technique when wheat gluten is used (Marchal *et al.*, 2003). These conclusions are drawn on the basis of experiments carried out during three harvests with Muscat musts.

7.10.3 Industrial flotation experiments

Trials were also conducted on an industrial scale starting with 500 hL of must and with a 80 hL flotation tank. An aspiration system collected the flotation foam (60 hL) on the top of the flotation tank (Fig. 7.3). Particles in the foam were removed with a rotary filter and the clear must (35 hL) was obtained. In the following example, two identical tanks, containing exactly the same must, were treated with bentonite (20 g/hL) plus silica gel (10 g/hL) plus gluten (20 g/hL; hereafter called 'gluten') or with bentonite (20 g/hL) plus silica gel (10 g/hL) plus fish glue soluble gelatine (20 g/hL; hereafter called 'FG'; Marchal *et al.*, 2003). The industrial experiment was repeated and the musts obtained with the same treatment were blended. So, 1000 hL of must clarified by flotation with gluten as protein fining agent were obtained, and also 1000 hL of must clarified by flotation using fish gelatine as protein fining agent.

Measurements were made just after flotation (between the flotation tank and the temporary storage tank). Turbidities decreased by more than 95% for both treatments when compared with the control must without flotation (1446 NTU). The final turbidity of the must treated by flotation with gluten as a fining protein was 20% higher than that of the must treated by flotation with FG as a fining protein. The same treatments were examined with the laboratory-scale device with the same must. Results confirmed the good efficaciousness of gluten (–95.4% of the initial NTU) compared to that of FG (–97.5% of the initial NTU). Turbidities were also measured in temporary storage tanks (500 hL), 14 hours after the industrial clarification by flotation. The turbidity of the must treated by flotation with gluten was 14% lower than that of the must treated by flotation with FG. During these 14 hours, a static settling occurred in the tank. Thus, results of static settling (when the must was fined and clarified by sedimentation of flocculates) indicated a higher efficiency for fining with gluten, even if the time required was longer. One can hypothesize that small flocculates, present in the must after gluten fining, have higher density than flocculates formed with fish gelatine. According to this hypothesis: (i) during the flotation (approximately 15 minutes on an industrial scale), the higher mass will reduce the velocity of the flocculates rising through the liquid and thus a perfect clarification will not be achieved; (ii) during static settling,

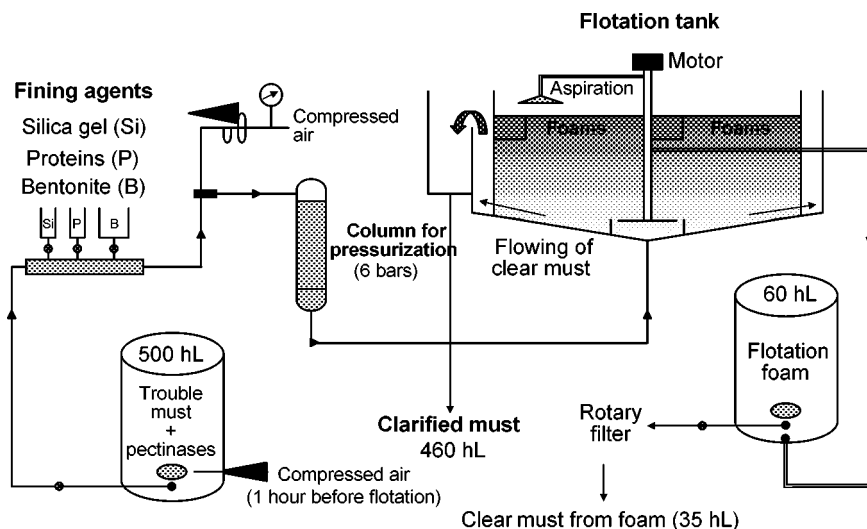


Fig. 7.3 Schematic of an industrial flotation device.

the sedimentation velocity will be higher for the same reason. At the production scale, clarification occurred in two steps: during flotation, FG was more efficient than gluten and gave lower turbidities; during the second step, the sedimentation of particles led to lower turbidities when gluten was added to the must before flotation (always combined with bentonite and silica gel in this study). It may be possible to improve the efficacy of gluten fining by reducing the flow rate of the pump that fills the flotation tank; then flocculates will have more time to reach the surface. Theoretically, this may result in clearer musts.

7.10.4 Flotation with lupine proteins

At the laboratory scale, turbidities of the musts fined with lupine proteins or gluten are nearly equal after flotation. Glutens are composed of numerous gliadins and glutenins (Gupta and Shepherd, 1990). Gliadins present differences in their amino acid composition, the consequence of which are a difference of wheat protein hydrophobicities (Popineau and Pineau, 1987) and *p* values. These biochemical characteristics are also induced by industrial processes (partial enzymatic hydrolysis for the gluten, no hydrolysis for lupine proteins). They are responsible for protein and wine phenolic compound interactions, leading to flocculation and clarification.

On the basis of the very complex molecular composition of wheat gluten (Popineau & Pineau, 1987; Gupta and Shepherd, 1990; Shewry *et al.*, 1994; Babiker *et al.*, 1996; Popineau and Denery-Papini, 1996) and lupine proteins (Duranti *et al.*, 1988, 1992), it is not possible at present to explain the efficiency of these protein extracts during fining and clarification by flotation.

7.10.5 Research of residual gluten proteins in must and wine by immunodetection

There is considerable interest in the possibility that some gluten remains in wine after treatment due to the intolerance of individuals with coeliac disease to this protein. Susceptible individuals with coeliac disease (non-tropical sprue) show lesions of the intestinal mucosa after consumption of the gliadins of wheat. The drastic decrease in absorptive surface is the cause for the malabsorption accompanied by weight loss and growth disturbances (Sturgess *et al.*, 1991; Cornell *et al.*, 1992). It is particularly important to insist on the detection of possible residual gluten, because it is already known that trace amounts of animal fining proteins may remain in the wine, especially if the wine has not been subsequently settled, racked and filtered (Douet *et al.*, 1999; Weber *et al.*, 2007; Rolland *et al.*, 2008). Residual gluten proteins in musts and wines fined with gluten have been assayed for using an enzyme-linked immunosorbent assay (ELISA) method and using a dot-blot method with chemoluminescence coloration (Marchal *et al.*, 2002b). The gluten content is calculated with regard to two standard curves (1–20 mg/L). Standard curves were obtained with gliadins extracted from the gluten used for the oenological experiments and also with a blend of gliadins extracted from 30 different wheat gluten samples. The two curves were superimposable. By using an ELISA immunotechnique, gluten-antibodies were not able to recognize their antigens in musts and wines treated with 20 g/hL gluten, these values corresponding to the detection limits of the ELISA method used. This result clearly indicates the absence of detectable residual gluten proteins in the must and wine (less than 1 mg/L, that is the limit of the kit detection). For the Codex Alimentarius, a food is considered gluten-free when it contains less than 10 mg/L gluten. Hence, white musts and wines clarified with wheat gluten derivatives may be considered as gluten-free. These results make the use of gluten for wine fining more attractive.

7.11 Other fining agents

7.11.1 Polyvinyl polypyrrolidone (PVPP)

PVPP is a high molecular weight fining agent composed of crosslinked monomers of polyvinylpyrrolidone. It essentially complexes with phenolic and polyphenolic components in wine by adsorption (Lin and Vine, 1990). It removes bitter compounds and browning precursors in both red and white wines. PVPP has also been recently tested in wines to eliminate odours described as mushrooms, mouldy, camphoric or earthy due to the presence of geosmin and ochratoxin A (OA). OA is a mycotoxin mainly produced by *Aspergillus carbonarius*; it is responsible for off-flavour, but it also poses risks for the human health because of its nephrotoxic, teratogenic, immunotoxic and neurotoxic activity. Higher toxin levels are found in red wines than white wines. Addition of PVPP to a Semillon wine (containing 56 mg grape-derived protein/L and *ca.* 8 µg OA/kg) did not significantly diminish OA in this treated white

wine compared to wine settled without fining agent (Leong *et al.*, 2006). In the same trial, addition of bentonite at 2.5 g/L resulted in a 67% reduction in OA concentration. Geosmin (*trans*-1,10-dimethyl-*trans*-9-decalol) is another compound identified as the responsible for a strong earthy odour in red and white wines (Darriet *et al.*, 2000, 2001, 2002) although detected at low concentration (up to 300 ng/L), due to its extremely low olfactory perception threshold (10 ng/L in water, 60–65 ng/L in white wine, 80–90 ng/L in red wine). The treatments with PVPP, zeolite and bentonite did not lead to any significant change in geosmin concentration in either red or white wine (Lisanti *et al.*, 2009). The reduction of geosmin and OA contamination of wine is a very complex problem; PVPP treatment does not bring solution.

7.11.2 Kieselsol

Kieselsol is a generic name for aqueous suspensions of silicon dioxide, and it is a by-product of the glass industry. Kieselsols are produced in Germany and are sold as 30% colloidal solutions. The primary use of kieselsol is for clarification and as a replacement for tannins during gelatine fining of white wines. Kieselsols are negatively charged and electrostatically bind to and adsorb positively charged proteins and initiate flocculation and settling. Several different kieselsol formulations are available at a variety of pH levels. Gelatine should be added first. Kieselsol is also used in combination with bentonite to improve the settling of flocculates.

7.11.3 Isinglass

Isinglass is a positively charged fining agent derived from the air bladder of fish. It is essentially available as flocculated isinglass without fishy odours. Isinglass is used principally in white still and sparkling wines to improve clarity without significantly modifying tannin levels. Usage levels are typically from 1–3 g/hL (Marchal *et al.*, 2002c, 2003).

7.11.4 Casein

Casein is nearly insoluble in must and white wine (in fact, the right word is 'caseins' as the oenological product contains all of the cow milk caseins). Potassium caseinate is water-soluble and is sometimes preferred by winemakers for this reason. Sodium caseinate is usually not used because it increases the sodium content of wine. Casein is a positively charged protein that flocculates in acidic media such as wine. When added to wine, casein adsorbs and mechanically removes suspended materials as it settles (Marchal *et al.*, 2002c). Casein is also used to remove undesirable odours, to bleach colour and to clarify white wines. It is sometimes used as a substitute for carbon in colour modification of juice and white wine. Usage levels are typically 10–30 g/hL.

7.11.5 Carbon or charcoal

The addition of carbon to juice or wine helps to diminish colour and oxidized flavours. Usage levels are typically 10–50 g/hL. The problem is that activated carbons are non-specific adsorptive agents made from wood. The sponge-like carbon binds with weakly polar molecules, especially those containing benzene rings, i.e. proteins and grape phenolics. Stripping of wine is often a problem with carbon because of its low selectivity, and great care has to be taken with its use. For example, a Pinot noir sparkling base wine treated with carbon consistently gives a sparkling wine with bad foaming properties when compared with the same sparkling wine elaborated without discoloration treatment (Marchal *et al.*, 2002a). Carbon also contains a large quantity of air, and oxidation sometimes follows carbon addition if the carbon is not quickly and thoroughly removed.

7.12 Equipment for the addition of fining agents to wine

The efficiency of fining is dependent upon the agent, the method of preparation and addition, the quantity employed, the pH, the temperature, the age of the wine and also previous treatments undergone by the wine. Fining is a surface action performed by the agent (adsorption); therefore, the method of hydration and addition of the agent is also of importance. Four common methods of adding fining agents are: (i) through a ‘Y’ on the suction side of a positive displacement pump while mixing; (ii) through an ‘in-line’ proportioning pump; (iii) through a ‘T’ into a Guth-type tank mixer; and (iv) added slowly in slurry form to a barrel using a dowel to stir in a figure-8 motion through the bung hole.

7.13 Wine fining: general conclusion and practical recommendations

Carefully controlled laboratory fining trials must be performed before any agent is added to wines in the cellar. This is an essential rule that must be systematically applied. In evaluating fining trials, the winemaker must observe, note and record how each fining agent alters clarity, lees production, lees compaction, colloidal stability and colour. To avoid ‘bad surprises’, a final analysis of protein stability should also be performed after fining and just prior to bottling.

Many different fining agents can often be used to achieve the same goal. If a winemaker is not satisfied with the results of the fining trials, it is always possible to test another fining agent. To achieve consistent results, it is essential that the preparation methods, temperature, mixing and timing are the same between laboratory fining trials and winery applications. Fining agents (especially bentonite and proteins) must be prepared just prior to use if possible. The efficaciousness of these fining agents can be considerably reduced by improper preparation. Fining agents need to be prepared in exactly the same way for both laboratory and winery use. Preparation equipment, temperatures and timing are critical. The difference

between using a blender in the laboratory and a paddle mixer in the winery could lead to overfining of the wine.

The winemaker should also carefully observe the impact of wine or must fining on gustatory and olfactory characteristics by rigorous sensory evaluation: the nose characteristics in general, body (front, middle, and finish), astringency, bitterness, the ageing potential. The ultimate aim of fining is to improve the wine's clarity not detract from its quality.

7.14 Acknowledgements

The authors would like to thank:

- Australia's grape growers and winemakers for their support through their investment body the Grape and Wine Research and Development Corporation, with matching funds from the Australian government.
- The Association Recherche Oenologique Champagne et Université, for their moral and financial support for more than 20 years.
- Our past and current colleagues and students for their input into our understanding of this area.

7.15 References

- Achaerandio I, Pachova V, Güell C and López F (2001), Protein adsorption by bentonite in a white wine model solution: Effect of protein molecular weight and ethanol concentration. *American Journal of Enology and Viticulture*, **52**, 122–126.
- Armstrong DE and Chesters G (1964), Properties of protein-bentonite complexes as influenced by equilibration conditions. *Soil Science*, **98**, 39–52.
- Babiker EFE, Fujisawa N, Matsudomi N and Kato A (1996), Improvement of the functional properties of gluten by protease digestion or acid hydrolysis followed by microbial transglutaminase treatment. *Journal of Agricultural and Food Chemistry*, **44**, 3746–3750.
- Bamforth CW (1999), Beer haze. *Journal of the American Society of Brewing Chemists*, **57**, 81–90.
- Bayly FC and Berg HW (1967), Grape and wine proteins of white wine varieties. *American Journal of Enology and Viticulture*, **18**, 18–32.
- Berg HW and Akiyoshi M (1961), Determination of protein stability in wine. *American Journal of Enology and Viticulture*, **12**, 107–110.
- Bézier A, Lambert B and Baillieu F (2002), Study of defense-related gene expression in grapevine leaves and berries infected with *Botrytis cinerea*. *European Journal of Plant Pathology*, **108**, 111–120.
- Blade WH and Boulton R (1988), Adsorption of protein by bentonite in a model wine solution. *American Journal of Enology and Viticulture*, **39**, 193–199.
- Boulton R, Singleton VL, LF B and Kunee RE (1996), Principles and Practices of Winemaking. Gaithersburg, MD: Aspen Publishers, Inc.
- Boyes S, Strübi P and Dawes H (1997), Measurement of protein content in fruit juices, wine and plant extracts in the presence of endogenous organic compounds. *Lebensmittel-Wissenschaft und-Technologie*, **30**(8), 778–785.
- Brown AE and Adikaram NKB (1983), A role for pectinase and protease inhibitors in fungal rot development in tomato fruits. *Phytopathologische Zeitschrift*, **106**, 239–251.

- Cabaroglu T, Razungles A, Baumes R and Gunata Z (2003), Effect of fining treatments on the aromatic potential of white wines from Muscat Ottonel and Gewurztraminer cultivars. *Sciences Des Aliments*, **23**, 411–423.
- Calderon P, Van Buren JP and Robinson WB (1968), Factors influencing the formation of precipitates and hazes by gelatin and condensed hydrolysable tannins. *Journal of Agricultural and Food Chemistry*, **16**, 479–482.
- Churchman GJ (1999), *Procedures for the efficient recycling of bentonite used for the removal of proteins from wine*, Final Report to Grape and Wine Research and Development Corporation, Project CSS 98/1, Adelaide, SA, available at: <http://www.gwrdc.com.au/webdata/resources/project/CSS981.pdf> (accessed March 2010).
- Cilindre C, Castro AJ, Clément C, Jeandet P and Marchal R (2007), Influence of *Botrytis cinerea* infection on Champagne wine proteins (characterized by two-dimensional electrophoresis /immunodetection) and wine foaming properties. *Food Chemistry*, **103**, 139–149.
- Cilindre C, Jegou S, Hovasse A, Schaeffer C, Castro AJ, Clement C, Van Dorsselaer A, Jeandet P and Marchal R (2008), Proteomic approach to identify champagne wine proteins as modified by *Botrytis cinerea* infection. *Journal of Proteome Research*, **7**, 1199–1208.
- Cornell H, Wieser H and Belitz HD (1992), Characterization of the gliadin-derived peptides which are biologically active in coeliac disease. *Clinica Chimica Acta*, **213**, 37–50.
- Dambrouck T, Marchal R, Marchal-Delahaut L, Parmentier M, Maujean A and Jeandet P (2003), Immunodetection of proteins from grapes and yeast in a white wine. *Journal of Agricultural and Food Chemistry*, **51**, 2727–2732.
- Darriet P, Pons M, Lamy S and Dubourdieu D (2000), Identification and quantification of geosmin, an earthy odorant contaminating wines. *Journal of Agricultural Food Chemistry*, **48**, 4835–4838.
- Darriet P, Lamy S, La Guerche S, Pons M, Dubourdieu D, Blancard D, Steliopoulos P and Mosandl A (2001), Stereodifferentiation of geosmin in wine. *European Food Research and Technology*, **213**, 122–125.
- Darriet P, Pons M, Henry R, Dumont O, Findeling V, Cartolaro P, Calonne A and Dubourdieu D (2002), Impact odorants contributing to the fungus type aroma from grape berries contaminated by powdery mildew (*Ucinula necator*): incidence of enzymatic activities of the yeast *Saccharomyces cerevisiae*. *Journal of Agricultural Food Chemistry*, **2002**, 50, 3277–3282.
- Davin A and Sahraoui A (1993), Débourage rapide des moûts de raisin par flottation à l'aide de bulles générées au sein du liquide par dépressurisation. *Revue Française d'Oenologie*, **140**, 53–64.
- Dawes H, Boyes S, Keene J and Heatherbell D (1994), Protein instability of wines: Influence of protein isoelectric point. *American Journal of Enology and Viticulture*, **45**, 319–326.
- Déchaudat C (1995), Clarifier des moûts avec des bulles de gaz. *La Vigne*, June, 50–51.
- Derckel J-P, Legendre L, Audran J-C, Haye B and Lambert B (1996), Chitinases of the grapevine (*Vitis vinifera* L.): five isoforms induced in leaves by salicylic acid are constitutively expressed in other tissues. *Plant Science*, **119**, 31–37.
- Derckel J-P, Audran J-C, Haye B, Lambert B and Legendre L (1998), Characterisation, induction by wounding and salicylic acid, and activity against *Botrytis cinerea* of chitinases and β -1,3-glucanases of ripening grape berries. *Physiologia Plantarum*, **104**, 56–64.
- Douet JP, Médina B and Castroviejo M (1999), Protéines résiduelles issues de colle de sang après traitement d'un vin blanc sec. *Journal International des Sciences de la Vigne et du Vin*, **33**, 177–185.
- Dubourdieu D and Canal-Llaubères RM (1989), Influence of some colloids (polysaccharides and proteins) on the clarification and stabilization of wines. In Lee TH, Williams PJ and Davidson DM (eds), *Proceedings of the Seventh Australian Wine Industry Technical Conference*, Winetitles, Adelaide, SA, 180–185.
- Duncan B (1992), Varietal differences in white grape protein: Implications for bentonite fining. *Australian and NZ Wine Industry Journal*, **7**, 189–193.

- Dupin IVS, Stockdale VJ, Williams PJ, Jones GP, Markides AJ and Waters EJ (2000), *Saccharomyces cerevisiae* mannoproteins that protect wine from protein haze: evaluation of extraction methods and immunolocalization. *Journal of Agricultural and Food Chemistry*, **48**, 1086–1095.
- Duranti M, Guerrieri N, Takahashi T and Cerletti P (1988), The legumin-like storage protein of *Lupinus albus* seeds. *Phytochemistry*, **27**, 15–23.
- Duranti M, Guerrieri N and Vecchio G (1992), The legumin precursor from white lupin seed. Identity of subunits, assembly and proteolysis. *European Journal of Biochemistry*, **206**, 941–947.
- Ferrarini R, Zironi R, Celotti E and Buiatti S (1992), Premiers résultats de l'application de la flottation dans la clarification des moûts de raisin. *Revue Française d'Oenologie*, **138**, 29–42.
- Ferreira RB, Monteiro S, Picarra-Pereira MA, Tanganho MC, Loureiro VB and Teixeira AR (2000), Characterization of the proteins from grapes and wines by immunological methods. *American Journal of Enology and Viticulture*, **51**, 22–28.
- Ferreira RB, Picarra-Pereira MA, Monteiro S, Loureiro VB and Teixeira AR (2002), The wine proteins. *Trends in Food Science and Technology*, **12**, 230–239.
- Ferreira RB, Monteiro S, Picarra-Pereira MA and Teixeira AR (2004), Engineering grapevine for increased resistance to fungal pathogens without compromising wine stability. *Trends in Biotechnology*, **22**, 168–173.
- Fogler HS (1992), *Elements of chemical reaction engineering*. Englewood Cliffs, NJ: Prentice-Hall Inc.
- Gerbaux V (1996), *Détermination de l'état de sursaturation et effet des polysaccharides sur la cristallisation du bitartrate de potassium dans les vins*, Doctoral dissertation, University of Toulouse, France.
- Giannakis C, Bucheli CS, Skene KGM, Robinson SP and Scott NS (1998), Chitinase and β -1,3-glucanase in grapevine leaves: a possible defence against powdery mildew infection. *Australian Journal of Grape and Wine Research*, **4**, 14–22.
- Girbau-Sola T, Lopez-Tamames E, Bujan J and Buxaderas S (2002), Foam aptitude of Trepát and Monastrell red varieties in Cava elaboration 1 base wine characteristics. *Journal of Agricultural and Food Chemistry*, **50**, 5596–5599.
- Girbau T, Stummer BE, Pocock KF, Baldock GA, Scott ES and Waters EJ (2004), The effect of *Uncinula necator* (powdery mildew) and *Botrytis cinerea* infection of grapes on the levels of haze-forming pathogenesis-related proteins in grape juice and wine. *Australian Journal of Grape and Wine Research*, **10**, 125–133.
- Gougeon RD, Reinholdt M, Delmotte L, Miéché-Brendlé J, Chézeau JM, Le Dred R, Marchal R and Jeandet P (2002), Direct observation of polylysine side-chain interaction with smectites interlayer surfaces through ^1H – ^{27}Al heteronuclear correlation NMR spectroscopy. *Langmuir*, **18**, 3396–3398.
- Gougeon RD, Soulard M, Reinholdt M, Miéché-Brendlé J, Chézeau JM, Le Dred R, Marchal R and Jeandet P (2003), Polypeptide adsorption onto a synthetic montmorillonite: A combined solid-state NMR, X-ray diffraction, thermal analysis and N_2 adsorption study. *European Journal of Inorganic Chemistry*, **7**, 1366–1372.
- Gupta RB and Shepherd KW (1990), Two step one dimensional SDS–PAGE analysis of LGM subunits of glutenin. I – Variation and genetic control of the subunits in hexaploid wheats. *Theoretical and Applied Genetics*, **80**, 65–74.
- Hayasaka Y, Adams KS, Pocock KF, Baldock GA, Waters EJ and Hoj PB (2001), Use of electrospray mass spectrometry for mass determination of grape (*Vitis vinifera*) juice pathogenesis-related proteins: a potential tool for varietal differentiation. *Journal of Agricultural and Food Chemistry*, **49**, 1830–1839.
- Hsu J-C and Heatherbell DA (1987a), Isolation and characterization of soluble proteins in grapes, grape juice and wine. *American Journal of Enology and Viticulture*, **38**, 6–10.
- Hsu J-C and Heatherbell DA (1987b), Heat-unstable proteins in wine. I. Characterization

- and removal by bentonite fining and heat treatment. *American Journal of Enology and Viticulture*, **38**, 11–16.
- Jacobs AK, Dry IB and Robinson SP (1999), Induction of different pathogenesis-related cDNAs in grapevine infected with powdery mildew and treated with ethephon. *Plant Pathology*, **48**, 325–336.
- Jayasankar S, Li Z and Gray DJ (2003), Constitutive expression of *Vitis vinifera* thaumatin-like protein after *in vitro* selection and its role in anthracnose resistance. *Functional Plant Biology*, **30**, 1105–1115.
- Jones PR, Gawel R, Francis IL and Waters EJ (2008), The influence of interactions between major white wine components on the aroma, flavour and texture of model white wine. *Food Quality and Preference*, **19**, 596–607.
- Koch J and Sajak E (1959), A review and some studies on grape protein. *American Journal of Enology and Viticulture*, **10**, 114–123.
- Kunz W, Henle J and Ninham BW (2004), ‘Zur Lehre von der Wirkung der Salze’ (about the science of the effect of salts): Franz Hofmeister’s historical papers. *Current Opinion in Colloid and Interface Science*, **9**, 19–37.
- Kwon SW (2004), Profiling of soluble proteins in wine by nano-high-performance liquid chromatography/tandem mass spectrometry. *Journal of Agricultural and Food Chemistry*, **52**, 7258–7263.
- Ledoux V, Dulau L and Dubourdieu D (1992), Interprétation de l’amélioration de la stabilité protéique des vins au cours de l’élevage sur lies. *Journal International des Sciences de la Vigne et du Vin*, **26**, 239–251.
- Lee T (1986), Protein instability: nature, characterization and removal by bentonite. In Lee (ed.), *Physical Stability of Wine*, Australian Society of Viticulture and Oenology, Adelaide, SA, 23–39.
- Lefebvre S, Gerland C, Maury C and Gazzola M (2000), Nouvelles colles végétales: origine, propriétés et performances. *Revue Française d’Oenologie*, **184**, 28–32.
- Leiper KA, Stewart GG, McKeown IP, Nock T and Thompson MJ (2005), Optimising beer stabilisation by the selective removal of tannoids and sensitive proteins. *Journal of the Institute of Brewing*, **111**, 118–127.
- Leong Su-Lin L, Hocking AD and Scott ES (2006), The effect of juice clarification, static or rotary fermentation and fining on ochratoxin A in wine. *Australian Journal of Grape and Wine Research*, **12**, 245–251.
- Leske PA, Bruer NGC and Capdeboscq V (1995), An evaluation of some characteristics of commercial bentonites. *Wine Industry Journal*, **10**, 73–77.
- Lin TY and Vine RP (1990), Identification and reduction of ellagic acid in muscadine grape juice. *Journal of Food Science*, **55**(1607–9), 1613.
- Lisanti MT, Piombino P, Gambuti A, Genovese A and Moio L (2009), Oenological treatments for the removal of geosmin responsible for earthy off-flavour in wine. *Bulletin de l’OIV*, **82**(935–937), 71–82.
- Lubbers S, Leger B, Charpentier C and Feuillat M (1993), Effet colloïde-protecteur d’extraits de parois de levures sur la stabilité tartrique d’une solution modèle. *Journal International des Sciences de la Vigne et du Vin*, **27**, 13–22.
- Lubbers S, Charpentier C and Feuillat M (1996), Etude de la rétention des composés d’arôme par les bentonites en moût, vin et milieux modèles. *Vitis*, **35**, 59–62.
- Madigan D, Byrne H, Matthews S, Kelly R, McEnroe C and Harmey D (2000), Studies on the effects of common process variables on the colloidal stability of beer. *Journal of the American Society of Brewing Chemists*, **58**, 160–164.
- Main GL and Morris JR (1994), Color of Seyval blanc juice and wine as affected by juice fining and bentonite fining during fermentation. *American Journal of Enology and Viticulture*, **45**, 417–422.
- Manteau S, Lambert B, Jeandet P and Legendre L (2003), Changes in chitinase and thaumatin-like pathogenesis-related proteins of grape berries during the Champagne winemaking process. *American Journal of Enology and Viticulture*, **54**, 267–272.

- Marchal R, Sinet C and Maujean A (1993), Etude des gélamines œnologiques et du collage des vins de base champenois. *Bulletin de l'Office International de la Vigne et du Vin*, **751–752**, 691–725.
- Marchal R, Barret J and Maujean A (1995), Relation entre les caractéristiques physico-chimiques d'une bentonite et son pouvoir d'adsorption. *Journal International des Sciences de la Vigne et du Vin*, **29**, 27–42.
- Marchal R, Bouquelet S and Maujean A (1996), Purification and partial biochemical characterization of glycoproteins in a Champenois Chardonnay wine. *Journal of Agricultural and Food Chemistry*, **44**, 1716–1722.
- Marchal R, Seguin V and Maujean A (1997), Quantification of interferences in the direct measurement of proteins in wines from the Champagne region using the Bradford method. *American Journal of Enology and Viticulture*, **48**(3), 303–309.
- Marchal R, Berthier L, Legendre L, Marchal-Delahaut P, Jeandet P and Maujean A (1998), Effects of *Botrytis cinerea* infection on the must protein electrophoretic characteristics. *Journal of Agricultural and Food Chemistry*, **46**, 4945–4949.
- Marchal R, Liger-Belair G, Berthier L, Brissonnet F, Jeandet P, Maujean A, Robillard B, Morard C, Viaux L and Duteurtre B (1999), Progress in the understanding of Champagne wine foaming properties. In Campbell GM, Webb C, Pandiella S and Niranjan K (eds), *Bubbles in Food*, American Society of Cereal Chemists, Eagan Press, St Paul, MN, 305–315.
- Marchal R, Chaboche D, Douillard R and Jeandet P (2002a), Influence of lysozyme on Champagne base wine foaming properties, *Journal of Agricultural and Food Chemistry*, **50**, 1420–1428.
- Marchal R, Marchal-Delahaut L, Lallement A and Jeandet P (2002b), Wheat gluten used as clarifying agent of red wines. *Journal of Agricultural and Food Chemistry*, **50**, 177–184.
- Marchal R, Marchal-Delahaut L, Michels F, Parmentier M, Lallement A and Jeandet P (2002c), Wheat gluten used as clarifying agent of musts and white wines. *American Journal of Enology and Viticulture*, **53**, 308–314.
- Marchal R, Weingartner S, Voisin C, Jeandet P and Chatelain F (2002d), Utilisation de modèles mathématiques pour optimiser les doses de bentonite gonflée et non gonflée lors du collage des vins blancs. Partie I: Clarification et stabilisation colloïdale. *Journal International des Sciences de la Vigne et du Vin*, **36**, 169–176.
- Marchal R, Lallement A, Jeandet P and Estabiet G (2003), Clarification of Muscat musts using wheat proteins and the flotation technique. *Journal of Agricultural and Food Chemistry*, **51**, 2040–2048.
- Marchal R, Warchol M, Cilindre C and Jeandet P (2006), Evidence for protein degradation by *Botrytis cinerea* and relationships with alteration of synthetic wine foaming properties, *Journal of Agricultural and Food Chemistry*, **54**, 5157–5165.
- Martínez-Rodríguez AJ and Polo MC (2003), Effect of the addition of bentonite to the tirage solution on the nitrogen composition and sensory quality of sparkling wines. *Food Chemistry*, **81**, 383–388.
- Mesquita PR, Piçarra-Pereira MA, Monteiro S, Loureiro VB, Teixeira AR and Ferreira RB (2001), Effect of wine composition on protein stability. *American Journal of Enology and Viticulture*, **52**, 324–330.
- Miedl M, Garcia MA and Bamforth CW (2005), Haze formation in model beer systems. *Journal of Agricultural and Food Chemistry*, **53**, 10161–10165.
- Milisaavljevic, D (1963), Prévention des troubles protéiques du vin par l'emploi des bentonites dans les moûts. *Annales de Technologie Agricole*, **12**, 315–330.
- Miller GC, Amon JM, Gilson RL and Simpson RF (1985), Loss of wine aroma attributed to protein stabilization by bentonite or ultrafiltration. *Australian Grapegrower Winemaker*, **256**, 49–50.
- Moine-Ledoux V and Dubourdieu D (1999), An invertase fragment responsible for improving the protein stability of dry white wines. *Journal of the Science of Food and Agriculture*, **79**, 537–543.

- Moine-Ledoux V and Dubourdiou D (2007), Role of mannoproteins in wine stabilization. In Jeandet P, Clément C and Conreux A (eds), *Macromolecules and Secondary Metabolites of Grapevine and Wine*, Paris, London, New York: Intercept/Lavoisier, 393–401.
- Monteiro S, Picarra-Pereira MA, Teixeira AR, Loureiro VB and Ferreira RB (2003a), Environmental conditions during vegetative growth determine the major proteins that accumulate in mature grapes. *Journal of Agricultural and Food Chemistry*, **51**, 4046–4053.
- Monteiro S, Barakat M, Picarra-Pereira MA, Teixeira AR and Ferreira RB (2003b), Osmotin and thaumatin from grape: a putative general defense mechanism against pathogenic fungi. *Phytopathology*, **93**, 1505–1512.
- Morant I (1990), Evaluation of bentonites available on the Australian market. *Technical Review*, **67**, 16–18, 31–33.
- Moretti RH and Berg HW (1965), Variability among wines to protein clouding. *American Journal of Enology and Viticulture*, **16**, 69–78.
- Muhlack R, Nordestgaard S, Waters EJ, O'Neill BK, Lim A and Colby CB (2006), In-line dosing for bentonite fining of wine or juice: contact time, clarification, product recovery and sensory effects. *Australian Journal of Grape and Wine Research*, **12**, 221–234.
- Murphey JM, Spayd SE and Powers JR (1989), Effect of grape maturation on soluble protein characteristics of Gewürztraminer and White Riesling juice and wine. *American Journal of Enology and Viticulture*, **40**, 199–207.
- Nordestgaard S, Chuan YP, O'Neill B, Waters E, Deans L, Policki P and Colby C (2006), In-line dosing of white wine for bentonite fining with centrifugal clarification. *American Journal of Enology and Viticulture*, **58**, 283–285.
- Oh HI, Hoff JE, Armstrong GS and Haff LA (1980), Hydrophobic interaction in tannin-protein complexes. *Journal of Agricultural and Food Chemistry*, **28**, 394–398.
- Paetzold M, Dulau L and Dubourdiou D (1990), Fractionnement et caractérisation des glycoprotéines dans les moûts de raisins blancs. *Journal International des Sciences de la Vigne et du Vin*, **24**, 13–28.
- Pastorello EA, Farioli L, Pravettoni V, Ortolani C, Fortunato D, Giuffrida MG, Garoffo LP, Calamari AM, Brenna O and Conti A (2003), Identification of grape and wine allergens as an endochitinase 4, a lipid-transfer protein, and a thaumatin. *Journal of Allergy and Clinical Immunology*, **111**, 350–359.
- Peng Z, Waters EJ, Pocock KF and Williams PJ (1996a), Red wine bottle deposits, I: a predictive assay and an assessment of some factors affecting deposit formation. *Australian Journal of Grape and Wine Research*, **2**, 25–29.
- Peng Z, Waters EJ, Pocock KF and Williams PJ (1996b), Red wine bottle deposits, II: cold stabilisation is an effective procedure to prevent deposit formation. *Australian Journal of Grape and Wine Research*, **2**, 30–34.
- Peng Z, Pocock KF, Waters EJ, Francis IL and Williams PJ (1997), Taste properties of grape (*Vitis vinifera*) pathogenesis-related proteins isolated from wine. *Journal of Agricultural and Food Chemistry*, **45**, 4639–4643.
- Pocock KF and Waters EJ (1998), The effect of mechanical harvesting and transport of grapes, and juice oxidation, on the protein stability of wines. *Australian Journal of Grape and Wine Research*, **4**, 136–139.
- Pocock KF and Waters EJ (2006), Protein haze in bottled white wines: how well do stability tests and bentonite fining trials predict haze formation during storage and transport? *Australian Journal of Grape and Wine Research*, **12**, 212–220.
- Pocock KF, Hayasaka Y, Peng Z, Williams PJ and Waters EJ (1998), The effect of mechanical harvesting and long-distance transport on the concentration of haze-forming proteins in grape juice. *Australian Journal of Grape and Wine Research*, **4**, 23–29.
- Pocock KF, Hayasaka Y, McCarthy MG and Waters EJ (2000), Thaumatin-like proteins and chitinases, the haze-forming proteins of wine, accumulate during ripening of grape (*Vitis vinifera*) berries and drought stress does not affect the final levels per berry at maturity. *Journal of Agricultural and Food Chemistry*, **48**, 1637–1643.

- Pocock KF, Høj PB, Adams KS, Kwiatkowski MJ and Waters EJ (2003), Combined heat and proteolytic enzyme treatment of white wines reduces haze forming protein content without detrimental effect. *Australian Journal of Grape and Wine Research*, **9**, 56–63.
- Pocock KF, Alexander GM, Hayasaka Y, Jones PR and Waters EJ (2006), Sulfate – a candidate for the missing essential factor that is required for the formation of protein haze in white wine. *Journal of Agricultural and Food Chemistry*, **55**, 1799–1807.
- Pollnitz AP, Capone DL, Caldersmith MC and Sefton MA (2003), The effect of various wine bottle closures and fining agents on flavour and aroma compounds in wine. In: SM Bell, KA de Garis, CG Dundon, RP Hamilton, SJ Partridge and GS Wall (eds), *Grapes growing at the Edge; Managing the Wine Business; Impacts on Wine Flavour*, Adelaide, SA: Australian Society of Viticulture and Oenology, 59–63.
- Popineau Y and Denery-Papini S (1996), Les protéines de réserve du grain de blé. In Godon (ed.), *Protéines Végétales*, Paris: Lavoisier, 120–172.
- Popineau Y and Pineau F (1987), Investigation of surface hydrophobicities of purified gliadins by hydrophobic interaction chromatography, reverse-phase high performance liquid chromatography and apolar ligand binding. *Journal of Cereal Science*, **5**, 215–234.
- Rankine BC (1963), Principles and pitfalls of winery analysis. *Australian Wine Brewing and Spirit Review*, **81**(6), 22, 24 and **81**(7), 17, 18, 20.
- Rankine BC (1989), *Making Good Wine*, Sydney, NSW: Pan Macmillan.
- Renault A-S, Deloire A and Bierne J (1996), Pathogenesis-related proteins in grapevines induced by salicylic acid and *Botrytis cinerea*. *Vitis*, **35**, 49–52.
- Ribéreau-Gayon J and Peynaud E (1935), La clarification des vins. *Revue Viticole*, **82**, 8–13.
- Ribéreau-Gayon J, Peynaud E, Sudraud P and Ribéreau-Gayon P (1977), *Sciences et Techniques du Vin, Volume IV : Clarification et Stabilisation*. Paris: Dunod.
- Robert N, Roche K, Lebeau Y, Breda C, Boulay M, Esnault R and Buffard D (2002), Expression of grapevine chitinase genes in berries and leaves infected by fungal or bacterial pathogens. *Plant Science*, **162**, 389–400.
- Robinson SP, Jacobs AK and Dry IB (1997), A class IV chitinase is highly expressed in grape berries during ripening. *Plant Physiology*, **114**, 771–778.
- Rolland JM, Apostolou E, De Leon MP, Stockley CS and O’Hehir RE. (2008), Specific and sensitive enzyme-linked immunosorbent assays for analysis of residual allergenic food proteins in commercial bottled wine fined with egg white, milk, and nongrape-derived tannins. *Journal of Agricultural and Food Chemistry*, **56**, 349–354.
- Sahraoui A. (1991), *Etude de la flottation par détente de liquide saturé sous pression. Application à la clarification des moûts*, Doctoral dissertation, University of Toulouse, France.
- Salzman RA, Tikhonova I, Bordelon BP, Hasegawa PM and Bressan RA (1998), Coordinate accumulation of antifungal proteins and hexoses constitutes a developmentally controlled defense response during fruit ripening in grape. *Plant Physiology*, **117**, 465–472.
- Saulnier L and Brillouet JM (1989), An arabinogalactan-protein from the pulp of grape berries. *Carbohydrate Research*, **188**, 137–144.
- Saywell LG (1934), Clarification of wines. *Industrial and Engineering Chemistry*, **26**, 981–982.
- Sbornik M, Rakoski J, Mempel M, Ollert M and Ring J (2007), IgE-mediated type-I-allergy against red wine and grapes. *Allergy*, **62**, 1339–1340.
- Schad SG, Trcka J, Vieths S, Scheurer S, Conti A, Brocker EB and Trautmann A (2005), Wine anaphylaxis in a German patient: IgE-mediated allergy against a lipid transfer protein of grapes. *International Archives of Allergy and Immunology*, **136**, 159–164.
- Senee J, Robillard B and Vignes-Adler M (1999), Films and foams of Champagne wines. *Food Hydrocolloids*, **13**, 15–26.
- Shewry PR, Miles MJ and Tatham AS (1994), The prolamin storage proteins of wheat and related cereals. *Progress in Biophysics and Molecular Biology*, **61**, 37–59.
- Siebert K (1999a), Protein-polyphenol haze in beverages. *Food Technology*, **53**, 54–57.

- Siebert KJ and Lynn PY (2003), Effects of alcohol and pH on protein-polyphenol haze intensity and particle size. *Journal of the American Society of Brewing Chemists*, **61**, 88–98.
- Siebert KJ, Carrasco A and Lynn PY (1996a), Formation of protein-polyphenol haze in beverages. *Journal of Agricultural and Food Chemistry*, **44**, 1997–2005.
- Siebert KJ, Troukhanova NV and Lynn PY (1996b), Nature of polyphenol-protein interactions. *Journal of Agricultural and Food Chemistry*, **44**, 80–85.
- Smart RE and Coombe BC (1983), Water relations of grapevines. In: Kozlowski TT (eds), *Water Deficits and Plant Growth. Vol. 7, Additional Woody Crop Plants*, Academic Press, New York, 137–196.
- Somers TC and Ziemelis G (1973), Direct determination of wine proteins. *American Journal of Enology and Viticulture*, **24**, 47–50.
- Sturgess RP, Ellis HJ and Ciclitira PJ (1991), Cereal chemistry, molecular biology and toxicity in coeliac disease (Progress reports), *Gut*, **32**, 1055–1060.
- Tattersall DB, Pocock KF, Hayasaka Y, Adams K, van Heeswijck R, Waters EJ and Høj PB (2001), Pathogenesis related proteins – their accumulation in grapes during berry growth and their involvement in white wine heat instability. Current knowledge and future perspectives in relation to winemaking practices. In Roubelakis-Angelakis KA (ed.), *Molecular biology and biotechnology of the grapevine*, Kluwer Academic, Dordrecht, the Netherlands, 183–201.
- Vassilopoulou E, Zuidmeer L, Akkerdaas J, Tassios I, Rigby NR, Mills ENC, van Ree R, Saxoni-Papageorgiou P and Papadopoulos NG (2007), Severe immediate allergic reactions to grapes: part of a lipid transfer protein-associated clinical syndrome. *International Archives of Allergy and Immunology*, **143**, 92–102.
- Vincenzi S, Mosconi S, Zoccatelli G, Dalla Pellegrina C, Veneri G, Chignola R, Peruffo A, Curioni A and Rizzi C (2005), Development of a new procedure for protein recovery and quantification in wine. *American Journal of Enology and Viticulture*, **56**, 182–187.
- Waters EJ, Wallace W and Williams PJ (1991), Heat haze characteristics of fractionated wine proteins. *American Journal of Enology and Viticulture*, **42**, 123–127.
- Waters EJ, Wallace W and Williams PJ (1992), Identification of heat-unstable wine proteins and their resistance to peptidases. *Journal of Agricultural and Food Chemistry*, **40**, 1514–1519.
- Waters EJ, Wallace W, Tate ME and Williams PJ (1993), Isolation and partial characterization of a natural haze protective factor from wine. *Journal of Agricultural and Food Chemistry*, **41**, 724–730.
- Waters EJ, Peng Z, Pocock KF, Jones GP, Clarke P and Williams PJ (1994), Solid-state ¹³C nmr investigation into insoluble deposits adhering to the inner glass surface of bottled red wine. *Journal of Agricultural and Food Chemistry*, **42**, 1761–1766.
- Waters EJ, Peng Z, Pocock KF and Williams PJ (1995), Proteins in white wine, II: Their resistance to proteolysis is not due to either phenolic association or glycosylation. *Australian Journal of Grape and Wine Research*, **1**, 94–99.
- Waters EJ, Shirley NJ and Williams PJ (1996), Nuisance proteins of wine are grape pathogenesis-related proteins. *Journal of Agricultural and Food Chemistry*, **44**, 3–5.
- Waters EJ, Hayasaka Y, Tattersall DB, Adams KS and Williams PJ (1998), Sequence analysis of grape (*Vitis vinifera*) berry chitinases that cause haze formation in wines. *Journal of Agricultural and Food Chemistry*, **46**, 4950–4957.
- Waters EJ, Alexander G, Muhlack R, Pocock KF, Colby C, O'Neill BN, Høj PB and Jones PR (2005), Preventing protein haze in bottled wine. *Australian Journal of Grape and Wine Research*, **11**, 215–225.
- Weber P, Steinhart H, Paschke A (2007), Investigation of the allergenic potential of wines fined with various proteinogenic fining agents by ELISA. *Journal of Agricultural and Food Chemistry*, **55**, 3127–3133.
- Yokotsuka K, Ebihara T and Sato T (1991), Comparison of soluble proteins in juice and wine from Koshu grapes. *Journal of Fermentation and Bioengineering*, **71**, 248–253.

- Zalewska-Sobczak J, Borecka H and Urbanek H (1981), Comparison of pectinase, xylanase and acid protease activities of virulent and less virulent isolates of *Botrytis cinerea*. *Phytopathologische Zeitschrift*, **101**, 222–227.
- Zoecklin B (1988), *Bentonite fining of juice and wine*. Publication 463–014, Virginia Cooperative Extension Service.

8

Micro-oxygenation, oak alternatives and added tannins and wine quality

W. J. du Toit, Stellenbosch University, South Africa

Abstract: Micro-oxygenation is a technique whereby controlled amounts of oxygen are introduced into wine to induce certain positive changes, such as an increase in colour, colour stabilization, changes in mouthfeel and reduction in reductive aromas. This technique has not been investigated scientifically in detail in the past, and a number of recent publications have shown its potential to achieve some of these changes. Micro-oxygenation could be used as an alternative to barrel ageing when used in combination with oak products such as staves, chips or exogenous tannins. However, the composition and the effect these products have on the wine depends on a number of factors such as their manufacture, composition and dosage used.

Key words: micro-oxygenation, oxygen, oak staves, oak chips, exogenous tannins.

8.1 Introduction

Micro-oxygenation is a process that was developed to introduce oxygen (O_2) into wine in a controlled manner. This was due to the positive role that small amounts of O_2 could play in the development of the flavour, aroma and phenolic composition of especially red wines during the ageing process. The storage of wine in barrels is an integral part of the ageing regime of many wines, but the cost of oak wood has led to other cheaper ageing techniques, such as micro-oxygenation, alternative oak treatments and tannins additions, being used more in recent years. This chapter will focus on the general effects that micro-oxygenation has on red wine's chemical and microbial composition and how it can be used to bring about

positive changes in the wine. The use of alternative oak treatments, such as chips, staves and exogenous tannins, will also be dealt with together with how these additions to wine can complement and enhance the ageing process. Attention will also be given to how these products are produced and how the winemaker can use them to obtain good-quality wines. Results generated in the field of micro-oxygenation and staves addition to red wine by the Department of Viticulture and Oenology, Stellenbosch University, will also be discussed.

8.2 Basic oxidation reactions and substrates of oxidation in wine

The atmosphere consists roughly of 21% O_2 . It is thus of little surprise that wine can never be completely protected from O_2 . Molecular O_2 is not very reactive in its triplet ground state. However, the addition of electrons from reduced metal ions such as Fe^{2+} can overcome this, which can lead to the O_2 being reduced to superoxide, peroxide and hydroxyl radicals. The formation of peroxide can lead to the formation of hydrogen peroxide (H_2O_2), which is a stronger oxidizing agent than O_2 (Danilewicz, 2003). The hydroperoxyl radicals are not reactive enough to extract hydrogen ions from a number of compounds, but phenolics are the exception. This is due to their good hydrogen donating properties, which leads to the formation of phenolate anions, which can be oxidized to a semi-quinone free radical, which is further oxidized to the corresponding quinone. One of the by-products of the oxidation of phenolic is H_2O_2 , as mentioned before. Hydrogen peroxide can react with ethanol to form acetaldehyde or ethanal. The main substrates for oxidation in wine are thus phenolic compounds. Phenolic compounds basically consist of a six carbon benzene ring with $-OH$ groups attached to it.

In wine, two main groups of phenolic compounds can be found: those originating from the grapes and those originating from oak wood. Phenolics from the grapes consist of non-flavanoids and flavonoids. The former, which have a simpler structure than the flavonoids, consist mainly of hydroxybenzoic and hydroxycinnamic derivatives such as caffeic acid, p-coumaric acid and furanic acids and are the main phenolic compounds in white wines (Margalit, 1997; Monagas *et al.*, 2005).

The second main group of grape-derived phenolics is the flavonoids. This group of molecules basically consists of two phenolic rings attached to a pyran ring. The most important flavonoids in wine are the anthocyanins, flavanols and flavonols. The latter, occurring as kaempferol, quercetin and myricetin, do not play a large role in the sensorial development of wine, due to their low concentrations. However, this is not the case with the other two main groups of phenolic compounds in wine, anthocyanins and flavanols. Flavanols consist of flavan-3-ols (catechins) and flavan-3,4-diols. In young red wines, these compounds can occur as monomers and dimers but, as wines age, they become more polymerized. This leads to these compounds forming condensed or so-called grape tannins when the polymerization has reached a certain extent (Du Toit *et al.*, 2006a).

The other main types of phenolic compounds in red wines are anthocyanins. These compounds are mainly responsible for the colour of red wine. Different types occur in wine, depending on the OH and OCH₃ constitution of the B-ring of the molecule, and they are esterified with glucose at the C₃ position of the molecule. This leads to the occurrence in wine of cyanidin, peonidin, delphinidin, petunidin and malvidin-3-monoglucoside, which can also be acylated with a cinnamic acid derivate (Ribéreau-Gayon *et al.*, 2000; Monagas *et al.*, 2005).

Another important group of phenolic compounds are those originating from oak wood, which can come into contact with the wine in the form of barrels, staves, chips or exogenous tannin additions. These non-volatile oak compounds consist mainly of oak or hydrolyzable tannins. Hydrolyzable tannins contain a polyhydric alcohol of which the hydroxyl-groups have been esterified with gallic acid or hexahydroxydiphenic acid. These tannins are normally not very stable in wine and can undergo acid hydrolyses to yield gallic and ellagic acids. Hydrolyzable tannins play an important role in the oxidation-reduction potential of a wine, due to them being more easily oxidized than grape-derived tannins. The other main substrates for oxidation in wine are ascorbic acid, ethanol, tartaric acid and SO₂. According to Devatine *et al.* (2007), 60% of O₂ in wine can be consumed by phenolics, 20% by ethanol and 10% by SO₂. Sulphur dioxide reacts with H₂O₂, thus preventing it, to a certain extent, from further oxidizing other compounds in wine.

8.3 Basic phenolic reactions in red wine involving oxygen

In a young red wine, such as after the completion of alcoholic fermentation, a large proportion of anthocyanins normally exist in their free monomeric forms. The compound mainly responsible for the red colour of red wine is the flavylium ion, which has a positive charge on the C-ring. These ions exist in equilibrium with the colourless carbinol base, the blue quinonic base, the yellow chalcone and colourless flavenesulfonate (Hrazdine and Franzese, 1974). In a young red wine, up to 50% of the anthocyanins can exist in the colourless form. In young red wine, free anthocyanins often associate through copigmentation with compounds such as amino acids, catechins and even other anthocyanins, which help to increase colour. However, during ageing, copigmentation starts to play a less important role and anthocyanins become more incorporated into a polymeric form of colour. During fermentation, reactions can take place between the nucleophilic C6 or C8 carbons of (+)-catechin, (–)-epicatechin or procyanidins and the electrophilic C4 carbon of the anthocyanin molecule. The resulting colourless flavene can be oxidized with O₂ addition to the corresponding flavylium ion and further into yellow xanthylum salt (Liao *et al.*, 1992; Santos-Buelga *et al.*, 1999; Ribéreau-Gayon *et al.*, 2000; Du Toit *et al.*, 2006a).

As mentioned before, acetaldehyde, originating from the oxidation of phenolic compounds, is carboxylated to the electrophile ethanal in an acid media such as wine. This compound will then react with the nucleophilic C6 and C8 positions of a flavanol molecule. After the loss of water, this moiety can react with the

electrophilic C8 of an anthocyanin, with subsequent protonation leading to the formation of a coloured pigment. It also seems that the position C6 of the anthocyanin can also be involved with this reaction. Trimeric and tetrameric pigments have been identified, but only position C8 of the anthocyanin molecule can be involved in this reaction, with the polymerization ceasing when the anthocyanin forms the two terminal products of the chain. Cinnamic acid derivatives and vinylphenols can also associate with anthocyanins. This is due to an electrophilic cyclo-addition of the ethylenic bond of the 4-vinylphenol molecule with C4 and C5 carbons of the anthocyanin, with subsequent oxidation leading to a pyrane ring. Grape-derived cinnamic acid's reaction with anthocyanins also leads to hydroxyphenyl pyranoanthocyanin, such as the formation of Pinotin A in Pinotage wines. Other pyranoanthocyanins are Vitisins A and B, which are formed due to the addition of pyruvate and acetaldehyde, respectively (Cano-López *et al.*, 2006; Du Toit *et al.*, 2006a). These pyranoanthocyanins and ethyl linked pigments contribute to the increase in the brownish, red and purple colour, respectively, of older red wine during ageing and are also more resistant towards SO₂ bleaching, pH changes and oxidation than monomeric anthocyanins (Morata *et al.*, 2003; Schwarz *et al.*, 2003).

These reactions normally take place when wine is matured in an oak barrel, where a small amount of O₂ is introduced by racking, topping up of the barrels and permutation of air through the staves. It is well known that when red wine is aged in an oak barrel compared to a steel tank with minimal O₂ pick-up, those aged in the barrel often have a higher colour density (Ribéreau-Gayon *et al.*, 2000).

In South African Shiraz and Pinotage red wines stored in different barrels, it was found that the wines normally achieve maximum colour density after 6–9 months of barrel age. During this period of time, the total red pigments decreased, but the percentage of pigments in the red form increased from 15 to 45%. All these reactions led to an increase in colour density, a stabilization of colour, as well as a decrease in free anthocyanins (Fourie, 2005).

During barrel ageing, it is mainly the acetaldehyde-induced polymerization that contributes to the polymerization of flavanols with each other. However, direct C4–C8 and C4–C6 polymerization reactions between procyanidin molecules produce products that are more reactive towards proteins and are hence more astringent than those formed from acetaldehyde-induced condensation reactions. Both types of reactions produce procyanidins with a limit of 8 or 10 flavan units. The interaction of anthocyanin molecules with procyanidins can also influence the taste of wine because they can form the terminal subunits, thus preventing further polymerization (Ribéreau-Gayon *et al.*, 2000; Monagas *et al.*, 2005). The tannin constitution of red wine will thus also differ due to O₂ addition to wine. Oak tannins play an important role in these reactions, as the oxidation of hydrolyzable tannins yields more acetaldehyde than from grape-derived tannins (Vivas and Glories, 1996).

8.4 When does oxygen come into contact with wine?

Oxygen comes into contact with wine throughout the whole production process.

When grapes are crushed, the juice can easily pick up 3–5 mg/L O_2 . Oxygen can also be added during fermentation, especially in red wine production. This is normally done to enhance membrane stability of the yeast towards high alcohol concentrations, as well as phenolic development. For the former, a dosage of 5 mg/L O_2 is optimal to achieve this when added at the end of the cell growth phase. Oxygen can be added with a splash racking, as well as artificially by pumping air or oxygen into the must. Other winemaking operations such as pumping (about 2 mg/L), transfer from tank to tank (up to 6 mg/L), filtration (4–7 mg/L), racking (3–5 mg/L), centrifugation (up to 8 mg/L), bottling (0.5–3 mg/L) and barrel ageing (20–45 mg/L/year), add more O_2 to the wine. During barrel ageing, the humidity of the wood and the thickness and the grain of the staves all play a role (Du Toit *et al.*, 2006a). Oxygen can also be introduced in a controlled manner to wine by a process called micro-oxygenation, which will now be discussed in more detail.

8.5 Micro-oxygenation

When wine is stored in stainless steel tanks, very little O_2 comes in contact with it, as would have happened had it been stored in a barrel. Patrick Ducournau and Thierry Lemaire (Parish *et al.*, 2000) developed this technique of micro-oxygenation where O_2 is bubbled into wine to overcome this limitation. The O_2 is supposed to bring about positive changes to the wine, according to some of the producers of micro-oxygenation machines. Some of these claims have been proven, but more research is needed to justify them scientifically; benefits claimed include an increase in colour density and decreases in astringency, sulphur off-odours, green, vegetative aromas and production costs. Stainless steel tanks in conjunction with micro-oxygenation and alternative wood products can be used to simulate an oak barrel. Wine is also supposed to become ready for market sooner when micro-oxygenation is used. Although micro-oxygenation has been used worldwide in wineries since the 1990s, only recently have scientific manuscripts in peer-reviewed journals been published on this subject (Du Toit *et al.*, 2006b). The reason for this is probably that the large-scale tanks, volumes of wine and micro-oxygenation equipment necessary to perform experiments on this subject make establishment of replicated trials cost- and space-prohibitive (Du Toit *et al.*, 2006b).

Micro-oxygenation is achieved by filling a known volume with gas and transferring it via a low-pressure circuit into the wine. The O_2 is normally dispersed in the form of small bubbles into the wine with a sparger made of either stainless steel or ceramic. The addition of the O_2 should, however, be controlled effectively to prevent too high additions of O_2 , which could lead to accumulation of O_2 inside or in the headspace of the wine, with subsequent microbial and/or oxidative spoilage. The O_2 dosage should thus be just lower than the wine's ability to consume it. Dosages normally recommended depend on the stage of winemaking, phenolic composition of the wine and desired outcome of the treatment. Oxygen additions during micro-oxygenation are normally in the range of 2–5 mg O_2 /L/

month after malolactic fermentation for colour and phenolic development (some oenologists even recommend up to 10 mg O₂/L/month). Oxygen transfer can be influenced by the concentration of CO₂ present in the wine (Devatine *et al.*, 2007).

However, it is recommended that O₂ be added at around 2–4 mg O₂/L/day before malolactic fermentation for a few days, a system which is apparently very effective in treating especially astringent, harsh tasting press fraction red wine, as well as leading to lower colour loss during malolactic fermentation (Llaudy *et al.*, 2006; Pérez-Magariño *et al.*, 2007). For the removal of sulphur off-odours, smaller dosages can probably be used. Literature is not very clear on how these dosages were decided upon, but they could simulate those quantities coming into contact with a wine on a monthly basis if it was stored in an oak barrel (20–45 mg O₂/L/year, which equates to about 1–4 mg O₂/L/month). According to Moutounet (2003), the maximum amount of O₂ a red wine can consume without acquiring an unwanted oxidation aroma is 50 mL/L, although the capacity of certain red wines for O₂ may be higher (Singleton, 1987).

8.5.1 Effect of micro-oxygenation on the chemical composition of red wine

A number of studies have investigated the effect of micro-oxygenation on the phenolic composition and colour development of red wine. It is known that during malolactic fermentation, the colour and phenolic composition of red wine often decreases, which is not solely due to an increase in pH. Micro-oxygenation has been proposed as a possible means to stabilize colour to decrease this loss during malolactic fermentation (Ribéreau-Gayon *et al.*, 2000; Pérez-Magariño *et al.*, 2007). It seems that in certain red wines, micro-oxygenation led to lower monomeric anthocyanins, an increase in colour density, and colour hue, with an increase in ethyl linked pigments and proanthocyanins, which could lead to more stable colour (Cano-López *et al.*, 2006, 2008; Du Toit *et al.*, 2006b; Fell *et al.*, 2007; Tao *et al.*, 2007). Ethyl linked compounds, consisting of anthocyanins and flavanols, are often unstable and can break down to form reactive ethyl–flavanol units, which may react again with dimers or other anthocyanins to form larger polymers (Cano-López *et al.*, 2008). However, micro-oxygenation does not always seem to lead to an increase in colour density and a loss in monomeric anthocyanins, as certain authors found a decrease in red colour with O₂ addition and higher levels of monomer anthocyanins in microoxygenated-treated wines (Atanasova *et al.*, 2002; Du Toit *et al.*, 2006b; Llaudy *et al.*, 2006). Higher concentrations of free anthocyanins in wines receiving micro-oxygenation could possibly be due to the instability of anthocyanin–ethyl–flavanol compounds where the ethyl link is severed, thereby releasing the free anthocyanins (Llaudy *et al.*, 2006). These tendencies probably depend on the phenolic composition of the wine, its age, and SO₂ concentration, and when and how much O₂ has been added. Tao *et al.* (2007) found that high levels of SO₂, such as 100–200 mg/L led to lower levels of colour pigment formation, with subsequent bleaching of the colour by the SO₂. However, this process is reversible with O₂ additions.

Electrochemical micro-oxygenation, in contrast to classical micro-oxygenation, led to an increase in bound SO_2 , which correlated with the production of acetaldehyde (Fell *et al.*, 2007). Du Toit *et al.* (2006b) also found the difference between the colour and modified colour densities in some wines receiving micro-oxygenation becoming smaller, indicating a resistance of the colour towards the bleaching effect of SO_2 . This was, however, not the case with all wines, and it seems that it is more effective in young than older red wines. Micro-oxygenation in one study was also found to be more effective in forming anthocyanin-derived pigments in red wines with a higher phenolic content when applied before malolactic fermentation (Cano-López *et al.*, 2008). It should also be kept in mind that often manuscripts will indicate a statistical increase in colour density as measured with a spectrophotometer due to O_2 additions, but this increase often amounts to very little in overall winemaking terms and can not be distinguished with the naked eye, raising the question whether it would be observed by the eventual consumer of the wine.

Some of our own research also focussed on the effect of adding O_2 to red wine during extended skin maceration after alcoholic fermentation. This technique is also starting to be used more often in South Africa. Winemakers claim that it reduces a large percentage of the so-called 'green tannins' in the wine. When one does this, one should be careful of volatile acidity development due to the possible growth of acetic acid and other spoilage bacteria. It is probably better to not place the sparger inside the yeast lees when applying micro-oxygenation, as the capacity of yeast lees to react with O_2 is actually higher than that of phenolic compounds (Du Toit *et al.*, 2006a).

As mentioned before, micro-oxygenation could also influence the tannin composition and structure of red wine, due to ethyl linkages between catechin moieties (Drinkine *et al.*, 2005). The effect of micro-oxygenation on the tannin and total phenolic content of red wine is not clear, with contradictory findings occurring in literature. Du Toit *et al.* (2006b) and Cano-López *et al.* (2008) found very little change in tannin and total phenol concentrations during micro-oxygenation treatment. Llaudy *et al.* (2006) found no significant differences between proanthocyanidin, total phenols measured at 280 nm, or in the DMAC index, although the astringency of the treated wine was 35% less. In contrast to this, Tao *et al.* (2007) found an increase in tannin concentration as measured with bovine serum albumin (BSA) or methyl cellulose precipitation during micro-oxygenation in wine with lower SO_2 concentrations. Where higher SO_2 concentrations were present, this did not occur, which is probably due to larger tannin colour pigment moieties which can react with the BSA or methylcellulose at lower SO_2 concentrations. All these changes can, depending on the level of monomeric flavanols, lead to either a reduction or increase in mean degree of polymerization (mDP) during micro-oxygenation treatment, which would also affect the astringency of the wine (Monagas *et al.*, 2005; Tao *et al.*, 2007). Red wine receiving micro-oxygenation can go through structuring and harmonization phases (Parish *et al.*, 2000). The former leads to the wine actually becoming more astringent and harsh and can last from 1 to 6 months. During the subsequent harmonization phase, a decrease in

astringency is observed, with the wine becoming more complex. These observations are, however, only observations and should be investigated further (Du Toit *et al.*, 2006b).

The ideal time to terminate the micro-oxygenation is when the wine has reached optimal softness, as over-oxidation might lead to a mDP of procyanidins that is too high, with a resulting increase in bitterness and with tannins becoming too dry on the taste (Parish *et al.*, 2000; Nikfardjam and Dykes, 2003; Cano-López *et al.*, 2006; Du Toit *et al.*, 2006b). The anthocyanin-to-tannin ratio of red wine should also influence the effectiveness of micro-oxygenation, with high enough concentrations of anthocyanins being available to assist with reducing astringency of tannins by polymerization through micro-oxygenation (Du Toit *et al.*, 2006b; Tao *et al.*, 2007).

When the available anthocyanins and flavanols begin to be depleted due to ethanal induced polymerization, acetaldehyde can begin to accumulate in the wine (Carlton *et al.*, 2007). Other phenolic compounds due to decrease during micro-oxygenation include quercetin (Fell, 2007) and catechin (Du Toit *et al.*, 2006b; Pérez-Magariño *et al.*, 2007). Polymeric phenols as measured at 280 nm peak with high-performance liquid chromatography (HPLC) ranged from 783 mg/L, 947 mg/L, 1022 mg/L and 1132 mg/L in, respectively, the control, 1.5 mg/L/O₂, 3 mg/L/O₂ and barrel-matured South African Pinotage wines which underwent micro-oxygenation for 24 weeks. This can be an indication that the 3 mg/L/O₂ dosage was on a par with that O₂ coming in contact with the wine matured in the barrel (Du Toit *et al.*, 2006b). Electrochemical micro-oxygenation is still in its infancy, but has been found to alter certain monomeric anthocyanins and free quercetin (Fell *et al.*, 2007). Cano-López *et al.* (2007) found red wines that underwent micro-oxygenation treatment to have higher colour intensity and SO₂-resistant pigments after 6 months bottle ageing. It is, however, not well known if these differences would be observed by the general customer.

Du Toit *et al.* (2006b) found a tasting panel to be able to statistically distinguish between the control and 3 mg/L/month treatment after 8 weeks, but not the 1.5 mg/L/month treatment from the control at this stage. However, after 12 weeks of micro-oxygenation treatment, they could distinguish between all three treatments. The same author also found a panel to least prefer a Pinotage wine in which the micro-oxygenation treatment has been applied for too long a period of time, leading to an increase in *Brettanomyces* counts as stated in the following section with an increase in medicinal and oxidized flavours.

Micro-oxygenation was also found to reduce unwanted sulphur compounds such as methyl and ethyl mercaptans to below their sensory threshold limits (McCord, 2003). Micro-oxygenation can also lead to vegetative, green wines becoming fruitier, but does not seem to reduce the hexanol and Z-3-hexenol concentrations, thought to be associated with these flavours and the herbaceous notes (Sullivan, 2002; Hernandez-Orte *et al.*, 2009).

Little research has been conducted on the effect of micro-oxygenation on the volatile composition of wine. McCord (2003) investigated the effect of micro-oxygenation on extractable wood components, but did not find any significant

differences between the control and treated wines. Heras *et al.* (2008) found wines treated with micro-oxygenation to extract less furfural, *cis*-whiskey lactones and eugenol, but enhanced the extraction of vanillin and guaiacol from American oak barrels. Micro-oxygenation when applied before malolactic fermentation led to larger differences in certain volatile compounds such as terpenes and ethyl esters just after the treatment, as well as vanillin concentrations after 4 months ageing in barrels. However, these differences became smaller with prolonged ageing (Hernandez-Orte *et al.*, 2009).

8.5.2 Effect of micro-oxygenation on the microbiology of red wine

Very little research has been conducted on the effect of micro-oxygenation on the microbial composition during ageing, an issue often raised by winemakers. *Brettanomyces* and acetic acid bacteria are both well known spoilage microorganisms of wine: the growth of both can be favoured by O₂ addition (Du Toit *et al.*, 2005). *Brettanomyces* are well known to produce volatile phenols such as 4-ethyl-phenol and 4-ethyl-guaiacol, which can render medicinal, horse sweat character at excessive concentrations in the wine. Acetic acid bacteria can transform ethanol into acetic acid, thereby raising the volatile acidity concentration of the wine (Du Toit and Lambrechts, 2002; Du Toit and Pretorius, 2002).

With selective plating, Du Toit *et al.* (2006b) found acetic acid bacteria to survive up to 20 weeks in wines which had undergone micro-oxygenation (10² cfu/mL) and were stored in barrel (10⁴ cfu/mL), whereas they disappeared after 4 weeks in the same control wine from a initial number of 10⁵ cfu/mL. Acetic acid bacterial numbers thus decreased in all the wines in general, with no increase in volatile acidity concentration (Du Toit *et al.*, 2006b). However, it should be kept in mind that acetic acid bacteria can go into a viable, but not culturable state in

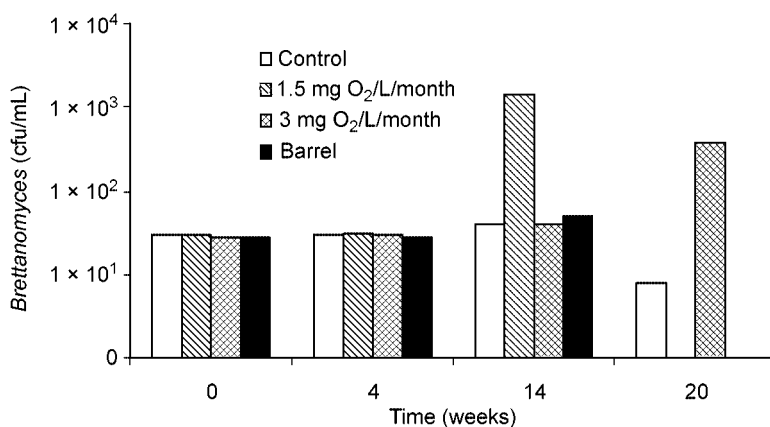


Fig. 8.1 *Brettanomyces* numbers in South African Pinotage during ageing in a control tank, a tank receiving 1.5 mg O₂/L/month micro-oxygenation, a tank receiving 3 mg O₂/L/month and a barrel (Du Toit *et al.*, 2006b).

conditions of low oxygen and high SO_2 concentrations, giving a possible under-estimation of their numbers (Du Toit *et al.*, 2005).

In Fig. 8.1 the *Brettanomyces* counts during the course of this experiment can be seen. It is clear that the counts stayed basically at 10^1 cfu/mL for the first 4 weeks, but then increased to more than 10^3 cells/mL in the 1.5 mg O_2 /L/month treatment after 14 weeks. At this time the free SO_2 concentration decreased in this tank to 18 mg/L and it was then increased to 35 mg/L, leading to a decrease in *Brettanomyces* cell counts in this tank. The counts in the 3 mg O_2 /L/month tank increased, with a correlating decrease in the free SO_2 (17 mg/L) at week 20 (Du Toit, 2006b).

Therefore, micro-oxygenation does not seem to support the growth of acetic acid bacteria but possibly supports their survival. With the permeation of O_2 through oak staves (Vivas *et al.*, 2003), small amounts of O_2 coming into contact with wine could hence support the survival of acetic acid bacteria and *Brettanomyces*. The judicious use and monitoring of SO_2 in combination with micro-oxygenation is important to prevent this, especially in the case of *Brettanomyces*. The addition of SO_2 to wines that have received O_2 and in which *Brettanomyces* growth has occurred is an option for the winemaker to prevent unwanted further spoilage. Overly high SO_2 concentrations could, however, lead to inhibition of favourable phenolic polymerization.

8.6 Recent micro-oxygenation research at the Department of Viticulture and Oenology, Stellenbosch University

We decided to investigate the effect of micro-oxygenation on South African wine quality and composition over a few years. The aims were firstly to ascertain the usage of micro-oxygenation in South African cellars and secondly to investigate the changes brought about by micro-oxygenation in red wines under a more scientific study. South African wine producers will sometimes use micro-oxygenation for insufficient periods of time. Some wine producers only add 0.5 mg O_2 /L/month for a few weeks. It is very doubtful if this will bring about positive changes to the wine, especially in red wines rich in phenolic compounds. Some wine producers were not concerned too much about establishing a specific dosage of O_2 for a wine; they preferred to give a constant dosage of 4 mg/L/month and then decide when to terminate the treatment by tasting the wine regularly.

We also decided to investigate the effect of adding different dosages of O_2 with micro-oxygenation in combination with oak staves and ascertain its effect on the phenolic, colour, sensory and volatile composition of different South African red wines. A previous study by Du Toit *et al.* (2006b) concluded that micro-oxygenation can be used in conjunction with oak staves to simulate an oak barrel, but the type of wood used is very important. We thus decided to first establish a stave dosage which would mimic that of an oak barrel and then to investigate the effect of using different types of staves at this dosage.

8.6.1 Method

A 2006 Cabernet Sauvignon and a 2007 Pinotage were used for these experiments. Both wines were prepared from standard practices, and treatments started soon after the completion of malolactic fermentation. American oak wood, each time from the same batch, was used in these experiments. The Department of Viticulture and Oenology, Stellenbosch University developed a method, in conjunction with a major South African cooper (Radoux), to toast oak wood destined for oak barrels or as staves in roughly the same manner. This entails toasting the staves destined for alternative products on both sides, as the wines in a barrel are in contact with the toasted sides of the barrel staves.

The internal surface area of a 225 L barrel was used as a standard to determine what surface area of oak staves should be added to the first wine (2006 Cabernet Sauvignon). Four 1000 L tanks were then filled with this wine and to two of them American oak staves were added at 40% of the internal surface contact area of a 225 L barrel and to two of them 100% of this surface contact area. The 40% was the recommendation of the supplier to roughly simulate a new oak barrel, as extraction was thought to be more in staves than from a barrel, with the 100% contact surface area used as a control. To one each of the 40% and 100% stave tanks micro-oxygenation (using equipment of Parsec) was applied to simulate the oxygen pick-up which would occur in a barrel at 4–5 mg/L/month O_2 (treatments 40%mox and 100%mox). To the other two tanks, no oxygen was added (treatments 40%C and 100%C). The same wine was also stored in two new 225 L barrels made from the same American wood as used to produce the staves. The same wine was also matured in an old third fill 225 L barrel (Table 8.1).

A similar experiment was also performed on a 2007 Pinotage, but in this experiment all four 1000 L tanks had a stave dosage of 40%. To two of the tanks micro-oxygenation was applied at 4–5 mg O_2 /L/month and to two others, none. One micro-oxygenation tank and one control tank received staves produced and toasted from wood normally used to produce alternative oak products. The other two tanks received staves made from the same batch of wood used to produce the new oak barrel used in the experiment. All oak used in these experiments was American oak (Table 8.1).

During the treatment of the wine, its colour and other phenolic characteristics were analyzed with a spectrophotometer. These included colour density, modified colour density (where the bleaching effect of SO_2 is negated by acetaldehyde addition), total red pigment, SO_2 -resistant pigments, total tannins, total anthocyanins and total phenolics. HPLC analyses for other individual phenolic compounds, such as catechin, cinnamic acid derivatives, individual anthocyanins, dimers B1 and B2, polymeric pigments, polymeric phenolics, etc. were also performed as described in Du Toit *et al.* (2006b). Certain volatile oak compounds, such as eugenol, lactones, furfural, etc. were also analyzed with gas chromatography–mass spectroscopy (GC–MS), as well as standard wine analyses such as SO_2 , alcohol, volatile acidity (VA), etc.

Table 8.1 Different oak and micro-oxygenation treatments for the 2006 Paarl Cabernet Sauvignon and Paarl 2007 Pinotage wines (100% and 40%: oak staves were added to the wine at 100% or 40% of the internal surface of a 225 L barrel

Cultivar and code	Treatment
<i>2006 Cabernet Sauvignon</i>	
40% C	Oak staves, produced from the same batch of oak and toasted in the same manner (medium) as the new American oak barrel, were added to the wine at 40% of the internal surface of a 225 L barrel. No micro-oxygenation (C) was applied.
100% C	Oak staves, produced from the same batch of oak and toasted in the same manner (medium) as the new American oak barrel, were added to the wine at 100% of the internal surface of a 225 L barrel. No micro-oxygenation (C) was applied.
40% mox	Oak staves, produced from the same batch of oak and toasted in the same manner (medium) as the new American oak barrel, were added to the wine at 40% of the internal surface of a 225 L barrel. Micro-oxygenation (mox) was applied at 4 mg O ₂ /L/month.
100% mox	Oak staves, produced from the same batch of oak and toasted in the same manner (medium) as the new American oak barrel, were added to the wine at 100% of the internal surface of a 225 L barrel. Micro-oxygenation (mox) was applied at 4 mg O ₂ /L/month.
NB	New barrel (225 L, American oak, medium toast).
<i>2007 Pinotage</i>	
N C	Oak staves produced from wood normally used to produce alternative oak products such as staves and toasted in an oven were added to the wine at 40% of the internal surface of a 225 L barrel. No micro-oxygenation (C) was applied.
B C	Oak staves produced from wood normally used to produce barrels and toasted in the same manner (medium) were added to the wine at 40% of the internal surface of a 225 L barrel. No micro-oxygenation (C) was applied.
N mox	Oak staves produced from wood normally used to produce alternative oak products such as staves and toasted in an oven were added to the wine at 40% of the internal surface of a 225 L barrel. Micro-oxygenation (mox) was applied at 4 mg O ₂ /L/month.
B mox	Oak staves produced from wood normally used to produce barrels and toasted in the same manner (medium) were added to the wine at 40% of the internal surface of a 225 L barrel. Micro-oxygenation (mox) was applied at 4 mg O ₂ /L/month.
NB	New 225 L American oak barrel made from the same batch of wood used to produce the B staves (225 L, American oak, medium toast).
OB	Old third fill barrel.

C, control, no O₂ addition; mox, micro-oxygenation at 4 mg O₂/L/month; NB, new barrel; OB, old barrel.

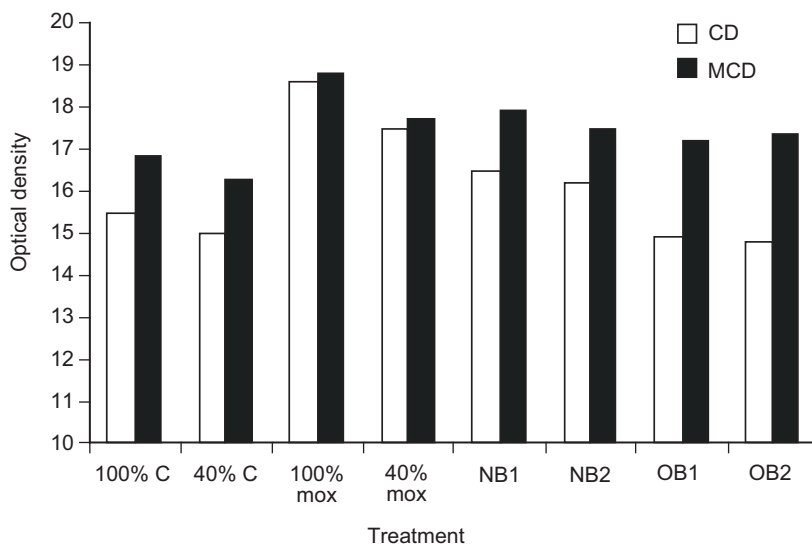


Fig. 8.2 Colour density (CD) and modified colour density (MCD) of the 2006 Cabernet Sauvignon wines after 180 days of treatments (100% and 40%, oak staves were added to the wine at 100% or 40% of the internal surface of a 225 L barrel; mox, micro-oxygenation at 4 mg/L/month; C, control, no O₂ addition; NB, new barrel; OB, old barrel).

8.6.2 Results

During the course of the experiment, the colour density (CD) became higher in the 2006 Cabernet Sauvignon wines to which oxygen was added than in the control. Wines to which a 100% oak stave contact was added also had a higher CD. The CD of the 40% oak treatment was roughly the same as that of a new barrel after 180 days of treatment (Fig. 8.2). Surprisingly enough, the modified colour densities (MCD) were also higher in the wines receiving oxygen and stored in barrels. The difference in the CD and MCD was smaller in the wines receiving oxygen and in the new barrels, due to the oxygen addition and higher concentrations of hydrolyzable tannins being released into the wine, which catalyzes pigment formation (Vivas *et al.*, 2001). The smaller difference in CD and MCD in these treatments indicated the colour becoming less prone to the bleaching effect of SO₂. This was also reflected in the anthocyanin concentration of the wines (Fig. 8.3), where the wines receiving higher concentrations of oak contact, and especially oxygen, had a lower value of anthocyanins, with the old barrel having the highest concentration of anthocyanins.

The colour fractions after 180 days also yielded interesting results (data not shown). Wines which had received the oxygen and 100% oak contact had the lowest proportion of free anthocyanins, with the highest proportion of polymeric pigment, thus rendering them more resistant towards the bleaching effect of SO₂, which was on par with that of one of the wines matured in the new oak barrel. The colour composition of the other barrel, however, differed, showing the variability that could occur between barrels from the same producer. Anthocyanins are

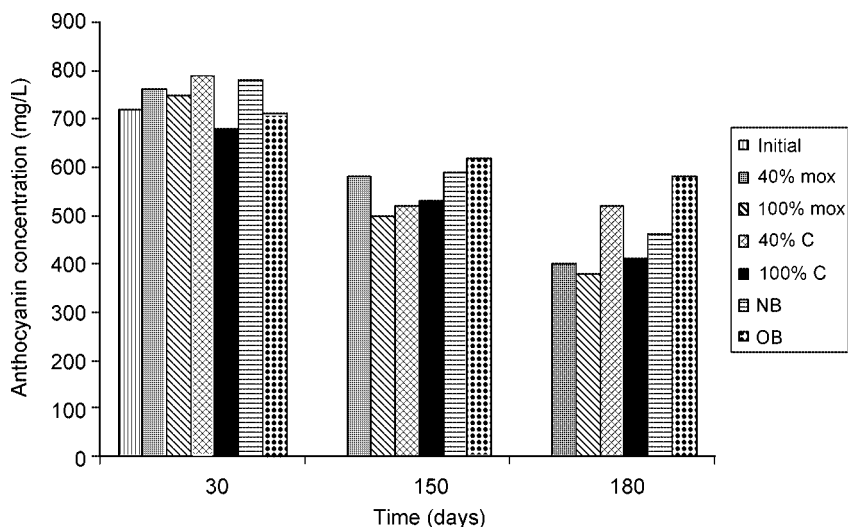


Fig. 8.3 Total anthocyanin concentrations of the 2006 Paarl Cabernet Sauvignon wines during the course of the different micro-oxygenation and oak treatments (100% and 40%, oak staves were added to the wine at 100% or 40% of the internal surface of a 225 L barrel; mox, micro-oxygenation at 4 mg O₂/L/month; C, control, no O₂ addition; NB, new barrel; OB, old barrel; Initial, initial anthocyanin concentration of the wine at the start of the trial).

normally incorporated into polymeric pigments in older red wine, which showed that the ageing development was accelerated in this wine.

Total red pigments were also lower in the stave and new barrel treated wines (ranging from 22 to 23.6 OD units) compared to that stored in an old barrel (27.8 OD units) after 180 days. However, the percentage of pigments in the red form was higher in these wines (around 80% for the micro-oxygenated wines, 75% for the new barrel wines, 70% for the control treatments in tanks) than those stored in old barrels (63%), explaining the higher colour density of these wines. This is typical of what happens with red wine in a new barrel, where colourless anthocyanins are incorporated into coloured pigments. This percentage was however slightly higher in the O₂ treated wines than in the wines matured in the oak barrel, which could be due to slightly excessive oxygen addition added to this wine. This highlights the need for further research regarding the role and addition of O₂ in wine.

Different colour analyses on the 2007 Pinotage wine revealed a slight increase during the course of the experiment and that wines stored in new barrels or receiving micro-oxygenation, yielded slightly higher colour densities (as in the N treatment), but these differences were very small (ranging from 20.3–21.2) in general winemaking terms. This correlates with previous findings (Du Toit *et al.*, 2006b) that have shown that micro-oxygenation often does not increase the colour drastically when used after malolactic fermentation on a wine with an already high colour density (> 18). Total red pigments were slightly lower in the wine receiving micro-oxygenation and very little difference was observed in the degree of red colour between the different treatments, although it increased from 25% to around

Table 8.2 Concentrations of different phenolic compounds (mg/L) of the 2006 Paarl Cabernet Sauvignon red wine after 200 days of different micro-oxygenation and oak treatments (100% and 40%: oak staves were added to the wine at 100% or 40% of the internal surface of a 225 L barrel)

Compound	Treatment				
	40% C	40% mox	100% C	100% mox	NB
Gallic acid	43.8	42.7	45.6	43.5	49
Gallocatechin	14.1	15.2	4.7	12.5	15
Epigallocatechin	6.6	9.5	5.8	8.5	10.6
Catechin	61.3	58.5	48.6	54.7	60
B1	71.1	66.5	56.9	55.9	67.2
Epicatechin	72.9	68.8	61.6	30.1	45
B2	64.9	56.6	57.9	50.8	77.2
Epicatechingallate	6.9	7.3	6.5	21.7	6.3
Total anthocyanins	206.5	159.7	144.4	118.3	135
Malvidin-3-gluc	106.6	86.3	86.6	65.8	76.4
Polymeric pigments	73.3	85.1	70.6	99	35.2
Polymeric phenols	1757.3	1801.8	1478.2	1913.4	746

Table 8.3 Concentrations of different phenolic compounds (mg/L) of the 2007 Paarl Pinotage red wine after 180 days of different micro-oxygenation and oak treatments

Compounds	Treatment					
	N C	N mox	B C	B mox	NB	OB
Gallic acid	8.3	3.6	9.9	4.6	10.1	25.1
Gallocatechin	2.7	7.5	2.6	3.6	9.5	7.1
Epigallocatechin	10.2	10.7	7.7	7.2	10.7	7.6
Catechin	38.2	39.2	32.4	29.0	43.3	50.3
B1	46.8	62.4	47.5	97.0	76.3	51.1
Epicatechin	35.3	0.0	33.9	31.5	12.0	10.7
B2	7.3	3.3	5.8	11.8	2.0	11.2
Epicatechingallate	3.6	11.7	0.0	0.0	13.2	12.1
Total anthocyanins	121.6	93.5	116.5	97.0	88.9	176.9
Malvidin-3-gluc	58.4	44.4	55.5	46.6	41.8	86.1
Vitisin A	3.6	3.6	3.5	3.6	3.7	3.9
Polymeric pigments	128.7	124.2	128.2	132.5	124.9	125.5
Polymeric phenols	1183.5	1101.2	1227.9	1198.9	1113.9	1018.5

Mox, micro-oxygenation applied at 4 mg O₂/L/month; C, control tanks, receiving no O₂; NB, wine matured in new barrel; OB, wine matured in old barrels; N refers to staves added to the wine from wood normally used for alternative oak products and B to staves made from wood normally used to produce barrels (oak staves were added at the equivalent of 40% of the internal surface of a 225 L barrel).

45% from day 45 to 180 in all the treatments (data not shown). Anthocyanin concentrations started to decrease after 45 days in the micro-oxygenated wines and were about 25% lower in the micro-oxygenated wines after 180 days than in the other treatments. Colour hue did not differ significantly between all the treatments in both the Cabernet Sauvignon and the Pinotage (data not shown).

Table 8.4 Free SO₂ concentrations (mg/L)^a of the 2006 Paarl Cabernet Sauvignon red wine during the course of different micro-oxygenation and oak treatments. Values in brackets indicate the amount of SO₂ added to the wine after the analyses

	Free SO ₂ concentration (mg/L) at:					
	30 days	50 days	75 days	100 days	130 days	160 days
100% mox	26	21	17 (10)	17 (10)	17 (10)	20
40% mox	27	24	22 (10)	21 (10)	21 (10)	25
100% C	35	35	28	24 (10)	25 (10)	33
40% C	35	35	28	24	25 (10)	33
NB	31	29	25	29	25 (10)	28
OB	39	40	37	36	34	30

^aInitial SO₂ concentration: 35 mg/L.

100% and 40%, oak staves were added to the wine at 100% and 40% of the internal surface of a 225 L barrel; mox, micro-oxygenation at 4 mg O₂/L/month; C, control, no O₂ addition; NB, new barrel; OB, old barrel.

In Tables 8.2 and 8.3 the HPLC analyses of certain individual phenolic compounds performed on the 2006 Cabernet Sauvignon and 2007 Pinotage are shown. Micro-oxygenation seems to have led to lower concentrations of epicatechin and the dimers B1 and B2. Micro-oxygenation also led to lower concentrations of malvidin-3-glucoside and total anthocyanins, although not to the same degree as with a new barrel. Higher oak dosages also decreased the malvidin-3-glucoside concentration. The same tendencies were also observed for delphinidin-3-glucoside and other anthocyanins (data not shown). Micro-oxygenation increased the polymeric pigment and phenol concentration, which was surprisingly the lowest in this case in the wine matured in the new barrel. This can be explained by the Bayer reaction, where oxygen addition to phenolics leads to the production of small amounts of acetaldehyde, which can form a bridge between phenolic compounds. It is believed that aldehydes originating from oak can also have this function, explaining why the same happened to a certain extent in the control wines receiving no oxygen. The Pinotage was also analyzed after 180 days for these compounds with HPLC, but no clear tendency could be observed, except for the micro-oxygenation and oak treatments leading to lower anthocyanin and gallic acid concentrations (Table 8.3).

Free SO₂ concentrations also decreased quicker in the wine stored in the new barrel, where 100% oak stave contact was used, and especially where oxygen was added in the Cabernet Sauvignon (Table 8.4). This is due to the production of H₂O₂, which can oxidize the SO₂. Aldehydes extracted from the oak wood could also possibly have reacted with the SO₂. Similar trends were observed in the 2007 Pinotage (results not shown).

The 40% oak stave additions yielded concentrations closer to that of the barrel treated wines than that of the 100% oak contact for certain volatile oak-derived compounds, such as *trans*- and *cis*-lactones, guaiacol and eugenol. As expected, lower concentrations of these compounds were observed in general in the old barrels. Both 4-ethyl-guaiacol and especially 4-ethylphenol concentrations were

Table 8.5 Concentrations ($\mu\text{g/L}$) of certain oak volatiles of 2006 Paarl Cabernet Sauvignon red wine during different micro-oxygenation and oak treatments. Samples were drawn after 200 days of treatment

Compound	Treatment				
	40% C	100% C	40% mox	100% mox	NB
Furfural	77.6	131.4	78.4	102.6	55.8
5-Methylfurfural	6	10	9	39.6	4.8
Guaiacol	144.8	250	150.2	205.6	109.6
<i>trans</i> -Oak lactone	149.6	223	98.8	229.2	179.2
<i>cis</i> -Oak lactone	102.2	226.6	108.8	250.4	136.4
o-Cresol	<5	<5	<5	5	<5
4-Ethylguaiacol	17.2	22.6	19.2	26.6	16.6
p-Cresol	11.2	7.4	8	27	7.8
Eugenol	19.2	37	17.2	26.6	18.2
4-Ethylphenol	100.2	88	108.4	100.8	86

100% and 40%, oak staves were added to the wine at 100% and 40% of the internal surface of a 225 L barrel; mox, micro-oxygenation at 4 mg $\text{O}_2/\text{L/month}$; C, control, no O_2 addition; NB, new barrel; OB, old barrel.

higher in one of the old barrel treated wines, indicating possible *Brettanomyces* infection (results not shown).

The concentrations of certain important oak volatiles, such as guaiacol, eugenol, furfural, and 4-methylfurfural, were slightly higher in the 40% treated wine compared to the new barrel after 180 days of treatment (Table 8.5). However, the new barrel had higher concentrations of lactones than the 40% treated wines, but lower than the 100% treated wines. This could be due to the wood not being completely homogeneous and due to slight differences in the toasting technique of the barrels and staves, but it gave some indication that a 40% stave contact area more closely resembled a new barrel than 100% oak stave contact area.

In Fig. 8.4 the development of *cis*-oak lactone can be seen in the 2007 Pinotage. It is clear that the concentration of this compound increased over time. *Cis*-oak lactone is a potent oak volatile, which is responsible for the typical oak wood or coconut flavour of red wines. It is clear that the concentration of this compound were closest to that of the wine matured in the new barrel when staves normally used for barrel production were used in the experiment. The barrel, however, yielded higher concentrations of eugenol (clove or spicy character) compared to any of the stave treatments (data not shown). These differences can be explained by the fact that not the same wood is used for alternative oak products (such as staves) and barrel production. Although we also tried to toast the staves in the barrel treatment in the same manner as those used for new barrel production, small differences in toasting intensities could also have contributed to the differences observed. The staves used in the new barrel treatment were toasted in an oven. These results indicate that a 40% stave addition is closer to a barrel than a 100%, but that the quality and origin of the wood is also very important to simulate an oak barrel when alternative oak

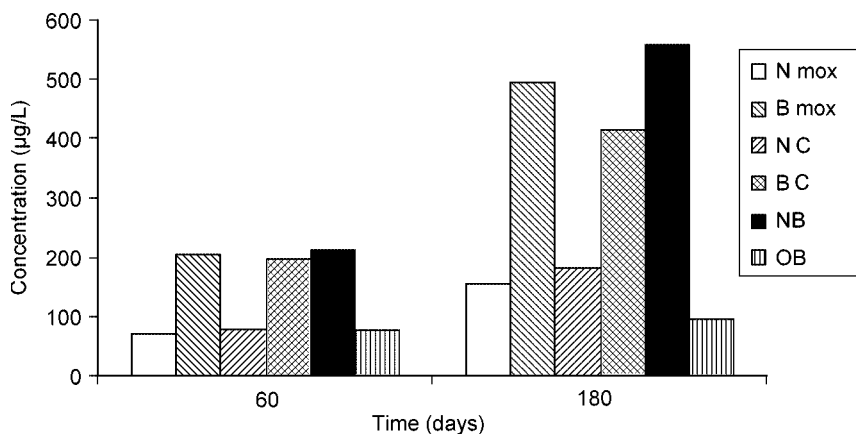


Fig. 8.4 Levels of *cis*-oak lactones in 2007 Paarl Pinotage during the course of the different micro-oxygenation and oak treatments (N refers to staves added to the wine made from wood normally used for alternative oak products and B to staves made from wood normally used to produce barrels. Mox refers to micro-oxygenation applied at 4 mg O₂/L/month and C to the control tanks receiving no O₂).

products such as staves are used in combination with micro-oxygenation to simulate an oak barrel.

A tasting panel could distinguish in general between the 100% oak contact and that of the barrel treatments, but not between the 40% oak contact and the new barrel, giving another indication that 40% is closer to the dosage simulating the barrel to the largest degree. The panel also, in general, preferred the barrel and 40% oak contact wines over the 100% oak contact wines, stating that the latter wines were over wooded.

More research, however, needs to be conducted on these aspects of micro-oxygenation, especially its effect on the taste and aroma of wine when used in combination with oak staves.

8.7 A few recommendations when using micro-oxygenation

When micro-oxygenation is applied, it is critical that all the oxygen added to the wine dissolves to prevent build up of oxygen in the wine. The factors that mainly influence this are the dosage added, the length of the tank, the wine's temperature, size of the O₂ bubbles and SO₂ concentration. The first thing to consider if micro-oxygenation is to be given is the composition and the quality of the wine. As mentioned before, it seems that micro-oxygenation is more effective in younger red wines, probably just after the completion of malolactic fermentation. This is especially true for the colour development of a young red wine. In older red wines (more than 6 months after malolactic fermentation), this process can probably still be considered to change the phenolic composition of the wine, but it is doubtful if it will influence the colour drastically (Du Toit *et al.*, 2006b). In these wines,

micro-oxygenation can probably be used effectively to lower unwanted sulphur odours. If very high phenolics occur in the wine, especially press wine, the O_2 can be added before malolactic fermentation, where 2–6 mg O_2 /L/day is added for a few days.

The dosage of micro-oxygenation to be given is not easy to establish. Indications are that about 40 mg O_2 /L/year comes into contact with the wine in a new oak barrel (Ribéreau-Gayon *et al.*, 2000). This boils down to a dosage of about 3–4 mg O_2 /L/month when micro-oxygenation is applied. However, wine is seldom treated with micro-oxygenation for 12 months and the wine will thus have less oxygen exposure than in an oak barrel. This can be overcome by increasing the O_2 dosage, but care should be taken not to over-oxidize the wine. Certain South African winemakers feel that if the wine was exposed to too much O_2 at an early stage it does not last as well during subsequent ageing, with a dry, dusty mouthfeel being experienced in these wines. Some also feel that they would rather start and stick with a constant dosage, say 4 mg O_2 /L/m, and stop the treatment when the wine is ready. Others will start off with higher dosages and lower the dosage if unwanted flavours are observed. More research is clearly necessary to establish which approach is more suitable.

One can also consider the phenolic composition of the wine when treatment starts. The first thing to assess is the colour density (the sum of 420 + 520 + 620 nm in a 10 mm cuvette). When the colour density is lower (< 10) and the modified colour density (where the bleaching effect of SO_2 is removed) > 15 , micro-oxygenation could be suitable for improving the colour. It is, however, doubtful if it will be effective in increasing the colour density visually in wines with a high initial colour density (> 18). One should thus also consider the modified colour density. If a large difference exist between this and the colour density, micro-oxygenation will probably also be effective in improving the colour. Oxygen can be introduced into the wine at 3–5 mg/L/day before malolactic fermentation or at 3–5 mg/L/month after malolactic fermentation when the wine has a high phenolic content (40–60 AU at 280 nm). The ideal tannin-to-anthocyanin ratio for effective ageing and polymerization is thought to be around 1 to 4, but this must still be proven scientifically. For the removal of sulphur compounds, lower dosages can probably be used.

We also noted that often winemakers in South Africa will use micro-oxygenation only for 2 to 3 weeks at a dosage of 0.5–1 mg O_2 /L/month. It is very doubtful if this low dosage for such a short period of time will lead to any significant changes in the wine. Red wine can actually accommodate more O_2 than one might normally think, if applied in a controlled manner.

Wines with higher turbidity can also handle more oxygen. One can also use indices such as the HCl index, but in our experience this index has been shown not to always give sensible values in young South African red wines.

The most effective way to determine if the addition of O_2 should be increased, decreased or terminated is still by taste. The taste of the wine should become softer over time during micro-oxygenation, without it becoming too polymerized and thus harsh and dried out. The wine should be tasted daily if treated before

malolactic fermentation or at least once a week when applied after malolactic fermentation. As soon as acetaldehyde or crushed apple skin flavours are observed, the dosage should be lowered or the treatment even terminated. Measuring the amount of acetaldehyde weekly has also been proposed as an assessment of possible over-oxidation (Carlton *et al.*, 2007)

Most micro-oxygenation systems have a dosage setting, but this is not effective if the tank or barrel is not tall enough and the bubbles simply go through the wine to the top without dissolving. Tanks should be tall enough, preferably more than 2.2 m. The sparger should also produce small enough bubbles, which will dissolve more easily into the wine. The sparger should also be cleaned regularly by leaving it in a strong base solution to remove accumulation of tartaric salts and phenolic compounds. The plastic pipe leading from the dosage system should also be cleaned of liquid before the treatment starts. The wine's temperature should be around 18 °C; much lower temperatures than this can lead to O₂ accumulation in the wine and much higher temperatures will not allow sufficient O₂ dissolving in the wine. The tanks should be filled completely as a headspace on top of the wine can lead to O₂ accumulation and even acetic acid bacterial growth. If a small headspace does exist, an inert gas such as N₂ or CO₂ should be blown regularly on the wine to displace O₂.

Free SO₂ levels and volatile acidity levels should also be checked regularly during the treatment. Free SO₂ levels should not be too high, thereby preventing favourable phenolic polymerization, but also not too low to allow unwanted microbial development. Levels should be maintained at about 20 mg/L free. It must, however, also be kept in mind that the effectiveness of SO₂ is also linked to the pH of the wine, with more molecular SO₂ occurring at a lower pH. Preliminary research showed that when we added 4 mg O₂/L/month to a wine, the free SO₂ had to be adjusted every month with about 5–10 mg/L SO₂. Levels around 20 mg/L free SO₂ should assist in preventing the growth of *Brettanomyces*, but if *Brettanomyces* infection is suspected micro-oxygenation should not be applied. In such wines, the free SO₂ level should be adjusted to about 40 mg/L to protect the wine, with other preventative steps such as filtration, etc. My research has shown that *Brettanomyces* cells take up SO₂ very quickly (within 1 hour), but it is only the free form of the SO₂ that kills this microorganism (Du Toit *et al.*, 2005).

One should also be careful when taking samples for analyses from tanks receiving micro-oxygenation that they do not pick up excess O₂. When samples are taken from a tank for analysis of SO₂, O₂ etc., it is important to realize that wine can quickly take up O₂ in the sampling procedure, which may give a wrong impression of the dissolved concentration of O₂ in the wine. Research done at our Department has shown that if a sample of wine is drawn into a sampling bottle from the sampling tap without prior displacement of air in the sampling bottle with CO₂, up to 1 mg/L of oxygen can be dissolved in the wine. This can increase up to 4 mg/L if the wine was drawn from the bottom port into another container before aliquoting it to the sampling bottle. These aspects should be kept in mind when collecting wine samples for analyses, or when a winemaker wants to keep a control sample in a bottle when micro-oxygenation will be used in a tank. Excess O₂ pick-

up in the samples could affect SO₂ and possibly microbial measurements of the wine if it is not analyzed immediately. It is thus possible to add more O₂ to the control sample than the treated wine if done incorrectly!

It is thus clear that micro-oxygenation can be used in an effective manner to bring about positive changes to red wine if used correctly, especially in younger red wines. The person responsible for this is still the oenologist, who must decide on an O₂ dosage and time of treatment. However, many unknowns still exist regarding the usage of micro-oxygenation, and these must be addressed in future research.

8.8 Alternative oak treatments

In recent years, winemakers worldwide have shown a renewed interest in so-called alternative oak products. This is due to the high production costs involved when wines are stored in barrels. Barrels also have other drawbacks such as taking up large space in a cellar, being a possible source of microbial infection when used more than once and losing their aromatic volatiles after repeated use. Alternative oak products, such as oak powder, chips and staves, are still prohibited in certain wine producing countries, but have been used for several years in others countries such as South Africa, Australia and Chile (Rodríguez-Bencomo *et al.*, 2008). However, the wide array of alternative oak treatments available to the winemaker has led to a need for further research on the effect of these products on the composition and quality of the wine, especially in light of the fact that the same quality wood is often not used for barrel and alternative oak product production.

8.8.1 Production of alternative oak treatments

Alternative oak treatments, such as oak powder and oak chips, are often produced from parts of the wood not being used for barrel production, such as off cuts (Campbell *et al.*, 2005). This leads to a perception that alternative oak products cannot yield the same quality results as those barrels in which wines are aged (Ribéreau-Gayon *et al.*, 2000). The challenge thus exists for the producers of alternative oak products to produce these products in such a way as to simulate an oak barrel or at the very least contribute to a wine's quality. Different factors must thus be taken into account by the producer, including the origin, species, type of toast, chip size, etc. Oak chips should be consistent in chemical composition, particle size and batch characteristics (Bowyer *et al.*, 2007).

It is a well known fact that the species of oak and its origin influence the composition of the oak. French oak, such as *Quercus robur*, occurring in the Limousin forests in France, for instance, contains more hydrolyzable tannins and lower aromatic compounds, while *Quercus petraea* from Alliers and Nevers are often lower in hydrolyzable tannins and higher in volatile compounds. *Quercus alba* from North America also has higher concentrations of flavoursome compounds, such as lactones and vanillin. However, the chemical composition of oak

originating from these areas can differ between forests from the same area and even between single trees in the same forest. Differences between these species seem to be reflected throughout the production process, which includes seasoning and toasting of the oak, and these differences should also be reflected in alternative oak products (Peuch *et al.*, 1999; Ribéreau-Gayon *et al.*, 2000; Doussot *et al.*, 2002; Garde-Cerdán and Ancín-Azpilicueta, 2006; Prida and Puech, 2006). Pérez-Coello *et al.* (2000) found American oak chips to impart higher ratio of *cis* to *trans* β -methyl- γ -lactone than those from Central France, Alliers and Vosges. Oak from Eastern Europe is also becoming more popular, and little research has been conducted on the composition of alternative oak products from these regions. Wines treated with chips originating from America had significantly higher amounts of lactones, eugenol and *iso*-eugenol than those treated with Hungarian chips, but vanillin levels were higher in the latter treatment. American oak chips sometimes impart double the amount of the more aromatic *cis* β -methyl- γ -lactone to wine, but in French chips the *trans* isomer may also occur in significant concentrations (Guchu *et al.*, 2006; Rodríguez-Bencomo *et al.*, 2008). The ratio of *cis* to *trans* lactones seems to stay the same between the oak and the treated wines. However, there has not been a large amount of data published on the effect of staves on wine's composition.

After the wood has been cut into staves, it needs to be seasoned, a process during which green tannins leach out and the flavour profile of the wood changes. The natural seasoning process, where the staves are left outside, can last from 2 to 3 years. Artificial seasoning, whereby the staves are kept in a ventilated oven, is a much shorter process, and is often used for staves destined for alternative oak production. However, it seems that this type of seasoning yields inferior results compared to natural seasoning, due to higher concentrations of bitter coumarins and astringent tannins found in these products.

Toasting of oak wood also brings about changes to the volatile and tannin characteristics of the wood. This generally leads to furanic compounds, volatile phenols and oak lactones being formed through thermal degradation of carbohydrates, lignin and dehydration of fatty acids, respectively (Garde-Cerdán and Ancín-Azpilicueta, 2006). Toasting also decreases the unwanted sawdust, plank or sappy character, sometimes associated with alternative oak treatments (Ribéreau-Gayon *et al.*, 2000; Chatonnet and Dubourdieu, 1998). The toasting time and temperature is thus very important when oak chips are produced. Toasting for 5 to 6 hours at 185–190 °C leads to maximum concentrations of vanillin and syringaldehyde formation in oak chips but after 24 hours of toasting, vanillin concentrations seemed to decrease. Toasting also leads to an increase in guaiacol, 4-methylguaiacol, vanillin and furanic aldehydes, especially at higher temperatures. It seems that this increase is more pronounced when the chips are heated in the presence of air compared to an anaerobic environment (Campbell *et al.*, 2005). Contradictory results exist regarding the effect of toasting temperature on the lactone concentrations, with certain authors stating that toasting at higher temperatures can lead to different lactone concentrations. In staves, increases in the *cis-trans* ratio have been observed (Ribéreau-Gayon *et al.*, 2000), but higher

toasting levels seem to lead to decreases in chips, especially in American oak containing more of the *cis* isomer (Rodríguez-Bencomo *et al.*, 2008). A toasting level of medium to medium plus thus seems to lead to the highest concentrations of lactones and vanillin, although this would probably also depend on the cooper's toasting regime and starting material.

8.8.2 Size of oak chips, extraction rates and oak staves

The size of the chips also plays an important role, as chips with a smaller size should impart more flavour to the wine due to a higher surface-to-volume ratio. Different size chips exist in the wine industry, with even oak dust being available. The latter is normally used during fermentation to rapidly impart an oak flavour to the wine.

The effect of toasting is also more pronounced in smaller size chips, due to the exposure of a larger surface area to air, especially in smaller chips where heat penetration to the centre of the chip takes places rapidly. Vanillin concentrations, after up to one hour of toasting, can be twice as high in the outer few mm in smaller chips (Campbell *et al.*, 2005). A study conducted by Martinez *et al.* (2001) tested chips with diameters of < 0.1; 0.5–1; 3–5 and 7–14 mm and concluded that the 3–5 mm size chips yielded the highest vanillin and syringaldehyde concentrations. However, it seems that when the size is <5 mm, losses due to evaporation may occur, explaining the higher concentrations of vanillin observed when cubes are used in comparison to powder. Vanillin can also be transformed into vanillin alcohol by microbial activity. The extraction of furfural and lactones does not seem to be influenced to such a large extent by chip size, although guaiacol has been shown to be more quickly extracted in wine when larger chips were used (Arapitsas *et al.*, 2004; Bautista-Ortín *et al.*, 2008).

It is a well known fact that oak chips often lead to a more rapid extraction of oak flavours than barrels. Oak chips are often only kept in contact with the wine for a period of time ranging from a few weeks to a few months and not years, which is the case when barrels are being used. In general, phenolic aldehydes, lactones and volatile phenols are extracted into wine during the first month. Furanic aldehydes can be biologically reduced to alcohols (Pérez-Prieto *et al.*, 2003). Ageing of wine with oak chips in tanks leads to it having high concentrations of lactones after 3 months, which does not increase drastically with further oak contact.

Oak chips are also being used more in combination with used barrels, to replenish the flavour compounds being extracted from the used barrels. When chips are used in combination with used barrels, the extraction rate was faster than with new barrels where no chips were added, reaching a maximum between 3 and 6 months. After this period of time, extraction can still continue in new barrels, due to the time taken for the wine to penetrate the staves and the flavour compounds to permeate back into the wine (Campbell *et al.*, 2005; Bautista-Ortín *et al.*, 2008). Chips or oak blocks can thus be used to replenish used barrels, but the micro-oxygenation effects of older barrels are much less than those of new ones.

The quantity of oak chips to be used differs, but general dosages for white and

red wines are 0.5–2 and 2–6 g/L, respectively, depending on the size of the chip and the allotted contact time.

Oak staves are also used more by winemakers worldwide, who often claim that they yield better results than oak chips. Very little research has been conducted on this aspect, but it has been proven that fermentation in contact with toasted oak staves leads to the formation of furfurylthiol, which imparts a strong coffee aroma to the wine. Staves are also used for a few weeks to a few months, often in association with micro-oxygenation (Tominaga *et al.*, 2000).

It also seems that under certain conditions, ageing with oak chips and staves can lead to a more rapid polymerization of certain phenolic compounds, during tank, barrel and bottle ageing (Del Alamo-Sanza *et al.*, 2004; Del Alamo-Sanza and Nevares-Domínguez, 2006).

8.9 Exogenous tannins in winemaking

Exogenous tannins have been available to the wine producer for a number of years. These tannins normally either consist of grape- or oak-derived tannins, which can be added at different stages of the winemaking process. Exogenous tannins are normally extracted by hot water, steam or ethanol (Keulder, 2006).

8.9.1 Composition of commercial tannins

The composition of commercial tannins can differ to a large extent, and there does not seem to be stringent requirements regarding the composition of these tannins internationally. Commercial tannins can be extracted from oak wood rich in tannins, nutgalls and grape seeds. Grape-derived tannins are normally recognized as grape-derived if the total flavanol concentration is more than 50 mg/g or the proanthocyanin tannin concentration more than 0.5 mg/g. Tannins originating from nutgalls should have a digallic acid concentration between 4 and 8 mg/g and those from oak should have a scopoletine concentration >4 µg/g (Keulder, 2006). Keulder (2006) found the absorbance at 280 nm and the Folin-Ciocalteu value (which both indicate total phenolics) to differ from 0.13–0.43 AU and 573–790 mg/L between eight different commercial tannins when dissolved at 1 g/L in an artificial wine medium. The condensed tannin concentration differed from 113–568 mg/L and the hydrolyzable tannin concentration ranged from 0.51–956 mg/L. However, it should be kept in mind that most of these tannins included grape- and oak-derived tannins, hence the large differences in tannin concentrations. Sarneckis *et al.* (2006) found total tannin concentrations to differ between 138 and 763 mg/L epicatechin equivalents in a 0.5 g/L tannin solution of different commercial tannins, correlating with the difference in compositions of these products. The properties of these tannins thus depend on the composition of the tannin, its origin and the extraction methods used (Zoecklein, 2005). Exogenous tannins normally consist of either grape- or oak-derived tannins or a mixture of the two.

8.9.2 Effect of commercial tannins addition on wine

There are many reasons why commercial tannins should be added to wine, according to the commercial suppliers of these products. These include: to counteract unripe grape tannins, limit laccase activity, help stabilize colour compounds, modify structural/textural mouthfeel properties of the wine, act as substrate for micro-oxygenation, precipitate proteins and increase the ageing potential (Zoecklein, 2005). However, very few of these claims have been confirmed (Parker *et al.*, 2007) and very few studies have been conducted on this aspect.

Keulder *et al.* (2006) conducted a large experiment stretching over 2 years on exogenous tannin additions to red grape musts. Five different commercial tannins were added according to the supplier's recommendations at two concentrations to grape must obtained from Merlot and Shiraz blocks in 2 consecutive years. These tannins were also added to grape must obtained from very rotten Cabernet Sauvignon grapes in one of these years. A wide range of colour, anthocyanins, tannin, total phenols and colour stability analyses were conducted with spectrophotometric and HPLC methods during these experiments. The Shiraz was also monitored during the following year of bottle maturation. The different tannin additions showed small differences regarding the above-mentioned analysis at the middle and end of fermentation, but these differences diminished over time in all the Merlot and Shiraz wines. However, the addition of some of the commercial tannins often substantially increased the gallic acid fraction of all the wines, which is probably due to hydrolyses of the oak tannins. However, tannin addition made a larger difference in the Cabernet Sauvignon must, which was made from rotten grapes. Some additions led to statistically higher total red pigments in the wine after fermentations, which is probably due to inhibition of laccase. In a study by the same author, it was found that oxygen additions *per se* after alcoholic fermentations to red wine led to higher colour development than exogenous tannin additions.

Some of these findings were correlated by Main and Morris (2007), who found that tannin additions led to an increase in polymerization, ionized pigments and yellow colour, but these increases had very little commercial significance. Parker *et al.* (2007) also found that tannin additions before or after alcoholic fermentations of Shiraz did not lead to significant colour changes or stabilization. The reason for this may be that the dosage normally recommended by the suppliers (ranging from 20–200 mg/L) is too low compared to the natural phenolic composition of healthy fruit, which can be up to 2000 mg/L (Ribéreau-Gayon *et al.*, 2000). It is also not well known if these tannins are actually incorporated into the final wine when added prior to alcoholic fermentation. It thus seems that exogenous tannin addition can be considered in terms of colour stabilization when *Botrytis* infection occurred, but may not always be necessary when healthy grapes with a high phenolic content are used.

Exogenous tannin addition could, however, influence the taste of wine. Keulder (2006) found that a tasting panel were able to distinguish wines with tannin additions 1 year after the treatment, but only if the tannins were added at high dosages (500–1000 mg/L). Parker *et al.* (2007) found that when tannin was added at 200 mg/L, a panel could not distinguish it from the control, irrespective if it was

added before or after alcoholic fermentation or just before the tasting. It is thus clear that the merit of exogenous tannin additions should be reassessed; especially its effect on the sensorial characteristics of the wine.

8.10 Future trends

The increased cost of oak barrels and the demand to bring red wine to the market more quickly could possibly see an increase in the use of micro-oxygenation and alternative oak products. Most wine consumers nowadays drink their wine quite quickly, without ageing it for a prolonged period of time. Micro-oxygenation could thus be employed to achieve this, but winemakers must realize that it is not a magic formula which can cure a bad quality wine. The use of micro-oxygenation in combination with alternative oak treatments will probably also increase, especially for mid-quality wines. However, in certain wine producing countries, alternative oak products are not allowed, which enables competitive producers from other countries to put wine on the shelf at a cheaper price. This will probably lead to some countries where these products are not yet allowed to change their viewpoint.

However, certain aspects regarding the use of alternative oak products should be addressed with further research. These include the establishment of specifications for the composition and use of these products, to ensure consistent quality of these products to the wine producer and eventual consumer. Whether the correct dosages of alternative oak treatments currently being used are optimal and their effect on wine's chemical and microbial composition will also have to be investigated. Many aspects of O₂ addition to red wine, such as the correct dosage, are still unclear and should also be investigated further. However, the addition of O₂ and alternative oak treatments to wine should also be done with care, as excessive oxidation and alternative oak additions can lead to a reduction in quality.

8.11 References

- Arapitsas P, Antonopoulos A, Stefanou E and Dourtoglou V G (2004), 'Artificial aging of wines using oak chips', *Food Chem*, **86**, 563–570.
- Atanasova V, Fulcrand H, Cheynier V and Moutounet M (2002), 'Effect of oxygenation on polyphenol changes occurring in the course of wine-making', *Anal Chim Acta*, **458**, 15–27.
- Bautista-Ortin A B, Lencina A G, Cano-López M, Pardo-Minguez F and López-Roca J M (2008), 'The use of oak chips during the ageing of a red wine in stainless steel tanks or used barrels: effect of the contact time and size of the oak chips on aroma compounds' *Aust J Grape Wine Res*, **14**, 63–70.
- Bowyer P K, Bompas C and Murat M L (2007), 'The aromatic science of oak and oak staves' *Aust NZ Grapegr Winemak*, **519**, 61–65.
- Campbell J I, Sykes M, Sefton M A and Pollnitz A P (2005), 'The effects of size, temperature, and air contact on the outcome of heating oak fragments' *Aust J Grape Wine Res*, **11**, 348–354.

- Cano-López M, Pardo-Minguez F, López-Roca J M and Gómez-Plaza E (2006), 'Effect of microoxygenation on anthocyanin and derived pigment content and chromatic characteristics of red wines' *Am J Enol Vitic*, **57**, 325–331.
- Cano-López M, Pardo-Minguez F, López-Roca J M and Gómez-Plaza E (2007), 'Chromatic characteristics and anthocyanin profile of a micro-oxygenated red wine after oak or bottle maturation' *Eur Food Res Technol*, **225**, 127–132.
- Cano-López M, Pardo-Minguez F M, Schamauch G, Saucier C, Teissedre P-L, López-Roca J M and Gómez-Plaza E (2008), 'Effect of microoxygenation on color and anthocyanin-related compounds of wines with different phenolic contents' *J Agric Food Chem*, **56**, 5932–5941.
- Carlton W, Gump B, Fugelsang K, Hasson A S and Kenneth (2007), 'Monitoring acetaldehyde concentrations during microoxygenation of red wine by headspace solid-phase microextraction with on-fiber derivatization' *J Agric Food Chem*, **55**, 5620–5625.
- Chatonnet P and Dubourdieu D (1998), 'Identification of substances responsible for the 'sawdust' aroma in oak wood' *J Sci Food Agric*, **76**, 179–188.
- Danilewicz J C (2003), 'Review of reaction mechanisms of oxygen and proposed intermediate reduction products in wine: central role of iron and copper' *Am J Enol Vitic*, **54**, 73–85.
- Del Alamo-Sanza M and Navares-Domínguez I (2006), 'Wine aging in bottle from artificial systems (staves and chips) and oak woods: Anthocyanin composition' *Anal Chim Acta*, **563**, 255–263.
- Del Alamo-Sanza M, Fernandez-Escudero J A and Castro-Torío R (2004), 'Changes in phenolic compounds and color parameters of red wine aged in oak chips and in oak barrels' *Food Sci Technol Int*, **104**, 233–241.
- Devatine A, Chiciuc I, Poupot C and Mietton-Peuchot M (2007), 'Micro-oxygenation of wine in the presence of dissolved carbon dioxide' *Chem Eng Sci*, **62**, 4579–4588.
- Doussot F, De Jea B and Quideau Sardon P (2002), 'Extractives content in cooperage oak wood during natural seasoning and toasting; influence of tree species, geographic location, and single-tree effects' *J Agric Food Chem*, **50**, 5955–5961.
- Drinkine J, Glories Y and Saucier C (2005) '(+)-Catechin-aldehyde condensations: competition between acetaldehyde and glyoxylic acid' *J Agric Food Chem*, **53**, 7552–7558.
- Du Toit W J and Lambrechts M G (2002), 'The enumeration and identification of acetic acid bacteria from South Africa red wine fermentations' *Int J Food Microbiol*, **74**, 57–64.
- Du Toit W J and Pretorius I S (2002), 'The occurrence, control and esoteric effect of acetic acid bacteria in winemaking' *Ann Microbiol*, **52**, 155–179.
- Du Toit W J, Pretorius I S and Lonvaud-Funel A (2005), 'The effect of sulfur dioxide and oxygen on the viability and culturability of a strain of *Acetobacter pasteurianus* and a strain of *Brettanomyces bruxellensis* isolated from wine' *J Appl Microbiol*, **98**, 862–871.
- Du Toit W J, Marais J, Pretorius I S and du Toit M (2006a), 'Oxygen in must and wine: a review' *S Afr J Enol Vitic*, **27**, 76–94.
- Du Toit W J, Lisjak K, Marais J and du Toit M (2006b), 'The effect of microoxygenation on the phenolic composition, quality and aerobic wine-spoilage microorganisms of different South African red wines' *S Afr J Enol Vitic*, **27**, 57–67.
- Fell F, Dykes S I, Nicolau L and Kilmartin P A (2007), 'Electrochemical microoxidation of red wine' *Am J Enol Vitic*, **58**, 443–450.
- Fourie B P (2005), *The influence of different barrels and oak derived products on the colour evolution and quality of red wines* M.Sc. thesis, Stellenbosch University.
- Keulder D B (2006), *The influence of commercial tannin additions on wine composition and quality* M.Sc. thesis, Stellenbosch University.
- McCord J (2003), 'Application of toasted oak and microoxygenation to ageing of Cabernet Sauvignon wines' *Aust NZ Grapegr Winemak*, **47**, 43–53.
- Garde-Cerdán T and Ancín-Azpilicueta C (2006), 'Review of quality factors on wine ageing in oak barrels' *Trends Food Sci Technol*, **17**, 438–447.
- Guchu E, Diaz-Maroto M, Perez-Coello M, Gonzalez-Vinas M and Cabezudo-Ibanez M

- (2006), 'Volatile composition and sensory characteristics of Chardonnay wines treated with American and Hungarian oak chips' *Food Chem*, **99**, 350–359.
- Heras M O, Rivero-Pérez D, Pérez-Magariño S, González-Huerta C and González-Sanjosé M L (2008), 'Changes in the volatile composition of red wines during aging in oak barrels due to microoxygenation treatment applied before malolactic fermentation' *Eur Food Res Technol*, **226**, 1485–1493.
- Hernandez-Orte P, Lapena A C, Escudero A, Astrain J, Baron C, Pardo I, Polo L, Ferrer S, Cacho J and Ferreira V (2009), 'Effect of micro-oxygenation on the evolution of aromatic compounds in wines: Malolactic fermentation and ageing in wood' *Food Sci Technol*, **42**, 391–401.
- Hrazdine G and Franzese A J (1974), 'Structure and properties on the acylated anthocyanins from *Vitis* species' *Phytochemistry*, **13**, 225–229.
- Laudy M D, Canals R, Gonzalez-Manzano S, Canals J M, Santos-Buelga C and Zamora F (2006), 'Influence of microoxygenation treatment before oak aging on phenolic compounds composition, astringency, and color of red wine' *J Agric Food Chem*, **54**, 4246–4252.
- Liao H, Cai Y and Haslam E (1992), 'Polyphenols interactions. Anthocyanins: Copigmentation and colour changes in young red wines' *J Sci Food Agric*, **59**, 299–305.
- Main G L and Morris J R (2007), 'Effect of macerating enzymes and postfermentation grape-seed tannin on the colour of Cynthiana wines' *Am J Enol Vitic* **58**, 365–372.
- Margalit Y (ed.) (1997), *Concepts in Wine Chemistry*. Wine Appreciation Guild, San Francisco, CA.
- Martinez R G, López H G, De la Serrana Mir M V, Alarcon M N, Herrera M O, Vique O C and Martinez C L (2001), 'Study of vanillin, syringaldehyde and gallic acid content in oak wood and wine spirits mixtures: influence of heat treatment and chips size' *J Wine Res* **12**, 175–182.
- Monagas M, Bartolome B and Gomez-Cordoves C (2005), 'Updated knowledge about the presence of phenolic compounds in wine' *Crit Rev Food Sci Nutr*, **45**, 85–118.
- Morata A, Gomez-Cordoves C, Colomo B and Suarez J A (2003), 'Pyruvic acid and acetaldehyde by different strains of *Saccharomyces cerevisiae*: Relationship with vitisin A and B formation in red wines' *J Agric Food Chem*, **51**, 7402–7409.
- Moutounet M (2003), 'La técnica de microoxigenación'. In ACE Enología, available from: www.acenologia.com.
- Nikfardjam M and Dykes S (2003), 'Microoxygenation research at Lincoln University Part 3: Polyphenolic analysis of Cabernet Sauvignon wine under the application of microoxygenation' *Aust NZ Grapegr Winemak*, **468**, 41–44.
- Parker M, Smith P A, Birse M, Francis I L, Kwiatkowski M J, Lattey K A, Liebich B and Herderich M J (2007), 'The effect of pre- and post-ferment additions of grape derived tannin on Shiraz wine sensory properties and phenolic composition' *Aust J Grape Wine Res* **13**, 30–37.
- Parish M, Wollan D and Paul R (2000), 'Microoxygenation – a review' *Aust NZ Grapegr Winemak Ann Tech*, **438a**, 47–50.
- Pérez-Coello M, Gonzáles-Vinas M A, Garcia-Romero E, Cabezudo M D and Sanz J. (2000), 'Chemical and sensory changes in white wines fermented in the presence of oak chips' *Int J Food Sci Tech*, **35**, 23–32.
- Perez-Magariño S, Sanchez-Iglesias M, Ortega-Heras M, Gonzalez-Huerta C and Gonzalez-Sanjosé M L (2007), 'Color stabilization of red wines by microoxygenation treatment before malolactic fermentation' *Food Chem*, **101**, 881–893.
- Pérez-Prieto L J, López-Roca J M, Martínez-Cutillas A, Pardo-Minguez F and Gómez-Plaza E (2003), 'Extraction and formation dynamic of oak-related volatile compounds from different volume barrels to wine and their behaviour during bottle storage' *J Agric Food Chem*, **51**, 5444–5449.
- Peuch J-L, Feuillat F and Mosedale J R (1999), 'The tannins of oak heartwood: structure, properties, and their influence on wine flavor' *Am J Enol Vitic*, **50**, 469–479.

- Prida A and Peuch J-L (2006), 'Influence of geographical origin and botanical species on the content of extractives in American, French, and East European oak woods' *J Agric Food Chem*, **54**, 8115–8126.
- Ribéreau-Gayon P, Glories Y, Maujean A and Dubourdieu D (2000), *Handbook of Enology, Volume 2: The chemistry of wine stabilization and treatments*. Wiley, Chichester, UK.
- Rodríguez-Bencomo JJ, Ortega-Heras M, Perez-Magariño S, Gonzalez-Huerta and Gonzalez-San Jose M L (2008), 'Importance of chip selection and elaboration process on the aromatic composition of finished wines' *J Agric Food Chem*, **56**, 5102–5111.
- Santos-Buelga C, Francia-Aricha E M, Du Pascual-Teresa S and Rivas-Gonzalo J C (1999), 'Contribution to the identification of the pigments responsible for the browning of anthocyanin-flavanol solutions' *Eur Food Res Tech*, **209**, 411–415.
- Sarneckis C J, Dambergs R G, Jones P, Mercurio M, Herderich M J and Smith P A (2006), 'Quantification of condensed tannins by precipitation with methyl cellulose: development and validation of an optimised tool for grape and wine analysis' *Aust J Grape Wine Res*, **12**, 39–49.
- Schwarz M, Jerz G and Winterhalter P (2003), 'Isolation and structure of Pinotin A, a new anthocyanin derivative from Pinotage wine' *Vitis*, **42**, 105–106.
- Singleton V L (1987), 'Oxygen with phenols and related reactions in musts, wines, and model systems: observations and practical implications.' *Am J Enol Vitic*, **38**, 69–77.
- Sullivan P (2002), *The effects of microoxygenation on red wines* Masters thesis, California State University.
- Tao J, Dykes S I and Kilmartin P A (2007), 'Effect of SO₂ concentration on polyphenol development during red wine microoxygenation' *J Agric Food Chem*, **55**, 6104–6109.
- Tominaga T, Blanchard L, Darriet P and Dubourdieu D (2000), 'A powerful aromatic volatile thiol, 2-furanmethanethiol, exhibiting roast coffee aroma in wines made from several *Vitis vinifera* grape varieties' *J Agric Food Chem*, **48**, 1799–1802.
- Vivas N and Glories Y (1996), 'Role of oak wood ellagitannins in the oxidation process of red wines during aging' *Am J Enol Vitic*, **47**, 103–107.
- Vivas de Gaulejac N V, Vivas N, Nonier M, Absalon C and Bourgeois G (2001), 'Study and quantification of monomeric flavan-3-ol and dimeric procyanidin quinonic forms by HPLC/ESI-MS Applications to red wine oxidation' *J Sci Food Agri*, **81**, 1172–1179.
- Vivas N, Viva N, Debèda N, Vivas de Gaujelac N and Nonier M F (2003), 'Mise en évidence du passage de l'oxygène au travers des douelles constituants les barriques par l'utilisation d'un dispositif original de mesure de la porosité du bois. Premiers resultats' *Sci Aliments*, **23**, 655–678.
- Zoecklein B W (2005), 'Enology notes #103'. Available at: <http://www.fst.vt.edu/extension/enology/EN/103.html> (accessed March 2010).

9

Alternatives to cork in wine bottle closures

J. Goode, www.wineanorak.com, UK

Abstract: There now exist several alternatives to cork as wine bottle closures. The emergence of alternative closures with consistent physical properties has focused attention on the science of post-bottling wine chemistry. While much research in this area remains undone, it is clear that oxygen transmission by the closure has a significant impact on how wines develop after bottling. The role of closure oxygen transmission is an area of active research, and the potential of matching closure type to wine style is now being discussed by winemakers. As well as oxygen transmission, closures differ in their capacity to scalp flavour compounds from wine. Other issues relating to closure choice include cost and consumer acceptance. While cork remains the dominant closure for wine, screwcaps and synthetic corks have now taken significant slices of the bottled wine market (estimated 2.5 billion and 3.7 billion units used annually).

Key words: closure types, synthetic corks, screwcaps, oxygen transmission.

9.1 Introduction

Forty years ago, all wine bottles were sealed with natural cork. There was no closures debate, simply because there was no other practical way to seal wine bottles. The wine trade certainly knew about the issue of musty taint ('corked' wine), but seemed to tolerate it. The cork industry was under no pressure to do a better job, because they had no competitors. The problem of cork taint really only became the subject of serious discussion in the 1980s, a move that may have been prompted by rising levels of cork taint. Indeed, some commentators suggest that

the rate cork taint increased in the 1980s and 1990s for two key reasons. First of all, the Portuguese revolution in 1974 led to many of the privately owned, carefully managed cork forests falling into common ownership (Portugal is the world's leading manufacturer of corks). The result was that the traditional management of these forests was abandoned in favour of a short-term strategy looking for maximum gain. Various pesticides were introduced into the forests, and cork was harvested before it should have been. The damage caused led to a drop in quality of the raw materials used for cork production, and the chemicals employed in the forests may have encouraged the production of taint compounds in the bark. The second factor is the growth of the popularity of bottled wines, which led to a large increase in demand for cork stoppers in the 1980s. Could it be that this increased demand led to a drop in quality? It is a possibility. A third factor could simply be that the wine industry has become better at spotting cork taint. The quality of wines has been rising, and these days even cheap wines are a lot cleaner and fruitier than they used to be. People are fussier about wine quality and, against the background of clean fruity wines, cork taint is much more obvious, and less likely to be tolerated, even by generally undemanding consumers.

The raised awareness of the problem of cork taint led to two responses. First, people began looking for alternatives to cork. Second, the cork industry started to look for ways of eradicating 2,4,6-trichloroanisole (TCA). The first response, the search for alternative wine bottle closures, is the topic of this chapter.

9.2 The key property of closures: oxygen transmission

It was only when wine producers began exploring alternatives to cork that discussion began about the technical requirements of wine bottle closures. So, we begin with the crucial question, and one for which we do not yet have a clear answer: what would an ideal closure look like in terms of performance? Results from comparative closure studies have shown that perhaps the most important property of closures is their level of oxygen transmission. Do we want our closure to give a completely hermetic (air-tight) seal that does not allow any oxygen transmission at all? That is, would a sealed glass ampoule be the perfect closure? Or do we want some oxygen transmission? If yes to the latter, then how much?

There are two ways in which a closure can transmit oxygen. The first is by permeation, which refers to the pressure-driven flow of molecules across a boundary. In wine bottles, this is caused by a pressure difference between the environment and the headspace of the wine, and its rate is determined by the steepness of the pressure gradient, the viscosity of the gas and the resistance of the closure. This permeation does not occur during isothermal conditions, which is why a steady temperature is good for wine storage. The second mechanism of oxygen transfer is by diffusion. This relies on a concentration gradient for oxygen, and can occur against a pressure gradient. This is why, somewhat counter-intuitively, oxygen can diffuse into the headspace of a sparkling wine against a high pressure gradient (which is caused by the carbon dioxide). The speed of this

diffusion relates to the steepness of the concentration gradient and the porosity of the medium, which can be different depending on the molecule diffusing. No good data are currently available for the diffusion of oxygen through wine closures.

As a natural product, cork is variable, and corks therefore have variable transmission rates. Because we lack good data on the physical sealing properties of a good cork – both permeability and diffusivity – we can not extrapolate from this and suggest the ideal properties that an alternative closure should possess.

In discussions of desired closure performance, we need to distinguish between two types of wine. On the one hand, there are wines destined for early consumption – the vast majority of wines made these days. They need to drink well on release and hold their quality for as long as it takes them to get through the supply chain. On the other hand, there are fine wines which are destined for ageing. In some cases, the expectation is that these wines will improve in bottle for many years, and then hold this quality for perhaps as long again. In other cases, the wines are not expected to improve dramatically, but they are supposed to retain drinkability for several years as they loiter on restaurant wine lists or in consumers' cellars. The closure requirements will differ for these different types of wines.

Thus we arrive at a slightly more nuanced pair of questions: what do we want from an alternative closure to ensure successful wine ageing for fine wine styles, and what is the ideal closure for a short-rotation wine? Then, if we dig a bit deeper, it emerges that red and white wines respond differently to oxygen; with their high phenolic content, red wines can absorb a lot more oxygen than white wines without showing an oxidised character. Indeed, red wines appear to need some limited oxygen exposure during the winemaking process in order to develop optimally. And white wines that are differently treated in their early stages will show different levels of resistance to oxidation: those that are protected from oxygen exposure all the way from the crusher to the bottle will be more fragile than those that have seen more oxygen during the winemaking process, for example through deliberate pre-fermentation juice oxidation or barrel fermentation. Add to this the effects of different concentrations of free sulphur dioxide at bottling and wine pH, and the picture that emerges is that different wines are likely to have different requirements from closures. It seems remarkable that cork has done so well as a 'one-size-fits-all' closure.

All this is irrelevant if the ideal closure is actually one that seals hermetically, with zero oxygen transmission. The assumption being made is that some limited oxygen transmission by the closure is desirable and has a role to play in wine development and ageing. This has been a hotly contested point, but the evidence now seems to be clear that limited oxygen transmission is indeed a good thing, and the way forward with closures is matching the right level of oxygen ingress with the right wine style: an extension of winemaking beyond bottling.

Until very recently, the claim by many commentators was that wine ageing was a reductive process that occurred in the absence of any oxygen. Thus an ideal closure in this scenario would be one that seals the tightest. However, studies on a Chardonnay wine bottled with corks and screwcaps, and also a small amount in sealed glass ampoules in the absence of oxygen showed that after 4 years the wine

in the ampoule showed significantly higher ratings for 'struck flint/rubber', terms used to describe reductive characters (caused by volatile sulphur compounds) (Skouroumounis *et al.*, 2005). This wine had developed even in the absence of any oxygen ingress through the closure. This is to be expected, because oxidation/reduction reactions will occur in wine even without any oxygen ingress. For some wine styles no extrinsic oxygen may be necessary for satisfactory development and ageing: contrary to popular opinion, the wines will not be locked in time. However, even if this is the case, the ampoule studies show that reduction problems are likely to be encountered in the absence of any oxygen ingress, and there is currently much debate about whether the tight seal afforded by tin-lined screwcaps is currently causing reduction problems in wines sealed this way because of the low redox potential that develops. On the balance of evidence, it seems that we can conclude that a very small level of oxygen transmission by the closure is necessary for satisfactory wine development, and that the precise level of such transmission has a marked effect on the character of the wine. Different wine styles are likely to have different responses to oxygen ingress, and choice of closure is therefore an important winemaking decision.

To date, the most important closure study has been that of the Australian Wine Research Institute (AWRI), whose important closure trial began in 1999 (Godden *et al.*, 2001, 2005; Casey 2002; Francis *et al.*, 2003). This trial used a battery of measurements to test oxygen transmission, including detailed sensory analysis and measurement of free and total sulphur dioxide concentrations. The study included 14 different closures, including a number of synthetic corks, two different screwcaps, two natural corks and a range of technological corks (such as Amorim's TwinTop and Sabaté's Altec). Of these, the synthetic corks allowed the most oxygen ingress, with the results that all the wines had oxidised within the first few years. The two corks used in the study showed varying degrees of oxygen transmission, but on average performed better than the synthetics. The TwinTops and Altecs provided fairly consistent tight seals, but there were problems with all the cork-based closures in that many of them showed TCA taint. The screwcaps provided the tightest seals, but there were reduction problems. Perhaps the most significant observation at the 63-month time point is that wines sealed with different closures look quite different, and these differences track fairly closely the degree of oxygen ingress, as deduced from the remaining concentrations of free sulphur dioxide.

Other trials have looked at closure performance. Lopes *et al.* (2005, 2007) have studied a range of closures using non-destructive colorimetry, involving an indigo carmine dye. Currently, synthetic closure manufacturer Nomacorc is partnering several academic institutions and other industrial partners to undertake a significant oxygen-in-wine study, based on the use of a luminescence-based probe. Other, as yet unpublished, data on closure oxygen transmission have been reported at conferences by Jim Peck of G3 Enterprises based on the use of MOCON measurements. The hope is that soon winemakers will have a better idea of the requirements of various wines in terms of closure oxygen transmission, and also the actual physical properties of the various alternative closure types.

9.3 The various closure types

9.3.1 Synthetic corks

Synthetic corks were until recently the most commonly used alternative to natural cork. Their origin was in California, although the UK market had a role in developing them. 'Metal Box UK made a synthetic insertable closure for wine from the early 70s, but it was not promoted and did not take off', explains Richard Gibson, now of Scorpex, but who in a previous role as Head of Technical Services with Southcorp was instrumental in first bringing wines sealed with alternative closures to the UK. 'I understand that the moulds used by Metal Box went to the USA, where they were used to make the Cellukork closure by Lermer Packaging. This closure was used by [Sonoma winery] St Francis in the early 1990s for various wines.' In 1992, St Francis released a few thousand cases of wine commercially sealed with Cellukork. From reading news reports in the early 1990s, it seems that at the time winemakers considered synthetics to be suitable for all wine types, even those meant for longer cellaring. Gibson adds that another synthetic cork, Tage, was also made from the mid-1980s in France, but again was not widely promoted. The first synthetic cork manufacturer to really make a big impact was SupremeCorq. 'They came on the radar screen late 1993', says Gibson. 'I met with them in the USA in 1994, and internal trials were commenced at Southcorp in that year.' The UK market proved receptive. 'The first commercial packaging was in 1995 with a 1994 Safeway Semillon Chardonnay, using a yellow SupremeCorq. This project came about through discussion/contact between Liz Robertson [then head of Safeway's wine department] and me. Liz was a big influence in the whole alternative closure movement. Synthetics fanned out quickly to the UK and Australian domestic product after that.'

Synthetic corks have some advantages over other alternative closures. Like corks, as an in-neck closure, they are robust. Producers who already have a bottling line do not need new equipment to use them, nor do they need different bottles. Synthetics also have good consumer acceptance – while they are plastic, they are the most-similar alternative to the cork, and conservative markets that will not accept screwcaps are penetrable by synthetics. Marketing departments like them because they can be made in various colours, many with four-colour printing options for branding. Alternatively, if a natural look is desired, wood grain patterns are an option with some manufacturers. Crucially, they are also cost-effective – one of the cheapest ways of sealing a bottle of wine. As manufactured, closures synthetics have consistent performance, lacking the natural variability of cork.

There are two rather different types of synthetic cork. The first type is injection moulded, with the key proponents of this style being SupremeCorq and Integra. As the name suggests, these are made using moulds which are then injected with a plastic copolymer. This process leaves a slightly marbled finish on the closure. One advantage of this form of manufacture is that the closures come ready 'chamfered'. This is the term used to describe the rounded end of these plastic corks which is designed to make insertion and reinsertion easier.

Extruded synthetics are the second style, and these closures come in two types.

Market leader Nomacorc and US company Neocork products are made by extruding a fine honeycomb plastic core which is then coated by a smooth cylindrical layer and chopped up into cork-sized pieces; Australian manufacturer NuKorc had a different single-extrusion process where heat was used to produce a smooth outside to the core of the closure (NuKorc went into administration in 2010). Extruded closures thus have a rather spongy looking core and a smooth outside. A disadvantage of extruded synthetics is that chamfering, if required, is an extra manufacturing step.

Both types of synthetic need to have a coating applied to them. The exact nature of the coating of each brand is a commercial secret, but commonly involves a silicone- or paraffin-based material, and it is crucial to closure performance: without it, extraction forces would be much higher. Natural corks are also coated in this way.

The challenge for synthetics is (a) to seal bottles tightly enough to prevent oxidation of the wine while still being (b) easy enough to extract, (c) easy enough to take off the corkscrew and (d) easy enough to reinsert into the bottle, should the contents not be drunk in a single session. The early synthetics were problematic, failing on all four objectives, but much work has gone into improving them and the new generation are better closures than their predecessors. Still, it is hard to answer the question about how synthetics perform, because good data on the current crop are not easily available, and it is unfair to lump them all together – there are subtle differences between all the different products.

This is where we hit the main drawback of synthetics, although for most wine styles it is not terribly relevant. The influential AWRI closure trial included a range of seven different synthetics: Aegis (moulded), Auscork (moulded), Betacorque (moulded), ECORC (extruded), Integra (moulded), Nomacorc (extruded) and NuKorc (extruded) (Godden *et al.*, 2005). The results from this trial were a bit of a blow to synthetic manufacturers in that they showed that the levels of oxygen transmission by synthetics was much higher than those for both screwcaps and natural cork (which showed some variation). While some of the synthetics performed noticeably better than others, most failed to protect the Semillon wine used in the trial to avoid it oxidising unacceptably by the 24-month time point. Further studies have shown that while corks transmit oxygen through the glass–cork interface, transmission by synthetics is largely through the closure. The plastics that the synthetics are made of are permeable to oxygen to a degree; it is not just a matter of the tightness of the seal.

Two things need to be said about the AWRI results: first, these were the 1999 versions of these closures, and the 2009 models would probably do a better job. Second, the vast majority of wines are made to be drunk in their youth, and a closure that is only good for 2 or 3 years is therefore perfectly adequate. Issues such as visual appearance, extraction force and off-screw force, boring as they seem, are very important and are often ignored at the expense of oxygen transmission.

There is a further potential drawback of synthetics, which is flavour scalping, although it is not clear how much of a ‘real world’ problem this is. The process of flavour scalping is a problem in the food industry, where aroma components are

absorbed by packaging. Aware from other studies that TCA could be absorbed from the wine by the cork, scientists at the AWRI investigated potential flavour scalping of aromatic wine components by a range of closure types (Capone *et al.*, 2003). The wine used was the same as in the AWRI closure trial, a 1999 Clare Valley Semillon, spiked with the flavour compounds they wanted to study. Closure performance was compared with that of the same wine in a sealed glass ampoule. After 2 years, the concentrations of many of the flavour compounds had changed significantly, partly through absorption by the closures but also through chemical modification independent of closure. Screwcaps did not absorb any flavour compounds, performing similarly to the sealed glass ampoules. Relatively non-polar volatile compounds were absorbed by the corks, technical corks and synthetic closures, while the more polar compounds were not absorbed by any closures. Of these closures, technical corks absorbed a little more than the natural corks, and synthetics absorbed a lot more than either. The conclusion was that synthetic corks are responsible for considerable flavour scalping, and even natural corks are capable of absorbing certain wine aroma components in limited amounts. But is this a problem for synthetics? It is hard to say. Many synthetic producers claim that their closure does not scalp. But the issue is not whether sophisticated analytical techniques can detect scalping; rather it is whether any scalping that may occur makes a sensory difference to the wine in question. Indeed, if corks scalp some flavour compounds, and we like the way wines age under a good cork, then the absence of any scalping by a closure is not automatically a good thing.

The current situation is therefore that for most commercial wines synthetics represent a good, cost-effective closure solution. They are cheap, reliable and have good acceptance by consumers. It seems wise, though, to avoid using them on any wines likely to hang around in the supply chain for more than a couple of years because of their higher levels of oxygen diffusion than corks and screwcaps. However, this higher level of oxygen diffusion is something that can be used creatively by producers for commercial red wines, making them show well early in their life when they are destined to be drunk.

9.3.2 Screwcaps

Screwcaps as an alternative to cork for bottling wine were born in 1959, when French company La Bouchage Mecanique introduced the Stelcap-vin as an alternative to cork. Their Stelcap closure had already proved successful for a range of spirits and liqueurs. The rights to manufacture this closure were acquired by Australian Consolidated Industries Ltd (ACI) in 1970. It was renamed Stelvin® for the Australian market and trials at the ACI laboratories took place. The results were reported for the first time in 1976 (Eric *et al.*, 1976). Four closures (three screwcaps with different wadding materials and a cork for comparison) were tested on six wines (three white and three red). The conclusions were that screwcaps were ideal for sealing wine bottles, but only if they had the right wadding material and if a satisfactory seal was obtained between bottle and cap. Interestingly, the best-performing Stelvin closure at the time had agglomerated cork as part of the



Fig. 9.1 A Stelvin screwcap from 1977, sealing a Riverina College Rhine Riesling Spätlese, Clare Valley.

wadding, although this cork was not in contact with the wine. This trial continued through to 1979. Although there was an industry push to move to screwcaps, this lost momentum, largely through fears about consumer acceptance, and also perhaps because awareness of the shortcomings of cork, particularly its high taint rate, were not as widespread at the time. The chief problem was that screwcaps were seen by consumers as fit for cheap wines only. Fig. 9.1 shows a screwcap from a 1977 Riverina College Rhine Riesling, from Australia's Clare Valley.

But screwcaps were not finished. As dissatisfaction with cork gradually increased, sporadic and in some cases sophisticated attempts to introduce screwcaps to the marketplace occurred. In the UK, when Penfolds launched the Bin 2 Shiraz Mataro under screwcap in 1996, it contained a neck tag soliciting responses from consumers. But what was needed was a more united push; a critical mass willing to effect change. In the meantime, the first synthetic corks were beginning to muscle in as a competitor to cork. Around this time, the issue of cork taint was getting increasing exposure in the press and, although it is impossible to measure the impact of this coverage, it probably primed some of the more aware consumers about various closure issues and may have had an impact on the way cork alternatives were later received.

A significant step forwards for the screwcap was an initiative taken in the late 1990s by Phil Laffer, chief winemaker with Orlando Wyndham, whose brands include the highly successful Jacob's Creek. Indeed, you could say that Laffer was largely responsible for kicking off the current screwcap resurgence. 'I was desperately keen to solve the screwcap issue', he recalls, 'but rather than do it from

the bottom I wanted to start from the top'. This is because the previous attempts to introduce screwcaps failed in part because these new closures were associated with cheap wines. The image was wrong. If screwcaps could be reintroduced for expensive wines, this hurdle would be successfully negotiated, because it would then be straightforward to use them for cheaper wines also.

His bright idea – one that proved a successful strategy – was to get a major retailer on board. 'We needed a retailer to work with us', he explains. 'Without this we would have had resistance from retailers'. Laffer chose as the wine a Richmond Grove Riesling from the 1997 and 1998 vintages. This is a premium Riesling brand from Watervale in the Clare Valley, which is only sold in Australia. He negotiated a promotion with Vintage Cellars, supplying half the wine screwcapped and half in cork. The screwcapped wine was enormously successful and sold out way ahead of the cork-sealed wine. It looked like the tide was finally changing, in Australia at least.

The next significant step in this story took place in the Clare Valley, South Australia. It was as recently as the 2000 vintage when Clare winemakers, frustrated by the poor performance of cork, banded together to make a stand on the issue. Clare is famous for its Rieslings, and these wines are made in a style that shows up any cork-related faults particularly transparently. The Clare winemakers had to overcome a significant logistical obstacle before they could offer their wines in screwcap: at the time, there was no Australian supplier who could offer bottles and caps of the required style and quality. As a result, they had to drum together enough like-minded producers willing to adopt screwcaps to generate an order for 250 000 bottles from Pechiney in France, which was the threshold needed to make this possible. With a collaborative effort, they managed it, and the combined shift was large enough to make headlines, for what at the time seemed a very brave move. The Clare initiative started the ball moving, and prompted the New Zealand winemakers to form the NZ Screwcap Initiative a year later. 'We were inspired and encouraged by the success of the Clare Valley Riesling move to screwcaps in 2000', reports one of the founder members, Michael Brajkovich MW of New Zealand's Kumeu River Wines, 'and like them we realized that we could achieve much more with a combined effort than we could ever do individually.' The situation in New Zealand is such that from an estimated 1% of wines sealed by screwcaps in 2001, bottles sealed with this closure are now the vast majority.

The large unanswered question concerning screwcaps is the issue of reduction. In this context, 'reduction' refers to a group of volatile sulphur compounds that can, in certain contexts, become wine aroma faults. In particular, in a low redox potential environment higher sensory threshold disulfides can shift to lower threshold mercaptans post-bottling. In the AWRI trial, there were specific aroma defects in the Semillon wines sealed with the metal-lined screwcaps, which tasters described as 'struck flint' or 'rubbery', and which were thought to derive from disulfides. It needs to be emphasised that these volatile sulphur compounds are produced by yeasts during fermentation and, in particular, are a result of stress such as low nitrogen availability or temperature shock. They are not caused by screwcaps; it is just that the low redox environment post-bottling where metal-lined caps are used is less forgiving of wines with a tendency towards reduction.



Fig. 9.2 Stelvin Lux screwcap.

At this point, we need to take a slight detour to consider the composition of the liners used in screwcaps. It is these, in tight apposition to the rim or lip of the bottle, that create the seal. Pechiney, manufacturer of the Stelvin cap offer two different liners. The first is composed of a polyethylene wadding with an inert Saranex™ layer (referred to as ‘Saranex only’), while the second differs because it has a thin tin layer (sandwiched between the polyethylene and a thin PVDC outer skin) that acts as an oxygen barrier. This oxygen barrier does not have to be tin; it could be another metal, such as the aluminium that was used in the AusCap screwcaps in the AWRI trial. Virtually all (if not entirely all) of the screwcaps used in Australia and New Zealand have this metal layer, and thus have extremely low oxygen transmission. In Europe, and particularly Switzerland where screwcaps have been used widely for a long time, it is mainly the saranex-only liner that is used, which has a higher rate of oxygen transmission. Typically, tin/saran-lined screwcaps show an oxygen transmission rate (OTR) of 0.000 002 mL/oxygen per day in air; the Saranex liner shows an OTR of 0.001 mL/oxygen per day. The average OTR of natural cork is about 0.0005 mL/oxygen per day (Jim Peck, personal communication). It is important to remember that not all screwcaps are equal.

Of course, reduced characters are not unique to screwcapped wines, but these problems can in many cases be alleviated by careful winemaking. However, this is a controversial issue, and the fact that it may take time to get this right, and with the increasing number of screwcapped bottles being sold it is likely that screwcaps will come under intense scrutiny over the next few years.

Longevity of screwcaps does not seem to be a problem for wine ageing. ‘Screwcaps are “guaranteed” to last at least 10 years if the bottle and application

machinery are correct. In practice we have seen screwcaps last for 25–30 years without problem,’ says Brajkovich. He adds, ‘Given that corks should really be replaced after about 30 years, the use of screwcaps for long term red wine ageing is definitely within the normal performance specifications of other closures, and time will tell just how good they are at performing the job.’ He thinks that screwcaps will turn out to be ideal for red wines as well as whites.

New developments with screwcaps include the Stelvin Lux and the Novatwist. The Stelvin Lux (Fig. 9.2) is designed to look more like a traditional capsule and does not roll on, but instead is screwed on using a preformed plastic thread inside the cap. The Novatwist, produced by Novembal (a Tetrapak subsidiary) is a plastic, long-skirted screwcap which, together with a PET bottle, forms a totally and easily recyclable package.

Recent estimates are that screwcaps now have over 10% of the bottled wine market. This total market is estimated to be some 18.3 billion annually (S. Cashin, Normacorc, personal communication), worldwide. Of this, screwcaps are now around 2.5 billion (Anne Seznac, Guala Closures, personal communication), with synthetic corks around 3.7 billion and natural cork accounting for most of the remainder. This is a dramatic rise from the situation 5 years ago, when screwcaps were estimated to be around 200 million worldwide.

9.3.3 Diam closures

Diam, made by Oeneo Bouchage, sits somewhere between natural cork and alternative closures, in that it is an in-neck closure with both cork and synthetic components (Fig. 9.3). It is made up of small granules of cork that have been cleaned with supercritical CO₂, combined with synthetic microspheres. Oeneo, previously known as Sabaté, is a French company who developed this technology in conjunction with the Supercritical Fluids and Membranes Laboratory of the Commissariat à l’Energie Atomique (CEA), a French government organisation. Named the ‘Diamond’ process, it uses supercritical CO₂ for selective extraction of volatile compounds from cork. The first work on this process was initiated in 1997 and it was patented in 1999, the patent being shared by Guy Lumia and Christian Perre of the CEA, and Jean-Marie Aracil, at the time Sabaté’s head of R&D.

Supercritical CO₂ is used for other purposes, such as extracting caffeine from coffee and manufacturing components of perfumes. If you pressurise a gas, at a certain point it becomes a liquid. If you then juggle the parameters of pressure and temperature, at a specific combination of these – known as the critical point – the interface between the two disappears, and you then have the penetration power of a gas and the extraction power of a liquid. For CO₂, this point is at 31.1 °C and 73 bars of pressure – a conveniently low temperature, even though the pressure is on the high side (73 times atmospheric pressure).

Sabaté’s interest in developing the Diamond process was motivated by problems they experienced with their own technical cork, Altec. Launched in 1995, the Altec represented a novel approach to alternative closures. The problem it addressed was the dissatisfaction of wine producers with the performance of the



Fig. 9.3 Diam closures.

inexpensive corks that they were using for their mass-market wines. Because of perceived consumer resistance to plastic corks and screwcaps – particularly strong in the French market – Sabaté devised a manufacturing process that produced what is in effect a hybrid closure: part cork, part synthetic, but which looks like a natural product.

This manufacturing process involves taking raw cork and fragmenting it into tiny particles. These are then sorted and most of the lignin – the hard, woody material that surrounds the lenticels (the tiny gas exchange pores in the cork) – is discarded. This cork ‘flour’ is then blended with proprietary polymer microspheres and the whole lot stuck together with a binding agent. The resulting closures are consistent, and in theory should have lower risk of TCA taint than normal corks, because the lignin-rich material that surrounds the lenticels is considered to harbour the majority of the TCA contamination.

Initially, the Altec was tremendously successful. But some years after its widespread adoption, reports began coming in that the Altec, which had been initially marketed as being taint-free, was causing unacceptably high levels of taint. The situation was particularly bad in the USA, where four wineries blamed Altec for tainting large numbers of their wines. This led to expensive legal action.

More bad news came from the results of the closures study conducted by the AWRI. When the 24-month results were published, they reported that each of the bottles sealed with Altec closures was found to have a TCA-like aroma, and follow-up chemical analysis found detectable concentrations of TCA in each of the Altec samples analysed.

The Diamond process has proved highly successful. Wines sealed with prototype Diamond-treated Altecs showed no detectable TCA, both by sensory and chemical analysis. These new taint-free closures are being marketed as ‘Diam’, and have been adopted enthusiastically by a number of clients.

Diam is still a relatively new closure, so it is hard to give solid assurances about its stability and performance over time. Currently, they say it will be good for 60 months. But on the basis of the almost identical Altec, which has been around a lot longer, there seems no reason to doubt that it will not be suitable for wines destined for longer ageing. Thus it could turn out that Diam could supplant natural cork as

the closure of choice for wines destined for long ageing, assuming that in the mean time synthetics or screwcap liners are not produced with the ideal oxygen transmission properties.

9.3.4 Zork closures

Zork is perhaps the most alternative of alternative closures, and consists of three main bits. The first is the visible bit, the cap, which provides a tamper-evident clamp that locks onto the band that surrounds the outside rim of most standard wine bottles. The second is the foil, which provides an oxygen barrier similar to a screw cap, and is found inside the cap – it is made of aluminium. The third component is the internal plunger, which creates a ‘pop’ on extraction and reseals after use, made of polyethylene.

In February 2006 Zork published the 18-month results from the independent analysis of Zork Version 1 as carried out by Provisor (a national research facility supporting the Australian wine industry; www.provisor.com.au). The trial compared bottles of Yalumba’s Pewsey Vale Riesling 2003 sealed with both Zork V1 and a tin-lined screwcap. Sensory analysis showed that the screwcap-sealed bottle was consistently higher in the flint/rubber attribute, which is consistent with other trials. The screwcap was also higher in fruit-type aromas and flavours. Zork V1 was higher in the attribute ‘oxidised’, but the report points out that free SO₂ concentrations remain high enough that the wine is not spoiled by oxidation.

Initially, the concentrations of free and total SO₂ decreased more rapidly under Zork than under screwcap, but this was thought to be because of oxygen present in the plunger (part of the Zork closure that goes into the neck of the bottle). After this, the decline is a bit greater than that of screwcap in rate. The initial loss of freshness because of oxygen in the plunger is complete at about 60 days.

The OTR has been measured at 0.0078 mL O₂/closure per day. This is a bit higher than that measured for tin-lined screwcaps, and only a little less than that of synthetic corks. Since the Provisor trial began, there have been some tweaks to the Zork design, with version 4 the latest, principally in the design of the plunger which has been modified to cope with different bottle bore dimensions.

Provisor calculate the shelf life of wines sealed with Zork to be 5.1 years. This is based on the assumption that significant oxidation occurs when the free SO₂ concentration declines below 10 mg/L. The lowest 95% confidence limit is estimated at 3.2 years. Thus Zork is fine for most commercial applications, but is not really recommended for wines that will be aged for more than a few years. Provisor acknowledge that this could be different for different wines. Zork is quite expensive as a closure, and therefore the only reason I suspect that producers would consider this is because of its striking appearance, which could be incorporated into the brand design.

9.3.5 Vino-Lok closures

Alcoa’s Vino-Lok® looks modern enough, but at its heart it is a bit ‘retro’, harking



Fig. 9.4 Vino-Lok, with the aluminium cap removed showing the glass plug in the bottle neck.

back to the old days when glass stoppers were widely used (called ‘stoppels’, these were forerunners to the cork for sealing bottled wines). And, until fairly recently chemistry labs were full of bottles sealed with glass stoppers. The difference here with Vino-Lok is that the seal is not one of ground glass against ground glass, but it is actually formed by a plastic ‘O’ ring (made of ethylene vinyl acetate) that acts as an interface between the bottle rim and stopper. It is quite a complicated closure solution because the glass stopper is held in place by an aluminium cap which has to be removed before the stopper can be extracted, and a special bottle neck is needed to accommodate this closure. It is also expensive, and this could be the biggest barrier to its widespread acceptance. Fig. 9.4 illustrates the appearance of Vino-Lok.

No data are currently available on the oxygen transmission properties of this seal. It should be pointed out that these are dependent on the characteristics of the O ring, which forms the oxygen barrier. Vino-Lok has already met with some success in the German and Austrian markets. In the USA, there is a variant of Vino-Lok on the market called VinTegra™.

9.3.6 The crown cap closure

The crown cap is widely used for beer, and is also used by sparkling wine producers to store wines before the dosage stage when the bottles are finally sealed with corks. Yet it has failed to win many converts as an alternative closure for wine. Perhaps this is an image issue; it might also reflect the vulnerability of the seal to knocks, less of an issue with smaller, lighter bottles that will be rotated fairly

quickly, but probably more of a problem with bigger bottles that spend longer in the supply chain. The seal of a crown cap is dependent on the lining material. Crown caps are currently being used to seal the high-end sparkling wines from Domaine Chandon in Australia's Yarra Valley. Crown caps could have quite a future as alternative closures for sparkling wines, given that it has not been easy to adapt screwcaps for fizz, but consumer acceptance is currently a problem.

9.4 Conclusions and future trends

From all this, the picture that emerges is that there is no such thing as a perfect closure. Winemakers are faced with many options, but still too little is known about the precise details of post-bottling wine chemistry to give definitive advice on closure choice based on firm data. Current work by Jim Peck of G3 (personal communication, July 2009) on screwcap liners has shown that it is possible to engineer liners with specified levels of oxygen transmission by means of microperforations in the tin layer of the liner. In addition, Peter Gago and his team at Penfolds (personal communication, September 2009) are working on a series of prototype glass-on-glass closures for use with Penfolds' high end red wine Grange. Gago's intention is to compare a hermetic seal with one where specified oxygen transmission levels are engineered by means of sintering the glass. Such a closure would be expensive, but on a wine retailing at A\$300, cost of closure is not a huge issue.

But what is needed are more data on exactly what oxygen transmission level we would like from the ideal closure. Does this depend on the wine? A large multicentre project looking at oxygen in wine, spearheaded by synthetic cork manufacturer Nomacorc, is currently underway. The hope is that studies such as these should assist winemakers by providing more detailed information of the role of the closure in post-bottling wine development.

9.5 References

- Capone D, Sefton M, Pretorius I and Hoj P (2003), Flavour scalping by wine bottle closures. *Wine Industry Journal*, **18**, 16–20.
- Casey J (2002), A commentary on the AWRI closure report. *Australian and New Zealand Grape Grower and Winemaker*, **457**, p 65–69.
- Eric B, Leyland DA and Rankine B C (1976), 'Stelvin' – evaluation of a new closure for table wines. *Australian Grape Grower and Winemaker*, **148**, 58–59.
- Francis L, Lattey K and Smyth H (2003), 'Reduced' aroma in screw-cap bottled white wines. *AWRI Technical Review*, **142**(10), 51–53
- Godden P, Francis L, Field J, Gishen M, Coulter A, Valente P, Hoj P and Robinson E (2001), Wine bottle closures: physical characteristics and effect on composition and sensory properties of a Semillon wine. 1. Performance up to 20 months post-bottling. *Australian Journal of Grape and Wine Research*, **7**, 62–105.
- Godden P, Lattey K, Francis L, Gishen M, Cowey G, Holdstock M, Robinson E, Waters E, Skouroumounis G, Sefton M, Capone D, Kwiatkowski M, Field J, Coulter A, D'Costa,

- N and Bramley B (2005), Towards offering wine to the consumer in optimal condition – the wine, the closures and other packaging variables: a review of AWRI research examining the changes that occur in wine after bottling. *Wine Industry Journal*, **20**, 20–30.
- Lopes P, Saucier C and Glories Y (2005), Nondestructive colorimetric method to determine the oxygen diffusion rate through closures used in winemaking. *Journal of Agricultural and Food Chemistry*, **53**, 6967–6973.
- Lopes P, Saucier C, Teissedre P L and Glories Y (2007), Main routes of oxygen ingress through different closures into wine bottles. *Journal of Agricultural and Food Chemistry*, **55**, 5167–5170.
- Skouroumounis G K, Kwiatkowski M J, Francis I L, Oakey H, Capone D L, Duncan B, Sefton M A and Waters E J (2005), ‘The impact of closure type and storage conditions on the composition, colour and flavour properties of a Riesling and a wooded Chardonnay wine during five years storage’. *Australian Journal of Grape and Wine Research*, **11**, 369–377.

Current issues in organic winemaking: consumer expectations, producer attitudes and oenological innovation

D. Rauhut, Forschungsanstalt Geisenheim (Geisenheim Research Center (GRC)), Germany; and C. Micheloni, AIAB – Italian Association for Organic Agriculture, Italy

Abstract: This chapter takes its origin from the collaboration among 11 research institutes and associations of the organic sector that participated in the ORWINE project. The acronym stands for ‘Organic viticulture and winemaking: development of environment and consumer friendly technologies for organic wine quality improvement and scientifically based legislative framework’ and it is a STREP (Specific Targeted Research or Innovation Project) financed by the EU Commission within the VI Framework Program. It ran from 2006 until 2009.

Key words: organic winemaking, SO₂ reduction, sulphites, low-input processing, consumer expectations.

10.1 Introduction

10.1.1 The EU ORWINE project

The EU ORWINE project was focused on the following topic: ‘Organic viticulture and winemaking: development of environment and consumer friendly technologies for organic wine quality improvement and scientifically based legislative framework’. It is a Specific Targeted Research or Innovation Project (STREP) financed by the EU Commission within the VI Framework Program. It ran from 2006 until 2009 and was organised and conducted by a collaboration of 11 partners (research institutes and organic associations).

The overall coordinator was Cristina Micheloni from the Italian Association for Organic Agriculture (AIAB) and the scientific coordinator was Roberto Zironi from the Department of Food Science, Udine University (UNIUD), Italy. The other partners of the project were the Technical Institute for Organic Agriculture (ITAB), Institut Français de la Vigne et du Vin (IFV, former abbreviation ITV) and Institut National de la Recherche Agronomique (INRA) in France; the Federal Association of Organic Wine Producers (ECOVIN) and the Forschungsanstalt Geisenheim/Geisenheim Research Center (GRC, former abbreviation SRIG – specifically the research group of Doris Rauhut) in Germany; the Research Institute of Organic Agriculture (FiBL) in Switzerland; the Università Cattolica del Sacro Cuore and Vinidea in Italy; the IFOAM EU regional group in Belgium.

The project aimed at developing the scientific basis of the legislative framework for the definition of European organic winemaking (Micheloni, 2009). To reach the goal, several activities were run in the 3 years of the project time:

- the identification of practices applied in organic viticulture and winemaking in typical EU and AC (Associated Countries) wine producing areas and the evaluation of their impact on environment and wine quality;
- the identification of buying motives and organic wine markets needs in order to address labelling provisions and communication strategies;
- the development of innovative winemaking technologies fitting to the organic concept and improving organic wine quality with particular regard to sulphites reduction;
- the on-farm testing of innovative technologies in order to assess their practicality and acceptability;
- the development of a code of best organic winemaking practices and recommendations for the development of EU regulation 2092/91 and policy support measures for the organic wine.

A participatory approach with stakeholder involvement at national and EU levels were followed during the whole project, ensuring a high acceptance of the proposed legislative framework.

In the following pages, a summary of the ORWINE studies is reported, but the complete documentation may be downloaded from www.orwine.org and specifically:

- public report about consumers research by Hanna Stolz and Otto Schmid, FiBL;
- analysis of market needs by Monique Jonis, ITAB;
- public report about producers investigation by Cristina Micheloni, AIAB and Gianni Trioli, Vinidea;
- public report of the final ORWINE recommendation for EU rules on organic wine by Monique Jonis, ITAB and Cristina Micheloni, AIAB;
- ‘Code of good organic viticulture and wine-making’ by Gianni Trioli and Uwe Hofmann and other members of the ORWINE consortium (Trioli and Hoffman, 2009).

10.1.2 The concept of organic wine

As organic wine is not defined unequivocally we first need to define what is organic wine or, more precisely, what consumers, producers and traders refer to when using the term 'organic wine'. As the definition is not totally shared by the different categories, we need to look at it from the different points of view.

Consumers' opinion

Information about consumers' opinion comes from a focus group study run by the ORWINE project in selected case study countries, Italy, France, Germany and Switzerland, on two target groups: (i) organic consumers and (ii) quality wine consumers. The main findings of the consumer study are that consumers purchase wine according to geographical origin, grape variety and price. The price is often a benchmark for consumers regarding the quality of the wine on offer, even though some consumers doubt whether cheap wines are automatically poor-quality wines or, conversely, whether expensive wines are always high-quality wines.

Regarding the image of organic wine, it may be described as highly positive in terms of production and processing, with the main emphasis on the issue of pesticide treatments, which are prohibited in organic grape production. Organic wine, in contrast to conventional wine, is expected to come from small-scale production facilities linked to specific knowledge systems. In addition, organic wines are considered to be purer wines with no additives or harmful residues compared to conventional wines. Thus, they are also considered as healthier and more salubrious by those consumers with experience of organic wine. Nevertheless, organic wine has a rather poor image regarding flavour and quality. Only very few consumers stated that they appreciate the more 'unique' taste of organic wine. The consumer study indicates clearly that communication strategies have to take account of this negative image regarding taste in particular. However, one problem is that wine shops, which are the purchasing place of high-quality wines, do not usually offer organic wines. Thus, consumers tend to believe that no organic premium wines exist.

Concerns about the use of the main additives and processing aids commonly used in conventional wine processing (sulphites, wood chips, selected bacteria, yeasts and enzymes) result in levels of acceptance that differ between consumers. Three positions were identified in this regard: (i) requests to ban substances that might have a possible negative impact on health, regardless of whether they are necessary for making a good wine; (ii) requests to prohibit additives or processing aids that might potentially affect wine flavour and/or its naturalness or tradition; or (iii) requests to allow the same substances to be used as in conventional production, but introduce lower concentrations for organic wine than for conventional wine, and introduce a declaration of their use in organic wine processing. A combination of the three strategies would match most consumers' interests.

Labelling preferences were also investigated and four different positions were identified:

- (1) a complete list of ingredients, additives and processing aids and methods to appear on the label with an indication of the quantities of additives and processing aids;

- (2) a reduced list of ingredients with the declaration of specific additives and processing techniques;
- (3) a declaration of those substances which were not used;
- (4) no declaration other than the organic label which stands for unambiguous cellar regulations.

The first suggestion would be the most transparent solution. However, it would entail unfair competition between conventional and organic wine if conventional wines were not subject to the same duty of declaration. Besides this, a complete list would be too much for consumers and wine bottle tabs alike.

Concerning the thorny issue of sulphites, the results can be interpreted in two ways: a maximum concentration should be set which, on the one hand, ensures no health risks but which, on the other hand, guarantees a good wine. There was no clear indication from the consumer study that sulphites should be completely banned for organic wine production (with few exceptions), although for most consumers sulphites were seen as critical. However, a lower maximum sulphite concentration might be a solution, which committed organic consumers would understand.

Market demands

From the qualitative market study run within the ORWINE project, some major trends in market demands and expectations can be summarised. In the conventional wine sector, two major wine consumption trends can be observed: a decrease in main European producing countries like France, Italy and Spain and an increase or a stabilisation in consuming countries like Northern European countries, Japan, USA. However, for organic wines, the consumption trend is totally different, with a general increase in all countries, even if it is higher in the consuming countries of Northern Europe than in the producing countries.

Concerning organic market development (for all major products), a general increasing growth trend can be observed in the four study countries (France, Italy, Germany and Switzerland) connected to the increasing demand of consumers for organic food and the development of organic farming in most of the European countries. Nevertheless, the growth rates and the market evolution differ strongly from one country to another.

The importance of import and export markets of wines varies considerably between the case study countries. Some countries, such as Italy and France, are strongly oriented to export. Other countries, like Germany and Switzerland, are more importers of wine. These trends also apply to organic wines. In the attempt to identify the major barriers to organic market development, four potential obstacles to the development of the organic wine market emerge as most significant.

- Low consumer knowledge about organic wines and organic wine production – this lack of knowledge seems to concern wine production in general.
- Problem of business to consumer marketing – this was ranked the third most significant obstacle in Switzerland, Italy and Germany, but seems to be less important in France.

- Strong competition between conventional wines and organic wines – when quality is equal, organic wines are less competitive because of their higher price. This is particularly true in France, where it was cited as the single most important obstacle to organic wine market development, and in Switzerland, where it was ranked third.
- High prices of organic wines – an important factor in France and Switzerland, where it ranked second, but less important in Germany and Italy.

Concerning organic wine consumption, the main consumer trends are the following:

- a wine consumption trend strongly oriented to young and fresh wines, well-balanced wines and organic and local wines;
- specialty wines, wines from disease-resistant and new varieties, labelled wines and dry wines should remain as popular as they are today;
- tannic wines, woody wines and old wines are expected to become less popular with consumers.

Producers' attitude and potentiality

After examining the ORWINE project outcomes (in this case a web-based survey that involved about 500 EU organic wine producers and about 80 in-depth interviews), we can understand what producers mean by organic wine and why they oriented themselves towards such a production strategy. The picture of organic viticulture in Europe, as derived from the analysis, shows that it is practised with only a few differences from conventional quality wine production in similar areas and in similar farming systems. Usually the farm size is small to medium, with a few exceptions like Spain and France. Interviewed producers are to be considered 'technically mature' as they have frequently been running their vineyards organically for more than 10 years. They often decided to convert for technical reasons (as well as for environmental and marketing ones), as they considered organic viticulture to be an instrument for improvement of wine quality. In countries such as Spain, Hungary and Slovakia, organic management has been introduced more recently.

All producers are specialised and for 94% of them organic wine is their main product. A large majority of them process exclusively their own grape and bought-in juice and wine is a limited exception; variety choice is very similar to conventional viticulture except for the use of resistant hybrids in Germany, Austria and Switzerland. Organic grapevines have only recently become available and, as a consequence, are rarely used.

Concerning agronomic operations, soil management is considered to be a main issue by a large majority of producers, while trellising and pruning systems show no significant differences from conventional viticulture. Fertilisation is based on green manures in Southern countries and on compost in the North, while the use of commercial fertilisers is relevant only in Italy and Spain. Irrigation is relevant only in Switzerland.

Pest and diseases are a constant issue, having significant impact on grape

quality at least every three years and even more frequently in Northern countries. In the South, the problem arises less frequently, with exceptions for specific pests in highly infected areas (corky bark virus, etc.). The use of copper makes downy mildew manageable and it never appears as a major problem. Production level is often lower compared to conventional producers of similar wine types in the same area, but it is usually a quality choice and not a limitation due to the organic method applied.

Wine processing is performed in cellars with a generally good level of technology and where temperature control and management devices in particular are available. In general, a limited quantity of adjuvants and additives is used, even if a wide range of products are used. The products are considered essential but are always used in specific cases and with a good understanding of the mode of action.

10.2 Organic wine: a synthesis attempt

Combining consumers' request with market needs and producers' actual and potential production strategies, it may be concluded that organic wine is:

good wine that maintains the characteristics of the grape and of the specific place and farming system where it comes from and without the use of additives, processing aids and techniques that may have a negative impact on human health (producer and consumer), on environment and on the integrity of the product. Some essential (to quality) additives of non-natural origin may be used but in limited amounts and exclusively in case of real need.

The above proposed definition must be considered not as 'perfect' but as a working tool able to give guidance to professionals and researchers as to how to produce and improve organic winemaking.

10.2.1 Demand and need to decrease treatments and additives in organic winemaking

Since 1991, when the first harmonised EU rules were adopted (EEC 2092/91), the organic sector has grown remarkably. The mainly market-driven policy encouraged farmers to convert to organic production and existing organic producers to improve their organic production techniques. New oenological treatments and developments, as well as technical progress triggered the need for adjustments to the legal framework and the extension of the legislation's scope by regulation EC 834/2007 (Fladl and Hulot, 2009).

Organic farming is focused on low-input farming and limiting the use of additives and treatments that have a negative impact on the environment and that affect human health, like preservatives from the group of sulphiting agents which can cause adverse reactions and allergic responses in sensitive people (Food Additives and Ingredients Association, <http://www.faia.org.uk/foodallergy.php>).

These demands are in line with a trend among consumers toward healthier living, which is linked to consumption of healthy food and drink. Following these trends, the product should be less processed, less preserved and more natural. The EU, as part of its allergens labelling regulation for food and beverages, stipulated that all wines sold in the EU must include a statement that wine 'contains sulphites' if there is any detectable presence of sulphite in the final product.

Due to their allergenic potential, it is now compulsory that the presence of additives derived from milk and eggs (fining agents such as casein, ovalbumin, etc.) be reported on the label (EEC 1333/2008). They could, to a certain extent, be replaced by other fining agents, like gelatine, isinglass or plant proteins, since their use is negatively perceived by consumers and market actors. A general trend towards lower SO₂ concentrations also in the production of conventional wines is noticeable in Commission Regulation (EC) No 606/2009 of 10 July 2009 laying down a reduction of 10 mg/L of total sulphite in the wine categories.

The studies within the ORWINE-project indicated that the following points are expected from organic wines and from a new regulation for organic wine production:

- clear difference between conventional and organic wine production; not only in the production of grapes;
- good and high quality
 - for each wine style, region and vintage
 - to keep national specialities – typical wines of certain regions
 - long ageing potential and stability
 - no use of genetically modified organisms (GMO) or products (derivatives) of GMO origin
 - no additives, processing aids and techniques that may have a negative impact on human health, on the environment and/or on the authenticity of the product; bio-conform!
 - additives and processing aids from agricultural should mainly come from organic production
 - some essential (to maintain quality) additives of non-natural origin may be used but in limited amounts and exclusively in case of real need
 - lower maximum concentrations for crucial substances (e.g. SO₂).

Wine from grapes (but not from other fruit) was excluded from the scope of EEC Reg. 2092/91; therefore organic wine producers have developed specific approaches for processing their wines in ways they consider to be compliant with organic-farming principles. These private initiatives in the different wine producing countries led to the development of standards more restrictive than the legal requirements for conventional wine, with limits on the use of additives and technical processes at all steps of wine processing, from grape picking to wine bottling and storage. They were developed by producer groups (Germany, France, Austria), organic-farming associations connected with certifiers (Austria, Germany, Greece, Italy, Switzerland), certifiers (Spain) and representative national platforms for the organic wine sector (Spain, Switzerland). In this last case, the

Table 10.1 Overview of the different statuses of additives regarding standards and European organic regulation

	Allowed in Annex VII of EC Reg. 889/2008	Not mentioned in Annex VII of EC Reg. 889/2008
Allowed by all private standards	SO₂ gas , enzyme preparations, ascorbic acid , selected micro-organisms (if non-GMO origin): dry yeasts , lactic acid bacteria and fresh lees , casein, isinglass and egg-white albumin, potassium carbonates , L-(+) tartaric acid , citric acid, bentonite , perlite, diatomaceous earth, charcoal, tannins , silicon dioxide (gel or colloidal solution), inert gases (argon, O₂ , N₂ , CO₂)	Ammonium sulphate
Allowed by at least one private standard	Potassium metabisulphite , potassium alginate, gelatine , gum arabic, potassium bitartrate, Ca-carbonates, cellulose , β-glucanase enzymes	Diammonium hydrogen phosphate (DAP) , diammonium sulphite, yeast ghosts , metatartaric acid , Aleppo pine resin, potassium hydrogen tartrate, double calcium malate and tartrate salt, copper sulphate , potassium caseinate, thiamine
Not mentioned in any private standards		Plant proteins , yeast mannoproteins , urease enzymes, copper citrate
Not mentioned or forbidden by at least one private standard		Wood chips, Lysozyme
Forbidden by all private standards		Sorbic acid , potassium ferrocyanide, dimethyl dicarbonate , calcium phytate, polyvinylpolypyrrolidone (PVPP)

The additives tested during the ORWINE project or for which a factsheet has been written are shown in bold face.

Source: Jonis and Micheloni, 2009 (modified).

participation of local and national public authorities gives a quasi-official status to the standards. They are mandatory for all organic winemakers in Switzerland, whereas in Spain the organic producers are free to follow or ignore the national standards. Limits have also been introduced in the consuming European countries. For example, one concerns SO₂ rates in wine at consumption (UK, NL).

Comparison of these standards was mainly focused on additives and processing aids, although techniques and general considerations were also taken into account.

Substances reported in the private standards can be divided into two categories:

- ingredients, additives and processing aids, which are already allowed for fruit wines in Annex VI of EU Regulation (EEC) 2092/91 (taken up in Annex VIII of EC Reg. 889/2008) and which are mentioned by most of the private standards;
- substances (additives and processing aids) that are not mentioned in the organic processing positive list (often simply because they are used exclusively for wine, which is so far outside the scope of the regulation) and are either allowed by most of private standards or not mentioned or forbidden – these ‘new substances’ have been evaluated by a staff of experts and factsheets have been compiled for each of them.

Table 10.1 reports the different statuses of ingredients, additives and fining agents used in winemaking as mentioned in national and private standards for organic winemaking. It is an overview of the existing situation and served as one of the starting points for the discussion of substances to be considered as compliant with the organic concept. All the additives and ingredients presented in Table 10.1 are allowed by the Common Market Organisation (EU Regulation 1493/1999) for conventional wine production (Jonis and Micheloni, 2009). The majority of stakeholders were against national or regional adaptation (except in the case of exceptional climatic conditions or ‘special’ wines) and preferred to regulate all issues at EU level and possibly harmonised with other international organic regulations. Both additives (and processing aids) and physical techniques have to be regulated under the new organic regulation. On the basis of scientific knowledge in oenology, experiences in winemaking and experimental research work, the analysis of private standards and the needs and experiences of wine producers, it can be clearly stated that a ‘zero-input’ organic winemaking can be possible only in very limited cases and seasons and is very seldom suitable for high-quality wines. Such a policy risks discouraging producers who would like to convert their farm to organic-wine production. ‘Zero-input’ regulation similar to that applied in USA, for instance, might expose a significant number of European producers – in certain regions and vintages – to unacceptable levels of risk.

First, therefore, it was important to identify and to evaluate those additives and processing aids which were accepted by organic wine producers, consumers, expert evaluation and laboratory experiments within the ORWINE-project and/or at pilot-winery experiments. Table 10.2 gives a summary of the evaluation procedure (Jonis and Micheloni, 2009). Two categories have been chosen: additives and processing aids which received only one positive evaluation and those with at least one negative evaluation (allergic properties have been considered as a negative evaluation).

In the agreement, it was proposed that a positive list of additives and processing aids should be created with, if necessary, limitations on the amount which can be used. Furthermore, microbiological products under the generic term of ‘microorganisms and derivates’ allowed by Regulation EEC 2092/91 were to a great extent positively evaluated by ORWINE studies and experts for organic winemaking. It

Table 10.2 Summary of additives and processing aids evaluation

Categories	Positive evaluation	At least one negative evaluation
Already allowed for organic processing	Selected yeasts and lactic acid bacteria, enzymes, ascorbic acid, potassium alginates, gum arabic, isinglass, potassium tartrate, potassium bicarbonates, calcium carbonates, tartaric acid, citric acid, bentonite, charcoal, CO ₂ , argon, N ₂ , diatomaceous earth, perlite, cellulose, wood and grape tannins, caramel	SO ₂ gas, gelatine, potassium metabisulphite, casein, egg-white (ovalbumin), lactalbumin, potassium caseinates
Not allowed in organic but allowed by most of the standards	Thiamine, copper sulphate, diammonium hydrogen phosphate, yeast ghosts (walls)	Ammonium sulphate, diammonium sulphite, metatartaric acid
Not allowed in organic and by the majority of standards or not mentioned	Ca-tartrate, plants proteins, yeasts mannoproteins, wooden chips, Aleppo pine resin	Sorbic acid, potassium ferrocyanide, DMDC, calcium phytate, PVPP, Lysozyme, plant proteins, ions exchange resins
Still not allowed by European regulation on wines, but will be allowed in the new regulation		Malic acid, lactic acid

Source: Jonis and Micheloni, 2009 (modified).

should also be pointed out that spontaneous alcoholic fermentation as well as spontaneous malolactic fermentation is favoured among some of the organic wine producers.

For some ‘derivates of microorganisms or other organisms’, like mannoproteins or Lysozyme, apprehension was expressed and, as a consequence, it was decided to submit them to a broader evaluation. There is no doubt that the use of GMO or their derivates is strictly rejected and not allowed in any of the private standards for organic food production.

All additives from agricultural production have to be taken from organic origin (e.g. sucrose or rectified concentrated musts (RCM)). No additives, processing aids and techniques that may have a negative impact on human health (e.g. allergic responses in sensitive individuals), on the environment and/or on the authenticity of the product should be used for organic wine production. All additives should be ‘bio-conform or bio-compatible’.

This definition is also related to the production process of certain substances and should in addition take into account CO₂ emissions or energy consumption during the whole production process. Therefore, exceptions could be made if certain substances are not available in an ecological quality and in an adequate

amount for organic wine production (Hofmann, 2009). A negative evaluation was mainly expressed for the use of sorbic acid (potassium sorbate), polyvinylpyrrolidone (PVPP), ammonium bisulphite, potassium ferrocyanide, dimethyldicarbonate (DMDC) and carboxymethylcellulose (CMC).

10.2.2 The real hot issue: SO₂ use and limitations

There has been an intensive debate concerning the controversial issue of whether to lower SO₂ use and to what extent and for which wine categories. The majority of the private standards on organic winemaking allow the use of SO₂ at significantly lower concentrations than the CMO limitations for conventional wines. The extent to which sulphites are lowered depends on the wine growing country, the wine style and wine production habits. Maximum allowed concentrations of sulphites are, for example, higher in the private standards of France, Germany and Switzerland than in those of Italy and Greece.

Producer surveys and analysis of organic wines from different exhibitions and competitions in Europe demonstrated that SO₂-concentrations of organic wines are in general low, but there tends to be an easily noticeable difference in that German and Austrian wines, for example, have a slightly increased total sulphite content compared to Italian wines.

Therefore, it can be predicted that producers will have different ideas on SO₂ reduction. The majority of wine producers of Germany and Austria are in favour of no limitation of sulphite use in comparison to conventional wines, whereas producers from France and Switzerland mainly agreed to a reduction of 20–30% of SO₂ limits. The majority of organic winemakers from Italy and Spain approved a reduction of SO₂ to 50 % of conventional limits.

The following three scenarios can be proposed after a controversial discussion among wine producers and scientists of the different wine growing areas in Europe (Jonis and Micheloni, 2009):

- *Scenario 1.* SO₂ not allowed in organic winemaking.
- *Scenario 2.* No specific limitation on SO₂ use in organic winemaking (limits as from CMO for conventional wines).
- *Scenario 3.* A step-wise limit on SO₂ use that must be significant compared with that in conventional winemaking and that allows the sustainable production of 'good' organic wine. Furthermore the progressive decrease should be based on yearly monitoring by EU member states.

It is recommended that the development of scenario 3 should be promoted with simultaneous consideration of the following points:

- only for main wine categories;
- special wines should be excluded;
- the reduction process should be based on large-sample analysis conducted by member states;
- SO₂ reduction should be accompanied by yearly evaluation;

- technical support of producers to maintain wine quality;
- regional specification should be considered (climatic conditions, specific tradition, etc.) as well as the ageing potential of wines;
- member states should have the opportunity for a derogation of SO₂ limits in years with extreme climatic situations or specific circumstances.

10.2.3 Evaluation of traditional and modern innovative techniques and practices

In organic winemaking, several traditional and also some modern physical techniques are commonly used and, in general, preferred, e.g. pressing technologies, aeration, oxygen addition, use of inert gases (nitrogen, CO₂, argon), thermal treatments, centrifugation/flotation and filtration. In general, it could be concluded that concentration techniques that are allowed by Reg. (CE) 479/2008 received mainly negative evaluations for organic production:

- reverse osmosis (RO) of musts;
- evaporation of musts;
- cryo-concentration of musts and wines.

Modern techniques and practices which are not allowed by Reg. (CE) 479/2008 and which have received general, negative evaluations are:

- ion-exchange resins for musts and wines (except for concentrated rectified musts; in order to avoid unfair conditions among producers from different countries in Europe);
- wine separation techniques, like spinning cone column, reverse osmosis on wine, distillation, etc.;
- ultra- and nano-filtration of wines.

Further techniques that resulted in more controversial evaluation are, for example, the physical acidification of musts and tartaric acid stabilisation with bipolar electrodialysis and acidification of wines with addition of lactic or malic acid (max. 4 g/L).

The ORWINE consortium has reached the decision to consider a negative list for technologies or technical processes for organic winemaking. This will also offer organic wine producers new developments and technical innovations which will eventually be recommended by the International Organisation of Vine and Wine (OIV) and/or which are already allowed and successfully used in other countries such as USA, Chile or South Africa.

Enrichment is a practice allowed in the EU for conventional wines. Depending on the regions, addition of dry sugar or concentrated must is allowed up to a certain concentration. Evaporation, cryo-concentration and RO are also allowed for the same purpose. The new wine regulation has limited the increase in percent alcohol by enrichment to a maximum of 3% in Zone A, 2% in Zone B and 1.5% in Zone C. In exceptional years this limit can be increased by 0.5%.

Regarding organic winemaking, there was no consensus among experts, pro-

ducers and consumers concerning the issue of whether enrichment is acceptable. In Italy, France, Spain and Portugal there is a clear preference (around 60% of producers in favour of limitation or prohibition of enrichment) for stricter limits to enrichment of organic wines, while the situation is the opposite in Germany (95% in favour of no further limitation), Austria (63 % in favour of no further limitation), Switzerland (52% in favour of no further limitation). What gets a unanimous consensus is that if enrichment is needed, it must be done using organic materials (sugar, concentrated must, rectified concentrated must). On the other hand several physical practices are in general seen by a majority as not compatible with the organic wine concept.

Nevertheless, the principle and the level of limitations remain controversial. The reason for limiting enrichment in organic winemaking is linked to the concept of 'integrity' or 'full respect of the raw material' that is required by consumers and often used by organic producers as a claim for their 'authenticity' and the highly 'natural' character of their products.

10.2.4 Strategies to improve organic wine quality and to lower the use of SO₂

One main task of the ORWINE consortium was to work out strategies to improve organic wine quality with regard to lower SO₂ additions and final total SO₂ concentrations. The experience of different research units, focusing research on selected topics and trials with pilot wineries in different wine growing areas of Europe, made it possible to develop protocols on important organic wine production processes for white and red wine production. The protocols were developed on the basis of sound grape processing. For red wine the main emphasis was on:

- production process differentiated between wine pH value greater than and less than 3.4;
- fermentation management with the use of a 'pied de cuve' with selected dried yeasts, well-balanced nutrient supplements according to the level of yeast available nutrients (YAN), malolactic fermentation (MLF) with starter cultures;
- management of spontaneous fermentation (alcoholic and malolactic);
- yeasts–lactic acid bacteria co-inoculation.

and for white wine:

- production process defined for three options: without MLF, MLF during or after alcoholic fermentation;
- fermentation management with the use of a 'pied de cuve' with selected dried yeasts, well-balanced nutrient supplements according to the level of yeast available nutrients (YAN), malolactic fermentation with starter cultures;
- management of spontaneous fermentation (alcoholic and malolactic);
- hyperoxygenation;
- ascorbic acid and tannins.

All protocols are available from the 'code of good organic viticulture and

Table 10.3 Overview of SO₂ reduction strategies from laboratory and pilot winery testing

Production step	Practice	Laboratory or experimental scale results	Tested in pilot wineries	Remarks/effect
Grape processing	Reductive wine-making – addition of tannins and ascorbic acid	Positive	Yes	Reduction of oxygen activity in juice and consequently good alternative to SO ₂ addition for musts protection
Juice processing	Hyper-oxygenation	Positive	Yes	Stabilisation of grape juice by oxidation of unstable phenolic compounds; indicated for some varieties
Alcoholic fermentation	Use of selected yeasts	Positive	Yes	Reduction of the dominance of indigenous microorganisms or high SO ₂ producing strains
	Activated yeasts starter culture	Positive	Yes	Prevention of microbial contamination by giving competitive advantage to selected yeast strains
	Thiamine supplementation	Positive	Yes	Reduction of the formation of SO ₂ binding compounds – higher free SO ₂ available
	Diammonium hydrogen phosphate (DAP) addition	Positive	Yes	Good yeast activity and reduction of the risks of microbial contamination
	Ammonium sulphate addition	Negative		Can be metabolised by yeasts to produce SO ₂ , better to use DAP
	Use of Lysozyme	Positive	Yes	Reduction of lactic acid bacteria contamination; useful in high pH juices and wines, where SO ₂ is less effective
Malolactic fermentation	Use of selected bacteria	Positive	Yes	Correct management of MLF
	Co-inoculation yeasts–lactic acid bacteria	Positive	Yes	Prevention of microbial contamination, reduced biogenic amines formation even with low SO ₂
Wine storage and preparation	On lees storage	Positive	Yes	Limitation of oxidations; risk of biogenic amine pollution if MLF is incomplete or occurs spontaneously

Table 10.3 *continued*

Production step	Practice	Laboratory or experimental scale results	Tested in pilot wineries	Remarks/effect
	Bipolar membrane treatment	Positive		Effective in pH reduction, and higher (more favourable) free total SO ₂ ratio, but not allowed and expensive; tartaric acid addition is equally effective
	Cross-flow microfiltration	Positive		Physical elimination of microorganisms
	Flash-pasteurisation	Positive		Physical elimination of microorganisms
	Glutathione (GSH) supplementation/preservation	Positive		Antioxidant effect, not allowed (useful: strategies for the preservation of natural GSH contents)

Source: Jonis and Micheloni, 2009 (modified).

winemaking'. The protocols have to be adapted to the wine style of the organic wine producers and the available facilities as well as being compatible experience on wine ageing, the style of wine and the wine growing region. The protocols allow a lower use of SO₂, within the range specified by most of the private standards for organic wine production. High-quality organic wine with a lower use of SO₂ can only be achieved if microbial contamination is avoided through the whole production process and if optimised alcoholic and malolactic fermentation and oxidation management can be achieved.

Within the ORWINE project, certain research topics were selected with a specific focus to evaluate processes and additives as well as innovative treatments which are not yet allowed to be applied to the production of organic wine. The main outcomes are outlined below.

- Lowering of SO₂, is technically possible with certain techniques and alternative additives which are in tune with the organic principles and are admitted by both CMO and organic regulations, as well as accepted by the majority of organic winemakers.
- The effect of certain oenological tools on SO₂ is influenced by grape variety, grape must composition and wine type.
- Potential for implementing SO₂ reducing strategies is higher in pre-fermentative steps than during wine storage and ageing.
- Strategies to reduce SO₂-binding compounds offer a more favourable ratio between free and total sulphur dioxide, and lead to a better efficiency of each SO₂ addition.
- The choice of an adequate yeast strain with a lower natural production of SO₂

and lower requirements for nutrient supplements is a key instrument to lower SO_2 concentrations in wine.

- An optimised combination of oenological tools is necessary (see and consult the ‘code of good organic viticulture and winemaking’ and available protocols for optimised management strategies to improve wine quality with lower input) to maintain and to avoid a negative effect on wine quality.

Table 10.3 offers a selection of SO_2 reduction strategies and additives tested within ORWINE project (Jonis and Micheloni, 2009).

10.3 Harmonisation process

An intensive debate about new rules for organic wine production has already started, launched by the European Commission in 2009. Apart from the different needs of the various wine growing regions the discussion also includes aspects relevant to climate change and investigates how organic wine production can be adapted to its serious effects (higher sugar values in the must, lower nutrient composition in grapes and musts, lower acidity and higher pH-values, etc. in certain vintages and wine growing regions). Organic winemaking should combine tradition, innovation and science to benefit the shared environment and promote fair relationships and a good quality of life for all involved (http://www.ifoam.org/growing_organic/definitions/doa/index.html). Furthermore, cycles, oenological treatments and strategies should be adapted to local conditions and should guarantee that the typical character and quality of wines from different winegrowing regions can be maintained.

10.3.1 How to deal with the production needs of ‘special wines’?

‘Special wines’ represent a small part of the global production of organic wines, but they can often be very important and/or emblematic of a region (for example, Sauternes or Porto) or correspond to local consumption habits (for example, Retsina wines in Greece). The reason for excluding ‘special wines’ from the SO_2 reduction proposal while including them in the general regulation for organic wine is that they are obtained in a very traditional and specific way. Setting SO_2 limits at the EU level would hamper their specialty status. They are important cultural and niche market products, but the total quantity of all ‘special wines’ produced in the EU is very limited. Even if their SO_2 concentration is often very high, the amount consumed is very low.

10.3.2 Links with non-European organic regulations

The National Organic Programme (NOP) in the USA requires that wines to which sulphites have been added cannot be labelled as ‘organic wines’. This presents a real challenge for European organic wine producers. Some have already started to produce wine categories with no sulphites for the US market.

It would be desirable for the EU to negotiate the equivalence of its rules, as soon as they are introduced, with the requirements of the export countries. One difficulty might be that the *Codex Alimentarius Guidelines* for organically produced food – the official worldwide reference for governmental bodies (along with IFOAM Basic Standards for the private sector) – has not yet considered wine processing, although adding wine-processing sulphites to the list has been discussed at several meetings.

As wine is the most regulated food product in the EU, it is advisable to organise the organic certification system taking into consideration at least the operational AOC certification. This means verifying if, and to what extent, the certification procedures and documents of the two systems can be unified, leading to simplification and cost reductions.

During producers meetings, some concern was expressed as well on the potential problems linked to the fulfilment of AOC sensorial parameters while decreasing or even avoiding SO₂ use. A few cases were analysed where organic low-SO₂ wines had difficulties in meeting specific AOC sensorial characters even if judged positively in terms of global quality. Such issues should be considered and dealt with at AOC committee level (Jonis and Micheloni, 2009).

10.4 Future trends

As the ORWINE project was limited in time, not all research questions could be studied or completed in a sufficient form. The list below offers some key areas where future research is necessary:

- Additional experiments are needed to evaluate the longevity and stability of wines prepared with specific oenological tools to lower SO₂ addition.
- Eventual SO₂ limits for organic wine should take into account the normal operating conditions of different winemaking areas, as well as climate, seasonal events, 'winemaking habits', etc.
- Assessment of the impact of new technologies on both the quality and the authenticity of organic wines, as well as their environmental impact (e.g., energy use, bio-diversity), including the applicability for different types of wineries.
- Further research is needed to determine the links between techniques used in the vineyard and grape quality (for example, how to avoid the nitrogen deficiency in the musts) in order to improve the quality of the grapes and so reduce the need of oenological additives.
- Investigation of the authenticity of wines: the ORWINE studies have shown that wines from farm protocols and SO₂-lowered wines could mostly be differentiated concerning flavour and taste. The overall appreciation of the wines was, however, equal. Now the important and still unsolved question is which vinification technology really produces the wine with highest 'authenticity'. 'Authenticity' is a frequently used term in (organic) winemaking and especially wine selling. This topic requires more scientific attention.

- Some organic producers are making high-quality wines with very few additives and processing aids. It would be interesting to study this approach, in order to develop practical and technical tools to help other producers to improve low-input winemaking.
- During the ORWINE project, the work on the evaluation of additives and processing aids highlighted the necessity to increase knowledge of the origin and mode of production of the discussed supplements and then to develop a scientific protocol to evaluate the compatibility of an additive for organic production (Jonis and Micheloni, 2009).

10.5 Acknowledgements

The authors gratefully acknowledge the financial support of the Commission of the European Communities, under Priority Area 1.2 (Organic Viticulture and Wine Processing) of the Sixth Framework Program for Research, Technological Development and Demonstration within the Integrated Project No. 022769 (Organic viticulture and winemaking: the development of environment and consumer friendly technologies for organic wine quality improvement and a scientifically based legislative framework). The information in this report does not necessarily reflect the Commission's views and in no way anticipates the Commission's future policy in this area.

The authors thank all ORWINE partners and participants, especially Roberto Zironi, Piergiorgio Comuzzo and Sergiu Scobioala (UNIUD; Italy), Monique Jonis (ITAB; France), Philippe Cottureau (IFV; France), Jean-Michel Salmon (INRA, France), Gianni Trioli (VINIDEA; Italy), Uwe Hofmann (ECOVIN, Germany), Maik Werner (GRC; Germany), Markus van der Meer, Dominique Levite, Otto Schmid, Franco Weibel (FIBL; Switzerland), Ettore Capri and Georgios Fragoulis (Universita Cattolica del Sacro Cuore; Italy), Lena Wietheger and Marco Schlüter (IFOAM EU Group, Belgium) and others.

Furthermore the authors are also grateful to winemakers in the pilot wineries for their support and willingness to test the ORWINE protocols on specific procedures for organic wine production.

10.6 References

- Fladl M and Hulot J F (2009), The new regislative framework for organic farming. In: IFOAM EU GROUP (ed.), *The New EU Regulaton for Organic Food and Farming: (EC) No 834/2007, Background, Assessment, Interpretation, Dossier*, Brussels, Belgium, 14–18.
- Food Additives and Ingredients Association, <http://www.faia.org.uk/foodallergy.php>.
- Hofmann U (2009), Ökologische Weinbereitung, *Das deutsche Weinmagazin*, 12/13 Juni, 34–37.
- IFOAM, http://www.ifoam.org/growing_organic/definitions/doa/index.html.
- Jonis M and Micheloni C (2009), Proposal and recommendations for elaborating rules for

organic wine in EU Regulation on organic production and labelling of organic products (Commission Reg. EC 884/2008 based on Council Regulation EC 834/2007 repealing Regulation EEC 2092/91), available at: http://www.orwine.org/intranet/libretti/d%205.8%20final%20version%201st%20july%20_263_01_0_.pdf (accessed March 2010).

Micheloni C (2009), ORWINE: a further step towards an EU regulation on organic wine-making. In: IFOAM EU GROUP (ed.), *The New EU Regulation for Organic Food and Farming: (EC) No 834/2007, Background, Assessment, Interpretation, Dossier*, Brussels, Belgium, 52–54.

Trioli G and Hofmann U (2009), *Code of Good Organic Viticulture and Wine-making* (ed.: Hofmann, U. ECOVIN – Federal Association of Organic Wine-Producer, Wormserstraße 162, D-55276 Oppenheim-Germany). This code is available in five languages and can be downloaded from the ORWINE website: www.orwine.org.

Part II

Managing sensory quality

Yeast selection for wine flavour modulation

P. Marullo, SARCO Laffort Inc., France; and D. Dubourdieu, INRA-Université Bordeaux, France

Abstract: Wine has very rich and complex aromas but is paradoxically obtained from grape juice that has very little odour itself. This aromatic transformation is mainly due to the action of yeasts during the alcoholic fermentation. First, yeasts produce secondary metabolites that give to wines its basic ‘winy’ characteristic. Second, some enzymatic activities may transform non-odorous compounds of certain grape varieties into varietal aromas, giving the specific characteristics of ‘*cépage*’. Depending on the yeast strain, the concentration of aromatic molecules is determined, modifying the organoleptic perception of wine. The enzymatic activities of yeast metabolism and their genetic control have been widely studied, allowing genetic selection approaches as well as molecular engineering for selecting more specialized commercial starters.

Key words: *Saccharomyces cerevisiae*, wine aroma, fermentation, genetic determinism, breeding, genetic engineering, esters, volatile thiols, off-flavours.

11.1 Introduction

Since the 1970s, winemakers have had the use of yeast starter (*Saccharomyces cerevisiae*) to carry out grape juice fermentation. The first aim of yeast starter selection is to avoid stuck and sluggish fermentations by inoculating must with strains able to rapidly convert sugar to ethanol. The development of active dry yeast (ADY) greatly facilitated the yeast inoculation and today, most of the world’s wine is made with commercial starters (Degré, 1993). Winemakers soon observed that some strains gave better organoleptic results than others, suggesting that the yeast strains have an important impact on wine quality. Louis Pasteur (1876)

himself had imagined that ‘The taste and properties of the wine could depend on the special nature of yeasts which develop during the fermentation of the grapes.’ Yeast strains were therefore selected according to their organoleptic properties. This selection was primarily oriented towards avoiding the abnormal production of off-flavour compounds (acetic acid, hydrogen sulphide, volatile phenols). Successively, the positive organoleptic role of yeast has been investigated and strains producing more fruity and complex wines selected. Initially, this was empirically based on the selection of aromatic vats fermented by indigenous yeasts. In the past 20 years, the influence of yeast metabolism on chemical composition of wines has been widely investigated. Today, the identification of several key aromas and the rapid determination of their concentration in wines pave the way to developing rational yeast selection programs in this field.

The wine yeast organism presents a huge genetic and phenotypic variability. Most of the fermentation traits and aromatic properties of yeast strains are inheritable (i.e. genetically determined) and may be manipulated by genetic approaches. With the advent of molecular biology and the determination of the complete genome sequence of *S. cerevisiae*, the pathways of yeast secondary by-products were dissected, allowing identification of the main enzymes (and genes) involved in aroma production during the alcoholic fermentation. The molecular manipulation of genes encoding for these enzymes is nowadays an important challenge for identifying the key levers of aroma modulation. Nevertheless, these molecular strategies do not diminish the importance of the classical genetics approaches (breeding and clonal selection) that allow efficient exploitation of the fantastic genetic diversity of natural wine yeast.

This chapter presents the key pillars for the achievement of a relevant and efficient wine starter selection. First, an overview of genetic selection strategies is provided, giving particular attention to the multigenetic determinism of traits as well as the genomic complexity of wine yeast. Second, recent insights are presented into yeast secondary metabolism involved in fermentative aromas. Although much progress has been made, the current knowledge of the metabolic pathways and their genetic regulation is in part insufficient to understand the aromatic variability of yeast strains. Third, the yeast bioconversion of grape odourless precursors in varietal aromas is presented. The biochemical mechanisms involved in this bioconversion remain largely unexplored and likely constitute a promising field of investigation. To give an overall vision of possible applications for wine aroma improvement, some illustrative examples of yeast selection are given along with the text.

11.2 Key issues in efficient wine yeast selection

11.2.1 Combining fermentation qualities and reducing off-flavour production

The first requirement for wine starter is its ability to achieve alcoholic fermentation. Ethanol resistance, completion of the fermentation, high temperature

resistance, cryophilic properties and high fermentation rate were in the past the primary objectives of yeast strain selection. Although new aromatic selection criteria must now be taken into account, these fermentation requirements must be a priority in all cases. In fact, some commercial starters exhibiting remarkable aromatic properties fail to ferment in harsh conditions. This often impairs the full expression of their aromatic properties and compromises their commercial success. Therefore, the classical criteria of yeast strain selection must be maintained and aromatic properties of wine yeast must be added to the fermentative ones. The second main objective for wine starter selection is to avoid the over-production of off-flavour metabolites that may compromise wine quality. The most important metabolites among these are acetic acid and hydrogen sulphide. Other off-flavours of lower odour impact may also be limited during strain selection; for example, the lack of volatile phenol production may be a decisive criterion of selection for white wine starters. Once these primary objectives are guaranteed, other aromatic criteria may be improved to give a particular specificity to a yeast strain.

11.2.2 Tailoring a yeast strain in accordance with a wine type

Around the world, more than 100 yeast starters are offered by several distributors, which could confuse the choice for the winemaker. Therefore, the efficient selection of a commercial starter needs to respond to a specific wine type. This can be defined in accordance to the aromatic specificity of grape variety used and may respond to a technical scheme of vinification (sparkling wines, fruity white wines for immediate consumption, yeast for 'barrel-aged' red wines, etc.). According to these objectives, the fermentation properties as well as the aromatic qualities of selected yeast must be different. Regarding fermentation properties, yeast adaptation to extreme temperature conditions is a good example: white wine requires cryophilic strains, while red wine vinification requires temperature-resistant strains.

Regarding aromatic properties, ester production illustrates the specificity of yeast strains. For example, wines intended for immediate consumption (<12 months) (popular premium) require the full expression of fruit aromas by using yeast strains producing substantial amounts of esters. Due to their relatively short life, esters do not allow a long-term expression of grape varieties. Therefore, winemakers wishing to better exploit the potentiality of 'terroir' and 'cepage' need to choose a strain producing smaller amount of esters. In other cases, the yeast strain may help the winemaker to produce wines with higher varietal typicity by revealing specific aromas from grape variety precursors as illustrated by the Sauvignon blanc example.

11.3 Selection of natural yeast isolates: methods and limits

Historically, the selection of yeasts for winemaking consisted of identifying from grape juice or wine *Saccharomyces* strains that can ferment grape juice efficiently and produce good-quality wines. As *Saccharomyces* yeasts are rarely present on

grapes, their isolation from natural grape flora is generally carried out in fermenting grape juice (Frezier and Dubourdieu, 1992; Martini, 1993; Sipiczki *et al.*, 2001; Caridi *et al.*, 2002; Cocolin *et al.*, 2004; Valero *et al.*, 2007). Once a relevant number of strains has been isolated, they are oenologically characterized for their fermentation and aromatic characteristics. The fermentation traits can be evaluated easily by carrying out small-scale fermentations in synthetic media and eventually in grape juice. Organoleptic traits are more complex to assess. The more rational approach consists of measuring the chemical compounds found in wine and selecting strains according to quantitative values. Today, through the use of gas chromatography/mass spectrometry/olfactometry, it is possible to measure the yeast production of chemicals determining a particular aroma in wine without the use of sensorial analysis. This allows yeast strain selection programs the chance to obtain appropriate candidates. Because many wine aromas are determined by chemicals present in trace amounts, efficient selection programs require a strong interaction between analytical chemists and wine microbiologists. Nevertheless, the analytical chemistry approach may present some limits. In fact, although many odour-active wine aroma compounds have been identified, some complex notes remain unexplained by a simple molecule. Moreover, the laboratory conditions of microvinifications are far different from the cellar reality (oxygen transfer, temperature control, grape must quality). Therefore, once some yeast strains are selected by chemical analysis, the real aromatic contribution of a few candidates also needs to be evaluated by sensorial analysis in different grape juice matrices. In any case, the success of a selection program is strongly dependent on the oenological relevance of the tests used for strain screening.

The likelihood of identifying a strain expressing many optimal properties for winemaking is very low. Different refinements have therefore been proposed to overcome this obstacle. Molecular engineering and classical genetics strategies are the major ones. Recently, an alternative strategy consists of mixed culture with 'aromatic' and 'vigorous' yeasts in the same vat. This mixed culture may be achieved with various *Saccharomyces* strains or by mixing *S. cerevisiae* and non-*Saccharomyces* species. Although recently developed by some commercial companies, the contribution of each partner to the mix is in most cases unpredictable and the industrial reliability of this approach remains to be demonstrated by solid scientific data.

11.4 Metabolic engineering

Metabolic engineering consists of using DNA recombinant techniques to modify the input/output of yeast metabolites in order to enhance the wine quality. Two main approaches are possible. The first, named self-cloning, consists of delivering a strain its genome consisting of only a *S. cerevisiae* sequence. This may be achieved either by replacing at a particular locus an allelic form of a gene (allele replacement) (Guillaume *et al.*, 2007; Marullo *et al.*, 2007a) or by modulating the expression level of a yeast target gene by using *S. cerevisiae* promoter and

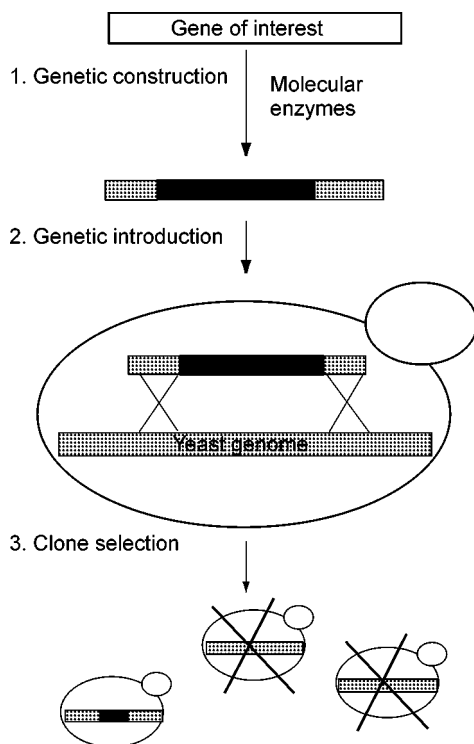


Fig. 11.1 Molecular cloning methodology for wine yeast selection. The figure summarizes the basic method of molecular cloning used for yeast genetic modification. A target gene can be modified *in vitro* by molecular enzymes and then introduced in yeast (transformation). The correct insertion of such a gene at an appropriate locus is ensured by the recombination enzyme of the yeast itself. Clones showing a correct insertion are selected by resistance or auxotrophic markers (freely adapted from Pretorius, 2000).

terminator sequences (Pretorius, 2000). The second is the introduction of heterologous DNA sequences (coming from another organism) to confer new biological functions to yeast.

The main advantage of these techniques is to modify precisely a particular trait without altering the other yeast characteristics. The basic steps of molecular cloning and transformation are illustrated in Fig. 11.1. These techniques are exhaustively described by many books and reviews (Kingsman *et al.*, 1991; Pretorius, 2000) and are detailed in Chapter 2.

Although very efficient, the GM (genetically modified) strategy requires prerequisite knowledge of metabolic pathways and requires the identification of enzymes to be modified. Although rational and efficient in some cases, the current tools for metabolic engineering have some limitations. For example, fine tuning is required to get an appropriate level of metabolite production, and the modified expression of genes involved in aroma formation does not always give satisfactory results. In fact, enhancement of gene expression has given noteworthy but exagger-

ated results in terms of wine sensory attributes (e.g. ester over-production) (Lilly *et al.*, 2006a). New promoter sequences giving more nuanced expression levels are therefore necessary. Moreover, the same genetic construction (for example the overexpression of enzymes involved in glycerol production pathway) does not give the same quantitative results according to strain background (Remize *et al.*, 1999). In fact, the modification of an enzyme activity in metabolic pathway needs to be considered not only locally but integrated in a global metabolic flux. The relative activity/quantity of other enzymes strongly influence this local modification, and the concentration of a specific metabolite may vary importantly according to the yeast background used. Finally, the most important obstacle for metabolic engineering is linked to the fact that the consumer refuses to drink wines obtained from genetically modified organisms. Several reports (Pretorius, 2000; Pretorius and Bauer, 2002; Dequin, 2001; Husnik *et al.*, 2006) have been published on this subject in order to rationalize the ecological and health impact of GM yeasts in wine. Today, the health impact of GM yeasts seems to be reduced by using genetic construction without antibiotic resistance genes. Some projects have even been proposed to improve the health security of wines (Coulon *et al.*, 2006). However, public opinion still remains mainly hostile to use of GM yeasts for winemaking. Regarding the ecological impact, several ecological studies have proven the very limited dissemination of industrial starters in the cellar vicinity (Valero *et al.*, 2005). Nevertheless, the risk of a non-total destruction of a GM starter after the vinification period remains an ethical problem that might not be rectified.

11.5 Conventional genetic strategies

11.5.1 Random mutagenesis

The greater part of yeast metabolic functions have been characterized by random mutagenesis using appropriate selective screening. In contrast to haploid laboratory strains of *S. cerevisiae*, wine yeasts are not well adapted to random mutagenesis. In fact, they are usually diploid or aneuploid, which impairs the selection of recessive mutations. Some protocols of spore mutagenesis have been proposed to overcome this obstacle (Romano *et al.*, 1983; Sipiczki *et al.*, 2004), but they are in practice poorly used. Among the selection programs developed for wine yeast, amino acid auxotrophic mutants were obtained for lowering the production of some higher alcohols (Rous *et al.*, 1983) or for limiting H₂S production (see Section 11.9). The development of such auxotrophic mutants in grape juice constitutes a considerable limit for industrial application due to possible nutritional deficiencies in grape must. More generally, the multigenic determinism of most oenological traits and the difficulty to mute diploid strains explain the sporadic use of random mutagenesis for wine starter selection.

11.5.2 Yeast breeding

Homo- and heterothallism

Cells of the *Saccharomyces* genus proliferate by multilateral budding. When

nutritional starvation occurs, diploid cells undergo meiosis generating four haploid spores wrapped into an ascus. When released from the ascus, the spores can germinate giving a haploid cell that proliferates vegetatively (single spore culture). Haploid cells possess a specific sexual type (α or a) determined by a gene occupying the *MAT* locus which can be expressed in two allelic forms (*MAT* α or *MAT* a) (Herskowitz *et al.*, 1992). Haploids of opposite sexual types can mate producing an asexual diploid cell (a/α). In contrast, the conjugation of cell belonging to the same sexual type is not possible. Yeasts are termed heterothallic when their single spore culture possess a stable sexual type, either α or a . Yeasts are termed homothallic when some of individuals of a single spore culture can modify their sexual status by switching their mating type. In this case, a single culture contains haploid cells with opposite sexual types. These cells rapidly conjugate, originating a diploid strain (autodiploidization) homozygous for all the genes but the sexual type. Most *S. cerevisiae* strains in nature are homothallic, whereas laboratory strains are generally heterothallic. The homo/heterothallic trait is determined by the *HO* gene which encodes for an endonuclease able to switch the allelic form of *MAT* locus (Herskowitz *et al.*, 1992).

Hybridization

The first studies on yeast hybridization were carried out by Winge and Laustsen (1938). Hybridization can be carried out between heterothallic yeast strains (by conjugation of haploid cells), as direct spore-cell mating (by hybridization between spores obtained from a homothallic strain and cells of a heterothallic strain), and between homothallic yeast strains (by spore conjugation) (Wickner, 1991; Pretorius, 2000; Rainieri and Pretorius, 2000). The conjugation of haploid cells of selected strains is a simple and fast method. Nevertheless, this technique is restricted by the limited availability of heterothallic strains of oenological interest. The cross of homothallic strains is the most widely used method, but requires a precise methodology due to their genetic complexity. In fact, natural wine yeasts present many heterozygous loci in their genomes. Moreover, they are frequently aneuploids (Bidenne *et al.*, 1992) and have a low spore viability (Vezinhet and Galzy, 1969; Mortimer *et al.*, 1994; Johnston *et al.*, 2000). In order to improve the cross-efficiency and to rationalize the choice of cross-partners, it is strongly recommended to start breeding programs with a suitable collection of diploid, fully homozygous strains with a high sporulation frequency and high spore viability. This collection is simply obtained by sporulating natural isolates or commercial starters (Rainieri and Pretorius, 2000; Marullo *et al.*, 2004). Moreover, in some cases, this first sporulation step leads to the elimination of deleterious mutations, readily improving the specific strain (Mortimer *et al.*, 1994; Ramirez *et al.*, 1999). Once suitable spore clones have been obtained, spores of respective progenitors are paired by micromanipulation and the conjugation is assessed by microscopic observation.

This time-consuming procedure requires a level of technical expertise that has probably limited its potential for wide development. To overcome the homothallism obstacle, several solutions have been proposed. First, homothallism can be converted

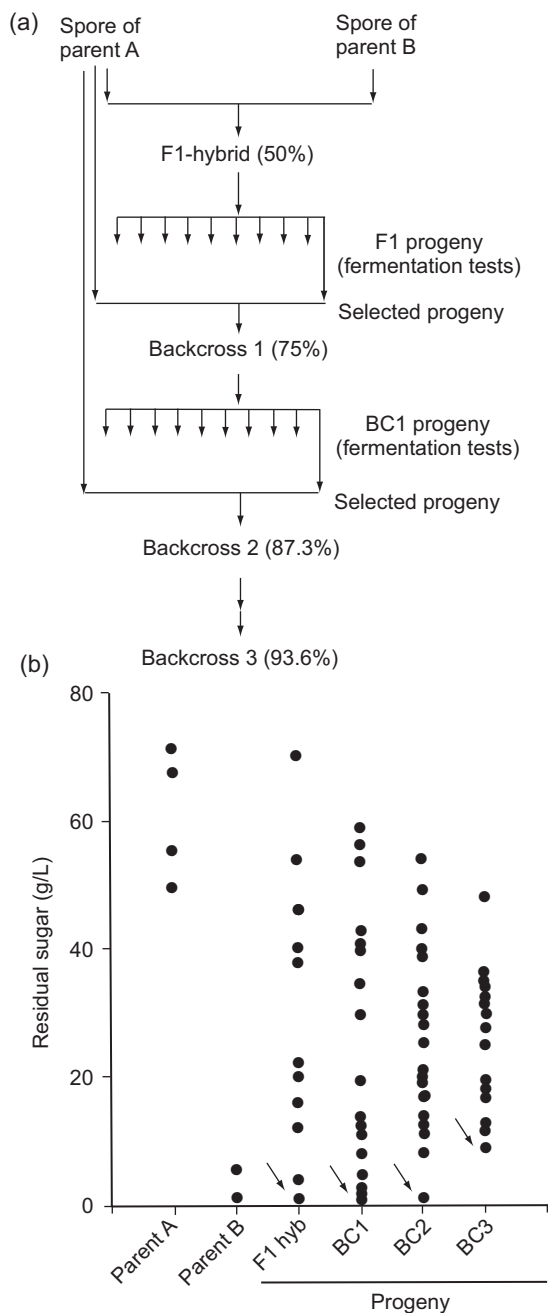


Fig. 11.2 Backcross program driven by physiological tests. (a) The diagram illustrates a backcross strategy used for improving fermentation properties of a commercial yeast starter (parent A). A strain donor presenting optimal fermentation properties is crossed with spores of parent A giving the F1-hybrid. Many progeny clones of the F1 hybrid are tested for their

to heterothallism by allele introgression (Bakalinsky and Snow, 1990), recombinant DNA strategies (Walker *et al.*, 2003, 2005) as well as by selecting ho-mutants (Briza *et al.*, 1986). Second, hybrids formed in a mixed culture may be selected by flow cytometry and cell sorting (Bell *et al.*, 1998). Finally, the use of natural (non-GM) dominant markers is also possible (Ramirez *et al.*, 1998).

The species of *Saccharomyces stricto sensu* genus share the same mating system and are therefore able to give interspecific hybrids both in laboratory (Naumov *et al.*, 1994, 2000; Naumova *et al.*, 2003) and natural conditions (Masneuf *et al.*, 1998; Le Jeune *et al.*, 2007; Gonzalez *et al.*, 2008). According to the biological species concept, these hybrids are sterile and cannot generate viable spore clones. Inter-specific hybridization of wine yeast species has been used in the past to combine technological properties of both species in a unique cell (Zambonelli *et al.*, 1997; Masneuf *et al.*, 2002; Solieri *et al.*, 2008). A recent review details the main features of inter-specific hybrids in wine (Sipiczki, 2008).

Breeding programs assisted by measuring trait values

The main limiting factor of yeast selection by breeding programs is not the homothallism but the quantitative nature of traits to be optimized. In fact, most oenological traits showed a continuous distribution of values due to their polygenetic control (Romano and Suzzi, 1985; Marullo *et al.*, 2004). Practically, only few spore clones derived from the sporulation of a heterozygous strain may inherit optimal technological values. Therefore, it is necessary to test many spore clones to obtain progenitors showing suitable properties. According to the general concepts of quantitative genetics, these properties are genetically determined by several genes (loci), each of them showing various allelic forms that influence the value of investigated trait. These loci are named QTL (quantitative trait loci), and each of them contributes to a small part of the trait variation observed among a progeny. Some QTL may have an important effect on trait values, while others have a minor impact. In a few cases, the segregation of one major QTL may be interpreted as a monogenic segregation. For example, bimodal segregation of volatile acidity (Giudici and Zambonelli, 1992), hydrogen sulphide (Romano *et al.*, 1985) or acetoin production (Romano *et al.*, 1995) allowed for easy selection

fermentation properties. In this case, the residual sugar at the end of fermentation is measured (see (b)). A selected progeny was then backcrossed with spores of parent A giving the backcross hybrid 1. This strain showed statistically 75% of genome identity with the parent A. By recursive crosses and fermentation test, a backcross hybrid presenting 93.6% of genome identity is obtained. This technique allows improving the fermentation properties of parent A but conserving the major part of its genome and consequently preserving its overall organoleptic properties. (b) The residual sugar measured at the end of the fermentation is shown for parent A and parent B (four repetitions for each strain). The parent A shows stuck fermentation while the donor parent B achieved the alcoholic fermentation. The values of residual sugars of progeny clones derived from hybrid F1 and backcross hybrid 1, 2 and 3 are also shown. Progeny clones marked by arrows were selected as parent during the backcross experiment. Data presented were obtained by fermenting a synthetic grape juice containing 260 g/L of sugars at 28 °C (freely adapted from Gourraud *et al.*, 2008).

in breeding programs. In other cases, an epistatic effect of a major QTL may mask polygenic control (Marullo *et al.*, 2006a). Nevertheless, in most cases, QTLs show additive effects and may interact together (epistasis) in a complex fashion. The measurement of trait heritability and the analysis of dominant recessive relations existing in a collection of industrial strains is the first requirement for starting a breeding program (Marullo *et al.*, 2004). Once verified, several unlinked traits may be improved from an F1-hybrid obtained from divergent progenitors showing complementary characteristics. It was demonstrated that many traits could be fixed step-by-step, using appropriate crosses with segregants from an F1-hybrid. Indeed, when two copies of the main 'enhancer alleles', controlling a specific trait, are present in a hybrid, the trait is considered to be 'fixed' and should not segregate widely in the related hybrid progeny. Other remaining unsuitable traits can be optimized in the next step, if the resulting hybrid carries one copy of each of the desirable recessive alleles. Consequently, this method is useful for optimizing most traits by successive crosses and segregation cycles (Marullo *et al.*, 2006a).

One limit of this technique is the availability of reliable and oenologically relevant tests for assessing trait value. Moreover, some crucial organoleptic properties of wine yeast remain difficult to measure by analytical approaches and must be evaluated by sensorial analysis. This is, for example, the case for mouthfeel properties that could vary according to yeast strain. Some commercial starters present a large range of organoleptic properties that have been empirically identified by winemakers. Therefore, instead of constructing *de novo* a new starter and risking unexpected defects, it is possible to conserve the overall organoleptic properties of an elite strain and specifically improve its (measurable) defects by a backcross approach (Fig. 11.2). This strategy allows conserving the previous acquisitions of strain selection in order to continuously improve wine starters. Recently, this technique was applied to improve a red wine starter much appreciated for its organoleptic properties but showing non-optimal fermentation properties (Gourraud *et al.*, 2008). By a phenotype-driven program, a strain for fermentation properties sharing more than 93% of genome identity with the genuine starter was obtained (Marullo *et al.*, 2009).

Breeding programs assisted by molecular markers

Although efficient, breeding programs suffer from the lack of knowledge of the loci/genes determining the desired trait value. In agronomical science, quantitative genetics concepts and QTL-mapping approaches allow the identification of numerous QTL in plants and animals (Frary *et al.*, 2000; Glazier *et al.*, 2002), despite the large genomes of agronomic species (50 to 2000-fold larger than the yeast genome). This task has been largely facilitated by using molecular genetic markers to tag loci controlling economically relevant traits in plants and animals (Lander and Botstein, 1989). Ten years ago, QTL-mapping strategies were applied to *S. cerevisiae* using the powerful genomic tools of DNA-microarrays (Winzeler *et al.*, 1998). Thanks to the compactness of the yeast genome and its sequence annotation, QTLs (Marullo *et al.*, 2007b; Labat *et al.*, 2008), genes (Steinmetz *et al.*, 2002; Hu *et al.*, 2007; Sinha *et al.*, 2008), and even single nucleotide polymorphisms

(SNP) (Deutschbauer and Davis, 2005; Marullo *et al.*, 2007a) that control quantitative traits, have been identified. These advances are bridging the gap between genotypes and phenotypes related to industrial yeast selection purposes. Once suitable alleles are identified, they may be introduced into a strain of interest by backcrossing suitable alleles in a recipient variety, called 'elite', which is generally one of the best ones on the market. This strategy, generally assisted by molecular markers, is routinely used in agronomy for plant and race improvement (Hospital and Charcosset, 1997) and has been recently applied to wine yeast selection. The aromatic properties of wine yeast such as hydrogen sulphide production and phenolic-off-flavour production have been improved without the use of physiological tests by following molecularly the presence of appropriate alleles (Marullo *et al.*, 2007b). The replacement of trait measurement by molecular genotyping allows the development of wide screening and increases the genetic selection power.

11.6 Mixed cultures as an alternative strategy

It is well known that many of non-*Saccharomyces* species initiate spontaneous alcoholic fermentation of the juice, but are very soon overtaken by the growth of *S. cerevisiae* that dominates the mid and final stages of the reaction. Many ecological studies have demonstrated the substantial biomass development (10^4 – 10^7 cell/mL) of non-*Saccharomyces* species during both spontaneous and *S. cerevisiae*-inoculated wine fermentations (Jolly *et al.*, 2003a,b; Combina *et al.*, 2005; Domizio *et al.*, 2007; Fleet, 2008; Zott *et al.*, 2008). It was assumed by many authors that this biomass might be sufficient to impact the chemical composition of the wine and that the contribution of these 'wild' yeasts to overall wine character was significant. Independently, pure culture experiments have reported that some non-*Saccharomyces* species have interesting properties relative to *Saccharomyces* ones. For example, they may produce low amounts of undesirable compounds such as acetic acid (Bely *et al.*, 2008) or produce metabolites of interest such as esters (Rojas *et al.*, 2003; Hernandez-Orte *et al.*, 2008). Recently, some studies have demonstrated that non-*Saccharomyces* species may release relevant amounts of varietal aromas from odourless precursors from grapes such as monoterpenes (Hernandez-Orte *et al.*, 2008) and volatile thiols (Zott *et al.*, 2007).

Grape juice co-inoculation with *S. cerevisiae* and selected non-*Saccharomyces* species has been tested in order to study their impact on wine flavour (Soden *et al.*, 2000; Jolly *et al.*, 2003a; Rojas *et al.*, 2003; Bely *et al.*, 2008). The use of mixed culture is necessary to ensure the achievement of alcoholic fermentation due to the insufficient ethanol tolerance of non-*Saccharomyces* species, although the relative amounts of many metabolites are changed, and only a few studies have assessed the sensorial impact of this mixed culture (Soden *et al.*, 2000). Moreover, the main problem of mixed culture is to ensure the effective establishment of both partners in natural grape juice that constitutes an unpredictable 'microbiological soup'. This area of wine biotechnology is constantly growing and may constitute a future way to modulate wine aroma. However, oenological treatments (cold soaking, pre-

fermentation maceration, sulphur dioxide addition, etc.) allowing the correct establishment of mixed culture must be thoroughly investigated. Therefore the industrial application of this approach remains very limited and requires important care in vinification procedures. A recent review presents the main features of this topic (Fleet, 2008).

11.7 Yeast by-products affecting wine aromas: glycerol

During the alcoholic fermentation of grape sugars, wine yeast produces ethanol, carbon dioxide and a number of metabolic by-products that influence positively or negatively the aroma of wines. The biochemical pathways of these compounds are interconnected and derived from nitrogen, sugar and sulphate metabolism (Fig. 11.3). Most of the enzymes (and their relative genes) controlling these reactions have been molecularly identified by previous academic works as detailed in this section and their genetic regulation is often well known. Nevertheless, the metabolic fine tuning explaining the huge variability between yeast strains has not been fully understood.

This section aims to give an oenological point of view of the aromatic impact of the main secondary by-products of fermenting yeast for winemaking. Once their specific metabolic pathways are detailed, special attention is given to assessing the relevance of selection strategies for controlling their amounts in wines.

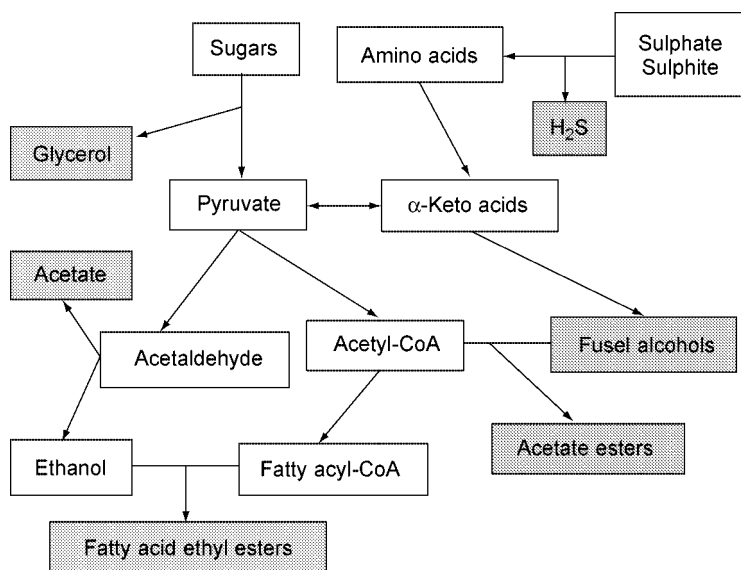


Fig. 11.3 Diagram showing the interaction of the aroma-influencing molecules produced by yeast metabolism during fermentation.

11.7.1 Organoleptic contribution of glycerol

Glycerol is a non-volatile compound that has no aromatic properties but has a sugary flavour similar to glucose. Due to its high concentration in wine (between 4 and 9 g/L), it was thought that glycerol affected wine flavour, giving an impression of fullness and softness. The benefits of increasing glycerol production to improve the sensory characteristics of wines lacking in body have been emphasized in the past (Noble and Bursick, 1984; Pretorius, 2000). Nevertheless, its real contribution is certainly lower, and some winemakers place too much importance on the organoleptic role of this compound. In fact, in many wines, the addition of 3–6 g/L of glycerol is not discernible for the majority of tasters (Ribéreau-Gayon *et al.*, 2000d; Gawel *et al.*, 2007). Moreover, the over-production of glycerol may be in some cases associated with the over-production of undesirable secondary metabolites such as acetic acid and acetoin. Therefore, the selection of yeast strains showing improved production of glycerol should not be a priority for organoleptic purposes.

11.7.2 Metabolic pathways and genetic control

Glycerol is quantitatively the most important by-product of alcoholic fermentation after ethanol and carbon dioxide. Its production mainly occurs during the consumption of the first 50 g/L of sugars through the glyceropyruvic fermentation. At this stage, the only way to ensure the re-oxidation of the NADH cofactor is by reducing dihydroxyacetone in glycerol (Ribéreau-Gayon *et al.*, 2000a). Glycerol is formed by two enzymatic steps: reduction of dihydroxyacetone phosphate to generate glycerol-3-phosphate, which is then dephosphorylated. Each of the two enzymes involved is encoded by two highly homologous genes: *GPD1* and *GPD2* for NAD-dependent glycerol-3-phosphate dehydrogenase (GPD), and *GPP1* and *GPP2* for glycerol-3-phosphatase (GPP) (Fig. 11.4). These isoenzymes have different physiological roles: the Gpd1p–Gpp2p combination is mainly involved in glycerol production during osmotic stress, whereas the Gpd2p–Gpp1p combination is primarily involved in adjusting the NADH/NAD redox balance under anaerobic conditions (Albertyn *et al.*, 1994; Ansell *et al.*, 1997). The first step in glycerol synthesis is the most important as GPD activity controls the amount of glycerol produced (Remize *et al.*, 1999, 2001). The expression regulation of this pathway was largely investigated and is partially controlled by the HOG MAP-kinase pathway mediated by several transcriptional regulators; for reviews see Hohmann (2002).

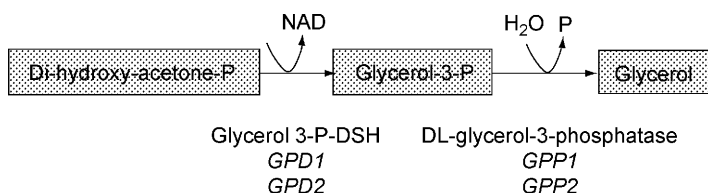


Fig. 11.4 Glycerol biosynthesis. This diagram shows the main metabolites, enzymes and relative genes involved in the *S. cerevisiae* pathway.

Table 11.1 The amount of initial sugar in grape must is strongly correlated with the final amount of glycerol and acetic acid in wine

	Wine 1	Wine 2	Wine 3	Wine 4	Wine 5
Initial sugar (g/L)	224	268	318	324	348
Fermented sugar (g/L)	211	226	211	179	152
Acetic acid (g/L)	0.26	0.45	0.62	0.84	1.12
Succinic acid (g/L)	0.26	0.25	0.26	0.26	0.28
Glycerol (g/L)	4.77	5.33	5.7	5.95	7.09

Source: Ribéreau-Gayon *et al.*, 2000d.

11.7.3 Factors affecting glycerol production

The amount of glycerol usually formed by *S. cerevisiae* in wine varies between 4 and 9 g/L. Some winemaking conditions affect this production. First of all, the initial sugar concentration of musts strongly affects glycerol concentration (Table 11.1). Several studies have also demonstrated that increases in temperature as well as sulphur dioxide additions resulted in greater glycerol production (Ough *et al.*, 1972; Gardner *et al.*, 1993; Ribéreau-Gayon *et al.*, 2000d). Nevertheless, the influence of environmental factors remained minor and the predominant factor for glycerol variability in wine was attributed to the yeast strain. The glycerol production of 19 industrial wine strains ranged from 6.4–8.9 g/L in controlled conditions (Remize *et al.*, 2000b). Interestingly, this study also demonstrated that the production of acetate and succinate, which differed substantially according to strain, was not correlated with glycerol.

11.7.4 Glycerol over-production: an example of central metabolism alteration and its aromatic consequences

The utilization of *S. cerevisiae* strains over-producing glycerol was proposed for reducing ethanol production rather than to increase sensorial quality of wines. This strategy was particularly efficient in terms of decreasing ethanol content and did not affect fermentation properties of yeasts (Remize *et al.*, 1999). Nevertheless, the modification of central metabolism has detrimental organoleptic consequences generating an accumulation of acetate (1 g/L), succinate, acetoin and 2,3 butanediol (Remize *et al.*, 1999). This is due to the increased utilization of NADH through glycerol pathway. To prevent the acetate accumulation, the major gene involved in acetate production *ALD6* (Remize *et al.*, 2000a) was disrupted in a *GPD1*-overexpressing strain (Eglinton *et al.*, 2002; Cambon *et al.*, 2006). In such a mutant, the acetate produced was similar to that of the control strain. However, this engineered strain accumulated acetoin at considerable levels due to its inefficient reduction of 2,3-butanediol (Cambon *et al.*, 2006). The negative sensorial impact of acetoin required a novel genetic modification. Recently the overexpression of *BDH1* gene was shown to limit the acetoin production in a glycerol over-producing strain by enhancing its conversion in 2,3-butanediol (Ehsani *et al.*, 2007).

11.8 Yeast by-products affecting wine aromas: acetic acid

11.8.1 Organoleptic contribution of acetic acid

Acetic acid is a volatile acid of wine resulting from bacterial spoilage and yeast metabolism. Beyond a certain limit, which varies according to wine, the acetic acid has a highly undesirable organoleptic effect. In healthy grape must with moderate sugar concentration (<220 g/L), acetic acid is above 0.3 g/L (Ribéreau-Gayon *et al.*, 2000a). However, higher concentrations are sometimes produced, depending on must composition, and on the winemaking process, substantially affecting wine quality. The yeast strain is another very important factor influencing acetic acid production, and noticeable variations are found between strains. Commercial starters with reduced acetate production would therefore be of high value in oenology.

11.8.2 Metabolic pathway and genetic control of acetic acid

In *S. cerevisiae*, acetate is produced as an intermediate of the pyruvate dehydrogenase (PDH) by-pass (Remize *et al.*, 2000a), which converts pyruvate into acetyl-CoA in a series of reactions catalyzed by pyruvate decarboxylases (PDC), acetaldehyde dehydrogenases (ALD) and acetyl-CoA synthetase. This pathway is the sole source of cytosolic acetyl-CoA, which is required for anabolic processes such as lipid biosynthesis. The reaction catalyzed by ALD also generates reducing equivalents, which are required for a variety of synthetic pathways and redox reactions, in the form of NAD(P)H. Five ALD isoforms have now been identified in *S. cerevisiae*. Three of them are cytosolic (encoded by *ALD6*, *ALD2*, and *ALD3*) and the other two are mitochondrial (encoded by *ALD4* and *ALD5*) (Navarro-Avino *et al.*, 1999). The major isoforms involved in wine fermentation are Ald6p and Ald5p. However, Ald4p and an unknown alternative pathway may compensate for the absence of these enzymes (Remize *et al.*, 2000a; Saint-Prix *et al.*, 2004). Moreover, it was suggested that the Ald2p and Ald3p proteins may contribute to acetate production in high sugar must (Pigeau and Inglis, 2005) due to their connection with osmotic stress metabolism (Blomberg, 2000). Nevertheless, deletion experiments (Saint-Prix *et al.*, 2004) as well as expression profiling of *ALD2* and *ALD3* in high sugar conditions (Pigeau and Inglis, 2007) are not in agreement with this hypothesis.

Although the acetate formation pathway has been described in detail, the genetic determinism explaining the variation of acetate production among yeast strains remains unclear. In fact, most yeast strains have functional *ALD* genes but present wide acetate production discrepancies, suggesting that other genetic causes have not yet been elucidated. Due to its central role in the control of redox balance, acetate is dependent on the metabolic flux equilibrium of NAD(P) cofactors. For example, the *ALD* isoform compartmentalization and cofactor specificity (NAD⁺ or NADP⁺) (Saint-Prix *et al.*, 2004), the redox relationship between acetate and glycerol production under high-osmolarity conditions (Blomberg, 2000), as well as the relationship between acetate over-production and

cell growth (Albers *et al.*, 1996; Marullo *et al.*, 2007a) make acetic acid production a typical quantitative trait. Recently, new strategies based on quantitative genetics approaches were carried out to identify QTL controlling acetic acid production (Marullo *et al.*, 2007a). A major QTL located on the chromosome IV was found, explaining the acetate over-production of a wine yeast starter. This over-production is due to a deleterious mutation in the *ASP1* gene and is caused by an inefficient assimilation of asparagine. Although not fully verified in grape juice conditions, this study suggested that complex interactions exist between strain genetic polymorphism and environment conditions.

11.8.3 Factors influencing acetic acid production

Factors influencing acetic acid formation are well identified. As for glycerol, the quantity of initial sugar is an important variable (Table 11.1). Other factors are anaerobiosis, extreme pH (<3.1 or >4.0), as well as amino acid or nitrogen deficiencies in the must (Monk and Cowley, 1984, Bely *et al.*, 2003). The moment of nitrogen addition may also play an important role, especially for botrytized musts with high sugar concentrations (Bely *et al.*, 2003). In red winemaking, high temperatures (>28 °C) also increase the acetic acid concentration, especially during cell growth. For dry white and rosé, excessive must clarification can also induce an increased production of acetic acid by yeast. This effect is essentially due to a depletion in long-chain unsaturated fatty acids that have a particular importance in acetic acid metabolism (Delfini and Costa, 1993; Ribéreau-Gayon *et al.*, 2000a).

The genetic nature of fermenting strain is certainly the factor having the major impact on acetic acid. Under the same winemaking conditions, the production level among several strains may vary ten-fold (between 0.1 and 1 g/L) (Giudici and Zambonelli, 1992; Romano *et al.*, 1994; Remize *et al.*, 2000b; Marullo *et al.*, 2004). The selection of individuals showing a low volatile acidity production is relatively easy to perform by appropriate fermentation screening. Biometric studies have been demonstrated that nearly 10 % of *S. cerevisiae* strains produced low levels of acetate and are suitable for starter selection (Giudici and Zambonelli, 1992). Moreover, this trait could be improved by using simple meiotic segregation (Giudici and Zambonelli, 1992, Marullo *et al.*, 2004) as well as by using hybridization (Giudici and Zambonelli, 1992, Marullo *et al.*, 2006a). The other way to reduce acetic acid consists of obtaining mutant strains for *ALD6* genes. Nevertheless, the examples given by molecular engineering are not encouraging, as the reduction of acetate metabolism in these strains generates an over-production of acetoin (Cambon *et al.*, 2006).

The use of non-*Saccharomyces* species was also proposed to reduce acetic acid production during alcoholic fermentation. In fact, some species such as *Torulaspora delbrueckii* were shown to produce very low amounts of acetic acid, even in high sugar concentrations. This particular aptitude might be due to a different mechanism of adaptation regarding osmotic stress. Although this species cannot be used alone to carry out industrial fermentations, its use in co-cultures

with *S. cerevisiae* can reduce strong volatile acidity, especially in high sugar musts (Bely *et al.*, 2008).

11.9 Yeast by-products affecting wine aromas: hydrogen sulphide

11.9.1 Hydrogen sulphide impact

Unpleasant odours may sometimes develop in wine during the alcoholic fermentation, due to formation of sulphides and thiols derived from the sulphured amino acid metabolism of yeast. Hydrogen sulphide (H_2S) is the most highly odour-impacting molecule of this family contributing to the 'reduced' defect of wines. This compound has particularly unpleasant smell (rotten eggs, garlic, etc.) with a perception threshold of 0.8 $\mu\text{g/L}$. Hydrogen sulphide production depends on the sulphate and sulphite uptake as well as sulphur-containing amino acid biosynthesis. The regulatory control of this pathway and its interconnection with nitrogen metabolism explain the huge variation of H_2S production according to yeast strain and winemaking conditions. The dissection of the H_2S pathway and its regulation have been extensively studied in the past. Lowering the quantities of this compound is a major objective for wine starter selection, and several approaches have been proposed to this end.

11.9.2 Metabolic pathway and genetic control of hydrogen sulphide

Rather than fully describing all the enzymatic reactions of sulphur uptake, the purpose of this section is to give an overview of pathway regulations in order to understand the variations of hydrogen sulphide production in *S. cerevisiae*. Readers with a deep interest in sulphur-containing amino acid biosynthesis might consult more specialized reviews (Thomas and Surdin-Kerjan, 1997).

In yeast, reduced sulphate is used in the synthesis of organic sulphur metabolites, mostly cysteine, methionine and S-adenosylmethionine (AdoMet). Sulphate reduction can be divided into three distinct regulatory subgroups of enzymatic activities: the sulphate activation and reduction group (i), sulphite reduction reaction (ii), and amino acid biosynthesis from sulphide (iii) (Fig. 11.5). The first reaction group activates sulphate ions into adenylyate compounds allowing the sulphate reduction by means of NADPH oxidation (Thomas and Surdin-Kerjan, 1997). From sulphate transport through sulphite production, two transporters (Sul1p, Sul2p) and three enzymes (Met3p, Met14p, Met16p) are involved.

The second reaction of sulphate assimilation is catalyzed by the sulphite reductase complex that reduced sulphite (SO_3^{2-}) in sulphide (S^{2-}). Mutation of this enzyme accompanied by methionine supplementation of the grape juice is not a viable strategy for the elimination of reduced sulphide formation. First, the fact that both cysteine and methionine are reactive chemically under these reductive fermentation conditions leads to a host of other equally undesirable sulphur-

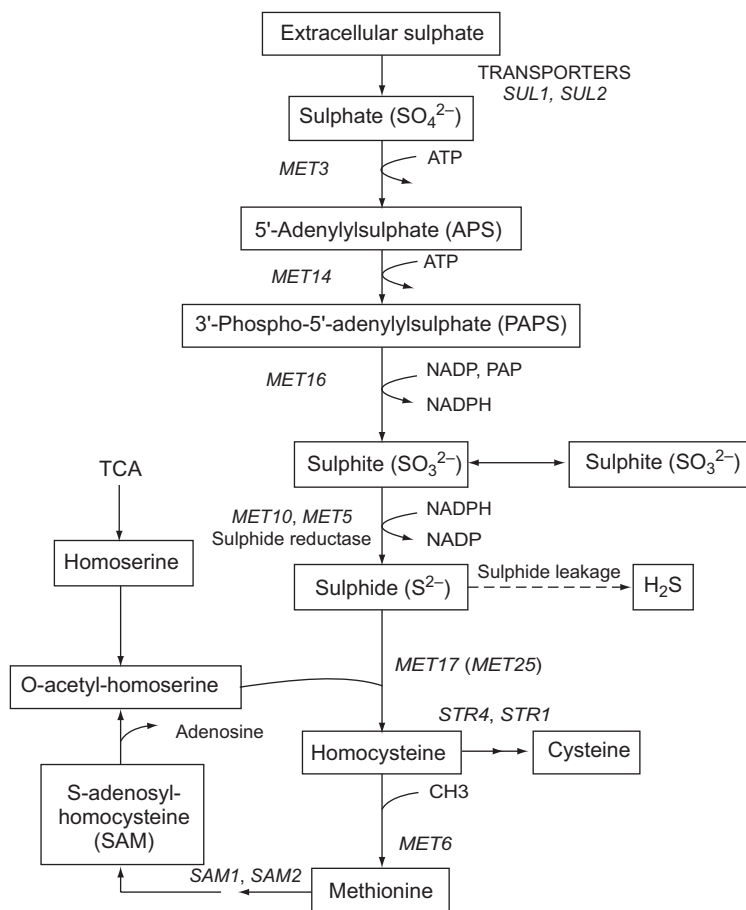


Fig. 11.5 Sulphur amino acid biosynthesis pathway and sulphide leakage. This diagram shows the main metabolites, enzymes and relative genes involved in the *S. cerevisiae* pathway.

containing spoilage compounds, i.e. methionol. Second, because mutation in sulphite reductase may generate SO_2 over-production (Hansen and Kielland-Brandt, 1996), the use of such mutant may modify the microbiological flora of wines and especially the lactic bacteria that are important factors of wine quality. During fermentation, H_2S production is due to the inefficient incorporation of sulphide into sulphur amino acid pathways (Jiranek *et al.*, 1995a, 1996). Therefore, the activation of sulphur amino acids by adding sulphide-binding precursors is an efficient means to reduce H_2S production.

The third part of the sulphate pathway consists of sulphide incorporation into sulphur-containing amino acids. The first enzyme of this subgroup, Met17p (also referenced as Met25p), is the homocysteine synthase. Homocysteine is then methylated in methionine (Met6p) or converted into cysteine by β -addition

(Cys4p) and γ -elimination (Cys3p). Several studies have identified *MET17* and *CYS4*, both located downstream of sulphite reductase, as controlling genes in the pathway affecting sulphide leakage (Spiropoulos *et al.*, 2000). The overexpression of *MET17* may significantly decrease the H_2S production in brewing (Omura and Shibano, 1995) and winemaking (Spiropoulos and Bisson, 2000) due to the activation of sulphur incorporation in the amino acid pool. Nevertheless, this overproduction effect of *MET17* is background-dependent and the constitutive activity of this enzyme varies importantly according to yeast strain (Ono *et al.*, 1999; Linderholm *et al.*, 2006). Likewise, the overexpression of *CYS4* in four native isolate strains did not lead to any significant reduction of H_2S , although overexpression of *CYS4* reduced H_2S in a brewing strain (Tezuka *et al.*, 1992). These findings suggest that simple increases in enzyme activity of reactions that consume reduced sulphide are in part insufficient to reduce sulphide formation in a reliable manner.

The control of the sulphate reduction pathway is multifaceted and depends on two main regulatory inputs. On the one hand, feedback inhibition and gene repression by end products, especially methionine derivatives, ensure that the activity of the sulphate reduction pathway is tightly regulated to match the metabolic demand for methionine and cysteine (Jiranek *et al.*, 1995a; Thomas and Surdin-Kerjan, 1997). As a matter of fact, most of the genes (*SUL1*, *SUL2*, *MET3*, *MET5*, *MET6*, *MET10*, *MET14*, *MET16* and *MET17*) are repressed in methionine-rich conditions. On the other hand, the sulphite reductase genes are also regulated by general amino acid control (Thomas and Surdin-Kerjan, 1997; Spiropoulos *et al.*, 2000). Under conditions of limitation for any one of several amino acids (lysine, histidine, arginine, isoleucine, leucine, valine, serine, phenylalanine, tryptophan, proline, methionine), and if nitrogen is otherwise plentiful, roughly 35 genes involved in amino acid biosynthesis (including *MET10*) are induced via general amino acid control through the transcription factor *Gcn4p*. This double genetic regulation has practical consequences as detailed further.

11.9.3 Factors affecting hydrogen sulphide production

Fermentation temperature (Karagiannis and Lanaridis, 1999), juice turbidity (Karagiannis and Lanaridis, 1999), and acidity (Vos and Gray, 1979) significantly affect H_2S concentrations, but the major factor influencing H_2S production is the free amino nitrogen (FAN) content of the must (Vos and Gray, 1979; Jiranek *et al.*, 1995a). Generally speaking, the higher the must FAN, the lower the H_2S production. The supplementation of nitrogen during the fermentation at the moment that reduced odours emerge is a practical application of these findings (Ribéreau-Gayon *et al.*, 2000b). Nevertheless, the addition effect of ammonium is dependent on the methionine concentration present in the must. Indeed, it has also been shown that both sulphate and sulphite can lead to H_2S production under oenological conditions (Eschenbruch, 1974; Giudici and Kunkee, 1994). The relative importance of these two precursors might be explained by the induction of sulphite reductase by *Gcn4p* and its inhibition by methionine (Spiropoulos *et al.*, 2000).

When methionine is present in sufficient concentrations (above its repression threshold), all the enzymes of the sulphate assimilation pathway including sulphite reductase are strongly repressed. In such conditions, sulphites imported from the must became the major source of H_2S (Jiranek *et al.*, 1995a). Nitrogen addition in early stages of fermentation induce sulphide uptake by inducing sulphur amino acid metabolism. Conversely, if methionine is present at concentrations below its repression threshold, the sulphate assimilation pathway is active and becomes the most important source of hydrogen sulphide. In these conditions, addition of ammonium sulphate combined with the depletion of a particular amino acid in the must would induce sulphite reductase, enhancing H_2S production. Therefore, the balance between methionine and ammonium might be a central parameter controlling H_2S during wine fermentation (Spiropoulos *et al.*, 2000).

11.9.4 Yeast strain influence on hydrogen sulphide production

Yeast strain has a major impact on H_2S production, and the amount released during the alcoholic fermentation may vary 10 times according to the yeast strain (Jiranek *et al.*, 1995b; Spiropoulos *et al.*, 2000). Although many studies have focused on *S. cerevisiae* (Romano *et al.*, 1985; Giudici *et al.*, 1993; Giudici and Kunkee, 1994; Jiranek *et al.*, 1995b; Spiropoulos *et al.*, 2000; Marullo *et al.*, 2004), this variation was also found in other species belonging to the non-*Saccharomyces* group (Mendes-Ferreira *et al.*, 2002). The rapid evaluation of H_2S production may be achieved by plating several strains on media containing bismuth sulphite (basically BiGGY medium) that were initially developed for the selective cultivation and differentiation of *Candida* spp. (Nickerson, 1953). The ability of strains to produce low amounts of H_2S is evaluated by the blackening of colonies that correspond of the reaction of bismuth ions with sulphides (Romano *et al.*, 1985; Jiranek *et al.*, 1995b). Although widely used, these media may be only partially informative because they only reflect the part of H_2S produced from sulphite and not from the sulphate route. Therefore, the correlation of BiGGY medium and oenological conditions is not systematic and the use of this media can lead to a misevaluation of the real H_2S production potentiality of strains (Spiropoulos *et al.*, 2000; Linderholm *et al.*, 2008).

A more reliable way consists of measuring the release of H_2S during fermentation by head space trapping (Tominaga *et al.*, 2004) or by using lead acetate strips (Giudici and Kunkee, 1994). Getting low H_2S producing strains is a basic goal for yeast selection and several approaches have been tried to obtain suitable strains. Screening of natural isolates (Giudici and Kunkee, 1994; Spiropoulos *et al.*, 2000), meiotic recombination approaches (Romano *et al.*, 1985; Marullo *et al.*, 2004) as well as breeding programs (Romano *et al.*, 1985; Marullo *et al.*, 2006a) have been routinely performed by many research institutes. Although monogenic segregation of H_2S has been previously reported (Romano *et al.*, 1985), this trait is usually continuously distributed among a large progeny (Marullo *et al.*, 2004, 2006a). This is consistent with the coexistence of different genes influencing H_2S production.

Recently, a functional genomics approach was carried out to identify new genes

involved in H₂S production. By screening a commercially available set of deletion mutants (4478 strains), Linderholm *et al.* (2008) looked for genes reducing H₂S production. Among 88 genes showing a difference in the BiGGY agar test, 16 mutants were significantly different for H₂S production during wine fermentation. Taking apart the genes of sulphite reductase as well genes directly involved in sulphate pathway, nine genes involved in acetaldehyde metabolism as well as histone elongation were found to indirectly affect H₂S production. Although the role of these genes remains unclear, this recent work underlines the complex genetic determinism of this trait and the necessity to use quantitative genetics approaches for yeast selection. Recently a QTL-detection program identified a genetic locus controlling H₂S production. Fine mapping narrowed a chromosomal region of 11 kilobases in which none of the candidate genes has been found (Marullo *et al.*, 2007b). Nevertheless, the identification of this region allows yeast selection programs by using molecular markers as detailed in the first section of this chapter.

11.10 Yeast by-products affecting wine aromas: higher alcohols

11.10.1 Aromatic impact overview of higher alcohols and general directions for yeast selection

Food fermentation by yeast is accompanied by the formation of higher alcohols (aliphatic and aromatic) known as fusel alcohols (Table 11.2). This name derives

Table 11.2 Main fusel alcohols of wine with their relative α -keto acids and amino acid precursors

Fusel alcohols	Aromatic descriptor (perception threshold)	Concentration in wine (mg/L)	Amino acid precursors	Related α -keto acids
3-Methyl butanol (isoamyl alcohol)	Pungent, balsamic (30 mg/L)	80–300	Valine	α -Keto isovalerate
2-Phenyl ethanol	Rose (10 mg/L)	10–100	Phenyl-alanine	Phenyl pyruvate
3-Methylthio-propanol (methionol)	Cooked cabbage (1.2 mg/L)	0–5	Methionine	4-Methylthio-2-oxobutanoate
2-Methyl butanol (active amyl alcohol)	Roasted onion (30 mg/L)	30–100	Leucine	α -Keto isocaproate
2-Methyl propanol (isobutanol)	Green notes (40 mg/L)	50–150	Isoleucine	3-Methyl-2-oxobutanoate

Source: Ribéreau-Gayon *et al.*, 2000b.

from the German word fusel (bad liquor) because these molecules are enriched during the spirits distillation process. At low concentrations (less than 300 mg/L), these compounds make a positive contribution to the flavours and aromas of wine (Ribéreau-Gayon *et al.*, 2000). At higher concentrations, they impart off-flavours, contributing particularly to excessive solvent and amylic notes. Most aliphatic higher alcohols, such as isoamylic alcohols, have heavy solvent-like odours and smell unpleasant. Therefore, the yeast selection of strains over-producing such higher alcohols is not a relevant direction *per se*. However, these compounds are the metabolic precursors of acetate esters that are important odour-impacting molecules capable of influencing wine aroma. On the other hand, 2-phenyl ethanol, which has a rose-like fragrance, brings to wine a more interesting note that may contribute to overall bouquet complexity. As a general consequence, a limited higher alcohol production (with the exception of 2-phenyl ethanol) should be among the selection criteria of wine yeast.

11.10.2 Metabolic pathways and genetic control of higher alcohols in yeast.

Yeasts produce higher alcohols by both anabolic and catabolic pathways. Propan-1-ol is formed from the condensation of pyruvate and acetyl CoA (Henschke and Jiranek, 1993), while other alcohols are derived from α -keto acids decarboxylation (Table 11.2). A *de novo* synthesis of α -keto acids from glucose, producing higher alcohols during fermentation, was described by Dickinson *et al.* (1997). However, the most important pathway involved the amino acid catabolism and was first proposed a century ago by Ehrlich (Ehrlich, 1907). Amino acids that are assimilated by the Ehrlich pathway (valine, leucine, isoleucine, methionine and phenylalanine) are taken up in a sequential manner throughout the fermentation (Henschke and Jiranek, 1993). After the initial transamination reaction, the resulting α -keto acid is decarboxylated into a fusel aldehyde and then oxidized or reduced into acids or alcohols, respectively (Fig. 11.6). The genes of enzymes involved in the Ehrlich pathway as well as their broad transcriptional regulation were identified as related in a recent mini-review (Hazelwood *et al.*, 2008).

Transamination step of Ehrlich pathway

The initial transamination step is controlled by four enzymes in *S. cerevisiae* (Bat1p, Bat2p, Aro8p and Aro9p). The mitochondrial and cytosolic branched-chain amino acid amino-transferases are encoded *BAT1* and *BAT2* genes, respectively (Eden *et al.*, 2001). The Aro8p and Aro9p were initially characterized as the aromatic amino acid aminotransferases I and II; however, they showed broader substrate specificity than originally described (Urrestarazu *et al.*, 1998; Hazelwood *et al.*, 2008). Their role in wine fermentation remains unexplored. The deletion effect of *BAT1* and/or *BAT2* in fermentation conditions showed a drastic reduction of isobutanol and in a minor part of isoamyl alcohol and 2-methyl butanol (Eden *et al.*, 2001; Yoshimoto *et al.*, 2002; Lilly *et al.*, 2006b).

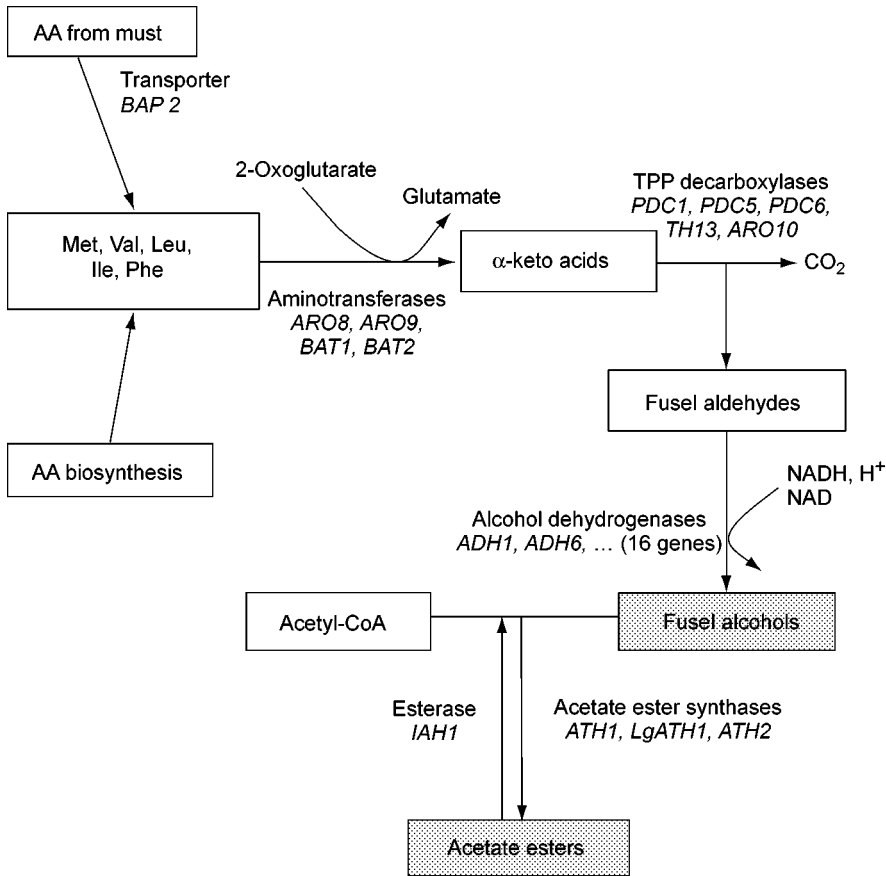


Fig. 11.6 The Ehrlich pathway and acetate ester production. This diagram shows the main metabolites, enzymes and relative genes involved in the *S. cerevisiae* pathway.

Decarboxylation in the Ehrlich pathway

In *S. cerevisiae*, the decarboxylation step of the Ehrlich pathway is supported by at least five TPP-dependent decarboxylases encoded by *PDC1*, *PDC5*, *PDC6*, *ARO10* and *THI3* (Hazelwood *et al.*, 2008). According to the branched amino-acid metabolized, one or more of these genes are specially affected: leucine catabolism is encoded by *THI3* (Dickinson *et al.*, 1997), valine by one of the three genes *PDC1*, *PDC5* and *PDC6* (Dickinson *et al.*, 1998), while isoleucine may be catabolized by one of each TPP-decarboxylase genes (Dickinson *et al.*, 2000). Methionine seems to be metabolized only by Aro10p, and phenylalanine by all of TPP-decarboxylases except Thi3p (Hazelwood *et al.*, 2008).

Reduction of fusel aldehydes in higher alcohols

The last step of the Ehrlich pathway consists of the oxidation or the reduction of fusel aldehyde in related fusel acid or alcohol. During alcoholic fermentation, most

of the Ehrlich pathway is routed in the reduction way and the formation of the fusel acid plays only a minor role (Hazelwood *et al.*, 2008). It was suggested that this reduction may have a relevant impact on redox metabolism and that it allows recycling excess NADH formed by cell growth. Other authors suggested that, physiologically, oxidative deamination provides the yeast with a mechanism for obtaining nitrogen when its pool has become depleted (Vollbrecht and Radler, 1973). Finally, it has been suggested that higher alcohol production might act as a detoxification process for the intracellular medium of α -keto acids and aldehydes, or as a means of regulating the metabolism of amino acids (Ribéreau-Gayon *et al.*, 2000a). The fine molecular determinism of this reduction has not yet been investigated, and the *S. cerevisiae* genome has at least 16 alcohol dehydrogenases to catalyze this reaction (Dickinson *et al.*, 2003) (Fig. 11.6).

11.10.3 Genetic regulation of the Ehrlich pathway during the alcoholic fermentation

The transcriptional network regulation of the *S. cerevisiae* Ehrlich pathway remains to be fully explained. In the oenological context, some expression studies showed that some transaminases, the TPP-decarboxylases and alcohol dehydrogenases have similar expression profiles. Recently, it was shown that *BAT2*, *PCD1* and *ADH1* shared similar expression profiles during alcoholic fermentation, while the mitochondrial aminotransferase encoded by *BAT1* was differently regulated (Molina *et al.*, 2007). These differential expressions showed that *BAT1* is highly expressed during exponential growth while *BAT2* is preferentially expressed during the stationary phase (Lilly *et al.*, 2006b). Nevertheless, according to the genetic background of strain used and fermentation conditions (temperature, amino acid content) the relative expression of genes involved in higher alcohols biosynthesis differs. For example, one study showed that *PDC1* and *ADH1* were co-regulated with *BAT1* while *BAT2* showed the same profiles of other Ehrlich pathway genes (*PDC6* and *ADH6*) (Lilly *et al.*, 2006b). In any case, the relation between these transcriptional differences and the variation of higher alcohol production between strains remains to be better understood.

11.10.4 Factors affecting higher alcohol production

Winemaking variables that influence higher alcohols include the degree of grape ripeness (Cabrera *et al.*, 1988), the grape variety (Rankine, 1967; Cottrell and McLellan, 1986), pH (Rankine, 1967), the nitrogen content (Ough and Bell, 1980; Henschke and Jiranek, 1993; Yoshimoto *et al.*, 2002), aeration (Bertrand, 1975) and temperature (Rankine, 1967; Bertrand, 1975; Cottrell and McLellan, 1986; Molina *et al.*, 2007). The effect of this last variable is different according to the chemical compounds. For example, 2-methyl butanol (active amyl alcohol) and 3-methyl-butanol (isoamyl alcohol) are higher in cold conditions while 2-phenyl ethanol is higher in warm fermentations (Molina *et al.*, 2007).

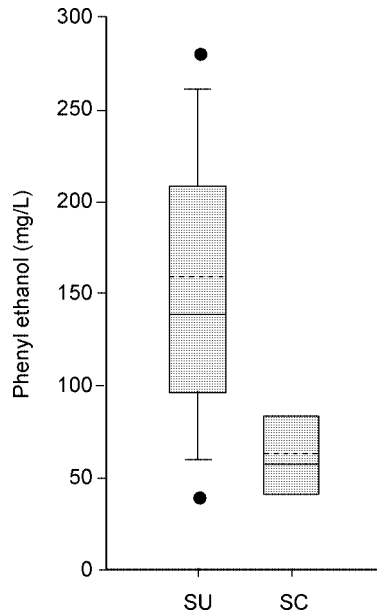


Fig. 11.7 Distribution of 2-phenyl ethanol production of *S. cerevisiae* (SC) and *S. uvarum* (SU). The 2-phenyl ethanol production of 6 *S. cerevisiae* and 29 *S. uvarum* strains was measured at the end of the alcoholic fermentation of a white grape must. The data are presented as a box plot. Full and dashed lines represent, respectively, median and quartiles delimitation, dotted points represent the 5%-tail values. The *S. uvarum* group produces higher amounts of this compound but also shows noticeable production variability (source: Isabelle Masneuf, unpublished data).

11.10.5 Contribution of *Saccharomyces* and non-*Saccharomyces* strains

The nature of the yeast strain also affects the higher alcohol production both at the inter-specific and intra-specific level. A part of the inter-specific differences reported is explained by the fact that the production of higher alcohols is correlated with the ethanol produced (Bertrand, 1975). Consequently, non-ethanol-resistant species were usually described as low producers of fusel alcohols (Rankine, 1967; Bertrand, 1975; Cabrera *et al.*, 1988). At the intra-specific level, *S. cerevisiae* strains have shown significantly different production of higher alcohols (Rankine, 1967; Bertrand, 1975; Cabrera *et al.*, 1988; Patel and Shibamoto, 2003) explaining an important part of trait variation (Cottrell and McLellan, 1986; Cabrera *et al.*, 1988). The case of 2-phenyl ethanol has relevant oenological consequences in wine quality. Compared to *S. cerevisiae*, *S. uvarum* strains (*S. bayanus* var. *uvarum*) produce substantially more 2-phenyl ethanol (Antonelli *et al.*, 1999; Masneuf *et al.*, 2002). This aromatic peculiarity of *S. uvarum* has a noteworthy consequence on the typicity of some French white wines (e.g. Sancerre or Alsace) that often present a marked rose note in their bouquet. In these cool regions, *S. uvarum* strains or *S. cerevisiae* × *S. uvarum* hybrids are frequently found among spontaneous fermentations (Masneuf *et al.*, 2002; Le Jeune *et al.*, 2007), enhancing

significantly the 2-phenyl ethanol concentration of these wines. Despite this interspecific influence, a strong intraspecific variation of 2-phenyl ethanol has also been found among *S. uvarum* and *S. cerevisiae* (Fig. 11.7) assessing the basis for selection programs (Masneuf *et al.*, 2002).

11.10.6 Genetic modification of higher alcohols pathways for aroma profiling

Molecular biology experiments have demonstrated that the simple overexpression of *BAT1* and/or *BAT2* genes may modulate the aromatic profiles of wines by over-producing isobutanol or isoamyl alcohol (Lilly *et al.*, 2006b). However, this modification poorly influences the level of 2-phenyl ethanol that is the sole relevant higher alcohol to be enhanced. In order to modify specifically this higher alcohol, the modulation of the *ARO8* and *ARO9* expression should be tested. Although not very relevant for winemaking, the higher alcohol enhancement should have an indirect impact on the production of a related ester. For example, the overexpression of *BAT1* significantly increases isoamyl acetate (banana aroma) by enhancing the level of its precursor (isoamyl alcohol) (Lilly *et al.*, 2006b).

11.11 Yeast by-products affecting wine aromas: esters

11.11.1 Aromatic impact overview of esters and general directions for yeast selection

Although volatile esters are only present in small quantities compared to those of higher alcohols, they have a higher impact on wine flavour. Two main classes of esters can be defined as: (i) acetic esters of higher alcohols and (ii) ethyl esters of fatty acids. Acetic esters of higher alcohols (e.g. isoamyl acetate, phenyl ethyl acetate) contribute positively to the aroma complexity of neutral wine. However, their rather unusual odours (banana, acid drops and apple) may mask some varietal aromas at their concentration in wines. Ethyl esters of fatty acids have less prominent odours (soap, candle wax) which contribute to the aroma finesse (especially for white wines). Volatile esters are present in a total concentration of a few mg/L and have nearly equivalent perception thresholds (Table 11.3). From an oenological point of view, it is important to note that the concentration of these molecules generally decreases during ageing by chemical hydrolysis (Garofalo and Piracci, 1994). Therefore, the presence of esters is not a guarantee of aroma intensity for wines destined for long ageing for which varietal aromas are more suitable. The level of ester production by yeast strain is a classical trait for starter selection, and several works have been conducted in this area. However, the presence of different esters can have a synergistic effect on the individual flavours (Meilgaard, 1975). Because most esters are present in concentrations around their threshold value, minor changes in their concentrations may have dramatic effects on wine flavour. It is quite difficult to evaluate this fragrance balance simply by

Table 11.3 Main esters produced by yeasts in wine

Ester family	Aromatic descriptor (perception threshold) ^a	Concentration in wine (mg/L) ^b
<i>Acetic esters</i>		
Isoamyl acetate	Banana (0.03 mg/L)	0.1–5.0
2-Phenyl ethyl acetate	Rose (0.25 mg/L)	0.005–5.0
Ethyl acetate	Solvent-like ‘acescence’ (160 mg/L)	
<i>Ethyl esters of fatty acids</i>		
Ethyl butyrate (C4)	Papaya, apple (20 µg/L)	0.050–0.350
Ethyl caproate (C6)	Green apple (5 µg/L)	0.10–0.15
Ethyl caprylate (C8)	Soap (2 µg/L)	0.10–3.00
Ethyl caprate (C10)	Candle wax (200 µg/L)	0.01–0.3

^aIn white wines according to Molina *et al.* (2007).

^bIn wines according to Guillaume Antalick (unpublished data).

making a list of the produced compounds. Therefore, global analysis of the impact of these must be achieved by using sensorial approaches.

11.11.2 Metabolic pathways and genetic control of esters production in *S. cerevisiae*

Ester formation consists of the condensation of an alcohol and an activated acyl-CoA molecule. Hence, the ester production rate is influenced by the concentration of these two co-substrates. In wine, recent studies (Molina *et al.*, 2007) confirm old investigations of brewers about the correlation between substrate concentrations and corresponding ester formation (Saerens *et al.*, 2008). This enzyme-catalyzed reaction depends on the total ester synthase activity which differs according to yeast strains (Saerens *et al.*, 2008; Van Laere *et al.*, 2008). In order to decipher the genetic causes of this variation, several studies were initiated to identify the genes involved in ester biosynthesis. These reactions and relative enzymes are summarized in Figs 11.6 and 11.8.

Acetic esters of higher alcohols

The acetic esters of higher alcohols resulting from the condensation of acyl-CoA and a higher alcohol are catalyzed by alcohol acetyltransferases (AATase, EC 2.3.1.84) (Fig. 11.6). *Saccharomyces cerevisiae* has two AATases: Atf1p and Atf2p encoded by *ATF1* and *ATF2* genes, respectively (Mason and Dufour, 2000). Moreover, *S. bayanus* species presented another AATase, the *Lg-Atf1p*, encoded by the *Yscatf1* gene that shares a strong homology with *ATF1* gene (Yoshimoto *et al.*, 1998). The metabolic function of such enzymes was primarily investigated in a brewery context, but knowledge reported may be easily extrapolated to wine. Of all known ester synthases, the *Atf1p* is the most important for the production of acetate esters. Depending on the genetic background, deletion analysis of *ATF* genes has shown that *Atf1p* is the main enzyme involved in the production of isoamyl acetate (80%), of phenyl ethyl acetate (75%) and of C3 to C8 acetate esters

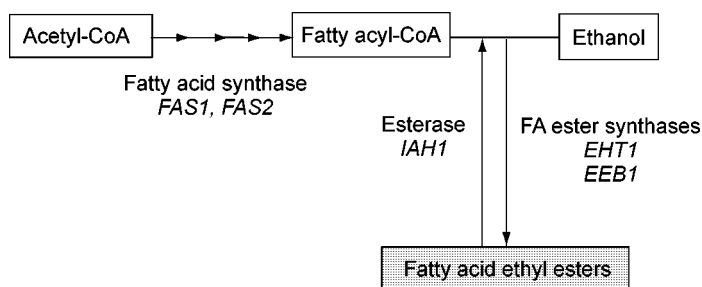


Fig. 11.8 Ethyl ester biosynthesis in *S. cerevisiae*. This diagram shows the main metabolites, enzymes and relative genes involved in the *S. cerevisiae* pathway.

(60–90%) (Fukuda *et al.*, 1998; Lilly *et al.*, 2000; Verstrepen *et al.*, 2003). Although having a lower impact, the other enzyme Atf2p also contributes to these esters' formation (Verstrepen *et al.*, 2003, 2006a). The *ATF* overexpression effect on acetate ester production was measured in beer (Fukuda *et al.*, 1998; Verstrepen *et al.*, 2003) and in wine contexts (Lilly *et al.*, 2000; Lilly *et al.*, 2006a). The high production level of these enzymes dramatically increases acetate esters (5–30-fold according the acetate ester type), confirming the hypothesis that the limiting factor of acetate esters production is the alcohol acetyl-transferase activity (Malcorps *et al.*, 1991). Moreover, the acetic ester concentrations are also influenced by the esterase activity, mainly due to the *Iah1p* protein. Fukuda *et al.* (1998) demonstrated that the relative expression levels of *ATF1* and *IAH1* modify the net rate of ester accumulation and modulate the isoamyl acetate/isoamyl alcohol balance. The comprehension of the genetic control of acetate esters opens the doors to genetic engineering approaches for modifying the wine aroma profile. Nevertheless, the fine tuning of acetate ester production is not yet optimized as discussed in the next section.

Ethyl esters of fatty acids

The second class of esters in wine is dominated by medium-chain fatty acids ethyl esters including ethyl butyrate (C4), ethyl caproate (C6), ethyl caprylate (C8) and ethyl caprate (C10) (Table 11.3). Biosynthesis of these is due to the esterification of ethanol with medium-chain fatty acyl-CoA (MCFA) by the fatty acid ethyl ester synthases (Saerens *et al.*, 2006) (Fig. 11.8). Two main enzymes, encoded by *EHT1* and *EEB1*, have recently been described as being responsible for the production of ethyl esters (Mason and Dufour, 2000; Saerens *et al.*, 2006, 2008). Deletion of these genes showed that *Eeb1* is the main enzyme, while *Eht1* plays only a minor role (Saerens *et al.*, 2006). Nevertheless the overexpression of *EHT1* in a commercial wine strain slightly increases ethyl esters (C6, C8, C10) producing more fruity wines (Lilly *et al.*, 2006a).

The rate of MCFA ethyl ester formation might be dependent on two factors: the concentration of two substrates and the total activity of enzymes involved in their synthesis/hydrolysis. As enhancing fatty acid ethyl synthase activity by overexpression only slightly affects ethyl ester production, the enzyme activity

appears not to be the limiting factor for ethyl ester production. In contrast, several studies demonstrated that the increasing level of MCFA directly affects the ethyl ester concentrations in wines (Torija *et al.*, 2003; Beltran *et al.*, 2008). To date, the role of genes involved in short-chain fatty acid (C6–C10) biosynthesis has not yet been investigated regarding ethyl ester production. Nevertheless, the genetic enhancement of these metabolic precursors by the overexpression of *FAS1* and *FAS2* gene was demonstrated (Furukawa *et al.*, 2003) and has been suggested to have a positive effect on the amount of their relative esters (Furukawa *et al.*, 2003; Saerens *et al.*, 2008).

11.11.3 Factors affecting ester production

Fermentation temperature is an important factor influencing ester formation and has been widely studied in the brewery context. Higher temperatures were historically correlated with an increase in esters production in the range 10–20 °C. This general observation may be explained by a more active amino acid metabolism that directly increases the rate of alcohol production and therefore ester production (Saerens *et al.*, 2008; Van Laere *et al.*, 2008). For winemaking, the general practice is to lower the fermentation temperature in order to increase ester production in white wines (Ribéreau-Gayon *et al.*, 2000a). Nevertheless, this apparent contradiction is due to the fact that high temperatures in a brewery context (10–15 °C) are considered as low temperatures for winemaking. Moreover, the maximum production peak of each ester may change with temperature. For example, phenylethyl acetate has a maximum peak at 20 °C while isoamyl acetate shows a maximal peak at 15 °C (Killian and Ough, 1979; Saerens *et al.*, 2008). This differential production profile of esters was proposed for modulating the aroma profile of wines (Molina *et al.*, 2007), and the difference in boiling points of each ester may contribute to this discrepancy (Killian and Ough, 1979). The effect of temperature on fatty acid-derived ethyl ester production is different from that of acetate esters. In fact, MCFA production decreases with higher temperatures and therefore ethyl ester production follows the same trend (Torija *et al.*, 2003; Beltran *et al.*, 2008).

The influence of nitrogen composition on ester production has also been widely investigated. Generally, higher ester production is correlated with higher amounts of yeast assimilable nitrogen (YAN) (Guitart *et al.*, 1999; Vilanova *et al.*, 2007; Carrau *et al.*, 2008). Nevertheless, the complex interaction of amino acid composition of grape must (Henschke and Jiranek, 1993) and differential nitrogen assimilation of yeast strains (Julien *et al.*, 2000) makes it difficult to draw clear conclusions about the role of nitrogen in ester production.

The yeast strain is certainly one of the major factors affecting ester production, and it is observed at both intra- and inter-specific levels (Bertrand, 1975; Soles *et al.*, 1982; Lema *et al.*, 1996; Antonelli *et al.*, 1999; Miller *et al.*, 2007). Not only is the average ester production impacted, but also the relative proportions of the individual esters differs dramatically from strain to strain. This strong variation has enabled exploitation in genetic selection programs. For example, the genetic segregation of a diploid strain may also be used for obtaining strains with a suitable

Table 11.4 Chemical analysis of Semillon wine fermented by the commercial strain Zymaflore X16 (Laffort, France) (parental strain) and eight spore progenies. An important variation in aromatic compounds production was observed. Data are given in mg/L and are the mean of triplicates

Fermentative aromas	Parental strain	Msp1	Msp2	Msp3	Msp4	Msp5	Msp6	Msp7	Msp8
2-Phenyl ethanol	60.7	40*	110.2*	38.4	31.8*	54.7	49.8	57	39.9*
2-Phenyl ethyl acetate	1.9	0.9*	2.3	2	1.5	2.2	1.9	3.7*	2.2
Isoamyl acetate	17.5	21.5*	30.3*	53.6*	20.6*	20.8*	23.4*	38*	27*
Hexyl acetate	12.8	16.5*	19.7*	30.1*	13.6	16.4*	16.1*	15.5	21.4*
Ethyl butyrate (C4)	1.1	1.6	1.7	2.1*	1.3	1.3	1.7	1.2	1.9*
Ethyl caproate (C6)	2.5	4.5*	2.9*	7.4*	2.8	3.1*	3.7*	2.1	1.6*
Ethyl caprylate (C8)	5.2	6.7*	3.8	4.9	5.3	6.1	6.1	4.3	3*
Ethyl caprate (C10)	1.1	0.8	0.3*	0.6	1.0	1.6*	1.2	1.0	0.6*

* Indicates when the value is significantly (ANOVA, $\alpha = 0.05$) different from the parental strain.

Source: Philippe Marullo (unpublished data).

ester production (Table 11.4). Nevertheless, the selection of yeast strains with an appropriate ester level is difficult to predict and also depends on vinification conditions. In fact, the yeast strain background, together with fermentation variables such as temperature, nitrogen and aeration, may considerably change the final ester balance (Carrau *et al.*, 2008).

11.11.4 Limits in the genetic modification of ester production

It has been demonstrated that the overexpression of *Atf1p* resulted in a strong increase in acetate esters (Lilly *et al.*, 2006a). Nevertheless, the expression systems used failed to give strains with high, but not excessive, amounts of esters. Moreover, due to the high levels of ethanol produced during the fermentation, the concentration of ethyl acetate dramatically increases in winemaking, compromising the use of such strategies to improve the acetate ester production of wine yeast strains (Lilly *et al.*, 2006a). In fact, ethyl acetate is responsible for the 'acescence' of wine and has a perception threshold of 160 mg/L, 200 times lower than that of acetic acid (Ribéreau-Gayon *et al.*, 2000d). The deleterious effect of ethyl acetate can be in part reduced by ageing (Lilly *et al.*, 2006b) but, after 6 months of bottle ageing, the ethyl acetate concentrations (140–180 mg/L) still affect the wine flavour, giving wines a hot flavour which reinforces the impression of bitterness on the aftertaste (Ribéreau-Gayon *et al.*, 2000d). Additional efforts are now necessary to evaluate the potential for specifically enhanced suitable aromas such isoamyl acetate and phenylethyl acetate. One possible way should be the increase of higher alcohol precursors by modulating the expression pattern of the Ehrlich pathway.

11.12 Varietal aromas resulting from grape precursor biotransformation

Odoriferous compounds from grape reflecting the particular variety, climate and soil play a decisive role in the quality and regional character of wines. These compounds are responsible for the varietal aroma of wines. Paradoxically, except for Muscat grape varieties, many ‘simple-flavoured’ grape varieties are practically odourless. Nevertheless, their relative wines show characteristic aromas that are relatively specific to the grape variety. This is true for the most of grape varieties: Merlot, Cabernet Sauvignon, Cabernet Franc, Sauvignon blanc, Semillon, Chardonnay, Chenin blanc and the different Pinot varieties. The revelation of these varietal compounds during the fermentation suggests that yeast metabolism modifies the must composition by releasing aromatic molecules from non-odorous precursors. The yeast strain influence in varietal aroma has been clearly demonstrated for certain compound (i.e. volatile thiols), while for other aroma families, the yeast impact seems less relevant for winemaking (terpenes, norisoprenoids) or remains to be clearly demonstrated (fruity aromas of red wines). To date, the molecular determinisms (i.e. gene, enzyme and regulation pathways) responsible for varietal aromas released from grape precursors have been in part identified for volatile thiols. For other compounds, genes and enzymatic activities involved in the influence of aroma revelation remain to be discovered and/or better evaluated. The yeast selection programs in relation to varietal aroma enhancement are also discussed in this section.

11.12.1 Phenolic acids

Organoleptic impact of volatile phenols

Although only trace amounts are present in must, wine contains two types of volatile phenols (ethyl- and vinyl-) that may negatively impact its flavour. Ethyl-4-phenol production is mainly due to microbial spoilage (i.e. *Brettanomyces*) (Chatonnet *et al.* 1995). Vinyl-4-phenol has an unpleasant odour of pharmaceutical and gouache paint; vinyl-4-guaiacol confers notes of clove. Beyond a certain concentration (720 µg/L for a 1:1 mixture of vinyl-4-phenol/vinyl-4-guaiacol), these compounds may be responsible for certain sensory faults (Ribéreau-Gayon *et al.*, 2000b). In beer, these are commonly known as phenolic-off-flavour (POF) character. The same note found in wine is usually called ‘phenolic’ or ‘pharmaceutical’. Vinyl-4-phenol and vinyl-4-guaiacol are derived from the decarboxylation by yeast of the phenolic acids of the hydroxycinnamic series (*para*-coumaric and ferulic acid, respectively). This decarboxylation occurs during the alcoholic fermentation and is inhibited by certain grape phenols contained in red grapes (Chatonnet *et al.*, 1993). Consequently POF affects principally white wines.

Factors affecting vinyl phenol precursors in grape must

The content of vinyl phenols is due to the concentration of certain phenolic acids

precursor in grape must. The concentration of hydroxycinnamic acids may vary according to grape variety, ripening and pre-fermentation operations (Ribéreau-Gayon *et al.*, 2000a). Clarification, oxidation and extraction conditions of grape juice may also influence the content of these precursors. The use of pectolytic enzymes may also increase the content of vinyl phenol precursors. In fact, certain pectolytic enzymes have a cinnamoyl esterase activity that may enhance the concentration of hydroxycinnamic acids in grape juice (Ribéreau-Gayon *et al.*, 2000a; Reynolds *et al.*, 2007b). The use of pectinase purified for cinnamate esterase activity prevents the risk of increasing the concentration of hydroxycinnamic acids during grape must clarification.

*Cinnamate decarboxylase and related gene in *S. cerevisiae**

The conversion of hydroxycinnamic acids in vinyl phenols is due the cinnamate decarboxylase (CD) activity of *S. cerevisiae*. This activity is constitutive and occurs only during the alcoholic fermentation. Some yeast strains (Chatonnet *et al.*, 1993; Shinohara *et al.*, 2000) do not produce POF and are called (POF-). Such strains do not have a cinnamate decarboxylase and have been selected for beer fermentation. They are also suitable for wine starter dedicated to white grape varieties (Chatonnet *et al.*, 1993; Marullo *et al.*, 2006a).

The genetic determinism of 'POF-' character of *S. cerevisiae* is due to the *PAD1* gene (YDR538w) that encode for phenylacrylic acid decarboxylase (Meaden and Taylor, 1991). The selection of POF-strain may be carried out by screening the sensitivity of strains to cinnamic acid (Clausen *et al.*, 1994). Another approach consists of screening the CA activity in batch culture by analyzing the vinyl phenols formation (Chatonnet *et al.*, 1993). As the POF-character segregates as a monogenic trait, it is easy to manipulate during breeding programs (Marullo *et al.*, 2006a). To refine this approach, it is possible to replace phenotypic tests by molecular marker selection. Recently, the *PAD1* gene of the Zymaflore VL1 strain (Laffort, France) was sequenced. The aspartate to glycine substitution at the amino acid position 213 (*pad1-D213G* allele) was proposed to be responsible for the POF-character of Zymaflore VL1. This allele was then used to drive selection programs with molecular markers (Marullo *et al.*, 2007b).

11.12.2 Terpenes and their glycoconjugate precursors

Aromatic impact of terpenic compounds

Terpenes are a large family of widespread compounds in the plant kingdom. They are constituted by a 10-carbon skeleton and may occur in different chemical forms such as hydrocarbons, aldehydes, acids and alcohols. The most odoriferous terpenes are mono-alcohols such as linalool, α -terpineol, nerol, geraniol, hotrienol and citronellol, which have a floral aroma of rose and lily-of-the-valley. The olfactory perception thresholds of these compounds are rather low, less than a few hundred $\mu\text{g/L}$ (Table 11.5). They can be present in grapes in free, odour-active forms, especially in Muscat varieties (Alexandria, Frontignan) as well as some

Table 11.5 Main monoterpenes and composition of wines made from various grape varieties

Monoterpene	Aromatic descriptor	Perception threshold ^a (µg/L)	Concentration (µg/L) in wine			
			Muscat of Alexandria	Muscat of Frontignan	Gewürztraminer	Albariño
Linalool	Rose	50	455	473	6	80
α-Terpineol	Lily-of-the-valley	400	78	87	3	37
Citronellol	Citronella	18	nd	nd	12	nd
Nerol	Rose	400	94	135	43	97
Geraniol	Rose	130	506	327	218	58
Hotrienol	Linden	110	nd	nd	nd	127

^a In wines.

nd = not detected.

Source: Ribéreau-Gayon *et al.*, 2000c.

Alsatian and German varieties (Gewürztraminer, Riesling, Pinot gris, etc.). Consequently, the typical floral aroma of these wines is also present in the grape juice (Ribéreau-Gayon *et al.*, 2000b). Moreover, terpenes may also be present in grapes in sugar-bound conjugate forms (glycoconjugates). Among the numerous glycosidic forms existing in grape, β -glucosides may be hydrolyzed during the alcoholic fermentation, releasing odour-active terpenols participating in wine aroma (Günata *et al.*, 1985). The aromatic contribution of glycoconjugate hydrolysis in wine aroma has been investigated in various grape varieties. In floral grape varieties, this glycosidic fraction is quantitatively important and contributes (if hydrolyzed) to enhanced varietal aroma of Muscat wines. This is especially true for Gewürztraminer that contains 20-fold more glycosidic forms than free forms (Günata, 1984). In non-floral varieties (Semillon, Viognier, Chenin blanc, and Chardonnay), free active forms occur at trace concentrations, and the hydrolysis of sugar-bound terpenols might contribute a floral aroma in these varieties. Nevertheless, from an oenological point of view, the enhancement of terpenic aromas in such neutral varieties runs the risk of reducing their typicity by masking other varietal flavours.

β -Glucosidase activity

Grapes contain β -glucosidases that convert odoriferous terpenols from their glycosylated precursors. However, these endogenous grape β -glucosidases are strongly inhibited by glucose and have a very limited impact on the hydrolysis of the sugar-bound fraction (Ribéreau-Gayon *et al.*, 2000c). Therefore, the biohydrolysis of glycoconjugates seems to be mainly due to yeast metabolism. Indeed, *S. cerevisiae* exhibits periplasmic glycosidases that may release terpenols from glycosidic precursors. Their optimal activity occurs at pH $\approx$ 5 and they are strongly inhibited by ethanol (Darriet *et al.*, 1988; Delcroix *et al.*, 1994; Mateo and Di Stefano, 1997; Ribéreau-Gayon *et al.*, 2000c; Tate and Reynolds, 2006).

Consequently, the real contribution of yeast has been controversial for many years. Recently, the contribution of yeast β -glucosidases during the alcoholic fermentation was evaluated by measuring terpene production in synthetic media containing the glycosidic precursor fraction (Ugliano *et al.*, 2006; Loscos *et al.*, 2007). This strategy only allows yeast hydrolysis to be taken into account and does not consider eventual free aromas present in grape juice. By comparing residual glycosides after the fermentation, it was demonstrated that just under 30% glycoconjugates are converted into related aroma compounds by yeast during the fermentation (Ugliano *et al.*, 2006). Moreover due to ethanol inhibition, this hydrolysis only occurs in the first steps of the reaction (Mateo and Di Stefano, 1997). Although significant, small variations (<20%) in aroma production were found among the different yeast strains used. Finally, aroma quantities produced are often lower than the perception threshold of such compounds (Ugliano *et al.*, 2006; Loscos *et al.*, 2007). Taken together, these findings emphasize the fact that glycoside hydrolysis plays only a secondary role in the aroma of wines. Moreover, the small phenotypic variation among strains considerably lowers the direct impact yeast strains may have in the formation of specific key aromas from glycosylated precursors. Therefore, the screening of yeast strains for β -glucosidase activity does not seem to be a very efficient strategy for yeast strain selection. Recently, it was suggested that the overall change of the relative proportions of terpenes observed between strains may modify the aromatic notes of wines (Loscos *et al.*, 2007). Although conceivable, this hypothesis does not allow a rational strategy of selection based on measurement of the concentration of key molecules, but instead needs to use sensorial analysis approaches (Tate and Reynolds, 2006; Reynolds *et al.*, 2007a,b).

Modulation of yeast terpenes biosynthesis

Yeasts like *S. cerevisiae* may produce by *de novo* synthesis small amounts of terpenes during alcoholic fermentation (Carrau *et al.*, 2005). Yeast may also modulate the terpenes content during alcoholic fermentation both for *Saccharomyces* and non-*Saccharomyces* species (King and Dickinson, 2000). Nevertheless, under winemaking conditions, the concentration of *de novo*-produced terpenes by yeast is much lower (50-fold) than the perception threshold of such compounds. Therefore, the intrinsic terpene production of yeast cannot be an efficient means of impacting wine aroma. Some mutations in the sterol pathway may strongly enhance this scarce production. Indeed, during sterol metabolism yeast produces the phosphorylated form of geraniol, geranyl diphosphate (GDP), and an intermediate of farnesyl diphosphate (FDP). It was reported that leaky FDP synthase (*Erg20p*) mutant laboratory strains of *S. cerevisiae* were able to produce relevant amounts of geraniol, linalool and citronellol (Chambon *et al.*, 1991; Javelot *et al.*, 1991), probably through the accumulation of GDP and subsequent bioconversions. This property was introduced into a wine yeast genetic background by repeated back-crosses (Javelot *et al.*, 1991). The resulting strain had a satisfactory terpene production and aroma profile. Nevertheless, the modification of the sterol synthesis pathway affected the ethanol tolerance, limiting its use for winemaking.

Recently, recombinant strategy has been proposed to induce terpene production by expressing plant monoterpenes synthases in wine yeast strains (Herrero *et al.*, 2008). Such GM strains seem to be not affected in their ethanol tolerance and produced more terpenes than the genuine wine yeast.

11.12.3 Volatile thiols

Aromatic impact of volatile thiols

Volatile thiols are important aroma components of certain wine varieties, such as Sauvignon blanc and Scheurebe (Tominaga *et al.*, 1996, 1998a,b; Guth, 1997; Darriet *et al.*, 1995), and have been isolated from wines made with many different cultivars of *Vitis vinifera*, including Gewürztraminer, Riesling, Colombard, Petit Manseng, Semillon, Cabernet Sauvignon and Merlot (Tominaga *et al.*, 1998a,b, 2000; Aznar *et al.*, 2001; Murat *et al.*, 2001). The first of the volatile thiols to be described was 4-methyl-4-sulfanylpentan-2-one (4MSP) (Darriet *et al.*, 1995), with aroma descriptions of 'blackcurrant' and 'broom tree' as well as 'cats' urine' (at higher concentrations). The 4MSP compound has a very low perception threshold (0.1 ng/L and 3.0 ng/L in water and wine, respectively). The concentrations found in wine, up to 30 ng/L (Darriet *et al.*, 1995; Tominaga *et al.*, 2000), indicate its important organoleptic contribution. In Scheurebe wine, 4MSP was the most important varietal odorant of the 42 compounds identified (Guth, 1997). Several other odorous volatile thiols have been identified in Sauvignon blanc wines (Tominaga *et al.*, 1996, 1998a); there are 3-sulfanylhexas-1-ol (3SH), 3-mercaptohexyl acetate (3SHA), 4-methyl-4-sulfanylpentan-2-ol (4MSPOH). The aroma of 3SH also has grapefruit and passion fruit characteristics. Its olfactory threshold is about 60 ng/L; it is always present in Sauvignon blanc wines at

Table 11.6 Volatile thiols and their cysteinylated precursors

Volatile thiols	Aromatic descriptor	Perception threshold (ng/L)	Related cysteinylated precursors
4-Methyl-4-sulfanylpentan-2-one (4MSP)	Box tree, passion fruit	0.8 ^a	S-4-(4-methylpentan-2-one)-L-cysteine
3-Sulfanylhexas-1-ol (3SH)	Grapefruit, passion fruit	60 ^a	S-3-(hexan-1-ol)-L-cysteine
3-Sulfanylhexasyl acetate (3SHA)	Box tree, broom	4 ^a	—
4-Methyl-4-sulfanylpentan-2-ol (4MSPOH)	Citrus zest	55 ^a	S-4-(4-methylpentan-2-ol)-L-cysteine
3-Sulfanylpentan-1-ol	Citrus, sulphur	950 ^b	ni
3-Sulfanylhexas-1-ol	Grapefruit	60 ^b	ni
3-Sulfanylheptan-1-ol	Grapefruit	35 ^b	ni
2-Methyl-3-sulfanylpentan-1-ol	Raw onion, sweet	nd	ni

^a In wines according to Dubourdieu *et al.* (2006).

^b In wines according to Sarrazin *et al.* (2007).

nd = not determined; ni = not identified.

concentrations of several hundred ng/L, sometimes several mg/L. The 3-mercaptohexyl acetate (3SHA) resulting from the acetylation of 3SH by yeast also contributes to wine aroma and has a complex odour, evoking not only boxwood but also grapefruit peel and passion fruit. Its olfactory threshold is 4 ng/L, and certain Sauvignon blanc wines can contain as much as several hundred ng/L (Tominaga *et al.*, 1996). The organoleptic role of 4MSPOH, which has an aroma like citrus zest, is more limited. Its concentration in wines seldom exceeds its olfactory threshold (55 ng/L), but this level can be reached in some wines. More recently, volatile thiols associated with noble botrytized grapes were described; these compounds confer to Sauternes wines their characteristic 'agrumes' notes (Tominaga *et al.*, 1998a, Sarrazin *et al.*, 2007). All of these molecules are shown in Table 11.6.

Identification of S-conjugate cysteine precursors

The concept of aroma precursors associated with Sauvignon blanc was first suggested by Peynaud (1980). 'Although the initial flavour (of the must) perceived in the mouth is relatively small, about 30 seconds after swallowing, one suddenly perceives the Sauvignon-like aroma by retro-olfaction.' This aroma recurrence or aftertaste could be explained by the action of the mouth enzymes on the aroma precursors present in the must. The evidence of a non-volatile and non-glycosylated precursor of the 4MSP in Sauvignon grapes has been shown (Darriet, 1993). The chemical nature of such molecules was then investigated by Takatoshi Tominaga. Indeed, 4MSP, 4MSPOH and 3SH were generated *in vitro* from non-volatile Sauvignon must extracts by the enzyme action of a cell-free extract obtained from the gastrointestinal bacterium *Eubacterium limosum* (Tominaga *et al.*, 1995). The specificity of the cysteine conjugate β -lyase (EC 4.4.1.13) of this bacterium (Larsen and Stevens, 1986) strongly suggested the existence of a precursor for these volatile thiols having a structure of S-cysteine conjugate. This hypothesis was verified formally by GC–MS identification (Tominaga *et al.*, 1998c) for 3SH, 4MSP and 4MSPOH; all these molecules sharing the same S-cysteine conjugate structure (Table 11.6).

Yeast β -lyase activity and related genes

During alcoholic fermentation of a grape juice, concentration of odorous thiols increases. This enhancement of varietal flavour related to yeast metabolism is due to degradation of cysteinylated precursors, leading to formation of corresponding volatile thiols. The wine yeast *S. cerevisiae* has been proposed to cleave thiol precursors during grape juice by an S-conjugate β -lyase. This enzyme is part of a large enzyme family, the carbon–sulphur lyases, in which a carbon–sulphur bond is cleaved in a β -elimination reaction. Enzymes of this type have been isolated from rats, humans, zebra fish and *Arabidopsis thaliana* (Lash *et al.*, 1990; Perry *et al.*, 1993; Mosca *et al.*, 1994; Cooper *et al.*, 2002; Papineau *et al.*, 2005). It has been shown that β -lyase enzyme activity results in the formation of a free thiol and an intermediate that spontaneously degrades to pyruvate and ammonia. Putative yeast genes sharing homology with vertebrate β -lyase enzymes were identified by bioinformatics. The identified genes (*BNA3*, *IRC7* and *CYS3*) were then deleted in

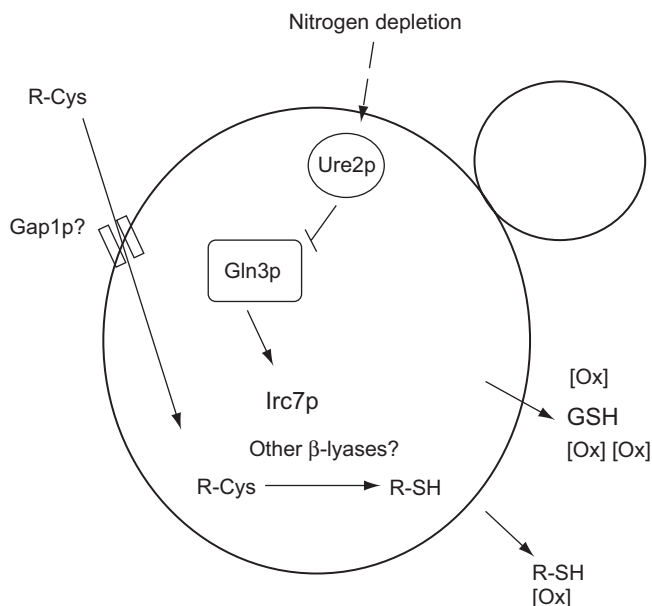


Fig. 11.9 Current model for volatile thiols release in winemaking conditions. The figure shows the major mechanism of volatile thiols release during the alcoholic fermentation. Cysteinyllated precursors R-Cys are taken up by yeast by amino acid transporters including Gap1p (Subileau *et al.*, 2008b) and then cleaved into volatile thiols by the cystathionine β -lyase *Irc7p* and probably other unidentified β -lyases. Its related gene *IRC7* is transcriptionally regulated by the Gln3p/Ure2p regulator pairs and is activated by the depletion of a nitrogen-rich source according to the NCR model (Thibon *et al.*, 2008). The relative amount of volatile thiols in wines is also regulated by the presence of reduced glutathione that contributes to preserving thiols from oxidation [Ox] (Lavigne-Cruège and Dubourdieu, 2002).

a wine yeast strain to test their effects on 4MSP release in a synthetic grape juice containing a synthetic 4MSP-precursor. In such experiments, each mutant released 2–3 fold less 4MSP than control, suggesting that all these genes may contribute to cleaving 4MSP-precursors in related aroma (Howell *et al.*, 2005). Nevertheless, these data were obtained by using 12 500 times more precursors than are naturally present in the must. Recently, Thibon *et al.* (2008) showed that using grape juice precursor concentration, only the *IRC7* gene is responsible for the bioconversion of 4MSP from its S-conjugate precursor (Fig. 11.9). This gene encodes for a cystathionine β -lyase (*Irc7p*) which is likely to be involved in cysteine–homocysteine interconversion. During the alcoholic fermentation, this enzyme contributes to 90% and 40% of 4MSP and 3SH release, respectively. Therefore, concerning 3SH release, other enzymes catalyzing the precursor cleavage are still unidentified. The yeast metabolism also influences the acetylation of 3SH into its acetate 3SHA, and some strains producing high amounts of esters are able to produce important amounts of 3MHA. Like other acetate esters, this conversion may be forced by enhancing the expression level of *ATF1* or by selecting strains producing important amounts of esters (Swiegers and Pretorius, 2007).

Genetic control of thiols bioconversion

The important variation in volatile thiol release should be due to the relative abundance of β -lyase among yeast strains (Thibon *et al.*, 2008). Nevertheless, it is likely that this is determined by many other genetic factors as suggested by the quantitative distribution observed among F1-hybrid progenies. The first insights into the regulation of volatile thiols were proposed recently by investigation of the role of nitrogen catabolic repression (NCR) in the volatile thiol bioconversion. The deletion of *URE2* gene enhanced the volatile thiols release three-fold, while the deletion of *GLN3* strongly impaired their release. These authors demonstrated that NCR directly enhances the cleavage activity by inducing the expression level of *IRC7* gene (Thibon *et al.*, 2008). Interestingly, these authors demonstrated that *URE2* deletion modifies the enantiomer balance of 3SH by activating specifically stereoselective enzymes such as *Irc7p*. The modification of enantiomer ratios might modify the aroma of wine due to their different aromatic notes (Tominaga *et al.*, 2006).

As their structure is similar to that of amino acids, uptake of aroma precursors from must into yeast cytoplasm might be driven by amino acid transporters. The addition of urea in the medium decreases volatile thiols release indirectly, suggesting that the NCR control of precursor uptake. Moreover, the deletion of the general amino acid transporter *GAP1* reduced 3SH release in a laboratory yeast background (Subileau *et al.*, 2008b). Nevertheless, the deletion of *URE2* did not modify precursor uptake from the must (Thibon *et al.*, 2008). Supplementary investigations are necessary to clarify the role NCR plays in precursor uptake. Moreover, as for other aroma compounds, the influence of yeast strain background seems to be crucial.

Factors influencing volatile thiols in wines

The powerful aromatic properties of these molecules have prompted recent studies to investigate how their production can be enhanced and controlled in winemaking. Viticultural practices such as nitrogen application (Choné *et al.*, 2006) and pre-fermentative operations such as skin contact (Peyrot des Gachons *et al.*, 2002) control the amount of thiols precursors in grape must. Fermentation conditions such as temperature (Howell *et al.*, 2004; Masneuf-Pomarède *et al.*, 2006) also influence the final concentration of these aromas in wine. However, the most important factor influencing the liberation of thiols is the yeast strain (Dubourdieu *et al.*, 2006). Indeed, a strong variability in the liberation of thiols (about ten-fold) was found among *S. cerevisiae* strains (Murat *et al.*, 2001; Howell *et al.*, 2004; Dubourdieu *et al.*, 2006; Masneuf-Pomarède *et al.*, 2006), *S. uvarum* strains and related inter-specific hybrids (Masneuf *et al.*, 2002) as well other non-*Saccharomyces* species (Fig. 11.10).

Genetic application for strains selection

Certain starters (VL3c and EG8), which winemakers have empirically noted produce Sauvignon blanc wines with pronounced varietal character, seem to be the most effective at enhancing volatile thiols (Table 11.6). Due to the important

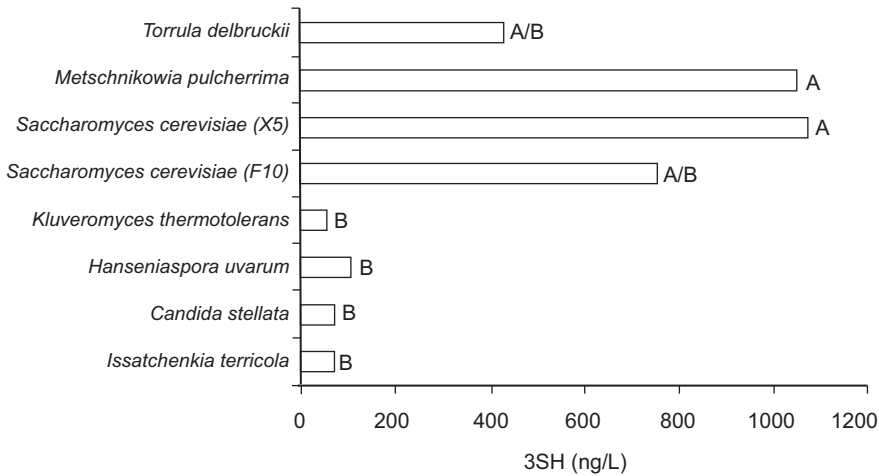


Fig. 11.10 Liberation of 3SH in synthetic must medium by non-*Saccharomyces* yeast. The 3-sulfanylhexasan-1-ol production of six non-*Saccharomyces* species as well as two *S. cerevisiae* strains (F10 and X5) was measured in a synthetic medium containing the usual concentration of the relative cysteine S-conjugate precursor. Data shown are the mean of duplicates and were compared by a multiple comparison test (Newman-Keuls; $p < 0.05$) (source: Zott *et al.*, 2007).

aromatic impact of these molecules, strains producing high concentration of volatile thiols have been selected. The identification of the precursor forms of volatile thiols leads to testing the ability of *S. cerevisiae* strains by adding synthetic precursors to model mediums. The evaluation of the volatile thiol production for numerous yeast strains in controlled conditions allows selecting new starters for industry. This may be done by simple clonal selection or by breeding programs. Some *S. bayanus* var. *uvarum* strains also produce important quantities of volatile thiols. Nevertheless, these strains also produce high levels of β -phenylethyl alcohol and its acetate, which can hide the specific aromas of Sauvignon blanc wines. In order to improve the aroma revelation by yeast selection and breeding, inter-specific hybrids of *S. cerevisiae* $\times$ *S. bayanus* var. *uvarum* were obtained (Masneuf *et al.*, 2002). Some of these revealed high amounts of 4MSP without producing an excess amount of the β -phenyl ethanol and its acetate. Moreover, this hybrid was able to ferment at lower temperature than *S. cerevisiae* parents. Another selection program was recently carried out by exploiting the repression role of Ure2p in thiols release (Thibon *et al.*, 2008). Laboratory strains carrying deleterious *ure2* mutation were used as donor strains in backcross programs according to a classical marker-assisted selection strategy. Wine yeast strains producing high concentrations of volatile thiols were selected as recipient parents. After four backcross steps, progenies carrying *ure2* alleles showed a higher volatile thiol production than those carrying the *wt* one (Marullo *et al.*, 2006b).

GM approaches were also proposed to enhance volatile thiols production. The *Escherichia coli* *tnaA* gene, encoding a tryptophanase with strong cysteine- β -lyase activity, was cloned and overexpressed in a commercial wine yeast strain under the

control of the yeast *PGK1* regulatory sequences. This modified strain expressing carbon–sulphur lyase activity released up to 25 times more 4MSP and 3SH in model fermentations than the control host strain, producing very intense passion fruit aromas (Swiegers *et al.*, 2007). Although efficient and demonstrative, this strategy has not been applied for commercial purposes due to the GM reticence of many consumers.

Perspectives for selecting strains producing higher amounts of volatile thiols

The identification of genes involved in enzymatic cleavage of aroma precursors does not constitute a sufficient input for the comprehension of the thiols bioconversion mechanism. The link between thiols release and NCR may in part explain the wide diversity found among strains. Moreover, mechanisms of precursor uptake from the must need to be clarified. First, only a small part (nearly 0.5–5%) of precursors is converted into the free aroma compounds (Peyrot des Gachons *et al.*, 2000; Thibon *et al.*, 2008). Second, other unidentified precursors of volatile thiols might exist as recently proposed (Subileau *et al.*, 2008a). The untransformed pool of precursor in related aroma constitutes an important reservoir of typicity to be exploited. Finally, the oxidation of aroma formed at the end of the fermentation may strongly influence the final amount of thiols in wine, and some yeast metabolites may modulate this oxidation. For example, there is a close relationship between glutathione content at the end of the fermentation and amounts of volatile thiols after few months of ageing (Lavigne-Cruège and Dubourdieu, 2002). As wine yeast strains present a noticeable variation in glutathione release (Lavigne *et al.*, 2007), the selection of strains producing high amounts of glutathione might be useful for preserving volatile thiol oxidation.

11.12.4 Red wine aromas derived from inodorous precursors

According to the model of volatile thiols, other inodorous precursors present in grape may play a decisive role in the typicity of non-floral varieties of red wines (e.g. Syrah, Merlot, Cabernet Sauvignon). Empirical observations in the winery indicate that the red- and black-berry characteristics of red wines result from aromatic development throughout alcoholic fermentation. The intensity of these aromas may be enhanced by pre-fermentation maceration (Girard *et al.*, 1997, 2001) supporting the idea that red grape skins contain important factors explaining the aromatic differences between red and white wines. Recently, the existence of inodorous constituents in red grape skins, extractible by ethanol that produce specific red- and black-berry aromas in red wines has been suggested (Pineau *et al.*, 2007). The yeast activity was shown to transform the inodorous precursors into aromatic compounds during the alcoholic fermentation. These findings pave the way for discovering a new class of molecules that might have an organoleptic impact on red wine quality. Because yeast activity is required for the production of such odours, a strain effect should be possible as shown by other authors for Pinot noir and Ruché wines (Delfini *et al.*, 2001).

11.13 Conclusions and future trends

11.13.1 Selection methods

The increasing identification of target genes involved in the winemaking process paves the way for designing by molecular approaches yeast starters. Molecular engineering offers the possibility of precisely changing specific characteristics of any existing wine yeasts by a rational approach. This technique is confronted by a double obstacle. First, because the resulting modified strain is a GMO, a lot of reticence exists regarding a commercial use. Recently, a wine starter able to carry out the malolactic fermentation was commercialized (ML01) in a few countries (Moldavia, the USA and Canada). Careful observation of consumers' reactions in the next years might give a trend for the possible use of GM in wine. Second, in some cases, the molecular innovation proposed (i.e. gene overexpression) produce wines with exaggerated character. The over-production of one type of aroma risks the masking of the complex balance of wine which is not compatible with wine quality.

To date, hybridization remains the most effective method for improving and combining traits under polygenic control (such as aromatic properties), particularly if the molecular nature of the mechanisms involved has not been elucidated. The inclusion or elimination of a specific property can be achieved fairly quickly by breeding programs. Recently, the identification of the genetic determinism of complex trait by quantitative genetics approach offers a new perspective for selecting yeast strains by using molecular-assisted selection programs. Due to the small size of yeast genome and its good level of annotation, the identification of many genetic loci controlling major technological traits for winemaking should be rapidly carried out, increasing the selection power of wine starters.

11.13.2 Selection criteria

This chapter has been focused on the main aroma categories found in wine and attempts to give to the reader a critical review of yeast strain impact on wine aroma. Three categories were explored. First, a suitable yeast starter may not be selected, due to major off-flavour defects, such as volatile acidity and hydrogen sulphide over-production. Second, starters may be selected according to their relative production in fermentative aromas (higher alcohols and esters). Third, yeasts may be selected for their interaction with the grape juice matrix for which they are designated (revelation of varietal aromas). This last item is certainly the wider field of investigation for modern oenology, and the many key molecules impacting grape-specific aromas remain to be discovered. Moreover, depending upon the compound family, the effect of yeast strains may be largely anecdotal (terpenes) or very relevant (volatile thiols). The list of yeast properties proposed in this chapter is not exhaustive, and some organoleptic selection criteria have not been described here, especially because few explanations have been proposed to date. Among future criteria, the impact of yeast strain on wine colour as well as mouthfeel could be mentioned. Although empirically verified, the causality of strain difference has not been clearly explained and constitutes future and relevant challenges.

11.14 References

- Albers E, Larsson C, Liden G, Niklasson C and Gustafsson L (1996), Influence of the nitrogen source on *Saccharomyces cerevisiae* anaerobic growth and product formation. *Appl. Environ. Microbiol.*, **62**, 3187–3195.
- Albertyn J, Hohmann S, Thevelein J M and Prior B A (1994), GPD1, which encodes glycerol-3-phosphate dehydrogenase, is essential for growth under osmotic stress in *Saccharomyces cerevisiae*, and its expression is regulated by the high-osmolarity glycerol response pathway. *Mol. Cell. Biol.*, **14**, 4135–4144.
- Ansell R, Granath K, Hohmann S, Thevelein J M and Adler L (1997), The two isoenzymes for yeast NAD⁺-dependent glycerol 3-phosphate dehydrogenase encoded by *GPD1* and *GPD2* have distinct roles in osmoadaptation and redox regulation. *Embo J.*, **16**, 2179–2187.
- Antonelli A, Castellari L, Zambonelli C and Carnacini A (1999), Yeast influence on volatile composition of wines. *J. Agric. Food Chem.*, **47**, 1139–1144.
- Aznar M, Lopez R, Cacho J F and Ferreira V (2001), Identification and quantification of impact odorants of aged red wines from Rioja. GC-olfactometry, quantitative GC-MS, and odor evaluation of HPLC fractions. *J. Agric. Food Chem.*, **49**, 2924–2929.
- Bakalinsky A and Snow R (1990), Conversion of wine strains of *Saccharomyces cerevisiae* to heterothallism. *Appl. Environ. Microbiol.*, **56**, 849–857.
- Bell P J, Deere D, Shen J, Chapman B, Bissinger P H, Attfield P V and Veal D A (1998), A flow cytometric method for rapid selection of novel industrial yeast hybrids. *Appl. Environ. Microbiol.*, **64**, 1669–1672.
- Beltran G, Novo M, Guillamon J M, Mas A and Rozes N (2008), Effect of fermentation temperature and culture media on the yeast lipid composition and wine volatile compounds. *Int. J. Food Microbiol.*, **121**, 169–177.
- Bely M, Rinaldi A and Dubourdieu D (2003), Influence of assimilable nitrogen on volatile acidity production by *Saccharomyces cerevisiae* during high sugar fermentation. *J. Biosci. Bioeng.*, **96**, 507–512.
- Bely M, Stoeckle P, Masneuf-pomarède I and Dubourdieu D (2008), Impact of mixed *Torulaspora delbrueckii*-*Saccharomyces cerevisiae* culture on high-sugar fermentation. *Int. J. Food Microbiol.*, **122**, 312–320.
- Bertrand A (1975), *Recherche sur l'analyse des vins par chromatographie en phase gazeuse*, Thesis, University of Bordeaux, France.
- Bidenne C, Blondin B, Dequin S and Vezinhet F (1992), Analysis of the chromosomal DNA polymorphism of wine strains of *Saccharomyces cerevisiae*. *Curr. Genet.*, **22**, 1–7.
- Blomberg A (2000), Metabolic surprises in *Saccharomyces cerevisiae* during adaptation to saline conditions: questions, some answers and a model. *FEMS Microbiol. Lett.*, **182**, 1–8.
- Briza P, Winkler G, Kalchhauser H and Breitenbach M (1986), Dityrosine is a prominent component of the yeast ascospore wall. A proof of its structure. *J. Biol. Chem.*, **261**, 4288–4294.
- Cabrera M J, Moreno J, Ortega J M and Medina M (1988), Formation of ethanol, higher alcohols, esters, and terpenes by five yeast strains in musts from Pedro Ximenez grapes in various degrees of ripeness. *Am. J. Enol. Vitic.*, **39**, 283–287.
- Cambon B, Monteil V, Remize F, Camarasa C and Dequin S (2006), Effects of *GPD1* overexpression in *Saccharomyces cerevisiae* commercial wine yeast strains lacking *ALD6* genes. *Appl. Environ. Microbiol.*, **72**, 4688–4694.
- Caridi A, Cufari J A and Ramondino D (2002), Isolation and clonal pre-selection of enological *Saccharomyces*. *J. Gen. Appl. Microbiol.*, **48**, 261–267.
- Carrau F M, Medina K, Boido E, Farina L, Gaggero C, Dellacassa E, Versini G and Henschke P A (2005), De novo synthesis of monoterpenes by *Saccharomyces cerevisiae* wine yeasts. *FEMS Microbiol. Lett.*, **243**, 107–115.
- Carrau F M, Medina K, Farina L, Boido E, Henschke P A and Dellacassa E (2008),

- Production of fermentation aroma compounds by *Saccharomyces cerevisiae* wine yeasts: effects of yeast assimilable nitrogen on two model strains. *FEMS Yeast Res.*, **8**, 1196–1207.
- Chambon C, Ladeveze V, Servouse M, Blanchard L, Javelot C, Vladescu B and Karst F (1991), Sterol pathway in yeast. Identification and properties of mutant strains defective in mevalonate diphosphate decarboxylase and farnesyl diphosphate synthetase. *Lipids*, **26**, 633–636.
- Chatonnet P, Dubourdieu D, Boidron J and Lavigne V (1993), Synthesis of volatile phenols by *Saccharomyces cerevisiae* in wines. *J. Sci. Food Agric.*, **62**, 191–202.
- Chatonnet P, Dubourdieu D and Boidron J N (1995), The influence of *Brettanomyces/Dekkera* sp. yeasts and lactic acid bacteria on the ethyl phenol content of red wines. *Am. J. Enol. Vitic.*, **46**, 463–468.
- Choné X, Lavigne-cruège V, Tominaga T, Van Leeuwen C, Castagnède C, Saucier C and Dubourdieu D (2006), Effect of vine nitrogen status on grape aromatic potential: flavour precursor (S-cysteine conjugates), glutathione and phenolic content in *Vitis vinifera* L. c.v. Sauvignon blanc grape juice. *Int. J. Sci. Vigne Vin*, **40**, 1–6.
- Clausen M, Lamb C J, Megnet R A and Doerner P W (1994), *PAD1* encodes phenylacrylic acid decarboxylase which confers resistance to cinnamic acid in *Saccharomyces cerevisiae*. *Gene*, **142**, 107–112.
- Cocolin L, Pepe V, Comitini F, Comi G and Ciani M (2004), Enological and genetic traits of *Saccharomyces cerevisiae* isolated from former and modern wineries. *FEMS Yeast Res.*, **5**, 237–245.
- Combina M, Elía A, Mercado L, Catania C, Ganga A and Martinez C (2005), Dynamics of indigenous yeast populations during spontaneous fermentation of wines from Mendoza, Argentina. *Int. J. Food Microbiol.*, **99**, 237–243.
- Cooper A J, Bruschi S A and Anders M W (2002), Toxic, halogenated cysteine S-conjugates and targeting of mitochondrial enzymes of energy metabolism. *Biochem. Pharmacol.*, **64**, 553–564.
- Cottrell T H E and McLellan M R (1986), The effect of fermentation temperature on chemical and sensory characteristics of wine from white grape cultivars grown in New York state. *Am. J. Enol. Vitic.*, **37**, 190–194.
- Coulon J, Husnik J I, Inglis D L, Van Der Merwe G K, Lonvaud A, Erasmus D J and Van Vuuren H J J (2006), Metabolic engineering of *Saccharomyces cerevisiae* to minimize the production of ethyl carbamate in wine. *Am. J. Enol. Vitic.*, **57**, 113–124.
- Darriet P (1993), *Recherches sur l'arôme et les précurseurs d'arômes du Sauvignon*. Thesis, University of Bordeaux, France.
- Darriet P, Boidron J N and Dubourdieu D (1988), L'hydrolyse des hétérosides terpeniques du Muscat à Petit Grain par les enzymes périplasmiques de *Saccharomyces cerevisiae*. *Conn. Vigne Vin*, **22**, 189–195.
- Darriet P, Tominaga T, Lavigne V, Boidron J and Dubourdieu D (1995), Identification of a powerful aromatic component of *vitis vinifera* L. var Sauvignon wines: 4-mercapto-4-methylpentan-2-one. *Flavour Fragr. J.*, **10**, 385–392.
- Degré R (1993), Selection and commercial cultivation of wine yeast and bacteria. In Fleet, G H (ed.), *Wine Microbiology and Biotechnology*. Paris, France, Harwood Academic, 421–427.
- Delcroix A, Gunata Z, Sapis J-C, Salmon J-M and Bayonove C (1994), Glycosidase activities of three enological yeast strains during winemaking: effect on the terpenol content of Muscat wine. *Am. J. Enol. Vitic.*, **45**, 291–296.
- Delfini C and Costa A (1993), Effects of the grape must lees and insoluble materials on the alcoholic fermentation rate and the production of acetic acid, pyruvic acid, and acetaldehyde. *Am. J. Enol. Vitic.*, **44**, 86–92.
- Delfini C, Cocito C, Bonino M, Schellino R, Gaiaj P and Baiocchi C (2001), Definitive evidence for the actual contribution of yeast in the transformation of neutral precursors of grape aromas. *J. Agric. Food. Chem.*, **49**(11), 5397–5408.

- Dequin S (2001), The potential of genetic engineering for improving brewing, wine-making and baking yeasts. *Appl. Microbiol. Biotechnol.*, **56**, 577–588.
- Deutschbauer A M and Davis R W (2005), Quantitative trait loci mapped to single-nucleotide resolution in yeast. *Nat. Genet.*, **37**, 1333–1340.
- Dickinson J R, Harrison S J, Dickinson J A and Hewlins M J E (2000), An investigation of the metabolism of isoleucine to active amyl alcohol in *Saccharomyces cerevisiae*. *J. Biol. Chem.*, **275**, 10937–10942.
- Dickinson J R, Harrison S J and Hewlins M J E (1998), An investigation of the metabolism of valine to isobutyl alcohol in *Saccharomyces cerevisiae*. *J. Biol. Chem.*, **273**, 25751–25756.
- Dickinson J R, Lanterman M M, Danner D J, Pearson B M, Sanz P, Harrison S J and Hewlins M J E (1997), A ¹³C nuclear magnetic resonance investigation of the metabolism of leucine to isoamyl alcohol in *Saccharomyces cerevisiae*. *J. Biol. Chem.*, **272**, 26871–26878.
- Dickinson J R, Salgado L E J and Hewlins M J E (2003), The catabolism of amino acids to long chain and complex alcohols in *Saccharomyces cerevisiae*. *J. Biol. Chem.*, **278**, 8028–8034.
- Domizio P, Lencioni L, Ciani M, Di Blasi S, Pontremolesi C and Sabatelli M P (2007), Spontaneous and inoculated yeast populations dynamics and their effect on organoleptic characters of Vinsanto wine under different process conditions. *Int. J. Food Microbiol.*, **115**, 281–289.
- Dubourdieu D, Tominaga T, Masneuf I, Peyrot Des Gachons C and Murat M L (2006), The role of yeasts in grape flavor development during fermentation: the example of sauvignon blanc. *Am. J. Enol. Vitic.*, **57**, 81–88.
- Eden A, van Nederveelde L, Drukker M, Benvenisty N and Debourg A (2001), *Involvement of branched-chain amino acid aminotransferases in the production of fusel alcohols during fermentation in yeast*. *Appl. Microbiol. Biotechnol.*, **55**, 296–300.
- Eglinton J M, Heinrich A J, Pollnitz A P, Langridge P, Henschke P A and De Barros Lopes M (2002), Decreasing acetic acid accumulation by a glycerol overproducing strain of *Saccharomyces cerevisiae* by deleting the *ALD6* aldehyde dehydrogenase gene. *Yeast*, **19**, 295–301.
- Ehrlich F (1907), Über die Bedingungen der Fuselölbildung und über ihren Zusammenhang mit dem Eiweissaufbau der Hefe. *Ber. Dtsch. Chem. Ges.*, **40**, 1027–1047.
- Ehsani M, Julien A and Dequin S (2007), Reduction of acetoin in *S. cerevisiae* overproducing glycerol during alcoholic fermentation. In Romano P, Capece A and Granchi L (eds), *26th International Specialized Symposium on Yeasts*, 3–7 June, Sorrento, Italy.
- Eschenbruch R (1974), Sulfite and sulfide formation during winemaking, a review. *Am. J. Enol. Vitic.*, **25**, 157–161.
- Fleet G H (2008), Wine yeasts for the future. *FEMS Yeast Res.*, **8**, 979–995.
- Frary A, Nesbitt T C, Grandillo S, Knaap E, Cong B, Liu J, Meller J, Elber R, Alpert K B and Tanksley S D (2000), fw2.2: a quantitative trait locus key to the evolution of tomato fruit size. *Science*, **289**, 85–88.
- Frezier V and Dubourdieu D (1992), Ecology of yeast strain *Saccharomyces cerevisiae* during spontaneous fermentation in a Bordeaux winery. *Am. J. Enol. Vitic.*, **43**, 375–380.
- Fukuda K, Yamamoto N, Kiyokawa Y, Yanagiuchi T, Wakai Y, Kitamoto K, Inoue Y and Kimura A (1998), Balance of activities of alcohol acetyltransferase and esterase in *Saccharomyces cerevisiae* is important for production of isoamyl acetate. *Appl. Environ. Microbiol.*, **64**, 4076–4078.
- Furukawa K, Yamada T, Mizoguchi H and Hara S (2003), Increased ethyl caproate production by inositol limitation in *Saccharomyces cerevisiae*. *J. Biosci. Bioeng.*, **95**, 448–454.
- Gardner N, Rodrigue N and Champagne C P (1993), Combined effects of sulphites, temperature, and agitation time on production of glycerol in grape juice by *Saccharomyces cerevisiae*. *Appl. Environ. Microbiol.*, **59**, 2022–2028.

- Garofalo A and Piracci A (1994), Évolution des esters des acides gras pendant la conservation des vins. Constantes d'équilibre et énergies d'activation. *Bull. OIV*, **67**(757–758), 225.
- Gawel R, Van Sluyter S and Waters E (2007), The effect of ethanol and glycerol on the body and other sensory characteristics of Riesling wines. *Aust. J. Grape Wine Res.*, **13**, 38–45.
- Girard B, Kopp G T, Reynolds A G and Cliff M (1997). Influence of vinification treatments on aroma constituents and sensory descriptors of Pinot noir wines. *Am. J. Enol. Vitic.*, **48**, 198–206.
- Girard B, Yuksel D, Cliff M A, Delaquis P, and Reynolds A G (2001). Vinification effects on the sensory, colour and GC profiles of Pinot noir wines from British Columbia. *Food Res. Int.*, **34**, 483–499.
- Giudici P and Kunkee R E (1994), The effect of nitrogen deficiency and sulphur-containing amino acids on the reduction of sulphate to hydrogen sulphide by wine yeasts. *Am. J. Enol. Vitic.*, **45**, 107–112.
- Giudici P and Zambonelli C (1992), Biometric and genetic study on acetic production for breeding of wine yeast. *Am. J. Enol. Vitic.*, **43**, 370–374.
- Giudici P, Zambonelli C and Kunkee R E (1993), Increased production of n-propanol in wine by yeast strains having an impaired ability to form hydrogen Sulphide. *Am. J. Enol. Vitic.*, **44**, 17–21.
- Glazier A M, Nadeau J H and Aitman T J (2002), Finding genes that underlie complex traits. *Science*, **298**, 2345–2349.
- Gonzalez S S, Barrio E and Querol A (2008), Molecular characterization of new natural hybrids of *Saccharomyces cerevisiae* and *S. kudriavzevii* in brewing. *Appl. Environ. Microbiol.*, **74**, 2314–2320.
- Gourraud C, Mansour C, Marullo P and Dubourdieu D (2008), Le breeding dirigé, ou comment renouveler une souche de référence inadaptée aux conditions modernes de vinification. *Rev. oenol.*, **128**, 11–15.
- Guillaume C, Delobel P, Sablayrolles J M and Blondin B (2007), Molecular basis of fructose utilization by the wine yeast *Saccharomyces cerevisiae*: a mutated HXT3 allele enhances fructose fermentation. *Appl. Environ. Microbiol.*, **73**, 2432–2439.
- Guitart A, Hernandez-orte P, Ferreira V, Pena C and Cacho J (1999), Some observations about the correlation between the amino acid content of musts and wines of the Chardonnay variety and their fermentation aromas. *Am. J. Enol. Vitic.*, **50**, 253–258.
- Günata Z (1984), *Recherches sur la fraction liée de nature glycosidique de l'arôme du raisin: importance des terpénylglycosides, action des glycosidases*. Science et techniques. Thesis, Sciences et techniques du Languedoc, Montpellier, France.
- Günata Z, Bayonove C, Baumes R and Cordonnier R (1985), The aroma of grapes. Extraction and determination of free and glycosidically bound fractions of some grape aroma components. *J. Chromatogr.*, **331**, 83–90.
- Guth H (1997), Identification of character impact odorants of different white wine varieties. *J. Agric. Food Chem.*, **45**, 3022–3026.
- Hansen J and Kiehlbrandt M C (1996), Inactivation of MET10 in brewer's yeast specifically increases SO₂ formation during beer production. *Nat. Biotechnol.*, **14**, 1587–1591.
- Hazelwood L A, Daran J-M, Van Maris A J A, Pronk J T and Dickinson J R (2008), The Ehrlich pathway for fusel alcohol production: a century of research on *Saccharomyces cerevisiae* metabolism. *Appl. Environ. Microbiol.*, **74**, 2259–2266.
- Henschke P A and Jiranek V (1993), Yeasts-metabolism of nitrogen compounds. In Fleet, G H (ed.) *Wine Microbiology and Biotechnology*. Paris, France, Harwood Academic, 77–154.
- Hernández-Orte P, Cersosimo M, Loscos N, Cacho J, García-Moruno E and Ferreira V (2008), The development of varietal aroma from non-floral grapes by yeasts of different genera. *Food Chem.*, **107**, 1064–1077.
- Herrero O, Ramon D and Orejas M (2008), Engineering the *Saccharomyces cerevisiae*

- isoprenoid pathway for de novo production of aromatic monoterpenes in wine. *Metab. Eng.*, **10**, 78–86.
- Herskowitz I, Rine J and Strathern J N (1992), Mating type determination and mating type interconversion in *Saccharomyces cerevisiae*. In Jones E W, Pringle J R and Broach J R (eds), *The Molecular and Cellular Biology of the Yeast Saccharomyces*. Plainview, NY, Cold Spring Harbor Lab. Press, 583–656.
- Hohmann S (2002), Osmotic stress signaling and osmoadaptation in yeasts. *Microbiol. Mol. Biol. Rev.*, **66**, 300–372.
- Hospital F and Charcosset A (1997), Marker-assisted introgression of quantitative trait loci. *Genetics*, **147**, 1469–1485.
- Howell K S, Swiegers J H, Elsey G M, Siebert T E, Bartowsky E J, Fleet G H, Pretorius I S and De Barros Lopes M A (2004), Variation in 4-mercapto-4-methyl-pentan-2-one release by *Saccharomyces cerevisiae* commercial wine strains. *FEMS Microbiol. Lett.*, **240**, 125–129.
- Howell K S, Klein M, Swiegers J H, Hayasaka Y, Elsey G M, Fleet G H, Hoj P B, Pretorius I S and De Barros Lopes M A (2005), Genetic determinants of volatile-thiol release by *Saccharomyces cerevisiae* during wine fermentation. *Appl. Environ. Microbiol.*, **71**, 5420–5426.
- Hu X H, Wang M H, Tan T, Li J R, Yang H, Leach L, Zhang R M and Luo Z W (2007), Genetic dissection of ethanol tolerance in the budding yeast *Saccharomyces cerevisiae*. *Genetics*, **175**, 1479–1487.
- Husnik J I, Volschenk H, Bauer J, Colavizza D, Luo Z and Van Vuuren H J J (2006), Metabolic engineering of malolactic wine yeast. *Metab. Eng.*, **8**, 315–323.
- Javelot C, Girard P, Colonna-Ceccaldi B and Vladescu B (1991), Introduction of terpene-producing ability in a wine strain of *Saccharomyces cerevisiae*. *J. Biotechnol.*, **21**, 239–252.
- Jiranek V, Langridge P and Henschke P A (1995a), Regulation of hydrogen sulphide liberation in wine-producing *Saccharomyces cerevisiae* strains by assimilable nitrogen. *Appl. Environ. Microbiol.*, **61**, 461–467.
- Jiranek V, Langridge P and Henschke P A (1995b), Validation of bismuth-containing indicator media for predicting H₂S-producing potential of *Saccharomyces cerevisiae* wine yeasts under enological conditions. *Am. J. Enol. Vitic.*, **46**, 269–273.
- Jiranek V, Langridge P and Henschke P A (1996), Determination of sulphite reductase activity and its response to assimilable nitrogen status in a commercial *Saccharomyces cerevisiae* wine yeast. *J. Appl. Bacteriol.*, **81**, 329–336.
- Johnston J R, Baccari C and Mortimer R K (2000), Genotypic characterization of strains of commercial wine yeasts by tetrad analysis. *Res. Microbiol.*, **151**, 583–590.
- Jolly N P, Augustyn O P H and Pretorius I S (2003a), The effect of non-*Saccharomyces* yeast on fermentation and wine quality. *S. Afr. J. Enol. Vitic.*, **24**, 55–62.
- Jolly N P, Augustyn O P H and Pretorius I S (2003b), The occurrence of non-*Saccharomyces cerevisiae* yeast species over three vintages in four vineyards and grape musts from four production regions of the Western Cape, South Africa. *S. Afr. J. Enol. Vitic.*, **24**, 35–42.
- Julien A, Roustan J-L, Dulau L and Sablayrolles J-M (2000), Comparison of nitrogen and oxygen demands of enological yeasts: technological consequences. *Am. J. Enol. Vitic.*, **51**, 215–222.
- Karagiannis S and Lanaridis P (1999), The effect of various vinification parameters on the development of several volatile sulphur compounds in Greek white wines of the cultivars Batiki and Muscat of Hamburg. *Am. J. Enol. Vitic.*, **50**, 334–342.
- Killian E and Ough C (1979), Fermentation esters-formation and retention as affected by fermentation temperature. *Am. J. Enol. Vitic.*, **30**, 301–305.
- King A and Dickinson J R (2000), Biotransformation of monoterpene alcohols by *Saccharomyces cerevisiae*, *Torulaspora delbrueckii* and *Kluyveromyces lactis*. *Yeast*, **16**, 499–506.

- Kingsman A J, Mellor E J, Dobson M J and Kingsman S M (1991), Recombinant DNA techniques. In Tuite M and Oliver S (eds), *Biotechnology Handbooks*, vol. 4 *Saccharomyces*, New York and London, Springer, 149–168.
- Labat L, Durrens P, Hubert C, Arveiller B, Marullo P and Dubourdiou D (2008), Combined use of backcrosses, TILING microarrays and SSCP genotyping for tracking industrial quantitative trait loci in *Saccharomyces cerevisiae*, 12th International Congress on Yeasts, 11–15 August, Kiev, Ukraine.
- Lander E S and Botstein D (1989), Mapping mendelian factors underlying quantitative traits using RFLP linkage maps. *Genetics*, **121**, 185–199.
- Larsen G L and Stevens J L (1986), Cysteine conjugate beta-lyase in the gastrointestinal bacterium *Eubacterium limosum*. *Mol. Pharmacol.*, **29**, 97–103.
- Lash L H, Nelson R M, Van Dyke R A and Anders M W (1990), Purification and characterization of human kidney cytosolic cysteine conjugate beta-lyase activity. *Drug. Metab. Dispos.*, **18**, 50–54.
- Lavigne-Cruège V and Dubourdiou D (2002), Role of glutathione on development of aroma defects in dry white wines. 13th International enology symposium, 9–12 June, Montpellier, France.
- Lavigne V, Pons A and Dubourdiou D (2007), Assay of glutathione in must and wines using capillary electrophoresis and laser-induced fluorescence detection. Changes in concentration in dry white wines during alcoholic fermentation and aging. *J. Chromatogr. A*, **1139**, 130–135.
- Le Jeune C, Lolier M, Demuyter C, Enry C, Legras J, Aigle M and Masneuf-Pomarede I (2007), Characterization of natural hybrids of *Saccharomyces cerevisiae* and *Saccharomyces bayanus* var. *uvarum*. *FEMS Yeast Res.*, **7**, 540–549.
- Lema C, Garcia-Jares C, Orriols I and Angulo L (1996), Contribution of *Saccharomyces* and non-*Saccharomyces* populations to the production of some components of Albarino wine aroma. *Am. J. Enol. Vitic.*, **47**, 206–216.
- Lilly M, Lambrechts M G and Pretorius I S (2000), Effect of increased yeast alcohol acetyltransferase activity on flavor profiles of wine and distillates. *Appl. Environ. Microbiol.*, **66**, 744–753.
- Lilly M, Bauer F F, Lambrechts M G, Swiegers J H., Cozzolino D and Pretorius I S (2006a), The effect of increased yeast alcohol acetyl transferase and esterase activity on the flavour profiles of wine and distillates. *Yeast*, **23**, 641–659.
- Lilly M, Bauer F F, Styger G, Lambrechts M. G and Pretorius I S (2006b), The effect of increased branched-chain amino acid transaminase activity in yeast on the production of higher alcohols and on the flavour profiles of wine and distillates. *FEMS Yeast Res.*, **6**, 726–743.
- Linderholm A L, Olineka T L, Hong Y and Bisson L F (2006), Allele diversity among genes of the sulphate reduction pathway in wine strains of *Saccharomyces cerevisiae*. *Am. J. Enol. Vitic.*, **57**, 431–440.
- Linderholm A L, Findleton C L, Kumar G, Hong Y and Bisson L F (2008), Identification of genes affecting hydrogen sulphide formation in *Saccharomyces cerevisiae*. *Appl. Environ. Microbiol.*, **74**, 1418–1427.
- Loscos N, Hernandez-Orte P, Cacho J and Ferreira V (2007), Release and formation of varietal aroma compounds during alcoholic fermentation from non-floral grape odorless flavour precursors fractions. *J. Agric. Food Chem.*, **55**, 6674–6684.
- Malcorps P, Cheval J, Jamil S and Dufour J P (1991), A new model for the regulation of ester synthesis by alcohol acetyltransferase in *Saccharomyces cerevisiae* during fermentation. *J. Am. Soc. Brew. Chem.*, **49**, 47–53.
- Martini A (1993), Origin and domestication of the wine yeast. *J. Wine Res.*, **4**, 165–176.
- Marullo P, Bely M, Masneuf-Pomarede I, Aigle M and Dubourdiou D (2004), Inheritable nature of enological quantitative traits is demonstrated by meiotic segregation of industrial wine yeast strains. *FEMS Yeast Res.*, **4**, 711–719.
- Marullo P, Bely M, Masneuf-Pomarede I, Pons M, Aigle M and Dubourdiou D (2006a),

- Breeding strategies for combining fermentative qualities and reducing off-flavor production in a wine yeast model. *FEMS Yeast Res.*, **6**, 268–279.
- Marullo P, Thibon C, Dubourdieu D and Tominaga T (2006b), *Use of ure2 mutant yeasts for increasing the release of aromatic volatile thiols by yeast during fermentation*. WO 2008/068635.
- Marullo P, Aigle M, Bely M, Masneuf-Pomarède I, Durrrens P, Dubourdieu D and Yvert G (2007a), Single nucleotide resolution QTL mapping of an physiological trait from derived wild *Saccharomyces cerevisiae* strains. *FEMS Yeast Res.*, **7**, 941–952.
- Marullo P, Yvert G, Bely M, Aigle M and Dubourdieu D (2007b), Efficient use of DNA molecular markers to construct industrial yeast strains. *FEMS Yeast Res.*, **7**, 1295–1306.
- Marullo P, Mansour C, Dufour M, Albertin W, Sicard D, Bely M and Dubourdieu D (2009), Genetic improvement of thermo-tolerance in wine *Saccharomyces cerevisiae* strains by a backcross approach. *FEMS Yeast Res.*, **9**, 1148–1160.
- Masneuf-Pomarède I, Mansour C, Murat M, Tominaga T and Dubourdieu, D (2006), Influence of fermentation temperature on volatile thiols concentration in sauvignon blanc wines. *Int. J. Food Microbiol.*, **108**, 385–390.
- Masneuf I, Hansen J, Groth C, Piskur J and Dubourdieu D (1998), New hybrids between *Saccharomyces sensu stricto* yeast species found among wine and cider production strains. *Appl. Environ. Microbiol.*, **64**, 3887–3892.
- Masneuf I, Murat M, Naumov G, Tominaga T and Dubourdieu D (2002), Hybrids *Saccharomyces cerevisiae* × *Saccharomyces bayanus* var. *uvarum* having a high liberating ability of some sulphur varietal aromas of *Vitis vinifera* Sauvignon blanc wines. *Int. J. Sci. Vigne Vin*, **36**, 205–212.
- Mason A B and Dufour J P (2000), Alcohol acetyltransferases and the significance of ester synthesis in yeast. *Yeast*, **16**, 1287–1298.
- Mateo J J and Di Stefano R (1997), Description of the β -glucosidase activity of wine yeasts. *Food Microbiol.*, **14**, 583–591.
- Meilgaard (1975), Flavor chemistry of beer; Part I: Flavour interaction between principal volatiles. *MBAA Technical Quarterly*, **12**, 107–117.
- Meaden P G and Taylor N R (1991), Cloning of a yeast gene which causes phenolic off-flavours in beer. *J. Inst. Brew.*, **97**, 353–357.
- Mendes-Ferreira A, Mendes-Faia A and Leao C (2002), Survey of hydrogen sulphide production by wine yeasts. *J. Food Prot.*, **65**, 1033–1037.
- Miller A C, Wolff S R, Bisson L F and Ebeler S E (2007), Yeast strain and nitrogen supplementation: dynamics of volatile ester production in Chardonnay juice fermentations. *Am. J. Enol. Vitic.*, **58**, 470–483.
- Molina A M, Swiegers J H, Varela C, Pretorius I S and Agosin E (2007), Influence of wine fermentation temperature on the synthesis of yeast-derived volatile aroma compounds. *Appl. Microbiol. Biotechnol.*, **77**, 675–687.
- Monk P R and Cowley P J (1984), Effect of nicotinic acid and sugar concentration of grape juice and temperature on accumulation of acetic acid yeast fermentation. *J. Ferment. Technol.*, **62**, 515–521.
- Mortimer R K, Romano P, Suzzi G and Polsinelli M (1994), Genome renewal: a new phenomenon revealed from a genetic study of 43 strains of *Saccharomyces cerevisiae* derived from natural fermentation of grape musts. *Yeast*, **10**, 1543–1552.
- Mosca M, Cozzi L, Breton J, Speciale C, Okuno E, Schwarcz R and Benatti L (1994), Molecular cloning of rat kynurenine aminotransferase: identity with glutamine transaminase K. *FEBS Lett.*, **353**, 21–24.
- Murat M L, Tominaga T and Dubourdieu D (2001), Assessing the aromatic potential of Cabernet Sauvignon and Merlot musts used to produce rose wine by assaying the cysteinylated precursor of 3-mercaptohexan-1-ol. *J. Agric. Food Chem.*, **49**, 5412–5417.
- Naumov G I, Naumova E S and Sancho E D (1994), Sibling species of the *Saccharomyces sensu stricto* complex in Spain. *Microbiologia*, **10**, 403–412.

- Naumov G., Masneuf I, Naumova E S, Aigle M and Dubourdieu D (2000), Association of *Saccharomyces bayanus* var. *uvarum* with some French wines: genetic analysis of yeast populations. *Res. Microbiol.*, **151**, 683–691.
- Naumova E S, Bulat S A, Mironenko N V and Naumov G I (2003), Differentiation of six sibling species in the *Saccharomyces sensu stricto* complex by multilocus enzyme electrophoresis and UP-PCR analysis. *Antonie Van Leeuwenhoek*, **83**, 155–166.
- Navarro-Avino J P, Prasad R, Miralles V J, Benito R M and Serrano R (1999), A proposal for nomenclature of aldehyde dehydrogenases in *Saccharomyces cerevisiae* and characterization of the stress-inducible *ALD2* and *ALD3* genes. *Yeast*, **15**, 829–842.
- Nickerson W J (1953), Reduction of inorganic substances by yeasts. Extracellular reduction of sulfite by species of *Candida*. *J. Infect. Dis.*, **93**, 43–56.
- Noble A C, and Bursick G F (1984), The contribution of glycerol to perceived viscosity and sweetness in white wine. *Am. J. Enol. Vitic.* **35**, 110–112.
- Omura F and Shibano Y (1995), Reduction of hydrogen Sulphide production in brewing yeast by constitutive expression of *MET25* gene. *J. Am. Soc. Brew. Chem.*, **53**, 58–62.
- Ono B I, Hazu T, Yoshida S, Kawato T, Shinoda S, Brzvwczy J and Paszewski A (1999), Cysteine biosynthesis in *Saccharomyces cerevisiae*: a new outlook on pathway and regulation. *Yeast*, **15**, 1365–1375.
- Ough C S and Bell A A (1980), Effects of nitrogen fertilization of grapevines on amino acid metabolism and higher-alcohol formation during grape juice fermentation. *Am. J. Enol. Vitic.*, **31**, 122–123.
- Ough C S, Fong D and Amerine M A (1972), Glycerol in wine: determination and some factors affecting. *Am. J. Enol. Vitic.*, **23**, 1–5.
- Papineau D, Walker J J, Mojzsis S J and Pace N R (2005), Composition and structure of microbial communities from stromatolites of Hamelin Pool in Shark Bay, Western Australia. *Appl. Environ. Microbiol.*, **71**, 4822–4832.
- Patel S and Shibamoto T (2003), Effect of 20 different yeast strains on the production of volatile components in Symphony wine. *J. Food Compos. Anal.*, **16**, 469–476.
- Perry S J, Schofield M A, Macfarlane M, Lock E A, King L J, Gibson G G and Goldfarb P S (1993), Isolation and expression of a cDNA coding for rat kidney cytosolic cysteine conjugate beta-lyase. *Mol. Pharmacol.*, **43**, 660–665.
- Peynaud E (1980), *Le Goût du Vin*. Paris, Dunod.
- Peyrot Des Gachons C, Tominaga T and Dubourdieu D (2000), Measuring the aromatic potential of *Vitis vinifera* L. Cv. Sauvignon blanc grape by assaying S-cysteine conjugates, precursors of the volatile thiols responsible for their varietal aroma. *J. Agric. Food Chem.*, **48**, 3387–3391.
- Peyrot Des Gachons C, Tominaga T and Dubourdieu D (2002), Localization of S-cysteine conjugates in the berry: effect of skin contact on aromatic potential of *Vitis vinifera* L. Cv. Sauvignon blanc must. *Am. J. Enol. Vitic.*, **53**, 144–146.
- Pigeau G M and Inglis D L (2005), Upregulation of *ALD3* and *GPD1* in *Saccharomyces cerevisiae* during Icewine fermentation. *J. Appl. Microbiol.*, **99**, 112–125.
- Pigeau G M and Inglis D L (2007), Response of wine yeast (*Saccharomyces cerevisiae*) aldehyde dehydrogenases to acetaldehyde stress during Icewine fermentation. *J. Appl. Microbiol.*, **103**, 1576–1586.
- Pineau B (2007), *Contribution à l'étude de l'arôme fruitéspec ifique des vins rouges de Vitis Vinifera L. cv. Merlot noir et Cabernet-Sauvignon*. Thesis n°1484, University of Bordeaux II, France.
- Pretorius I S (2000), Tailoring wine yeast for the new millennium: novel approaches to the ancient art of winemaking. *Yeast*, **16**, 675–729.
- Pretorius I S and Bauer F F (2002), Meeting the consumer challenge through genetically customized wine-yeast strains. *Trends Biotechnol.*, **20**, 426–432.
- Rainieri S and Pretorius I S (2000), Selection and improvement of wine yeasts. *Ann. Microbiol.*, **50**, 15–31.
- Ramirez M, Perez F and Regodon J A (1998), A simple and reliable method for hybridization

- of homothallic wine strains of *Saccharomyces cerevisiae*. *Appl. Environ. Microbiol.*, **64**, 5039–5041.
- Ramirez M, Regodon J A, Perez F and Rebollo J E (1999), Wine yeast fermentation vigor may be improved by elimination of recessive growth-retarding alleles. *Biotechnol. Bioeng.*, **65**, 212–218.
- Rankine (1967), Formation of higher alcohols by wine yeast and relationship to taste thresholds. *J. Sci. Food Agric.*, **18**, 583–589.
- Remize F, Roustan J L, Sablayrolles J M, Barre P and Dequin S (1999), Glycerol overproduction by engineered *Saccharomyces cerevisiae* wine yeast strains leads to substantial changes in by-product formation and to a stimulation of fermentation rate in stationary phase. *Appl. Environ. Microbiol.*, **65**, 143–149.
- Remize F, Andrieu E and Dequin S (2000a), Engineering of the pyruvate dehydrogenase bypass in *Saccharomyces cerevisiae*: role of the cytosolic Mg(2+) and mitochondrial K(+) acetaldehyde dehydrogenases Ald6p and Ald4p in acetate formation during alcoholic fermentation. *Appl. Environ. Microbiol.*, **66**, 3151–3159.
- Remize F, Sablayrolles J M and Dequin S (2000b), Re-assessment of the influence of yeast strain and environmental factors on glycerol production in wine. *J. Appl. Microbiol.*, **88**, 371–378.
- Remize F, Barnavon L and Dequin S (2001), Glycerol export and glycerol-3-phosphate dehydrogenase, but not glycerol phosphatase, are rate limiting for glycerol production in *Saccharomyces cerevisiae*. *Metab. Eng.*, **3**, 301–312.
- Reynolds A G, Schlosser J, Power R, Roberts R, Willwerth J and De Savigny C (2007a), Magnitude and interaction of viticultural and enological effects. I. Impact of canopy management and yeast strain on sensory and chemical composition of Chardonnay Musque. *Am. J. Enol. Vitic.*, **58**, 12–24.
- Reynolds A G, Schlosser J, Sorokowsky D, Roberts R, Willwerth J and De Savigny C (2007b), Magnitude of viticultural and enological effects. II. Relative impacts of cluster thinning and yeast strain on composition and sensory attributes of Chardonnay Musque. *Am. J. Enol. Vitic.*, **58**, 25–41.
- Ribéreau-Gayon P, Dubourdieu D, Donèche B and Lonvaud A (2000a), Biochemistry and alcoholic fermentation and metabolic pathways. *Handbook of Enology, Vol. 1, Microbiology of Wine*, London, UK, Wiley, 51–74.
- Ribéreau-Gayon P, Glories Y, Maujean A and Dubourdieu D (2000b), Chemical nature, origin and consequences of the main organoleptic defects. *Handbook of Enology, Vol. 2, The Chemistry of Wine: Stabilization and Treatments*, London, UK, Wiley, 233–284.
- Ribéreau-Gayon P, Glories Y, Maujean A and Dubourdieu D (2000c), Varietal aroma. *Handbook of Enology, Vol. 2, The Chemistry of Wine: Stabilization and Treatments*, London, UK, Wiley, 205–230.
- Ribéreau-Gayon P, Glories Y, Maujean A and Dubourdieu D (2000d), Alcohols and other volatile compounds. *Handbook of Enology, Vol. 2, The Chemistry of Wine: Stabilization and Treatments*, London, UK, Wiley, 51–64.
- Rojas V, Gil J V, Piñaga F and Manzanares P (2003), Acetate ester formation in wine by mixed cultures in laboratory conditions. *Int. J. Food Microbiol.*, **86**, 181–188.
- Romano P and Suzzi G (1985), Sensitivity of *Saccharomyces cerevisiae* vegetative cells and spores to antimicrobial compounds. *J. Appl. Bacteriol.*, **59**, 299–302.
- Romano P., Soli M. G And Suzzi G (1983), Procedure for mutagenizing spores of *Saccharomyces cerevisiae*. *J. Bacteriol.*, **156**, 907–908.
- Romano P, Soli G, Suzzi G, Grazia L and Zambonelli C (1985), Improvement of a wine *Saccharomyces cerevisiae* strain by a breeding program. *Appl. Environ. Microbiol.*, **50**, 1064–1067.
- Romano P, Suzzi G, Turbanti L and Polsinelli M (1994), Acetaldehyde production in *Saccharomyces cerevisiae* wine yeasts. *FEMS Microbiol. Lett.*, **118**, 213–218.
- Romano P, Suzzi G, Mortimer R and Polsinelli M (1995), Production of high levels of acetoin

- in *Saccharomyces cerevisiae* wine yeasts is a recessive trait. *J. Appl. Bacteriol.*, **78**, 169–174.
- Rous C, Snow R A and Kunkee R (1983), Reduction of higher alcohols by fermentation with leucine-auxotrophic mutant of wine yeast. *J. Inst. Brew.*, **89**, 274–278.
- Saerens S M, Verstrepen K J, Van Laere S D, Voet A R, Van Dijk P, Delvaux F R and Thevelein J M (2006), The *Saccharomyces cerevisiae* *EHT1* and *EEB1* genes encode novel enzymes with medium-chain fatty acid ethyl ester synthesis and hydrolysis capacity. *J. Biol. Chem.*, **281**, 4446–4456.
- Saerens S, Verstrepen K, Thevelein J and Delvaux F (2008), Ethyl ester production during brewery fermentation: a review. *Cerevisia*, **33**, 82–90.
- Saint-prix F, Bonquist L and Dequin S (2004), Functional analysis of the *ALD* gene family of *Saccharomyces cerevisiae* during anaerobic growth on glucose: the NADP⁺-dependent Ald6p and Ald5p isoforms play a major role in acetate formation. *Microbiol.*, **150**, 2209–2220.
- Sarrazin E, Shinkaruk S, Tominaga T, Bennetau B, Frerot E and Dubourdieu D (2007), Odorous impact of volatile thiols on the aroma of young botrytized sweet wines: identification and quantification of new sulfanyl alcohols. *J. Agric. Food Chem.*, **55**, 1437–1444.
- Shinohara T, Kubodera S and Yanagida F (2000), Distribution of phenolic yeasts and production of Phenolic off-flavors in wine fermentation. *J. Biosci. Bioeng.*, **90**, 90–97.
- Sinha H, David L, Pascon R C, Clauder-munster S, Krishnakumar S, Nguyen M, Shi G, Dean J, Davis R W, Oefner P J, Mccusker J H and Steinmetz L M (2008), Sequential elimination of major-effect contributors identifies additional quantitative trait loci conditioning high-temperature growth in yeast. *Genetics*, **180**(3), 1661–1670.
- Spiczki M (2008), Interspecies hybridization and recombination in *Saccharomyces* wine yeasts. *FEMS Yeast Res.* **8**(7), 996–1007.
- Spiczki M, Romano P, Lipani G, Miklos I and Antunovics Z (2001), Analysis of yeasts derived from natural fermentation in a Tokaj winery. *Antonie Van Leeuwenhoek*, **79**, 97–105.
- Spiczki M, Romano P, Capece A and Paraggio M (2004), Genetic segregation of natural *Saccharomyces cerevisiae* strains derived from spontaneous fermentation of Aglianico wine. *J. Appl. Microbiol.*, **96**, 1169–1175.
- Soden A, Francis I L, Oakey H and Henschke P A (2000), Effects of co-fermentation with *Candida stellata* and *Saccharomyces cerevisiae* on the aroma and composition of Chardonnay wine. *Aust. J. Grape Wine Res.*, **6**, 21–30.
- Soles R, Ough C S and Kunkee R (1982), Ester concentration differences in wine fermented by various species and strains of yeasts. *Am. J. Enol. Vitic.*, **33**, 94–98.
- Solieri L, Antunez O, Perez-Ortin J E, Barrio E and Giudici P (2008), Mitochondrial inheritance and fermentative: oxidative balance in hybrids between *Saccharomyces cerevisiae* and *Saccharomyces uvarum*. *Yeast*, **25**, 485–500.
- Spiropoulos A and Bisson L F (2000), MET17 and hydrogen sulphide formation in *Saccharomyces cerevisiae*. *Appl. Environ. Microbiol.*, **66**, 4421–4426.
- Spiropoulos A, Tanaka J, Flerianos I and Bisson L F (2000), Characterization of hydrogen sulphide formation in commercial and natural wine isolates of *Saccharomyces*. *Am. J. Enol. Vitic.*, **51**, 233–248.
- Steinmetz L M, Sinha H, Richards D R, Spiegelman J I, Oefner P J, Mccusker J H and Davis R W (2002), Dissecting the architecture of a quantitative trait locus in yeast. *Nature*, **416**, 326–330.
- Subileau M, Schneider R, Salmon J M and Degryse E (2008a), New insights on 3-mercaptohexanol (3MH) biogenesis in Sauvignon Blanc wines: Cys-3MH and (E)-hexen-2-al are not the major precursors. *J. Agric. Food Chem.*, **56**, 9230–9235.
- Subileau M, Schneider R, Salmon J M and Degryse E (2008b), Nitrogen catabolite repression modulates the production of aromatic thiols characteristic of Sauvignon Blanc at the level of precursor transport. *FEMS Yeast Res.*, **8**, 771–780.

- Swiegers J H and Pretorius I S (2007), Modulation of volatile sulphur compounds by wine yeast. *Appl. Microbiol. Biotechnol.*, **74**, 954–960.
- Swiegers J H, Capone D L, Pardon K H, Elsey G M, Sefton M A, Francis I L and Pretorius I S (2007), Engineering volatile thiol release in *Saccharomyces cerevisiae* for improved wine aroma. *Yeast*, **24**, 561–574.
- Tate D and Reynolds A G (2006), Validation of a rapid method for measuring β -glucosidase activity in fermenting Muscat grape musts. *Am. J. Enol. Vitic.*, **57**, 60–68.
- Tezuka H, Mori T, Okumura Y, Kitabataki K and Tsumura Y (1992), Cloning of a gene suppressing hydrogen sulphide production by *Saccharomyces cerevisiae* and its expression in a brewing yeast. *J. Am. Soc. Brew. Chem.*, **50**, 130–133.
- Thibon C, Marullo P, Claisse O, Cullin C, Dubourdieu D and Tominaga T (2008), Nitrogen catabolic repression controls *Saccharomyces cerevisiae* volatile thiols release during wine fermentation. *FEMS Yeast Res.*, **8**, 1076–1086.
- Thomas D and Surdin-Kerjan Y (1997), Metabolism of sulphur amino acids in *Saccharomyces cerevisiae*. *Microbiol. Mol. Biol. Rev.*, **61**, 503–532.
- Tominaga T, Masneuf I and Dubourdieu D (1995), Mise en évidence d'un S-conjugué de la cystéine, précurseurs d'arôme du Sauvignon. *Int. J. Sci. Vigne Vin*, **29**, 227–232.
- Tominaga T, Darriet P and Dubourdieu D (1996), Identification de l'acétate de 3-mercaptohexanol, composé à forte odeur de buis, intervenant dans l'arôme des vins de Sauvignon. *Vitis*, **35**, 207–210.
- Tominaga T, Furrer A, Henry R and Dubourdieu D (1998a), Identification of new volatile thiols in aroma of *Vitis vinifera* L. var. Sauvignon blanc wines. *Flavour Fragr. J.*, **13**, 159–162.
- Tominaga T, Murat M and Dubourdieu D (1998b), Development of a method for analyzing the volatile thiols involved in the characteristic aroma of wines made from *Vitis vinifera* L. cv Sauvignon blanc. *J. Agric. Food Chem.*, **46**, 1044–1048.
- Tominaga T, Peyrot Des Gachons C and Dubourdieu D (1998c), A new type of flavor precursors in *Vitis vinifera* L. cv Sauvignon blanc: S-Cysteine conjugates. *J. Agric. Food Chem.*, **46**, 5215–5219.
- Tominaga T, Baltenweck-Guyot R, Gachons C P D and Dubourdieu D (2000), Contribution of volatile thiols to the aromas of white wines made from several *Vitis vinifera* grape varieties. *Am. J. Enol. Vitic.*, **51**, 178–181.
- Tominaga T, Masneuf I and Dubourdieu D (2004), Powerful aromatic volatile thiols in wines made from several *Vitis vinifera* grape varieties and their releasing mechanism. In Shahidi F and Weerasinghe D (eds), *Nutraceutical beverages: chemistry, nutrition, and health effects*. Washington, DC, American Chemical society, 314–337.
- Tominaga T, Niclass Y, Frérot E and Dubourdieu D (2006), Stereoisomeric distribution of 3-Mercaptohexan-1-ol and 3-Mercaptohexyl acetate in dry and sweet white wines made from *Vitis vinifera* (Var. Sauvignon blanc and Semillon). *J. Agric. Food Chem.*, **54**, 7251–7255.
- Torija M J, Beltran G, Novo M, Poblet M, Guillamon J M, Mas A and Rozes N (2003), Effects of fermentation temperature and *Saccharomyces species* on the cell fatty acid composition and presence of volatile compounds in wine. *Int. J. Food Microbiol.*, **85**, 127–136.
- Ugliano M, Bartowsky E J, McCarthy J, Moio L and Henschke P A (2006), Hydrolysis and transformation of grape glycosidically bound volatile compounds during fermentation with three *Saccharomyces* yeast strains. *J. Agric. Food Chem.*, **54**, 6322–6331.
- Urrestarazu A, Vissers S, Iraqui I and Grenson M (1998), Phenylalanine- and tyrosine-auxotrophic mutants of *Saccharomyces cerevisiae* impaired in transamination. *Mol. Gen. Genet.*, **257**, 230–237.
- Valero E, Schuller D, Cambon B, Casal M and Dequin S (2005), Dissemination and survival of commercial wine yeast in the vineyard: a large-scale, three-years study. *FEMS Yeast Res.*, **5**, 959–969.
- Valero E, Cambon B, Schuller D, Casal M and Dequin S (2007), Biodiversity of *Saccharo-*

- myces* yeast strains from grape berries of wine-producing areas using starter commercial yeasts. *FEMS Yeast Res.*, **7**, 317–329.
- Van Laere S D, Verstrepen K J, Thevelein J M, Vandijck P and Delvaux F R (2008), Formation of higher alcohols and their acetate esters. *Cerevisia*, **33**, 65–81.
- Verstrepen K J, Van Laere S D, Vanderhaegen B M, Derdelinckx G, Dufour J P, Pretorius I S, Winderickx J, Thevelein J M and Delvaux F R (2003), Expression levels of the yeast alcohol acetyltransferase genes *ATF1*, *Lg-ATF1*, and *ATF2* control the formation of a broad range of volatile esters. *Appl. Environ. Microbiol.*, **69**, 5228–5237.
- Vezinhet F and Galzy P (1969), Remarks on the sporulation of *Saccharomyces cerevisiae* Hansen. *Antonie Van Leeuwenhoek*, **35**, Suppl: H23–4.
- Vilanova M, Ugliano M, Varela C, Siebert T, Pretorius I S and Henschke P A (2007), Assimilable nitrogen utilisation and production of volatile and non-volatile compounds in chemically defined medium by *Saccharomyces cerevisiae* wine yeasts. *Appl. Microbiol. Biotechnol.*, **77**, 145–157.
- Vollbrecht D and Radler F (1973), The formation of higher alcohols by amino acid auxotrophic mutants of *Saccharomyces cerevisiae*. I. The conversion of amino acids to higher alcohols. *Arch. Mikrobiol.*, **94**, 351–358.
- Vos P J A and Gray R S (1979), The origin and control of hydrogen sulphide during fermentation of grape must. *Am. J. Enol. Vitic.*, **30**, 187–197.
- Walker M E, Gardner J M, Vystavelova A, McBryde C, De Barros Lopes M and Jiranek V (2003), Application of the reuseable, KanMX selectable marker to industrial yeast: construction and evaluation of heterothallic wine strains of *Saccharomyces cerevisiae*, possessing minimal foreign DNA sequences. *FEMS Yeast Res.*, **4**, 339–347.
- Walker M, Vystavelova A, Pedler S, Eglinton J and Jiranek V (2005), PCR-based gene disruption and recombinatory marker excision to produce modified industrial *Saccharomyces cerevisiae* without added sequences. *J. Microbiol. Methods*, **63**, 193–204.
- Wickner R (1991), Method of classical genetics. In Tuite M and Oliver S (eds), *Biotechnology Handbooks Saccharomyces*, New York and London, Springer, 101–145.
- Winge O and Laustsen O (1938), Artificial species hybridization in yeast, *Compt. rend. travaux lab. Carlsberg, série physiologique*, **22**, 235.
- Winzeler E A, Richards D R, Conway A R, Goldstein A L, Kalman S, Mccullough M J, Mccusker J H, Stevens D A, Wodicka L, Lockhart D J and Davis R W (1998), Direct allelic variation scanning of the yeast genome. *Science*, **281**, 1194–1197.
- Yoshimoto H, Fujiwara D, Momma T, Ito C, Sone H, Kaneko Y and Tamai Y (1998), Characterization of the *ATF1* and *Lg-ATF1* gene encoding alcohol acetyltransferases in the bottom fermenting yeast *Saccharomyces pastorianus*. *J. Ferment. Bioeng.*, **86**, 15–20.
- Yoshimoto H, Fukushima T, Yonezawa T and Sone H (2002), Genetic and physiological analysis of branched-chain alcohols and isoamyl acetate production in *Saccharomyces cerevisiae*. *Appl. Microbiol. Biotechnol.*, **59**, 501–508.
- Zambonelli C, Passarelli P, Rainieri S and Giudici P (1997), Technological properties and temperature response of interspecific *Saccharomyces* hybrids. *J. Sci. Food Agric.*, **78**, 7–12.
- Zott K, Masneuf-Pomarede I, Tominaga T, Thibon C, Dubourdieu D and Lonvaud-Funel A (2007), Impact of non-*Saccharomyces* yeast on volatile thiols formation. *Proceeding of 8th International Symposium of Enology of Bordeaux*, Bordeaux, France, TEC & DOC Dunod, 85.
- Zott K, Miot-Sertier C, Claisse O, Lonvaud-Funel A and Masneuf-Pomarede I (2008), Dynamics and diversity of non-*Saccharomyces* yeasts during the early stages in winemaking. *Int. J. Food Microbiol.*, **125**, 197–203.

12

***Brettanomyces/Dekkera* off-flavours and other wine faults associated with microbial spoilage**

L. Conterno, Fondazione E. Mach – IASMA Research and Innovation Centre, Italy; and T. Henick-Kling, Washington State University, USA

Abstract: This chapter briefly summarises the history and the worldwide impact of *Brettanomyces* off-flavour in wines. The production of volatile compounds by *Brettanomyces bruxellensis*, defined metabolic pathways and the relation of these compounds to wine off-flavours are described. *Brettanomyces/Dekkera* physiology is discussed based on factors affecting yeast growth, nutritional requirements and metabolites not solely related to wine 'Brett' off-flavour. Other faults attributable to *Brettanomyces/Dekkera* presence are described, such as mousy taint and the relationships of *B. bruxellensis* to biogenic amines production. Other microbially related wine spoilage problems (e.g. excess acetic acid, excess diacetyl, geranium taint, mannitol) associated with microbial species other than *Brettanomyces* are reviewed. Detection and identification of microorganisms in the winery and in wine with specific reference to *Brettanomyces* are described. This chapter also provides a definition of hygiene principles, and a description of specific technologies which can be used to guarantee wine microbial stability (filtration, preservatives, ozone, pasteurisation, DMDC). Brief descriptions of new technologies are included.

Keywords: *Dekkera/Brettanomyces bruxellensis*, wine spoilage, 4-ethylphenols, off-flavour related metabolism, winery hygiene, microbial detection.

12.1 Introduction

Yeasts of the species *Dekkera/Brettanomyces bruxellensis* have been indicated as

a major cause of yeast contamination affecting wine quality. These yeasts can produce a range of metabolites with strong negative sensory impact. Wine defects or taints are undesirable characteristics caused by physical, chemical and/or microbial off-flavour generation. Such flavour taints have a large economic impact on the wine industry (Loureiro and Malfeito-Ferreira, 2003; Henick-Kling *et al.*, 2005). The generic effect of *Brettanomyces* contamination, usually named phenolic off-flavour, has been described with various attributes such as 'leathery', 'pharmaceutical', 'burnt plastic', 'Band-Aid®', 'smoky', 'barnyard' (Licker *et al.*, 1999; Henick-Kling *et al.*, 2000, 2001; Romano *et al.*, 2009) unified under the term 'Brett character' or simply 'bretty'. The chemicals most widely identified with 'bretty' off-odours are the 4-ethylguaiacols (4EGs) and the 4-ethylphenols (4EPs) (Chatonnet *et al.*, 1990). Recent investigations have been carried out on the sensory impact of these ethyl phenols by Romano *et al.* (2009). Additional bretty aroma characteristics and impact chemicals have been described in wine, and the sensory or microbial finding has not always agreed with the presence of the 4-ethylphenols (Arvik, personal communication, Conterno *et al.*, 2003). The presence of 4EP and 4EG cannot be used alone to identify possible *Brettanomyces* contamination in wines. Clearly, additional chemical markers are needed together with simple and inexpensive microbiological methods for the early detection of *Brettanomyces* yeast in wine. In addition, the sensory contribution of various strains of *Brettanomyces* and of other compounds needs to be investigated further (Licker *et al.*, 1999; Brock *et al.*, 2006; Romano *et al.*, 2008, 2009).

B. bruxellensis is not the only microorganism capable of producing 4-ethylphenols (EPs); in fact, some lactic bacteria such as *Lactobacillus brevis*, *Lb. plantarum*, and *Pediococcus pentosaceus* are also capable of producing EPs (Cavin *et al.*, 1993). However, in more recent studies, this capacity has been tested only in synthetic media (Couto *et al.*, 2006). EPs can also be produced by the yeast *Pichia guilliermondii* which is not capable of growing in wine, but might grow in early stages of fermentation during grape cold soaks or on damaged grapes before harvest (Dias *et al.*, 2003; Barata *et al.*, 2006).

The precursors of EPs are common wine constituents, present in various concentrations, depending on the grape variety, climatic conditions and winemaking techniques (Table 12.1). The concentration of these specific phenolic compounds also depends on cinnamoyl esterase, found in many non-purified commercial enzyme preparations, which may increase the amount of hydroxycinnamic acids in the wine (Gerbaux *et al.*, 2002). *Brettanomyces* is most frequently found in the winery and on cellar equipment, barrels, floors and floor drains (Peynaud and Domercq, 1956; Renouf *et al.*, 2007a), and the winery is therefore considered the main source of *Brettanomyces* in wines (Torok *et al.*, 1996). This yeast has been isolated from musts and from equipment including tanks, barrels and bottling lines, and also in the air of the winery (Alguacil *et al.*, 1998; Connell *et al.*, 2002; Renouf *et al.*, 2007a). Connell *et al.* (2002) detected from three to 100 colonies/250 L of air sampled from near the bottling room and in the crushing rooms, respectively.

Barrels provide an excellent environment for *Brettanomyces* growth. Its capability to produce the enzyme β -glucosidase that can metabolise a sugar, cellobiose,

Table 12.1 p-Coumaric, caffeic and ferulic acid contents of several wines worldwide

	p-Coumaric acid (mg/L)			Caffeic acid (mg/L)			Ferulic acid (mg/L)		
	Average	Range	No. of samples	Average	Range	No. of samples	Average	Range	No. of samples
<i>White wine</i>									
Albarino	5.08	1.45–8.11	6	0.64	0.09–1.46	6	0.54	0.31–0.64	6
Alvarinho	2.60	1.30–3.31	8	1.91	0.25–4.06	8	0.81	0.54–1.13	8
Chardonnay	2.47	1.00–5.00	10	9.07	2.90–18.30	7	<1		1
Codega	0.53	0.13–2.10	8	0.45	0.132–1.359	8	<0.01	0–0.02	6
Gewürztraminer	2.25	1.50–3.00	2	4.00		1			
Gouveio	0.65	0.27–1.64	13	0.27	0.196–0.457	13	0.03	0–0.26	11
Loureiro	0.61	0.21–0.88	6	0.59	0.22–0.89	6	0.36	0.30–0.48	6
Malvasia	0.51	0.13–0.93	12	0.19	0.092–0.794	12	0.01	0–.03	8
Posip	5.94		1	5.03			1.72		1
Riesling	2.63	1.90–4.00	5	3.80	3.79–3.80	2	4.42		1
Rukatac	2.79		1	2.41			3.87		1
Sauvignon blanc	1.38	1.00–2.00	4	1.70	1.30–2.10	2			
Seyval blanc	2.28	2.25–2.30	2	1.51	1.50–1.51	2	<1		1
Vidal	2.11	1.20–3.01	2	3.41	2.20–4.61	2	<1		1
Zinfandel	3.80	0.47–3.04	19	3.97	2.70–5.70	1			
<i>Red wine</i>									
Beaujolais	3.30		1						
Cabernet Franc	3.70	3.70	2	17.00	4.00–30.00	2	1.78	0.70–2.86	2
Cabernet Sauvignon	4.35	3.00–8.00	8	10.22	4.90–12.95	9	0.85	0.70–1.00	2
Gamay	3.50	2.50–4.50	2	4.75		1	<1		1
Grenache	2.56	0.90–5.20	8	6.78	2.40–16.85	8			
Malbec	4.70		1						
Merlot	3.96	1.10–6.40	8	8.06	3.15–5.00	4	0.80	0.60–1.00	2
Pinot Noir	4.53	1.00–5.30	7	18.98	4.70–22.00	3	0.85	0.70–1.00	2
Shiraz/Syrah	5.65	4.60–6.10	6	10.61	4.65–27.00	5			

Note: Range and average data were calculated from data found in literature from: Okamura and Watanabe, 1981; Somers *et al.*, 1987; Hermann, 1989; Roggero and Archier, 1989; Ritter *et al.*, 1994; Frankel *et al.*, 1995; Soleas *et al.*, 1997; Goldberg *et al.*, 1998, 1999, 2000; Ramos *et al.*, 1999; Gorinstein *et al.*, 2000; Andrade *et al.*, 2001; Budic-Leto and Lovric, 2002; Gutierrez *et al.*, 2005; Fernandez-Pachon *et al.*, 2006.

found in toasted wood (Shantha Kumara *et al.*, 1993), a higher resistance to SO₂, and the lack of other microbial competitors at this winemaking stage favour barrel colonisation. The repeated use of contaminated barrels perpetuates *Brettanomyces* spoilage over successive vintages (Chatonnet *et al.*, 1992, 1993). *Brettanomyces* can also be found in new barrels and can enter the winery in this way. It is likely that new barrels are contaminated with *Brettanomyces* from contaminated water and soil during the wood ageing and barrel making processes. More recently, thanks to the development of media and methods capable of detecting organisms in a wide microbial population, *Brettanomyces* has been detected in vineyards and on grapes, establishing the grape as an important source for winery contamination (Agnolucci *et al.*, 2007; Renouf *et al.*, 2007b). *Brettanomyces* is also found in water and vineyard soils. Thus, there are many sources for *Brettanomyces* in the vineyard and winery environment and it enters the winery every year. This emphasises the importance of good winery hygiene as the main tool to avoid spoilage.

Brettanomyces yeasts have been detected in many substrates including fermented beverages like beer, cider, white wine, sparkling wine, sherry wines and especially red wines (Ibeas *et al.*, 1996; Rodrigues *et al.*, 2001). *B. bruxellensis* has been isolated from wines worldwide: Australia (Heresztyn, 1986), Brazil (Zardetto *et al.*, 1959), Chile (Saavedra *et al.*, 2005; Contreras *et al.*, 2008), France (Barret *et al.*, 1955; Peynaud *et al.*, 1956; Chatonnet *et al.*, 1992; Larue *et al.*, 1991; Renouf *et al.*, 2007b), Germany (Roder *et al.*, 2007), UK (Tucknott *et al.*, 1976), Italy (Florenzano, 1951; Agnolucci *et al.*, 2009), New Zealand (Wright and Parle, 1974), Portugal (Velázquez *et al.*, 1993), South Africa (van der Walt and van Kerken, 1959; van Zyl, 1962), Spain (Arjun, 1979; Mateo *et al.*, 1991; Ibeas *et al.*, 1996), USA (Hock, 1990; Larue *et al.*, 1991; Chatonnet *et al.*, 1992) and Uzbekistan (Mavlani, 1969). The precise evolutionary relationships among isolates from different regions has not been established yet, but the numerous studies and surveys carried out worldwide, together with the isolation of *Brettanomyces* from wines produced worldwide are testimony to the magnitude of the problem and the winemakers' fear of this yeast. *D. bruxellensis* has been present in wineries for centuries, and in certain styles of wine a slight 'Brett character' is accepted as contributing what is considered desirable complexity to the aroma (Fugelsang, 1977). The winemaking trends have changed in favour of the production of wines that contain more residual sugar, higher pH, longer maceration periods, delayed SO₂ additions, increased use of barrel ageing and no filtration – all factors that are favourable to *D. bruxellensis* growth – and it seems that the frequency and severity of *D. bruxellensis* contamination has increased in the past 10 years. It may also be possible that a change in consumer sensory preference has caused more awareness of a problem that has actually been present for decades.

12.2 *Brettanomyces/Dekkera* off-flavours and their related metabolism

4-Ethylphenol and 4-ethylguaiacol were the first volatile compounds identified in

wines with bretty off-odours (Chatonnet *et al.*, 1992). Red wines were found with up to 6047 µg/L (average 440 µg/L) and 1561 µg/L (average 82 µg/L) of 4EP and 4EG, respectively. Fariña *et al.* (2007) found 1120 µg/L of 4EP in a Tannat wine together with 120 µg/L of 4EG. Hesford *et al.* (2004) showed the presence of 4-ethyl catechol (4EC) in wine produced after *B. bruxellensis*-induced caffeic acid degradation. Taking into account that the sensory threshold for 4EC is about 50 µg/L (Hesford *et al.*, 2004), 75% of samples, analysed in a study from Larcher *et al.* (2008), were below 89 µg/L, while only 10% were over 163 µg/L. A defective wine, characterised by the maximum values for both 4EP and 4EG, contained 1610 µg/L of 4EC.

Chatonnet *et al.* (1992), based on their findings, hypothesised that ethyl phenol production is a two-step metabolic pathway whereby hydroxycinnamic acid is first decarboxylated to the correspondent vinylphenol, and then further reduced by a vinylphenol reductase to the correspondent ethyl phenol (Fig. 12.1). The enzyme hydroxycinnamate decarboxylase was described and purified in *B. anomalus* by Edlin *et al.* (1998). Chatonnet's hypothesis, also supported by the study of Edlin *et al.* (1995) was not verified until Godoy *et al.* (2008) purified two enzymes, a p-coumarate decarboxylase (CD; its mass was 41 kDa in native conditions) and a vinylphenol reductase (VR; in natural conditions its mass was 118 kDa), involved in the hydroxycinnamic acid metabolism. Both enzymes were purified and characterised from *B. bruxellensis*. In their study, the CD enzyme had optimal activity at a temperature of 40 °C and a pH of 6.0. For p-coumaric acid, the $K_{\max}$ and $V_{\max}$ values were 1.22 ± 0.08 mM and 98 ± 0.15 µmol/min mg, respectively. The VR enzyme had a $K_{\max}$ value of $>3.37 \pm 2.05$ mM and $V_{\max}$ 107.62 ± 50.38 µmol/min mg with NADPH used as a cofactor. Both enzymatic activities were stable at pH 3.4, but in the presence of ethanol, the CD activity decreased drastically while the VR activity was more stable.

Furthermore, the CD enzyme was purified, and its activity was revealed to be common to all *D. bruxellensis* isolates. This was not the case with VR, which was detected at a low activity in the microorganisms analysed. The VR activity was also correlated to the yeast random amplified polymorphic DNA (RAPD) patterns (Godoy *et al.*, 2009). Moreover, the CD activity measured with cell extract was detected only if the cells were pre-incubated in presence of hydroxycinnamic acid. On the other hand, VR activity did not require induction, although it was higher in cell extracts from cells pre-incubated with hydroxycinnamic acid (Harris *et al.*, 2009). The results from this study showed strong differences among strains in their ability to produce EPs, as was previously observed by Conterno *et al.* (2005). In this latter study, however, the CD activity expressed by cells did not need induction and, besides the strain difference, was also shown to be remarkably reduced when the cell density was $<1 \times 10^7$ cfu/mL.

The production of vinyl phenols has been defined as phenotype POF+ (phenolic off-flavour) which has been observed also in brewing. Goodey and Tubb (1982) reported how a single dominant gene (*PofI*) controls the phenotype POF+. Clausen *et al.* (1994) have cloned and sequenced *PadI* that encodes for the phenylacrylic acid decarboxylase responsible for the POF1 phenotype. It is a single copy gene present at the left end of Chromosome IV of *Saccharomyces*

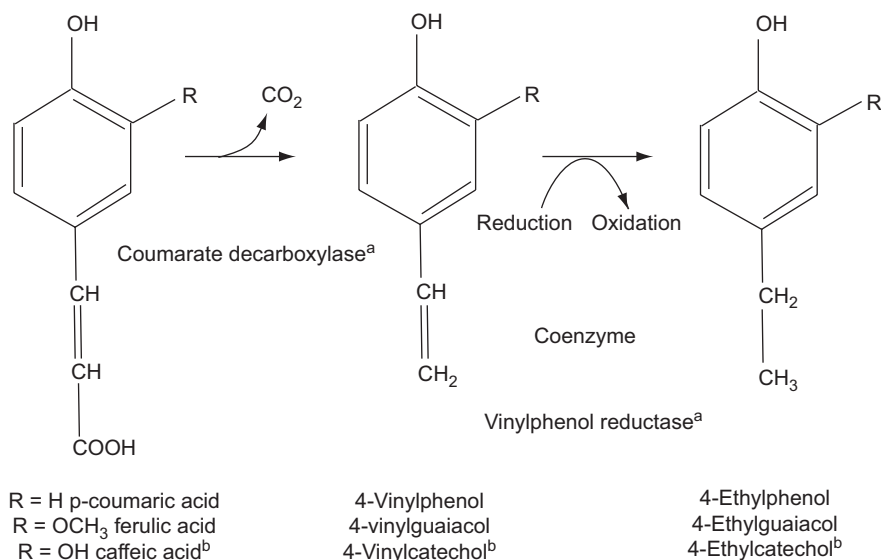


Fig. 12.1 Metabolic pathways leading to the production of 4-ethylphenols from *Brettanomyces bruxellensis* (^aafter Godoy *et al.*, 2008; ^bafter Hesford *et al.*, 2004).

cerevisiae as was also confirmed by Shinohara *et al.* (2000). Cavin *et al.* (1997) and Barthelmebs *et al.* (2000) studied the p-coumarate decarboxylase gene in *Lactobacillus*. Unfortunately, no information is available in *B. bruxellensis* genetic regulation of ethyl phenol production. Measurable amounts of 4EP were produced over 0.2 g/L of glucose and fructose, and were associated with biomass (Barata *et al.*, 2008). Production rates of 4EP observed in red wines varied from 349 to 1882 µg/L per one log CFU/mL (Barata *et al.*, 2008).

Research by Licker *et al.* (1999), looking at the odour-active compounds in wines with Brett flavour through sensory assessment and gas chromatography–olfactometry analysis, indicated that Brett aroma in wine is a complex mixture of odour-active compounds including acids, alcohols, aldehydes, ketones, esters and phenolics. Isovaleric acid and a second, as yet unidentified, compound were predominant odour-impact compounds. Guaiacol, 4EG and 4EP along with other non-identified odour-active compounds were also found. Brett odours have been attributed to the presence of ethyl phenols produced by *Brettanomyces* together with other less characterised compounds (Herestyn, 1986; Licker *et al.*, 1999; Brock *et al.*, 2006).

The growth of *Brettanomyces/Dekkera* in wine has been associated with various forms of spoilage including cloudiness, film formation (less common) and a range of off-flavours (Chatonnet *et al.*, 1995; Henick-Kling *et al.*, 1998). Depending on the conditions and precursors available, these yeasts can produce acetic acid, contributing to wine's volatile acidity (Arvik and Henick-Kling, 2002; Aguilar Uscanga *et al.*, 2003); isovaleric acid, described as a rancid, cheesy aroma; and volatile phenols (Edlin *et al.*, 1995; Dias *et al.*, 2003; Coghe *et al.*, 2004). The

latter is determined by several factors, including the substrate and enzyme activity (e.g. if the substrate is p-coumaric acid, it will be converted by an enzyme to form the vinyl derivative, 4-vinyl phenol, which will be further reduced by another enzyme to form the ethyl form or 4EP. It is the ethyl derivatives that are typically associated with 'Bretty character', as the vinyl derivatives (precursor of ethyl derivatives) can be produced by many other wine microbes as already mentioned (Cavin *et al.*, 1993). Excluding the influence of grape variety, the enzymatic activity or capability of the various organisms indicated differs, depending on the wine conditions, and therefore the precursor amounts and the concentrations of these volatile phenols vary in different wines. Besides the ethyl phenols, *Brettanomyces* can produce other volatile compounds (Licker *et al.*, 1998; Brock *et al.*, 2006) contributing to the whole Brett character. The sensory impact seems to be also influenced by yeast strain (Brock *et al.*, 2006).

A wide range of compounds in wine spoiled by *B. bruxellensis* has been identified, including amides (e.g. N-methyl-oxo-butylacetamide, N-methyl-butyl acetamide) several higher alcohols (e.g. isoamylalcohol, phenylethanol, hexanol, 3-methylthiopropanol) and their acetic esters, polyols (e.g. butandiol, propandiol), carboxylic acids and their ethyl esters. Synthesis of carboxylic acids (from C3 to C10) and ethyl esters (from C6 to C10) by *B. bruxellensis* together with the production of volatile phenols has been observed (Romano *et al.*, 2008) in dry wine. The quantities varied between 210 and 1340 µ/L for total ethyl esters, and between 5.23 and 7.23 mg/L for the total carboxylic acids. The amounts of these carboxylic acids (up to 0.8 mg/L isobutyric acid and 0.7 mg/L isovaleric acid) and ethyl esters (up to 0.4 mg/L ethyl octanoate and 0.8 mg/L ethyl decanoate) has been quantified. It has been also ascertained that isobutyric acid and isovaleric acid possess a masking effect on the perception of the ethyl phenols 4EP and 4EG, whose detection threshold values are three times higher (290 µg/L vs 92 µg/L) if wines are supplemented with these carboxylic acids (Romano *et al.*, 2009).

12.3 *Brettanomyces/Dekkera* taxonomy and phylogenetic relationships with other wine yeasts

Today, there are five accepted species of *Brettanomyces*: *B. anomalus*, *B. bruxellensis*, *B. custersianus*, *B. naardensis* and *B. nanus*. *B. bruxellensis* was the first to be named in 1921 by Kufferath and van Laer (1921). Custers described its physiology in 1940 (Custers, 1940). Because van der Walt and van Kerken (1960) reported ascospore formation in a *B. bruxellensis* strain, *Dekkera* become the teleomorphic genus in 1964 (van der Walt, 1964). The anamorph–teleomorph relationship was demonstrated by Smith *et al.* (1990) together with the numerous synonyms.

The first molecular phylogenetic studies of the species in the *Dekkera* and *Brettanomyces* were done analysing the genomic maps of their mitochondrial DNA (Clark-Walker *et al.*, 1987), on the restriction analysis of the gene coding for

the small subunit of rRNAs (Molina *et al.*, 1993) and on the partial base sequence analysis of 18S and 26S rDNA of the teleomorph species of the genus *Dekkera*.

Investigations carried out using 18S rRNA gene sequencing analysis (Cai *et al.*, 1996) showed that *B. anomalus* differed at two nucleotide position from its teleomorph *D. anomala*. The same analysis reveals that *B. bruxellensis* differed at seven nucleotide positions from *D. bruxellensis*. The two species *B. anomalus* and *B. bruxellensis* (and their teleomorph) were distinct, but the low level of 18S rRNA sequence divergence (about 0.1–0.4%) revealed a close relationship. The data from 18S rRNA were in accordance with partial small- and large-subunit rRNA sequences (Boekhout *et al.*, 1994; Yamada *et al.*, 1994) and rDNA RFLP analysis (Molina *et al.*, 1993). Although *D. custersiana* and *D. naardensis* also form a significant phylogenetic group with the other *Dekkera* species, a high sequence divergence has been found.

Monosaccharide composition of purified cell walls was compared to the 18S rRNA gene sequence phylogenetic tree and *D. bruxellensis* showed the same glucose–mannose pattern of the Hemiascomycetes (Schweigkofler *et al.*, 2002). Recently, 212 hemiascomycetous yeast sugar transporters were analysed and classified into nine phylogenetic clusters for which the function of at least one member was experimentally identified (Palma *et al.*, 2007). Due the limited number of sequences available, phylogenetic relationships of the sugar porters of *Dekkera* species in relation to the other Hemiascomycetes has not been determined.

A recent study by Kurtzman and Robnett (2003) demonstrates that phylogenetic trees can be a valuable means of investigating the evolutionary history of species and for defining and naming biologically meaningful species groupings. However, in situations where complicated evolutionary processes are hypothesised to have played a role in species evolution, phylogenetic trees can be inappropriate (Børsting *et al.*, 1997; Nguyen *et al.*, 2000; Casaregola *et al.*, 2001; de Barros Lopes *et al.*, 2002; James *et al.*, 2005). In such situations, phylogenetic networks had provided a useful alternative. The phylogenetic network methods neighbour net and consensus networks were used by Woolfit *et al.* (2007) in the genome sequencing survey of *D. bruxellensis*. In this analysis of almost 400 protein-coding genes, *D. bruxellensis* alone diverged from three other groups (one represented by *Candida albicans* and *Debariomyces hansenii*, a second grouping *Schizosaccharomyces pombe*, *Aspergillus nidulans* and *Yarrowia lipolytica*, and a third including *S. cerevisiae*, *Kluiveromyces lactis*, *C. glabrata* and *A. gossypii*). As in previous analysis, this approach was unable to determine whether the yeast diverged in the Hemiascomycetes lineage shortly before *C. albicans* and *Db. hansenii* or if these three species formed together a divergent lineage.

The whole genome of *B. bruxellensis* has not been fully sequenced to date. Although *B. bruxellensis* partial sequence information were diffused in 2007 (Woolfit *et al.*, 2007). *B. bruxellensis* CBS 2499 was sequenced using a shotgun sequencing approach based on Sanger technology (Sanger *et al.*, 1977; Staden, 1979; Roach *et al.*, 1995). It is reasonable to expect that, because of the availability of new faster and less expensive sequencing tools, the whole genome sequence will be available soon.

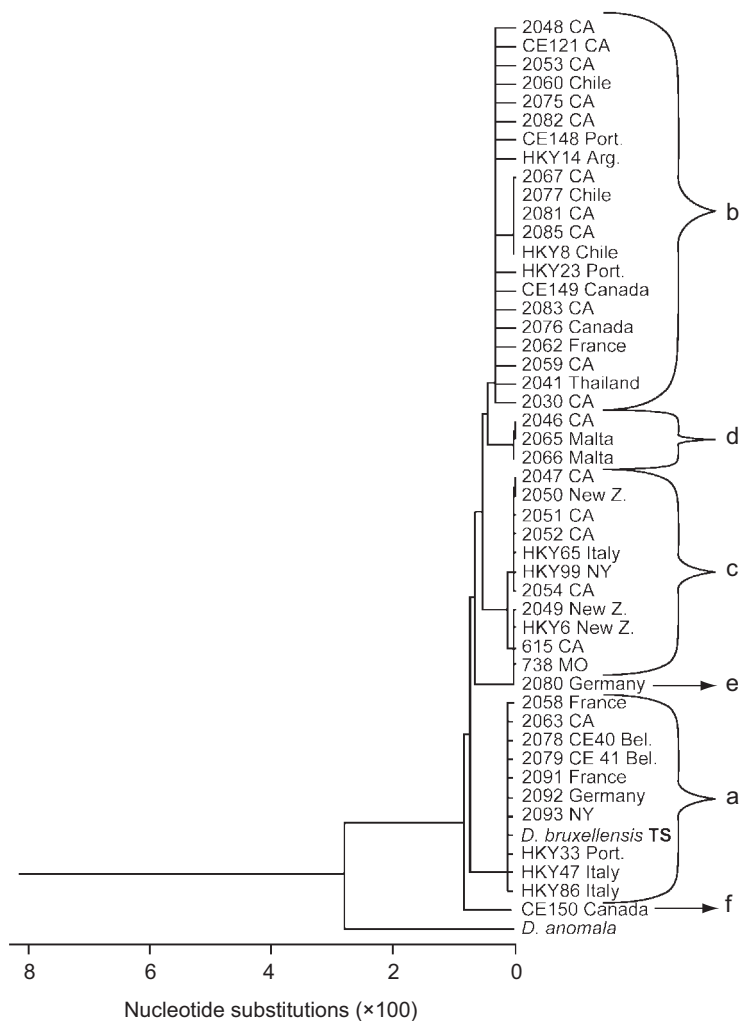


Fig. 12.2 Similarity among aligned sequences of a portion of 26S rRNA gene. Different letters (from a to f) are labelling the different groups of similar sequence (Arg., Argentina; Bel., Belgium; CA, California; NY, State of New York; MO, State of Missouri; New Z., New Zealand; Port., Portugal; TS, type strain sequence) (from Conterno *et al.*, 2006).

More recent studies by Egli *et al.* (1998), Mitrakul *et al.* (1999) and Egli and Henick-Kling (2001) also included karyotyping and RAPD analysis. Hellborg and Piškur (2009) examined the karyotype of 30 isolates which showed from four to at least nine chromosomes. The same study also indicated that some genes were present in double copy and clearly pointed out that this yeast is not a simple haploid. Partial 26S rRNA gene sequence comparison of a broad group of *Brettanomyces* isolates showed that they could be grouped into distinct genetic groups (Fig. 12.2), and these genetic groups can be related to physiological

characteristics (Conterno *et al.*, 2006). Only two subgroups were distinguished by the analysis of the nuclear rDNA locus (26S domain D1/D2, and a fragment from the mitochondrial 15S locus) carried out for 30 other *B. bruxellensis* strains (Hellborg and Piškur, 2009).

12.4 *Brettanomyces/Dekkera* physiology

D. bruxellensis adopts several strategies to survive in the physiologically challenging environments of grape must and wine. Like *S. cerevisiae*, it is Crabtree positive and fermentation metabolism preferentially occurs in the presence of high glucose under aerobic conditions (Kurtzman and Fell, 1998; Barnett and Entian, 2005; Piškur *et al.*, 2006). *D. bruxellensis* is ethanol tolerant, facultative anaerobic and petite positive (i.e. it can survive without mitochondrial DNA). These features evolved independently in each of the two lineages, possibly in response to similar selective pressures (Woelfit *et al.*, 2007). Other differences characterise the two species. For example, *D. bruxellensis* utilises glucose less efficiently and grows much more slowly than *S. cerevisiae* (Silva *et al.*, 2004) but can assimilate a wider variety of alternative carbon sources (Conterno *et al.*, 2006). This peculiarity alone possibly explains the typical ecological succession observed during the winemaking process. *S. cerevisiae* dominates throughout the primary fermentation phase and then is replaced by *D. bruxellensis*, when ethanol concentration is usually high and only small amounts of residual sugars and nitrogen remain such as during wine barrel ageing (Renouf *et al.*, 2006).

Studies on the factors influencing the production of compounds responsible for the characteristic flavours associated with *Brettanomyces* have been carried out (Heresztyn, 1986; Ciani and Ferraro, 1997; Rodrigues *et al.*, 2001), but very little research has been performed on the physiology and genetics controlling the growth of the yeast itself. Rodrigues *et al.* (2001) investigated media components to develop a differential medium. Since those tests were done with a limited number of strains, and different strains and different growth conditions were used in each test little information can be gathered on the diversity among *Brettanomyces* yeasts.

12.4.1 Carbon and nitrogen source requirement

B. bruxellensis can utilise several carbon sources, including ethanol (Dias *et al.*, 2003). Selected cultures were tested in minimal media for growth on different carbon and nitrogen sources (Conterno *et al.*, 2006). Most of the isolates could grow on the hexose monosaccharides glucose, fructose and galactose and the disaccharides sucrose, maltose, cellobiose and trehalose. Other sugars such as arabinose (pentose monosaccharide), lactose (disaccharide) and raffinose (trisaccharide) did not support the growth of most of the isolates. However, in some conditions, *B. bruxellensis* showed both α - and β -glycosidase activity (Mansfield *et al.*, 2002; Shantha Kumara *et al.*, 1993). The sugar alcohols adonitol (ribitol),

glycerol and mannitol also did not support the growth of many *B. bruxellensis* isolates. Few of the isolates tested by Conterno *et al.* (2006) grew on the organic acids lactate, malate, succinate and citrate. About 25% of the isolates could grow on ethanol as a sole carbon source while less than 10% grew on soluble starch. None of these isolates was able to grow on cellulose or tartaric acid as sole carbon source under the conditions applied.

B. bruxellensis was able to grow with similar behaviour in synthetic media with low glucose or fructose concentrations (Barata *et al.*, 2008). Under the same conditions, production of 4-ethylphenol occurred at sugar concentration equal to or higher than 0.2 g/L, and was shown to be correlated with the growth rate and with the amount of biomass produced. Growth and ethyl phenol production was also observed in synthetic wine media in absence of both glucose and fructose, and also depending on the yeast strain (Conterno *et al.*, 2007).

Thirty-five *B. bruxellensis* strains used the amino acid arginine as a carbon source, but none of them could use proline as a carbon source. All of the isolates required biotin and thiamine for growth. In the same study, all of the isolates grew on ammonium, proline and arginine as a nitrogen source (Conterno *et al.*, 2006). Most of the same isolates grew on nitrate when tested as a sole nitrogen source. Five strains of *B. bruxellensis* were able to grow and produce EPs in synthetic wine media with only 50 mg/L of total amino acids (Conterno *et al.* 2007). *B. bruxellensis* consumed the amino acids present in wine with strain-related preference for aspartic acid, arginine, tryptophan, phenylalanine cysteine, histidine, alanine and proline, and also appear to release glutamic acid and, to a lesser extent, glutamine (Silva *et al.*, 2007).

12.4.2 Aerobic vs anaerobic conditions (Custers effect)

'Custers effect' was introduced by Scheffers (1966) to describe the negative Pasteur effect; in other words, the inhibition of alcoholic fermentation in anaerobiosis. Among many other yeast species, the Custers effect has also been observed in *Brettanomyces* spp (Custers, 1940; Wikén *et al.*, 1961, Scheffers, 1966), depending on the glucose concentration (higher at 0.3% than at 2% of glucose) (Carrascosa *et al.*, 1981), producing acetic acid from glucose with the parallel reduction of NAD⁺ (Scheffers and Misset, 1974, Scheffers and Nanninga, 1977). In particular, growth of *B. bruxellensis* in batch cultures with 1.5% of glucose and 1% of yeast extract has been described (Wijsman *et al.*, 1984a): (i) in aerobiosis, glucose is used to produce biomass and CO₂, together with ethanol and acetic acid in equal ratio, ethanol which is subsequently oxidised to acetic acid used to produce biomass and CO₂; (ii) the shift from aerobiosis to anaerobiosis brings all glucose consumption (ethanol, biomass, acetic acid and CO₂ production) to a stop, followed by a new onset after 7 to 8 hours with the production of biomass and ethanol; (iii) the shift from anaerobiosis to aerobiosis resumes the aerobic pattern. Production of glycerol did not occur in any condition (Wijsman *et al.*, 1984b). Gaunt *et al.* (1988) provided evidence already reported by Scheffers (1961) that organic hydrogen acceptors (acetaldehyde, acetone and acetoin) alleviate the

Custers effect by restoring the redox balance. In the Barnett and Entian review (2005), it is clearly explained how in yeast under the Custers effect the alcoholic fermentation occurs at a faster rate in aerobic, compared with anaerobic, conditions because of the glycolysis inhibition by the high ratio of NADH:NAD⁺; the consumption of NAD⁺ caused by the acetic acid production decreases the amount of coenzyme necessary in the glycolysis pathway.

Low biomass and ethanol production were observed in minimal medium (5% glucose, 0.2% (NH₄)₂SO₄, 0.1% yeast extract, pH 4; Aguilar-Uscanga *et al.*, 2003), in non-aerated conditions in comparison with aerated batch growth (air supply 300 L/h = oxygen transfer rate of 145.6 mg O₂/L/h). In the same experiment, little glycerol was produced in both an aerated and a non-aerated bioreactor (0.36 and 0.03 g/L, respectively). In the same study the effect on aeration on biomass production was evaluated and the optimum aeration rate was 60 L/h (oxygen transfer rate 43 mg O₂/L/h).

During wine fermentation, *B. bruxellensis* growth rate and biomass were higher in semi-aerobic or anaerobic conditions than under strict anaerobiosis (Ciani *et al.*, 2003). However, in all cases, the sugar fermentation was completed (wine with 20 g/L of residual sugar).

12.4.3 Acetic acid production

Production of acetic acid by *Dekkera/Brettanomyces* has been demonstrated (Freer, 2002). Full aerobiosis leads to a high production of acetic acid by *B. bruxellensis* (Wijsman *et al.*, 1984; Ciani and Ferraro, 1997) while semi-aerobiosis caused the production of ethanol in concentrations similar to those produced by *S. cerevisiae* (Table 12.2) but, with higher amount of glucose in anaerobiosis, the lowest amount of acetic acid was produced (Freer, 2002). The production of acetic acid from glucose and/or from ethanol as sole energy sources was shown to be strain dependent (Freer, 2002; Freer *et al.*, 2003). Acetic acid was produced at a high concentration (9.4 g/L) under aerated conditions, while under non-aerated conditions, only 0.025–0.15 g/L of acetic acid were synthesised in a minimal culture medium (5% glucose, 0.2% (NH₄)₂SO₄, 0.1% yeast extract, pH 4) (Aguilar-Uscanga *et al.*, 2003). During wine fermentation (20 g/L of residual sugar),

Table 12.2 Acetic acid and ethanol production ratio in synthetic grape juice and natural grape juice (in parentheses) from *Brettanomyces bruxellensis* versus *Saccharomyces cerevisiae* (ratio Bb/Sc) in relation to aeration

	Aerobiosis, dissolved oxygen (40–80%)	Semi-aerobiosis, dissolved oxygen (6.5–35%)	Anaerobiosis (under N ₂)
Ethanol (Bb/Sc)	0.12 (0.085)	0.57 (0.97)	0.40 (0.76)
Acetic acid (Bb/Sc)	2.74 (4.06)	0.50 (1.86)	0.27 (1.83)

Source: data from Ciani and Ferraro (1997).

B. bruxellensis produced less acetic acid, acetaldehyde and ethyl acetate under strict anaerobiosis or anaerobiosis than in semi-aerobic conditions (Ciani *et al.*, 2003).

12.4.4 Temperature, pH, ethanol and other factors influencing growth

In a study carried out by Conterno *et al.* (2006), all the isolates (in a group of 35) grew at pH 2.5 and 94% of them grew at pH 2.0. A little over a third of the isolates grew at 37 °C and a little less than a third grew at 10 °C. Three of the isolates could grow at both 37 °C and 10 °C. A very recent study carried out by Brandam *et al.* (2008) showed the effect of the temperature on *B. bruxellensis* growth. Temperature between 15 and 32 °C did not influence the final biomass concentration or the cell density, but a temperature of 35 °C caused a reduction of biomass (by 65% or 75% respectively). The increase of temperature from 15 to 32 °C had a positive effect on the growth rate. Cell viability was reduced by 40% in culture grown at 35 °C. Higher residual glucose concentrations and lower ethanol and acetic acid production at 35 °C was also observed in the same study.

In synthetic media with 10% ethanol, 20g/L of glucose and pH 3.5, cell death of *B. bruxellensis* occurred in 24 hours of incubation at 36 and 44 °C and no variation of 4-ethylphenol concentration was detected (Barata *et al.*, 2008). Thirty-five *B. bruxellensis* strains tolerated at least 10% ethanol (Conterno *et al.*, 2006). Growth and EPs production was also observed at 12% ethanol for one *D. bruxellensis* strain (Dias *et al.*, 2003; Silva *et al.*, 2004). Several strains of *B. bruxellensis* were capable of growing and produce EPs at 13 and 14% ethanol (Conterno *et al.*, 2007), and some also at 15% ethanol (Benito *et al.*, 2009).

Published results on other growth parameters show conflicting results. Rose and Harrison (1971) found that pyridoxine had a positive affect on the growth of *B. bruxellensis* while Aguilar-Uscanga *et al.* (2000) did not observe any influence on cell growth after adding 0.1 and 0.2 mg/L of pyridoxine. According to Aguilar-Uscanga *et al.* (2000), 340 mg/L of potassium and 7 mg/L of magnesium are enough to support *B. bruxellensis* growth.

12.5 Other defects associated with the presence of *Brettanomyces/Dekkera*

12.5.1 Mousy off-flavour

Dekkera/Brettanomyces has been related to mousy off-flavour in wine since 1956 by Peynaud and Domercq. In a study on fermented beverages, *Dekkera anomala* was the only yeast isolated from apple cider able to spoil the beverage with a mousy off-flavour. Three strains of *B. bruxellensis* previously called *B. intermedius* and *B. lambicus* grown in a chemically defined medium produced mousy taint in wine (Heresztyn, 1986). Grbin and Henschke (2000) showed the four species *Dekkera anomala*, *D. bruxellensis*, *Brettanomyces naardensis* and *B. custersianus* to be able to produce the mousy off-flavour.

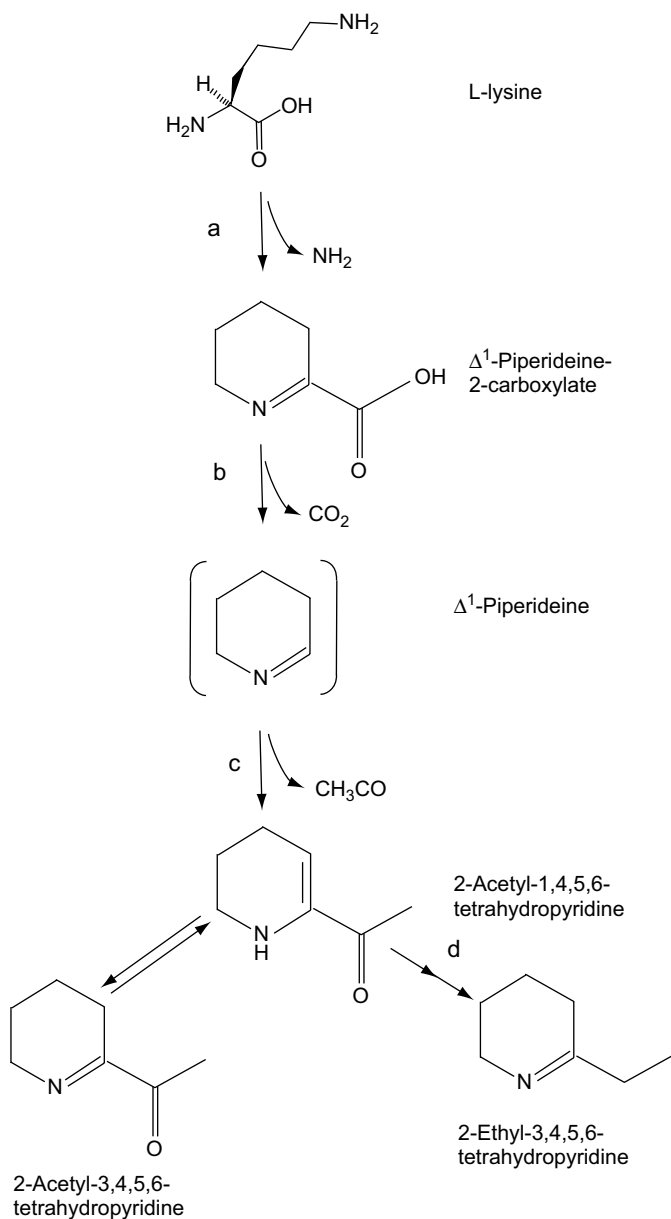


Fig. 12.3 Proposed pathway of biosynthesis of 2-acetyl-1,4,5,6-tetrahydropyridine, 2-acetyl-3,4,5,6-tetrahydropyridine, and 2-ethyl-3,4,5,6-tetrahydropyridine (adapted from Grbin *et al.*, 2007) (a, L-lysine α -aminotransferase or α -oxydase; b, Δ^1 -piperideine-2-carboxylic acid decarboxylase; c, acylase condensation reaction; d, dehydrogenase).

Mousiness or mousy off-flavour is generally perceived as an aftertaste, described as mouse urine. The three compounds primarily identified as responsible for mousiness are 2-ethyltetrahydropyridine (ETHP), 2-acetyltetrahydropyridine (ATHP) and 2-acetylpyrroline (APY) (Tucknott, 1977; Strauss and Heresztyn, 1984; Heresztyn, 1986; Grbin and Henschke, 2000). Heterofermentative wine lactic bacteria have the widespread capability to produce mousy off-flavour and mousy N-heterocycles (Costello *et al.*, 2001). Their precursors and biosynthesis have been studied in *Lactobacillus hilgardii* by Costello and Henschke (2002). Ethanol, amino acids and minerals are necessary for the formation of the mousy off-flavour. For a detailed review on mousy off-flavour in wine, see Snowdon *et al.* (2006).

D. anomala produces EHP and ATP in the presence of L-lysine and APY in presence of ornithine. Grbin *et al.* (2007) observed that the L-lysine concentration did not influence the amount of EHP, but a dose-dependent near linear response was recorded at increasing concentrations of the same amino acid up to 100 mg/L. However, a ten-fold increase of L-lysine from 100 to 1000 mg/L did not correspond to the same increase in ATP. Supplementation with 100 mg/L of ornithine led to production of only 0.5 µg/L of APY, driving the conclusion that *Dekkera* yeast are not responsible for APY; its presence in wine should therefore be attributed to bacteria. Grbin *et al.* (2007) also proposed a pathway for the ATP and EHP production (Fig. 12.3): L-lysine would be transformed in Δ^1 -piperidine in a deamination (likely by an α -oxidase) followed by decarboxylation through the intermediate Δ^1 -piperidine-2-carboxylate; a condensation reaction (acylase via acetyl-CoA or others) would incorporate the acetaldehyde molecule necessary to produce ATP; EHP would be produced from ATP by reduction. As also reported by Snowdon *et al.* (2006), *Dekkera/Brettanomyces* may be able to produce ATP in absence of lysine due to its capability to biosynthesise the amino acid lysine.

While ethanol appears to be the precursor to the side chain of both ATP and APY in *Dekkera/Brettanomyces* (Heresztyn, 1986), no studies have been found on the effect of metal ions as reported for lactic acid bacteria (Costello and Henschke, 2002). The importance of oxygen was originally suggested by Tucknott (1977), and oxygen was suggested to be the trigger for mousy off-flavour biosynthesis by yeast (Grbin, 1998). However, further studies are needed to clarify the relationship between oxygen and mousy off-flavour related biosynthesis. Ammonium phosphate is needed by *Brettanomyces* to produce mousy taint under both aerobic and anaerobic conditions (Lay, 2003).

12.5.2 Biogenic amines

Wine also contains biogenic amines, such as histamine, tyramine, putrescine, cadaverine, spermidine and spermine. These are formed mainly during the malolactic fermentation from amino acids present in the must (Buteau *et al.* 1984, Vidal-Carou *et al.*, 1990; Soufleros *et al.*, 1998; Lonvaud-Funel, 2001) but can also be present in the fruit. More than 20 biogenic amines have been identified in

wine as very recently reviewed by Anli and Bayram (2009). Large amounts of these biogenic amines can cause problems for some sensitive consumers. These health problems are mainly caused by histamine and tyramine. Histamine can cause headaches, hypertension and digestive problems while tyramine is associated with migraine and hypertension (Sillas-Santos, 1996; Shalaby, 1996). Wines have been reported to have an amine concentration ranging from a few mg/L to about 50 mg/L.

Amines are thought to result from malolactic fermentation since the concentration of amines has been found to increase substantially in some wines after this secondary fermentation (Cilliers and van Wyk, 1985; Bertrand *et al.*, 1989). Biogenic amines in wine appear to be largely formed by *Pediococcus* species (Coton *et al.*, 1998) and *Lb. brevis* (Arena and Manca de Nadra, 2001; Lonvaud-Funel, 2001). *Oenococcus oeni* production of biogenic amines is strain dependent (Leitao *et al.*, 2000; Guerrini *et al.*, 2002). In two separate fermentation trials carried out on Aglianico grape must, *B. bruxellensis* produced the largest amount of biogenic amines, more than *S. cerevisiae* and other wine yeast species (Caruso *et al.*, 2002; Granchi *et al.*, 2005). In both studies, the performances of 10 *B. bruxellensis* strains were evaluated revealing a strain-dependent variability for the production of 2-phenylalanine (67% and 50% of the total amount) and agmatine (about 15 and 38% of the total). The average total biogenic amines production was 15 mg/L and 20 mg/L, respectively. Production of ethanolamine, methylamine, tryptamine, putrescine, and cadaverine was also detected (Caruso *et al.*, 2002; Granchi *et al.*, 2005). Strain-dependent biogenic amine production in synthetic media was observed among five CBS *B. bruxellensis* strains (Vigentini *et al.*, 2008). These strains produced detectable amounts of spermidine (up to 1.2–1.3 mg/L), putrescine (up to 0.4 mg/L) and cadaverine (up to 0.15 mg/L). Spermidine concentration decreased from 1.2–1.3 mg/L to 0.7–0.8 mg/L after 60 days. In a study carried out on synthetic wine media, the production of biogenic amines in different amount was referred to seven *B. bruxellensis* defined by specific RAPD–PCR (polymerase chain reaction) and mtDNA restriction profiles (Agnolucci *et al.*, 2009). None of the studied group produced more than 0.3 mg/L of phenylethylamine, and the total production was rated between 1.76 and 4.68 mg/L.

12.6 Other faults associated with microbial spoilage

During winemaking, the microbial species most represented is usually *S. cerevisiae*, responsible for alcoholic fermentation, followed by *O. oeni*, which is desired for malolactic fermentation in white and red wines, for its capability to convert malic acid into lactic acid and for changes in mouthfeel and wine aroma. These particular microbial species are beneficial mainly because these specific yeasts and bacteria produce desirable wine aromas and tastes. However, they may become spoilage microorganisms when their activity leads to undesirable metabolic end-products (hydrogen sulphide, acetic acid, acetaldehyde) or if it occurs at unwanted stages (e.g. refermentation or malolactic fermentation in bottled wine). *O. oeni* may also

Table 12.3 Summary of major wine defects caused by microbial activity

Wine spoilage	Microbial species responsible	Reference
'Lactic souring'	<i>Lactobacillus hilgardii</i> , <i>Lb. fructivorans</i>	Lonvaud-Funel, 1999; Manca de Nadra and Strasser de Saad, 1995; Swiegers <i>et al.</i> , 2005
Bitterness (acrolein from glycerol metabolism)	<i>Lb. hilgardii</i> , <i>Pediococcus parvulus</i> , <i>Oenococcus oeni</i>	Boulton <i>et al.</i> , 1996; Claisse and Lonvaud- Funel, 2001
Ropiness (polysaccharides that create viscosity)	<i>O. oeni</i> , <i>P. damnosus</i> , <i>P. pentosaceus</i> , <i>Lb. spp</i>	van Vuuren and Dicks, 1993; Waling <i>et al.</i> , 2000; Gindreau <i>et al.</i> , 2001
Mousy off-flavour	Heterofermentative <i>Lb. spp</i> , <i>B. bruxellensis</i>	Fugelsang, 1997; Snowdon <i>et al.</i> , 2006; Costello, and Henschke, 2002; Grbin, and Henschke, 2000
Phenolic-off flavour	<i>B. bruxellensis</i> , <i>B. anomalus</i> , <i>Lb. plantarum</i> , <i>Lb. brevis</i> , <i>Pediococcus sp</i>	Cavin <i>et al.</i> , 1993; Chatonnet <i>et al.</i> , 1995; Oelofse <i>et al.</i> , 2008
Production of ethyl carbamate precursors	<i>Lb. brevis</i> , <i>Lb. hilgardii</i> , <i>O. oeni</i>	Sponholz, 1993; Liu <i>et al.</i> , 1996; Arena <i>et al.</i> , 1999; Mira de Orduña <i>et al.</i> , 2001; Tonon <i>et al.</i> , 2001a,b
Biogenic amines		
Histamine	<i>Pediococcus sp</i> , <i>Lb. hilgardii</i> and <i>O. oeni</i>	Lonvaud-Funel and Joyeux, 1994;
Tyramine	<i>Lb. brevis</i> and <i>Lc. mesenteroides</i>	Leitao <i>et al.</i> , 2000; Moreno-Arribas <i>et al.</i> , 2000;
Putresceine	<i>O. oeni</i> , <i>Lb. hilgardii</i>	Arena <i>et al.</i> , 2001 Moreno-Arribas <i>et al.</i> , 2003; Moreno-Arribas and Lonvaud-Funel, 1999; Guerrini <i>et al.</i> , 2002
Film-forming yeast (acetaldehyde)	<i>Candida spp</i> , <i>Pichia spp</i> , <i>Metchnikowia spp</i> , <i>Debaryomyces spp</i>	Sponholtz, 1993; Rojas <i>et al.</i> , 2001; Romano <i>et al.</i> , 2003
Undesired re-fermentation, haze, acetic acid	<i>Zygosaccharomyces spp</i>	Sousa <i>et al.</i> , 1996, 1998
Destruction of wine acidity by total consumption of malic acid	<i>Schizosaccharomyces pombe</i>	Carre <i>et al.</i> , 1983; Peinado <i>et al.</i> , 2004
Acetoin, ethyl acetate, isoamyl alcohol and isobutanol	<i>Saccharomyces ludwigii</i>	Thomas, 1993

produce exo-polysaccharides and can result in the formation of ropiness in wine (Bauer *et al.*, 2003).

Activity of other yeast or bacteria species, particularly post-fermentation, is usually detrimental to the wine's sensory qualities. In fact, other bacteria such as *Lactobacillus* spp and *Pediococcus* spp are often a problem for the wine industry. Wine spoilage can produce ropy wine, haze, sediments, bitterness and off-flavours (Henick-Kling, 1993; Walling *et al.*, 2005a,b; Werning *et al.*, 2006) and biogenic amines (Arena and Manca de Nadra, 2001). Lactic acid bacteria (LAB) belonging to *Lb. brevis*, *Pediococcus damnosus*, *P. parvulus* and *P. pentosaceus* are usually considered potential spoilage agents in wine (Henick-Kling, 1993; Du Plessis *et al.*, 2004).

Because of the large number of microbial species involved and the large number of compounds produced in different types of wine spoilage, we will only provide a brief description of these faults (Table 12.3) and recommend reading the reviews on yeast and bacteria related wine spoilage by du Toit and Pretorius (2000) and Loureiro and Malfeito-Ferreira (2003).

12.6.1 Acetic acid

Healthy, undamaged grapes contain very little to no acetic acid, but during alcoholic fermentation different yeasts form various amounts of acetic acid, and a small amount is formed during malolactic fermentation. Bacteria fermenting citric acid in the wine produce large amounts of this acetic acid. In stuck fermentations, heterofermentative lactic bacteria convert residual sugar into lactic and acetic acid. Increase of acetic acid has been observed in stuck fermented wine in which lactobacilli were found responsible (Huang *et al.*, 1996). Vinegar bacteria (*Acetobacter* and *Gluconobacter*) can produce large quantities of acetic acid by ethanol oxidation but only in the presence of air. The ecology, growth and metabolic behaviour of acetic acid bacteria on the grape, in musts and in wine was reviewed by Drysdale and Fleet (1988) and more recently by Bartowsky and Henschke (2008). In South African red wine fermentations, volatile acidity increases caused by acetic acid bacteria, *Gluconobacter oxydans*, *Acetobacter liquefaciens* and *A. pasteurianus*, was studied (Du Toit and Lambrechts, 2002). Anaerobic conditions that generally prevail in wine, together with the addition of free sulphur dioxide, are usually considered sufficient to prevent acetic acid bacterial spoilage. However, Joyeux *et al.* (1984a,b) showed that acetic acid bacteria are able to survive and even grow during the winemaking process due to momentary aeration introduced by wine agitation or raking. An amount of molecular sulphur dioxide equal to 0.8 mg/L was shown to be enough to prevent *A. aceti* growth in wine (Beech and Thomas, 1985) while amounts equal to 1.2 mg/L were effective against viability of *A. pasteurianus* (du Toit *et al.*, 2005). Furthermore, acetic acid bacteria can produce other compounds, apart from acetic acid, that can influence wine quality (Drysdale and Fleet, 1989; Bartowsky and Henschke, 2008).

12.6.2 Lactic acid

Lactic acid in wines confers a slightly sour taste as it does in fermented vegetables and milk. Very little to no lactic acid comes with the fruit, but all wines contain some. After primary fermentation (L-(+)-lactic) is formed from malic acid during malolactic fermentation (Henick-Kling, 1993; Lonvaud-Funel, 1999; Swiegers *et al.*, 2005). Lactic souring (Peynaud, 1984), also described as ‘acidification’ (Sponholz, 1993) or ‘piqûre lactique’ (Lonvaud-Funel, 1999), is taking place when lactic bacteria attack the sugar like glucose and pentoses. D-(–)-lactic acid would therefore be produced from pyruvic acid together with acetic acid depending on the homofermentative or heterofermentative nature of the lactic acid bacteria acting species present. Although more often *Lactobacillus hilgardii* and *Lb. fructivorans* are considered the species involved, *O. oeni* have also been considered as in some cases responsible for this type of spoilage (Lonvaud-Funel, 1999). The risk of production of D-(–)-lactic acid from hexoses has to be considered also in relation to environmental condition. In fact, the capability of LAB to metabolise sugars is influenced by wine pH (Davis *et al.*, 1988; Henick-Kling, 1986; Henick-Kling *et al.*, 1991).

12.6.3 Mannitol taint

Heterofermentative lactic acid bacteria can produce mannitol from fructose by enzymatic reduction of fructose-6-phosphate (Pilone *et al.*, 1991). Lactic acid bacteria capable of producing mannitol are *Lb. brevis* (Sponholz, 1993), *Lb. kunkee* (Edwards *et al.*, 1998a,b) and some *O. oeni* strains (Pilone *et al.*, 1991).

12.6.4 Bitterness

Another well-known case of wine spoilage caused by lactic acid bacteria metabolism is attributable to the formation of acrolein. Some lactic acid bacteria convert glycerol to 3-hydroxypropionaldehyde. This compound can be reduced to 1,3-propanediol, oxidised to 3-hydroxypropionic acid or chemically dehydrated to acrolein (Fugelsang and Edwards, 2007). Acrolein reacts with phenolics to form a bitter compound. Several strains of *O. oeni*, *P. parvulus* and *Lactobacillus* spp showed a capacity to produce 3-hydroxypropionaldehyde, therefore indirectly producing acrolein (Davis *et al.*, 1988). *Leuconostoc mesenteroides* and *Lb. hilgardii* have also been found responsible for bitter taste due to glycerol metabolism (Sponholz, 1993).

12.6.5 Ropiness

Wines with an increased viscosity (ropy) have been observed during the winemaking process or in already bottled wine. *L. mesenteroides* (Sponholz, 1993), *P. parvulus* (Lonvaud Funel *et al.*, 1993), *P. pentosaceus* (Manca de Nadra and Strasser de Saad, 1995) and some *O. oeni* and *Lactobacillus* spp strains have been found to be able to produce this wine defect by producing hexopolysaccharide (Walling *et al.*,

2001, 2005a,b; Dols-Lafargue *et al.*, 2008). The capability to produce hexopolysaccharides is controlled by a single transmembrane glucosyltransferase encoded by *gtf* gene (Walling *et al.*, 2001, 2005a,b; Werning *et al.*, 2006; Dols-Lafargue *et al.*, 2008). The extracellular polysaccharides production by *pediococci* was induced by ethanol (Lonvaud Funel *et al.*, 1993; Manca de Nadra and Strasser de Saad, 1995). LAB responsible for ropiness in wine showed a strong resistance to pH, ethanol and SO₂ (Lonvaud-Funel and Joyeux, 1988); the correlation between the gene encoding for the glucosyltransferase (the protein controlling ropy polysaccharide synthesis) and LAB stress resistance has been demonstrated (Dols-Lafargue *et al.*, 2008). Although never isolated from ropy wine, *L. mesenteroides* showed an ability to produce hexopolysaccharides (MonTERSINO *et al.*, 2008).

12.6.6 Film-forming yeasts

Development of film-forming yeasts (many species of *Pichia*, *Candida*, *Metchnikowia* and *Debaryomyces*) is strictly correlated to oxygen exposure and, therefore, to wine which is not sufficiently protected against contact with air. These yeasts oxidise ethanol, glycerol and acids, giving wines with unacceptably high contents of acetaldehyde, esters and acetic acid. Acetaldehyde and esters are the major undesired products in this wine spoilage (Sponholz, 1993; Rojas *et al.*, 2001).

12.6.7 Other faults associated with yeast

Schizosaccharomyces pombe, *Zygosaccharomyces bailii* and *Z. fermentati* are well known for their tolerance of high ethanol concentrations (>10%) and occur in winery environments (Romano and Suzzi, 1993; Fleet, 2000). Surprisingly, they are rarely reported as contributors to grape juice fermentations and the reasons for this require more investigation. All three species have the ability to utilise malic acid, which is a positive attribute in many winemaking instances. The yeast species *Z. bailii* is highly resistant to ethanol and can metabolise the acetic acid produced during wine fermentation, either in the presence or absence of residual sugar concentrations (Sousa *et al.*, 1996, 1998). Due to these features, the yeast is able to grow at the final stages of wine fermentation. In fact, *Z. bailii* is often responsible for causing re-fermentation in bottled wines leading to the formation of deposit and haze which lower the visual quality of the wine.

Romano *et al.* (2003) revealed that the flavour profiles created by the predominant yeasts of the early fermentation phases, *Hanseniaspora uvarum* and *Candida stellata*, were characterised by a high production of acetoin and ethyl acetate (higher in *C. stellata* than in *H. uvarum*) and in general a low production of higher alcohols. *Z. fermentati* produced high amounts of 2,3-butanediol.

Sch. pombe and *Saccharomyces ludwigii*, although dangerous spoilage microorganisms, are not regarded as common wine contaminants (Kunkee and Bisson, 1993; Fugelsang, 1997). *Sa. ludwigii* was shown to be a high producer, in

particular, of acetoin, ethyl acetate, isoamyl alcohol and isobutanol. Its remarkable tolerance to sulphite makes it a frequent isolate in wineries where high sulphite doses or sulphited grape juices are used (Thomas, 1993). *Sch. pombe* can cause total depletion of wine malic acid together with of formation of large amounts of sulphur-containing compounds (Carre *et al.*, 1983) and acetaldehyde (Peinado *et al.*, 2004).

12.6.8 Geranium off-odour

The geranium off-odour, due to the presence in wine of 2-ethoxyhexa-3,5-diene, is reminiscent of crushed geranium leaves and originates from reduction of sorbic acid carried out by lactic acid bacteria. The reduction product sorbinol will isomerise, under wine condition, to form 3,5-hexadiene-2-ol that, reacting with ethanol, will generate 2-ethoxyhexa-3,5-diene (Crowell and Guymon, 1975; Sponholz, 1993; du Toit and Pretorius, 2000) which has a sensory threshold of about 0.1 mg/L (Hühn *et al.*, 1999). According to Edinger and Splittstoesser (1986) no strains of *Pediococcus* or *Lactobacillus* are capable of producing sorbinol by reduction of sorbic acid.

12.6.9 Tartaric acid degradation

Certain strains of *Lb. plantarum* and *Lb. brevis* are capable of metabolising tartaric acid under aerobic conditions (Radler and Yannissis, 1972). The enzyme involved in this phenomenon has been indicated to be a tartrate dehydratase, which converts tartrate to oxaloacetate. Oxaloacetate will follow two different pathways, depending on which bacteria are carrying out the transformation (Radler and Yannissis, 1972). In fact, *Lb. plantarum* (facultative homofermentative) will break oxaloacetate into pyruvic acid and carbon dioxide, subsequently reducing half of the pyruvic acid to lactic acid, and split the other half to acetic acid and carbon dioxide. *Lb. brevis* (heterofermentative) will metabolise oxalacetic acid partly to succinic acid and partly to acetic acid and carbon dioxide. In both cases, acetic acid production may affect the wine volatile acidity.

12.6.10 Cork taint and other related taints

Corky, mouldy, earthy taints occasionally occur in wines, and significantly depreciate their quality and acceptability to the consumer (see Chapter 13). The main component responsible for this problem is 2,4,6-trichloroanisole, but other anisoles, guaiacol, geosmin and pyrazines may also be involved in this taint (Lee and Simpson, 1993). They are produced by microbial metabolism during production and storage of corks used for bottling the wine. When the cork is in contact with the wine, these taint-causing compounds are leached into the wine. The microbial ecology and biochemistry of this problem are quite complex and not well understood (Silva Pereira *et al.*, 2000). Filamentous fungi that grow within the lenticels of the cork as part of the natural microflora or as contaminants are mainly

responsible for the problem, but yeasts and bacteria have also been implicated (Lee and Simpson, 1993; Jager *et al.*, 1996; Silva Pereira *et al.*, 2000; Alvarez-Rodríguez *et al.*, 2002).

12.7 Detection and methods to prevent and defeat microbial spoilage faults

12.7.1 Methods to detect microbial spoilage

It is difficult to detect *Brettanomyces* in wine because of its slow growing nature and its presence in low cell numbers. Low microbial cell numbers are also typical for other microbial species (yeast and bacteria) after the primary fermentation processes (alcoholic and malolactic fermentation) have been completed. In wine ageing in tanks and barrels and in bottled wine, microbial cells can remain viable but not growing for months before growth is initiated. During this time, cells may be in viable but not-culturable status (VBNC) and therefore escape the traditional detection method (Millet and Lonvaud-Funel, 2000; Barer, 1997). When discovered with conventional plating methods, *Brettanomyces* cells may already be in a growing state and the wine is often already sensory damaged.

A study was carried out by Heard and Fleet in 1986 investigating several carbon sources and other main constituents with the aim to make the selective media for *Brettanomyces*. In the past decades, several media have been proposed to plate and to enrich *Brettanomyces* (Gerbaux *et al.*, 2000; Rodrigues *et al.* 2001; Conterno *et al.*, 2003; Renouf and Lonvaud-Funel, 2007). These media contain specific compounds used to select or indicate the presence of *Brettanomyces* yeast together with compounds essential for common yeast growth: vitamins, in particular pyridoxine (Rose and Harrison, 1971); ethanol may be added because of the ethanol tolerance of *B. bruxellensis* (Medawar *et al.*, 2003); chloramphenicol (200 mg/L) is sometimes added to prevent bacterial growth (Couto *et al.*, 2005); cycloheximide can be added to prevent *Saccharomyces* growth in doses starting from 20 up to 100 mg/L together with p-coumaric acid as the precursor of 4-ethylphenol (Kurtzmann and Fell, 1998; Rodrigues *et al.*, 2001; Couto *et al.*, 2005); bromocresol green may also be added as differential compound to indicate medium acidification (Rodrigues *et al.*, 2001). The more recent formulation studied by Renouf and Lonvaud-Funel (2006) is an enrichment media that showed a good success in recovering *Dekkera/Brettanomyces* yeast from the grape surface, therefore revealing its utility in detecting the yeast where presented in very low numbers. Simplicity and low cost are the major advantages of using traditional plating methods. However, all plating techniques are slow due the long incubation time (up to 15 days) together with the requirement for the positive results to be further confirmed, and the possibility of false negative results is a major disadvantage.

False negative results are due to the possible VNBC status of the yeast. For these reasons, molecular biology protocols have been studied. Restriction fragment length polymorphism (RFLP) has been applied together with PCR of rRNA genes

(Molina *et al.*, 1993; Mittrakul *et al.*, 1999), nested PCR (Quesada and Cenis, 1995; Ibeas *et al.*, 1996), denaturing gradient gel electrophoresis (DGGE) of PCR amplified portions of the 26S rRNA gene (Ibern-Gomez *et al.*, 2001; Cocolin *et al.*, 2004), enzyme-linked immunosorbent assay (ELISA) (Kuniyuki *et al.*, 1984). Specific sequences of the 5.8S rRNA gene internal transcribed spacer region (ITS 1 and 2), together with the sequence region beyond the D1–D2 domains region of the 26S rRNA gene (Cai *et al.*, 1996; Esteve-Zarzoso *et al.*, 1999; Kurtzman and Robnett, 2003), are mostly used to identify spoilage yeast, also taking into account the availability of a large database of over 132 yeast species (Arias *et al.*, 2002). In order to detect intraspecific differences RAPD–PCR of total DNA (Mittrakul *et al.*, 1999; Phister and Mills 2003), amplified fragment length polymorphisms (AFLPs) (de Barros Lopes *et al.*, 1999; Curtin *et al.*, 2007), sequencing a portion of the 26S rRNA gene (Conterno *et al.*, 2006), PCR–DGGE (Renouf and Lonvaud-Funel, 2007), restriction enzyme analysis (REA)–PCR (Miot-Sertier and Lonvaud-Funel, 2007) and RAPD–PCR with OPA primers (Martorell *et al.*, 2006; Agnolucci *et al.*, 2009) have been applied. The use of the above listed approach allowed the identification of up to eight genotypes among wine strains.

PNA (peptide nucleic acid) molecular probes targeting rDNA are used coupled with fluorescence in situ hybridisation (FISH), or *in situ* hybridisation (ISH) (Stender *et al.*, 2001; Connell *et al.*, 2002). Molecular probes appear to be the fastest form of identification, about 2 hours after isolation, but the equipment needed is expensive. There are many forms of identification available in laboratories, but there has yet to be a method of determination that is fast, specific and inexpensive that can be used in the winery (Stender *et al.*, 2001). Most of these identification tests are currently based on rRNA gene sequences; they are very accurate but not feasible in a winery, due to equipment cost and the need for highly skilled technicians. In addition, even with high specificity and reliability, they do not meet the requirement for a rapid and sensitive method. In this regard, the real-time PCR or qPCR has been shown to satisfy the need for a real-time detection of *B. bruxellensis* also at low density (Delaherche *et al.*, 2004; Tessoniere *et al.*, 2009). At this time, the quantification limit has been assessed at 31 cfu/mL.

12.7.2 Prevention and remediation in relation to *Brettanomyces bruxellensis*

Sanitation is an important part of good winemaking practices that allows for the development of wine with strong varietal and regional flavour expression. In addition, it serves as a safeguard to preserve the wine quality during ageing. Poor cellar hygiene can lead to the growth of microorganisms that can spoil grape must and wine. Moreover, good sanitation practices can also prevent the development of microbial populations that are responsible for production of biogenic amines in wine, such as lactic acid bacteria and yeast that include *B. bruxellensis*. Even though it is not clear yet whether biogenic amines are the result of yeast or lactic acid bacteria (*pediococci* and *lactobacilli*) metabolism, preventing the growth of these populations will ensure the storage quality of the finished wine and serve as

a protective hurdle from the potential development of biogenic amines. Many species of non-*Saccharomyces* yeast, *Lactobacillus* and *Pediococcus* may also be particularly difficult to eradicate due to their capability to form biofilms that serve to protect microorganisms from sanitiser treatments (Timke *et al.*, 2004; Storgards *et al.*, 2006; Joseph *et al.*, 2007). All microorganisms, including mould, bacteria and yeast, are entering into the wine by several routes: grapes (environmental contamination), winery equipment and winery personnel (du Toit and Pretorius, 2000; Loureiro and Malfeito Ferreira, 2003). The selective pressure of the winery ecosystem defines their survival. Typically, the microorganisms that survive the winemaking process, such as acetic and lactic acid bacteria and yeast, are the microorganisms with a high tolerance to low pH, ethanol and low temperature (du Toit and Pretorius, 2000; Loureiro and Malfeito Ferreira, 2003). Although no foodborne pathogens survive in wine (Møretrø and Daeschel, 2004) and the only fungal toxin found in wine is ochratoxin A, originating from fungal infection of the grapes (some strains of *Aspergillus*) (Martinez-Rodríguez and Carrascosa, 2009), good cellar sanitation is considered as a critical factor to ensure wine stability (Fugelsang, 1997; Marriot, 1999; Worobo, 2005). However, few scientific studies have been published which recommend specific winery sanitation procedures (Gerland, 1995; Caboulet *et al.*, 2004). In a study on the influence of viticultural and oenological practices on the yeast population during winemaking, lower numbers of *Zygosaccharomyces*, *Pichia*, and *Candida* were attributed to better hygiene and sanitation practices in the winery without additional details for their conclusion (Regueiro *et al.*, 1993). Other than good sanitation practices, many factors have been studied with the specific aim to find a more effective way to avoid *Dekkera/Brettanomyces* wine spoilage.

Compounds with antimicrobial action against B. bruxellensis

Sulphur dioxide (SO₂) is the main preservative used in the wine industry (Ribéreau-Gayon *et al.*, 2006). SO₂ microbial activity has to be considered in relation to its molecular form (Macris and Markakis, 1974). The amount of molecular SO₂ is related to the free form which depends on pH. The microbial activity is also impacted by temperature and ethanol concentration (Boulton *et al.*, 1996). Controversial observations have been reported in the published literature on the effect of SO₂ on *B. bruxellensis*. In fact, some studies found *B. bruxellensis* sensitive to free SO₂ concentration higher than 30 mg/L (Chatonnet *et al.*, 1992; Gerbaux *et al.*, 2002) while others (Van der Walt and Van Kerken, 1960; Froudière and Larue, 1988) observed no *D. bruxellensis* growth in presence of more than 30 mg/L. These differences could be related to the concentration of the molecular form, as also discussed by Oelofse *et al.* (2008) and also to differences among strains. In fact, *Dekkera/Brettanomyces bruxellensis* strains, as many other microbial species, exhibit various degrees of sensitivity to SO₂. A study carried out on 35 *D. bruxellensis* strains revealed that at pH 3.4 about 48% of the strains tolerated more than 20 mg/L of SO₂: six tolerated up to 30 mg/L and 11 up to 50 mg/L of SO₂ (Conterno *et al.*, 2006). Similar strain variability was observed by Barata *et al.* (2008). The effect on cultivability of one *B. bruxellensis* strain has been studied

(du Toit *et al.*, 2005), finding the effective dose of 0.25 mg/L of molecular SO₂. Barata *et al.* (2008) observed the minimal inhibitory concentration of molecular sulphur dioxide to be 0.98 mg/L in red wine with 13.8% (v/v). Recently, Benito *et al.* (2009) observed that, in absence of other limiting factors and at pH 3.5, one strain of *D. bruxellensis* arrested the metabolic conversion of p-coumaric acid in 4EP when the free SO₂ concentration ranged between 9.6 and 20.2 mg/L; they defined the addition of more than 20 mg/L as a 'preventive effect'. As these authors also pointed out, to ensure protection, especially in wooden barrels, the amount of free SO₂ should be checked periodically. The relationship between strain resistance and strain enzymatic activity at different doses of free SO₂ must be established in relation to wine pH and composition.

Sorbic acid had a preventive effect (in relation to hydroxycinnamic acid-related enzymes) at doses higher than 900 mg/L in the absence of other limiting factors at pH 3.6 (Benito *et al.*, 2009). However, a legal dose of 200 mg/L has been established as reported in International Oenological Codex (IOC, 2009). Moreover, sorbic acid can be metabolised by lactic acid bacteria and its activity is pH dependent. Therefore, it can hardly be used to control growth of *Brettanomyces* under winemaking conditions.

Dimethyl dicarbonate (DMDC) is a preservative also known under the trade name Velcorin®. It is authorised for use in USA and Europe with the maximum limit of 200 mg/L (200 mg/kg in Australia) as reported by Costa *et al.* (2008). Appropriate dosing apparatus is necessary for DMDC usage, and the results depend to some degree on the volume of liquid treated, the cell density and the microbial species present (Costa *et al.*, 2008). The dose of 150 mg/L of DMDC inhibits *B. bruxellensis* cells contained in 200 mL of grape juice (Renouf *et al.*, 2008). However, DMDC undergoes rapid hydrolysis (Van der Riet *et al.*, 1989; Delfini *et al.*, 2002). Therefore its antimicrobial action will not last, as confirmed by Renouf *et al.* (2008). According to a recent study, the minimum inhibitory concentrations (MIC) of DMDC in 50 mL of wine (12% ethanol, pH 3.5) inoculated with 500 and 10⁶ CFU/mL of *B. bruxellensis* were 100 and 300 mg/L respectively (Costa *et al.*, 2008). However, the same authors, observed the combined effect of 100 mg/L of DMDC and 25 mg/L of potassium metabisulphite to provoke the loss of viability of about 1 × 10⁶ CFU/mL of *D. bruxellensis*.

Benzoic acid and nicotinic acid (not authorized in wine) 'preventive effects' have been assessed at doses of 150–200 mg/L and up to 25 mg/L, respectively (Benito *et al.*, 2009). Ferulic acid was found to completely inhibit cell growth at 1 mM (Harris *et al.*, 2008).

Antimicrobial peptides. The antimicrobial action of lactoferrin-derived peptides has been tested against several strains of *D. bruxellensis*. The tested strains showed varying sensitivity to the 15-residue peptide (LfcinB17-31), which showed strain-dependent growth inhibition in wine (Enrique *et al.*, 2008). A *Pichia membranifaciens*-produced protein was recently observed to possess antifungal activity (Santos *et al.*, 2009). The killer toxin (PMKT2), a protein with an apparent molecular mass of 30 kDa and an isoelectric point of 3.7, showed, at pH 4.5, optimal killer activity at temperatures up to 20 °C. PMKT2 was able to inhibit

B. bruxellensis while *S. cerevisiae* was fully resistant, indicating that PMKT2 could be used in wine fermentations to avoid the development of the spoilage yeast without deleterious effects on the desired fermentative yeast. In simulated winemaking conditions, PMKT2, as well as *P. membranifaciens* CYC 1086, was able to inhibit *B. bruxellensis*.

Effect of physical treatment against Dekkera/Brettanomyces bruxellensis

Pulsed electric field (PEF) technology was evaluated to control *D. anomala* and *D. bruxellensis* in the winery. The non-exponential kinetics of inactivation by PEF of these microorganisms in both products has been described by mathematical equations based on the Weibull distribution. An optimum treatment of 186 kJ/kg at 29 kV/cm has been established, which permitted a 99.9% reduction of the spoilage flora of must and wine (Puertolas *et al.*, 2009).

Filtration has been evaluated as practice to remove *Brettanomyces* from wine (Renouf *et al.*, 2007c), establishing the need of K100 (1 µm) filter sheet to stabilise the wine against non-*Saccharomyces* and reducing the population to less than 10 CFU/mL. Recent studies indicate that some strains of *Brettanomyces* can pass through 1 µm pore size membranes, and 0.6 µm membranes should be used to ensure complete removal of *Brettanomyces* (Edwards, personal communication). In relation to possible reduction of cell size occurring when the yeasts are in a viable but not culturable status (Millet and Lonvaud-Funel, 2000), the use of a filter with small pore size (0.45 µm) would be more effective as verified by Calderón *et al.* (2004). This practice should be considered a pre-bottling wine treatment or a remediation treatment in case of detection of increase of non-*Saccharomyces* yeast cells in barrel ageing wine. Excessive filtration or incorrectly carried out filtration though can cause a loss of wine aroma and texture (Flores *et al.*, 1991).

Barata *et al.* (2008) observed in wine complete loss of viability within 12 hours for 10 strains of *D. bruxellensis* exposed at 36 °C. Other studies in relation to temperature inactivation showed the inactivation of a population of *B. bruxellensis* 10⁶ cfu/mL at 37.5 °C for 6 minutes or at 41 °C for 0.6 minutes (Couto *et al.*, 2005). However, possible side effects on wine aroma and flavour of thermal treatment, create some concern.

Alternative methods such as the use of 3–6 g/L of chitosan (Gómez-Rivas *et al.*, 2004), fining agents, high hydrostatic pressure (Puig *et al.*, 2003; Mok *et al.*, 2006) and grape must and wine UV treatment (Henick-Kling *et al.*, 2005) have also been investigated. None of these methods so far has been found satisfactory.

The use of reverse osmotic pressure appears to offer the potential for wine remediation in cases where no viable microorganisms are present. In one study, total ethylphenols concentration was reduced by 77%, but not without reduction of some other aromatic compounds (Ugarte *et al.*, 2005).

Barrel sanitation

Barrel sanitation is an important practice to limit the spread of *B. bruxellensis* in the winery. It is unfortunately very difficult to achieve. The yeast and the microbial

population survive in the barrel wood and are protected from the action of SO₂ (Swaffieeld *et al.*, 1997; Laureano *et al.*, 2004). Laureano *et al.* (2004) studied the effect of several treatments for barrel sanitation including the use of hot water (70 °C and 90 °C), a 200 mg/L SO₂ solution at pH 3 and steam. The most effective was a combination of the following operations: cold water rinse, followed by a 70 °C hot water rinse and steaming under low pressure (0.5 kg/cm) for 10 minutes. However, even after the described treatment *D. bruxellensis* cells were still recovered. Ozonated water reduced 99% of *B. bruxellensis* population in the study carried out by Coggan (2003); however, Cantacuzene *et al.* (2003) found that the simple use of ozonated water was not enough to reduce the yeast population but a treatment with ozone gas coupled with hot water at 82 °C for 2 minutes was necessary. More recently, laboratory tests with high-power ultrasound showed some possibility (Yap *et al.*, 2007). In laboratory tests, more than 97% of a population of about 5×10^6 cfu/mL was killed (ultrasound power at 50 watts for 90–120 s).

12.8 Conclusions

Nowadays, *Dekkera/Brettanomyces bruxellensis* is the yeast responsible for the more dreaded red wine microbial spoilage, the phenolic off-flavour. Discoveries concerning its original physiological and genetical nature have recently sparked the interest of the wider scientific community, not just the one specialising in oenology. In fact, the scientific literature published on *B. bruxellensis* over the past three years alone (2007–2009) represents approximately 30% of the records published on this topic since 1960. The more recent research findings discussed in this chapter have greatly improved understanding of this original biological system, its ecology, the major phenotypic and genotypic feature characterising different strains, the phenolic off-flavour pathway, the methods to detect, prevent and correct *B. bruxellensis* spoilage faults. Together with other wine defects associated with other yeast and bacteria, the study of the diffusion of *B. bruxellensis* in the wineries has also helped to increase the attention paid to hygiene practices and to the study of alternative technologies to prevent microbial instability in wine, and to better control the associated microbial ecosystem. However, the reason for metabolising hydroxycinnamyc acid and a more complete knowledge on the yeast genome and metabolome have still to be revealed and are needed to allow a better control of *B. bruxellensis* diffusion in the winery.

12.9 References

- Agnolucci M, Scarano S, Rea E, Toffanin A and Nuti M (2007), Detection of *Dekkera/Brettanomyces bruxellensis* in pressed sangiovese grapes by real time PCR, *Italian Journal of Food Science*, **19**, 153–164.
- Agnolucci M, Vigentini I, Capurso G, Merico A, Tirelli A, Compagno C, Foschino R and Nuti M (2009), Genetic diversity and physiological traits of *Brettanomyces bruxellensis*

- strains isolated from Tuscan Sangiovese wines, *International Journal of Food Microbiology*, **130**, 238–244.
- Aguilar Uscanga MG, Delia ML and Strehaiano P (2000), Nutritional requirements of *Brettanomyces bruxellensis*: Growth and physiology in batch and chemostat cultures, *Canadian Journal of Microbiology*, **46**, 1046–1050.
- Aguilar-Uscanga MG, Delia ML and Strehaiano P (2003), *Brettanomyces bruxellensis*: Effect of oxygen on growth and acetic acid production, *Applied Microbiology Biotechnology*, **61**(2), 157–162.
- Aguilar Uscanga MG, Abarca BIE, Rodríguez JG and Garcia RC (2007), Carbon sources and their effect on growth, acetic acid and ethanol production by *Brettanomyces bruxellensis* in batch culture, *Journal of Food Process Engineering*, **30**(1), 13–23.
- Alguacil M, Fidalgo M, Jimenez J, Lozano J, Neva M and Perdigonos F (1998), Detección de *Dekkera/Brettanomyces* en instalaciones de vendimia mediante PCR, *Alimentos Equipos y Tecnología*, **10**, 81–85.
- Alvarez-Rodriguez ML, López-Ocana LL, López-Coronado JM, Rodriguez E, Martinez MJ, Larriba G and Coque JR (2002), Cork taint of wines: role of filamentous fungi isolated from cork in formation of 2,4,6-trichloroanisole by O-methylation of 2,4,6-trichlorophenol, *Applied and Environmental Microbiology*, **68**, 5860–5869.
- Anli RE and Bayram M (2009), Biogenic amines in wines, *Food Reviews International*, **25**, 86–102.
- Andrade P, Oliveira B, Seabra R, Ferreira M, Ferreres F and Garcia-Vihuela C (2001), Analysis of phenolic compounds in Spanish Albarino and Portuguese Alvarinho and Loureiro wines by capillary zone electrophoresis and HPLC, *Electrophoresis*, **22**, 1568–1572.
- Arena ME and Manca de Nadra MCM (2001), Biogenic amine production by *Lactobacillus*, *Journal of Applied Microbiology*, **90**, 158–162.
- Arena ME, Saguir FM and Manca de Nadra MC (1999), Arginine, citrulline, and ornithine metabolism by lactic acid bacteria from wine, *International Journal of Food Microbiology*, **52**, 155–161.
- Arias CR, Burns JK, Friedrich LM, Goodrich RM and Parish ME (2002), Yeast species associated with orange juice: evaluation of different identification methods, *Applied and Environmental Microbiology*, **68**, 1955–1961.
- Arjun BMV (1979), Yeast spoilage of fortified wines with emphasis on sherry, *Journal of Applied Bacteriology*, **47**, R10–R11.
- Arvik T and Henick-Kling T (2002), Overview: *Brettanomyces bruxellensis* occurrence, growth, and effect on wine flavor, Practical Winery and Vineyard, May/June, 117–123.
- Barata A, Nobre A, Correia P, Malfeito-Ferreira M and Loureiro V (2006), Growth and 4-ethylphenol production by the yeast *Pichia guilliermondii* in grape juices, *American Journal of Enology and Viticulture*, **57**, 133–138.
- Barata A, Pagliara D, Piccininno T, Tarantino F, Ciardulli W, Malfeito-Ferreira M and Loureiro V (2008), The effect of sugar concentration and temperature on growth and volatile phenol production by *Dekkera bruxellensis* in wine, *FEMS Yeast Research*, **8**(7), 1097–1102.
- Barer MR (1997), Viable but non-culturable and dormant bacteria: time to resolve an oxymoron and a misnomer?, *Journal of Medical Microbiology*, **46**, 629–631.
- Barnett JA and Entian DE (2005), A history of research on yeasts 9: regulation of sugar metabolism, *Yeast*, **22**, 835–894.
- Barret A, Bidan P and Andre L (1955), Several accidents in wine production caused by film yeasts, *Compt Rendu Academie Agriculture France*, **41**, 426–430.
- Barthelmebs L, Divies C and Cavi, JF (2000), Knockout of the p-coumarate decarboxylase gene from *Lactobacillus plantarum* reveals the existence of two other inducible enzymatic activities involved in phenolic acid metabolism, *Applied and Environmental Microbiology*, **66**, 3368–3375.
- Bartowsky EJ and Henshke PA (2008), Acetic acid bacteria spoilage of bottled red wine – a review, *International Journal of Food Microbiology*, **125**, 60–70.

- Bauer R, Nel HA and Dicks LMT (2003), Pediocin PD-1 as a method to control growth of *Oenococcus oeni* in wine, *American Journal of Enology and Viticulture*, **54**, 86–91.
- Beech FW and Thomas S (1985), Action antimicrobienne de l'anhydride sulfureux, *Bulletin de L'OIV*, **58**(652–653), 564–581.
- Benito S, Palomero F, Morata A, Calderon F and Suarez-Lepe JA (2009), Factors affecting the hydroxycinnamate decarboxylase/vinylphenol reductase activity of *Dekkera/Brettanomyces*, application for *Dekkera/Brettanomyces* control in red wine making, *Journal of Food Science*, **74**, M15–M22.
- Bertrand JR, Vasseur JJ, Gouyette A, Rayner B, Imbach JL, Paoletti C and Malvy C (1989), Mechanism of cleavage of apurinic sites by 9-aminollipicine, *Journal of Biological Chemistry*, **264**, 14172–14178.
- Boekhout T, Kurtzman CP, Odonnell K and Smith MT (1994), Phylogeny of the yeast genera *hanseniaspora* (*Anamorph kloeckera*), *Dekkera* (*Anamorph Brettanomyces*) and *eeniella* as inferred from partial 26S ribosomal DNA nucleotide sequences, *International Journal of Systematic Bacteriology*, **44**, 81–86.
- Børsting C, Hummel R, Schultz ER, Rose TM, Pedersen MB, Knudsen J and Kristiansen K (1997), *Saccharomyces carlsbergensis* contains two functional genes encoding the acyl-CoA binding protein, one similar to the *ACB1* gene from *S-cerevisiae* and one identical to the *ACB1* gene from *S-monacensis*, *Yeast*, **13**, 1409–1421.
- Boulton RB, Singleton VL, Bisson LF, Kunkee RE (eds) (1996), Principles and practices of winemaking, Aspen Publishers Inc, Gaithersburg, MD.
- Brandam C, Castro-Martinez C, Delia ML, Ramon-Portugal F and Strehaiano P (2008), Effect of temperature on *Brettanomyces bruxellensis* metabolic and kinetic aspects, *Canadian Journal of Microbiology*, **54**, 11–18.
- Brock PE, Conterno L, Lavin E, Acree T and Henick-Kling T (2006), Identification of aroma compounds from different strains of *Brettanomyces bruxellensis* in model wine, *American Journal of Enology and Viticulture*, **57**, 529–529.
- Budic-Leto I and Lovric T (2002), Identification of phenolic acids and changes in their content during fermentation and ageing of white wines Posip and Rukatac, *Food Technology and Biotechnology*, **40**, 221–225.
- Buteau C, Duitschaever CI and Ashton GC (1984), High performance liquid chromatographic detection and quantification of amines in must wine, *Journal of Chromatography*, **284**, 201–210.
- Caboulet D, Poupault P, Morard JY, Sadirac L and Lemarchand E (2004), Influence de l'hygiène au cours des phases fermentaires sur l'élaboration et la qualité du vin, *Revue Française d'Oenologie*, **44**, 12–18.
- Cai JP, Roberts IN and Collins MD (1996), Phylogenetic relationships among members of the ascomycetous yeast genera *Brettanomyces*, *Debaryomyces*, *Dekkera*, and *Kluyveromyces* deduced by small-subunit rRNA gene sequences, *International Journal of Systematic Bacteriology*, **46**, 542–549.
- Calderón F, Morata A, Uthurry C and Suárez JA (2004), Aplicaciones de la ultrafiltración en la industria enológica, Últimos avances tecnológicos, *Tecnología del Vino*, **16**, 49–54.
- Cantacuzene NO, Dormedy ES, Smilanick JL, Fugelsang KC, Wample RL, Bacon DJ and Dormedy DF (2003), Treating *Brettanomyces* in oak cubes with gaseous and aqueous ozone, ASEV 54th Annual Meeting, Reno (NV), June 18–20, *American Journal of Enology and Viticulture*, **54**(3), 224A.
- Carrascosa JM, Viguera MD, De Castro IN and Scheffers WA (1981), Metabolism of acetaldehyde and clusters effects in the yeast *brettanomyces anstiens*, *Journal of Microbiology*, **47**, 209–216.
- Carre E, Lafon-Lafourcade S and Bertrand A (1983), Biological deacidification of dry white wines by the fermentations of malic acid by yeasts, *Connaissance de la Vigne et du Vin*, **17**, 43–54.
- Caruso M, Fiore C, Contursi M, Salzano G, Paparella A and Romano P (2002), Formation

- of biogenic amines as criteria for the selection of wine yeasts, *World Journal of Microbiology and Biotechnology*, **18**, 159–163.
- Casaregola S, Nguyen HV, Lapathitis G, Kotyk A and Gaillardin S (2001), Analysis of the constitution of the beer yeast genome by PCR, sequencing and subtelomeric sequence hybridization, *International Journal of Systematic Bacteriology*, **51**, 1607–1618.
- Cavin JF, Andioc V, Etievant PX and Devies C (1993), Ability of wine lactic acid bacteria to metabolize phenol carboxylic, *American Journal of Enology and Viticulture*, **44**, 76–80.
- Cavin JF, Barthelmebs L, Guzzo J, VanBeeumen J, Samyn B, Travers JF, Divies C (1997), Purification and characterization of an inducible p-coumaric acid decarboxylase from *Lactobacillus plantarum*, *FEMS Microbiology Letters*, **147**, 291–295.
- Chatonnet P, Boidron JN and Pons M (1990), Maturation of red wines in aol barrels evolution of some volatile compounds and their aromatic impact, *Science des Aliments*, **10**, 565–587.
- Chatonnet P, Dubourdieu D, Boidron JN and Pons M (1992), The origin of Ethylphenols in wines, *Journal of the Science of Food and Agriculture*, **60**, 165–178.
- Chatonnet P, Boidron JN and Dubourdieu D (1993), Influence of the ageing and sulfuring conditions of red wines aged in barrels on their content in acetic acid and ethylphenols, *Journal International des Sciences de la Vigne et du Vin*, **27**, 277–298, 314–315.
- Chatonnet P, Dubourdieu D and Boidron JN (1995), The influence of *Brettanomyces/Dekkera* sp yeasts and lactic acid bacteria on the ethylphenol content of red wines, *American Journal of Enology and Viticulture*, **46**, 463–468.
- Ciani M and Ferraro L (1997), Role of oxygen on acetic acid production by *Brettanomyces/Dekkera* in winemaking, *Journal of the Science Food and Agriculture*, **75**, 489–495.
- Ciani M, Maccarelli F and Fatichenti F (2003), Growth and fermentation behaviour of *Brettanomyces/Dekkera* yeasts under different conditions of aerobiosis, *World Journal of Microbiology and Biotechnology*, **19**, 419–422.
- Cilliers JD and van Wyk CJ (1985), Histamine and tyramine content of South African wine, *South African Journal for Enology and Viticulture*, **6**, 35–40.
- Claisse O, and Lonvaud-Funel A (2001), Détection de bacteries lactiques produisant du 3-hydroxypropionaldéhyde (précurseur d'acroléine) à partir du glycérol par tests moléculaires, *Lait*, **81**, 173–181.
- Clark-Walker GD, Hoebe P, Plazinska A, Smith DK and Wimmer EH (1987), Application of mitochondrial DNA analysis to yeast systematic, *Studies in Mycology*, **30**, 259–266.
- Clausen M, Lamb CJ, Megnet R and Doerner PW (1994), PAD1 encodes phenylacrylic acid decarboxylase which confers resistance to cinnamic acid in *Saccharomyces cerevisiae*, *Gene*, **142**, 107–112.
- Cocolin L, Rantsiou K, Iacumin L, Zironi R and Comi G (2004), Molecular detection and identification of *Brettanomyces/Dekkera bruxellensis* and *Brettanomyces/Dekkera anomalous* in spoiled wines, *Applied and Environmental Microbiology*, **70**, 1347–1355.
- Coggan M (2003), *Ozone in wineries*. Available at: http://www.ciprocess.co.uk/pdfs/Ozone_In_Wineries.pdf (accessed March 2010).
- Coghe S, Benoot K, Delvaux F, Vanderhaegen B and Delvaux F (2004), Ferulic acid release and 4-vinylguaiaicol formation during brewing and fermentation: indications for feruloyl esterase activity in *Saccharomyces cerevisiae*, *Journal of Agricultural and Food Chemistry*, **52**, 602–608.
- Connell L, Stender H and Edwards CG (2002), Rapid detection and identification of *Brettanomyces* from winery air samples based on peptide nucleic acid analysis, *American Journal of Enology and Viticulture*, **53**, 322–324.
- Conterno L, Capraro BR and Henik-Kling T (2003), Comparison of media to isolate *Brettanomyces bruxellensis* from wine, ASEV 54th Annual Meeting, Reno (NV), June 18–20, *American Journal of Enology and Viticulture*, **54**(3), 224A.
- Conterno L, Hesford F and Henik-Kling T (2005), Volatile phenols off-odor production by different strain of *Brettanomyces bruxellensis*, ASEV 56th Annual Meeting, Seattle (WA), June 22–24, *Abstract in American Journal of Enology and Viticulture*, **56**(3), 291.

- Conterno L, Joseph CML, Arvik TJ, Henick-Kling T and Bisson LF (2006), Genetic and physiological characterization of *Brettanomyces bruxellensis* strains isolated from wines, *American Journal of Enology and Viticulture*, **57**, 139–147.
- Conterno L, Lasik G, Tomasino E, Schneider K, Hesford F and Henick-Kling T (2007), Influence of sugar and nitrogen sources on growth and phenolic off-flavor production by *Brettanomyces bruxellensis* isolated from wine, *American Journal of Enology and Viticulture*, **58**(3), 411A.
- Contreras A, Salinas F, Ganga A and Martínez C (2008), Polymerase chain reaction confirmatory method for microbiological detection of *Brettanomyces bruxellensis* in wines, *Journal of Rapid Methods and Automation in Microbiology*, **16**, 308–319.
- Costa A, Barata A, Malfeito-Ferreira M and Loureiro V (2008), Evaluation of the inhibitory effect of dimethyl dicarbonate (DMDC) against wine microorganisms, *Food Microbiology*, **25**, 422–427.
- Costello PJ and Henschke PA (2002), Mousy off-flavor of wine: precursors and biosynthesis of the causative N-heterocycles 2-ethyltetrahydropyridine, 2-acetyltetrahydropyridine, and 2-acetyl-1-pyrroline by *Lactobacillus hilgardii* DSM 20176, *Journal of Agricultural and Food Chemistry*, **50**, 7079–7087.
- Costello PJ, Lee TH and Henschke PA (2001), Ability of lactic acid bacteria to produce N-heterocycles causing mousy off-flavour in wine, *Australian Journal of Grape and Wine Research*, **7**, 160–167.
- Coton E, Rollan G, Bertrand A and Lonvaud funnel A (1998), Histamine-producing lactic acid bacteria in wines: early detection, frequency, and distribution, *American Journal of Enology and Viticulture*, **49**, 199–204.
- Couto MMB, Reizinho RG and Duarte FL (2005), Partial 26S rDNA restriction analysis as a tool to characterise non-*Saccharomyces* yeasts present during red wine fermentations, *International Journal of Food Microbiology*, **102**, 49–56.
- Couto JA, Campos FM, Figueiredo A R, and Hogg T A (2006), Ability of lactic acid bacteria to produce volatile phenols, *American Journal of Enology and Viticulture*, **57**, 166–171.
- Crowell EA and Guymon IF (1975), Wine constituents arising from sorbic acid addition and identification of 2-ethoxyhexa-3,5-diene as a source of geranium-like off-odor, *American Journal of Enology and Viticulture*, **26**, S:97–102.
- Curtin CD, Bellon JR, Henschke PA, Godden P and Lopes MADB (2007), Genetic diversity of *Dekkera bruxellensis* yeasts isolated from Australian wineries, *FEMS Yeast Research*, **7**, 471–481.
- Custers MTJ (1940), *Onderzoekingen over het gistgeslacht Brettanomyces*. Thesis, De Technische Hoogeschool te Delft.
- Davis CR, Wibowo D, Fleet GH and Lee TH (1988), Properties of wine lactic acid bacteria, their potential enological significance, *American Journal of Enology and Viticulture*, **39**, 137–142.
- de Barros Lopes M, Rainieri S, Henschke PA and Landgridge A (1999), AFLP fingerprinting for analysis of yeast genetic variation, *International Journal of Systematic Bacteriology*, **49**, 915–924.
- de Barros Lopes, Bellon M, Shirley JR, Neil J and Ganter PF (2002), Evidence for multiple interspecific hybridization in *Saccharomyces sensu stricto* species, *FEMS Yeast Research*, **1**, 323–331.
- Delaherche A, Claisse O and Lonvaud-Funel A (2004), Detection and quantification of *Brettanomyces bruxellensis* and ‘ropy’ *Pediococcus damnosus* strains in wine by real-time polymerase chain reaction, *Journal of Applied Microbiology*, **97**, 910–915.
- Delfini C, Gaia P, Schellino R, Strano M, Pagliara A and Ambro S (2002), Fermentability of grape must after inhibition with dimethyl dicarbonate (DMDC), *Journal of Agricultural and Food Chemistry*, **50**, 5605–5611.
- Dias S, Sancho T, Stender H, Querol A, Malfeito-Ferreira M and Loureiro V (2003), Identification of yeasts isolated from wine-related environments and capable of producing 4-ethylphenol, *Food Microbiology*, **20**, 567–574.

- Dols-Lafargue M, Lee HY, Le Marrec C, Heyraud A, Chambat G and Lonvaud-Funel A (2008), Characterization of *gtf*, a glucosyltransferase gene in the genomes of *Pediococcus parvulus* and *Oenococcus oeni*, two bacterial species commonly found in wine, *Applied and Environmental Microbiology*, **74**(13), 4079–4090.
- du Plessis HW, Dicks LMT, Pretorius IS, Lambrechts MG and du Toit M (2004), Identification of lactic acid bacteria isolated from South African brandy base wines, *International Journal of Food Microbiology*, **91**, 19–29.
- du Toit M and Pretorius IS (2000), Microbial spoilage and preservation of wine: using weapon from nature's own arsenal – a review, *South African Journal for Enology and Viticulture*, **21**(special issue), 74–96.
- du Toit WJ and Lambrechts MG (2002), The enumeration and identification of acetic acid bacteria from South African red wine fermentations, *International Journal of Food Microbiology*, **74**, 57–64.
- du Toit WJ, Pretorius IS and Lonvaud-Funel A (2005), The effect of sulphur dioxide and oxygen on the viability and culturability of a strain of *Acetobacter pasteurianus* and a strain of *Brettanomyces bruxellensis* isolated from wine, *Journal of Applied Microbiology*, **98**, 862–871.
- Drysdale GS and Fleet GH (1988), Acetic acid bacteria in winemaking: a review, *American Journal of Enology and Viticulture*, **39**, 143–154.
- Drysdale GS and Fleet GH (1989), The effect of acetic acid bacteria upon the growth and metabolism of yeasts during the fermentation of grape juice, *Journal of Applied Bacteriology*, **67**, 471–481.
- Edinger WD and Splittstoesser DF (1986), Production by lactic acid bacteria of sorbic alcohol, the precursor of the geranium odor compound, *American Journal of Enology and Viticulture*, **37**, S. 34–38.
- Edlin D, Narabad A, Dickinson J and Lloyd D (1995), The biotransformation of simple phenolic compounds by *brettanomyces anomalus*, *FEMS Microbiology Letters*, **125**, 311–315.
- Edlin D, Narbad A, Gasson MJ, Dickinson JR and Lloyd D (1998), Purification and characterization of hydroxycinnamate decarboxylase from *Brettanomyces anomalus*, *Enzyme and Microbial Technology*, **22**, 232–239.
- Edwards CG, Hacy KM and Collins MD (1998a), Identification and characterization of two lactic acid bacteria associated with sluggish/stuck fermentation, *American Journal of Enology and Viticulture*, **43**, 445–448.
- Edwards CG, Haag KM, Collins RA, Hutson RA and Huang YC (1998b), *Lactobacillus kunkei* sp nova spoilage organism associated with grape juice fermentation, *Journal of Applied Microbiology*, **84**, 698–702.
- Egli CM and Henick-Kling T (2001), Identification of *Brettanomyces/Dekkera* species based on polymorphism in the rRNA internal transcribed spacer region, *American Journal of Enology and Viticulture*, **52**, 241–247.
- Egli CM, Edinger W, Mittrakul C and Henick-Kling T (1998), Dynamics of indigenous and inoculated yeast populations and their effect on the sensory character of Riesling and Chardonnay wines, *Journal of Applied Microbiology*, **85**, 779–789.
- Enrique M, Marcos JF, Yuste M, Martinez M, Valles S and Manzanares P (2008), Inhibition of the wine spoilage yeast *Dekkera bruxellensis* by bovine lactoferrin-derived peptides, *International Journal of Food Microbiology*, **127**, 229–234.
- Esteve-Zarzoso B, Belloch C, Uruburu F and Querol A (1999), Identification of yeasts by RFLP analysis of the 5.8S rRNA gene and the two ribosomal internal transcribed spacers, *International Journal of Systematic Bacteriology*, **49**, 329–337.
- Fariña L, Boido E, Carrau F and Dellacassa E (2007), Determination of volatile phenols in red wines by dispersive liquid–liquid microextraction and gas chromatography-mass spectrometry detection, *Journal of Chromatography A*, **1157**, 46–50.
- Fernandez-Pachon M, Villano D, Troncoso A and Garcia-Parrilla M (2006), Determination of the phenolic composition of sherry and table white wines by liquid

- chromatography and their relation with antioxidant activity, *Analytica Chimica Acta*, **563**, 101–108.
- Fleet GH (2000), *Schizosaccharomyces*. In: Robinson RK, Batt CA, Patel PD (eds.), *Encyclopedia of Food Microbiology*, vol. 3. Academic Press, London, UK, 1984–1989.
- Florenzano G (1951), Diffusione e significato enologico dei lieviti *Brettanomyces*. *Atti dell'Accademia Italiana della Vite e del Vino*, **3**, 236–249.
- Flores JH, Heatherbell DA, Henderson LA and McDaniel MR (1991), Ultrafiltration of wine: effect of ultrafiltration on the aroma and flavor characteristics of White Riesling and Gewürztraminer wines, *American Journal of Enology and Viticulture*, **42**, 91–96.
- Frankel E, Waterhouse A and Teissedre P (1995), Principal phenolic phytochemicals in selected California wines and their antioxidant activity in inhibiting oxidation of human low-density lipoproteins, *Journal of Agricultural and Food Chemistry*, **43**, 890–894.
- Freer SN (2002), Acetic acid production by *Dekkera/Brettanomyces* yeasts, *World Journal of Microbiology and Biotechnology*, **18**, 271–275.
- Freer SN, Dien B and Matsuda S (2003), Production of acetic acid by *Dekkera/Brettanomyces* yeasts under conditions of constant pH, *World Journal of Microbiology and Biotechnology*, **19**, 101–105.
- Froudière I and Larue F (1988), Condition de survie de *Brettanomyces* (*Dekkera*) dans le moût de raisin et le vin, *Connaissance de la Vigne et du Vin*, **2**, 296–303.
- Fugelsang KC (1997), Winery Sanitation. In Fugelsang KC (ed.), *Wine Microbiology*, The Chapman and Hall Enology Library, New York, 159–168.
- Fugelsang KC and Edwards CG (2007), Wine spoilage. In Fugelsang KC and Edwards CG (eds), *Wine Microbiology: Practical Application and Procedures*, Springer, New York, 162–181.
- Fugelsang KC and Edwards CG (2007), *Wine Microbiology: Practical Application and Procedures*, Springer, New York.
- Gaunt, DM, Degn H and Lloyd D (1988), The influence of oxygen and organic hydrogen acceptors on glycolytic carbon dioxide production in *Brettanomyces anomalus*, *Yeast*, **4**, 249–255.
- Gerbaux V, Jeudy S and Monamy C (2000) Etude des phenols volatils dans les vins de Pinot noir en Bourgogne, *Bulletin de l'OIV*, **73**(835–836), 581–599.
- Gerbaux V, Vincent B and Bertrand A (2002), Influence of maceration temperature and enzymes on the content of volatile phenols in Pinot noir wines, *American Journal of Enology and Viticulture*, **53**, 131–137.
- Gerland C (1995), *Optimisation des procédures de nettoyage*, *Revue des Œnologues*, **21**(77S) 41–43.
- Gindreau E, Wailling E, and Lonvaud-Funel A (2001), Direct polymerase chain reaction detection of rosy *Pediococcus damnosus* strains in wine, *Journal of Applied Microbiology*, **90**, 535–542.
- Godoy L, Martinez C, Carrasco N and Ganga MA (2008), Purification and characterization of a p-coumarate decarboxylase and a vinylphenol reductase from *Brettanomyces bruxellensis*, *International Journal of Food Microbiology*, **127**, 6–11.
- Godoy L, Garrido D, Martinez C, Saavedra J, Combina M and Ganga MA (2009), Study of the coumarate decarboxylase and vinylphenol reductase activities of *Dekkera bruxellensis* (*anamorph Brettanomyces bruxellensis*) isolates, *Letters in Applied Microbiology*, **48**, 452–457.
- Goldberg D, Tsang E, Karumanchiri A and Soleas G (1998), Quercetin and p-coumaric acid concentrations in commercial wines, *American Journal of Enology and Viticulture*, **49**, 142–151.
- Goldberg D, Karumanchiri A, Soleas G and Tsang E (1999), Concentrations of selected polyphenols in white commercial wines, *American Journal of Enology and Viticulture*, **50**, 185–193.
- Goldberg D, Dam J, Carey M and Soleas G (2000), Cultivar-specific patterns of polyphenolic

- constituents in wines from the Finger Lakes Region of New York State, *Journal of Wine Research*, **11**, 155–164.
- Gómez-Rivas L, Escudero-Abarca BI, Aguilar-Uscanga MG, Hayward-Jones PM, Mendoza P and Ramírez M (2004), Selective antimicrobial action of chitosan against spoilage yeasts in mixed culture fermentations, *Journal of Industrial Microbiology and Biotechnology*, **31**, 16–22.
- Goodey AR and Tubb RS (1982), Genetic and biochemical analysis of the ability of *Saccharomyces cerevisiae* to decarboxylate cinnamic acids, *Journal of General Microbiology*, **128**, 2615–2620.
- Gorinstein S, Caspi A, Zemser M and Trakhtenberg S (2000), Comparative contents of some phenolics in beer, red and white wines, *Nutritional Research*, **20**, 131–139.
- Granchi L, Romano P, Mangani S, Guerrini S and Vincenzini M (2005), Production of biogenic amines by wine microorganisms, *Bulletin de l'OIV*, **78**, 595–609.
- Grbin P R (1998), *Physiology and metabolism of Dekkera/Brettanomyces yeast in relation to mousy taint production*. Ph.D. Thesis, The University of Adelaide, Adelaide, SA.
- Grbin PR, and Henschke PA (2000), Mousy off-flavor production in grape juice and wine, *Australian Journal of Grape and Wine Research*, **6**, 255–262.
- Grbin PR, Herderich M, Markides A, Lee TH and Henschke PA (2007), The role of lysine amino nitrogen in the biosynthesis of mousy off-flavor compounds by *Dekkera anomala*, *Journal of Agricultural and Food Chemistry*, **55**, 10872–10879.
- Guerrini S, Mangani S, Granchi L and Vincenzini M (2002), Biogenic amine production by *Oenococcus oeni*, *Current Microbiology*, **44**, 374–378.
- Gutierrez I, Lorenzo E and Espinosa A (2005), Phenolic composition and magnitude of copigmentation in young and shortly aged red wines made from the cultivars, Cabernet Sauvignon, Cencibel, and Syrah, *Food Chemistry*, **92**, 269–283.
- Harris V, Ford CM, Jiranek V and Grbin PR (2008), *Dekkera* and *Brettanomyces* growth and utilisation of hydroxycinnamic acids in synthetic media, *Applied Microbiology and Biotechnology*, **78**, 997–1006.
- Harris V, Ford CM, Jiranek V and Grbin PR (2009), *Survey of enzyme activity responsible for phenolic off-flavour production by Dekkera and Brettanomyces yeast*, *Applied Microbiology and Biotechnology*, **81**, 1117–1127.
- Heard GM and Fleet GH (1986), Evaluation of selective media for enumeration of yeasts during wine fermentation, *Journal of Applied Bacteriology*, **60**, 477–481.
- Hellborg L, and Piškur J (2009) Complex nature of the genome in a wine-spoilage yeast, *Dekkera bruxellensis*, *Eukariotic Cells*, **8**, 1739–1749.
- Henick-Kling T (1986), *Growth and metabolism of Leuconostoc oenos and Lactobacillus plantarum in wine*. Ph.D. Thesis, The University of Adelaide, Adelaide, SA.
- Henick-Kling T. (1993), Malolactic fermentation. In Fleet GH (ed.), *Wine Microbiology and Biotechnology*, Harwood Academic, Chur, Switzerland, 289–326.
- Henick-Kling T, Cox DJ and Olsen EB (1991), Production de l'énergie durant la fermentation malolactique, *Revue Française d'Oenologie*, **132**, 63–66.
- Henick-Kling T, Edinger W, Daniel P and Monk P (1998), Selective effects of sulfur dioxide and yeast starter culture addition on indigenous yeast populations and sensory characteristics of wine, *Journal of Applied Microbiology*, **84**, 865–876.
- Henick-Kling T, Egli C, Licker J, Mitrakul C and Acree TE (2000), *Brettanomyces* in wine. In *Proceedings of the 5th International Symposium on Cool Climate Viticulture and Oenology*, Melbourne, SA, 16–20 January.
- Henick-Kling T, Egli C, Licker J and Acree TE (2001), *Brettanomyces* in wine. In Henick-Kling T (ed.), *Proceedings of the 30th Annual New York Wine Industry Workshop*, NY State Agricultural Experiment Station, Geneva, NY, 67–79.
- Henick-Kling T, Conterno L, Arvik T, Porret N, Gafner J, Gadoury DM, Wilcox WF and Seem RC (2005), *Brettanomyces* yeasts: occurrence and effect on wine quality, *Proceeding 14th International Enology Symposium*, Porto Carras, Greece, May 8–12.

- Heresztyn T (1986), Metabolism of volatile phenolic compounds from hydroxycinnamic acids by *Brettanomyces* yeast, *Archives of Microbiology*, **146**, 96–98.
- Hermann K (1989), Occurrence and content of hydroxycinnamic and hydroxybenzoic acid compounds in foods, *Critical Reviews in Food Science and Nutrition*, **28**, 315–345.
- Hesford F, Schneider K, Porret N and Gafner J (2004), Identification and analysis of 4-ethyl catechol in wines tainted by *Brettanomyces* off-flavor, ASEV 55th Annual Meeting, San Diego, CA, June 29–30, *American Journal of Enology and Viticulture*, **55**, 304A.
- Hock S (1990), Coping with *Brettanomyces*, *Practical Winery and Vineyard*, January/February, 26–31.
- Huang YC, Edwards CG, Peterson JC, Haag KM (1996) Relationship between sluggish fermentations and the antagonism of yeast by lactic acid bacteria, *American Journal of Enology and Viticulture*, **47**, 1–10.
- Hühn T, Sponholz WR and Pulver D (1999), The influence of microorganisms in winemaking, *Scientific and Technical Information Edition CDR 3*, November, 41–84, available at: <http://www.beverages.ch/service/Microorganisms.PDF> (accessed April 2010).
- Ibeas JI, Lozano I, Perdigones F and Jimenez J (1996), Detection of *Dekkera-Brettanomyces* strains in sherry by a nested PCR method, *Applied and Environmental Microbiology*, **62**, 998–1003.
- Ibern-Gomez M, Andres-Lacueva C, Lamuela-Raventos RM, Lao Luque C, Buxaderas S and de la Torre-boronat MC (2001), Differences in phenolic profile between oak wood and stainless steel fermentation in white wines, *American Journal of Enology and Viticulture*, **52**, 159–164.
- Jager J, Diekmann J, Lorenz D and Jakob L (1996), Cork-borne bacteria and yeasts as potential producers of off-flavours in wine, *Australian Journal of Grape and Wine Research*, **2**, 35–41.
- James SA, Bond CJ, Stratford M and Roberts N (2005), Molecular evidence for the existence of natural hybrids in the genus *Zygosaccharomyces*, *FEMS Yeast Research*, **5**, 747–755.
- Joseph CML, Kumar G, Su E and Bisson LF (2007), Adhesion and biofilm production by wine isolates of *Brettanomyces bruxellensis*, *American Journal of Enology and Viticulture*, **58**, 373–378.
- Joyeux A, Lafon-Lafourcade S and Ribéreau-Gayon P (1984a), Evolution of acetic acid bacteria during fermentation and storage of wine, *Applied and Environmental Microbiology*, **48**, 153–156.
- Joyeux A, Lafon-Lafourcade S and Ribéreau-Gayon P (1984b), Metabolism of acetic acid bacteria in grape must consequences on alcoholic and malolactic fermentation, *Sciences des Aliments*, **4**, 247–255.
- Kufferath H and van Laer MH (1921), Études sur les levures de limbic. Leur action chimique sur les milieu de culture, *Bulletin des Sociétés Chimiques Belges*, **30**, 270–276.
- Kuniyuki AH, Rous C and Sanderson JL (1984), Enzyme Linked immunosorbent assay (ELISA) detection of *Brettanomyces* contaminants in wine production, *American Journal of Enology and Viticulture*, **35**, 143–145.
- Kunkee RE and Bisson LF (1993), Wine-making yeasts. In Rose AH and Harrison JS (eds), *The Yeasts: Yeast Technology*, Academic Press, London, UK, **5**, 69–127.
- Kurtzman CP and Fell JW (1998), *The Yeasts, a Taxonomic Study*, 4th edn, Elsevier Science B.V., Amsterdam, the Netherlands.
- Kurtzman CP and Robnett CJ (2003), Phylogenetic relationships among yeasts of the ‘*Saccharomyces* complex’ determined from multigene sequence analyses, *FEMS Yeast Research*, **3**, 417–432.
- Larcher R, Nicolini G, Bertoldi D and Nardin T (2008), Determination of 4-ethylcatechol in wine by high-performance liquid chromatography–coulometric electrochemical array detection, *Analytica Chimica Acta*, **609**, 235–240.
- Lagunas R, Dominguez C, Busturia A and Saez MJ (1982), Mechanisms of appearance of the Pasteur effect in *Saccharomyces cerevisiae* inactivation of sugar transport systems, *Journal of Bacteriology*, **152**, 19–25.

- Larue F, Rozes N, Froudiere I, Couty C and Perreira GP (1991), Influence of development of *Dekkera/Brettanomyces* in wine musts, *Connaissance de la Vigne et du Vin*, **25**, 149–165.
- Laureano P, D'Antuono I, Barata A, Malfeito-Ferreira M and Loureiro V (2004), Efeito de diferentes tratamentos de sanificação na população de *Dekkera bruxellensis* isolada da madeira de barricas, *Enologia – Revista da Associação Portuguesa de Enologia*, **43**(44), 3–8.
- Lay H (2003), Research on the development of 'mousy taint' in wine and model solutions, *Mitteilungen Klosterneuburg*, **53**, 243–250.
- Lee TH and Simpson RF (1993), Microbiology and chemistry of cork-taints in wine. In Fleet GH (ed.), *Wine Microbiology and Biotechnology*, Harwood Academic, Chur, Switzerland, 353–372.
- Leitao MC, Teixeira HC, Crespo MTB and San Romao MV (2000), Biogenic amines occurrence in wine amino acid decarboxylase and proteolytic activities expression by *Oenococcus oeni*, *Journal of Agricultural and Food Chemistry*, **48**, 2780–2784.
- Licker JL, Acree TE, Waterhouse AL and Ebeler SE (1999), What is 'Brett' (*Brettanomyces*) flavour? A preliminary investigation. In Waterhouse AL and Ebeler SA (eds), *Chemistry of Wine Flavour*, American Chemical Society, Washington DC, 96–115.
- Liu S-Q, Pritchard GG, Hardman MJ and Pilone GJ (1996), Arginine catabolism in wine lactic acid bacteria: Is it via the arginine deiminase pathway or the arginase-urease pathway? *Journal of Applied Bacteriology*, **81**, 486–492.
- Lonvaud-Funel A (1999), Lactic acid bacteria in the quality improvement and depreciation of wine, *Antonie van Leeuwenhoek*, **76**, 317–331.
- Lonvaud-Funel A (2001), Biogenic amines in wines: role of lactic acid bacteria, *FEMS Microbiology Letters*, **199**, 9–13.
- Lonvaud-Funel A and Joyeux A (1988), Une altération bactérienne des vins: la maladie des vins filants, *Science des Aliments*, **8**, 33–49.
- Lonvaud-Funel A and Joyeux A (1994), Histamine production by wine lactic acid bacteria: Isolation of a histamine producing strain of *Leuconostoc oenos*, *Journal of Applied Bacteriology*, **77**, 401–407.
- Lonvaud-Funel A, Guilloux Y and Joyeux A (1993), Isolation of a DNA probe for identification of glucan producing *Pediococcus damnosus* in wine, *Journal of Applied Bacteriology*, **74**, 41–47.
- Loureiro V and Malfeito-Ferreira M (2003), Spoilage yeast in the wine industry. A review, *International Journal of Food Microbiology*, **86**, 23–50.
- Manca de Nadra MC and Strasser de Saad AM (1995), Polysaccharide production by *Pediococcus pentosaceus* from wine, *International Journal of Food Microbiology*, **27**, 101–106.
- Mansfield A K, Zoecklein B W and Whiton R S (2002), Quantification of glycosidase activity in selected strains of *Brettanomyces bruxellensis* and *oenococcus oeni*, *American Journal of Enology and Viticulture*, **53**, 303–307.
- Martinez-Rodriguez AJ and Carrascosa AV (2009), HACCP to control microbial safety hazards during winemaking: Ochratoxin A, *Food Control*, **20**, 469–475.
- Marriot NG (1999), Beverage plant sanitation: winery sanitation. In Marriott NG (ed.), *Principles of food sanitation*, Aspen, Gaithersburg, MD, 312–317.
- Macris BJ and Markakis P (1974), Transport and toxicity of sulfur dioxide in *Saccharomyces cerevisiae* var *ellipsoideus*, *Journal of the Science of Food and Agriculture*, **25**, 21–29.
- Martorell P, Barata A, Malfeito-Ferreira M, Fernandez-Espinar MT, Loureiro V and Querol A (2006), Molecular typing of the yeast species *Dekkera bruxellensis* and *Pichia guilliermondii* recovered from wine related sources, *International Journal of Food Microbiology*, **106**, 79–84.
- Mateo JJ, Jimenez M, Huerta T and Pastor A (1991), Contribution of different yeasts isolated from Monastrell grapes to the aroma of wine, *International Journal of Food Microbiology*, **14**, 153–160.

- Mavilani MI (1969), Yeast microflora of wine-making district of Uzbekistan, *Antonie Van Leeuwenhoek*, **35**, suppl:D3.
- Medawar W, Strehaiano P and Delia ML (2003), Yeast growth: lag phase modelling in alcoholic media, *Food Microbiology*, **20**, 527–532.
- Miot-Sertier C and Lonvaud-Funel A (2007), Development of a molecular method for the typing of *Brettanomyces bruxellensis* (*Dekkera bruxellensis*) at the strain level, *Journal of Applied Microbiology*, **102**, 555–562.
- Millet V and Lonvaud-Funel A (2000), The viable but non-culturable state of wine micro-organisms during storage, *Letters in Applied Microbiology*, **30**, 136–141.
- Mira de Orduña R, Liu S-Q, Parchett ML and Pilone GJ (2001), Ethyl carbamate precursor citrulline formation from arginine degradation by malolactic wine lactic acid bacteria, *FEMS Microbiology Letters*, **183**, 31–35.
- Mitrakul CM, Henick-Kling T and Egli CM (1999), Discrimination of *Brettanomyces/Dekkera* yeast isolates from wine by using various DNA fingerprinting methods, *Food Microbiology*, **16**, 3–14.
- Mok C, Song KT, Park YS, Lim S, Ruan R and Chen P (2006), High hydrostatic pressure pasteurization of red wine, *Journal of Food Science*, **71**, 265–269.
- Molina FI, Shen P and Jong SC (1993), Validation of the species concept in the genus *Dekkera* by restriction analysis of gene coding for ribosomal RNA, *International Journal of Systematic Bacteriology*, **43**, 32–35.
- Monterisino S, Prieto A, Munoz R and De Las Rivas B (2008), Evaluation of exopolysaccharide production by *leuconostoc mesenteroides* strains isolated from wine, *Journal of Food Science*, **73**(4), M196–M199.
- Moreno-Arribas V and Lonvaud-Funel A (1999), Tyrosine decarboxylase activity of *Lactobacillus brevis* IOEB 9809 isolated from wine and *L. brevis* ATCC 367, *FEMS Microbiology Letters*, **180**, 55–60.
- Moreno-Arribas V, Torlois S, Joyeux A, Bertrand A and Lonvaud-Funel A (2000), Isolation, properties and behaviour, of tyramine-producing lactic acid bacteria from wine, *Journal of Applied Microbiology*, **88**, 584–593.
- Moreno-Arribas MV, Polo MC, Jorganes F, and Muñoz R (2003), Screening of biogenic amine production by lactic acid bacteria isolated from grape must and wine, *International Journal of Food Microbiology*, **84**, 117–123.
- Móretró T and Daeschel MA (2004), Wine is bactericidal to foodborne pathogens, *Journal of Food Science*, **69**, M251–M257.
- Nguyen HV, Pulvirenti A and Gaillardin C (2000), Rapid differentiation of the closely related *Kluyveromyces lactis* var. *lactis* and *K-marxianus* strains isolated from dairy products using selective media and PCR/RFLP of the rDNA non transcribed spacer 2, *Canadian Journal of Microbiology*, **46**, 1115–1122.
- Oelofse A, Pretorius IS and du Toit M (2008), Significance of *Brettanomyces* and *Dekkera* during winemaking: a synoptic review, *South African Journal of Enology and Viticulture*, **29**, 128–144.
- OIV (2009), *International Oenological Codex*, International Organization of Vine and Wine, Paris, available at: http://news.reseau-concept.net/images/oiv_uk/Client/CODEX_2009_EN.pdf (accessed April 2010).
- Okamura and Watanabe (1981), Determination of phenolic ciannamates in white wine and their effect on wine quality, *Agricultural and Biological Chemistry*, **45**, 2063–2070.
- Palma M, Goffeau A, Spencer-Martins I and Barret SV (2007), A phylogenetic analysis of the sugar porters in hemiascomycetous yeasts, *Journal of Molecular Microbiology and Biotechnology*, **12**, 241–248.
- Peinado RA, Moreno JA, Munoz D, Medina M and Moreno J (2004), Gas chromatographic quantification of major volatile compounds and polyols in wine by direct injection, *Journal of Agricultural and Food Chemistry*, **52**, 6389–6393.
- Peynaud E (1984), *Knowing and Making Wine* (2nd edn), Wiley, New York.

- Peynaud ME and Domercq S (1956), Sur les *Brettanomyces* isolés de raisins et de vin, *Archiv für Mikrobiologie*, **24**, 266–280.
- Peynaud ME, Lafon M and Ribéreau-Gayon MJ (1956), Respiration et fermentation comparées des différents genres et espèces de levures de vins, *Bulletin de Vins de la Société de Chimie Biologique*, **38**, 12–14.
- Phister TG and Mills DA (2003), Real-time PCR assay for detection and enumeration of *Dekkera bruxellensis* in wine, *Applied and Environmental Microbiology*, **69**, 7430–7434.
- Pilone GJ, Clayton MG and Vanduivenboden RJ (1991), Characterization of wine lactic acid bacteria single broth culture for tests of heterofermentation, mannitol from fructose and ammonia from arginine, *American Journal of Enology and Viticulture*, **42**, 153–157.
- Piškur J, Rožypec E, Polakova S, Merico A and Compagno C (2006), How did *Saccharomyces* evolve to become a good brewer?, *Trends in Genetics*, **22**, 183–186.
- Puertolas E, López N, Condon S, Raso J and Alvarez I (2009), Pulsed electric fields inactivation of wine spoilage yeast and bacteria, *International Journal of Food Microbiology*, **130**, 49–55.
- Puig A, Vilavella M, Daoudi L, Guamis B and Minguez S (2003), Microbiological and biochemical stabilization of wines by application of high pressure processing, *Bulletin de l'OIV*, **76**, 596–617.
- Quesada MP and Cenis JL (1995), Use of random amplified polymorphic DNA (RAPD PCR) in the characterization of wine yeasts, *American Journal of Enology and Viticulture*, **46**, 204–208.
- Radler F and Yannissis C (1972), Decomposition of tartrate by lactobacilli, *Archiv für Mikrobiologie*, **82**(3), 219.
- Regueiro LA, Lema Costas C and Lopez Rubio JE (1993), Influence of viticultural and enological practices on the development of yeast population during winemaking, *American Journal of Enology and Viticulture*, **44**, 405–408.
- Renouf V and Lonvaud-Funel A (2007), Development of an enrichment medium to detect *Dekkera/Brettanomyces bruxellensis*, a spoilage wine yeast on the surface of grape berries, *Microbiological Research*, **162**, 154–167.
- Renouf V, Miot-Sertier C, Strehiaiano P and Lonvaud-Funel A (2006), The wine microbial consortium: a real terroir characteristic, *Journal International des Sciences de la Vigne et du Vin*, **40**, 209–216.
- Renouf V, Strehiaiano P and Lonvaud-Funel A (2007a), Yeast and bacteria analysis of grape, wine and cellar equipments by PCR-DGGE, *Journal International des Sciences de la Vigne et du Vin*, **41**, 51–61.
- Renouf V, Lonvaud-Funel A and Coulon J (2007b), The origin of *Brettanomyces bruxellensis* in wines: a review, *Journal International des Sciences de la Vigne et du Vin*, **41**, 161–173.
- Renouf V, Perello MC, De Revel G and Lonvaud-Funel A (2007c), Microbiology of bottled wines: impacts of the filtration. *American Journal of Enology and Viticulture*, **58**, 379–386.
- Renouf V, Strehiaiano P and Lonvaud-Funel A (2008), Effectiveness of dimethyldicarbonate to prevent *Brettanomyces bruxellensis* growth in wine, *Food Control*, **19**, 208–216.
- Ribéreau-Gayon P, Dubourdieu D, Donèche B and Lonvaud A (2006) *Handbook of enology, Vol 1, The Microbiology of Wine and Vinifications*, Wiley, Chichester, UK.
- Ritter G, Gotz L and Dietrich H (1994), Untersuchung der phenolischen Substanzen in Rheingauer Rieslingweinen, *Viticulture and Enology Science*, **49**, 71–77.
- Roach JC, Boysen, C, Wang K and Hood L (1995), Pairwise end sequencing: a unified approach to genomic mapping and sequencing, *Genomics*, **19**, 345–353.
- Roder C, König H and Fröhlich J (2007), Species-specific identification of *Dekkera/Brettanomyces* yeasts by fluorescently labeled DNA probes targeting the 26S rRNA, *FEMS Yeast Research*, **7**, 1013–1026.
- Rodrigues F, Gonçalves G, Pereira-da-Silva S, Malfeito-Ferreira M and Loureiro V (2001), Development and use of a new medium to detect yeasts of the genera *Dekkera/Brettanomyces*, *Journal of Applied Microbiology*, **90**, 588–599.

- Roggero J and Archier P (1989), Mise au Point d'une méthode de dosage des phénols simples des vins application à des vins d'origines et d'âges différents, *Connaissance de la Vigne et du Vin*, **23**, 25–37.
- Rojas V, Gil JV, Piñaga F and Manzanares P (2001), Studies on acetate ester production by non-*Saccharomyces* wine yeasts, *International Journal of Food Microbiology*, **70**, 283–290.
- Romano P and Suzzi G (1993), Higher alcohol and acetoin production by *Zygosaccharomyces* wine yeasts, *Journal of Applied Bacteriology*, **75**, 541–545.
- Romano P, Granchi L, Caruso M, Borra G, Palla G, Fiore C, Ganucci D, Caligiani A and Brandolini V (2003), The species-specific ratios of 2,3-butanediol and acetoin isomers as a tool to evaluate wine yeast performance, *International Journal of Food Microbiology*, **86**, 163–168.
- Romano A, Perello MC, de Revel G and Lonvaud-Funel A (2008), Growth and volatile compound production by *Brettanomyces/Dekkera bruxellensis* in red wine, *Journal of Applied Microbiology*, **104**, 1577–1585.
- Romano A, Perello MC, Lonvaud-Funel A, Sicard G and de Revel G (2009), Sensory and analytical re-evaluation of 'Brett character', *Food Chemistry*, **114**, 15–19.
- Ramos R, Andrade P, Seabra R, Pereira C, Ferreira M and Faia M (1999), A preliminary study of non-colored phenolics in wines of varietal white grapes (codega, gouveio and malvasia fina): effects of grape variety, grape maturation and technology of winemaking, *Food Chemistry*, **67**, 39–44.
- Rose AH and Harrison JS (1971), *The Yeast*, vol. 2, London, UK, Academic Press.
- Saavedra D, Godoy P, Narváez C, Zepeda S, Urrutia M, Brevis P and Reyes E (2005), *Brettanomyces*, un contaminante en nuestras viñas/*Brettanomyces*, a contaminant in our vineyards, *Cienc trab*, **7**, 93–96.
- Sanger F, Nicklen S and Coulson AR (1977), DNA sequencing with chain terminating inhibitors, *Proceedings of the National Academy of Sciences of the United States of America*, **74**, 5463–5467.
- Santos A, Mauro MS, Bravo E and Marquina D (2009), PMKT2, a new killer toxin from *Pichia membranifaciens*, and its promising biotechnological properties for control of the spoilage yeast *Brettanomyces bruxellensis*, *Microbiology-SGM*, **155**, 624–634, Part 2.
- Scheffers WA (1961), On the inhibition of alcoholic fermentation in *Brettanomyces* yeasts under anaerobic conditions, *Experientia*, **17**, 40–42.
- Scheffers WA (1966), Stimulation of fermentation in yeasts by acetoin and oxygen, *Nature*, **210**, 533–534.
- Scheffers WA and Misset O (1974), The Custers effect in *Brettanomyces intermedius*, *Proceedings of the Fourth International Symposium on Yeasts*, Vienna, Austria, Part I: 39–40.
- Scheffers WA and Nanninga GL (1977), The anaerobic inhibition of fermentation (Custers effect) in *Brettanomyces*, *EUCHEM Conference on Metabolic Reactions in the Yeast Cell in Aerobic and Anaerobic Conditions*, Helsinki, Finland, June 13–16, 53–55.
- Schweigkofler W, Lopandic K, Molnar O and Prillinger H (2002), Analysis of phylogenetic relationships among Ascomycota with yeast phases using ribosomal DNA sequences and cell wall sugars, *Organisms Diversity & Evolution*, **2**, 1–17.
- Shalaby AR (1996), Significance of biogenic amines to food safety and human health, *Food Research International*, **29**, 675–690.
- Shantha Kumara HMC, De Cort S and Verachtert H (1993), Localization and characterization of d-glucosidase activity in *Brettanomyces lambicus*, *Applied and Environmental Microbiology*, **59**, 2352–2358.
- Shinohara T, Kubodera S and Yanagida F (2000), Distribution of phenolic yeasts and production of phenolic off-flavors in wine fermentation, *Journal of Bioscience and Microbiology*, **90**, 90–97.
- Sillas-Santos MH (1996), Biogenic amines: Their importance in foods, *International Journal of Food Microbiology*, **29**, 213–231.

- Silva P, Cardoso H and Gerós H (2004), Studies on the wine spoilage capacity of *Brettanomyces D Dekkera* spp., *American Journal of Enology and Viticulture*, **55**, 65–72.
- Silva BM, Silva LR, Valentaño P, Seabra RM, Andrade PB, Trujillo ME and Velazquez E (2007), HPLC determination of free amino acids profile of Dão red wine: effect of *Dekkera bruxellensis* contamination, *Journal of Liquid Chromatography & Related Technologies*, **30**, 1371–1383.
- Silva Pereira C, Figueiredo Marques JJ and San Romao MV (2000), Cork taint in wine: scientific knowledge and public perception a critical review, *Critical Reviews in Microbiology*, **26**, 147–162.
- Soleas G, Dam J, Carey M, and Goldberg D (1997), Toward the fingerprinting of wines: cultivar-related patterns of polyphenolic constituents in Ontario wines, *Journal of Agricultural and Food Chemistry*, **45**, 3871–3880.
- Somers T, Verette E and Pocock K (1987), Hydroxycinnamate esters of *Vitis vinifera*: changes during white vinification and effects of exogenous enzymic hydrolysis, *Journal of Science and Food Agriculture*, **40**, 67–78.
- Souffleros E, Barrios ME and Bertrand A (1998), Correlation between the content of biogenic amines and other wine compounds, *American Journal of Enology and Viticulture*, **49**, 266–278.
- Sousa MJ, Miranda L, CorteReal M and Leao C (1996), Transport of acetic acid in *Zygosaccharomyces bailii*: effects of ethanol and their implications on the resistance of the yeast to acidic environments, *Applied and Environmental Microbiology*, **62**, 3152–3157.
- Sousa MJ, Miranda L, CorteReal M and Leao C (1998), Mechanisms underlying the transport and intracellular metabolism of acetic acid in the presence of glucose in the yeast *Zygosaccharomyces bailii*, *Microbiology-SGM*, **144**, 665–670.
- Smith MT, Yamazaky M and Poot GA (1990), *Dekkera*, *Brettanomyces* and *Eniella*: electrophoretic comparison of enzymes and DNA-DNA homology, *Yeast*, **6**, 299–310.
- Snowdon EM, Bowyer MC, Grbin PR and Bowyer PK (2006), Mousy off-flavor: A review, *Journal of Agriculture and Food Chemistry*, **54**, 6465–6474.
- Sponholz WR (1993), *Wine spoilage by microorganisms*. In Fleet GH (ed.), *Wine Microbiology and Biotechnology*, Harwood Academic, Chur, Switzerland, 395–420.
- Staden R (1979), Strategy of DNA Sequencing employing computer programs, *Nucleic Acids Research*, **6**, 2601–2610.
- Stender H, Kurtzman C, Hyldig-Nielsen JJ, Sorensen D, Broomer A; Oliveira K, Perry-O’Keefe H, Sage A, Young B and Coull J (2001), Identification of *Dekkera bruxellensis* (*Brettanomyces*) from wine by fluorescence in situ hybridization using peptide nucleic acid probes, *Applied and Environmental Microbiology*, **67**, 938–941.
- Storgards E, Tapani K, Hartwall P, Saleva R and Suihko ML (2006), Microbial attachment and biofilm formation in brewery bottling plants, *Journal of the American Society of Brewing Chemists*, **64**, 8–15.
- Strauss C R and Heresztyn (1984), T. 2-Acetyl tetrahydropyridines a cause of the ‘mousy’ taint in wine, *Chemistry and Industry (London)*, 109–110.
- Swaffield CH, Scott JA and Jarvis B (1997), Observations on the microbial ecology of traditional alcoholic cider storage vats, *Food Microbiology*, **14**, 353–361.
- Swiegers JH, Bartowsky EJ, Henschke PA and Pretorius IS (2005), Yeast and bacterial modulation of wine aroma and flavour, *Australian Journal of Grape and Wine Research*, **11**, 139–141.
- Tessonniere H, Vidal S, Barnavon L, Alexandre H and Remize F (2009), Design and performance testing of a real-time PCR assay for sensitive and reliable direct quantification of *Brettanomyces* in wine, *International Journal of Food Microbiology*, **129**, 237–243.
- Thomas CE (1993), On old wine and new bottles: the transformation of ethical emphasis in higher education, *Quest*, **45**, 133–138.
- Timke M, Wolking D, Ngoc Quynh WL, Altendorf K and Lipski A (2004), Microbial composition of biofilms in a brewery investigated by fatty acid analysis, fluorescence in

- situ hybridisation and isolation techniques, *Applied Microbiology and Biotechnology*, **66**, 100–107.
- Tonon T, Bourdineau JP and Lonvaud-Funel A (2001a), Catabolisme de l'arginine par *Oenococcus oeni*: aspects énergétiques et génétiques, *Lait*, **81**, 139–150.
- Tonon T, Bourdineau JP and Lonvaud-Funel A (2001b), The *arcABC* gene cluster encoding the arginine deiminase pathway of *Oenococcus oeni*, and arginine induction of a CRP-like gene, *Research in Microbiology*, **152**, 653–661.
- Torok T, Mortimer R, Romano P, Suzzi G and Polsinelli M (1996), Quest for wine yeast – an old story revised, *Journal of Industrial Microbiology*, **17**, 303–313.
- Tucknott OG (1977), Taints in fermented juice products: mousy taint in cider. In *Annual Report*, Long Ashton Research Station, University of Bristol, 132.
- Tucknott, OG and Davies PA (1976), Production of the mousy taint by microorganisms. In *Annual Report*, Long Ashton Research Station, University of Bristol, 140.
- Ugarte P, Agosin E, Bordeu E and Villalobos JI (2005), Reduction of 4-ethylphenol and 4-ethylguaiacol concentration in red wines using reverse osmosis and adsorption, *American Journal of Enology and Viticulture*, **56**, 30–36.
- van Vuuren HJJ, and Dicks LMT (1993), *Leuconostoc oenos*: A review, *American Journal of Enology and Viticulture*, **44**, 99–112.
- van der Walt JP (1964), *Dekkera*, a new yeast genus of the *Saccharomycetaceae*, *Antonie van Leeuwenhoek*, **30**, 273–280.
- van der Walt JP and van Kerken A (1959), *The wine yeasts of the Cape. Part II. The occurrence of Brettanomyces intermedius and Brettanomyces schanderlii in South African table wines*, *Antonie van Leeuwenhoek*, **25**, 145–151.
- van der Walt J.P and van Kerken (1960) *The wine yeast of the Cape Part IV. Ascospore formation in the yeast genus Brettanomyces*, *Antonie van Leeuwenhoek* **26**, 292–296
- Van der Riet W B, Botha A and Pinches SE (1989), The effect of dimethylcarbonate on vegetative growth and ascospores of *Byssichlamysfulva* suspended in apple juice and strawberry nectar, *International Journal of Food Microbiology*, **8**, 95–102.
- van Zyl JA (1962), *Turbidity in South African dry wines caused by the development of the Brettanomyces yeasts*, South Africa Department Agriculture Technical Service Science Bulletin 381.
- Velázquez E, Cruz-Sánchez JM, Mateos PF, Monte E and Chordi A (1993), Prolewine, a program for wine-making yeast identification, *Binary*, **5**, 58–61.
- Vidal-Carou MC, Codonylsalcedo R and Marinefont A (1990), Histamine and tyramine in Spanish wines relationships with total sulphur dioxide level volatile acidity and malolactic fermentation intensity, *Food Chemistry*, **35**, 217–227.
- Vigentini I, Romano A, Compagno C, Merico A, Molinari F, Tirelli A, Foschino R and Volonterio G (2008), Physiological and oenological traits of different *dekkera/Brettanomyces bruxellensis* strains under wine-model conditions, *FEMS Yeast Research*, **8**, 1087–1096.
- Walling E, Gindreau E and Lonvaud-Funel A (2001), La biosynthèse d'exopolysaccharide par des souches de *Pediococcus damnosus* isolées du vin, mise au point d'outils moléculaires de detection, *Lait*, **81**, 289–300.
- Walling E, Gindreau E and Lonvaud-Funel A (2005a), A putative glucan synthase gene dos detected in exopolysaccharide-producing *Pediococcus damnosus* and *Oenococcus oeni* strains isolated from wine and cider, *International Journal of Food Microbiology*, **88**, 53–62.
- Walling E, Dols-Lafargue M and Lonvaud-Funel A (2005b), Glucose fermentation kinetics and exopolysaccharide production by ropy *Pediococcus damnosus* IOEB8801, *Food Microbiology*, **22**, 71–78.
- Werning ML, Ibarburu I, Duenas MT, Irastorza A, Navas J and López P (2006), *Pediococcus parvulus* *gtf* gene encoding the GTF glycosyltransferase and its application for specific PCR detection of beta-D-glucan-producing bacteria in foods and beverages, *Journal of Food Protection*, **69**, 161–169.

- Wijnsman MR, van Dijken JP, van Kleeff BHA and Scheffers WA (1984), Inhibition of fermentation and growth in batch cultures of the yeast *Brettanomyces intermedius* upon a shift from anaerobic conditions (Custers effect), *Antonie van Leeuwenhoek*, **50**, 183–192.
- Wikén T, Scheffer WA and Verhaar AJM (1961), On existence of a negative Pasteur effect in yeast classified in genus *Brettanomyces* Kufferath et Van laer, *Journal of Microbiology and Serology*, **27**, 401.
- Woolfit M, Rozpedowska E, Piškur J and Wolfe KH (2007), Genome survey sequencing of the wine spoilage yeast *Dekkera (Brettanomyces) bruxellensis*, *Eukaryotic Cell*, **6**, 721–733.
- Worobo RW (2005), An introductory overview of the risks of wine spoilage. How organisms get in wine and sanitation techniques, *Wine Spoilage Organisms and Winery Sanitation Vinquiry Seminar on Wine Quality*, Rohnert Park CA, May 11.
- Wright JM and Parle JN (1974), *Brettanomyces* in the New Zealand wine industry, *New Zealand Journal of Agricultural Research*, **17**, 273–278.
- Yap A, Jiranek V, Grbin P, Barnes M and Bates D (2007), Studies on the application of high-power ultrasonics for barrel and plank cleaning and disinfection, *Australian and New Zealand Wine Industry Journal*, **22**, 96–104.
- Yamada Y, Matsuda M, Maeda K and Mikata K (1994), The phylogenetic relationships of species of the genus *Dekkera van der Walt* based on the partial sequences of 18S and 26S ribosomal RNAs (*Saccharomycetaceae*), *Bioscience Biotechnology and Biochemistry*, **58**, 1803–1808.
- Zardetto de Toledo O, Teixeira C, Gonzales C and Verona O (1959), Preliminary study upon yeast microflora of grape and grape must in the district of S. Paulo (Brasil), *Annals of Microbiology*, **9**, 22–34.

13

Reducing cork taint in wine

**R. Jung and V. Schaefer, Forschungsanstalt Geisenheim
(Geisenheim Research Center (GRC)), Germany**

Abstract: This chapter deals with the sources, prevention and treatment of the musty and mouldy off-flavours in wine which consumers usually associate with the use of natural corks as wine stoppers (cork taint). However, these off-flavours can also be found in wines that were bottled with alternative closures. Therefore, the chapter focuses on the quality management of the production of natural cork stoppers and on the avoidance of musty and mouldy off-flavours in the wineries.

Key words: cork taint, musty and mouldy off-flavours, halogenated anisoles, cork quality management.

13.1 Introduction: cork taint

Cork taint is one of the most unpleasant and most frequent organoleptic defects in wines. It is caused by aromatic compounds that are present in the cork material and are transferred into the wine after bottling (Sefton and Simpson, 2005). It is mostly described as a musty, mouldy or earthy smell. Sometimes its sensorial perception is also described as burnt rubber, smoky or even camphor, which impacts negatively the aroma of the affected wine.

Referring to Pena Neira *et al.* (2000), 0.5–2 % of all bottled wines in Europe and 1–5.5% of all wines bottled in Australia are affected by this problem. The different descriptions of the cork taint-related aromas reflect its different sources and also the individual sensitivity of the wine consumer and the influence of the wine style on the perception of cork taint (Prescott *et al.*, 2005; Mazzoleni and Maggi, 2007).

The first descriptions of cork taint as a wine contamination go back to the early

20th century (Ribéreau-Gayon *et al.*, 2006), but it was not until the beginning of the 1980s that a chemical compound, namely 2,4,6 trichloroanisole (TCA), was finally identified as a main cause of cork taint in wine (Tanner *et al.*, 1981; Buser *et al.*, 1982).

Based on this discovery, a number of further compounds with similar negative flavour attributes were discovered in mouldy and musty smelling wines that were not affected by TCA (e.g. geosmin, 2-methyl-isoborneol, octane-3-one, pyrazines, etc.) (Mazzoleni *et al.*, 1994; Kugler and Rapp, 1997; Hesford and Schneider, 2002; Sefton and Simpson, 2005).

In addition, cork taint-related flavours were found in wines that were barrel samples or not closed with natural cork stoppers, indicating that natural cork stoppers are not the only source of mouldy off-flavours (Chatonnet *et al.* 1994, 2004; Hesford and Schneider, 2002). Therefore, depending on the compound causing the cork taint, the consumer has a different impression of the problem.

13.2 Compounds causing musty–mouldy off-flavours

13.2.1 Chlorinated anisoles

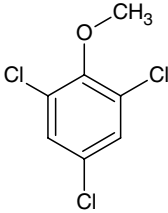
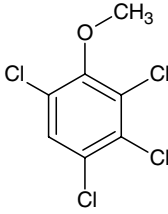
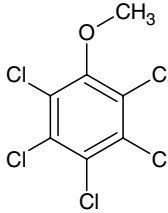
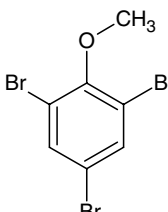
The chemical compound 2,4,6-trichloroanisole (TCA) was identified as the main cause of classical cork taint at the beginning of the 1980s (Tanner *et al.*, 1981; Buser *et al.*, 1982). In a study by Soleas and co-workers (2002), it was reported that 51% of wines affected by fungal taints had elevated concentrations of TCA (above 2 ng/L), showing that the chlorinated anisoles are responsible for the majority of musty off-flavours. However, 49% of the affected wines did not suffer from a contamination with TCA indicating that their spoilage might be attributed to the occurrence of other musty-smelling compounds.

TCA is a compound with an intense smell and has a very low perception threshold value. In technical literature, the ranges of perception threshold values of TCA vary substantially depending upon the matrix to which it has been added. The perception threshold value for TCA in drinking water is below 1 ng/L (Curtis *et al.*, 1972; Griffiths, 1974), whereas the threshold value in wine is about 2–5 ng/L, although some authors report higher detection thresholds (Table 13.1). However, it is known, that the threshold values in wine depend strongly on the kind of wine, the wine style and the experience of the panellist (Prescott *et al.*, 2005; Mazzoleni and Maggi, 2007).

Regarding the formation of TCA, it is well-known that TCA results from the microbiological methylation of chlorophenols (Fig. 13.1). A source of chlorophenols is the hypochloride treatment of the cork raw material during the production process, which has become unusual nowadays.

Another source for TCA is the microbiological formation of TCA by mould fungi such as *Penicillium* spp. whereby a phenol due to the reactions of the pentose phosphate pathway and the shikimic acid pathway is formed over the preliminary stage shikimic acid (Pfeifer, 2002). Afterwards, this phenolic compound is chlorinated chemically in the presence of hypochloride.

Table 13.1 Structure and detection thresholds of different haloanisoles

Compound	Detection threshold
2,4,6 Trichloroanisole (TCA) 	In water: 30–300 pg/L (Curtis <i>et al.</i>, 1972; Griffiths, 1974) In wine: – dry white wine, 4.0 ng/L (Ribéreau-Gayon <i>et al.</i>, 2006) – dry white wine, 2.0 ng/L (Hervé, E. <i>et al.</i>, 2000) – wine, 1.4–4.6 ng/L (Sefton and Simpson, 2005) – Sauvignon blanc, 210 ng/L (inexperienced panel), and 17.4 ng/L (experienced panel) (Suprenant and Butzke, 1997) – white wine, 10.0 ng/L (Pfeifer, 2002) – red wine, 60.0–100 ng/L (Pfeifer, 2002) – red wine: 22.0 ng/L (Alvarez-Rodriguez <i>et al.</i>, 2002)
2,3,4,6 Tetrachloroanisole (TeCA) 	In water: 4 ng/L (Curtis <i>et al.</i>, 1972) In wine: – white and red wines, 10–15 ng/L – sparkling wines, 5 ng/L (Chatonnet <i>et al.</i>, 2004)
Pentachloroanisole 	In wine: 4000 ng/L (Ribéreau-Gayon <i>et al.</i>, 2006)
2,4,6 Tribromoanisole (TCA) 	In water: 8–20 pg/L (Saxby <i>et al.</i>, 1992; Whitfield <i>et al.</i>, 1997) In wine: – model medium, 3.4 ng/L – standard wine, 7.9 ng/L (Chatonnet <i>et al.</i>, 2004)

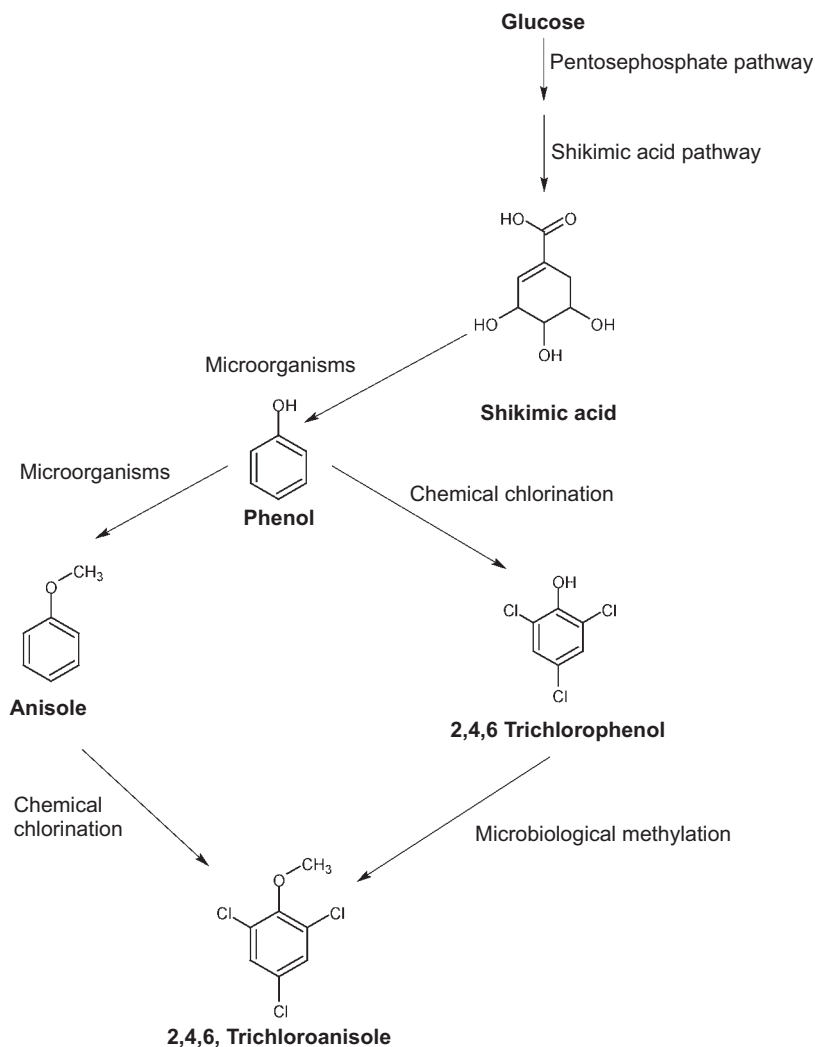


Fig. 13.1 Biosynthesis of 2,4,6 trichloroanisole (TCA) (Maujean *et al.*, 1985 Pfeifer, 2002).

Trichloroanisole can also be formed by the dismantling of chlorine-containing chemicals, for example pentachlorophenol (PCP) (e.g. wood preserver ‘Raco’; pesticide ‘Dowicide’), which can actually be considered to be the main source for halophenols in our environment due to their widespread use in the last decades (Daniels and Lake *et al.*, 2007; Rudy and Scholten, 2007; Simpson and Sefton, 2007).

In the course of the chemical production of PCP, every PCP formulation contains 2,4,6 trichlorophenol (TCP) and 2,3,4,6 tetrachlorophenol (TeCP) as impurities. As well as the TCP, the TeCP and also PCP can be methylated by

microorganisms and thus transferred into the corresponding anisoles which means an enormous decrease of the perception thresholds of those compounds. The dismantling of chlorophenols into chloroanisoles is mainly conducted by mould fungi like *Penicillium* spp. and *Trichoderma*-strains (Fiedler *et al.*, 1996).

13.2.2 Tribromoanisole

At the beginning of the 1980s, 2,4,6 tribromoanisole (TBA) was found as a musty-smelling contamination in maritime sediments and seafood. It was assumed that TBA was formed by a microbiological methylation of 2,4,6 tribromophenol (Watanabe *et al.*, 1983; Allard *et al.*, 1987). An additional chemical formation of TBA was reported in connection with the chlorination of drinking water given that the water contained bromine ions and phenol (Sweetman and Simmons, 1980).

After the use of pentachlorophenol-containing preparations was forbidden at the end of the 1980s, this compound was replaced by tribromophenol-containing preparations as fungicides and wood preservative, especially in the packaging industry. At the same time, the use of TBP as a flame protection agent for wood (e.g. pallets) and in synthetic materials (e.g. paint) increased. The use of tribromophenol in the food sector is not without problems, since, just as with the trichlorophenol, an anisole (TBA) can be formed by microbiological dismantling of tribromophenol (Whitfield *et al.*, 1997; Rudy and Scholten, 2007).

It has already been proven in various studies that the cause for musty off-flavours in tainted food and wines was not TCA but TBA (Whitfield *et al.*, 1997; Hesford and Schneider, 2002; Chatonnet *et al.*, 2004). Polyethylene (PE) packaging, in particular, showed a high permeability for TBA (Whitfield *et al.*, 1997). Thus in different investigations, TBA contamination was demonstrated from plastic wine stoppers, from the plastic sealing compound of crown caps and screw caps and from natural corks, as well as from filter layers, wooden pallets, cardboard boxes and plastic packaging (Hesford and Schneider, 2002; Rudy and Scholten, 2007). The perception threshold value for TBA is listed as between 3.4 and 7.9 ng/L depending on the tested medium (Chatonnet *et al.*, 2004).

13.2.3 2-Methoxy-3,5-dimethylpyrazine (MDMP)

2-Methoxy-3,5-dimethylpyrazine (MDMP) has recently been reported to be a contaminant responsible for a cork taint-like aroma defect called 'fungal must'.

The first description of this taint in wine was made by Duncan (1995) whose panellists were able to distinguish 'fungal must' from other kinds of cork taint, which was also confirmed by gas chromatography/olfactometry (GCO) assessments of extracts from cork stoppers affected by this taint. These extracts showed an intensely musty odorous zone under GCO analysis, not corresponding to TCA or other cork taint-related compounds in the GC-aromagram.

The odour threshold values of MDMP are quite similar to the detection thresholds of TCA. The isolation and identification of MDMP was accomplished by Simpson *et al.* (2004) who determined an odour threshold of 0.4 ng/L MDMP

for water and a threshold of 2.1 ng/L for a light dry white wine. Little is known about the formation of MDMP. The first report of MDMP as a cause of a musty smell was made by Mottram *et al.* (1984) who found MDMP in a machine cutting emulsions used in engineering workshops. They were able to isolate an aerobic, Gram-negative bacterium, but were unable to characterise it with known bacterial species.

Various bacteria (e.g. *Pseudomonas* spp.) are capable of producing pyrazine-related off-flavours, and also moulds (e.g. *Aspergillus flavus*) have been reported to produce similar compounds (Mottram *et al.*, 1984). MDMP was also reported to occur as an important aroma compound of coffee (Czerny and Grosch, 2000), where it was supposed to have a bacterial origin.

Regarding the occurrence of MDMP on corks, Simpson *et al.* (2004) assumed that a possible source of MDMP could be the cutting punches that produce the cork cylinders. Moreover, they did not rule out the possibility that contamination could occur somewhere else during the cork processing or storage by MDMP-producing bacteria growing on the cork material. Furthermore, they suggested that the origin of MDMP might not necessarily be bacterial.

13.2.4 Guaiacol

Another compound which is known to contribute to cork taint-like aroma defects in wine is guaiacol. Guaiacol causes a phenolic and medicinal taint in an affected wine (Simpson *et al.*, 1986). In a study by Simpson *et al.* (1986), the authors found concentrations of 0.07–2.63 mg/L guaiacol in tainted wines and amounts of 0.13–2.14 mg/L guaiacol in the associated corks. In this case, the guaiacol was transferred from the corks into the wine after bottling.

Its flavour threshold has been reported to be 0.030 mg/L in an aqueous solution containing 12% ethanol (Riboulet, 1982). An aroma threshold of 0.020 mg/L in a dry white wine was reported by Simpson and colleagues (1986), whereas Kennison *et al.* (2007) reported elevated guaiacol contents in wines made of grapes which had been exposed to straw-derived smoke for 1 hour, and postulated aroma thresholds for guaiacol for water (5.5 µg/L) as well as for white (95 µg/L) and red wines (75 µg/L).

According to Alvarez-Rodriguez *et al.* (2003) guaiacol-mediated cork taint can be attributed to the degradation of vanillic acid by bacterial strains (e.g. *Bacillus subtilis*, *Streptomyces* spp.) growing on cork. Other sources for guaiacol in wine can be via oak barrel maturation or the grapes themselves (Sefton and Simpson, 2005).

13.2.5 Geosmin

Geosmin (*trans*-1,10-dimethyl-*trans*-9-decalol) is a metabolite of various actinomycetes and fungi. It has been found to be responsible for an earthy taint in different foods like apple juice (Zierler *et al.*, 2004), cereals, fish, vegetables (Whitfield, 1998) and wine (Darriet *et al.*, 2000). Darriet *et al.* (2000) identified

geosmin in different French and Algerian wines and postulated flavour thresholds of 10 ng/L in water, 80–90 ng/L in a neutral red wine and 60–65 ng/L in a neutral white wine. The same authors found geosmin to be relatively stable in wine, thus invalidating previous findings indicating a rapid degradation of geosmin in acidic solutions (Gerber *et al.*, 1965).

Due to its microbiological formation, contamination with geosmin can occur on cork stoppers or in barrels and thus be transferred into the wine (Kugler and Rapp, 1997; Darriet *et al.*, 2000). Its main source, however, can be linked to the effect of rot on grapes; especially secondary rot species such as *Penicillium* spp. are capable of producing elevated amounts of geosmin on infected grapes. Geosmin is stable during alcoholic fermentation and is thus transferred into the wine (La Guerche *et al.*, 2006).

13.2.6 2-Methylisoborneol, 1-octen-3-one, 1-octen-3-ol

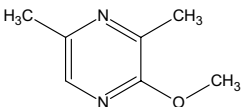
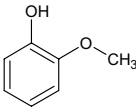
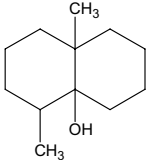
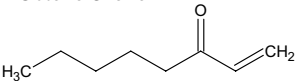
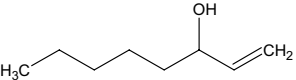
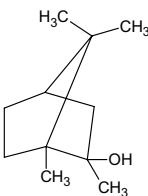
Along with geosmin, 2-methylisoborneol (MIB), 1-octen-3-one and 1-octen-3-ol are compounds that are closely linked with the growth of moulds (Sefton and Simpson, 2005). These compounds can appear during the production process of cork stoppers due to growth of mould on the cork material. 1-octen-3-one and 1-octen-3-ol are typical mould metabolites (Kugler and Rapp, 1997), known to be responsible for a typical mushroom, fungal and mouldy flavour (Hesford and Schneider, 2002; Zierler *et al.*, 2004; La Guerche *et al.*, 2006).

MIB is a metabolite of *Botrytis cinerea*, some *Penicillium* spp. and some *Streptomyces* spp. MIB and 1-octen-3-one have also been found in musts made from rotten grapes but not in the corresponding wines, indicating that they are not stable during alcoholic fermentation (La Guerche *et al.*, 2006). Findings of both compounds in bottled wine can therefore be linked to the cork stopper and the growing of mould on the cork during the manufacturing process. 1-octen-3-ol, however, has also been found on rotten grapes and the musts made from them, and is stable during alcoholic fermentation so that an occurrence of this compound in a wine can be caused by mould growth on the grapes as well as by contaminated cork stoppers (La Guerche *et al.*, 2006). The flavour thresholds of these compounds are listed in Table 13.2. The compounds listed above (Tables 13.1 and 13.2) can be considered to be causes of the most cork taints. Nevertheless, not all compounds that can be involved in cork taint-like wine defects can be listed here.

13.3 Quality management and control methods for wine corks: introduction

For corks made from natural cork or technical cork stoppers (agglomerated corks, Champagne corks), an exact determination of the quality was hardly possible in former times. The evaluation of corks, as well as their classification into quality grades, was mainly based on the judgement regarding their visual aspects. This way, corks with a smooth surface, which is low in pores, were generally preferred.

Table 13.2 Structure and detection thresholds of musty–mouldy non-haloanisole compounds

Compound	Detection threshold
2-Methoxy-3,5-dimethylpyrazine 	In water: 0.4 ng/L (Simpson <i>et al.</i> , 2004) In wine: – dry white wine, 2.1 ng/L (Simpson <i>et al.</i> , 2004)
Guaiacol 	In water: 5.5 µg/L In wine: – white wine, 95 µg/L – red wine, 75 µg/L (Kennison <i>et al.</i> , 2007)
Geosmin 	In water: 10 ng/L (Darriet <i>et al.</i> , 2000) In wine: – white wine, 60–65 ng/L (Darriet <i>et al.</i> , 2000) – red wine, 80–90 ng/L (Darriet <i>et al.</i> , 2000)
1-Octene-3-one 	In water: 0.003 µg/L In wine: – red wine, 0.07 µg/L (La Guerche <i>et al.</i> , 2006)
1-Octene-3-ol 	In water: 2 µg/L In wine: – red wine, 40 µg/L (La Guerche <i>et al.</i> , 2006)
2-Methylisoborneol 	In water: 0.012 µg/L (La Guerche <i>et al.</i> , 2006) In wine: – red wine, 0.04 µg/L (La Guerche <i>et al.</i> , 2006)

Conclusions regarding the cork quality were also drawn partly from the formation and the number of annual rings. There is no doubt that experienced specialists actually succeed in grading into different quality classes in this way but, nevertheless, methods of measuring which allowed an objective examination of the cork quality were lacking in the past.

The fundamental question concerns which criteria really allow conclusions regarding the ‘inner quality’ of cork stopper material. Many cork companies, bottling companies and research institutes were, therefore, looking for suitable methods of testing, which – in most cases, without any exchange of experience with other parties – were used as a guideline for an internal company quality control. Thus, over the last few years, a large number of individual evaluation criteria were created. Additionally, the fact that different devices were being used for measuring and different reference scales for evaluation of the measurements resulted in a total confusion.

For the cork-producing industry, quality management (QM) and quality controls start with the production of the raw material and end with the finishing of the stoppers. Cork distributors and wineries can carry out tests only on finished cork stoppers before using them as bottle closures; they then take the responsibility for the bottled product that is sold to the customer. Quality control measurements that can be applied directly to cork stoppers give the fundamental information based on which lot is accepted or rejected. At the second step, receiving the closures and using them in the bottling process, quality controls and management are demanded.

In this regard, the QM of cork stoppers used as closures for wine can be divided into two steps:

- test procedures to evaluate the quality of cork stoppers;
- handling and processing of corks during bottling and storage.

13.4 Test procedures to evaluate the quality of cork stoppers

Generally, suitable laboratory equipment and a skilled staff are absolutely necessary to carry out tests on cork closures. This is particularly the case for the sensorial tests, where a correct evaluation can only be made by experienced persons. There is an extreme need of material and time for the tests. Because natural cork is a very heterogeneous material, large numbers of random samples have to be tested, in order to provide secure statements concerning quality.

Therefore, it makes sense to carry out these tests with a large sample – as early as possible, i.e. already when producing the corks. Testing small delivery lots to the extent described would be completely uneconomic and hardly feasible. In this context, the practice by which the wine producers dictate the control measurements of the different tests to the cork suppliers as a part of their delivery conditions helps to minimise the quantity of examinations. Nevertheless, the bottling company or winery will have to do their own tests to verify the test results of the delivering company.

A sensible sequence of tests will enable several tests to be carried out with one cork. Destructive tests should only be carried out at the end of the test chain. For

control of incoming goods, the following sequence of control measurements could be recommended:

1. Determination of the dimensions and annual rings.
2. Weighing of the corks.
3. Carrying out the sensory test (in screw-lid glasses) – in case of no complaints, further tests can follow.
4. Drying of corks to their weight constancy (thus the initial humidity, the absorption of humidity and the dry weight can be established).
5. Evaluation of the corks regarding irregularities and faults.

The aforementioned tests should be carried out by using representative samplings. In addition, for the determination of the tightening quality and elasticity, tests of the cork-elasticity can be done using special equipment.

13.4.1 Catalogue of methods for testing of corks

Starting in 1992 a 'Cork Research Team' under the leadership of the 'Geisenheim Research Center', consisting of scientists from different research institutes, wine producers and the cork industry, tried to standardise cork-quality measuring methods. After these methods had been tried and tested in practice, in 1996 the first edition of the *Methods of Testing for the Determination of the Cork Quality/Geisenheim Test Methods* was published.

The methods of testing – including the selection of samplings – were written in the form of QM-working instructions. They contain exact indications regarding the purpose and measuring principles of the tests, the means for testing, the equipment which is necessary for the test and for carrying out the measurements, as well as hints for the evaluation of the measurements.

An important condition for the selection of the different working methods, besides the security of the statements, was the practicability. Only methods which are minimally susceptible to interference and which also allow the testing of extensive samplings can be used in practice. Today, this is an important condition for testing the natural material cork under quality-orientated production conditions in the cork commerce and at the wine producers'. Standardised production procedures according to ISO Standards require an exact specification of all the materials used, which is also necessary for product liability reasons.

The *Methods of Testing for the Determination of the Cork Quality* are now available in the 3rd edition (Jung and Zürn, 2007). Generally, the 'Test Procedures to Evaluate the Quality of Cork Stoppers' can be divided into 'Standard Test Procedures' which are obligatory and 'Additional Test Procedures' which can be done facultatively.

13.5 Standard test procedures

13.5.1 Sampling

The first important step when undertaking quality control measurements with a

natural product is the correct sampling of a representative number of corks from the lot. It is only with analysis of a correct number of samples that the quality of the whole lot can be adequately described.

Depending on the kind of 'fault', the quantity of samples to be analysed can vary significantly. Concerning sensory deviations of the bottled products, a higher quantity of samples is necessary than for physical aspects, e.g. the lengths of corks. As a guide, ISO 4707, ISO 3951 and ISO 2859 are recommended to be used for taking samples and defining the test quantity. After sampling, the AQL (acceptable quality level) also has to be defined for each test, again depending on the importance of the possible fault.

13.5.2 Visual control of corks

Visual control enables conclusions regarding 'outer' quality of new but also of used corks to be drawn. The visual control of new corks includes mainly comparison with a representative sample. Faults like cuts, holes, dust, areas of wooden, non-elastic cork tissue or so-called 'green wood' (cork tissue with increased moisture without sufficient elasticity) can be found by experience or by comparison with negative samples.

13.5.3 Cork dimensions

In order to guarantee an optimal tightening performance, the dimensions of the corks must fall within a fixed range of tolerances. Corks that are outwith the tolerance field in terms of diameter or length need not be accepted by the client. Digital systems allow the definition of cork diameters with several single measurements to be determined with great precision. Deviations from the specifications given by the producers can be easily recognised. Table 13.3 gives an overview of cork dimensions and possible variations in diameters and lengths.

Table 13.3 Dimensions for cork stoppers

Product/ parameter	Natural cork	Agglomerated cork	1 + 1 Twin top cork	Cork for sparkling wine
Length (as average)	Length, ± 0.5 mm Standard dev., 0.2–0.4	Length, ± 0.5 mm Standard dev., 0.2–0.4	Length, ± 0.5 mm Standard dev., 0.2–0.4	48 mm + 0.3 mm/ –0.5 mm (as average) ± 1.0 mm (as single value)
Diameter (as average)	Diameter, ± 0.4 mm Standard dev., 0.1–0.3	Diameter, ± 0.4 mm Standard dev., 0.1–0.3	Diameter, ± 0.4 mm Standard dev., 0.1–0.3	31/30.5 mm ± 0.2 mm (as average) ± 0.4 mm (as single value)

13.5.4 Sensory examination of corks

The sensory evaluation of cork stoppers is the central test to decide whether to accept or refuse a delivered cork lot. Very often analytical measurements only give one result for a lot of single cork stoppers within one sample. With the sensory test described in the procedure, a wide range of single samples can be checked, giving a good and much more detailed impression of the cork lot.

To get an impression of the sensory properties of corks, single samples are put into screw lid glasses with a humid atmosphere. This is a 'quick test', which is carried out in order to find out whether corks are neutral in taste or if they transmit an off-flavour to the wine. By defining the odour impression into three categories, the testers have to decide whether to accept or reject a cork lot and also whether a repeat test is necessary. Besides the higher quantity of single results, a further advantage of the sensory test is the control of *all* odour-active compounds. This is in contrast to analytical measurements which can focus only on defined, mainly single, substances or molecules.

13.5.5 Dry weight and specific weight of corks

The dry weight of the cork and the specific weight are the most important parameters for evaluating the homogeneity of a delivery lot. As the dry weight of corks (and the specific weight) has a direct influence on the physical behaviour/elasticity of corks, a delivery lot should not consist of corks whose weights differ too much. The more uniform the dry weight of the delivery lot, the more uniform the closing features.

In the test procedure for examination of dry weights, average numbers for dry weights of the different types of cork stoppers are documented. This can be a guide to judging single cork lots but also enables comparison with following delivered lots and with the offered sample.

Table 13.4 documents the average dry weights of different kinds of cork stoppers in relation to an average price for the corks expressed in Euro. The table, which was created in cooperation with the cork industry, also shows the values for the standard deviations in dry weights in order to indicate the homogeneity of a cork lot.

13.5.6 Humidity of the cork

The humidity content has a decisive effect on the elasticity and the microbiological stability of corks. Evaluation of cork humidity, therefore, is of crucial importance for evaluating the closing features of corks. Different measurement methods exist to measure cork humidity. The most exact way is the weighing and drying to constant weight. Having the cork dimensions, the correct way to define the cork humidity is to calculate the humidity by volume (% v/v).

13.5.7 Proof of hydrogen peroxide in corks

The hydrogen peroxide used for the 'bleaching' of corks is normally neutralised

Table 13.4 Dry weights and specific weights for cork stoppers of different qualities or price classes (€/1000 pieces)*

Product	Average dry weight (g)	Standard deviation	Specific weight (g/cm ³)
Natural cork 38 × 24 mm (17.19 cm ³)			
100–130 €	2.65–3.15	0.15–0.45	0.15–0.18
131–180 €	2.65–2.95	0.15–0.35	0.15–0.17
>181 €	2.65–2.85	0.15–0.35	0.15–0.17
Natural cork 45 × 24 mm (20.36 cm ³)			
120–150 €	3.15–3.65	0.15–0.45	0.15–0.18
151–200 €	3.15–3.55	0.15–0.35	0.15–0.17
>201 €	3.15–3.45	0.15–0.35	0.15–0.17
1 + 1 Twin top cork (medium quality)			
39 × 23.5 mm (16.92 cm ³)	4.30–4.80	<0.25	
44 × 23.5 mm (19.08 cm ³)	4.50–5.50	<0.25	0.24–0.28
Agglomerated cork (medium quality)			
38 × 23 mm (15.79 cm ³)	4.30–5.30	<0.30	0.27–0.33
Sparkling wine corks (medium quality)			
30.5/31 × 48 mm (35.07–36.23 cm ³)	8.70–9.70	<0.25	0.24–0.28

*Calculation on the basis of data from companies of the German Cork Industry with certification 'Geisenheimer Prüfsiegel'.

during the production procedure. The test is used to check for residues of this oxidising substance, which could become problematic in the bottled wine by reducing the SO₂ content, leading directly or indirectly, to an increased oxidation of wine components.

13.5.8 Test bottling

A 'test bottling' enables the tightening capability of corks to be assessed under practice conditions in bottles. The test can be based on an average sample of daily production, but should be executed daily, in order to avoid mistakes concerning the cork-coating. Further information about the sealing effect of the stoppers in the bottles can be achieved by using different storage temperatures.

13.5.9 Control of extraction forces

The sufficiently close fitting-in of wine and Champagne corks in the bottle necks is crucial for the tightness of the bottles and, at the same time, it describes the elastic

restoring procedure of the corks in the bottle necks. Also, the effect of surface treatment, which is used for every cork, can be controlled by measuring extraction forces for still wine stoppers or torque forces for sparkling wine stoppers.

Concerning still wine stoppers, forces between 200 and 400 N (Newton) are usual and accepted by consumers. If cork stoppers can be extracted below 200 N, very often the stopper can be turned in the bottle neck by insertion of the corkscrew and therefore does not have enough expansion to seal the bottle neck. Extraction forces higher than 400 N can cause complaints from consumers who are not able to extract the corks. Torque forces to turn sparkling wine closures can also be measured and are defined as varying between 2 and 4 Nm (Newton-metres).

13.6 Additional test procedures

This section describes additional tests which can be useful for cork producers in the production control when manufacturing and finishing corks; these tests are also interesting for extended quality control measurements in wineries and for bottling companies.

13.6.1 Determination of the content of cork dust

The pores of corks (lenticels) contain particles of dust. Normally, these are eliminated during the production of corks or special dedusting processes. Nevertheless, during the production – especially of lower quality corks with wooden tissues and more lenticels – new dust may emerge from rubbing, and this dust may get into the bottles. Especially in the storage containers and conveyance systems of the corking machines, cork dust may be rubbed off. Through the test, the corks' initial dust content can be determined. Corks, therefore, are introduced in bottles and, after adding liquid and shaking of the bottles, removed dust can be separated by filtration.

13.6.2 Phenolic compounds in corks

Phenols are natural compounds in natural cork stoppers or any wooden material. If these compounds are absorbed into the bottled wine, this can be negative for its sensory characteristics or at least bring some variation into a lot of wine bottles. By soaking corks in alcoholic solution and measuring the colour in the soaking solution, an impression of the phenolic composition of the used corks can be achieved. Also, a measurement of total phenols in the liquid can indicate the possible influence of the used cork stoppers.

13.6.3 Test of cork coating/surface treatment

A simple test enables the adhesive strength of the cork coating to be tested in order to avoid problems, particularly concerning leaking of bottles. An incorrect cork

coating can lead to ingress of wine between the bottle neck and the cork, resulting ultimately in a significant loss of elasticity, because the absorbed wine replaces the compressed air in the cork tissue. Additional leakage can occur after or parallel to this moisturising process. The test to control the cork coating works by standing corks upright in a coloured liquid and examining the capillarity effect on the surface.

13.6.4 Testing of the cork elasticity

The elasticity and the resilience of corks can be tested in different ways. Corks with a less homogeneously developed cork tissue or strongly lignified corks have different physical properties and elastic behaviour than corks made out of homogeneous cork tissue. Tests on cork elasticity can reveal these properties. Generally, so-called ‘universal test machines’ are used for this kind of test. To test corks under realistic conditions, it is possible to integrate a four-jaw-corker into these machines. Measurements of compression forces, initial diameters and diameters after compression and time of resilience help to gain an understanding of the structure, the elastic behaviour and the homogeneity of a cork lot.

13.7 Handling and processing of corks and bottles during bottling and storage

Quality control methods are essential not only in the production of cork closures but also in their correct use. Using different kinds of closures also requires the use of different bottling technologies. For a long time, the use of cork stopper closures and their ‘proper handling’ were not discussed but, with the increase of synthetic closures, bottlers became more aware of the special characteristics and technical demands created by using ‘stoppers’ of different structures and physical behaviours in the bottling process.

The following steps help to avoid problems involved in using stoppers made from cork (but also synthetic) materials. These ‘recommendations’ are also written in the ‘Geisenheim Test Methods’ (Part II).

13.7.1 Receiving incoming goods

The control of incoming goods has to be carried out by the client, directly after receiving the corks, in order to avoid later complaints. This includes the checking of simple and obvious facts. The following controls should be carried out immediately the corks are received:

- visual control of the packaging regarding damages;
- checking of the delivery quantity, number of lots;
- checking of the dimensions of the corks;
- checking of the printing on the corks;
- comparison of the corks with a specimen.

13.7.2 Storing conditions for corks

Corks should be used as soon as possible after being delivered, and they should be stored in rooms that are neutral in odours until they are used. The humidity content of the air in the storeroom is also important for the quality of the corks. The following have to be taken into consideration:

- storage time not longer than 6 months after receipt of delivery;
- no strange odours – clean air in the storeroom;
- frost-free storage, humidity content between 40 and 60%;
- immediate re-closure of opened cork-bags;
- collection of leftover corks after bottling (also from the storage containers of the corking machine) into closeable bags, or ask the cork supplier for a new, sterile packaging, respectively.

13.7.3 Bottles: filling and storage conditions

Together with the closures, the bottle neck, its inner diameter and also the bottle volume are of great importance for the closure quality and shelf life of the bottled product. Therefore, bottles have to be controlled and bottling and storage conditions have to be exactly determined.

Control of the bottles

For a tight closure, the inner course of the bottle neck is of crucial importance. When carrying out a sampling with the bottles intended for filling, the inner diameters of the bottle necks have to be measured by means of an adjusted inside touch-sensitive device. The results of this measurement have to be compared to the norm-values and bottles should be rejected if the deviation from the norm is significant.

Determination of the nominal and the full-to-the-brim volume of the bottles indicates which empty spaces are left after the filling of the bottles. The question of what cork to use – 38 mm, 45 mm or an even longer cork – can only be answered if the difference between the filling level and the lower edge of the cork is known. It is only in this way that the presence of a sufficient empty space for an eventual warming-up expansion of the wine can be ascertained.

Some bottles have indications made by the producers showing at what distance from the upper edge of the bottle neck the nominal volume is reached. Such indications are desirable and very helpful. Nevertheless, here, as well, the correctness of these indications should be checked by own measurements.

Filling level, headspace in bottles

During the filling procedure, it has to be taken care that – on observing the nominal volume – an empty space is left between the level of the wine and the cork. This headspace is necessary as a buffer for the expansion of the volume (dilation) of the wine caused through warming-up which, otherwise, entails a ‘pushing’, i.e. a leaking of the bottles.

The extent of the expansion of the volume depends directly on the difference between filling, storage or transport temperatures, but it is also influenced by the alcohol and residual sugar content of the wine. As a rule of thumb, one can assume that an expansion of 0.2–0.4 mL/L is associated with a warming-up by 1 °C. This means that, for instance, a wine which was filled at 10 °C and, which on transport was warmed up to 40 °C (which can be easily reached with summer temperatures), will expand its volume by about 6 mL (dry wine with about 10% alcohol by volume) to about 12 mL (wine containing residual sugar with about 14% alcohol by volume). An empty space of 10 mm – measured between the lower edge of the cork and the level of the wine (in a bottle standing upright) – corresponds roughly to only 3 mL of volume.

A warming-up of the wine to 20 °C filling temperature reduces the possible margin between filling and storage temperature and, thus, the risk of an excessive expansion of the volume. In any case, the filling temperature of 20 °C is necessary for the setting of the correct nominal filling quantity, according to the regulations laid down by the law.

Sealing of bottles, corking

During the closing procedure, the cork is first compressed in the corks of the corking machine until it has reached a diameter of 15.8 mm, and then it is pushed into the bottle neck. Depending on the elasticity of the corks and on the dimensions, but also on the structure of the surface of the corks and on the pushing-in speed, the so-called closing pressure is built up through the ‘piston effect’ of the corks, through compression of the air in the bottle neck. This pressure, which for a short period of time can reach 4–5 bar, in most cases reduces itself to 1–2 bar after some minutes and, in the course of the following days, it is slowly decomposed completely.

This decomposition can be accelerated by rinsing the headspace of the bottles with carbonic acid. For this procedure, CO₂ is blown into the bottle necks immediately before corking. By using 0.08–0.4 g CO₂ per bottle, the air is displaced from the empty spaces in the bottles. As carbonic acid dissolves itself in the wine very quickly after corking, the closing pressure is decomposed in this way.

Due to the very low quantity of carbonic acid, no sensory influence on the wine can be stated. Moreover, the decomposition of hydrogen oxide out of the headspaces of the bottles entails a reduction of sulphurous acid (SO₂) in the wine during the first period of storage.

Through the so-called ‘evacuation’ of the headspaces in the filled bottles, the air pressure in the head space can also be reduced right before corking. By using these vacuum corkers, the closing pressure becomes significantly lower. When using this technology to reduce inner bottle pressures after insertion of the stopper, it is important to ensure that if a real ‘vacuum’ in the bottle neck is taking place.

Storage of bottles

After the closing procedure, the cork needs time to expand and to press itself against the inner wall of the bottle neck and to tighten the bottle. This expanding phase, which depends on the elasticity of the corks but also on the form of the bottle

neck and its variation, lasts only some minutes. If the bottles are laid on their sides immediately after corking, very often the restoring phase of the corks is not yet finished. Through the closing pressure, wine can then be pressed between the cork and the glass of the bottle neck. In this way, a liquid layer is built up, which later leads to leaking bottles, due to its capillarity effect.

It is, therefore, absolutely essential to plan for a sufficient restoring phase after corking (at least 5 minutes). In small companies, this can be achieved through keeping the bottles standing on a buffer table or a bottle trolley for a short period of time; in large companies this can possibly be done through a prolongation of the running time on the transportation lines before packaging.

With cool storage, the wines develop more slowly after bottling than they do at higher temperatures. Steady temperatures of 10–15 °C, if possible, are regarded as optimal for the storage of bottled wines. The humidity content of the air in the storage cellar is important as well. A relative humidity content of 60% should not be exceeded with labelled bottles; otherwise, soaked paper of labels and cartons showing mould growth may occur.

Traditionally, wine bottles are stored on their sides (horizontally). However, numerous tests over the past few years have shown that an upright (vertical) storage has significant advantages. For instance, the losses of wine during storage time are clearly lower and the bottles are in most cases 'cleaner'. A drying out of the corks does not take place. Moreover, in upright standing bottles humidity also increases during storage. This is achieved through the saturated atmosphere in the headspace of the bottles.

In sensory comparisons between wines being stored in bottles standing upright and those stored on their sides, the wines which were stored upright were often preferred, due to their higher homogeneity from bottle to bottle. This can be explained by the more or less intensive extraction of cork components in horizontally stored wines from the naturally differing cork stoppers.

Tests concerning the sulphurous acid content showed that the tightening features of corks in bottles being stored upright were just as good as those of corks in bottles stored horizontally. A difference in the consumption of sulphurous acid, which could be measured as a consequence of an increased permeability of oxygen, could not be stated through the comparative tests. Unfortunately, in most companies, the bottle storage facilities are designed for storing the bottles on their sides, and they cannot be changed without a reduction in capacity.

13.8 Prevention of musty–mouldy off-flavours in the cellar environment

To reduce the occurrence of musty–mouldy off-flavours in wine, an important step, beside the quality management of the cork stopper production process, is the prevention of these off-flavours throughout the winemaking process. As already mentioned in the previous sections, the compounds responsible for musty off-flavours are mostly products of microbiological activities. Therefore, it is important

to avoid the growth of undesirable microorganisms, firstly on the grapes themselves and secondly on the oenological equipment. Residues of wine and lees have to be carefully removed from tanks, barrels and pipes. In particular, dead spots, for example in ball valves, are quite capable of being sources of microbiological spoilage. The growth of moulds, however, can cause contamination with geosmin and other mould-related compounds (see Section 13.2).

13.8.1 Cellar contamination with halogenated anisoles

Regarding the formation of musty and mouldy off-flavours in a cellar environment, precursor compounds with relatively high perception thresholds (e.g. 2,4,6-tribromophenol) are quite often broken down into very intensive smelling compounds (e.g. 2,4,6-tribromoanisole) (Chatonnet *et al.*, 2004). This leads to the most frequent problem in the cellar environment nowadays.

Contamination with halophenols and their corresponding anisoles, which are found quite frequently in cellar buildings, suggests that a high percentage of wines are already contaminated with haloanisoles before bottling. Barker *et al.* (2001) showed that wine corks are capable of being contaminated with TCA via air after only a short exposure of 24 hours. Whitfield *et al.* (1997) reported a high permeability of polyethylene packaging for TBA and thus demonstrated a contamination of packaged food (raisins) after a short exposure of the packages to a contaminated atmosphere. In other studies (Hesford and Schneider, 2002; Rudy and Scholten, 2006), elevated amounts of halogenated anisoles (e.g. TCA, TBA) were found in the sealing compound mass of screw caps, plastic wine stoppers, filter sheets, plastic and paper packaging material and wooden pallets.

The sources of these halophenols like TCA, TeCA (2,3,4,6 tetrachloroanisole), PCA and TBA in the cellar are mainly wooden parts like roof timber or wooden ceilings and also barrels which have been treated with halophenols as wood preservative (Chatonnet *et al.*, 1994). In particular, buildings that were built in the 1970s and 1980s tend to suffer from an elevated chloroanisole content in the air. This is due to the fact that the use of chlorophenols as protection agent for timber was quite usual until the end of the 1980s. Until the mid-1980s, PCP-containing formulations for the mould protection of wine barrels were also common (Rudy and Scholten, 2006).

Another source of halophenols is the use of tribromophenol as a replacement for chlorophenols as a wood preserver and as flame protection agent for wood (e.g. wooden pallets), different kinds of paint and plastics. High humidity in the cellar especially contributes to a microbial transfer of the used halophenol into an anisole. In this context, the use of high-pressure water blasters in the cellar is not without risk; very well cleaned cellars often suffer from this problem, especially if linked with insufficient ventilation (Rudy and Scholten, 2006). Chatonnet and co-workers (2004) reported a contamination with TBA of 12 ng/L in wines stored in oak barrels. In the same study, high levels of contamination were discovered in the oak barrels themselves (143–224 ng/g wood) and also in the atmosphere of the cellar (46–162 ng/g adsorbent) with even higher amounts of the precursor TBP. In this case, the wine, the fining agents as well as oenological equipment can be

Table 13.5 Contamination with halogenated anisoles in a German winery (wine or air; analysed by Geisenheim Research Center Section of Microbiology and Biochemistry)

Type of sample	2,4,6 Trichloroanisole (ng/L)	2,4,6 Tribromoanisole (ng/L)	2,4,6 Tribromophenol (ng/L)
Müller-Thurgau wine dry; no off-flavour	nd	nq	No data
Müller-Thurgau wine dry; musty-mouldy	nq	3.7	No data
Steel tank cellar (adsorbent)	nd	nq	18
Wood barrel cellar (adsorbent)	nd	nq	11
Storage room (adsorbent)	nd	20.0	20

nd = not detectable (<1 ng/L); nq = not quantifiable, traces.

Table 13.6 Contamination with halogenated anisoles in a German winery (air), effect of the TCA source removal (analysed by Geisenheim Research Center Section of Microbiology and Biochemistry)

Sampling of air	2,4,6 Trichloroanisole (ng/L)	2,4,6 Trichlorophenol (ng/L)	2,4,6 Tribromoanisole (ng/L)	2,4,6 Tribromophenol (ng/L)
Pressing room (absorbent)	1.3	nq	nq	<10
Storage room (absorbent)	1.7	<10	nq	73
Cellar (absorbent)	6.8	55	nq	22
Filling room (absorbent)	44.3	60	1.0	46
Filling room (absorbent)	40.4	72	1.2	59
Filling room (absorbent) after removal of the contamination sources	2.6	nd	nq	<10
Filling room (absorbent) after removal of the contamination sources	2.5	nd	nq	<10

nd = not detectable (<1 ng/L); nq = not quantifiable, traces.

contaminated if they come in contact with this spoiled air. A significant problem is that humans become accustomed to this musty odour, so that winemakers will not realise the problem because they do not perceive the odour (Rudy and Scholten, 2006).

Table 13.5 shows some data from a study at the Geisenheim Research Institute. A German winemaker had frequent problems with musty off-flavours in white wines bottled in screw cap bottles. An analysis of the air inside the rooms of the

Table 13.7 Concentrations of halogenated anisoles in different contaminated wood samples (analysed by Geisenheim Research Center Section of Microbiology and Biochemistry)

Sample type	2,4,6 Trichloroanisole (ng/L)	2,4,6 Trichlorophenol (ng/L)	2,4,6 Tribromoanisole (ng/L)	2,4,6 Tribromophenol (ng/L)
Roof timber filling room (wood)	10.1	>200	3.1	102
Wooden ceiling filling room (wood)	37.0	>5000	nd	nq
Wooden wall casing (wood)	23	>1000	1.5	22
Wall filling room (render)	50	>1000	nq	nq

nd = not detectable (<1 ng/L); nq = not quantifiable, traces.

Table 13.8 Concentrations of halogenated anisoles in different wine samples from a German winery (analysed by Geisenheim Research Center Section of Microbiology and Biochemistry)

Sample type	2,4,6 Trichloroanisole (ng/L)	2,4,6 Trichlorophenol (ng/L)	2,4,6 Tribromoanisole (ng/L)	2,4,6 Tribromophenol (ng/L)
Pinot blanc barrel sample	nq	23	nd	nq
Silvaner barrel sample	nd	10	nd	21
Superior table wine barrel sample	nd	<10	nd	nq
Pinot blanc bottled	2.2	<10	nd	10
Silvaner bottled	2.3	12	nd	17
Superior table wine bottled	2.3	14	nd	16

nd = not detectable (<1 ng/L); nq = not quantifiable, traces.

winery revealed a contamination with TBA in a non-ventilated storage room (20 ng/L adsorbent), where the screw caps had been stored along with wooden pallets and other packaging material.

Due to the fact that all wine treatment agents were stored in non-contaminated rooms, the vector for the contamination in this case was the screw caps which had been contaminated via the atmosphere of the storage room. The source of the TBA in this case was TBP-treated wooden pallets.

In another case, white wines of a German winery suffered from elevated TCA contents after bottling. Due to the fact that the winery has been using plastic stoppers for 5 years without problems, an investigation of the sources of the suddenly appearing contamination was conducted. By investigation of

different cellar materials and also by sampling the air, the source could be detected in the filling room (Tables 13.6 and 13.7). A wooden wall casing and a wooden ceiling were found to contain high amounts of TCA and TCP. Regarding the sudden appearance of the problem, it was found that a newly installed boiler in the filling room was raising the room temperature. Additionally, the humidity was raised by the steam disinfection of the bottling machine and, together with insufficient ventilation, caused an increased growth of mould on the wooden parts. However, this also increased the formation of TCA from TCP by moulds along with the increased degassing of those compounds due to the raised temperature.

Table 13.8 shows that the wines did not contain TCA in the barrel but did so after bottling. Due to the fact that no contaminated material had been used for the bottling, it can be assumed that the contamination in this case was due to the contact the wine had with the contaminated air by pumping it into the filling tank. To improve the situation, all wooden parts in the filling room were removed and the room was ventilated. Another analysis of the air showed that the situation had improved significantly (Table 13.6).

13.8.2 Avoidance of haloanisole contamination in the cellar environment

Due to the results of this and other studies (Hesford and Schneider, 2001; Chatonnet *et al.*, 2004; Rudy and Scholten, 2006) regarding the avoidance of a halophenol and haloanisole contamination in a winery, the following recommendations can be made.

- No use of halophenol-treated wood in the cellar environment, meaning:
 - avoidance of wooden pallets especially in wet rooms;
 - use of halophenol-free plastic or stainless steel pallets;
 - removing of contaminated timber from the cellar;
 - removing of contaminated barrels, tanks.
- Demanding halophenol- and haloanisole-free materials for use in the cellar (certification of the materials), including packaging materials, barrels, silicon bungs, etc.:
 - no use of halophenol-containing paints and plastic in the cellar;
 - avoiding the use of chlorine-containing detergents in the cellar environment;
 - taking care of a good ventilation to lower the concentration of the contaminants in the air and also to lower the humidity;
 - periodical analysis of the atmosphere of the winery to check if the air is free of halophenols and haloanisoles.

Additionally it has to be recommended that:

- all wine treatment agents as well as wine stoppers are stored in a neutral, well-ventilated room and are removed from wooden pallets.

13.9 Methods to reduce musty off-flavours in contaminated wines

Regarding the removing of musty off-flavours from tainted wines, different approaches have been suggested to remove or to deodorise the off-flavour. As an example, only slightly affected wines were blended with unaffected wines to lower the concentration of the responsible compounds below the detection threshold. This strategy, however, cannot be recommended due to the fact that it risks spoiling an even greater amount of wine. Roasted wheat or mustard seeds (already used for other aroma defects), have also been tried out to deodorise musty-smelling wines but with no satisfactory results.

13.9.1 Adsorbent charcoal

Some producers report quite satisfying results with the use of activated charcoal as fining agent on the removal of musty off-flavours. The maximum allowable dose used to treat a wine is 100 g/hL inside the EU. Depending on the strength of the off-flavour, high amounts of charcoal have to be used, which also leads to a diminishing of the fruitiness of the treated wines. This means, therefore, that only wines that are just slightly affected by musty off-flavours can be deodorised by the use of charcoal; otherwise, the amount of charcoal needed is so high that the wine will lose its aroma.

13.9.2 Polyethylene

Whitfield *et al.* (1997) reported a high affinity of polyethylene (PE) film for halogenated anisoles. Capone and colleagues (1999) confirmed an absorbance of chloroanisoles by paraffin wax and an even higher absorbance of chloroanisoles by polyethylene film from tainted wines. They reported that a greater number of chlorine atoms located in the anisole molecule leads to increased efficiency of the anisole uptake. They reported the reduction of TCA and TeCA by 90% and 97%, respectively in 2 L of wine stored in a polyethylene wine bladder after 4 days storage time. Furthermore, they stated that treating a litre of chloroanisole-tainted wine with a polyethylene film with dimensions $500 \text{ cm}^2 \times 50 \text{ }\mu\text{m}$ thickness was capable of removing 85% of TeCA.

Concerning the effect of treatment with a polyethylene piece of the same size on the aroma of non-contaminated wines, they stated that no discernible effect could be reported for red and oak fermented white wines, whereas a lessening of the floral and fruity character of a Riesling wine was found. Nevertheless, their findings indicated that the use of polyethylene film for the removal of chloroanisoles from wine offers an effective treatment for tainted wines. Regarding the effect of this treatment on the aroma compounds of the wines, it has to be stated that especially fruity wines like Riesling suffer from contact with polyethylene.

13.9.3 Inert yeast particles

Fernandez and colleagues (2007) tested the effect of high adsorptive yeast cells on

chloroanisole-tainted wines. In laboratory trials, they tested the treatment of wines that were spoiled with 5 ng/L TCA and 25 ng/L TeCA and used doses of 0, 100, 200, 400 and 800 mg/hL yeast cells. The treatment time was 48 hours and the wines were stirred three times per day. After the treatment, they reported a reduction of 27% TCA, 55% TeCA and 72% PCA for the wines treated with 400 mg/hL yeast cells. Regarding the different doses used to treat the wines, they showed that the removal of the anisoles was proportional to the amount of adsorbent used. In addition, in large-scale tests, the same effects were also observed, which proved that the wine aroma was not negatively affected by the treatment.

13.9.4 Filtration with special filter sheets

In a study at the Geisenheim Research Center, a special depth filter sheet (Fibrafix TX-R®) was tested, which was claimed to be able to remove TCA and TBA from wine, without negatively affecting the wine flavour (Jung *et al.*, 2008). The use of these special filter sheets is similar to conventional filter sheets: they are rinsed before filtration and sterilisation can be conducted with steam (125 °C). A flow rate of 350 L/m²/h is recommended by the manufacturer. The filter layer consists of refined and bleached cellulose as well as a small quantity of polyamidamine (below 3%) to increase the wet-strength. A special aluminium silicate serves as an absorbent in the filter sheets, binding TCA and/or TBA, and removes those compounds from the wine. Additionally Eder *et al.* (2008) demonstrated the removal of geosmin by these filter sheets.

Filtration effect on TCA/TBA contents

Throughout the different preliminary and main test phases conducted by the study of Jung *et al.* (2008), different wines were contaminated with different amounts of TCA and TBA. Afterwards, the filtration trials were performed with 20 × 20 cm filter sheets in a Seitz stainless-steel sheet filter. The filtration surface in all tests was 0.2 m², and the filtration rate was around 60 L per hour.

Figures 13.2 and 13.3 show the effect of the filtration on wines spiked with different amounts of TCA and TBA. Regardless of the amount of TCA before the filtration, a reduction of TCA as well as TBA was achieved, whereas a filtration with a conventional filter sheet had no effect on TCA removal (Fig 13.4.).

Sensorial effect of the filtration

The sensorial tests conducted with 15 trained panellists confirmed a TCA-reducing effect of the filtration. The panellists were unable to differentiate between a wine contaminated with 20 ng/L TCA filtered with the special filter sheet and a non-contaminated wine filtered with the same sheet in triangle testing. A comparison of a non-contaminated wine filtered with a conventional filter sheet and the same wine filtered with the special filter sheet revealed that the wines could be differentiated by the panellists, but no preference could be ascertained. An additional analysis of the aroma compounds of these wines revealed only slight changes (Fig. 13.5).

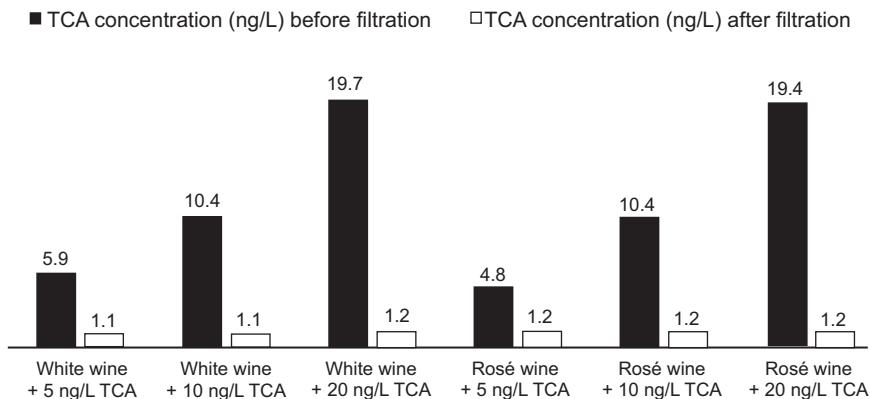


Fig.13.2 Removal of TCA (2,4,6 trichloroanisole) with special filter sheets (Jung *et al.*, 2008).

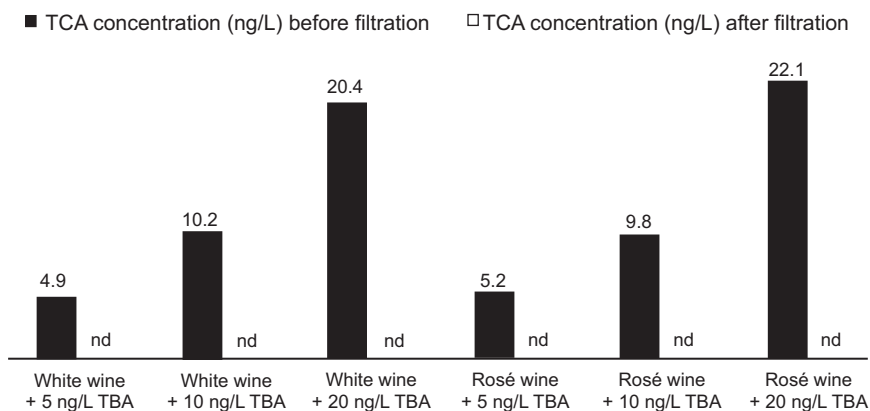


Fig. 13.3 Removal of TBA (2,4,6 tribromoanisole) with special filter sheets (Jung *et al.*, 2008); nd = not detectable.

Capacity of the filter sheets

Based on technical data provided by the manufacturer claiming a possible volume of 2000 L/m² to be filtered, regarding the filtration surface of 0.2 m² used in the test trials, the point where the TCA/TBA absorbance of the sheet should be depleted had to be after about 400 L of wine. Figure 13.6 shows results of a test filtration of 500 L of Müller-Thurgau wine, contaminated with 20 ng/L TCA. After a filtration volume of 400 L, the TCA concentration increased above 5 ng/L and the wines were sensorially described as being ‘musty–mouldy’, which corroborated the data given by the manufacturer.

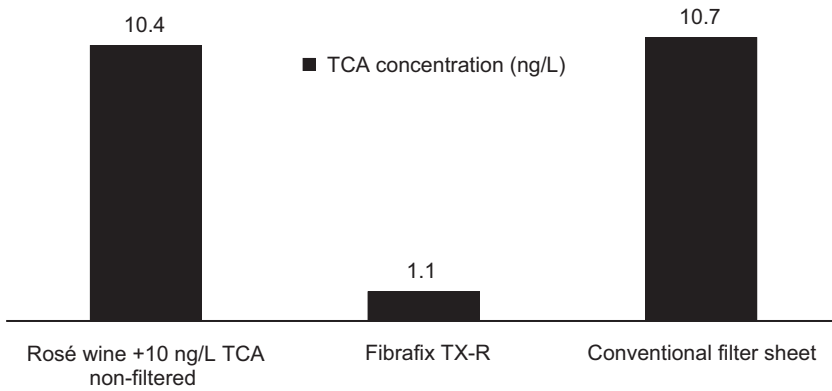


Fig. 13.4 Comparison of TCA (2,4,6 trichloroanisole) removal using a conventional filter sheet vs TX-R filter sheet (Jung *et al.*, 2008).

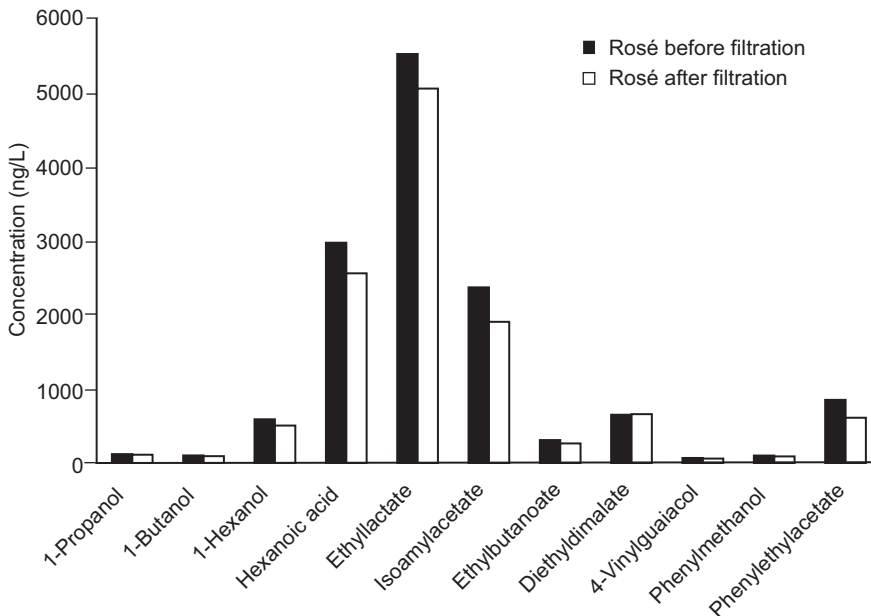


Fig. 13.5 Effect of filtration on the contents of aroma compounds in rosé wine before and after filtration (Jung *et al.*, 2008).

Aluminium ion migration

Given the fact that the active ingredient in the filter sheets is an aluminium silicate, it was necessary to check whether there is a migration of aluminium ions into the wine during the filtration process. Four samples out of the filtration test mentioned above along with an unfiltered sample were analysed by a certified analytical laboratory. The data (see Fig. 13.7) showed only very small changes in the aluminium concentration, indicating that product safety is not affected by the

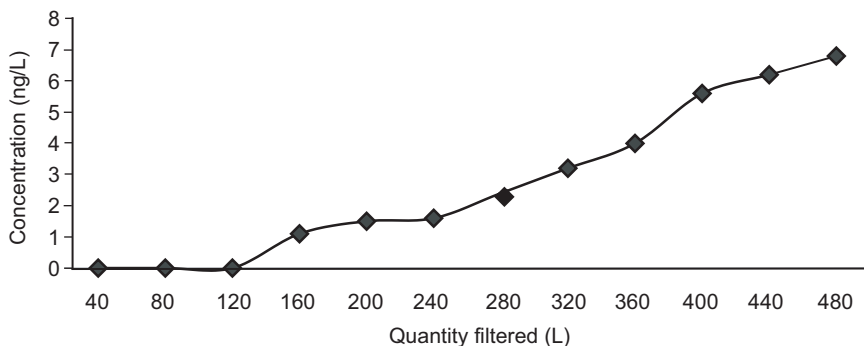


Fig. 13.6 Efficiency capacity of 2,4,6 trichloroanisole (TCA) removal (contamination 20 µg/L TCA; filter-surface 0.2 m²; rate 60 L/h) (Jung *et al.*, 2008).

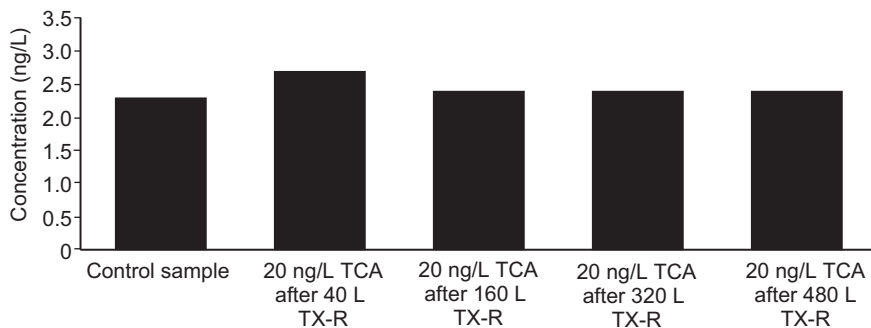


Fig. 13.7 Aluminium ion migration during filtration of a white wine with Filtrox Fibrifix TX-R® (filtration surface 0.2 m²) (Jung *et al.*, 2008).

migration of aluminium ions from the filter sheets into the wine in light of a legally permitted (in Germany) maximum of 8 mg/L aluminium.

13.10 References

- Allard A-S, Remberger M and Neilson A H (1987), Bacterial o-methylation of halogen-substituted phenols. *Appl. Environ. Microbiol.* **53**, 839–845.
- Alvarez-Rodriguez, M., Lopez-Ocana L., Lopez-Coronado J.M., Rodriguez E., Martinez M.J., Larriba G and Coque J-J R (2002), Cork taint of wine: role of the filamentous fungi isolated from cork in the formation of 2,4,6-trichloroanisole by O-methylation of 2,4,6-trichlorophenole. *Appl. Environ. Microbiol.* **68**, 5860–5869.
- Alvarez-Rodriguez M L, Belloch C, Villa M, Uruburu F, Larriba G and Coque J-J R (2003), Degradation of vanillic acid and production of guaiacol by microorganisms isolated from cork samples. *FEMS Microbiol. Lett.* **220**, 49–55.
- Barker D A, Capone D L, Pollnitz A P, McLean H J, Francis I L, Oakey H and Sefton M A (2001), Absorption of 2,4,6-trichloroanisole by wine corks via the vapour phase in an enclosed environment. *Aust. J. Grape Wine Res.* **7**, 40–46.

- Buser H., Zanier C and Tanner H (1982), Identification of 2,4,6-trichloroanisole as a potent compound causing cork taint. *J. Agric. Food Chem.* **30**, 359–362.
- Capone D L, Skouroumounis G K, Barker D A, McLean H J, Pollnitz A P and Sefton M A (1999), Absorption of chloroanisoles from wine by corks and by other materials. *Aust. J. Grape Wine Res.* **5**, 91–98.
- Chatonnet P, Guimberteau G, Dubourdieu D and Boidron J N (1994), Nature et origin des odeurs de moisi dans les caves. Incidence sur la contamination des vins. *J. Int. Sci. Vigne Vin.* **28**, 131–151.
- Chatonnet P, Bonnet S, Boutou S and Labadie M (2004), Identification and responsibility of 2,4,6-tribromoanisole in musty corked odors in wine. *J. Agric. Food Chem.* **52**, 1255–1262.
- Curtis R F, Land D G, Griffiths M N, Gee M G, Robinson D, Peel J L, Dennis C and Gee J M (1972), 2,3,4,6-tetrachloroanisole: association with musty taint in chickens and microbiological formation. *Nature* **235**, 223–224.
- Czerny M and Grosch W (2000), Potent odorants of raw Arabica coffee. Their changes during roasting. *J. Agric. Food Chem.* **48**, 868–872.
- Daniels-Lake B, Prange K, Gaul S, Mc Rae K and Antueno R (2007), A musty off flavor in Nova Scotia potatoes is associated with 2,4,6-trichloroanisole released from pesticide-treated soils and high soil temperature. *J. Amer. Soc. Hort. Sci.* **132**, 112–119.
- Darriet P, Pons M, Lamy S and Duboudieu D (2000), Identification and quantification of geosmin, an earthy odorant contaminating wines. *J. Agric. Food Chem.* **48**, 4835–4838.
- Duncan B (1995), Closure quality control at Southcorp Wines. In Leske P A and Eglinton J M (eds), *Proceedings of the ASVO Oenology Seminar on Corks and Closures*. Australian Society of Viticulture and Oenology, Adelaide, SA, 29–30 (cited by Simpson *et al.* 2004).
- Eder R, Hütterer E-M, Weingart G and Brandes W (2008), Verringerung der Gehalte an 2,4,6-Trichloroanisole und Geosmin in Wein mittels Spezialfilterschichten. *Mitteilungen Klosterneuburg* **58**, 12–16.
- Fernandez O, Fauveau C, Pellerin P, Puech C, Vuchot S. Transl.: Früh P (2007), Verwendung von hochadsorptiven Hefezellwänden zur Entfernung von Kork- und Mufftönen, sowie zur Verringerung des Ochratoxin A-Wertes. Available at: http://www.keller-mannheim.de/fileadmin/pdf/getraenke_deutsch/downloads/DSM-Artikel_Hochadsorptive_Hefezellwaende.pdf (accessed March 2010).
- Fiedler H., Hilpert M., Hub M and Hutzinger O (1996), *Handbuch Altlasten und Grundwasserschadensfälle Stoffbericht Pentachlorphenol*. Landesanstalt für Umweltschutz, Baden-Württemberg, Karlsruhe, Germany. p 9 ff.
- Gerber NN and Lechevalier H A (1965), Geosmin an earthy-smelling substance isolated from *Actinomyces*. *Appl. Microbiol.* **13**, 935–938.
- Griffiths N M (1974), Sensory properties of the chloroanisoles. *Chem. Senses Flavor* **1**, 187–195.
- Hervé E, Price S, Burns G and Weber P (2000), Chemical analysis of TCA as a quality control tool for natural corks. Available at: <http://www.corkqc.com/currentresearch/CorkTaint/ETS%20CQC-SPME.pdf> (accessed March 2010).
- Hesford F and Schneider K (2002), Entstehung von Korkton im Wein. *Schweiz. Z. Obst-Weinbau* 16/02, 415–417.
- Jung R and Zürn F (2007), Geisenheimer Testmethoden 3. Auflage (Teil 1 und 2), Hrsg. Forschungsanstalt Geisenheim Fachgebiet Kellerwirtschaft Deutscher Kork-Verband e.V., Bielefeld 01/2001.
- Jung R, Schaefer V, Christmann M, Hey M, Fischer S and Rauhut D (2008), Removal of 2,4,6-trichloroanisole (TCA) and 2,4,6-tribromoanisole (TBA) from wine. *Mitteilungen Klosterneuburg* **58**, 58–67.
- Kennison K R, Wilkinson K L, Williams H G, Smith J H and Gibberd M R (2007), Smoke-derived taint in wine: Effect of postharvest smoke exposure of grapes on the chemical composition and sensory characteristics of wine. *J. Agric. Food Chem.* **55**, 10897–10901.
- Kugler D and Rapp A (1997), Bildung und Entwicklung von Inhaltsstoffen in Korkborke

- während des Herstellungsprozesses von Flaschenkorken. *Dtsch. Lebensm. Rundsc.* **93**, 174–177.
- La Guerche S, Dauphin B, Pons M, Blancard D and Darriet P (2006), Characterisation of some mushroom and earthy off-odors microbially induced by the development of rot on grapes. *J. Agric. Food Chem.* **54**, 9193–9200.
- Maujean A, Millery P and Lemaesquier H (1985), Explications biochimiques et metaboliques de la confusion entre gout de bouchon et gout de mois. *Rev. Fr. Oenol.* **24**, 55–62.
- Mazzoleni V, Caldentey P, Careri M, Mangia A and Colagrande O (1994), Volatile components of cork used for production of wine stoppers. *Am. J. Enol. Vitic.* **45**(4), 401–406.
- Mazzoleni V and Maggi L (2007), Effect of wine style on the perception of 2,4,6-trichloroanisole, a compound related to cork taint in wine. *Food Res. Int.* **40**, 694–699.
- Mottram D S, Patterson R L S and Warrilow E (1984), 2,6-Dimethyl-3-methoxypyrazine: a microbiologically-produced compound with an obnoxious musty odour. *Chem. Ind. June*, 448–449.
- Pena-Neira A, Fernandez de Simon B, Garcia-Vallejo M C, Hernandez T, Cadahia E and Suarez J A (2000), Presence of cork-taint responsible compounds in wines and their cork stoppers. *Eur. Food Res. Technol.* **211**, 257–261.
- Pfeifer, O (2002), *Spurenanalyse halogenierter Phenylmethylether (Anisole) in der aquatischen Umwelt*. Dissertation, Universität Ulm, Germany, 12–20.
- Prescott J, Norris L, Kunst M and Kim S (2005), Estimating a ‘consumer rejection threshold’ for cork taint in white wine. *Food Qual. Pref.* **16**, 345–349.
- Ribéreau-Gayon P, Glories Y, Maujean A and Dubourdiou D (2006), *Handbook of Enology Volume 2. The Chemistry of Wine Stabilization and Treatments*, 2nd edn. Wiley, Chichester, UK.
- Riboulet J-M (1982), *Contribution a l’etude chimique et microbiologique des “gouts de bouchon” dans les vins*. These, Universite de Bordeaux II, France. (cited by Simpson *et al.*, 1986).
- Rudy H and Scholten G (2006), Korkgeschmack ohne Kork? *Der Deutsche Weinbau* 15/06, 24–27.
- Rudy H and Scholten G (2007), Diagnose: Korkgeschmack!? *Das Deutsche Weinmagazin* 3/07, 16–18.
- Saxby M J, Reid W J and Wragg G S (1992), *Index of Chemical Taints*. Leatherhead Food, Leatherhead, UK.
- Sefton M and Simpson R (2005), Compounds causing cork taint and the factors affecting their transfer from natural cork closures to wine – a review. *Aust. J. Grape Wine Res.* **11**, 226–240.
- Simpson R F, Amon J M and Daw A J (1986), Off-flavour in wine caused by guaiacol. *Food Technol. Aust.* **38**(1), 31–33.
- Simpson R F, Capone D L and Sefton M A (2004), Isolation and identification of 2-methoxy-3,5-dimethylpyrazine, a potent musty compound from wine corks. *J. Agric. Food Chem.* **52**, 5425–5430.
- Simpson R and Sefton M (2007), Origin and fate of 2,4,6-trichloroanisole in cork bark and wine corks. *Aust. J. Grape Wine Res.* **13**, 106–116.
- Soleas G J, Yan J, Seaver T and Goldberg D M (2002), Method for the gas chromatographic assay with mass selective detection of trichloro compounds in corks and wines applied to elucidate the potential cause of cork taint. *J. Agric. Food Chem.* **50**, 1032–1039.
- Suprenant A and Butzke C E (1997), Implications on sensory quality control of cork stoppers. In: Henick-Kling T, Woly TE and Harkness EM (eds), *Proceedings of the Fourth International Symposium on Cool Climate Viticulture and Enology*. New York State Agricultural Experimental Station, Geneva, NY, 70–74.
- Sweetman J A and Simmons M S (1980), The production of bromophenols resulting from the chlorination of waters containing bromide ion and phenol. *Water Res.* **14**, 287–290.
- Tanner H, Zanier C and Buser H (1981), 2,4,6-Trichloroanisole: Eine dominierende Komponente des Korkgeschmacks. *Schweiz. Z. Obst-Weinbau* **117**, 97–103.

- Watanabe I, Kashimoto T and Tatsukawa R (1983), Polybrominated anisoles in marine fish, shellfish and sediments in Japan. *Arch. Environ. Contam. Toxicol.* **12**, 615–619.
- Whitfield F, Hill J and Shaw K (1997), 2,4,6-tribromoanisole: a potential cause of mustiness in packaged food. *J. Agric. Food Chem.* **45**, 889–893.
- Whitfield F B (1998), Microbiology of food taints. *Int. J. Food Sci. Technol.* **33**, 31–51.
- Zierler B, Siegmund B and Pfannhauser W (2004), Determination of off-flavour compounds in apple juice caused by microorganisms using headspace solid phase microextraction-gas chromatography-mass spectrometry. *Anal. Chim. Acta* **520**, 3–11.

14

Ladybug (Coccinellidae) taint in wine

A. Botezatu and G. Pickering, Brock University, Canada

Abstract: Ladybug taint is a wine fault caused by the unintentional introduction of Coccinellidae beetles into the winemaking process, and is characterized by the presence of undesirable peanut, green pepper, asparagus, earthy, green beans and musty aromas and flavors in affected wine. This chapter reviews the causes of the taint, implications for quality, detection thresholds and tolerances. The chapter also discusses remediation options and future research directions.

Key words: ladybug taint, *Harmonia axyridis*, multicolored asian ladybeetle (MALB), *Coccinella septempunctata*, Coccinellidae, methoxypyrazines, wine fault, wine quality.

14.1 Introduction

The multicolored asian ladybeetle (MALB) *Harmonia axyridis* (Pallas) (Coleoptera: Coccinellidae) was first introduced in North America in 1916 from Asia (Japan and Korea), and on multiple occasions since, as a form of biocontrol against aphids and some small soft-bodied pests (Gordon, 1985; Chapin and Brou, 1991; Koch, 2003; Koch *et al.*, 2004). While it has proven very efficient in this respect, it has also extended its range into non-target regions and crops, including grape vineyards. In the vineyard, MALB adults aggregate on grape clusters during the harvest period, and are transported with the grapes into the winery (Galvan *et al.*, 2006). If present during crushing or subsequent winemaking operations, they can produce off-aromas and flavors in the finished wine known as *ladybug taint* (LBT) (Pickering *et al.*, 2004, 2007b, 2008b).

While there is anecdotal evidence that MALB have been adversely affecting



Fig. 14.1 Multicolored Asian ladybeetles, showing a variety of color morphs and attraction to grape and vines. The black markings on the yellow pronotum – often M-shaped – (center photo) are characteristic of the species (photographs courtesy of Kevin Ker and Ryan Brewster).

wine quality since the last century, the first widely reported link was noted in 2001 in some North American wine regions, where winemakers reported an unpleasant aroma and taste in wines from that vintage described as ‘burnt peanut butter’ and reminiscent of crushed ladybeetles (Pickering *et al.*, 2004). These reports coincided with the observation of large numbers of MALB in vineyards at the time of harvest (Pickering *et al.*, 2004) (Fig. 14.1).

These observations in 2001 marked the beginning of a concerted effort by winegrowers and the research community to investigate the association between MALB and wine quality, and to develop interventions to mitigate deleterious effects. These efforts have gained momentum in recent years, with the increasing awareness of LBT amongst the wine community and consumers, and the widening distribution of MALB in winemaking regions of the world.

14.2 Quality implications

Ladybug taint affects both the aroma and flavor profile of wine. This was first described by Pickering *et al.* (2004), where MALB were added to white and red grape must, and the subsequent wines evaluated using descriptive analysis. White wines were characterized by higher intensities of peanut, bell pepper and asparagus aromas and flavors, while red wines were rated higher for peanut, asparagus/bell pepper and earthy/herbaceous aromas and flavors. Sweet, acid and bitter tastes were affected in red wines, and lower intensities of fruit and floral descriptors were seen in both white and red wines. These effects are more pronounced at higher levels of MALB addition (Pickering *et al.*, 2004) (Fig. 14.2).

Similar sensory profiles have since been reported by Pickering *et al.* (2005) as well as Ross and Weller (2007). Interestingly, these taint characteristics appear to be relatively stable over time, with Pickering *et al.* (2005) finding similar sensory profiles in tainted wines produced with MALB, at bottling and after 10 months ageing. Dead MALB may also have the capacity to taint wine. Pickering *et al.* (2008b) added MALB to red must at various stages post-mortem to simulate incorporation of dead beetles with grapes during vintage operations; a phenomenon reported by growers after insecticide is applied to MALB-infested vineyards.

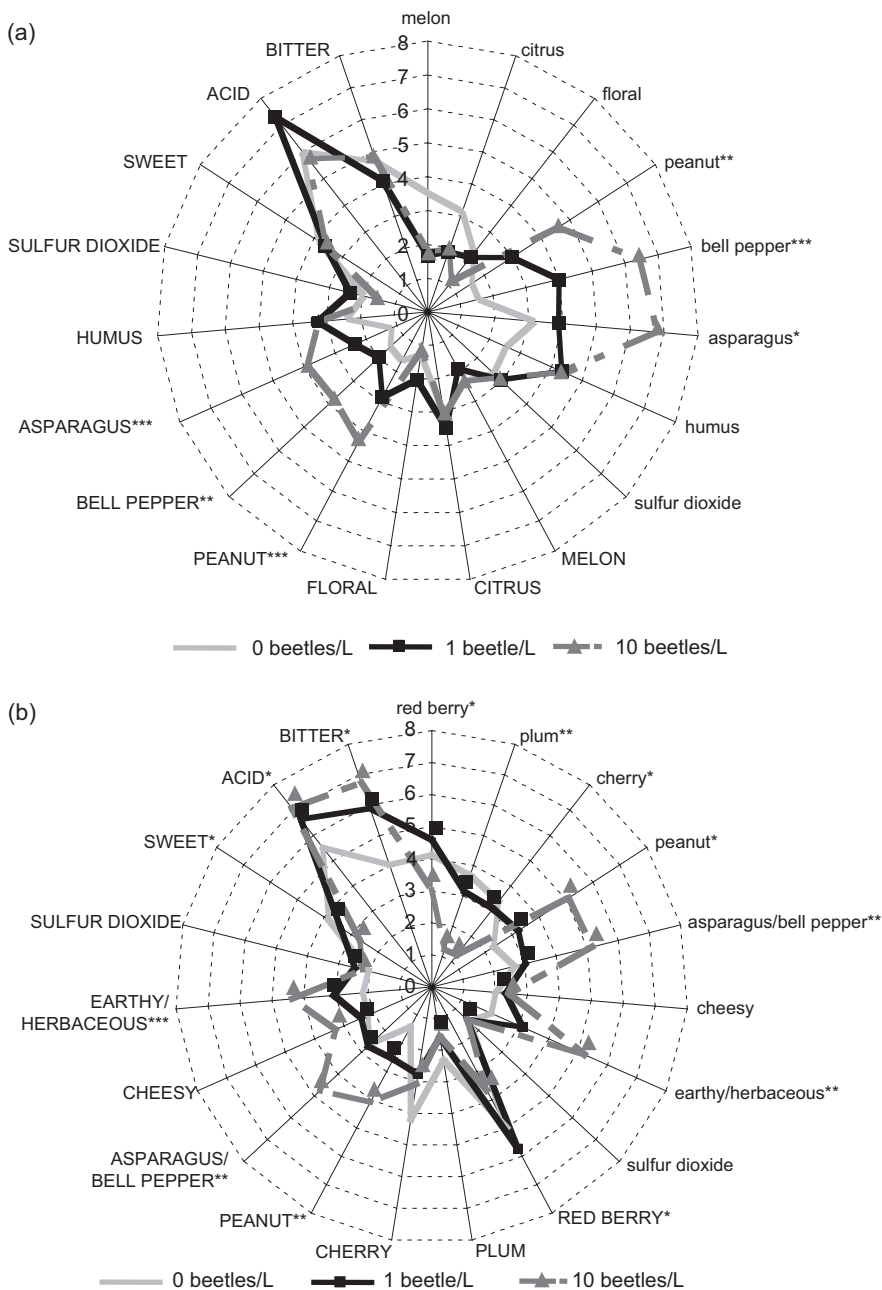


Fig. 14.2 Sensory intensity scores for white (a) and red (b) wine aroma (lowercase) and flavor (uppercase) attributes for three levels of addition of *Harmonia axyridis* beetles (MALB) to juice. Values are the mean ratings of six judges and triplicate assessments. *, ** and *** indicate significant differences at 0.05, 0.01 and 0.001, respectively (adapted from Pickering *et al.*, 2004).

Taint characteristics were more pronounced when live beetles were added; however, dead beetles were still able to elicit some negative impact on wine quality at 1-day post-mortem, but not at 3 days post-mortem and beyond.

Basic physical and chemical measures of wine quality, including color, pH, titratable acidity, ethanol, residual sugar and volatile acidity, are not significantly affected by the presence of MALB during wine production (Pickering *et al.*, 2008a). Ladybug taint is capable of profoundly affecting wine quality by directly influencing ortho-nasal aroma, retro-nasal aroma and taste modalities, as well as masking varietal and other desirable sensory attributes of white and red wine. Elucidation of the compound(s) responsible for LBT may assist with targeting interventions aimed at remediating the problem.

14.3 Causal compound(s)

Coccinellidae can emit a potent cocktail of odor-active compounds that may serve various behavioral functions, including defense, aggregation and mate-attraction. Of these compounds, alkyl-methoxypyrazines (MPs) have come under the closest scrutiny for their purported role in LBT. Multidimensional gas chromatography olfactometry–mass spectrometry (GCO–MS) has been used to determine the characteristic odorants emitted by MALB, and showed that 2-isopropyl-3-methoxypyrazine (IPMP) (Fig. 14.3c) was the most abundant and potent of the alkyl MPs released (Cai *et al.*, 2007). Sensory analysis correlated IPMP with aromas such as ‘peanuts’, ‘potatoes’, ‘peas’ and ‘earthy’ (Cai *et al.*, 2007). Three other pyrazines were also identified (2-isobutyl-3-methoxypyrazine – IBMP, 2-*sec*-butyl-3-methoxypyrazine – SBMP and 2,5 dimethyl-3-methoxypyrazine – DMMP) (Fig. 14.3b, c and d), and their aromas were defined, respectively, as

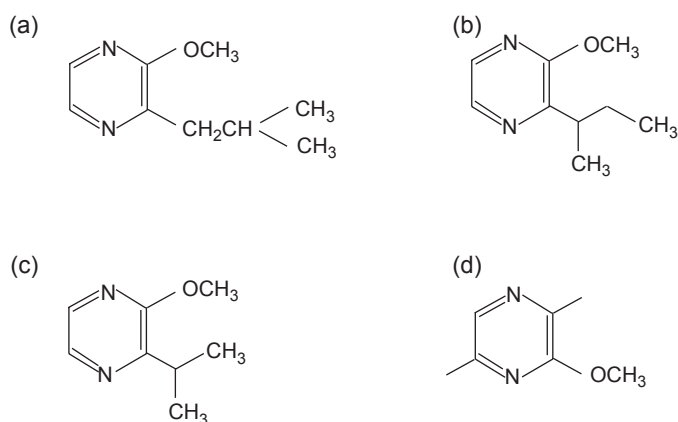


Fig. 14.3 Chemical structures of IBMP, SBMP, IPMP and DMMP: (a) 3-isobutyl 2-methoxypyrazine or IBMP; (b) 3-*sec*-butyl 2-methoxypyrazine or SBMP; (c) 3-isopropyl-2-methoxypyrazine or IPMP; (d) 2,5 dimethyl-3-methoxypyrazine.

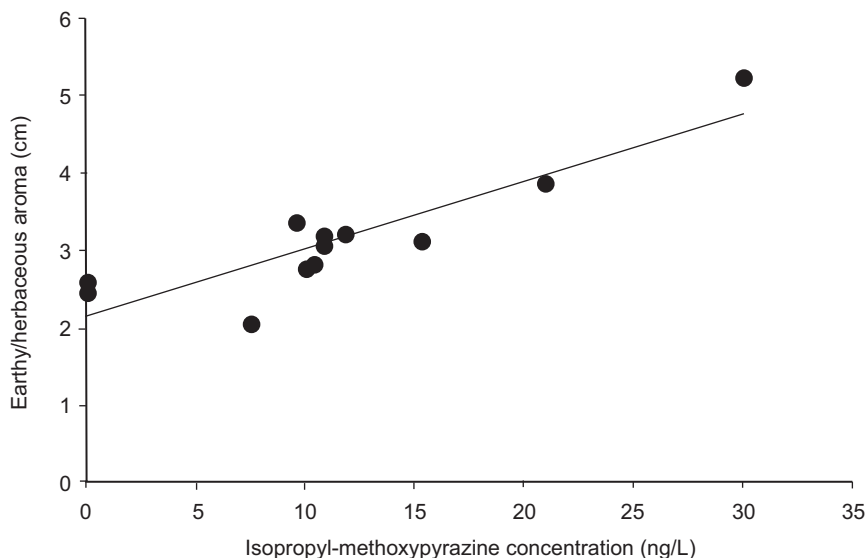


Fig. 14.4 Intensity of earthy/herbaceous aroma and isopropyl-methoxypyrazine concentration in 12 experimental red wines (adapted from Pickering, 2005).

‘peanut, potato, earthy, spicy’, ‘nutty, potato, peanut’ and ‘mouldy, earthy’ (Cai *et al.*, 2007), characteristics that are consistent with descriptors used for LBT-affected wines (Pickering *et al.*, 2004, 2005, 2007a,b; Ross and Weller, 2007). Cudjoe *et al.* (2005) also identified IPMP, IBMP and SBMP in the headspace above MALB, with IPMP the most prevalent.

The first direct evidence that MPs were responsible for LBT in wine came from Pickering *et al.* (2005), where IPMP concentrations increased in white and red wines fermented with MALB. The control wines (no added beetles) contained approximately 8 ng/L IPMP, and increased to 38 ng/L (white) and 30 ng/L (red) when wines were fermented with 10 beetles/L juice; concentrations well above the sensory threshold in wine (Pickering *et al.*, 2005). Later studies have also shown IPMP concentration in wine to increase in the presence of MALB (Pickering *et al.* 2008a,b). Additionally, the intensity of the earthy/herbaceous dimensions of LBT strongly correlates with IPMP concentration in red wine (Pickering *et al.*, 2005; Fig. 14.4). Finally, red wine spiked with 15 ng/L IPMP displays a very similar sensory profile to wine produced with the addition of MALB to juice (Pickering *et al.*, 2006b; Fig. 14.5).

One study failed to find an association between IPMP and MALB in wine. Galvan *et al.* (2008) analyzed wines produced from Frontenac grapes that had been artificially infested with MALB at levels of one, three and eight live beetles/kg grapes and from Leon Millot grapes harvested from vineyards with two different degrees of ‘natural’ MALB infestation (20% and 50% of the clusters infested with one or more *Harmonia axyridis* adults). No differences in concentration of IPMP were found between wines with different infestation

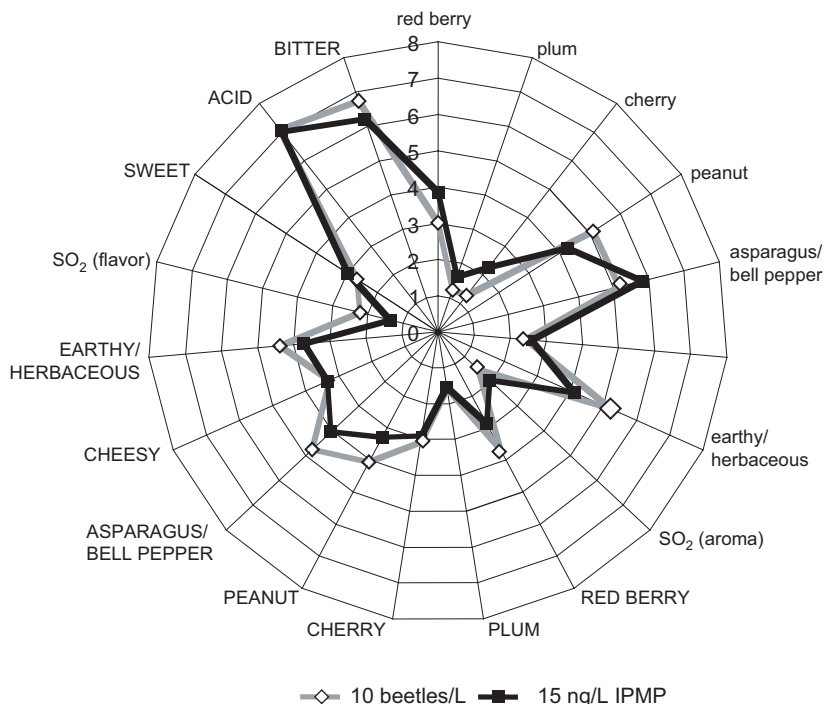


Fig. 14.5 Similarity of sensory profiles between red wine produced with the addition of 10 *Harmonia axyridis* (MALB) beetles/L juice and red wine spiked with 15 ng/L isopropyl-methoxypyrazine (IPMP). Lowercase signifies aroma attributes, and uppercase signifies flavor (adapted from Pickering *et al.*, 2006b).

levels. The authors noted that sensory analysis of the wines showed that the addition of MALB affected the sensory profiles. The discrepancy between the findings of this and previous studies in relation to IPMP may be related to the different analytical techniques employed.

An interesting dimension to the role of MPs in LBT is that this same class of compounds is produced by some wine grapes, and found in some varietal wines. MPs can confer typicality to some wines, such as Sauvignon blanc, Cabernet Sauvignon, Cabernet Franc and Carmenère, or they can have a negative impact on quality, by contributing ‘greenness’ and ‘earthiness’ (Allen *et al.*, 1991; Lacey *et al.*, 1991; Roujou de Boubée *et al.*, 2000, 2002; Sala *et al.*, 2005; Belancic and Agosin, 2007; Escudero *et al.*, 2007). However, in contrast with LBT, IBMP is normally the predominant grape-derived MP in wine, both with respect to concentration and odor-impact (Godelmann *et al.*, 2007).

DMMP, tentatively identified in the hemolymph of MALB (Cai *et al.*, 2007), is not known to be produced by grapevines. If it is confirmed in MALB-affected wine, it may prove to be a useful ‘diagnostic’ for quality control purposes.

14.4 Threshold and tolerances

An important question for both grape-growers and winemakers is ‘how much is too much?’ That is, what densities of MALB in the vineyard should trigger concern and intervention, and what concentrations of IPMP are necessary before LBT becomes noticeable in wine?

MPs in wine are very odor-active. While most of wines’ aromatic and volatile components are encountered in the $\mu\text{g/L}$ to mg/L range (Ferreira *et al.*, 2000; Ortega *et al.*, 2001), MP concentrations are measured in ‘parts per trillion’ or ng/L (Lacey *et al.*, 1991), and contribute to aroma at these trace concentrations (Maga, 1989; Kotseridis *et al.*, 1998; Romero *et al.*, 2006; Pickering *et al.*, 2007a). The detection threshold for IPMP in red wine is 1–2 ng/L (Maga, 1989; Pickering *et al.*, 2007a) and 0.3–1.6 ng/L in white wine (Pickering *et al.*, 2007a). Similar thresholds have been reported in grape juices (Pickering *et al.*, 2008c). Factors mediating IPMP threshold in wine include mode of evaluation (ortho- vs retro-nasal assessment), wine style and familiarity with the odorant (Pickering *et al.*, 2007a). Cudjoe *et al.* (2005) calculated the concentrations of MPs in individual MALB as approximately 27.5 $\mu\text{g/bug}$ for IPMP, 3.2 $\mu\text{g/bug}$ for IBMP and 2.6 $\mu\text{g/bug}$ for SBMP. How do these values translate into tolerance limits in the vineyard?

Pickering *et al.* (2007b) calculated that 1530 beetles/t of white grapes or 1260 beetles per tonne of red grapes were necessary before LBT would be detected in the subsequent wines. While these values may appear high, much greater densities of MALB have been observed in vineyards at harvest in some years. The authors caution that these limits may vary depending on grape variety and the processing methods used, and thus they recommend a more conservative limit of 200–400 beetles/t of grapes. This latter range is consistent with the limits on MALB placed by some wineries on their growers in North America. Using different methodology, Galvan *et al.* (2007) have derived similar values, estimating the threshold at which 10% of the population would be able to detect LBT as 1.9 beetle/kg grapes (equivalent to 1900 beetles/t) or 0.27 beetle/grape cluster for Frontenac grapes. In agreement, Ross *et al.* (2007) estimated the threshold to be 1.8 MALB/kg (1800 beetles/t) in Concord grapes.

There is no scientific evidence to support the position of ‘zero tolerance for ladybugs’ advocated by some wineries and wine agencies. Coccinellidae have long been part of the fauna found in, or in close proximity to, grape vineyards; they only pose a threat to wine and juice quality in years when densities become excessive.

14.5 Other Coccinellidae species

Cudjoe *et al.* (2005) sought to determine the relative MP composition of three Coccinellidae species common to North America: *Coccinella septempunctata* (seven-spot), *Harmonia axyridis* (MALB) and *Hippodamia convergens*. They detected IPMP in all three species, with the highest concentration (0.81 $\mu\text{g/mg}$) in

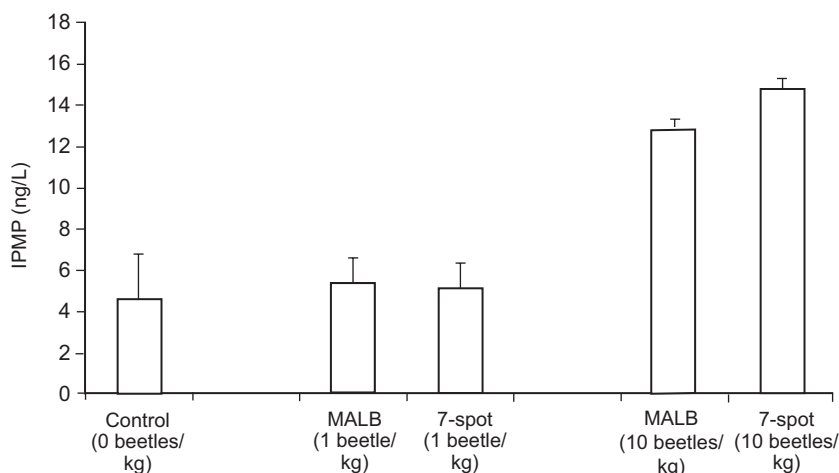


Fig. 14.6 3-Isopropyl-2-methoxypyrazine (IPMP) concentration in Cabernet Sauvignon wine produced with the addition of *Harmonia axyridis* (MALB) or *Coccinella septempunctata* (7-spot) added at grape crush.

Hippodamia convergens and the lowest in *Coccinella septempunctata* (0.008 µg/mg). SBMP and IBMP were above the limits of detection only in MALB and *Hippodamia convergens* (Cudjoe *et al.*, 2005).

The previous finding suggested that MP concentrations were not sufficiently high in non-MALB species to concern winegrowers. However, Pickering *et al.* (2010b) investigated the possible effects of seven-spot on wine quality under industrial conditions, by adding 0, 1 or 10 beetles/kg to Vidal and Cabernet Sauvignon grapes at crush and comparing MP concentrations and sensory profiles in the finished wines. As another control, MALB were added to separate batches of fruit at the same densities and processed in an identical manner. Regardless of species, the addition of beetles at the higher density resulted in similar changes to the sensory profiles of the wines, consistent with ladybug taint. The sensory results are supported by the IPMP concentrations, which were similar in both MALB- and seven-spot-spiked wines (Fig. 14.6). The inconsistency between this result and that of Cudjoe *et al.* (2005) may be due to procedural differences (live beetles in the Pickering study versus frozen and thawed beetles in the Cudjoe study) as well as the different analytical techniques employed.

Given that MALB or seven-spot may be the most prevalent Coccinellidae species in the majority of the world's wine regions, these results should widen the concern over the presence of ladybeetles in vineyards at harvest. However, it is unclear whether seven-spot reach the vineyard densities required to affect wine quality as often or as rapidly as MALB.

14.6 Post-harvest prevention and remediation

Given the significant impact LBT can have on the quality of wine, there has been



Fig. 14.7 Custom-designed shaker tables are effective at removing MALB from hand-harvested grapes.

considerable interest in finding preventive and remedial options for the taint, that will leave the desirable components of wine unaffected. Both conventional and unconventional approaches have been investigated.

After harvest, MALB can be partially removed from the grape clusters prior to de-stemming and crushing by using shaker tables or by soaking the grapes in water (where beetles float to the surface and can be removed). Both approaches are limited to hand-harvested fruit, and water treatment has the further disadvantage of diluting sugar concentration and potentially reducing quality. Shaker tables, however, have proven very effective, although throughput is limited (Fig. 14.7).

Clarification of white wine juice has proven beneficial. Kotseridis *et al.* (2008) have shown that juice clarification before fermentation can lead to a reduction in IPMP concentration of up to 50% in the subsequent wines compared to wines produced from unclarified juice. They also showed that the longer the settling period, the greater the decrease in IPMP for both naturally and bentonite clarified juices.

The possible effect of yeast strain in mediating IPMP concentrations and LBT in wine was investigated with Cabernet Sauvignon, where juice was spiked with 30 ng/L IPMP and fermented with commercially available yeast strains (Lalvin BM45, Lalvin EC1118, Lalvin ICV-D21 and Lalvin ICV-D80) (Pickering *et al.*, 2008d). Unexpectedly, IPMP concentration in wine fermented with BM45 *increased* by approximately 29% after fermentation, while remaining unaffected in wine made from the other strains. The highest intensity scores for taint-related attributes were in the D80 fermented wines (Fig. 14.8), while the lowest were in the D21 wines. These differences are likely due to the masking effects of other aroma and flavour attributes. The study recommends D21 as a good option for juice obtained from underripe fruit and fermentation of juice affected by LBT (Pickering *et al.*, 2008d).

Pickering *et al.* (2006a) have examined the effects of various commercial fining agents on reducing both the concentration of IPMP in LBT-affected wines and the sensory characteristics associated with LBT. Bentonite, activated charcoal, oak chips, deodorized oak chips and ultraviolet and white light were investigated. For white wine, activated charcoal reduced IPMP concentration by 34%, but did not decrease LBT intensity, highlighting the lack of selectivity on wine odorants of

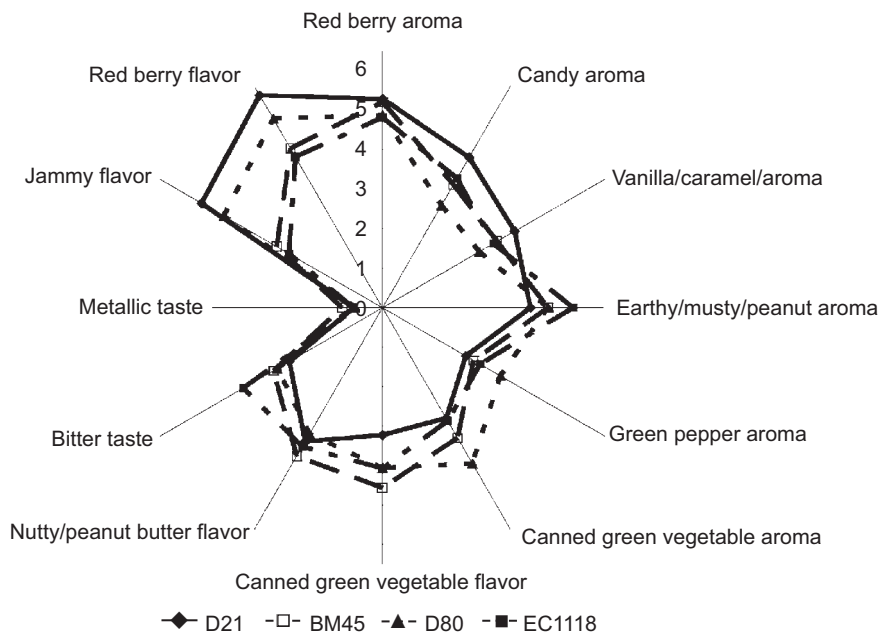


Fig. 14.8 Sensory attribute intensity scores for Cabernet Sauvignon wine made from juice spiked with 30 ng/L 3-isopropyl-2-methoxypyrazine and fermented by four different yeast strains (adapted from Pickering *et al.*, 2008d).

charcoal. Treatment with oak chips significantly reduced the asparagus flavor intensity and produced a trend of lower intensities for all LBT attributes. A trend for red wines to be more responsive to treatments than white wines was evident, particularly for flavor. As with white wine, reduced intensity for all LBT-related attributes was observed with the oak chip addition (Pickering *et al.*, 2006a). As LBT intensity was not affected by deodorized oak chips, the researchers ascribed the efficacy of oak chips to perceptual masking. Since not all wine styles lend themselves to fermentation and/or maturation with oak, other processing or remedial options were advocated.

Blake *et al.* (2009) investigated the effect of closure and packaging on IPMP and other odorants in Riesling and Cabernet Franc wines. All three MPs were affected by the closure or packaging type to some extent. Storage in Tetrapak cartons was the most effective packaging for lowering IPMP concentrations, with a 23% and 41% reduction in Riesling and Cabernet Franc, respectively. The authors speculate that this may have been due to migration of IPMP to the aluminium surface layer of the carton, with subsequent adsorption on the oxide layer. The study also reported a 10–21% decrease in IPMP concentration after 18 months bottle ageing in wines closed with synthetic molded closures (Blake *et al.*, 2009). The adsorptive capacity of synthetic corks for MPs has also been demonstrated by Pickering *et al.* (2010a). Soaking wine for 140 hours with synthetic closures resulted in decreases of 70%, 77% and 89% for IPMP, IBMP

and SBMP, respectively. Sensory analysis of the treated wine was not performed due to volume restrictions; however, given the limited number of additives and treatments permitted in the winemaking process, this novel application of synthetic closures or related polyethylene material warrants further investigation.

A more unorthodox approach to remediating LBT has been tested by Wilson *et al.* (2003) with the application of irradiation to tainted wine. In their pilot study, a dose of 100 Gy was reported to decrease the perceived intensity of LBT; however the authors do not elaborate on how other sensory attributes of the wines were affected by the irradiation treatment, particularly with respect to the likely oxidative damage to flavor from such a treatment.

14.7 Conclusions and future trends

Ladybug taint can appear in wine when *Harmonia axyridis* (MALB) and other Coccinellidae species are incorporated into grape juice prior to fermentation. The taint is caused by IPMP, originating in the hemolymph of the beetles. The sensory profile of a wine affected by LBT is characterized by 'asparagus', 'bell pepper', 'green beans', 'potatoes', 'herbaceous', 'musty' and 'earthy' descriptors, and varietal attributes are diminished. Several methods for taint remediation, including traditional fining agents, have been investigated, with mixed results. Juice clarification prior to fermentation and use of oak chips are helpful, while some packaging/closure materials, including polyethylene, show promise for reducing IPMP concentrations in affected wines. Development of remedial agents with high specificity to IPMP is needed; until such time, prevention of LBT by excluding Coccinellidae from wine processing is critical. As the same class of compounds responsible for LBT are also found in wine from grape sources, research into chemical markers that delineate MALB-affected wine would be advantageous. MP ratios and some alkaloid species are candidates for such an endeavor.

14.8 Abbreviations

DMMP	2,5 dimethyl-3-methoxypyrazine
IBMP	2-isobutyl-3-methoxypyrazine
IPMP	2-isopropyl-3-methoxypyrazine
GCO–MS	gas chromatography–olfactometry–mass spectrometry
LBT	ladybug taint
MALB	multicolored asian ladybeetle (<i>Harmonia axyridis</i> (Pallas))
MPs	3-alkyl-2-methoxypyrazines
SBMP	2- <i>sec</i> -butyl-3-methoxypyrazine

14.9 References

- Allen M, Lacey, M Harris, R and Vance Brown W (1991), 'Contribution of methoxypyrazines to Sauvignon blanc wine aroma', *American Journal of Enology and Viticulture*, **42**(2), 109–112.
- Belancic A and Agosin E (2007), 'Methoxypyrazines in grapes and wines of *Vitis vinifera* cv. Carmenere', *American Journal of Enology and Viticulture*, **58**, 462–469.
- Blake A, Kotseridis Y, Brindle ID, Inglis D, Sears, M and Pickering, GJ (2009) 'Effect of closure and packaging type on 3-alkyl-2-methoxypyrazines and other impact odorants of Riesling and Cabernet Franc wines', *Journal of Agriculture and Food Chemistry*, **57**, 4680–4690.
- Cai L, Koziel J and O'Neal M (2007), 'Determination of characteristic odorants from *Harmonia axyridis* beetles using *in vivo* solid-phase microextraction and multidimensional gas chromatography-mass spectrometry-olfactometry', *Journal of Chromatography A*, **1147**, 66–78.
- Chapin JB and Brou, VA (1991), '*Harmonia axyridis* (Pallas), the third species of the genus to be found in the United States (Coleoptera: Coccinellidae)', *Proceedings of the Entomological Society of Washington*, **93**, 630–635.
- Cudjoe E, Wiederkehr T and Brindle I (2005), 'Headspace gas chromatography-mass spectrometry: a fast approach to the identification and determination of 2-alkyl-3-methoxypyrazine pheromones in ladybugs', 2005, *The Analyst*, **130**, 152–155.
- Escudero A, Campo E, Farina L, Cacho J and Ferreira V (2007), 'Analytical characterization of the aroma of five premium red wines. Insights into the role of odor families and the concept of fruitiness of wines', *Journal of Agricultural and Food Chemistry*, **55**(11), 4501–4510.
- Ferreira V, López R and Cacho J (2000), 'Quantitative determination of the odorants of young red wines from different grape varieties', *Journal of the Science of Food and Agriculture*, **80**, 1659–1667.
- Galvan T, Burkness EC and Hutchinson WD (2006), 'Influence of berry injury on infestation of the Multicolored Asian Lady Beetle in wine grapes', *Plant Health Progress*, available at: <http://www.plantmanagementnetwork.org/pub/php/brief/2006/wine/> (accessed March 2010).
- Galvan T, Burkness E, Vickers Z, Stenberg P, Mansfield A and Hutchison W (2007), 'Sensory-based action threshold for Multicolored Asian Lady Beetle-related taint in winegrapes', *American Journal of Enology and Viticulture*, **58**, 518–522.
- Galvan T, Kells S and Hutchinson W (2008), 'Determination of 3-alkyl-2-methoxypyrazines in lady beetle-infested wine by solid-phase microextraction headspace sampling', *Journal of Agricultural and Food Chemistry*, **56**(3), 1065–1071.
- Godelmann R, Limmert S and Kuballa T (2007), 'Implementation of headspace solid-phase-microextraction-GC-MS/MS methodology for determination of 3-alkyl-2-methoxypyrazines in wine', *European Food Research and Technology*, **227**, 449–461.
- Gordon RD (1985), 'The Coleoptera (Coccinellidae) of America north of Mexico', 1985, *Journal of the New York Entomological Society*, **93**(1), 912.
- Koch RL (2003), 'The Multicolored Asian Lady Beetle, *Harmonia axyridis*: A review of its biology, uses in biological control, and non-target impacts', *Journal of Insect Science*, **3**, 32.
- Koch R, Burkness EC, Burkness WSJ and Hutchinson WD (2004), 'Phytophagous preferences of the Multicolored Asian Lady Beetle (Coleoptera:Coccinellidae) for autumn-ripening fruit', *Journal of Economic Entomology*, **97**, 539–544.
- Kotseridis Y, Anocibar Belouqui A, Bertrand A and Doazan JP (1998), 'An analytical method for studying the volatile compounds of Merlot noir clone wines', *American Journal of Enology and Viticulture*, **49**(1), 44–48.
- Kotseridis Y, Spink M, Brindle I, Pickering GJ, Blake A, Sears M, Chen X, Soleas G and Inglis D (2008), 'Quantitative analysis of 3-alkyl-2-methoxypyrazines in juice and wine

- using stable isotope labelled internal standard assay', *Journal of Chromatography A*, **1190**, 294–301.
- Lacey M, Allen M, Harris R and Brown W (1991), 'Methoxypyrazines in Sauvignon Blanc grapes and wines', *American Journal of Enology and Viticulture*, **42**(2), 103–108.
- Maga J (1989), 'Sensory and stability properties of added methoxypyrazines to model and authentic wines', in Charalambous B (ed.), *Flavors and Off-Flavors*, Elsevier, Amsterdam, the Netherlands, 61–70.
- Ortega C, Lopez R, Cacho J and Ferreira, V (2001), 'Fast analysis of important wine volatile compounds. Development and validation of a new method based on gas chromatography flame ionisation detection analysis of dichloromethane microextracts', *Journal of Chromatography A*, **923**, 205–214.
- Pickering GJ, Lin J, Riesen R, Reynolds A, Brindle A and Soleas G (2004), 'Influence of *Harmonia axyridis* on the sensory properties of white and red wine.', *American Journal of Enology and Viticulture*, **55**(2), 153–159.
- Pickering GJ, Yong L, Reynolds A, Soleas G, Riesen R and Brindle I (2005), 'The influence of *Harmonia axyridis* on wine composition and aging.', *Journal of Food Science*, **70**(2), S128–S135.
- Pickering GJ, Lin J, Reynolds A, Soleas G and Riesen, R (2006a), 'The evaluation of remedial treatments for wine affected by *Harmonia axyridis*', *International Journal of Food Science Technology*, **41**(1), 77–86.
- Pickering GJ, Lin Y and Ker K (2006b), 'Origin and remediation of Asian Lady Beetle (*Harmonia axyridis*) taint in wine.' In Dris R (ed.), *Crops: Growth, Quality and Biotechnology. III. Quality Management of Food Crops for Processing Technology*. WFL Publisher, Meri-Rastilan tie 3 C, Helsinki, Finland, 785–794. ISBN 952-91-8601-0.
- Pickering GJ, Karthik A, Inglis D, Sears N and Ker K (2007a), 'Determination of ortho- and retro-nasal detection thresholds for 2-isopropyl-3-methoxypyrazine in wine', *Journal of Food Science*, **72**(7), S468–S472.
- Pickering GJ, Ker K and Soleas GJ (2007b), 'Determination of the critical stages of processing and tolerance limits for *Harmonia axyridis* for 'ladybug taint' in wines', *Vitis*, **46**(2), 85–90.
- Pickering GJ, Spink M, Kotseridis Y, Brindle ID, Sears M and Inglis D (2008a), 'The influence of *Harmonia axyridis* morbidity on 2-isopropyl-3-methoxypyrazine in 'Cabernet Sauvignon' wine.', *Vitis*, **47**(4), 227–230.
- Pickering GJ, Spink M, Kotseridis Y, Brindle ID, Sears M and Inglis D (2008b), 'Morbidity of *Harmonia axyridis* mediates ladybug taint in red wine.', *Journal of Food, Agriculture and Environment*, **6**(3), 133–137.
- Pickering GJ, Karthik A, Inglis D, Sears M and Ker K (2008c), 'Detection thresholds for 2-isopropyl-3-methoxypyrazine in Concord and Niagara grape juice.', *Journal of Food Science*, **73**(6), S262–S266.
- Pickering GJ, Spink M, Inglis D, Brindle ID, Sears M, Beh A-L (2008d), 'Yeast Strain Affects 3-isopropyl-2-methoxypyrazine concentration and sensory profile in Cabernet Sauvignon Wine', *Australian Journal of Grape and Wine Research*, **4**(3), 230–237.
- Pickering GJ, Blake AJ, Soleas GJ and Inglis D (2010a), 'Remediation of wine with elevated concentrations of 3-alkyl-2-methoxypyrazines using cork and synthetic closures.', *Journal of Food, Agriculture and Environment*, **8**(2), 97–101.
- Pickering GJ, Inglis D and Carter N (2010b), 'Multiple *Coccinellidae* species can elicit ladybug taint in wine', 2010, *American Journal of Enology and Viticulture* (in preparation).
- Romero R, Chacon J, Garcia E and Martinez J (2006), 'Pyrazine content in four red grape varieties cultivated in a warm climate', *Journal International des Science de la Vigne et du Vin*, **40**(4), 203–207.
- Ross CF and Weller K (2007), 'Sensory evaluation of suspected *Harmonia axyridis*-tainted red wine using untrained panelists.', *Journal of Wine Research* [Online], **18**(3), 187–193.
- Ross C, Ferguson H, Keller M, Walsh D, Weller K and Spayd S (2007), 'Determination of

- ortho-nasal aroma threshold for Multicolored Asian Lady Beetle in Concord grape juice', *Journal of Food Quality*, **30**, 855–863.
- Roujou de Boubée D, Van Leeuwen C and Dubordieu D (2000), 'Organoleptic impact of 2-methoxy-3-isobutylpyrazine on red Bordeaux and Loire wines. Effect of environmental conditions on concentrations in grapes during ripening', *Journal of Agricultural and Food Chemistry*, **48**(10), 4830–4834.
- Roujou de Boubée D, Cumsille A, Pons M and Dubourdieu D (2002), 'Location of 2-methoxy-3-isobutylpyrazine in Cabernet Sauvignon grape bunches and its extractability during vinification', *American Journal of Enology and Viticulture*, **53**(1), 1–5.
- Sala C, Busto O, Guasch J and Zamora F (2005), 'Contents of 3-alkyl-2-methoxypyrazines in musts and wines from *Vitis vinifera* variety Cabernet Sauvignon: influence of irrigation and plantation density', *Journal of the Science of Food and Agriculture* [Online], **85**(7), 1131–1136.
- Wilson K, Boreham D and Moran G (2003), 'Applications of radiation within the wine industry', *Canadian Undergraduate Physics Journal*, **1**, 17–19.

Understanding and controlling non-enzymatic wine oxidation

P. A. Kilmartin, The University of Auckland, New Zealand

Abstract: The role of oxygen in wine is examined, with a focus upon polyphenol-mediated oxidation processes, and the changes in wine chemistry that follow. A distinction is drawn between benefits of oxygenation ascribed to red wines, such as improved colour, mouthfeel and removal of unwanted aromas, and negative effects on white wine quality, particularly related to wine browning and a loss of fruity aromas. Methods for determining the susceptibility of a wine to oxidation are also examined, along with the use of small molecule antioxidants (sulphur dioxide, glutathione and ascorbic acid) in winemaking.

Key words: wine oxidation, polyphenols, antioxidants, browning.

15.1 Introduction

The addition of oxygen to red wines during ageing is known to improve quality, at least initially, with effects including the removal of unwanted aromas, colour stabilisation and improvements in mouthfeel. Predicting which wines will benefit from oxygen exposure, and the correct dosage to employ, is difficult, and the underlying processes are still not well understood. On the other hand, oxygen additions rarely improve white wines where the preservation of fruity aromas is sought. Losses of aroma compounds, browning and the appearance of phenolic precipitates are particular problems associated with oxidation of white wines.

In this chapter, the science behind wine oxidation processes will be summarised, with a focus upon polyphenol-mediated processes, leading to a consideration

of oxidation effects in red and white wines and their impact on wine aroma and colour. Attention will also be given to ways of monitoring a wine's susceptibility to oxidation, the use of antioxidants and the role of winemaking processes in controlling wine oxidation. Insights provided by the placement of electrodes in wine (to monitor or modify oxidative substrates) will be included, reflecting the author's own interests in this area.

A number of the issues surrounding wine oxidation are examined in Volume 1 of this title; for example, volatile aroma compounds (Chapter 1), wine mouthfeel, texture and flavour (Chapter 2), wine colour (Chapter 3). In the current volume (Volume 2), oxidation issues such as microoxygenation (Chapter 8) and flavour deterioration (Chapter 16) are discussed. The topic of non-enzymatic wine oxidation has also been the subject of a number of reviews that can be consulted for further details and perspectives (Singleton, 2001; Cheynier, 2002; Danilewicz, 2003; du Toit *et al.*, 2006b; Waterhouse and Laurie, 2006; Li *et al.*, 2008).

15.2 Oxygen in wine

An initial consideration with non-enzymatic wine oxidation is how much exposure to O_2 a wine receives at various stages in the winemaking process. Larger oxygen doses are supplied through pump-overs, but slower rates of O_2 ingress are expected for wines in barrels or subject to micro-oxygenation in storage tanks; a certain amount of O_2 will persist even after bottle storage. Air-saturated wine can take up to 6 mL/L (8.6 mg/L) of O_2 at room temperature, with greater solubility at a lower temperature or when pure oxygen is used in place of air. Oxygen concentrations in wine can be measured using various electrochemical techniques such as the Clark electrode (Vidal *et al.*, 2004; Laurie *et al.*, 2008), while colorimetric methods have been developed to measure oxygen in wine bottles (Lopes *et al.*, 2005).

The build up of dissolved oxygen has been investigated following a wide range of winemaking operations (Castellari *et al.*, 2004). Racking at 15–20 °C led to O_2 increasing to a median value of 0.37 mg/L, but this value increased three-fold when the temperature of racking was lowered to 10 °C. On the other hand, agitation of wine during barrel ageing (*bâtonnage*) did not increase dissolved O_2 significantly, while the process of mixing wines from different casks raised the concentration to a median of 1.75 mg/L. Filtration typically led to increases of less than 0.3 mg/L, although centrifugation could enrich O_2 by 1.2 mg/L, and tartaric acid stability at –5 °C by 1.27 mg/L. Continuous stabilisation systems could saturate the wine with O_2 . Wine filling procedures gave quite variable concentrations of O_2 depending upon the technology used, with a median O_2 addition of 0.8 mg/L in this trial. For wines subject to micro-oxygenation in stainless steel tanks, a significant impact on dissolved O_2 occurred only with a supply rate of 5 mL/L/month, when concentrations of O_2 similar to barrel storage were recorded (c. 0.04 mg/L), although this value went up considerably during the cooler months of the year when the storage temperature was not controlled. Measurements made by a further group showed that pump-overs can introduce several mg/L of O_2 , while the concentration of

dissolved O_2 during controlled micro-oxygenation was difficult to measure, but values of 0.2–0.25 mg/L were determined for a red wine undergoing micro-oxygenation at 5 mL/L/month (Waterhouse and Laurie, 2006; Laurie *et al.*, 2008).

The presence of oxygen in the wine at bottling, and in the small ullage (headspace) above the wine (which can contain a few mg of O_2), has been examined in some detail over a period of several months after bottling (Vidal and Moutounet, 2006). These results show that the headspace can function as a reservoir for oxygen that gradually passes into the wine, pointing to continued oxidation processes for some time after bottling. Low but distinct concentrations of O_2 were present even in bottles of wine several years old. Further measures of headspace O_2 concentrations have shown that, in the absence of a vacuum or inert gas system, enough oxygen can remain following the bottling process to add a few mg/L of O_2 to the wine (particularly with screwcap closures). This amount is equivalent to several months of the oxygen addition expected through the closure (Kontoudakis *et al.*, 2008). Contributions will also come from oxygen contained within the natural cork structure, and released after bottling, and from oxygen entering the wine through the closure at a slow rate – less than 1 μ L of O_2 a day was found to enter through many wine bottle closures after the first 12 months of storage (Lopes *et al.*, 2006, 2007).

15.3 Polyphenol oxidation

The action of polyphenol oxidase (PPO) enzymes in juice has been well characterised, and shown to be very rapid in the presence of oxygen and prior to protection of the juice with sulphur dioxide (Singleton, 1987). PPO enzyme activity leads to the formation of reactive quinones that can combine with glutathione, various aroma compounds and other polyphenols to produce brown products (Cheynier *et al.*, 1989a, 1993). Oxygenation of red grape musts by additions of pure O_2 has been shown to have a number of effects on the finished wine, including a loss of phenolics such as anthocyanins and hydroxycinnamic acids, leading to a lower colour intensity and sensory differences (Castellari *et al.*, 1998).

Hyperoxidation of musts has also been employed to decrease phenolic concentration through their precipitation and thus lower the susceptibility of the finished wine to browning (Cheynier *et al.*, 1989b; Schneider, 1998; Ho *et al.*, 2000). However, issues remain about how the aroma profile of the wine will be modified for better or worse. On the other hand, the late harvesting of grapes, as occurs with icewines, can lead to wines with very low concentrations of polyphenols, and lower subsequent browning potential, but with a high aroma intensity further enhanced by freeze-concentration (Kilmartin *et al.*, 2007). The important area of enzymatic oxidation of grape juice is outside the scope of this chapter and will not be considered further here – it is only under certain conditions without SO_2 that these enzymes are expected to play a role in wine browning and only for a limited period of time (Traverso-Rueda and Singleton, 1973). On the other hand, the non-enzymatic uptake of oxygen by wine is thought to lead to similar products and

subsequent chemistry as with PPO enzymatic oxidation, with the exception that PPO does not produce the hydrogen peroxide associated with non-enzymatic oxidation (Singleton, 1987).

Vernon Singleton at UC Davis made a key finding regarding the susceptibility of wine to oxidation. The oxidation of ethanol to acetaldehyde proceeds at a significant rate only through the coupled oxidation of readily-oxidisable polyphenols such as caffeic acid (typical of white wine hydroxycinnamic acids) and catechin (a flavanol at high concentrations in red wine) (Wildenradt and Singleton, 1974). In the absence of polyphenols, seen to be the main initial substrates of wine oxidation, ethanol and tartaric acid are very stable against oxidation. A product of polyphenol oxidation was a strong oxidant, presumed to be hydrogen peroxide, which could oxidise other substances in wine such as ethanol to produce more volatile aldehydes, although any SO_2 present could compete for the hydrogen peroxide.

Polyphenol oxidation was observed in model experiments to proceed more rapidly at a higher pH, ascribed to the more rapid oxidation of the phenolate ion form, through to a semiquinone radical and ultimately to a reactive quinone (Fig. 15.1) (Singleton, 1987). A very small percentage of phenolate ions are expected at wine pH (pKa values are typically around 9–10), but several times more will be present at a pH of 4 compared to a pH of 3, consistent with the manner in which high-pH wines are more oxidisable. An interesting finding coming from the early model solution studies involving quinones was the formation of polyphenol oligomers and the regeneration of easily oxidisable catechol groups; the resulting products tend to absorb light at higher wavelengths and contribute to wine browning. As a result, more units of oxygen are taken up than expected for a given number of original polyphenol molecules.

The triplet spin state of O_2 limits the type of molecules it can directly react with; it develops oxidising power only after it is activated in some way (Singleton, 1987; Danilewicz, 2003). In the process of being reduced through to water, and the gain of four electrons per O_2 molecule, oxygen passes through several reactive species including the hydroperoxyl radical ($\text{HO}_2^\bullet$), hydrogen peroxide (H_2O_2) and the hydroxyl radical ($\text{OH}^\bullet$). It is here that certain catalytic metals, particularly iron and copper, are thought to play a key role by complexing O_2 and facilitating the oxidation process, including the formation of reactive oxygen species and polyphenol oxidation (Pohl, 2007). Through a calculation of reduction potentials for oxygen, polyphenols and iron species under wine conditions, the whole process can be mediated by the catalytic action of the $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox couple (Fig. 15.2)

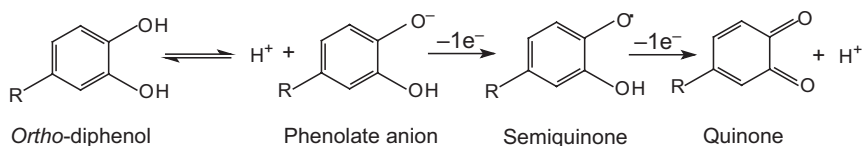


Fig. 15.1 Polyphenol forms before and after oxidation.

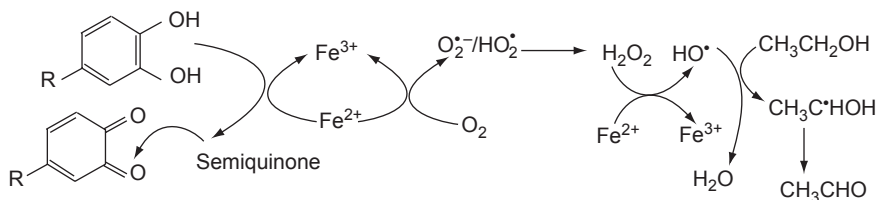


Fig. 15.2 Overall wine oxidation processes; adapted from Danilewicz (2003).

(Danilewicz, 2003). Ferrous ions convert H₂O₂ to the very reactive OH[·] (the Fenton reaction), a strong oxidant which will react with most organic compounds, including ethanol (CH₃CH₂OH), leading to the production of acetaldehyde.

Danilewicz has further demonstrated that copper has a role in facilitating the redox cycling of iron, to create a synergistic effect that increases the rate of polyphenol oxidation (Danilewicz, 2007). This idea has some experimental backing in that through the use of ion exchange procedures, it has been shown that the Fe and Cu concentration of white wines can be greatly lowered leading to less susceptibility to browning in an accelerated oxidation test (Benitez *et al.*, 2002). However, while the lower metal concentration was seen as favourable for oxidative stability, the lower browning seemed mainly to be due to a loss of polyphenol concentration, assayed with a chelating resin, while the aroma intensity was also substantially lowered following the treatments. The chemical oxidation of flavanols and procyanidin dimers using H₂O₂, and trapping of quinone products using sulphonation, also proceeded more effectively in the presence of a Cu as opposed to a Fe catalyst (Vivas de Gaulejac *et al.*, 2001).

The ease with which different polyphenols can be oxidised has been determined by measuring the electrode potential at which oxidation occurs for each compound in a model wine solution (Kilmartin *et al.*, 2001). Polyphenols containing an *ortho*-diphenol (catechol) or *ortho*-triphenol (galloyl) group showed the lowest oxidation–reduction potentials (i.e. they are the strongest reducing agents, a property often associated with antioxidants). The polyphenols tested included the common wine polyphenols gallic acid, catechin and epicatechin, caffeic acid and quercetin. Red and white wines gave a positive current response at the same potentials, indicating that polyphenols with these diphenol structures are the main initial substrates of wine oxidation (Kilmartin *et al.*, 2002). Higher potentials were needed to oxidise malvidin anthocyanins, and compounds with more isolated phenolic groups, such as coumaric acid and resveratrol, and the *meta*-diphenol group on the flavonoid A-ring.

While measurements of oxidation–reduction potentials indicate which compounds will be most easily oxidised, they do not fully explain polyphenol reactivity and stability. For example, malvidin anthocyanins degrade faster in wine than catechol-containing hydroxycinnamic acids, despite the catechol group being much easier to oxidise (Tao *et al.*, 2007; De Beer *et al.*, 2008). It seems that further reactions involving the various solution forms of the anthocyanins, including the formation of bridged structures with other polyphenols, are major forms of loss in

preference to direct anthocyanin oxidation. Nevertheless, a model solution study showed an increased loss of both malvidin-3-glucoside and catechin when oxygen was present (Jordao *et al.*, 2006).

The production of acetaldehyde (CH_3CHO) has long been seen as an important product of coupled polyphenol oxidation steps (Fig. 15.2). Waterhouse has emphasised that hydroxyl radicals can react with almost all wine components to produce a variety of aldehydes and ketones, and numerous other products that can be involved in wine ageing (Waterhouse and Laurie, 2006). This concept has been extended to the effect of $\text{H}_2\text{O}_2/\text{Fe}$ catalysed oxidation processes acting upon glycerol (present in wine at 5–20 g/L), particularly involving the hydroxyl radical ($\text{OH}^\bullet$), where the generation of glyceraldehyde and dihydroxyacetone has been demonstrated (Laurie and Waterhouse, 2006a).

Acetaldehyde and other aldehydes, products of yeast activity as well as non-enzymatic wine oxidation, then have important roles to play in wine ageing processes. These include linking reactions between flavonoids such as the anthocyanins, one of several mechanisms for the polymerisation of wine polyphenols. Further mechanisms include direct linkages and reactions involving other small molecules such as pyruvate (to form the vitisins), with many new products identified by the group of Veronique Cheynier (Fulcrand *et al.*, 2006). The evolution of red wine pigments has been examined in some detail in recent studies, including the formation and degradation of various classes of pigment, providing considerable insight into changes in wine colour with age (Monagas *et al.*, 2005a,b; Alcalde-Eon *et al.*, 2006; García-Puente Rivas *et al.*, 2006). Some of the newly formed compounds are more stable than the monomeric anthocyanins from which they are derived, and a high proportion are resistant to sulphite bleaching. The role of oxidation here, alongside the production of acetaldehyde to facilitate condensation processes, includes the oxidation of some of the products through to coloured forms (Asenstorfer *et al.*, 2003). Further, less well understood processes involve semiquinone and quinone forms of oxidised polyphenols (Monagas *et al.*, 2005a).

The role of further aldehydes in wine ageing processes has been recently demonstrated through the addition of glyceraldehydes to red wines. This addition increased the 520 nm absorbance by several percent due to the formation of condensed structures between flavonoids, or to the unbleaching of anthocyanins with SO_2 attached (Laurie and Waterhouse, 2006a). In further model solution studies, glyceraldehyde additions to (epi)catechin and malvidin-3-glucoside led to the formation of new condensed products, demonstrating the potential role for glyceraldehydes as a mediator in flavonoid polymerisation (Laurie and Waterhouse, 2006b).

In addition to polyphenols, a role for yeast lees as a competing substrate for oxygen in wine has been demonstrated. During alcoholic fermentation, yeast cells require oxygen for their metabolic activity, and the reactive oxygen species that are produced can potentially oxidise wine polyphenols (Salmon, 2006). The consumption of oxygen in a must has been found to follow an exponential trend (first-order kinetics) over several minutes due to metabolic yeast activity. At the end of

fermentation, phenolic compounds take part in oxygen consumption (with a change to second-order kinetics) (Silva and Lambri, 2006). After alcoholic fermentation is complete, there are several winemaking approaches which maintain contact between the ageing wine and non-viable yeast lees which can consume O_2 for up to 3 years. The consumption of O_2 by yeast lees has been ascribed to a mild oxidation of the membrane lipids of the yeast lees, leading to lipid peroxides and further products that may add to the wine flavour (Salmon *et al.*, 2000).

The relative reactivity of O_2 towards yeast lees and wine polyphenols has been compared; yeast lees on their own exhibited an initial rapid consumption of O_2 before reaching a steady state, whereas O_2 consumption by wine polyphenols was slow, constant and prolonged (Salmon *et al.*, 2002). However, the O_2 consumption capacities of both yeast lees (a dominant role) and wine polyphenols decreased as they aged in a model wine medium. Moreover, when yeast lees and wine polyphenols were combined, the observed O_2 consumption was less than the expected sum of the components on their own, pointing to an interaction between polyphenols and yeast cell walls. The oxygen consumption kinetics of a Cabernet Sauvignon wine also changed considerably after filtering to remove yeast lees – unfiltered wines removed 9 mg/L of O_2 within 2 days, while filtered wines took 6 days (Dykes, 2007; Dykes and Kilmartin, 2007).

In accelerated ageing trials of wines on yeast lees at 28 °C, there were no changes in proteins and free fatty acids relating to added O_2 , apart from a lower concentration of ergosterol in wines stored in the presence of O_2 (Fornairon-Bonnefond and Salmon, 2003). The addition of yeasts to a Sherry-type white wine was also found to delay browning in an accelerated ageing trial (Lopez-Toledano *et al.*, 2006).

15.4 Oxidation of aroma compounds

A further general effect of wine oxidation is upon wine aroma (du Toit *et al.*, 2006b), ranging from the removal of sulphur-containing ‘reduced’ compounds, and other aroma losses, through to the production of new ‘oxidised’ aromas. Many sulphur-containing compounds produce unwanted aromas in wines (Mestres *et al.*, 2000), and wine oxygenation processes such as racking provide the first means for their removal, with copper fining also used for this purpose. On the other hand, some positive contributions to wine aroma are made by certain sulphur-containing compounds which might be lost through wine oxidation, such as the role of dimethyl sulfide in enhancing the berry fruit note (Segurel *et al.*, 2004; Escudero *et al.*, 2007), and the contribution of varietal thiols such as 3 mercaptohexanol (3MH) to the grapefruit/passionfruit aromas of Sauvignon blanc and other wines (Tominaga *et al.*, 1998).

The removal of sulphur-containing compounds can occur through addition of sulphides to the quinones formed from polyphenol oxidation (Fig. 15.3a). Reactive quinones, and/or hydrogen peroxide, was suggested to cause significant losses of 3MH following oxygen exposure to red wines in barrels or catechin-containing

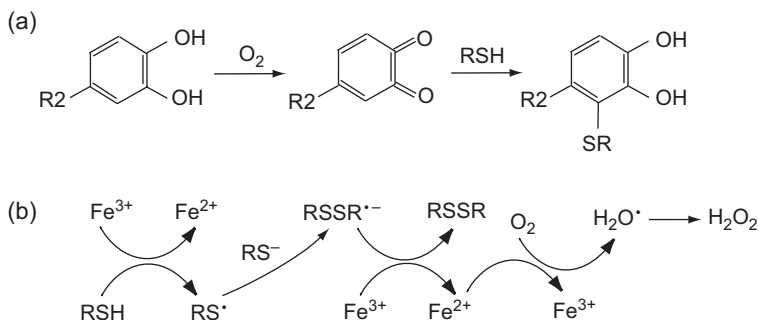


Fig. 15.3 Mechanisms for the oxidation of sulphur-containing compounds in wine; (a) polyphenol-mediated oxidation; (b) metal catalysed thiol oxidation; adapted from Danilewicz *et al.* (2008).

model solutions, with protection afforded by added SO_2 or anthocyanins (Blanchard *et al.*, 2004). The interaction of the model polyphenol, 4-methyl catechol, with 3MH was also shown to lead to greater consumption of O_2 than either compound alone (Danilewicz *et al.*, 2008), pointing to a polyphenol-mediated oxidation mechanism for the loss of the thiol.

Direct oxidation of thiols (mercaptans) through to disulphide forms (not removable by copper fining) is also expected (Rauhut *et al.*, 1996; Mestres *et al.*, 2000). Higher concentrations of dimethyl disulphide and diethyl disulphide, but lower concentrations of ethyl mercaptan and 2-mercaptoethanol, in older wines, were seen as indicative of oxidation processes (Fedrizzi *et al.*, 2007). Increases in dimethyl sulphide with age were ascribed to the degradation of S-methyl methionine, as reported previously (Marais, 1979). In the case of the thiol-containing amino acid cysteine, a rapid reaction in the presence of O_2 , Fe(II) and Cu(II) was seen, ascribed to the metal catalysed oxidation of thiols presented in Fig. 15.3b (Danilewicz *et al.*, 2008). The conversion of disulphides back to thiols due to the reducing ability of sulphites in wine has also been demonstrated in model solution studies (Bobet *et al.*, 1990). This result points to the potential for the release of thiol compounds during wine ageing with lower sensory thresholds than the disulphides they originate from. The release of mercaptans from the hydrolysis of thioacetic esters during bottle storage has also been described (Rauhut *et al.*, 1996). However, few experimental studies on the effects of oxidation upon sulphur-containing compounds have been undertaken, partly owing to the demands of reliable analytical methods, and more research is needed in this area. In one micro-oxygenation study, the addition of O_2 was seen to lower concentrations of methyl mercaptan and ethyl mercaptan, but no evidence for the formation of disulphides was found (McCord, 2003).

The development of new, unwanted aromas from wine oxidation is also an important issue in wine ageing (Escudero *et al.*, 2002; Silva Ferreira *et al.*, 2003). Some of the compounds which have been linked to sensory terms for oxidised wines, such as 'farm-feed' and 'woody-like', include 3-(methylthio)propionaldehyde (methional), phenylacetaldehyde, 1,1,6-trimethyl-1,2-dihydronaphthalene

(TDN – giving the ‘kerosene’ odour in aged Riesling) and 4,5-dimethyl-3-hydroxy-2(5-H)-furanone (sotolon) (Silva Ferreira *et al.*, 2003). This was backed up by an accelerated ageing study on white wines saturated with O₂ and stored at 45 °C over 3 months, which showed a decrease in terpenes and norisoprenoids (floral aromas) and an increase in phenylacetaldehyde (honey-like), methional (farm-feed) and TDN (Silva Ferreira *et al.*, 2002). On the other hand, treatment of the wine with SO₂ decreased the aroma degradation. The concentration of sotolon has been linked to the concentration of dissolved oxygen in bottled wine, pointing to the importance of cork O₂ permeation in the development of this oxidised aroma (Lavigne *et al.*, 2008).

While the formation of various aldehydes has been suggested for the formation of off-odours in oxidised wines, and the formation of a ‘cooked-vegetable’ nuance, the content of acetaldehyde itself has at times not increased significantly during oxidation experiments, and many of the important odorants in the oxidised aroma remain to be identified (Escudero *et al.*, 2002). The odour properties of several oxidation-related volatile aldehydes in wines were recently examined (Cullère *et al.*, 2007). The aldehydes were generally present at higher concentrations in oxidised white wines compared to young white wines, and in red wines aged for a long time compared to young red wines. The highest aldehyde concentrations lay above the respective odour activity values (OAVs), with methional and phenylacetaldehyde scoring particularly highly. For the younger wines, the concentrations of the aldehydes were typically below the OAVs. Synergistic effects were seen with particular combinations of aldehydes, including a negative influence on the sensory properties of wine through the addition of 2-alkenals, whereas branched aldehydes provided dried fruit notes typical of aged red wines.

15.5 Measures of wine oxidation status

Uptake of O₂ by wine is difficult to measure, and too slow for a measurement of the total oxygen uptake ability of a wine without resorting to accelerated oxygenation at higher temperatures or pH (alkaline oxygen uptake). This is because a wine saturated with oxygen will typically take 1–2 weeks to consume the 8–9 mg/L of O₂ added, and wines can take up multiple saturations of oxygen proportional to the concentration of polyphenols present (Singleton, 1987). Of greater interest has been the development of methods to assess the susceptibility of white wines to suffer browning problems during the expected wine storage period, often assessed by the absorbance of the wine at 420 nm or using CIELAB colour space values.

An accelerated test for the browning of white wines was outlined by Singleton, looking to standardise a variety of related tests which had been employed previously (Singleton and Kramling, 1976). In this procedure, wine treated with bentonite to remove proteins (which tend to precipitate brown pigments), but not total phenols, was sealed in tubes with an O₂ headspace of at least 25% and left for 5 days at 55 °C before measuring the absorbance at 420 nm. Some questions remain whether the chemical reactions at the elevated temperature are different

from those for slower browning under normal wine storage. Using a similar accelerated test, the oxidation and browning of wines has been shown to lead to a loss of antioxidant capacity in accelerated tests at 55 °C on white wines, while higher SO₂ concentrations lessened this decline (Sioumis *et al.*, 2005). An examination of phenolic composition during the accelerated test established a correlation between the rate of browning and epicatechin concentrations.

Measures of total phenols, such as the result obtained using the Folin–Ciocalteu (FC) reagent (itself an oxidising agent), are seen as important because they quantify the main substrates of wine oxidation (Singleton *et al.*, 1999). Among potential interferants that can also give a response, glutathione and ascorbic acid have been examined either as being directly oxidised by the FC reagent or through the regeneration of quinone forms to allow further FC oxidation. SO₂ has also been known to amplify the reaction of the FC reagent with phenols (for catechol but not more isolated phenol groups), likely also through a rapid regeneration of oxidisable phenols, unless acetaldehyde is added to prevent this augmentation.

A number of methods have been described in which electrodes are placed in wine to make a measure of oxidation status. The measurement of the potential at an inert platinum electrode with no current flowing (the open-circuit potential commonly referred to as the redox potential of wine) has a long history in enology (Dikanovic-Lucan and Palic, 1995; Vivas *et al.*, 1996), and is still being used in wine research to measure the state of oxidation or reduction of a wine (Del Alamo *et al.*, 2006). One idea behind the measure is that the various organic species in wine which can move to and from oxidised and reduced forms (e.g. polyphenols and their quinone forms) would adjust to match this measured potential value, in a similar manner to the way many metal ions which can exist in multiple oxidation states (e.g. iron and copper) adjust their relative concentrations to a common redox potential. However, this approach has been called into question by the interpretation of the electrode measure in wine as a mixed potential largely coupling ethanol oxidation (which occurs very readily at platinum electrodes) with oxygen or proton reduction (Kilmartin, 1988; Tomlinson and Kilmartin, 1997; Kilmartin and Zou, 2001; Durliat and Comtat, 2005). Further, given that polyphenols are the major substrates of interest for wine oxidation, but are not typically oxidised at platinum electrodes at the redox potential values commonly seen in wine, this raises questions about the significance of this measure for wine ageing. At best, it provides a broad measure of the oxygen content of a wine, which can be measured more accurately using other approaches. Statements about the measured 'redox potential', based upon values taken during fermentation or wine storage, can largely be recast in terms of concentrations of dissolved oxygen. However, it is interesting to note that a recent application of redox potential measurements at platinum electrodes to teas has provided an index of the extent of 'fermentation' from green tea polyphenols to the theaflavin forms found in black teas (Chen *et al.*, 2007). In this case, and in the absence of ethanol, the decrease in the relative proportion of polyphenol versus oxidised quinone forms during tea fermentation is likely responsible for the increase in redox potential values. Potentiometric measures at platinum electrodes during the course of redox titrations have also

been considered in enology to quantify the reducable and oxidisable species present (Vivas *et al.*, 2004; Durliat and Comtat, 2005; Martinez *et al.*, 2005).

An inert electrode that is much less responsive to ethanol than platinum, but is very suitable for the oxidation of polyphenols (of great interest for wine oxidation status), is glassy carbon. By maintaining the glassy carbon electrode at a certain potential, a measure of the more easily oxidisable polyphenols can be obtained, to provide a total phenols measure. Analysis of wine using flow injection analyses and detection at a glassy carbon electrode was used to provide a measure of the 'antioxidant power' of wines (Mannino *et al.*, 1998). Polyphenol compounds which can be oxidised at +0.4 V (versus a Ag/AgCl reference electrode) are included in this measure expressed as catechin equivalents, giving lower values than recorded using the FC reagent, which responds to a wider range of polyphenol compounds.

The electrochemical technique of cyclic voltammetry at a glassy carbon electrode has been applied to diluted wines to provide such a measure of easily oxidisable polyphenols (Kilmartin *et al.*, 2001, 2002; Zou *et al.*, 2002; De Beer *et al.*, 2004). The peak current seen at around 0.4 V (Ag/AgCl) is directly proportional to the quantity of catechol- or galloyl-type phenolic groups present, and has been used in further studies to characterise wines and to assess their polyphenol content (Castaignede *et al.*, 2003; Piljac *et al.*, 2004; Ksenzhek *et al.*, 2007). For example, following an accelerated ageing test at 15 or 45 °C, a decline in the cyclic voltammetry current was noted for white wines initially saturated with O₂ through partial removal of oxidisable species (Rodrigues *et al.*, 2007).

We can also note here that an alternative accelerated browning method was developed based upon subjecting the wine to a potential of 1.5 V applied between two platinum electrodes, and monitoring the absorbance at 420 nm (Palma *et al.*, 2000). Tests on fino sherry wine showed a basic similarity in the development of phenolic compounds using the accelerated browning technique as for samples evolved for 8 to 10 months under exposure to air.

15.6 White wine oxidation

Oxidation issues for white wines centre around the browning associated with the coupling of oxidised polyphenols and with losses in varietal aroma. White wines are characterised by lower concentrations of total polyphenols (typically 200–500 mg/L), a lack of red-coloured anthocyanins and predominance of hydroxycinnamic acids. Only low concentrations of flavonoids such as the catechins or quercetin glycosides are typically found in white wines, but their presence is important in the context of wine browning, and higher concentrations are expected in musts exposed to longer skin contact times and harder pressings (Maggu *et al.*, 2007).

The browning of white wines usually shows a lag time and a slow phase before speeding up, pointing to an autocatalytic process involving more readily oxidisable polyphenol oligomeric products. Browning in a white wine has been related more closely to the flavonoid content than to total phenols or concentrations of

hydroxycinnamic acids (Singleton, 1987). A greater capacity to brown was seen in an accelerated test on a Chardonnay wine when 200 mg/L of catechin was added to the wine (Singleton and Kramling, 1976). While PPO prefers caffeic acid as a substrate over catechin, model studies showed that the oxidised caffeic acid quinone preferred to couple with catechin more so than with caffeic acid, leading to catechin quinones which then proceed to form condensation products (Cheynier *et al.*, 1989a). Rapid browning tests on 80 white wines (at 50 °C for 3 weeks) showed a correlation between browning and the concentration of catechin phenolics, but no correlation between browning and copper content, with pH or free SO₂ (Simpson, 1982). On the other hand, a comparison of browning rates with wine composition for 32 white wines did not show a direct correlation between total phenol or iron concentration (Fernandez-Zurbano *et al.*, 1995). The chemical variables that did affect browning were total SO₂, which was inversely related to browning, while a higher pH tended to accelerate wine oxidation. This is not to say that metals are not involved in browning – it might be that as a catalyst only a certain concentration of iron is required to have its effect – but rather the concentration of polyphenols and iron did not dictate the susceptibility to oxidation.

The products involved in the browning of white wine polyphenols have been difficult to determine, but some model solution studies have indicated possible pathways. Glyoxylic acid, formed by the oxidation of tartaric acid, has been shown to crosslink two catechin molecules, when catalysed by metal ions, to form yellow-coloured xanthylum pigments (George *et al.*, 2006). Iron was found to be a more efficient catalyst than copper, while the addition of the hydroxycinnamate caffeic acid inhibited the accumulation of these pigments. This raises questions about the relevance of xanthylum pigments to the oxidation of white wines. The xanthylum cation pigments generated from analogous epicatechin solutions were found to be nearly twice as intense in colour as those generated from catechin-containing solutions (Labrousse *et al.*, 2005). Similar pigments were found in reactions between catechin and dihydroxymaleic acid, a product of tartaric acid oxidation by hydroxyl radicals (Clark, 2008). In the presence of yeasts, the production of coloured compounds in the reaction between catechin and glyoxylic acid was lowered, through the competition of yeasts for oxygen and their ability to retain brown products (Merida *et al.*, 2006). The oxidation of caffeic acid in a model wine solution in the presence of an Fe(II) catalyst has been shown to produce dihydroxybenzaldehyde (Lutter *et al.*, 2007). When the dihydroxybenzaldehyde reacted with catechin, various colourless and yellow/red compounds were produced, including the bridging of two catechin molecules.

Wines stored for 12 months showed a decrease in polyphenols and colour change from pale yellow to yellow–brown which was greater in a wine subjected to a variable temperature regime in southwest Spain than one stored at a constant temperature (Recamales *et al.*, 2006). Storage in a horizontal or a vertical position had no effect on the compounds studied. A 5-year closure trial of Chardonnay and Riesling wines conducted at the Australian Wine Research Institute (AWRI) showed a number of interesting trends in the white wines when natural cork,

screwcap and synthetic closures were compared (Skouroumounis *et al.*, 2005a). The higher rate of O₂ ingress through the synthetic closure was seen as explaining the lower concentrations of SO₂, higher browning and more advanced oxidised aromas. Lower concentrations of 1,1,6-trimethyl-1,2-dihydronaphthalene (TDN – the ‘kerosene’ aroma in aged Riesling) in the Riesling wines were ascribed to absorption into the closure rather than to oxidation. At the opposite end, with limited O₂ ingress, were wines stored in glass ampoules, in which the rate of browning and SO₂ loss was the least, with screwcaps and some of the wines under cork performing similarly well. The wines under screwcap and in glass ampoules were lowest in oxidised characters and gave a discernable ‘struck flint/rubber’ aroma which was related to low oxygen ingress combined with the presence of certain sulphur-containing precursors at bottling. These and other studies from the AWRI have demonstrated how much wine development is influenced by closure type.

15.7 Red wine oxidation

15.7.1 Implications of micro-oxidation

Red wines are characterised by higher concentrations of total phenols (1000–5000 mg/L) and are also rich in the flavonoids responsible for wine colour (anthocyanins) and astringency (flavanol oligomers and polymers). The development of red wine polyphenols with age has been examined in a number of studies where the effects of oxygen are known to contribute (Alcalde-Eon *et al.*, 2006; Garcia-Puente Rivas *et al.*, 2006), with periodic O₂ saturations leading to a loss of sulphites and monomeric phenolics and an increase in red polymeric pigments (Castellari *et al.*, 2000). There has also been considerable interest in the technique of micro-oxygenation (MOX) developed in the mid-1990s for the ageing of red wines (Oenodev, 2008), with three principal aims: enhancing wine colour, the removal of unwanted reduced and vegetal aromas and lowering astringency. While few scientific studies were published on red wine MOX during the first decade of its application, several studies have recently appeared and the main findings relating to non-enzymatic wine oxidation will be summarised here (see Chapter 8 for more information about micro-oxygenation).

Consistent trends relating to the effect of oxidation on wine colour development have been noted in the recently published MOX studies. These include a greater loss of monomeric anthocyanins as a result of the application of MOX versus control wines and a corresponding greater formation of polymeric pigments (resistant to SO₂ bleaching), often with an increase in wine colour density, or at least a lower loss in wine colour compared to the control wine (Atanasova *et al.*, 2002; Castel *et al.*, 2001; McCord, 2003; Cano-Lopez *et al.*, 2006, 2007, 2008; du Toit *et al.*, 2006a; Lesica and Kosmerl, 2006; Perez-Magariño *et al.*, 2006; Sartini *et al.*, 2007; Tao *et al.*, 2007; De Beer *et al.*, 2008). The changes observed in pigment classes have also included greater formation of ethyl-bridged compounds

associated with increased acetaldehyde concentrations (Atanasova *et al.*, 2002; Cano-Lopez *et al.*, 2006), although during later bottle storage the concentrations of these compounds could subsequently decline (Cano-Lopez *et al.*, 2007). The accumulation of acetaldehyde in wines subject to MOX has been recorded (Cano-Lopez *et al.*, 2008), a trend also seen with an electrochemical micro-oxygenation (ELMOX) approach as a build up of bound SO_2 (Fell *et al.*, 2007).

It was also observed that MOX can be applied for several weeks before a marked increase in measured acetaldehyde occurs (Carlton *et al.*, 2007). The suggestion was made that acetaldehyde produced initially would bind anthocyanins and flavanols, but a point would be reached when excess acetaldehyde would become apparent, marking a possible practical way to monitor MOX development. Other phenolic compounds which were lost more rapidly as a result of MOX included flavonols such as quercetin and flavanols such as catechin and epicatechin (McCord, 2003; du Toit *et al.*, 2006a; Sartini *et al.*, 2007), but little change was seen in hydroxycinnamic acids such as caftaric acid and caffeic acid (De Beer *et al.*, 2008), particularly while adequate concentrations of free SO_2 were maintained in the wines (Tao *et al.*, 2007). The rate of oxygenation has also been shown to affect the rate at which ageing processes, such as changes in wine polyphenols, are accelerated as a result of MOX (Dykes, 2007; Dykes and Kilmartin, 2007).

One of the benefits ascribed to MOX is the lowering of vegetative characters in wines (Moutounet *et al.*, 1996). In the limited number of studies in which the effects of MOX on wine aroma have been considered, little influence of MOX was found upon concentrations of esters, short-chain fatty acids and terpene compounds, meaning that the common observation that MOX lowered the perception of vegetal aromas remains to be established (Heras *et al.*, 2008). It was only for MOX in the barrel that an influence was seen in a lesser extraction of volatile compounds from the wood.

Defining in chemical terms the decline in astringency often associated with red wine MOX has proved equally difficult. In one study which included sensory profiling, both the berry/plum intensity and overall wine quality fell in the MOX wines, but no change was recorded in the astringency measure (De Beer *et al.*, 2008). Some changes in the MDP of the tannins have also been observed, looking for any indication that MOX will lead to smaller tannin polymers with a lesser degree of astringency, but inconsistent trends were observed between the wines tested in one study (Cano-Lopez *et al.*, 2008), while a further study showed only small differences between MOX and control wines in tannin MDP values where a greater effect was seen with SO_2 over O_2 additions (Dykes, 2007; Dykes and Kilmartin, 2007).

15.7.2 Implications of bottle closure type

Turning from MOX studies to red wines in the bottle, in a 3-year closure trial on Cabernet Sauvignon wines, a number of differences were observed depending upon the ullage volume (4, 16 or 64 mL) of air for wines stored under screwcap, alongside a synthetic closure and natural cork (Kwiatkowski *et al.*, 2007). The wine with the greatest (64 mL) air headspace showed significant losses of SO_2 soon

after bottling, along with a greater decline of total anthocyanins and increase in non SO_2 -bleachable pigments, coupled with higher 'oxidised' aroma scores. By contrast, the wine under screwcap with the smallest (4 mL) air headspace showed the smallest loss of SO_2 and lowest development of non SO_2 -bleachable pigments, but also recorded higher 'struck flint/rubber' aromas, although this was not a dominating character. Wines under cork, screwcap with 16 mL air headspace and synthetic closures showed intermediate values. These results again illustrated how different closures and bottling procedures can lead to quite different wines, and that red wines can still develop with relatively low levels of oxygen ingress. On the other hand, the question whether the 'struck flint/rubber' aromas can be lessened through the use of increased oxygen concentrations at bottling, or whether these are being masked by other (e.g. 'oxidised') aromas, was left open.

Red wine storage trials have been extended to the use of polyethyleneterephthalate (PET) bottles, including PET with a polyester copolymer oxygen scavenger incorporated into the plastic material which decreased the O_2 permeability more than 10-fold (Giovanelli and Brenna, 2007). Tests conducted at 20 and 30 °C over 170 days showed that the SO_2 , anthocyanin and flavanol retention was greatest in the PET container with the oxygen scavenger included, and even better than storage in glass containers, ascribed to the ability of the PET with scavenger to remove O_2 already dissolved in the wine. The use of maturation vessels constructed out of high-density polyethylene (HDPE) has also been introduced as a mean of oxygenating wine, in which the rate of O_2 ingress can be controlled by the thickness of the tank (Flecknoe-Brown, 2006).

15.8 Influence of wine antioxidants

15.8.1 Sulphur dioxide

The means to influence or control wine oxidation processes begin with the rate at which O_2 is supplied to the wine, ranging from the mild oxygenation occurring during barrel ageing and MOX, to the more limited entry of O_2 into wine stored in bottles. A further avenue available to winemakers is to make use of antioxidant compounds, either to remove O_2 from the wine or to reverse or otherwise alter oxidative processes (remembering that antioxidants can potentially play a variety of roles, some direct and others more indirect, in their antioxidant effect). Among the compounds which need to be considered here are the bisulphite (SO_2) added almost universally in modern winemaking (but with limits sought due to associated health issues), glutathione naturally present in the grape, and ascorbic acid which is used as a winemaking additive in some countries.

The role of SO_2 was initially seen mainly in terms of inhibition of polyphenol oxidase activity (Singleton *et al.*, 1985), or the fast scavenging of hydrogen peroxide (Danilewicz, 2003), but not through a direct reaction with oxygen. SO_2 can be oxidised by O_2 in model wine solutions only in the presence of catalytic metals (Fe and Cu), in which case the oxidation of ethanol is also increased resulting in a build up of acetaldehyde-bound SO_2 (Danilewicz, 2007; Danilewicz

et al., 2008). However, the oxidation of catechol-containing polyphenols was a much faster process, leading to greater losses of SO_2 in the process, but with minimal ethanol oxidation. An inhibitory role for polyphenols in SO_2 autoxidation was thus proposed. SO_2 was not found to be a scavenger of the superoxide ion at concentrations typically found in wine, while superoxide was effectively scavenged by ascorbic acid and polyphenols (Vivas *et al.*, 1997).

The role of SO_2 is also understood to include the rapid reduction of oxidised polyphenols (Cheynier *et al.*, 1989a, 1993). This was seen in the synergistic activity of SO_2 and catechin in both the Folin–Ciocalteu and Randox (ABTS) total antioxidant assays (Saucier and Waterhouse, 1999). With both of these assays, a minimal response was seen with SO_2 on its own, but a significant enhancement of the response due to catechin was obtained when SO_2 was present, which was explained by the ability of SO_2 to recycle the catechin–quinone oxidation products back to catechin, allowing it to react again.

In a study on the loss of volatile esters in wine, with the aim of making use of less free SO_2 during storage (35 versus 55 mg/L), the addition of glutathione or caffeic acid slowed the decrease of several esters, whereas SO_2 concentrations did not affect volatile losses (Roussis *et al.*, 2007). In a further study involving selected aroma compounds in a model wine medium (isoamyl acetate, ethyl hexanoate and linalool), protection was afforded by SO_2 , and by mixtures of glutathione with either caffeic acid or gallic acid at concentrations similar to those in wine (Roussis and Sergianitis, 2008). SO_2 was also seen to increase the inhibitory action of caffeic acid and gallic acid upon the loss of various esters and terpenes in model wine studies (Lambropoulos and Roussis, 2006). In these studies, we see the polyphenols caffeic acid and gallic acid being added to provide protection against the loss of certain aroma compounds, which may result from their preferential reaction with oxygen compared to other vulnerable substrates – a quite different effect from their role in browning reactions and in other oxidative processes which are promoted by the presence of such polyphenols.

15.8.2 Glutathione

Protection of grape must from oxidation is afforded by the presence of high concentrations (50–100 mg/L) of free glutathione in crushed grapes. The formation of S-glutathionyl caftaric acid formed via the oxidation of caftaric and coutaric acids serves to limit the browning of musts (Singleton *et al.*, 1985), making the glutathione to caftaric acid ratio important in terms of browning susceptibility (Cheynier and Van Hulst, 1988). Furthermore, ratios of remaining caftaric acid to the glutathionyl adduct have been suggested as an index of the enzymatic oxidation to which a must was exposed (Singleton, 1987). A combination of white must oxidation and glutathione additions was found to lessen the decline in phenolics and browning capacity of the oxygenation treatment and maintain typical grape variety flavour (Vaimakis and Roussis, 1996). Concentrations of glutathione in pressed grape juice were found to be much lower in the absence of added SO_2 and ascorbic acid, but were quite stable during frozen storage (du Toit *et al.*, 2007). On

the other hand, the pool of glutathione in the juice and subsequent wine is sensitive to the pressing conditions used (Maggu *et al.*, 2007), with lower concentrations found in a more oxidative compared to a reductive juice treatment.

Glutathione potentially has a similar protective role in wines given that it can react with oxidised polyphenols (quinones) in preference to varietal aroma compounds such as 3-mercaptohexanol, or other polyphenols (Dubourdieu *et al.*, 2001). Wine yeasts have a role in determining the concentration of glutathione which remains in the wine, with protection of glutathione afforded by the presence of yeast lees (Lavigne *et al.*, 2007). At the same time, relatively little change occurs in either caftaric acid or S-glutathionyl caftaric acid during fermentation and wine storage, although in a recent red wine bottling study, losses of caftaric acid after 170 days of storage at 20 °C were accompanied by an increase in the concentration of S-glutathionyl caftaric acid, a result which could not be explained but which may involve glutathione addition to caftaric acid quinones formed by chemical oxidation processes (Giovanelli and Brenna, 2007). The analogous addition of cysteine to oxidised caffeic acid to form S-cysteinecaffeic acid has been demonstrated using sodium periodate as a chemical oxidant (Bassil *et al.*, 2005), which illustrates the generality of this reaction. The loss of the terpenes linalool and α -terpineol was shown to be inhibited by the addition of 20 mg/L of glutathione to Muscat wines which were kept open at 20 °C for 3 days (Papadopoulou and Roussis, 2001).

15.8.3 Ascorbic acid

In the presence of iron as a catalyst, ascorbic acid is readily oxidised by oxygen (Danilewicz, 2003). This provides a direct removal of O₂ that is not ascribed to SO₂ or glutathione, which are instead seen to act indirectly by reversing polyphenol oxidation processes. Ascorbic acid is present in grapes, but quickly falls to low concentrations in the must upon exposure to oxygen (Singleton, 1987). The addition of ascorbic acid to wines has been considered, particularly in the context of lessening the amount of SO₂ that then needs to be applied, but the issue has a controversial history given that pro-oxidative effects have been observed following the initial antioxidant activity. While the formation of hydrogen peroxide has been implicated in the pro-oxidative effect, which would then require adequate SO₂ concentrations to protect against the effects of H₂O₂, the breakdown of ascorbic acid is known to lead to quite a range of products.

One role ascribed to added ascorbic acid has been the reduction of quinones formed from polyphenol oxidation (Singleton, 1987; Peng *et al.*, 1998; Danilewicz, 2003). Photoreactions of various quinones with ascorbic acid have been described (Goerner, 2004), along with thermally activated reactions of quinones with ascorbate through to semiquinones which can react with oxygen (quinone recycling of biological significance) (Roginsky *et al.*, 1999). In a study on the reduction of *p*-benzoquinone by ascorbic acid, a higher reaction rate was seen at a pH of 4.5 versus 2 due to the higher reactivity of ascorbate compared to the undissociated acid, but the reaction was limited to aqueous solutions (Isaacs and Van Eldik,

1997). However, the reduction of quinones arising from the oxidation of typical wine polyphenols still remains to be confirmed.

In early model solution studies containing ethanol, ascorbic acid was shown to produce acetaldehyde very rapidly, which could be slowed but not completely eliminated through the addition of SO_2 , becoming a promoter of oxidation rather than an antioxidant (Wildenradt and Singleton, 1974). The addition of ascorbic acid and SO_2 to sparkling wines at disgorgement was examined over a period of 11 months (Marks and Morris, 1993). The incidence of browning was the lowest in the wines to which SO_2 had been added (also with the highest acetaldehyde concentrations), and while ascorbic acid treatments lowered browning compared to the controls in most cases, it gave no additional benefit to the SO_2 treatments. However, there was no extra loss of SO_2 in the 11-month trial in the wines to which ascorbic acid was added. In the case of accelerated ageing tests and heating for 2 days at 50 °C in the presence of O_2 , the treatments containing ascorbic acid browned to a much greater degree than the controls, likely through the generation of a potent oxidant such as H_2O_2 under the extreme conditions of this test. In further white wine and model solution studies (containing catechin and caffeic acid), the addition of ascorbic acid was found to lessen browning initially, but then led to greater SO_2 consumption and browning in accelerated trials undertaken at 45 °C and in wines stored in bottles for several months to 5 years (Peng *et al.*, 1998), indicating a change from an antioxidative to a pro-oxidative effect at a certain point in time.

Model solution studies at 45 °C have shown that ascorbic acid will accelerate the browning of catechin (measured at 440 nm) once a lag period of some days has passed, this period being dependent upon the concentration of O_2 present in the headspace (Bradshaw *et al.*, 2001). Some questions were raised whether hydrogen peroxide was the ascorbic acid oxidation product responsible for the onset of browning, as addition of H_2O_2 did not increase browning to the same extent and led to different products according to high-performance liquid chromatography (HPLC). Further studies confirmed that the end of the lag period corresponded to the complete oxidation of ascorbic acid, consistent with a crossover point from antioxidant to pro-oxidative activity, and that xanthylum cations were responsible for the increased absorbance at 440 nm after the lag period, but more so than if ascorbic acid had not been used (Bradshaw *et al.*, 2003). The addition of SO_2 was shown to extend the lag period in model solution studies of catechin browning induced by ascorbic acid, due to the additional time required to oxidise the ascorbic acid completely (Bradshaw *et al.*, 2004). In the process, the SO_2 was heavily consumed, suggesting that when ascorbic acid is used in winemaking, a higher, rather than a lower, concentration of SO_2 may be required, while SO_2 alone can effectively inhibit oxidative browning.

The AWRI closure trial on Chardonnay and Riesling, referred to above, included an examination of the effect of ascorbic acid additions upon wine ageing in the bottle (Skouroumounis *et al.*, 2005b). After 3 years, the addition of ascorbic acid led to either no difference in aroma, or less oxidised and more fruity aromas. Further, wines without ascorbic acid additions were browner (visual assessment

and CIELAB scores) in the case of the Chardonnay wines, with no difference in browning for the Riesling wines. SO_2 concentrations were also very similar regardless of ascorbic acid additions, or else slightly higher in the wines to which ascorbic acid had been added. In this study, the addition of ascorbic acid thus complemented the use of SO_2 , rather than manifesting a pro-oxidative effect, which might only occur once ascorbic acid has been largely depleted.

15.9 Conclusion

Oxygen remains both a friend and a foe to the winemaker. The full development of quality red wines relies on O_2 to provide a faster rate of maturation with regard to colour and mouthfeel and to mitigate against unwanted off-odours. With white wines the extension of shelf life can largely be defined in terms of resistance to oxidation, although the control of unwanted sulphur-containing compounds may weigh against seeking completely anaerobic conditions. Among the choices available to the winemaker are the rate and extent of oxygen exposure at all stages in winemaking, and the addition of small-molecule antioxidants (SO_2 , glutathione and ascorbic acid). Further approaches, but which bring other effects to wine flavour that need to be considered, include keeping the wine on yeast lees during ageing to exploit the preferential uptake of O_2 by the lees, and the use of various fining agents, particularly to remove oxidative substrates (e.g. polyphenols and sulphur-containing compounds). The key role of metal catalysts in wine oxidation raises the possibility of making use of procedures to lower wine metal content (especially for iron and copper) or to explore the use of chelating agents to lessen the catalytic oxidative effect of the metals.

Several chemical principles underlying the non-enzymatic oxidation of wine have been established over the last 30 years, and this understanding informs the strategies taken by winemakers to control wine oxidation and adds to their accumulated knowledge of years (and centuries) of practical experience. At the same time, keeping wine research on-track with regard to industry needs, and creating the most effective dialogue between researchers and winemakers, is particularly important and should be given more attention, because a number of questions relating to non-enzymatic wine oxidation remain unanswered and are in need of further research. Some of this work includes moving out from the simpler chemical systems in which wine oxidation principles have been most clearly established (and which chemists like to work with) to the more complex wine matrix with its additional influences.

Some areas where further research is required include the effect of oxygen on sulphur-containing compounds, to determine how much reactions involving O_2 influence the concentrations of these compounds as a wine ages. A further research area concerns the identification and quantification of oxidative changes, to polyphenols and other compounds, which lead to changes in wine astringency. It is recognised that in both of these areas good analytical tools are being developed (aroma and tannin measures), which can be applied more widely to bridge the big

gap in understanding of how observed chemical changes impact upon the sensory appreciation of aroma and astringency. For example, the greater formation of pigmented polymers is a well-established consequence of red wine MOX (and may or may not be sufficient to impact on the visual assessment of the wine), but the capping of tannin oligomers and polymers with anthocyanin units may have an important role in lowering wine astringency.

With wine aromas, the synergistic and masking effects of sulphur-containing compounds and newly formed aldehydes, in relation to fruity, 'reduced' and 'oxidised' aromas, require more systematic study. In addition to combinations of aroma compounds, more attention in sensory studies should be given to the inclusion of polyphenols (and oxidised polyphenols), for their effects on the volatility of aroma compounds (Aronson and Ebeler, 2004). Likewise metal-catalysed oxidation studies using model solutions can benefit from the inclusion of multiple alcohol and aroma substrates. The principles developed in future studies in this direction may well explain a number of the sensory changes which occur with both red and white wines as a result of oxidative processes.

Alongside chemical changes in the wine, a greater understanding of local O₂ concentrations in both the tank and wine bottle is still needed. Spatial gradients of dissolved oxygen, and parameters such as bubble size, need to be examined and modelled more completely to better understand the impact of O₂ additions to large-scale winery tanks of various shapes and sizes (Dykes, 2007; Dykes and Kilmartin, 2007). The concentration of O₂ in the wine above a MOX diffuser may be quite different than near the outer edges of the tank. Likewise for bottled wines, does the O₂ at bottling dominate over ingress through the closure during the first year of bottle ageing and, in the longer term, is the 1 µL or less a day of O₂ that may enter the wine through the most effective cork and screwcap closures significant to wine development in the longer term?

The ultimate goal with this accumulated knowledge remains the better design of storage vessels and application of winemaking practices, along with the optimum use of both O₂ and added antioxidants to improve the quality of wines at the point at which they are enjoyed by the consumer.

15.10 References

- Alcalde-eon C, Escribano-Bailon M T, Santos-Buelga C and Rivas-Gonzalo, J C (2006), Changes in the detailed pigment composition of red wine during maturity and ageing. *Anal. Chim. Acta*, **563**, 238–254.
- Aronson J and Ebeler S E (2004), Effect of polyphenol compounds on the headspace volatility of flavors. *Am. J. Enol. Vitic.*, **55**, 13–21.
- Asenstorfer R E, Markides A J, Iland P G and Jones G P (2003), Formation of vitisin A during red wine fermentation and maturation. *Aust. J. Grape Wine Res.*, **9**, 40–46.
- Atanasova V, Fulcrand H, Cheynier V and Moutounet M (2002), Effect of oxygenation on polyphenol changes occurring in the course of winemaking. *Anal. Chim. Acta*, **458**, 15–27.
- Bassil D, Makris D P and Kefalas P (2005), Oxidation of caffeic acid in the presence of L-cysteine: isolation of 2-S-cysteinylcaffeic acid and evaluation of its antioxidant properties. *Food Res. Int.*, **38**, 395–402.

- Benitez P, Castro R and Barroso C G (2002), Removal of iron, copper and manganese from white wines through ion exchange techniques: effects on their organoleptic characteristics and susceptibility to browning. *Anal. Chim. Acta*, **458**, 197–202.
- Blanchard L, Darriet P and Dubourdieu D (2004), Reactivity of 3-mercaptohexanol in red wine: Impact of oxygen, phenolic fractions, and sulfur dioxide. *Am. J. Enol. Vitic.*, **55**, 115–120.
- Bobet R A, Noble A C and Boulton R B (1990), Kinetics of the ethanethiol and diethyl disulfide interconversion in wine-like solutions. *J. Agric. Food Chem.*, **38**, 449–452.
- Bradshaw M P, Prenzler P D and Scollary G R (2001), Ascorbic acid-induced browning of (+)-catechin in a model wine system. *J. Agric. Food Chem.*, **49**, 934–939.
- Bradshaw M P, Cheynier V, Scollary G R and Prenzler P D (2003), Defining the ascorbic acid crossover from anti-oxidant to pro-oxidant in a model wine matrix containing (+)-catechin. *J. Agric. Food Chem.*, **51**, 4126–4132.
- Bradshaw, M P, Scollary G R and Prenzler P D (2004), Examination of the sulfur dioxide-ascorbic acid anti-oxidant system in a model white wine matrix. *J. Sci. Food Agric.*, **84**, 318–324.
- Cano-Lopez M, Pardo-Minguez F, Lopez-Roca J M and Gomez-Plaza E (2006), Effect of microoxygenation on anthocyanin and derived pigment content and chromatic characteristics of red wines. *Am. J. Enol. Vitic.*, **57**, 325–331.
- Cano-Lopez M, Pardo-Minguez F, Lopez-Roca J M and Gomez-Plaza E (2007), Chromatic characteristics and anthocyanin profile of a micro-oxygenated red wine after oak or bottle maturation. *Eur. Food Res. Technol.*, **225**, 127–132.
- Cano-Lopez M, Pardo-Minguez F, Schmauch G, Saucier C, Teissedre P-L, Lopez-Roca J M and Gomez-Plaza E (2008), Effect of micro-oxygenation on color and anthocyanin-related compounds of wines with different phenolic contents. *J. Agric. Food Chem.*, **56**, 5932–5941.
- Carlton W K, Gump B, Fugelsang K and Hasson A S (2007), Monitoring acetaldehyde concentrations during micro-oxygenation of red wine by headspace solid-phase microextraction with on-fiber derivatization. *J. Agric. Food Chem.*, **55**, 5620–5625.
- Castaignede V, Durliat H and Comtat M (2003), Amperometric and potentiometric determination of catechin as model of polyphenols in wines. *Anal. Lett.*, **36**, 1707–1720.
- Castel C, Morand A, Pujol G, Peyron D and Naudin R (2001), Influence on phenolic composition and sensory characteristics of microoxygenation on grape pomaces and during aging of red wines in Burgundy. *Ind. Bevande* **30**, 271–276.
- Castellari M, Arfelli G, Riponi C and Amati A (1998), Evolution of phenolic compounds in red winemaking as affected by must oxygenation. *Am. J. Enol. Vitic.*, **49**, 91–94.
- Castellari M, Matricardi L, Arfelli G, Galassi S and Amati A (2000), Level of single bioactive phenolics in red wine as a function of the oxygen supplied during storage. *Food Chem.*, **69**, 61–67.
- Castellari M, Simonato B, Tornielli G B, Spinelli P and Ferrarini R (2004), Effects of different enological treatments on dissolved oxygen in wines. *Ital. J. Food Sci.*, **16**, 387–396.
- Chen P C, Chang F S, Chen I Z, Lu F M, Cheng T J and Chen R L C (2007), Redox potential of tea infusion as an index for the degree of fermentation. *Anal. Chim. Acta*, **594**, 32–36.
- Cheynier V (2002), Oxygen in wine and its role in phenolic reactions during ageing. In Allen M, Bell S, Rowe N and Wall G (eds), *Uses of Gases in Winemaking*. Adelaide, SA, Australian Society of Viticulture and Enology, 23–27.
- Cheynier V F and van Hulst M W J (1988), Oxidation of trans-caftaric acid and 2-S-glutathionylcaftaric acid in model solutions. *J. Agric. Food Chem.*, **36**, 10–15.
- Cheynier V, Basire N and Rigaud J (1989a), Mechanism of trans-caffeoyltartaric acid and catechin oxidation in model solutions containing grape polyphenoloxidase. *J. Agric. Food Chem.*, **37**, 1069–1071.
- Cheynier V, Rigaud J, Souquet J M, Barillere J M and Moutounet M (1989b), Effect of pomace contact and hyperoxidation on the phenolic composition and quality of Grenache and Chardonnay wines. *Am. J. Enol. Vitic.*, **40**, 35–42.

- Cheyrier V, Masson G, Rigaud J and Moutounet M (1993), Estimation of must oxidation during pressing in Champagne. *Am. J. Enol. Vitic.*, **44**, 393–9.
- Clark AC (2008), The production of yellow pigments from (+)-catechin and dihydroxyfumaric acid in a model wine system. *Eur. Food Res. Technol.*, **226**, 925–931.
- Cullère L, Cacho J and Ferreira V (2007), An assessment of the role played by some oxidation-related aldehydes in wine aroma. *J. Agric. Food Chem.*, **55**, 876–881.
- Danilewicz J C (2003), Review of reaction mechanisms of oxygen and proposed intermediate reduction products in wine: central role of iron and copper. *Am. J. Enol. Vitic.*, **54**, 73–85.
- Danilewicz J C (2007), Interaction of sulfur dioxide, polyphenols, and oxygen in a wine-model system: central role of iron and copper. *Am. J. Enol. Vitic.*, **58**, 53–60.
- Danilewicz J C, Seccombe J T and Whelan J (2008), Mechanism of interaction of polyphenols, oxygen, and sulfur dioxide in model wine and wine. *Am. J. Enol. Vitic.*, **59**, 128–136.
- De Beer D, Harbertson J F, Kilmartin P A, Roginsky V, Barsukova T, Adams D O and Waterhouse A L (2004), Phenolics: A comparison of diverse analytical methods. *Am. J. Enol. Vitic.*, **55**, 389–400.
- De Beer D, Joubert E, Marais J and Manley M (2008), Effect of oxygenation during maturation on phenolic composition, total antioxidant capacity, colour and sensory quality of Pinotage wine. *S. Afr. J. Enol. Vitic.*, **29**, 13–25.
- Del Alamo M, Nevares I and Carcel L M (2006), Redox potential evolution during red wine aging in alternative systems. *Anal. Chim. Acta*, **563**, 223–228.
- Dikanovic-Lucan Z and Palic A (1995), Redox potential in the wine making process: A review. *Bull. O.I.V.*, **68**, 762–79.
- Du Toit W J, Lisjak K, Marais J and Du Toit M (2006a), The effect of micro-oxygenation on the phenolic composition, quality and aerobic wine-spoilage microorganisms of different South African red wines. *S. Afr. J. Enol. Vitic.*, **27**, 57–67.
- Du Toit W J, Marais J., Pretorius I S and Du Toit M (2006b), Oxygen in must and wine: A review. *S. Afr. J. Enol. Vitic.*, **27**, 76–94.
- Du Toit W J, Lisjak K, Stander M and Prevo D (2007), Using LC-MSMS to assess glutathione levels in South African white grape juices and wines made with different levels of oxygen. *J. Agric. Food Chem.*, **55**, 2765–2769.
- Dubourdieu D, Moine-Ledoux V, Lavigne-Cruege V, Blanchard L and Tominaga T (2000), Recent advances in white wine aging: the key role of the lees. In Rantz J M (ed.), *Proceedings of the ASEV 50th Anniversary Annual Meeting*, Seattle, WA, American Society for Enology and Viticulture 345–352.
- Durliat H and Comtat M (2005), Critical evaluation of potentiometric redox titrations in enology. *Anal. Chim. Acta*, **545**, 173–181.
- Dykes S I (2007), *The effect of oxygen dosage rate on the chemical and sensory changes occurring during micro-oxygenation of New Zealand red wine*. PhD thesis Chemistry Department, Auckland, The University of Auckland.
- Dykes S I and Kilmartin P A (2007), Micro-oxygenation – optimising the maturation process. *Wine Ind. J.*, **22**, 31–45.
- Escudero A, Asensio E, Cacho J and Ferreira V (2002), Sensory and chemical changes of young white wines stored under oxygen. An assessment of the role played by aldehydes and some other important odorants. *Food Chem.*, **77**, 325–331.
- Escudero A, Campo E, Farina L, Cacho J and Ferreira V (2007), Analytical characterization of the aroma of five premium red wines. Insights into the role of odor families and the concept of fruitiness of wines. *J. Agric. Food Chem.*, **55**, 4501–4510.
- Fedrizzi B, Magno F, Badocco D, Nicolini G and Versini G (2007), Aging effects and grape variety dependence on the content of sulfur volatiles in wine. *J. Agric. Food Chem.*, **55**, 10880–10887.
- Fell A J, Dykes S I, Nicolau L and Kilmartin P A (2007), Electrochemical microoxidation of red wine. *Am. J. Enol. Vitic.*, **58**, 443–450.

- Fernandez-Zurbano P, Ferreira V, Pena C, Escudero A, Serrano F and Cacho J (1995), Prediction of oxidative browning in white wines as a function of their chemical composition. *J. Agric. Food Chem.*, **43**, 2813–2817.
- Flecknoe-Brown A (2006), 'Diffusive' methods of oxygenating wine: simpler, better, lower-cost barrel replacement. *Aust. NZ Grapegrow. Winemak.*, **54**, 64–60.
- Fornairon-Bonnefond C and Salmon J-M (2003), Impact of oxygen consumption by yeast lees on the autolysis phenomenon during simulation of wine aging on lees. *J. Agric. Food Chem.*, **51**, 2584–2590.
- Fulcrand H, Duenas M, Salas E and Cheynier V (2006), Phenolic reactions during winemaking and aging. *Am. J. Enol. Vitic.*, **57**, 289–297.
- Garcia-Puente Rivas E, Alcalde-Eon C, Santos-Buelga C, Rivas-Gonzalo JC and Escribano-Bailon M T (2006), Behaviour and characterization of the color during red wine making and maturation. *Anal. Chim. Acta*, **563**, 215–222.
- George N, Clark A C, Prenzler P D and Scollary G R (2006), Factors influencing the production and stability of xanthylum cation pigments in a model white wine system. *Aust. J. Grape Wine Res.*, **12**, 57–68.
- Giovanelli G and Brenna O V (2007), Oxidative stability of red wine stored in packages with different oxygen permeability. *Eur. Food Res. Technol.*, **226**, 169–179.
- Goerner H (2004), Photoreactions of p-benzo-, p-naphtho- and p-anthraquinones with ascorbic acid. *Photochem. Photobiol. Sci.*, **3**, 933–938.
- Heras M O, Rivero-Perez M D, Perez-Magarino S, Gonzalez-Huerta C and Gonzalez-SanJose M L (2008), Changes in the volatile composition of red wines during aging in oak barrels due to microoxygenation treatment applied before malolactic fermentation. *Eur. Food Res. Technol.*, **226**, 1485–1493.
- Ho P, Rogerson F S S, Watkins S J, Silva M D C M, Hogg T A and Vasconcelos I (2000), Effect of skin contact and oxygenation of musts on the composition of white port wines. *Sci. Aliments*, **19**, 687–699.
- Isaacs N S and van Eldik R (1997), A mechanistic study of the reduction of quinones by ascorbic acid. *J. Chem. Soc., Perkin Trans.*, **2**, 1465–1467.
- Jordao A M, Ricardo-da-silva J M and Laureano O (2006), Effect of oak constituents and oxygen on the evolution of malvidin-3-glucoside and (+)-catechin in model wine. *Am. J. Enol. Vitic.*, **57**, 377–381.
- Kilmartin P A (1988), *The measurement of the redox potential of wine*. BSc(hons), thesis. Chemistry Department, Wellington, New Zealand, Victoria University.
- Kilmartin P A and Zou H (2001), The effect of electrode material on the measured redox potential of red and white wines. *Electroanalysis*, **13**, 1347–1350.
- Kilmartin P A, Zou H and Waterhouse A L (2001), A cyclic voltammetry method suitable for characterizing antioxidant properties of wine and wine phenolics. *J. Agric. Food Chem.*, **49**, 1957–1965.
- Kilmartin P A, Zou H and Waterhouse A L (2002), Correlation of wine phenolic composition versus cyclic voltammetry response. *Am. J. Enol. Vitic.*, **53**, 294–302.
- Kilmartin P A, Reynolds A G, Pagay V, Nurgel C and Johnson R (2007), Polyphenol content and browning of Canadian icewines. *J. Food, Agric. Environ.*, **5**, 52–57.
- Kontoudakis N, Biosca P, Canals R, Fort F, Canals J and Zamora F (2008), Impact of stopper type on oxygen ingress during wine bottling when using an inert gas cover. *Aust. J. Grape Wine Res.*, **14**, 116–122.
- Ksenzhek O, Petrova S and Kolodyazhny M (2007), Redox spectra of wines. *Electroanalysis*, **19**, 389–392.
- Kwiatkowski M J, Skouroumounis G K, Lattey K A and Waters E J (2007), The impact of closures, including screw cap with three different headspace volumes, on the composition, colour and sensory properties of a Cabernet Sauvignon wine during two years' storage. *Aust. J. Grape Wine Res.*, **13**, 81–94.
- Labrousche F, Clark A C, Prenzler P D and Scollary G R (2005), Isomeric influence on the oxidative coloration of phenolic compounds in a model white wine: Comparison of

- (+)-catechin and (–)-epicatechin. *J. Agric. Food Chem.*, **53**, 9993–9998.
- Lambropoulos I and Roussis I G (2006), Inhibition of the decrease of volatile esters and terpenes during storage of a white wine and a model wine medium by caffeic acid and gallic acid. *Food Res. Int.*, **40**, 176–181.
- Laurie V F and Waterhouse A L (2006a), Oxidation of glycerol in the presence of hydrogen peroxide and iron in model solutions and wine. potential effects on wine color. *J. Agric. Food Chem.*, **54**, 4668–4673.
- Laurie V F and Waterhouse A L (2006b), Glyceraldehyde bridging between flavanols and malvidin-3-glucoside in model solutions. *J. Agric. Food Chem.*, **54**, 9105–9111.
- Laurie V F, Law R, Joslin W S and Waterhouse A L (2008), In situ measurements of dissolved oxygen during low-level oxygenation in red wines. *Am. J. Enol. Vitic.*, **59**, 215–219.
- Lavigne V, Pons A and Dubourdieu D (2007), Assay of glutathione in must and wines using capillary electrophoresis and laser-induced fluorescence detection. *J. Chromatogr., A*, **1139**, 130–135.
- Lavigne V, Pons A, Darriet P and Dubourdieu D (2008), Changes in the sotolon content of dry white wines during barrel and bottle aging. *J. Agric. Food Chem.*, **56**, 2688–2693.
- Lesica M and Kosmerl T (2006), Microoxidation of Pinot Noir wine. *Acta Agric. Slov.*, **87**, 461–475.
- Li H, Guo A and Wang H (2008), Mechanisms of oxidative browning of wine. *Food Chem.*, **108**, 1–13.
- Lopes P, Saucier C and Glories Y (2005), Nondestructive colorimetric method to determine the oxygen diffusion rate through closures used in winemaking. *J. Agric. Food Chem.*, **53**, 6967–6973.
- Lopes P, Saucier C, Teissedre P-L and Glories Y (2006), Impact of storage position on oxygen ingress through different closures into wine bottles. *J. Agric. Food Chem.*, **54**, 6741–6746.
- Lopes P, Saucier C, Teissedre P-L and Glories Y (2007), Main routes of oxygen ingress through different closures into wine bottles. *J. Agric. Food Chem.*, **55**, 5167–5170.
- Lopez-Toledano A, Mayen M, Merida J and Medina, M (2006), Yeasts used to delay browning in white wines. *Food Chem.*, **97**, 498–504.
- Lutter M, Clark A C, Prenzler P D and Scollary G R (2007), Oxidation of caffeic acid in a wine-like medium: production of dihydroxybenzaldehyde and its subsequent reactions with (+)-catechin. *Food Chem.*, **105**, 968–975.
- Maggu M, Winz R, Kilmartin P A, Trought M C T and Nicolau L (2007), Effect of skin contact and pressure on the composition of Sauvignon Blanc must. *J. Agric. Food Chem.*, **55**, 10281–10288.
- Mannino S, Brenna O, Buratti S and Cosio M S (1998), A new method for the evaluation of the 'antioxidant power' of wines. *Electroanalysis*, **10**, 908–912.
- Marais J (1979), Effect of storage time and temperature on the formation of dimethyl sulfide and on white wine quality. *Vitis*, **18**, 254–60.
- Marks A C and Morris J R (1993), Ascorbic acid effects on the post-disgorgement oxidative stability of sparkling wine. *Am. J. Enol. Vitic.*, **44**, 227–31.
- Martinez S, Valek L, Piljac J and Metikos-Hukovic M (2005), Determination of wine antioxidant capacity by derivative potentiometric titration with electrogenerated chlorine. *Eur. Food Res. Technol.*, **220**, 658–661.
- McCord J (2003), Application of toasted oak and micro-oxygenation to ageing of Cabernet Sauvignon wines. *Aust. NZ Grapegrow. Winemak.*, **474**, 43–53.
- Merida J, Lopez-Toledano A and Medina M (2006), Influence of aerobic and anaerobic conditions and yeasts on the reaction between (+)-catechin and glyoxylic acid. *Eur. Food Res. Technol.*, **222**, 451–457.
- Mestres M, Busto O and Guasch J (2000), Analysis of organic sulfur compounds in wine aroma. *J. Chromatogr., A*, **881**, 569–581.
- Monagas M, Bartolome B and Gomez-Cordoves C (2005a), Updated knowledge about the presence of phenolic compounds in wine. *Crit. Rev. Food Sci. Nutr.*, **45**, 85–118.

- Monagas M, Gomez-Cordoves C and Bartolome B (2005b), Evolution of polyphenols in red wines from *Vitis vinifera* L. during aging in the bottle. *Eur. Food Res. Technol.*, **220**, 607–614.
- Moutounet M, Ducournau P, Chassin M and Lemaire T (1996), Apparatus for supplying oxygen to wines and its technological significance. *Fifth International Symposium of Oenology*, Bordeaux, France, 15–17 June, 1995, 411–414.
- Oenodev (2008), www.oenodev.com.
- Palma M, Barroso C G and Perez-Bustamante J A (2000), Evaluation of similarities between natural and accelerated browning of fino sherry wines by chemometric techniques. *Analyst (Cambridge, U. K.)*, **125**, 1151–1154.
- Papadopoulou D and Roussis I G (2001), Inhibition of the decline of linalool and alpha-terpineol in Muscat wines by glutathione and n-acetyl-cysteine. *Ital. J. Food Sci.*, **13**, 413–419.
- Peng Z, Duncan B, Pocock K F and Sefton M A (1998), The effect of ascorbic acid on oxidative browning of white wines and model wines. *Aust. J. Grape Wine Res.*, **4**, 127–135.
- Perez-Magarino S, Sanchez-Iglesias M, Ortega-Heras M, Gonzalez-Huerta C and Gonzalez-SanJose M L (2006), Color stabilization of red wines by microoxygenation treatment before malolactic fermentation. *Food Chem.*, **101**, 881–893.
- Piljac J, Martinez S, Stipcevic T, Petrovic Z and Metikos-Hukovic M (2004), Cyclic voltammetry investigation of the phenolic content of Croatian wines. *Am. J. Enol. Vitic.*, **55**, 417–422.
- Pohl P (2007), What do metals tell us about wine? *TrAC, Trends Anal. Chem.*, **26**, 941–949.
- Rauhut D, Kurbel H, Dittrich H H and Grossmann M (1996), Properties and differences of commercial yeast strains with respect to their formation of sulfur compounds. *Wein-Wissenschaft, Wiesbaden*, **51**, 187–192.
- Recamales A F, Sayago A, Gonzalez-Miret M L and Hernanz D (2006), The effect of time and storage conditions on the phenolic composition and colour of white wine. *Food Res. Int.*, **39**, 220–229.
- Rodrigues A, Silva Ferreira A C, Guedes De Pinho P, Bento F and Geraldo D (2007), Resistance to oxidation of white wines assessed by voltammetric means. *J. Agric. Food Chem.*, **55**, 10557–10562.
- Roginsky V A, Barsukova T K and Stegmann H B (1999), Kinetics of redox interaction between substituted quinones and ascorbate under aerobic conditions. *Chem. Biol. Interact.*, **121**, 177–97.
- Roussis I G and Sergianitis S (2008), Protection of some aroma volatiles in a model wine medium by sulphur dioxide and mixtures of glutathione with caffeic acid or gallic acid. *Flavour Fragrance J.*, **23**, 35–39.
- Roussis I G, Lambropoulos I and Tzimas P (2007), Protection of volatiles in a wine with low sulfur dioxide by caffeic acid or glutathione. *Am. J. Enol. Vitic.*, **58**, 274–278.
- Salmon J-M (2006), Interactions between yeast, oxygen and polyphenols during alcoholic fermentations: practical implications. *LWT – Food Sci. Technol.*, **39**, 959–965.
- Salmon J M, Fornairon-Bonnefond C, Mazauric J P and Moutounet M (2000), Oxygen consumption by wine lees: impact on lees integrity during wine ageing. *Food Chem.*, **71**, 519–528.
- Salmon J M, Fornairon-Bonnefond C and Mazauric J P (2002), Interactions between wine lees and polyphenols: influence on oxygen consumption capacity during simulation of wine aging. *J. Food Sci.*, **67**, 1604–1609.
- Sartini E, Arfelli G, Fabiani A and Piva A (2007), Influence of chips, lees, and micro-oxygenation during aging on the phenolic composition of a red Sangiovese wine. *Food Chem.*, **104**, 1599–1604.
- Saucier C T and Waterhouse A L (1999), Synergetic activity of catechin and other antioxidants. *J. Agric. Food Chem.*, **47**, 4491–4494.
- Schneider, V (1998), Must hyperoxidation: a review. *Am. J. Enol. Vitic.*, **49**, 65–73.

- Segurel M A, Razungles A J, Riou C, Salles M and Baumes R L (2004), Contribution of dimethyl sulfide to the aroma of Syrah and Grenache Noir wines and estimation of its potential in grapes of these varieties. *J. Agric. Food Chem.*, **52**, 7084–7093.
- Silva A and Lambri M (2006), Oxygen measures and consumption in must and wine. *Anal. Chim. Acta.*, **563**, 391–395.
- Silva Ferreira A C, De Pinho Paula G, Rodrigues P and Hogg T (2002), Kinetics of oxidative degradation of white wines and how they are affected by selected technological parameters. *J. Agric. Food. Chem.*, **50**, 5919–5924.
- Silva Ferreira A C, Hogg T and Guedes De Pinho P (2003), Identification of key odorants related to the typical aroma of oxidation-spoiled white wines. *J. Agric. Food Chem.*, **51**, 1377–1381.
- Simpson R F (1982), Factors affecting oxidative browning of white wine. *Vitis*, **21**, 233–239.
- Singleton V L (1987), Oxygen with phenols and related reactions in musts, wines, and model systems: observations and practical implications. *Am. J. Enol. Vitic.*, **38**, 69–77.
- Singleton V L (2001), A survey of wine aging reactions, especially with oxygen. In Rantz J M (ed.) *Proceedings of the ASEV 50th Anniversary Annual Meeting*. Seattle, WA, American Society for Enology and Viticulture, 323–336.
- Singleton V L and Kramling T E (1976), Browning of white wines and an accelerated test for browning capacity. *Am. J. Enol. Vitic.*, **27**, 157–160.
- Singleton V L, Salgues M, Zaya J and Trousdale E (1985), Caftaric acid disappearance and conversion to products of enzymic oxidation in grape must and wine. *Am. J. Enol. Vitic.*, **36**, 50–56.
- Singleton V L, Orthofer R and Lamuela-Raventos R M (1999), Analysis of total phenols and other oxidation substrates and antioxidants by means of Folin–Ciocalteu reagent. *Methods Enzymol.*, **299**, 152–178.
- Sioumis N, Kallithraka S, Tsoutsouras E, Makris D P and Kefalas P (2005), Browning development in white wines: dependence on compositional parameters and impact on antioxidant characteristics. *Eur. Food Res. Technol.*, **220**, 326–330.
- Skouroumounis G K, Kwiatkowski M J, Francis I L, Oakey H, Capone D L, Duncan B, Sefton M A and Waters E J (2005a), The impact of closure type and storage conditions on the composition, colour and flavour properties of a Riesling and a wooded Chardonnay wine during five years' storage. *Aust. J. Grape Wine Res.*, **11**, 369–384.
- Skouroumounis G K, Kwiatkowski M J, Francis I L, Oakey H, Capone D L, Peng Z, Duncan B, Sefton M A and Waters E J (2005b), The influence of ascorbic acid on the composition, colour and flavour properties of a Riesling and a wooded Chardonnay wine during five years' storage. *Aust. J. Grape Wine Res.*, **11**, 355–368.
- Tao J, Dykes S I and Kilmartin P A (2007), Effect of SO₂ concentration on polyphenol development during red wine micro-oxygenation. *J. Agric. Food Chem.*, **55**, 6104–6109.
- Tominaga T, Murat M-L and Dubourdieu D (1998), Development of a method for analyzing the volatile thiols involved in the characteristic aroma of wines made from *Vitis vinifera* L. Cv. Sauvignon Blanc. *J. Agric. Food Chem.*, **46**, 1044–1048.
- Tomlinson J W and Kilmartin P A (1997), Measurement of the redox potential of wine. *J. Appl. Electrochem.*, **27**, 1125–1134.
- Traverso-rueda S and Singleton V L (1973), Catecholase activity in grape juice and its implications in winemaking. *Amer. J. Enol. Vitic.*, **24**, 103–109.
- Vaimakis V and Roussis I G (1996), Must oxygenation together with glutathione addition in the oxidation of white wine. *Food Chem.*, **57**, 419–422.
- Vidal J C and Moutounet M (2006), Monitoring of oxygen in the gas and liquid phases of bottles of wine at bottling and during storage. *J. Int. Sci. Vigne Vin*, **40**, 35–45.
- Vidal J C, Toitot C, Boulet J C and Moutounet M (2004), Comparison of methods for measuring oxygen in the headspace of a bottle of wine. *J. Int. Sci. Vigne Vin*, **38**, 191–200.
- Vivas De Gaulejac N, Vivas N, Nonier M-F, Absalon C and Bourgeois G (2001), Study and quantification of monomeric flavan-3-ol and dimeric procyanidin quinonic forms by HPLC/ESI-MS. Application to red wine oxidation. *J. Sci. Food Agric.*, **81**, 1172–1179.

- Vivas N, Glories Y, Bertrand A and Zamora F (1996), Principal method for the measurement of oxidoreduction potentials in wines. *Bull. O.I.V.*, **69**, 617–633.
- Vivas N, Saint Cricq De Gaulejac N and Glories Y (1997), Influence of SO₂ and ascorbic acid on the scavenger effect of tannins, measured on superoxide anion. Application to red wines. *Vitis*, **36**, 91–96.
- Vivas N, Nonier M F and De Gaulejac N V (2004), Titrimetric method based on potentiometric titration to evaluate redox couples in wine and polyphenols. *Vitis*, **43**, 205–208.
- Waterhouse A L and Laurie V F (2006), Oxidation of wine phenolics: a critical evaluation and hypotheses. *Am. J. Enol. Vitic.*, **57**, 306–313.
- Wildenradt H L and Singleton V L (1974), Production of aldehydes as a result of oxidation of polyphenolic compounds and its relation to wine aging. *Amer. J. Enol. Vitic.*, **25**, 119–126.
- Zou H, Kilmartin P A, Inglis M J and Frost A (2002), Extraction of phenolic compounds during vinification of Pinot Noir wine examined by HPLC and cyclic voltammetry. *Aust. J. Grape Wine Res.* **8**, 163–174.

Ageing and flavour deterioration in wine

A. W. Linsenmeier, D. Rauhut and W. R. Sponholz,
Forschungsanstalt Geisenheim (Geisenheim Research Center
(GRC)), Germany

Abstract: This chapter is divided into two parts: the aroma changes of wines due to storage (the normal ageing) and the so-called untypical ageing (UTA) note, an off-flavour that can appear rapidly after bottling. In the first case (concerning ageing), there are a lot of compounds involved such as aldehydes, esters and terpenes; however, the untypical ageing note is mainly due to the character impact compound o-aminoacetophenone. Nevertheless, additional compounds that are involved in untypical ageing are also discussed. The chemical formation processes for the aroma compounds and the factors influencing the formation are described.

Key words: ageing, untypical ageing (UTA), storage, stress, o-aminoacetophenone (AAP).

16.1 Introduction: ageing

Ageing is a process in wine evolution leading to profound changes in chemical composition and aroma. The term ‘ageing’ is used with various connotations. The ageing of wine is often associated with a negative development of the sensory characteristics. However, ageing can also improve wine quality; this is sometimes called ‘maturation’. Premium quality wines, particularly red wines, with a complex aromatic structure can gain in quality from maturation time. In many winemaking regions, ‘wood ageing’, the oxidative ageing of red wines in wooden barrels, is traditional practice. This wood ageing can be followed by an anaerobic bottle ageing time to improve the sensorial characteristics of the wine. Wood maturation is often combined with an ageing ‘*sur lie*’, which is also a common

practice for Champagnes (see Chapter 6). Further examples where the ageing process plays an important role in winemaking technologies are the production of Ports and Sherries. In the latter case, there are even two different ageing procedures: oxidative ageing for the *oloroso*-style wines and biological ageing with the influence of flor yeasts, resulting in *fino* Sherries.

Briefly, ageing causes a degradation of aroma compounds mainly associated with fruit and floral aromas and a formation of molecules leading to new aroma characteristics – the aroma of aged wines. Red wines lose their colour and white wines become yellow–brown (see Chapter 3 in Volume 1). Whether these changes improve wine quality or not depends on the consumer's preferences. Whereas earlier (Roman era, Middle Ages, etc.) aged wines had a great reputation and still have for many consumers, nowadays the young fruity wine type is generally preferred. However, the consumer's expectations are important: for a white wine stored for 5 years, ageing flavours are acceptable or even positive, but for a young wine these flavours are judged as negative. This is why especially prematurely aged wines with a rapid loss of fruitiness and the oxidative spoilage of wines without developing pleasant ageing aromas cause serious problems for the wine industry. The first section of this chapter mainly deals with these changes during bottle ageing. The second part of this chapter deals with the untypical ageing (UTA) which in fact is not an ageing process but an off-flavour, and is therefore separated from the ageing section.

16.2 Sensory changes during storage/ageing

During storage, the ageing process leads to the loss of fruit and floral aromas or, in general, a loss of volatiles that are characteristic of young wines (Fig. 16.1). The period between bottling and the loss of fruitiness may differ drastically from wine to wine, even from bottle to bottle (Marais and Pool, 1980). In rare cases, especially for complex red wines, the fruity aromas can increase for a significant period due to ageing before they decrease (Ribéreau-Gayon *et al.*, 2000). With ageing time, the colour of white wine becomes more intense, while the colour of red wine decreases. Various authors have reported that aroma degradation occurs before chromatic degradation (Singleton and Kramling, 1976; Singleton *et al.*, 1979; Escudero *et al.*, 2002; Silva Ferreira *et al.*, 2003b). Wines that benefit from bottle ageing show increased harmoniousness and decreased harshness; the acidity of white wines and the astringency of red wines decrease (Simpson, 1979; Fernández de Simón *et al.*, 2006). They develop a desirable bouquet with increased complexity and new pleasant ageing aromas. Wines ageing this way may reach a peak in quality after storage for 2–7 years, depending on the wine cultivar and initial quality (Simpson, 1979; Ribéreau-Gayon *et al.*, 2000). On the other hand, many wines do not improve with bottle storage. Particularly fruity white wines and Muscat-type wines (Chisholm *et al.*, 1995) suffer loss of fruitiness without developing positive ageing aromas, or even show Sherry-like aromas of oxidation (Escudero *et al.*, 2002; Lambropoulos and Roussis, 2007; Hernanz *et al.*, 2009).

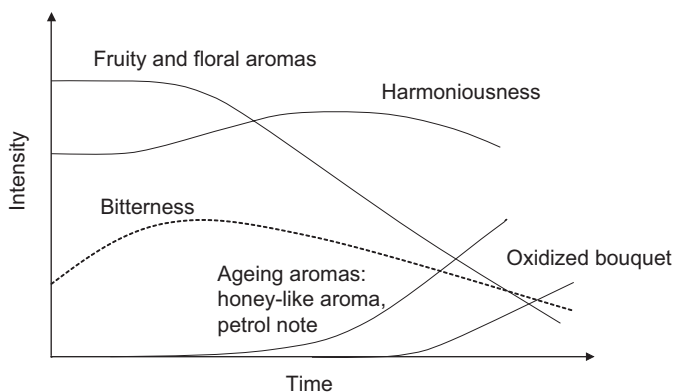


Fig. 16.1 Sensory changes during wine ageing.

For these reasons, the plot in Fig. 16.1 is certainly a simplification of sensory changes during storage. The aroma attributes of (negatively) aged white wines are described as ‘honey-like’, ‘farm-feed’, ‘hay’, ‘woody-like’, ‘rotten food’ (Silva Ferreira *et al.* 2002b, 2003b). The Vidal blanc cultivar has been found to develop vegetative aromas like asparagus or straw (Chisholm *et al.*, 1995), while Riesling wines are known to develop a kerosene aroma also called petrol note (Marais, 1979). For red wines, flavour deterioration is described as prunes, figs, overripe fruit (Pons *et al.*, 2008).

The phenomenon of bottle sickness is related to wine ageing. Due to bottling, there is a temporary loss of fruit and floral aromas. This effect is reversible and, after a short time of storage, the wines normally show the original flavour profiles. Wildenradt and Singleton (1974) proposed that bottle sickness is attributed to the production of a small amount of acetaldehyde by oxygen contact at bottling. Subsequent formation of compounds that result in a reversible chemical inequilibrium is assumed to occur in this situation.

16.3 Aromatic compounds related to flavour deterioration

Flavour deterioration due to ageing is the result of both degradation and formation. The degradation of aromatic compounds causes the loss of the fruity and floral aromas in young wines. The formation of compounds up to concentrations above their thresholds leads to new perceptible flavours. The wine aroma is often very intense and complex. More than 50 different aroma compounds above the flavour threshold can be found. Many researchers have tried to identify the aroma compounds related to flavour deterioration. Table 16.1 and Fig. 16.2 show the compound groups and certain representatives involved in ageing. In some cases, one or very few impact compounds can be described such as 1,1,6-trimethyl-1,2-dihydronaphthalene (TDN) which contributes to the petrol note in aged Riesling wines (Marais *et al.*, 1992). However, interactions between different aroma

Table 16.1 Odour-active compounds affected by ageing

Group	Compounds		
Aldehydes	Acetaldehyde (IV)	Alkenal (V)	Branched aldehyde (VI)
Esters	Isoamylacetate (VII)	Diethylsuccinate (VIII)	Ethyl ester
Terpenes	Linalool (IX)	Nerol (X)	α -Terpineol (XI)
C-13 norisoprenoids	β -Damascenone (XIII)	TDN (1,1,6-trimethyl-1,2-dihydronaphthalene) (XIV)	Vitispirane
Furanoic derivatives	Furfural (XVIII)	Furfuryl alcohol (XIX)	HMF (5-hydroxymethyl furfural (XX))
Alcohols	2-Methyl-1-propanol (I)	Benzylalcohol (II)	2-Phenylethanol (III)
Fatty acids	Octanoic acid (XII)	Decanoic acid	
Phenols	4-Vinylguaiacol (XV)	4-Vinylphenol (XVI)	Vanillin (XVII)
Sulphur compounds	Dimethyl sulphide		
Lactone	Whiskeylactone (XXI)	4-Ethoxycarbonyl- γ -butyrolactone	

volatiles may enhance or suppress the flavour impact of the single compounds. This could be demonstrated for compounds with relatively low odour activity values (OAV) which can play a more important role than compounds with high OAV (Escudero *et al.*, 2004), or for branched aldehydes that mask unpleasant alkenals (Culleré *et al.*, 2007). Furthermore, low amounts of a substance can be positive while higher amounts can be negative for aroma quality, and even sub-threshold amounts (OAV <1) can influence sensory perception due to synergic effects (Culleré *et al.*, 2007).

Studies are often conducted on heated and/or oxidized wines. Such extreme conditions are used to increase the concentrations of minor aroma compounds and to simulate a long storage time, but it is questionable whether these treatments reflect normal long-term storage.

16.3.1 Aldehydes

Volatile aldehydes represent a large group of compounds subjected to concentration changes during ageing. As aldehydes can be formed from alcohols by oxidation, aroma deterioration due to oxidation is often related to the synthesis of aldehydes (Culleré *et al.*, 2007). Acetaldehyde was one of the first aroma compounds attributed to the aroma changes resulting from contact with oxygen

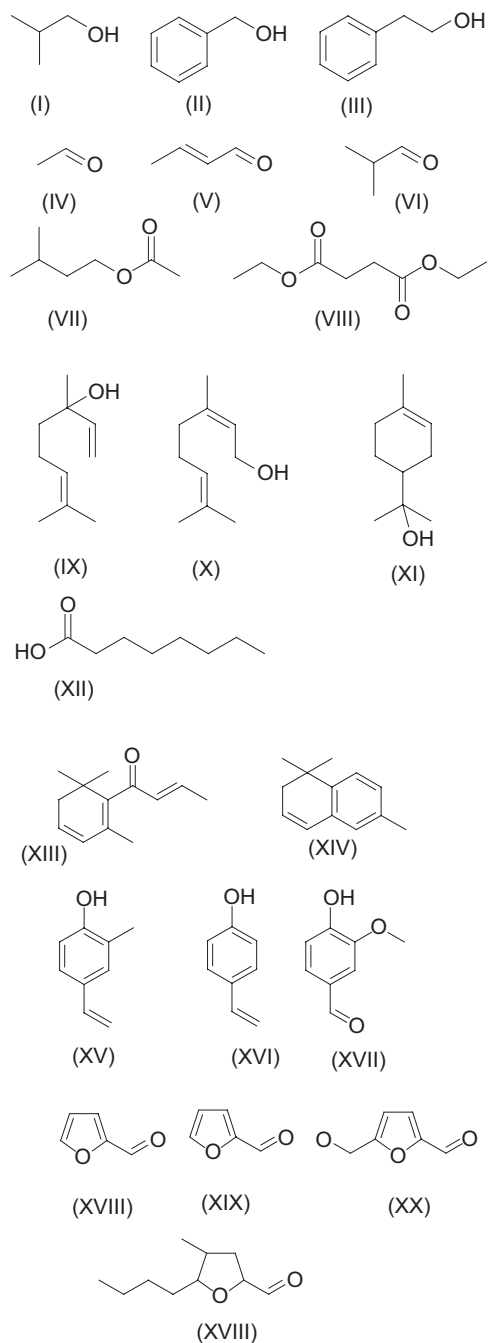


Fig. 16.2 Structural formulae of odour active compounds. Individual compounds are identified in Table 16.1.

(Wildenradt and Singleton, 1974; Baró and Quiro, 1977; Ribéreau-Gayon *et al.*, 2000; Culleré *et al.*, 2007). Sulphur dioxide inhibits this aldehyde-forming reaction, but even in presence of SO₂, some acetaldehyde can be formed (Wildenradt and Singleton, 1974). However, it has been shown that the contribution of acetaldehyde to the aroma of oxidation-spoiled white wine is not so relevant under normal cellar conditions. During the storage of red wines, even a decrease in acetaldehyde can be found, due to an interaction with phenolic compounds. Increases have been attributed to abnormal conditions of wine exposure to air or have been found, for example, in the case of Sherries aged with the solera system (Escudero *et al.*, 2002; Silva Ferreira *et al.*, 2002a; Culleré *et al.*, 2007).

Generally, volatile aldehydes increase during ageing (Wildenradt and Singleton, 1974; Marais and Pool, 1980), but the exact role played by aldehydes in the aroma of wine is not always clearly understood (Ferreira *et al.*, 2004). Gas chromatographic–olfactometric (GC–O) studies showed that wines stored at high temperatures and high levels of dissolved oxygen suffered a rapid oxidative spoilage, which was related to the presence of strong-smelling aldehydes, particularly methional and phenylacetaldehyde (Simpson, 1978b; Escudero *et al.*, 2000; Aznar *et al.*, 2003; Silva Ferreira *et al.*, 2003b; Ferreira *et al.*, 2004; Culleré *et al.*, 2007). Sensory investigations showed that pleasant descriptors were negatively correlated to phenylacetaldehyde (Aznar *et al.*, 2003). The honey-like aroma of oxidized wines can be explained by the formation of phenylacetaldehyde. (Silva Ferreira *et al.*, 2002b). The odour properties of (E)-2-nonenal are particularly important, since it can be responsible for ‘sawdust’ or ‘plank’ off-flavour (Chatonnet and Dubourdieu, 1998; Ferreira *et al.*, 2004). The concentrations of (E)-2-nonenal were clearly highest in red wines, particularly in those with considerable age (Ferreira *et al.*, 2004). Furthermore, branched aliphatic aldehydes (methylpropanal, 2-methylbutanal, 3-methylbutanal) were found in high concentrations in Ports, aged red wines and Sherries (Culleré *et al.*, 2007).

The kind of ageing (oxidation or long-term ageing) influences the type of aldehydes that are formed. Four different patterns can be found: young wines have lowest levels in (E)-2-alkenals and branched aldehydes. The second pattern represents wines oxidized in the laboratory that are richer in (E)-2-alkenals (and in methional and phenylacetaldehyde), but not in branched aliphatic aldehydes. In contrast, Sherries contain higher quantities of branched aldehydes, methional and phenylacetaldehyde, but not of (E)-2-alkenals. The fourth pattern corresponds to aged red wines and Port wines, characterized by large amounts of both (E)-2-alkenals and branched aldehydes (Culleré *et al.*, 2007).

The individual aldehydes caused masking and synergic sensory effects (Culleré *et al.*, 2007). The addition of sub-threshold amounts of (E)-2-octenal or (E)-2-nonenal to neutral red wine caused a significant sensory effect. (E)-2-octenal, at low or higher concentrations, was correlated with negative properties (dirty, closed room and earthy notes). In the case of (E)-2-nonenal, the addition of low concentrations of this compound caused the impression of aged wine with brandy-like notes. Higher amounts of this compound gave earthy, mouldy, papery, dusty and rancid notes to the wine (Culleré *et al.*, 2007). The presence of aliphatic

branched aldehydes masked the unpleasant odour linked to the (E)-2-alkenals and increased the sweet orange and dried fruit notes (Culleré *et al.*, 2007) as is the case for aged red wines and Ports (Culleré *et al.*, 2007).

Further aldehydes formed during ageing include hexenal and hexanal (Marais and Pool, 1980; Roufet *et al.*, 1986; Chisholm *et al.*, 1995). These compounds are often not present above their thresholds, so they will not contribute to the aroma of the aged wines but, together with other aldehydes, they can be considered as markers for oxidation.

16.3.2 Esters

Volatile esters are important compounds of the wine aroma. The acetate esters are of particular importance, but ethyl esters of straight-chain fatty acids with an even number of carbon atoms, such as ethyl hexanoate, ethyl octanoate, or ethyl decanoate, are also considered as highly positive flavour attributes of young wine aroma and introduce floral and fruity flavour notes (Díaz-Maroto *et al.*, 2005). As a result, changes in ester composition with ageing have often been reported to affect the aroma composition of wines (Simpson, 1979; Marais and Pool, 1980; Chisholm *et al.*, 1995). The variation in their concentrations during ageing is different; esters may increase or decrease during storage due to chemical esterification or hydrolysis towards chemical equilibria (Simpson, 1979; Ramey and Ough, 1980; Zoecklein *et al.*, 1999). The concentrations of ethyl esters of straight-chain fatty acids, derived from the lipid metabolism of yeast, and the acetates also, generally decrease during wine ageing (Díaz-Maroto *et al.*, 2005). The initially-produced esters are slowly hydrolyzed during storage until an equilibrium is reached with the corresponding acids and alcohols (Chisholm *et al.*, 1995). On the other hand, the branched fatty acid ethyl esters are more or less stable or can even increase during the ageing of wine (Díaz-Maroto *et al.*, 2005).

In particular, during wine ageing, a decrease can be found in the concentrations of isoamyl acetate, hexyl acetate, 2-phenylethyl acetate, ethyl butyrate, ethyl caprate, ethyl caprylate, ethyl caproate, ethyl 4-hydroxybutyrate, ethyl hexanoate and hexyl hexanoate (Simpson, 1979; Marais and Pool, 1980; Zoecklein *et al.*, 1999; Perez-Coello *et al.*, 2003). Ethyl esters hydrolyze more slowly than acetate esters, so the apple aroma of ethyl butyrate and ethyl 2-methylbutyrate will not be lost very quickly (Ramey and Ough, 1980; Chisholm *et al.*, 1995).

On the other hand, an increase during storage is found for ethyl acetate, diethyl succinate, ethyl lactate, ethyl monosuccinate and diethyl malate (Marais and Pool, 1980; Rapp *et al.*, 1985; Perez-Coello *et al.*, 2003). Opposite tendencies, however, may also occur, so that ethyl esters of caproic, caprylic and capric acids can also increase during ageing (Marais and Pool, 1980).

16.3.3 Terpenes

For some cultivars (e.g. Muscat), monoterpene alcohols are impact compounds to the wine aroma, whereas for other cultivars (e.g. Riesling) they are important for

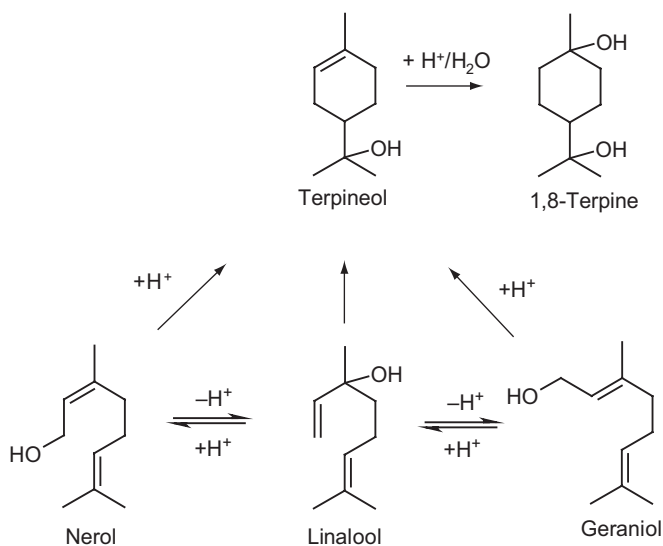


Fig. 16.3 Rearrangements of monoterpenes during bottle ageing (according to Rapp *et al.*, 1985).

the aroma of young wines (Rapp and Mandery, 1986). During storage, due to oxidation, monoterpene alcohols, such as linalool, geraniol and diendiol-1 (3,7-dimethylocta-1,5-dien-3,7 diol), decrease to terpene dioxides and dihydic alcohols that have higher thresholds (Marais, 1978; Simpson, 1979; Simpson and Miller, 1983; Marais and Pool, 1980; Chisholm *et al.*, 1995). For these reasons, Muscat-like wines with intense floral aroma mainly due to the high concentrations of monoterpenes in general do not improve with ageing. Some changes in the concentration of individual free monoterpenes result from biochemical rearrangement and hydrolysis; linalool can be formed from nerol and geraniol, while α -terpineol may be generated by the conversion of linalool, nerol and geraniol (Wilson *et al.*, 1984; Rapp *et al.*, 1985; Zoecklein *et al.*, 1999) (Fig. 16.3). In aged wines, increased concentrations of hotrienol, nerol oxide, linalool oxide and α -terpineol have been observed (Williams *et al.*, 1980; Chisholm *et al.*, 1995; Zoecklein *et al.*, 1999); furthermore, 1,8-terpine is formed during ageing from α -terpineol (Rapp *et al.*, 1985; Rapp and Mandery, 1986) (Fig. 16.3). Rapp and Mandery (1986) proposed α -terpineol as ageing indicator because they found a linear increase of α -terpineol during storage time.

The major portion of the monoterpenes occur as bound forms, mostly as non-volatile monoterpene glycosides. Some of the terpenoids in aged wine are derived from these glycosides but also from other compounds such as hydroxylated linalool (Simpson and Miller, 1983; Günata *et al.*, 1986; Zoecklein *et al.*, 1999). The storage of wines not only decreases the free monoterpenes, but also the bound forms of linalool, α -terpineol, nerol, geraniol, furan linalool; total glycosides are reduced to an average of 60% (Zoecklein *et al.*, 1999).

16.3.4 C₁₃ norisoprenoid compounds

Some of the compounds resulting from carotenoid degradation are important contributors to the aroma of aged wines. An important example of these is 1,1,6-trimethyl-1,2-dihydronaphthalene (TDN), which is responsible for the kerosene-like or petrol note in Riesling wines (Marais *et al.*, 1992; Zoecklein *et al.*, 1999). Apart from TDN, vitispiranes may contribute to the bottle-aged character (Simpson, 1978a,b, 1979; Simpson and Miller, 1983; Zoecklein *et al.*, 1999). In Cava sparkling wines, TDN and vitispirane have been found to increase with storage time (Franciolo *et al.*, 2003). Furthermore, theaspirane, ionene, toluene and m-xylene were identified as thermal degradation products of β -carotene and related isoprenoids (Simpson and Miller, 1983). A number of TDN precursors are present in Riesling wines. They are formed by an acid-catalyzed rearrangement of glycosylated norisoprenoids and possibly non-glycosylated compounds (Winterhalter *et al.*, 1990; Zoecklein *et al.*, 1990).

In cool-climate wine regions, TDN in Riesling wines can normally only be perceived after 4–6 years of bottle ageing. In warm-climate wine producing regions, TDN can arise much earlier and in considerably higher quantities, so that these wines are frequently rejected by consumers. Grapes exposed to direct sunlight also yield wines containing higher TDN concentrations than shaded grapes (Marais *et al.*, 1992). Higher N fertilization has been found to decrease the TDN concentration in aged wines as well as in young wines (Linsenmeier and Löhnertz, 2007). Furthermore, there are minor effects of different Riesling clones and yeast strains on TDN formation (Sponholz and Hühn, 1997).

The intensity of β -damascenone (cooked apple aroma) shows a large decrease upon ageing (Rapp *et al.*, 1985; Chisholm *et al.*, 1995). In general, β -damascenone is positively correlated to pleasant flavour descriptors (Aznar *et al.*, 2003), but β -damascenone is also considered as off-flavour with a strong odour of prunes in red wines (Pons *et al.*, 2008).

16.3.5 Furanic compounds

Numerous furan derivatives have been shown to increase in concentration with storage time: e.g., 2-ethoxyfuran and ethyl 2-furoate and 2-furfural in particular, plus minor quantities of 2-acetal-furan, 5-ethoxymethyl-2-furfural, 5-methyl-2-furfural, 2,2'-methylenebisfuran and 5-hydroxymethyl furfural (HMF) (Simpson and Miller, 1983; Fernández de Simón *et al.*, 2006). The volatile thiol 2-furanmethanethiol is formed from the precursor furfural, and is usually encountered in barrel-fermented white wines, but the furfural and 2-furanmethanethiol concentration may also increase in proportion to the bottle ageing time of Champagne (sparkling) wine (Tominaga *et al.*, 2003). However, this increase in furanic compounds due to storage is by no means ubiquitous. Fernández de Simón *et al.* (2006) observed that furfural, HMF, furfuryl alcohol, and 5-methylfurfural increased during the first year of bottle maturation but slightly decreased during the second year.

Furfural and their derivatives usually go into the wine during barrel storage.

However, furfural can also be formed in wine during bottle ageing, from other hexoses and pentoses (Fernández de Simón *et al.*, 2006). The presence of higher amounts of furfural in older wines is consistent with its formation from carbohydrates, a reaction which is catalyzed by heat and acid, but oxygen seems not to be involved (Simpson, 1979; Simpson and Miller, 1983).

In fortified wines such as Ports and Sherries, the furanone sotolon is found in high concentrations (Cutzach *et al.*, 1998; Lavigne *et al.*, 2008). It is also known to increase due to botrytis infection of the grapes (Sponholz and Hühn, 1994). Sotolon increases with ageing, but usually the concentration remains below the perception threshold (Pons *et al.*, 2008). It is considered as a good 'marker' for the defective ageing phenomenon which affects white wines during bottle ageing: higher concentrations are produced due to elevated oxygen concentrations such as an uncontrolled permeability to oxygen of cork stoppers (Lavigne *et al.*, 2008).

16.3.6 Higher alcohols and fatty acids

Changes in the concentration of higher alcohols and fatty acids are linked with the corresponding ester. An increase of higher alcohols due to ageing can be partially explained by acid-catalyzed ester hydrolysis (Rapp *et al.*, 1985) as well as fatty acids may increase or decrease during storage due to chemical esterification or hydrolysis (Ramey and Ough, 1980; Rapp *et al.*, 1985; Zoecklein *et al.*, 1999). The higher alcohols n-butanol, hexanol and free benzyl alcohols generally increased, whereas amyl alcohols and 2-phenyl ethanol showed decreases. In other studies, the higher alcohols remain practically constant (Marais and Pool, 1980). Hexanoic acid and octanoic acid show no changes during the storage of wines (Marais and Pool, 1980).

16.3.7 Volatile phenols

Volatile phenols in wines are formed as a result of either yeast or bacterial metabolism or of the hydrolysis of oak wood-derived phenols. For example, 4-vinyl guaiacol and 4-vinyl phenol can originate by enzymatic or thermal decarboxylation from cinnamic acids such as p-coumaric and ferulic acids (Zoecklein *et al.*, 1999).

The concentrations of vanillin and syringaldehyde decrease during bottle ageing (Zoecklein *et al.*, 1999; Fernández de Simón *et al.*, 2006). However, the associated chemical reactions allow both formation and degradation, and this is why guaiacol and eugenol can show an increase during the first year of bottling and a decrease during the second year (Fernández de Simón *et al.*, 2006).

Apart from 2-furanmethanethiol, further volatile thiols such as benzenemethanethiol and ethyl 3-mercaptopropionate have been found above their perception thresholds in aged Champagnes. The benzenemethanethiol and ethyl 3-mercaptopropionate concentrations of the wines peaked after 13–15 years and then decreased. There was no correlation between the 3-mercaptohexan-1-ol concentrations and bottle ageing time (Tominaga *et al.*, 2003).

Finally, the lactone γ -nonalactone is considered important for the flavour of aged red wine with a strong odour of prune (Pons *et al.*, 2008). The *cis* and *trans* isomers of whiskey lactone (β -methyl- γ -octalactone), an oak lactone, can increase in bottle maturation; they may be generated from precursor forms that have been extracted from wood barrel. In longer storage, a substantial decrease can be observed (Fernández de Simón *et al.*, 2006).

16.3.8 Sulphur compounds

Sulphur compounds can have noticeable effects on wine flavour. Many authors have observed that dimethyl sulphide (DMS) increases upon ageing; its odour threshold has been determined to be 25 $\mu\text{g/L}$ in wine (Simpson, 1979; Spedding and Raut, 1982; Goniak and Noble, 1987; Chisholm *et al.*, 1995). Loscos *et al.* (2008) identified S-methylmethionine in grapes as a precursor of DMS in wine. Ségurel *et al.* (2005) and Dagan (2006) demonstrated that the potential precursor of DMS is dependent on several factors and increases intensively with overripening.

Other sulphur compounds such as 2-methyl-tetrahydrothiophen-3-one and benzothiazole have been found in aged Riesling wines (Simpson and Miller, 1983). Thioacetic acid-S-esters such as thioacetic acid-S-methyl ester and thioacetic acid-S-ethyl ester can hydrolyze during storage and ageing of wines to the corresponding thiols, methanethiol and ethanethiol which have very low threshold values. Therefore, increased levels of thioacetic acid-S-esters can be the cause for reductive off-flavours in bottle-aged wines if methanethiol and/or ethanethiol are released (Rauhut, 1996, 2003). Thiols can be oxidized to disulphides which have other threshold values and odour impressions. This process is mainly dependent on the ingress of oxygen during storage of wine. Anocibar Belouqui (1998) found bis(2-hydroxyethyl)disulphide, precursor of 2-sulphanylethanol in certain wines.

16.4 Chemical reactions of ageing

Wine ageing is a result of oxidation and non-oxidative processes. For white wine, oxidative spoilage is a long-known problem in winemaking. Nowadays, in general, a reductive vinification occurs with low levels of oxygen in the wines. In these cases, non-oxidative reactions causing flavour deterioration play an important role. These are esterification and ester hydrolysis, etherification, Maillard reactions and glycoside hydrolysis.

16.4.1 Oxidation processes

Reactive oxygen species. Excessive contact of wine with oxygen causes rapid flavour deterioration. Prior to oxidation, the stable O_2 is formed to reactive oxygen species such as the superoxide anion O_2^- , the perhydroxyl radical ($\text{OOH}^\bullet$) or a free oxygen radical $\text{O}^\bullet$. Ferrous iron (Fe^{2+}) and copper ions (Cu^+) are oxidized to Fe^{3+} or Cu^{2+} , respectively, forming the superoxide anion O_2^- from O_2 . The superoxide



Fig. 16.4 Oxidation of alcohol to aldehyde.

anion can form the perhydroxyl radical ($\text{OOH}^\bullet$), having a much higher reactivity (Vanderhaegen *et al.*, 2006). Oxidation in wine also occurs due to a coupled auto-oxidation of certain phenolic substances (caffeic acid, catechin, myricetin), ascorbic acid, melanoidins, reductones or enediols. The oxidation of phenol to a quinone (or ascorbic to dehydroascorbic, etc.) co-produces strong oxidants such as hydrogen peroxide (Wildenradt and Singleton, 1974).

Aldehydes are formed by the *oxidation of alcohols* (see Fig. 16.4). Ethanol, the most important alcohol in wine, is oxidized to acetaldehyde by a coupled auto-oxidation (Wildenradt and Singleton, 1974). There are numerous higher alcohols in wine such as 2-methyl-propanol, 2-methyl-butanol and 2-phenyl-ethanol, and the corresponding aldehydes are also found in wines, but it is questionable if the direct oxidation of alcohols is the main pathway for these aldehydes. Another important aldehyde of wine ageing is methional. The direct peroxidation of methionol to methional has been mentioned by several authors, as a decrease of methionol can be observed with ageing (Ferreira *et al.*, 1997; Silva Ferreira *et al.*, 2002b), but methional can be formed by Strecker degradation, too (Culleré *et al.*, 2007).

16.4.2 Esterification and ester hydrolysis

Esters are formed by chemical condensation between alcohol and organic acids. In wine, esterification and ester hydrolysis are acid-catalyzed equilibrium reactions (Fig. 16.5). The ester profile is also influenced by enzymes with esterase activity in yeasts (Vanderhaegen *et al.*, 2006). Elevated concentrations of ethyl esters of straight-chain fatty acids and acetates are formed during fermentation due to the lipid metabolism of yeasts (Díaz-Maroto *et al.*, 2005). In general, these esters (isoamyl acetate, hexyl acetate, 2-phenylethyl acetate, ethyl butyrate, ethyl caprate) decrease to their equilibrium in wines, while other esters such as diethyl succinate or ethyl lactate are formed during ageing (Chisholm *et al.*, 1995).

16.4.3 Maillard reactions

The diverse reactions between reducing sugars and an amino group-containing substance, together with the following reactions, are classified as Maillard reactions (Vanderhaegen *et al.*, 2006). These reactions can be divided into three



Fig. 16.5 Esterification of acetic acid in ethanol results in an ester. The esterification is a reversible reaction.

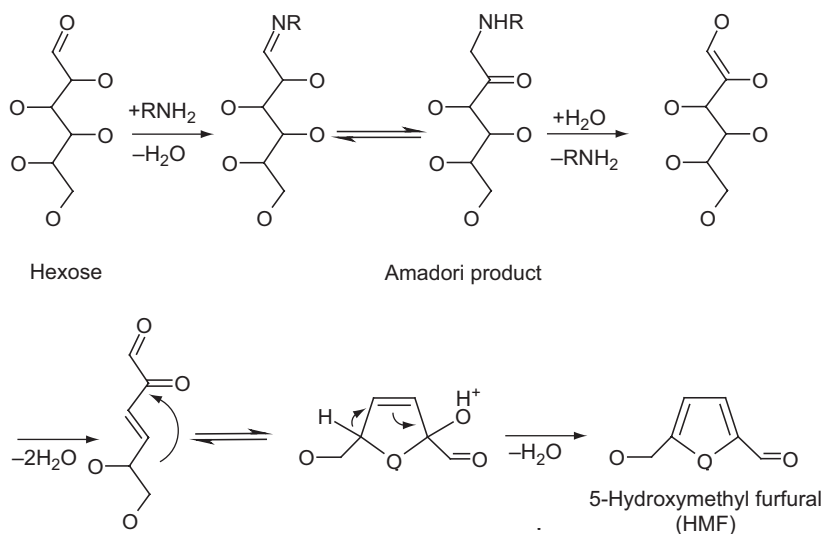


Fig. 16.6 Formation of 5-hydroxymethyl furfural (HMF) (according to Vanderhaegen *et al.*, 2006).

mechanisms: first, the Amadori or Heyns rearrangement which generates cyclic compounds such as furfurals, pyrazines or thiazoles (Fig. 16.6); second, the Strecker degradation, where amino acids in the presence of dicarbolic compounds are formed to odour-active aldehydes such as methional and phenylacetaldehyde (Fig. 16.7). This reaction involves transamination, followed by decarboxylation of the subsequent α -ketoacid, resulting in an aldehyde with one carbon atom less than the amino acid (Vanderhaegen *et al.*, 2006). Finally, the intermediary of the Maillard reaction, together with other compounds, can produce melanoidins (browning substances and roast/burnt flavouring compounds).

Sotolon is formed at a constant rate during ageing, and its concentration is highly correlated with the concentrations of sugar and sugar derivatives (HMF, furfural). Therefore, sotolon is suggested to be formed by Maillard reactions as a result of condensation of molecules such as butane-2,3-dione (diacetyl) and hydroxyacetaldehyde (Silva Ferreira *et al.*, 2003a).

Methional increases in wines spiked with methionine, which suggests that methional can be formed via the Strecker degradation of methionine (Culleré *et al.*, 2007). It was concluded that the Strecker mechanism is the main pathway for the formation of methional and not direct oxidation (Silva Ferreira *et al.*, 2002b). Nevertheless, dissolved oxygen plays a fundamental role in the formation of methional, due to the formation of α -dicarbonyl compounds, which are reactants in the Strecker reaction (Silva Ferreira *et al.*, 2002b). A chemical degradation of the amino acid threonine into 2-ketobutyric acid is possible due to the acidic conditions in wine (Silva Ferreira *et al.*, 2003a).

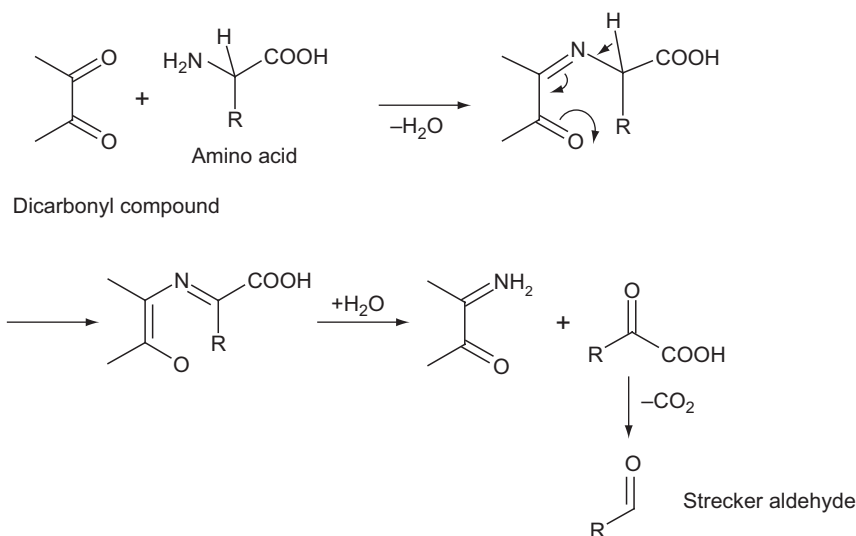


Fig. 16.7 Strecker degradation of amino acids (according to Vanderhaegen *et al.*, 2006).

16.4.4 Glycoside hydrolyzation

Monoterpenes in must are found in free forms and glycosidically bound, where the major portion of terpenes in must consists of glycosidically bound forms. For example, linalool is bound 80–85%, geraniol, linalool oxide and α -terpineol up to 95% (Günata *et al.*, 1986; Francis *et al.*, 1992; Linsenmeier and Löhnhertz, 2006). During fermentation, only small amounts of the bound forms were hydrolyzed, so in wine there are still considerable parts of glycosidically bound monoterpenes (Rapp *et al.*, 1984; Rapp and Mandery, 1986; Günata *et al.*, 1986; Linsenmeier and Löhnhertz, 2006). Furthermore, norisoprenoids and volatile phenols are glycosidically bound, too (Winterhalter *et al.*, 1990; Versini *et al.*, 1988; Mateo and Jiménez, 2000). Glycosides are non-volatile and therefore of no sensory significance, but the hydrolysis of glycosides generates aroma volatiles (Francis *et al.*, 1992). An increase in terpenes during ageing might be a result of chemical hydrolysis. (E)- β -damascenone might also result from a hydrolyzation during ageing, as the precursors are linked to sugars (Bureau *et al.*, 2000).

Glycosides can be hydrolyzed by enzymes (Günata *et al.*, 1985, 1990) or by acid (di Stefano, 1982; Rapp *et al.*, 1985). The enzymatic hydrolysis can occur with grape glycosidases and with yeast glycosidases (di Stefano, 1989; Günata *et al.*, 1990). This form of hydrolysis does not change the monoterpene distribution and produces a varietal flavour (Mateo and Jiménez, 2000).

While monosaccharide can directly be hydrolyzed by β -glycosidase, the enzymatic hydrolysis of disaccharides is done by various enzymes. In the first step, α -L-rhamnosidase, α -L-arabinosidase or β -D-apiosidase make the cleavage of the

terminal sugar and β -glycosides are released which, in the second step, are hydrolyzed by β -glucosidases (Günata *et al.*, 1998). Grapes have an enzyme with β -glycosidase activity, but it is not stable and only shows low activity in must and wines (Aryan *et al.*, 1987; Günata *et al.*, 1990; Lecas *et al.*, 1991). Furthermore, only low activity of α -L-rhamnosidase, α -L-arabinosidase or β -D-apiosidase has been detected in grapes (Günata *et al.*, 1990). Günata *et al.* (1988) found endoglycosidase in grape skins which are able to hydrolyze disaccharide-glycosides directly, nevertheless, their impact on the release of aroma is not yet clear.

The yeast has enzyme showing β -glycosidase activity, but in general it is glucose inhibited. Some *Saccharomyces* strains have β -glycosidase activity which is quite independent of glucose, but it is inhibited by ethanol (Mateo and Jiménez, 2000). Furthermore the temperature and pH in wines were not optimal for the enzyme activity of β -glucosidases (Delcroix *et al.*, 1994). The major part of the glycosides in grape and must were not hydrolyzed during fermentation (Günata *et al.*, 1986).

Grape enzymes are quite instable and showed low activity; yeast enzymes were inhibited by glucose (Mateo and Jiménez, 2000). Acidic hydrolysis occurs during ageing due to the low pH in the wines, and it is fairly slow in typical winemaking conditions. After hydrolysis, monoterpenoids can be rearranged to other compounds (Sefton, 1998; Mateo and Jiménez, 2000). Francis *et al.* (1992, 1996) were able to show that the products of acidic hydrolysis are typical for the aged character (of red wines). They hydrolyzed an extract of glycosides of grape skin and must and donated the hydrolysate in a neutral white wine.

16.4.5 Aldol condensation

Aldolic condensation between two carbonyl compounds followed by cyclization does not normally occur in wine. In 'flor Sherry', for example, sotolon can be formed during ageing by aldolic condensation between acetaldehyde and 2-ketobutyric acid produced from threonine due to an enzymatic reaction, which is only possible because of the presence of a yeast 'flor' (Silva Ferreira *et al.*, 2003a).

Acetalization of aldehydes is another form of aldol condensation reaction whereby cyclic acetals (R-dioxolane) originate from a condensation reaction between 2,3-butanediol and an aldehyde (acetaldehyde, isobutanol, 3-methylbutanol, respectively) (Vanderhaegen *et al.*, 2006).

16.5 Factors influencing the ageing process and future trends in research

Oxygen ingress into wine is the main factor influencing the oxidative process of wine ageing. The smallest amounts of oxygen contact for wines stored in barrels are found for stainless steel, followed by fiberglass, wood and polyethylene as cooperage material (Troost, 1980). Furthermore, keeping the barrels topped up is good oenological practice to prevent the oxidation of the wine. During bottle

ageing, the type of closure has an enormous influence on the oxygen ingress into the wine. For centuries, bottles have nearly always been sealed with natural cork, and in the year 2000 even (still) 97% of the stopper material was cork. However, in the last decade alternative sealing systems appeared on the market, so nowadays about 50% of the sealing systems are natural and technical cork, 30% are synthetic corks and about 20% are metallic screw caps (Mas *et al.*, 2002; Schuessler *et al.*, 2008). The oxidation of the bottle-stored wines was highest with synthetic corks (Mas *et al.*, 2002; Lopes *et al.*, 2006; Jung, 2007). In general, screw caps were the most airtight closures (Lopes *et al.*, 2006; Jung, 2007), but they also resulted in notably higher oxidation rates compared with the other sealing systems (Mas *et al.*, 2002). The oxygen ingress of natural and technical corks is intermediate between those of screw caps and plastic corks, but it also depends on the quality and the length of the corks. Higher quality and longer corks can lead to the slowest oxygen ingress (Lopes *et al.*, 2006; Jung, 2007). Recently, the glass cork stopper has been available as an innovative sealing system which is as airtight as the screw cap (Jung, 2007).

Bottles were traditionally stored in horizontal position, and it has been mentioned that the vertical storage position leads to higher oxygen ingress, especially for natural and technical corks which might dry out in vertical position (Jung, 2007). In fact, the corks of bottles stored upright are drier than those of bottles in horizontal position, but also for the upright storage the humidity of the corks remained high enough to preserve the elasticity of the corks (Jung, 2007). Consequently, the influence of the storage position on the ageing of wines has not been found to be consistent. Whereas Mas *et al.* (2002) and Jung (2007) detected higher oxidation for vertically stored bottles combined with low-quality corks and plastic corks, there was no influence for wines sealed with screw caps. In contrast to this, Lopes *et al.* (2006) have not found an impact of the storage position regardless of the sealing closure.

Further packaging factors concerning ageing due to oxygen ingress were filling height, the gas composition in the headspace and bottling line procedure (Godden *et al.*, 2005; Jung, 2007).

The oxidation process during wine ageing is mainly influenced by the wine's redox potential. Therefore, the concentration of dissolved oxygen, but also promoting compounds such as metal ions and inhibiting compounds, the antioxidants, play an important role. It is by no means trivial to conclude from chemical analysis that the redox potential is an indicator of the wine's future dynamic of ageing. In this sense, the potentiometrically-measured 'resistance to oxidation' might be a useful instrument. It has been shown that the sensory degradation is better explained by the 'resistance to oxidation' than by wine age or the degree of browning. Linalool, for example, decreases with lower "resistance to oxidation" (Silva Ferreira *et al.*, 2003c).

Sulphite is certainly a very effective antioxidant added to wine during winemaking. Polyphenol compounds are important antioxidants originating from the grapes. Polyphenols can react with free radicals producing more stable phenoxy radicals due to the delocalization of free radicals over the aromatic ring.

But there are also polyphenols which are pro-oxidants due to their ability to transfer electrons to transition metal ions (Vanderhaegen *et al.*, 2006). The phenolic acids caffeic acid and gallic acid in wines inhibit the decrease of volatile esters and terpenes (Lambropoulos and Roussis, 2007).

Reductones such as ascorbic acid are further antioxidant compounds. The reducing power of the reductones is due to the endiol group, which can generate carbonyls. Further reductones are melanoidins produced by the Maillard mechanism. Chelating agents for metal ions can inhibit oxidation processes, too. Some phenols, amino acids and melanoidins are able to chelate transition metal ions (Vanderhaegen *et al.*, 2006).

Storage temperature is a key factor influencing wine ageing (Marais and Pool, 1980; Silva Ferreira *et al.*, 2002b; Hernanz *et al.*, 2009). Higher temperatures accelerate the ageing process. Like all chemical reactions, the ageing process follows the generalization of the Arrhenius equation saying that the reaction rate doubles for every 10 °C increase in temperature. Wine stored for 12 months at 30 °C or even 20 °C lost the young wine bouquet, while, at storage temperatures of 0 °C and even 10 °C, the bouquet was practically unchanged (Marais and Pool, 1980). Consequently, it is very important to store the wines at low temperature because in wine production a too high storage temperature is the main cause for prematurely aged wines. For the madeirization process, the wines are stored at higher temperatures but, apart from these special wines, higher storage temperatures can be used to improve wine quality due to faster maturation.

The pH of the wines also has substantial effects on ageing (Marais, 1978; Marais and Pool, 1980; Silva Ferreira *et al.*, 2002b). Lower pH may lead to higher concentrations of esters, aldehydes, linalool oxide and norisoprenoid compounds such as TDN or vitispirane (Marais, 1978; Silva Ferreira *et al.*, 2002b).

The illumination of bottles during storage can also affect the wines, resulting in a decrease of fruity aroma and an increase of thiol-related aroma. This off-flavour called light-struck (also sunstruck or goût de lumière) is due to sulphur compounds such as dimethyl sulphide, dimethyl disulfide, hydrogen sulphide, methanethiol and methyl mercaptan (Maujean *et al.*, 1978; Dozon and Noble, 1989; La Follette *et al.*, 1993; D'Auria *et al.*, 2002; Jung *et al.*, 2007). Light photoactivates pantothenate, riboflavin or its derivate which catalyze the degradation of sulphur-containing amino acids such as methionine and cysteine (Maujean and Seguin, 1983).

16.5.1 Future trends

There is a lack of published information about the influence of viticulture on the ageing process. Actual trials have investigated the impact of stress on the vine (UV radiation, water stress, etc.) on ageing. Another ongoing research topic seeks to find methods allowing one to predict the shelf life of bottled wines.

16.6 Untypical ageing (UTA) off-flavour

16.6.1 Introduction

The untypical ageing (UTA; syn. atypical ageing (ATA)) off-flavour is a defect in white wines which develop a bouquet characterized by the loss of fruitiness as well as odour taints like acacia blossom or naphthalene. UTA is of great economic importance for the grape and wine production industry; a wine with strong UTA flavour cannot be sold, and the reputation of a winery can be strongly damaged when the wine develops this off-flavour after being sold to customers. Up to now, the UTA flavour has only been observed in white wines.

The process of untypical ageing is completely different from that of oxidative spoilage, old age of wines or premature ageing of wines; therefore the expression 'premature ageing' should not be used for UTA. Moreover, 'untypical ageing' is not even an ageing process, but an off-flavour occurring without ageing, i.e. usually very soon after the bottling of the wine.

UTA was first discovered in Germany in the early 1990s (Pohl, 1992; Jakob, 1993). The sensory attributes of UTA wines were described as: acacia blossom, naphthalene, polish smells, wax, furniture varnish and dirty dish cloth, along with a loss of fruitiness, bitterness, phenolic flavours and light colour. Recently, this off-flavour has been reported not only in many European countries but also in the USA, Australia and New Zealand (Rapp *et al.*, 1993; Henick-Kling *et al.*, 2008).

16.6.2 Chemical formation of o-aminoacetophenone (AAP) and oenological untypical ageing inhibition

AAP was identified as a character impact compound for the UTA off-flavour using gas chromatography–mass spectrometry (GC–MS) and GC–O (Rapp *et al.*, 1993). Besides anthranilic acid methyl ester (methyl anthranilate), AAP is also responsible for the foxy note of *Vitis labruscana* grapes (Nelson *et al.*, 1977; Acree *et al.*, 1990). In *V. vinifera* grapes, AAP can be found in concentrations <0.2 µg/L (Hoenicke, 2002). Sensory evaluations by Christoph *et al.* (1995) showed that the threshold in water is 0.2 µg/L. In white wine, the threshold is generally 0.5–1.0 µg/L, although sometimes even a concentration of 0.3 µg/L can be perceived. In red wines, concentrations >1.5 µg/L are necessary to produce changes in the aroma. Small concentrations have often been judged as positive; in aromatic white wines, this is true even up to 1 µg/L.

AAP is mainly formed in wines during storage. Immediately after fermentation, there are only low concentrations of AAP in wines; it is formed after sulphiting. In unsulphited wines, AAP is not found (Christoph *et al.*, 1998; Hoenicke, 2002). The formation mechanism of AAP in wines is still not known. It seems to be that AAP is formed from the precursor indoleacetic acid (IAA) due to sulphiting (Fig. 16.8). The hypothesis is that the oxidation of sulphite induces the formation of superoxide radicals which results in the cleavage of the indole ring and the formation of AAP from IAA (Christoph *et al.*, 1998; Hoenicke, 2002). In accordance with this hypothesis, it has been found that the donation of antioxidants to the wine before

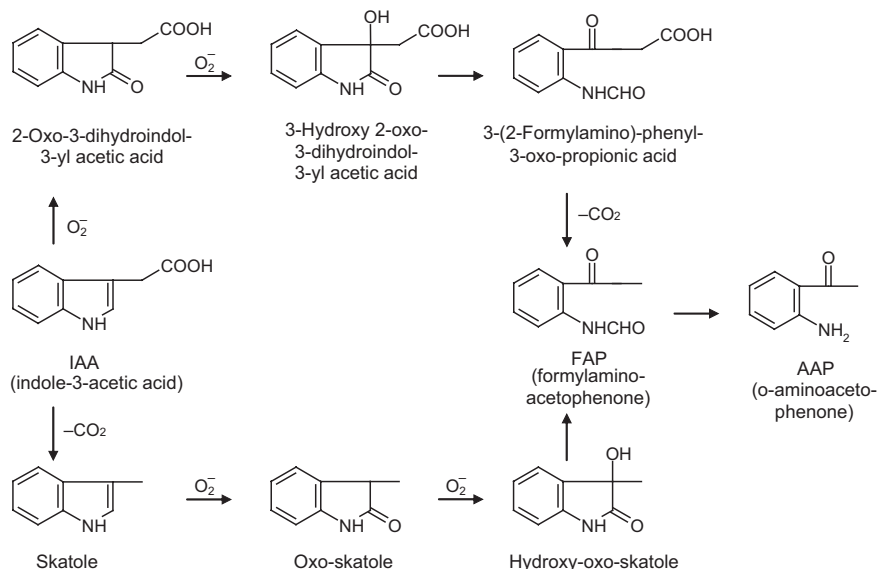


Fig. 16.8 Proposed formation of o-aminoacetophenone (AAP) by co-oxidation of indoleacetic acid (IAA) (according to Hoenicke *et al.*, 2002, modified).

sulphiting can inhibit the formation of AAP. Further compounds that can be co-oxidized to AAP due to sulphiting are, for example, tryptophan, indole lactic acid and kynurenine, but the formation rate of AAP is much lower than that from IAA (Christoph *et al.*, 1998, 1999; Hoenicke, 2002). AAP can also be formed microbially by yeasts from tryptophan, but these are normally only small amounts below the threshold (Ciolfi *et al.*, 1995; Rapp *et al.*, 1995; Sponholz *et al.*, 1997). There is also a minor influence of yeast strain on AAP formation (Sponholz *et al.*, 1997; Hühn *et al.*, 1999).

The oenological possibilities to prevent or even to eliminate an UTA off-flavour appear to be limited. Fining agents cannot remediate UTA-tainted wines. The use of high amounts of activated carbon also is not recommended – the AAP concentration will be reduced but, during storage, it nonetheless increases. Also, positive wine aromas decrease with the use of activated carbon, so the UTA aroma can even increase following carbon treatment due to minor masking effects (Rapp *et al.*, 1998; Kettern, 2000). Low temperatures during fermentation and the dosage of fermentation additives cannot decrease the AAP formation but may reduce the UTA intensity due to masking effect by other aromas (Bader, 1996; Köhler *et al.*, 1996). Low storage temperatures (<10 °C) can strongly reduce but not prevent AAP formation (Hoenicke, 2002). A heavy pressing or the on-skin fermentation of white grapes result in wines with reduced AAP concentrations, probably due to the higher polyphenol levels, but these wines were nonetheless of inferior sensory quality (Köhler *et al.*, 1996). The addition of antioxidants remains the only suitable oenological method to reduce the UTA risk. In particular, the use of ascorbic acid

before or shortly after the sulphating, is an effective method to decrease AAP formation (Geßner *et al.*, 1998, 1999).

16.6.3 o-Aminoacetophenone and other aroma compounds responsible for untypical ageing

AAP is the aroma compound mainly responsible for the UTA off-flavour. As described above, the threshold is strongly influenced by masking effects of other wine aroma compounds. Fan *et al.* (2007) found that wines spiked with increasing AAP up to 1 µg/L showed UTA-associated aromas (mothball, medicinal, painty). Christoph *et al.* (1995) also found highly significant correlations between AAP-spiked wines (up to 4 µg/L) and the sensory perception of UTA, with $r^2 = 0.67$ (white wine) and $r^2 = 0.88$ (red wine), but small concentrations were often judged as positive; in aromatic wines this is true even up to 1 µg/L.

When comparing analytical AAP concentrations with sensorially-perceived UTA values, various contradictory results can be observed: when AAP is above the threshold, strong relationships between AAP and UTA have been found (Linsenmeier *et al.*, 2007a; Rauhut *et al.*, 2007; Schmarr *et al.*, 2007). Generally, low UTA intensities at relatively high AAP values can be attributed to the masking effects of the wine matrix. When AAP is below the threshold, no correlation is found between AAP and UTA (Fig.16.9), and in some cases UTA intensity was actually high (Seiter, 2000; Hoenicke, 2002; Linsenmeier, 2007; Henick-Kling *et al.*, 2008; Pflieginger *et al.*, 2008).

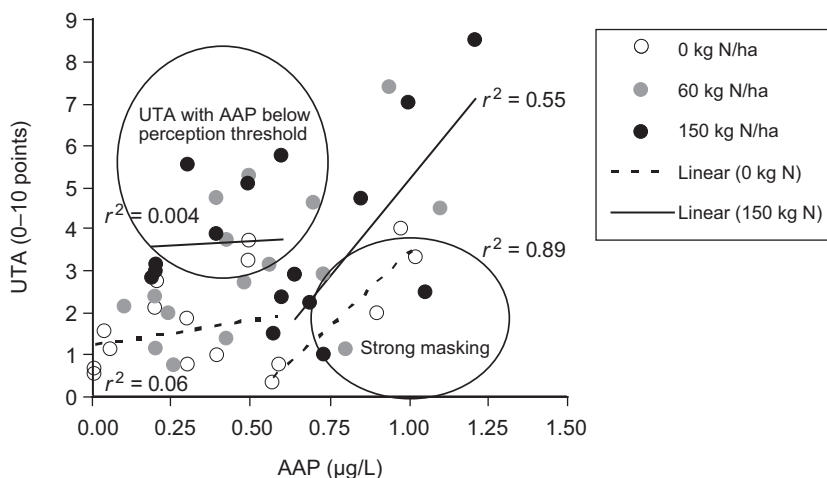


Fig. 16.9 How UTA (untypical ageing) is influenced by the o-aminoacetophenone (AAP) concentration in wine (wines are from vintages 1996–1999 and from three N fertilization levels, data from Linsenmeier *et al.* (2007a)). Regression lines for the 0 and 150 kg N/ha treatments were included separated for below and above perception threshold of AAP (0.6 µg/L).

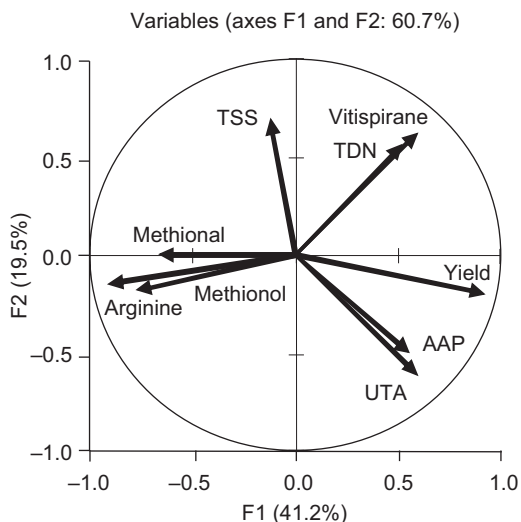


Fig. 16.10 Principal components analysis of various aroma compounds (o-aminoacetophenone (AAP), vitispirane, methional, methionol, 1,1,6-trimethyl-1,2-dihydronaphthalene (TDN)) and indicators (total soluble solids (TSS), yield, arginine) discussed for UTA (untypical ageing). Data from an N fertilization trial (Linsenmeier, 2007); $n = 24$ (TDN and vitispirane); $n = 48$ (remainder)).

Measurement faults in both AAP and UTA analysis notwithstanding, there are likely other aroma compounds responsible for UTA. This matches the different descriptors of UTA; e.g. the floral aroma (acacia blossom) of AAP on one hand and the naphthalene note on the other hand, which might be caused by another substance (Fischer and Sponholz, 2000; Seiter, 2000). For these reasons, further UTA off-flavours have been postulated repeatedly, although no relationship has thus far been found. Still, Christoph *et al.* (1995) have excluded TDN, vitispirane, methyl anthranilate, anthranilic acid ethyl ester (ethyl anthranilate) and indole, because the corresponding amounts found in wines tainted with UTA were far below the odour threshold. The odour threshold of these compounds is ca 100 µg/L, but only 5 µg/L of ethyl anthranilate and 15 µg/L of indole (Christoph *et al.*, 1995) were measured. Rauhut and Kürbel (2002) found high concentrations of methional correlated with high AAP concentrations in certain UTA wines. Henick-Kling *et al.* (2008) proposed TDN as an indicator for UTA in wines, but other trials showed no correlation between UTA and TDN or methional, respectively (Fig.16.10). Geßner *et al.* (1999) did not find a typical UTA impression when skatole, indole, guaiacol and ethyl anthranilate were added to white wines. The threshold of skatole in wine is about 1 µg/L. Only wines spiked with skatole from 3 µg/L upwards were clearly tainted. Thus, skatole concentrations <0.1 µg/L (Hühn *et al.*, 2002) are unlikely to cause UTA. Ciolfi *et al.* (1995) found odour-active substances similar to AAP, and Dubourdieu and Lavigne-Cruege (2002) proposed sotolon as an UTA-active substance, but the importance of these substances for UTA has not yet been examined.

Coulter *et al.* (2007) detected indole in specific off-flavour wines in levels up to 350 µg/L. Indole seems to be the cause of an off-flavour in a small number of certain vintage wines which were affected by a plastic/chemical-like character.

Thus, the research into UTA off-flavours other than AAP is still an active field. Furthermore, in sensory tests a clear distinction should be made between the flowery type of UTA due to the aroma compound AAP and other attributes because these different off-flavours might possibly have different causes.

16.6.4 Viticultural factors – untypical ageing prevention

There is a general consensus that UTA is caused by various stress reactions in the vine. In the first publications about UTA, drought, high yield and reduced fertilization were mentioned as triggering factors (Pohl, 1992; Jakob, 1993). Inept oenological practices and pesticides have been excluded as UTA causes (Pohl, 1992; Christoph *et al.*, 1995; Rapp and Versini, 1995). Although there is no doubt that UTA is caused by viticultural factors, there is a lack of information as to what viticultural factors coupled with weather conditions cause UTA. The sensorial impression of an UTA flavour is the result of a complex interaction between UTA aroma compounds (mainly AAP), other off-flavour compounds that can increase UTA and positive aroma compounds masking an UTA. Viticultural factors can influence both UTA-causing aroma compounds as well as UTA-masking compounds. Figure 16.11 summarizes the complex relationship between the sensory UTA note and the viticultural risk factors and how they are linked with aroma compounds and precursors. Several studies dealing with UTA have been conducted without measuring aroma compounds, so it is not clear if the results were due to UTA-influencing aroma compounds, UTA-masking compounds or both. Furthermore, these aroma compounds cannot be found, in general, in the grape berries, but are produced during and after fermentation from precursors. This is the reason why it is difficult to determine in which physiological way the formation of UTA is influenced in the plant. One problem of many trials is that UTA is often determined in wines produced with forced ageing (storage at higher temperature for a few days) to save time. Still, it is not sure if the UTA caused by forced ageing is the same as in normally aged wines. Thus, UTA formation in wine continues to be a vital research field.

There is a strong *influence of the vintage* on UTA. In Germany, for example, wines from the vintages 1983, 1989, 1991, 1993, 1997, 1999, 2003 have been especially tainted with UTA off-flavours. Statistically, these UTA years can be correlated with dry weather conditions. According to Sponholz *et al.* (1997), water deficiency in the period between *véraison* and 60 °Oe leads to UTA vintages. Schwab *et al.* (1996) found a high UTA risk at a soil moisture of less than 30% available field capacity in sandy soil (about 30–40 mm available plant water). Nevertheless, from 1994–1999, the statistical influence of soil moisture (between *véraison* and 60 °Oe) on AAP was only 10%, and soil moistures of 20% available field capacity in loamy soil (about 50 mm available plant water) resulted in a vintage with very low UTA in 1996 and in a pronounced UTA vintage in 1999 (Linsenmeier *et al.*, 2007a, 2008). When comparing weather data with UTA

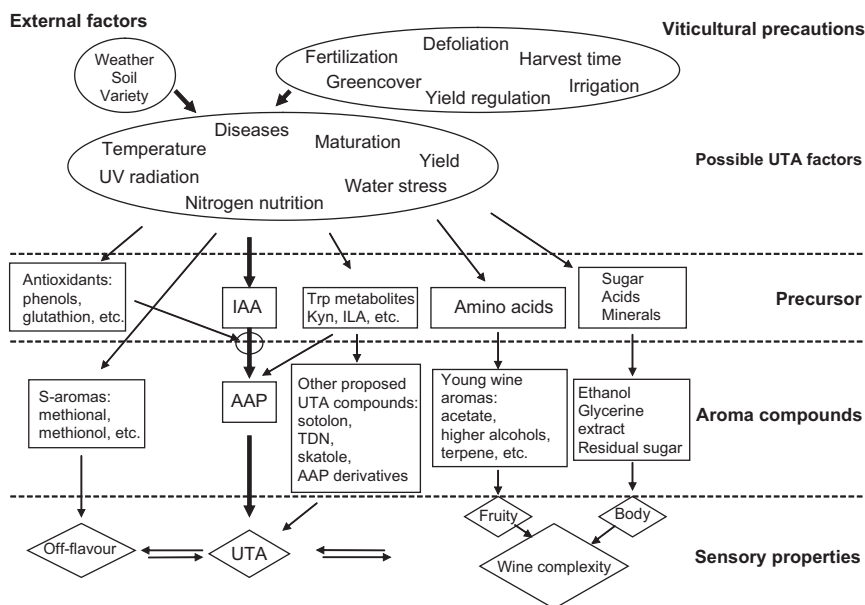


Fig. 16.11 Influence of viticultural factors on untypical ageing (UTA) and its aroma compounds. Abbreviations: AAP, o-aminoacetophenone; IAA, indoleacetic acid; ILA, indolelactic acid; Kyn, kynurenine; TDN, 1,1,6-trimethyl-1,2-dihydronaphthalene; Trp, tryptophan.

values, it must be considered that the variable 'vintage' in reality represents an inseparable mix of variables; apart from weather conditions, other factors such as yield, soluble solids, N supply and plant disease can be regarded as being part of the variable 'vintage'. For example, in years with widespread plant diseases as in 1999 (*Botrytis*, sour rot), the harvest date was early, which is also considered as an UTA risk. Furthermore, vintages with high yields or of low potential wine quality have proved to accelerate UTA in wines.

The early opinion that all kinds of stress of the vine plant induce UTA cannot be substantiated. This was shown by N fertilization trials. Studies of the influence of N fertilization on the formation of UTA have given inconsistent results. In four vintages, Schwab *et al.* (1996) twice observed lower UTA values in the wines with N fertilization; in one year, however, UTA was highest when the fertilization was moderate and in another year no UTA could be observed, no matter which amount of N was applied. In a long-term N fertilization experiment, Müller (1999) did not detect any differences in the occurrence of UTA in wines. Seiter (2000), however, observed that N supply consistently inhibited the formation of UTA. For irrigated vines, Henick-Kling *et al.* (2008) found in two of three vintages (with dry weather conditions) that there was less UTA in treatments without N fertilization compared with soil or foliar fertilization. It was only in a year without water stress that N deficiency resulted in more wines affected by UTA. Nevertheless, they concluded that, to avoid UTA, a sufficient N apply is important.

UTA can appear very quickly after the bottling of the wine but, as mentioned previously, often this off-flavour occurs later. The *dynamics of UTA perception* of vineyards fertilized differently is shown in Fig. 16.12. In this trial, the wines that were only 1 year old either clearly showed higher UTA, when the vines had an N deficiency (1999) or they showed no N fertilization effect on UTA (in 1998) (Linsenmeier *et al.*, 2007a). However, after 2–3 years of storage, the influence of N fertilization shifted and wines from fertilized vines showed higher UTA values than wines produced from N-deficient vines. This different dynamics depending on N supply can be one reason for diverging results appearing in N fertilization studies. It is easier to understand these contradictory results concerning the influence of N fertilization when the underlying aroma compounds are measured. This is surely not as simple as suggested: aroma analysis can be difficult (especially of AAP); second, there are a lot of possible aroma compounds that occur in the wine; and third, it is not even certain if all aroma compounds involved in the UTA off-flavour are known. The influence of N fertilization on AAP concentration in wine was measured by Seiter (2000) and Linsenmeier *et al.* (2007b). Seiter (2000) found higher AAP concentrations with N deficiency. On the other hand, Linsenmeier *et al.* (2007b) found a clearly positive influence of N fertilization on the formation of AAP concentration in wines (Fig. 16.9). This was confirmed by an N fertilization trial using container and fertigation technique (Pfliehringer *et al.*, 2008). As the results shown in Figs 16.9 and 16.12 were from a long-term trial, where the treatments with no N fertilization showed a clear N-deficiency stress (small growth, light green leaves, minor yield), it must be concluded that plant stress does not inevitably result in an UTA off-flavour, but that stress can also protect wines from UTA and AAP formation.

AAP is the main aroma compound responsible for UTA. In a 4-year trial with three N fertilization levels, the correlation between AAP concentration in wines and UTA intensity was highly significant, but variation was also high (Fig. 16.9), partly due to different N supply. The regression curve of the unfertilized treatment was situated below that of the highly fertilized treatment (150 kg/ha every year). As a consequence, at the same AAP concentrations, the wines from unfertilized (stressed) treatments were found to exhibit lower UTA intensity than wines from fertilized treatments. Furthermore, the regression curves of the vintages with higher N supply and without water stress were situated below the dry years with lowered N supply (Linsenmeier *et al.*, 2007a; not shown). This partly explains why the appearance of UTA is stronger in dry years.

Two different masking effects due to fertilization could be observed. Young wines profited from N fertilization, probably by accumulating fruity aromas. Better N supply led to an increase of sensory fruitiness as well as of fruity aromas measured analytically (Rapp and Versini, 1996; Linsenmeier *et al.*, 2005). Aged wines often showed higher UTA levels with N fertilization. In older wines, the fruity aromas were reduced and N deficiency increased the masking capacity of UTA, due to phenols and higher alcohols formed in stress situations in the zero treatments. This might explain the contradiction between our findings and former descriptions of positive effects of N fertilization for the prevention of UTA.

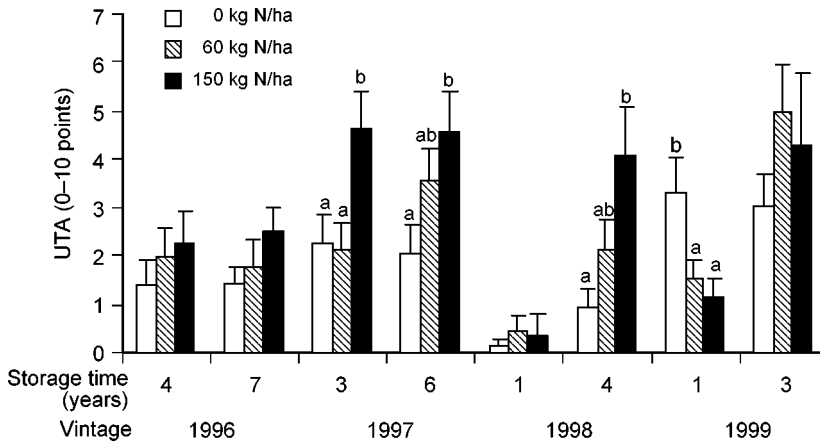


Fig. 16.12 Dynamic of untypical ageing (UTA) perception in a N fertilization trial ($n=4$, error bars = standard error; data from Linsenmeier, 2007). Significant differences between treatments are indicated by different letters above columns ($\alpha = 5\%$).

Drought is considered as an important viticultural UTA risk factor (Pohl, 1993; Rapp and Versini, 1995; Löhnertz, 1996). Nevertheless, there is a considerable lack of published information about the influence of the water stress (irrigation) of vine plants on UTA. In a trial with Silvaner in 1991, the UTA value in wines actually increased due to irrigation. The AAP concentration only increased with irrigation and soil treatment, but not with green cover (Schwab *et al.*, 1996; Rapp *et al.*, 1998). In contrast to this, in a 3-year trial, Henick-Kling *et al.* (2008) found that wines from irrigated treatments tended to have less UTA. In a trial with Chardonnay in 2002, Reynolds *et al.* (2007) observed a UTA risk for non-irrigated treatment and for irrigation until the lag phase or *véraison*. They found no UTA risk for irrigation until berry set or for full season irrigation. Pflieghinger *et al.* (2008) found that the UTA risk (as well as the AAP formation) increased with a drought from *véraison* onwards, especially when the vines had enough water before *véraison*. However, there was no increased UTA risk when the vines had a low water supply for the entire growing season, compared to those that always had enough water.

In wines made from grapes grown with permanent *green cover*, UTA occurred more often (Wohlfahrt, 1995; Schwab *et al.*, 1996; Schwab, 1998; Seiter, 2000; Löhnertz *et al.*, 2002), or showed no effect; but AAP was not influenced by the soil treatment (Schwab *et al.*, 1996; Rapp *et al.*, 1998). On a dry experimental site, Schorr (2003) also observed no differences between permanent green cover and clean cultivation. A permanent green cover is a potential water and nutrition competitor with the vine, and therefore it should influence UTA appearance much like the combination of N deficiency and drought stress. It must be kept in mind, however, that N deficiency is an UTA risk in young wines but, in old wines, it results in minor UTA symptoms, while lack of irrigation is also not consistently reported as a UTA risk. Therefore, it is not surprising that a green cover can also result in both less and more UTA in wines.

The negative influence of *UV radiation* has been shown. A reduction of UV radiation using foliar protection or the application of adsorbent oils on the vine plants resulted in lower AAP concentrations in wine (Hühn *et al.*, 1999, 2002; Sponholz *et al.*, 2001). On the other hand, further trials with UV adsorbent application also showed no influence on AAP (Cuperus, 2007).

Fruit zone leaf removal has various implications: the photosynthetic performance may be slightly reduced and the translocation of N from the foliage is prevented. Moreover, although there is a better light microclimate of the canopy and the fruit zone, there is also a higher UV radiation. In a 3-year trial, a defoliation of 50% resulted in wines with more UTA, but AAP was not influenced (Schwab and Peternel, 2001; Hoenicke, 2002). Consequently, in this case, fruit zone leaf removal only had a minor masking effect on UTA.

Although *excessive yield* is considered as an important UTA risk (Löhnertz, 1996; Schwab *et al.*, 2001; Walg, 2003), this has not been clearly shown. In two trials with different varieties, only once were higher yields implicated in higher UTA values. AAP concentration was not influenced by yield (Rapp and Versini, 1995; Schwab *et al.*, 1996; Schwab and Peternel, 2001). In a 6-year N fertilization trial, the differences in yield could not explain the high variance of UTA between the fertilization treatments of 1 year. Nevertheless, years with higher yield on average also showed higher UTA values. Considering all years, the AAP concentration was not affected by yield but, in many years, with increasing yield, the AAP concentration in wines was higher (Linsenmeier *et al.*, 2007a, b).

The choice of *rootstock* can play a minor role in UTA formation. According to Peternel (1998), the rootstock SO4 tended to produce more UTA than 5BB, which is more tolerant to stress.

Red wine varieties do not show UTA. This can be explained by the higher phenol concentrations that reduce AAP formation. Besides, due to the more complex aromas in red wines, the perception value of AAP is clearly higher than in white wines (Christoph *et al.*, 1995; Schwab *et al.*, 1999). The wines of white vine varieties can all get UTA. Due to masking effects, aromatic varieties such as Pinot gris, Riesling and Müller-Thurgau are not as affected as wine varieties with neutral aromas (Wohlfahrt, 1995; Bader, 1996; Schwab *et al.*, 1996). These results are more empirical in character; a trial was not conducted in this respect. In all *V. vinifera* wines, AAP concentrations <0.3 µg/L can be found, while in *V. labruscana* varieties, AAP is partly responsible for the 'foxy' note and can be found in concentrations of 6 µg/L (Rapp *et al.*, 1998).

Early harvest is a considerable UTA risk. Wines from grapes that are harvested late can have minor AAP concentrations (Köhler *et al.*, 1995) but, even when the AAP concentration of early and late harvest are equal, the wines from late harvest appear to have less UTA, because they have more body (Köhler *et al.*, 1995; Schwab *et al.*, 1999).

Authors have often remarked that *UTA risk factors were cumulative or additive*: a permanent green cover is especially negative when the harvest is too early, and high yield together with a dry year increase the risk of UTA.

The UTA risk factors listed above often show contradictory results. In trials,

most risk factors can induce as well as reduce UTA, or they do not influence UTA. As mentioned previously, there is still a lack of valid information about UTA formation: in most of the trials the treatments were not replicated. An aroma analysis was rarely conducted.

16.6.5 Indoleacetic acid, antioxidants and indicator precursors

IAA, a phytohormone responsible for plant growth, is supposed to be the main precursor of AAP. Total IAA in grapes is not influenced by stress, but seems to be correlated with amino acids, particularly tryptophan (Linsenmeier *et al.*, 2004). Nevertheless, despite the proposed formation of AAP from the degradation of IAA, AAP in wines is not correlated with IAA in the grapes (Hoenicke, 2002; Linsenmeier, 2007); however, when IAA is added to the wine, the AAP formation increases linearly (Hoenicke, 2002). Antioxidant concentration correlates negatively with UTA (Hoenicke *et al.*, 2002; Linsenmeier *et al.*, 2007a), but only slight correlations were found with AAP (Linsenmeier *et al.*, 2007b).

Low amino acid concentrations in grapes coincide with stress for the vines and have therefore to be considered as indicator for UTA risk. The correlation between amino acids (or total N) in must and the production of AAP was negative over several years (Fig 16.13), whereas positive relationships were found within individual vintages. The overall negative correlation was caused by variation between years: in the years exhibiting high amino acid concentrations in the must, less AAP was found in the wines. The positive correlation within years reflects a reverse effect of N fertilization in this trial. Other compounds in must or wines such as tryptophan metabolites (anthranilic acid, indole lactic acid, tryptophol) or minerals also did not correlate with AAP and UTA (Linsenmeier, 2007). As consequence, a suitable indicator in vines, must or wine to predict the AAP formation or the UTA risk has not been found.

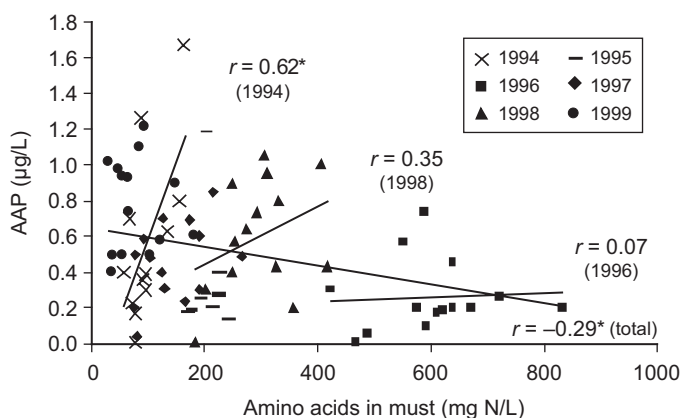


Fig. 16.13 Influence of N fertilization (0–150 kg N/ha, not shown) and vintage on the relationship between amino acids in must and the o-aminoacetophenone (AAP) concentration in the corresponding wine (data from Linsenmeier (2007); *significance at $\alpha = 5\%$).

16.6.6 Future trends

Our knowledge about the chemical formation of AAP in wine is still rather limited. Neither IAA (the proposed precursor) nor antioxidants (regulation of the transformation of IAA to AAP) correlate with AAP formation. Are there other precursors? Is there another regulation of the transformation, or is there another method to measure antioxidants correlating better with AAP transformation? The viticultural causes of UTA formation have not been sufficiently explained; many of the trials listed above were more empirical in character and only show tendencies without dealing with replicates and aroma analysis. Moreover, results were weak and contradictory. More recent trials have dealt with UV radiation and water deficiency. Investigations have repeatedly shown discrepancies between the (supposed) main aromatic substance AAP and the sensory description of UTA. It is also noteworthy that UTA is also detected in wines with AAP below the perception threshold. This can be explained by the existence of other aroma compounds than AAP producing an UTA. Sensory evaluations should clearly distinguish between the 'classical UTA' attribute (acacia blossom due to AAP) and other off-flavours due to different aroma compounds. Up to now, no further aroma substances have been found; therefore, aroma analysis is the focus of present studies.

16.7 References

Ageing section (16.1–16.5)

- Anocibar Belouqui A (1998), *Contribution a l'étude des composés soufrés volatils des vins rouges*, Dissertation, Université de Bordeaux II France.
- Aryan A, Wilson B, Strauss C and Williams PJ (1987), 'The properties of glucosidases of *vitis vinifera* L and a comparison of their β -glucosidase activity with that of possible applications in enology', *Am J Enol Vitic*, **38**, 182–188.
- Aznar M, López R, Cacho J and Ferreira V (2003), 'Prediction of aged red wine aroma properties from aroma chemical composition. Partial least squares regression models', *J Agric Food Chem*, **51**, 2700–2707.
- Baró ALJ and Quiros JA (1977), 'Les conditions de formation des aldehydes dans les vins', *Bull OIV*, **554**, 242–253.
- Bureau SM, Baumes RL and Razungles AJ (2000), 'Effects of vine or bunch shading on the glycosylated flavor precursors in grapes of *Vitis vinifera* L. cv. Syrah', *J Agric Food Chem*, **48**, 1290–1297.
- Chatonnet P and Dubourdieu D (1998), 'Identification of substances responsible for the 'sawdust' aroma in oak wood', *J Sci Food Agric*, **76**, 179–188.
- Chisholm MG, Guiher LA and Zaczekiewicz SM (1995), 'Aroma characteristics of aged Vidal blanc wine', *Am J Enol Vitic*, **46**, 56–62.
- Culleré L, Cacho J and Ferreira V (2007), 'An assessment of the role played by some oxidation-related aldehydes in wine aroma', *J Agric Food Chem*, **55**, 876–881.
- Cutzach I, Chatonnet P and Dubourdieu D (1998), 'Rôle du sotolon dans l'arôme des vins doux naturels, influence des conditions d'élevage or du vieillissement', *J Int Sci Vigne Vin*, **32**, 223–233.
- Dagan L (2006), *Potential aromatique des raisins de Vitis Vinifera L. cv. Petit Manseng et Gros Manseng. Contribution à l'arôme des vins de pays Côtes de Gascogne*, Dissertation, University of Montpellier II.

- D'Auria M, Emanuele L, Mauriello G and Raccioppi R (2002), 'On the origin of the goût de lumière in champagne', *J Photochem Photobiol A: Chem*, **158**, 21–26.
- Delcroix A, Günata YZ, Sapis JC, Salmon JM and Bayonove CL (1994), 'Glycosidase activities of three enological yeast strains during winemaking: Effect on the terpenol content of Muscat wine', *Am J Enol Vitic*, **45**, 291–296.
- Díaz-Maroto MC, Schneider R and Baumes R (2005), 'Formation pathways of ethyl esters of branched short-chain fatty acids during wine aging', *J Agric Food Chem*, **53**, 3503–3509.
- Dozon N and Noble AC (1989), 'Sensory study of the effect of fluorescent light on a sparkling wine and its base', *Am J Enol Vitic*, **40**, 265–271.
- Escudero A, Hernandez-Orte P, Cacho J and Ferreira V (2000), 'Clues about the role of methional as character impact odorant of some oxidized wines', *J Agric Food Chem*, **48**, 4268–4272.
- Escudero A, Asensio E, Cacho J and Ferreira V (2002), 'Sensory and chemical changes of young white wines stored under oxygen. An assessment of the role played by aldehydes and some other important odorants', *Food Chem*, **77**, 325–331.
- Escudero A, Gogorza B, Melús MA, Ortín N, Cacho J and Ferreira V (2004), 'Characterization of the aroma of a wine from Maccabeo. Key role played by compounds with low odor activity values', *J Agric Food Chem*, **52**, 3516–3524.
- Fernández de Simón B, Cadahía E, Hernández T and Estrella I (2006), 'Evolution of oak-related volatile compounds in a Spanish red wine during 2 years bottled, after aging in barrels made of Spanish, French and American oak wood', *Anal Chim Acta*, **563**, 198–203.
- Ferreira V, Culleré L, López R and Cacho J (2004), 'Determination of important odor-active aldehydes of wine through gas chromatography-mass spectrometry of their O-(2,3,4,5,6-pentafluorobenzyl)oximes formed directly in the solid phase extraction cartridge used for selective isolation', *J Chromatogr A*, **1028**, 339–345.
- Franciolo S, Torrens J, Riu-Aumateli M, López-Tamames E and Buxaderas S (2003), 'Volatile compounds by SPME-GC as age markers of sparkling wines', *Am J Enol Vitic*, **54**, 158–162.
- Francis IL, Sefton MA and Williams PJ (1992), 'Sensory descriptive analysis of hydrolysed precursor fractions from Semillon, Chardonnay, and Sauvignon Blanc grape juices', *J Sci Food Agric*, **59**, 511–520.
- Francis IL, Tate ME and Williams PJ (1996), 'The effect of hydrolysis conditions on the aroma released from Semillon grape glycosides', *Aust J Grape Wine Res*, **2**, 70–76.
- Godden P, Lattey K, Francis L, Gishen M, Cowey G, Holdstock M, Robinson E, Waters E, Skouroumounis G, Sefton M, Capone D, Kwiatkowski M, Field J, Coulter A, D'Costa N and Bramley B (2005), 'Towards offering wine to the consumer in optimal condition – the wine, the closures and other packaging variables: a review of AWRI research examining the changes that occur in wine after bottling', *Aust N Z Wine Ind J*, **20**, 20–30.
- Goniak OJ and Noble AC (1987), 'Sensory study of selected volatile sulfur compounds in white wine', *Am J Enol Vitic*, **38**, 223–227.
- Günata YZ, Bayonove CL, Baumes RL and Cordonnier RE (1985), 'The aroma of grapes. I. Extraction and determination of free and glycosidically bound fractions of some grape aroma components', *J Chromatogr*, **331**, 83–90.
- Günata YZ, Bayonove CL, Baumes RL and Cordonnier RE (1986), 'Stability of free and bound fractions of some aroma components of grapes cv. Muscat during wine processing: preliminary results', *Am J Enol Vitic*, **37**, 112–114.
- Günata YZ, Blondeel C, Vallier MJ, Lepoutre JP, Sapis JC and Watanabe N (1988), 'An endoglycosidase from grape skin of cv. M. Alexandria hydrolyzing potentially aromatic disaccharide glycosides', *J Agric Food Chem*, **46**, 2748–2753.
- Günata YZ, Dugelay I, Sapis JC, Baumes C and Bayonove C (1990), 'Action des glycosidases exogènes au cours de la vinification: libération de l'arôme à partir de précurseurs glycosidiques', *Conn Vigne Vin*, **24**, 133–144.

- Günata YZ, Bitteur S, Brillouet JM, Bayonove CL and Cordonnier RE (1998), 'Sequential enzymic hydrolysis of potentially aromatic glycosides from grape', *Carbohydr Res*, **184**, 139–149.
- Hernanz D, Gallo V, Reclames FÁ, Meléndez-Martínez AJ, González-Miret ML and Heredia FJ (2009), 'Effect of storage on the phenolic content, volatile composition and colour of white wines from the varieties Zalema and Colombard', *Food Chem*, **113**, 530–537.
- Hühn T, Sponholz WR and Großmann M (1999), 'Freisetzung unerwünschter Aromastoffe aus Pflanzenhormonen bei der alkoholischen Gärung', *Vitic Enol Sci*, **54**, 105–113.
- Jung R (2007), 'Einflussfaktoren auf die Lagerfähigkeit von Wein', *Deutsches Weinbaujahrbuch*, **58**, 184–192.
- Jung R, Hey M, Hoffmann D, Leiner T, Patz CD, Rauhut D, Schuessler C and Wirsching M (2007), 'Lichteinfluss bei der Lagerung von Wein', *Mitteilungen Klosterneuburg*, **57**, 224–231.
- La Follette G, Stambor J and Aiken J (1993), 'Chemical and sensory considerations in sur lies production of chardonnay wines. III. Occurrence of sunstruck flavor', *Vitic Enol Sci*, **48**, 208–210.
- Lambropoulos I and Roussis IG (2007), 'Inhibition of the decrease of volatile esters and terpenes during storage of a white wine and a model wine medium by caffeic acid and gallic acid', *Food Res Int*, **40**, 176–181.
- Lavigne V, Pons A, Darriet P and Dubourdieu D (2008), 'Changes in sotolon content of dry white wines during barrel and bottle aging', *J Agric Food Chem*, **56**, 2688–2693.
- Lecas M, Günata YZ, Sapis JC and Bayonove CL (1991), 'Purification and some properties of β -glucosidase from grape', *Phytochemistry*, **30**, 451–454.
- Linsenmeier A and Löhnertz O (2006), 'Erhöht Stickstoffdüngung das Aromapotenzial beim Wein?', *Die Winzer-Zeitschrift*, **7**, 36–38.
- Linsenmeier A and Löhnertz O (2007), 'Changes in norisoprenoid levels with nitrogen fertilization in aged *Vitis vinifera* var. Riesling wines', *S Afr J Enol Vitic*, **28**, 17–24.
- Lopes P, Saucier C, Teissedre P-L and Glories Y (2006), 'Impact of storage position on oxygen ingress through different closures into wine bottles', *J Agric Food Chem*, **54**, 6742–6746.
- Loscos N, Segurel M, Dagan L, Sommerer N, Marlin T and Baumes R (2008), 'Identification of S-methylmethionine in Petit Manseng grapes as dimethyl sulphide precursors in wine', *Anal Chim Acta*, **621**, 24–29.
- Marais J (1978), 'The effect of pH on esters and quality of Colombar wine during maturation', *Vitis*, **17**, 296–403.
- Marais J (1979), 'Effect of storage time and temperature on the formation of dimethyl sulphide and on white wine quality', *Vitis*, **18**, 254–260.
- Marais J and Pool HJ (1980), 'Effect of storage time and temperature on the volatile composition quality of dry white table wines', *Vitis*, **19**, 151–164.
- Marais J, van Wyk CJ and Rapp A (1992), 'Effect of sunlight and shade on norisoprenoid levels in maturing Weisser Riesling and Chenin blanc grapes and Weisser Riesling wines', *S Afr J Enol Vitic*, **13**, 23–32.
- Mas A, Puig J, Lladó N and Zamora F (2002), 'Sealing and storage position effects on wine evolution', *J Food Sci*, **67**, 1374–1378.
- Mateo JJ and Jimenénz M (2000), 'Monoterpenes in grape juice and wines', *J Chromatogr A*, 557–567.
- Maujean A and Seguin N (1983), 'Contribution à l'étude des "goûts de lumière" dans le vin de Champagne. III. – Les réactions photochimiques responsables des "goûts de lumière" dans le vin de Champagne', *Sci Aliment*, **3**, 589–601.
- Maujean A, Haye M and Feuillat M (1978), 'Contribution à l'étude des "goûts de lumière" dans le vin de champagne. II. – Influence de la lumière sur le potentiel d'oxydoreduction. Correlation avec la teneur en thiols du vin', *Conn Vigne Vin*, **12**, 277–290.
- Nelson RR, Acree TE, Lee CY and Butts RM (1977), 'Methyl anthranilate as an aroma constituent of American wine', *J Food Sci*, **42**, 57–59.

- Perez-Coello MS, González-Vinas MA, Garcia-Romero E, Diaz-Maroto MC and Cabezero MD (2003), 'Influence of storage temperature on the volatile compounds of young white wines', *Food Control*, **14**(5), 301–306.
- Pons A, Lavigne V, Eric F, Darriet P and Dubourdieu D (2008), 'Identification of volatile compounds responsible for prune aroma in prematurely aged red wines', *J Agric Food Chem*, **56**, 5285–5290.
- Ramey DD and Ough CS (1980), 'Volatile ester hydrolysis or formation during storage of model solutions and wines', *J Agric Food Chem*, **28**, 928–934.
- Rapp A and Mandery H (1986), 'Wine aroma', *Experientia*, **42**, 873–884.
- Rapp A, Mandery H and Ullemeyer H (1984), 'Neue Ergebnisse über die Aromastoffe des Weines', in Lemperle E and Rasenberger H (eds), *Proc 7th Oenol Symp*, Rom, 167–196.
- Rapp A, Güntert M and Ullemeyer H (1985), 'Changes in flavor substances during storage in bottles of white wines of Riesling variety', *Z Lebens Unters Forsch*, **180**, 109–116.
- Rauhut D (1996), *Qualitätsmindernde schwefelhaltige Stoffe im Wein. Vorkommen, Bildung, Beseitigung*, Dissertation, Justus-Liebig-Universität Gießen, Geisenheimer Berichte Band 24.
- Rauhut D (2003), 'Impact of volatile sulfur compounds on wine quality' in Davidian JC, Grill D, De Kok LJ, Stulen I, Hawkesford M, Schnug E and Rennenberg H (eds), *Sulfur Transport and Assimilation in Plants. Regulation, Interaction and Signaling*, Leiden, the Netherlands, Backhuys, 121–131.
- Ribéreau-Gayon P, Glories Y, Maujean A and Dubourdieu D (2000), *Handbook of Enology. Volume 2: The Chemistry of Wine Stabilization and Treatments*, Chichester, UK, Wiley.
- Rouflet M, Bayonove CL and Cordonnier RE (1986), 'Changes in fatty acids from grape lipidic fractions during crushing exposed to air', *Am J Enol Vitic*, **37**, 202–205.
- Schuessler C, Steinschulte M and Jung R (2008), 'Untersuchung zur Sauerstoffdurchlässigkeit alternativer Weinverschlüsse', *Proc 15th Int Enol Symp Int Assoc Enol Manag Wine Market*, Chaintré, France, Oenoplurimedia, 54–56.
- Sefton MA (1998), 'Hydrolytically-released volatile secondary metabolites from a juice sample of *Vitis vinifera* cvs Merlot and Cabernet Sauvignon', *Aust J Grape Wine Res*, **4**, 30–38.
- Ségurel M, Razungles A, Riou C, Trigueiro MGL and Baumes R (2005), 'Ability of possible precursors to release DMS during wine aging and in the condition of heat alkaline treatment', *J Agric Food Chem*, **53**, 2637–2645.
- Silva Ferreira AC, Barbe JC and Bertrand A (2002a), 'Heterocyclic acetals from glycerol and acetaldehyde in Port wines: evolution with aging', *J Agric Food Chem*, **50**, 2560–2564.
- Silva Ferreira AC, Guedes de Pinho P, Rodrigues P and Hogg T (2002b), 'Kinetics of the oxidative degradation of white wines and how they are affected by selected technological parameters', *J Agric Food Chem*, **50**(21), 5919–5924.
- Silva Ferreira AC, Barbe JC and Bertrand A (2003a), '3-Hydroxy-4,5-dimethyl-2(5H)-furanone: a key odorant of the typical aroma of oxidative aged port wine', *J Agric Food Chem*, **51**, 4356–4363.
- Silva Ferreira AC, Olivera C, Hogg T and Guedes de Pinho P (2003b), 'Identification of key odorants related to the typical aroma of oxidation-spoiled white wines', *J Agric Food Chem*, **51**, 1377–1381.
- Silva Ferreira AC, Olivera C, Hogg T and Guedes de Pinho P (2003c), 'Relationship between potentiometric measurements, sensorial analysis, and some substances responsible for aroma degradation of white wines', *J Agric Food Chem*, **51**, 4668–4672.
- Simpson RF (1978a), '1,1,6-Trimethyl-1,2-dihydrionaphthalene: an important contributor to the bottle aged bouquet of wine', *Chem Ind*, **1**, 37.
- Simpson RF (1978b), 'Aroma and compositional changes in wine with oxidation, storage and ageing', *Vitis*, **17**, 274–287.
- Simpson RF (1979), 'Aroma composition of bottle aged white wine', *Vitis*, **18**, 148–154.
- Simpson RF and Miller GC (1983), 'Aroma composition of aged Riesling wine', *Vitis*, **22**, 51–63.

- Singleton VL and Kramling TE (1976), 'Browning of white wines and accelerated test for browning capacity', *Am J Enol Vitic*, **27**, 157–160.
- Singleton VL, Trousdale E and Zaya J (1979), 'Oxidation of wines. I. Young white wines periodically exposed to air', *Am J Enol Vitic*, **30**, 49–54.
- Spedding DJ and Raut P (1982), 'The influence of dimethyl sulphide and carbon disulphide in the bouquet of wines', *Vitis*, **21**, 240–246.
- Sponholz WR and Hühn T (1994), '4,5-dimethyl-3-hydroxy-2(5H) furanon (sotolon), an indicator for Botrytis infection?', *Vitic Enol Sci*, **49**, 37–39.
- Sponholz WR and Hühn T (1997), 'Factors influencing the ageing of Riesling wines – clonal material and used yeast strain', *Vitic Enol Sci*, **52**, 103–108.
- Stefano R, di (1982), 'Occurrence of linalool precursors in white Muscat of Piedmont', *Vigne Vini*, **9**, 45–47.
- Stefano R, di (1989), 'Free and glycoside terpenes and actinidols changes during must and wine storage at various pH', *Riv Vitic Enol*, **42**, 11–23.
- Tominaga T, Guimbertau G and Dubourdieu D (2003), 'Role of certain volatile thiols in the bouquet of aged champagne wines', *J Agric Food Chem*, **51**, 1016–1020.
- Troost G (1980), *Handbuch der Getränketechnologie. Technologie des Weines*, Stuttgart, Germany, Eugen Ulmer.
- Vanderhaegen B, Neven H, Verchtert H and Derdelinckx G (2006), 'The chemistry of beer aging – a critical review', *Food Chem*, **95**, 357–381.
- Versini G, Dalla Serra A, Dell'Eva M, Scienza A and Rapp A (1988), 'Evidence of some glycosidically bound new monoterpenes and norisoprenoids in grapes', in Schreier P (ed.), *Bioflavour 187. Analysis, Biochemistry Biotechnology*, Berlin, Germany, Walter de Gruyter, 161–172.
- Wildenradt HL and Singleton VL (1974), 'The production of aldehydes as a result of oxidation of polyphenolic compounds and its relation to wine aging', *Am J Enol Vitic*, **25**, 119–126.
- Williams PJ, Strauss CR and Wilson B (1980), 'Hydroxylated linalool derivatives as precursors of volatile monoterpenes of muscat grapes', *J Agric Food Chem*, **28**, 766–771.
- Wilson B, Strauss CR and Williams PJ (1984), 'Changes in free and glycosidically bound monoterpenes in developing muscat grapes', *J Agric Food Chem*, **32**, 919–924.
- Winterhalter P, Sefton MA and Williams PJ (1990), 'Volatile C13-norisoprenoid compounds in Riesling wine are generated from multiple precursors', *Am J Enol Vitic*, **41**, 277–283.
- Zoecklein BW, Hackney CH, Duncan SE and Marcy JE (1999), 'Effect of fermentation, aging and thermal storage on total glycosides, phenol-free glycosides and volatile compounds of White Riesling (*Vitis vinifera* L.) wines', *J Ind Microb Biotech*, **22**, 100–107.

UTA section (16.6)

- Acree TE, Lavin EH, Nishida R and Watanabe S (1990), 'o-Aminoacetophenone, the "foxy" smelling component of labruscana grapes', in Bessière Y and Thomas AF (eds), *Flavour Science and Technology*, Chichester, UK, Wiley, 49–52.
- Bader W (1996), 'Das Rätsel um die UTA – Lösung in Sicht?', *Der Deutsche Weinbau*, **9**, 16–21.
- Christoph N, Bauer-Christoph C, Geßner M and Köhler HJ (1995), 'Die "Untypische Alterungsnote im Wein", Teil I: Untersuchungen zum Auftreten und zur sensorischen Charakterisierung der "Untypischen Alterungsnote"', *Rebe und Wein*, **48**, 350–356.
- Christoph N, Bauer-Christoph C, Geßner M and Köhler HJ, Simat TJ and Hoenicke K (1998), 'Bildung von 2-Aminoacetophenon und Formylaminoacetophenon im Wein durch Einwirkung von schwefeliger Säure auf Indol-3-essigsäure', *Vitic Enol Sci*, **53**, 79–86.
- Christoph N, Geßner M, Simat TJ and Hoenicke K (1999), 'Off-flavor compounds in wine and other food products formed by enzymatical, physical, and chemical degradation of tryptophan and its metabolites', *Adv Exp Med Biol*, **467**, 659–669.

- Ciolfi G, Garolfo A and Stefano R di (1995), 'Identification of some o-aminoacetophenones as secondary metabolites of *Saccharomyces cerevisiae*', *Vitis*, **34**, 195–196.
- Coulter AD, Capone DL, Baldock GA, Cowey GD, Francis, IL, Hayasaka Y, Holdstock MG, Sefton MA, Simos CA and Travis B (2007), 'Taints and off-flavours in wine – case studies of recent industry problems', *Proc Thirteenth Aust Wine Indust Tech Conf*, Glen Osmond SA, Australia, 73–80.
- Cuperus FS (2007), *Influence of UV-B radiation on quality determining compounds in must and wine and suspension cell cultures of Vitis vinifera*, Dissertation, Eidgenössische Technische Hochschule ETH Zürich.
- Dubourdieu D and Lavigne-Cruege V (2002), 'Role of glutathione on the development of aroma defects in dry wines', in Trogus H, Gafner J and Sutterlin A (eds), *Proceedings from the 13th International Enology Symposium*, International Association of Enology, Management and Wine Marketing, Breisach, Germany, 331–347.
- Fan W, Tsai IM and Qian MC (2007), 'Analysis of 2-aminoacetophenone by direct-immersion solid-phase microextraction and gas chromatography-mass spectrometry and its sensory impact in Chardonnay and Pinot gris wines', *Food Chem*, **105**, 1144–1150.
- Fischer U and Sponholz R (2000), 'Die sensorische Beschreibung der Untypischen Alterungsnote', *Der Deutsche Weinbau*, **3**, 16–21.
- Geßner M, Köhler HJ, Christoph N and Curschmann K (1998), 'Möglichkeiten zur Vermeidung der Untypische Alterungsnote', *Der Deutsche Weinbau*, **18**, 18–21.
- Geßner M, Köhler HJ and Christoph N (1999), 'Die "Untypische Alterungsnote" im Wein, Teil VIII: Auswirkung von Inhaltsstoffen und Antioxidantien auf die Bildung von o-Aminoacetophenon', *Rebe und Wein*, **52**, 264–267.
- Henick-Kling T, Gerling C, Martinson T, Acree T, Lakso A and Chiang L (2008), 'Studies on the origin and sensory aspects of atypical aging in white wines', *Proc 15th Int Enol Symp Int Assoc Enol Manag Wine Market*, Chaintre, France, Oenoplurimedia, 31–36.
- Hoenicke K (2002), *Untersuchungen zur Bildung von 2-Aminoacetophenon im Wein und Entstehung der 'Untypischen Alterungsnote' (UTA)*, Dissertation, Universität Hamburg.
- Hoenicke K., Simat, T. J., Steinhart H, Christoph N, Geßner M and Köhler HJ (2002), 'Untypical aging off-flavor' in wine: formation of 2-aminoacetophenone and evaluation of its influencing factors', *Anal Chim Acta*, **458**, 29–37.
- Hühn T, Sponholz WR, Bernath K, Friedmann A, Hess G, Muno H and Fromm W (1999), 'The influence of high-energy short-wave radiation and other environmental factors on the genesis of compounds affecting the wine quality in *Vitis Vinifera* L., c.v. Müller-Thurgau', *Vitic Enol Sci*, **54**, 101–104.
- Hühn T, Cuperus S, Pflieghinger M, Sponholz WR, Bernath K, Platzwahl W, Großmann M, Amadò R, Galli J and Friedmann A (2002), 'Einfluß von Umwelt- und Substrateffekten auf die Bildung von wertgebenden Weininhaltsstoffen', in Trogus H, Gafner J and Sutterlin A (eds), *Proceedings from the 13th International Enology Symposium*, International Association of Enology, Management and Wine Marketing, Breisach, Germany, 313–328.
- Jakob K (1993), 'Naphtalinton – läßt er sich vermeiden?', *Der Deutsche Weinbau*, **13**, 23.
- Kettern W (2000), 'UTA-Vermeidung durch sachgerechte Kellerwirtschaft', *Die Winzerzeitschrift*, **15**, 30–32.
- Köhler HJ, Christoph N, Geßner M and Bauer-Christoph C (1995), 'Die "Untypische Alterungsnote" im Wein, Teil III: Zusammenhänge zwischen dem Auftreten der "Untypischen Alterungsnote" und dem Reifestadium der Trauben (Lesetermin)', *Rebe und Wein*, **48**, 424–430.
- Köhler HJ, Christoph N, Bauer-Christoph C, Geßner M and Curschmann K (1996), 'Die "Untypische Alterungsnote" im Wein, Teil V: Einfluß kellerwirtschaftlicher Maßnahmen auf die Ausprägung der UTA', *Rebe und Wein*, **7**, 213–216.
- Linsenmeier A (2007), *Einfluss der Stickstoffversorgung der Rebe (Vitis vinifera L. cv. Riesling) auf den untypischen Alterungsston*, Dissertation, Justus-Liebig-Universität Gießen, Geisenheimer Berichte Band 60.

- Linsenmeier A, Muno H, Schubert S and Löhnertz O (2005), 'Effect of different nitrogen fertilization on aroma quality of Riesling', *Proc GESCO (Groupe Étude des Systèmes de Conduite de la Vigne)*, **2**, 724–730.
- Linsenmeier A, Rauhut D, Kürbel H, Löhnertz O and Schubert S (2007a), 'Untypical ageing off-flavour and masking effects due to long-term nitrogen fertilization', *Vitis*, **46**, 33–38.
- Linsenmeier A, Rauhut D, Kürbel H, Schubert S and Löhnertz O (2007b), 'Ambivalence of the influence of nitrogen supply on o-aminoacetophenone in 'Riesling' wine', *Vitis*, **46**, 91–97.
- Linsenmeier A, Rauhut D and Löhnertz O (2008), 'N-Düngung – Freund oder Feind des UTA?', *Deutsches Weinbaujahrbuch*, **59**, 134–141.
- Löhnertz O (1996), 'UTA und Rebenernährung. Streß macht alt', *Deutsches Weinmagazin*, **18**, 18–23.
- Löhnertz O, Schultz HR, Hünneke B and Linsenmeier A (2002), 'Weinbauliche Maßnahmen zur Vermeidung von UTA', in Trogus H, Gafner J and Sutterlin A (eds), *Proceedings from the 13th International Enology Symposium*, International Association of Enology, Management and Wine Marketing, Breisach, Germany, 215–228.
- Müller E (1999), '15 Jahre Stickstoffdüngungsversuche: Erfahrungen und Konsequenzen', *Das Deutsche Weinmagazin*, **19**, 27–31 and **20**, 29–32.
- Peternel M (1998), 'Stickstoff und Qualität. Zusammenfassung des Vortrags bei den 40. Veitshöchheimer Weinbautage', *Rebe und Wein*, **3**, 91.
- Pflieghinger M, Muskat S, Löhnertz O and Schubert S (2008), 'Einfluss verschiedener Faktoren auf die Bildung von AAP in Weinen von Reben aus Großcontainern', 47. Arbeitstagung des Forschungsrings des Deutschen Weinbaus bei der DLG, Bad Kreuznach, Germany.
- Pohl H (1992), 'Probleme mit einem Fehlton', *Der Badische Winzer*, **8**, 395–396.
- Pohl H (1993), 'Woher kommt die untypische Alterungsnote beim Wein?', *Der Badische Winzer*, **8**, 316–317.
- Rapp A and Versini G (1995), 'Fehlaroma: Die untypische Alterungsnote', *Der Deutsche Weinbau*, **18**, 18–22.
- Rapp A and Versini G (1996), 'Influence of nitrogen compounds in grapes on aroma compounds of wines', *Vitic Enol Sci*, **51**, 193–203.
- Rapp A, Günter, M and Ullemeyer H (1985), 'Über Veränderungen der Aromastoffe während der Flaschenlagerung von Weißweinen der Rebsorte Riesling', *Z Lebensm Unters Forsch*, **180**, 109–116.
- Rapp A, Versini G and Ullemeyer H (1993), '2-Aminoacetophenon: Verursachende Komponente der "untypischen Alterungsnote" ("Naphthalinton", "Hybridton") bei Wein', *Vitis*, **32**, 61–62.
- Rapp A, Versini G, Engel H and Ullemeyer H (1998), 'Fremde und unerwünschte Aromastoffe des Weines: Die untypische Alterungsnote', *Proc Intervitis Interfructa. 5. Int Symp*, 11–12 May, Stuttgart, Germany, 270–289.
- Rauhut D and Kürbel H (2002) 'Böckserbildung und/oder untypischer Alterungsston: eine mögliche Differenzierung', in Trogus H, Gafner J and Sutterlin A (eds), *Proceedings from the 13th International Enology Symposium*, International Association of Enology, Management and Wine Marketing, Breisach, Germany, 371–383.
- Rauhut D, Kürbel H, Fischer S and Beisert B (2007), 'Simultaneous analysis of N- and S-compounds as screening for the identification of off-flavours in wine', in Hofmann T, Meyerhof W and Schieberle P (eds), *Recent Highlights in Flavor Chemistry and Biology*. Garching, Germany, Deutsche Forschungsanstalt für Lebensmittelchemie, 314–317.
- Reynolds AG, Lowrey WD, Tomek L, Hakimi J and de Savigny C (2007), 'Influence of irrigation on vine performance, fruit composition, and wine quality of chardonnay in a cool, humid climate', *Am J Enol Vitic*, **58**, 217–228.
- Schmarr HG, Ganß S, Sang W and Potouridis T (2007), 'Analysis of 2-aminoacetophenone in wine using a stable isotope dilution assay and multidimensional gas chromatography-mass spectrometry', *J Chromatogr A*, **1150**, 78–84.

- Schorr T (2003), *Einfluss unterschiedlicher Bodenpflegesysteme auf den Bodenwasser- und -stickstoffhaushalt, die Wuchsleistung der Rebe und die Most- und Weinqualität*, Dissertation, Albert-Ludwig-Universität Freiburg im Breisgau, Germany.
- Schwab A (1998), 'Extensivierung von Bodenpflege und Düngung und deren Auswirkung auf die Most- und Weinqualität', *Deutsches Weinbaujahrbuch*, **49**, 93–101.
- Schwab A and Peternel M (2001), 'Ungenügende Traubenreife: Ursache der "Untypischen Alterungsnote (UTA)" im Wein', *Rebe und Wein*, **9**, 14–17.
- Schwab A, Peternel M, Köhler HJ and Heigel KP (1996), 'Die "Untypische Alterungsnote" im Wein, Teil IV: Beeinflussung durch weinbauliche Maßnahmen', *Rebe und Wein*, **49**, 181–187.
- Schwab A, Christoph N, Köhler HJ, Geßner M and Simat TJ (1999), 'Einfluß weinbaulicher Maßnahmen auf die Ausprägung der Untypischen Alterungsnote bei Weißweinen. Teil I: Einfluß des Lesezeitpunktes', *Vitic Enol Sci*, **54**, 114–120.
- Seiter P (2000), 'Der Einfluß von Stickstoffdüngung und Bodenpflege auf die Stickstoffversorgung der Rebe und die Weinqualität. Eine Studie zum Problem des "Untypischen Alterungstons"', Dissertation, Albert-Ludwig-Universität Freiburg im Breisgau.
- Sponholz WR, Hühn T, Engelmann A and Sieben A (1997), 'Mögliche Einflüsse weinbaulicher Parameter auf die Ausbildung des "Untypischen Alterungstons" bei Rieslingweinen', *Vitic Enol Sci*, **52**, 41–50.
- Sponholz WR, Hühn T and Grossmann M (2001), 'Einfluß von Umwelt- und Substrateffekten auf die Bildung von wertbestimmenden Inhaltsstoffen', *Proc Intervitis Interfructa*, **6**, *Int Symp*, 14–16 May, Stuttgart, Germany, 98–112.
- Walz O (2003), 'Stress vermeiden', *Das Deutsche Weinmagazin*, **4**, 30–34.
- Wohlfahrt P (1995), 'Untypische Alterungsnote: Erfahrungen aus den vergangenen drei Jahren', *Der Badische Winzer*, **8**, 383–388.459

Biogenic amines and the winemaking process

M. V. Moreno-Arribas, Institute of Industrial Fermentations (CSIC), Spain; and A. Y. Smit and M. du Toit, Stellenbosch University, South Africa

Abstract: In spite of the low quantities of biogenic amines present in wines compared to other fermented foods, these compounds represent one of the most important risks for wine quality and safety. In addition, biogenic amines may constitute an economic problem in commercial transactions of wines. The aim of this chapter is to present an overview of the current knowledge about the incidence of biogenic amines in wines, the factors that affect their formation, their toxicological impact on humans, the main analytical tools for their identification and quantification in wines or the molecular approaches for the decarboxylase positive detection of microorganisms. Finally, the most suitable preventive measurements during the wine chain production are emphasised.

Key words: biogenic amines, winemaking, lactic acid bacteria, toxicology, technological factors and practices, molecular and analytical approaches, control and prevention.

17.1 Introduction

Biogenic amines are a group of biologically active compounds that are widespread in nature. The term ‘amine’ is used for basic nitrogenous compounds of low molecular weight that are produced within the normal metabolism of humans, animals, plants or microorganisms. In foods and beverages, biogenic amines are formed mainly by the decarboxylation of the corresponding precursor amino acids. This reaction is catalysed by substrate-specific enzymes, decarboxylases, of the microbial flora of the food or wine environment. They are generated either as the

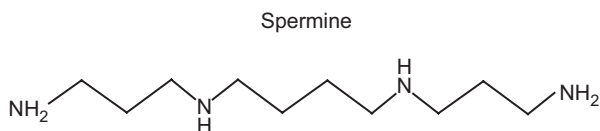
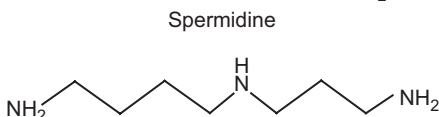
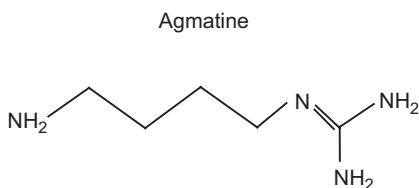
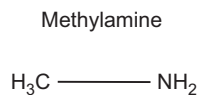
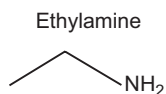
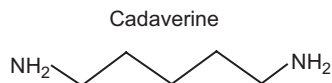
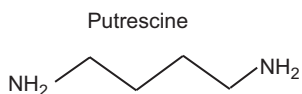
result of endogenous decarboxylase-positive microorganisms in grapes or by the growth of contaminating decarboxylase-positive microorganisms in the wine. The microorganisms associated with wine that could have an impact on the production of biogenic amines are the yeasts, lactic acid bacteria (LAB), acetic acid bacteria (AAB) and fungi. The main biogenic amines associated with wine are histamine, tyramine, putrescine and cadaverine, and their presence serves as marker molecules for the quality of the wine. The presence of these amines in wine is currently a great discussion point, as many European countries have informal legal limits for the amount of histamine, which impacts the import and export of wines to EU countries. The importance of biogenic amines in the safety and quality of food and beverages can be seen as several reviews on this topic have been published in the last 5 years (Ancín-Azpilicueta *et al.*, 2008; Smit *et al.*, 2008). This chapter will focus on the incidence of biogenic amines in wines, the factors that affect their formation, their toxicological impact on humans, methods to detect the biogenic amines or the decarboxylase-positive microorganisms. Finally, the preventative measurements will be discussed.

17.2 Incidence of biogenic amines in wines and health effects

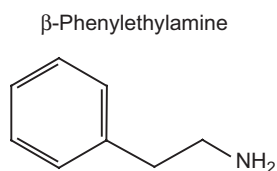
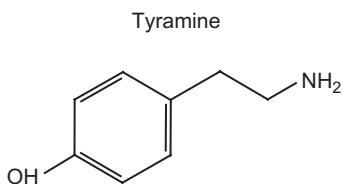
Biogenic amines are produced mainly from the decarboxylation of precursor amino acids and, depending on their volatile status, they can be divided into volatile and non-volatile amines. The biogenic amines associated with wine can be separated further according to their chemical structure into aromatic (tryptamine, phenylethylamine), aliphatic (cadaverine, putrescine, spermine, spermidine) or heterocyclic (histamine, tryptamine) biogenic amines (Smit *et al.*, 2008; Moreno-Arribas and Polo, 2009) (Fig. 17.1). The volatile amines can also be produced from the amination or transamination of aldehydes or ketones found in wine (Ough *et al.*, 1981). Biogenic amines in wine can have different origins such as the grapes, the alcoholic and/or malolactic fermentation (MLF) process, barrel ageing or storage where wine has been exposed to decarboxylase-positive microorganisms. These microorganisms normally occur due to poor grape quality, uncontrolled fermentations and poor cellar hygiene practices (Zee *et al.*, 1983; ten Brink *et al.*, 1990; Shalaby, 1996; Leitão *et al.*, 2005).

Most countries have investigated the presence and concentration of biogenic amines in their wines. These include Greece, Spain, Hungary, the USA (Oregon), South Africa, Argentina, Canada, Portugal, Italy and France (Glória *et al.*, 1998; Soufleros *et al.*, 1998, 2007; Mafra *et al.*, 1999; Gerbaux and Monamy, 2000; Hajós *et al.*, 2000; Herbert *et al.*, 2005; Bover-Cid *et al.*, 2006; mo Dugo *et al.*, 2006). The concentrations of biogenic amines in wine differ substantially from none to highly-elevated concentrations of 330 mg/L (see Table 17.1). Due to the origin of putrescine from grapes or the yeast, it is the most abundant in wine (Arena and Manca de Nadra, 2001). Histamine levels are currently used as marker for the safety and quality of wine and this will impact import and exports. The upper limits used for histamine (mg/L) in European countries are: Switzerland and Austria

Aliphatic amines



Aromatic amines



Heterocyclic amines

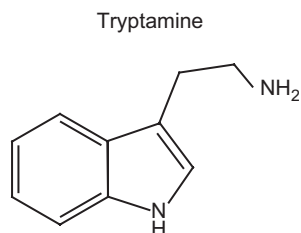
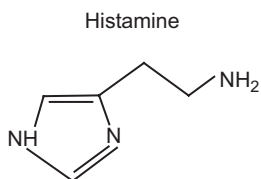


Fig. 17.1 Chemical structures of biogenic amines relevant to winemaking (Smit *et al.*, 2008).

Table 17.1 Concentration (mg/L) of biogenic amines in wines

Wines	Histamine	Tyramine	Putrescine	Cadaverine	2-Phenylethyl-amine	Spermine	Ethylamine	Reference
South African red wines	0–28 (R)	0–16 (R)	0–331 (R)	0.4–3 (R)	–	–	–	Unpublished data
South African white wines	0	0–9 (R)	0–42 (R)	0–0.3 (R)	–	–	–	Unpublished data
Spanish red wines	0–25 (R)	0–19 (R)	0–55 (R)	0–14 (R)	0–16.26 (R)	–	0–7.92 (R)	Marcobal <i>et al.</i> (2005a)
Turkish red wines	0–1.97 (R)	0–0.29 (R)	0–5.92 (R)	0–3.94 (R)	0–3.87 (R)	0–0.75 (R)	–	Anli <i>et al.</i> (2004)
Rioja young red wines	8.72 (A)	4.98 (A)	32.97 (A)	0.61 (A)	–	–	–	Vazquez–Lasa <i>et al.</i> (1998)
Tokaj wines	0.02–0.04 (M)	0.05–0.31 (M)	2.14–2.55 (M)	0.07–0.19 (M)				Hajós <i>et al.</i> (2000)
French wines	8.9 (A)	3.1 (A)	7.7 (A)	0.2 (A)	1.7 (A)	0.1 (A)		Bauza <i>et al.</i> (1995)
Portuguese red wines	0–24 (R)	0–4 (R)	0–28 (R)	0–8 (R)				Leitão <i>et al.</i> (2005)
Chardonnay young wines	0.67 (A)	2.09 (A)	3.04 (A)	0.51 (A)	–	0.85 (A)	–	González-Marco and Ancín Azpilicueta (2006)

A, average; M, mean; R, range.

Source: adapted from Ancín-Azpilicueta *et al.*, 2008.

(10), France (8), Belgium (5–6), Finland (5), Holland (3) and Germany (2) (Lehtonen, 1996). These biogenic amines will not only impact on wine flavour, but will also have a physiological effect on sensitive individuals.

17.2.1 Pharmacological and toxicological aspects

Histamine, tyramine, putrescine, cadaverine, β -phenylethylamine and tryptamine are a group of biologically active compounds that have important physiological effects (Table 17.2) in humans. They may exert either psychoactive or vasoactive effects. Psychoactive amines affect the nervous system by acting on neural transmitters, while vasoactive amines act on the vascular system (Lovenberg, 1973). Tyramine, tryptamine and β -phenylethylamine are included in the presser amine group, which consists of vasoactive amines. They cause a rise in blood pressure by constricting the vascular system, increasing the heart rate and forcing the heart to contract. In contrast, histamine reduces the blood pressure by causing vasodilatation (Smith, 1981).

Table 17.2 The physiological effects of biogenic amines

Biogenic amines	Physiological effects
Histamine	Liberates adrenaline and noradrenaline Excites the smooth muscles of the uterus, the intestine and the respiratory tract Stimulates both sensory and motor neurons Controls gastric acid secretion
Tyramine	Peripheral vasoconstriction Increases cardiac output Causes lacrimation and salivation Increases respiration Increases blood sugar level Releases noradrenaline from the sympathetic nervous system Causes migraine
Putrescine and cadaverine	Hypotension Bradycardia Dyspnoea Lockjaw Paresis of the extremities Potentiates the toxicity of other amines
β -Phenylethylamine	Releases noradrenaline from the sympathetic nervous system Increases the blood pressure Causes migraine Schizophrenia
Tryptamine	Increases the blood pressure

Source: Downing, 2003.

It is important to emphasise that harmful effects resulting from the consumption of food and beverages rich in biogenic amines can be expected only when these amines gain access to the bloodstream. This is normally prevented by the action of a defence mechanism of the body that is situated in the digestive tract (Taylor, 1985).

Histamine

Histamine is a normal constituent of the body, where it mediates several important functions. Firstly, histamine is a mediator of allergic reactions (Taylor, 1985). Special granules of the mast cells and blood basophils contain large amounts of histamine, which is released into the bloodstream in response to an allergic reaction. The release of histamine by mast cell degranulation and the intake of food containing histamine can exert the same effects (Ten Brink *et al.*, 1990).

Histamine exerts its effects by binding to receptors on cellular membranes. Two types of histamine receptors, known as the H_1 and H_2 receptors, exist in humans and other related species (Taylor, 1986). Both these receptors are found in the cardiovascular system. The interaction of histamine with these receptors results in the dilatation of peripheral blood vessels, capillaries and arteries (Stratton *et al.*, 1991). Furthermore, capillary permeability is increased, which leads to the leakage of plasma into the tissues and to haemoconcentration (Joosten, 1988). Histamine also causes an increase in the rate and strength of the heartbeat (Joosten, 1988).

Histamine receptors are also found in various secretory glands. Gastric acid secretion is regulated by histamine through H_2 receptors located on the parietal cells (Soll and Wollin, 1977). Secretion by the pancreas, the intestine and the bronchi is stimulated by histamine (Joosten, 1988). Histamine can also liberate adrenaline and noradrenaline from the suprarenal gland and, as a result, directly stimulate the heart (Joosten, 1988; Shalaby, 1996). Mediated by H_1 receptors, histamine also excites the smooth muscles of the uterus, the intestine and the respiratory tract (Taylor, 1986). In addition, histamine is probably a neurotransmitter in the central nervous system. It is responsible for sensory and motor neuron stimulation, which is mediated by H_1 receptors (Taylor, 1986).

The most frequent foodborne intoxications caused by biogenic amines are related to histamine. Histamine poisoning is a chemical intoxication with a short incubation period (Taylor, 1986). The onset of the symptoms usually begins several minutes to a few hours after the ingestion of histamine-containing food. Not many incidences of histamine poisoning have been recorded. The symptoms of food allergy and those of histamine poisoning are very similar, and physicians occasionally make a faulty diagnosis (Ten Brink *et al.*, 1990). Moreover, both allergic symptoms and histamine poisoning are responsive to antihistamine treatment (Becker *et al.*, 2001).

Histamine can directly stimulate the heart, cause the smooth muscles of the uterus, intestine and respiratory tract to contract or relax, stimulate both sensory and motor neurons and control gastric acid secretion. Thus, it is not surprising that histamine poisoning is characterised by a wide variety of symptoms. Characteristic symptoms affecting the cutaneous (i.e. skin) system include rash, urticaria, edema

and localised inflammation (Taylor, 1986). Gastrointestinal involvement is characterised by nausea, vomiting, diarrhoea and abdominal cramps (Taylor, 1985). Other symptoms include hypotension, headache, heart palpitations, tingling, flushing and burning sensations in the mouth (Gilbert *et al.*, 1980; Taylor, 1985).

Shock, bronchospasm, suffocation and severe respiratory distress have been reported in severe cases (Franzen and Eysell, 1969). However, the illness usually has a mild character and the symptoms do not last long (Joosten, 1988).

Tyramine

Unlike histamine, tyramine is not an important metabolite in the body and it is generally present at a very low concentration. Tyramine mainly acts indirectly by releasing noradrenaline from the sympathetic nervous system. The most prominent effect of this action is an increase in blood pressure as a result of the vascular system being constricted and the cardiac output being increased (Stockley, 1973; Joosten, 1988). Tyramine also dilates the pupils and palpebral tissue, causes lacrimation and salivation and increases respiration. If present in high concentrations, the blood sugar level is also increased (Franzen and Eysell, 1969; Joosten, 1988).

The importance of tyramine in foods is principally due to its toxicological implication. Tyramine is toxic in itself, but, apart from that, it reacts with monoamine oxidase inhibitor (MAOI) drugs to give rise to a hypertensive crisis (Shalaby, 1996). One of the functions of MAO (monoamine oxidase) in the intestine and liver is to destroy potentially harmful amines, specifically tyramine, before they reach the blood. The use of MAOI drugs for the treatment of depression eliminates this detoxification mechanism. Thus, high concentrations of the presser amine derived from food accumulate in the blood. The affected patients then become very susceptible to the toxic actions of tyramine, resulting in a hypertensive crisis (Blackwell, 1963). Such attacks may take place as long as 3 weeks after drug withdrawal. This is the time needed to restore the MAO enzyme (Smith, 1981).

Putrescine and cadaverine

Both putrescine and cadaverine seem to have a much lower pharmacological activity than histamine and tyramine. Putrescine and cadaverine and the polyamines spermidine and spermine are indispensable components of living cells. These amines are important in the regulation of nucleic acid function and protein synthesis, and probably also in the stabilisation of membranes (Bardócz *et al.*, 1993; Halász *et al.*, 1994). Toxic effects are observed only after ingestion of very large amounts of putrescine and cadaverine. Intoxication symptoms that have been reported are hypotension, bradycardia, dyspnoea, lockjaw and paresis of the extremities (Franzen and Eysell, 1969). The most important effect of the presence of these compounds in wine is their ability to potentiate the toxicity of other amines (Taylor, 1986). Furthermore, it should be mentioned that secondary amines, such as putrescine and cadaverine, can react with nitrite to form the heterocyclic carcinogenic nitrosamines, nitrosopyrrolidine and nitrosopiperidine (Huis in't Veld *et al.*, 1990).

Phenylethylamine

Pharmacologically, phenylethylamine resembles tyramine, as it also is a neurosympathomimetic amine. In some instances, hypertensive crisis incidents were related to the ingestion of foods containing phenylethylamine (Joosten, 1988). In addition to its sympathomimetic action, phenylethylamine has also been associated with migraine (Sandler *et al.*, 1974) and psychiatric disorders such as schizophrenia (O'Reilly *et al.*, 1991; Buckland *et al.*, 1997). The presence of phenylethylamine in food is less frequent than that of tyramine.

Tryptamine

Tryptamine is also a neurosympathomimetic amine. However, it can also exert its effect directly by stimulating smooth muscles and thereby increasing the blood pressure (Joosten, 1988).

17.2.2 Detoxification of biogenic amines

Biogenic amines do not usually represent any health hazard to individuals. Many foods contain small amounts of amines, which can easily be tolerated. A fairly efficient detoxification system exists in the intestinal tract of mammals (Taylor, 1985). The detoxification system is composed of two distinct enzymes, MAO and diamine oxidase (DAO) (ten Brink *et al.*, 1990). Mono- and diamine oxidases are present in eukaryotes and were also described for bacteria (Voigt and Eitenmiller, 1978; Murooka *et al.*, 1979; Ishizuka *et al.*, 1993; Yamashita *et al.*, 1993). These enzymes convert amines into non-toxic products, which are then excreted. For example, histamine can be metabolised by one of two enzymatic pathways (Fig. 17.2). In the first pathway, the ring structure of histamine is methylated by histamine N-methyltransferase (HMT) to form N-methylhistamine. This product can be further oxidised by MAO to form N-methyl imidazole acetic acid. In the second pathway, histamine is oxidised by DAO to form imidazole acetic acid (Stratton *et al.*, 1991). Tyramine is excreted as *p*-hydroxyphenylacetic acid, after oxidation by MAO (Smith, 1981). Putrescine and cadaverine can be degraded by DAO and possibly also by MAO (Franzen and Eysell, 1969). Phenylethylamine and tryptamine are destroyed by MAO (Sandler *et al.*, 1974; Joosten, 1988).

This detoxification system is capable of metabolising normal dietary intakes of biogenic amines. However, it apparently fails to eliminate the large amounts of biogenic amines ingested with spoiled foods. The consumption of a high dose of amines saturates the amine-metabolising capacity of the human body (Ten Brink *et al.*, 1990). One or more substances known as potentiators can also suppress the activity of the detoxifying enzymes and thereby lower the efficiency of detoxification. Potentiators can be classified either as foodborne putrefactive amines or as pharmacological agents (Stratton *et al.*, 1991). Drugs known to suppress these enzymes are antihistamines (Barth and Lorenz, 1978), antimalarials (Cohn, 1965) and other medications, such as monoamine oxidase inhibitors used as antidepressants (Horwitz *et al.*, 1964), and isoniazid, an antituberculosis agent (Kahana and Todd, 1981). Some amines, specifically putrescine and cadaverine, inhibit the

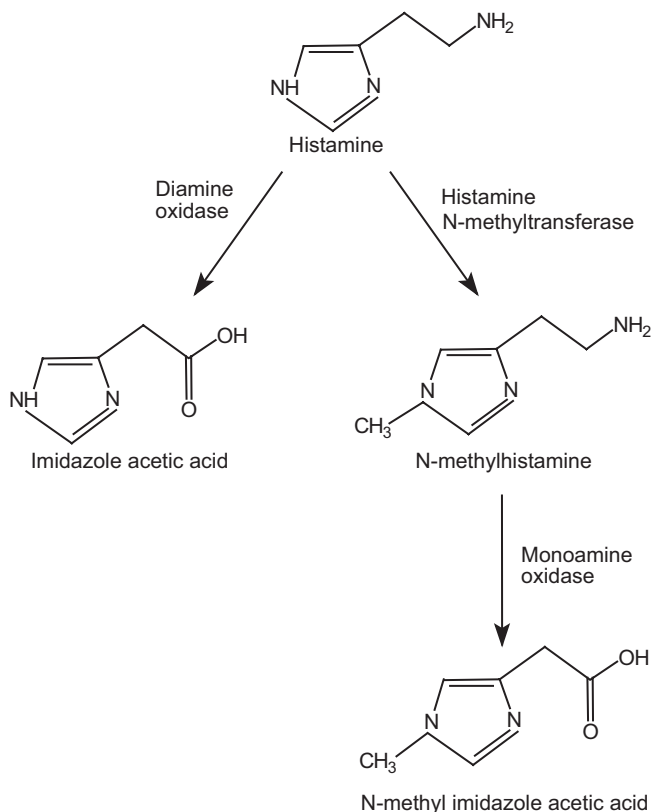


Fig. 17.2 Metabolic pathways for the detoxification of histamine (Downing, 2003).

detoxifying enzymes, DAO and HMT (Hui and Taylor, 1985). Other biogenic amines that may act as potentiators include tyramine, tryptamine and phenylethylamine. Tyramine can inhibit MAO (Voigt and Eitenmiller, 1978), while tryptamine inhibits DAO (Stratton *et al.*, 1991). Phenylethylamine is a DAO and HMT inhibitor (Hui and Taylor, 1985). In addition, alcohol also increases sensitivity to biogenic amines (ten Brink *et al.*, 1990). It is also possible that the detoxification system may be genetically deficient (Halász *et al.*, 1994).

17.3 Formation of biogenic amines during the winemaking process

Wine is an ideal substrate for amine production, as its manufacturing process involves not only available free amino acids, but also the possible presence of decarboxylase-positive microorganisms and some favourable environmental conditions (i.e. external pH) that affect the growth of microorganisms as well as the

activity of decarboxylase enzymes (Lonvaud-Funel, 1999). Hence, knowledge of the concentrations of biogenic amines in wines is necessary to assess the health hazards that may arise from drinking wines. Recently, knowledge concerning the presence of amines in wines has been reviewed (Ferreira and Pinho, 2006; Ancín-Azpilicueta *et al.*, 2008; Smit *et al.*, 2008).

Usually, red wines present greater concentrations of biogenic amines than the white ones. These greater values in red wines are attributed to the presence of LAB and the MLF. White wines, in general, contain smaller concentrations of amino acids and lower pH, due to the absence of maceration during the alcoholic fermentation. Obviously, other types of wines and special vinifications, as is the case with sparkling wines and biologically aged wines, are also susceptible to the problem of the formation of biogenic amines, although this theme has been the subject of very few studies in the literature (Bodmer *et al.*, 1999; Moreno-Arribas and Polo, 2008).

Origin of biogenic amines during wine manufacture

Amines in wine may have two different sources of origin, namely the grape and raw materials, and the different fermentation processes. However, data are complex and published results are often contradictory. In several studies, biogenic amines have been suggested as an indicator of hygienic quality or of poor manufacturing practices (Zee *et al.*, 1983). Indeed, some amines such as putrescine and cadaverine are associated with poor sanitary conditions of grapes (Leitão *et al.*, 2005). Some polyamines, such as putrescine, may already be present in grape berries (Halász *et al.*, 1994; Bover-Cid *et al.*, 2006; Del Prete *et al.*, 2008) or be produced during fermentation processes, ageing or storage, when wine is exposed to the undesirable activity of decarboxylase-positive microorganisms. Contamination may also occur from poor sanitary conditions of both grapes and processing equipment. Although yeasts may also contribute to the final levels of biogenic amines in wines (Caruso *et al.*, 2002; Torrea and Ancín, 2002), most biogenic amine contamination of wine is believed to take place during MLF due to the presence of LAB strains that produce enzymes which decarboxylate the corresponding precursor amino acids (Marcobal *et al.*, 2006a) (Table 17.3). In relation to amine formation during MLF, many authors reported that LAB actively produce biogenic amines (Lonvaud-Funel and Joyeux, 1994; Moreno-Arribas *et al.*, 2000; Landete *et al.*, 2005a,b), but other authors did not (Buteau *et al.*, 1984). Amino acids are mostly the direct precursors of amines and, as a consequence, the higher the content of free amino acids, the higher the probability of biogenic amine production (Martín-Álvarez *et al.*, 2006).

Marcobal *et al.* (2006a) examined the changes in biogenic amines during the industrial manufacture of 55 batches of red wines. The origin of these amines in relation to must, alcoholic fermentation, MLF, sulphur dioxide addition and wine ageing as well as the interactions between amines and their corresponding amino acids and pH were statistically described in samples from the same batches throughout the elaboration process. These authors concluded that some amines can be produced in the grape or the musts (such as putrescine, cadaverine and

Table 17.3 Microorganisms associated with biogenic amines production in winemaking

Species	Role	Biogenic amine and reported metabolic activity	Reference
<i>Saccharomyces cerevisiae</i>	Fermenting wine yeast	Histamine	Goñi and Ancín Azpilicueta (2001); Torrea and Ancín (2002); Caruso <i>et al.</i> (2002); Granchi <i>et al.</i> (2005)
<i>Brettanomyces bruxellensis</i>	Spoilage yeast	Agmatine, phenylethylamine, ethanolamine	Caruso <i>et al.</i> (2002); Granchi <i>et al.</i> (2005)
<i>Kloeckera apiculata</i> , <i>Candida stellata</i> , <i>Metschnikowia pulcherrima</i>	Natural wild yeast	Agmatine, phenylethylamine, ethanolamine	Caruso <i>et al.</i> (2002); Granchi <i>et al.</i> (2005)
<i>Botrytis cinerea</i>	Azsu wines fungi	Tyramine, putrescine, cadaverine, phenylethylamine, spermidine	Hajós <i>et al.</i> (2000); Sass-Kiss <i>et al.</i> (2000)
<i>Lactobacillus</i> spp., <i>Pediococcus</i> spp.	Fermenting and spoilage lactic acid bacteria	Histamine (histidine decarboxylase)	Farías <i>et al.</i> (1993); Moreno-Arribas <i>et al.</i> (2003); Landete <i>et al.</i> (2005a); Lucas <i>et al.</i> (2005); Constantini <i>et al.</i> (2006)
		Tyramine (tyrosine decarboxylase)	Moreno-Arribas <i>et al.</i> (2000); Lucas <i>et al.</i> (2003); Constantini <i>et al.</i> (2006); Marcobal <i>et al.</i> (2006b); Landete <i>et al.</i> (2007b,c)
		Putrescine (ornithine decarboxylase)	Arena and Manca de Nadra (2001); Moreno-Arribas <i>et al.</i> (2003)
		Phenylethylamine	Moreno-Arribas <i>et al.</i> (2000); Landete <i>et al.</i> (2007c)
<i>Oenococcus oeni</i>	Malolactic fermentation	Histamine (histidine decarboxylase)	Lonvaud-Funel and Joyeux (1994); Coton <i>et al.</i> (1998a, b); Guerrini <i>et al.</i> (2002); Landete <i>et al.</i> (2005a, 2007b)
		Putrescine (ornithine decarboxylase)	Guerrini <i>et al.</i> (2002); Marcobal <i>et al.</i> (2004)

phenylethylamine) or can be formed by yeasts during alcoholic fermentation (ethylamine and phenylethylamine), although quantitatively only very low levels are reached in these stages. They also found an increase in the concentrations of histamine, tyramine and putrescine during MLF that coincide with the decrease in amino acid precursors. This increase is observed for most of the amines and, especially, for histamine and tyramine in the samples at the first stage of MLF. This indicates that the biosynthesis of decarboxylase amino acid enzymes can take place when the bacterial population is at the end of the growth phase and during the exponential phase, in agreement with recent previous *in vitro* studies (Pessione *et al.*, 2005). Furthermore, the formation of these amines was significantly correlated with the disappearance of their corresponding amino acids, which clearly supports the hypothesis that malolactic bacteria are mainly responsible for the accumulation of these amines in wines.

Similar data were reported by other authors. For example, in French wines, histamine, tyramine and putrescine have been shown to increase during MLF to a very variable extent (Bauza *et al.*, 1995; Soufleros *et al.*, 1998). In other works, only the formation of histamine (Herbert *et al.*, 2005; Landete *et al.*, 2005b) or of tyramine (Vidal-Carou *et al.*, 1990) was associated with LAB and MLF or wine storage. In contrast, other authors have pointed out that biogenic amines, and in particular histamine, decrease during MLF (Buteau *et al.*, 1984). The fact that wine LAB have a variable capacity to produce biogenic amines from amino acids explains the variability in the data of biogenic amine production obtained by the different authors.

It is well known that among the species and strains of wine LAB, some are practically unable to produce biogenic amines whereas others are characterised for their high capacity of production. It is often possible to detect amino-biogenic strains among heterofermentative lactobacilli (*Lactobacillus hilgardii* and *Lactobacillus brevis*) (Moreno-Arribas *et al.*, 2000), although among histamine-producing strains, some *Pediococcus* (Bodmer and Gafner, 2000; Landete *et al.*, 2005a) and *Oenococcus oeni* strains have been isolated (Coton *et al.*, 1998a). The latter species has also been found to be a putrescine producer (Marcobal *et al.*, 2004, 2006c). Although it has been demonstrated that oenococci could produce histamine and putrescine, this property seems to be variable and is only present in some strains, which explains why in other studies, strains of biogenic amine-producing *O. oeni* have not been found (Constantini *et al.*, 2006; Moreno-Arribas and Polo, 2008).

Generally, most of the studies agree that during wine storage and ageing, biogenic amines concentrations undergo few variations, being observed as a slight decrease or stabilisation of these compounds during the ageing of wines in oak barrels (Jiménez-Moreno *et al.*, 2003; Marcobal *et al.*, 2006a). The synthesis of biogenic amines during wine ageing in barrels would be due to the presence of residual populations of LAB with decarboxylase activity. It would seem that some strains survive longer in wine than those that do not have this activity (Lonvaud-Funel, 1999). Additionally, biogenic amines could be formed in the absence of viable LAB in wine, as a notable enzyme activity still exist for several months of oak ageing (Coton *et al.*, 1998b).

Table 17.4 Summary of the information available about technological factors affecting the formation of biogenic amines in grapes and wines

Technological factors	References
<i>Viticultural factors</i>	
Grape varieties	Halász <i>et al.</i> (1994); Glòria <i>et al.</i> (1998); Hajós <i>et al.</i> (2000); Cecchini <i>et al.</i> (2005); Landete <i>et al.</i> (2005b); Bover-Cid <i>et al.</i> (2006); Soufleros <i>et al.</i> (2007); Del Prete <i>et al.</i> (2008)
Vine nitrogen fertilization	Spayd <i>et al.</i> (1994); Soufleros <i>et al.</i> (2007)
Vintage and region production	Sass-Kiss <i>et al.</i> (2000); Herbert <i>et al.</i> (2005); Martín-Álvarez <i>et al.</i> (2006)
<i>Oenological factors</i>	
Skin maceration techniques	Bauza <i>et al.</i> (1995); Martín-Álvarez <i>et al.</i> (2006)
Wine composition and physiochemical parameters	Vidal-Carou <i>et al.</i> (1990); Lonvaud-Funel and Joyeux (1994); Rollán <i>et al.</i> (1995); Moreno-Arribas and Lonvaud-Funel (1999, 2001); Landete <i>et al.</i> (2006); Martín-Álvarez <i>et al.</i> (2006); Marcobal <i>et al.</i> (2006a); Mangani <i>et al.</i> (2005); Arena <i>et al.</i> (2007)
Factors related to wine ageing conditions	Vazquéz-Lasa <i>et al.</i> (1998); Moreno and Ancín Azpilicueta (2004); Martín-Álvarez <i>et al.</i> (2006); Marcobal <i>et al.</i> (2006a); Alcaide-Hidalgo <i>et al.</i> (2007); Hernández-Orte <i>et al.</i> (2008)

Technological factors and practices

Regarding oenological practices and technological factors that may influence biogenic amine formation in wines, special attention should be paid to some oenological practices frequently used to enhance wine complexity (flavour and aroma), but that can also increase the concentration of the precursor amino acids, such as the ageing of wines with lees or longer maceration times (Martín-Álvarez *et al.*, 2006) (Table 17.4). Yeasts can indirectly play an important role on the production of biogenic amines by LAB, since they affect the amino acid composition during the alcoholic fermentation (Soufleros *et al.*, 1998) or during autolysis (Moreno-Arribas *et al.*, 1998; Villamiel *et al.*, 2008). These amino acids can act as precursors for the formation of biogenic amines during MLF and during wine ageing (Martín-Álvarez *et al.*, 2006; Marques *et al.*, 2008). Yeast and bacteria lees can also be a source for decarboxylase enzymes that could be involved in amine production (Marcobal *et al.*, 2004).

The use of malolactic starter cultures with specific characteristics such as very reduced or null capacity to produce biogenic amines is the most popular strategy to control amine production during winemaking. *O. oeni* commercial starters are selected on the basis of their oenological properties and among them the absence of amino acid decarboxylase enzymes is an important requirement. Research performed in *in vitro* studies (Moreno-Arribas *et al.*, 2003) and in industrial

conditions during winemaking (Marcobal *et al.*, 2006a) has shown the viability of using bacteria starters without the genes involved in biogenic amine production to prevent the formation of amines in wines. Moreover, it has been shown that the co-inoculation of *O. oeni* starters simultaneously to the alcoholic fermentation, as opposed to the conventional inoculation once the alcoholic fermentation has been completed, is more effective to avoid the production of biogenic amines (Smit, 2007; van der Merwe, 2007).

The use of different clarification substances and oenological adjuvants, such as bentonite or PVPP, at the normal dose used in winemaking, can affect the concentration of biogenic amines in wines, since these compounds have the ability for the adsorption of some amines (Alcaide-Hidalgo *et al.*, 2008). Also, the type of container employed during MLF (stainless steel or oak barrel) seems to affect biogenic amine content of wines (Alcaide-Hidalgo *et al.*, 2007). Some alternative techniques for wine ageing based on the use of wood chips or oenological tannins can also influence the amine production, depending on the product type and on the dose of these compounds applied to the wines (García-Ruiz *et al.*, 2008, 2009).

Biogenic amine formation is also determined by wine parameters, such as pH, ethanol and SO₂, which have an important effect on the diversity of microorganisms, decarboxylase enzyme activity and decarboxylase gene expression (Lonvaud-Funel, 1999; Leitão *et al.*, 2000; Mazzoli *et al.*, 2009).

Molecular approaches for understanding biogenic amines formation during winemaking

Molecular methods can provide tools for rapid, specific and reliable detection of amino acid decarboxylase genes of wine microorganisms (specifically LAB) at early stages of winemaking. Thus, a molecular approach could ensure early risk assessment and control measures to prevent the accumulation of biogenic amines in wine. Molecular methods are independent of culture conditions, and can therefore be used to detect potential biogenic amine producers even under circumstances where the LAB had lost the ability to produce biogenic amines. Such instances have been reported after prolonged storage or during subcultivation in the laboratory (Lonvaud-Funel and Joyeux, 1994; Leitão *et al.*, 2000; Lucas *et al.*, 2008).

Polymerase chain reaction (PCR) methods detecting bacterial genes related to biogenic amine production pathways are well described in the literature, and have been reviewed by Landete *et al.* (2007a). Oligonucleotide primer sequences for the amplification of internal fragments of the genes encoding histidine-, tyrosine-, ornithine- and lysine decarboxylase by PCR have been designed by a number of research groups and are briefly discussed here with reference to wine LAB. Table 17.5 summarises the published primer sequences suitable and potentially suitable to detect the decarboxylase genes of wine LAB.

Lonvaud-Funel and Joyeux (1994) isolated the first histidine decarboxylase (*hdc*) enzyme of a wine lactic acid bacterium (*O. oeni* 9204). This decarboxylase was studied and purified by Coton *et al.* (1998b). Using the gene sequence of *O. oeni* 9204 and also other HDC⁺ strains, detection systems based on PCR were

Table 17.5 Oligonucleotide primers for the detection of biogenic amine-producing wine lactic acid bacteria

Primer name	Primer 5'→3' sequence	Reference
<i>Primers designed for the detection of histidine decarboxylase</i>		
CL1	CCWGGWAAWATWGGWAATGGWTA	Le Jeune <i>et al.</i> (1995)
CL2	GAWGCWGTWGTGCATATTWATTGWCC	Le Jeune <i>et al.</i> (1995)
JV16HC	AGATGGTATTGTTTCTTATG	Le Jeune <i>et al.</i> (1995)
JV17HC	AGACCATACACCATAACCTT	Le Jeune <i>et al.</i> (1995)
JV17	AGACCATACACCATAACCTTG	Coton <i>et al.</i> (1998a,b)
CL1mod	CCAGGWAACATTGGTAATGGATA	Landete <i>et al.</i> (2005a)
HDC3	GATGGTATTGTTTCKTATGA	Coton and Coton (2005)
HDC4	CAAACACCAGCATCTTC	Coton and Coton (2005)
PHDC1	CCGTGCGGAAACAAAGAAT	Constantini <i>et al.</i> (2006)
PHDC2	CCAAACACCAGCATCTTCA	Constantini <i>et al.</i> (2006)
HIS1-F	GGNATNGTNWSNTAYGAYMGNGCNGA	De las Rivas <i>et al.</i> (2006)
HIS1-R	ATNGCDATNGCNSWCCANACNCCRTA	De las Rivas <i>et al.</i> (2006)
hdcAf	ATGAAGCCAGGACAAGTTGG	Lucas <i>et al.</i> (2008)
hdcAr	AATTGAGCCACCTGGAATTG	Lucas <i>et al.</i> (2008)
<i>Primers designed for the detection of tyrosine decarboxylase</i>		
P2-for	GAYATIATIGGIATIGGIYTIGAYCARG	Lucas and Lonvaud Funel (2002)
P1-rev	CCRTARTCIGGIATIGCRAARTCIGTRTG	Lucas and Lonvaud Funel (2002)
41	CAYGTNGAYGCNGCNTAYGGNGG	Marcobal <i>et al.</i> (2005b)
42	AYRTANCCCATYTTTRTGNGGRTC	Marcobal <i>et al.</i> (2005b)
Pt3	TACACGTAGATGCTGCATATG	Constantini <i>et al.</i> (2006)
Pt4	ATGGTTGACTATGTTTTAAAGAA	Constantini <i>et al.</i> (2006)
p0303	CCACTGCTGCATCTGTTTG	Lucas <i>et al.</i> (2003)
TD5	CAAATGGAAGAAGAAGTAGG	Coton <i>et al.</i> (2004)
TD2	ACATAGTCAACCATRRTTGAA	Coton <i>et al.</i> (2004)
TDC-F	TGGYTNGTNCNCARACNAARCAITA	De las Rivas <i>et al.</i> (2006)
TDC-R	ACRTARTCNACCATRTTRAARTCNGG	De las Rivas <i>et al.</i> (2006)
tdcf	CAAATGGAAGAAGAAGTTGG	Nannelli <i>et al.</i> (2008)
tdcr	GAACCATCAGCAACAATGTG	Nannelli <i>et al.</i> (2008)
<i>Primers designed for the detection of ornithine decarboxylase</i>		
3	GTNTTYAAYGCNGAYAARACNTAYTTYGT	Marcobal <i>et al.</i> (2004)
16	TACRCARAATACTCCNGGNGGRTANGG	Marcobal <i>et al.</i> (2004)
4	ATNGARTTNAGTTCRCAYTTYTCNGG	Marcobal <i>et al.</i> (2005b)
15	GGTAYTGTTYGAYCGGAAWAWCAYAA	Marcobal <i>et al.</i> (2005b)
AODC1	GMTCGTGAAATYAARCKG	Constantini <i>et al.</i> (2006)
AODC2	KGRGTTTCMGCYGGRTAT	Constantini <i>et al.</i> (2006)
PUT1-F	TWYMAYGCNGAYAARACNTAYYYTGT	De las Rivas <i>et al.</i> (2006)
PUT1-R	ACRCANAGNACNCCNGNGGRTANGG	De las Rivas <i>et al.</i> (2006)
odcf	TGCACTTCCATATCCTCCAGD	Nannelli <i>et al.</i> (2008)
odcr	GAATTTCTGGAGCAAATCCA	Nannelli <i>et al.</i> (2008)
<i>Primers designed for the detection of agmatine deiminase</i>		
AGDIfor	GAACGACTAGCAGCTAGTTAT	Lucas <i>et al.</i> (2007)
AGDIrev	CCAATAGCCGATACTACCTTG	Lucas <i>et al.</i> (2007)

continued

Table 17.5 *continued*

Primer name	Primer 5'→3' sequence	Reference
agdif	ATGCCCCGGTGAATTTGAA	Nannelli <i>et al.</i> (2008)
agdir	TTGCGCTGGTTTAGCACC	Nannelli <i>et al.</i> (2008)
<i>Primers designed for the detection of lysine decarboxylase (needs to be evaluated for wine LAB)</i>		
CAD1-F	TTYGAYWCNGCNTGGGTNCCNTAYAC	De las Rivas <i>et al.</i> (2006)
CAD1-R	CCRTGDATRTCNGTYTCRAANCCNGG	De las Rivas <i>et al.</i> (2006)
CAD2-F	CAYRTNCCNGGNCAYAA	De las Rivas <i>et al.</i> (2006)
CAD2-R	GGDATNCCNGGNGGRTA	De las Rivas <i>et al.</i> (2006)

K = G or T; R = A or G; W = A or T; Y = C or T; S = C or G; M = A or C; D = A, G, or T; N = A, G, C or T.

developed by researchers to screen for potential histamine-producing bacteria of different genera in wine (Le Jeune *et al.*, 1995; Coton *et al.*, 1998a; Coton and Coton, 2005; Landete *et al.*, 2005a; Lucas *et al.*, 2008). In two reports (Lucas *et al.*, 2005, 2008) the *hdc* genes were detected on plasmids in the genomes of *L. hilgardii* and *O. oeni*, respectively. The authors proposed that the instability of such plasmids could lead to the loss of decarboxylase activity during subcultivation in laboratory medium.

The *hdc* gene was found to be present in a high proportion of wine LAB strains, despite the actual levels of histamine produced in wine being generally low (Lonvaud-Funel and Joyeux, 1994; Le Jeune *et al.*, 1995; Coton *et al.*, 1998a; Guerrini *et al.*, 2002; Landete *et al.*, 2005a, 2007b; Lucas *et al.*, 2008). In contrast, other studies have found the frequency of histamine producers in wine to be very low (Moreno-Arribas *et al.*, 2003). Among the LAB strains found capable of contributing to histamine production in wine are *O. oeni*, *L. hilgardii*, *Pediococcus parvulus*, *Lactobacillus mali* and *Leuc. mesenteroides* (Lehtonen, 1996; Moreno-Arribas *et al.*, 2003; Landete *et al.*, 2005a; Lucas *et al.*, 2005).

Moreno-Arribas and Lonvaud-Funel (1999) isolated a tyramine-producing LAB strain from wine, which was identified as *L. brevis* (IOEB 9809). The tyrosine decarboxylase (*tdc*) gene of this strain was purified and characterised (Lucas and Lonvaud-Funel, 2002; Lucas *et al.*, 2003) and a rapid PCR assay was developed to detect *L. brevis* strains carrying the *tdc* gene (Lucas and Lonvaud-Funel, 2002). Downing (2003) determined that the described primers could also be used to identify tyrosine-producing *L. hilgardii* strains. More primer sets designed by other authors for the detection of the *tdc* gene in wine LAB are listed in Table 17.5. Furthermore, the amino acid phenylalanine is also a substrate for *tdc*; producing phenylethylamine in a secondary reaction (Landete *et al.*, 2007c). The ability of wine LAB to form phenylethylamine seems to be rare, with *L. brevis* and *L. hilgardii* being the only strains to date associated with high levels of phenylethylamine in wines (Moreno-Arribas *et al.*, 2000; Landete *et al.*, 2007c).

A number of wine LAB strains have since been identified as tyramine producers, of which most belong to the species *L. brevis* and *L. hilgardii* (Moreno-Arribas

et al., 2000; Downing, 2003). *Leuc. mesenteroides* was also found to be a tyramine producer (Moreno-Arribas *et al.*, 2003). Very few *O. oeni* strains seem to have the genetic ability to produce tyramine (Moreno-Arribas *et al.*, 2000; Guerrini *et al.*, 2002; Moreno-Arribas *et al.*, 2003; Landete *et al.*, 2007b). Until recently, *O. oeni* strains were shown able to produce tyramine only in laboratory media and cider (Choudhury *et al.*, 1990; Gardini *et al.*, 2005; Garai *et al.*, 2007). However, Rosi *et al.* (2009) showed that a number of indigenous *O. oeni* strains could produce biogenic amines, including tyramine, during MLF in synthetic medium and in wine (the presence of the decarboxylase genes was not determined by molecular means in all studies).

Putrescine is reported to be the most abundant biogenic amine in wine, both qualitatively and quantitatively (Glória *et al.*, 1998; Soufleros *et al.*, 1998; Vasquez-Lasa *et al.*, 1998; Soleas *et al.*, 1999; Landete *et al.*, 2005b; Bover-Cid *et al.*, 2006; González Marco and Ancín Azpilicueta, 2006; Alcaide-Hidalgo *et al.*, 2007). Marcobal *et al.* (2004) first reported the presence of the ornithine decarboxylase (*odc*) gene in the genome of *O. oeni* (BIFI-83) and designed the first complete set of primers to amplify this gene. However, putrescine can be produced in wine via two independent pathways; via the *odc* pathway or via the breakdown of arginine, a major amino acid in wine (Mangani *et al.*, 2005). Arginine is degraded to agmatine via arginine decarboxylase. Agmatine is converted to putrescine via the agmatine deiminase (AgDI) pathway (Arena and Manca de Nadra, 2001; Lonvaud-Funel, 2001). PCR detection protocols for the AgDI pathway have been described by Lucas *et al.*, 2007 and Nannelli *et al.*, 2008. Wine LAB capable of producing putrescine via one or both of these pathways include strains of *O. oeni*, *Lactobacillus plantarum*, *L. hilgardii*, *L. brevis* and *P. parvulus* (Arena and Manca de Nadra, 2001; Marcobal *et al.*, 2004; Nannelli *et al.*, 2008). Table 17.5 lists the primer sets that were designed to detect *odc* and AgDI in wine LAB.

De Las Rivas *et al.* (2006) described the first oligonucleotides for the PCR detection of lysine decarboxylase encoding genes in foodborne bacteria (Table 17.5). The possibility exists to extend this PCR method to detect cadaverine-producing bacteria in wine.

Multiplex amplification methods were developed to reduce the quantity of reagent, labour costs and time, because more than one gene species (target amines) can be detected simultaneously using this approach, making it suitable for the routine screening of wine LAB. Multiplex PCR assays for the detection of decarboxylases in fermented food and beverages, including wine, have been described by Coton and Coton (2005), De las Rivas *et al.* (2005, 2006) and Marcobal *et al.* (2005b).

To better understand biogenic amine formations during wine fermentation and the whole wine elaboration process, quantitative real-time PCR (qRT-PCR) can be employed as a screening tool. For example, qRT-PCR can be used to detect and count decarboxylase-positive LAB during MLF.

Molecular technologies for microbial community analysis in food were reviewed by Rudi *et al.* (2002). These technologies include a DNA-based approach for separate detection of viable and dead bacteria multiplex qRT-PCR methods for

the absolute quantification of bacteria, and a DNA array-based detection method with the possibility of detecting several different targets simultaneously to analyse complex microbial communities and as new bacterial fingerprinting tool. Such technologies are currently being developed for use in wine. Lucas *et al.* (2008) developed a method based on qRT-PCR to screen and quantify histamine-producing bacteria in wine. Nannelli *et al.* (2008) described a qRT-PCR method to detect the presence of tyramine and putrescine-forming bacteria in wines and to quantify the producing cells.

Molecular methods could provide unambiguous detection of the presence of biogenic amine-producing wine microorganisms. Although these methods cannot indicate the final concentrations of biogenic amines that will develop in wine, the risk of biogenic amine spoilage could be linked to *hdc*⁺ population densities (Lucas *et al.*, 2008). Altogether, molecular techniques such as uniplex, multiplex and qRT-PCR provide promising alternatives to traditional culture methods to determine and understand biogenic amine formation during winemaking.

17.4 Methods of detection and quantification of biogenic amines in wines

Detection and quantification of biogenic amines can either focus on the decarboxylase-positive microorganisms to assess the potential risk or determine the levels of the biogenic amines present to ensure the safety and quality. Methods available currently are qualitative or quantitative.

17.4.1 Qualitative methods

Screening of decarboxylase-positive microbes using selective media

One of the first methods used to detect biogenic amine-producing microbes was using selective and/or differential media (adding different amino acid precursors) that included a pH indicator which would change colour due to the formation of the amine. This was first applied in the detection of histamine-producing lactobacilli (Sumner *et al.*, 1985). In the last 10 years, research has improved the selective screening medium to correlate well with chemical analytical methods (Bover-Cid and Holzapfel, 1999; Downing, 2003). To evaluate the potential risk of biogenic amine-producing microbes it is advisable to determine the actual levels with a qualitative method as false positives have been previously detected (Morena-Arribas *et al.*, 2003).

Enzymatic detection

The first enzymatic method was developed to determine histamine in fish. This method was problematic in must and wine matrix, after which Landete *et al.* (2004) developed a direct enzymatic assay that correlated well with high-performance liquid chromatography (HPLC) analysis. Enzyme-linked immunosorbent assay

(ELISA) was tested in wine for the first time by Marcobal *et al.* (2005a) and showed good correlation with HPLC as well. An enzyme sensor array measuring three biogenic amines simultaneously in wine was evaluated by Lange and Wittmann (2002). This method is simple and rapid as expensive instrumentation is not needed.

Thin-layer chromatography

This method was used to determine biogenic amine contents of foods (Halász *et al.*, 1994; Shalaby, 1995). Through derivatisation, the method was improved to separate the most important wine-related biogenic amines, namely histamine, tyramine, putrescine and phenylethylamine (García-Moruno *et al.*, 2005).

Polymerase chain reaction (PCR) detection of decarboxylase genes

Assessing the potential risk of a product at an early stage of production will assist in managing the fermentation process better and therefore minimise the spoilage. This can be done by detecting the different biogenic amine decarboxylase-positive microorganisms in your product. Molecular methods using gene-specific primers are being exploited for rapid and early detection compared to culture-dependent methods. The drive to sequence genomes of yeasts and bacteria that are important to winemaking has widened the biogenic amine decarboxylase gene primers available to detect the specific genes. These primers can also be applied with qRT-PCR to allow for screening and quantification of biogenic amine-producing LAB (Landete *et al.*, 2007a; Lucas *et al.*, 2008; Nannelli *et al.*, 2008).

The histidine-, tyrosine-, ornithine- and lysine decarboxylase gene-specific primers for different wine LAB have been developed (Le Jeune *et al.*, 1995; Coton *et al.*, 1998b; Downing, 2003; Marcobal *et al.*, 2005b; De las Rivas *et al.*, 2006). Multiplex PCR reactions have been developed to assess more than one gene in a single reaction to improve the throughput of screening (Coton and Coton, 2005; De las Rivas *et al.*, 2005; Marcobal *et al.*, 2005b).

17.4.2 Quantitative methods

Due to their chemical properties and low concentrations in wine, the detection and quantification is not a simple procedure, but requires clean-up from the matrix, concentration and derivatisation. Concentration is mostly done by solid phase extraction (SPE) or liquid–liquid extraction (García-Villar *et al.*, 2006).

Gas chromatography

Fernandes and Ferreira (2000) developed a very sensitive gas-chromatography/mass spectrometry (GC–MS) method to detect diamines, polyamines and aromatic amines after derivatisation with *o*-heptafluorobutyl (OPA).

Liquid chromatography

This type of chromatography is currently the preferred analytical method, including HPLC, liquid chromatography (LC) and micellar liquid chromatography

(MLC). Various HPLC methods have been developed or optimised, especially derivatisation OPA (Lehtonen, 1996; Soleas *et al.*, 1999; Vidal-Carou *et al.*, 2003; Marcobal *et al.*, 2005a), fluorenylmethylchloroformate (Bauza *et al.*, 1995), 1,2-naphthoquinone-4-sulfonate (García-Villar *et al.*, 2007) and many others (for review see Önal *et al.*, 2007; Smit *et al.*, 2008), and these have increased the detection levels and stability. The other varying factor that has been investigated is the detection system used, for example HPLC-ultra violet (Lehtonen, 1996; Vidal-Carou *et al.*, 2003; Gómez-Alonso *et al.*, 2007; Önal, 2007), HPLC-fluorescence (Loukou and Zotou, 2003) and reverse-phase-HPLC (Alberto *et al.*, 2002; Marcobal *et al.*, 2005a; mo Dugo *et al.*, 2006; Pereira *et al.*, 2008).

Gil-Agustí *et al.* (2007) used MLC as a direct injection method as no sample treatment is required. A variation of this method, micellar electrokinetic chromatography with a fluorescent detector, can be used to detect and quantify biogenic amines and their amino acids (Nouadje *et al.*, 1997).

Liquid chromatography has recently been re-evaluated as an analytical tool after the first method was applied by Ibe *et al.* (1991) to discriminate 16 biogenic amines. This method was refined by Hlabangana *et al.* (2006) (ion-pair-LC), Hernández-Borges *et al.* (2007) (nano-LC) and Millán *et al.* (2007) and Smit (2007) (LC-MS).

Capillary electrophoresis

Capillary electrophoresis (CE) is advantageous compared to the other methods mentioned due to high resolution and short run times (Smit *et al.*, 2008). The sensitivity of CE analysis can be problematic when UV detection is used, and this was improved by coupling the CE to MS (electrospray; ion-trap; time-of-flight) (Santos *et al.*, 2004; Chiu *et al.*, 2006; Simó *et al.*, 2008). Other methods to optimise or improve CE analyses that were investigated include conductometric detection (Kvasnicka and Voldrich, 2006) and clean and concentrate samples with solid-phase-extraction (Acre *et al.*, 1998) and high-performance CE (Kovács *et al.*, 1999) and in-capillary derivatisation (García-Villar *et al.*, 2006).

17.5 Methods and tools to prevent the presence of biogenic amines in wines

As we have seen, the formation of biogenic amines in wines is a complex process in which diverse factors are involved, mainly related to the fermentation conditions used in the wine cellars and some environmental factors. Careful monitoring of the vinification process, especially in critical stages (i.e. MLF), could be highly valuable to prevent the action of microorganisms with a capacity to produce biogenic amines and in this way to control the presence and concentration of these compounds during the elaboration of wine.

In wines that have the potential to contain biogenic amines, it is of special interest to act before these are formed, with a fast method of detection of the responsible bacteria, to eliminate certain bacteria strains and to replace them with

known commercial starter cultures that do not contain the respective decarboxylase genes. After wine has matured, it is difficult to eliminate or to reduce the amines present. In most cases, amines do not alter the flavour or the smell of the product, and therefore it is difficult to detect. Because of this, the detection of the presence of genes that encode for decarboxylase enzymes by PCR is advisable in wines before MLF. The test of choice is the qRT-PCR, which should be repeated at critical points during fermentation to ensure that the potential risk is managed to ensure a quality product. This technique is also a useful tool to quantify the number of cells with the capacity to synthesise biogenic amines. According to Lucas *et al.* (2008), LAB begin to produce histamine when the population is in the order of 10^3 cells/mL. If the presence of decarboxylase-positive bacteria is detected before MLF, it is a good indicator for the winemaker to inoculate with commercial malolactic starter cultures in order to out-compete the natural flora. It is also advisable to check the presence of biogenic amine producer strains after MLF, in order to avoid prolonged contact with yeast lees and to ensure microbial stability during wine storage and ageing.

Additionally, at present, the more practical option to control the potential biogenic amine formation is based on the inhibition of the growth of decarboxylase-positive indigenous bacteria and other microorganisms responsible for this alteration. Sulphur dioxide (SO_2) is the antimicrobial agent traditionally used in wine cellars to eliminate yeasts and bacteria that are not desirable in subsequent phases. Industrial studies carried out during wine maturation indicated that the addition of SO_2 to red wines, to the concentration allowed to reduce the bacterial population, prevents the formation of biogenic amines during wine ageing (Marcobal *et al.*, 2006a). These conclusions encourage the addition of SO_2 immediately after the degradation of malic acid, to eliminate as quickly as possible the LAB and therefore prevent the formation of biogenic amines. Due to the high pH values of many wines, the use of SO_2 can be less effective. For that reason, in some cases, the concentration of amines can increase in wines during the ageing. In fact, different studies have shown that red wines with high contents in histamine (>10 mg/L) are characterised by pH values above 3.7 (Landete *et al.*, 2005b; Marcobal *et al.*, 2006a; Martín-Álvarez *et al.*, 2006). Other alternatives focused on 'natural antimicrobial agents' have been developed to reduce the use of sulphite in wines. Among them, lysozyme (EC 3.2.1.17) has recently been introduced into the wine industry and offers important advantages for controlling MLF in wine (Pilatte *et al.*, 2000; Bartowsky, 2003). As wine pH increases, the antimicrobial activity of lysozyme increases as well, which makes it attractive for preventing spoilage in wines with high pH (Gao *et al.*, 2002; Delfini, 2004).

17.6 Future trends

Biogenic amines have the potential to become one of the marker molecules in future to assess wine quality and safety. Methods have been developed to assess both the levels of biogenic amines and the potential risk possessed by the natural

wine microflora. The quantitative analytical methods have been improved to such an extent that they are rapid, sensitive and accurate and they can be used on a routine basis. In future, another technology that might be used as a screening tool is infrared spectroscopy. A preliminary study indicated that this technology has great potential (Smit, 2007). The development of the qRT-PCR method will improve the detection and quantification of the potential risk with regard to spoilage microbes.

The management of the fermentation processes will be crucial to minimise the risks such as grape skin maceration, pH management and the addition of preservatives at critical points. As MLF is one of the biggest risk phases in the winemaking process, careful consideration should be given to the inoculation with commercial starter cultures and the timing of inoculation as this can lower the levels of certain biogenic amines.

To ensure wine quality and safety with regard to biogenic amines it is necessary to control the whole wine value chain from the grapes to the bottling of the final product.

17.7 References

- Acre L, Ríos A and Valcárcel M (1998), Direct determination of biogenic amines in wine by integrating continuous flow clean-up and capillary electrophoresis with indirect UV detection. *J Chromatogr A*, **803**, 249–260.
- Alberto M R, Arena M E and Manca de Nadra M C (2002), A comparative survey of two analytical methods for identification and quantification of biogenic amines. *Food Control*, **13**, 125–129.
- Alcaide-Hidalgo J, Moreno-Arribas M V, Martín-Álvarez P J and Polo M C (2007), Influence of malolactic fermentation, postfermentative treatments and ageing with lees on nitrogen compounds of red wines. *Food Chem*, **103**, 572–581.
- Alcaide-Hidalgo J, Moreno-Arribas M V, Polo M C and Pueyo E (2008), Partial characterization of peptides from red wines. Changes during malolactic fermentation and aging with lees. *Food Chem*, **107**, 622–630.
- Ancín-Azpilicueta C, González-Marco A and Jiménez-Moreno N (2008), Current knowledge about the presence of biogenic amines in wine. *Crit Rev Food Sci Nutr*, **48**, 257–275.
- Anli R E, Vural N, Yilmaz S and Vural Y H (2004), The determination of biogenic amines in Turkish red wines. *J Food Comp Anal*, **17**, 53–62.
- Arena M E, Manca de Nadra M C (2001), Biogenic amine production by *Lactobacillus*. *J Appl Microbiol*, **90**, 158–162.
- Arena M E, Fiocco D, Manca de Nadra M C, Pardo I and Spano G (2007), Characterization of a *Lactobacillus plantarum* strain able to produce tyramine and partial cloning of a putative tyrosine decarboxylase gene. *Curr Microbiol*, **55**, 205–210.
- Bardóczi S, Grant G, Bron D S, Ralph A and Pusztai A (1993), Polyamines in food – Implications for growth and health. *J Nutr Biochem*, **4**, 66–71.
- Barth H and Lorenz W (1978), Structural requirements of imidazole compounds to be inhibitors or activators of histamine methyltransferase: investigation of histamine analogues and H₂-receptor antagonists. *Agents Actions*, **8**, 359–365.
- Bartowsky E (2003), Lysozyme and winemaking. *Aust J Grape Wine Res*, **473a**, 101–104.
- Bauza T, Blaise A, Daumas F and Cabanis J C (1995), Determination of biogenic amines and their precursor amino acids in wines of the Vallée du Rhône by high-performance liquid chromatography with precolumn derivatization and fluorimetric detection. *J Chromatogr A*, **707**, 373–379.

- Becker K, Southwick K, Reardon J, Berg R and MacCormack J N (2001), Histamine poisoning associated with eating tuna burgers. *J Am Med Assoc*, **285**, 1327–1330.
- Blackwell B (1963), Hypertensive crisis due to monoamine oxidase inhibitors. *Lancet*, **2**, 849–851.
- Bodmer S and Gafner H (2000), Histamin in wein. *Schweizerische Zeitschrift für Obst- und Weinbau*, **21**, 546–547.
- Bodmer S, Imark C and Kneubühl M, (1999), Biogenic amines in foods: histamine and food processing. *Inflamm Res*, **48**, 296–300.
- Bover-Cid S and Holzapfel W H, (1999), Improved screening procedure for biogenic amine production by lactic acid bacteria. *Int J Food Microbiol*, **53**, 33–41.
- Bover-Cid S, Izquierdo-Pulido M, Mariné-Font A and Vidal-Carou M C (2006), Biogenic, mono-, di- and polyamine contents in Spanish wines and influence of a limited irrigation. *Food Chem*, **96**, 43–47.
- Buckland P R, Marshall R, Watkins P and McGuffin P (1997), Does phenylethylamine have a role in schizophrenia? LSD and PCP up-regulate aromatic L-amino acid decarboxylase mRNA levels. *Mol Brain Res*, **49**, 266–270.
- Buteau C, Duitschaever C L and Ashton G C (1984), A study of the biogenesis of amines in a Villard Noir wine. *Am J Enol Vitic*, **35**, 228–236.
- Caruso M, Fiore C, Contrusi M, Salzano G, Paparella A and Romano P (2002), Formation of biogenic amines as criteria for the selection of wine yeast. *World J Microbiol Biotech*, **18**, 159–163.
- Cecchini F, Morassut M and Moruno E G (2005), Influence of cultivar on biogenic amines content in wines. *XIV International GESCO Viticulture Congress*, Geisenheim, Germany, 23–27 August, 845–850.
- Chiu T C, Lin Y W, Huang Y F and Chang H T (2006), Analysis of biologically active amines by CE. *Electrophoresis*, **23**, 4792–4807.
- Choudhury N, Hansen W, Engesser D, Hammes W P and Holzapfel W H (1990). Formation of histamine and tyramine by lactic acid bacteria in decarboxylase assay medium. *Lett Appl Microbiol*, **11**, 278–281.
- Cohn V H (1965), Inhibition of histamine methylation by antimalarian drugs. *Biochem Pharmacol*, **14**, 1886–1888.
- Constantini A, Cersosimo M, Del Prete V and García-Moruno E (2006), Production of biogenic amines by lactic acid bacteria: screening by PCR, thin-layer chromatography, and high-performance liquid chromatography of strains isolated from wine and must. *J Food Prot*, **69**, 391–396.
- Coton E and Coton M (2005), Multiplex PCR for colony direct detection of Gram-positive histamine- and tyramine-producing bacteria. *J Microbiol Meth*, **63**, 296–304.
- Coton E, Rollan G, Bertrand A and Lonvaud-Funel A (1998a), Histamine-producing lactic acid bacteria in wines; early detection, frequency and distribution. *Am J Enol Vitic*, **49**, 199–204.
- Coton E, Rollan G C and Lonvaud-Funel A (1998b), Histidine decarboxylase activity of *Leuconostoc oenos* 9204: purification, kinetic properties, cloning and nucleotide sequence of the HDC gene. *J Appl Microbiol*, **84**, 143–151.
- Coton M, Coton E, Lucas P and Lonvaud A (2004), Identification of the gene encoding a putative tyrosine decarboxylase of *Carnobacterium divergens* 508. Development of molecular tools for the detection of tyramine producing bacteria. *Food Microbiol*, **21**, 125–130.
- De las Rivas B, Marcobal A and Muñoz R (2005), Improved multiplex-PCR method for the simultaneous detection of food bacteria producing biogenic amines. *FEMS Microbiol Lett*, **244**, 367–372.
- De las Rivas B, Marcobal A, Carrascosa A V and Muñoz R (2006), PCR detection of foodborne bacteria producing the biogenic amines histamine, tyramine, putrescine, and cadaverine. *J Food Prot*, **69**, 2509–2514.
- Del Prete V, Constantini A, Cecchini F, Morassut M and García-Moruno E (2008),

- Occurrence of biogenic amines in wine: The role of grapes. *Food Chem*, **112**, 474–481.
- Delfini C (2004), Resistance screening assay of wine lactic acid bacteria on lysozyme: efficacy of lysozyme in unclarified grape musts. *J Agric Food Chem*, **52**, 1861–1866.
- Downing L, (2003), *Characterisation of biogenic amine-encoding genes in lactic acid bacteria isolated from South African wine*. Master Thesis, Stellenbosch University, Stellenbosch, South Africa.
- Fariás M E, Manca de Nadra M, Rollán G C and Strasser de Saad A (1993), Histidine decarboxylase activity in lactic acid bacteria from wine. *J Int Sci Vigne Vin*, **27**, 191–199.
- Fernandes J O and Ferreira M A (2000), Combined ion-pair extraction and gas chromatography-mass spectrometry for the simultaneous determination of diamines, polyamines and aromatic amines in Port wine and grape juice. *J Chromatogr A*, **886**, 183–195.
- Ferreira I M and Pinho O (2006), Biogenic amines in Portuguese traditional foods and wines. *J Food Protect*, **69**, 2293–2303.
- Franzen F and Eysell K (1969), *Biologically Active Amines in Man*. Pergamon Press, New York.
- Gao Y C, Zhang G, Krentz S, Dariu S, Power J and Lagarde G (2002), Inhibition of spoilage lactic acid bacteria by lysozyme during wine alcoholic fermentation. *Aust J Grape Wine Res*, **8**, 76–83.
- Garai G, Dueñas M T, Irastorza A, Martín-Álvarez P J and Moreno-Arribas M V (2007), Biogenic amine production by lactic acid bacteria isolated from cider. *Lett Appl Microbiol*, **45**, 473–478.
- García-Moruno E, Carrascosa A V and Muñoz R (2005), A rapid and inexpensive method for the determination of biogenic amines from bacterial cultures by thin-layer chromatography. *J Food Prot*, **68**, 625–629.
- García-Ruiz A, Bartolomé B, Martínez-Rodríguez A, Pueyo E, Martín-Álvarez P J and Moreno-Arribas M V (2008), Potential of phenolic compounds for controlling lactic acid bacteria growth in wine. *Food Control*, **19**, 835–841.
- García-Ruiz A, Bartolomé B, Cueva, C, Martín-Álvarez P J and Moreno-Arribas M V (2009), Inactivation of oenological lactic acid bacteria (*Lactobacillus hilgardii* and *Pediococcus pentosaceus*) by wine phenolic compounds *J Appl Microbiol*, **107**, 1042–1053.
- García-Villar N, Saurina J and Hernández-Cassou S (2006), High-performance liquid chromatographic determination of biogenic amines in wines with an experimental design optimization procedure. *Anal Chim Acta*, **575**, 97–105.
- García-Villar N, Hernández-Cassou S and Saurina J (2007), Characterization of wines through the biogenic amine contents using chromatographic techniques and chemometric data analysis. *J Agric Food Chem*, **55**, 7453–7461.
- Gardini F, Zaccarelli A, Belletti N, Faustini F, Cavazza A, Maruscelli M, Mastrocola D and Suzzi G (2005), Factors influencing biogenic amine production by a strain of *Oenococcus oeni* in a model system. *Food Control*, **16**, 609–618.
- Gerbaux V and Monamy C (2000), Biogenic amines in Burgundy wines. Contents and origin in wines. *Rev Fr Oenol*, **183**, 25–28.
- Gil-Agustí M, Carda-Brochm S, Monferrer-Pons L and Esteve-Romero J (2007), Simultaneous determination of tyramine and tryptamine and their precursor amino acids by micellar liquid chromatography and pulsed amperometric detection in wines. *J Chromatogr A*, **1156**, 288–295.
- Gilbert R T, Hobbes G, Murray C K, Craick-Shank J G and Young S E J (1980), Scombrototoxic fish poisoning: Feature of the first 50 incidents to be reported in Britain (1976–79). *Br Med J*, **281**, 71–72.
- Glória M B, Watson B T, Simon-Sarkadi L and Daeschel M A (1998) A survey of biogenic amines in Oregon Pinot noir and Cabernet Sauvignon wines. *Am J Enol Vitic*, **49**, 279–282.
- Gómez-Alonso S, Hermosín-Gutiérrez U and García-Romero E (2007), Simultaneous HPLC analysis of biogenic amines, amino acids, and ammonium ion as animoenone derivatives in wine and beer samples. *J Agric Food Chem*, **55**, 608–613.

- Goñi D T and Ancín Azpilicueta C (2001), Influence of yeast strain on biogenic amine content in wines: relationship with the utilization of amino acids during fermentation. *Am J Enol Vitic*, **52**, 185–190.
- González-Marco A and Ancín-Azpilicueta C (2006), Amine concentrations in wine stored in bottles at different temperatures. *Food Chem*, **99**, 680–685.
- Granchi L, Romano P, Mangani S, Guerrini S and Vincenzini M (2005), Production of biogenic amines by wine microorganisms. *Bull OIV* **78**, 595–609.
- Guerrini S, Mangani S, Granchi L and Vincenzini M (2002), Biogenic amine production by *Oenococcus oeni*. *Curr Microbiol*, **44**, 374–378.
- Hajós G, Sass-Kiss A, Szerdahelye E and Bardocz S (2000) Changes in biogenic amine content of Tokaj grapes, wines, and Aszu-wines, *J Food Sci*, **65**, 1142–1144.
- Halász A, Baráth Á, Simon-Sarkadi L and Holzapfel W (1994), Biogenic amines and their production by microorganisms in food. *Trends Food Sci Technol*, **5**, 42–49.
- Herbert P, Cabrita M J, Ratola N, Laureano O and Alves A (2005) Free amino acids and biogenic amines in wines and musts from the Alentejo region. Evolution of amines during alcoholic fermentation and relationship with variety, sub-region and vintage. *J Food Eng*, **66**, 315–322.
- Hernández-Borges J, D'Orazio G, Aturki Z and Fanali S (2007), Nano-liquid chromatography analysis of dansylated biogenic amines in wine. *J Chromatogr A*, **1147**, 192–199.
- Hernández-Orte P, Lapeña A C, Peña-Gallego A, Astrain J, Baron C, Pardo I, Polo L, Ferrer S, Cacho J and Ferreira V (2008), Biogenic amine determination in wine fermented in oak barrels. Factors affecting formation. *Food Res Int*, **41**, 697–706.
- Hlabangana L, Hernandez-Cassou S and Saurine J (2006), Determination of biogenic amines in wines by ion-pair liquid chromatography and post-column derivatization with 1,2-naphthoquinone-4-sulphonate. *J Chromatogr A*, **1130**, 130–136.
- Horwitz D, Lovenberg W, Engelman A and Sjoerdsma A (1964), Monoamine oxidase inhibitors, tyramine and cheese. *J Am Med Assoc* **188**, 90–92.
- Hui J Y and Taylor S L (1985), Inhibition of *in vivo* histamine metabolism in rats by foodborne and pharmacologic inhibitors of diamine oxidase, histamine N-methyl transferase and monoamine oxidase. *Toxicol Appl Pharmacol*, **8**, 241–249.
- Huis in't Veld J H J, Schaafsma G J, Silla H and Smith J E (1990), Health aspect of food biotechnology. In: Zeuthen P, Cheftel J C, Ericksson C, Gormley T R, Link P and Paulus K (eds), *Processing and Quality of Foods. Food Biotechnology: Avenues to healthy and nutritious products*, Vol. 2. Elsevier Applied Science, London and New York, 2.73–2.97.
- Ibe A, Saito K, Nakazato M, Kikuchi Y, Fujinuma K and Nishima T (1991), Quantitative determination of amines in wine by liquid chromatography. *J Assoc Off Anal Chem*, **74**, 695–698.
- Ishizuka H, Harinouchi S and Beppu T (1993) Putrescine oxidase of *Micrococcus rubens*: primary structure and *Escherichia coli*. *Gen Microbiol*, **139**, 425–432.
- Jiménez-Moreno N, Goñi D T and Ancín-Azpilicueta C (2003), Changes in amine concentrations during aging of red wine in oak barrels. *J Agric Food Chem*, **51**, 5732–5737.
- Joosten H M L J (1988), The biogenic amine contents of Dutch cheese and their toxicological significance. *Neth Milk Dairy J*, **42**, 25–42.
- Kahana L M and Todd E (1981) Histamine intoxication in a tuberculosis patient on isoniazid. *Can Dis Weekly Rep*, **7**, 79–80.
- Kovács A, Simon-Sarkadi L and Ganzler K (1999) Determination of biogenic amines by capillary electrophoresis. *J Chromatogr A*, **836**, 305–313.
- Kvasnicka F and Voldrich M (2006), Determination of biogenic amines by capillary zone electrophoresis with conductometric detection. *J Chromatogr A*, **1103**, 145–149.
- Landete J M, Ferrer S and Pardo I (2004), Improved enzymatic method for the rapid determination of histamine in wine. *Food Addit Contam*, **21**, 1149–1154.
- Landete J M, Ferrer S and Pardo I (2005a), Which lactic acid bacteria are responsible of histamine production in wine?. *J Appl Microbiol*, **99**, 580–586.

- Landete J M, Ferrer S, Polo L and Pardo I, (2005b), Biogenic amines in wines from three Spanish regions. *J Agric Food Chem*, **53**, 1119–1124.
- Landete J M, Pardo I and Ferrer S (2006), Histamine, histidine, and growth-phase mediated regulation of the histidine decarboxylase gene in lactic acid bacteria isolated from wine. *FEMS Microbiol Lett*, **260**, 84–90.
- Landete J M, de las Rivas B, Marcobal A and Muñoz R (2007a), Molecular methods for the detection of biogenic amine-producing bacteria on foods. *Int J Food Microbiol*, **117**, 258–269.
- Landete J M, Ferrer S and Pardo I, (2007b), Biogenic amine production by lactic acid bacteria, acetic bacteria and yeast isolated from wine. *Food Control*, **18**, 1569–1574.
- Landete J M, Pardo I and Ferrer S (2007c), Tyramine and phenylethylamine production among lactic acid bacteria isolated from wine. *Int J Food Microbiol*, **115**, 364–368.
- Lange J and Wittmann C (2002) Enzyme sensor array for the determination of biogenic amines in food samples, *Anal Bioanal Chem*, **372**, 276–283.
- Lehtonen P (1996), Determination of amines and amino acids in wine – a review. *Am J Enol Vitic*, **47**, 127–133.
- Leitão MC, Teixeira HC, Barreto Crespo M T and San Romão M V (2000), Biogenic amines occurrence in wine. Amino acid decarboxylase and proteolytic activities expression by *Oenococcus oeni*. *J Agric Food Chem*, **48**, 2780–2784.
- Leitão M C, Marques A P and San Romão M V (2005), A survey of biogenic amines in commercial Portuguese wines. *Food Control*, **16**, 199–204.
- Le Jeune C, Lonvaud-Funel A, ten Brink B, Hofstra H, van der Vossen J M B M (1995), Development of a detection system for histidine decarboxylating lactic acid bacteria based on DNA probes, PCR and activity test. *J Appl Bacteriol*, **78**, 316–326.
- Lonvaud-Funel A (1999), Lactic acid bacteria in the quality improvement and depreciation of wine. *Antonie Van Leeuwenhoek*, **76**, 317–331.
- Lonvaud-Funel A (2001), Biogenic amines in wines: role of lactic acid bacteria. *FEMS Microbiol Lett*, **199**, 9–13.
- Lonvaud-Funel A and Joyeux A (1994), Histamine production by wine lactic acid bacteria: isolation of a histamine-producing strain of *Leuconostoc oenos*. *J Appl Bacteriol*, **77**, 401–407.
- Loukou Z and Zotou A (2003), Determination of biogenic amines as dansyl derivatives in alcoholic beverages by high-performance liquid chromatography with fluorimetric detection and characterization of the dansylated amines by liquid chromatography-atmospheric pressure chemical ionization mass spectrometry. *J Chromatogr A*, **996**, 103–113.
- Lovenberg W (1973), Some vaso- and psychoactive substances in food: Amines, stimulants, depressants and hallucinogens. In: *Toxicants Occurring Naturally in Foods*, National Academy of Science, Washington DC, 170–188.
- Lucas P and Lonvaud-Funel A (2002), Purification and partial gene sequence of the tyrosine decarboxylase of *Lactobacillus brevis* IOEB 9809. *FEMS Microbiol Lett*, **211**, 85–89.
- Lucas P, Landete J, Coton M, Coton E, Lonvaud-Funel A (2003), The tyrosine decarboxylase operon of *Lactobacillus brevis* IOEB 9809: characterization and conservation in tyramine-producing bacteria. *FEMS Microbiol Lett*, **229**, 65–71.
- Lucas P M, Wolken W A M, Claisse O, Lolkema J S and Lonvaud-Funel A, (2005), Histamine-producing pathway encoded on an unstable plasmid in *Lactobacillus hilgardii* 0006. *Appl Environ Microbiol*, **71**, 1417–1424.
- Lucas P M, Blancato V S, Claisse O, Magni C, Lolkema J S and Lonvaud-Funel A (2007), Agmatine deiminase pathway genes in *Lactobacillus brevis* are linked to the tyrosine decarboxylation operon in a putative acid resistance locus. *Microbiology*, **153**, 2221–2230.
- Lucas P M, Claisse O, Lonvaud-Funel A (2008), High frequency of histamine-producing bacteria in the enological environment and instability of the histidine decarboxylase production phenotype. *Appl Environ Microbiol*, **74**, 811–817.
- Mafra I, Herbert P, Santos L., Barros P and Alves A (1999), Evaluation of biogenic amines

- in some Portuguese quality wines by HPLC fluorescence detection of OPA derivatives. *Am J Enol Vitic*, **50**, 128–132.
- Mangani S, Geurrini S, Granchi L and Vincenzini M (2005), Putrescine accumulation in wine: role of *Oenococcus oeni*. *Curr Microbiol*, **51**, 6–10.
- Marcobal A, de las Rivas B, Moreno-Arribas M V and Muñoz R (2004), Identification of the ornithine decarboxylase gene in the putrescine-producer *Oenococcus oeni* BIFI-83. *FEMS Microbiol Lett*, **239**, 213–220.
- Marcobal A, Polo M C, Martín-Álvarez P J and Moreno-Arribas M V (2005a), Biogenic amines content of red Spanish wines. Comparison of a Direct ELISA and an HPLC method for the determination of histamine in wines. *Food Res Int*, **38**, 387–394.
- Marcobal A, de las Rivas B, Moreno-Arribas M V and Muñoz R (2005b), Multiplex-PCR method for the simultaneous detection of acid lactic bacteria producing histamine, tyramine and putrescine, three major biogenic amines. *J Food Prot*, **68**, 874–878.
- Marcobal A, Martín-Álvarez P J, Polo M C, Muñoz R and Moreno-Arribas M V (2006a), Formation of biogenic amines throughout the industrial manufacture of red wine. *J Food Prot*, **69**, 391–396.
- Marcobal A, Martín-Álvarez P J, Moreno-Arribas M V and Muñoz R (2006b), A multifactorial design for studying factors influencing growth and tyramine production of the lactic acid bacteria *Lactobacillus brevis* CECT 4669 and *Enterococcus faecium* BIFI-58. *Res Microbiol*, **157**, 417–424.
- Marcobal A, de las Rivas B, Moreno-Arribas M V and Muñoz R (2006c), Evidence for horizontal gene transfer as origin of putrescine production in *Oenococcus oeni* RM83. *Appl Environ Microbiol*, **72**, 7954–7958.
- Marques A P, Leitao M C and San Romao M V (2008), Biogenic amines in wines: Influence of oenological factors. *Food Chem*, **107**, 853–860.
- Martín-Álvarez P J, Marcobal A, Polo C and Moreno-Arribas M V (2006), Technological factors influencing biogenic amine production during red wine manufacture. *Eur Food Res Technol*, **222**, 420–424.
- Mazzoli R, Lamberti C, Coisson J D, Purroti M, Arlorio M, Giuffrida M G, Gounta C and Pessione E (2009), Influence of ethanol, malate and arginine on histamine production of *Lactobacillus hilgardii* isolated from an Italian red wine. *Amino Acids*, **36**, 81–89.
- Millán S, Sampedro M C, Unceta N, Goicolea M A and Barrio R J (2007), Simple and rapid determination of biogenic amines in wine by liquid chromatography-electrospray ionization ion trap mass spectrometry. *Anal Chim Acta*, **584**, 145–152.
- mo Dugo G, Vilasi F, La Torre G L and Pellicanò T M (2006), Reverse phase HPLC/DAD determination of biogenic amines as dansyl derivatives in experimental red wines. *Food Chem*, **95**, 672–676.
- Moreno-Arribas V and Lonvaud-Funel A (1999), Tyrosine decarboxylase activity of *Lactobacillus brevis* IOEB 9809 isolated from wine and *L. brevis* ATCC 367. *FEMS Microbiol Lett*, **180**, 55–60.
- Moreno-Arribas V and Lonvaud-Funel A (2001), Purification and characterization of tyrosine decarboxylase of *Lactobacillus brevis* IOEB 9809 isolated from wine *FEMS Microbiol Lett*, **195**, 103–107.
- Moreno-Arribas M V and Polo C (2008), Occurrence of lactic acid bacteria and biogenic amines in biologically aged wines. *Food Microbiol*, **25**, 875–881.
- Moreno-Arribas M V and Polo C (2009), Amino acids and biogenic amines. In: Moreno-Arribas M V and Polo M C (eds), *Wine Chemistry and Biochemistry*. Springer, New York, 163–189.
- Moreno-Arribas V, Pueyo E, Polo M C and Martin-Alvarez P J (1998) Changes in the amino acid composition of the different nitrogenous fractions during the aging of wine with yeasts. *J Agric Food Chem*, **46**(10), 4042–4051.
- Moreno-Arribas V, Torlois S, Joyeux A, Bertrand A and Lonvaud-Funel A (2000), Isolation, properties and behaviour of tyramine-producing lactic acid bacteria from wine. *J Appl Microbiol*, **88**, 584–593.

- Moreno-Arribas M V, Polo M C, Jorganes F and Muñoz R (2003), Screening of biogenic amine production by lactic acid bacteria isolated from grape must and wine. *Int J Food Microbiol*, **84**, 117–123.
- Moreno N J and Ancín Azpilicueta C A (2004), Influence of wine turbidity on the accumulation of biogenic amines during ageing. *J Sci Food Agric*, **84**, 1571–1576.
- Murooka Y, Doi N and Harada T (1979), Distribution of membrane-bound monamine oxidase in bacteria. *Appl Environ Microbiol*, **38**, 565–569.
- Nannelli F, Claisse O, Gindreau E, de Revel G, Lonvaud-Funel A and Lucas P M (2008), Determination of lactic acid bacteria producing biogenic amines in wine by quantitative PCR methods. *Lett Appl Microbiol*, **47**, 594–599.
- Nouadje G, Siméon N, Dedieu F, Nertz M, Puig Ph and Couderc F (1997), Determination of twenty eight biogenic amines and amino acids during wine aging by micellar electrokinetic chromatography and laser-induced fluorescence detection. *J Chromatogr A*, **765**, 337–343.
- Önal A (2007), A review: Current analytical methods for the determination of biogenic amines in foods. *Food Chem*, **103**, 1475–1486.
- O'Reilly R, Davis B A, Durden D A, Thorpe L, Machnee H and Boulton A A (1991), Plasma phenylethylamine in schizophrenic patients. *Biol Psychiatry*, **30**, 145–150.
- Ough C S, Dault C E and Crowell E A (1981), Identification of new volatile amines in grapes and wines. *J Agric Food Chem*, **29**, 938–941.
- Pereira V, Pontes M, Câmara J S and Marques J C (2008), Simultaneous analysis of free amino acids and biogenic amines in honey and wine samples using in loop ortho-phthalaldehyde derivatization procedure. *J Chromatogr A*, **1189**, 435–443.
- Pessione E, Pessione A, Lamberti C, Coisson D J, Riedel K, Mazzoli R, Bonetta S, Eberl L and Giunta C (2009) First evidence of a membrane-bound, tyramine and beta-phenylethylamine producing, tyrosine decarboxylase in *Enterococcus faecalis*: a two-dimensional electrophoresis proteomic study. *Proteomics*, **9**(10), 2695–2710.
- Pilatte E, Nygaard M, Gao Y C, Krentz S, Power J and Lagarde G (2000), Studies on effects of lysozyme on different strains of *Oenococcus oeni*. Applications in control of malolactic fermentation. *Rev Fr Oenol*, **185**, 26–29.
- Rollán G C, Coton E and Lonvaud-Funel A (1995), Histidine decarboxylase activity of *Leuconostoc oenos* 9204. *Food Microbiol*, **12**, 455–461.
- Rosi I, Nannelli F and Giovani G (2009), Biogenic amine production by *Oenococcus oeni* during malolactic fermentation of wines obtained using different strains of *Saccharomyces cerevisiae*. *Food Sci Technol*, **42**, 525–530.
- Rudi K, Nogva H K, Moen B, Nissen H, Bredholt S, Mørseth T, Naterstad K and Holck A (2002), Development and application of new nucleic acid-based technologies for microbial community analyses in foods. *Int J Food Microbiol*, **78**, 171–180.
- Sandler M, Youdim M B H and Hanington E (1974), A phenylethylamine oxidising defect in migraine. *Nature*, **250**, 335–336.
- Santos B, Simonet B M, Rios A and Valcarcel M (2004), Direct automatic determination of biogenic amines in wine by flow injection-capillary electrophoresis-mass spectrometry. *Electrophoresis*, **25**, 3427–3433.
- Sass-Kiss A, Szerdahelyi E and Hajós G (2000), Study of biologically active amines in grapes and wines by HPLC. *Chromatogr Suppl*, **51**, 316–320.
- Shalaby A R (1995), Multidetector, semiquantitative method for determining biogenic amines in foods. *Food Chem*, **52**, 367–372.
- Shalaby A R (1996), Significance of biogenic amines to food safety and human health. *Food Res Int*, **29**, 675–690.
- Simó C, Moreno-Arribas M V and Cifuentes A (2008), Ion-trap versus time-of-flight mass spectrometry coupled to capillary electrophoresis to analyze biogenic amines in wine. *J Chromatogr A*, **1195**, 150–156.
- Smit A Y (2007), *Evaluating the influence of winemaking practices on biogenic amine production by wine microorganisms*. Master Thesis, Stellenbosch University, Stellenbosch, South Africa.

- Smit A Y, Du Toit W J and Du Toit M (2008), Biogenic amines in wine: understanding the headache. *S Afr J Enol Vitic*, **29**, 109–127.
- Smith T A (1981), Amines in food. *Food Chem*, **6**, 169–200.
- Soleas G J, Carey M and Goldberg D M (1999), Method development and cultivar-related differences of nine biogenic amines in Ontario wines. *Food Chem*, **64**, 49–58.
- Soll A H and Wollin A (1977), The effects of histamine prostaglandin E2 and secretin on cyclic AMP in separated canine fundic mucosal cells. *Gastroenterology*, **72**, 1166.
- Soufleros E, Barrios M L and Bertrand A (1998) Correlation between the content of biogenic amines and other wine compounds. *Am J Enol Vitic*, **49**, 266–278.
- Soufleros E H, Bouloumpasi E, Zotou A and Loukou Z (2007), Determination of biogenic amines in Greek wines by HPLC and ultraviolet detection after dansylation and examination of factors affecting their presence and concentration. *Food Chem* **101**, 704–716.
- Spayd S E, Wample R L, Evans R G, Stevens R G, Seymour B J and Nagel C W (1994) Nitrogen fertilization of white Riesling grapes in Washington. Must and wine composition. *Am J Enol Vitic*, **45**, 34–42.
- Stockley I H (1973), Monoamine oxidase inhibitors. Part 1. Interactions with sympathomimetic amines. *Pharmacol J*, **210**, 590–594.
- Stratton J E, Hutkins R W and Taylor S L (1991), Biogenic amines in cheese and other fermented foods: A review. *J Food Prot*, **54**, 460–470.
- Sumner S S, Speckhard M W, Somers E B and Taylor S L (1985), Isolation of histamine-producing *Lactobacillus buchneri* from Swiss cheese implicated in a food poisoning outbreak. *Appl Environ Microbiol*, **50**, 1094–1096.
- Taylor S L (1985), *Histamine poisoning associated with fish, cheese and other food*. Monograph VPH/FOS/85.1. World Health Organization, Geneva.
- Taylor S L (1986), Histamine food poisoning: Toxicology and clinical aspects. *Crit Rev Toxicol*, **17**, 91–128.
- ten Brink B, Damink C, Joosten H M L J and Huis in 't Veld J H J (1990) Occurrence and formation of biologically active amines in foods. *Int J Food Microbiol*, **11**, 73–84.
- Torrea D and Ancín C (2002), Content of biogenic amines in a Chardonnay wine obtained through spontaneous and inoculated fermentations. *J Agric Food Chem*, **50**, 4895–4899.
- Van der Merwe H, (2007), *The evaluation of malolactic fermentation starter cultures under South African winemaking conditions*. Master Thesis, Stellenbosch University, Stellenbosch, South Africa.
- Vazquez-Lasa M B, Iñiguez-Crespo M, González-Larraina M and González-Guerrero A (1998), Biogenic amines in Rioja wines. *Am J Enol Vitic*, **49**, 229.
- Vidal-Carou M C, Codony-Salcedo R and Mariné-Font A (1990), Histamine and tyramine in Spanish wines: relationships with total sulfur dioxide level, volatile acidity and malolactic fermentation intensity. *Food Chem*, **35**, 217–227.
- Vidal-Carou M C, Lahoz-Portolés F, Bover-Cid S and Mariné-Font A (2003), Ion-pair high-performance liquid chromatographic determination of biogenic amines and polyamines in wine and other alcoholic beverages. *J Chromatogr A*, **998**, 235–241.
- Villamiel M, Polo M C and Moreno-Arribas M V (2008), Nitrogen compounds and polysaccharides changes during the biological ageing of sherry wines. *LWT-Food Sci Technol*, **41**, 1842–1846.
- Voigt M N and Eitenmiller R R (1978) Role of histidine and tyrosine decarboxylases and mono-diamine oxidases in amine build-up in cheese. *J Food Prot*, **41**, 182–186.
- Yamashita M, Sakaue M, Inwata N, Sugino H and Murooka Y (1993), Purification and characterization of monoamine oxidase from *Klebsiella aerogenes*. *J Ferm Bioeng*, **76**, 289–295.
- Zee J A, Simard R E, Heureux L L and Tremblay J (1983), Biogenic amines in wines. *Am J Enol Vitic*, **34**, 6–9.

Managing the quality of icewines

A. J. Bowen, Brock University, Canada

Abstract: This chapter discusses the various factors which affect icewine quality and the regulations for its production. Research initiatives are highlighted whenever possible. Topics covered include viticultural, harvest, oenological and sensory considerations such as cultural practices, fermentation kinetics, cultivar and yeast strain selection, important chemical quality variables and icewine sensory profiles. Authentication of icewine and future trends are briefly discussed.

Key words: icewine, wine quality, viticultural practices, fermentation kinetics, chemical composition.

18.1 Introduction

Icewine is a rare, intensely sweet late harvest wine that was discovered accidentally, as so many great things are, in 1794 in Franconia, Germany when an early frost froze Riesling grapes still hanging on the vine in late November. To salvage the grapes they were pressed frozen and the drips of concentrated juice were fermented into wine to create the first *eiswein*. Historically, *eiswein* (or icewine) was made only in very small quantities when weather permitted and was cherished with family and friends. Everything changed when Canada came onto the icewine scene in the late 1980s. With its reliably cold winters, icewine could be produced every year and production increased with demand. Now, Canada produces millions of cases of this sweet specialty for local and international consumption every year. In 2007, 1.17 million litres of icewine were produced in Ontario (VQAO, 2008).

With so much icewine being produced, there is now the need to understand all aspects of icewine production from a viticultural, oenological and sensory perspective to ensure a consistent, high-quality wine. This chapter will outline what icewine is, how it is produced and what regulates its production. It will include a section on viticulture, cultural practices, cultivar selection and harvest parameters to maintain quality. An overview of oenological considerations such as yeast strain selection, methods of inoculation, fermentation kinetics and chemical variables important to quality will be discussed. Research initiatives in these areas will also be highlighted. The impact of yeast strain, harvest date and geographic origin on the sensory profile of icewine will be mentioned. Finally, there will be a brief section on authentication of icewine and what the future trends might be in icewine production and research.

18.2 Definitions of icewine

Icewine is a sweet late harvest dessert wine produced from pressing grapes that have naturally frozen on the vine, leaving water behind as ice crystals. Frozen grapes are harvested and pressed at -7°C (EU regulations) to -8°C (Canada) typically between December and January. The resultant must is concentrated in sugar, acids and aroma and flavour compounds, giving the wine its rich, balanced and intense flavour profile. Icewine is produced in many countries around the world at the northern limits of grapegrowing where winter conditions are cold enough to allow the grapes to freeze on the vine before harvest. Canada is the world's largest producer of icewine, but other countries including Germany, Austria, the USA, Slovenia, Luxembourg, Croatia, the Czech Republic and Hungary produce icewine as well (Schreiner, 2001). Icewine is reportedly even produced in small quantities from Vidal blanc in Sweden close to 60°N latitude (Dr A Reynolds, personal communication 2009). Icewine is made in all wine production regions of Canada spanning from Nova Scotia on the east coast to British Columbia on the west coast. The bulk of icewine production is in Ontario, principally the Niagara Peninsula, where warm summers and cold winters allow for optimal conditions to grow and harvest grapes for icewine.

Countries that produce icewine use different names to describe it, but all refer to the same product. In North American it is *icewine*, in Germany and Austria it is *eiswein*; Slovenia refers to it as *ledenno vino*, the Czech Republic as *ledové vino*; and it is known as *jégbor* in Hungary (Schreiner, 2001). Regardless of the country of origin, there are strict regulations to adhere to in order to label a wine icewine. The International Organization of Vine and Wine (OIV) signed an agreement with Germany, Austria and Canada on 23 June 2000 which established international regulations for icewine production (OIV, 2003). A further amendment in 2003 by the OIV gave a definition of icewine and regulations that must be followed for its production. This agreement states that icewine (*eiswein* or *vin de glace*) can only be made from grapes frozen in the vineyard at temperatures lower than or equal to -7°C . The frozen grapes must be harvested and pressed at these

Table 18.1 Regulations for icewine production

Regulatory organization	Maximum temperature at harvest (°C)	Minimum soluble solids (°Brix)	Minimum residual sugar (g/L)	Minimum titratable acidity (g/L)	Maximum volatile acidity (g/L)	Ethanol (% v/v)
OIV	−7	25.3	–	–	2.1	Min. 5.5
Germany and Austria	−7	29.58	–	–	2.1	Min. 5.5
VQA-O	−8	35.0	125	6.5	2.1	7–14.9
VQA-BC	−8	35.0	100	6.5	2.1	7–14.9

OIV, International Organization of Vine and Wine; VQA, Vintners Quality Alliance of Ontario (O) and British Columbia (BC).

same temperatures and contain a minimum of 25.3 °Brix (110 °Oechsle). The minimum alcoholic strength of the finished wines must be 5.5% by volume and the maximum limit of volatile acidity is 2.1 g/L expressed as acetic acid. It also stipulates that all grapes for icewine come from the same region. With the exception of the Mosel in Germany which follows these regulations, Germany and Austria require that the minimum °Brix of the must for icewines be 29.6 (125 °Oechsle in Germany or 27.0 Klosterneuburger in Austria) which is the same requirement as *Beerenauslese* (a sweet late harvest, Botrytis-affected dessert wine).

Canada has its own, even more stringent, regulations set forth by the Vintners Quality Alliance (VQA) Act of Ontario and similar rules in British Columbia (Ontario, 1999; British Columbia, 2005). In the VQA Act, the term icewine is trademark protected and can only be used for wines made from grapes naturally frozen in the vineyard in specified viticultural areas, at temperatures ≤ −8 °C after the 15 November of the vintage year. The grapes must be harvested and pressed at these same temperatures. The minimum allowed soluble solids concentration of the juice must be 35 °Brix with no single pressing being less than 32 °Brix. The finished wine is required to have a residual sugar concentration of not less than 125 g/L (Ontario) or 100 g/L (British Columbia) and a titratable acidity of not less than 6.5 g/L (expressed as tartaric acid). Other styles of dessert wines can be made from late harvest grapes with must concentrations less than 35 °Brix; late harvest, select late harvest and special select late harvest have minimum soluble solids concentrations of 22, 26, 30 °Brix, respectively. Table 18.1 summarizes the Canadian and European regulations for icewine production.

It is believed, anecdotally, that the freeze and thaw cycles to which grapes destined for icewine are subjected give the wine its unique flavour profile and cannot be duplicated by freezing alone. The production of icewine is a very expensive, labour-intensive and risky undertaking. Grapes are left hanging on the vine long past commercial harvest, which results in the loss of yield from dehydration, rot and wind, as well as animal predation. The grapes destined for icewine must be netted to protect them from predators and to prevent loss from clusters falling on the ground. Since the water is left behind as ice crystals, the yield from icewine grapes is only 15–20% that of table wine, requiring substantially

more acreage for the same yield (Pickering, 2006). Icewine must is typically sold by the litre with one litre of Vidal icewine juice selling for \$15.76 CDN and one litre of *Vitis vinifera* icewine juice selling for \$24.16 CDN at 35 °Brix (GGO, 2008). There is also always the risk that the grapes will rot before the temperatures are cold enough for an icewine harvest or that a mild winter will prevent icewine all together.

Due to the reasons stated above, there are winegrowers around the world that do not have the climate to be able to freeze grapes in the vineyard before harvest or do not want to take the risk; however, they still want to make an icewine-style wine. Before the OIV regulations were in place, wine producers had free rein to make dessert wines from cryoextracted grapes and call it icewine, often at a fraction of the price. Now that there are regulations in place to ensure that the concentration in the must from which icewines are produced occurs naturally and not from artificially freezing (cryoextraction) the grapes, a quality standard has been set. This standard ensures that a bottle labelled icewine is in fact the result of naturally frozen grapes being pressed and fermented into a unique, world renowned dessert wine.

18.3 Viticulture

There are major viticultural considerations necessary for producing high-quality icewine. These include cultivar selection, cultural practices including, site selection, disease management and predation.

Although icewine has been made in Europe since the 18th century, it was made only in very small quantities and often by chance. Therefore, virtually no research has been published on the subject. Despite the fact that icewine has become an internationally recognized product, there is very little published literature on the subject and virtually nothing on viticultural practices. Most information is anecdotal and based on what has worked in the past rather than developing practices unique to improving the quality of icewine. Current research at the Cool Climate Oenology and Viticulture Institute at Brock University, St Catharines, ON, Canada is for the first time (to the author's knowledge) looking at the effects of viticultural practices on icewine quality, specifically the effect of harvest date and crop level on icewine chemical and sensory properties.

18.3.1 Cultivar selection

Icewines have been made from a wide range of cultivars, dependent predominantly on region and personal preference. Riesling and Vidal blanc are the most common choices for icewine production in North America, but other cultivars include Cabernet Franc, Gewürztraminer, Sauvignon blanc, Chenin blanc, Chardonnay, Pinot blanc, Fürmint, Ehrenfelser and Kerner. The choice of cultivar for icewine production is important as it will affect its quality. Characteristics of a good icewine cultivar are that it has thick skin, is late maturing, possesses a high natural acidity and is winter hardy.

First, it is important to have a thick-skinned cultivar that can withstand rot, wind, rain and freeze–thaw events. A grape berry destined for icewine goes through a series of freeze and thaw cycles that impose physical and mechanical stress on its structure. Physical stress results from the perforation of the berry's cellular components, such as cell walls and vacuoles, by the jagged edges of ice crystals formed inside the berry when subjected to freezing temperatures. As the ambient temperature warms up, the ice crystals melt causing cellular components previously segregated to mix. The result is the integrity of the berry is compromised which can lead to enzymatic activity and changes in its chemical and sensory profile. Mechanical stress is exerted on the skin of the berry throughout and after the growing season in late harvested grapes, the result of water movement and expansion from heavy rains and ice formation. If the berry skin cannot support the expansion due to the influx of water or freezing, it splits, releasing its contents, and it becomes susceptible to oxidation and infection by a host of pathogens. Therefore, thick-skinned cultivars are essential to maintain berry integrity enabling the juice to be released at pressing and not before.

Second, late maturing cultivars are advantageous as they have a shorter hang time from regular harvest to icewine harvest in December to January. The longer the hang time of grapes destined for icewine after they mature, the greater the potential for loss of yield due to dropped fruit, rot, desiccation and predation. A cultivar with a long growing season, such as Riesling, which is typically harvested mid- to late October, would in general hold up better and yield more juice than an early maturing cultivar such as Chardonnay, which is harvested mid-September. The strength of the actual cluster will also affect the ability of the cultivar to hang until icewine harvest. Having a strong peduncle and rachis will better enable the cluster to hang on the vine, preventing loss of yield from dropped fruit during strong winds, and rain or snowstorms.

Third, the cultivar should possess a high natural acidity to balance the residual sweetness in the wine. The sensory profile of icewine often describes the acid–sugar balance as being an important determinant of the wine's quality. Without the acid backbone, icewine can be perceived as too sweet, unbalanced or flabby and not age-worthy. As grapes mature, the sugar concentration increases and the acid concentration decreases; however, certain cultivars such as Riesling, Chenin blanc and Sauvignon blanc retain a high natural acidity at full maturity. In Canada, acidification is permitted in icewine up to 4.0 g/L, enabling icewine to be made from low-acid cultivars like Gewürztraminer. This is not the case in Europe where acidification is not permitted in any quality wine; therefore, the natural acidity of the cultivar is even more important.

Fourth, the winter hardiness of the cultivar is crucial to ensure continued success. Due to the cold temperatures to which grape vines are subjected in order to make icewine, it is important that winter hardy cultivars are selected and planted; therefore, most icewine cultivars are of western European origin (*Proles Occidentalis*). Tempranillo or Zinfandel may have the potential to make wonderful icewines but, if the vine cannot survive the sub-zero temperatures of the winter, then there is not much purpose to make it an icewine from a viticulture or business perspective.



Fig. 18.1 Vidal icewine cluster on the vine in January.

German and Austrian *eiswein* is usually produced from a blend of several cultivars at the discretion of the winemaker and does not have to be listed on the bottle. It is believed that the blend of different flavour profiles create the best end product (Schreiner, 2001). In contrast, Ontario icewines are usually made from a single cultivar. In Ontario, although the majority of icewines are made from Vidal blanc and Riesling, there are many other cultivars from which icewine can be produced as mandated by the VQA of Ontario (Ontario, 1999). British Columbia produces icewine from Riesling, Pinot blanc, Ehrenfelser and Kerner. Vidal blanc is not a heavily planted cultivar in this region. In British Columbia, it is permitted to make icewine from a blend of permitted cultivars; however, the majority of its icewine is from a single cultivar. Ontario accounts for 75% of icewine production in Canada, and Vidal blanc and Riesling are its two predominant cultivars. In Ontario, Vidal blanc is the most planted hybrid producing 8994 tonnes of fruit in 2007 (GGO, 2008); it is the workhorse of the Canadian icewine industry with 75% of Canadian icewine produced from this cultivar.

Vidal blanc is a white French–American hybrid consisting of 75% *V. vinifera* genetic background, from a cross between Ugni blanc and Rayon d’Or (Seibel 4986) (Galet, 1998). A key viticultural feature is its winter hardiness. The cultivar has large cylindrical clusters, shown in Fig. 18.1, with medium-sized, thick-skinned berries that are disease resistant (Galet, 1998). It is a high-acid cultivar prone to overcropping, which enables it to produce large yields for icewine production. Cluster thinning is essential for table wine production. Vidal blanc is

a late maturing cultivar harvested usually in mid- to late October in eastern North America (Chisholm *et al.*, 1994). The propensity of Vidal blanc to produce well-balanced, concentrated icewines has pushed it to international acclaim and recognition.

Riesling is considered by many to be the best choice of cultivar for producing icewines of the highest quality and ageability due to its high natural acidity. It is the noble grape of Germany, known for producing a wide range of wine styles from bone dry to ultra sweet, both clean and *Botrytis*-affected. Riesling has all the characteristics of the ideal icewine grape; it is late maturing, high-acid and thick-skinned providing some disease resistant and winter hardiness (for *V. vinifera*).

18.3.2 Viticultural practices

Site selection

While it may seem obvious, in order to consistently produce icewine, the vineyard must be located in a site that will reach temperatures below -8°C after the traditional harvest period. The site should also be suitable for growing *V. vinifera* grape cultivars or French–American hybrids such as Vidal blanc. Most soil types are suitable for icewine production, but sites that are susceptible to water stress will not be as productive due to restrictions in yield.

Disease management

In premium winegrape production, control of disease in the vineyard is essential; icewine is no exception. It is commonly known that the quality of the fruit affects the quality of the wine. The aim of most viticulturists is to keep disease pressure low throughout the growing season in order to produce disease-free fruit. This is especially important for late harvest wines that have long hang time and more exposure to climatic elements. Top-quality icewines are produced from ‘clean’ fruit that is not infected with *Botrytis* (grey rot, not noble rot) or other pathogens.

Grapes destined for icewine require even more diligence in the vineyard than most table wine grapes with respect to a fungicide spray control regime. Since the grapes will remain on the vine for the entire duration of the infectious periods for many grapevine diseases, there usually needs to be extra applications of fungicides to reduce disease incidence late in the season, especially for *Botrytis* control. This is particularly true with the *V. vinifera* cultivars in Ontario where pressure from disease can be extremely high in many years, particularly in the cases of powdery mildew (*Erysiphe necator* (formerly *Uncinula necator*)) and downy mildew (*Plasmopara viticola*) as well as *Botrytis cinerea* and other bunch rots (i.e. sour rot complex). These diseases can, of course, have a deleterious impact on the quality and quantity of icewine grapes similar to table winegrape production. Disease can result in reduction in yields, desiccation, poor fruit composition (i.e. lower sugars, higher acids) and high volatile acidity. Occurrence of disease and rot can also result in a diminution of fresh fruit character that is typical in icewines. Fermentations can be very sluggish with highly diseased fruit (Fleet, 2003).

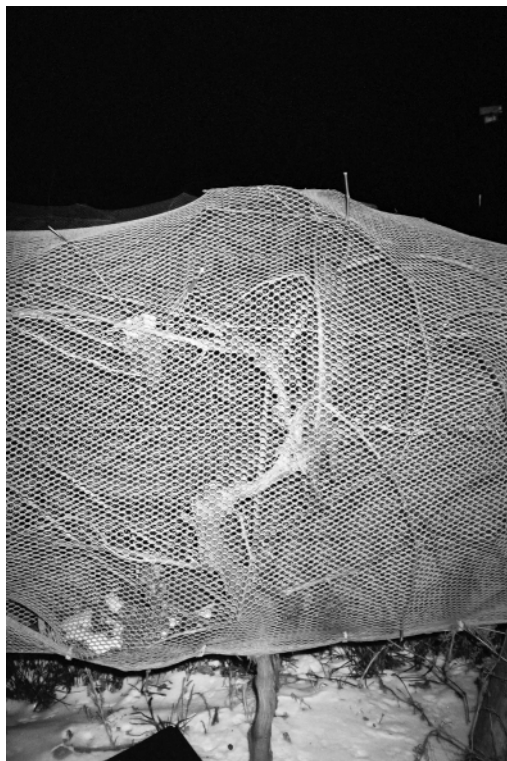


Fig. 18.2 Netting is placed over the rows to protect the grapes from predation.

Bird predation

Birds are a tremendous threat to success in icewine production. The first commercial icewine harvest in Ontario was in 1984; however, the first attempt to make icewine was in 1983. Unfortunately, the birds beat the winemakers to the grapes and the entire crop was lost! Once grapes are near maturity, birds seek out fruit *en masse*. The species that is the most problematic is the European starling (*Sturnus vulgaris*) in which flocks of hundreds or even thousands can devour an entire vineyard of grapes destined for icewine in a short period of time. As the cold winter season progresses, bird predation increases dramatically as birds are willing to make near-suicidal attempts to feed on grapes, once most of their other food supplies diminish. The fact that the grapes are tremendously sweet also increases their desire to eat them. The most common practice to eliminate bird predation in vineyards destined for icewine is through the use of netting which can result in nearly 100% protection. The netting is wrapped around the canopy and fruiting zone so that birds cannot gain access above or below the fruit (Fig. 18.2). The netting also serves another purpose of catching any fruit that falls off the vine during the pre-harvest period and trapping it in the net. The netting can be thought of as a 'sausage' filled with frozen clusters of grapes. By mid- to late winter, most

of the grapes are not hanging on the vine but are actually caught in the net. Netting is associated with high capital costs as well as labour costs for installation and removal. Even though the cost of netting materials has decreased, these costs do play a major factor in icewine production.

In addition to netting, other methods of bird control are used for increased protection. These include visual repellents (hawk kites, balloons, streamers) and acoustical repellents such as propane-powered cannons ('bird bangers'), and electric sound devices ('squawkers'). The propane cannons fire loud, unexpected blasts that cause birds to flee. The electric sound devices include random noises irritating to birds as well as reproduced distress calls. Irritating noises can make the birds feel uncomfortable or insecure or render communication between each other impossible leading them to flee the area. The recorded distress calls mimic certain bird species that also discourage birds from gathering or nesting nearby. Timers can be set on all of these devices so that they can be active during the feeding periods but not an annoyance to humans living close to vineyards during the night.

18.4 Harvest considerations

Harvest considerations are crucial to obtain a juice concentrated in sugar, acids and flavour components. Yield calculations for icewine are based on October estimates for regular harvest (Ziraldo and Kaiser, 2007) because it becomes more difficult to accurately estimate crop late in the season as the individual berries are at different stages of desiccation due to climatic conditions. Yield estimations of 150 L/tonne, or approximately 7 tonnes/acre for Vidal blanc, and 125 L/tonne or 5 tonnes/acre for Riesling are used by one of Ontario's largest icewine producers as a guideline for maximum yield and quality (Ziraldo and Kaiser, 2007). However, anecdotally it is not uncommon for growers to crop vines up to 10 tonnes/acre to increase the volume of icewine juice they can sell to the wineries, especially for Vidal blanc.

Perhaps the most important harvest decision is choosing when to harvest the grapes. The single most important variable that determines the °Brix value of the resultant icewine juice is harvest temperature. Icewine is one of the only wine styles for which the soluble solids concentration at maturity is irrelevant, because it is the temperature that is critical to freezing the water inside the berry and concentrating all its components. As 'hang time' increases for late harvest wines, desiccation of the fruit causes the berry to shrivel concentrating the acid and sugars in the berry. As a result, colder temperatures are required in order to freeze the berry to achieve the desired soluble solids concentration in the must later in the season. Higher yields of icewine juice are usually obtained in December because there is less desiccation of the fruit; therefore, the air temperature does not have to be as cold to achieve the desired °Brix value. However, it is believed that the freeze and thaw cycles are critical to developing the sensory profile for which icewines are known. Some producers will have several icewine harvests from mid-December to late January to achieve a balance between flavour profile and yield. It is for this reason that icewines

are rarely harvested during the first cold snap of winter in late November or early December.

The optimal harvest and pressing temperature range for icewine is -9°C to -11°C giving a soluble solids concentration in the must between 38 and 40 °Brix, titratable acidity between 10 and 12 g/L and a pH between 3.1 and 3.3 (Zirald and Kaiser, 2007). If temperatures are higher, the water is not sufficiently frozen in the berry, in addition to the legal issues pertaining to harvesting and pressing at -8°C or colder. If temperatures are considerably lower, the berry is too frozen, increasing the potential °Brix values and pressing time but decreasing the yield and profitability.

It is important to remember that the concentration of the pressed icewine juice is independent of the soluble solids concentration in the must ('must density'). Instead, it is primarily dependent on the temperature at pressing. An equation can be used to determine the effect of harvest temperature on juice soluble solids (Würdig *et al.*, 1975):

$$\text{specific gravity of must (at } 20^{\circ}\text{C)} = [21 + (17 \times \Delta T) \times 10^{-3}] + 1 \quad [18.1]$$

$$\text{where } \Delta T = 0.0^{\circ}\text{C} - (\text{pressing temperature } ^{\circ}\text{C}) \quad [18.2]$$

Based on this equation, pressing temperatures of -8°C and -14°C would result in icewine musts with 35 and 55 °Brix, respectively, by converting specific gravity to °Brix. Obviously, there will be higher must yield at -8°C than at -14°C due to the frozen water remaining in the berries. Due to the loss in yield when pressing at colder temperatures ($<-12^{\circ}\text{C}$), and the problems associated with fermenting a must with such a high soluble solids concentration, it is optimal to press icewine grapes at temperatures that achieve must-soluble solids concentrations between 38 and 40 °Brix.

18.4.1 Hand harvest

Whether to pick by hand or machine harvest is another important decision. Traditionally, icewine was hand picked during the night once temperatures reached at least -8°C or lower. Vineyard owners would call upon friends and family, and the crew of people dressed for the cold conditions would begin opening the nets and place picking bins under the vines and collect the frozen fruit. Normally, shears are used to harvest wine grapes, but this is not the case with icewine harvests. Since the grapes and vines are frozen, icewine grapes can be easily plucked from the vines so harvesters can readily pull them off the vines with their hands. The harvest is labour-intensive, mainly due to the fact that the nets are tedious to work with. Also, harvesting crews may have to trek through 30 cm or more of snow in the vineyard which can be difficult. Figure 18.3 shows picking bins of hand-harvested Vidal blanc icewine grapes ready to be processed. With increased production throughout Ontario, the cost of hand picking became excessive as the availability and reliability of labour diminished. Vineyards were fighting over picking crews and had exhausted the goodwill of family and friends to pick during the frigid nights. Some of these difficulties were overcome through the use of mechanization.



Fig. 18.3 Hand-harvested Vidal icewine grapes.

18.4.2 Mechanical harvest

Contrary to what many people think, icewine grapes can be mechanically harvested. Advances in mechanical harvesting technology have enabled grape harvesters to increase the efficiency of the harvest while having no impact on the quality of the grapes or the grapevines. In fact, machine harvesting results in cleaner fruit with a reduced amount of leaves and stems due to the harvester's blowers (Ziraldó and Kaiser, 2007). Mechanical harvesting is particularly useful when large amounts of icewine grapes need to be picked within a short harvest window and there is a shortage of harvest crews. One of the major concerns about mechanical harvesting is damage to the frozen shoots of the vines, which can be quite brittle, by the shakers in the picking head of the harvester. Luckily, modern grape harvesters can be extremely gentle on the dormant vines.

There are various ways in which winegrowers use the harvesters for icewine grapes. Some growers will manually open the nets with a crew of people as the harvester moves through the vineyard. The harvester moves down the row, but the shaker is not used. The harvester's role is basically to collect the fruit that the crew is putting on the conveyers in the harvest tunnel of the grape harvester. Other growers will run the harvester over the vine rows with the nets in place. Below -10°C , the frozen berries, like hard marbles, are literally shaken out of the nets into the harvester. Some of the disadvantages to this method of harvest are that partially frozen berries will get caught in the nets and there is some loss of fruit due to grapes falling out the nets prior to the harvester reaching them. In either case, the nets are opened by hand just prior to the harvester going down the rows.

18.4.3 Pressing

Pressing of icewine grapes is often done in the vineyard and the juice sold to the winery. This minimizes the time between harvest and pressing which is crucial when considering the strict temperature guidelines. If grapes have to be transported, the temperature could rise above the required -8°C and pressing would not be allowed or feasible to achieve desired $^{\circ}\text{Brix}$ value. The two most important features of a good press for icewine grapes are: (i) high pressure to quickly separate the juice from the ice within the berries; and (ii) produce as clean a juice as possible (Pickering, 2006). The most-used press for icewine (in Canada) is the hydraulic basket press, although traditional basket press and membrane presses are also employed. The benefit of the hydraulic basket press is the high pressure it can reach which efficiently squeezes the juice out the berry, in order to maximize yield and minimize time.

18.5 Oenology

This section will give an overview of research initiatives aiming to develop a better understanding of the fermentation of icewine. It will cover fermentation kinetics, yeast strain selection and inoculation techniques.

18.5.1 Fermentation kinetics

The high sugar concentration of icewine must puts commercial yeast strains under extreme hyperosmotic stress leading to metabolic changes that affect cell growth, fermentation ability and sensory profile. Hyperosmotic stress causes cell shrinkage, reduced peak cell concentration and biomass and elevated levels of glycerol and volatile acidity in icewines (Pitkin *et al.*, 2002; Nurgel *et al.*, 2004b). It has been shown that when yeast cells are under hyperosmotic stress they activate the high osmolarity glycerol (HOG) response causing the cell to produce glycerol to balance the high external osmotic pressure. The HOG response is essential to yeast survival in a high-sugar environment and is dependent on the expression of genes for glycerol synthesis such as glycerol-3-phosphate dehydrogenase (*GPD1*). In order for the yeast to continue to produce glycerol, they must generate the cofactor NADH. It is thought that yeast generate high concentrations of acetic acid to restore the redox balance of the cell through cytosolic NAD^{+} -dependent aldehyde dehydrogenases (*ALD*) that reduce NAD^{+} to NADH while oxidizing acetaldehyde to acetic acid. There are three cytosolic aldehyde dehydrogenases known in *Saccharomyces cerevisiae*; two are NAD^{+} -dependent (*ALD2* and *ALD3*) and one is NADP^{+} -dependent (*ALD6*). *ALD3* has previously been shown to be induced under saline hyperosmotic stress conditions in *S. cerevisiae* (Norbeck and Blomberg, 2000).

Aldehyde dehydrogenases (encoded by *ALD2*, *ALD3*, *ALD4*, *ALD5*, *ALD6*) and glycerol-3-phosphates (encoded by *GPD1* and *GPD2*) were studied to determine their role in the production of acetic acid and glycerol in icewine fermentations

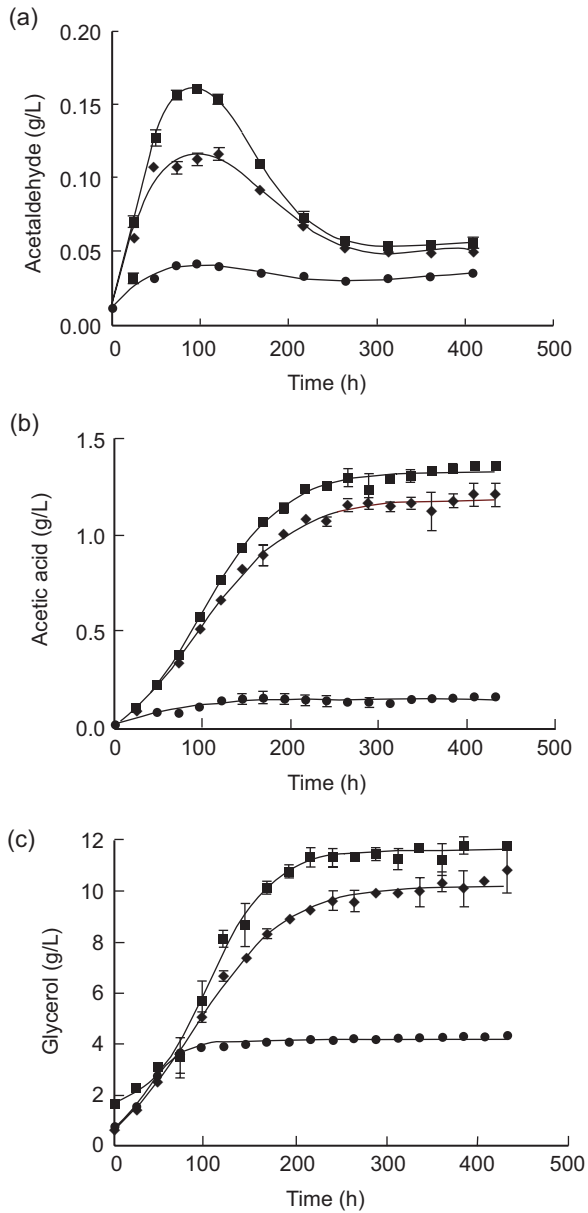


Fig. 18.4 Yeast metabolite production of icewine juice (■), chaptalized diluted icewine juice (♦), and diluted icewine juice (●) inoculated with K1-V1116 yeast. The production of acetaldehyde (a), acetic acid (b), and glycerol (c) were measured daily in duplicate throughout fermentation (Pigeau and Inglis, 2007).

(Pigeau and Inglis, 2005, 2007). Three fermentation conditions were investigated; icewine juice (38.8 °Brix), diluted icewine juice (21.3 °Brix) and chaptalized diluted icewine juice (35.6 °Brix). Figure 18.4 shows the yeast metabolite production of acetaldehyde, acetic acid, glycerol under the three fermentations conditions investigated. Only the expression of *ALD3* varied among fermentation conditions, its expression was 6.2-fold higher in the icewine juice (38.8 °Brix) compared with the diluted icewine juice (21.3 °Brix). Peak expression of *ALD3* occurred in the undiluted and chaptalized icewine juice as the cells reached stationary phase and lowest expression coincided with acetic acid concentration levelling off. In the diluted icewine juice, the expression of *ALD3* remained low throughout the fermentation, indicating that it is the aldehyde dehydrogenase responsible for acetic acid production of yeast cells under hyperosmotic stress. *GPD1* expression also varied among fermentation conditions; its expression was 2.6- and 2.1-fold higher in the icewine fermentation and chaptalized dilute icewine fermentation, respectively, when compared to the diluted icewine fermentation. *GPD2* expression was not upregulated in the icewine fermentations and therefore it can be concluded that it is the glycerol-3-phosphate encoded by *GPD1* that is upregulated to produce higher glycerol concentrations in icewine fermentations to balance the external osmotic pressure of the cell.

Understanding the mechanism by which acetic acid is produced in icewine is important from a quality standpoint due to the relatively low sensory threshold of this compound. There are legal restrictions for the maximum allowed concentrations of volatile acidity (mainly acetic acid) in table wine (1.3 g/L) and icewines (2.1 g/L). The legal limit for icewines was increased to 2.1 g/L after it was understood that the higher concentration of acetic acid in these wines was a yeast by-product from being placed in a hyperosmotic environment and not from other sources such as bacterial contamination from *Acetobacter*. However, elevated concentrations above the legal requirements are not recommended from both the sensory and wine spoilage perspectives.

High glycerol concentrations are desired in icewines, and they contribute to the viscosity of the wine if they are >25.8 g/L (Noble and Bursick, 1984). Knowing the balance between glycerol and volatile acidity production is an important quality parameter in icewine fermentation. Yeast growth has been found to be negatively correlated with icewine juice soluble solids concentration, suggesting that metabolic resources are being diverted towards combating osmotic stress. This is supported by the positive correlation of glycerol and acetic acid production with the soluble solids concentration of the icewine juice (Pigeau *et al.*, 2007). The over-production of glycerol during an icewine fermentation shifts sugar metabolism away from ethanol production and towards acetic acid production leading to elevated concentrations of volatile acidity and lower than desired ethanol concentration in the finished wine (Pigeau *et al.*, 2007).

The initial soluble solids concentration of the icewine must is a quality determinant; a must at 35 °Brix will place the yeast under less osmotic stress than a must at 42 °Brix, thereby resulting in less accumulation of acetic acid and glycerol in the former than the latter. Pigeau *et al.* (2007) found that increasing the soluble solids

Table 18.2 The chemical composition of Riesling icewine juice prior to fermentation at four different initial sugar concentrations (°Brix)

Chemical variable	40 °Brix	42 °Brix	44 °Brix	46 °Brix
Titrateable acidity (g/L)	9.42 ± 0.14	9.73 ± 0.04	10.23 ± 0.03	10.75 ± 0.35
Acetic acid (g/L)	0.11 ± 0.02	0.15 ± 0.01	0.16 ± 0.01	0.21 ± 0.01
Glycerol (g/L)	4.85 ± 0.05	5.40 ± 0.04	5.80 ± 0.02	5.90 ± 0.27
pH	3.15	3.15	3.14	3.15
Glucose + fructose (g/L)	447 ± 1	467 ± 1	487 ± 5	508 ± 2

Source: Pigeau *et al.*, 2007.

Table 18.3 The chemical composition of Riesling icewine juice after fermentation at four different initial sugar concentrations (°Brix)

Chemical variable	40 °Brix	42 °Brix	44 °Brix	46 °Brix
Titrateable acidity (TA; g/L)	11.8 ± 0.1	12.4 ± 0.5	13.2 ± 0.4	13.1 ± 0.3
Acetic acid (g/L)	1.79 ± 0.10	1.99 ± 0.07	2.09 ± 0.06	2.11 ± 0.08
TA represented by acetic acid (%)	19	20	20	20
Glycerol (g/L)	15.95 ± 0.76	15.49 ± 0.81	16.50 ± 0.77	14.97 ± 0.13
Residual glucose + fructose (g/L)	235.1 ± 2.2	286.8 ± 8.1	345.1 ± 0.9	396.9 ± 6.4
Ethanol (% v/v)	11.9	10.0	7.7	6.3

Source: Pigeau *et al.*, 2007.

concentration of icewine must from 40 to 46 °Brix decreased yeast growth, sugar consumption rate, the total amount of sugar consumed and the total concentration of ethanol produced. The chemical composition of the Riesling icewine juices and finished wines at the four different soluble solids concentrations can be seen in Tables 18.2 and 18.3, respectively. They further concluded that, theoretically, Riesling icewine juice >52.5 °Brix would not ferment and that 42 °Brix would be the maximum soluble solids concentration to produce an icewine with 10 % (v/v) alcohol.

18.5.2 Yeast strain selection

The major determinant of yeast strain selection for icewine is the ability of the yeast to ferment in a high-sugar environment. For this reason, commercial yeast strains are most commonly used for icewine fermentation. Yeast undergo hyperosmotic stress because of the high sugar environment of the icewine must. This can often lead to sluggish or stuck fermentations with less than the desired alcohol of 10% (v/v). Although indigenous yeast populations such as species of *Candida*, *Rhodotorula*, *Cryptococcus*, *Hanseniaspora* and *Saccharomyces* may be present on the icewine grape bloom, they may not be sufficient to ferment the juice into icewine (Subden *et al.*, 2003; Nurgel *et al.*, 2004b). Nurgel *et al.* (2004b) identified

the indigenous yeast population from spontaneous Vidal blanc and Riesling icewine fermentations which produced 10.4 and 6.4% alcohol (v/v), respectively. While both fermentations shared many similar indigenous yeast species such as *Kloeckera apiculata*, *Rhodotorula glutinis*, and *Candida pulcherrima*, the major differences in the ability of the yeast to ferment the icewine to 10% alcohol in the Vidal blanc spontaneous fermentation, were due to the presence of *S. cerevisiae*. In the same study, both Vidal blanc and Riesling icewine musts inoculated with commercial *S. cerevisiae* strains were able to ferment to a desired alcohol concentration (10% v/v). It is, therefore, generally advised that most icewines be inoculated with a commercial yeast strain.

Other factors to consider when selecting a yeast strain are alcohol tolerance of the yeast, optimal fermentation temperature, hydrogen sulphide and volatile acidity production, assimilable nitrogen and oxygen requirements. The impact of seven yeast strains on fermentation rate, acetic acid and glycerol production, and sensory characteristics during icewine fermentation in real and synthetic icewine must was studied (Erasmus *et al.*, 2004). The seven yeast strains were Vin13, N96, Vin7, EC1118, K1-V1116, 71B and Zymaflore ST. The N96 fermentation was the first to reach the desired alcohol of 11% v/v after 17 days followed by EC1118, K1-V1116, Vin13 and ST; two strains, Vin7 and 71B, were not able to ferment to the desired alcohol. Acetic acid and glycerol production were both different among yeast strains; ST produced the least and Vin7 the most acetic acid and glycerol. Vin13, N96, EC1118 and K1-V1116 did not differ in acetic acid concentration and Vin13, EC1118 and N96 did not differ in glycerol concentration. Sensory analysis showed EC1118 and N96 were associated with ratings of higher quality versus ST which was associated with sulphur-like aromas. From this study, it was recommended that ST, N96 and EC1118 are the yeast strains most suitable for icewine fermentations. Table 18.4 provides a summary of some recommended yeast strains for icewine fermentations. In Canada, the most common yeast strains for icewine fermentation are K1-V1116 followed by EC1118.

Lalvin ICV K1V-V1116 is a *S. cerevisiae cerevisiae* yeast strain isolated in 1972 at INRA in Montpellier, France. This was the first competitive factor yeast to go into commercial production and has become one of the most popular dehydrated yeast strains. Relevant attributes for icewine production include its ability to start fermentation rapidly, alcohol tolerance to 18%, large fermentation temperature range (10–35 °C) and ability to ferment at low temperatures, low production of hydrogen sulphide and volatile acidity, and low to average requirement of yeast-assimilable nitrogen. The yeast produces high concentration of floral esters such as isoamyl acetate, hexyl acetate and phenyl ethyl acetate that contribute to the aromas of the finished wines. It is known for its ability to ferment under difficult conditions and is recommended for the production of icewine (Anon, 2008).

Lalvin EC1118 (Prise de Mousse) is a *S. cerevisiae bayanus* isolated from the Champagne region of France. It has been used for a wide variety of fermentations including icewines, cider and Champagnes, and has also been widely used to restart stuck fermentations. EC-1118 has a strong competitive factor, and has the ability to ferment at a wide range of temperatures (15–25 °C). It ferments rapidly

Table 18.4 Commercial yeast strains for icewine fermentations

Yeast strain ^a	Sensory effect ^b	Fermentation speed	Competitive factor	Alcohol tolerance (°C)	Nitrogen needs	Temperature range (°C)
ICV-K1 (V1116)	Esters	Fast	Yes	18	Low	10–35
EC1118	Neutral	Fast	Yes	18	Low	10–30
R2	Esters	Moderate	Yes	16	Medium	5–32
DV10	Neutral	Fast	Yes	18	Low	10–35
43	Neutral	Fast	Neutral	18+	Low	13–35
R-HST	Neutral, mouthfeel	Moderate	Yes	15	Medium	10–30
ProDessert® BA11	Esters, mouthfeel	Moderate	Sensitive	16	High	10–25
Zymaflore ST	Enhanced varietal character	Slow	Sensitive	15	High	15.5–20
Zymaflore VL2	Enhanced varietal character, esters, mouthfeel	Moderate	Yes	15.5	Medium	13–20

^aSupplier is Lallemand except for Zymaflore ST and VL2.

^bSource: Scott Laboratories Fermentation Handbook, 2005.

at a wide pH range, has good flocculation, alcohol tolerance to 18% alcohol and low production of both hydrogen sulphide and volatile acidity (Anon, 2008).

Once the yeast strain has been selected, it is also important to consider the amount of yeast to add to the fermentation. Since it is well documented that yeast in an icewine fermentation have smaller cell size, less biomass and a lower peak concentration than in a table wine fermentation, it seems intuitive that they would require a higher initial concentration of inoculum. Kontkanen *et al.* (2004) investigated the effect of inoculum rate on icewine fermentations at two concentrations, 20 and 50 g dry weight/hL of K1-V1116 yeast and the two methods of inoculation, direct and stepwise acclimation of the yeast cells to the high sugar concentration, over a 30-day period. Regardless of inoculation method, the 20 g dry weight/hL fermentations did not achieve the desired 10% alcohol, with only 7.8% and 8.1% alcohol (v/v) produced for the direct and step-wise acclimation methods, respectively. In contrast, at 50 g dry weight/hL, both inoculation methods finished fermentation with 12.1% and 10.5% alcohol (v/v) for the stepwise and direct inoculum, respectively. It is therefore recommended to inoculate icewine with a dose of 50 g dry weight/hL of must using a stepwise acclimation method to condition the yeast cells to the high soluble solids concentration. The stepwise acclimatization resulted in higher cell biomass and viability of the yeast cells allowing it to ferment the icewine must to the desired alcohol concentration.

To acclimatize the yeast according to Kontkanen *et al.* (2004), they are rehydrated in 40 °C water for 15 minutes with gentle swirling every 5 minutes. The

temperature of the starter culture is then decreased to 30 °C. Diluted icewine juice (20 °Brix) is added in equal parts to the starter culture for a soluble solids concentration of the starter culture of approximately 10 °Brix, and placed in a 25 °C water bath for one hour stirring every 30 minutes. Next, an equal portion of room temperature icewine juice is added to the starter culture for a concentration of 20 °Brix, which is returned to the water bath for 2 hours, stirring every 30 minutes. After this 3 hour and 15 minute acclimatization period, the yeast cells are now conditioned to a 20 °Brix environment and are ready for inoculation.

In addition to conditioning the yeast to the high-sugar environment, yeast nutrient addition is a common practice in Canada to ensure that the icewine must has the required micronutrients to complete fermentation. Common additions include diammonium phosphate (DAP) and GO-FERM®. DAP provides yeast with yeast-assimilable nitrogen source in the form of amino acids and ammonia. GO-FERM® (Lallemand) is a commercially available inactive yeast supplement enriched with vitamins (pantothenate and biotin) and minerals (magnesium, zinc and manganese) that is added during yeast rehydration. Kontkanen *et al.* (2004) found that the addition of micronutrients (GO-FERM®) during yeast rehydration reduced the overall fermentation time from 500 to 350 hours, increased yeast cell biomass with cells reaching peak values earlier in the fermentation and reduced the amount of acetic acid produced per set amount of sugar consumed.

18.6 Chemical analysis of icewines

Chemical variables are always monitored in winemaking; they can provide valuable information with respect to final wine quality. Common chemical variables monitored in wine production are soluble solids concentration (°Brix), pH, titratable acidity (TA) nitrogen concentration (g N/L) in juice and residual sugar, ethanol, pH, TA, and sulphur dioxide (free and bound) in wine. In icewine, it is of course also important to monitor volatile acidity and glycerol concentrations because of the hyperosmotic stress placed on the yeast.

Several studies have investigated the chemical variables of icewine in an effort to characterize and set quality limits for this wine style. They have listed the range and median scores of many of the chemical variables without an explanation of how the values actually relate to icewine quality. VQA regulations in Canada and OIV regulations in the EU do regulate acceptance limits for many chemical variables in icewine, such as minimum soluble solids concentration (35 °Brix), ethanol concentration (between 7 and 14.9% alcohol v/v), residual sugar (125 g/L) and TA (6.5 g/L) of the finished wine, volatile acidity (2.1 g/L) and total sulphur dioxide (400 mg/L), perhaps this being considered the quality standard. Chemical variables in icewine vary based on cultivar, vintage and viticultural area (Cliff *et al.*, 2002; Nurgel *et al.*, 2004a; Soleas and Pickering, 2007). The chemical compositions of 345 Vidal icewine juices from the Niagara Peninsula, Ontario are shown in Table 18.5. The most pertinent chemical variables in relation to icewine quality will be discussed in this section.

Table 18.5 Chemical composition of 345 icewine must from the 2003 and 2004 vintages in the Niagara Peninsula, ON, Canada

Cultivar	Soluble solids (°Brix)	Amino acid nitrogen (mg/L)	Ammonia nitrogen (mg/L)	Total yeast-assimilable nitrogen (mg/L)	pH	Titrateable acidity (g/L tartaric acid)
Riesling	38.0 ± 2.1	381.7 ± 132.6	79.7 ± 38.6	461.3 ± 166.5	3.22 ± 0.18	9.0 ± 1.7
Vidal blanc	39.3 ± 1.7	497.7 ± 105.2	57.3 ± 18.7	555.0 ± 120.2	3.38 ± 0.16	10.5 ± 1.5
Cabernet Franc	39.4 ± 2.0	292.9 ± 116.7	22.3 ± 8.5	315.3 ± 123.0	3.79 ± 0.24	5.7 ± 1.2

Source: Ziraldo and Kaiser, 2007 (p.183) data compiled by the research group of D. L. Inglis, CCOVI, Brock University, ON, Canada.

18.6.1 Soluble solids

Much has already been discussed in this chapter regarding soluble solids concentration in icewine must. Although this is one of the major quality variables in icewine, it is also legally regulated that the minimum concentration be 35 °Brix. These limits are set forth by the OIV and VQA. Increasing soluble solids concentration increases the hyperosmotic stress placed on the yeast, which positively correlates to acetic acid and glycerol concentrations, and is negatively correlated to ethanol production. It is therefore recommended for quality icewine to have a must concentration between 35 and 40 °Brix. The average soluble solids concentration in survey of 335 Ontario icewines across five vintages was 39.9 °Brix ± 2.1 (Soleas and Pickering, 2007).

18.6.2 Titrateable acidity

It is important to have a relatively high TA in icewine to balance the residual sugar in order to achieve the sugar–acid balance by which this wine style is characterized. As in all wines, the TA of the must will depend on cultivar; Riesling has a higher TA than Vidal blanc, for example. The mean TA of 14 Riesling icewines and 28 Vidal icewines was 11.8 ± 2.3 g/L and 9.9 ± 1.6 g/L, respectively (Nurgel *et al.*, 2004a). A survey of 297 Vidal icewine musts from the Niagara Peninsula, Ontario had a mean TA of 10.5 ± 1.5 g/L with a 2 g/L increase in the TA of the finished wines (1.27 g/L from acetic acid and 0.7 g/L from succinic acid) (Inglis *et al.*, 2006). There is an increase in the TA of icewine from must to wine; acetic acid production by the yeast accounts for this increase. Pigeau *et al.* (2007) found that 87–100% of the increase in TA could be attributed to acetic acid production by yeast under hyperosmotic stress. The higher the initial soluble solids concentration of the must, the higher the final TA of the finished wine as a result of acetic acid production; the increase in TA for 40, 42, 44 and 46 °Brix must was 87, 90, 80 and 100%, respectively, in the finished wines.

18.6.3 Nitrogen

A study of 297 Vidal blanc musts from the Niagara Peninsula found that ammonia nitrogen ranged from 6–118 mg N/L with a mean of 57 ± 19 mg N/L and amino nitrogen excluding proline ranged from 182–738 mg N/L with a mean of 497 ± 105 mg N/L (Inglis *et al.*, 2006). This is well above the minimum total nitrogen requirements of 140 mg N/L to prevent stuck or sluggish fermentations in table wines (Agenbach, 1977) and the 190 mg N/L suggested as the optimal concentration to limit acetic acid formation in musts with high soluble solids concentrations (Bely *et al.*, 2003). However, nitrogen availability has not been shown to contribute to sluggish fermentations in icewine (Pigeau and Inglis, 2005). After 5 days of fermentation, yeast in icewine juice consumed only 28% of the assimilable amino acids and ammonia as compared to 90% in a dilute icewine juice condition (Martin *et al.*, 2008). Nitrogen concentration is not a limiting factor in icewine fermentations and therefore quality; yeast fermenting icewine require only the minimum amount of assimilable nitrogen to complete fermentation.

18.6.4 Ethanol and residual sugar

It is difficult to discuss ethanol concentration without any mention of residual sugar since they are inversely correlated. The higher the ethanol concentration the less residual sugar will remain in the finished wine. Legal limits dictate the amount of ethanol (7–14.9%) and residual sugar (minimum 125 g/L) permitted in icewine (VQAO, 1999). Soleas and Pickering (2007) found the average alcohol percentage and residual sugar concentration of 348 Canadian icewines to be 10.6% alcohol (v/v) and 207 g/L, respectively. Similar average ethanol and residual sugar concentrations were reported by Cliff *et al.* (2002) with 10.1% alcohol (v/v) and 202.2 g/L respectively, and Nurgel *et al.* (2004a) with 10.2% alcohol (v/v) and 214 g/L, respectively. Due to the high residual sugar concentration of icewines, antimicrobial agents such as sulphur dioxide and sorbate are added prior to bottling. These additives provide protection against microbial instability and secondary fermentation in bottle. The mean sulphur dioxide and sorbic acid concentrations in 348 Canadian icewines were 179 ± 72 mg/L and 140 ± 54.4 mg/L respectively (Soleas and Pickering, 2007). Although none of the above values is linked directly to quality, all wines were from commercial wineries and available for purchase. These three separate studies, representing a range of vintages, cultivars and regions, found a similar trend between ethanol and residual sugar concentrations, so it can be safely assumed that this is perhaps the optimum range based on winemaker and buyer preferences.

18.6.5 Volatile acidity

Volatile acidity comprises the acetic acid and ethyl acetate, two compounds that impart negative sensory characteristics to wine at high concentrations and which can also be an indicator of microbial spoilage. Although acetic acid has been discussed at length, it is important to mention it again because of the legal

regulations of volatile acidity that must be adhered to when making icewine. The maximum permitted concentration of volatile acidity in icewine is 2.1 g/L, which is higher than the 1.3 g/L limit for table wine, due to yeast production of acetic acid as a by-product of fermentation under hyperosmotic stress. In terms of quality, it is important that volatile acidity concentrations remain below the sensory thresholds so that off-aromas and flavours are not perceived by the consumer.

A survey of 348 icewines from a variety of cultivars, vintages and regions found volatile acidity concentrations to range from 0.11–2.35 g/L with a mean of 1.12 g/L, which is well below the 2.1 g/L limit set by VQA and OIV, and below the sensory threshold of acetic acid of 3.2 g/L (Cliff and Pickering, 2006). Similar results were found in surveys of 297 Vidal blanc icewines from Ontario (Inglis *et al.*, 2006) and 51 commercial icewines, of which 40 were from Ontario and 11 from British Columbia (Nurgel *et al.*, 2004a), where the mean acetic acid concentrations were 1.27 ± 0.09 g/L and 1.3 ± 0.5 g/L, respectively. Cliff and Pickering (2006) determined the sensory thresholds of acetic acid to be 3.185 g/L and ethyl acetate to be 0.198 g/L in icewine. The only study to quantify ethyl acetate concentration in icewine found the mean concentration to be 0.24 ± 0.07 g/L (Nurgel *et al.*, 2004a), which is above the sensory threshold. The authors therefore proposed that legal limits on VA be subdivided to consider both acetic acid and ethyl acetate concentration. Since only 51 wines were evaluated for ethyl acetate concentration in the study by Nurgel *et al.* (2004a), further research should be conducted on a larger subset of wines to determine if ethyl acetate widely occurs above its sensory threshold in icewines. Currently, it is acetic acid production and not ethyl acetate that is of concern in icewine fermentations since ethyl acetate is produced from the esterification of ethanol in the presence of acetic acid. Keeping acetic acid levels in check should limit the production of ethyl acetate in icewines. The role of ethyl acetate in icewines still needs to be elucidated.

18.6.6 Glycerol

Glycerol is produced as a stress response to regulate the osmotic imbalance of the yeast cell in high-sugar environments. Anecdotally, glycerol has been thought to contribute to the viscosity of wine. However, the concentration range of glycerol found in table wines is below the 25.8 g/L of glycerol required to elicit a sensory response (Nobel and Bursick, 1984). Glycerol has, however, been shown to contribute to the perceived sweetness of white table wine with a detection threshold of 5.2 g/L (Noble and Bursick, 1984). In icewine, glycerol concentrations have been found above the stated detection threshold of 5.2 g/L, with a mean of 10.8 ± 0.6 g/L found in a survey of 297 Vidal icewines (Inglis *et al.*, 2006) and a mean of 12.4 ± 3.1 g/L in 51 commercial icewines from Ontario and British Columbia (Nurgel *et al.*, 2004a), which are far below the concentrations required to change the perceived viscosity of the wine. In terms of quality, glycerol concentration is of interest because high concentration in icewine may affect the sugar–acid balance by increasing the perceived sweetness of the finished wines.

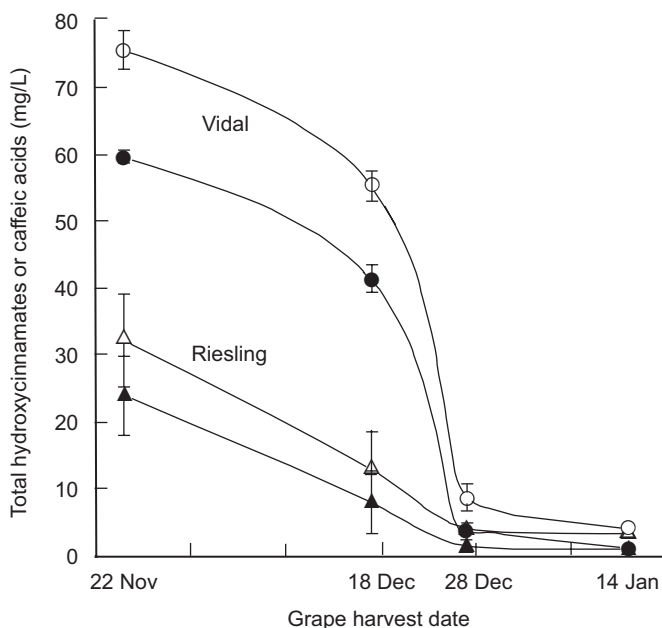


Fig. 18.5 Total hydroxycinnamates (open symbols) and total caffeic acids (filled symbols) in Vidal (circles) and Riesling (triangles) icewine from four different harvest dates (Kilmartin *et al.*, 2007).

18.6.7 Polyphenols

The polyphenol content of white wines relates to the browning potential of the juice or wine when in contact with oxygen. Degree of browning can be measured by spectrophotometer at 420 nm (A_{420}) and provides an estimate of the amount of brown pigment present in the juice or wine (Iland *et al.*, 2004). Vidal blanc icewines have been shown to have higher A_{420} values than Riesling; mean A_{420} values for Vidal blanc and Riesling icewines were 0.596 and 0.365, respectively (Nurgel *et al.*, 2004a) and 0.45 and 0.33, respectively (Cliff *et al.*, 2002). It is hypothesized that the difference is due to the higher pH and phenolic concentration of the Vidal blanc (Nurgel *et al.*, 2004a). Kilmartin *et al.* (2007) found the concentration of total hydroxycinnamates and total caffeic acids to substantially decrease with later harvest date in both Riesling and Vidal blanc icewines (Fig. 18.5). Climatic events and/or hang time of the fruit in the later harvest dates is likely the explanation for the loss of polyphenols. However, very little is known about the actual polyphenol content of icewines and how they affect icewine quality. Further research will be required to understand the role of polyphenols in icewines.

18.6.8 Volatile compounds

The volatile composition of icewine is not well researched. To date, only a handful

of papers have reported information on icewine volatile composition and none has been able to relate the findings to icewine sensory properties or the identification of key compounds that could be important quality parameters. Most papers have focused on differences in icewine volatile composition based on geographic location, vintage and cultivar (Cliff *et al.*, 2002; Giraudel *et al.*, 2007; Setkova *et al.*, 2007).

Cliff *et al.* (2002) compared 34 volatile compounds in Ontario and British Columbia icewines by gas chromatography–mass spectroscopy (GC–MS). They found only eight compounds to be significantly different between regions. British Columbia icewines had higher concentrations of diethyl hydroxybutanediate and ethyl isopentylsuccinate but lower concentration of 2-methyl-1-propanol, 3-methyl-1-butanol, benzeneethanol, ethyl lactate, 1,3-propandiol diacetate, and epoxylinallol than Ontario icewines. However, no single impact compound was found by gas chromatography–olfactometry (GC–O) and it was concluded that icewine aroma was a complex interaction between several volatile compounds.

Another study to look at the volatile composition of icewine developed a method using rapid headspace solid-phase microextraction–gas chromatographic–time-of-flight–mass spectrometry (SPME–GC–TOF–MS) for qualitative profiling of Canadian and Czech icewines (Giraudel *et al.*, 2007; Setkova *et al.*, 2007). They surveyed over 130 different icewines from both countries combined and identified 201 peaks in the chromatograms as potential icewine aroma compounds; a subset of 58 peaks was then evaluated in all samples. Using principal component analysis (PCA), the 58 volatile compounds were able to differentiate icewines based on cultivar, geographic location (Ontario, British Columbia and Czech), vintage and winemaking style (oak versus stainless steel). However, again in this study no key impact compounds were identified for icewines quality or correlated to icewine sensory profiles. The determination of volatile compounds important to icewine quality is an area of research still in need of elucidation.

18.7 Sensory properties of icewine

Icewine is described as a wine style with intense aroma and flavours. However, very little is known regarding what affects the sensory properties of icewines. It is hypothesized that the freeze and thaw cycles icewine grapes endure will affect the sensory properties of the wines. Bowen *et al.* (2008a) found that Vidal blanc icewines from later harvest dates had higher intensity ratings for all significantly different attributes. In Riesling, early harvest dates were associated with fresh fruit descriptors and later harvest dates were associated with higher intensity ratings for dried fruit and nutty attributes.

Principal component analysis of the Vidal blanc and Riesling harvest date wines showed that 70 and 91%, respectively, of the variation in the wines was explained by factors 1 and 2 (Fig. 18.6). All aroma and flavour attributes in Vidal blanc were associated with later harvest dates (H3 and H4) and factor 2 separated early harvest dates (H1 and H2) from late harvest dates (H3 and H4). In Riesling

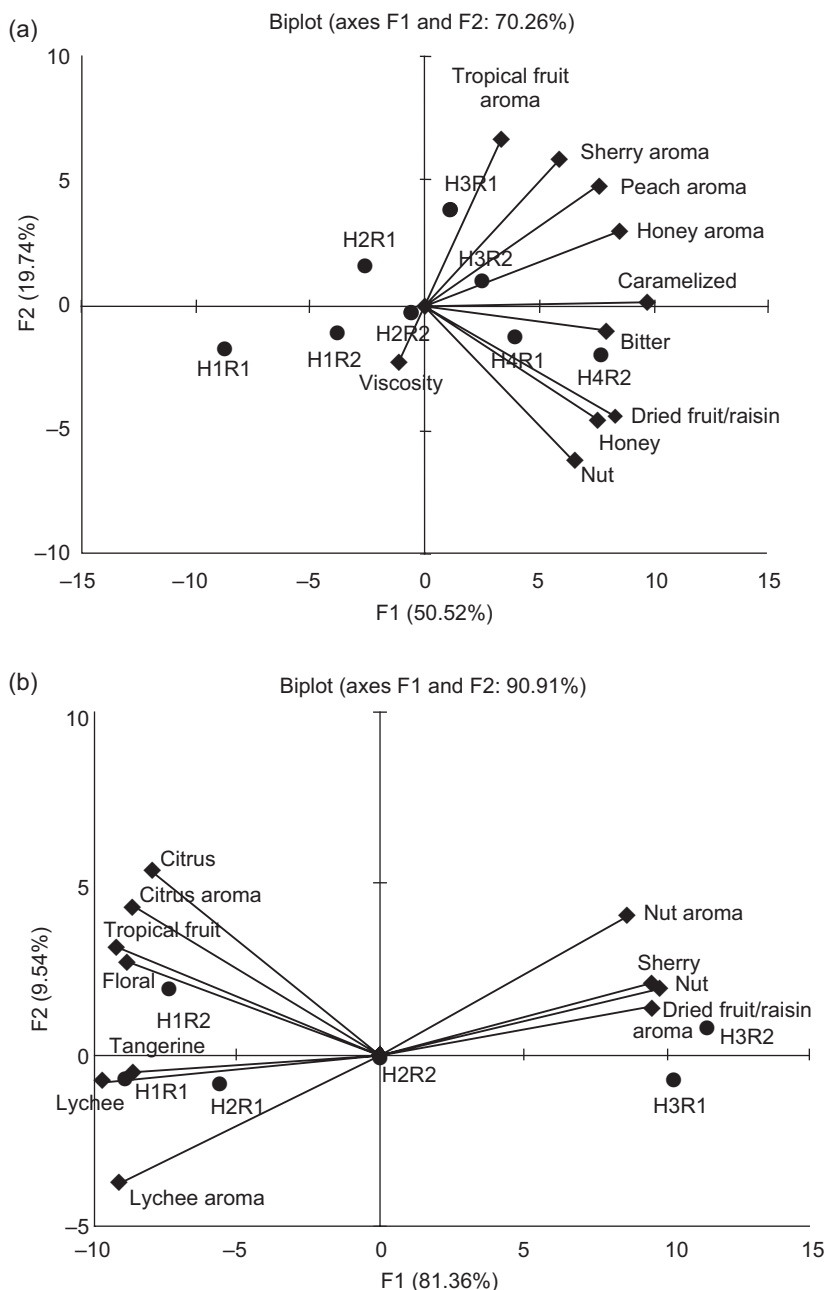


Fig. 18.6 Sensory maps of significantly different aroma and flavour attributes in Vidal (a) and Riesling (b) icewines from four harvest dates 2004–05. Harvest 1 (H1), 19 December 2004; harvest 2 (H2), 29 December 2004; harvest 3 (H3), 18 January 2005; harvest 4 (H4), 11 February 2005 (Vidal only). The R1 and R2 designations refer to replicates (Bowen *et al.*, 2008a).

harvest date wines, all attributes were heavily loaded on factor 1 with fresh fruit and tropical attributes associated with harvest 1 (H1) and dried fruit and nutty attributes associated with harvest 3 (H3); harvest 2 (H2) was not well explained by the PCA. Therefore, harvest date has been shown to affect the sensory profiles of Vidal blanc and Riesling icewines. Chemical analysis (gas chromatography–olfactometry–mass spectrometry; GC–O–MS) is still being completed to correlate to these sensory results to determine what compounds may be responsible for changes in the sensory profiles of these wines. The potential significance of these findings is that they provide information to winemakers about how time of harvest will affect the sensory profile of the resultant icewine. Therefore, depending on the desired sensory profile of icewine, it could dictate whether grapes are picked in December for fresh and tropical fruit aroma and flavours or January for more nut and dried fruit characteristics. Perhaps both profiles are desired in an icewine, then grapes can be picked at different times throughout the icewine harvest season (December to February), fermented in separate tanks and blended before bottling to achieve the desired sensory profile.

Crop level has also been shown to affect the sensory profiles of icewines. Vidal blanc icewines made from three different crop levels – control (unthinned), thin (to basal cluster) at fruit set and thin at *véraison* – were found to differ in their sensory profiles (Bowen *et al.*, 2008b). All significantly different sensory attributes were found to be associated with the thinned treatments in 2003 and 2004 experimental icewines. The thinned treatments were described as having higher intensity ratings for attributes such as peach, apple/pear, dried fruit and caramelized aromas and tropical fruit, tangerine and apple/pear flavours. As with the harvest date wines, previously mentioned, chemical analysis using GC–O–MS is underway to correlate the sensory and analytical results to determine what odour-active compounds are responsible for the changes in the sensory profiles of these wines. The potential significance of these findings would be that cluster thinning icewines grapes intensifies the sensory profile of the resultant wines.

Sensory analysis has also been used to determine the effect of yeast inoculation procedure on icewine aroma and flavour profiles. Kontkanen *et al.* (2005) found Vidal icewines produced by direct inoculation with K1-V1116 without micronutrients were described with raisin, butter and spicy aromas whereas those with micronutrient were described as sweet in taste with honey and orange flavours. Icewines produced by the stepwise acclimatization without micronutrient were described with peach and terpene aromas and with micronutrient were described with pineapple and alcohol aromas and alcohol and honey flavours.

The sensory properties of icewines vary among geographic location and vintage. Nurgel *et al.* (2004a) found Ontario icewines ($n = 11$) to have higher intensity ratings for apricot, raisin, honey and oak aromas compared to British Columbia icewines ($n = 9$). However, the British Columbia wines had higher intensities for pineapple and oxidized aromas than Ontario icewines. Cliff *et al.* (2002) compared the sensory profiles of icewines from Ontario ($n = 9$), British Columbia ($n = 13$) and Germany ($n = 3$). Ontario icewines were associated with the highest fruit and floral aromas and golden copper colour. British Columbia

icewines were rated high for sweetness, body/viscosity and intensity of aftertaste. German icewines were associated with a nutty/oily character and had the highest acidity. Principal component analysis showed separation for the Canadian and German wines but not between Ontario and British Columbia wines.

Unfortunately, none of these studies can conclude anything about the impact of sensory properties on icewine quality. They provide information that the profile of icewines can change as the result of a variety of factors; however, it is unclear whether they have a positive or negative effect on quality. In sensory terms, quality is a hedonic term that can only be assessed by a large number of people ideally in a consumer preference study. The results from such a study could then be correlated to the sensory and chemical results from descriptive analytical analysis of the same wines. Current research at Brock University is attempting to characterize Vidal blanc and Riesling icewine sensory and chemical compositional analysis.

18.8 Authentication

Icewine is a luxury item that can demand a high price. Unfortunately, this also means that counterfeit icewine on the export marketplaces has become inevitable. Taiwan and China are the main countries known for producing counterfeit icewine (Schreiner, 2001). The counterfeiters produce icewine packaging to imitate legitimate labels and bottles of Canadian producers or portray Canadian symbols to mimic the authentic packaging (Pickering, 2006). Among other issues of counterfeit icewine, the quality reputation of real icewine is at stake, since the illegitimate icewines are sweetened table wine at best. Fake icewines can be made via several different methods: (i) cryoextraction, freezing grapes from regular harvest and pressing them frozen; (ii) chaptalization, sugar or juice concentrate addition to raise sugar concentration of the juice; or (iii) by taking sweet wine or alcohol and flavouring it with juice extracts. All of the above techniques are used and then marketed as icewine.

Gas chromatographic methods are currently being used to monitor icewine for the fraudulent versions. The alcohol source of the wine can be determined using gas chromatography–isotope ratio mass spectrometry (GC–IRMS) to find out if the alcohol is from grapes, grain, molasses or synthetic alcohol based on the ratio of carbon-13 (^{13}C) to carbon-12 (^{12}C). This ratio can tell if the plant is a C_4 or C_3 plant; *V. vinifera* is a C_3 plant whereas sugar cane and other *Gramineae* are C_4 plants. GC–IRMS compares the alcohol source in the product being tested against the alcohol source of the reference sample. Ethanol is extracted from the sample and run on a GC where it is combusted into carbon dioxide and the resulting ratio of ^{13}C to ^{12}C is measured and compared to the reference sample. The Liquor Control Board of Ontario (LCBO) analyzes real icewines of several different cultivars within a vintage to determine its isotopic ratio of ^{12}C and ^{13}C . Real icewines from a given year must be within one standard deviation to be sold as VQA icewine; fake icewines usually differ by 3 to 4 standard deviations (Gedeon, 2005). A database of the ^{13}C to ^{12}C ratios of icewines from different vintages and

regions can be compiled to allow for the wine of a given year to be identified as authentic or counterfeit.

However, not everyone has the analytical capability to monitor icewine via GC–IRMS. Kodak has come up with another solution. In 2007, they produced a TRACELESS system that forensically embeds markers into the wine label by silk-screening to ensure the authenticity of the label. They are teaming up with Stanpac to offer the service to Ontario wineries to protect their icewine for the export market. Although this method provides no information on the contents inside the bottle, it is at least a way that wineries can ensure the product for export and sale in foreign markets is their product as the label will have unique markers to identify origin.

Ideally, the hope is to find chemical marker(s) that can be used to distinguish the icewine made from grapes frozen on the vine from those that were not. Preliminary results from Dr Reynolds' research group at Brock University show that there are sensory (Bowen *et al.*, 2008a), volatile compound (Ballester *et al.*, 2007; Bowen *et al.*, 2008b) and polyphenol differences (Kilmartin *et al.*, 2007) in the profile of icewines from later harvest dates in Vidal icewines. Perhaps one or more of these will prove to be key compounds important for icewine authentication. Regardless of the method, the end goal is to produce a technique to distinguish true icewine from its imposters.

18.9 Future trends

The Asian market is the largest export market for icewine and, due to the importance of red colour in that culture, research into red cultivars for icewine is of increasing interest. Currently, in Ontario, red icewines are routinely made from the cultivar Cabernet Franc. In fact, Cabernet Franc is the third most used cultivar after Vidal blanc and Riesling for icewine production. Unfortunately, getting the colour extraction from the juice without also extracting the astringent compounds tends to make red icewines more pink than red. Research designed to improve colour extraction into red icewine while maintaining quality would be beneficial, and profitable, to the industry. Currently nothing is known about the sensory profiles of Cabernet Franc icewines and very little on their chemical composition. It would therefore be useful to characterize this cultivar from a sensory and chemical perspective. Information gathered could be used to maintain consistency and quality red icewines.

The majority of icewine research has focused on Riesling and Vidal blanc and wines produced in Canada. To ensure the quality of this international wine style, it will be necessary to expand research into other cultivars, such as those used in Europe, and also to characterize icewine from regions other than Canada.

Icewine research has come a long way in the past 15 years, but there are still many unknowns to understand all the factors related to maintain the highest quality. However, the major quality issue for icewine in the future will be authentication. Without authentication, it will be difficult to protect the reputation

and demand for this unique wine style from counterfeit versions of inferior quality been marketed as legitimate. Therefore, future trends in icewine research should include initiatives that increase our understanding of all the variables that affect icewine in an effort to develop a method of authentication. There has been a research focus to determine the viticultural and oenological factors that affect the sensory and chemical profiles of icewine in an effort to develop markers which characterize icewine. This is a good initial step; however, there are still many unanswered questions to be addressed before icewine can be characterized and the quality protected.

18.10 References

- Agenbach W A (1977), A study of must nitrogen content in relation to incomplete fermentations, yeast production and fermentation activity, in *Proceeding of the South African Society of Enology and Viticulture*, Stellenbosch, SA, 66–87.
- British Columbia (2005), Provincial government, Agri-Food Choice and Quality Act, 'Wines of marked quality regulation' B.C. Reg. 79/2005 includes amendments up to B.C. Reg. 216/2008.
- Anon (2008) Lallemand Inc., www.lallemandwine.com.
- Ballester J, Bowen A, Stranges M, Reynolds A and Brindle I (2007), 'Chemical markers for genuine Vidal icewines: volatile compounds composition', in *Proceedings of the Bacchus at Brock International Wine Conference*, St Catharines, ON, Brock University Press.
- Bely M, Rinaldi A and Dubourdieu D (2003), 'Influence of assimilable nitrogen on volatile acidity production by *Saccharomyces cerevisiae* during high sugar fermentation', *J Biosci Bioeng*, **96**, 507–512.
- Bowen A J, Reynolds A G and Lesschaeve I (2008a), 'The influence of harvest date on the sensory profiles of icewines from the Niagara Peninsula', *Am J Enol Vitic*, **59**, 346A.
- Bowen A J, Reynolds A G and Lesschaeve I (2008b), 'Odour potency of aroma compounds in Niagara icewines', *Am J Enol Vitic* (abstract; in press).
- Chisholm M G, Guiher L A, Vonah T M and Beaumont J L (1994), 'Comparison of some French–American hybrid wines with White Riesling using gas chromatography–olfactometry', *Am J Enol Vitic*, **45**, 201–212.
- Cliff M A and Pickering G J (2006), 'Determination of odour detection thresholds for acetic acid and ethyl acetate in ice wine', *J Wine Res*, **17**, 45–52.
- Cliff M, Yuksel D, Girard B and King M (2002), 'Characterization of Canadian ice wine by sensory and compositional analyses', *Am J Enol Vitic*, **53**, 46–53.
- Erasmus D J, Cliff M, and van Vuuren H J J (2004), 'Impact of yeast strain on the production of acetic acid, glycerol, and the sensory attributes of icewine', *Am J Enol Vitic*, **54**, 371–378.
- Fleet G H (2003), 'Yeast interactions and wine flavour', *Int J Food Microbiol*, **86**, 11–22.
- Galet P (1998), *Grape Varieties and Rootstock Varieties*, Chaintre, France, Oenoplurimedia.
- Gedeon J (2005), 'Counterfeit icewine puts the chill on Canadian sales', *Wine Business Monthly Online*, 2.
- Giraudel J L, Setkova L, Pawliszyn J and Montury M (2007), 'Rapid headspace solid-phase microextraction-gas chromatographic-time-of-flight mass spectrometric method for qualitative profiling of ice wine volatile fraction III. Relative characterization of Canadian and Czech ice wines using self-organizing maps', *J Chromatogr A*, **1147**, 241–253.
- Grape Growers of Ontario (2008), 60th Annual report for the year ending 31 January, 2008, Vineland Station, ON.
- Iland P, Bruer N, Edwards G, Weeks S and Wilkes E (2004), *Chemical Analysis of Grapes and Wine: Techniques and Concept*, Campbelltown, Patrick Iland Wine Promotions PTY

- Ltd.
- Inglis D L, Pigeau G, Quai J, Pistor M and Kaiser K (2006), 'Chemical composition of Vidal icewine juice and nitrogen usage during fermentation', in Creasy G L (ed.), *Sixth International Cool Climate Symposium for Viticulture and Oenology – Wine Growing for the Future*, Auckland, New Zealand, New Zealand Society for Viticulture and Oenology.
- Kilmartin P A, Reynolds A G, Pagay V, Nurgel C and Johnson R (2007), 'Polyphenol content and browning of Canadian icewines', *J Food Agric Environ*, **5**(3&4), 52–57.
- Kontkanen D, Inglis D L, Pickering G J, and Reynolds A G (2004), 'Effect of yeast inoculation rate, acclimatization, and nutrient addition on icewine fermentation', *Am J Enol Vitic*, **55**, 363–370.
- Kontkanen D, Pickering G J, Reynolds A G, and Inglis D L (2005), Impact of yeast conditioning on the sensory profile of Vidal icewines. *Amer J Enol Vitic*, **56**, 298A.
- Martin S, van der Merwe, G Shelp B and Inglis D (2008), 'The impact of hyperosmotic stress on yeast nitrogen usage during icewine fermentation', *Am J Enol Vitic*, (abstract; in press).
- Noble A C and Bursick G F (1984), 'The contribution of glycerol to perceived viscosity and sweetness in white wine', *Am J Enol Vitic*, **35**, 110–112.
- Norbeck J and Blomberg A (2000), 'The level of cAMP-dependent protein kinase A activity strongly affects osmo-tolerance and osmo-instigated gene expression changes in *Saccharomyces cerevisiae*', *Yeast*, **16**, 121–137.
- Nurgel C, Pickering G J and Inglis D L (2004a), 'Sensory and chemical characteristics of Canadian ice wines', *J Sci Food Agric*, **84**, 1675–1684.
- Nurgel C, Inglis D L, Pickering G J, Reynolds A G and Brindle I (2004b), 'Dynamics of indigenous and inoculated yeast populations in Vidal and Riesling icewine fermentation', *Am J Enol Vitic*, **55**, 435A.
- OIV (2003), *Definition of icewine*, Resolution OENO 6/2003, Paris, France, Organisation Internationale de la Vigne et du Vin.
- Ontario (1999), Provincial government, *Vintners Quality Alliance Act of 1999*, Regulation 406/00, Toronto, ON, accessed through e-laws: www.e-laws.gov.on.ca.
- Pickering, G (2006), 'Icewine – the frozen truth', in Creasy G L (ed.), *Sixth International Cool Climate Symposium for Viticulture & Oenology – Wine Growing for the Future*, Auckland, New Zealand, New Zealand Society for Viticulture and Oenology.
- Pigeau G M and Inglis D L (2005), 'Upregulation of ALD3 and GPD1 in *Saccharomyces cerevisiae* during icewine fermentation', *J Appl Microbiol*, **99**, 112–125.
- Pigeau G M and Inglis D L (2007), 'Response of wine yeast (*Saccharomyces cerevisiae*) aldehyde dehydrogenases to acetaldehyde stress during icewine fermentation', *J Appl Microbiol*, **103**, 1576–1586.
- Pigeau G M, Bozza E, Kaiser K and Inglis D L (2007), 'Concentration effect of Riesling icewine juice on yeast performance and wine acidity', *J Appl Microbiol*, **103**, 1691–1698.
- Pitkin C, Kontkanen D and Inglis D (2002), 'The effects of varying soluble solids concentration in icewine juice on metabolite production by *Saccharomyces cerevisiae* K1-1116 during fermentation', in Cullen C, Pickering G and Phillips R (eds), *Proceeding of the Inaugural Brock University Wine Conference – Bacchus to the Future*, St Catharines, ON, Brock University Press, 179.
- Schreiner, J (2001), *Icewine – the Complete Story*, Toronto, ON, Warwick Publishing.
- Scott Laboratories, Inc. (2005), *Scott Laboratories Fermentation Handbook*, Petaluma, California, Scott Laboratories, Inc.
- Setkova L, Risticvic, R and Pawliszyn J (2007), 'Rapid headspace solid-phase microextraction-gas chromatographic-time-of-flight mass spectrometric method for qualitative profiling of ice wine volatile fraction II. Classification of Canadian and Czech ice wines using statistical evaluation of the data', *J Chromatogr A*, **1147**, 224–240.
- Soleas G J and Pickering G J (2007), 'Influence of variety, wine style, vintage and viticultural area on selected chemical parameters of Canadian icewine', *J Food Agric Environ*, **5**(3&4), 97–102.

- Subden R E, Husnik J I, van Twest R, van der Merwe G and van Vuuren H J J (2003), 'Autochthonous microbial population in a Niagara peninsula icewine must', *Food Res Int*, **36**, 747–751.
- Vintners Quality Alliance of Ontario (2008), *Vintage Report 2007*, Toronto, ON, available at: http://www.vqaontario.com/Library/AnnualReports/vqa_annual_report_07.pdf (accessed March 2010).
- Wüldig G, Schlotter H A and Mueller T (1975), 'Versuche zur Eisweinbereitung' *Weinwirtsch (Neustadt/Wstr.)*, **111**, 982–984.
- Ziraldo D and Kaiser K (2007), *Icewine – Extreme Winemaking*, Toronto, ON, Key Porter Books Limited.

Managing the quality of sparkling wines

S. Buxaderas and E. López-Tamames, University of Barcelona, Spain

Abstract: The main characteristic of sparkling wines is that they contain supersaturated CO₂ gas (minimum 3 bars) and form bubbles and foam when poured into the flute. The bubbles arise from induced homogeneous or heterogeneous nucleation which gives rise to a liquid film surrounding the CO₂ gas. This film is formed by wine compounds and confers upon bubbles their individuality, keeping them stable for some time on the wine surface. The variety of the base wine used and the ageing time the wine has been in contact with yeast give sparkling wines their characteristic colour and aroma. To avoid browning, some technological practices require special care. Yeast autolysis also plays an important role in the development of the yeasty, sweet and toasty aroma notes.

Key words: sparkling wine, foam, bubble, foam sensory descriptors, colour, phenolic compounds, aroma, volatile compounds, sensory descriptors, ageing, quality.

19.1 Types of sparkling wines: definitions and characteristics

The presence of carbon dioxide (CO₂) is the main characteristic differentiating sparkling wines from the so-called still wines. Although the consumer might think that this is the only difference between these wines and the others, the CO₂ gas content of sparkling wines entails the application of different winemaking conditions and specific technological practices. In Europe, a place with a great tradition in the making of fizzy wine, the amount of CO₂ is the differential characteristic between sparkling wines whose gas pressure allows foam formation and bubbly wines that do not form foam due to their lower gas pressure (not less than 1 bar and not more than 2.5 bar) (semi-sparkling wines or pearl wines: *vins pétillant* in

France; *vinos de aguja* in Spain; *vinhos verdes* in Portugal; *Moscato d'Asti* in Italy, etc.). According to European regulations, sparkling wine shall be the wine which, when the container is opened, releases CO₂ derived exclusively from fermentation. In the case of a bubbly wine, with CO₂ derived wholly or partially from an addition of that gas, the labelling shall be aerated sparkling wine. For a sparkling wine designation, the EU requires that the product has an excess pressure, due to CO₂ in solution, of not less than 3 bars when kept at a temperature of 20 °C in closed containers, and for which the total alcoholic strength of the base wine (*cuvée*) intended for their preparation shall not be less than 8.5% by volume. However, for a sparkling wine to be classed as a quality sparkling wine, it shall have a minimum of excess pressure of 3.5 bars and a minimum alcoholic degree of the base wine of 9% by volume (EC, 2008).

The elaboration of sparkling wine involves two production stages: preparation of the base wine or *cuvée* (pressing of the grapes, clarification and fermentation of the must, and clarification and stabilization of the wine), and the sparkle stage (bottling and *tirage* process, second fermentation, wine ageing in contact with yeast lees, elimination of yeast or riddling, disgorging and addition of the expedition liquor) (Buxaderas and López-Tamames, 2003). The base wine is a dry wine that results from the blending or *coupage* of dry wines to which sugar (sucrose or concentrated must) and a yeast solution have been added and left to undergo a second alcoholic fermentation either in sealed bottles or in pressurized tanks that trap the CO₂ gas produced during fermentation (*charmat*, *granvas* or bulk method). The first case or bottle fermentation is a common practice in European countries such as France (since the seventeenth century) or Spain (since the nineteenth century) with long-standing traditions in the elaboration of sparkling wines. However, to carry out the second fermentation in a bottle is a more delicate elaboration practice requiring more dedication from the winemaker. For this reason, the method of choice in many countries is the bulk method, even among those with a long traditional sparkling wine production history. Nevertheless, the bottle fermentation method is commonly used to elaborate sparkling wines whose quality is guaranteed by a designation of origin and geographical indications as per EU directives (EC, 2008). The best known sparkling wines within this premium category are Champagne (located in the Epernay and Reims area in France), Cava (mainly produced in the Penedès region, Catalonia, Spain) and Talento wines (North Italy). Bottle fermentation was initially called *Champenoise methode*, but inside the EU this denomination is used exclusively in the French region of Champagne. In Italy and Spain, this method is known as the classic (*metodo classico* or *Spumanti Metodo Classico*) or traditional method (*método tradicional*), respectively.

19.2 Description of the organoleptic characteristics of sparkling wines

The organoleptic quality of all wines depends on aroma and colour. In the case of sparkling wines, it also depends on the wine's capacity to create foam. Even

though the first two aspects noticed by the consumer are colour and foam, it is the latter which grants sparkling wines a social role that differs from that of still wines. The joyful effervescence of the bubbles that rise to the wine's surface and create foam is the reason why sparkling wines are consumed especially in toasts and celebrations. However, once the carbon gas has caused the relatively fleeting foam effect, colour and aroma become the main characters again and will determine the choice of sparkling wine selected.

19.2.1 Foam

Food foams are considered gas bubble dispersions in a continuous liquid or semi-solid phase, which contains a soluble tensoactive agent. Frequently, the gas is CO₂ and the continuous phase is a dissolution or aqueous suspension of proteins (Cheftel *et al.*, 1989). Comelles *et al.* (1991) proposes a definition of foam in oenology: a binding of small bubbles, consisting of a near sphere-shaped liquid film that surrounds the endogenous CO₂ gas produced in the second fermentation.

The bubbles rise up through the liquid mass and accumulate on the wine's surface or in the liquid/air interphase. The bubble can therefore be interpreted as the foam's unitary cell, the interior of which holds CO₂ surrounded by a liquid or wine film, creating a double layer (see Fig. 19.1). The inner layer leads to the film's CO₂/liquid interphase and the outer layer to the liquid/air phase. This film of liquid gives the bubble its spherical shape and keeps it stable for some time upon air exposure and separated from the other bubbles, preventing bubble coalescence and bubble rupture and flattening.

Physical characteristics

The structure of the bubbles that form in the wine base and the appearance of foam on its surface are phenomena that require supersaturation of CO₂ and the presence of compounds in the wine capable of constituting the film that confers to bubbles their individuality. No foam is produced in liquids that contain CO₂ under pressure. In carbonated drinks with a low amount of dry extract such as mineral water or in some soft drinks, CO₂ is found in large cavities which ascend through the liquid due to their lower density and, once they have reached the surface of the liquid, breakdown immediately releasing the gas. The presence of compounds with tensoactive properties in sparkling wines is what gives bubbles their nearly spherical shape (see Fig. 19.1). In addition, tensoactive compounds found in wine reduce surface tension which makes it easier for bubbles to ascend to the surface.

In an aqueous medium, the polar group of molecule in the tensoactive compound dissolves easily, whereas the hydrophobic group is absorbed at the liquid/air interphase of the wine's surface or at the gas/liquid interphase of the bubble's film. In the first case, tensoactive compounds reduce the liquid's superficial tension, thus making it easier for the bubbles to overcome the attraction exerted by the molecules inside the wine on their ascent to the surface (Jordan and Napper, 1988). In the second case, tensoactive compounds concentrate on the bubble's film, pushing the hydrophobic group towards the gas, which favours the entry of

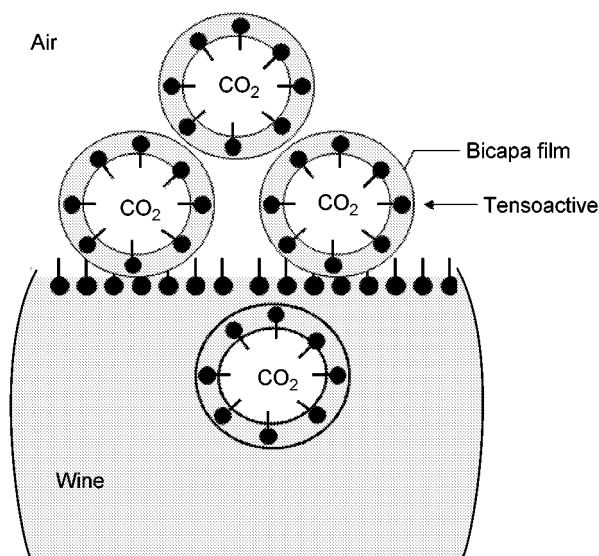


Fig. 19.1 Schematic of bubbles as foam's unitary cell, the interior of which hold CO_2 surrounded by a liquid or wine film, creating a double layer stabilized by tensoactive compounds at the liquid/air or CO_2 interphase.

gas and prevents CO_2 from dissolving in the liquid mass. Because of this, many bubbles increase in size during their journey towards the wine's surface. Codrington (1986) estimated the size of a bubble to be between 30 and 120 μ .

Sparkling wines may contain various compounds with tensoactive properties, but alcohol, glycerol and tartaric acid are always present and most abundant. These three elements reduce the superficial tension of pure water from 72 mN/m to 47.3 mN/m when the ethanol concentration is 12%, and to 68.0 mN/m and 57.0 mN/m when the pure water contains 0.4% glycerol or tartaric acid, respectively. When these three compounds are found in the wine at the previously said concentrations, a synergic effect is triggered and the superficial tension decreases to 42.7 mN/m. According to Comelles *et al.* (1991), the superficial tension of sparkling wines varies between 45 and 48 mN/m. However, ethanol does not behave as a true tensoactive compound. As the amount of ethanol increases, the superficial tension of pure water decreases, but when the amount of ethanol exceeds 12% a critical micellar concentration is reached and the superficial tension of water stops decreasing. A critical micellar concentration is defined as the quantity of tensoactive compound that saturates the liquid's surface, and therefore a larger amount of the substance has to adopt a micellar structure in order to remain in the liquid. This is exactly what happens in most tensoactive compounds but, in the case of ethanol, when the amount present exceeds 12%, no micelles are formed due to the small distance between the polar and the hydrophobic poles. Thus, ethanol is dissolved in the base of the liquid where it can act as a solvent for the hydrophobic groups of other molecules. This explains why wines that contain an ethanol concentration of

more than 12% are less sparkling than those that do not exceed the critical micellar concentration of ethanol (López-Barajas *et al.*, 1997). In fact, some authors have used surface tension to define the foamability of alcoholic drinks and to model solutions (Maujean *et al.*, 1990; Maeda *et al.*, 1991; Dussaud *et al.*, 1994).

Bubble formation in sparkling wines occurs as a result of two main mechanisms. The first mechanism has to do with microbubbles that form in the wine when it is subjected to shaking during handling and transportation (Bidan *et al.*, 1986; Jordan and Napper, 1988; Maujean, 1989). As the pressure in the bottle decreases after it is uncorked, microbubbles will keep growing until they burst, thereby giving the nuclei the chance to form new microbubbles. These new microbubbles will in turn grow and burst, giving rise to more microbubbles that will then form a chain of bubbles (Maujean, 1989). This process is referred to as induced homogeneous nucleation. Bubble formation is also caused by the presence of endogenous particles such as cork dust, by macromolecules that retain gas such as clarifiers (bentonite) and by traces of yeast cells (Senée *et al.*, 1999), but mainly by the presence of microcavities similar to those in potassium bitartrate microcrystals or those found in defective glass walls. When the microcavity is protected by a lipophilic layer that prevents it from getting filled with wine, it can become full of gas due to capillarity which gives rise to a newly-formed bubble that continues to expand in volume until it is subjected to ascending gravitational forces (Casey, 1988). Nonetheless, in order for a microcavity to nest an embryonic bubble it is necessary for it to have a curvature radius such that the pressure of the bubble impedes the diffusion and re-dissolution of the CO₂ in the wine. According to Casey (1988), the minimum curvature radius the microcavity should have is defined by Laplace's law where:

$$P = 2\gamma/R_c$$

where P = the differential pressure, γ = the superficial tension and R_c = the critical radius.

When the bubble has grown into an adequate size to leave the microcavity, a possibility is created for another bubble to grow (induced heterogeneous nucleation). The main difference between both types of nucleation is that in homogeneous nucleation one single bubble is produced, whereas in heterogeneous nucleation the microcavity can continue to produce hundreds of bubbles over some time (Jordan and Napper, 1988). For this reason, it is assumed that the train of bubbles that rises through the wine is caused by induced heterogeneous nucleation, as new bubbles are constantly and repeatedly being formed.

Bubbles which remain stable on the wine's surface or at the liquid/air interphase either gather to form a layer that covers the surface or frequently concentrate in the superficial area of the wine around or in contact with the glass wall. This first monolayer will be displaced by a second layer, and so on successively. This agglomeration of different bubble layers is known as foam, and the concentration of bubbles around the wine's circumference delimited by the glass rim is known as the crown.

The foam produced by a sparkling wine when it is poured into a glass consists

of two phases. The first phase consists of tumultuous foam that rises several centimetres along the glass wall due the rapid decompression that takes place after the bottle is uncorked. This foam explosion lasts just a few seconds and is mainly composed of microbubbles that have formed and increased in size during the shaking, transportation and uncorking of the bottle (Guggenberger and Kleber, 1964; Würdig and Müller, 1979; Casey, 1988).

Soon after, the foam is reduced to a few millimetres and a second but longer lasting phase starts. Initially one, two or several bubble layers distributed across the glass surface are observed. These layers are generated by several bubble trains that originate from the microcavities. When the sparkling wine temperature rises due to the existing room temperature, the ethanol in the surface of the glass starts to evaporate in larger amounts in the centre of the glass than around its circumference (Tuinier *et al.*, 1996). This causes the superficial tension of the wine to be higher in the centre than around the circumference; consequently, a bubble gradient is established from the centre towards the crown, where it is easier for bubbles to reach the surface since the superficial tension in this area is lower (Tuinier *et al.*, 1996). In this second phase, the intensity of the foam gradually goes down over time as the wine loses the CO₂ that has dissolved due to the increase in temperature and the physical dragging effect caused by effervescence.

Chemical characteristics

Foam duration is directly related to bubble stability, and stability is itself dependent on the composition of the film that supports it. The film acts as an elastic barrier (Casey, 1995) formed not only by tensoactive compounds but also by other wine compounds such as proteins, polysaccharides and fatty acids, among others, that give the film its elastic and viscous properties (Bramforth, 1985). The viscosity of the bubble's wall surface, which is part of the foam, is related to the viscosity of the wine and affects its sparkling properties (Bourne, 1982; Maeda *et al.*, 1991).

When the bubble is located on the wine's surface, it loses liquid from the film over time due to gravity and differences in pressure and/or evaporation (Cheftel *et al.*, 1989; Maujean, 1989). Hence, time is what reduces the film's thickness due to drainage of the liquid. This gravitational effect or drainage makes the bubble's wall thinner and more fragile and, consequently, more prone to rupture or collapse. According to Kinsella (1981), a high viscosity attenuates the destructive effects of collisions between bubbles. Moreover, drainage velocity decreases with increased liquid viscosity of the bubble's wall (Brissonnet and Maujean, 1991; Robillard *et al.*, 1993; Marchal *et al.*, 1996). In brief, a high colloid content in the wine will translate into a more viscous film resulting in a slow loss of the liquid, which will in turn lead to increased bubble half-life and increased foam stability (Jordan and Napper, 1988; Dickinson, 1994).

Three types of studies have been conducted in an attempt to establish the wine compounds that play a role in foam formation. Some studies assess whether there is an increase of foam properties upon addition of different amounts of wine compounds (proteins, polysaccharides, etc.) to a solution or model wine (12% ethanol hydroalcoholic solution with added tartaric acid and glycerol) (Molan *et*

al., 1982; Robillard *et al.*, 1993; Dussaud *et al.*, 1994; Tuinier *et al.*, 1996). This type of study presents two pitfalls. Added proteins (albumins) and polysaccharides (galacturonans) create patterns that are not completely similar to wine proteins and polysaccharides. Also, the compound's ability to favour foam in a model solution does not take into consideration the interaction and participation of other wine components.

A second type of study aims to correlate the amounts of different compounds analyzed in different wines with certain physical measurements of the foam in the same wines using, normally, a method based on the one proposed by Bikerman (1938). This method, known as *gas-sparging*, causes foam formation by injecting a gas flow (N_2 , air and CO_2) into the wine. The Mosalux device (Maujean *et al.*, 1990) has been the most widely used gas-sparging system since 1990 and the first to have the capacity to carry out a normalized and computerized determination of the physical properties of foam proposed by Poinsaut (1991), Robillard *et al.* (1993) and Gallart *et al.* (1997): HM, height of the initial tumultuous foam, which represents the first explosion of foam or foamability that occurs when the sparkling wine is poured onto the glass; HS, the height at which foam remains stable after the tumultuous foam is reduced, which would correspond to the height of the crown; TS, the time the foam remains stable after the gas flow is interrupted, a datum intended to estimate the time bubbles remain on the wine surface or to predict the Bikerman coefficient (Σ), a measurement that determines the half-life of a stable bubble (Maujean *et al.*, 1990; Pueyo *et al.*, 1995; Andrés-Lacueva *et al.*, 1996a, 1997; López-Barajas *et al.*, 1997, 1998, 1999; Lao *et al.*, 1999; Girbau-Solà *et al.*, 2002a,b). These types of studies establish a series of compounds that can influence the foaming properties of wine, but make no distinction between cause–effect or the degree of influence that each of the compounds has, neither do they provide any information regarding the participation of other compounds that are not analyzed.

Lastly, there is a third type of study that analyzes the compounds that form part of the separate foam of the wine (Brissonnet and Maujean, 1991, 1993; Gallart *et al.*, 2002). These studies are based on the assumption that the wine compounds giving rise to the film coating of the bubble are more concentrated in the isolated foam of the wine than they were in the wine before foam formation was triggered. A Mosalux device (Maujean *et al.*, 1990) was used to achieve foam formation, as this device was provided with a system for adding CO_2 into the wine contained in a graduated probe. The foam created is separated by means of a suction pump and the components in the foam and initial wine determined. Both concentrations are then compared. The results yielded by these studies are not conclusive, given that a compound can intervene in the bubble's film without this meaning that an enriched content is detected analytically in the wine's separate foam. However, components with higher concentrations in the foam than in the initial wine do in fact intervene in bubble formation and stability.

Table 19.1 includes a summary of the compounds that have been related to foam in the different studies published, making reference to the type of study through the type of sample used: model wine, base wine or sparkling wine, isolated foam. In the majority of the works shown in Table 19.1, the compounds have been related to the

Table 19.1 Compounds related to foam properties in sparkling wines

Compounds	References	Type of sample	Results
Proteins	Gomerieux, 1989	Model wine	Increase foam height
	Molan <i>et al.</i> , 1982	Model wine	Increase foam
	Robillard <i>et al.</i> , 1993	Model wine and 2 BW ^a	Increase foam stability
	Maujean <i>et al.</i> , 1990	31 BW ^a	Increase height foam
	Malvy <i>et al.</i> , 1994	1 BW ^a	Increase height and stability foam
	Pueyo <i>et al.</i> , 1995	9 BW and 1 Cava series ^b (10)	Reduce height collar
	Andrés-Lacueva <i>et al.</i> , 1996a	48 BW ^a	Increase stability foam
	Andrés-Lacueva <i>et al.</i> , 1997	6 Cava series ^b (48)	Increase height foam
	López-Barajas <i>et al.</i> , 1998	36 BW ^a	Reduce stability foam
	Lao <i>et al.</i> , 1999	18 BW ^a	Increase height foam
	Girbau-Solà <i>et al.</i> , 2002a	15 BW ^a	Reduce stability foam
	Brissonnet and Maujean, 1991, 1993	Separated foam	Increase foam and collar height
			Increase foam height
Poly-saccharides	Gomerieux, 1989	Model wine	Increase stability foam
	Andrés-Lacueva <i>et al.</i> , 1996a	48 BW ^a	Increase height foam
	Andrés-Lacueva <i>et al.</i> , 1997	6 Cava series ^b (48)	Increase height foam
	López-Barajas <i>et al.</i> , 1998	36 BW ^a	Reduce stability foam
	Lao <i>et al.</i> , 1999	18 BW ^a	Increase height foam
Polyphenols	Andrés-Lacueva <i>et al.</i> , 1997	6 Cava series ^a (48)	Increase height foam
	Lao <i>et al.</i> , 1999	18 BW ^a	Reduce stability foam
	Girbau-Solà <i>et al.</i> , 2002a	15 BW ^a	Reduce stability foam
			Reduce foam height
Lipids	Dussaud <i>et al.</i> , 1994	Model wine and 3 BW ^a	No influence at ethanol concentrations >5%
	Maujean <i>et al.</i> , 1990	1 BW ^a	C ₈ and C ₁₀ increase collar height and reduce stability foam
	Gallart <i>et al.</i> , 2002	14 BW 3 separated foam	C ₈ , C ₁₀ , C ₁₂ reduce height foam
Glycerol			Ethyl esters increase height foam
	Robillard <i>et al.</i> , 1993	Model wine and 2 BW ^a	Increase foam

continued

Table 19.1 *continued*

Compounds	References	Type of sample	Results
Organic acids	Robillard <i>et al.</i> , 1993	Model wine and 2 BW ^a	Tartaric acid: increase foam
	Pueyo <i>et al.</i> , 1995	9 BW and 1 Cava series ^b (10)	Tartaric acid: increase collar height
	Andrés-Lacueva <i>et al.</i> , 1996a	48 BW ^a	Malic acid: increase foam height Lactic acid: reduce foam height Citric and galacturonic acids: reduce foam stability time
	Andrés-Lacueva <i>et al.</i> , 1997	6 Cava series ^b (48)	Malic acid: increase foam height and reduce foam stability
	López-Barajas <i>et al.</i> , 1998	36 WB ^a	Malic acid: increase foam height and reduced foam stability time Lactic acid: reduced foam height and increased foam stability time Citric and galacturonic acids: reduce foam stability time
	Girbau-Solà <i>et al.</i> , 2002a	15 BW ^a	Malic acid: increase foam height

^aBW, base wines.^bCava series (*n*), total number of bottles of Spanish sparkling wine (Cava) analyzed during ageing.

physical parameters obtained by the Mosalux device (Maujean *et al.*, 1990) or similar gas-sparging methods (Pueyo *et al.*, 1995; Moreno-Arribas *et al.*, 2000). Proteins are the most studied compounds with respect to wine foamability. In most works, proteins favour the height of the initial tumultuous foam (HM), and in some of these works proteins are also believed to increase the mid-life of the bubble (Σ). The work conducted by Pueyo *et al.* (1995) is an exception to this because the authors observed a negative correlation between the protein content and the height of stable foam (HS), whilst in Cava (Spanish sparkling wine) a positive correlation with the time the foam remains stable was observed (TS). Brissonnet and Maujean (1991) compared foam proteins with those of wine and observed that foam proteins were more hydrophobic than those of wine, which led them to believe that the participation of proteins in foam formation is dependent on their hydrophobicity but not on their molecular weight or isoelectric point. In contrast, Robillard *et al.* (1993) are of the opinion that the positive charge conferred by proteins to wine pH allows them to migrate to the wine/air interphase and to stabilize foam. The association between the capacity of proteins and their molecular weight to form foam has only been demonstrated in studies conducted using model wine or beer (Molan *et al.*, 1982; Maeda *et al.*, 1991; Lusk *et al.*, 1995). Nonetheless, the study

conducted by López-Barajas *et al.* (1998) showed that some compounds that increase foamability (HM), such as proteins, do not seem to increase the time the foam remains stable (TS). These authors justify the results because they found that both physical measurements of foam (HM and TS) are negatively correlated ($r = -0.5113$; $p = 0.002$), and hence some compounds that are active agents on foamability may not prolong the time the bubble remains stable. As in proteins, these authors mention other compounds as positive for foamability and negative for the time the bubble remains stable: total content of polysaccharides and malic acid.

With regard to polysaccharides, some authors agree in considering these macromolecules as foaming agents that increase foamability (Andrés-Lacueva *et al.*, 1996a; Lao *et al.*, 1999); in contrast, other authors mention a negative correlation between the polysaccharide content in sparkling wines and stability time (TS) (Andrés-Lacueva *et al.*, 1997). According to Senée *et al.* (1999), a large number of polysaccharides interact with proteins (proteoglycans) acquiring tensoactive properties making them more soluble at the gas/liquid interphase, thereby decreasing liquid drainage. Only one study describing a connection between the molecular size of polysaccharides and foaming properties has been found (López-Barajas *et al.*, 2001). Polysaccharides of M_r 62 000–48 000 and 13 000–11 000 increase foamability (HM) and polysaccharides of M_r 3000–2000 increase bubble stability (Σ).

There appears to be consensus among published works in relation to the influence of the phenolic compounds on foam. Phenols reduce foam stability in wine, and only one study carried out on sparkling wines found a positive correlation between these compounds and foam height (Andrés-Lacueva *et al.*, 1997). However, proteins, polysaccharides and phenols are compounds that interact among themselves and consequently it is difficult to evaluate the isolated effect of each of them on foam.

Conflicting results have been published on the influence of lipid and fatty acid content on wine foaming properties. Dussaud *et al.* (1994) added a mixture of lipids to wine which did not alter its foam capacity. Maujean *et al.* (1990) added different amounts of two main fatty acids, C_8 and C_{10} to the wine and found that both fatty acids increased stable foam height (HS), had no effect on foamability (HM) and reduced stability time (TS), even though the effect of C_8 was found to be concentration-dependent. Pueyo *et al.* (1995) claim that linoleic acid and palmitic acid favour foam stability and increase height, respectively. The study by Gallart *et al.* (2002) studied free fatty acids and ethanol-esterified fatty acids. These authors found a negative correlation between fatty acids C_8 , C_{10} and C_{12} and foamability (HM), whilst the corresponding ethyl esters showed a positive correlation. On the other hand, they found a higher concentration or enrichment of fatty acids C_6 , C_8 , C_{10} and C_{16} and their esterified forms in the foam isolated from three different wines compared to that found in the initial wines. Because of this, the authors believe that the influence of fatty acids on foamability depends on the proportion between the esterified amount and the free amount, or on the esterified proportion (the esterification ratio, ERC_n): quotient between the ethyl ester concentration (EE_n) and the fatty acid concentration (C_n). So, the higher the coefficient

the greater the foamability; thus, it appears that it is the esterified forms of fatty acids that increase foam height (HM). This could explain the variability of the results yielded by previous studies which only examined total fatty acids.

Different authors agree in pointing to malic acid as an enhancer of the height reached by the initial foam (Andrés-Lacueva *et al.*, 1996a, 1997; López-Barajas *et al.*, 1998; Girbau-Solà *et al.*, 2002a), but conversely they also refer to this acid as a foam destabilizer (Andrés-Lacueva *et al.*, 1997; López-Barajas *et al.*, 1998). It appears that whilst malic acid increases foam height and reduces stability, lactic acid exerts the opposite effect, which suggests that a malolactic fermentation of wine, which reduces acidity and reduces foam height, simultaneously favours foam stability as the concentration of lactic acid increases (Andrés-Lacueva *et al.*, 1996a; López-Barajas *et al.*, 1998). The presence of other acids such as citric and galacturonic acid does not favour foam stability (Andrés-Lacueva *et al.*, 1996a; López-Barajas *et al.*, 1998).

In view of the results shown in Table 19.1, it can be concluded that the foaming capacity of wines depends on equilibrium amongst all the compounds that favour its formation and stability and those that do not (Bissonnet and Maujean, 1991; Viaux *et al.*, 1994; Pueyo *et al.*, 1995). López-Barajas *et al.* (1999) developed mathematical equations that calculated foam parameters (HM and Σ) from the physico-chemical characteristics of wines. The author used 96 wines from different harvests, varieties, wineries, and from different stages of elaboration (cloudy and filtered wines) and, using the 12 variables that are usually analyzed in the wineries, obtained values for foamability and foam stability. Some of the variables or compounds included in the equation are or are related to the components listed in Table 19.1 (proteins, organic acids, absorbance at 280 nm, glycerol and alcohols).

19.2.2 Phenolic compounds and colour

Many of the phenolic compounds found in grape skin are principally responsible for wine colour. These compounds are secondary metabolites synthesized by the plant to protect itself from stress situations, such as a fungal attack, lack of water, ultraviolet radiation and temperature changes (Deloire *et al.*, 1998). In the case of wine, phenolic compounds provide colour but also affect other sensory properties (taste, astringency and smell).

Phenolic compounds are a large chemical family characterized by having at least one aromatic ring with one or more hydroxyls as substituents. They are classified in two main groups: non-flavonoids and flavonoids. Non-flavonoids are distributed in the skin and pulp of the grape and have only one (called phenolic acids, such as cinnamic and benzoic acids) or two phenolic nuclei linked together by an unsaturated chain (stilbenes). These compounds do not provide flavour or aroma, although they are the precursors of volatile phenols with unpleasant odours (Chatonnet *et al.*, 1992), and they give the wine a yellowish or even brown colour when oxidized.

Flavonoids can cause the acid pH of wine to acquire a yellow or red colour and

are found only in the skin and pedicel of the grape. The common structure of flavonoids is two benzene rings linked together by an oxygen heterocycle. Depending on the structure of the heterocycle, there are several different groups: flavonols, anthocyanin, catechin or procyanidins.

Traditionally, flavonols were considered the cause of the yellow colour of white wines. However, the fact that these phenols are located in the skin and white wines ferment without the grape's skin discredits the idea that flavonols are the only cause of the yellow colouring. (+)-catechin is the most abundant flavonoid in white wines, but when contact between the juice and skins is scarce (+)-catechin may not even be detected when determining the content of phenolic compounds. In white wines, the total content of flavonoids quantified by high-performance liquid chromatography (HPLC) represents a maximum of 20% of phenolic compounds, whereas hydroxycinnamic acids (free and esterified forms) reach close to 50% or even 60% (Baderschneider and Winterhalter, 2001; Chamkha *et al.*, 2003). For this reason, Díez de Bethencourt *et al.* (1993) believe that the colour of white wines is more dependent on the oxidation of benzoic and cinnamic acids (whose compounds upon combination with some metals such as iron turn yellowish in colour) than on flavonoid content.

Phenolic compounds have been extensively studied, either to describe different varieties, their influence on wine organoleptic properties such as colour, or to assess their biochemical antimicrobial, anticarcinogenic and antioxidant properties. However, most studies have focused on red wines, a few on white wines and very few on phenolic compounds found in sparkling wines (Ibern-Gómez *et al.*, 2000; Chamkha *et al.*, 2003; Pozo-Bayón *et al.*, 2003). The total concentration of phenolic compounds in white wine ranges between 50 and 350 mg/L (expressed as gallic acid); a much lesser quantity than that found in red wines (800–4000 mg/L) (Cheynier *et al.*, 1998). This difference is not only due to the fact that red grape must is fermented as it macerates with grape skins, but also because the red grape contains a greater quantity of phenolic compounds, red anthocyanin in particular, which is not found in the skin of the white grape.

Table 19.2 includes concentration intervals of phenolic compounds identified in two of the world's most distributed types of sparkling wines: Champagne (France) and Cava (Spain). Although data in Table 19.2 are limited to the results derived from three studies (Ibern-Gómez *et al.*, 2000; Chamkha *et al.*, 2003; Pozo-Bayón *et al.*, 2003), they provide the phenolic compound concentration present in the main grape varieties used to make sparkling wines: Chardonnay and Pinot noir for Champagne, Cava, *Sekt* (Germany) and *Cap classique* (South Africa), and Macabeo, Xarel-lo, Parellada, Garnacha, Trepat and Monastrell for Cava. Muscat varieties are used to make Asti *Spumante* (Italy) and Riesling varieties for German *Sekt*. In the study conducted by Chamkha *et al.* (2003), the results of phenolic compounds found in Champagne are compared with those found in Cava, as well as with those already published on white wines. Even though most sparkling wines are made with different varieties of white grapes (*blanc de blanc*) or red grapes vinified into white (*blanc de noir*), the second fermentation can cause notable differences in the concentration of phenolic compounds between base wines and

sparkling wines. Thus, the second fermentation boosts and increases free hydroxycinnamic acids formation through forms esterified with tartaric acid (Singleton and Trousdale, 1983; Chatonnet *et al.*, 1993).

There is limited published information on the phenolic concentration of white sparkling wines, but even less on rosé wines, in which the skins and must have been kept in contact until achieving the desired pinkish or rose colour. However, their colour points to the fact that, in contrast to white sparkling wines, rosé sparkling wines, like red wines, also contain anthocyanins.

As shown in Table 19.2, the most abundant phenolic compounds found in sparkling wines are hydroxycinnamic acids, especially caftaric acid and tyrosol. Tyrosol is a non-carboxylic phenol, which is a by-product of juice fermentation from tyrosine deamination (Singleton and Trousdale, 1983; Betés-Saura *et al.*, 1996). The flavonoids detected are (+)-catechin and, in a lesser proportion, quercetin-3-glucuronide. The red grape varieties used to make sparkling wines shown on Table 19.2, when vinified into white, do not appear to have a phenolic compound content much different from that of white grape wines.

19.2.3 Volatile compounds and aroma

The manufacturing of sparkling wines using the *sur lie* method involves two main different fermentative steps. The first step is similar to most white wine fermentations giving rise to the base wine. In the secondary fermentation, this base wine is mixed with a combination of yeasts, sucrose and bentonite, also known as *tirage* liquor, bottled and allowed to ferment and age. At the end of the ageing period, yeast cells are removed from the bottles for commercialization. During this ageing process, cell self-degradation takes place under the action of the cell's own enzymes. This very slow process is observed only after cell death. Its effect is to enrich the wine in soluble, low molecular weight compounds as well as in colloids (Guilloux-Benatier and Chassagne, 2003).

The ageing period is necessary to give these wines their characteristic aroma and flavour, which are the result of yeast autolysis and the concomitant release of cell components and their breakdown products in the wine (Charpentier and Freyssinet, 1989; Leroy *et al.*, 1990; Martínez-Rodríguez and Polo, 2000; Barcenilla *et al.*, 2003). Moreover, the special ageing of sparkling wines gives them a more complex volatile profile since autolytic and enzymatic reactions may take place. It is thought that the base wine, autolysis of the lees in the second fermentation, and the ageing time in contact with these lees, are the most important factors that affect the sensory quality of sparkling wines. Several authors have proposed the exchange of nitrogenous and glucidic constituents between the yeast biomass and autolysis medium as markers of autolysis: amino acids being aroma precursors (Feuillat, 1998; Marchand *et al.*, 2000; Martínez-Rodríguez *et al.*, 2001) and mannoprotein binding compounds (Voilley *et al.* 1990; Lubbers *et al.*, 1994; Feuillat, 2003; Fleet, 2003; Chalier *et al.*, 2007).

Aroma is also of paramount importance for producers of high-quality sparkling wines such as champagne (Vannier *et al.*, 1999). In this regard, these wines'

Table 19.2 Concentration of phenolic compounds determined in sparkling wines (Champagne and Cava)

	Chamkha <i>et al.</i> , 2003		Ibern-Gómez <i>et al.</i> , 2000 Macabeo, Parellada, Xarel-lo, Chardonnay <i>blanc de blanc</i> <i>n</i> = 6 (mg/l) ^a	Pozo-Bayón <i>et al.</i> , 2003		
	Chardonnay <i>blanc de blanc</i> <i>n</i> = 2 (mg/l) ^a	Pinot noir <i>blanc de noir</i> <i>n</i> = 2 (mg/l) ^a		Trepat <i>blanc de noir</i> <i>n</i> = 2 (mg/l) ^a	Monastrell <i>blanc de noir</i> <i>n</i> = 2 (mg/l) ^a	Garnacha rosé sparkling wine <i>n</i> = 2 (mg/l) ^a
<i>Non-carboxylic</i>						
Tyrosol	28–39	10–21	8.5–25.9	12.0–18.3	11.0–20.2	9.5–12.6
Tryptophol	–	–	–	0.82–1.78	0.08–0.45	1.42–3.68
<i>Hydroxybenzoic acids</i>						
Gallic	0.39–0.84	0.17–0.75	0.21–0.58	0.47–0.67	0.44–1.00	1.14–1.46
Protocatechuic	0.12–0.36	0.24–0.59	0.66–3.92	0.44–0.84	0.55–1.31	0.42–0.74
<i>p</i> -Hydroxybenzoic	0.05–0.13	0.04–0.20	–	0.12–0.20	0.10–0.16	0.01–0.03
Vanillic	0.19–0.37	0.18–0.27	–	0.42–0.44	0.40–1.62	0.41–0.55
Syringic	–	–	–	0.65–0.75	0.61–1.07	0.78–0.98
<i>Hydroxycinnamic acids</i>						
<i>trans</i> -Caffeic	1.2–1.7	1.64–2.10	0.48–1.76	0.66–1.20	0.07–0.33	2.52–3.48
<i>trans-p</i> -Coumaric	0.30–0.67	0.12–0.57	0.10–1.10	0.27–0.39	0.10–0.16	1.37–1.77
<i>cis-p</i> -Coumaric			–	0.28–0.42	0.10–0.22	0.45–0.59
<i>Hydroxycinnamates</i>						
<i>cis</i> -Caftaric	27–33	9–31	0.43–1.48	0.11–0.21	0.13–0.43	0.26–0.52
<i>trans</i> -Caftaric			9.33–26.47	1.45–2.89	1.82–3.32	7.08–11.06
<i>cis</i> -Coutaric	0.07–0.42	0.08–0.54	2.31–14.18	1.31–2.27	1.53–2.39	0.95–2.77
<i>trans</i> -Coutaric			2.27–16.18	0.26–0.38	0.42–0.76	0.46–1.58
<i>trans</i> -Fertaric	0.6–1.0	0.49–0.80	0.46–1.31	0.06–0.14	0.08–0.26	0.14–0.48

2-S-glutathionylcaftaric	2.1–3.7	3.1–8.3	1.46–4.55	–	–	–
<i>Stilbenes</i>						
<i>trans</i> -Resveratrol glucoside	–	–	0.07–0.16 ^b	0.29–0.39	nd ^c	nd
<i>cis</i> -Resveratrol glucoside	–	–	0.16–0.49 ^b	0.38–0.66	0.19–0.27	nd
<i>cis</i> -Resveratrol	–	–	0.10–0.21 ^b	0.17–0.29	nd ^c	0.09–0.23
<i>trans</i> -Resveratrol	0.05–0.12	0.05–0.18	0.04–0.05 ^b	–	–	–
<i>Flavonoids</i>						
(+)-Catechin	0.60–2.30	0.22–5.10	0.73–3.93	0.72–1.38	0.45–1.35	0.13–1.81
(–)-Epicatechin	0.60–1.20	0.28–1.98	–	–	–	–
Quercetin	0.06–0.15	0.05–0.21	0.05–0.17	–	–	–

^aThe external standard method was used to determine the concentrations. *cis*- and *trans*-caftaric acids, *cis*- and *trans*-coutaric acids, *trans*-fertaric acid, and *trans*- and *cis*-resveratrol glucosides were quantified using the free forms. ^bThese values have been determined by the same research group (Andrés-Lacueva *et al.*, 2002). ^cnd, non-detected.

flavour is empirically described by Cava makers as having yeasty, sweet or toasty notes. This pattern could result from the larger increase in citrus and apple intensity in red varieties and in vanilla/butter aroma in white varieties during the 18 months the wine is in contact with the yeast, following the secondary fermentation (De la Presa-Owens *et al.*, 1998). Although several volatile compounds, such as 1,2-dihydro-1,1,6-trimethylnaphthalene (TDN) (Winterhalter, 1991), vitispirane isomers and diethyl succinate have been used as Cava ageing markers (Francioli *et al.*, 2003; Riu-Aumatell *et al.*, 2006; Bosch-Fuste *et al.*, 2007), distinctive compounds involved in the flavour of aged sparkling wine have not yet been well characterized.

Ethyl esters of aliphatic acids (mainly ethyl hexanoate, ethyl octanoate, and ethyl decanoate) are an important family in the volatile profile of sparkling wines. Other characteristic esters were ethyl-9-decenoate, ethyl lactate and diethyl succinate. The first ethyl ester has been reported in sparkling wine samples (Riu-Aumatell *et al.*, 2006), whilst ethyl lactate and diethyl succinate are post-fermentative volatiles formed during the ageing of Cava in contact with lees from the second fermentation. The concentration of diethyl succinate increases during ageing, being considered a marker of the evolution of Cava during cellar storage (Francioli *et al.*, 2003; Riu-Aumatell *et al.*, 2006).

It is interesting to monitor and compare the sensory profile and aroma of the compounds in base wines and their corresponding Cavas subjected to different ageing periods in order to obtain quality parameters and to describe the correct sensory evolution of these special wines. Table 19.3 shows the volatile profile of base wines (Torrens *et al.*, 2008) and their corresponding sparkling wines (unpublished data provided by Freixenet S.A.). The amount of acetate esters (hexyl acetate, phenylethyl acetate, octyl acetate, etc.) decreases upon ageing with lees, and some of these esters cannot be detected in sparkling wine (Table 19.3). According to the literature, these acetate esters decrease during Cava production (Riu-Aumatell *et al.*, 2006). The decrease observed in some esters could be caused by the synthesis–hydrolysis activity of esterases released by yeast during the days following alcoholic fermentation (Pérez-Serradilla and Luque de Castro, 2008).

The bibliography on terpenoid compounds and lactones is discordant. The amount of the varietal compounds during bottle storage changes due to acid-catalyzed reactions (Rapp, 1998). A revision of the literature shows that some yeast-secreted mannoproteins could interact with volatile compounds, leading to sensory changes in the wine (Chalier *et al.*, 2007). Terpenes such as citronellol, lactones such as decalactone and C_{13} -norisoprenoids such as damascenone are described in aged *sur lie* sparkling wines. The presence of these compounds is desirable in wines because of the floral and fruity notes they provide (Pérez-Serradilla and Luque de Castro, 2008). The enzymatic activity of the lees probably causes the release of several compounds from their glycosidic precursors (Bautista *et al.*, 2007). In this sense, benzylic alcohol was found in Macabeo and Chardonnay grapes in the form of a glucopyranoside (García *et al.*, 2003) and the aroma potential of the variety of base wine could play an important role. As previously described, the enzymatic activity of yeast lees could cause the release of these

Table 19.3 Volatile compounds (mg/L) determined in base wine ($n = 18$) and Cava ($n = 36$) and their sensory notes

Compound	Base wine ^a	Cava sparkling wine ^b	Odour description
Ethyl isobutyrate	0.030	nd	Sweet, rubber ^c
Isobutyl acetate	0.054	nd	Fruit, apple, banana
Ethyl butyrate	0.410	0.498	Apple
Ethyl isovalerate	nd	0.033	Fruit ^c
Isoamyl acetate	3.086	0.155	Banana ^c
Ethyl hexanoate	0.868	0.876	Apple peel, fruit ^c
Hexyl acetate	0.312	nd	Fruit, herb ^c
<i>cis</i> -3-Hexenyl acetate	0.029	nd	Green, banana ^c
Ethyl lactate	7.532	81.290	Fruit ^c
Hexanol	0.938	1.268	Resin, flower, green ^c
<i>cis</i> -3-Hexenol	0.217	0.315	Grass ^c
Ethyl octanoate	1.170	1.049	Fruit, fat ^c
Furfural	nd	0.576	Bread, almond, sweet ^c
Octyl acetate	0.009	nd	Fruit ^c
Benzaldehyde	0.083	nd	Almond, burnt sugar ^c
Linalool	0.004	0.005	Flower, lavender ^c
Isobutyric acid	0.563	nd	Rancid, butter, cheese ^c
Butyric acid	0.060	nd	Rancid, cheese, sweat ^c
Ethyl decanoate	0.261	0.066	Grape ^c
Diethyl succinate	nd	8.113	Wine, fruit ^c
Isoamyl octanoate	0.036	nd	
Isovaleric acid	0.074	nd	Sweat, acid, rancid ^c
Methionol	3.249	3.469	Sweet, potato ^c
Citronellol	0.004	nd	Rose ^c
2-Phenylethyl acetate	0.255	n.d.	
β -Damascenone	0.129	0.065	Apple, rose, honey ^c
Hexanoic acid	6.262	6.162	Sweat ^c
Benzyl alcohol	nd	0.104	
Isoamyl decanoate	0.042	nd	
2-Phenylethanol	12.936	18.890	Honey, spice, rose, lilac ^c
Octanoic acid	8.691	6.331	Sweat, cheese ^c
<i>g</i> -Decalactona	0.002	nd	Peach, fat ^c
4-Vinylguayacol	0.148	nd	Spicy, peppery ^d
4-Vinylphenol	0.050	nd	Almond shell ^c
Acetal (1,1-diethoxyethane)	nd	0.724	Fruit, cream ^c
Diacetyl	n.d.	0.017	Butter ^c
Acetoin (3-hydroxybutanoate)	nd	3.754	Butter, cream ^c
Acetaldehyde	47.206	33.222	Pungent, solventy ^d
Ethyl acetate	43.944	32.583	Pineapple ^c
Isobutanol	20.500	18.186	Wine, solvent, bitter ^c
Isoamyl alcohol	149.222	129.808	Whiskey, malt, burnt ^c
Vitispirane isomer 1	nd	0.123	Camphor ^d
Vitispirane isomer 2	nd	0.104	Camphor ^d
1,2-Dihydro-1,1,6-trimethylnaphthalene (TDN)	nd	0.066	Kerosene ^d

Source: ^aTorrens *et al.*, 2008; ^bFreixenet S.A.; ^cAcree T and Heinrich A (<http://www.flavornet.org/flavornet.html>); ^dGenovese *et al.*, 2007.

volatile compounds during wine fermentation, ageing or commercialization (Bautista *et al.*, 2007).

Compounds such as TDN and vitispirane that are not detected in wine are, however, detected in Cava (Table 19.3), as previously reported in the literature (Francioli *et al.*, 2003; Riu-Aumatell *et al.*, 2006). In contrast, increased amounts of compounds such as furfural, benzylic alcohol, acetal, diacetyl and acetoin are identified in Cava. TDN and vitispirane play an important role in the varietal character of the wine due to a very low odour threshold (20 ppb). Their appearance is caused by carotenoid degradation in the presence of oxygen, or by exposure to the sun (Genovese *et al.*, 2007).

Furans are compounds derived from sugar degradation. They are partially responsible for the yeasty and toasty notes of aged wines (Câmara *et al.*, 2004, 2006; Bosch-Fusté *et al.*, 2008; Campo *et al.*, 2008). Maillard's reaction is specifically responsible for furfural formation and its yeasty aroma (Comuzzo *et al.*, 2006; Genovese *et al.*, 2007). This compound was described in the literature as having a bread, almond or sweet taste, but it was defined in the work conducted by Comuzzo *et al.*, (2006) as cheesy or pungent, as determined by gas chromatography–olfactometry (GC–O). Moreover, oxidation-related aldehydes have been associated with the aroma of some sparkling wines (Culleré *et al.* 2007).

Sulphur compounds play an important role in the flavour of sparkling wines (Tominaga *et al.*, 2003). The sensory importance of 2-methyl-3-furanthiol and 2-furan methanethiol, formed by Maillard reaction when food is cooked, is well-known. The formation mechanism of these compounds has been previously studied, particularly in cooked meat, as well as in a model medium containing cysteine and pentose (Marchand *et al.*, 2000). The presence of these volatile thiols has also been reported in certain wines over the years (Blanchard *et al.*, 2001). 2-furfurylthiol has a strong aroma of roasted coffee and has been shown to contribute to the roast-coffee aroma of certain wines such as Champagnes.

Sulphur-containing flavour compounds have different sensory properties depending on the position of the sulphur atom in the molecules and their concentrations. These compounds may be found in several foods such as coffee, meat, passion fruit, onions, Iberian ham or wine (Swiegers and Pretorius, 2007). Most of them make a negative sensory contribution with descriptors such as cabbage, rotten egg, sulphurous, garlic, onion, and rubber. However, a few sulphur compounds contribute with positive aromas, such as strawberry, passion fruit, grapefruit, and box tree (Tominaga *et al.*, 2003). Despite their sensory importance, the low levels of these compounds in wine make their analytical determination very difficult. However, some sulphur-containing compounds are described (Campo *et al.*, 2008) in Cava sparkling wines (4-mercapto-4-methyl-2-pentanone, 3-mercaptohexanol, 2-methyl-3-furanthiol, and 2-methyl-3-furanmethanethiol).

19.3 Factors affecting sensory quality

The organoleptic characteristics, foam, colour and aroma of a sparkling wine

depend on its composition. In turn, the composition of wine depends on multiple factors. Some of these factors are independent of the oenologist, such as environmental and climatic factors that control the quality of the harvest, whilst other factors are susceptible to modification, such as the grape varieties used and the yeast strains selected, as well as the oenological practices employed. All these factors modulate wine composition and, in turn, define the foam, colour and aroma that the final product will display. In addition, there are other factors such as storage conditions alien to the producer that will also have an impact on the quality observed by the consumer. In the case of foam it is true that its quality will also be conditioned by how the sparkling wine is serviced (washing and drying of glasses) (Liger-Belair *et al.*, 2008); hence, this organoleptic quality is also dependent of factors over which the oenologist has no control. Factors that go uncontrolled by the producer and that affect the quality of the commercial product are important because they have an impact on the consumer's opinion and can damage the efforts made by the wine producer during the winemaking process. For this reason and with the aim of compensating the negative effects of factors that have nothing to do with the oenologist, on occasions sparkling wines are served in glasses scratched intentionally at the bottom by the glassmaker. These scratches are microcavities that have diameters big enough to act as homogenous nucleation centres. This artificially triggered effervescence ensures foam formation, although some authors consider that bubbles arising from scratches are much more vigorous and chaotic than those arising from residual tiny cellulose fibres from cleaned glasses (Liger-Belair *et al.*, 2008). On the other hand, the main difference between the taste of sparkling wine and still wine is due to the presence of a continuous bubbling process. Some authors believe that flow patterns triggered by ascending bubbles are indeed suspected to play a major role in flavour and gas release, and can thus influence the taste of sparkling wine (Liger-Belair *et al.*, 2008).

19.3.1 Foam quality

Given that foam formation in wine is dependent upon its chemical composition and also on the equilibrium amongst numerous compounds, all factors involved in wine composition do have an effect on foam quality. For this reason, some works have been published relating foamability and foam quality to several of these factors, both intrinsic and extrinsic, such as the grape variety used (Andrés-Lacueva *et al.*, 1996a,b, 1997; López-Barajas *et al.*, 1998; Moreno-Arribas *et al.*, 2000), year of wine harvest (Senée *et al.*, 1999; Andrés-Lacueva *et al.*, 1996a), clarifying treatments, filtration and wine stabilization, wine blends to constitute the base wine and the ageing time in contact with yeast residues from the second fermentation.

Conditioning factors

There are some factors that are greater than all others in terms of their degree of influence over foam properties: geographical area of grape-growing region, viticulture techniques, grape picking methods, pressing type, vinification and

growing practices (Maujean *et al.*, 1990; Andrés-Lacueva *et al.*, 1996a, 1997). Of the studies carried out, the most outstanding are those by López-Barajas *et al.* (1997) and de Girbau-Solà *et al.* (2002a) in which the authors list the foam properties of wines with the maturation indices of the grapes used in their elaboration (ratio between soluble solids (°Brix) and titratable acidity (grams of tartaric acid per litre of juice)). According to López-Barajas *et al.* (1997), and as subsequently confirmed by Girbau-Solà *et al.* (2002a), the appropriate maturation indexes for foamability and stability are in the range 4–5.5. These authors consider that maturation indices above 5.5 provide the wine with a less optimal alcoholic content for the formation of foam than wine produced from grapes with a maturation index within the stated range. Other authors have compared the foaming properties of monovarietal wines to assess whether differences in the chemical composition of the grape variety provided different foaming characteristics (López-Barajas *et al.*, 1997, 1998, Andrés-Lacueva *et al.*, 1996a, 1997; Girbau-Solà *et al.*, 2002a,b). The reason for wanting to determine the foam capacity of wines as a function of their variety of origin is to enable the use foaming properties as an additional criterion when choosing the wines that will constitute the base wine (*coupage*), which will undergo a second fermentation. López-Barajas *et al.* (1998) showed that there was concordance between the physical parameters of foam from six wines obtained using the Mosalux device (HM, Σ , TS) and the values of those same parameters calculated using the values obtained for each monovarietal wine used in the blend or *coupage*. These authors conclude that the foam capacity of wine depends on the foaming properties of each of the wines used in the blend.

Foam changes from cloudy wines to base wines (Maujean *et al.*, 1990; Poinssaut, 1991; Robillard *et al.*, 1993; López-Barajas *et al.*, 1998) occur because active compounds for foam formation are modified during ranking, clarification, tartrate stabilization and filtrations. The oenological processes that entail the cleansing and stability of the wine eliminate colloids that play a role in its foamability and/or stability. These treatments should be the least drastic possible as, on the one hand, they ensure the wine's stability and, on the other hand, they damage the foam capacity of the wine.

The second fermentation that wine is subjected to in order to obtain sparkling wine causes a decrease in foamability of around 50% after the ninth month of ageing, but in some wines this second fermentation increases wine stability (Girbau-Solà *et al.*, 2002b). Once again, the precipitation of colloids pulled behind with yeast residues and/or with the bentonite that is frequently added to the titrating liquor may account for the decreased foamability seen in the sparkling wine when compared to the base wine (Maujean *et al.*, 1990; Andrés-Lacueva *et al.*, 1997; Vanrell *et al.*, 2007). Despite this decrease in foamability, several authors agree in that the greater the foam capacity of the base wine, the greater the foaming properties of the sparkling wine will be (Maujean *et al.*, 1990; Girbau-Solà *et al.*, 2002b). On the other hand, the increase in foam stability observed during the ageing period may be due to the hydrolysis of proteins and polysaccharides, thus obtaining shorter chains that will probably dissolve

much more easily in the film of the bubbles, thereby increasing their stability (Moreno-Arribas *et al.*, 2000).

In fact, sparkling wines considered to be of the highest quality, for which the best quality base wines have been chosen, have been allowed to age for two or even more years. It is to be assumed that a long ageing period does not favour foamability, but it will indeed favour the stability of the foam. Most likely, winemakers prefer foam to remain apparent for as long as possible, even if it is at the expense of less abundant foam.

Foam sensory descriptors

After having selected the best grapes, carefully elaborated the wine so that it maintains the best quality foam, colour and aroma, and once the wine has undergone a second fermentation and ageing, the oenologist expects the consumer to enjoy the best quality foam in the flute (Bidan *et al.*, 1986; Liger-Belair *et al.*, 2001). A small-sized bubble is expected to remain longer on the surface giving rise to a more elegant effervescence. This depends largely on the surface of the flute, on the concentration of CO₂ and on the temperature and depth of the liquid (Casey, 1995). Because of this, and in order to maintain the effervescence and foam for as long as possible, sparkling wines are served cool and in flute-type wine glasses.

There are no explicit rules to define and establish criteria to be followed in order to evaluate the effervescence and foam of sparkling wines. Most wine tasting forms include a section for visual inspection, using a hedonic scale (a lot/little, large/small, etc.) to assess effervescence and foam formation (Hardy, 1991). No evaluation criteria to grant more objective and normalized scores have yet been developed. Some authors have attempted to define what is understood by 'quality foam', what the necessary criteria to evaluate the foam are and what working methodology would be suitable in order to obtain reproducible results in wine tasting sessions (Obiols *et al.*, 1998). These authors conduct sessions with expert panellists to define the criteria for quality foam and effervescence and, in so doing, choose the appropriate descriptors for their evaluation (Obiols *et al.*, 1998; Gallart *et al.*, 2004). During the course of the sessions, these authors established some working conditions in relation to the temperature of sparkling wines and the wine glass used, the washing and drying of the wine glasses, the method used to open the bottle and the way in which the sparkling wine is poured into the glass (Obiols *et al.*, 1998). Once the standard working conditions had been established, the expert panellists decided on the criteria to be followed for the evaluation of the descriptors which, after some previous sessions, appeared to be discriminatory towards the quality (Gallart *et al.*, 2004). Table 19.4 lists the descriptors proposed by Gallart *et al.* (2004): initial foam, foam area, foam collar, bubble size, effervescence speed and global impression. Table 19.4 also shows the quality criteria for each descriptor and the quantification each of them receives depending on the criterion applied. Descriptors are evaluated in a value range between 1 and 3, except for the 'global impression' descriptor for which the authors propose a scoring range between 1 and 4. The lowest score (1) corresponds to the lowest quality and the highest score (3 to 4) to the highest quality. The descriptor 'global impression' was used upon the

Table 19.4 Discriminant descriptors used in foam sensory analysis of sparkling wines

Descriptors	Sensory analysis	
	Categories	Scores
Initial foam (foam formed immediately after pouring the sparkling wine into the glass)	Abundant	3
	Normal	2
	Poor	1
Foam area (proportion of wine surface area covered by bubbles)	Total	3
	Partial	2
	None	1
Foam collar (bubble lace formed around the wine surface, following the glass circumference)	Total	3
	Partial	2
	None	1
Bubble size (increases with height in the glass)	Small	3
	Medium	2
	Large	1
Effervescence speed (speed at which bubbles rise to the wine surface)	Fast	3
	Medium	2
	Slow	1
Global impression ^a (overall foam quality)	Very good	4
	Good	3
	Acceptable	2
	Bad	1

^a Global impression is a descriptor evaluated by the panelist at the end of each sample evaluation.

Source: Gallart *et al.* (2004).

judges' recommendation. The work conducted by Gallart *et al.* (2004) shows that in describing the overall quality of the foam, this descriptor reflected whether the other descriptors were representative of foam quality.

19.3.2 Colour changes during the elaboration process

The grape phenolic compounds that remain in the wine following the elaboration process condition its colour. Many factors, such as grape variety, climate and viticulture, as well as oenology practices modify the concentration and type of phenol in wines (Singleton and Trousdale, 1983; Jackson and Lombard, 1993; De la Presa-Owens *et al.*, 1995a,b; Romero-Pérez *et al.*, 1996). Additionally, phenolic compounds combine with other components found in wine such as sugars and proteins and are easily oxidized, leading to brown pigments. This last feature is one of the most important aspects in the elaboration of young white wines and base wines. The reason for this is that browning modifies the yellow colour that is so characteristic of these wines; and thus the higher the degree of oxidation the higher the browning, sometimes leading to an undesirable brownish colour (Singleton and Cilliers, 1995).

Thanks to the addition of sulphur dioxide, which blocks the polyphenol oxidase

enzyme (PPO, EC 1.10.3.1), the oxidation of phenolic compounds is a result of non-enzymatic reactions, favoured by an early appearance of quinones in the wine must before the addition of sulphur dioxide (Macheix *et al.*, 1991; Rossi and Singleton, 1996). These quinones, called primary quinones, are yellow in colour, sometimes leaning towards a brownish or brick colour. Quinones are very unstable, oxidant and strongly electrophilic substances (lacking in electrons) that can play a role in reactions of nucleophilic molecule aggregation (exceeding in electrons) and in oxide reduction reactions. Caffeoyl tartaric *o*-quinone is abundant in white wines for two reasons: caftaric acid is the most abundant cinnamic derivative and the main PPO substrate. Caffeoyl tartaric *o*-quinone can be used to piggyback nucleophile molecules such as glutathione (a tripeptide abundant in grapes), giving rise to 2-S-glutathionylcaftaric acid or GRP (*grape reaction product*). 2-S-glutathionylcaftaric acid (2-SGC) is a colourless compound. Because of this, glutathione can help avoid non-enzymatic browning, blocking caffeoyl tartaric *o*-quinone by preventing its action on other substrates. This is possible provided there is more glutathione than caftaric acid and that it can be combined with all the caffeoyl tartaric *o*-quinone that may result from the combination. Nonetheless, 2-SGC can itself become oxidized due to a coupled oxidation reaction of caffeoyl tartaric *o*-quinone, giving rise to 2-SGC-derived *o*-quinone. Caffeoyl tartaric *o*-quinone can also trigger this type of reactions with flavonoids. On the one hand, the aggregation reaction will allow the caftaric acid-derived quinone to attach to carbon 6 or 8 of the A nucleophilic ring; on the other hand, a coupled oxidation could occur leading to the formation of the flavonoid quinone and regeneration of the caftaric acid (Cheynier and Van Hulst, 1988; Singleton, 1988; Cilliers and Singleton, 1990). Both, products resulting from the coupling reaction between caffeoyl tartaric *o*-quinone and flavonoids or 2-SGC-derived quinone give rise to polymers of a brownish colour.

Wine cellars that make sparkling wines have a series of measures to avoid oxidation of phenolic compounds and resulting browning. In the Champagne and Cava regions, it is customary to pick grapes in boxes that hold a maximum of 20–40 kg to avoid rupturing the grape's skin, given that the breaking of the grape favours quinone formation due to PPO action. Extreme crushing of the grape is avoided and the juice is extracted following a soft pressing, such as that obtained from a pneumatic press. The must is sulphited immediately after the pressing because it facilitates the process of settling and keeps PPO blocked. Following alcoholic fermentation, the wine remains in the tank protected from oxygen by a layer of nitrogen that occupies the upper part of the tank. Each time the wine is racked to another tank, nitrogen is introduced to occupy this upper part. Must settling, racking, wine clarification, tartaric stabilization and the following filtration reduce the phenolic concentration. If clarification is performed with chemical agents, the loss of phenolic content depends on the chemical nature of the clarifying substance used (Puig-Deu *et al.*, 1999). During white vinification approximately 20% of phenolic compounds are lost, mainly hydroxycinnamic acids. The study conducted by Betés-Saura *et al.* (1996) demonstrated that, in the transition from must to wine, all determined phenolic compounds varied signifi-

cantly ($p < 0.001$), except for gallic acid, syringic acid, *trans*-ferulic acid and procyanidin B2, which do not present significant differences in must and wine.

During the ageing of sparkling wines, absorbance at 420 nm increases the higher the content of phenolic compounds is in the base wine used in the second fermentation (Ibern-Gómez *et al.*, 2000). If absorbance increases at 420 nm due to the oxidation of phenolic compounds, there should be a decrease in absorbances at 280 nm (total phenol index) and at 320 nm (hydroxycinnamate group index). As demonstrated by the study carried out by Ibern-Gómez *et al.* (2000), both absorbances (280 and 320 nm) decrease at the beginning of the ageing process up until month 12, then slightly increase, only to decrease again after month 27. Some authors believe that phenolic compounds can be absorbed by yeast cell walls (Mazauric and Salmon, 2006) and that, as the wall is broken down due to yeast autolysis, phenolic compounds are released again in the wine (Somers *et al.*, 1987; Cheynier *et al.*, 1990; Schneider, 1995). However, the varying value of these absorbances (280 and 320 nm) can also be due to other compounds. Yeast autolysis releases amino acids such as tyrosine and phenylalanine and nucleic acids that absorb at 280 nm (Ibern-Gómez *et al.*, 2000). Recently, a sugar-derived compound in Cavas has been determined, with maximum absorption at 280 nm and with a greater presence in sparkling wine, under CO₂ pressure (Bosch-Fusté *et al.*, 2009). This information points to the realization that interpreting the value of absorbance at 280 nm as a global measure of phenolic compound content leads to more error than originally thought when it was proposed. It is also true that not all phenolic compounds present in wine are determined when using HPLC.

Ageing sparkling wine in contact with lees is believed to stabilize the colour because phenolic compounds are combined with polysaccharides excreted by second fermentation yeasts (Escot *et al.*, 2001) and with mannoproteins from cell autolysis (Fornairon-Bonnefond *et al.*, 2002; Vidal *et al.*, 2003). The absorption of phenolic compounds in the cell walls or their combination with macromolecules would prevent their oxidation and would stabilize the wine's colour. In fact, a study performed by Girbau-Solà *et al.* (2002b) demonstrates that after ageing 12 Cavas (two *blanc de blanc*, five *blanc de noir* and five *rosé*) for 21 months, colour intensity (CI: sum of the absorbance at 420 and 520 nm, expressed in arbitrary units (a.u.) $\times 1000$) did not vary in five of them, but decreased by approximately 22% in the rest, and between 19 and 36% in the *rosé* sparkling wines. Interestingly, colour intensity did not vary in those sparkling wines whose base wine did not exceed 120 a.u. of colour intensity, except for in one *rosé* wine (190 a.u.). These results do not seem to coincide with the absorbance increases at 420 nm seen in six Cavas by Ibern-Gómez *et al.* (2000) after 15 months of ageing. However, there is still much to clarify about the autolytic process of dead yeast cells in the ageing conditions of sparkling wines (under CO₂ pressure). It has been proven that ageing increases the porosity of cell walls, the rupture of membranes and the release of cytoplasmic constituents such as hydrolytic enzymes and probably isomerases and oxidases (Charpentier and Feuillat, 1992; Cunier, 1997; Ibern-Gómez *et al.*, 2000; Guilloux-Benatier and Chassagne, 2003).

19.3.3 Aroma changes during the elaboration process

The different stages of elaboration hamper the quality of sparkling wines in the same way as in other wines. The critical quality points are: blending of varietal wines (*coupage*) that constitute the base wine (Torrens *et al.*, 2008), the yeast used in the second fermentation, the time that the wine stays in contact with lees and the type of expedition liquor added to the sparkling wine when preparing the finished product.

There are few studies on the evolution of the volatile fraction of sparkling wines during ageing, and there are differences of opinion among authors. Nevertheless, some aspects of these studies are noted as follows:

- (a) the increase of diethyl succinate and other ethyl esters of organic acids;
- (b) the decrease of acetates (mainly 2-phenylethyl, hexyl and isoamyl acetates);
- (c) the changes in the content of ethyl esters of fatty acids;
- (d) the increase of sulphur and nitrogenous compounds.

There are contradictory results concerning the evolution of ethyl esters during the contact with lees. Some authors (Margheri *et al.*, 1984) have observed a decrease of fatty acids, explained by the absorption of these compounds in the cellular walls of yeast, whilst Mazzoleni and Colagrande (1993) have described an increase of short-chain fatty acids during ageing. These apparent divergences could be explained by the time variations considered in each study and by the different types of sparkling wines used (elaborated with different varieties and techniques). In addition, an exhaustive characterization of the volatile fraction which entails the analysis of a high number of real samples has not been carried out in these works.

However, most studies concur that yeast autolysis plays an important role in the development of aroma properties. Sparkling wines elaborated using the traditional practice of *sur lie* ageing include a long time of contact between lees and wine (years). During this period, different compounds such as lipids, carbohydrates, amino acids, peptides and volatiles could be released in a process known as autolysis (Charpentier and Feuillat, 1992; Dos Santos *et al.*, 2000; Feuillat, 2003). This process plays a very important role in sparkling wines *bouquet* conformation (Fleet, 2003; Guilloux-Benatier and Chassagne, 2003). Lees are composed mainly of yeast cells (Fornairon-Bonnefond *et al.*, 2002), and it is the outer part of the yeast which is in constant contact with the wine. This cell wall makes up between 25 and 50% of cell volume and is structured in an inner network of ramified glucans and in an outer layer of mannoproteins (Caridi, 2006). These compounds confer to cells physico-chemical properties that enable yeasts and lees to interact with other compounds. The sorption capacity of the surface of yeast and lees in several organic substances has been proven (Jiménez and Ancín, 2007). Moreover, the sorptive properties of this biological material were proven by adding yeast cell walls to a model wine containing esters, higher alcohols and β -ionone, obtaining an increased retention in solution (Lubbers *et al.*, 1994). The yeast could act as a lipophilic matrix that could retain a large amount of aromas which, over time and by denaturation of the cell walls, could release aroma compounds to the wine medium.

There is still a lack of studies considering the evolution of sparkling wine aroma

during the shelf life (Escudero and Etiévant, 1999; Vasserot *et al.*, 2001). The effect of exposure to fluorescent light on sparkling wine and the still wine from which it was made was studied in green and flint (clear) bottles by Dozon and Noble (1989). The lightstruck aroma (*goûts de lumière*) produced on exposure of still and sparkling wines to fluorescent light was characterized by descriptive analysis. With increased time of exposure, a decrease in citrus aroma intensity occurred, while the intensity of cooked cabbage, corn nuts, wet dog/wet wool and soy/marmite aromas increased.

19.4 Quality control

Organoleptic quality of sparkling wines is mainly determined by a sensory analysis. The variety of sensory impressions that are absorbed by the senses of sight, smell and taste of expert wine tasters cannot be replaced by any instrument. Hence, even today the most effective means to assess organoleptic quality is usually by subjective evaluation. When defining sensory descriptors that characterize wine or when detecting aromatic defects, expert wine tasters are usually infallible. However, the subjectivity could be a problem, and it is especially evident when describing a group of sensations that are different to the aromatic patterns the wine taster is familiar with. On the other hand, experts can guarantee organoleptic quality of the finished product before it is launched commercially, though quality can be altered if distribution and storage conditions are inadequate.

Wine cellars would like to have objective parameters to complement the expert's subjective assessment. These parameters exist for hygienic quality, but organoleptic quality is more complex and only some indicators with limited application have been defined.

In order to evaluate the foaming properties of sparkling wines in a wine glass, a visual panoramic method based on the analysis of images obtained by video cameras has been proposed (Machet *et al.*, 1994). This method consists of three video cameras that record images of the wine's surface and the ascension of bubbles. These images are processed in a computer that provides information on foam collar width, number of bubbles on the surface, size of ascending bubbles and effervescence speed (Viaux *et al.*, 1994; Marchal *et al.*, 1996; Liger-Belair *et al.*, 2001). With this system, foaming behaviour of sparkling wines can be directly evaluated in a wine glass. This method of processed images represents an advantage over the Mosalux apparatus (Maujean *et al.*, 1990) (see the section on chemical characteristics above), given that the latter requires the use of a graduated cylinder where carbonic gas is injected in order to make the wine foam. Therefore, when applied to sparkling wines, these must first be degasified. The Mosalux method measures certain physical parameters of foam to infer the foaming behaviour of sparkling wine that will most likely occur in the wine glass. The advantages of the Mosalux method over the visual panoramic method are that it is simpler, cheaper and, according to data derived from Gallart *et al.* (2004), the physical measures of foam are correlated with the sensory descriptors of foam. The

foaming properties of 51 sparkling wines from different grape varieties, harvests and ageing times were evaluated by applying both the Mosalex gas-sparging method (Maujean *et al.*, 1990; Poinssaut, 1991; Gallart *et al.*, 1997) and sensory analysis (Obiols *et al.*, 1998; Gallart *et al.*, 2004). The foam's physical parameters and sensory descriptors are statistically correlated ($p > 0.05$); the greater the values of foamability (HM), average bubble lifetime or Bikerman's coefficient (Σ) and foam stability time (TS), the higher the scores obtained by the different descriptors (see Table 19.4).

The classic and simple parameter to measure colour is the colour index (CI: sum of the absorbance at 420 and 520 nm). In base wines it ranges between 80 and 140 a.u. depending on whether it is a *blanc de blanc* or a *blanc de noir*, while in rosé sparkling wines values are located around 200 a.u. However, there is a more complex method that provides greater information on the chromatic characteristics of wine called the CIE Lab method, whose procedure has been described and approved by the International Organization of Vine and Wine (OIV) during the III General Assembly, which took place in Logroño (Spain) in June of 2006 (OENO/SCMA/01/202 resolution). Chromatic characteristics of wine are defined by three attributes of visual sensation: tonality – colour itself (red, yellow, green or blue), luminosity, since a certain colour can be more or less luminous, and chromatism or level of colouring, referring to the higher or lower intensity of colour. Using a spectrophotometer and a specific computer program, trichromatic components X, Y, Z could be determined to calculate, with the appropriate mathematical algorithms, colorimetric coordinates: clarity (L^* related to luminosity), tone (H^* for which red/green colour components, a^* and yellow/blue components, b^* are used), and level of colouring or chromatism derived from previous data (Wiszecki and Stiles, 1982; Kowaliski, 1990; Sève, 1996; Echávarri *et al.*, 2000). The procedure recommended by the OIV was approved after confirming the method's reproducibility in two interlaboratory trials.

Quality control of aroma compounds continues to be rather difficult. The three procedures most commonly used are: analysis of the volatile and semi-volatile compounds (Francioli *et al.*, 2003; Bosch-Fusté *et al.*, 2007; Campo *et al.*, 2008); employment of sensory descriptive analysis (Vannier *et al.*, 1999; Escudero *et al.*, 2000); and sniffing methods (Comuzzo *et al.*, 2006; Campo *et al.*, 2008) such as gas chromatography (GC) in conjunction with olfactometry detection (GCO).

A primary target of the investigation of olfactory products is the qualitative and quantitative analysis of the chemical compounds that confer their characteristic aroma. The disparity in the chemical nature and the concentrations of the compounds that make up the aroma of sparkling wine makes their analysis difficult and laborious; bearing in mind that no analytical method is universally applicable and capable of giving a completely representative aromatic profile. Aroma analysis of sparkling wines is usually carried out by means of solvent extraction (Bosch-Fusté *et al.*, 2007). The best known method among new extraction methods aimed at avoiding the disadvantages posed by the most classic methods (economic and time costs, ecological damage, interferences and artefacts, etc.) is solid phase microextraction (SPME). This is a system designed in the mid-1990s for the

analysis of pollutants in residual waters that is being increasingly applied in wine analysis, mainly using polydimethylsiloxane (PDMS) or carboxen (CW) as fibre polymers (Comuzzo *et al.*, 2006; Bosch-Fusté *et al.*, 2007). Recently, another solventless method based on the sorptive capacity of polymers has been developed: stir bar sorptive extraction (SBSE). SBSE has been widely used, specifically in association with thermal desorption (TD). However, TD units are expensive devices and, although this approach presents a remarkable sensitivity, it is not the most indicated to analyze thermolabile compounds. For this type of analysis, Coelho *et al.* (2009) have developed a SBSE method with liquid desorption (LD) applied for the quantification of varietal and fermentative volatile compounds in sparkling wines with promising results.

The classic system of analysis of an aroma extract involves the separation of its components in a gas chromatograph (GC) equipped with a capillary column of high effectiveness, and coupled to a mass spectrometer (MS). When the compounds are separated in the column, they enter the spectrometer, where they are fragmented by accelerated electrons at 70 eV of potential (electronic impact – EI) and the fragments are subsequently separated in a magnetic field. Once detected and quantified by order of molecular weight, the fragments are filed in the same way as matrix files and, if necessary, the mass versus the abundance curve is represented in a bar chart (MS spectrum). With this type of chart, a digital library can be created that will later be used to identify these compounds in subsequent investigations. If mass spectrometry is used as the detection technique, the spectrum of a separated compound in the chromatograph is compared with commercial digital libraries of reported MS spectra. In order to determine the qualitative composition of wine aroma, these databases are needed. Of these libraries, the most widely used ones are the Wiley MS database and the MS spectrum databank of the National Institute of Standards and Technology (NIST).

There are several techniques such as aroma extract dilution analysis (AEDA) or the charm analysis that can be applied to gas chromatography in order to help determine if a volatile compound is odour-active in an aroma extract. These GCO techniques (Priser *et al.* 1997; Rapp, 1998; Escudero and Etiévant, 1999; Escudero *et al.*, 2000) provide detailed information on the impact of the aroma of an unknown compound on wine, allowing us to devote our efforts to its identification. Therefore, the effect of yeast autolysis on the organoleptic properties of wine should be re-evaluated using techniques such as GCO complemented with sensory descriptive analysis (De la Presa-Owens *et al.*, 1998; Vannier *et al.*, 1999; Escudero *et al.*, 2000). Chemometric analysis such as multivariate statistically procedures is needed to correlate instrumental results and sensory perception data (Vannier *et al.*, 1999; Noble and Ebeler, 2002; Campo *et al.*, 2005).

19.5 Conclusions

There are two main sparkling wine elaboration methods: the *granvas*, bulk or *charmat* method, where base wine fermentation takes place in an isobaric deposit,

and the *champanoise*, *traditional* or *classic* method, in which base wine fermentation takes place in a sealed bottle. Two key features differentiate the organoleptic qualities of these wines from any other type of wine. Firstly, the refermentation of the base wine, and secondly, wine ageing in contact with lees and under approximately 6 atm of CO₂ pressure. For *granvas*, ageing is usually short, lasting about 3 months, and results in lighter sparkling wines with younger aromas than those obtained with the *champanoise* method, in which ageing takes a long time (at least 9 or 12 months depending on the sparkling wine). Traditionally, ageing in contact with lees was only applied in some white and sparkling wines. This type of ageing is now much more frequent and is applied to boost the organoleptic properties of wine.

Refermentation of the base wine requires that the selected yeast meet a series of characteristics such as: capacity to be developed in a hydro alcoholic environment of close to 10° v/v and capacity to complete fermentation under 6 atm of CO₂ pressure until reaching the desired alcohol content for the sparkling wine. It is also advisable that yeast cells are able to agglomerate and precipitate with the aim of facilitating their release during the disgorgement phase.

Long ageing in contact with lees and in a CO₂ atmosphere confers upon the wine its peculiar organoleptic properties. The formation of foam is possible because it is a saturated CO₂ solution and because a series of compounds form the film surrounding the gas, resulting in bubbles. Research conducted on the formation of foam coincides in determining that the appearance of more or less foam does not depend on isolated compounds, but on the balance between many of the compounds found in wine. However, it seems the second fermentation and ageing does not favour foaming properties due, most likely, to the fact that the refermentation process increases alcohol content to values where alcohol loses tensoactive properties, to the diminution of the proportion of ethyl esters with respect to free fatty acids because the metabolism of yeasts generates new fatty acids and, lastly, to lee sedimentation, for which on many occasions bentonite is added to the *tirage* liquor, dragging cells and also macromolecules such as proteins. Despite this, some authors have confirmed that the greater the base wine's foaming capacity is, the greater the sparkling wine's foam properties are.

The yellow colour of sparkling wine generally slowly intensifies or slightly browns after 18 months of ageing. This probably occurs because phenolic compounds responsible for colour have an affinity to adhere to the cell walls of lees or even to precipitate with proteins. As cell walls degrade, a seemingly slow autolytic process, absorbed phenolic compounds are released. Some authors believe that yeast lees act by protecting wine from oxidation, but it is likely that this protection is limited and may be reduced when lees reach an advanced stage of decomposition.

The special ageing in contact with lees of some type of sparkling wines gives these wines a more complex volatile profile since autolytic and enzymatic reactions may take place. This flavour is empirically described by winemakers as presenting yeasty, sweet, and toasty notes, but the compounds that influence the sensory evolution, from base wine until aged sparkling wine, are not yet well

known. The analysis of the sparkling wine aroma by hyphenated techniques, such as SPME, SBSE, GCO and descriptive sensorial analysis, indicates that the sensory profile and the aroma compounds of sparkling wines depend on the quality of base wines and on the time and/or ageing characteristics. The most important ethyl esters (mainly fatty acid esters) and acetates, fruity compounds, decreased during *sur lie* ageing. This decrease could be caused by the synthesis–hydrolysis activity of esterases released by the lees. Moreover, some compounds with high sensory impact, such as the ethyl esters of dicarboxylic acids (ethyl succinate), norisoprenoids (1,1,6-trimethyl-1,2-dihydronaftalene TDN and vitispiranes), furan compounds (ethyl furoate, 2-acetyl-furan and furaneol) and mercaptanes (2-methyltiofen-3-one and furfurylthiol) appear to have the greatest effect on flavour characteristics. The biochemical mechanisms that take part in the adsorption and/or release capacities of aroma compounds interacting with the yeast cell-walls during ageing would be the most important factors determining the organoleptic quality of sparkling wines.

19.6 References

- Andrés-Lacueva C, López-Tamames E, Lamuela-Raventós RM, Buxaderas S and Torre-Boronat MC (1996a), 'Characteristics of sparkling base wines affecting foam behavior', *J Agric Food Chem*, **44**, 989–995.
- Andrés-Lacueva C, Gallart M, López-Tamames E and Lamuela-Raventós RM (1996b), 'Influence of variety and aging on foaming properties of sparkling wines (Cava) 1', *J Agric Food Chem*, **44**, 3826–3929.
- Andrés-Lacueva C, Lamuela-Raventós RM, Buxaderas, S and Torre-Boronat MC (1997), 'Influence of variety and aging on foaming properties of sparkling wine (Cava) 2', *J Agric Food Chem*, **47**, 2520–2525.
- Andrés-Lacueva C, Ibern-Gómez M, Lamuela-Raventós RM, Buxaderas S and de la Torre-Boronat MC (2002), 'Cinnamates and resveratrol content for sparkling wine characterization', *Am J Enol Vitic*, **53**, 147–150.
- Baderschneider B and Winterhalter P (2001), 'Isolation and characterization of novel benzoates, cinnamates, flavonoids, and lignans from Riesling wines and screening for antioxidant activity', *J Agric Food Chem*, **49**, 2788–2798.
- Barcenilla J M, Martín-Álvarez P J, Vian A and Gonzalez R (2003), 'Characterization of yeast strains on the basis of autolytic activity and volatile compounds', *Food Sci Technol Int*, **9**, 95.
- Bautista R, Fernández E and Falqué E (2007), 'Effect of the contact with fermentation-lees or commercial lees on the volatile composition of white wines', *Eur Food Res Technol*, **224**, 405–413.
- Betés-Saura C, Andrés-Lacueva C and Lamuela-Raventós RM (1996), 'Phenolics in white free run juices and wines from Penedès by high-performance liquid chromatography. Changes during vinification', *J Agric Food Chem*, **44**, 3040–3046.
- Bidan P, Feuillat M and Moulin PJ (1986), 'Vins mousseux, rapport de France', *Bull OIV*, **59**, 563–626.
- Bikerman JJ (1938), 'The unit of foaminess', *Trans Farad Soc*, **34**, 634–638.
- Blanchard L, Tominaga T and Dubourdieu D (2001), 'Formation of furfurylthiol exhibiting a strong coffee aroma during oak barrel fermentation from furfural released by toasted staves', *J Agric Food Chem*, **49**, 4833–4835.
- Bosch-Fusté J, Riu-Aumatell M, Guadayol JM, Caixach J, López-Tamames E and Buxaderas

- S (2007), 'Volatile profiles of sparkling wines obtained by gas chromatography-mass spectrometry (GC-MS) analysis', *Food Chem*, **105**, 428–435.
- Bosch-Fusté J, Sartini E, Flores-Rubio C, Caixach J, López-Tamames E and Buxaderas S (2009), 'Viability of total phenol index value as quality marker of sparkling wines Cavas', *Food Chem*, **114**, 782–790.
- Bosch-Fusté J, Vichi S, Riu-Aumatell M, Buxaderas S and López-Tamames E (2008), 'Determination of furan compounds in Cava sparkling wine', in Chassagne D (ed.), *Wine Active Compounds 2008*, Oenoplurimedia, 107–108.
- Bourne MC (1982), *Food Texture and Viscosity: Concept and Measurement*, New York, Academic Press.
- Bramforth CW (1985), 'The foaming properties of beer', *J Inst Brew*, **91**, 370–383.
- Brissonnet F and Maujean A (1991), 'Identification of some foam-active compounds in Champagne base wines', *Am J Enol Vitic*, **42**, 150–152.
- Brissonnet F and Maujean A (1993), 'Characterization of foaming proteins in Champagne base wines', *Am J Enol Vitic*, **44**, 297–301.
- Buxaderas S and López-Tamames E (2003), Wines production of sparkling wines, in Caballero B, Trugo L and Finglas P (eds), *Encyclopedia of Food Sciences and Nutrition*, London, UK, Elsevier Science, vol. **10**, 6203–6209.
- Câmara JS, Marques JC, Alves MA and Silva-Ferreira AC (2004), '3-Hydroxy-4,5-dimethyl-2(5H)-furanone levels in fortified Madeira wines: relationship to sugar content', *J Agric Food Chem*, **52**, 6765–6769.
- Câmara JS, Alves MA and Marques JC (2006), 'Changes in volatile composition of Madeira wines during their oxidative ageing', *Anal Chem Acta*, **563**, 188–197.
- Campo E, Ferreira V, Escudero A and Cacho J (2005), 'Prediction of the wine sensory properties related to grape variety from dynamic-headspace gas chromatography-olfactometry data', *J Agric Food Chem*, **53**, 5682–5690.
- Campo E, Cacho J and Ferreira V (2008), 'The chemical characterization of the aroma of dessert and sparkling white wines (Pedro Ximénez, Fino, Sauternes, and Cava) by gas chromatography-olfactometry and chemical quantitative analysis', *J Agric Food Chem*, **56**, 2477–2484.
- Caridi A (2006), 'Enological functions of parietal yeast mannoproteins', *Antonie van Leeuwenhoek*, **89**, 417–422.
- Casey JA (1988), 'Bubble size bubble frequency and rate of gas loss in sparkling wine', *Aust Grapegrow Winemak*, **295**, 19–22.
- Casey JA (1995), 'Effervescence in sparkling wines, the sequel', *Aust Grapegrow Winemak*, **378a**, 37–47.
- Chalier P, Angot B, Delteil D, Doco T and Gunata Z (2007), 'Interactions between aroma compounds and whole mannoprotein isolated from *Saccharomyces cerevisiae* strains', *Food Chem*, **100**, 22–30.
- Chamkha M, Cathala B, Cheynier W and Douillard R (2003), 'Phenolic composition of Champagnes from Chardonnay and Pinot noir vintages', *J Agric Food Chem*, **51**, 3179–3184.
- Charpentier C and Feuillat M (1992), 'Yeast autolysis', in Fleet GH (ed), *Wine Microbiology and Biotechnology*, Chur, Switzerland, Harwood Academic, 225–242.
- Charpentier C and Freyssinet M (1989), 'The mechanism of yeast autolysis in wine', *Yeast*, **5**, S181–S186.
- Chatonnet P, Dubourdieu D, Boidron JN and Pons M (1992), 'The origin of ethylphenols in wines', *J Sci Food Agric*, **60**, 165–78.
- Chatonnet P, Dubourdieu D, Boidron JN and Lavigne V (1993), 'Synthesis of volatile phenols by *Saccharomyces cerevisiae* in wines', *J Sci Food Agric*, **62**, 191–202.
- Cheftel JC, Cuq IL and Lorient D (1989), 'Propiedades funcionales de las proteínas', in Cheftel JC, Cuq, IL and Lorient D (eds), *Proteínas alimentarias*, Zaragoza, Spain, Acribia S.A., 88–102.
- Cheynier VF and Van Hulst MWJ (1988), 'Oxidation of *trans*-caftaric acid and 2-S-

- glutathionylcaftaric acid in model solution', *J Agric Food Chem*, **36**, 10–15.
- Cheyrier V, Rigaud J, Souquet JM, Duprat F and Mountounet M. (1990), 'Must browning in relation to the behavior of phenolic compounds during oxidation', *Am J Enol Vitic*, **41**, 346–349.
- Cheyrier V, Mountounet, M and Sarni-Manchado P (1998), 'Les composés phénoliques, in Flanzky C (ed), *Oenologie: fondements scientifiques et technologies*, Cachan, France, Lavoisier, 123–162.
- Cilliers JLL and Sinleton VL (1990), 'Nonenzymatic autoxidative reactions of caffeic acid in wines', *Am J Enol Vitic*, **41**, 84–86.
- Codrington I (1986), Effect of alcohol protein and fermentation rate on bubble size, in Lee TH and Lester D C (eds), *Production of Sparkling Wine by the Méthode Champenoise: Proceedings of a Seminar*, Adelaide, SA, Australian Society of Viticulture and Oenology, 157–168.
- Coelho E, Coimbra MA, Nogueira JMF and Rocha SM (2009), 'Quantification approach for assessment of sparkling wine volatiles from different soils, ripening stages, and varieties by stir bar sorptive extraction with liquid desorption', *Anal Chim Acta*, **635**, 214–221.
- Comelles F, Bosch MP and Castro JJ (1991), 'Aspectes físico-químics relacionats amb la capacitat escumant de diferents varietats de raïm', *ACE Revista d'Enologia*, **23**, 7–16.
- Comuzzo P, Tat L, Tonizzio A and Battistutta F (2006), 'Yeast derivatives (extracts and autolysates) in winemaking: release of volatile compounds and effects on wine aroma volatility' *Food Chem*, **99**, 217–230.
- Culleré L, Cacho J and Ferreira V (2007), 'An assessment of the role played by some oxidation-related aldehydes in wine aroma', *J Agric Food Chem*, **55**, 876–881.
- Cunier C (1997), 'Ceppi di lievito e composizione fenolica dei vini rossi', *Vignevini*, **7–8**, 39–42.
- De la Presa-Owens C, Lamuela-Raventós RM, Buxaderas S and de la Torre-Boronat MC (1995a), 'Differentiation and grouping characteristics of varietal grape musts from Penedès region (I)', *Am J Enol Vitic*, **46**, 283–291.
- De la Presa-Owens C, Lamuela-Raventós RM, Buxaderas S and de la Torre-Boronat MC (1995b), 'Characterization of Macabeo, Xarel-lo and Parellada white wines from the Penedès region (II)', *Am J Enol Vitic*, **46**, 529–541.
- De la Presa-Owens C, Schlich P, Davies H D and Noble A C (1998), 'Effect of méthode champenoise process on aroma of four *V. vinifera* varieties', *Am J Enol Vitic*, **49**, 289–294.
- Deloire A, Kraeva E, Dai GH, Renault AS, Richard J, Chatelain C, Charbonneau A and Andary C (1998), 'Les mécanismes de defense de la vigne, des utilisations possibles pour lutter contre les pathogènes', *Phytoma*, **510**, 46–51.
- Dickinson E (1994), 'Colloidal aspects of beverages', *Food Chem*, **51**, 343–347.
- Díez de Bethencourt C, Sanjosé-González ML and Junquera B (1993), 'El pardeamineto enzimático en uva y vino', *La Semana Vitivinícola*, **2423**, 171–181.
- Dos Santos AM, Feuillat M and Charpentier C (2000), 'Flor yeast metabolism in a model system similar to the cellar aging of the French "vin jaune"', evolution of some by-products nitrogen compounds and polysaccharides', *Vitis*, **39**, 129–134.
- Dozon NM and Noble AC (1989), 'Sensory study of the effect of fluorescent light on a sparkling wine and its base wine', *Am J Enol Vitic*, **40**, 265–271.
- Dussaud A, Robillard B, Carles B, Duteurtre B and Vignes-Adler, M (1994), 'Exogenous lipids and ethanol influences on the foam behavior of sparkling base wines', *J Food Sci*, **59**, 148–151, 167.
- EC (2008), Commission Directive 2008/479/EC of 29 April 2008 on the common organization of the market in wine, *Official Journal of the European Union*, **L148**, 6.06.2008, 1–61.
- Echávarri JF, Ayala F and Negueruela AI (2000), 'Influence du pas de mesure dans le calcul des coordonnées de couleur du vin', *Bull OIV*, **831–832**, 370–378.
- Escot S, Feuillat M, Dulau L and Charpentier C (2001), 'Release of polysaccharides by yeast

- and the influence of released polysaccharides on colour stability and wine astringency', *Aust J Grape Wine Res*, **7**, 153–159.
- Escudero A and Etiévant P (1999), 'Effect of antioxidants on the flavor characteristics and the gas chromatography/olfactometry profiles of Champagne extracts', *J Agric Food Chem*, **47**, 3303–3308.
- Escudero A, Charpentier M and Etiévant P (2000), 'Characterization of aged Champagne wine aroma by GC–O and descriptive profile analyses', *Sci Alim*, **20**, 331–346.
- Feuillat M (1998), 'Autolyse de levures', in: Flanzly C (ed.), *Oenologie: Fondements Scientifiques et Technologiques*, Paris, France, Lavoisier, 444–454.
- Feuillat M (2003), 'Yeast macromolecules: origin, composition and enological interest', *Am J Enol Vitic*, **54**, 211–213.
- Fleet GH (2003), 'Yeast interaction and wine flavour', *Int J Food Microbiol*, **86**, 11–22.
- Fornairon-Bonnefond C, Camarasa C, Mouneton M and Salmon JM (2002), 'New trends on yeast autolysis and wine ageing on lees: a bibliographic review', *J Int Sci Vigne Vin*, **36**, 49–69.
- Francioli S, Torrens J, Riu-Aumatell M, López-Tamames E and Buxaderas S. (2003), 'Volatile compounds by SPME–GC as age markers of sparkling wines', *Am J Enol Vitic*, **54**, 158–162.
- Gallart M, Francioli S, Viu-Marco A, López-Tamames E and Buxaderas S (1997), 'Foam measurements in wines: comparison of parameters obtained by gas sparging method', *J Agric Food Chem*, **45**, 4687–4690.
- Gallart M, López-Tamames E, Suberbiola G and Buxaderas S (2002), 'Influence of free fatty acids on wine foaming', *J Agric Food Chem*, **50**, 7042–7045.
- Gallart M, Tomás X, Suberbiola G, López-Tamames E and Buxaderas S (2004), 'Relationship between foam parameters obtained by the gas-sparging method and sensory evaluation of sparkling wines', *J Sci Food Agric*, **84**, 127–133.
- García E, Chacón JL, Martínez J and Izquierdo PM (2003), 'Changes in volatile compounds during ripening in grapes of Airén, Macabeo and Chardonnay white varieties grown in La Mancha Region (Spain)', *Food Sci Technol Int*, **9**, 33–41.
- Genovese A, Gambuti A, Piombino P and Moio L (2007), 'Sensory properties and aroma compounds of sweet Fiano wine', *Food Chem*, **103**, 1228–1236.
- Girbau-Solà MT, López-Tamames E, Buján J and Buxaderas S (2002a), 'Foam aptitude of Trepát and Monastrell red varieties in Cava elaboration, 1, base wine characteristics', *J Agric Food Chem*, **50**, 5596–5599.
- Girbau-Solà MT, López-Barajas M, López-Tamames E and Buxaderas S (2002b), 'Foam aptitude of Trepát and Monastrell red varieties in Cava elaboration, 2, second fermentation and aging', *J Agric Food Chem*, **50**, 5600–5604.
- Gomerieux T (1989), 'Degassing and maintenance of effervescence in champagne wines', *Rev Fr Oenol*, **116**, 15–20.
- Guilloux-Benatier M and Chassagne D (2003), 'Comparison of components released by fermented or active dried yeasts after aging on lees in a model wine', *J Agric Food Chem*, **51**, 746–751.
- Guggenberger J and Kleber W (1964), *Über den Mechanismus des Wildwredens von Bier*, in proceedings of the European Brewery Convention Congress 1963, Amsterdam, the Netherlands, Elsevier, 299–319.
- Hardy G (1991), Production methods in Champagne, in Henick-Kling T, Baroody E and Woodbury R (eds), *Proceedings of the 1989 New York Sparkling Wine Symposium*, Geneva, NY, New York State Agricultural Experimental Station, 23–27.
- Ibern-Gómez M, Andrés-Lacueva C, Lamuela-Raventós RM, Buxaderas S, Singleton VL and de la Torre-Boronat, MC (2000), 'Browning of Cava (sparkling wine) during aging in contact with lees due to the phenolic composition', *Am J Enol Vitic*, **51**, 29–36.
- Jackson DL and Lombard PB (1993), 'Environmental and management practices affecting grape composition and wine quality. A review', *Am J Enol Vitic*, **44**, 409–430.
- Jiménez Moreno N and Ancín Azpilicueta C (2007), 'Binding of oak volatile compounds by

- wine lees during simulation of wine ageing', *Lebensm.-Wiss Technol*, **40**, 619–624.
- Jordan AD and Napper DH (1988), Some aspects of the physical chemistry of bubble and foam phenomena in sparkling wines, in Lee T (ed.), *Proceedings of the Sixth Australian Wines Industry Technical Conference*, Adelaide, SA, Australian Industrial Publishers, 237–246.
- Kinsella JE (1981), 'Functional properties of proteins: possible relationships between structure and function in foams', *Food Chem*, **7**, 273–288.
- Kowaliski P (1990), *Vision et Mesure de la Couleur*, Paris, France, Masson.
- Lao C, Santamaría A, López-Tamames E, Buján J, Buxaderas S and de la Torre-Boronat MC (1999), 'Effect of grape pectic enzyme treatment on foaming properties of white musts and wines', *Food Chem*, **65**, 169–173.
- Leroy M.J, Charpentier M, Duteurtre B, Feuillat M and Charpentier C (1990), 'Yeast autolysis during Champagne aging', *Am J Enol and Vitic*, **41**, 21–28.
- Liger-Belair G, Lemaresquie, H, Robillard B, Duteurtre B and Jeandet P (2001), 'The secrets of fizz in Champagne wines: a phenomenological study', *Am J Enol Vitic*, **52**, 88–92.
- Liger-Belair G, Beaumont F, Vialatte MA, Jégou S, Jeandet P and Polidori G (2008), 'Kinetics and stability of the mixing flow patterns found in Champagne glasses as determined by laser tomography techniques: likely impact on Champagne tasting', *Anal Chim Acta*, **621**, 30–37.
- López-Barajas M, Viu-Marco A, López-Tamames E, Buxaderas S and de la Torre-Boronat MC (1997), 'Foaming in grape juices of white varieties', *J Agric Food Chem*, **45**, 2526–2529.
- López-Barajas M, López-Tamames E, Buxaderas S and de la Torre-Boronat MC (1998), 'Effect of vinification and variety on foam capacity of wine', *Am J Enol Vitic*, **49**, 397–402.
- López-Barajas M, Viu-Marco A, López-Tamames E, Buxaderas S, Tomás X and de la Torre-Boronat MC (1999), 'Prediction of wine foaming', *J Agric Food Chem*, **47**, 3743–3748.
- López-Barajas M, López-Tamames E, Buxaderas S, Suberbiola G and de la Torre-Boronat MC (2001), 'Influence of wine polysaccharides of different molecular mass on wine foaming', *Am J Enol Vitic*, **52**, 146–150.
- Lubbers S, Voilley A, Feuillat M and Charpentier C (1994), 'Influence of mannoproteins from yeasts on the aroma intensity of a model wine', *Lebensm.-Wiss Technol*, **27**, 108–114.
- Lusk LT, Goldstein H and Ryder D (1995), 'Independent role of beer proteins, melanoidins and polysaccharides in foam formation', *J Am Soc Brew Chem*, **53**, 93–103.
- Maeda K, Yokoi S, Kamada K and Kamimura M (1991), 'Foam stability and physicochemical properties of beer', *J Am Soc Brew Chem*, **49**, 14–18.
- Macheix JJ, Sapis JC and Fleuriet A (1991), 'Phenolic compounds and polyphenoloxidase in relation to browning in grapes and wines', *CR Food Sci Nutr*, **30**, 441–486.
- Machet F, Robillard B and Duteurtre B (1994), 'Application de l'analyse d'images à l'évaluation de la stabilité de la mousse de vins mousseux', *Sci Alim*, **13**, 73–87.
- Malvy J, Robillard B and Duteurtre B (1994), 'Influence des protéines sur le comportement de la mousse des vins de Champagne', *Sci Aliment*, **14**, 87–98.
- Marchal R, Bouquelet S and Maujean A (1996), 'Purification and partial biochemical characterization of glycoproteins in a Champenoise Chardonnay wine', *J Agric Food Chem*, **44**, 1716–1722.
- Marchand S, de Revel G and Bertrand A (2000), 'Approaches to wine aroma: release of aroma compounds from reactions between cysteine and carbonyl compounds in wine', *J Agric Food Chem*, **48**, 4890–4895.
- Margheri G, Gianotti L and Pellegrini R (1984), 'Quality factors of young light wines. Influence of assimilable nitrogen of musts and of aromatic compounds of wines', *Enotecnico*, **20**, 448–457.
- Martínez-Rodríguez AJ and Polo MC (2000), 'Characterization of the nitrogen compounds released during yeast autolysis in a model wine system', *J Agric Food Chem*, **48**, 1081–1085.
- Martínez-Rodríguez AJ, Carrascosa AV and Polo MC (2001), 'Release of nitrogen com-

- pounds to the extracellular medium by three strains of *Saccharomyces cerevisiae* during induced autolysis in a model wine system', *Int J Food Microbiol*, **68**, 155–160.
- Maujean A (1989), 'Histoire des Bulles', *Rev Fr Oenol*, **120**, 11–17.
- Maujean A, Poinsaut H, Dantan H, Brissonnet F and Cossiez E (1990), 'Study of the performance and quality of the foam in sparkling wines, II Perfecting of measuring technique for foaming ability, performance and stability of the foam in sparkling wines', *Bull OIV*, **711–712**, 405–426.
- Mazauric JP and Salmon JM (2006), 'Interactions between yeast lees and wine polyphenols during simulation of wine aging: II Analysis of desorbed polyphenol compounds from yeast lees', *J Agric Food Chem*, **54**, 3876–3881.
- Mazzoleni V and Colagrande O (1993), 'Composition of bottle-fermented Italian sparkling wines', *Ital J. Food Sci*, **5**, 129–137.
- Molan PC, Edwards M and Eschenbruch R (1982), 'Foaming in winemaker, II Separation and partial characterization of foam-inducing proteins excreted by a pure culture wine yeast', *Eur J Appl Microbiol Biotechnol*, **16**, 110–113.
- Moreno-Arribas V, Pueyo E, Nieto FJ, Martín-Álvarez PJ and Polo MC (2000), 'Influence of the polysaccharides and the nitrogen compounds on foaming properties of sparkling wines', *Food Chem*, **70**, 309–317.
- Noble AC and Ebeler SE (2002), 'Use of multivariate statistics in understanding wine flavor', *Food Rev Int*, **18**, 1–21.
- Obiols JM, de la Presa-Owens C, Buxaderas S, Bori JL and de la Torre-Boronat MC (1998), 'Protocol d'avaluació de la formació d'efervescència i escuma en vins escumosos', *ACE Revista d'Enologia*, **15**, 3–8.
- Pérez-Serradilla JA and Luque de Castro MD (2008), 'Role of lees in wine production: a review', *Food Chem*, **111**, 447–456.
- Poinsaut P (1991), 'Le Mosalux: appareil de mesure du pouvoir moussant d'un vin', *Rev Fr Oenol*, **59**, 35–43.
- Pozo-Bayón MA, Hernández MT, Martín-Álvarez PJ and Polo MC (2003), 'Study of low molecular weight phenolic compounds during the aging of sparkling wines manufacture with red and white grape varieties', *J Agric Food Chem*, **51**, 2089–2095.
- Priser C, Etiévant P, Nicklaus S and Brun O (1997), 'Representative Champagne wine extracts for gas chromatography olfactometry analysis', *J Agric Food Chem*, **45**, 3511–3514.
- Pueyo E, Martín-Álvarez PJ and Polo C (1995), 'Relationship between foam characteristics and chemical composition in wines and Cava (sparkling wines)', *Am J Enol Vitic*, **46**, 518–524.
- Puig-Deu M, López-Tamames, E, Buxaderas S and Torre-Boronat MC (1999), 'Quality of base and sparkling wines as influenced by the type of fining agent added pre-fermentation', *Food Chem*, **66**, 35–42.
- Rapp A (1998), 'Volatile flavour of wine: correlation between instrumental analysis and sensory perception', *Nahrung*, **42**, 351–363.
- Riu-Aumatell M, Bosch-Fusté J, López-Tamames E and Buxaderas S (2006), 'Development of volatile compounds of Cava (Spanish sparkling wine) during long ageing time in contact with lees', *Food Chem*, **95**, 237–242.
- Robillard B, Delpuech E, Viaux L, Malvy J, Vignes-Alder M and Duteurtre B (1993), 'Improvements of methods for sparkling base wine foam measurements and effect of wine filtration on foam behavior', *Am J Enol Vitic*, **44**, 287–392.
- Romero-Pérez AI, Lamuela-Raventós RM, Buxaderas S and de la Torre-Boronat MC (1996), 'Resveratrol and piceid as varietal markers of white wines', *J Agric Food Chem*, **44**, 1975–1978.
- Rossi JA and Singleton VL (1996), 'Contribution of grape phenols to oxygen absorption and browning of wines', *Am J Enol Vitic*, **17**, 231–239.
- Schneider V (1995), 'Evaluation of small amounts of flavonoid phenols in white wines by colourimetric assays', *Am J Enol Vitic*, **46**, 274–277.
- Senée J, Viaux L, Robillard B, Duteurtre B and Vignes-Adler M (1999), 'The endogenous

- particles of a sparkling wine and their influence on the foaming behaviour', *Food Hydrocolloid*, **12**, 217–226.
- Sève R (1996), *Physique de la couleur*, Paris, France, Masson.
- Singleton VL (1988), 'Wine phenols', in Linskens HF and Jackson JF (eds), *Wine Analysis*, Berlin, Germany, Springer-Verlag, 11–218.
- Singleton VL and Cilliers JJJ (1995), 'Phenolic browning: a perspective from grape and wine research', in *Enzymatic Browning and its Prevention*, ACS Symposium Series, **600**, 32–48.
- Singleton VL and Trousdale E (1983), 'White wines phenolics: varietal and processing differences as shown by HPLC', *Am J Enol Vitic*, **34**, 27–34.
- Somers TC, Vérette E and Pocock KF (1987), 'Hydroxycinnamate esters of *Vitis vinifera* changes during white vinification and, effects of exogenous enzymatic hydrolysis', *J Sci Food Agric*, **40**, 67–78.
- Swiegers JH and Pretorius IS (2007), 'Modulation of volatile sulfur compounds by wine yeast', *Appl Microbiol Biotechnol*, **74**, 954–960.
- Tominaga T, Guimbartau G and Dubourdieu D (2003), 'Role of certain volatile thiols in the bouquet of aged champagne wines', *J Agric. Food Chem*, **51**, 1016–1020.
- Torrens J, Urpí P, Riu-Aumatell M, Vichi S, López-Tamames E and Buxaderas S (2008), 'Different commercial yeast strains affecting the volatile and sensory profile of cava base wine', *Int J Food Microbiol*, **124**, 48–57.
- Tuinier R, Bisperink CGJ, Van der Berg C and Prins A (1996), 'Transient foaming behavior of aqueous alcohol solutions as related to their dilational surface properties', *J Colloid Interface Sci*, **179**, 327–334.
- Vannier A, Brun OX and Feinberg MH (1999), 'Application of sensory analysis to champagne wine characterisation and discrimination', *Food Qual Prefer*, **10**, 101–107.
- Vanrell G, Canals R, Esteruelas M, Fort F, Canals JM and Zamora F (2007), 'Influence of the use of bentonite as a riddling agent on foam quality and protein fraction of sparkling wines (Cava)', *Food Chem*, **104**, 148–155.
- Vasserot Y, Jacopin C and Jeandet P (2001), 'Effect of bottle capacity and bottle-cap permeability to oxygen on dimethylsulfide formation in champagne wines during aging on the lees', *Am J Enol Vitic*, **52**, 54–55.
- Viaux L, Morard C, Robillard B and Duteurtre B (1994), 'The impact of base wine filtration on Champagne foam behavior', *Am J Enol Vitic*, **45**, 407–409.
- Vidal S, Williams P, Doco T, Mountounet M and Pellerin P (2003), 'The polysaccharides of red wine: total fractionation and characterization', *Carbohydr Polym*, **54**, 439–447.
- Voilley A, Lamer C, Dubois P and Feuillat M (1990), 'Influence of macromolecules and treatments on the behavior of aroma compounds in a model wine', *J Agric Food Chem*, **38**, 248–251.
- Winterhalter P (1991), '1,1,6-Trimethyl-1,2-dihydronaphthalene (TDN) formation in wine, 1, Studies on the hydrolysis of 2,6,10,10-tetramethyl-1-oxaspiro[4.5]dec-6-ene-2,8-diol rationalizing the origin of TDN and related C-13 norisoprenoids in Riesling wine', *J Agric Food Chem*, **39**, 1825–1829.
- Wiszecki G and Stiles WS (1982), *Colour Science: Concepts and Methods, Quantitative Data and Formulae*, 2nd edn, New York, Wiley.
- Würidig G and Müller T (1979), 'Zum Phänomen des Überschaumens von Schaumwein', *Die Wienwirtschaft*, **19**, 653–657.

Extraction technologies and wine quality

A. Razungles, Montpellier SupAgro, France

Abstract: During the transformation of grapes into wine, the extraction of must from the grapes and the expression of the quality of the grapes in the wine have always been of great interest to winemakers. The quality and identity of wine are directly linked to fixed and volatile constituents contained particularly in the solid parts of the berries: skin, pulp and pips. This chapter aims to describe the processes used today to optimize extraction during vinification of white, rosé and red wines. The extraction techniques depend on chemical, biological and physical parameters, amongst which heat and matter transfers during maceration are the most important. However, higher or lower temperatures are used, not only to increase dissolution and diffusion of the compounds of interest, but also in order to break down the cell walls of the grape berries. Processes such as carbonic maceration or techniques based on the use of vats or specific materials are also described. Finally, different techniques for pressing, the last step in the extraction process, are presented.

Key words: grape, wine, extraction, procedure, winemaking, maceration, pressing.

20.1 Introduction

There is one stage among the various stages in the transformation of grapes into wine that is of particular importance: the extraction of compounds from the grape berries. This extraction process has been mentioned in numerous books (Peynaud, 1972; Ribéreau-Gayon *et al.*, 1976, 2004; Amerine *et al.*, 1980; Jackson, 1994; Flanzy, 1998; Flanzy *et al.*, 2009). Wine is, of course, the result of profound biological changes: alcoholic and malolactic fermentations generate numerous compounds that are of interest to the winemaker. However, the

expression of the quality of the grapes in the wine will depend on the extraction of fixed and volatile constituents in the solid parts of the berries. These are the compounds that will contribute the most to the identity of the wine in terms of grape variety, terroir and cultivation characteristics. Among these compounds, the most sought after are the aromas and their precursors, phenolic compounds and, to a lesser extent, the polysaccharides. The extraction of these substances is usually partial (for example, for Muscat grapes, 50% of the aromas, and 30–60% of the anthocyanins and tannins) (Boulet and Escudier, 1998), and will depend on numerous chemical, biological and physical parameters that contribute first to the dissolution of the compounds in the must and then to the diffusion of this same must out of the solid parts of the grape. Among the parameters affecting extraction, the choice of temperature and duration will have the most impact (Zimman *et al.*, 2002).

For red wines, higher temperatures and longer macerations will promote the extraction of compounds from the grape skins (Ribéreau-Gayon *et al.*, 1976). As was noted by Cheynier and Sarni-Manchado in Chapter 2 (Volume 1), anthocyanidins will be extracted quickly, while the extraction of proanthocyanidins will increase with maceration time. Furthermore, epigallocatechin units from the skin and pulp will be extracted more quickly than epicatechin gallate units, coming preferably from the seeds. The extraction of the latter continues until pressing. Furthermore, in traditional red winemaking, the initiation and control of alcoholic fermentation will indirectly participate in the successful management of extraction during maceration. Being obliged to interrupt maceration because fermentation has stopped is always an inconvenience.

The degree of ripeness of the grapes can also affect the kinetics of extraction due to the reduced physical resistance of the membranes. While there will be only slight differences in the extraction of phenolic compounds from grapes with different degrees of ripeness (Fig. 20.1), the colour of wines will be stronger with very ripe grapes, suggesting that complex interaction phenomena between the extracted phenolic compounds (derived pigments, copigmentation) are involved (Fournand *et al.*, 2006; Role *et al.*, 2009).

However, the extraction of grape constituents does not only concern substances that will improve a wine's quality: over-extraction or poorly managed extraction can result in defects developing in wines. The enologist has to adapt extraction conditions to the desired objectives and set limits that should not be exceeded to avoid an over-extraction of defect-inducing compounds (Gerbaux *et al.*, 2002).

The quality of the harvest can play a significant role in the extraction, or not, of such compounds. For example, a healthy harvest of undamaged grapes will enable a more intense extraction than a harvest that has been affected by *Botrytis cinerea*, for which it will be necessary to attempt to reduce the extraction of the laccase produced by the Botrytis. Moreover, a harvest affected by *Botrytis cinerea* will have undergone changes as a result of fungal activity (Boidron and Torres, 1978; Cordonnier, 1987) and will have less potential for extraction. The latter will be considerably reduced.

In all cases, the sorting of the harvest is an important stage in the elimination of

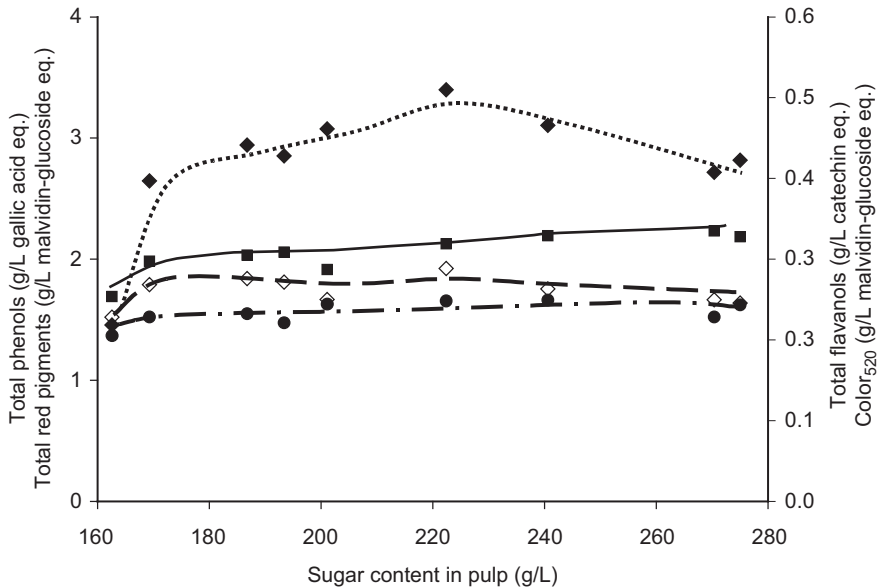


Fig. 20.1 Changes in the maximum extracted amounts of total phenols (■), total flavanols (●), total red pigments (◇) and colour at 520 nm (◆) in hydroalcoholic solution as a function of pulp sugar content (Fournand *et al.*, 2006).

impurities likely to transmit undesirable flavours or odours to the wine. There are now many manual and automatic sorting techniques, implemented before and/or after destemming, which enable a selection of only those berries that meet the winemaker's requirements. With mechanical harvests, these corrective sorting operations are difficult if the grapes are not kept whole. Mechanical harvesting requires meticulous management to preserve full potential in terms of quality. Such harvesting is not selective, and the result often has a semi-liquid form with whole and crushed grapes, mixed with leaf debris, stalks, vine shoots and fasteners, that will be macerated in the vats with the harvest. A perfectly trellised vineyard and careful adjustment and driving of the harvest machine should limit the presence of potentially harmful foreign bodies. Nowadays, if mechanical harvesting is optimized and most of the grapes are whole, it is usually possible for various physical sorting procedures to take place.

The phases of extraction concern all categories of wine: white, rosé and red. There is no doubt that most extraction techniques have been developed in the context of making red wine, but there has been a lot of work on white wine techniques, in particular relating to pressing and pre-fermentation maceration (skin-contact).

In this chapter, after setting out the various extraction factors, we will look at increasing the value of white grape harvests through extraction. We will quickly cover the special case of rosé wines since, in purely technical terms, they are very close to white wines. We will then consider the main extraction techniques used in the production of red wines.

20.2 Extraction factors

20.2.1 Chemical factors

In vinification, two chemical factors are considered responsible for the extraction of skin constituents: alcohol (ethanol) and sulphur dioxide. Their effects have been known for a long time and in the past have given rise to technical applications exclusively in the making of red wines (such as vinomaceration and maceration with sulphur dioxide).

Water

This might appear obvious, but the water in the grapes constitutes the primary extraction factor. Water is a polar solvent which will enable the dissolution and extraction of compounds which include sugars, organic acids, anthocyanins, etc. However, water is also responsible for the extraction of the cations, particularly potassium. This extraction is generally not desirable because it causes instabilities through the precipitation of tartaric acid salts and a reduction in wines' acidity. In addition to its role as a solvent, water acts as a driving force in pumping over, racking and punching down. It is the crushing of the grapes that results in the presence of water or free must which is useful during maceration. A mechanical harvest will generate a sufficient release of must to enable a homogenizing pumping over after the tank has been filled.

Ethanol

Ethanol is formed during alcoholic fermentation by yeasts and has a dual effect on extraction: it acts as an antiseptic, killing the cells in the skins, pulp and seeds; and then as a solvent, helping to dissolve numerous compounds, both polar (phenolic compounds) (Ribéreau-Gayon *et al.*, 1976) and lipophilic (certain aromas and precursors). It also weakens the waxy cuticle of the seeds, making them more easily extractable. These two properties make ethyl alcohol an ally of the winemaker in the extraction phases. However, for red wines made with over-mature grapes, when fermentation is nearly complete, high levels of alcohol can result in the over-extraction of certain compounds and the increased presence of less desirable compounds in sensory terms, such as the galloylated tannins of grape seeds, which have the particularity of being harder. This is the reason for the tendency for moderate extraction at the end of fermentation, particularly in the case of long macerations.

In the past, ethanol's extractive properties have been used to advantage in three procedures used in red winemaking: the 'super 4' technique consisting of the addition of wine to a harvest that has been destemmed and crushed as soon as the grapes are placed in vats, so that the grapes are submerged, from the start of maceration, in a liquid containing 4 % alcohol by volume. This technique is also the basis for continuous wine production, in which the harvest was continuously fed into a reactor that already contained fermenting grapes. The purpose of these techniques was two-fold: first, to stimulate the growth of the yeast flora responsible for fermentation (*Saccharomyces cerevisiae*) to the detriment of undesirable flora with

little resistance to alcohol; and second, to shorten the maceration phase. However, these techniques were developed in the days of mass production and are no longer in use today. The same is true of the Vinomaceration technique, developed in the early 1980s. This involved the total immersion of the harvested grapes in hot wine (70 °C) for 30 minutes. The immersion had the effect of enhancing the extraction of compounds in the skin (anthocyanins) through the combined actions of the ethanol and the heat. The length of the maceration was reduced to a contact time of 30 minutes, after which the wine was racked and then reheated. The harvest was then pressed and alcoholic fermentation took place at the liquid stage. This technique was too labour-intensive and has not been developed further.

Sulphur dioxide

Sulphur dioxide weakens skin and pulp cells through its antiseptic action. It results in greater extraction of compounds in the solid parts of the berry (Olivieri *et al.*, 1976; Amrani Joutei and Glories, 1995; Feuillat, 1997; Bakker *et al.*, 1998). However, the fact that this technique is only effective with substantial doses of sulphur dioxide (≥ 150 mg/L), quickly prevented it from becoming widespread. The impact of sulphur dioxide on extraction is very limited at the doses usually used in red wine vinification (Watson *et al.*, 1995 ; Sacchi *et al.*, 2005). Conversely, in the mechanical harvesting of white wine grapes, the use of this additive in the hoppers can result in the undesirable extraction of polyphenolic or potassium compounds. The addition of sulphur dioxide to only the juice in hoppers with a double-bottom can be effective against oxidation without causing an increase in the extraction of unwanted compounds.

20.2.2 Biological factors

Yeasts

Yeasts do not play a direct role in extraction. However, they participate indirectly through their production of ethanol and through the heat generated during fermentation. This last parameter (increase in temperature) is at the origin of two techniques sometimes used during red wine maceration: hot maceration of the pomace and hot final maceration, which we will come back to in Section 20.5.4.

Enzymes

Enzyme preparations producing pectinase activities, particularly polygalacturonase, and also cellulase and hemicellulase, used in red winemaking at the start of the maceration phase, can modify the extraction of various molecules. They are more effective when the grapes are crushed.

Effect on anthocyanins and tannins

The use of enzymes seems to have an intensifying effect on the colour of red wines during the first days of maceration (Villetaz, 1996; Cabeza *et al.*, 2009). This is not necessarily linked solely to the extraction of anthocyanins but also, most probably,

to co-pigmentation phenomena with monomer flavonols. Enzyme preparations do not seem to produce significant increases in levels of proanthocyanidins which diffuse into the must over time. The proanthocyanidins are located in the external layers of the hypodermis and are thus far from the first strata of skin cells that are exposed to enzymatic activities from the inside of the berries burst during crushing. The consequence is that colour cannot be stabilized during short macerations (Main and Morris, 2007). In these conditions, the benefit of increased colour diffusion during the first days of maceration decreases to become insignificant after 10 days (Ducasse *et al.*, 2010).

Effect on polysaccharides

With the exception of rhamnogalacturonan II (RGII), the other polysaccharides rich in arabinose and galactose are present in smaller quantities in red wines treated with pectolytic enzymes during the maceration phase. However, this will vary with the ripeness of the grapes (Ducasse *et al.*, 2010).

Effect on aromas and their precursors

It is hard to establish the role of enzymes in the extraction of skin aromas, or their precursors, in the context of traditional maceration, because there do not yet appear to be any studies of this subject. Enzymes have mostly been used to reveal the aromatic potential present in the grapes in the form of glycosides (Gunata *et al.*, 1993).

Effect on the extraction of juice

The use of pectolytic enzymes during skin maceration in white or rosé wines enables an increase in the quantity of free-run wine after four to six hours of maceration (Cayla, 2007). In tests on grape varieties such as Cinsault or Grenache, these gains in extraction reached as much as 16%. Furthermore, the addition of enzymes has the effect of improving the overall yield at pressing and, in some cases, can reduce the pressing time. In traditional red wine vinification, the use of these enzymes also generally increases juice yield (Sacchi *et al.*, 2005), although the use of extraction enzymes is less justified because the effect of the maceration period is more significant.

Special case of anaerobic metabolism

Anaerobic metabolism is desirable in the context of vinification by carbonic maceration (Flanzy, 1935; Flanzy *et al.*, 1987). The maceration of whole grapes in asphyxiating conditions in a carbon dioxide atmosphere causes a modification in the enzymatic activity endogenous to the grapes. This activity results, in a few days, in profound modifications in the composition of the berry and its texture. Indirectly, the extraction of skin constituents is facilitated. This technique will be described in Section 20.7.

20.2.3 Physical factors in extraction

Among the physical factors we will consider passive factors, such as transfer of

heat, and dynamic factors based on the transfer of matter. We will also consider the effect of changes in pressure.

Heat transfer

Increases in temperature play a dominant role in extraction phenomena at several levels (Olivieri *et al.*, 1976; Reynolds *et al.*, 2001), on the one hand because they accelerate chemical and biological extraction activity, and on the other because they reduce the viscosity of liquids and thus increase their diffusion, particularly in semi-solid environments such as pomace during maceration. High temperatures also have an indirect effect in that they break down skin cell membranes. Several procedures use this property (hot maceration, thermovinification, flash-release and thermo-release).

Similarly, very low temperatures can also be used to breakdown cellular structure (cryoflash process and supra-extraction process) or to enable selective extraction such as in the production of 'icewines', controlled by a selective cryoextraction procedure. To a lesser degree, cold temperatures that are above freezing (8–12 °C) are also used to delay the start of fermentation during skin maceration in white wines or pre-fermentation maceration in red wines.

Transfer of matter

During red wine vinification with traditional maceration, as carbon dioxide is emitted, the pomace rises in the vat and floats at the surface. This semi-submerged position does not optimize extraction by the must of the constituents in the solid parts of the grapes. To remedy this problem, the winemaker has two possible techniques: either to break up the cap by pumping the must below up over it (transfer of liquid = pumping over), or to submerge the cap by using vertical pressure (transfer of solids = punching down).

Transfer of liquid

Several techniques are used to draw out the pomace's constituents by the movement of liquid. The most common is pumping over, which consists of pumping the juices at the base of the vat up over the top of the cap, thereby 'washing' it. However, there is another technique, racking or '*délestage*' which requires more work, but is more effective. This involves, during red wine maceration, transferring all of the must from one vat to another. The unsupported pomace breaks up under its own weight on the bottom of the vat. All of the must is then pumped back into the vat over the top of the pomace. This technique is very efficient and enables a good diffusion of the skin constituents in the must.

Transfer of solids

Various procedures and equipment have been developed to enable the immersion of the pomace cap in the fermenting must and/or its breaking up (punching down) in order to enhance extraction of desirable constituents. As with the preceding techniques, punching down ensures very good extraction of skin constituents. However, used excessively, it can produce unbalanced wines in terms of flavour, with dry or bitter astringency.

These various transfer techniques will be looked at more closely in the section on traditional maceration. Whatever extraction factor is being considered, the amount of time for which it is applied or the frequency of transfer operations will be the deciding elements in the optimization of the extraction in accordance with the desired quality objectives.

The effect of changes in pressure

For winemakers, the term 'pressure' evokes the pressing operation. In fact, the application of pressure, formerly with the hands or feet, and later with a press, must be the oldest method of extraction that exists, or at least the first to be understood and used by man. We will return in more detail to the presses and pressing techniques that make up the final stage in the extraction process (Section 20.9.2). Pressure is also a parameter in racking techniques or, more simply, static draining from the solids: the must runs out of the pomace under the effect of the pressure transmitted by its own weight.

As with temperature, a sudden change in pressure can be used to amplify extraction phenomena, either by bringing into play a change of state of the water in the cells (flash-release principle), or by using the compression–reduction in pressure of the fermentation gases to make the must penetrate or pass through the cap like a sponge (maceration procedures in hermetically sealed vats).

20.3 Techniques applied to white wine vinification

20.3.1 Skin maceration

The location of aroma compounds and their precursors in the skins of the berries (Gunata *et al.*, 1985; Dubourdieu *et al.*, 1986; Wilson *et al.*, 1986; Baumes *et al.*, 1988; Razungles *et al.*, 1988) has resulted in winemakers adopting macerations with prolonged contact (4 to 18 hours) between the must and the solid parts of the grapes, in order to encourage the migration of aroma precursor constituents into the juice. This practice is most often used with grape varieties with known aromatic potential residing in the skins. Results obtained with a range of varieties show a significant increase in levels of various precursors, particularly glycoside aroma precursors (Table 20.1) (Dubourdieu *et al.*, 1986; Baumes *et al.*, 1988; Peyrot des Gachons *et al.*, 2002) or other glucosides as resveratrol glucosides (Clare *et al.*, 2004).

Durations and temperatures of maceration are very variable and depend on the grape variety and the available chilling facilities. However, it is strongly recommended that this maceration be carried out at a temperature close to 10 °C and for a short time (less than 18 hours), for two reasons. The first reason is to avoid the extraction of phenolic compounds likely to oxidize and result in browning. The second reason is to delay the start of fermentation after the maceration phase and settling. Skin maceration has another drawback in that it promotes the diffusion of cations contained in the skins. The levels of potassium are higher in musts resulting from skin maceration and will present greater instability with regard to tartaric acid

Table 20.1 Effect of skin maceration (4 hours, 18 °C) on the increase in volatile compounds in musts of white grape varieties compared with a standard directly pressed must (Baumes *et al.*, 1988)

Volatile compound	Grape variety		
	Bourboulenc	Chardonnay	Chenin
C6 compounds	+ 233%	+ 217%	+ 284%
Terpenols	+ 168%	+ 122%	+ 157%
Other alcohols	+ 228%	+ 238%	+ 104%
Esters	+ 120%	Absent	Absent
Carbonylated compounds	+ 120%	+ 140%	+ 133%
Lactones	+ 110%	+ 233%	+ 155%
Volatile phenols	+ 144%	+ 176%	+ 149%

Source: Baumes *et al.*, 1988.

salts. This precipitation is accompanied by a deacidification that is not always desirable in white wines. For the same reasons, destemming is recommended before skin maceration. Vats with emptying mechanisms or an automaceration facility (Section 20.5.3) may be used for skin maceration.

The wines resulting from such maceration often present more intense odour profiles but, in terms of balance, can be thicker, less fresh and less lively. This technique is suitable for grapes with high initial acidity.

20.3.2 'Icewines' and cryoselection

The pressing of partially frozen grapes results in a selective extraction of the sweetest juice, which is in the ripest grapes. Only these grapes, which will not have frozen, will be subject to crushing and extraction. The grapes which, at the same sub-zero temperature, have frozen, will not be pressed. This is an example of Raoult's law which states that the freezing point of an aqueous solution is inversely proportional to its solute content (in this case, sugar). This phenomenon has been used in the production of icewines which originally developed in the Rhine region. The frozen grapes are harvested in the night or on particularly cold days, and then immediately pressed. The technique produces musts with a very high sugar content, that are then fermented to make naturally sweet wines.

This technique was taken further with the development of cryoselection (Chauvet *et al.*, 1986). Grapes are handpicked at ambient temperature and then frozen in trays in a cold room. They are then pressed at a temperature that is established on the basis of two parameters: the desired sugar content of the must (Table 20.2) and the volume of unextracted must after pressing (Table 20.3). A decision is made on the basis of an appropriate balance between economic and qualitative factors. This technique can be used to produce sweet wines in years when weather conditions are poor but, in terms of quality and complexity, the wines obtained will not rival those made from selectively picked grapes that have been affected by noble rot.

Table 20.2 Effect of the temperatures used during the cryoselection process of several musts on the potent alcoholic grade (PAG), expressed in %alc. vol., of the selected moieties (Chauvet *et al.*, 1986)

Process temperature (°C)	PAG of musts before process	PAG of selected musts after process
–6	16	19.6
–8.5	16	20.7
–10	16.2	22.0
–11	16	23.2
–15	17	23.4

Source: Chauvet *et al.*, 1986.

Table 20.3 Effect of temperatures used during the cryoselection process of a must on the volume of selected and residual juices after pressing (Chauvet *et al.*, 1986)

Process temperature (°C)	Volume of selected juice (L)	Volume total: selected juice + not selected juice (L)
–6	81.5	103
–8.5	74.5	107.5
–11	67	102
–15	60	100

Source: Chauvet *et al.*, 1986.

20.3.3 Supra-extraction

The super-extraction technique was developed to replace crushing in white wine vinification (Defranoux *et al.*, 1989). It consists of lowering the temperature of the harvest to –4 °C, then letting it rise again to 10 °C before pressing. The start of gradual freezing alters the structure of the cell walls of the skins through the formation of ice crystals in the air spaces in the cellular tissue. On reheating, the must is released more easily and there is increased extraction of skin compounds. This technique has not seen much development in the context of white wines, but it did lead to the Cryo-flash® process that we will consider in red wine vinification (Section 20.5).

20.4 Techniques applied to rosé wine vinification

In the production of rosés, the winemaker is particularly concerned with obtaining the desired colour. Four extraction techniques are widely used: rosés obtained by direct pressing, rosés obtained by the ‘*saignée*’ method, traditional maceration rosés and carbonic maceration rosés (Flanzy *et al.*, 2009).

20.4.1 Direct pressing rosés

Direct pressing rosés are made in the same way as traditional white wines. The

grapes are brought to the press and pressed directly. For many grape varieties, this operation is in itself enough to extract a proportion of the anthocyanins. The quality of the juice will depend on the nature of the pressing system (Section 20.9.2). These rosés are generally very pale and have higher acidities than those obtained by other techniques (Cayla, 2009).

20.4.2 Saignée rosés

Saignée rosés are obtained after 2 to 6 hours of maceration of a harvest intended for the production of a red wine. This 'saignée' (10–15% of the total volume of juice) also enables a concentration of the red wine that will be produced from the same vat. However, the grapes will have been harvested primarily for the production of red wine, and will not always have desirable characteristics in terms of ripeness and concentration (high alcohol, low acidity) for rosé wines.

20.4.3 Maceration rosés

In the production of maceration rosés, there is extraction during maceration and during pressing. The procedure is similar to skin maceration in white wines, but the objectives include an additional parameter: the extraction of the anthocyanins. The maceration temperature is kept low to enable a subsequent settling phase and to slow down the diffusion of colour (Crachereau, 2009). This technique can be used with all the red grape varieties, but is recommended for varieties that do not easily release their colour such as Grenache Noir, Cinsault and Pinot Noir. The wines obtained tend to be more aromatic, deeper in colour, less acidic and more full-bodied than rosés made using direct pressing. When enzymatic treatments are applied during maceration, colour remains more stable during storage (Salinas *et al.*, 2003; Cayla, 2007).

20.4.4 Carbonic maceration rosés

Originally used for the production of red wines, the carbonic maceration technique will be described in another section (Section 20.7). Applied to the vinification of rosés, it has the advantage of producing wines with pleasant, fruity aromas, with very unobtrusive tannins. The harvest is placed in vats in whole bunches. Maceration time will vary and depends on the grape variety: from 36 hours if the optimal maceration temperature (32 °C) is obtained quickly, up to 48 hours or more when the temperature is lower. After pressing, static settling will be possible if the musts have not started fermenting. Musts that are already fermenting can be clarified using dynamic methods before continuing their fermentation.

20.5 Techniques applied to traditional red wine vinification

On a simplistic level, one could say that extraction is the primary objective in making red wines. However, there has to be something to extract. There is little

point in applying the best techniques to grapes that are of poor quality, mixed ripeness, under-concentrated or in some way defective. It is very fortunate that this notion of the technical quality of the harvest, based on a knowledge of grape varieties, terroirs and growing techniques, has at last come to be considered as an indispensable condition for good maceration, and thus for a good wine.

There have been several studies of the kinetics of extraction, mainly the extraction of phenolic compounds. The work of Ribéreau-Gayon *et al.* (1976) remains current and provides the basis for subsequent studies (Watson *et al.*, 1995; Auw *et al.*, 1996; Gomez-Plaza *et al.*, 2001; Reynolds *et al.*, 2001; Sacchi *et al.*, 2005). Techniques for the separation and identification of the various fractions of polyphenolic compounds now allow a more precise monitoring of extractions and control of macerations in accordance with desired objectives.

Anthocyanins and colour intensity peak after around six days of maceration. Values then decrease and gradually stabilize. This decrease has been attributed to various factors: an instability in these anthocyanidins forms which can condense into insoluble forms, or a refixation of the anthocyanins on the yeasts (Medina *et al.*, 2005) or the skins and pulp of the macerating grapes. However, the extraction of tannins continues for the entire maceration period. This initial research has enabled an understanding of the value of long macerations for wines for laying down, and, conversely, the need to reduce maceration times when the goal is to make a supple wine low in tannins.

There is one factor that plays a key role throughout maceration: ethanol. Alcoholic fermentation takes place during maceration. The ethanol produced by yeast increases in quantity, to such an extent that it enables a greater extraction of solid-phase compounds, particularly phenolic compounds in the skins and most of all in the seeds (Amrani Joutei and Glories, 1995).

Besides chemical, biological and thermal extraction factors, the transfer of compounds from the solid parts of the grape to the juice will also result from physical procedures involving either the juice (pumping-over, racking), or solid elements (punching down). However, in the latter case, the downward pressure on the cap will also cause movement in the liquid phase. The objective is to increase contact between the two phases, thereby optimizing dissolution–diffusion phenomena (Razungles *et al.*, 1998).

20.5.1 Destemming and crushing

In traditional red wine vinification, the objective of maceration is to extract the constituents of the skin and pulp. However, a degree of caution is required in order to avoid the extraction of undesirable constituents. These macerations involve comparatively long periods, and should be free of any foreign bodies or unhealthy berries likely to affect the final quality of the wine. For this reason effective sorting during both mechanical and manual harvests is strongly recommended.

The operations of destemming and crushing may be considered optional, but are carried out in most cases. In certain circumstances, when the grapes are not sufficiently rich in tannins, the stalks may be partly or fully included in the

maceration. They will provide the must with the polyphenols required for good colour stabilization. However, the stalks should be completely lignified to avoid any green flavours. Crushing has the advantage of enhancing extraction phenomena by releasing juices, which act as the extraction solvent, but may be unnecessary when grapes are harvested mechanically. And in certain cases the grapes will not be crushed at all, when the winemaker wishes to macerate whole berries in order to delay or reduce extraction phenomena.

Manufacturers offer a wide range of highly efficient equipment that is usually able to meet all requirements in the various operations of moving, sorting, separating and crushing grapes. Destemmer–sorters have progressed enormously and now enable very high percentages of whole grapes to be obtained (more than 90%). New equipment has also been developed making increasingly efficient sorting possible, particularly of mechanical harvests. For example, the Tribaie automatic sorter (made by Triviti Amos) and optical sorters (made by Bücher, Diemme, Pellenc, Pera). These devices, preceded by a destemmer, have a remarkable ability to select berries on the basis of quality, and are very fast. Destemmers can be incorporated into harvesters: stalks and foreign bodies are immediately separated and left on the ground.

20.5.2 Procedures for red and other wines based on displacement of the must

Pumping over

The pumping over procedure is nearly as old as maceration itself. Reynaud (1972) has demonstrated the benefits of pumping over on wines' colour intensity. More recently, Vivas and Vache (1991) confirmed the importance of the volume of daily pumping over in relation to wines' anthocyanins. The protocols adopted in pumping over to extract skin compounds are very varied in terms of duration, volumes and frequency (Vivas and Vache, 1991). These aspects also vary as a function of the maceration process and it is impossible to set down a procedure for general use. The same applies to all the other procedures described or cited below. Daily pumping over of a quantity of wine corresponding to the volume of the vat appears to be sufficient for an appropriate extraction of anthocyanins during the first half of the fermentation; for the second half, pumping over of half the volume of the vat every day can be sufficient. This is a long and repetitive operation. It is not surprising that winemakers have developed systems that free them from this tedious task.

Vats are now available that enable pumping over to be programmed, for example Gimar's Selector® system (Fig. 20.2). This vat not only enables pumping over to be fully automated, it also makes it possible to introduce controlled quantities of oxygen beneath the cap (oxyplus procedure), while at the same time controlling the fermentation process on the basis of densimetric measurements. This equipment can pump the must directly over the cap, or onto the sides of the vat for a gentler effect (with less mechanical impact). There is also a programmable

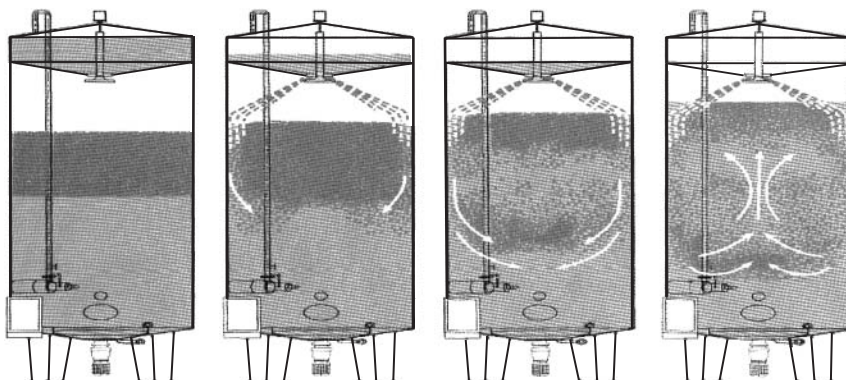


Fig. 20.2 Operating cycle of Gimar's Selector® system.

feature that can moisten the cap with some of the must, before submerging it with a sudden release of the contents of the upper vat. According to the manufacturer, this procedure causes less 'damage' to the cap, thereby reducing the creation of lees. Depending on the programming, the pumping over can be carried out gently, minimizing the formation of lees, or it can be more like a hydraulic punching down. Other manufacturers also offer automated systems, e.g. Atea's Vinicycle vat (Vivas and Vache, 1991).

At the end of the 1990s and early 2000s, the first pumping over systems using a submerged pump, such as the Turbopigeur® (Socma), appeared (Razungles, 2006). This device can be moved from vat to vat, avoiding the cumbersome pipes of traditional pumps. This pump was developed from boring equipment and is placed in a pipe which is pushed through the pomace cap (Fig. 20.3a). The must is sucked up from beneath the cap and discharged over the top via a deflector. The powerful pump makes it possible to submerge the cap very rapidly. Egretier has developed a similar device which differs from Socma's Turbopigeur in that it has horizontal outlets for the must (Fig. 20.3b). These outlets are located inside the cap, resulting in a hydraulic dispersal of the pomace in the cap, as well as pumping over.

In both cases, the wine does not have to circulate outside of the vat, thereby preventing possible losses and reducing labour. However, it becomes necessary to provide oxygen, which is essential for successful fermentation. This is done via a diffuser fixed to the base of the pipe containing the pump.

Use of carbon dioxide generated by fermentation as a pumping over 'driver'

One litre of must produces about 50 L of CO₂ during fermentation. This emission of gas and the high pressure that it creates was used by Duccellier (1963) in his hydraulic valve system. The problem with this method was the winemaker was no longer in total control and the pumping over rate was dictated by the fermentation process.

The 1980s saw the development of a gas pumping over technique which could

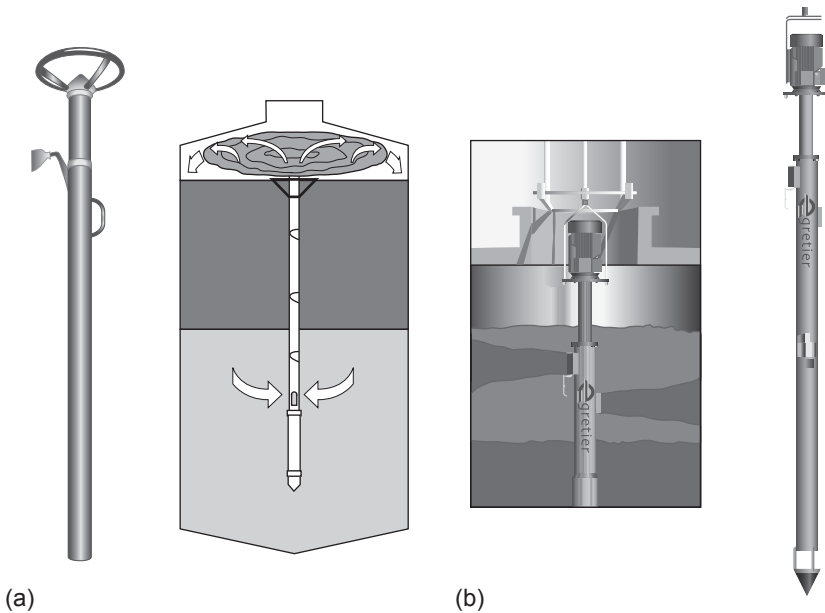


Fig. 20.3 Turbopigreur pumping over system by Socma (a) and pigeur pumping over system by Egretier (b).

also be considered as a mixing effect, similar to punching down. This procedure, described and studied by Coutanceau (1988) and Desseigne and Caussanel (1990), consists of causing a sudden release of CO_2 , which reaches supersaturation in a fermenting vat. This discharge is catalyzed by an injection of gas into the base of the vat, which has the effect of drawing up a column of liquid which breaks through the pomace and then spreads over the top. Several gas injection devices are available. This technique ultimately led to the design of a multipurpose vat: the Magyar. By way of example, a 50 hL vat stirred up every day by 0.9 m^3 of nitrogen produces an extraction identical to that obtained by punching down three times a day.

More recently, an ingenious procedure called the Ganymede system has been developed by Tec Inox (Fig. 20.4). This procedure uses a circular vat with a type of diaphragm half way up that consists of an inverted truncated cone, with a hole in the middle. This cone traps the carbon dioxide that is generated in the lower part of the vat. The accumulated gas can then be released when desired by transferring the contents of lower part of the vat to the upper part via a pipe fitted with a valve. The sudden release of the CO_2 under the pomace cap generates a column of liquid that pierces the cap and creates a pumping over effect. The manufacturer suggests various ways of using the vat. This extremely simple procedure has shown itself to be particularly effective at extracting skin compounds. It is as at least as effective as the extraction observed in a control maceration, and offers a considerable saving in terms of labour and total control of extraction phenomena, since pumping over can be initiated when desired or automated. As in the preceding examples of gas

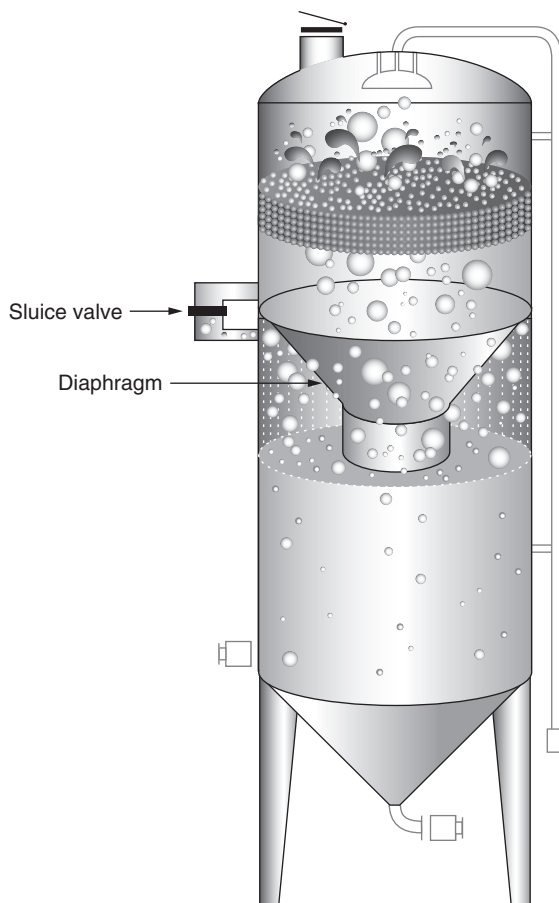


Fig. 20.4 Diagram of the Ganimede vat (Tec Inox).

pumping over, oxygen has to be provided during fermentation by a diffuser, to enable good biomass development and colour stabilization (Razungles, 2006).

Racking

Racking is a variation on the pumping over procedure. It consists of draining off all the free juice in a macerating vat. The pomace settles on the bottom of the vat and releases its liquid under its own weight, like a sponge full of water. It is then resubmerged using a powerful pump. This procedure results in a greater degree of extraction than pumping over. As with pumping over, the effectiveness of racking will depend on whether it is carried out at the beginning or end of alcoholic fermentation. Racking can be combined with a breaking down of the pomace cake if stainless steel cables are installed between the sides of the vat. When it descends, the pomace cap will be broken up by the cables. Racking is a good way of increasing extraction phenomena at a comparatively low cost.

20.5.3 Procedures based on displacement of the pomace: punching down

Punching down and crushing with the feet are very old techniques that can be found in Ancient Egyptian illustrations. Traditional punching down with the feet or with the use of a baton still takes place in artisanal winemaking in some regions, such as in Burgundy in France or in the Douro Valley in Portugal. This operation is so time-consuming and unpleasant that a wide range of devices and vats have been developed that enable the procedure to be totally or partially automated. Among these we will consider mobile devices that can be moved from vat to vat, stationary vats equipped with punching down systems and, finally, rotary vats. In most cases, and in the context of moderate use, these punching down systems enable an enhancement of skin constituent extraction phenomena, but the frequency of their use has to be adapted to the harvest in order to avoid excessive extraction.

Mobile punching down devices

The number of mobile punching down models available has increased in recent years. Nearly all of them use a pneumatic plunger with a disc or folding device (a plate or blade) at its extremity. The other (top) end of the plunger is attached to a support that may be the vat's vent or a rail, or a mobile trolley above the vats. This punching down device can make vertical, angular and rotational movements. It can be controlled manually or completely automated. In the latter case, almost the entire punching down operation can be automated for a series of open, adjacent vats. Other systems involve devices that assist punching down and have to be positioned manually by the operator. Each punching down, carried out once or twice a day (with eight to ten submersions of the cap), enables an increase in the extraction of tannins and anthocyanins comparative to a control vat with traditional maceration and two pumping overs each day (Feuillat and Peyron, 1998). These easy-to-use devices provide a fairly good reproduction of the traditional trampling with the feet technique.

Another punching down system involves the placing of an inflatable bladder under the top of the vat with a surface area less than half of that of the vat. When inflated, the bladder presses on the top of the vat and pushes down on the pomace cap just below, which becomes submerged in the must. The bladder is then repositioned over another section of the cap and the operation repeated.

Traditional vats equipped with punching down systems

The simplest punching down systems are adapted to traditional vats, usually stainless steel, which are equipped with two to four pneumatic plungers attached to the top of the vat (Guerin). These plungers have a plate at their extremity which pushes the pomace cap into the must as the plunger goes down. When it comes back up the plate is rotated to a vertical position so as not to lift the pomace. The plungers can be operated simultaneously or alternately to break up the cap as thoroughly as possible. Unlike the previous systems, the punching down is limited to the vertical displacement of the plunger.

Automaceration vats

The first automaceration vat models contained moving metal parts that could submerge the cap when operated. The trapezoidal Fabbri vat has rakes between two chains supported by wheels. When the system is activated, the rakes break up the top of the pomace cap and pushes it down into the must. Brauthite, Giboulot and Emonnot offer vats with a completely different system involving a worm screw in the centre of a horizontal cylinder, fixed to a central axis with spokes. The diameter of the screw is only slightly less than that of the vat, and when it begins to turn, the pomace cap is drawn down and submerged. Vaslin's Vinimatic® vat is also cylindrical and horizontal. However, unlike the two previous models, it is mounted on rollers and can be rotated. The interior wall has a fixed helical blade. When the vat is rotated, the pomace is trapped by the blade and submerged (Feuillat and Peyron, 1998). The possible automation and ease of use of these vats suggests that they should be used carefully and not excessively. They are also self-emptying and allow fully automated gradual emptying. However, these vats confine the must and need to be aerated, either with diffusers or pumping over, which also enables better temperature control. These four vat designs allow the maceration process to be shortened. They are also multipurpose and can be used for making rosé wines with maceration, or white wines with skin maceration.

These automaceration vats (VASLIN, EMMONOT, FABBRI) were tested in Burgundy in 1982 and 1983 (Thionnet *et al.*, 1984). They were compared to traditional maceration in control vats with the same capacity (50 hL) that were pumped over for 20 minutes twice a day. The punching down mechanisms in the automaceration vats were activated twice a day for 5 minutes. In all the tests, total phenolic compound extraction was greater in the automaceration vats. In the Vaslin vat, which brings the seeds back into suspension with each rotation, there was a particularly high extraction of phenolic compounds (30% greater than the control vat). Extraction of anthocyanins was also greater in the automaceration vats. The Fabbri vat extracted the most anthocyanins and the resulting wine was the easiest to clarify. Furthermore, the percentage of stable anthocyanins, combined with tannins, was higher in the automaceration vats than in the control vats, and the wines from these vats achieved better ratings on tasting than the control wines, particularly in years with high yields (Feuillat and Peyron, 1998).

20.5.4 Thermal treatments applied to traditional maceration

Winemakers have sought to optimize extraction phenomena for a great many years through the application of heat. There is a wide variety of techniques using heat, the most well known being thermovinification, flash-release or, more recently, thermo-release. Hot pre-fermentation maceration had already gained some popularity well before these techniques were developed. Chilling is also used to lower the temperature during pre-fermentation or before maceration, to develop new aromas, or to improve the extraction of skin constituents.

Cold pre-fermentation maceration

Cold pre-fermentation maceration, which also originated in Burgundy, is now

practised in many other regions to make wines with high added value. This procedure consists of keeping the crushed destemmed harvest at between 8 and 15 °C for 2–6 days. When the temperature rises again, fermentation starts spontaneously or after yeast has been added and maceration continues in the traditional manner. Wines made using this procedure are described as being fruitier with stronger colours and smoother tannins.

Several teams (Feuillat, 1997; Cuenat, 1998; Gerbaut *et al.*, 1998; Retali, 2004) have tested the procedure, comparing it to traditional maceration. Measurements of the presence of polyphenolic compounds (anthocyanins and tannins) were carried out, but were not, in nearly all of the cold maceration samples, significantly different from those found in the control wines. In certain tests carried out by Cuenat *et al.* (1996) and Cuenat (1998) on Pinot Noir and by Retali (2004) on Nielluccio, significant differences in the extraction of polyphenolic compounds were recorded. However, these differences could have been due to several factors: the effect of the colder temperatures; the long periods spent in heat exchangers to lower the temperature; or the increase in the overall length of maceration (+4 days and +8 days, respectively) as compared to the control macerations. On tasting, the perception of tannins was similar to that of the control wines. The cold pre-fermentation maceration wines did have darker colours and more developed fruity aromas (blackcurrant and small red fruit) than the control wines. No presence of varietal aromas or their precursors was detected in these studies, and the authors suggest the possibility of an enrichment in odour compounds linked to microbial activity of cryotolerant yeasts (*Hanseniaspora uvarum*) (Massoutier *et al.*, 1998). These yeasts may produce significant quantities of esters or have an effect on certain varietal aroma precursors before the strains of *Saccharomyces cerevisiae* become active (Charpentier and Feuillat, 1998). If this hypothesis is correct, then it would be advisable not to over sulphite the harvest in order to give these strains a chance to develop, although this would be balanced by an increased risk of oxidation or the development of moulds or harmful bacteria.

The Cryo-flash® procedure

The Cryo-flash procedure was developed in 2003 by Imeca (now Pera) and consists of chilling the layers of cells in the skin hypodermis of the grapes until they begin to freeze (−2.5 °C to −3 °C), thereby breaking them down through the development of intracellular micro-crystals of ice. The harvest is destemmed and crushed, and then drained to leave only 60% of the juice in contact with the solids. The latter are then continuously submerged in two successive baths of cold juice. The skins are rich in varietal aroma precursor compounds and polyphenolic compounds. The structural degradation caused by the micro-crystals of ice increases the diffusion into the juice of small-sized compounds such as odour compounds and their precursors (Fig. 20.5) (Crespy and Bessière, 2005; Bessière *et al.*, 2005). This diffusion was greater than that observed in a control sample of grapes that had been macerated without chilling. However, observations with an electron microscope show that the cells' pecto-cellulosic membranes seem to remain intact after the treatment, while the clusters of polyphenolic constituents

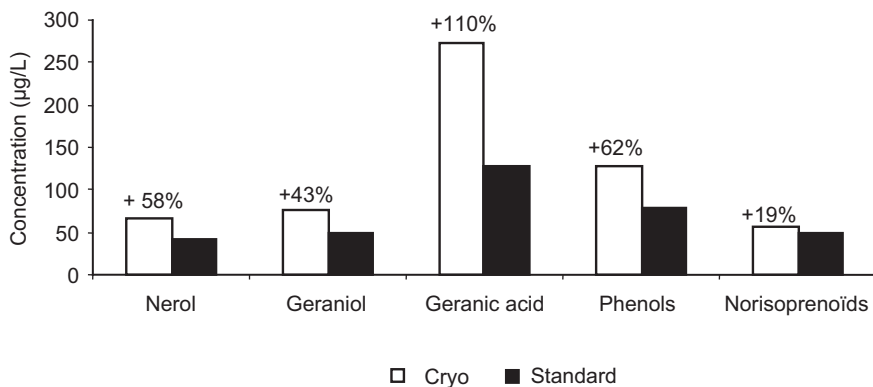


Fig. 20.5 Effect of Cryo-flash treatment (performed in laboratory conditions) on the extraction of glycosidic aroma precursors from Muscat berries (Bessière *et al.*, 2005).

have virtually disappeared. These observations back up the hypothesis of a partial degradation of the cell walls, making them more permeable to certain molecules. However, not all of the polyphenolic compounds are found in the juice: a large proportion remains trapped in the solid parts. A traditional maceration after the Cryo-flash is necessary to complete their extraction: after the freezing treatment, the pomace is returned to a vat with the initial juice and undergoes a traditional maceration that may or may not be preceded by a cold pre-fermentation maceration. The Cryo-flash procedure seems particularly attractive because chilling, unlike heating, does not alter most of the constituents that will determine the quality of the future wine. The only drawback is that the temperature monitoring seems to be difficult to control. As a result this technique is not widely used.

Hot pre-fermentation maceration

In Burgundy, due to the tendency of Pinot Noir to resist releasing its colour, small producers would often heat the grapes (Ferré, 1926). This technique involves heating the harvest in a cauldron with a double lining before it is placed in vats; the problem is that the heating is uneven. Less than 20 years ago, in the same region, some highly respected estates had tubular exchange systems that enabled the harvest to be brought to a temperature of 35 °C quickly and efficiently before it was placed in vats. There is no doubt that such hot pre-fermentation maceration extracts more colour, but maceration at temperatures in the region of 35–40 °C unfortunately provides very favourable conditions for oxidizing enzymatic activity. And since this activity has the particularity of destroying colour, the potential benefits can easily be cancelled out. Furthermore, the application of these temperatures for periods of 12–24 hours can result in the growth of ascetic bacteria likely to produce increases in volatile acidity. This growth is even more likely if the harvest is any way spoiled. In such cases a sizeable sulphite addition is necessary as a protective measure.

Another hot pre-fermentation maceration technique consists of applying a

temperature of between 70 and 75 °C for 6–7 hours. In these conditions, the levels of polyphenols obtained are the same as those of a traditional maceration (Cottureau and Desseigne, 2007). Other effects can also be observed: destruction of the alkyl methoxypyrazines, a softening associated with changes in the tannic structure and hydrolysis of the aroma glycosides, all of which should improve the characteristics of the future wine (Razungles, 2006).

Hot maceration of the pomace

During maceration, the hot maceration of the pomace procedure (Vivas *et al.*, 1991) consists of running off the fermenting juice when it reaches a density of about 1.060. Placed in an empty vat, the juice continues alcoholic fermentation in the liquid phase at a temperature below 25 °C. Left to itself, the temperature of the pomace quickly rises to 38–42 °C. After 48 hours the juice, which still contains highly active yeasts, is recombined with the pomace for alcoholic fermentation to be completed. The increase in temperature of the pomace for 48 hours has the effect of weakening the skin cells and enhancing the extraction of compounds. This technique can enable a 25% increase in total polyphenol extraction after 8 days of maceration.

Hot final maceration

Some researchers have examined the effects of applying higher temperatures during traditional maceration, after alcoholic fermentation has been completed (Sapis and Ribéreau-Gayon, 1979; Ribéreau-Gayon and Ribéreau-Gayon, 1980). This hot final maceration (40 °C) enhances the extraction of tannins and colour. It also offers better protection against oxidase breakdown. However, there would appear to be a risk of over-extraction resulting in flavour defects.

20.6 Very hot, short maceration applied to red wine vinification

20.6.1 Thermovinification

Heating the harvest as a means of improving extraction has been practised for many years in various regions of France, such as Burgundy or the Languedoc (Ferré, 1926), it has also developed in other countries, namely the USA, Romania and Italy. The technique was examined by French researchers during the 1970s (Flanzy and Bénard, 1969; Oliviéri *et al.*, 1976; Bourzeix and Bénard, 1978; Mourgues and Bénard, 1980). It is divided into two clearly separated stages: a pre-fermentation stage or extraction phase and then a fermentation a stage that takes place in the liquid phase (Oliviéri, 1991; Boulet and Escudier, 1995).

Pre-fermentation and fermentation stages

During the pre-fermentation phase, the grapes are heated to a high temperature to enhance the extraction of compounds contained in the solid parts of the grapes, particularly the skins. The temperature is the key factor here, causing degradation

of the cell walls. However, this technique also enables the inhibition of numerous enzymatic activities through heat denaturation of proteins. Oxidase activity is inhibited by heat treatment of the harvest. In particular, this technique inhibits laccase activity by *Botrytis cinerea*. This procedure is thus recommended for the vinification of grapes affected by grey rot.

By pursuing these two objectives of extracting skin constituents and inhibiting laccase activity, the process's temperature–duration parameters were established after numerous trials, as was the requirement of a minimum temperature of 65 °C to correctly extract skin colour. However, this temperature is insufficient to destroy the laccase activity of grapes with *Botrytis cinerea*. Conversely, at 70 °C, oxidase activity is almost completely inhibited. The most appropriate solution would appear to be the maintaining of the grapes at 70 °C for 30 minutes in order to optimize both the extraction of skin compounds and the destruction of oxidation enzymes, while at the same time reducing the energy required for heating (Olivieri *et al.*, 1976).

At the end of the hot maceration, the grapes are pressed and fermentation takes place in the liquid phase. Fermentation in heat-treated musts is very active, and maintenance of the temperature below 25 °C is desirable to avoid alcoholic fermentation being adversely affected by excessively high temperatures.

Procedures and equipment

Manually or mechanically harvested grapes intended for heat-treatment should be destemmed to avoid the extraction of stalk constituents. The grapes are then crushed, as the presence of a sufficient quantity of juice is necessary for the transfer of calories. Sulphites are then added (5–6 g/hL) to prevent any oxidation during the rise in temperature. The must is placed in a buffer vat which ensures a continuous supply to the heat exchangers. There are two types of heating system: scraped surface heat exchangers and systems using direct heating by the must.

Scraped surface heat exchangers for thermovinification were originally developed by Gasquet, and are now supplied by Ertia. The harvest, after being destemmed and crushed, is pumped into two coaxial scraped surface exchangers, installed in series. In the case of a healthy harvest, it is possible to carry out a partial draining before the heating operation in order to save energy and only heat part of the must. However, a minimum amount of must should be left in contact with the solid parts to promote heat transfer. The grapes are heated by hot water (85–105 °C) flowing in the other direction through the heat exchangers' walls. Paddles, supported by the exchangers' central axis, are slowly rotated. They ensure better movement of the grapes from one end to the other of the exchangers and suppress laminar flow that can cause overheating. The temperature must rise quickly (reaching 70 °C in less than 2 minutes), in order to pass through the 35–45 °C range rapidly, where enzymatic activity is a maximum. Having passed through the heat exchangers, the harvest is placed in a vat for a maceration of 20–30 minutes at 70 °C. This vat represents an interim stage before draining and pressing. The juices from draining and pressing are blended and cooled to ambient temperature.

Imeca (now Pera) has developed a heating system involving the immersion of

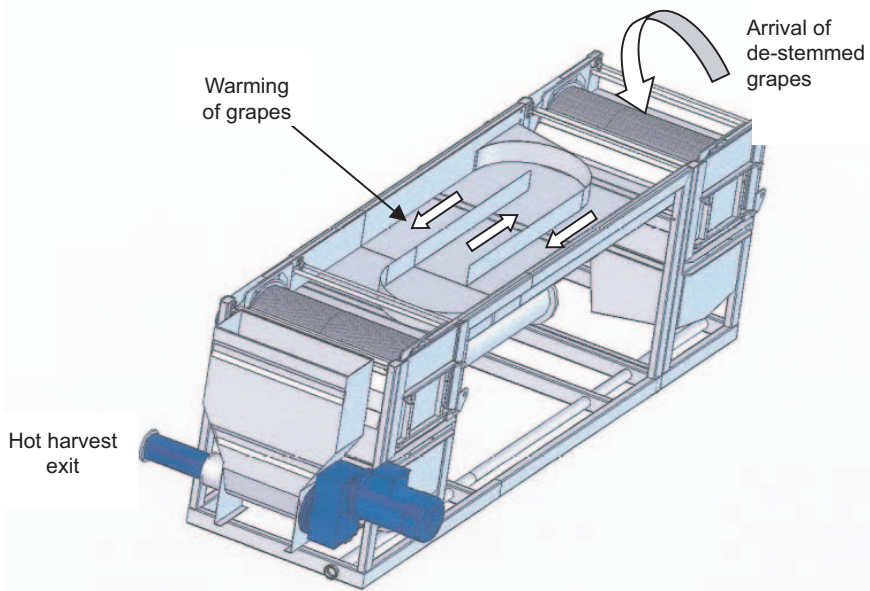


Fig. 20.6 Gulfstream thermal exchanger (Pera) for whole or de-stemmed grapes.

the harvest in its own must, which has been pre-heated. As in the previous procedure, the destemmed and crushed harvest is placed in a buffer vat before being transferred to a container of hot must. The berries are heated instantaneously as a result of the direct contact. The bottom of the container is inclined and has a worm screw which extracts the solid parts that have been brought to 70 °C. They are then placed in a maceration vat for 20–30 minutes. After this operation, the must is drained and the harvest is pressed as above. The musts are combined and cooled to ambient temperature.

This initial linear arrangement has now been replaced by a tower system (Thermocompact). The various stages of the buffer vat, heating of the harvest and draining take place in concentric ring-shaped containers. The buffer vat is high up in the centre of the tower. The grapes from this vat fall into a circular container filled with hot must. They are then transported by this liquid and circulate back through the buffer vat. This operation continues until the grapes reach 70 °C. They are then drained into a new ring container and placed in the maceration vat that makes up the lower part of the arrangement. Pera now offers a further development of this technique (Gulfstream procedure). The grapes are always transported by a liquid and are separated by a dandy roller before the maceration vat.

The Gulfstream system (Fig. 20.6) has many advantages compared to the initial linear system developed by Imeca: this arrangement is compact and does not take up much floor space, the harvest passes from one element to the next without pumping, the device is multipurpose and enables the heating of an entire harvest. This last factor, combined with the possibility of controlling the temperature of the must in a range from 5–80 °C enables a wide range of uses such as the heating of

an entire harvest or whole berries for hot or cold pre-fermentation macerations, or heating a harvest for carbonic maceration. In white winemaking, it also enables a harvest to be cooled with a view to skin maceration or cold pressing.

After cooling, musts that have undergone heat treatment are ready for alcoholic fermentation. However, as with the vinification of white and rosé wines, a pre-fermentation clarification of the musts enable fruitier wines to be obtained through the development of fermentation aromas. Since pectolytic enzymes are destroyed by heat, the natural clarification of musts by sedimentation is difficult, it is better to use physical procedures such as centrifugal decanters, rotary vacuum filters or flotation. This last technique seems particularly suited to the continuous processing of these fairly heavy musts. It consists in eliminating any particles in suspension in the must by injecting a pressurized gas (nitrogen) into the base of the vat. The particles are drawn towards the surface and skimmed off by a rotating rake located at the top of the vat. Alcoholic fermentation takes place at a controlled temperature, below 25 °C.

Effects on the composition of musts and wines

The destruction of nearly all enzymatic activity including oxidase activity ensures a definite improvement in quality in harvests with *Botrytis cinerea*. There is no doubt that colour is preserved in heat-treated wines, as is the aromatic potential, which is also subject to oxidation. However, the loss of a part of the aromatic potential that is generated by the fungus is in no way compensated for by heat-treatment.

Heating of the grapes results in an increase in extractible nitrogen compounds, particularly the peptides and amino acids that are easily absorbed by yeasts and bacteria. Musts from harvests that have been heated thus have an increased fermentation capacity. Even though a proportion of the yeasts and bacteria present in the harvest are destroyed by heat treatment, a microbial flora that has resisted the heat can develop after cooling and spoil the must. The addition of yeast is therefore strongly recommended, as is an appropriate chilling facility, since fermentation can be very active in these conditions.

Heating the grapes extracts more phenolic compounds than a traditional vinification with destemmed grapes; 50–60% of the tannins and 70–80% of the anthocyanins in the skins are extracted (Bourzeix and Bénard, 1978). Compared to traditional vinification, greater extraction is achieved, particularly of the anthocyanins. This results in reduced stability of some of the colour constituents because of the high anthocyanin/tannin ratio, and some of the anthocyanins cannot form stable complexes with the tannins. However, wines that have undergone heat treatment retain their colour better than their unheated counterparts (Boulet and Escudier, 1995).

Acid and neutral polysaccharides increase in equal proportions in heat-treated wines (Mourgues and Bénard, 1980), but without a real impact on the sensorial qualities of the wines. Heating also produces a greater extraction of organic acids and metals but, as alcohol levels increase during fermentation, the salts of these anions and cations become unstable and precipitate, occurring in the finished wine

in the same proportions as those found in wines produced by traditional maceration.

This technique, originally developed for harvests lacking concentration or affected by *Botrytis cinerea*, now enables the production of highly appreciated wines from balanced, healthy harvests. There are also savings in labour and vat facilities. It does, however, require specific skills in order to be managed with the necessary precision. Wines resulting from heat-treatment are often considered to be aromatic and to have stronger colours, they are softer in the mouth and intended for early drinking. Their tannic structure is not suited to laying down. This would not necessarily be the case if the heating was followed by a traditional maceration, as described previously.

20.6.2 Flash-release procedure

With the same objective of extracting desirable compounds contained in the solid parts of the grape (skins, pulp and seeds), researchers at the French National Institute for Agronomic Research (INRA), in collaboration with Aurore Développement and IMECA–Della Toffola, developed a prefermentation extraction procedure based on heating the harvest followed by instantaneous cooling under high vacuum. This rapid extraction procedure was named flash-release (Escudier *et al.*, 1993; Ageron *et al.*, 1995).

Procedures and equipment

As with thermovinification, flash-release is a continuous procedure. The destemmed and partially drained harvest is brought to a high temperature (90–95 °C) by steam injection as it passes through a chamber. Two parallel worm screws in the chamber move the harvest into a volumetric pump. During this transfer, steam is injected through vents located on the floor of the chamber, and in the axes of the screws. In this way the grapes are brought to the desired temperature in less than 2 minutes. Oxygen is excluded from the heating atmosphere. These conditions enable the avoidance of overheating and phenomena such as the Maillard reaction. Some heating devices only reach 80–85 °C, rather than the required temperature of 90 °C. In this case, a 15–30 minute hot maceration is recommended before carrying out the release operation to obtain similar qualitative results. The extraction benefits decrease markedly below 70–75 °C (Moutounet and Escudier, 2000).

After passage through the chamber, the harvest is pumped by a volumetric pump into a high vacuum enclosure (20–50 hPa). The vacuum is produced by a vacuum pump connected to a condenser (Fig. 20.7). The placing in a vacuum of the hot harvest results in the instantaneous vaporization of the water that is present; the evaporation causes the grapes to cool to boiling point (30–35 °C in the ambient vacuum created by the pump). The sudden expansion associated with the instantaneous evaporation of part of the water contained in the harvest breaks down the cellular structures and weakens the cell walls of the skins. The cooled harvest is removed from the vacuum vat by a second volumetric pump to be transferred either to a press, or into a maceration vat.

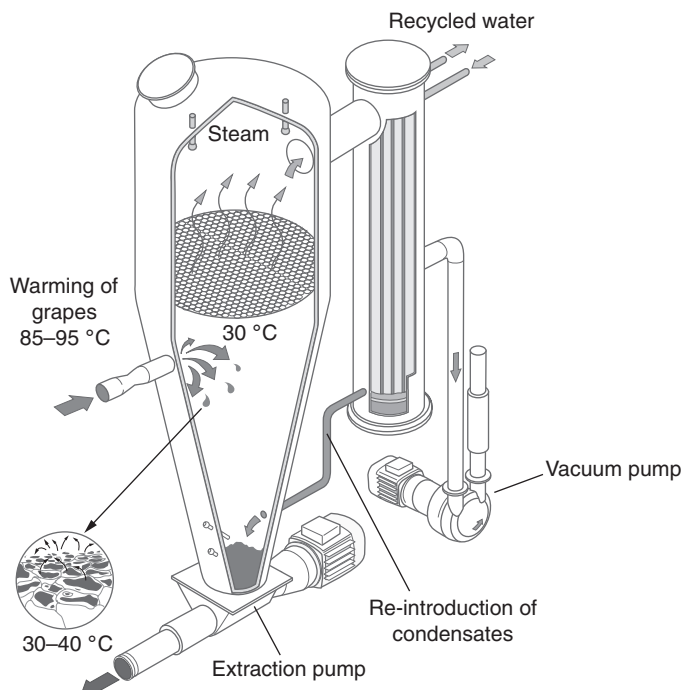


Fig. 20.7 Diagram of the flash-release process (Escudier *et al.*, 2008).

The flash-release procedure requires the incorporation of water in the form of ‘hot’ vapour, but there is no dilution, because this vapour is collected from the release operation in the condenser connected to the vacuum vat. This ‘cold’ vapour coming from the grapes is reheated by an evaporator before being injected into the chamber to heat the harvest. However, not all of the cold vapour, which represents 7–12% of the weight of the harvest, is recycled in the form of steam. The rest can be reincorporated, or not, into the harvest. The second option offers the possibility of a partial concentration of the harvest.

There are two options after the harvest leaves the vacuum vat: pressing or maceration. In the first option, the harvest is pressed when it leaves the vacuum vat. In this case, a short hot maceration (less than 10 minutes) is recommended before the release. The must from the pressing will have a tannin/anthocyanins ratio enabling better sensorial balance and good colour stabilization (Escudier *et al.*, 2006; 2008). The harvest is ‘thick’ and difficult to press. It is difficult to obtain the juice. The latter has the same characteristics as juice from heat-treated harvests.

The second option is to macerate the harvest when it leaves the vacuum vat. In this case, the kinetics of extraction of flash-treated harvests are faster than those observed in traditional maceration. Maximal extraction of anthocyanins and polyphenols occurs after 3–4 days (Boulet and Escudier, 1998). The pomace cap which forms with fermentation is suppler than that obtained in traditional maceration of a destemmed harvest. The skins are easily ‘washed’ and put back into

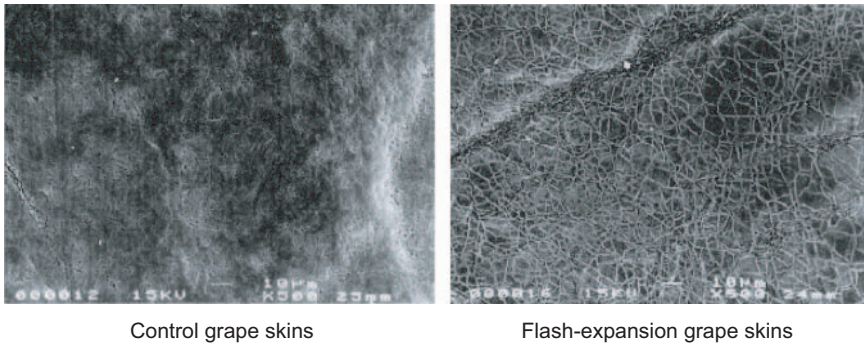


Fig. 20.8 Effect of flash-expansion (flash-release) on the skin surface of berries (using electron microscopy). On the right, cracks of $0.3\ \mu\text{m}$ in depth are visible (Ageron *et al.*, 1995).

suspension during pumping over. After 2 days of maceration with daily pumping over, the seeds separate from the skins and the pulp and settle on the bottom of the vat where they can easily be extracted if a less astringent wine is desired.

As in thermovinification, flash-release enables the treatment of harvests affected by *Botrytis cinerea*. Denaturation of the laccase enzyme produced by this fungus requires 1.2 seconds at a temperature of $91\ ^\circ\text{C}$. The flash-release process largely satisfies these parameters and a study has shown that laccase activity ceases after treatment.

Effects on wine quality

After processing, the grape skins break down (Fig. 20.8) (Boulet and Escudier, 1998). Rheological measurements show that the resistance of grape skins to piercing after flash-release and vinification is less than those from a control vinification (Ageron *et al.*, 1995). Electron microscope observation of skin cell walls that have undergone flash-release reveals a dense network of $0.3\ \mu\text{m}$ deep 'cracks', which are not present in the control grapes. The skin tissue of grapes after flash-release is highly separated. The degraded state of the cellular walls results in increased extraction of the cellular constituents of the skins, which can be used to advantage in red winemaking.

Flash-release treatment enables better extraction of all classes of phenolic compounds (hydroxycinnamic acids, flavonols, anthocyanins, catechins, proanthocyanidins). However, in the absence of post-treatment maceration, the levels of all these compounds decrease considerably during alcoholic fermentation. Wines that have been macerated after flash-release contain large quantities of flavonols, catechins, anthocyanins and proanthocyanidins (Fig. 20.9) (Morel-Salmi *et al.*, 2006; Escudier *et al.*, 2008). Flash-release treatment extracts 30–60% of polyphenols compared to a control wine (destemmed harvest macerated for 3 weeks) (Boulet and Escudier, 1998).

Flash-release also increases the extraction of polysaccharides rich in arabinose and galactose (PRAG) (about 30%), and that of type II rhamnogalacturonan from

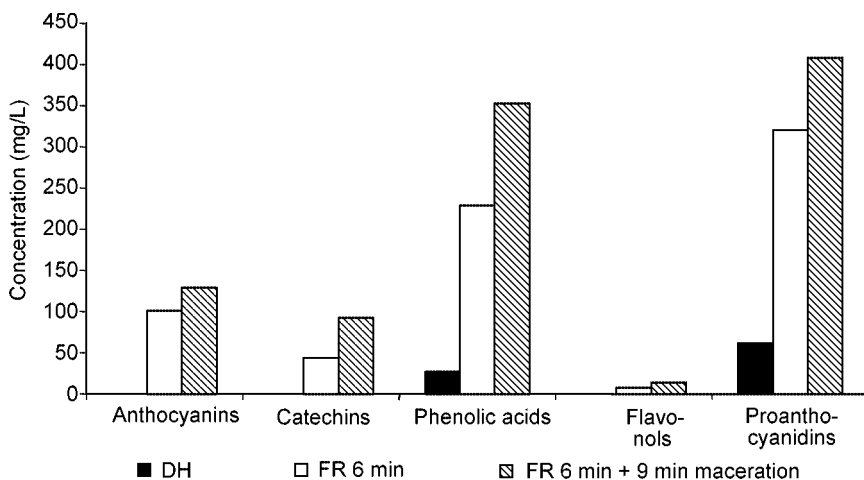


Fig. 20.9 Comparison of polyphenols levels in Grenache musts from a de-stemmed harvest (DH) and two from flash-release (FR) processing just before fermentation. FR, 6 min to reach a temperature of 95 °C (Escudier *et al.*, 2008).

the skin cell walls, although this extraction can only be achieved if flash-release is followed by a maceration of about 10 days. The use of pectinolytic enzymes on grapes that have undergone flash-release treatment results in a significant decrease in PRAG content (48%), but not in levels of type II rhamnogalacturonan, which tend to increase (Doco *et al.*, 2007).

The flash-release procedure was initially developed to extract fruit aromas. It also applies to the extraction of grape aromas and their precursors. The particularly short heating period does not damage the grapes' aromatic quality. A study shows increased levels of fatty acid ethyl esters and of β -ionone in Grenache Noir wines that have been subjected to flash-release (Kotséridis *et al.*, 2002). Lastly, since one can choose not to reincorporate all of the condensed water, it is possible to remove hexenal and hexenal-type aroma compounds which form during destemming of the grapes.

For all of these reasons, flash-release wines tend to present with more structure, colour and astringency. The wines are typically redder, full-bodied with a sensation of sweetness, and have greater length (Moutounet and Escudier, 2000). The style of wines does not become 'standardized' as a result of using the technique. In some cases, there can be considerable astringency which necessitates maturing with moderate additions of oxygen. The higher levels of phenolic compounds result in wines with a greater resistance to oxidation and a good capacity for ageing. Flash-release enables an enrichment of wines' aromatic potential, but is not recommended for white wines because it extracts high quantities of phenolic compounds. The wines produced also tend to be softer.

It is strongly recommended that only destemmed harvests are treated since stalks subjected to flash-release can release elements resulting in greenness and bitterness in the wine that is produced (Boulet and Escudier, 1998). The other

drawback is that the stalks fix the extracted phenolic compounds (Moutounet and Escudier, 2000).

20.6.3 The thermo-release procedure

The thermo-release procedure was developed by BRUNET ERTIA and is not yet widely practised. It consists in applying high pressure to the hot harvest followed by a sudden release with a return to atmospheric pressure. As in flash-release, the objective is to weaken the skin structure in order to increase extraction. The procedure is carried out in two pressurized chambers ('pressure bottles'), placed downstream of the scraped surface heat exchanger and upstream of a maceration vat (Debaud, 2006; Escudier *et al.*, 2008). Unlike flash-release, this procedure does not involve water vaporization and does not require the harvest to be cooled. Heating the harvest to 70–85 °C appears to be sufficient. The pressure applied can vary considerably: from one to five bars depending on the desired style of wine, the variety and ripeness of the grapes.

Comparative trials with traditional maceration showed a 30–40% increase in effective extraction in the must and the finished wine compared to the control. The wines obtained have greater aromatic intensity, structure and volume. The procedure was tested on Merlot, Syrah and Cabernet Sauvignon and enabled improvements in the colour and tannin extraction performance of a thermovinification line (Dubernet *et al.*, 2005). Work carried out by the French National Institute of Agronomic Research (INRA) in 1987 showed that the application of very high pressure (40–60 bars) to a harvest, at ambient temperature, followed by a release to atmospheric pressure, did not result in significant increases in extraction compared to the control. This confirms that temperature is the key factor in this extraction procedure, and not the pressure, even if it is high (Escudier *et al.*, 2008).

20.7 Vinification of red wines by carbonic maceration

Vinification by carbonic maceration (CM) uses the capacity of grapes to change from metabolism by respiration to a type of fermentation by anaerobic metabolism (AM) when the grapes are placed in an atmosphere that is rich in CO₂ and very low in oxygen (Flanzy, 1935). Specific research, with a view to using this type of metabolism to advantage in vinification, began to be carried out in the 1960s (Flanzy *et al.*, 1987; Flanzy, 1998).

20.7.1 Anaerobic maceration and carbonic maceration procedure

Anaerobic metabolism is induced by placing grapes in a vat from which almost all the oxygen has been removed (Chambroy and Flanzy, 1983). In such conditions, certain enzymatic survival systems remain active in the grapes (Garcia *et al.*, 1974; Sauvage *et al.*, 1991), particularly the enzyme alcohol dehydrogenase which plays

a key role in providing energy to the grape cells in anaerobiosis (Molina *et al.*, 1986; Romieu *et al.*, 1992; Tesnière *et al.*, 1994).

During anaerobic metabolism, a small proportion of the hexoses is converted into alcohol with the same yield as a yeast, but production remains limited (1–2% vol. of ethanol formed). The secondary fermentation products are found in the same proportions as those observed for alcoholic fermentation. When the temperature approaches 30 °C, malic acid is highly catabolized and is essentially converted into ethanol and succinic acid. During AM, various changes affect the nitrogen compounds: protein levels go down, while amino acids levels go up (Flanzy *et al.*, 1987). Grapes bathed in this atmosphere absorb CO₂ and other volatile fermentation products such as ethanol (Flanzy *et al.*, 1987). Conversely, grapes discharge CO₂ into the atmosphere that surrounds them, at least during the first part of AM.

At the bottom of the vat, grapes are steeped in juice from the crushing affect of the grapes' own weight. At this level, anaerobic metabolism is slowed by diffusion and inhibition phenomena (Buret *et al.*, 1976). This must begins alcoholic fermentation, which has the effect of maintaining a saturated CO₂ atmosphere. The growing production of alcohol in the must increases the exchanges with the whole submerged grapes.

When the grapes are being placed in vats, moderate sulphating of all batches is recommended (2–3 g per 100 kg of harvest) in order to inhibit the growth of acetic bacteria and moulds. It is also possible to acidify the must in the bottom of the vat in order to discourage unwanted bacteria. The success of CM also resides in placing whole bunches in the vat. Any operations likely to damage or crush the grapes must be avoided. Mechanical harvesting is thus not compatible with CM. The grapes are carefully placed in vats that should not be very deep, to avoid them crushing themselves under their own weight, and thereby releasing too much juice. The vat should be saturated with CO₂ before the grapes are introduced in order to accelerate AM. A continuous supply of CO₂ is recommended to make up for its absorption by the berries, at least until the must that has collected in the bottom of the vat has begun to ferment. The CO₂ generated by this fermentation will eventually compensate for the absorption by the berries.

The time between the filling of the vat and the point at which it becomes a net producer of CO₂ depends on the temperature of the grapes. A warm harvest (35 °C) will reach this point after six hours, while a harvest at 20 °C may take 40 hours. It is therefore advisable to place the grapes in vats at a temperature close to 28–30 °C, for example by harvesting in the afternoon. It is also strongly recommended that fermentation of the must in the bottom of the vat is started quickly by inoculation with selected yeasts. This will reduce the risk of bacterial growth, particularly *Acetobacter aceti* and moulds. The colder the grapes, the greater this risk becomes (Flanzy, 1998). If required, equipment is now available that can raise the temperature of an entire harvest, such as those proposed by Imeca/Pera, which heat the grapes by immersion in hot must. This implies transfers that in some cases can damage the berries. However, this is preferable to anaerobic metabolism at low temperatures (15–20 °C) which no longer serves any purpose (Flanzy, 1998).

Table 20.4 Comparison between extractions of phenolic compounds during a 7 day carbonic maceration (CM) and the same carbonic maceration followed by a 6 day maceration with pumping over (CM + PO)

	Grape variety					
	Syrah		Black Grenache		Black Carignan	
	CM	CM + PO	CM	CM + PO	CM	CM + PO
Total anthocyanins (mg/L)	584	571	176	219	464	452
Procyanidins (g/L)	3.22	3.38	2.19	2.47	2.69	2.91
Colour intensity (absorbance 420 + 520 + 620 nm/1 cm)	8.90	7.70	3.25	3.47	6.47	8.14

Source: INRA Montpellier–UMR SPO Polyphenols Interactions–Experimental Station Pech Rouge.

Maceration with AM can be very variable, depending on the ripeness and variety of grapes. These two parameters will affect the grapes' resistance to crushing. The length of maceration will also depend on the type of wine that one wishes to make: short maceration for supple wines or longer for more structured wines. Maceration times can range from 6 to 12 (or 15) days. For temperatures above 30 °C, shorter macerations (6–8 days) will be sufficient. However, at 25 °C, this may be extended to 12 days. Any longer than this and the winemaker will need to be extra vigilant, on account of the risk of the growth of lactic bacteria in the must in the bottom of the vat. In the event that alcoholic fermentation stops, the bursting of the grapes will continue to enrich this environment with sugar, which will become particularly susceptible to inoculation by lactic bacteria.

In some cases, when the objective is a wine with a capacity for laying down, it is possible, after AM lasting 6–10 days, to continue the maceration for an equivalent period, with daily pumping over as in traditional maceration. A trial with 6 further days of maceration with pumping over, after 7 days of AM, showed an increase of about 10% in the tannin content of the wines compared to a control with just CM (Table 20.4). This increase rises to 20% if pectolytic enzymes are added during the first pumping over (Flanzy, 1998). The anthocyanin content also increased in this trial, but on a more reduced scale (5%). In sensorial terms, the wines produced by a maceration with pumping over after AM generally show greater aromatic intensity, structure and mouthfeel than the control wines, but the impression of sweetness is reduced.

After CM, the must is drained to obtain the free-run juice and the grapes are pressed to obtain the press juice. The latter is generally still rich in sugar and superior to the free-run juice in terms of organoleptic quality. The two juices are usually combined to continue alcoholic fermentation in the liquid phase. However, malolactic fermentation tends to occur naturally in F due to the reduced level of malic acid, which is consumed during AM. In such cases, it is preferable to wait until the end of this fermentation before blending. Fermentation in the liquid phase can be carried out at temperatures around 20 °C. In such conditions, alcoholic and

malolactic fermentation can take place without the loss of the volatile compounds generated during AM.

20.7.2 Effects on wine quality

While they are not as rich in fixed compounds as wines made using traditional maceration, CM wines are characterized by aromatic richness, suppleness and harmonious flavours. The maceration conditions (temperature and duration) in the first stage, and subsequent traditional maceration, can result in CM wines with an appropriate structure for laying down, while at the same time preserving the aromatic characteristics associated with this technique. Several researchers have tried to establish aromatic constituents typical of AM, but without conclusive results (Dubois *et al.*, 1977; Versini and Tomasi, 1983; Dell'oro and Di Stefano, 1991; Salinas *et al.*, 1996), although, these wines were found to have higher levels of volatile phenols after being kept (Etievant *et al.*, 1989). Tested on Muscat, Syrah and Grenache, this technique enables the varietal character of wines' aromas to be highlighted (Bitteur *et al.*, 1996). In particular, it enables the development of the aroma characteristics of varieties considered to be of a more 'neutral' nature (Flanzy, 1998). CM is more developed in hot regions where it enables the production of good-quality white, rosé and red wines.

20.8 Traditional vinification of red wines with whole berries

The whole-berry vinification technique was developed with the arrival of the new destemmers which, combined with densimetric or optical sorting systems, now enable vats to be filled with whole berries, separated from their pedicel and any fragments of plant matter. The vats are filled by gravity or with pumps that do not damage the grapes (peristaltic or helical rotor pumps). This is generally followed by cold pre-fermentation maceration for several days at 8–10 °C. After this cold maceration stage, the juice present in the bottom of the vat begins to ferment as the temperature rises. Yeast inoculation is recommended to ensure healthy alcoholic fermentation. Given the small quantity of juice released by the grapes, punching down is preferred to pumping over for homogenization, and the beginning of a gentle extraction of skin compounds. After a few days of maceration, the vat will contain enough juice to allow pumping over rather than punching down, the frequency of which can be reduced. Traditional maceration continues until draining off and pressing. In some cases, the grapes are placed in barrels of new wood, which contribute to the quantities of tannins in the fermenting must, thereby helping to stabilize the anthocyanins in the presence of the ethanal produced by the yeasts.

Little scientific information is currently available on this technique and it is difficult to assess which of the various parameters involved (whole grapes, cold maceration, punching down) contribute most to the quality and characteristics of the wines obtained. The resulting wines tend to be dark in colour, with fairly typical, powerful fruity aromas. They are well-structured in the mouth and the

tannins produce a marked but pleasant astringency. This vinification technique has been highly successful and has gained popularity in many of the top wine-producing regions.

20.9 Draining and pressing

It would be impossible to end this chapter on extraction during vinification without mentioning pressing. This indispensable operation for extracting the juice from the grapes is as old as wine itself. It would be unrealistic to attempt to provide a detailed consideration of all pressing techniques, which would justify another entire chapter. We will therefore content ourselves in this section with a review of the latest advances in draining and pressing operations.

20.9.1 Draining

Draining cannot completely replace pressing, but can enable a large percentage of the juice to be extracted purely by gravity, under the harvest's own weight. Requiring little energy, this operation is particularly well-suited to the extraction of juice from fresh or heat-treated harvests. Draining also helps to save the pressing operation for the extraction of the residual juices. Drained juice generally contains fewer particles and dissolved polysaccharides than pressed juices because the frictional forces applied during harvesting are weaker than those applied by a press.

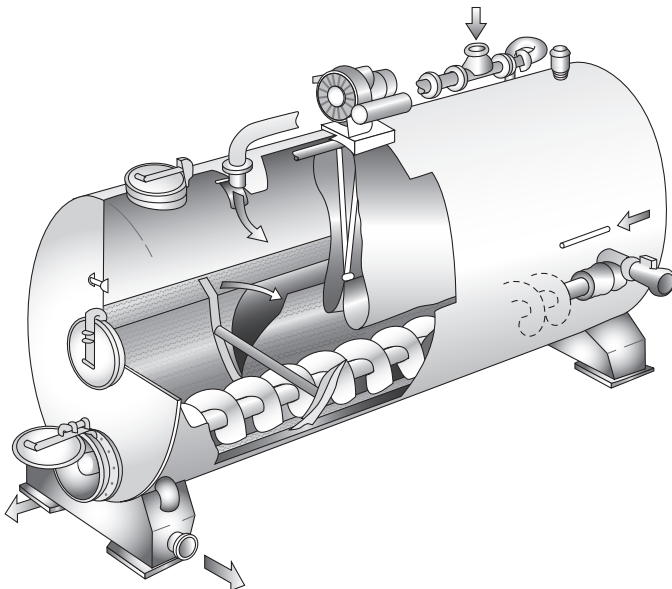


Fig. 20.10 Diagram of the Elite vat (Pera)

While dynamic drainers with inclined worm screws derived from continuous presses remain the most effective devices for the continuous treatment of red grape harvests after heat-treatment or flash-release, extraction by draining of white and rosé wine musts owe a great deal to the development of the self-emptying vats described earlier. These vats have enabled the widespread use of draining with a corresponding decrease in the proportion of the harvest that is pressed. Juice obtained by draining has excellent qualitative characteristics. However, the self-emptying vats have primarily been maceration vats for the production of red wines and some manufacturers now offer vats especially dedicated to draining, such as the 'Elite' vat made by Pera (Fig. 20.10). This vat has a horizontal shape to enable the juice to be drained off more quickly as the harvest is kept at a low level. The vat is fitted with two worm screws in its base in order to remove the drained harvest more easily. It also has an inflatable rubber balloon suspended from the top of vat which, when inflated, presses down on the harvest. This gentle pressure has a similar effect to a light pressing and increases juice extraction while preserving its characteristics of free-run juice (Desseigne and Grenier, 1998).

20.9.2 Pressing

Pressing constitutes the final stage of extraction in all forms of vinification. Particular care must be taken during this operation, because, if the press is not used appropriately, the quality of the juice or wines extracted can very quickly suffer. There are several types of press with different qualitative and flow characteristics. Each type should be used in accordance with objectives in terms of production and quality.

Continuous presses

Pera impulsion presses enable continuous pressing of a harvest after heat-treatment or flash-release. Smaller quantities of lees are generated compared to traditional continuous worm screw presses, and can be substantially reduced by flotation. This technique, also continuous, is suitable for musts that contain particularly large quantities of matter. Continuous band presses use another principle, combining the benefits of pneumatic and continuous presses (Ibarrola, 1989). However, these presses occupy a large floor area, and issues with the cleaning and repair of the bands have limited their development.

Discontinuous presses

The discontinuous presses consist of pneumatic presses and plate presses. The market is currently shared by two types of pneumatic press: open cage and closed cage presses. These are made by several manufacturers in various countries. Closed cage presses have the advantage that they retain the juice after filling, thereby allowing skin macerations for white and rosé wines. Pneumatic pressing has become standard in the vinification of these wines due to the small quantities of lees that it creates. However, transfer operations of the harvest during filling, particularly in the case of axial filling, can result in a partial crushing of the harvest,

seriously reducing the benefits of using these presses (Delteil, 1998). Plate presses are used mostly for pressing red wine pomace, but can be used in white wine vinification, provided operations to break up the pomace (*rebêchage*) are reduced, as they result in more lees, and also higher levels of phenolic compounds and potassium.

Pneumatic closed cage presses

Pneumatic presses are increasingly designed with closed cages and are currently the subject of three areas of research: automation of control, ease of cleaning and prevention of oxidation. Automation of the pressing cycles, originally controlled by pressure sensors, is increasingly controlled by more accurate and more efficient flow sensors. Traditional flow meters are installed on the drainage channels of the cages of large units, while simpler systems with floats can be installed in the troughs of smaller presses. These sensors provide additional information. The monitoring of pressings programmes and the separation of different presses is made easier.

The cleaning of the drains in closed cage presses remains a delicate and time-consuming operation. In some presses, flexible hose drains pass through the middle of the harvest to facilitate the removal of the juice, but these are difficult to clean and need to be dismantled on a regular basis. Fixed drains on the interior surface of the cages can be cleaned by backwashing jets attached to the ends of flexible hoses. Considerable research has been carried out on the streamlining of these drains and drainage outlets to facilitate their unblocking without manual intervention.

Protection against ambient oxygen at the time of pressing remains a major concern in white wine vinification, particularly in the case of grapes varieties that are sensitive to oxidation. Different manufacturers offer techniques to protect must against oxygen during pressing (Diemme, Pera: Enoxy process). Vaslin-Bücher has developed a press (Inertys® process) equipped with a hermetic mechanism that protects musts from oxygen (Fig.20.11). The mechanism relies on the hermetic sealing of the cage and the must drainage pipes. Changes in the effective volume of the cage during pressing are compensated by the injection or withdrawal of a neutral gas (nitrogen). The gas is stored in a balloon by means of a reversible pump. When the membrane is deflated in order to break up the pomace, the space made by the drained juice is replaced by nitrogen). This procedure appears to be effective in protecting musts against oxidation: the resulting musts keep a greenish colour compared to those from a press with a traditional membrane of the same size. The value of this type of protection with regard to the preservation of volatile thiols is the subject of several studies.

Plate presses

Interest has revived in vertical presses with horizontal plates for the pressing of fermented pomace in traditional red winemaking. Equipped with stainless steel cages, they enable a full, high-quality pressing of the pomace. During filling, the cage is placed directly beneath the vats with a fork-lift truck, thereby avoiding any

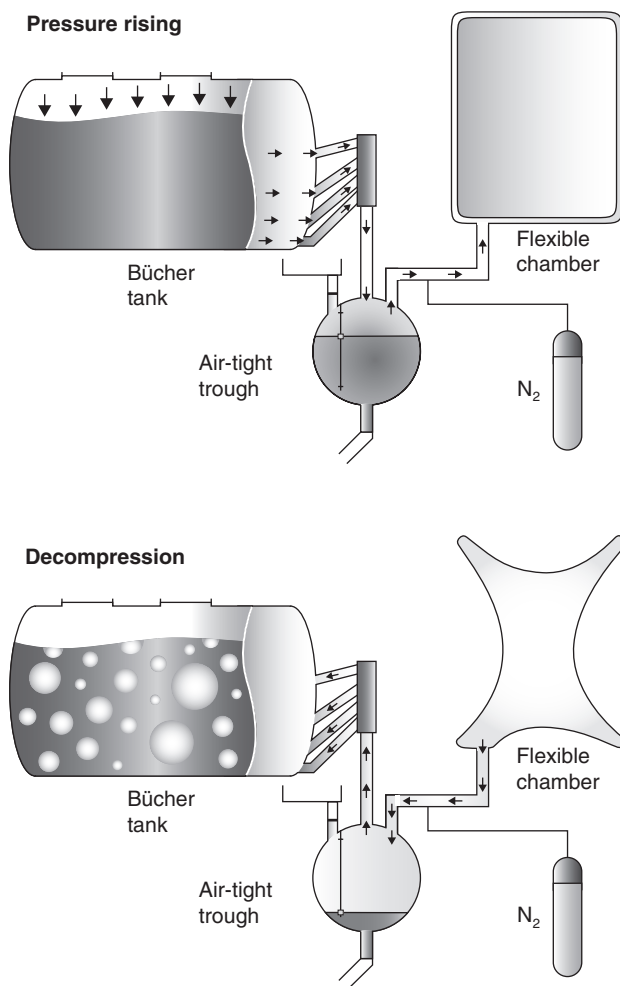


Fig. 20.11 Diagram of the Vaslin-Bücher pressing process showing extraction of must under vacuum and protection of pomace by nitrogen.

pumping of the pomace. The emptying of these presses is carried out in the same way with a fork-lift truck removing the cake of pressed pomace. The only drawback is that breaking up of the pomace is not possible with this type of press.

In order to overcome this problem and enable the use of plate presses for the extraction of musts from white grape harvests, Coquard, a manufacturer in the Champagne region, has designed two systems involving presses with vertical plates that allow a gentle breaking up of the pomace. The first press consists of sliding floor and a cage with two walls and a top that are removable. There are then two pressing plates mounted on jacks that face each other. Initial pressing results in the pomace being compressed into a parallelepiped which is then rotated

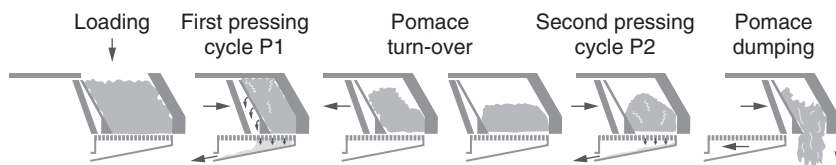


Fig. 20.12 Automatic horizontal basket press with vertical plates (Coquard).

through 90° by the floor to be broken up and compressed again after putting the side walls and ceiling of the cage back into position. The second press is simpler. The floor is fixed, but the pressing plates are inclined to the vertical, producing a trapezoidal pomace cake after pressing (Fig. 20.12). When the plates are withdrawn, this unstable shape tips to one side. For the second pressing, the pomace cake is then at a different angle and is broken up by the plates before being pressed. This press is particularly well suited to white winemaking and is authorized for pressing grapes in Champagne.

20.10 Conclusion

As we have seen over the course of this chapter, the extraction techniques that may be applied in the making of red, white and rosé wines are numerous and varied. Certain techniques and certain extraction factors may be common to several types of vinification. The choice of the most appropriate technique will be made on the basis of production objectives, the desired style of wine and the quality of the grapes to be treated.

Other factors, that we have not considered here, will also have to be taken into account. These concern the costs of implementing the procedures, and the cost of the equipment and its maintenance. All of the successive operations involved in the introduction of a new technique have to be considered in calculating these costs. Finally, energy consumption and the environmental impact of the equipment are becoming increasingly important elements. It would thus seem sensible to favour strategies that take into account the climate of the region where the wines are to be made, in order to take advantage of natural conditions, particularly the ambient temperature, thereby enabling energy savings.

We can therefore say that, in the short term, processing technologies should no longer be considered as procedures that have to be adapted to upstream and downstream technical constraints. They should enter as factors in the decision-making process in their own right, on the same basis as commercial factors and other factors relating to the nature of the wine and the harvest.

20.11 References

Ageron D, Escudier JL, Abbal P and Moutounet M (1995), Prétraitement des raisins par flash-détente sous vide poussé. *Rev. Fr. Œnol.*, **153**, 50–54.

- Amerine MA, Berg HW, Kunkel RE, Ough CS, Singleton VL and Webb AD (1980), *The Technology of Wine Making*. 4th edn, Avi Publishing, Westport, CT.
- Amrani Joutei K and Glories Y (1995), Tanins et anthocyanes : localisation dans la baie et mode d'extraction. *Rev. Fr. Œnol.*, **153**, 28–31.
- Auw JM, Blanco V, O'Keefe SF and Sims CA (1996), Effect of processing on the phenolic and color of Cabernet Sauvignon, Chambourcin and Noble wines and juices. *Am. J. Enol. Vitic.*, **47**, 279–286.
- Bakker J, Bridle P, Bellworthy SJ, Garcia-viguera C, Reader HP and Watkins SJ (1998), Effect of sulfur dioxide and must extraction on colour, phenolic composition and sensory quality of red table wine. *J. Sci. Food Agric.*, **78**, 297–307.
- Baumes R, Bayonove C, Barillère JM, Escudier JL and Cordonnier R (1988), La macération pelliculaire dans la vinification en blanc. Incidence sur la composante volatile des moûts. *Conn. Vigne Vin*, **22**, 209–223.
- Bessiere A, Pujol F, Crespy A (2005), La Cryo-flash: un nouvel outil d'aide à l'obtention de vins fruités. *Rev. Fr. Œnol.*, **215**, 30–35.
- Bitteur S, Tesnière C, Fauconnet A, Bayonove C and Flanzy C (1996), Anaérobiose carbonique du raisin Muscat. 2. Evolution de la répartition des terpénols libres et liés. *Sci. Aliments*, **16**, 37–48.
- Boidron JN, Torres P (1978), Influence de la pourriture grise sur les arômes du Muscat. *Progrès Agric. Vitic.*, **95**, 1–7.
- Boulet JC, Escudier JL (1995), Thermotraitement de la vendange: évolutions et conséquences. *Rev. Fr. Œnol.*, **153**, 45–49.
- Boulet JC and Escudier JL (1998), Flash-détente. In: Flanzy C (ed.), *Œnologie*, 1998, Lavoisier Tec et Doc, Paris, France, 797–805.
- Bourzeix M and Bénard P (1978), Influence du chauffage de la vendange sur les caractéristiques des vins obtenus. *C R Acad. Agric. Fr.*, **64**, 1190–1196.
- Buret M, Chambroy Y, Flanzy C (1976), Métabolisme anaérobie de la baie de raisin en milieu liquide. II. Etude expérimentale. *Sci. Aliments*, **25**, 45–72.
- Cabeza M.S., Merin M.G., Martin M.C., Sabate D.C., Audisio M.C., Morata de Ambrosini V.I (2009). Effect of a pectinase-surfacting preparation on extraction of pigments and total polyphenol from Malbec grape skins. *Am. J. Enol. Vitic.*, **60**(4), 477–483.
- Cayla L (2007), Enzymage préfermentaire: plusieurs intérêts. *Rosé.com*, **11**, 10–11.
- Cayla L (2009), Conditions d'extraction des moûts. In: Flanzy C, Masson G and Millo F (eds), *Le Vin Rosé*. Féret Ed., Bordeaux, France, 153–155.
- Chambroy Y and Flanzy C (1983), Echanges gazeux entre baies de raisin et une atmosphère ambiante appauvrie en oxygène. I. Etude générale et méthodologie. *Sci. Aliments*, **3**, 451–467.
- Charpentier C and Feuillat M (1998). Métabolisme des levures cryotolérantes : application à la macération préfermentaire du Pinot Noir en Bourgogne. *Rev. Fr. Œnol.*, **170**, 36–38.
- Chauvet S, Sudraud P and Jouan P (1986), La cryoextraction sélective des moûts. Premières observations et Perspectives. *Bull. OIV*, **667–668**, 1022–1043.
- Clare SS, Skurray G and Shalliker RA (2004), Effect of pomace-contacting method on the concentration of cis- and trans-resveratrol glucoside in wines. *Am. J. Enol. Vitic.*, **55**(4), 401–406.
- Cordonnier R (1987), Les effets de *Botrytis cinerea* sur la couleur et l'arôme. *Rev. Fr. Œnol.*, **108**, 21–29.
- Cottareau P and Desseigne JM (2007), Chauffage de la vendange et arômes fruités. *CR Entrepreneurs Viti-Vinicoles Rhône Méditerranée*, 20–22.
- Coutenceau C (1988) Le brassage gaz. *Vigne et Vin*, 71–73.
- Crachereau JC (2009), La saignée: point critique de l'élaboration des vins rosés et Clairets de Bordeaux. In: Flanzy C, Masson G and Millo F (eds), *Le Vin Rosé*. Féret Ed., Bordeaux, France, 157–159.
- Crespy A and Bessiere A (2005), La pellicule de la baie de raisin : anatomie de la partie du

- fruit et possibilités d'utilisation optimale en vinification par la cryo-flash IMECA. *Rev. Œnol.*, **117**, 34–38.
- Cuenat P (1998), Techniques de vinification du Pinot Noir en zone septentrionale. *Rev. Fr. Œnol.*, **170**, 53–57.
- Cuenat P, Lorenzini F, Bregy CA and Zufferey E (1996), La macération préfermentaire à froid du Pinot noir. *Rev. Suisse Vitic. Hortic. Arboric.*, **28**(4), 259–265.
- Debaud A (2006), Evolution des techniques de chauffage de la vendange. *Rev. Fr. Œnol.* (Cahier Technique), **216**, 20–26.
- Defranoux C., Gineys D., Joseph P (1989), Le potentiel aromatique du Chardonnay. Essai d'utilisation des techniques de cryoextraction et de supraextraction (procédé Kreyer). *Rev. Œnol.*, **29**, 29–34.
- Dell'Oro V., Di Stephano R (1991), Macerazione carbonica. Evoluzione degli acidi vanillico e siringico e degli antociani durante la conservazione di uve Barbera e Nebbiolo in atmosfera di anidride carbonica. *Vini Italia*, **453**, 95–108.
- Delteil D (1998), Le pressurage des raisins. In: Flanzky C (ed), *Œnologie*, 1998, Lavoisier Tec et Doc, Paris, France, 711–714.
- Desseigne JM and Caussanel JM (1990), Etude du remontage par diffusion de gaz. *Vigne et Vin*, 97–100.
- Desseigne JM and Grenier P (1998), Génie œnologique et matériel: évolution et dimensionnement. In: Flanzky C (ed), *Œnologie*, 1998, Lavoisier Tec et Doc, Paris, France, 1218–1262.
- Doco T, Williams P and Cheynier V (2007), Effect of flash release and pectinolytic enzyme treatments on wine polysaccharide composition. *J. Agric. Food Chem.*, **55**, 6643–6649.
- Dubernet M, De Parzia E, Dufort B and Fontaine P (2005), La thermo détente, une nouvelle dimension apportée à la thermovinification. *Rev. Fr. Œnol.*, **214**, 24–28.
- Dubois P, Etiévant P, Dekimpe J, Buret M, Chambroy Y and Flanzky C (1977), Etude sur l'arôme des vins de macération carbonique. *CR Acad. Agric.*, **63**, 1183–1189.
- Dubourdieu D, Ollivier C and Boidron JN (1986), Incidence des opérations préfermentaires sur la composition chimique et les qualités organoleptiques des vins blancs secs. *Conn. Vigne Vin*, **20**, 53–76.
- Ducasse MA, Canal Llaubères RM, De Lumley M, Williams P, Souquet JM, Fulcrand H, Doco T and Cheynier V (2010), Effect of macerating enzyme treatment on the polyphenol and polysaccharide composition of red wines. *Food Chem.*, **118**, 369–376.
- Ducellier G (1963), La valve hydrolique et quelques unes de ses applications. *Génie Chimique*, **89**, 53–59.
- Escudier JL, Moutounet M and Cogat PO (1993), *Produit alimentaire, obtention et application à la fabrication de jus de fruits ou de vins*. French Patent 9313287.
- Escudier JL, Bes M, Morel C, Micolajczak M, Sanson A and Martin M (2006), Vinification en rouge : macérations post-fermentaires, macérations carboniques, flash-détente sous vide. *Rev. Fr. Œnol.* (Cahier technique), **216**, 11–19.
- Escudier JL, Micolajczak M and Bes M (2008), Chauffage de la vendange: les technologies disponibles et les méthodes de vinification associées. In: Nouvelles technologies en œnologie, techniques et applications. *Rev. Fr. Œnol.* (Cahier Technique), **229**, 9–18.
- Etiévant P, Issanchou S, Marie S, Ducruet V and Flanzky C (1989), Sensory impact of volatile phenols on red wine aroma: influence of carbonic maceration and time of storage. *Sci. Aliments*, **9**, 19–33.
- Ferré L (1926), Autolyse de la matière colorante dans les raisins entiers soumis à l'action de la chaleur humide. Application à la vinification des vins rouges. *CR Acad. Agric. Fr.*, **12**, 370–381.
- Feuillat M (1997), Vinification du Pinot Noir en Bourgogne par macération préfermentaire à froid. *Rev. Œnol.*, **82**, 29–31.
- Feuillat M and Peyron D (1998), Influence du système de cuvaizon 'pigeage/remontage' sur la composition phénolique des vins. In: Flanzky C, (ed.), *Œnologie*, Lavoisier Tec et Doc, Paris, France, 773–779.

- Flanzy M (1935), Nouvelle méthode de vinification. *C R Acad. Agric. Fr.*, **21**, 935–938.
- Flanzy C (ed.) (1998), *Œnologie*, Tec et Doc Lavoisier, Paris, France.
- Flanzy M and Bénard P (1969), Influence de divers modes de vinification, en particulier du chauffage de la vendange, sur les caractéristiques des vins. *C.R. Acad. Agric. Fr.*, **55**, 1266–1274.
- Flanzy C, Flanzy M and Bénard P (1987), *La vinification par macération carbonique*. INRA, Paris, France.
- Flanzy C, Masson G and Millo F (2009), *Le Vin Rosé*. Féret Ed., Bordeaux, France.
- Fournard D, Vicens A, Sidhoum L, Souquet JM, Moutounet M and Cheynier V (2006), Accumulation and extractibility of grape skin tannins and anthocyanins at different advanced physiological stages. *J. Agric. Food Chem.*, **54**, 7331–7338.
- Garcia P, Buret M, Chambroy Y and Flanzy C (1974), Evolution de l'activité de la malate NADP oxydoréductase (EC 1.1.1.40) au cours du métabolisme anaérobie des baies de raisin. *CR Acad Agric. Fr.*, **60**, 1193–1198.
- Gerbaux V, Naudin R, Meurgues O and Monamy C (1998), Influence de différents procédés de macération sur la composition phénolique, l'activité laccase et la qualité organoleptique des vins de Pinot Noir. *Rev. Fr. Œnol.*, **170**, 47–52.
- Gerbaux V, Vincent B and Bertrand A (2002), Influence of maceration temperature and enzymes on the content of volatile phenols in Pinot noir wines. *Am. J. Enol. Vitic.*, **53**(2), 131–137.
- Gomez-Plaza ER, Gil-Munoz JM, Lopez-Roca A, Martinez-Cutillas A and Fernandez-fernandez JI (2001), Phenolic compounds and color stability of red wines. Effect of skin maceration time. *Am. J. Enol. Vitic.*, **52**, 266–270.
- Gunata Z, Bayonove C and Cordonnier R (1985), The aroma of grapes. Extraction and determination of free and glycosidically bound fractions of some grape aroma components. *J. Chromatogr.*, **331**(1), 83–90.
- Gunata Z, Dugelay I, Sapis JC, Bayonove C and Baumes R (1993), Role of enzymes in the use of the flavor potential from grape glycosides in winemaking. In: Schreier P and Winterhalter P (eds). *Progress in Flavour Precursor Studies*, Allured Publishing, Carol Stream, IL, 219–234.
- Ibarrola M (1989), Las prensas de bandas : una tecnologia punta en la elaboracion de vinos blancos de calidad. *Vitic./Enol. Prof.*, **1**, 61–65.
- Jackson RS (1994), *Wine Science. Principles and Applications*. Academic Press, San Diego, CA.
- Kotséridis Y, Escudier JL and Moutounet M (2002), Flash détente et qualité des vins. *Progr. Agric. Vitic.*, **20**, 438–442.
- Main GL and Morris JR (2007), Effect of macerating enzymes and postfermentation grape-seed tannins on the color of Cynthiana Wines. *Am. J. Enol. Vitic.*, **58**(3), 365–372.
- Massoutier C, Alexandre H, Feuillat M and Charpentier C (1998), Isolation and characterization of cryotolerant saccharomyces strains. *Vitis*, **37**(1), 55–59.
- Medina K, Boido E, Dellacassa E and Carrau F (2005), Yeasts interactions with anthocyanins during red wine fermentation. *Am. J. Enol. Vitic.*, **56**(2), 104–109.
- Molina I, Nicolas M and Crouzet J (1986), Grape alcohol deshydrogenase. I. Isolation and characterization. *Am. J. Enol. Vitic.*, **37**, 169–173.
- Morel-Salmi C, Souquet JM, Bes M and Cheynier V (2006), Effect of flash release treatment on phenolic extraction and wine composition. *J. Agric. Food Chem.*, **54**, 4270–4276.
- Mourgues J and Bénard P (1980), Le chauffage de la vendange et ses conséquences. *CR Acad. Agric. Fr.*, **66**, 823–827.
- Moutounet M and Escudier JL (2000), Prétraitements des raisins par flash-détente sous vide; influence sur la qualité des vins. *Bull. OIV*, **73**, 827–828.
- Olivieri C (1991), Aspects biologiques, physicochimiques et techniques des traitements thermiques des raisins entiers ou foulés. *Bull. OIV*, **604**, 477–497.
- Olivieri C, Salgues M and Planque J (1976), Chauffage de la vendange et SO₂. *Progr. Agric. Vitic.*, **12**, 411–419, and **13**, 438–441.

- Peynaud E (1972), *Connaissance et Travail du Vin*. Dunod, Paris, France.
- Peyrot des Gachons C, Tominaga T and Dubourdiou D (2002), Localisation of S-cysteine conjugates in the berry: Effect of skin contact on aromatic potential of *Vitis Vinifera* L. cv. Sauvignon Blanc must. *Am. J. Enol. Vitic.*, **53**(2), 144–146.
- Razungles A (2006), Vinification traditionnelle en rouge: traitement de la vendange, macération et extraction. *Rev. Fr. Œnol. (Cahier Technique)*, **216**, 1–10.
- Razungles A, Bayonove C, Cordonnier R and Sapis JC (1988), Grape carotenoids: changes during the maturation and localization in mature berries. *Am. J. Enol. Vitic.*, **39**(1), 44–48.
- Razungles A, Blouin J, Boulet JC, Escudier JL, Feuillat M, Flanzky C and Peyron D (1998), Vinification en rouge. In: Flanzky C (ed.), *Œnologie*, Lavoisier Tec et Doc, Paris, France, 753–809.
- Retali E (2004), Macération préfermentaire à froid: application à une vendange de Nielluccio. *Rev. Fr. Œnol.*, **209**, 16–18.
- Reynolds A, Cliff M, Girard B and Koop TG (2001), Influence of fermentation temperature on composition and sensory properties of Semillon and Shiraz wines. *Am. J. Enol. Vitic.*, **52**, 235–240.
- Ribéreau-Gayon P and Ribéreau-Gayon J (1980), La vinification en rouge par macération finale à chaud. *CR Acad. Agric.*, **66**, 207–215.
- Ribéreau-Gayon J, Peynaud E., Ribéreau-Gayon P and Sudraud P (1976), *Traité d'Œnologie Sciences et Techniques du Vin. Tome 3 – Vinifications Transformations du Vin*. Ed. Dunod, Paris, France.
- Ribéreau-Gayon P, Dubourdiou D, Donèche B and Lonvaud A (2004), *Traité d'Œnologie. 1- Microbiologie du Vin – Vinifications*, 5^{ème} edn, La Vigne – Dunod, Paris, France.
- Role L, Torchio F, Zeppa G and Gerbi V (2009), Relationship between skin break force and anthocyanin extractability at different ripening stages. *Am. J. Enol. Vitic.*, **60**(1), 93–97.
- Romieu C, Tesnière C, Than Ham L, Flanzky C and Robin JP (1992), An examination of the importance of anaerobiosis and ethanol in causing injury to grape mitochondria. *Am. J. Enol. Vitic.*, **43**, 129–133.
- Sacchi KL, Bisson LF and Adams DO (2005), A review of the effect of winemaking techniques on phenolic extraction in red wines. *Am. J. Enol. Vitic.*, **56**(3), 197–206.
- Salinas MR, Alonso GL, Navarro G, Pardo F, Jimeno J and Huerta MD (1996), Evolution of the aromatic composition of wines undergoing carbonic maceration under different aging conditions. *Am. J. Enol. Vitic.*, **47**, 134–144.
- Salinas MR, Garijo J, Pardo F, Zalacain A and Alonso GL (2003), Color, polyphenol, and aroma compounds in rosé wines after prefermentative maceration and enzymatic treatments. *Am. J. Enol. Vitic.*, **54**(3), 195–202.
- Sapis JC, Ribéreau-Gayon J (1979), Etude expérimentale de la vinification en rouge. *CR Acad. Agric. Fr.*, **65**, 472–478.
- Sauvage FX, Romieu C, Sarris J, Pradal M, Robin JP and Flanzky C (1991), Evolution de quelques activités enzymatiques au cours de la maturation du raisin. Influence d'un stress hypoxique après la vendange. *Rev. Fr. Œnol.*, **132**, 14–20.
- Tesnière C, Romieu C and Robin JP (1994), Grape *in vivo* protein synthesis: changes to response to anaerobiosis. *Am. J. Enol. Vitic.*, **45**, 267–272.
- Thionnet B, Peyron D, Feuillat M (1984), Influence des techniques de cuvaizon sur la teneur en composés phénoliques des vins de Bourgogne. *CR Acad. Agric. Fr.*, **70**, 395–401.
- Versini G and Tomasi T (1983), Confronto tra i componenti volatili dei vini rossi ottenuti con macerazione tradizionale e macerazione carbonica. Importanza differenziante del cinnamato di etile. *L'Enotecnico*, **19**, 595–600.
- Villetaz JC (1996), Utilisation des enzymes en œnologie pour l'extraction de la couleur et la révélation des arômes. *Bull. OIV*, **69**, 843–860.
- Vivas N and Vache L (1991), Interprétation des performances de la macération par l'observation des mouvements liquides et gazeux agissant sur le marc. *Rev. Œnolog.*, **60**, 51–55.

- Vivas N, Callede JP and Gautier P (1991), Conduite d'une vinification en rouge comportant deux phases séparées. *Rev. Fr. Œnol.*, **130**, 29–32.
- Watson BT, Price SF and Valladao M (1995), Effect of fermentation practices on anthocyanin and phenolic composition of Pinot noir wines. *Abstr. Am. J. Enol. Vitic.*, **46**, 404.
- Wilson B, Strauss CR and Williams PJ (1986), The distribution of free and glycosidically bound monoterpenes among skin, juice and pulp fractions in some white grapes varieties. *Am. J. Enol. Vitic.*, **37**(2), 107–111.
- Zimman A, Joslin WS, Lyon ML, Meier J and Waterhouse AL (2002), Maceration variables affecting phenolic composition in commercial-scale Cabernet Sauvignon winemaking trials. *Am. J. Enol. Vitic.*, **53**(2), 93–98.

Index

- AAP *see* o-aminoacetophenone
acetaldehyde, 228, 437
acetaldehyde dehydrogenases, 307
acetaldehyde-induced polymerisation, 229
acetic acid, 26, 307–9, 542–3
 factors influencing production, 308–9
 metabolic pathways and genetic control, 307–8
 organoleptic contribution, 307
acetic acid bacteria, 234
Acetobacter, 363
Acetobacter aceti, 620
2-acetylpyrroline, 360
2-acetyltetrahydropyridine, 360
acidity adjustment, 162
acrolein, 364
active dry yeast, 293
AdoMet, 309
ageing, 118–19, 164
agmatine, 510
albumins, 559
alcohol, 148–54, 171
 see also higher alcohols
alcoholic fermentation, 3–4, 36, 38
aldehyde dehydrogenases, 534–5
aldehydes, 171, 462, 464–5
aldol condensation, 473
alternative oak treatments, 246–9
 future trends, 251
 oak chips size, extraction rates and staves, 248–9
 production, 246–8
 wine quality, 226–51
aluminium silicate, 413
Amadori rearrangement, 471
amines, 361
amino acids, 170
ammonia phosphate, 13
Amorim's TwinTop, 258
amplified fragment length polymorphism, 51, 368
anion-selective membrane, 159–60
anthocyanins, 106, 172–4, 206, 228
antioxidants, 485
arabinases, 102
arginine, 510
aroma, 176–7
aroma extract dilution analysis, 580
aroma liberation, 119–21
artificial seasoning, 247
ascorbic acid, 448–50
ASM *see* anion-selective membrane
Aspergillus, 95, 101, 102
Aspergillus carbonarius, 214
Aspergillus niger, 102
Asti Spumante, 209
Atea's Vinicycle vat, 602
Atflp, 319, 322
ATHP *see* 2-acetyltetrahydropyridine
AusCap screwcaps, 264
autolysis, 166–70
automaceration vats, 606
autophagy, 169
AWRI1631, 51
 β -damascenone, 467
 β -glucanases, 121
 β -glucosidases, 40
Bacillus amyloliquefaciens, 102
bad liquor *see* fusel

- barrel ageing, 229, 230
- barrel sanitation, 371–2
- base wine, 554, 572
- bâtonnage, 170
- bâtonnés, 170
- bentonite, 189, 198–9, 557
- bentonite fining, 121
 - white wines, 198–206
 - bentonite, 198–9
 - bentonite fining and volume of lees, 202–4
 - bentonite preparation methods, 199
 - bentonite regeneration after wine fining, 204–5
 - improving bentonite efficiency by in-line dosing, 205–6
 - predictive assays for stability, 201
 - protein adsorption by bentonite particles, 200–1
 - sensory impact, 201–2
- benzenemethanethiol, 468
- benzoic acid, 370
- benzothiazole, 469
- BiGGY medium, 312
- Bikerman coefficient, 559
- Bin 2 Shiraz Mataro, 262
- biogenic amines, 40, 66, 80–2, 172
 - chemical structures, 496
 - concentrations in wines, 497
 - detection and quantification methods, 511–13
 - formation during winemaking process, 502–11
 - associated microorganisms in biogenic amines production, 504
 - molecular approaches, 507–11
 - origin, 503, 505
 - technological factors and practices, 506–7
 - future trends, 514–15
 - incidence in wines and health effects, 495–502
 - pharmacological and toxicological aspects, 498–501
 - histamine, 499–500
 - phenylethylamine, 501
 - putrescine and cadaverine, 500
 - tryptamine, 501
 - tyramine, 500
 - physiological effects, 498
 - prevention methods and tools, 513–14
 - primer sequences to detect the decarboxylase genes of wine LAB, 508–9
 - qualitative detection methods, 511–12
 - enzymatic detection, 511–12
 - polymerase chain reaction detection, 512
 - screening using selective media, 511
 - thin-layer chromatography, 512
 - quantitative detection methods, 512–13
 - capillary electrophoresis, 513
 - gas chromatography, 512
 - liquid chromatography, 512–13
 - technological factors affecting formation in grapes and wines, 506
 - winemaking process, 494–515
- biotechnology, 37, 44–7, 52
- biotin, 17
- bird bangers, 531
- bisulphite, 446
- Botrytis*, 127
- Botrytis cinerea*, 195–6, 590, 610, 612, 613, 615
- bottle ageing, 459–75
 - aromatic compounds related to flavour deterioration, 461–9
 - aldehydes, 462, 464–5
 - C₁₃ norisoprenoid compounds, 467
 - esters, 465
 - furanic compounds, 467–8
 - higher alcohols and fatty acids, 468
 - monoterpenes rearrangement during bottle ageing, 466
 - odour-active compounds affected by ageing, 462
 - odour active compounds structural formulae, 463
 - sulphur compounds, 469
 - terpenes, 465–6
 - volatile phenols, 468–9
 - chemical reactions of ageing, 469–73
 - 5-hydroxymethyl furfural formation, 471
 - acetic acid esterification in ethanol, 470
 - alcohol to aldehyde oxidation, 470
 - aldol condensation, 473
 - amino acids Strecker degradation, 472
 - esterification and ester hydrolysis, 470
 - glycoside hydrolysis, 472–3
 - Maillard reactions, 470–1
 - oxidation processes, 469–70
 - future trends, 475
 - influencing factors, 473–5

- sensory changes during storage/
 - ageing, 460–1
 - plot of changes during storage, 461
- bottles
 - filling and storage conditions, 403–5
 - control, 403
 - filling level and headspace, 403–4
 - sealing and corking, 404
 - storage, 404–5
- bovine serum albumin, 200–1
- Bradford method, 193
- breeding, 44
- Brettanomyces*, 105, 182, 234, 235
- Brettanomyces anomalus*, 353
- Brettanomyces bruxellensis*, 48, 67, 85
 - prevention and remediation, 368–72
 - barrel sanitation, 371–2
 - compounds with antimicrobial actions, 369–71
 - physical treatment effect, 371
- Brettanomyces bruxellensis* CBS 2499, 353
- Brettanomyces custersianus*, 358
- Brettanomyces/Dekkera*
 - methods to detect, prevent and defeat microbial spoilage faults, 367–72
 - microbial spoilage detection methods, 367–8
 - prevention and remediation, 368–72
- off-flavours in wines, 346–72
 - EPs precursors in various concentrations, 348
 - metabolic pathways, 351
 - related metabolism, 349–52
- other associated defects, 358–61
 - ATHP and ETHP biosynthesis pathway, 359
 - biogenic amines, 360–1
 - mousy off-flavour, 358–60
- other faults associated with microbial spoilage, 361–7
 - acetic acid, 363
 - bitterness, 364
 - cork taint and other related taints, 366–7
 - film-forming yeast, 365
 - Geranium off-odour, 366
 - lactic acid, 364
 - major wine defects, 362
 - mannitol taint, 364
 - other faults associated with yeast, 365–6
 - ropiness, 364–5
 - tartaric acid degradation, 366
 - physiology, 355–8
 - acetic acid and ethanol production ratio, 357
 - acetic acid production, 357–8
 - aerobic vs anaerobic conditions, 356–7
 - carbon and nitrogen source requirement, 355–6
 - temperature, pH, ethanol, and other influencing growth, 358
 - taxonomy and phylogenetic relationships with other wine yeast, 352–5
 - 26S rRNA gene sequence, 354
- Brettanomyces intermedius*, 358
- Brettanomyces lambicus*, 358
- Brettanomyces naardensis*, 358
- Brettanomyces taint*, 156–7
- bretty off-odours, 347, 350
- bromocresol green, 367
- burnt peanut butter, 419
- C₁₃ norisoprenoid compounds, 467
- Cabernet Sauvignon, 105, 116
- Cabernet Sauvignon wine, 438, 445
- cadaverine, 500
- caffeic acid, 435
- caffeoyl tartaric *o*-quinone, 575
- Candida stellata*, 10
- capillary electrophoresis, 513
- carbon, 216
- carboxen, 580
- Carignan, 117, 118
- casein, 215
- catechin, 435, 443
- cation-selective membrane, 159–60
- Cava, 564
- CE 479/2008, 282
- cell wall proteins, 168
- Cellukork closure, 259
- cellulases, 102
- centrifugation, 203
- Champagne, 176, 564
- champagne corks, 400
- champenoise method, 554, 581
- chaptalisation, 548
- charcoal, 216
- Chardonnay, 192, 202, 208–9, 527
- Chardonnay must, 9
- Chardonnay wine, 443, 449–50
- charmat method, 581
- Chenin blanc, 527
- chitin, 24, 168
- chloramphenicol, 367

- CIELAB colour space values, 440, 579
- cinnamyl esterase, 104, 105
- citric acid degradation, 76–7
- Clairette de Die, 209
- clarification, 20, 188–217
- clarification test, 125–6
- clarifiers, 557
- Clark electrode, 433
- closing pressure, 404
- Coccinella septempunctata*, 424–5
- Coccinellidae, 421, 424, 425
- cold soaking, 9
- cork stoppers
 - additional test procedures, 401–2
 - coating/surface treatment test, 401–2
 - cork dust content determination, 401
 - elasticity testing, 402
 - phenolic compounds, 401
 - alternatives to cork in wine bottle closures, 255–69
 - handling and processing of corks and bottles, 402–9
 - anisole concentrations in wine, 408
 - anisole concentrations in wood, 408
 - bottle filling and storage conditions, 403–5
 - cellar contamination with halogenated anisoles, 406–9
 - haloanisole contamination avoidance, 409
 - halogenated anisoles contamination, 407
 - musty–mouldy off-flavours prevention in cellar environment, 405–6
 - receiving incoming goods, 402
 - storage conditions, 403
 - TCA source removal effect on anisole contamination, 407
 - quality evaluation test procedures, 396–7
 - corks testing methods catalogue, 397
 - quality management and control methods, 394
 - standard test procedures, 397–401
 - average dry weights, 400
 - control of extraction forces, 400–1
 - cork diameters and length variations, 398
 - dimensions, 398
 - dry and specific weight, 399
 - humidity, 399
 - proof of hydrogen peroxide, 399–400
 - sampling, 397–8
 - sensory examination, 399
 - test bottling, 400
 - visual control, 398
- cork taint
 - compounds causing musty–mouldy off-flavours, 389–94
 - 2-methoxy-3,5-dimethyl-pyrazine, 392–3
 - 2-methylisoborneol, 1-octen-3-one, 1-octen-3-ol, 394
 - chlorinated anisoles, 389–92
 - geosmin, 393–4
 - guaiacol, 393
 - haloanisole structures and detection thresholds, 390
 - non-haloanisole structure and detection thresholds, 395
 - TCA biosynthesis, 391
 - tribromoanisole, 392
 - musty off-flavours reduction methods, 410–14
 - adsorbent charcoal, 410
 - aluminium ion migration, 414
 - conventional filter vs TX-R filter sheet, 413
 - effect of filtration in rosé wine, 413
 - filtration with special filter sheets, 411–14
 - inert yeast particles, 410–11
 - polyethylene, 410
 - TBA removal, 412
 - TCA efficiency capacity, 414
 - TCA removal, 412
 - reduction in wines, 388–414
- coupage, 572, 577
- CR 1881/2006, 179
- critical micellar concentration, 556
- critical point, 265
- cross-flow filtration, 139
- crown, 557
- crown cap, 268–9
- cryo-concentration, 282
- Cryo-flash procedure, 607–8
- cryoextraction, 548
- cryoselection, 597
- CSM *see* cation-selective membrane
- Custers effect, 356
- cuvee, 554
- cycloheximide, 46, 367
- Dekkera*, 352
 - off-flavours in wines, 346–72
- Dekkera anomala*, 358

- Dekkera bruxellensis*, 353, 355, 358
délestage, 595
 denaturing gradient gel
 electrophoresis, 50, 368
 depsidase, 104
 DGGE *see* denaturing gradient gel
 electrophoresis
 diacetyl, 76–7
 diafiltration, 151
 Diam closures, 265–7
 illustration, 266
 diamine oxidase, 501
 diammonium phosphate, 12, 13, 540
 Diamond process, 265, 266
 dihydroxybenzaldehyde, 443
 dimethyl dicarbonate, 370
 dimethyl methoxypyrazine, 421
 dimethyl sulphide, 439, 469
 direct spore-cell mating, 299
 distillation, 149
 DMMP *see* dimethyl methoxypyrazine
 DNA arrays, 45
 Dowicide, 391
 draining, 621–2
- EC 479/2008, 98
 EC 606/2009, 277
 EEC 1333/2008, 277
 EEC 2092/1991, 276, 277, 279
 EEC Regulation N° 337/79, 164–5
 EEC Regulation N° 882/87, 166
 eiswein *see* icewines
 electrochemical micro-
 oxygenation, 232, 233, 445
 electrodialysis, 158–62
 acidity adjustment with bipolar
 membrane, 162
 principle, 159–60
 process description, 160–1
 salt reduction, 161–2
 treatment on wine with elevated
 chloride, 162
 typical plant configuration, 161
 wine passing through alternating ASM
 and CSM, 159
 elite, 303
 Elite vat, 624
 encapsulated yeast, 28–9
 enrichment, 282
 enzyme-linked immunosorbent
 assay, 368, 511–12
 enzyme purification, 104–6
 enzymes, 93, 593–4
 activities, 101–4
 cellulases, 102
 glucanases, 102–3
 glycosidases, 102
 hemicellulases, 102
 lysozyme, 103–4
 pectin substrate and action of
 pectinases, 101
 pectinases, 101–2
 proteases, 103
 urease, 103
 advances in discovery, 104–9
 anthocyanin–vinyl phenol
 adducts, 106
 enzyme purification, 104–6
 HPLC polysaccharide profiles, 107
 rhamnogalacturonan degradation
 patterns, 108
 vinyl-4-phenol formation, 105
 volatile phenols concentration, 105
 wine polysaccharide
 composition, 106–9
 applications in winemaking, 99–104
 analytical methods, 101
 dose recommendations, 99–100
 list, 100
 mode and conditions of use, 99
 OIV resolutions relevant to
 use, 100
 definitions, 94–5
 effect on wine quality biological
 factors
 anthocyanins and tannins, 593–4
 aromas and their precursors, 594
 juice extraction, 594
 polysaccharides, 594
 monitoring performance, 124–6
 clarification test, 125–6
 clarification test using Imhoff
 cones, 126
 other tests, 126
 pectin test, 124–5
 pectin test with acidified
 alcohol, 125
 production methods, 95–8
 composition, 97–8
 fermentation process for industrial
 production, 97
 recovery, 96–7
 solid state fermentation, 96
 submerged liquid fermentation, 96
 regulatory aspects, 98–9
 association of enzyme producers, 99
 European regulation, 98
 other regulations, 98–9

- use in post-fermentation stages, 118–24
 - aroma enhancement in Gewüztraminer, 120
 - aroma liberation pathway, 120
 - changes in turbidity and filterability index, 124
 - filtration, 121–4
 - Kieselguhr earth filtration cycles of Gamay wine, 123
 - maturation, 118–21
 - red Merlot wine filterability, 122
 - white Sauvignon wine filterability, 122
- use in pre-fermentation stages, 109–18
 - clarification in whites, rosés and reds, 112–13
 - clarified juice and sediment volume on Semillon must, 113
 - extraction in red wine-making, 113–16
 - extraction in whites or rosés, 109–12
 - Grenache must clarification, 111
 - pectinase comparison, 110
 - pectinase hydrolysis, 114
 - pectinase treatment on Carignan must clarification, 118
 - pressing trials on Sauvignon grapes, 110
 - sensory attributes, 111
 - sensory attributes prior to thermotreatment, 117
 - thermovinification, 116–18
- used in industrial processes, 95
- wine quality, 93–127
 - future trends, 126
- Erlich pathway
 - in higher alcohols, 314–16
 - decarboxylation, 315
 - fusel aldehydes reduction, 315–16
 - transamination step, 314
- Escherichia coli* *tnaA* gene, 331
- ester hydrolysis, 470
- esterification, 470
- esters, 171, 318–22, 465
 - aromatic impact, 318
 - ethyl ester biosynthesis, 320
 - factors affecting production, 321–2
 - limits in genetic modification, 322
 - main classes
 - acetic esters of higher alcohols, 319–20
 - ethyl esters of fatty acids, 320–1
 - main esters produced by yeast in wine, 319
 - metabolic pathways and genetic control, 319–21
 - Semillon wine chemical analysis, 322
- ethanol, 68–9, 367, 592–3
- ETHP *see* 2-ethyltetrahydropyridine
- ethyl acetate, 322, 542–3
- ethyl carbamate, 103
- ethyl esters, 568
- 4-ethylguaiaicol, 179, 347
- 4-ethylguaiaicol (4-ethyl-2-methoxyphenol), 157
- 4-ethylphenol, 179, 347
- 2-ethyltetrahydropyridine, 360
- EU Regulation 1493/1999, 279
- EU Regulation 1622/2000, 182
- evaporation, 136, 282
- evaporative perstraction, 151–2
- exogenous tannins
 - commercial tannins
 - addition effect, 250
 - compositions, 249
 - future trends, 251
 - wine making, 249–51
 - wine quality, 226–51
- extraction technologies
 - and wine quality, 589–625
 - biological factors, 593–4
 - anaerobic metabolism, 594
 - enzymes, 593
 - yeasts, 593
 - chemical factors, 592–3
 - ethanol, 592–3
 - sulphur dioxide, 593
 - water, 592
- draining and pressing, 621–5
 - automatic horizontal basket press with vertical plates, 625
 - draining, 621–2
 - Elite vat diagram, 621
 - pressing, 622–5
 - Vaslin-Bücher pressing process diagram, 624
- phenolic compounds extraction
 - differences, 591
- physical factors, 594–6
 - effect of changes in pressure, 596
 - heat transfer, 595
 - transfer of liquid, 595
 - transfer of matter, 595
 - transfer of solids, 595–7
- red wines vinification by carbonic maceration, 617–20

- anaerobic maceration and carbonic maceration procedure, 617–20
- effects on wine quality, 620
- techniques applied to rosé wine vinification, 598–9
- carbonic maceration rosés, 599
- direct pressing rosés, 598–9
- maceration rosés, 599
- saignée rosés, 599
- techniques applied to traditional red wine vinification, 599–609
 - destemming and crushing, 600–2
 - procedures based on displacement of the must, 601–4
 - procedures based on displacement of the pomace, 605–6
 - thermal treatments applied to traditional maceration, 606–9
- techniques applied to white wine vinification, 596–8
- icewines and cryoselection, 597
- skin maceration, 596–7
- supra-extraction, 597
- traditional vinification of red wines with whole berries, 620–1
- very hot, short maceration applied to red wine vinification, 609–17
 - flash-release procedure, 613–17
 - thermo-release procedure, 617
 - thermovinification, 609–13
- Fenton reaction, 436
- fermentation, 554
- fermentor, 18
- ferulic acid, 370
- Fibrafix TX-R, 411
- filter sheets, 411–14
 - aluminium ion migration, 413–14
 - capacity, 412
 - filtration effect on TCA/TCB contents, 411
 - filtration sensorial effect, 411
- filterability index, 121, 123
- filtration, 121–4, 371
- fingerprint, 43, 44
- fining, 189
 - see also* bentonite fining; gelatine fining
 - use of gelatine in white wines, 206–7
 - white wines, 188–217
 - with plant proteins, 207–9
- flash-release procedure, 613–17
 - effects on wine quality, 615–17
 - flash-expansion effect on skin surface of berries, 615
- polyphenols levels in Grenache musts, 616
- procedures and equipment, 613–15
- process diagram, 614
- flavanols, 227
- flavonoids, 227, 563–4
- Flavourtech, 138
- flavylum ion, 228
- flotation technique, 209–14
- fluorescence in situ hybridisation, 368
- flux, 140
- fluxomics, 43
- foam, 555–63
 - chemical characteristics, 558–9, 561–3
 - destabiliser, 563
 - discriminant descriptors used in sparkling wines sensory analysis, 574
 - physical characteristics, 555–8
 - quality, 571–4
 - conditioning factors, 571–3
 - sensory descriptors, 573–4
- foaming, 177–8
- Folin-Ciocalteu reagent, 441, 447
- Folin-Ciocalteu value, 249
- footprint, 42–3, 44
- freeze concentration, 135–6
- Frontenac grapes, 422
- fructophilic index, 28
- fructose, 27
- fungal must, 392
- fungicides, 529
- furanic aldehydes, 248
- 2-furanmethanethiol, 468
- furanoic compounds, 467–8
- furfurylthiol, 249
- fusel alcohols, 313–14
- galactanases, 102
- galacturonans, 559
- Gamay, 121
- Ganimede vat, 603
- gas chromatography, 512
- gas chromatography–isotope ratio mass spectrometry, 548
- gas chromatography–mass spectrometry, 545
- gas chromatography–olfactometry, 351, 392, 545, 570
- gas chromatography–olfactometry–mass spectrometry, 421, 545
- gas-sparging method, 559, 561
- Geisenheim Test Methods, 397, 402
- gel electrophoresis, 42

- gel filtration chromatography, 191
 gelatine, 189, 204, 206–7
 gelatine fining, 207
 genetic engineering, 45–7
 genomics, 41
 geosmin, 215
 Gewürztraminer, 119, 120, 527
 Gimar's vat Selector system, 601
 global impression, 573
 glucan synthesis, 83–4
 glucanases, 102–3, 127
 Glucanex, 103
Gluconobacter, 363
 glucose, 27
 glucosyl-transferase, 84
 glutathione, 447–8
 2-S-glutathionylcaftaric acid, 575
 glutens, 208–9
 glycerol, 304–6, 543
 aromatic impacting molecules, 304
 biosynthesis, 305
 factors affecting production, 306
 grape must initial sugar
 concentration, 306
 metabolic pathways and genetic
 control, 305
 organoleptic contribution, 305
 overproduction, 306
 glycerol-3-phosphate, 305
 glycerol-3-phosphate
 dehydrogenase, 305, 534
 glycerol dehydrase pathway, 83
 glycoprotein, 192
 glycosidases, 102, 119
 glycoside hydrolysis, 472–3
 glycosylation, 192–3
 glyoxylic acid, 443
 GO-FERM, 540
 goût de lumière *see* light-struck
 grape reaction product, 575
 grape tannins, 227
 green tannins, 232
 green wood, 398
 Gulfstream procedure, 611
 glycerol degradation, 82–3

 halophenols, 406
 hand harvest, 532
Hanseniaspora uvarum, 9, 607
Hansenula, 102
Harmonia axyridis, 418, 422, 424
 haze, 189–90
 HCL index, 244
hdc genes, 509

 head space trapping, 312
 hemicellulases, 102
 heterothallic yeast, 299
 Heyns rearrangement, 471
 high-density polyethylene, 446
 high osmolarity glycerol, 534
 high-performance liquid
 chromatography, 233, 449, 511, 564
 higher alcohols, 313–18
 aromatic impact, 313–14
 Erich pathway and acetate ester
 production, 315
 Erich pathway genetic regulation
 during alcoholic fermentation, 316
 factors affecting production, 316
 genetic modification for aroma
 profiling, 318
 metabolic pathways and genetic
 control, 314–16
 S. cerevisiae and *S. uvarum*
 distribution, 317
 Saccharomyces and non-
 Saccharomyces contribution, 317
 wine fusel alcohols, 313
Hippodemia converges, 424–5
 histamine, 80, 361, 499–500
 metabolic pathways for
 detoxification, 502
 histamine N-methyltransferase, 501
 histidine decarboxylase, 80
 HOG MAP-kinase pathway, 305
 homothallic yeast, 299
 HPLC *see* high-performance liquid
 chromatography
 hybridisation, 48
 hydraulic basket press, 534
 hydrogen peroxide, 446
 hydrogen sulphide, 12, 309–13
 factors affecting production, 311–12
 impact, 309
 metabolic pathways and genetic
 control, 309–11
 sulphur amino acid biosynthesis
 pathway and sulphide leakage, 310
 yeast strain influence on
 production, 312–13
 hydrolases, 94
 hydrolysable tannins, 228
 hydroxyphenyl pyranoanthocyanin, 229
 hyperoxidation, 434

 IBMP *see* isobutyl methoxypyrazine
 icewines
 and cryoselection, 597

- authentication, 548–9
- chemical analysis, 540–5
 - ethanol and residual sugar, 542
 - glycerol, 543
 - icewine must chemical composition, 541
 - nitrogen, 542
 - polyphenols, 544
 - soluble solids, 541
 - titratable acidity, 541
 - total hydroxycinnamates and total caffeic acids, 544
 - volatile acidity, 542–3
 - volatile compounds, 544–5
- definitions, 523–6
- future trends, 549–50
- harvest considerations, 531–4
 - hand harvest, 532
 - hand harvested Vidal icewine grapes, 533
 - mechanical harvest, 533
 - pressing, 534
- oenology, 534–40
 - commercial yeast strains for icewine fermentation, 539
 - fermentation kinetics, 534–7
 - icewine juice yeast metabolite production, 535
 - yeast strain selection, 537–40
- quality management, 523–50
- regulations for icewine production, 525
- Riesling icewine juice chemical composition
 - after fermentation, 537
 - prior to fermentation, 537
- sensory properties, 545–8
 - Vidal and Riesling sensory maps, 546
- viticultural practices, 529–31
 - bird predation, 530–1
 - disease management, 529
 - netting to protect grapes from predation, 530
 - site selection, 529
- viticulture, 526–31
 - cultivar selection, 526–9
 - Vidal icewine cluster on vine, 528
- IFOAM EU, 272
- immunodetection, 214
- in situ hybridisation, 368
- inactive yeasts, 181
- indoleacetic acid, 485
- induced homogeneous nucleation, 557
- Inertys process, 623
- Integra, 259
- inter-specific hybridisation, 301
- interactomics, 43
- IPMP *see* 2-isopropyl-3-methoxypyrazine
- iron, 448
- irradiation, 428
- isinglass, 215
- ISO 2859, 398
- ISO 3951, 398
- ISO 4707, 398
- isobutyl methoxypyrazine, 421
- isoelectric point, 206
- 2-isopropyl-3-methoxypyrazine, 421
- jégbor *see* icewines
- juice clarification, 426, 428
- karyotyping, 48
- kieselsol, 207, 215
- Kloeckera*, 17
- Kloeckera apiculata*, 9
- Koshu, 190
- LAB *see* lactic acid bacteria
- lactic acid, 563
- lactic acid bacteria, 61, 360, 363, 507
 - factors affecting growth in
 - wine, 67–71
 - ethanol, 68–9
 - other compounds, 69–70
 - pH, 68
 - sulphur dioxide, 68
 - temperature, 70–1
- species variations, 71–4
 - main species in grape and wine, 71
 - species identified at beginning of alcohol fermentation, 72
 - species in grape must and wine, 71–2
- spontaneous growth, 61–71
 - arginine deiminase locus, 66
 - heterofermentation and mannitol production, 64
 - inhibition and activation by yeast, 61–3
- L-malic acid evolution in red wine, 70
- main nutrients for lactic acid bacteria, 63–7
- malolactic fermentation duration, 70
- yeast and lactic acid bacteria general evolution, 62
- wine quality improvement, 74–9
 - citric acid degradation, 76–7

- citric acid degradation pathway, 77
- deacidification by malic acid
 - degradation, 75–6
- malolactic reaction, 76
- methionine conversion into odorous molecules, 78
- methionine metabolism, 78
- oxidase activities and aroma compounds, 79
- wine spoilage, 79–85
 - acrolein production, 83
 - biogenic amines production, 80–2
 - biogenic amines production from amino acids, 81
 - gene clusters coding for histidine and tyramine decarboxylation pathways, 82
 - glucan synthesis and ropy wines, 83–4
 - glycerol degradation to acrolein and bitterness, 82–3
 - lactic spoilage, 79–80
 - new tools for detecting undesirable strains, 84–5
- lactic souring, 364
- Lactobacilli*, 26, 68
- Lactobacillus brevis*, 63, 509
- Lactobacillus fructivorans*, 69
- Lactobacillus hilgardii*, 63, 69, 509
- Lactobacillus plantarum*, 69
- lactone- γ -nonalactone, 469
- ladybug taint, 418–28
 - abbreviations, 428
 - causal compounds, 421–3
 - chemical structures, 421
 - earthy/herbaceous dimensions intensity, 422
 - red wine sensory profile, 423
 - future trends, 428
 - multicoloured Asian ladybeetles, 419
 - other Coccinellidae species, 424–5
 - IPMP concentration, 425
 - post-harvest prevention and remediation, 425–8
 - custom-designed shaker tables, 426
 - sensory attribute intensity scores, 427
 - quality implications, 419–21
 - wine sensory intensity scores, 420
 - threshold and tolerances, 424
- Laplace's law, 557
- LBT *see* ladybug taint
- lead acetate strips, 312
- ledeno vino *see* icewines
- ledové vino *see* icewines
- lees, 118–19, 165, 172, 182
 - ageing of red wines, 172–6
 - ageing of white wines, 170–2
 - definition and composition, 164–6
 - influence of decanting duration on composition, 165
 - intervention in sparkling winemaking, 166
 - intervention in winemaking, 165
- Leon Millot grapes, 422
- Leuconostoc mesenteroides*, 72, 510
- light-struck, 475
- lipids, 23–4, 178
- liquid chromatography, 512–13
- lysozyme, 100, 103–4, 127, 514
- macro-oxygenator, 15
- madeirisation process, 475
- magnesium, 16
- Maillard reaction, 470–1, 471, 570, 613
- MALB *see* multicoloured Asian ladybeetle
- malic acid, 563
- malic acid degradation, 75–6
- malolactic enzyme, 75
- malolactic fermentation, 60–1, 88, 231, 495
 - control by malolactic starters, 85–8
 - effects on wine quality, 60–89
 - lactic acid bacteria
 - species variations, 71–4
 - spontaneous growth, 61–71
 - wine quality improvement, 74–9
 - wine spoilage, 79–85
- Oenococcus oeni*
 - evolution of isolated strains, 74
 - intra-specific diversity and variations, 73–4
 - RAPD profiles, 74
 - selection during alcoholic fermentation, 72
- malolactic starters, 85–8
 - definition, 86
 - isolation and selection, 86–7
 - reasons for use, 85–6
 - use, 87–8
- mannoproteins, 174–5, 178–9
- maturation, 118–21, 459
 - ageing on lees, 118–19
 - aroma liberation, 119–21
- mechanical harvest, 196–7, 533
- membrane separation
 - evaporative perstraction, 154

- permeate distillation, 149–51
- techniques, 139–44
 - concentration by reverse osmosis, 143–4
 - membrane terminology, 139–40
 - nanofiltration, 142–3
 - reverse osmosis, 140–2
- treatment and recombination, 145–6
- membrane surface area, 140
- Memstar process, 154
- 3-mercaptopropionate, 468
- Merlot wines, 122
 - colour and phenolics extraction kinetics, 115
 - colour and tannin composition, 116
 - concentration, 114
- metabolic engineering, 47, 51–2
 - wine yeast and advances in yeast selection methods, 34–52
- metabolomics, 42
- methional, 440
- methionine, 78
- methionol, 20
- Méthode Traditionnelle, 176
- 2-methyl-tetrahydrothiophen-3-one, 469
- 4-methylguaiacol (2-methoxy-4-methylphenol), 155
- micro-oxygenation
 - future trends, 251
 - recent research, 235–43
 - Cabernet Sauvignon phenolic compounds concentrations, 240
 - cis-oak lactones in Pinotage, 243
 - colour density and modified colour density, 238
 - different oak and micro-oxygenation treatments, 237
 - free SO₂ concentrations, 241
 - oak volatiles concentrations, 242
 - Pinotage phenolic compounds concentrations, 240
 - results, 238–43
 - total anthocyanins concentrations, 239
 - workplan, 236
- red wine, 230–46
 - Brettanomyces* counts, 234
 - chemical composition effect, 231–4
 - microbiology effect, 234–5
 - wine quality, 226–51
- microbial spoilage, 361–7
 - wine faults, 367–72
 - methods to detect microbial spoilage, 367–8
 - prevention and remediation, 368–72
- miniflotator, 210
- MOCON measurements, 258
- molecular biology, 37
 - identification of wine yeasts, 47–51
- molecular weight cut-off, 140
- monoamine oxidase, 500, 501
- monoamine oxidase inhibitor, 500
- monogenic segregation, 301
- monoterpenoids, 473
- Mosalux device, 559, 561, 572, 578
- MS spectrum databank, 580
- Mucor*, 95
- multicoloured Asian ladybeetle, 418
- multilocus sequence typing, 51
- Muscadet, 170
- Muscat, 119, 209, 448
- Muscat Gordo Blanc wine, 190
- Muscat Ottonel, 202
- must clarification, 209–14
- must density, 532
- musty–mouldy off-flavours
 - avoidance in wineries, 388–414
 - causal compounds, 389–94
- N fertilisation, 481
- nanofiltration, 142–3
- natural corks, 261
- natural seasoning process, 247
- netting, 530–1
- nicostatine, 370
- nitrogen, 11–12, 14, 66, 170, 542
 - inorganic, 13–14
 - organic, 12–13
 - organic vs inorganic, 13
- nitrogen catabolic repression, 330
- non-enzymatic oxidation, 432–51
 - aroma compounds oxidation, 438–40
 - sulphur-containing compounds oxidation mechanisms, 439
 - influence of wine anti-oxidants, 446–50
 - ascorbic acid, 448–50
 - glutathione, 447–8
 - sulphur dioxide, 446–7
 - measures of wine oxidation status, 440–2
- oxygen in wine, 433–4
- polyphenol oxidation, 434–8
 - overall wine oxidation processes, 436
 - polyphenol forms before and after oxidation, 435
- red wine oxidation, 444–6

- bottle closure type
 - implications, 445–6
 - micro-oxidation implications, 444–5
 - white wine oxidation, 442–4
- non-flavonoids, 563
- Novozymes, 94
- NS35032, 112
- NS35040, 112
- nucleic acids, 177
- nucleotides, 177
- o*-aminoacetophenone
 - formation and oenological UTA
 - inhibition, 476–8
 - influence on UTA, 478
 - proposed formation by indoleacetic acid co-oxidation, 477
 - responsible for UTA, 478–80
- o*-heptafluorobutyl, 513
- oak ageing, 173
- oak barrel, 229, 235
- oak chips, 248
- oak chips treatment, 427
- oak staves, 235, 248–9
- oak tannins, 229
- ochratoxin A, 179, 214
- odour activity values, 440
- Oenococcus*
 - oeni*, 64, 76, 89, 505, 509, 510
 - evolution of strains isolated during malolactic fermentation, 74
 - intra-specific diversity and variations, 73–4
 - RAPD profiles, 74
 - selection during alcoholic fermentation, 72
- oenological yeasts, 4, 19
 - selection, 24–5
 - competitive factor, 25
 - lag phase and multiplication, 24–5
 - suitability for limiting conditions in must, 25
- omics, 41, 42
- organic farming, 276
- organic viticulture and wine-making, 271–88
- organic wine, 273, 276
- organic winemaking, 271–88
 - future trends, 287–8
 - harmonisation process, 286–7
 - links with non-European organic regulations, 286–7
 - special wine production needs, 286
- ORWINE project, 271–76
 - consumers' opinion, 273–4
 - market demands, 274–5
 - organic wine concept, 273–6
 - producers' attitude and potentiality, 275–6
 - synthesis attempt, 276–86
 - additives and processing aids evaluation, 280
 - additives different statuses, 278
 - demand and need to decrease treatments and additives, 276–81
 - quality improvement and SO₂ use reduction, 283–6
 - SO₂ reduction strategies, 284–5
 - SO₂ use and limitations, 281–2
 - traditional and modern techniques and practices evaluation, 282–3
- ornithine decarboxylase, 510
- ORWINE *see* organic viticulture and winemaking
- osmotic pressure, 139
- ossein gelatine, 207
- oxaloacetic acid, 76
- oxidation processes, 469–70
- oxygen, 14–16, 433–4, 450
- Oxysense probe, 258
- p*-benzoquinone, 448
- pantothenic acid, 17
- passage, 140
- PCP *see* pentachlorophenol
- PCR *see* polymerase chain reaction
- pectin test, 124–5
- pectinase purified, 324
- pectinases, 95, 100, 101–2, 104, 111, 113, 127
- pectolytic enzymes, 209
- Pediococci*, 68
- Pediococcus damnosus*, 84
- Pediococcus parvulus*, 84
- Pediococcus* spp., 80
- Penicillium* spp., 389
- pentachlorophenol, 391
- peptide nucleic acid, 368
- permeate, 140
- permeate flow, 140
- perstraction, 151–2
- petrol note, 461
- pharmaceutic, 323
- phase change techniques, 135–9
- phenolic, 323
- phenolic acids, 323–4
 - cinnamate decarboxylase and related gene in *S. cerevisiae*, 324

- factors affecting vinyl phenol
 - precursors in grape must, 323–4
 - volatile phenols organoleptic impact, 323
- phenolic compounds, 180, 563–5
- phenolic off-flavour, 105, 323, 347
- phenols, 172–4
- 2-phenyl ethanol, 314, 317–18
- phenylacetaldehyde, 440, 464
- phenylethylamine, 501
- phosphorus pentoxide, 16
- Pichia fermentans*, 10
- Pinot noir, 105, 216
- Pinotage wines, 229, 233
- piqûre lactique, 364
- plate presses, 623–5
- PMKT2, 370
- pneumatic closed cage presses, 623
- polydimethylsiloxane, 580
- polyethyleneterephthalate, 446
- polymerase chain
 - reaction, 48, 73, 85, 361, 507, 512
- polymerase chain reaction–denaturing gradient gel electrophoresis, 50
- polymerase chain reaction–restriction fragment length polymorphism, 49, 50
- polyphenol oxidase, 434, 446
- polyphenol oxidation, 434–8
- polyphenols, 173, 544
- polysaccharides, 106–9, 594
- polyvinyl polypyrrolidone, 214–15
- potassium, 16
- potassium caseinate, 215
- pre-fermentative cold maceration *see* cold soaking
- pressing, 534, 622–5
 - continuous presses, 622
 - discontinuous presses, 622–3
 - plate presses, 623–5
 - pneumatic closed cage presses, 623
- principal component analysis, 545
- Proles Occidentalis*, 527
- proteases, 103, 127
- protein clouding, 191–2
- protein haze formation, 197–8
- proteins, 172
 - see also* wine proteins
- proteolysis, 167–8
- proteomics, 41
- pulse field gel electrophoresis, 73
- pulsed electric field technology, 371
- putrescine, 81–2, 500, 510
- pyridoxine, 367
- quantitative real-time PCR, 510, 512, 515
- quantitative trait loci, 301, 308
- Quercus alba*, 246
- Quercus petrea*, 246
- Quercus robur*, 246
- quinones, 575
- racking, 595
- Raco, 391
- random amplification of polymorphic DNA, 51, 73
- random amplified polymorphic DNA, 350
- Radox total antioxidant assay, 447
- Raoult's law, 597
- RAPD *see* random amplification of polymorphic DNA
- reactive oxygen species, 175, 469–70
- rebêchage, 623
- red wines, 283, 419
 - ageing on lees, 172–6
 - oxidation, 444–6
 - bottle closure type implications, 445–6
 - micro-oxidation implications, 444–5
 - procedures based on displacement of the must, 601–4
 - carbon dioxide as pumping over driver, 602–4
 - Ganimede vat diagram, 604
 - Gimar's vat Selector system operating cycle, 602
 - pumping over, 601–2
 - racking, 604
 - Turbopigreur pumping over system, 603
- procedures based on displacement of the pomace, 605–6
 - automaceration vats, 606
 - mobile punching down devices, 605
 - traditional vats with punching down systems, 605
- techniques applied to vinification, 599–609
 - Cryo-flash treatment effect, 608
 - destemming and crushing, 600–1
- thermal treatments applied to
 - traditional maceration, 606–9
 - cold pre-fermentation maceration, 606–7
 - Cryo-flash procedure, 607–8
 - hot final maceration, 609
 - hot pre-fermentation maceration, 608–9

- pomace hot maceration, 609
 - traditional vinification with whole berries, 620–1
 - very hot short maceration applied to vinification, 609–17
 - flash-release procedure, 613–17
 - Gulfstream thermal exchanger, 611
 - thermo-release procedure, 617
 - thermovinification, 609–13
 - vinification by carbonic maceration, 617–20
 - anaerobic maceration and carbonic maceration procedure, 617–20
 - effects on wine quality, 620
 - phenolic compounds extraction comparison, 619
 - redox potential of wine, 441
 - reductones, 475
 - Regulation CE 2087/1997, 208
 - rehydration, 5
 - resistance to oxidation, 474
 - resolution OENO 4/2001, 182
 - restriction fragment length polymorphism, 367
 - resveratrol, 40
 - retentate, 140
 - reverse osmosis, 140–2, 282
 - reverse osmotic pressure, 371
 - Riesling, 119, 523, 526, 529
 - Riesling wine, 443, 449
 - ropy wines, 83–4
 - rosé wine vinification
 - applied techniques, 599–600
 - carbonic maceration rosés, 599
 - direct pressing rosés, 598–9
 - maceration rosés, 599
 - saignée rosés, 599
- S-cysteinecaffeic acid, 448
- S-glutathionyl caftaric acid, 447
- 18S rRNA gene sequencing analysis, 353
- Sabate's Altec, 258
- Saccharomyces*, 4
- Saccharomyces bayanus*, 36, 45
- Saccharomyces bayanus* var. *uvarum*, 331
- Saccharomyces cerevisiae*, 36, 592
 - cell wall degradation, 168–9
 - difficult fermentation conditions, 19
 - enzyme purification, 104
 - glucosidases activities, 102
 - inoculation, 8
 - interaction with LAB, 61, 63
 - intraspecies discrimination, 50
 - systems biology technology
 - application, 43–4
 - utilisation for overproducing glycerol, 306
 - viability, 16
 - yeast selection for wine flavour modulation, 293, 296, 298, 302, 303
- Saccharomyces paradoxus*, 36, 45
- Saccharomyces stricto sensu*, 301
- Saccharomyces uvarum*, 317
- Saccharomyces* yeasts, 295–6
- salt reduction, 161–2
- sanitation, 368
- Saranex, 264
- Sauvignon blanc, 119, 122, 438, 527
- S288C, 51
- Schizosaccharomyces pombe*, 365
- Scorpex, 259
- screwcaps, 261–5
 - Stelvi Lux screwcap, 264
 - Stelvi screwcap, 262
- sec*-butyl methoxypyrazine, 421
- self-cloning, 296
- Semillon, 112, 113, 202
- Semillon wine, 260
- shaker tables, 426
- Sherry-type white wine, 438
- Shiraz, 196
- shotgun sequencing approach, 353
- single nucleotide polymorphisms, 302
- smoke taint, 154–6
- sodium caseinate, 215
- sodium periodate, 448
- solid phase microextraction, 579–80
- solid state fermentation, 96
- sorbic acid, 370
- sotolon, 468, 471
- Sparkling Burgundy, 176
- sparkling wines, 189, 202, 209
 - ageing, 176–8
 - aroma, 176–7
 - foaming properties, 177–8
 - definitions and characteristics, 553–4
 - discriminant descriptors used in foam sensory analysis, 574
 - factors affecting sensory quality, 570–8
 - aroma changes during elaboration process, 577–8
 - colour changes during elaboration process, 574–6
 - foam quality, 571–4
 - organoleptic characteristics description, 554–70

- bubbles as foam's unitary cell, 556
- compounds related to foam
 - properties, 560–1
- foam, 555–63
- phenolic compounds and
 - colour, 563–5
 - volatile compounds and
 - aroma, 565–70
- phenolic compounds
 - concentration., 566–7
 - quality control, 578–80
 - quality management, 553–82
- special wines, 286
- spermidine, 361, 500
- spermine, 500
- spinning cone column, 149
- spinning cone distillation, 136–9
- Spumanti Metodo Classico, 554
- squawkers, 531
- stabilisation, 188–217
- Stelcap-vin, 261
- Stelvin, 261
- still wines, 553
- stir bar sorptive extraction, 580
- stoppels, 268
- Strecker degradation, 470
- stuck fermentations, 363
- submerged liquid fermentation, 96
- sulphate, 197, 198
- sulphite, 474–5
- sulphur compounds, 21–3, 469
- sulphur dioxide, 68, 80, 369, 446–7, 464, 514, 574–5, 593
- sulphur odours, 22
- sulphurous acids, 405
- sunstruck *see* light-struck
- super 4 technique, 592
- supercritical CO₂, 265
- superoxide anion, 469–70
- supra-extraction, 597
- SupremeCorq, 259
- sur lie* method, 459, 565
- surfactants, 177–8
- synthetic corks, 259–61
- syringaldehyde, 468
- systems biology, 44–7, 52
 - approach to wine yeast studies, 41–4
 - application of techniques, 43–4
 - revolutionary approach to
 - understanding yeast, 41–3
 - identification of wine yeasts, 47–51
- taint removal, 154–7
 - Brettanomyces* taint, 156–7
 - membrane separation and
 - adsorption, 156
 - smoke taint, 154–6
- tannin-to-anthocyanin ratio, 244
- tannins, 115–16, 158, 174–5, 206, 249
- tartrate, 172
- TBA *see* tribromoanisole
- TCA *see* 2,4,6-trichloroanisole
- technical corks, 261
- Tempranillo, 527
- terpenes, 324–7, 465–6
 - aromatic impact, 324–5
 - β -glucosidase activity, 325–6
 - biosynthesis modulation, 326–7
 - wines monoterpenes and
 - compositions, 325
- 2,3,4,6-tetrachloroanisole, 406
- 2,3,4,6-tetrachlorophenol, 391
- thermal gradient gel electrophoresis, 50
- thermo-release procedure, 617
- Thermocompact, 611
- thermovinification, 116–18, 609–13
 - effects on musts and wines
 - composition, 612–13
 - pre-fermentation and fermentation
 - stages, 609–10
 - procedures and equipment, 610–12
- thiamine, 17
- thin-layer chromatography, 512
- thiols, 179–80
- tin-lined screwcaps, 258
- tirage liquor, 565
- toasting, 247
- Torulaspora delbrueckii*, 10, 308
- TRACELESS system, 549
- trans-membrane pressure, 140
- transcriptomics, 41
- Tribaie automatic sorter, 601
- tribromoanisole, 392
- tribromophenols, 406
- 2,4,6-trichloroanisole, 256, 366, 389
- 2,4,6-trichlorophenol, 391
- Trichoderma harzianum*, 103
- 1,1,6-trimethyl-1,2-dihydronaphthalene (TDN), 444, 461, 467, 570, 582
- tryptophan, 485
- turbidity, 20, 124, 126
- Turbopigreur pumping over system, 602
- tyramine, 81, 500
- tyrosine decarboxylase, 81
- tyrosol, 565
- ultrafiltration, 157–8
- Ultrasymp 100 G, 94

- Uncinula necator*, 195
 universal test machines, 402
 untypical ageing off-flavour, 476–86
 AAP and other aroma compounds responsible for UTA, 478–80
 AAP formation and oenological UTA inhibition, 476–8
 correlation between amino acids in must and AAP production, 485
 dynamics of UTA perception of vineyards, 483
 future trends, 486
 indoleacetic acid, antioxidants and indicator precursors, 485
 influence of viticultural factors, 481
 prevention, 480–5
 principal components analysis of various aroma compounds, 479
URA3 gene, 46
 urease, 100, 103
 UTA *see* untypical ageing off-flavour
- vanillin, 248, 468
 Velorin, 370
 Vidal blanc, 526, 528–9
 vin de glace *see* icewines
 vinegar bacteria, 363
 Vinimatic vat, 606
 Vino-Lok closures, 267–8
 illustration, 268
 Vinomaceration technique, 593
 Vinoxym Process, 109
 VinTegra, 268
 Vintners Quality Alliance Act, 525
 vinyl-4-guaiacol, 323
 vinyl-4-phenol, 323
Vitis vinifera, 526
 vitispiranes, 467, 570, 582
 volatile acidity, 542–3
 volatile compounds, 544–5, 565–70
 volatile phenols, 468–9
 volatile thiols, 327–32
 aromatic impact, 327–8
 bioconversion genetic control, 330
 current model, 329
 genetic application for strains selection, 330–2
 influencing factors, 330
 S-conjugate cysteine precursors identification, 328
 selecting strains perspective, 332
 3SH liberation, 331
 volatile thiols and their cysteinylated precursors, 327
- Yeast β -lyase activity and related genes, 328–9
- water stress, 196
 wheat gluten, 208
 white wines, 283, 419
 ageing on lees, 170–2
 bentonite fining, 198–206
 effect of temperatures used during the cryoselection process
 potent alcoholic grade, 598
 volume of selected and residual juices after pressing, 598
 gelatine in wine fining, 206–7
 gelatine fining, 207
 nature of oenological gelatines, 206
 polyphenol-protein reaction, 206–7
 must clarification using flotation technique, 209–14
 flotation principle, 209–10, 210
 flotation trials at laboratory scale, 210–12
 flotation with lupine proteins, 213
 industrial flotation device schematic, 213
 industrial flotation experiments, 212–13
 laboratory flotation schematic, 211
 research on residual proteins in must and wine, 214
- new directions in
 stabilisation, clarification and fining, 188–217
 equipment for addition of fining agents, 216
 origin of wine proteins, 190–1
 practical recommendations, 216–17
 protein haze formation, 197–8
 white wines, proteins and haze, 189–90
 wine proteins characterisation, 191–3
- other fining agents, 214–16
 carbon or charcoal, 216
 casein, 215
 isinglass, 215
 kieselsol, 215
 polyvinyl polypyrrolidone, 214–15
 oxidation, 442–4
 protein levels, 193–7
 effect of grape fungal infection, 195–6
 mechanical harvesting, 196–7
 protein concentration and berry development, 194

- role of water stress, 196
 - wine protein quantification, 193–4
- skin maceration effect on volatile compounds in white grape musts, 597
- techniques applied to
 - vinification, 596–8
 - icewines and cryoselection, 597
 - skin maceration, 596–7
 - supra-extraction, 597
- wine fining with plant proteins, 207–9
 - gluten with other fining agents, 208–9
 - wheat gluten and other fining agents for white wine, 208
- Wiley MS database, 580
- wine bottle closures
 - alternatives to cork, 255–69
 - future trends, 269
 - key properties, 256–8
 - types, 259–69
 - crown cap closures, 268–9
 - Diam closures, 265–7
 - screwcaps, 261–5
 - synthetic corks, 259–61
 - Vino-Lok closures, 267–8
 - Zork closures, 267
- wine clarification, 208–9
- wine composition
 - membrane and other techniques for management, 133–62
 - convention and intervention, 134–5
 - electrodialysis, 158–62
 - some caveats, 134
 - ultrafiltration, 157–8
 - membrane separation techniques, 139–44
 - concentration by reverse osmosis, 143–4
 - grape juice or wine concentration, 144
 - membrane terminology, 139–40
 - nanofiltration, 142–3
 - reverse osmosis, 140–2
 - separation by nanofiltration membranes, 143
 - wine components and their molecular weights, 142
 - wine flows at high pressure, 141
 - membrane separation treatment and recombination, 145–6
 - selective removal of undesirable wine components, 146
 - wine concentration effects, 145
- phase change techniques, 135–9
 - evaporation, 136
 - freeze concentration, 135–6
 - fruit juice freeze concentration, 136
 - spinning cone column – model SCC10,000, 138
 - spinning cone column – rotating shaft, 137
 - spinning cone column – thin film of product descends, 137
 - spinning cone distillation, 136–9
- problem of excess alcohol, 148–54
 - alcohol reduction by membrane separation and evaporative perstraction, 153
 - alcohol-rich stream separated from counter-flow of strip water, 152
 - diafiltration, 151
 - distillation-based techniques, 149
 - evaporative perstraction, 151–2
 - membrane separation and permeate distillation, 149–51, 150
 - Memstar alcohol adjustment system, 153
 - Memstar process, 154
 - spinning cone column, 149
 - traditional options for lower wine alcohol, 148–9
- taint removal, 154–7
 - 4-ethylguaiacol (4-ethyl-2-methoxyphenol), 157
 - 4-methylguaiacol (2-methoxy-4-methylphenol), 155
 - Brettanomyces* taint, 156–7
 - membrane separation and adsorption, 156
 - smoke taint, 154–6
- volatile acidity removal, 146–8
 - permeate is separated by reverse osmosis stage, 147
 - pKa values for common wine acids, 147
- wine fermentation
 - ageing of red wines on lees, 172–6
 - effects of phenols and anthocyanins, 172–4
 - mannoproteins and tannins interactions, 174–5
 - redox phenomena, 175–6
 - ageing of sparkling wines, 176–8
 - aroma, 176–7
 - foaming properties, 177–8
 - ageing of white wines on lees, 170–2
 - biogenic amines and lees, 172

- lees interactions with wine aroma compounds, 171–2
- nitrogenous substances and polysaccharides, 170
- protein and tartrate stabilisation, 172
- volatile compounds, 170–1
- ageing on lees and use of speciality inactive yeasts, 164–82
- lees definition and composition, 164–6
 - influence of decanting duration on composition, 165
 - lees intervention in red winemaking, 165
 - lees intervention in sparkling winemaking, 166
- undesirable compounds removal from wine, 178–80
 - ochratoxin A, 179
 - phenolic compounds and browning pigments, 180
 - thiols, 179–80
 - volatile phenols, 179
- yeast autolysis, 166–70
 - compounds released, 169–70
 - compounds released by yeast, 169
 - covalently linked proteins in yeast cell wall, 168
 - mechanisms, 167–9
 - structural and ultrastructural changes in yeast cell, 167
- yeast specialities mimicking lees, 180–2
 - inactive yeasts, 181
 - yeast cell walls, 182
 - yeast extracts, 181
 - yeast mannoproteins, 181–2
- wine proteins, 189–90, 193–7
 - characterisation, 191–3
 - glycosylation, 192–3
 - isoelectric point, 191–2
 - molecular weight, 191
 - stability, 193
 - haze formation, 197–8
 - levels in white wines, 193–7
 - effect of grape fungal infection, 195–6
 - mechanical harvesting, 196–7
 - protein concentration and berry development, 194
 - quantification, 193–4
 - role of water stress, 196
 - origin, 190–1
- wine quality
 - alternative oak treatments, 246–9
 - oak chips size, extraction rates and staves, 248–9
 - production, 246–8
 - basic oxidation reactions and substrates, 227–8
 - basic phenolic reactions involving oxygen, 228–9
 - effects of malolactic fermentation, 60–89
 - control by malolactic starters, 85–8
 - future trends, 88–9
 - improving wine quality and lactic acid bacteria, 74–9
 - lactic acid bacteria species variations, 71–4
 - lactic acid bacteria spontaneous growth, 61–71
 - wine spoilage and lactic acid bacteria, 79–85
 - enzymes, 93–127
 - advances in enzyme discovery, 104–9
 - definitions and production methods, 94–8
 - enzyme applications in winemaking, 99–104
 - enzyme use in post-fermentation stages, 118–24
 - enzyme use in pre-fermentation stages, 109–18
 - future trends, 127
 - monitoring enzyme performance, 124–6
 - regulatory aspects, 98–9
 - exogenous tannins in winemaking, 249–51
 - commercial tannins addition effect, 250
 - commercial tannins compositions, 249
 - future trends, 251
 - metabolic engineering of wine yeast and advances in yeast selection methods, 34–52
 - biotechnology, systems biology and generation of new yeast strains, 44–7
 - future trends, 51–2
 - improving wine yeasts, 37–40
 - molecular biology and systems biology in wine yeasts identification, 47–51
 - systems biology approach to wine yeast studies, 41–4

- micro-oxygenation, 230–46
 - chemical composition effect, 231–4
 - microbiology effect, 234–5
 - recent research, 235–43
- micro-oxygenation, oak alternatives
 - and added tannins, 226–51
- oxygen contact with wine, 229–30
- yeast fermentation management, 3–29
 - difficult fermentation
 - conditions, 19–21
 - nutrient strategies, 11–17
 - preventing stuck and sluggish fermentations, 23–7
 - restarting stuck and sluggish fermentations, 27–9
 - sequential yeast inoculation strategies, 10
 - sulphur compounds and dealing with them, 21–3
 - yeast and fermentation management and wine quality, 4–5
 - yeast inoculation, 8
 - yeast inoculation rate, 8–9
 - yeast inoculation timing, 9–10
 - yeast rehydration and handling, 5–7
 - yeast storage, 10–11
- wine yeast selection
 - conventional genetic strategies, 298–303
 - random mutagenesis, 298
 - yeast breeding, 298–303
- flavour modulation, 293–333
 - metabolic engineering, 296–8
 - mixed cultures as an alternative strategy, 303–4
 - molecular cloning methodology, 297
 - natural yeast isolates selection, 295–6
- future trends
 - selection criteria, 333
 - selection methods, 333
- key issues, 294–5
 - combining fermentation analysis and reducing off-flavour production, 294–5
 - yeast strain tailoring, 295
- varietal aromas from grape precursor
 - biotransformation, 323–32
 - phenolic acids, 323–4
 - red wine aromas derived from
 - inodorous precursors, 332
 - terpenes and their glycoconjugate precursors, 324–7
 - volatile thiols, 327–32
- yeast by-products affecting wine
 - aromas, 304–22
 - acetic acid, 307–9
 - esters, 318–22
 - glycerol, 304–6
 - higher alcohols, 313–18
 - hydrogen sulphide, 309–13
- wine yeasts
 - biotechnology, systems biology and generation of new strains, 44–7
 - breeding and other traditional approaches, 44–5
 - genetic engineering potential, 45–7
 - metabolic engineering, 47
 - current targets, 37–40
 - levels of action for potentially improved yeasts, 39
 - metabolic engineering and advances in selection methods, 34–52
 - commercial strains, 35–6
 - future trends, 51–2
 - molecular biology and systems biology, 47–51
 - hybridisation and karyotyping, 48
 - intraspecies discrimination, 50–1
 - PCR-based interspecies discrimination, 48–50
 - RDN1 locus in *Saccharomyces cerevisiae*, 49
 - systems biology approach to studies, 41–4
 - application of techniques, 43–4
 - different ‘omics’ involved, 42
 - revolutionary approach, 41–3
- wines
 - see also* specific type of wine
 - ageing and flavour deterioration, 459–86
 - ageing, 459–60
 - ageing chemical reactions, 469–73
 - aromatic compounds related to flavour deterioration, 461–9
 - factors affecting ageing
 - process, 473–5
 - sensory changes during storage/ageing, 460–1
 - untypical ageing off-flavour, 476–86
 - biogenic amines and winemaking process, 494–515
 - Brettanomyces/Dekkera* off-flavour, 346–72
 - EPs precursors in various concentrations, 348

- methods to detect, prevent and defeat microbial spoilage faults, 367–72
- other associated defects, 358–61
- other faults associated with microbial spoilage, 361–7
- physiology, 355–8
- related metabolism, 349–52
- taxonomy and phylogenetic relationships with other wine yeast, 352–5
- cork taint reduction, 388–414
 - additional test procedures, 401–2
 - compounds causing musty–mouldy off-flavours, 389–94
 - cork stoppers quality evaluation procedures, 396–7
 - corks and bottles handling and processing, 402–9
 - musty off-flavours reduction, 410–14
 - quality management and control methods for wine corks, 394
 - standard test procedures, 397–401
- extraction technologies and wine quality, 589–625
 - draining and pressing, 621–5
 - extraction factors, 592–6
 - red wines vinification by carbonic maceration, 617–20
 - rosé wine vinification techniques, 598–9
 - traditional red wine vinification techniques, 599–609
 - traditional vinification of red wines with whole berries, 620–1
 - very hot, short maceration applied to red wine vinification, 609–17
 - white wine vinification techniques, 596–8
- ladybug taint, 418–28
 - causal compounds, 421–3
 - future trends, 428
 - multicoloured Asian ladybeetles, 419
 - other Coccinellidae species, 424–5
 - post-harvest prevention and remediation, 425–8
 - quality implications, 419–21
 - threshold and tolerances, 424
- monoterpenes and compositions, 325
- non-enzymatic oxidation, 432–51
 - aroma compounds oxidation, 438–40
 - influence of wine anti-oxidants, 446–50
 - measures of wine oxidation status, 440–2
 - oxygen in wine, 433–4
 - polyphenol oxidation, 434–8
 - red wine oxidation, 444–6
 - white wine oxidation, 442–4
 - volatile compounds in base wine, 569
- wood ageing, 459
- xanthylum pigments, 443
- xylanases, 102
- yeast, 593
 - breeding, 298–303
 - backcross program, 300
 - breeding programs assisted by measuring trait values, 301–2
 - breeding programs assisted by molecular markers, 302–3
 - homo- and heterothallism, 298–9
 - hybridisation, 299, 301
 - by-products affecting wine aromas, 304–22
 - handling, 5–7, 25–6
 - inoculation, 8
 - rate, 8–9
 - sequential strategies, 10
 - timing, 9–10
 - rehydration, 5–7, 25–6
 - selection, 34–52
 - storage, 10–11
- yeast assimilable nitrogen, 11, 12, 14, 321
- yeast autolysis, 166–70
 - compounds released, 169–70
 - mechanisms, 167–9
 - cell wall degradation, 168–9
 - proteolysis, 167–8
 - structural and ultrastructural changes in yeast cell, 167
- yeast extracts, 181
- yeast fermentation
 - difficult conditions, 19–21
 - high potential alcohol, 19–20
 - high sugar, 21
 - tall and narrow tanks, 21
 - under- or over-clarified white juice, 20
 - improved wine quality, 3–29
 - sequential yeast inoculation strategies, 10
 - sulphur compounds, 21–3

- yeast and fermentation management
 - and wine quality, 4–5
- yeast inoculation, 8
- yeast inoculation timing, 9–10
- yeast storage, 10–11
- nutrient strategies, 11–17
 - inorganic nitrogen, 13–14
 - laboratory-scale fermentor, 18
 - minerals, 16–17
 - needed quantity of nitrogen, 14
 - nitrogen, 11–12
 - optimal times to add nitrogen, 14
 - organic nitrogen, 12–13
 - organic vs inorganic nitrogen, 13
 - oxygen, 14–16
 - pilot scale fermentor, 18
 - Saccharomyces cerevisiae*
 - viability, 16
 - vitamins, 17
 - yeast inoculation and nutrient
 - strategy summary, 15
- preventing stuck and sluggish fermentations, 23–7
 - aerobic survival factors, 23–4
 - agrochemical and natural
 - inhibitors, 26–7
 - nutrition, 23
 - oenological yeast selection, 24–5
 - temperature management, 24
 - yeast rehydration and
 - handling, 25–6
- restarting stuck and sluggish fermentations, 27–9
 - encapsulated yeast, 28–9
- yeast inoculation rate, 8–9
 - comparison trial, 9
- yeast rehydration and handling, 5–7
 - rehydrated active dried yeast, 6
 - rehydrated active dried yeast
 - with micronutrients and sterols, 7
- yeast mannoproteins, 181–2
 - see also* mannoproteins
- zero-input regulation, 279
- Zinfandel, 527
- Zork closures, 267
- Zork VI, 267
- Zygosaccharomyces bailii*, 365
- Zygosaccharomyces fermentati*, 365